AGUSTA

A109E ROTORCRAFT FLIGHT MANUAL

ROTORCRAFT FLIGHT MANUAL

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LIST OF REVISED PAGES

Revision No.	Subject	R.A.I. Approved
—	Issue	Letter No. 96/2277/MAE dated 31-5-1996
—	Issue (Supersedes issue dated 31-5-1996)	Letter No. 96/3153/MAE dated 25-7-1996
—	Issue (Supersedes issue dated 25-7-1996)	Letter No. 97/3147/MAE dated 30-7-1997
1	Revised pages A-1/A-2, B-1/B-2, 1-33 and 5-1/5-2. Added Appendices 10 and 11.	Letter No. 97/5288/MAE dated 17-12-1997
2	Revised pages A-1/A-2, B-1, v, vi, 1-ii, 1-iii/1-iv, 1-2, 1-10, 1-11, 1-12, 1-13, 1-16B, 1-20, 1-21, 1-24, 1-29, 1-32, 1-39, 1-40, 2-i/2-ii, 2-3, 2-12, 2-16, 2-18, 2-19, 2-20, 2-21, 2-23, 2-24, 2-25, 2-26, 2-28, 2-29, 2-35, 2-37, 2-38, 2-39, 2-41, 2-42, 2-46, 3-i, 3-17, 3-22, 3-23, 3-24, 3-25, 3-26, 3-27, 3-28, 3-29, 3-30, 3-32, 3-34, 3-35, 3-38, 3-42, 3-47, 3-49, 3-51, 3-54, 3-55/3-56 , 4-1/4-2, 4-3, 4-4, 4-17, 4-18, 5-1, Appendix 2 pages 6 and 7 of 8, Appendix 5 pages 2, 3 and 6 of 18, Appendix 6 pages 4, 5, and 6 of 18. Added pages B-2, B-3/B-4, 2-20A/2-20B, 2-26A/2-26B, 2-42A/2-42B, 3-26A/3-26B, 3-30A, 3-30B, 3-30C/3-30D, 5-2 and Appendices 12 and 13.	Letter No. 98/401/MAE dated 29-1-1998
3	Revised pages A-1/A-2, B-1, B-2, B-3/B-4 and 5-2. Added Appendix 14.	Letter No. 98/809/MAE dated 19-2-1998
4	Revised pages A-1/A-2, B-1, B-2, B-3/B-4, 5-2 and Appendix 3 page 1, 3 and 4 of 4. Added Appendix 15 and Appendix 16.	Letter No. 98/1385/MAE dated 20-3-1998

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5	Revised pages A-1, B-1, B-2, B-3/B-4, 5-2, Appendix 7 page 4 of 5. Added page A-2, Appendix 17, 18 and 19.	Letter No. 98/2201/MAE dated 8-5-1998
6	Revised pages A-2, B-1, B-2, B-3/B-4, vii/viii and Appendix 15 page 4 of 11.	Letter No. 98/2576/MAE dated 29-5-1998
7	Revised pages A-2, B-1, B-2, B-3/B-4, vi, 5-2, Appendix 15 page 1, 2, 4 and 5 of 11. Added Appendix 20.	Letter No. 98/5857/MAE dated 26-11-1998
8	Revised pages A-2, B-1, B-2, B-3/B-4 and 5-2. Added Appendix 21.	Letter No. 98/6164/MAE dated 17-12-1998
9	Revised paes A-2, B-1, B-2, B-3/B-4, 1-42, 4-ii, 4-32, 4-33, 4-34 and 5-2. Added pages 4-35, 4-36 and Appendix 22.	Letter No. 99/1254/MAE dated 1-4-1999
10	Revised pages A-2, B-1, B-2, B-3/B-4, 4-31, 5-2, Appendix 12 pages 11, 18, 20, 24, 26, 31 and 36 of 88, Appendix 16 pages 1 thru 8 of 8 and Appendix 19 pages 1 thru 8 of 8. Added Appendix 23.	Letter No. 99/1608/MAE dated 23-4-1999
11	Revised pages A-2, B-1, B-2, B-3, B-4, v, 1-2, 1-9, 1-13, 1-19, 1-27, 1-29, 1-42, 2-6, 2-7, 2-10, 2-14 thru 2-20, 2-20A/2-20B, 2-23 thru 2-25, 2-28, 2-29, 2-33, 2-42, 2-46, 3-11, 3-13, 3-14, 3-17, 3-18. 3-23 thru 3-61/3-62, 4-18, 4-19, Appendix 1 pages 3 thru 5 of 5, Appendix 2 pages 3 and 4 of 8, Appendix 3 page 3 of 4, Appendix 8 pages 1 thru 8 of 8, Appendix 11 page 3 of 5, Appendix 12 pages 10 of 88, 12 thru 14 of 88, 17 of 88, 19 and 20 of 88, 22 and 23 of 88, 25 thru 27 of 88, 29 and 30 of 88, 32 thru 35 of 88, 38 of 88, 47 and 48 of 88, 50 thru 58 of 88, 60 thru 66 of 88, 68 thru 70 of 88, 72 thru 76 of 88, 78 thru 82 of 88, 84 thru 88 of 88, Appendix 13 pages 1 of 12, 6 of 12, 8 of 12, Appendix 15 pages 4 and 5 of 11, 7 of 11. Added page 1-8C/1-8D and 1-43/1-44.	Letter No. 99/2997/MAE dated 26-7-1999

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12	Revised pages A-2, B-1 thru B-4, 1-42, 4-32 thru 4-34, Appendix 10 page 1 to 9 of 9. Added pages A-3/A-4. Deleted pages 4-35 and 4-36.	Letter No. 99-8181-TMI/C dated 27-7-1999
13	Revised pages A-3/A-4, B-1, B-2, B-3, B-4, vi, 3-i, 3-ii, 3-6, 3-8, 3-9, 3-10, 3-11, 3-45, 3-46, 3-47, 3-48 and 5-2. Added pages 3-8A/3-8B, 3-48A thru 3-48L/3-48M and Appendix 24.	Letter No. 99/2951/MAE dated 30-7-1999
14	Revised pages A-3/A-4, B-1 thru B-4, 2-16, 3-48C, 3-48J, 3-48K, 4-i, 5-2, Appendix 10 pages 1 thru 11 of 11, Appendix 15 page 1 of 11, Appendix 19 pages 1 thru 9 of 9, Appendix 21 page 3 of 12 and Appendix 23 page 5 of 30.	Letter No. 99/3917/MAE dated 25-10-1999
15	Revised pages A-3/A-4, B-1 thru B-4, Appendix 10 pages 1 thru 12 of 12, Appendix 19 page 4 of 9, Appendix 21 page 3 of 12.	Letter No. 99/4350/MAE dated 1-12-1999
16	Revised pages A-3/A-4, B-1 thru B-4, 3-i, 3-ii, 3-6, 3-8, 3-8A/3-8B, 3-11, 3-12, 3-14, 3-45 thru 3-48, 3-48A thru 3-48U/3-48V. Added pages 3-12A/3-12B	Letter No. 99/4798/MAE dated 28-12-1999
17	Revised pages A-3/A-4, B-1 thru B-4, 5-1 and 5-2. Added Appendix 25, 26 and 27.	Letter No. 99/4800/MAE dated 28-12-1999
18	Revised pages A-3/A-4, B-1 thru B-4, 1-16B, 2-17, 2-22, 2-24, 2-25, 2-27, 2-28, 3-24, 3-25, 3-26, 3-28 and 5-2. Added pages B-5/B-6, 5-3/5-4 and Appendix 28 and 29.	Letter No. 00/504/MAE dated 16-2-2000
19	Revised pages A-3/A-4, B-1 thru B-5, 2-21, 2-25, 3-48A, 3-48B, 3-48C, 3-48D, 3-48E, 3-48J, 3-48M, 5-2 and 5-3, Appendix 15 page 1 of 11, Appendix 29 pages 4 and 5 of 15. Added Appendix 30.	Letter No. 00/851/MAE dated 16-3-2000

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20	Revised pages A-3, A-4, B-1 thru B-5/B-6 and 5-3/5-4. Added Appendix 31.	Letter No. 00-5638-TMI/C dated 31 March 2000
21	Revised pages A-4, B-1 thru B-5/B-6, 1-42, 2-14, 2-24, 4-6 and 5-3/5-4. Added Appendix 32.	Letter No. 00-6654-TMI/C dated 24 May 2000
22	Revised pages A-4, B-1 thru B-4, 2-16, 2-28, 3-11, 3-48K, 3-48M, 3-48N, 3-48Q, 3-48R, 3-48T, 4-ii and 5-3. Added pages 2-16A/2-16B, 5-4 and 5-5/5-6. Added Appendix 33 and 34.	Letter No. 00/2775/MAE dated 27 September 2000
23	Revised pages A-4, B-1 thru B-6, 1-i, 1-3, 1-4, 1-5, 1-9, 1-41, 3-24, 4-1/4-2, 4-11, 4-12, 4-14, 4-15, 4-17, 4-18, 4-20 thru 4-31, Appendix 5 pages 9 thru 18 of 18, Appendix 6 pages 9 thru 18 of 18, Appendix 11 page 2 and 3 of 5, Appendix 24 page 11 and 12 of 28, Appendix 28 page 2 and 8 of 108, Appendix 33 pages 25 thru 32 of 34, Appendix 34 pages 2, 4 thru 10 of 14. Added Appendix 35.	Letter No. 171035/SPA dated 11 July 2001
24	Revised pages A-4, B-1 thru B-6, 5-3 and 5-5/5-6, Appendix 21 page 3 of 12.	Letter No. 171100/SPA dated 7 August 2001
25	Revised pages A-4, B-1 thru B-6, 5-3 thru 5-5/5-6, Appendix 8 pages 2 and 3 of 8. Added Appendix 36.	Letter No. 171300/SPA dated 30 November 2001
26	Revised pages A-4, B-1 thru B-6, 5-3 and 5-5/5-6. Added Appendix 37.	Letter No. 171350/SPA dated 19 December 2001
27	Revised pages A-4, B-1, B-3 and B-5, 5-3 and 5-5/5-6. Added Appendix 38.	Letter No. 171301/SPA dated 30 November 2001
28	Revised pages A-4, B-1 thru B-6, Appendix 25 pages 1 thru 8 of 8 and Appendix 37 pages 1 thru 4 of 4.	Letter No.02/171178/SPA dated 5 April 2002

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Revision No.	Subject	E.N.A.C. Approved
29	Revised pages B-1 thru B-6, Appendix 25 pages 1, 2 and 4 thru 8 of 8. Added pages A-5/A-6.	Letter No.02/171289/SPA dated 22 May 2002
30	Revised pages A-5/A-6, B-1 thru B-6, 2-11 thru 2-14, 2-45, 3-36, 3-37 and Appendix 4 pages 1 thru 5 of 5. Added page 2-14A/2-14B.	Letter No.02/171525/SPA dated 23 September 2002
31	Revised pages A-5/A-6, B-1, B-3, B-5, B-6, 5-3 thru 5-5/5-6, Appendix 31 page 1 of 8. Added Appendix 39.	Letter No.02/171602/SPA dated 12 November 2002
32	Revised pages A-5/A-6, B-1 thru B-6, 5-3 thru 5-5/5-6, Appendix 7 page 1 thru 4 of 5 and Appendix 17 pages 2, 3 and 6 of 7. Added Appendix 40.	Letter No. 03/171104/SPA dated 11 March 2003
33	Revised pages A-5/A-6, B-1 thru B-6, 2-12, 5-2 and Appendix 17 pages 1 thru 10 of 10.	Letter No. 03/171105/SPA dated 11 March 2003
34	Revised pages A-5/A-6, B-1 thru B-6, 5-3 thru 5-5 and Appendix 21 pages 3, 4, 11 and 12 of 12. Added page 5-6, Appendix 41 and 42.	Letter No. 03/171177/SPA dated 2 May 2003
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35	Revised pages A-5, B-1 thru B-6, 1-15, 1-16A, 1-16B, 2-32, 2-39, 3-ii, 3-10, 3-11, 3-48D, 3-48E, 3-48M, 3-48N, 3-48P, 3-48Q, 3-48R, 3-48S, 3-48T, 3-48U, 3-48V, 3-49, 3-50, 5-3, 5-4, 5-5, 5-6, Appendix 12 (pages 8, 9, 10, 37, 50 and 51 of 88), Appendix 15 (pages 1 thru 9 of 9), Appendix 21 (pages 3 and 10 of 12), Appendix 41 (pages 2 thru 5 of 7), Appendix 42 (pages 1 thru 14 of 14). Added Appendix 43 (pages 1 thru 10 of 10) and Appendix 44 (Pages 1 thru 5 of 5).	Approval N° 2004-1960 dated 3 March 2004

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Revision No.	Subject	E.A.S.A. Approved
36	Revised pages A-6, B-1 thru B-6 and Appendix 10.	Approval N° 2004-5763 dated 2 June 2004
37	Revised pages A-6, B-1 thru B-6, 5-6, Appendix 7 pages 2 thru 5 of 5, Appendix 10 pages 2, 4, 5, 11, 13 and 14 of 15, Appendix 42 pages 1 and 2 of 14. Added Appendix 42 pages 6A/6B of 14.	Approval N° 2004-7476 dated 14 July 2004
38	Revised pages A-6, B-1 thru B-6, 5-3 thru 5-5. Added Appendix 45.	Approval N° 2005-2351 dated 14 March 2005
39	Revised pages A-6, B-1 thru B-6, 1-9, 1-10, 1-16B, 1-38, 1-41, 2-35, 2-46, 3-i, 3-18. Appendix 12 pages 2 thru 6, 15 thru 19, 26, 29 thru 32, 35, 48, 61, 63, 64, 68, 70 of 88. Appendix 24 pages 3, 11, 12, 15 thru 17, 24 thru 28 of 28. Appendix 35 page 3 of 5. Added pages B-7/(B-8 blank), 1-16C/(1-16D blank), Appendix 12 pages 18A, 18B, 34A, 34B, 62A, 62B, 70A/(70B blank) of 88, Appendix 24 pages 26A/(26B blank) of 28.	Approval N° 2005-2994 dated 11 April 2005
40	Revised pages A-6, B-1 thru B-7, Appendix 45 (page 7 of 90).	Approval N° 2005-3994 dated 3 May 2005
41	Revised pages A-6, B-1 thru B-7, v, vi, 2-12, 2-13, 2-14, 2-14A, 2-14B, 5-6, Appendix 10 (pages 3, 5, 7, 8, 9 of 15), Appendix 41 (pages 1 thru 8 of 8).	Approval N° 2005-5823 dated 8 June 2005
42	Revised pages A-6, B-1 thru B-7, 1-iii, 1-15, 2-i, 2-ii, 5-3 thru 5-5, Appendix 15 (pages 1 thru 16 of 16), Appendix 24 (pages 4 thru 7 of 28), Appendix 33 (pages 25 thru 32 of 34), Appendix 38 (pages 2, 3 of 3), Appendix 42 (pages 1 thru 16 of 16), Appendix 45 (pages 4, 14, 19, 45 thru 53 of 90). Added pages 1-34A/1-34B, 2-47/2-48, Appendix 24 (pages 12A thru 12D of 28) and Appendix 46.	Approval EASA. R.C.01098 and EASA. R.C.01099 dated 3 August 2005

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43	Revised pages B-1 thru B-7, 5-6, Appendix 10 (pages 4, 11, 12 of 15), Appendix 31 (pages 3, 7 of 8), Appendix 41 (pages 2, 7 of 8). Added pages A-7/(A-8 blank).	Approved under the Authority of DOA N° E.A.S.A. 21J.005 dated 12 August 2005
44	Revised pages A-7, B-1 thru B-7, 1-9, 5-1 thru 5-6, Appendix 5 (pages 1 thru 23 of 23), Appendix 12 (page 11 of 88). Added pages 5-7 and 5-8, Appendix 47, Appendix 48.	Approval EASA. R.C.01065, EASA. R.C.01252 dated 27 September 2005 and EASA. R.C.01251 dated 28 September 2005
45	Revised pages A-7, B-1 thru B-7/B-8, 1-9, 5-8, Appendix 46 (page 37 of 241).	Approval EASA. R.C.01307 dated 30 January 2006
46	Revised pages A-7, B-1 thru B-7, 5-8, Appendix 12 (pages 34A, 34B, 62A, 70 and 70A of 88), Appendix 46 (pages 1 and 16 of 241).	Approved under the Authority of DOA N° E.A.S.A. 21J.005 dated 29 May 2006
47	Revised pages A-7, B-1 thru B-7 and Appendix 44 (pages 1, 3 thru 5 of 5).	Approval EASA.R.C.01863 dated 18 December 2006
48	Revised pages A-7, B-1 thru B-7 and 5-8.	Approval EASA.R.C.01944 dated 18 December 2006
49	Revised pages A-7, B-1 thru B-7, v, vi, 3-7, 3-8 and 3-8A.	Approval EASA.R.C.01862 dated 9 February 2007
50	Revised pages A-7, B-1 thru B-7 and 5-8.	Approval EASA.R.A.01218 dated 12 February 2007
51	Revised pages A-7, B-1 thru B-7, 5-3 and Appendix 48 (page 1 of 3).	Approval EASA.R.A.01273 dated 22 March 2007
52	Revised pages A-7, B-1 thru B-7, 5-3, 5-6, 5-8. Added Appendix 49.	Approval EASA.R.C.02434, EASA.R.C.02435, EASA.R.C.02436 and EASA.R.C.02437 dated 25 September 2007
53	Revised pages A-7, B-1 thru B-7 and 5-8.	Approval EASA.R.C.02268 dated 22 November 2007

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Revision No.	Subject	E.A.S.A. Approved
54	Revised pages A-8, B-1 thru B-7, Appendix 2 (pages 1 thru 9 of 9).	Approval EASA.R.A.01446 dated 3 March 2008
55	Revised pages A-8, B-1 thru B-7 and 5-8.	Approval EASA.R.C.02849, EASA.R.C.02850 and EASA.R.C.02851 dated 26 May 2008
56	Revised pages A-8, B-1 thru B-7, 5-3, 5-5 and 5-6. Added Appendix 50.	Approval EASA.R.C.03167, dated 05 December 2008

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TABLE OF CONTENTS

INTRODUCTION

GENERAL

It is responsibility of the flight crew to be familiar with the contents of the RFM including all revisions and any temporary revision and supplement which are applicable at the time of flight.
The contents of this copy shall correspond with the revisions which are shown on the Log of Pages.

TERMINOLOGY

WARNINGS, CAUTIONS AND NOTES

Warnings, Cautions and Notes are used throughout this manual to emphasize important and critical instructions and are used as follows:

WARNING

An operating procedure, practice, etc., which, if not correctly followed, could result in personal injury or loss of life.

CAUTION

An operating procedure , practice, etc., which, if not strictly observed, could result in damage to, or destruction of, equipment.

NOTE

An operating procedure, condition, etc., which is essential to highlight.

USE OF PROCEDURAL WORDS

The concept of procedural word usage and intended meaning which has been adhered to in preparing this RFM is as follows:
″**Shall**″ or ″**Must**″ have been used only when application of a procedure is mandatory.
″**Should**″ has been used only when application of a procedure is recom-mmended.
″**May**″ has been used only when application of a procedure is optional.
″**Will**″ has been used only to indicate futurity, never to indicate a mandatory procedure.
″**Condition**″ has been used to determine if the item under examination presents external damage which could jeopardize its safe operation.
″**Secured**″ has been used to determine if the item under examination is correctly locked; mainly referred to doors and disconnectable items.
″**Security**″ has been used to determine if the item under examination is correctly positioned and installed.

ABBREVIATION

The use of capitol letters in the text, apart from normal grammatical usage indicates the actual wording of marking of indicators, controls or control positions on the helicopter.
Abbreviations and acronyms used throughtout this RFM are defined as follows:

– a.c.	:	Alternating current
– AEO	:	All Engines Operative
– A.F.C.S.	:	Automatic Flight Control System
– A/F	:	Airframe
– AGL	:	Above Ground Level
– ALS	:	Ambient Light Sensor
– AWG	:	Aural Warning Generator

– CAS	:	Calibrated Airspeed
– CVDR	:	Cockpit Voice Data Recorder
– CVR	:	Cockpit Voice Recorder
– DAU	:	Data Acquisition Unit
– d.c.	:	Direct current
– EAPS	:	Engine Air Particle Separator
– ECU	:	Engine Control Unit
– EDU	:	Electronic Display Unit
– FCU	:	Fuel Computer Unit
– FDR	:	Flight Data Recorder
– Hd	:	Density altitude
– Hp	:	Pressure altitude
– IAS	:	Indicated Air Speed
– IDS	:	Integrated Display System
– IFR	:	Instrumental on Flight Rules
– ISA	:	International Air Atmosphere
– LCF	:	Low Cycle Fatigue
– LH	:	Left hand
– MCL	:	Master Caution Light
– MPOG	:	Minimum Pitch on Ground
– MFR	:	Manufacturer
– MWL	:	Master Warning Light
– N1	:	Gas generator turbine speed
– N2	:	Power turbine speed
– NR	:	Rotor speed

– OAT : Outside Air Temperature

– OEI : One Engine Inoperative

– PAX : Passenger(s)

– PLA : Power Lever Angle (alias Engine Power Lever)

– PMS : Power Management Switch (alias engine mode switch)

– RFM : Rotorcraft Flight Manual

– RH : Right hand

– RPM : Revolutions per minute

– TAS : True Airspeed

– TRQ : Torque

– TOT : Turbine Outlet Temperature

– VFR : Visual Flight Rules

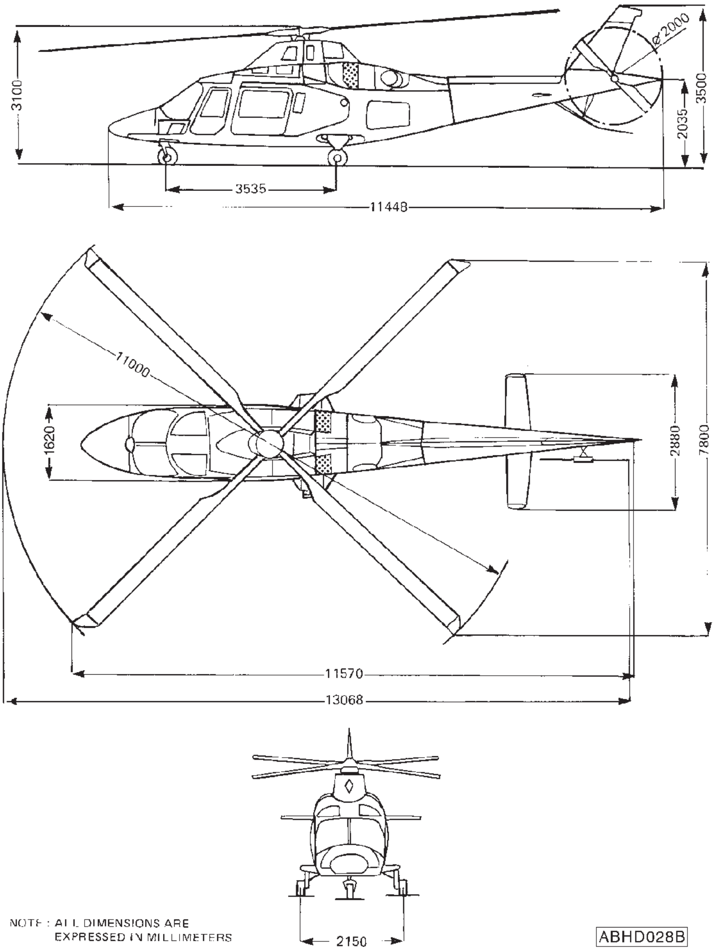
– V_{NE} : Never Exceed Speed

– V_{LO} : Maximum landing gear operating speed

– V_{LE} : Maximum landing gear extended speed.

– V_{MO} : Maximum operating limit speed.

– WOW : Weight on Wheels



Helicopter - Three Views.

LIST OF REVISED PAGES

Revision No.	Subject	F.A.A. Approved
—	Issue	26-8-1996
—	Issue (Supersedes issue dated 26-8-1996)	30-7-1997
1	Revised pages A-1/A-2, B-1, B-2, 1-33 and 5-1/5-2. Added Appendices 10 and 11.	21-5-1998
2	Revised pages A-1/A-2, B-1, B-2, v, vi, 1-ii, 1-iii/1-iv, 1-2, 1-10, 1-11, 1-12, 1-13, 1-16B, 1-20, 1-21, 1-24, 1-29, 1-32, 1-39, 1-40, 2-i/2-ii, 2-3, 2-12, 2-16, 2-18, 2-19, 2-20, 2-21, 2-23, 2-24, 2-25, 2-26, 2-28, 2-29, 2-35, 2-37, 2-38, 2-39, 2-41, 2-42, 2-46, 3-i, 3-17, 3-22, 3-23, 3-24, 3-25, 3-26, 3-27, 3-28, 3-29, 3-30, 3-32, 3-34, 3-35, 3-38, 3-42, 3-47, 3-49, 3-51, 3-54, 3-55/3-56, 4-1/4-2, 4-3, 4-4, 4-17, 4-18, 5-1, Appendix 2 pages 6 and 7 of 8, Appendix 5 pages 2, 3 and 6 of 18, Appendix 6 pages 4, 5, and 6 of 18. Added pages B-3/B-4, 2-20A/2-20B, 2-26A/2-26B, 2-42A/2-42B, 3-26A/3-26B, 3-30A, 3-30B, 3-30C/3-30D, 5-2 and Appendix 13.	21-5-1998
3	Revised pages A-1/A-2, B-1, B-2, B-3/B-4 and 5-2. Added Appendix 14.	21-5-1998
4	Revised pages A-1/A-2, B-1, B-2, B-3/B-4 and 5-2, Appendix 3 page 1, 3 and 4 of 4. Added Appendices 15 and 16.	21-5-1998
5	Revised pages A-1, B-1, B-2, B-3/B-4, 5-2, Appendix 7 page 4 of 5. Added Appendices 17, 18 and 19	15-7-1998

Revision No.	Subject	F.A.A. Approved
6	Revised pages B-1, B-2, B-3/B-4, vii/viii and Appendix 15 page 4 of 11. Added page A-2.	15-7-1998
7	Revised pages A-2, B-1, B-2, B-3/B-4, vi, 5-2, Appendix 15 page 1, 2, 4 and 5 of 11. Added Appendix 20.	13-1-1999
8	Revised pages A-2, B-1, B-2, B-3/B-4 and 5-2. Added Appendix 21 and approved Appendix 12.	13-1-1999
9	Revised pages A-2, B-1, B-2, B-3/B-4, 1-42, 4-ii, 4-32, 4-33, 4-34, 4-35, 4-36 and 5-2. Added page 4-37/4-38 and Appendix 22.	10-9-1999
10	Revised pages A-2, B-1, B-2, B-3, B-4, 4-31, 5-2, Appendix 12 page 11, 18, 20, 24, 26, 31 and 36 of 88, Appendix 16 pages 1 thru 8 of 8 and Appendix 19 pages 1 thru 8 of 8. Added Appendix 23.	10-9-1999
11	Revised pages A-2, B1, B-2, B-3, B-4, v, 1-2, 1-9, 1-13, 1-19, 1-27, 1-29, 1-42, 2-6, 2-7, 2-10, 2-14 thru 2-20, 2-20A/2-20B, 2-23 thru 2-25, 2-28, 2-29, 2-33, 2-42, 2-46, 3-11, 3-13, 3-14, 3-17, 3-18, 3-23 thru 3-61/3-62, 4-18, 4-19, Appendix 1 pages 3 thru 5 of 5, Appendix 2 pages 3 and 4 of 8, Appendix 3 page 3 of 4, Appendix 8 pages 1 thru 8 of 8, Appendix 11 page 3 of 5, Appendix 12 pages 10 of 88, 12 thru 14 of 88, 17 of 88, 19 and 20 of 88, 22 and 23 of 88, 25 thru 27 of 88, 29 and 30 of 88, 32 thru 35 of 88, 38 of 88, 47 and 48 of 88, 50 thru 58 of 88, 60 thru 66 of 88, 68 thru 70 of 88, 72 thru 76 of 88, 78 thru 82 of 88, 84 thru 88 of 88, Appendix 13 pages 1 of 12, 6 of 12, 8 of 12, Appendix 15 pages 4 and 5 of 11, 7 of 11. Added page 1-8C/1-8D and 1-43/1-44.	16-12-1999

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Revision No.	**Subject**	**F.A.A. Approved**
12	Revised pages B-1, B-2, B-3, B-4, 1-42, 4-32 thru 4-34, Appendix 10 page 1 to 9 of 9. Added pages A-3/A-4. Deleted pages 4-35 and 4-36.	16-12-1999
13	Revised pages A-3/A-4, B-1, B-2, B-3, B-4, vi, 3-i, 3-ii, 3-6, 3-8, 3-9, 3-10, 3-11, 3-45, 3-46, 3-47, 3-48 and 5-2 Added pages 3-8A/3-8B, 3-48A thru 3-48L/3-48M and Appendix 24.	16-12-1999
14	Revised pages A-3/A-4, B-1 thru B-4, 2-16, , 3-48C, 3-48J, 3-48K, 4-i, 5-2, Appendix 10 pages 1 thru 11 of 11, Appendix 15 page 1 of 11, Appendix 19 pages 1 thru 9 of 9, Appendix 21 page 3 of 12 and Appendix 23 page 5 of 30.	16-12-1999
15	Revised pages A-3/A-4, B-1 thru B-4, Appendix 10 pages 1 thru 12 of 12 and Appendix 19 page 4 of 9, Appendix 21 page 3 of 12. Added pages B-5/B-6 blank.	16-12-1999
16	Revised pages A-3/A-4, B-1 thru B-4, 3-i, 3-ii, 3-6, 3-8, 3-8A/3-8B, 3-11, 3-12, 3-14, 3-45 thru 3-48, 3-48A thru 3-48U/3-48V. Added pages 3-12A/3-12B.	23-03-2000
17	Revised pages A-3/A-4, B-1 thru B-5/B-6, 5-1 and 5-2. Added Appendix 25, 26, and 27.	23-03-2000
18	Revised pages A-3/A-4, B-1 thru B-5/B-6, 1-16B, 2-17, 2-22, 2-24, 2-25, 2-27, 2-28, 3-24, 3-25, 3-26, 3-28 and 5-2. Added pages 4-35/4-36, 5-3/5-4 and Appendix 28 and 29. Deleted pages 4-37/4-38.	23-03-2000

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19	Revised pages A-3, A-4, B-1 thru B-5/B-6, 2-21, 2-25, 3-48A thru 3-48E, 3-48J, 3-48M, 5-2 and 5-3/5-4, Appendix 15 page 1 of 11, Appendix 29 pages 4 and 5 of 15. Added Appendix 30.	27-03-2000
20	Revised pages A-4, B-1 thru B-5/B-6 and 5-3/5-4. Added Appendix 31.	9-06-2000
21	Revised pages A-4, B-1 thru B-5/B-6, 1-42, 2-14, 2-24, 4-6 and 5-3/5-4. Added Appendix 32.	28-07-2000
22	Revised pages A-4, B-1 thru B-4, 2-16, 2-28, 3-11, 3-48K, 3-48M, 3-48N, 3-48Q, 3-48R, 3-48T, 4-ii and 5-3. Added pages 2-16A/2-16B, 5-4 and 5-5/5-6. Added Appendix 33 and 34.	27-11-2000
23	Revised pages A-4, B-1 thru B-6, 1-i, 1-3, 1-4, 1-5, 1-9, 1-41, 3-24, 4-1/4-2, 4-11, 4-12, 4-14, 4-15, 4-17, 4-18, 4-20 thru 4-31, Appendix 5 pages 9 thru 18 of 18, Appendix 6 pages 9 thru 18 of 18, Appendix 11 pages 2 and 3 of 5, Appendix 24 pages 11 and 12 of 28, Appendix 28 pages 2 and 8 of 108, Appendix 33 pages 25 thru 32 of 34, Appendix 34 pages 2, 4 thru 10 of 14. Added Appendix 35.	3-10-2001
24	Revised pages A-4, B-1 thru B-6, 5-3 and 5-5/5-6, Appendix 21 page 3 of 12.	3-10-2001
25	Revised pages A-4, B-1 thru B-6, 5-3 thru 5-5/5-6, Appendix 8 pages 2 and 3 of 8. Added Appendix 36.	7-03-2002
26	Revised pages A-4, B-1 thru B-6, 5-3 and 5-5/5-6. Added Appendix 37.	7-03-2002

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27	Revised pages B-1 thru B-6, 5-3 and 5-5/5-6. Added page A-5/A-6 and Appendix 38.	7-03-2002
28	Revised pages A-5/A-6, B-1 thru B-6, Appendix 25 pages 1 thru 8 of 8 and Appendix 37 pages 1 thru 4 of 4.	14-06-2002
29	Revised pages A-5/A-6, B-1 thru B-6 and Appendix 25 pages 1, 2 and 4 thru 8 of 8.	30-09-2002
30	Revised pages A-5/A-6, B-1 thru B-6, 2-11 thru 2-14, 2-45, 3-36, 3-37 and Appendix 4 pages 1 thru 5 of 5. Added page 2-14A/2-14B.	23-10-2002
31	Revised pages A-5, B-1, B-3, B-5, B-6, 5-3, 5-5/5-6, Appendix 31 page 1 of 8. Added Appendix 39.	24-03-2003
32	Revised pages A-5, B-1 thru B-6, 5-3 thru 5-5/5-6, Appendix 7 pages 1 thru 4 of 5 and Appendix 17 pages 2, 3 and 6 of 7. Added Appendix 40.	7-11-2003
33	Revised pages A-5, B-1 thru B-6, 2-12, 5-2 and Appendix 17 pages 1 thru 10 of 10.	7-11-2003
34	Revised pages A-5, B-1 thru B-6, 5-3 thru 5-5 and Appendix 21 pages 3, 4, 11 and 12 of 12. Added page 5-6 , Appendix 41 and 42.	7-11-2003
35	Revised pages A-5, B-1 thru B-6, 1-15, 1-16A, 1-16B, 2-32, 2-39, 3-ii, 3-10, 3-11, 3-48D, 3-48E, 3-48M, 3-48N, 3-48P, 3-48Q, 3-48R, 3-48S, 3-48T, 3-48U, 3-48V, 3-49, 3-50, 5-3, 5-4, 5-5, 5-6, Appendix 12 (pages 8, 9, 10, 37, 50 and 51 of 88), Appendix 15 (pages 1 thru 9 of 9), Appendix 21 (pages 3 and 10 of 12), Appendix 41 (pages 2 thru 5 of 7), Appendix 42 (pages 1 thru 14 of 14). Added Appendix 43 (pages 1 thru 10 of 10) and Appendix 44 (Pages 1 thru 5 of 5).	16-04-2004

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Revision No.	Subject	F.A.A. Approved
36	Revised pages A-6, B-1 thru B-6 and Appendix 10.	27-6-2005
37	Revised pages A-6, B-1 thru B-6, 5-6, Appendix 7 pages 2 thru 5 of 5, Appendix 10 pages 2, 4, 5, 11, 13 and 14 of 15, Appendix 42 pages 1 and 2 of 14. Added Appendix 42 pages 6A/6B of 14.	27-6-2005
38	Revised pages A-6, B-1 thru B-6, 5-3 thru 5-5. Added Appendix 45.	27-6-2005
39	Revised pages A-6, B-1 thru B-6, 1-9, 1-10, 1-16B, 1-38, 1-41, 2-35, 2-46, 3-i, 3-18. Appendix 12 pages 2 thru 6, 15 thru 19, 26, 29 thru 32, 35, 48, 61, 63, 64, 68, 70 of 88. Appendix 24 pages 3, 11, 12, 15 thru 17, 24 thru 28 of 28. Appendix 35 page 3 of 5. Added pages B-7/(B-8 blank), 1-16C/(1-16D blank), Appendix 12 pages 18A, 18B, 34A, 34B, 62A, 62B, 70A/(70B blank) of 88, Appendix 24 pages 26A/(26B blank) of 28.	27-6-2005
40	Revised pages A-6, B-1 thru B-7, Appendix 45 (page 7 of 90).	27-6-2005
41	Revised pages A-6, B-1 thru B-7, v, vi, 2-12, 2-13, 2-14, 2-14A, 2-14B, 5-6, Appendix 10 (pages 3, 5, 7, 8, 9 of 15), Appendix 41 (pages 1 thru 8 of 8).	27-6-2005
42	Revised pages A-6, B-1 thru B-7, 1-iii, 1-15, 2-i, 2-ii, 5-3 thru 5-5, Appendix 15 (pages 1 thru 16 of 16), Appendix 24 (pages 4 thru 7 of 28), Appendix 33 (pages 25 thru 32 of 34), Appendix 38 (pages 2, 3 of 3), Appendix 42 (pages 1 thru 16 of 16), Appendix 45 (pages 4, 14, 19, 45 thru 53 of 90). Added pages 1-34A/1-34B, 2-47/2-48, Appendix 24 (pages 12A thru 12D of 28) and Appendix 46.	14-9-2005

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Revision No.	Subject	F.A.A. Approved
43	Revised pages B-1 thru B-7, 5-6, Appendix 10 (pages 4, 11, 12 of 15), Appendix 31 (pages 3, 7 of 8), Appendix 41 (pages 2, 7 of 8). Added pages A-7/(A-8 blank).	Not F.A.A. approved
44	Revised pages A-7, B-1 thru B-7, 1-9, 5-1 thru 5-6, Appendix 5 (pages 1 thru 23 of 23), Appendix 12 (page 11 of 88). Added pages 5-7 and 5-8, Appendix 47, Appendix 48.	17-10-2005
45	Revised pages A-7, B-1 thru B-7/B-8, 1-9, 5-8, Appendix 46 (page 37 of 241)	8-3-2006
46	Revised pages A-7, B-1 thru B-7, 5-8, Appendix 12 (pages 34A, 34B, 62A, 70 and 70A of 88), Appendix 46 (pages 1 and 16 of 241).	26-6-2007
47	Revised pages A-7, B-1 thru B-7 and Appendix 44 (pages 1, 3 thru 5 of 5).	26-6-2007
48	Revised pages A-7, B-1 thru B-7 and 5-8.	26-6-2007
49	Revised pages A-7, B-1 thru B-7, v, vi, 3-7, 3-8 and 3-8A.	5-03-2008
50	Revised pages A-7, B-1 thru B-7 and 5-8.	5-03-2008
51	Revised pages A-7, B-1 thru B-7, 5-3 and Appendix 48 (page 1 of 3).	5-03-2008
52	Revised pages A-7, B-1 thru B-7, 5-3, 5-6, 5-8. Added Appendix 49.	Not F.A.A. Approved
53	Revised pages A-7, B-1 thru B-7 and 5-8.	Not F.A.A. Approved
54	Revised pages A-8, B-1 thru B-7, Appendix 2 (pages 1 thru 9 of 9).	E.A.S.A. Approved on behalf of F.A.A. with Letter EASA D(2008)MMA/LNI/08.3798 dated 14-07-2008
55	Revised pages A-8, B-1 thru B-7 and 5-8.	Not F.A.A. Approved

NOTE: Revised text is indicated by a black vertical line in the outer margin of the page and the approval revision number is printed in the lower margin.

LOG OF PAGES

NOTE: The yellow pages apply to F.A.A. registered A109E helicopters only and supersede the corresponding pages of the Rotorcraft Flight Manual. The holders of the Rotorcraft Flight Manual who are affected shall place the attached pages in front of the corresponding pages of the basic Rotorcraft Flight Manual.

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IFR OPERATION

REQUIRED EQUIPMENT FOR IFR FLIGHT SINGLE PILOT

In addition to that required for VFR day and night flight, the following items must be operational before IFR flight is initiated.

— SAS1, SAS2, ATTD HOLD
— Pilot's flight instruments as follows:
  - Attitude director indicator (ADI)
  - Horizontal situation indicator (HSI) and course deviation indicator (CDI)
  - Attitude indicator - Stand-by
  - Instantaneous vertical speed indicator (I.V.S.I.)
  - Encoding altimeter
— Pilot's and stand-by intercommunication system
— VHF communication system
— VOR/ILS
— ATC Transponder
— Clock

NOTE

Additional equipment may be required for dedicated IFR operations.

Both helipilots (SAS1 and 2), ATTD HOLD mode, and force trim must be engaged for IFR flight.

NOTE

With anyone of the required IFR equipment inoperative, VFR flight is permitted.

NOISE CHARACTERISTICS

The following noise levels comply with ICAO Annex 16, Chapter 8 ''Noise requirements'' and FAR part 36 Appendix H.

Model:A109E	**Engine: Pratt & Whitney 206C**			**Gross Weight: 2850 Kg**		
Configuration	Level flyover EPNL (EPNdB)		Takeoff EPNL (EPNdB)		Approach EPNL (EPNdB)	
Clean aircraft No external kits installed	FAA 90.8	ICAO 90.9	FAA 91.4	ICAO 91.3	FAA 91.4	ICAO 91.4

PART I

R.A.I. Approved

SECTION 1

LIMITATION

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LIST OF ILLUSTRATIONS

List of illustrations (Contd.)

SECTION 1
LIMITATIONS

GENERAL

COMPLIANCE WITH THE OPERATING LIMITATIONS IN SECTION 1 OF THIS MANUAL IS MANDATORY.
THE HELICOPTER MUST ALSO BE OPERATED IN ACCORDANCE WITH THE APPROPRIATE OPERATING RULES.

BASIS OF CERTIFICATION

The helicopter is certified under FAR Part 27 with the exemption to para 27.1(a) for maximum gross weight increase to 2850 kg (6283 lb).

TYPE OF OPERATION

This helicopter, in the basic configuration, is approved for operation under day and night VFR, non-icing conditions.
The IFR configured helicopter is certified for IFR operation during day and night non-icing conditions.
No aerobatic manoeuvres are permitted.

VFR OPERATION

The operation is authorized at night under visual contact flying conditions. The orientation shall be maintained by visual reference to ground objects, solely as a result of lights on the ground or adequate celestial illumination.

FLIGHT WITH DOORS REMOVED

Flight with doors removed is prohibited.

IFR OPERATION

REQUIRED EQUIPMENT FOR IFR FLIGHT SINGLE PILOT

In addition to that required for VFR day and night flight, the following items must be operational before IFR flight is initiated.

— SAS1, SAS2, ATTD HOLD
— Pilot's flight instruments as follows:
  - Attitude director indicator (ADI)
  - Horizontal situation indicator (HSI) and course deviation indicator (CDI)
  - Attitude indicator - Stand-by
  - Instantaneous vertical speed indicator (I.V.S.I.)
  - Encoding altimeter
— Pilot's and stand-by intercommunication system
— Dual communication transceivers (VHF 1 and 2)
— VOR/ILS 1
— VOR/ILS 2 with additional course selector indicator (CSI)
— ADF
— ATC Transponder
— Clock

Both helipilots (SAS1 and 2), ATTD HOLD mode, and force trim must be engaged for IFR flight.

NOTE

With anyone of the required IFR equipment inoperative, VFR flight is permitted.

FLIGHT WITH ONE OF THE TWO HELIPILOTS INOPERATIVE

IFR flight may be initiated to fly to a repair facility providing IFR qualified second pilot occupies the copilot's station and the aircraft is fitted with dual controls and dual flight instruments.

Following one helipilot failure, climb rate must be limited to 500 FPM and cruise flight above 120 Kts requires the pilot to maintain hands on flight controls.

OPTIONAL EQUIPMENT

Refer to Section 5 and to pertinent Appendices for additional limitations, procedures and performance data with optional equipment installed.

FLIGHT CREW

The minimum flight crew consists of one pilot who shall operate the helicopter from the right crew seat.
The left crew seat may be used for an additional pilot when the approved dual controls and copilot's instruments are installed.

NUMBER OF SEATS

Eight (pilot included).

AIRSPEED LIMITATIONS (KIAS)

See Figure 1-1 for Airspeed limitations V_{NE} (Power on) and Figure 1-2 for Airspeed limitations V_{NE} (Power off).

Maximum landing gear operating speed V_{LO}	120 KIAS.
Maximum landing gear extended speed V_{LE}	120 KIAS.
Minimum airspeed in autorotation without close external references	60 KIAS.
Minimum IFR airspeed	50 KIAS.
Minimum speed during IFR approach	55 KIAS.

NOTE

For the steepest demonstrated approach slope see ″Approach Angle Vs Airspeed and Rate of Descent″ of Section 2.

AIRSPEED LIMITATION - VNE (POWER ON)

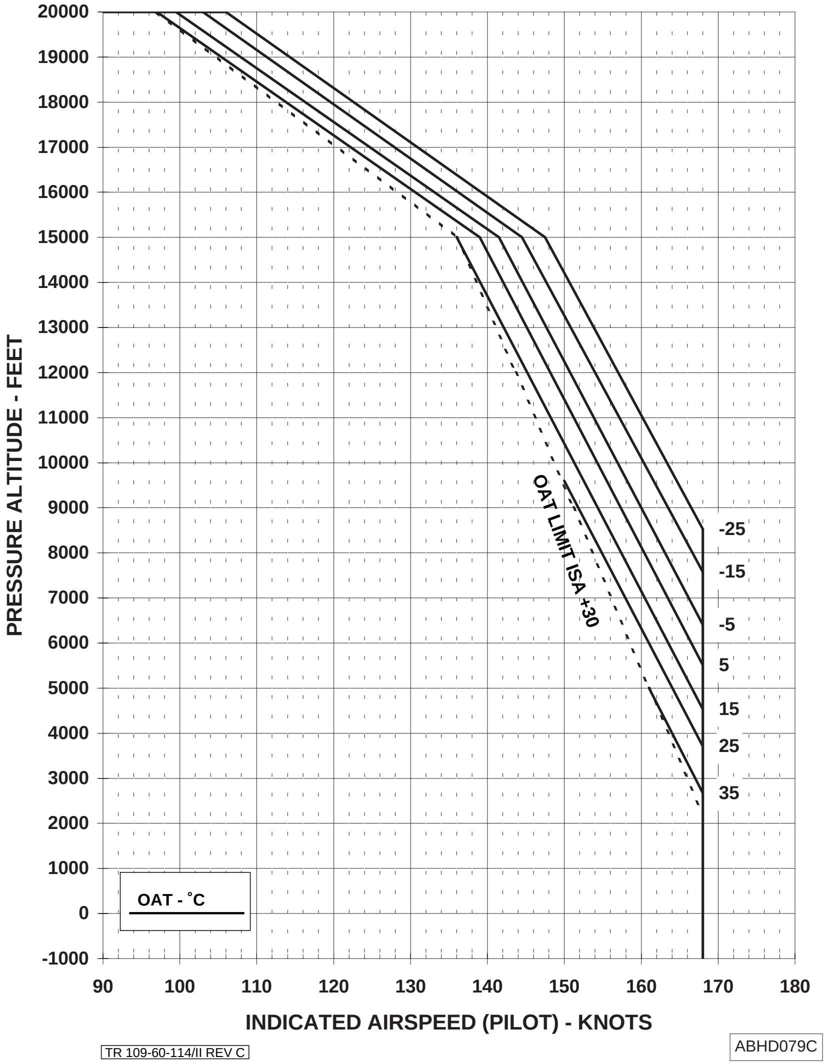


Fig. 1-1. Airspeed Limitations - V_{NE} (Power on).

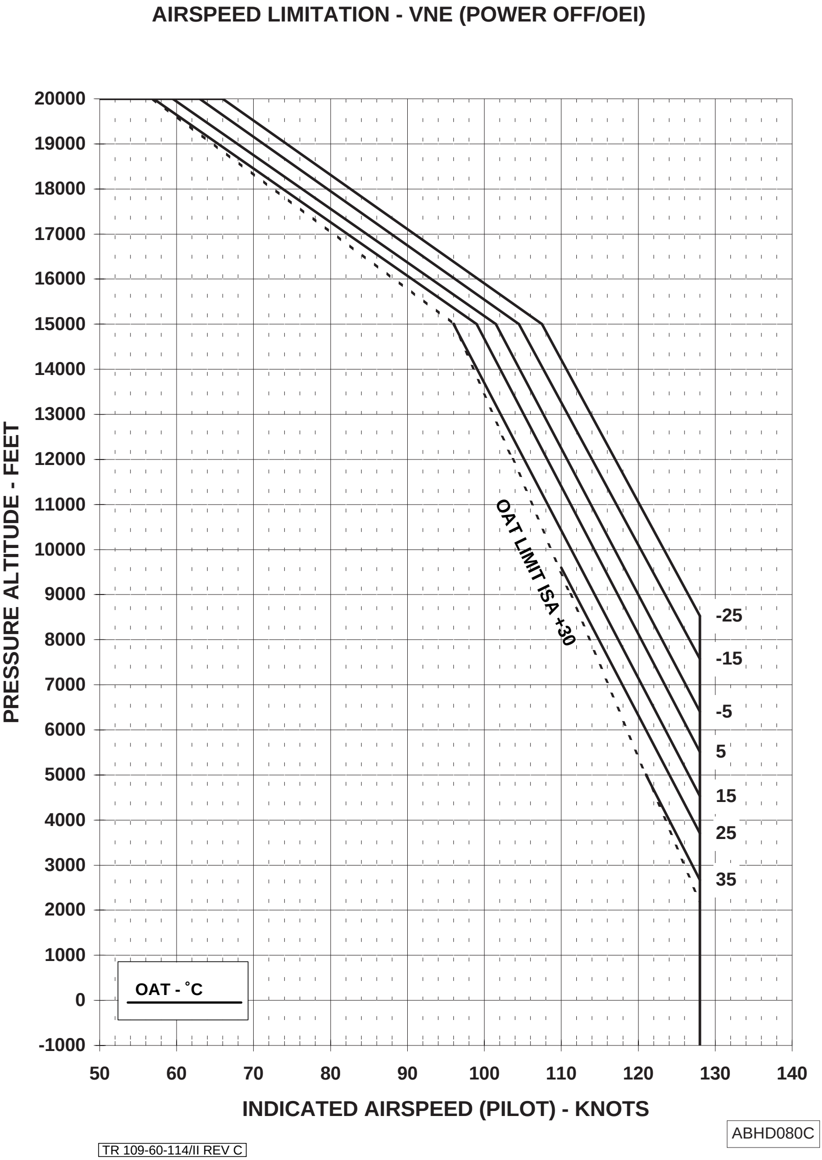


Fig. 1-2. Airspeed Limitations - V_{NE} (Power off/OEI).

GROUND SPEED LIMITATIONS

On concrete even surfaces:

Maximum speed for running takeoff and landing	: 40 Kts
Maximum taxiing speed (nose wheel unlocked)	
- straight	: 20 Kts
- turn	: 10 Kts

On unprepared or uneven surfaces:

Maximum speed for running takeoff and landing	: 20 Kts
Maximum taxiing speed (nose wheel unlocked)	
- straight	: 20 Kts
- turn	: 10 Kts.

WEIGHT LIMITATIONS

Maximum gross weight for takeoff and landing	: 2850 kg (6283 lb)

NOTE

The maximum takeoff and landing weight may be limited by performance data contained in Section 4.

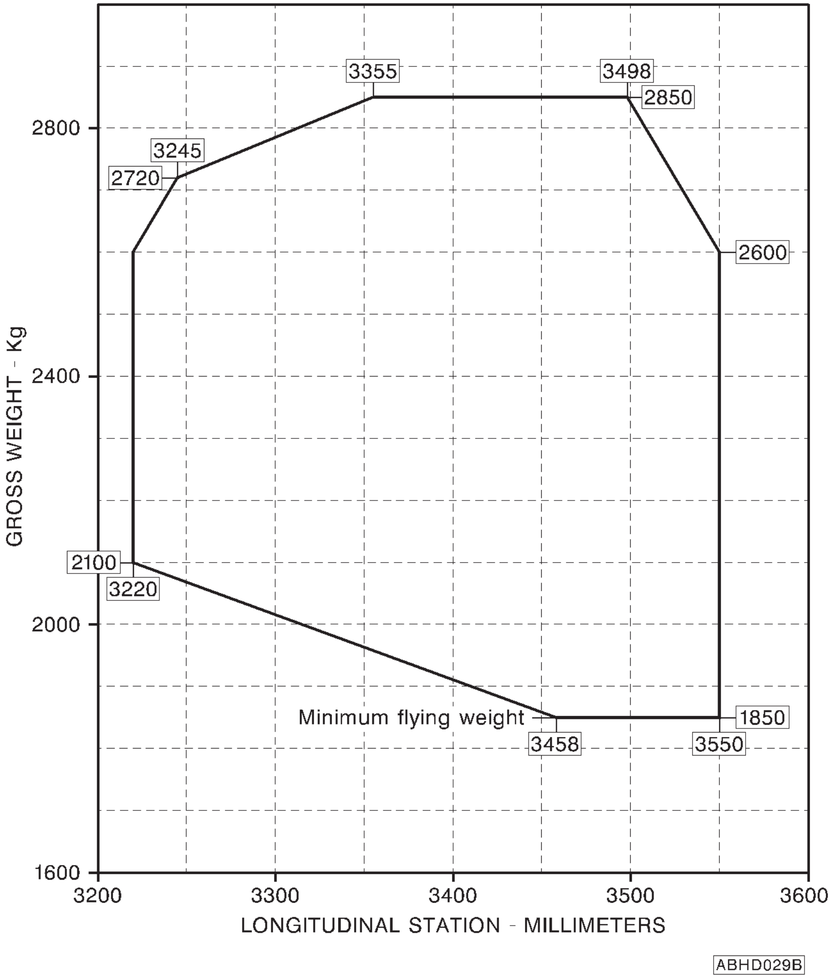
Minimum gross weight for flight	: 1850 kg (4078 lb)

CENTER OF GRAVITY LIMITATIONS

See figure 1-3 for Longitudinal CG limits and figure 1-4 for Lateral CG limits.

NOTE

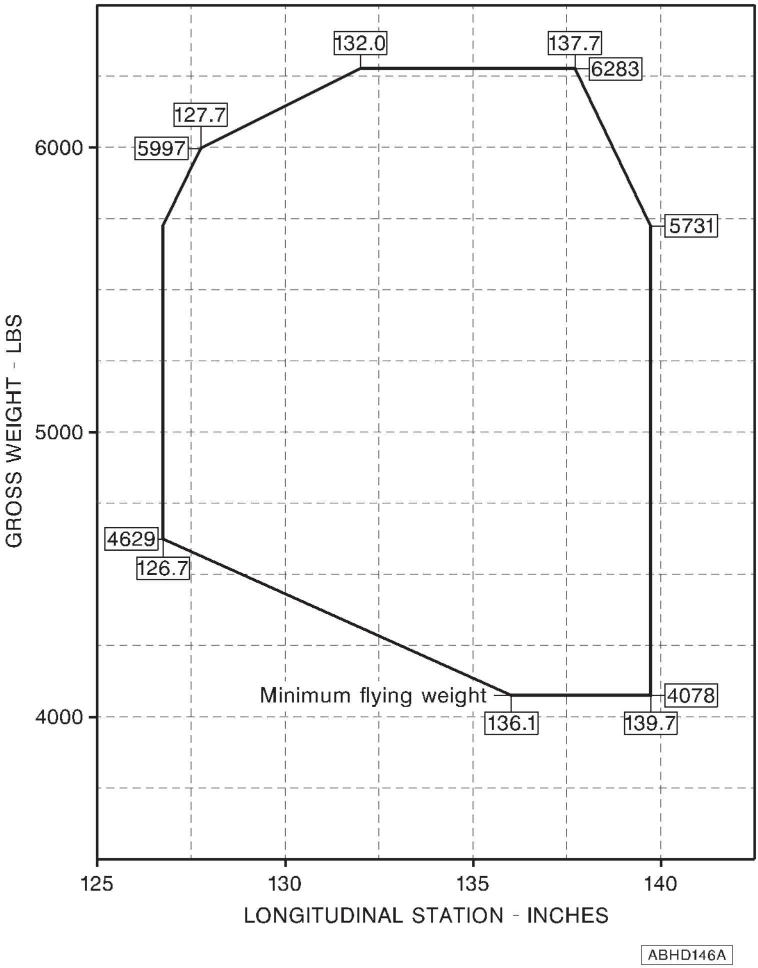
The Center of Gravity limits refer to landing gear extended configuration.
There is no significant CG change with landing gear retracted.



NOTE

Longitudinal Station ″0″ is 1835 mm forward of the front jack point.

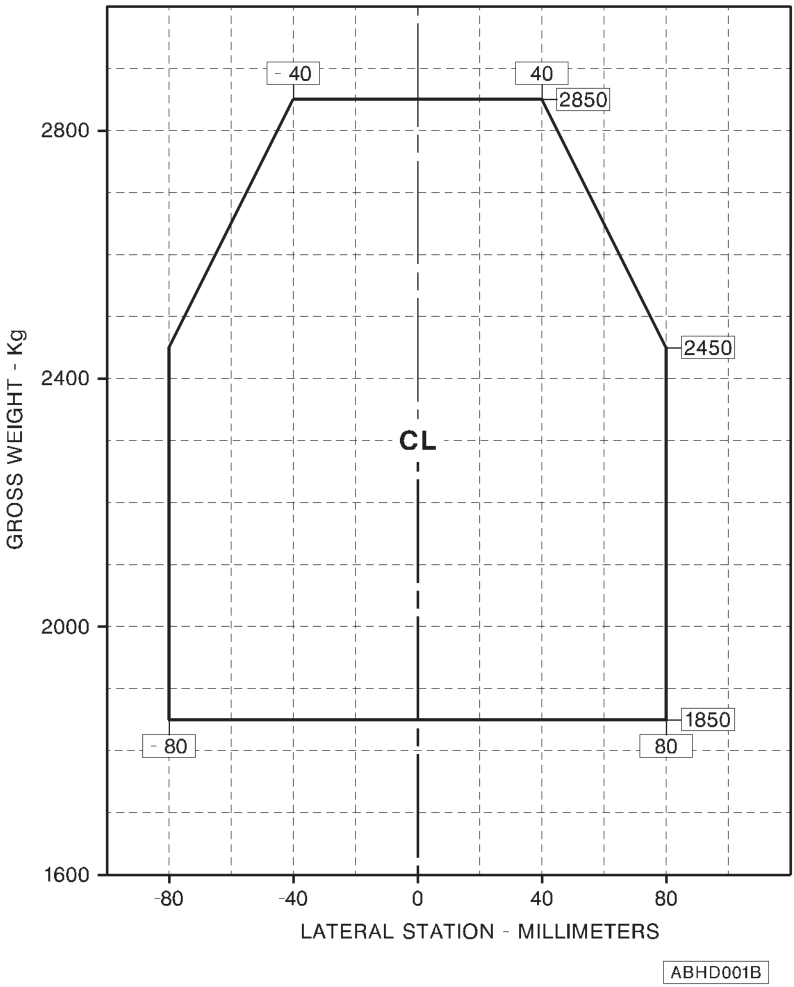
Fig. 1-3 (sheet 1 of 2). Longitudinal CG limits (metric).



NOTE

Longitudinal Station ″0″ is 72.2 in. forward of the front jack point.

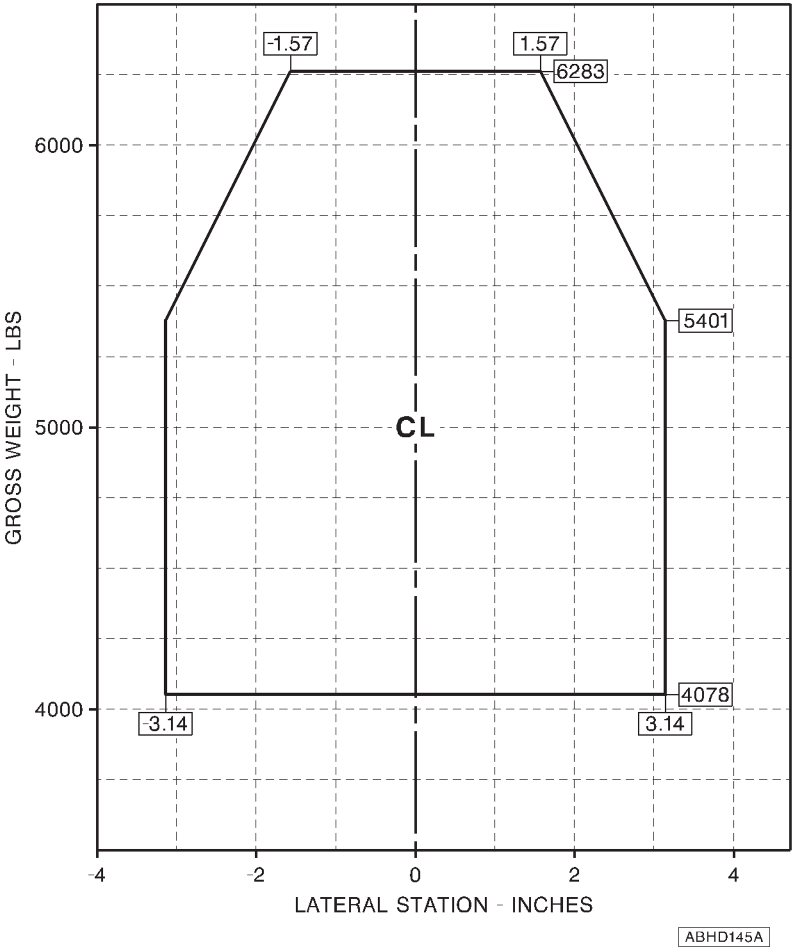
Fig. 1-3 (sheet 2 of 2). Longitudinal CG limits (english).



NOTE

The lateral station ″0″ is 450 mm inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

Fig. 1-4 (sheet 1 of 2). Lateral CG limits (metric).



NOTE

The lateral station ″0″ is 17.7 in. inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

Fig. 1-4 (sheet 2 of 2). Lateral CG limits (english).

BAGGAGE COMPARTMENT LIMITATIONS

Maximum load : 150 kg (330 lb)

NOTE

Refer to Section 6, Weight and Balance, for load distribution.

Maximum unit load : 500 kg/m^2 (102 lbs/sq ft)

ALTITUDE LIMITATIONS

Maximum pressure altitude for takeoff and landing:	: 15000 ft (4572 m)
Maximum operating pressure altitude:	: 20000 ft (6096m)

AMBIENT AIR TEMPERATURE LIMITATIONS

The minimum ambient air temperature for operation is -25°C (-13°F).
The maximum sea level ambient air temperature for operation is +45°C (113°F) and decreases with pressure altitude at the standard lapse rate of 2°C (3.6°F) every 1000 ft (305 m) up to 20000 ft (6096 m).

POWER PLANT LIMITATIONS

The helicopter is powered by two (2) Pratt & Whitney Canada Inc. PW206C turboshaft engine.

GAS GENERATOR (N1) RPM

NOTE

100% N1 corresponds to 58000 RPM.

All engines operative (AEO)

Continuous operation	: 50 to 97.4%
Maximum continuous	: 97.4%
Takeoff range	: 97.4% to 98.7%
Maximum	: 98.7%
Transient (20 sec)	: 103.4%

One engine inoperative (OEI)

Continuous operation	: 50 to 100.4%
Maximum continuous	: 100.4%
2.5 minute range	: 100.4 to 102.4%
Maximum	: 102.4%
Transient (20 sec)	: 103.4%

POWER TURBINE (N2) RPM

NOTE

100% N2 corresponds to 6000 RPM output shaft speed and to 39807 RPM power turbine speed.

All engines operative (AEO)

Minimum	: 99%
Continuous operation	: 99 to 101%
Takeoff and landing	: 101 to 102%
Maximum	: 102%
Transient (20 sec)	: 112.4%

One engine inoperative (OEI)

Minimum	: 90%
Takeoff and landing (real emergency and/or equivalent cat. ″A″ operations only)	: 90 to 99%
Continuous operation	: 99 to 101%
Takeoff and landing	: 101 to 102%
Maximum	: 102%
Transient (20 sec)	: 112.4%

TURBINE OUTLET TEMPERATURE (TOT)

All engines operative (AEO)

Maximum continuous	: 820°C
Takeoff (5 min) range	: 820°C to 863°C
Maximum	: 863°C

One engine inoperative (OEI)

Maximum continuous	: 885°C
2.5 min range	: 885°C to 930°C
Maximum	: 930°C
Transient (20 sec)	: 972°C
Starting (2 sec)	: 875°C

ROTOR LIMITATIONS

NOTE

100% rotor speed corresponds to 384 RPM.

Power On (AEO)

Minimum	: 99%
Continuous operation	: 99 to 101%
Takeoff and landing (below 60 KIAS and 1000 ft AGL)	: 101 to 102%.

NOTE

During an instrumental approach procedure the use of 102% NR is allowed up to 120 KIAS.

Maximum	: 102%

Power On (OEI)

Minimum	: 90%
Takeoff and landing	: 90 to 99%
Continuous operation	: 99 to 101%
Takeoff and landing (below 60 KIAS and 1000 ft AGL)	: 101 to 102%

NOTE

During an instrumental approach procedure the use of 102% NR is allowed up to 120 KIAS.

Maximum	: 102%

Power Off

Minimum	: 90%
Continuous operation	: 90 to 110%
Maximum	: 110%

TRANSMISSION LIMITATIONS

TORQUE (TRQ%)

NOTE

100% torque corresponds to 534 Nm (394 ft lb).

All engines operative (AEO)

Maximum continuous	: 100%
Transient (6 sec)	: 110%

One engine inoperative (OEI)

Maximum continuous	: 124%
2.5 min	: 142%
Transient (6 sec)	: 156%

NOTE

The One Engine Inoperative (OEI) rating use is intended for emergency use, when one engine becomes inoperative due to an actual malfunction. OEI operations for maintenance or training purposes shall be limited to the maximum continuous OEI power rating.

NOTE

Transient must not be used intentionally. OEI transient may be used only in case of real emergency, when one engine becomes inoperative due to an actual malfunction.

FUEL SYSTEM LIMITATIONS

Fuel pressure

Cautionary range	: 0 to 7 psi
Normal range	: 7 to 25 psi
Maximum	: 25 psi

Approved fuels

Type	Specification
JET A	ASTM D1655
JET A-1	ASTM D1655
JET A-2	ASTM D1655
JET B	ASTM D1655

Military Specification (Reference only)

Type	Specification
JP-4 (*) JP-5 (*) JP-8 (*)	MIL-T-5624 MIL-T-5624 MIL-T-83133

(*) Contains fuel system icing inhibitor (FSII). (for JP-8, MIL-T-83133C allows two grades. The grade meeting NATO code F-34 has FSII while the grade meeting code F-35 has no FSII without prior agreement).

NOTE

An approved fuel or any mixture of acceptable fuels may be used. However, changing to a fuel with a substantially different heating value or specific gravity may require maintenance in the form of engine fuel control (trimmer) adjustment.
Refer to Pratt & Whitney manuals.

NOTE

For operation below 4°C the use of anti-ice additive is authorized but not mandatory due to aircraft anti-ice fuel filter installation. For additive requirements and blending procedures refer to Pratt & Whitney manuals.

NOTE

The use of gasoline (avgas) must be restricted to emergency purpose only. Avgas shall not be used for more than 150 hours during any period between overhauls.
Refer to Pratt & Whitney manuals.

ENGINE LUBRICATION SYSTEM LIMITATIONS

Oil pressure

The oil pressure limits vary as function of the gas generator speed according to figure 1-5. These oil pressure limits must be respected when engine oil temperature is between 71 - 120°C.

Transient : 200 psi

NOTE

The engine can run with oil pressure up to 200 psi during or after a start or if the oil temperature drops significantly below 71°C. Oil pressure will decrease as oil temperature increases and is not expected to endure for more than 5 sec. The operation at an oil pressure above the normal range (up to 200 psi) is permitted for a period of 10 minutes.

Oil temperature

Maximum : 125°C

Approved lubricating oils (Type II, 5 cSt)

DESIGNATION	SPECIFICATION
BP Turbo Oil 2380 / EXXON Turbo Oil 2380	MIL-PRF-23699F
BP Turbo Oil 25 / EXXON Turbo Oil 25	MIL-PRF-23699F
AeroShell Turbine Oil 500	MIL-PRF-23699F
AeroShell Turbine Oil 560	MIL-PRF-23699F
Royco Turbine Oil 500	MIL-PRF-23699F
Royco Turbine Oil 560	MIL-PRF-23699F
Turbonycoil 525-2A	PWA 521
Mobil Jet Oil II	MIL-PRF-23699F
Castrol 5000	MIL-PRF-23699F

NOTE

Mixing of oils of different brands, type and Manufacturers is prohibited.

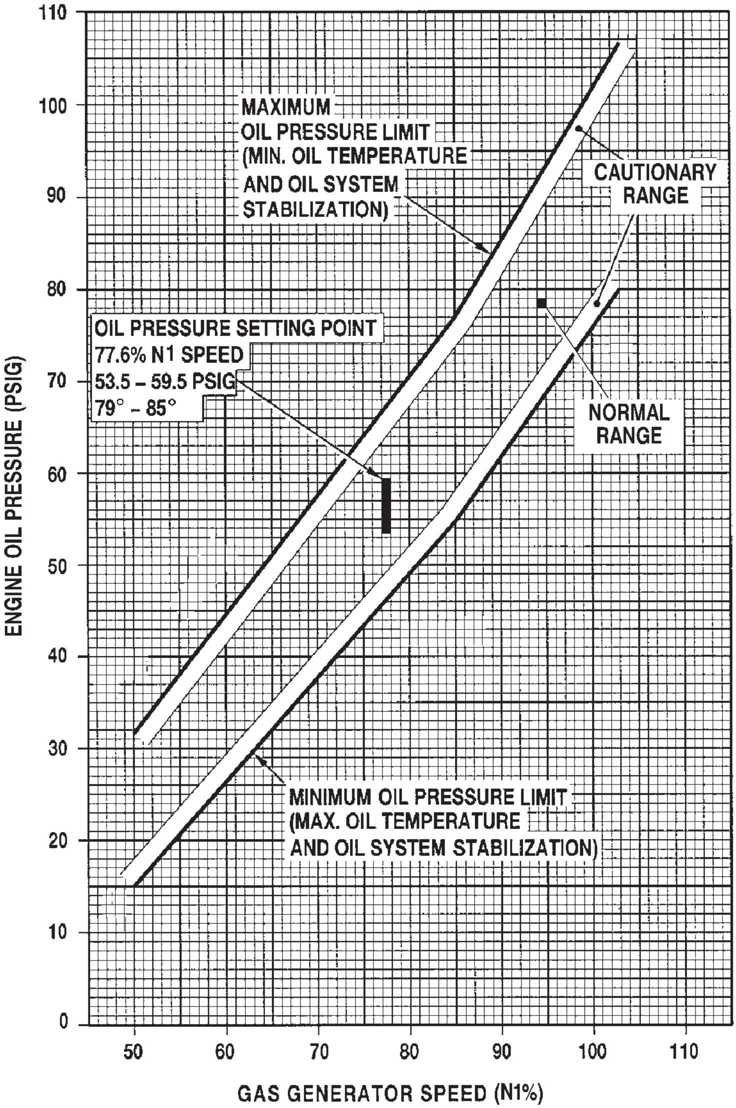


Fig. 1-5. Engine oil pressure limits

MAIN TRANSMISSION LUBRICATION SYSTEM LIMITATIONS

Oil pressure

Minimum	: 30 psi
Normal range	: 30 to 50 psi
Cautionary range	: 50 to 70 psi
Maximum	: 70 psi

Oil temperature

Maximum	: 115 °C

Approved lubricating oils

DESIGNATION	SPECIFICATION
BP Turbo Oil 2380 / EXXON Turbo Oil 2380	MIL-PRF-23699
BP Turbo Oil 2197	MIL-PRF-23699
Mobil Oil Jet II	MIL-PRF-23699
Mobil Oil Jet 254	MIL-PRF-23699
AeroShell Turbine Oil 500	MIL-PRF-23699
AeroShell Turbine Oil 555	DOD-L-85734
AeroShell Turbine Oil 560	DOD-L-85734
Castrol 5000	MIL-PRF-23699

Oils limited to ambient temperature above -40°C (-40°F).

NOTE

Mixing of oils of different brands, type and Manufacturers is prohibited.

TAIL ROTOR GEARBOX LUBRICANT LIMITATIONS

Approved lubricating oils

DESIGNATION	SPECIFICATION
BP Turbo Oil 2380 / EXXON Turbo Oil 2380	MIL-PRF-23699
BP Turbo Oil 2197	MIL-PRF-23699
Mobil Oil Jet II	MIL-PRF-23699
Mobil Oil Jet 254	MIL-PRF-23699
AeroShell Turbine Oil 500	MIL-PRF-23699
AeroShell Turbine Oil 555	DOD-L-85734
AeroShell Turbine Oil 560	DOD-L-85734
Castrol 5000	MIL-PRF-23699

Oils limited to ambient temperature above -40°C (-40°F).

NOTE

Mixing of oils of different brands, type and Manufacturers is prohibited.

ENGINE STARTER LIMITATIONS

The engine starter duty cycle is the following:

— 30 seconds on, 1 minute off
— 30 seconds on, 1 minute off
— 30 seconds on, 30 minutes off

GENERATOR LIMITATIONS

Helicopter equipped with emergency bus:

For each generator

Continuous operation : 0 to 160 A

Maximum	:	160 A
Transient (20 seconds)	:	200 A

Helicopter not equipped with emergency bus:

For each generator

Continuous operation	:	0 to 130 A
Maximum	:	130 A
Transient (20 seconds)	:	160 A

MAIN HYDRAULIC SYSTEM LIMITATIONS

FLUID PRESSURE

Minimum	: 1200 psi
Cautionary range	: 1200 to 1400 psi
Normal range	: 1400 to 1550 psi
Maximum	: 1550 psi

Approved fluids

Hydraulic fluid MIL-H-5606

WARNING

Operation of each hydraulic system must be checked before each flight in accordance with Section 2 of this manual.

UTILITY HYDRAULIC SYSTEM LIMITATIONS

NORMAL SYSTEM FLUID PRESSURE

Minimum	: 500 psi
Cautionary range	: 500 to 1140 psi
Normal range	: 1140 to 1550 psi
Maximum	: 1550 psi

EMERGENCY SYSTEM FLUID PRESSURE

Minimum	: 1140 psi
Maximum	: 1550 psi

Approved fluids

Hydraulic fluid MIL-H-5606

INSTRUMENT MARKINGS

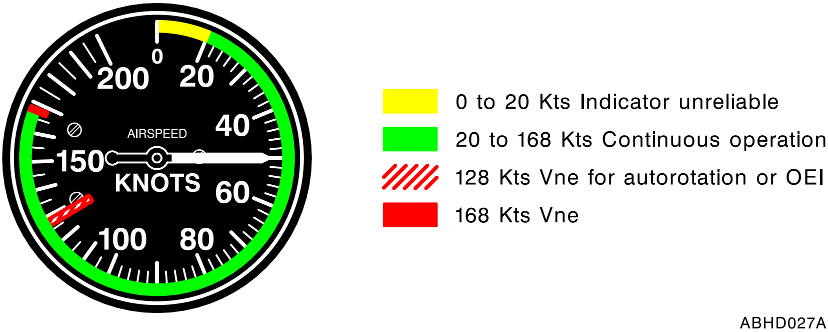


Fig. 1-6. Airspeed indicator

ELECTRONIC DISPLAY UNIT FORMATS

The EDUs present the following information:

EDU 1

- N1
- NR/N2
- TOT
- TRQ
- Caution, warning and advisory messages

EDU 2

- Engine oil pressure
- Engine oil temperature
- Transmission oil pressure
- Transmission oil temperature
- Fuel pressure
- Fuel quantity
- Fuel flow (provision only)
- Main hydraulic pressure
- Utility hydraulic pressure
- DC ammeter
- DC voltmeter
- AC voltmeter
- OAT
- Advisory and status messages

The main display formats are:

EDU 1	EDU 2
- START	- MAIN
- CRUISE: AEO / OEI / AUTOROTATION	- AUXILIARY

- REVERSIONARY: AEO / OEI / AUTOROTATION

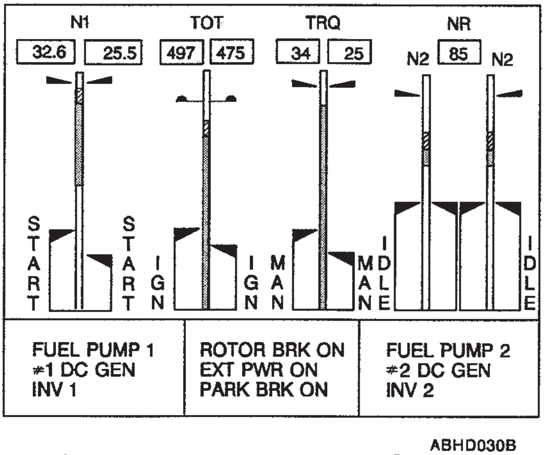


Fig. 1-7. EDU 1 - START Mode (AEO)

The START mode is displayed when selected by the START key on EDU 1.

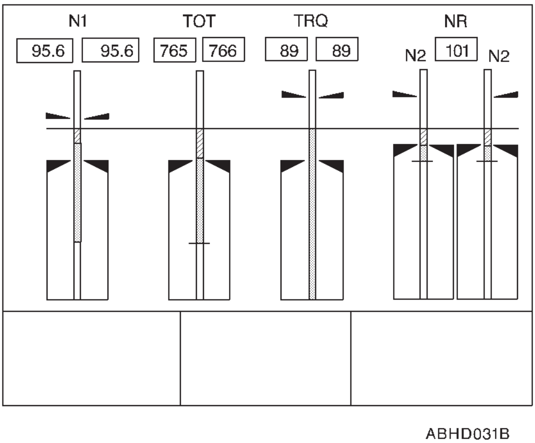


Fig. 1-8. EDU 1 - CRUISE Mode (AEO)

The CRUISE mode is displayed upon IDS initialization and/or when selected by the CRUISE key or automatically at the end of the start sequence. It is also displayed automatically on both EDUs in case of complete loss of secondary data.
The cruise (AEO) format is presented also in REVERSIONARY mode.

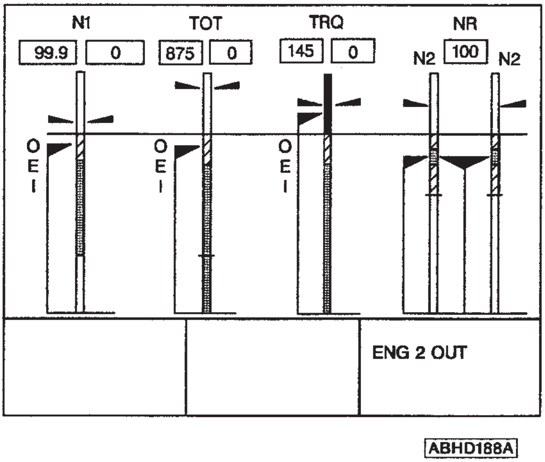


Fig. 1-9. EDU 1 - OEI Mode

The OEI mode is a subset of CRUISE mode and is automatically selected if one engine becomes inoperative.
The OEI format is presented also in REVERSIONARY mode.

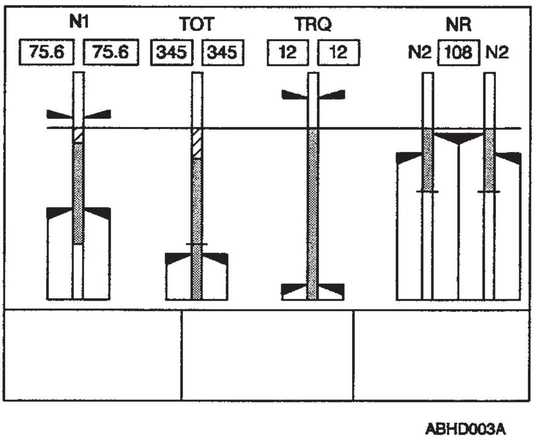


Fig. 1-10. EDU 1 - AUTOROTATION Mode

The AUTOROTATION mode is automatically selected when both engines fail or run at idle, both engine torque inputs are at zero or in presence of NR/N2 split.
The AUTOROTATION format is presented also in REVERSIONARY mode.

Intentionally blank

Fig. 1-11. Blank

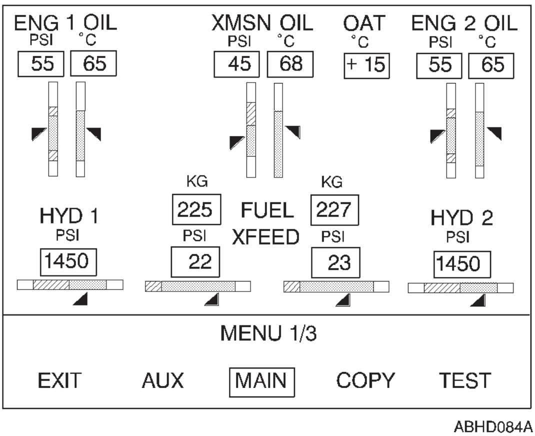


Fig. 1-12. EDU 2 - MAIN Mode

The MAIN mode is displayed upon IDS initialization and/or when selected by the MAIN key on EDU 2.

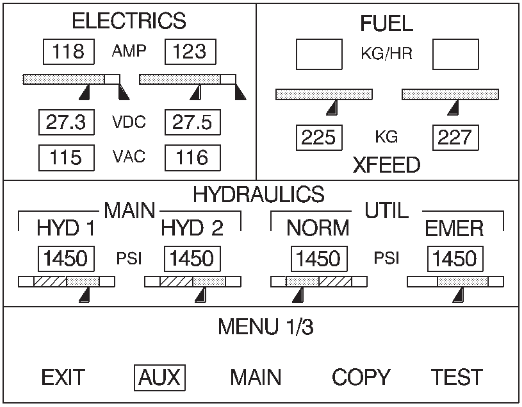


Fig. 1-13. EDU 2 - AUXILIARY Mode

The AUXILIARY mode can be displayed by selecting the AUX key on EDU 2. The AUXILIARY page reverts automatically to the MAIN page when any caution/warning message occurs.

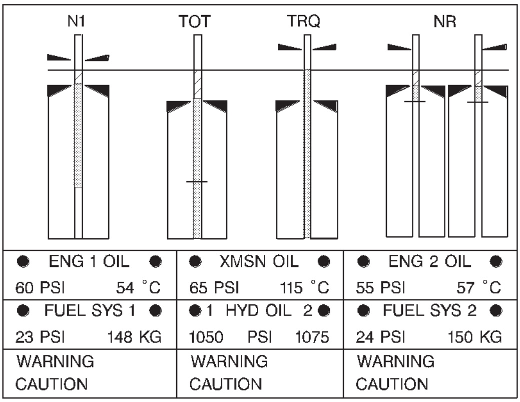


Fig. 1-14. EDU 1 and/or EDU 2 - REVERSIONARY Mode

Any EDU enters in the REVERSIONARY mode when it is manually commanded by pressing the ON/OFF switch to OFF on the failed EDU or automatically when a critical failure occurs and is detected by the healthy EDU.

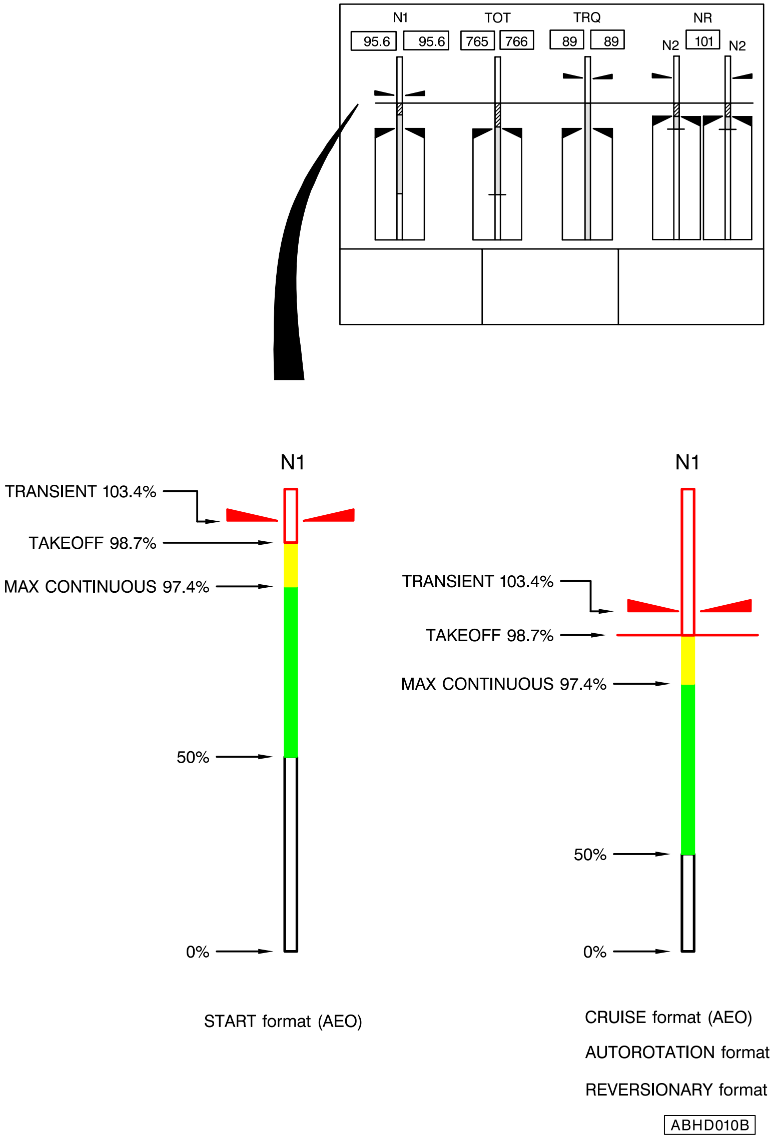


Fig. 1-15. EDU 1 - GAS GENERATOR SPEED (AEO)

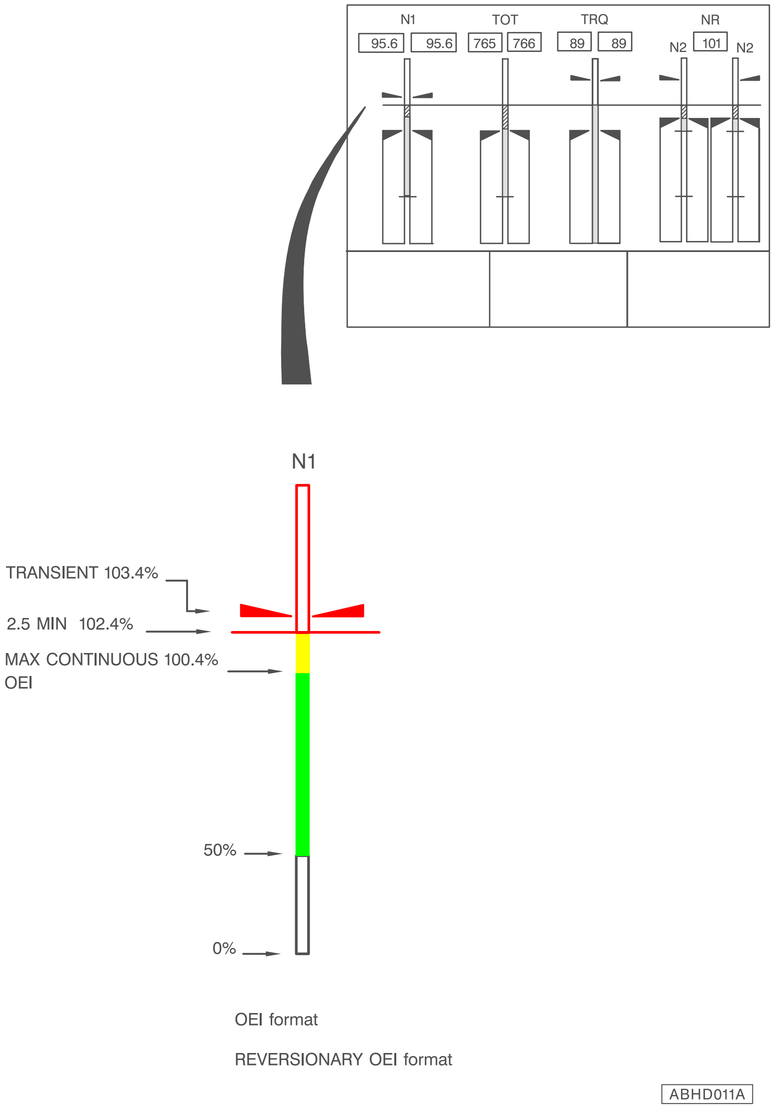


Fig. 1-16. EDU 1 - GAS GENERATOR SPEED (OEI)

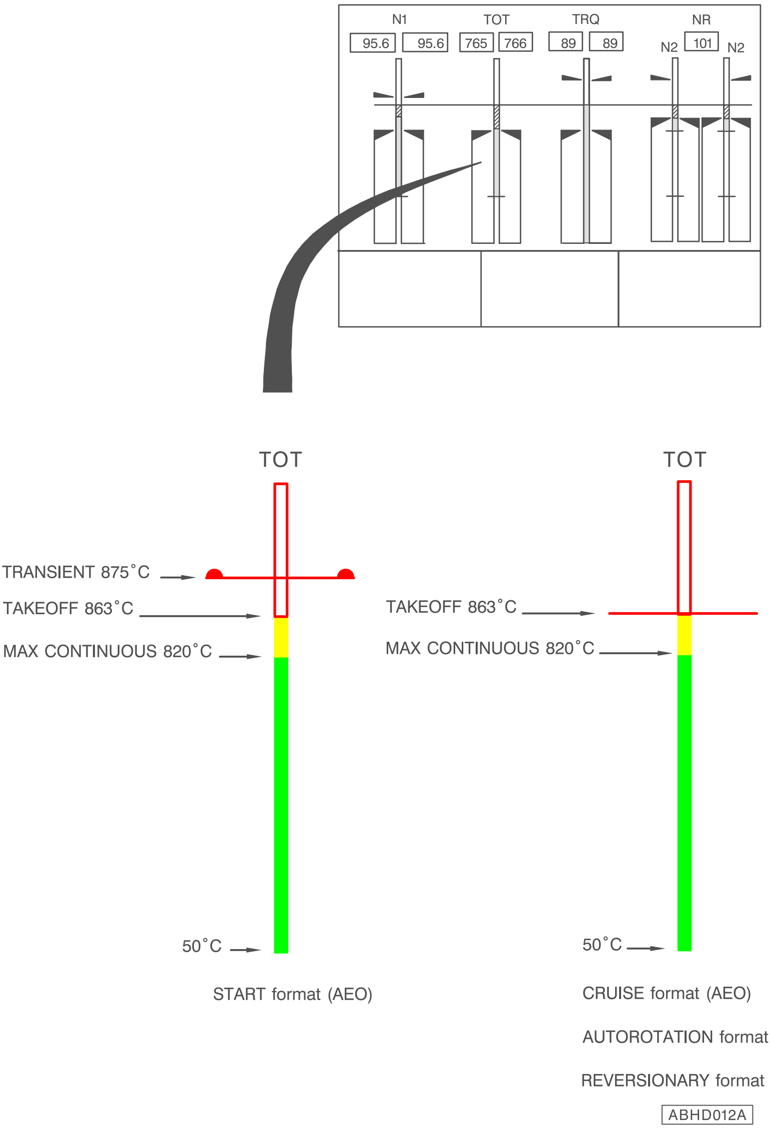
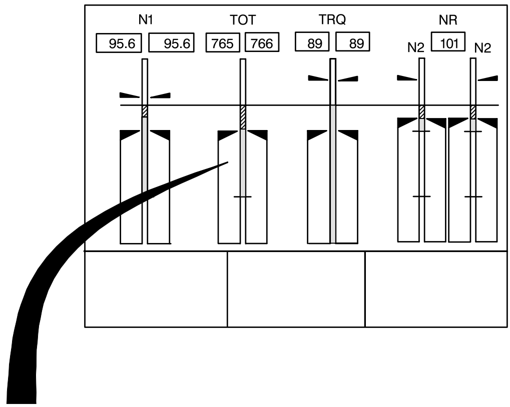
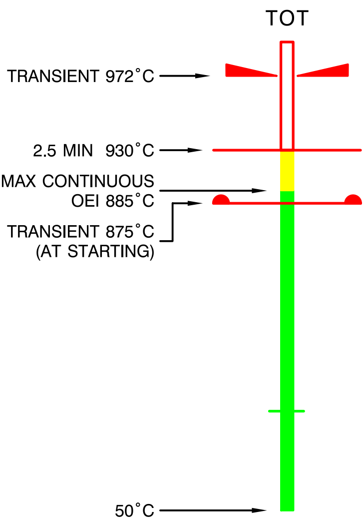


Fig. 1-17. EDU 1 - TURBINE OUTLET TEMPERATURE (AEO)





NOTE

The transient value at starting (half red dot at 875°C) is displayed on the scale of the engine being started, during the restart phase only.

OEI format

REVERSIONARY OEI format

ABHD013B

Fig. 1-18. EDU 1 - TURBINE OUTLET TEMPERATURE (OEI)

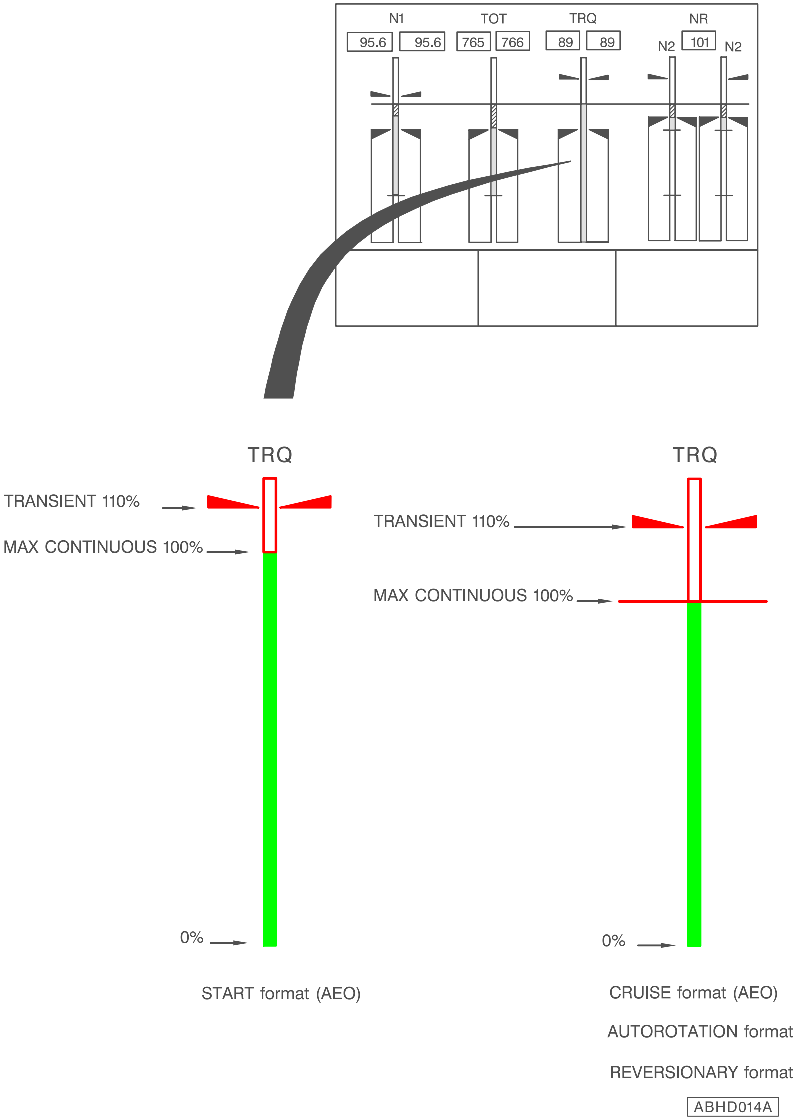
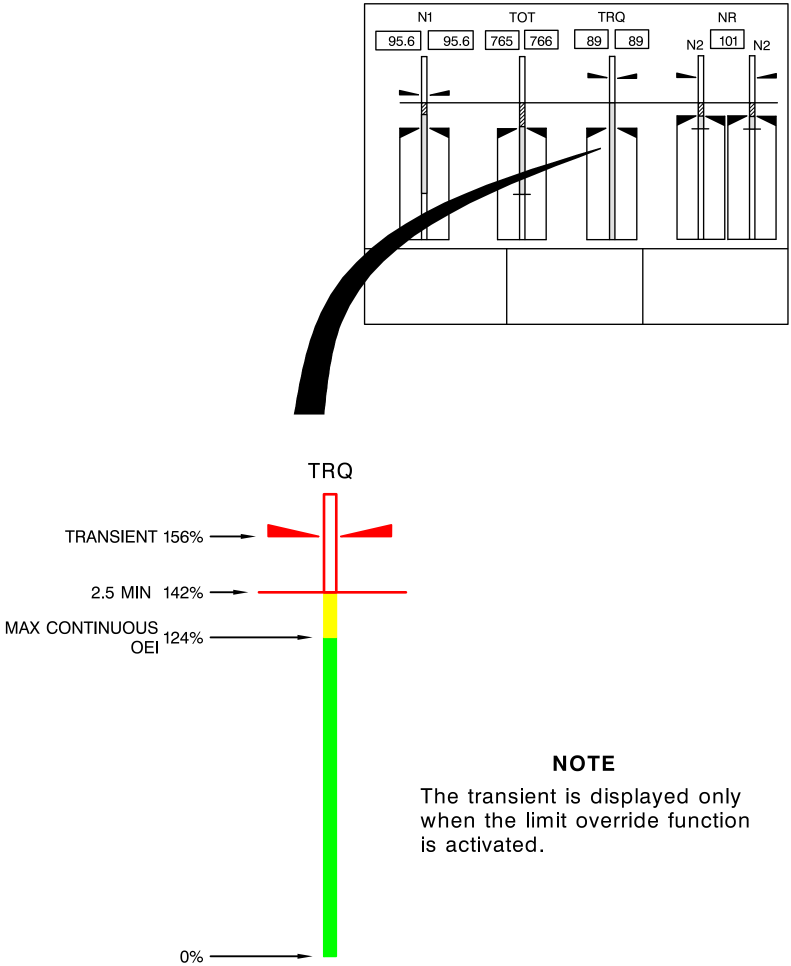


Fig. 1-19. EDU 1 - TORQUE (AEO)



NOTE

The transient is displayed only when the limit override function is activated.

OEI format

REVERSIONARY OEI format

ABHD189B

Fig. 1-20. EDU 1 - TORQUE (OEI)

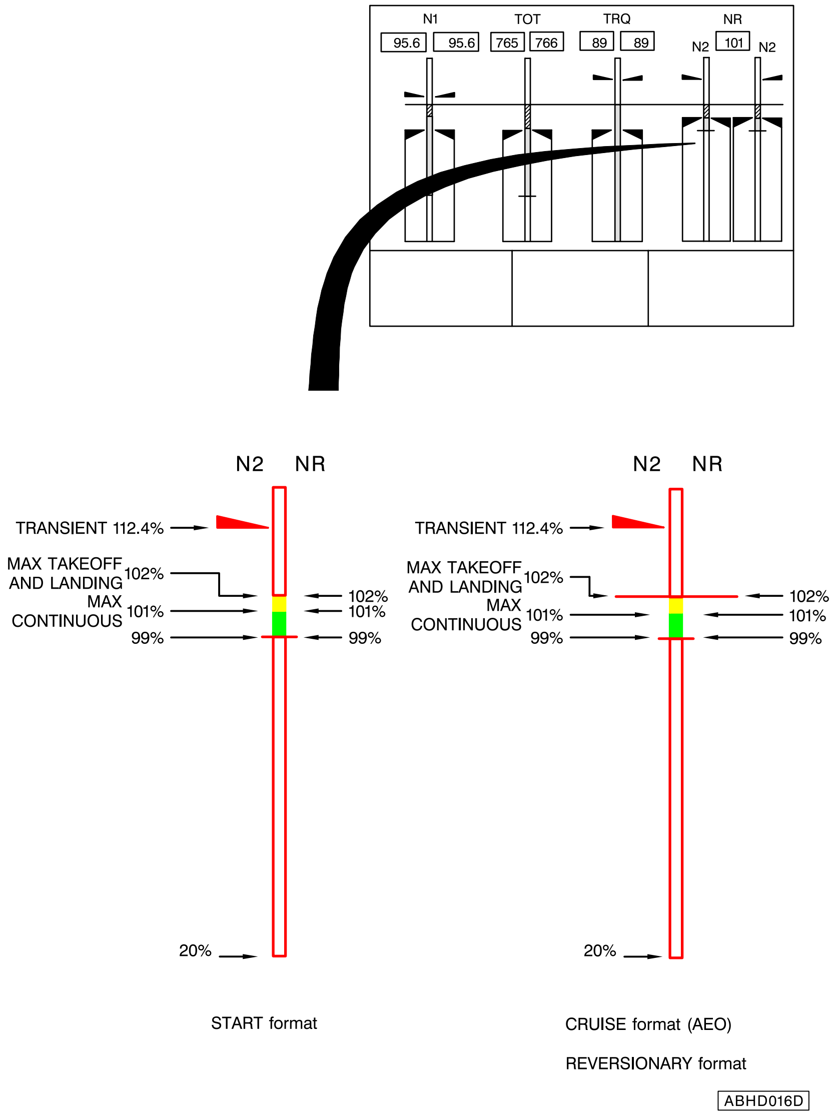


Fig. 1-21. EDU 1 - ROTOR POWER TURBINE SPEED (AEO)

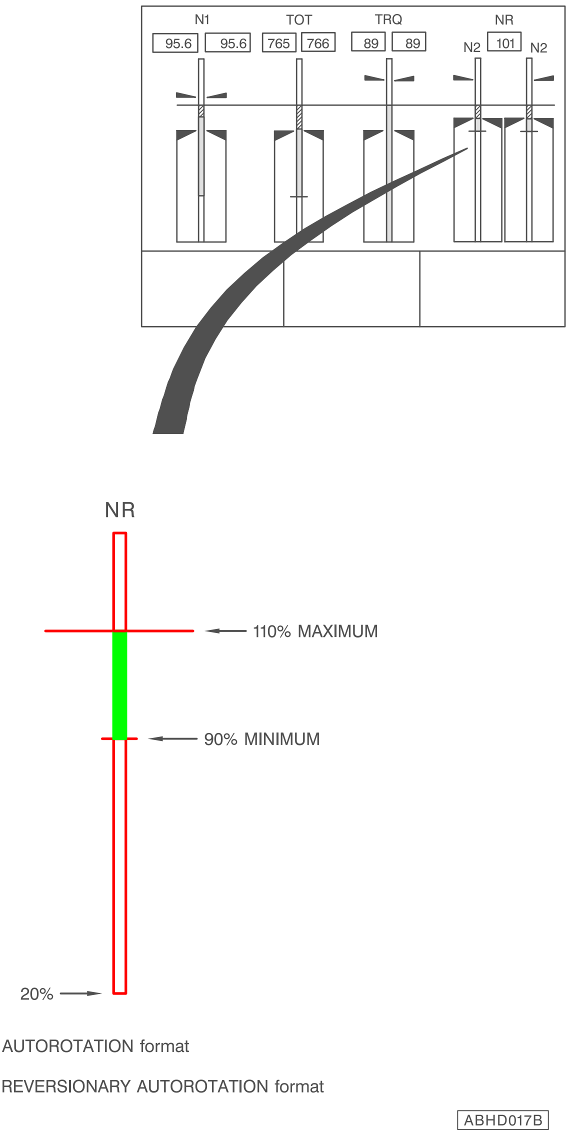


Fig. 1-22. EDU 1 - ROTOR SPEED (Power off)

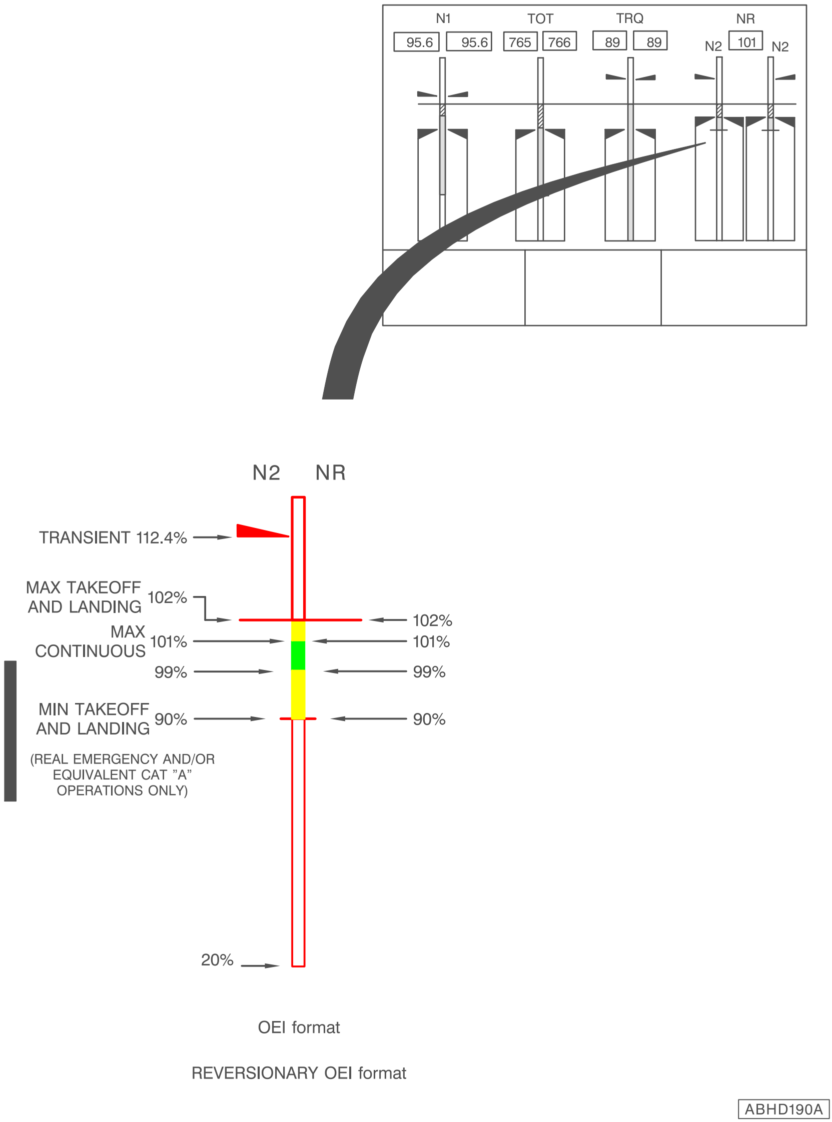


Fig. 1-23. EDU 1 - ROTOR POWER TURBINE SPEED (OEI)

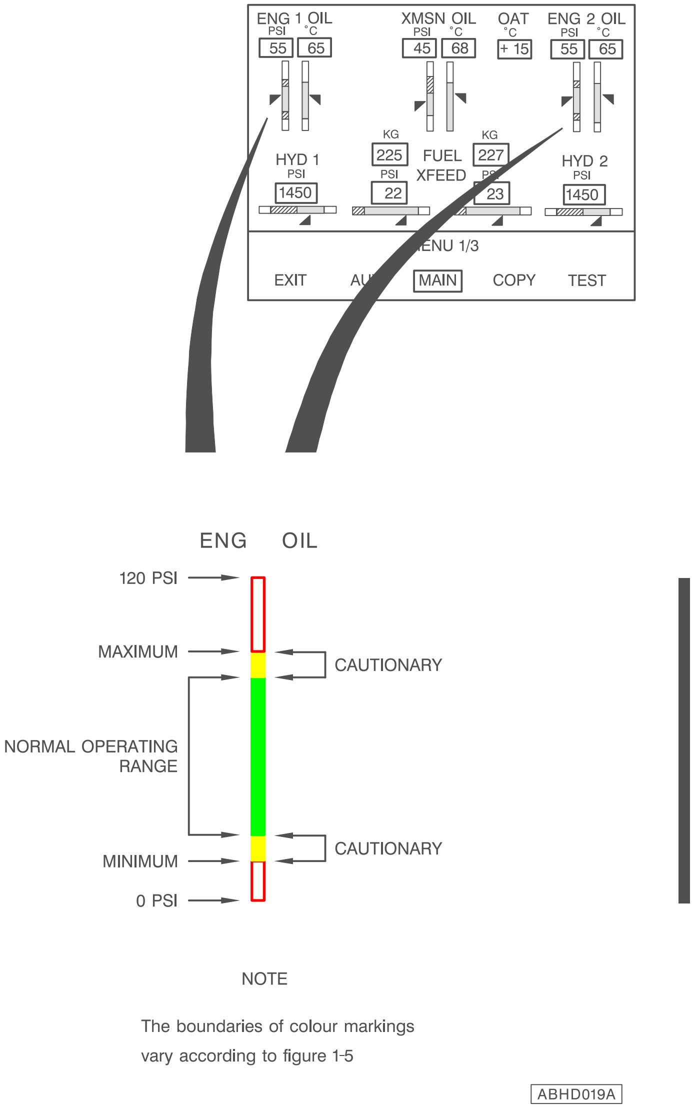


Fig. 1-24. EDU 2 (MAIN) - ENGINE OIL PRESSURE

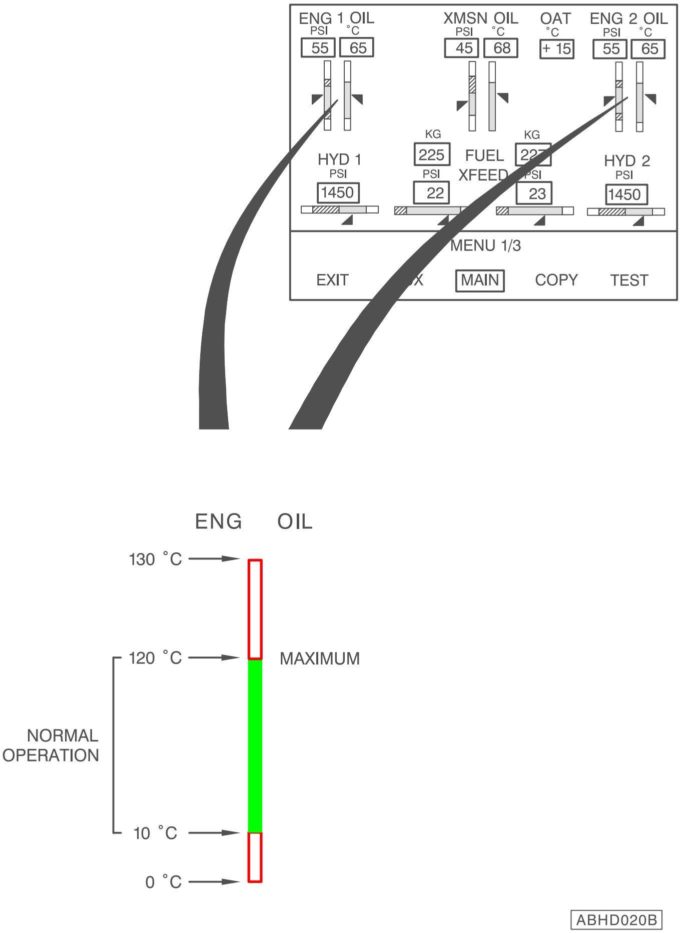
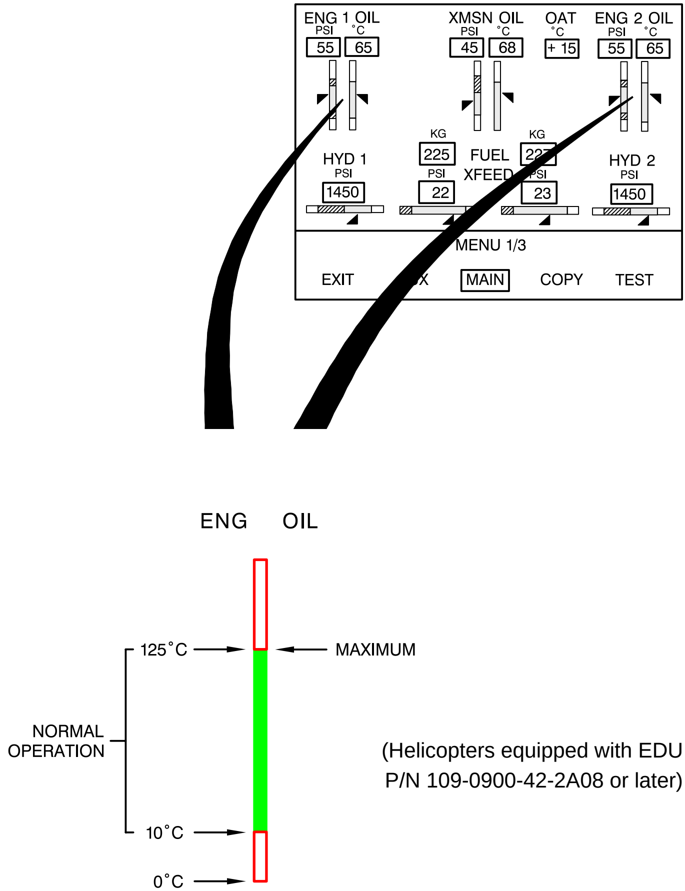


Fig. 1-25. EDU 2 (MAIN) - ENGINE OIL TEMPERATURE



(Helicopters equipped with EDU P/N 109-0900-42-2A08 or later)

ABHD599A

Figure 1-25A. EDU 2 (MAIN) - ENGINE OIL TEMPERATURE.

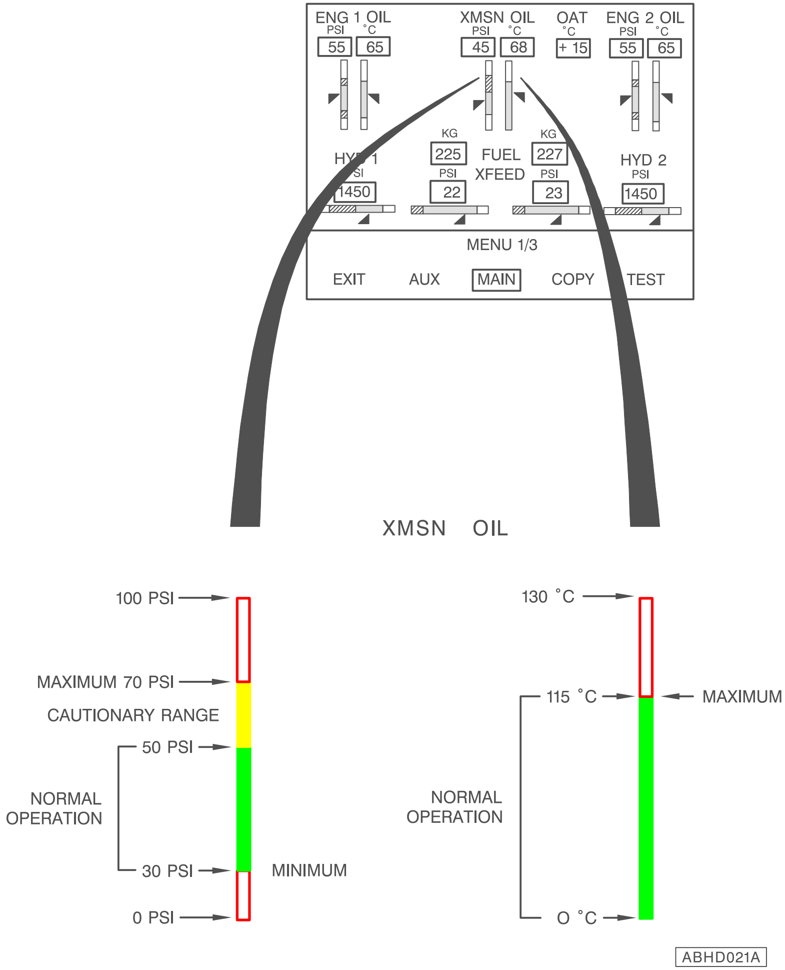


Fig. 1-26. EDU 2 (MAIN) - TRANSMISSION OIL (PRESSURE AND TEMPERATURE)

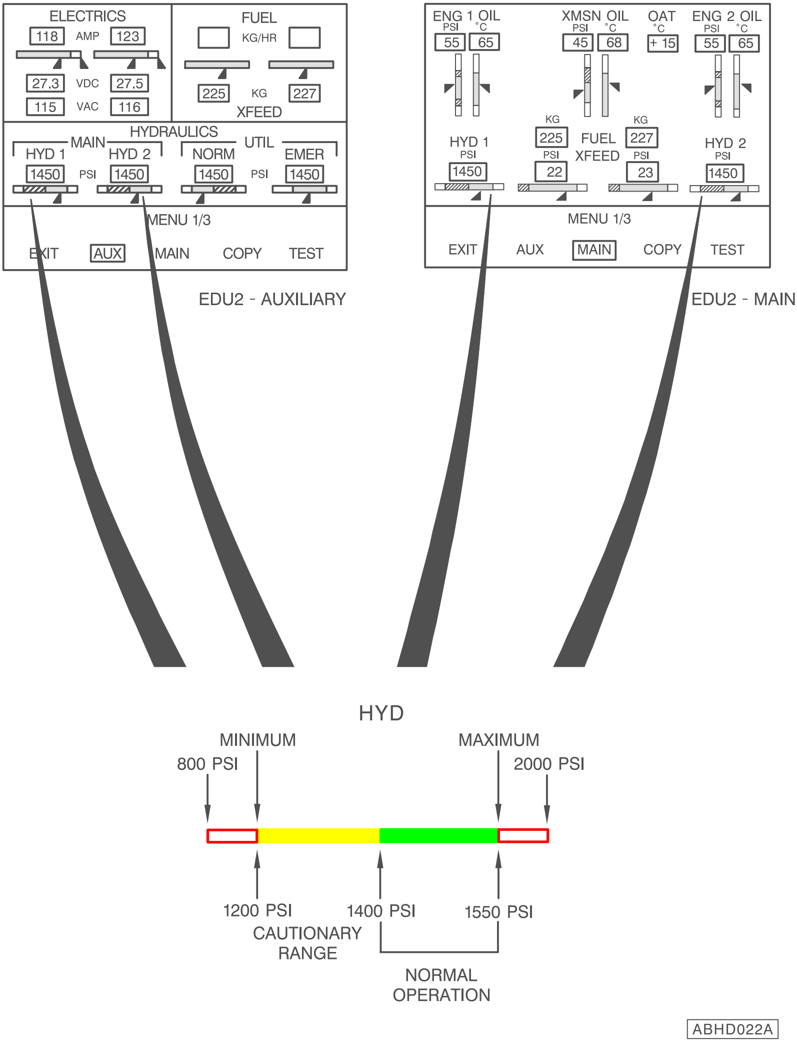


Fig. 1-27. EDU 2 (MAIN/AUXILIARY) - MAIN HYDRAULIC SYSTEM PRESSURE

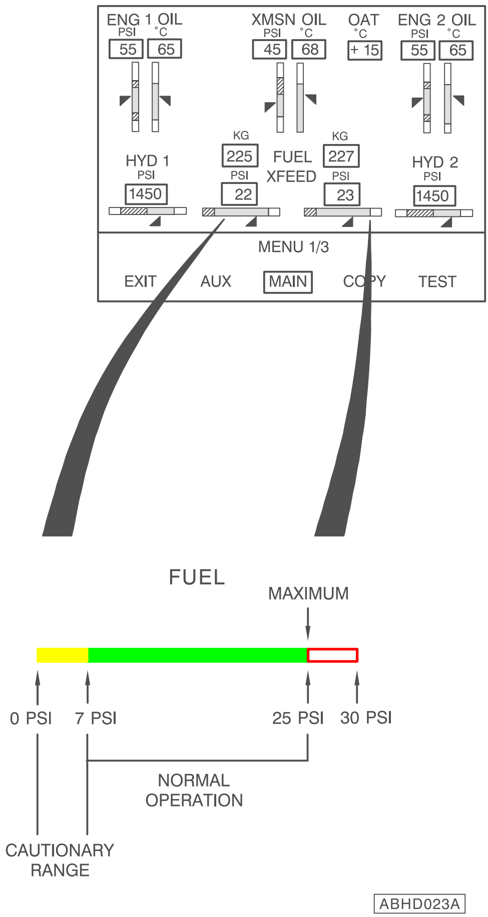


Fig. 1-28. EDU 2 (MAIN) - FUEL PRESSURE

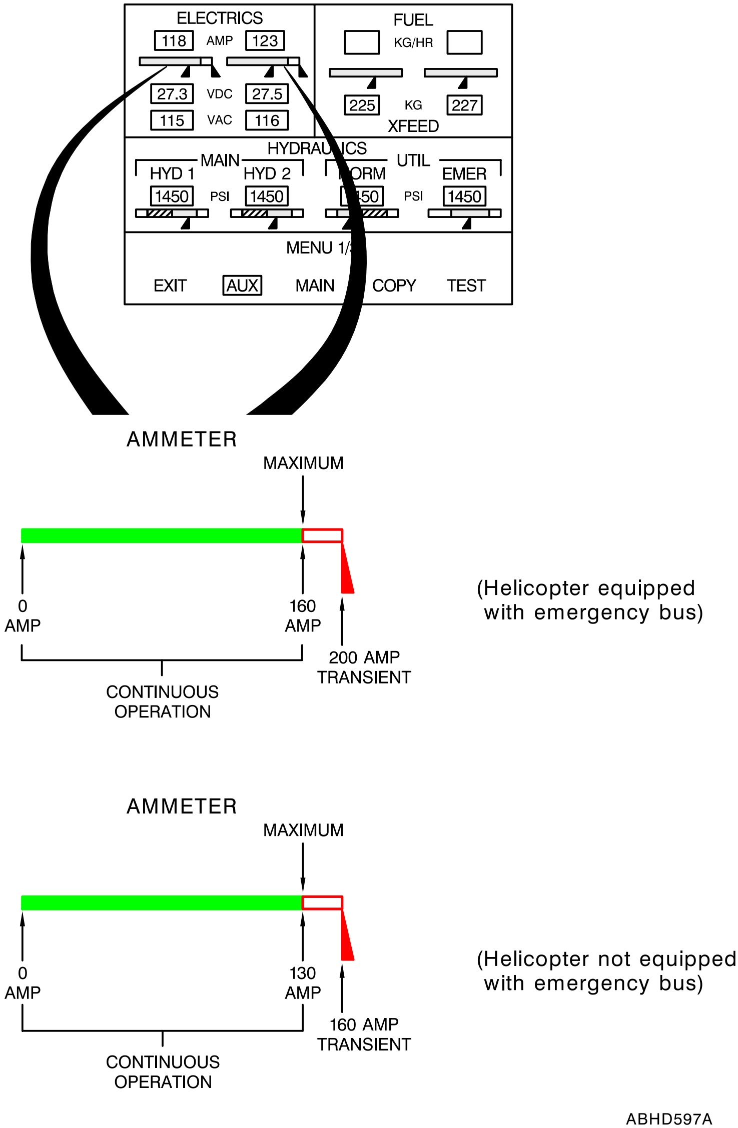


Fig. 1-29. EDU 2 (AUXILIARY) - AMMETER.

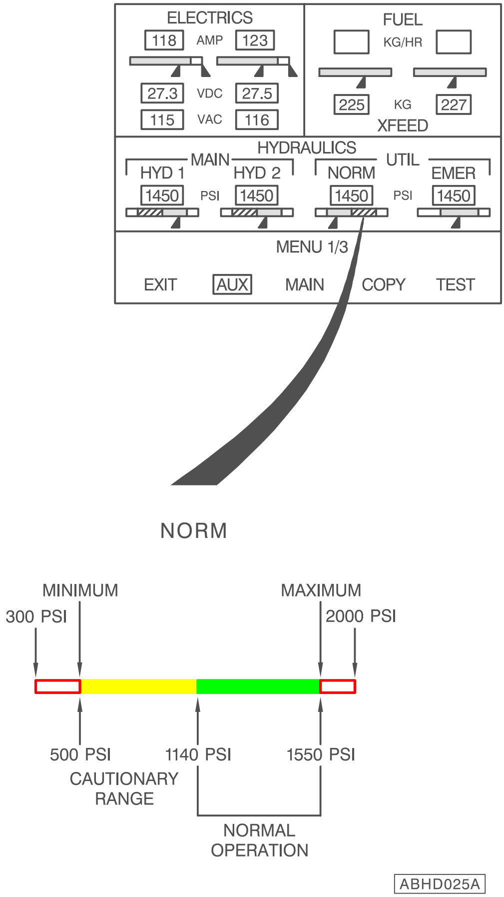


Fig. 1-30. EDU 2 (AUXILIARY) - UTILITY HYDRAULIC SYSTEM PRESSURE (NORMAL)

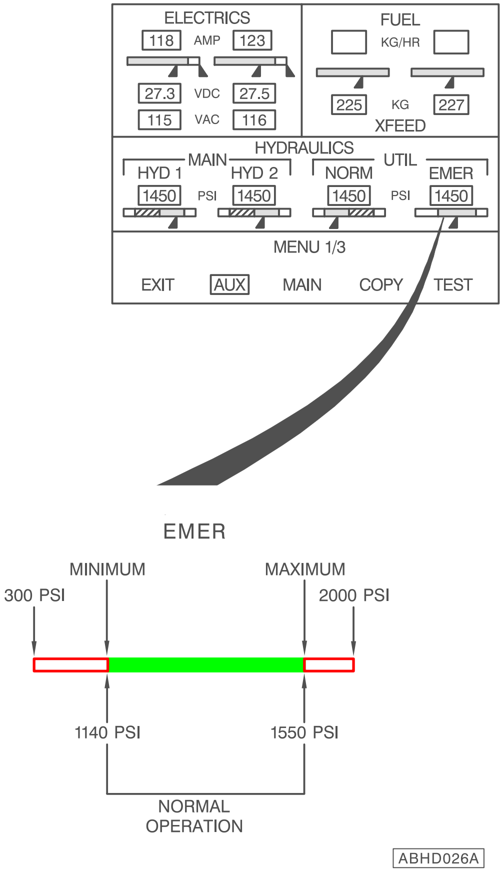


Fig. 1-31. EDU 2 (AUXILIARY) - UTILITY HYDRAULIC SYSTEM PRESSURE (EMERGENCY)

PLACARDS

THE MARKINGS AND PLACARDS INSTALLED ON THIS HELICOPTER CONTAIN OPERATING LIMITATIONS WHICH MUST BE COMPLIED WITH WHEN OPERATING THIS HELICOPTER. OTHER OPERATING LIMITATIONS WHICH MUST BE COMPLIED WITH WHEN OPERATING THIS HELICOPTER ARE CONTAINED IN THE R.A.I. APPROVED RFM

In clear view of the pilot

AIRSPEED LIMITATIONS V_{NE} – KIAS

OAT°C \ Hp–ft	–1000 TO SL	3000	6000	9000	12000	15000	18000	20000
45	168	165	–	–	–	–	–	–
35	168	168	158	–	–	–	–	–
25	168	168	161	152	143	–	–	–
15	168	168	164	155	146	136	111	–
5	168	168	167	157	148	139	114	97
–5	168	168	168	160	151	142	117	100
–15	168	168	168	164	154	145	120	103
–25	168	168	168	167	157	148	123	106

AUTOROTATION OR OEI:
REDUCE V_{NE} BY 40 KIAS

MAX FWD TOUCHDOWN SPEED: 40 Kts
V_{LO} AND V_{LE} : 120 KIAS

ABHD455C

In clear view of the pilot

IN ALTN POSITION
CLOSE VENTS AND
TURN HEATER/ECS OFF

In clear view of the pilot

IN ALTN POSN
MAINTAIN INST
ACCURACY BY
CLOSING
WINDOWS VENTS
AND TURNING
HEATER OFF

In clear view of the pilot, only with sliding windows P/N 109-0822-68 installed

ALT. STATIC
DECREASE ALTIMETER READINGS BY 300 ft

In clear view of the pilot

MAXIMUM LOAD 150 KG (330 LB)
MAX UNIT LOAD 500 KG/M^2 (102 LB/SQ.FT.)

Baggage compartment

MAXIMUM ALLOWABLE BAGGAGE LOAD					
ZONE	1	2	3	4	5
KG	150	140	132	123	108
MAX UNIT FLOOR LOADING 500 KG/M^2 (102 LB/SQ.FT.)					

Baggage compartment

ALL CARGO MUST BE SECURED
MAX LOAD PER TIE DOWN RING 91 KG (200 LB)

Baggage compartment

SECTION 2

NORMAL PROCEDURES

TABLE OF CONTENTS

LIST OF ILLUSTRATIONS

SECTION 2
NORMAL PROCEDURES

INTRODUCTION

This section contains instructions and procedures for operating the helicopter from the planning stage, through actual flight conditions, to securing the helicopter after landing.

Normal and standard conditions are assumed in these procedures. Pertinent data in other sections is referenced when applicable.

The instructions and procedures contained herein are written for the purpose of standardization and are not applicable to all situations.

The minimum and maximum limits, and the normal and cautionary operating ranges for the helicopter and its subsystems are indicated by instrument markings and placards. Refer to Section 1 for a detailed explanation of each operating limitation.

Each time an operating limitation is exceeded, an appropriate entry shall be made in the logbook (helicopter, engine, etc.). The entry shall state which limit was exceeded, the duration of time, the extreme value attained, and any additional information essential in determining the maintenance action required.

FLIGHT PLANNING

Each flight should be planned adequately to ensure safe operations and to provide the pilot with the data to be used during flight.

Essential weight and balance, and performance information should be compiled as follows:

— Check type of mission to be performed and destination.
— Select appropriate performance charts to be used from Section 4.
— Review the appropriate Supplements of this Rotorcraft Flight Manual for the optional equipment(s) installed.

TAKE-OFF AND LANDING DATA

Refer to Section 1 - LIMITATIONS and Section 4 - PERFORMANCE .

WEIGHT AND BALANCE DATA

Ascertain proper weight and balance of the helicopter as follows:

— Consult Section 6 - Weight and Balance.
— Ascertain weight of fuel, oil, payload, etc.
— Compute take-off and anticipated landing gross weights
— Check helicopter center of gravity (CG) locations.
— Check that the weight and CG limitations in Section 1 are not exceeded.

PREFLIGHT CHECKS

Preflight checks are intended as those checks to be performed by the pilot in order to ascertain that the helicopter is flightworthy and adequately equipped. They are therefore not meant as detailed mechanical inspections, but as a guide to check the condition of the helicopter.

″Check″ is intended as observing the helicopter to find any obvious damage.
″Damage″ is intended as any abnormal or out of limits condition.

If during preflight check these conditions are found, inspections shall be carried out before the flight, in order to ascertain the helicopter airworthiness.

Have care to brief passengers on relevant operational procedures and associated hazards.

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

The following procedure outlines the pilot walk-around and interior checks (Figure 2-1).

Main and tail rotor tie-down (if present)	: Removed.

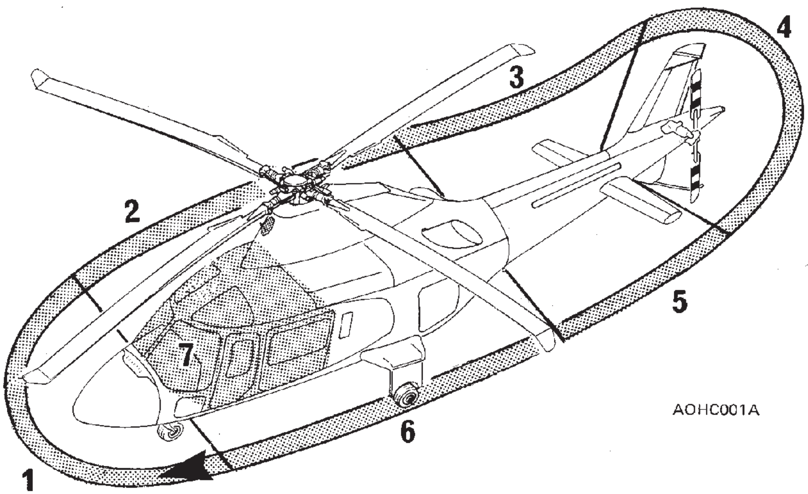
AREA N° 1 (Helicopter nose)

Nose exterior	: Condition.
Ventilation air intake	: Free of obstructions.
Landing lights	: Condition.
Nose landing gear	: Condition, shock strut extension, leaks, tire pressure.
Battery	: Connection secured.
Avionic equipment	: Secured.
Accumulators	: Condition and leaks. Discharge accumulators by pressing relevant red buttons. Door secured.

CAUTION

The discharge of accumulators causes loss of parking brakes.

Drains and breather	: Conditions.
Nose compartment access door	: Security.



AREA No. 1 : Helicopter nose
AREA No. 2 : Fuselage - RH side
AREA No. 3 : Tail boom - RH side
AREA No. 4 : Fins, 90° gearbox, tail rotor, tail skid
AREA No. 5 : Tail boom - LH side
AREA No. 6 : Fuselage - LH side
AREA No. 7 : Cabin interior

Figure 2-1. Preflight check sequence.

AREA N° 2 (Fuselage - rh side)

Windshield and roof transparent panel	:	Condition and cleanliness.
Pitot tube	:	Cover removed, condition and obstructions.
IDS external air temperature probe	:	Condition, free of obstructions.
Fuselage exterior	:	Condition.
Pilot cabin door	:	Condition, cleanliness, window secured.
Antenna(s)	:	Condition.
Passenger cabin door	:	Condition, cleanliness, window secured.
Passenger door jettison window	:	Security of window and seal retainer.
Cowling and fairings	:	Condition and security.
Servo hydraulic system valves and filter group	:	Check for leaks and by-pass indication (red button out: filter clogged). Door secured.
Hydraulic system tanks	:	Check fluid level and filler cap for security. Door secured.
Main rotor hub and blades	:	Condition and security.
Main rotor dampers	:	Check for correct charge indication.

Main rotor pitch change links	:	Condition and security.
Upper anticollision light	:	Condition.
Servo actuator	:	Condition and leaks.
Main transmission and accessories (visible area)	:	Condition and leaks.
Transmission oil cooler	:	Free of obstructions.
Transmission external oil filter	:	By-pass indication (white button out: filter clogged). Door secured.
Cooler blower air intake	:	Free of obstruction.
Cooler system belt	:	Condition and security.
Airframe (A/F) fuel filter	:	Condition and leaks.
Service step	:	Secured.
Engine air intake screen and chamber	:	Cover removed; free of damage and/or obstructions.
Engine area	:	Check for fuel and/or oil leaks.
Engine oil	:	Check gauge for oil presence (sight glass dark).
Engine oil filter impending bypass indicator	:	Check for correct indication.
Engine-transmission drive shaft	:	Condition.
Engine supports (visible area)	:	Condition.

Engine cowling	:	Condition, secured.
Fuel filler cap	:	Secured.
Main landing gear	:	Condition, shock strut extension, leaks, tire pressure.
Drain and vent lines	:	Free of obstructions.
Lower anticollision light	:	Condition.
Antenna(s) (if installed)	:	Condition.
Engine exhaust	:	Cover removed, condition.

AREA N° 3 (Tail boom - rh side)

Tail boom exterior	:	Condition.
Antenna(s) (if installed)	:	Condition.
Stabilizer	:	Condition and security.
Navigation light	:	Condition.

AREA N° 4 (Fins, tail gearbox, tail rotor and skid)

Upper tail fin	:	Condition.
Lower tail fin and skid	:	Condition.
Tail rotor long drive shaft bearing	:	Check condition verifying no mark slippage. Door secured.
Tail rotor gearbox	:	Check oil level, check for leaks and for filler cap secured.

Access door	: Secured.
Tail navigation light	: Condition.
Tail rotor hub and blades	: Condition, cleanliness and freedom of flapping.
Tail rotor pitch change mechanism	: Condition and security.

Area n° 5 (tail boom - lh side)

Tail boom exterior	: Condition.
Stabilizer	: Condition and security.
Navigation light	: Condition.
Antenna(s) (if installed)	: Condition.
Tail rotor long drive shaft bearings	: Check condition verifying no mark slippage.
Tail rotor drive shaft cover	: Secured.

AREA N° 6 (Fuselage - lh side)

Fuselage exterior	: Condition.
Engine exhaust	: Cover removed, condition.
Baggage compartment	: Cargo (if on board) properly secured.
Power relay box circuit breakers	: In. Access door secured.

NOTE

The above mentioned circuit breakers are accessible from baggage compartment through an inspection door.

Baggage door	:	Secured.
Lower avionic bay access door	:	Secured.
Tail rotor middle drive shaft bearings	:	Check condition verifying no mark slippage. Door secured.
Engine area	:	Check for fuel and/or oil leaks.
Engine oil	:	Check gauge for oil presence (sight glass dark).
Engine oil filter impending bypass indicator	:	Check for correct indication.
Engine-transmission drive shaft	:	Condition.
Engine supports (visible area)	:	Condition.
Engine cowling	:	Condition, secured.
Engine air intake screen and chamber	:	Cover removed; free of damage and/or obstructions.
Cooler system belt	:	Condition and security.
Main rotor hub and blades	:	Condition and security.
Main rotor dampers	:	Check for correct charge indication.
Main rotor pitch change links	:	Condition and security.

Upper anticollision light	: Condition.
Swashplate and driving scissors	: Condition and security.
Main transmission and accessories (visible area)	: Condition and leaks.
Transmission	: Filler cap secured.
Transmission oil	: Check correct level. Door secured.

NOTE

If rotor brake (if installed) has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

Servo actuators	: Condition and leaks. Door secured.
Cooler blower air intake	: Free of obstructions.
Airframe (A/F) fuel filter	: Condition and leaks.
Service step	: Secured.
Drain and vent lines	: Free of obstructions.
Lower avionic bay access door	: Secured.
Main landing gear	: Condition, shock strut extension, leaks, tire pressure.
Passenger door jettison window	: Security of window and seal retainer.

Passenger cabin door	:	Condition, cleanliness, window secured.
Cowling and fairings	:	Condition and security.
Copilot cabin door	:	Condition, cleanliness, window secured.
Antenna(s)	:	Condition.
Pitot tube	:	Cover removed, condition and obstructions.
Windshield and roof transparent panel	:	Condition and cleanliness.

AREA N°7 (Cabin interior)

Passenger jettison window	:	Security.
Cabin interior	:	Security of equipment and cargo.
First aid kit	:	On board.
Cabin fire extinguisher	:	Security.
Passenger doors	:	Security. Sliding windows closed, if installed.
Copilot door jettison handle and relevant safety latch	:	Correct position.
Copilot safety belt and inertia reel	:	Condition and belt fastened if seat is unoccupied.
Copilot seat	:	Security.

Copilot flight controls (if installed)	:	Condition and security.
Landing gear lever lock	:	Check in NORMAL position and lockwired.
Lower and lateral transparent panels	:	Cleanliness and integrity.
Copilot door	:	Security. Sliding window closed, if installed.
BATT RELAY circuit breaker	:	In.

NOTE

The above mentioned circuit breaker is accessible from the pilot pedal bay and thus cannot be seen from the on-board seated position.

CVDR TEST (if installed)

BATT	:	ON.
FDR and CVR fault lights	:	After 5 seconds, check both lights go off.
CVDR test	:	Press and hold TEST switch on the CVDR control panel for at least 5 seconds. Check green lamp TEST stays on (the TEST lamp stays on as long as the TEST button is pressed).
BATT	:	OFF.
Pilot door	:	Security. Sliding window closed, if installed
Pilot door jettison handle and relevant safety latch	:	Correct position.
Pilot safety belt and inertia reel	:	Condition.

Pilot seat	: Security.
Pilot flight controls	: Condition and security.
Lower and lateral transparent panels	: Cleanliness and integrity.
Instruments, panels and breakers	: Condition and legibility.
ELT switch on instrument panel (if installed)	: Check in ARM position.

For the following checks connect the d.c. supply.

Accumulators (HYDRAULIC UTILITY NORM/EMER)	: Check for correct precharge.
LH airframe (A/F) fuel filter	: Gently drain while respective fuel pump is operating. Check for by-pass indication, # 1 A/F F FLTR caution message on EDU 1, by pushing red button on filter.

NOTE

Fuel is pressurized, therefore drainage should be carried out by gently pushing red button. Failure to comply with this advice could lead to some fuel being squirted around.

LH fuel drain switch	: Raise the guard and operate the switch to drain the respective fuel sump. Check for fuel dripping from the drain and FUEL DRAIN 1 caution message displayed. Fully lower the guard to switch the drain valve off. Check FUEL DRAIN 1 caution message out and no more fuel dripping.

RH airframe (A/F) fuel filter : Gently drain while respective fuel pump is operating.
Check for by-pass indication, # 2 A/F F FLTR caution message on EDU 1, by pushing red button on filter.

NOTE

Fuel is pressurized, therefore drainage should be carried out by gently pushing red button. Failure to comply with this advice could lead to some fuel being squirted around.

RH fuel drain switch : Raise the guard and operate the switch to drain the respective fuel sump.
Check for fuel dripping from the drain and FUEL DRAIN 2 caution message displayed.
Fully lower the guard to switch the drain valve off. Check FUEL DRAIN 2 caution message out and no more fuel dripping.

Check following systems for correct operation:
— Navigation and anticollision lights.
— Landing light.

Disconnect the d.c. electrical supply.

PILOT'S PREFLIGHT CHECK

(Every flight)

Main and tail rotors tie-down (if present) : Removed.

Nose, fuselage, tail boom exterior, tail fins and skid : Check.

Stabilizers	: Condition and security.
Nose and main landing gear	: Check.
Access doors nose compartment, tail rotor gearbox, baggage compartment, lower avionic bay	: Security.
Cowlings, fairings and relevant access doors	: Condition and security, check for evidence of oil.
Windshield and roof transparent panels	: Condition and cleanliness.
Pitot tubes	: Free of obstructions. Covers removed.
Crew and passenger cabin doors	: Condition, cleanliness, windows secured, doors secured, latches in correct position. Sliding windows closed, if installed.

CVDR TEST (if installed)

BATT	: ON.
FDR and CVR fault lights	: After 5 seconds, check both lights go off.
CVDR test	: Press and hold TEST switch on the CVDR control panel for at least 5 seconds. Check green lamp TEST stays on (the TEST lamp stays on as long as the TEST button is pressed).
BATT	: OFF.

Cabin interior and baggage compartment	: Security of equipment and cargo.
Main and tail rotor blades	: Check.

Air intakes	:	Free of obstructions. Covers removed.
Engine exhausts	:	Check. Covers removed.
Fuel filler cap	:	Secured.
Crew safety belts and inertia reels	:	Condition, belt fastened.
Tail rotor gearbox	:	Check oil level, check for leaks and for filler cap secured.
Transmision oil	:	Check correct oil level. Door secured.

NOTE

If rotor brake (if installed) has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

ENGINE PRE-START CHECK

All switches	:	OFF or CLOSED.
Pedals and seat	:	Adjust.
Seat belt	:	Fasten and adjust.
Nose wheel lock	:	ON (lever up).
Parking brake	:	ON (pull out and turn).
Cyclic stick	:	Centered (or positioned to counter wind) and friction adjusted.

Collective lever	:	Full down and friction adjusted.
Circuit breakers	:	All engaged.
Engine power levers	:	OFF (full aft).
ENG 1 and 2 MODE switches	:	OFF (fully counterclockwise).
LD-SH (Load share) switch	:	As required.

NOTE

The LD - SH switch allows the pilot to keep matched engine torque or TOT, as required.

OEI MODE switch (if installed)	:	NORM, check.
OEI TNG switch (if installed)	:	Off position (centered), check.
Altimeter	:	Set.
STATIC source switch	:	NORM and protected.
LDG GEAR control lever	:	DOWN, check.
SERVO (Main hydraulic)	:	NORM.
SAS 1 and 2 switches	:	OFF.
AUTOTRIM switch	:	OFF.

NOTE

The AUTOTRIM switch should be turned off during ground operation to preclude undesired cyclic movements.

External power (if used)	:	Connect.

NOTE

Be sure that the external power source supplies not less than 28 volt.

NOTE

(Only for helicopters equipped with emergency bus).
The battery is automatically disconnect when an external power source is connected to the helicopter. Check that BATT OFF caution message is displayed on EDU 1 if the external power is used.

BATTery : ON.

GEN BUS 1 and 2	: ON.
GEN 1 and 2	: ON
Force trim	: ON.
Electronic Display Units (EDU 1 and EDU 2)	: Check on.

NOTE

Both EDUs are automatically activated when the helicopter is electrically powered: EDU 1 in CRUISE mode and EDU 2 in MAIN mode.

EDU 1 : Select MENU, press TEST key and check the following test sequences on both EDUs.

NOTE

The TEST function can also be selected on EDU 2.

EDU 1 : The test sequence shall display the CRUISE format with the following data:

N1	97.4%
TOT	820°C
TRQ	100%
NR/N2	101%

The caution and warning messages, stimulated by the DAU, shall be presented in the message area during test routine.

NOTE

During the test the DAU will activate the MASTER WARNING/MASTER CAUTION lights, the ENG FIRE warning message and it will also illuminate the engine power lever grips. If a failure is detected on engine fire and/or fuel low detectors, the caution message FIRE DET and/or F LOW FAIL will appear.

NOTE

It is possible to listen to only one of the ENG FIRE messages due to the short length of the EDU TEST.
The ENG FIRE message of the second engine can be heard, even during the test sequence, by pushing the MASTER WARNING RESET pushbutton.

NOTE

If the helicopter is configured with the engine control panel P/N 109-0900-55-101 on the central console, the test of the EDU will also activate the FIRE warning lights adjacent to the engine mode switches.

EDU 2 : The test sequence shall display the MAIN format with the following data:

ENG OIL pressure	50 PSI
ENG OIL temperature	100°C
XMSN OIL pressure	40 PSI
XMSN OIL temperature	100°C
FUEL pressure	20 PSI
Hydraulic oil pressure	1500 PSI

Fuel quantity, decreasing.

In the advisory area (lower part of the screen) the test sequence shall display:
EDU 1 and EDU 2 software identification number.
DAU-A and -B software identification number.
EDU 1 BIT PASS.
EDU 2 BIT PASS.
DAU-A BIT PASS.
DAU-B BIT PASS.

NOTE

In case of different result of TEST sequence refer to pertinent paragraph in Section 3.

EDU 1 and EDU 2	:	After 10 seconds automatically return to previous selected formats.
EDU 1	:	Select MENU and enter page 2. Verify CH-A and CH-B legends to be green and white respectively.
Aural Warning Generator Test	:	Set AWG switch on TEST and maintain. Check the aural message ″TEST OK″ and after about 6 seconds the vocal alarm operates in the following sequence: ROTOR LOW ENGINE 1 OUT ENGINE 2 OUT ENGINE 1 FIRE ENGINE 2 FIRE WARNING ROTOR HIGH LANDING GEAR 150 FEET

NOTE

The vocal alarm message ″WARNING″ is activated when # 1 ECU FAIL, # 2 ECU FAIL, XMSN OIL PRES, XMSN OIL HOT or BATT HOT are displayed on EDU 1.

NOTE

The test sequence also includes the ″200 FEET″ message. This message is not utilized by the system and it is only produced during the test, being a foreseen option of the Aural Warning Generator.

ADI stand-by	:	Flag retracted, erect (operate ″PULL TO CAGE″ knob, if necessary).
Fuel quantity	:	Check.
RPM switch (on collective control)	:	Set 100%.
Engine power levers	:	Set gently to FLIGHT position.

CAUTION

Avoid rapid movement of engine power levers in order to not damage the internal mechanism.

# 1 ENG GOV switch	:	AUTO.
# 2 ENG GOV switch	:	AUTO.
ENG TRIM toggle switches (on collective control)	:	Verify the operation, then leave the engine power levers in the FLIGHT position

NOTE

Each engine trim toggle switch controls the respective power lever from IDLE to FLIGHT position when in AUTO mode, and from IDLE to MAX position when in MANUAL mode.

NOTE

Both engine power levers should always be operated through the toggle switches located on the collective control stick. They shall be operated manually only in case of failure of the remote control (PLA MOTOR caution message active), or before starting, to position the levers to FLIGHT.

ENG 1 FUEL switch	:	ON - Fuel valve indicator, vertical.

FUEL PUMP 1 switch	:	ON - FUEL PUMP 1 caution message out, check pressure.
ENG 2 FUEL switch	:	ON - Fuel valve indicator vertical.
FUEL PUMP 2 switch	:	ON - FUEL PUMP 2 caution message out, check pressure.

NORMAL ENGINE START

Either engine may be started first.
Either engine may be started using either the auto or the manual mode.

NOTE

It is recommended the normal engine starts be made using the auto mode. For starting procedure in manual mode refer to Section 3.

Collective control	: Flat pitch, check.
Rotor brake (if installed)	: Disengaged (lever full forward).
EDU 1	: Select START page.

ENGINE 1 START

ENG 1 MODE switch	: IDLE.

NOTE

(only for helicopters equipped with emergency bus)
If the battery discharge control box is installed after few seconds the BATT DISCH warning message on EDU1 will be displayed.

NOTE

It is recommended to start the engine to IDLE, nevertheless, if necessary, it is possible to start to FLIGHT by setting the ENG MODE switch directly to FLT.

Gas Producer (N1)	: Note increasing and START legend vertically displayed.
Engine temperature (TOT)	: Note increasing and IGN legend vertically displayed.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

CAUTION

Following aborted start shutdown perform the following procedure before restarting:
- Allow 30 seconds fuel drain period
- Perform a 30 seconds DRY MOTORING RUN.
Refer to Section 1 for engine starter limitations and to DRY MOTORING RUN procedure in this Section.

Engine oil pressure : Check.

NOTE

During cold starting conditions the engine oil pressure can rise up to a transient of 200 PSI. The pressure decreases as oil temperature rises. The oil pressure limits in Section 1, Fig. 1-5 must be respected when main oil temperature is 71°C - 120°C.

CAUTION

Do not apply power until engine oil temperature reaches 10°C.

WARNING

If the main rotor has not begun to rotate when gas producer (N1) reaches 40%, abort the start by setting the ENG MODE switch to OFF.

Engine N°1 starter : Automatically deactivated when N1 is 50%. START and IGN legends automatically suppressed.

Main hydraulic system : When the main rotor begins to rotate, check rise in main hydraulic pressure.

Hydraulic utility system (normal and emergency) : When accumulators are discharged, as main rotor begins to rotate, check pressure rise in both systems and note the activation of MAIN UTIL CHRG and EMER UTIL CHRG caution messages. Note both caution messages are suppressed when systems are charged.

CAUTION

If MAIN UTIL CHRG and/or EMER UTIL CHRG caution messages are not displayed during systems charging, abort and correct trouble.

N°1 engine power turbine speed (N2) : Check stabilized to IDLE speed of 65% ± 1%.

NOTE

If the engine has been started directly to FLT (flight) the N2 will stabilize to 100%.

NOTE

Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

Engine and transmission oil : Check pressure and temperature.

ENG 1 MODE switch : FLT

NOTE

In the starting phase it is suggested to select FLIGHT mode as soon as possible in order to speed up the engine oil heating.

ENGINE 2 START

Repeat above procedure to start engine N°2.

CAUTION

Ensure that the second engine engages as the N2 reaches FLIGHT condition. A nonengaged engine shows positive N2 value and near zero torque. If a nonengagement occurs, shut down the nonengaged engine first. When the nonengaged engine has stopped, shut down the engaged engine. If a sudden (hard) engagement occurs, shut down both engines. Maintenance action is required.

Engine parameters	:	Check within limits.
External power	:	Disconnect (if used). Check BATT OFF caution message out (only for helicopters equipped with emergency bus).
INV 1 and 2 switches	:	ON.
RAD- MSTR switch	:	ON.
Clock	:	Set.
Rotor speed (NR)	:	100% check.

QUICK ENGINE START

This procedure can be followed whenever the situation requires to speed up the takeoff.

CAUTION

It is not advised to accomplish this procedure on battery.

Collective control	:	Flat pitch, check.
Rotor brake (if installed)	:	Disengaged (lever full forward).
EDU 1	:	Select START page.
ENG 1 MODE switch	:	FLT.
ENG 2 MODE switch	:	FLT, when engine 1 gas producer (N1) is above 10%.

CAUTION

Avoid to operate the ENG MODE switches simultaneously.

Gas producer (N1) : Note increasing and START legend vertically displayed.

Engine Temperature (TOT) : Note increasing and IGN legend vertically displayed.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

CAUTION

Following aborted start shutdown perform the following procedure before restarting:
- Allow 30 seconds fuel drain period
- Perform a 30 seconds DRY MOTORING RUN.

Refer to Section 1 for engine starter limitations and to DRY MOTORING RUN procedure in this Section.

Engine oil pressure : Check.

NOTE

During cold starting conditions the engine oil pressure can rise up to a transient of 200 PSI. The pressure decreases as oil temperature rises. The oil pressure limits in Section 1, Fig. 1-5 must be respected when main oil temperature is 71°C - 120°C.

CAUTION

Do not apply power until engine oil temperature reaches 10°C.

WARNING

If the main rotor has not begun to rotate when gas producer (N1) reaches 40%, abort the start by setting the ENG MODE switch to OFF.

Engine N°1 and N°2 starters	:	Automatically deactivated when N1 are 50%. START and IGN legends automatically suppressed.
Main hydraulic system	:	When the main rotor begins to rotate, check rise in main hydraulic pressure.
Hydraulic utility system (normal and emergency)	:	When accumulators are discharged, as main rotor begins to rotate, check pressure rise in both systems and note the activation of MAIN UTIL CHRG and EMER UTIL CHRG caution messages. Note both caution messages are suppressed when systems are charged.

CAUTION

If MAIN UTIL CHRG and/or EMER UTIL CHRG caution messages are not displayed during systems charging, abort and correct trouble.

Engine power turbine speed (N2)	:	Check both stabilized to Flight speed, 100%.

NOTE

Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

Engine and transmission parameters	:	Check within limits.
External power	:	Disconnect (if used). Check BATT OFF caution message out (only for helicopters equipped with emergency bus).
INV 1 and 2 switches	:	ON.
RAD-MSTR switch	:	ON.
Clock	:	Set.
Rotor speed (NR)	:	100% check.

FALSE START

DRY MOTORING RUN

NOTE

The following procedure is used to clear internally trapped fuel and vapor or if there is evidence of a fire within the engine.

ENG GOV switch	:	AUTO or MANUAL.
ENG MODE switch	:	OFF.
Power control lever	:	OFF.
ENG FUEL switch	:	OFF.
Fuel pump	:	OFF.
STARTER GEN IGN circuit breaker	:	Out.

NOTE

When STARTER GEN IGN circuit breaker is out, the pertinent ignition system is deactivated even though the Engine Control Unit will provide the IGN message to the EDU.

Starter pushbutton (on engine power lever)	:	Push and hold as necessary.

NOTE

To operate the starter it is also possible to utilize the ENG MODE switch by setting IDLE.

CAUTION

Refer to Section 1 for engine starting limitations.

Gas producer (N1)	:	Note increasing.
Starter pushbutton	:	Release.
STARTER GEN IGN circuit breaker	:	In.

SYSTEMS CHECK

Engine and transmission oil	:	Pressure and temperature within limits.

NOTE

The transmission oil pressure can be in the cautionary range (yellow band) provided that the oil temperature is below 65°C.

SERVO (Main Hydraulic)	:	NORM, check. Make small clockwise cyclic movements, collective and pedals movements. Pressure drops must be equal for both systems (N°1 and N°2) and they should not exceed 70 psi.

Set 2 OFF. SERVO 2 caution message displayed on EDU 1. Check operation of system N°1 with same cyclic, collective and pedal movements. Pressure drop should not exceed 70 psi and there should be no force increase, discontinuity or cyclic/collective coupling.

Repeat check with system N°1 OFF to check system N°2.

Then set NORM.

NOTE

Tail rotor boost pressure is furnished by system N°1. When system N°2 is being checked, it is normal that the rotor control pedals will be unboosted.

CROSS FEED switch : NORM (bar vertical)

FUEL PUMP 1 : OFF. Note fall in fuel pressure, activation of FUEL PUMP 1 caution message on EDU 1, automatic operation of cross-feed valve (bar horizontal), XFEED advisory message on EDU 2 activated and consequent new increase of fuel pressure.

NOTE

When FUEL PUMP 1 (2) is OFF and crossfeed valve is active, N°2 (1) FUEL quantity box shall appear in magenta.

FUEL PUMP 1 : ON. FUEL PUMP 1 caution message out and crossfeed valve automatically closed (bar vertical) and XFEED advisory message suppressed.

FUEL PUMP 2 : OFF. Note fall in fuel pressure, activation of FUEL PUMP 2 caution message on EDU 1, automatic operation of cross-feed valve (bar horizontal), XFEED advisory message on EDU 2 activated and consequent new increase of fuel pressure.

FUEL PUMP 1 : OFF. Note fall in fuel pressure, activation of FUEL PUMP 1 and 2 caution messages, cross-feed valve still open (bar horizontal) and XFEED message still present. Verify operation of engine driven fuel pumps.

FUEL PUMP 1 and 2 : ON. FUEL PUMP 1 and 2 caution messages out and crossfeed valve automatically closed (bar vertical) and XFEED message suppressed.

PITOT heat : ON (singly), check current peak on loadmeter and PITOT 1 (2) HEAT advisory message displayed; then OFF.

INVerter 1 : OFF. Check INV 1 caution message displayed.
Check for proper reading (115 V) on both AC systems on EDU 2 AUX display.

INVerter 1	: ON. Check INV 1 caution goes out.
INVerter 2	: OFF. Check INV 2 caution message displayed. Check for proper reading (115 V) on both AC systems on EDU 2 AUX display.
INVerter 2	: ON. Check INV 2 caution goes out.
Cyclic stick	: Friction fully unlocked, freedom of movement.
COMPASS switch	: MAG.

Set communication and navigation frequencies as required and check audio panel.

Marker beacon lights	: Test.
IVSI	: Needle near zero. Vertical speed selector as desired.
Altimeters	: Set and check.

Gyro Compass (HSI instrument):

Compass flag	: Retracted.
Compass heading	: Consistent with MAG COMP HDG.
Select heading	: As desired.
Select nav. course	: As desired.

ADI stand-by	: Erected and flag retracted. Verify the consistency with the ADI.
ADI	: Erected and flag retracted. Verify the consistency with the ADI stand-by.
Hydraulic system	: Check.

Flight director (if installed):

SBY button	: Press and check - SBY pushbutton: illuminated - ALL mode controller lights: on - GA and DH lights on ADI: on. - FD flag on ADI: in view.
GA button	: Press and check - Pushbutton (ON): illuminated. - GA light on ADI: on. - Pitch Cue: visible and centered. - Roll Cue: visible and centered. - Collective Cue: visible and below center.
HDG button	: Press and check - Pushbuton (ON): illuminated. - Roll Cue should follow movement of heading, select ″bug″ (HDG on HSI instrument).
FD/SBY switch (on cyclic stick)	: Press - FD should return to stand-by.

Radio Altimeter (if installed):

Indication	: Zero altitude.
OFF flag	: Retracted.
DH SET selector	: Set 50 ft.
TEST switch	: Press and maintain - DH light: out. - Needle 100 ft.
TEST switch	: Release - DH light: on.

HELIPILOT system:

SAS 2 switch	: SAS 2: - SAS 2 OFF caution message out. - ATTD OFF caution message out.

Cyclic stick motion must not cause HELIPILOT indicator motion.

SAS 2 switch	: OFF: - SAS 2 OFF caution message displayed.
SAS 1 switch	: SAS 1: - SAS 1 OFF caution message out. - ATTD OFF caution message out.

Cyclic stick motion must not cause HELIPILOT indicator motion.
Pedal motion will cause YAW HELIPILOT indicator motion.

SAS 1 switch	: OFF: - SAS 1 OFF caution message displayed.
SAS 1 switch	: SAS 1: - SAS 1 OFF caution message out.
SAS 2 switch	: SAS 2: - SAS 2 OFF caution message out.
ATTD HOLD switch	: ATTD HOLD: - ATTD OFF caution message out.
FD controller (if installed)	: Stand-by (SBY).

COUPLED/DECOUPLED switch (if installed)	:	As desired.
Autotrim (if installed):		
AUTOTRIM switch	:	AUTOTRIM.
HELIPILOT indicators	:	Check centered.
Ammeters	:	Within limits and sharing load.
Voltmeters	:	Within limits.
Cockpit lights	:	As required.
External lights	:	Check and leave as required.
PITOT 1 and 2 switches	:	ON.
PITOT 1(2) FAIL caution light (if installed)	:	Push to test.
PITOT 1 and 2 switches	:	Leave as required.

CAUTION

Turn Pitot heat on for flight in visible moisture and rain regardless of ambient temperature.

Power levers	:	Check FLIGHT position (no #1 (#2) PLA message displayed).
RPM switch	:	Set 102%.
NR/N2 speed	:	102% stabilized, check.
Parking brake	:	Off. Check PARK BRK ON caution message out.
Caution and warning messages	:	Check none.

TAXIING

Nose wheel lock	: OFF.
Collective and cyclic	: Increase slowly the collective then move the cyclic stick forward moderately to start movement.
Pedal brakes	: Check operation.
Pedal control	: As required to select the direction.
Collective and pedal brakes	: To reduce speed and to stop, lower the collective and apply pedal brakes.
Nose wheel lock	: ON.

NOTE

If the nose wheel is not aligned forward it will be self-centered and locked as soon as the helicopter lifts off.

TAKE-OFF

HOVER TAKE-OFF

Collective pitch	: Increase slowly and lift the helicopter to a hover.

CAUTION

The helicopter is free of ground resonance.
However if, for some reason, ground resonance should occur, lift the helicopter free of the ground immediately. If unable to become airborne, lower collective and shut-down engines.

Tail rotor pitch	: Apply as necessary to maintain direction.
Flight instruments	: Check.
Engine instruments	: Within limits.
Main hydraulic system instruments	: Within limits.
Utility hydraulic system instruments	: Within limits.

NOTE

In hover the helicopter has a slight inclination to the left. During lift-off correct as necessary.

Cyclic and collective	: Apply as necessary to accelerate and climb smoothly, reaching take-off power at approximately 60 knots.
Landing gear	: UP (by 200 ft AGL). Check light sequence: - Green light off - Red light on (retraction) - Red light off (retracted and locked)

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	: Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	: Pressure within limits.

FTR pushbutton (cyclic)	: Trim as desired for attitude reference changes during hover and climb out.
RPM switch	: Set 100%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering conditions.

NR/N2 speed	: 100% stabilized, check.

ROLLING TAKE-OFF

Collective and cyclic	: Apply collective and forward cyclic as necessary to obtain forward speed on the ground. Apply collective as necessary to become airborne. Accelerate to 60 Knots and to desired climb.
Tail rotor pitch	: Apply as necessary to maintain direction.
Flight instruments	: Check.
Engine instruments	: Within limits.
Main hydraulic system instruments	: Within limits.
Utility hydraulic system instruments	: Within limits.
Landing gear	: UP (by 200 ft AGL). Check light sequence: - Green light off - Red light on (retraction) - Red light off (retracted and locked)

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	:	Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	:	Pressure within limits.
FTR push-button (cyclic)	:	Trim as desired for attitude reference changes during climb out.
RPM switch	:	Set 100%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering condition.

NR/N2 speed	:	100% stabilized, check.

IN FLIGHT

NOTE

During IFR flight, the pilot is recommended to cross check the flight instruments indication for consistency, including the ADI stand-by indicator.

Collective	:	Adjust as necessary to keep engine parameters within limits.
Load Share (LD-SH) switch	:	As required.

NOTE

The LD-SH switch allows the pilot to keep matched engine torque or TOT, as required.

Airspeed	:	Maintain within limits shown on airspeed placard.
Landing light	:	OFF, if used. LANDING LT ON advisory message out.

Pitot heat : As required.

CAUTION

Turn Pitot heat on for flight in visible moisture and in rain regardless of ambient temperature.

Altitude : As desired.

NOTE

Refer to applicable operating rules for high altitude oxygen requirements.

Landing gear lever : DOWN when flying below 200 ft height AGL.

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

HYD UTIL CHRG caution message : Displayed during landing gear operation.
Out after the landing gear is locked.

Utility hydraulic systems : Pressure within limits.

HELIPILOT must be monitored and re-centered by depressing FTR button.

FLIGHT DIRECTOR OPERATION (IF INSTALLED)

Attitude mode COUPLED/DECOUPLED (if installed)

The Flight Director may be used either coupled or uncoupled.
A COUPLED/DECOUPLED switch is provided for that purpose.

When flying decoupled, the pilot flies the helicopter to satisfy Flight Director Commands as programmed on the Flight Director Control Panel.
When in this mode, the pilot should either center the commands or go to standby on his mode controller.
When flying coupled, the aircraft is automatically flown in pitch and roll to satisfy Flight Director Commands.

VOR operation

NAV MODE

CAUTION

VOR approaches in NAV mode are prohibited.
VOR approaches are permitted in VOR APP mode.

The NAV mode is designed for cross-country operation and localizer/glide slope approaches and any radial capture in this mode should be effectuated more than 15 NM from the station.
When flying near a VOR station the flight director senses the ″Cone of Confusion″ and automatically switches to Over Station Sensor Logic at which time course deviation signal is ignored and the system will fly course arrow heading for a fixed time to allow for station passage. This system can cause significant course deviation prior to termination of Over Station Sensor Logic; therefore it is recommended to use the HDG Mode near the station.

VOR APP MODE

The VOR APP Mode is designed for operating radial captures less than 15 NM from the station and to permit VOR approaches.
Very near to the station oscillations could occur around the tracked radial.
If the oscillations are undesirable, use HDG mode.

GO-AROUND MODE

CAUTION

When operating in GO-AROUND mode, use caution to avoid exceeding engine power limits when satisfying the collective command bar.

Missed approach - attitude hold mode

Apply power to approximately maximum continuous power.

For approach speeds below 80 KIAS no initial pitch-up attitude adjustment is required. Re-center HELIPILOT indicators with FTR button.

Approach angle vs airspeed and rate of descent (Sea level - ISA)

Kts IAS (CAS) / R/D Ft/min	60 (67)	70 (74)	80 (81)	100 (100)	120 (116)
500	4.2	3.8	3.5	2.8	2.4
750	6.3	5.7	5.2	4.2	3.7
1000	8.4	7.6	7.0	5.6	4.9

NOTE

The steepest demonstrated approach slope is 8.4 degrees.

APPROACH AND LANDING

RPM switch : Set 102%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering conditions.

NR/N2 speed	:	102% stabilized, check.
Landing gear lever	:	DOWN. Check light sequence: - Red light on (extension) - Red light off - Green lights on (extended and locked). LANDING GEAR caution message suppressed, if previously activated.

NOTE

When descending below 150 feet AGL vocal message ″ONE FIFTY FEET″ is active regardless landing gear status. This message is suppressed if AWG switch is set to REGRADE.

CAUTION

Check the lever to be correctly engaged in the DOWN position to avoid inadvertent retraction of the landing gear.

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	:	Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	:	Pressure within limits.
Nose wheel lock	:	ON (lever up).
Parking brake	:	OFF (ON if landing on a slope).
External light	:	As required.
Approach path	:	During the approach, reduce the airspeed gradually with the cyclic and apply collective as necessary so that the helicopter will arrive at a hover in ground effect in such a manner that, in the event of an engine failure along the approach, it is possible to make a safe landing.

After reaching a hover, descend slowly to the surface.
After ground contact lower the collective lever to the bottom stop, unless taxiing is desired, in which case maintain to control the helicopter.

CAUTION

The helicopter is free of ground resonance.
However if, for some reason, ground resonance should occur, lift the helicopter free of the ground immediately. If unable to become airborne, lower collective and shut-down engines.

Pedal brakes	: As necessary
Parking brake	: ON (OFF if ground taxiing is desired).
Nose wheel lock	: OFF if ground taxiing is desired.

NOTE

A run-on landing is possible on smooth surfaces. In this case the forward speed must be below 40 Knots and ground contact should be made at minimum vertical speed. Once ground contact is made, collective can be reduced, maintaining enough to control the helicopter. Forward speed can then be reduced to a taxiing speed or the helicopter may be brought to stop.

SHUTDOWN

Collective lever	: Check fully down.
Cyclic stick	: Centered and trimmed.
Pedals	: Centered.

NOTE

Do not apply collective in this phase and during subsequent rotor deceleration, particularly in windy conditions.
Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

ENG 1 and 2 MODE switches : Set to IDLE then to OFF.

NOTE

If necessary the engine may be shut-down directly from FLT.

Fuel pumps 1 and 2 : OFF. FUEL PUMP 1 and 2 caution messages displayed.

CAUTION

During shutdown check that the N1 speed decelerates freely. Note any abnormal noise or rapid rundown and take corrective action as required per Maintenance Manual.

WARNING

If there is any evidence of fire within the engine after shut-down, perform immediately a DRY MOTORING RUN as described in this Section.

ENG l and 2 FUEL switches : OFF. Fuel valve indicators horizontal.

CROSS FEED switch : CLOSED. Crossfeed indicator vertical.

Pitot heat : OFF.

Cockpit lights	:	OFF.
External lights	:	OFF.
Landing light	:	OFF, if used.
Miscellaneous switches	:	OFF.
RAD-MSTR (Radio Master) switch	:	OFF.
Power levers	:	OFF (full aft).

NOTE

Move both engine power levers to OFF before turning off the electrical power in order to avoid fuel spillage on the ground.

BATTery and GENerators 1 and 2	:	OFF.
INVerters 1 and 2	:	OFF.

NOTE

(with EDU software version 007 and DAU software version 006 only). Wait until N1 is 0% before turning off the electrical power to permit DCU to memorize the LCF counts.

POST FLIGHT CHECK

Landing gear lever lock : NORMAL and lockwired.

If conditions require, perform the following:

Pitot, intake and exhaust covers : Installed.

NOTE

The pitot tube covers must be installed at least 5 minutes after pitot heat has been switched off. The engines exhaust ducts covers must be installed at least 30 minutes after engine shut down. Refer to Maintenance Manual for additional information.

FLIGHT HANDLING CHARACTERISTICS

STEEP APPROACHES AND VERTICAL DESCENT MANOEUVRES

Low speed steep approaches (up to 20 kts) and vertical descent manoeuvres should be performed with a rate of descent not exceeding 900 ft/min.

SECTION 3

EMERGENCY AND MALFUNCTION PROCEDURES

TABLE OF CONTENTS

Table of Contents (Cont.)

LIST OF ILLUSTRATIONS

SECTION 3
EMERGENCY AND MALFUNCTION PROCEDURES

INTRODUCTION

The following procedures contain the indications of equipment or system failure or malfunction, the use of emergency features of primary and backup systems, and appropriate warnings and explanatory notes.

All corrective action procedures listed herein assume the pilot gives first priority to aircraft control and a safe flight path.

NOTE

The helicopter should not be operated following any emergency landing or shutdown until the cause of the malfunction has been determinated and corrective maintenance action taken.

DEFINITIONS

The following items indicate the degree of urgency in landing the helicopter.

LAND IMMEDIATELY	: The urgency of the landing is paramount. The primary consideration is to assure the survival of the occupants.
LAND AS SOON AS POSSIBLE	: Land without delay at the nearest suitable area (i.e., open field) at which a safe approach and landing is reasonably assured.

LAND AS SOON AS PRACTICAL : The duration of the flight and landing site are at the discretion of the pilot. Extended flight beyond the nearest approved landing area is not recommended.

WARNING SYSTEM

Many of the malfunctions described in this Section are indicated through the display of red warning messages or yellow caution messages on the EDU 1. Aircrew attention is drawn to the warning/cautions by flashing master warning/caution lights.
The red warnings are accompanied by an audio warning tone and by a vocal warning.
Green advisory and cyan status messages are also displayed on the EDU 2.

Whenever a warning or caution message activates, appropriate actions should be taken to deal with the indicated malfunction, after which the associated master warning/caution light(s) should be cancelled either by pressing the MASTER RESET pushbutton on the collective stick either by pressing the red and yellow PUSH TO RESET lights on the cockpit panel.
By cancelling the red or yellow master light(s) also the audio and vocal warnings are cancelled and the light(s) are reset for future indications.

The EDUs present a specific area partitioned in three columns, each one capable to display up to 3 lines of 14 characters each, for caution, warning, advisory and status messages.
The order of priority of the messages is:

WARNING	red color
CAUTION	yellow color
ADVISORY	green color
STATUS	cyan color

The ″last-in″ message goes always on the top of the relative screen area and displace the existing list down.
When the messages are listed on more pages, the scroll function can be activated through the rocker switch with vertical arrows positioned on the right side of the EDU control panel.

Warning and Caution messages remain presented until the causative condition has been corrected.
Advisory messages have precedence over status annunciations and remain presented until the causative condition no longer exists.
The exceedances requiring the pilot's attention (i.e. Transmission overtorque, Engine overspeed) are displayed as cautions on EDU 1.

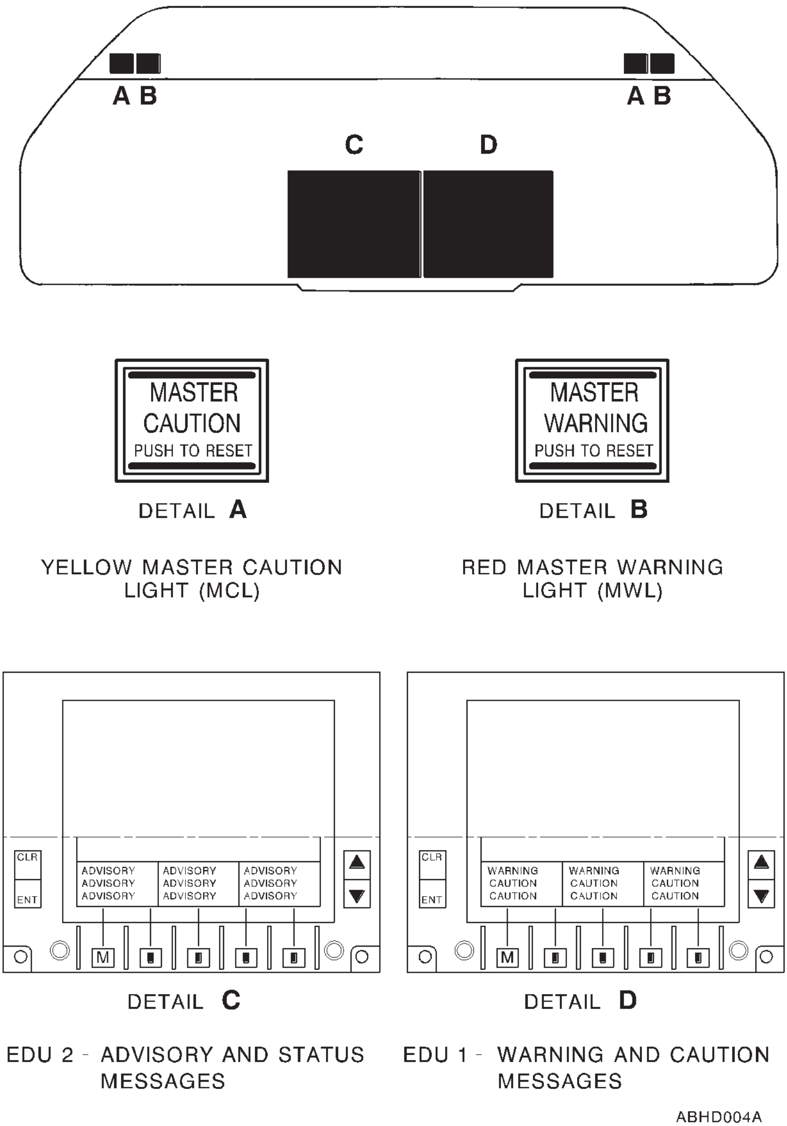


Figure 3-1 - Cockpit Layout of the Warning and Caution System.

MASTER CAUTION AND MASTER WARNING LIGHTS

Panel wording	Fault condition	Corrective action
MASTER CAUTION	See caution message on EDU 1.	Reset after malfunction acknowledgment and relative action.
MASTER WARNING	See warning message on EDU 1.	Reset after malfunction acknowledgment and relative action.

WARNING CAUTION AND AURAL MESSAGES

NOTE

Warning, caution and aural messages are generally accompanied by specific written messages on EDU 1.
Refer to warning and caution message list for fault description and corrective action.

Vocal alarm	EDU message
ROTOR LOW and cabin acoustic signal	ROTOR LOW
ROTOR HIGH	ROTOR HIGH
ENGINE ONE OUT	ENG 1 OUT
ENGINE TWO OUT	ENG 2 OUT
ENGINE ONE FIRE	ENG 1 FIRE
ENGINE TWO FIRE	ENG 2 FIRE
WARNING	XMSN OIL PRES
WARNING	XMSN OIL HOT
WARNING	BATT HOT
WARNING	# 1 ECU FAIL
WARNING	# 2 ECU FAIL
WARNING	# 1 OIL PRES
WARNING	# 2 OIL PRES
WARNING	BATT DISCH (*)
WARNING	ELECTRICAL (**)
LANDING GEAR	LANDING GEAR
ONE FIFTY FEET	-

(*) Only helicopters equipped with battery discharge control box.
(**) Only helicopters equipped with emergency bus.

NOTE

″ONE FIFTY FEET ″ intends to draw pilot attention that the helicopter is descending below 150 ft AGL.
The message is suppressed if the AWG switch is set to RE-GRADE.

WARNING MESSAGES (RED)

EDU message	Fault condition	Corrective action
ROTOR LOW	Rotor RPM low. Rotor RPM between 80% and 95.5% (power on) or 80% and 89.5% (power off).	Use collective to adjust RPM.
ROTOR HIGH	Rotor RPM high. Rotor RPM above or equal to 105.5% (power on) or 110.5% (power off).	Use collective to adjust RPM.
ENG 1 (2) OUT	N1 abnormally low, below 35%, on engine 1 (2). Probable engine failure.	Lower collective and adjust rotor RPM. Shut down affected engine; proceed to a suitable landing site and land. (See the pertinent paragraphs of this Section).
ENG 1 (2) OUT (*) On ground	Engine #1(2) in IDLE during take-off or probable engine failure.	Lower collective to MPOG. If EDU and aural message disappear, on affected engine set ENG MODE switch to FLT position and carry out normal TAKE-OFF procedure. If EDU and aural messages persist, carry out normal SHUTDOWN procedure.

(*) Only helicopters equipped with installation P/N 109-0012-11.

EDU message	Fault condition	Corrective action
ENG 1 (2) FIRE	Fire in engine compartment.	Shutdown affected engine. Land as soon as possible. (See the pertinent paragraphs of this Section).
XMSN OIL PRES	Transmission oil pressure below minimum limit.	Reduce power. Land as soon as possible.
XMSN OIL HOT	Transmission oil temperature above maximum limit.	Reduce power. Land as soon as possible.
# 1 (# 2) OIL PRES	Engine oil pressure below minimum or above maximum.	Check the oil pressure indication, if outside the upper or lower limit, set # 1 (# 2) engine to MANUAL mode and reduce power of the affected engine. Check again the oil pressure and, if necessary, shutdown the engine preceeding as per paragraph ″FAILURE OF ONE ENGINE″. Land as soon as practicable.
BATT DISCH	Output voltage of both generators below 26.5 V dc.	See table ″ELECTRICAL POWER FAILURE (Helicopters equipped with emergency bus)″ in this section.

NOTE

Only helicopters equipped with battery discharge control box.

EDU message	Fault condition	Corrective action
BATT HOT	Battery temperature exceeding limits.	Switch battery off. Land as soon as possible.
# 1 (# 2) ECU FAIL	Critical hardware failure of the engine control unit.	Automatic reversion to manual mode of the control of the affected engine. Land as soon as practical (See pertinent paragraph in this Section).
And only for helicopters equipped with emergency bus, also:		
ELECTRICAL	Failure of both generators.	See table ″ELECTRICAL POWER FAILURE (Helicopters equipped with emergency bus)″ in this Section.

NOTE

When the ELECTRICAL warning message is activated, # 1 and # 2 DC GEN cautions are suppressed.

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
# 1 (# 2) OIL HOT	Engine oil temperature above maximum.	Check the oil temperature indication, if above the maximum shut down the engine and proceed as per paragraph ″FAILURE OF ONE ENGINE″.
# 1 (# 2) OIL CHIP	Metal particles in the engine oil.	If possible reduce power. The engine should be shut down as soon as practical to prevent damage.
XMSN OIL CHIP	Metal particles in the main transmission oil.	Reduce power. Land as soon as possible.

EDU message	Fault condition	Corrective action
TGB OIL CHIP	Metal particles in the tail rotor gearbox oil.	Reduce power. Land as soon as possible.
FUEL PUMP 1 (2)	Fuel pumps 1 (2) failed.	Affected fuel pump OFF. Check the automatic operation of crossfeed valve (bar horizontal). XFEED advisory message on EDU 2 activated and consequent new increase of fuel pressure.

CAUTION

With the crossfeed valve open, the fuel contained in the lower tank 1 (2) (110 kg) is unusable.

# 1 (# 2) A/F F FLTR	Airframe fuel filter partially clogged.	Proceed with flight. Correct trouble prior to next flight.
# 1 (# 2) FUEL LOW	Fuel quantity is low.	When each engine is drawing fuel from its tank, the remaining flight duration is approximately 15 minutes from caution message activation. When both engines are drawing fuel from one tank, the remaining flight duration is approximately 6 minutes from caution message activation.
# 1 (# 2) F LOW FAIL	Fuel low sensor failure. No fuel low indication.	Keep under control the fuel quantity and proceed with flight.

EDU message	Fault condition	Corrective action
# 1 (# 2) FUEL FLTR	Engine fuel filter partially clogged.	Proceed with flight. Correct trouble before next flight.
FUEL DRAIN 1 (2)	Fuel drain valve 1 (2) open.	Close the valve before refuelling and/or taking off.
# 1 (# 2) DC GEN	See the applicable ″ELECTRICAL POWER FAILURE″ table in this Section.	
BUS TIE	See the applicable ″ELECTRICAL POWER FAILURE″ table in this Section.	
EXT PWR ON	External power connected to the aircraft. Door open.	Disconnect as necessary. Check door closed.
BATT OFF	Battery disconnected.	Check battery switch position. If the switch is ON and the message is still present, proceed with flight. Correct trouble before next flight.

NOTE

BATT OFF caution message is applicable only to helicopters equipped with emergency bus.
When the external power is connected, the battery is automatically disconnected and the BATT OFF caution message is activated.

SERVO 1 (2)	Failure of one of the two servo hydraulic systems.	Check if servo pressure is low. Land as soon as practical (refer to paragraph ″SERVO HYDRAULIC SYSTEM MALFUNCTION″ in this Section).

EDU message	Fault condition	Corrective action
# 1 (# 2) OVSPD	N2 overspeed control system in operation.	Check the engine instruments and warning devices. If RPM drops, shut down the affected engine and proceed as per paragraph ″FAILURE OF ONE ENGINE″. Land as soon as practical.

EDU message	Fault condition	Corrective action
# 1 (# 2) OVSP DET	Failure of the over-speed protection system in the ECU.	Proceed with flight. Make corrective action before next flight.
MAIN UTIL PRES	Failure of the main utility hydraulic system (pressure below the minimum).	Check main utility system pressure indicator. Proceed with flight. Emergency brake system (PARK AND EMERG BRAKE) may be required at landing.

WARNING

The utility hydraulic system services the landing gear extension system, the wheel brakes, the wheels lock and the rotor brake, if installed.
There is sufficient pressure in the emergency accumulator for one extension of the landing gear, after which there is sufficient pressure to operate the emergency brakes. (The parking brake handle is used for the emergency brake system and does not provide differential braking). The toe brakes are not connected to the emergency system.

EMER UTIL PRES	Failure of the emergency utility hydraulic system.	Proceed with flight. Correct trouble before next flight.
MAIN UTIL CHRG	Solenoid valve locked open. The main utility system is operative.	Proceed with flight. Correct trouble before next flight.

EDU message	Fault condition	Corrective action
EMER UTIL CHRG	Solenoid valve locked open. The emergency utility system is operative.	Proceed with flight. Correct trouble before next flight.

NOTE

MAIN UTIL CHRG and EMER UTIL CHRG do not activate the Master Caution Light.

CABIN DOOR	Pilot's and/or passenger's door not correctly closed.	Close door correctly before flight.
BAG DOOR	Baggage compartment door not correctly closed.	Close door correctly before flight.
PARK BRK ON	Parking brake ON.	Check if handle is set to off. If the message is still present, do not make a running landing and do not taxi.
LANDING GEAR	Landing gear retracted below 200 ft.	Extend landing gear.
SAS 1 OFF	No power to the system.	Check SAS 1 engage switch, breakers and VG 1 caution message.

EDU message	Fault condition	Corrective action
SAS 2 OFF	No power to the system.	Check SAS 2 engage switch, breakers and VG 2 caution message.
VG 1	Gyro not erected, loss of power to gyro.	Check breaker, refer to stand-by attitude indicator.
VG 2	Gyro not erected, loss of power to gyro.	Check breaker.
ATTD OFF	No attitude retention. Probable switch failure.	Check ATTD HOLD switch.
INV 1 (2)	See the applicable ″ELECTRICAL POWER FAILURE″ table in this Section.	
AWG FAIL	Aural warning system failure.	Proceed with flight. Correct trouble before next flight.
XMSN OVTRQ	Transmission overtorque. Dual engine operation: transmission torque equal or greater than 101%. One engine inoperative: transmission torque equal or greater than 143%.	Lower collective lever to maintain torque within limits and not exceeding the transient.

EDU message	Fault condition	Corrective action
# 1 (# 2) DCU	Data Collection Unit failure: TOT and torque trim values for engine calibration not available. Fault message active only on ground before engine starting.	Correct trouble before flight.
#1 (#2) ECU MAINT	Engine Control Unit non critical failure. Fault message active only on ground before engine starting.	Correct trouble before flight.
# 1 (# 2) FIRE DET	Engine fire detection system inoperative.	Land as soon as practical. Correct trouble before next flight.
# 1 (# 2) PMS	Power Management Switch failure.	Proceed with flight. Utilize the engine power lever to shutdown the engine, if necessary. Correct trouble before next flight.
# 1 (# 2) PLA	Engine Power Lever out of FLIGHT position detent. Caution message active in AUTO mode only.	Set the engine power lever correctly.

EDU message	Fault condition	Corrective action
#1 (#2) PLA MOTOR	Engine power lever remote control inoperative.	Proceed with flight. If manual mode operation is required, move engine power lever manually, paying attention to engine response. Correct trouble before next flight.
# 1 (# 2) GEN CTL	Generator control box breaker tripped out.	Correct trouble before operating the engine starter.
#1 (#2) OVSPD TEST	N2 overspeed control test circuit failure. Fault message active only on ground.	Correct trouble before next flight.
# 1 (# 2) ECU DATA	Loss of data from Engine Control Unit.	Check the engine instruments for correctness of backup indications. Correct trouble before next flight.

EDU message	Fault condition	Corrective action
# 1 (# 2) TOT LIMITER	TOT limiter function of Engine Control Unit not operative and failure in thermocouples system. Engine temperature matching inoperative. Possible loss of TOT reading.	Verify that LD-SH switch is set to TORQUE. If TOT reading is still present, proceed with flight paying attention not to exceed TOT limits. If TOT reading is missing, proceed with flight paying attention not to exceed 92% N1 and 10000 ft pressure altitude. Avoid as far as possible to bleed air from the engine (i.e. for heater, ECS, and EAPS). Correct trouble before next flight.
TRQ LIMITER	Torque limiter function of Engine Control Unit not operative or no communication between Engine Control Units. Engine torque matching inoperative.	Proceed with flight paying attention not to exceed torque limits. Verify that LD-SH switch is set to TOT. Correct trouble before next flight.
RPM SELECT	RPM switch inoperative.	Proceed with flight paying attention that RPM value will remain fixed at the value selected before failure. (See paragraph ″RPM SWITCH MALFUNCTION″).

EDU message	Fault condition	Corrective action
IDS	Failure of the Integrated Display System. Possible degradation in system function.	Try to identify the faulty information and take the pertinent corrective action. Refer to paragraph ″INTEGRATED DISPLAY SYSTEM FAILURE″ in this Section.
DAU MISCMP-P	Some primary data from one or both channels of Data Acquisition Unit invalid. Possible degradation in system function.	Try to identify the faulty information and take the pertinent corrective action. Refer to paragraph “INTEGRATED DISPLAY SYSTEM FAILURE” in this Section.

CAUTION LIGHT (YELLOW)

(If installed)

Panel wording	Fault condition	Corrective action
PITOT 1(2) FAIL	Associated pitot heater failure	Exit pitot icing conditions as soon as possible. Switch the affected system off.

ENGINE FAILURES

FAILURE OF ONE ENGINE

INDICATIONS:

EDU1	: ENG OUT warning message displayed and vocal alarm «ENGINE OUT» activated when N1 below 35% and/or N1 decreasing more than 20% per second.
Engine power turbine speed (N2)	: Rapidly decreasing.
Torque	: Rapidly decreasing.
Engine temperature (TOT)	: Rapidly decreasing.

PROCEDURE

Collective	: Reduce as required to maintain rotor RPM.
Airspeed	: As required for single engine flight.

CAUTION

Do not attempt to restart the engine if the cause of the failure has not been ascertained.

NOTE

If the engine restart is to be attempted refer to ENGINE RESTART IN FLIGHT procedure in this Section.

ENG MODE switch (affected engine)	: OFF (full counterclockwise).
Engine power lever (affected engine)	: OFF (full aft).

ENG FUEL switch (affected engine)	:	OFF. Fuel valve indicator horizontal.
Fuel pump (affected engine)	:	OFF. FUEL PUMP caution message displayed.
CROSS FEED switch	:	CLOSED: Crossfeed indicator vertical.
GEN switch (affected engine)	:	OFF. Check load on operating generator (See AUXILIARY page on EDU 2).

Land as soon as practical.

NOTE

Land maintaining some forward speed if terrain permits.

NOTE

It is recommended that single engine operation be carried out in auto mode.
In manual mode engine may only achieve take-off rating.

FAILURE OF TWO ENGINES

INDICATIONS:

EDU1	:	ENG 1 OUT and ENG 2 OUT warning messages displayed and vocal alarm «ENGINE ONE OUT» and «ENGINE TWO OUT» activated when N1 below 35% and/or N1 decreasing more than 20% per second.
Rotor speed	:	Rapidly decreasing. ROTOR LOW warning message displayed on EDU 1, vocal alarm «ROTOR LOW» and cabin acoustic signal activated.

Engine power turbine speed (N2)	: Rapidly decreasing (both engines).
Torque	: Rapidly decreasing (both engines)
Engine temperature (TOT)	: Rapidly decreasing (both engines).

PROCEDURE

Collective	: Reduce as required. Establish autorotation at 70 to 75 Kts speed.
Landing gear lever	: DOWN.

CAUTION

Do not attempt to restart the engine if the cause of the failure has not been ascertained.

NOTE

If time before landing permits to attempt engine restart refer to ENGINE RESTART IN FLIGHT procedure in this Section.

ENG 1 MODE and ENG 2 MODE switches	: OFF (full counterclockwise).
Engine power levers	: OFF (full aft).
ENG 1 FUEL and ENG 2 FUEL switches	: OFF. Fuel valve indicators horizontal.
Fuel pumps 1 and 2	: OFF. FUEL PUMP 1 and FUEL PUMP 2 caution messages displayed.

– CROSS FEED switch : CLOSED.
Crossfeed indicator vertical.

– GEN 1 and GEN 2 switches : OFF.

Perform an autorotative landing.

NOTE

At approximately 100 ft to 70 ft above the ground depending on weight, initiate a flare to reduce the rate of descent and the forward speed; at approximately 10 ft bring the helicopter to a near level attitude.
As the helicopter settles, apply collective at approximately 4 feet to cushion the touchdown.
Land with forward speed if the terrain permits.

ENGINE HOT START

A hot start is caused by an accumulation of fuel in the combustion chamber, and/or delay in light-off, causing flames to be emitted from the exhaust and/or an overtemperature of the TOT.

PROCEDURE

Abort the start of affected engine as follows:

ENG MODE switch : OFF (full counterclockwise).

NOTE

If engine fire occurs during a starting in manual mode, set the power lever of the affected engine to OFF.

Fuel pump : OFF.

ENG FUEL switch	: OFF.
CROSS FEED switch	: CLOSED.

Perform a DRY MOTORING RUN as described in Section 2.

ENGINE RESTART IN FLIGHT

Each engine may be started using either the auto or manual mode. It is recommended that normal engine starts be made using the auto mode.

CAUTION

When the cause of an engine flameout is suspected to be mechanical, do not attempt a restart.

Engine restart in flight with ECU operative (AUTO mode)

Airspeed	: 70 Kts.
ENG MODE switch (inoperative engine)	: OFF, check.
Engine power lever (inoperative engine)	: FLIGHT.
ENG GOV switch (inoperative engine)	: AUTO, check.
ENG FUEL switch (inoperative engine)	: ON.
FUEL PUMP switch (inoperative engine)	: ON.
CROSS FEED switch	: CLOSED, check.

GEN switches	: Both ON.

CAUTION

The pilot shall engage the starter and the ignition only when N1 is below 20%.

ENG MODE switch (inoperative engine)	: IDLE.

NOTE

It is recommended to start the engine to IDLE, neverthless, if necessary, it is possible to start to FLIGHT by setting the ENG MODE switch directly to FLT.

Gas producer (N1)	: Note increasing and START legend vertically displayied.
Engine temperature (TOT)	: Note increasing at light-off and IGN legend vertically displayed.

CAUTION

Until N1 is below 35% be ready to abort the starting sequence in case that the TOT overcomes the starting limit (half red dot).

NOTE

The engine governing logic becomes operative and sets the TOT limit to 650°C during start when N1 is in the range from 35% to 54%.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

Engine oil pressure : Check.

Engine starter : Automatically deactivated when N1 is 50%. START and IGN legends automatically suppressed.

Engine power turbine speed (N2) : Check stabilized to idle speed of 65% ± 1%.

NOTE

If the engine has been started directly to FLIGHT the N2 will stabilize to 100%.

CROSS FEED switch : NORM (bar vertical).

Engine restart in flight with ECU inoperative (MANUAL mode)

Airspeed : 70 Kts.

ENG MODE switch (inoperative engine) : OFF, check.

Engine power lever (inoperative engine) : OFF, check.

ENG GOV switch : MANUAL.

NOTE

In presence of an ECU failure the engine control system reverts and operates in manual mode independently from the ENG GOV switch position.
However it can be convenient to set the ENG GOV switch to MANUAL for congruence with the mode condition.
The ENG GOV switch shall mandatorily be on MANUAL, only when the manual mode is selected voluntarily (i.e. for training).

ENG FUEL switch (inoperative engine) : ON.

FUEL PUMP switch (inoperative engine) : ON.

CROSS FEED switch : CLOSED, check.

GEN switches : Both ON.

CAUTION

The pilot shall engage the starter and the ignition only when N1 is below 20%.

ENG MODE switch (inoperative engine) : IDLE.

Engine power lever (inoperative engine)	:	Move to FLIGHT.
Starting button	:	Push and hold. START and IGN legends vertically displayed beside N1 and TOT scales on the EDU panel.
Gas producer (N1)	:	Note increasing.
Engine temperature (TOT)	:	Note increasing at light-off, then keep it under control by moving the engine power lever as necessary to prevent exceeding the TOT transient.

NOTE

If engine hangs below idle (less than 54% N1), slowly move power lever forward beyond FLIGHT position, until the engine accelerates, monitoring TOT, N1 and NR. If the engine does not accelerate shutdown the engine by setting OFF the power lever and release the starting button.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limits

shutdown the engine by setting OFF the pertinent power lever and release the starting button.

Starting button	:	Release when gas producer (N1) reaches 50%. START and IGN legends suppressed.

Engine power control	:	Set the power as required by using either the engine power lever or the ENG TRIM toggle switch on the collective control.
Engine oil pressure	:	Check.
CROSS FEED switch	:	NORM (bar vertical)

ENGINE CONTROL UNIT MALFUNCTIONS

There are two levels of control system malfunctions: ″non-critical″ and ″critical″.
″Non-critical″ malfunction is defined every fault condition which allows the control system to continue operating in AUTO mode with minimal operational impact.
Major non-critical malfunctions (i.e. PMS failure, thermocouple failure, etc.) are announced by specific caution messages (i.e. #1 PMS, #1 TOT LIMITER, TRQ LIMITER, etc.) in order to make the pilot aware of the relative degradation and to let him take the necessary precautions.
All other non-critical malfunctions are announced by ECU MAINT message only on ground, before starting or after engine shutdown, while in flight they are automatically accommodated by the logic of the system to provide continued AUTO mode function.
″Critical″ malfunction is defined every fault condition, announced by ECU FAIL warning message and MAN legend, that cannot be accommodated and having as a consequence the ECU reversion to MANUAL mode.
In this case the FMM tracking system will maintain N1 ±2%.
Following an ECU FAIL the engine with the serviceable ECU will maintain rotor governing within its limits.
Both ECU malfunctions, ″non-critical″ and ″critical″, can be cleared by cycling the ENG GOV switch from AUTO to MANUAL and back to AUTO if the source of the fault has been eliminated.

NOTE

On ground it may be necessary to cycle the ENG GOV switch twice to clear the ECU malfunctions.

If the failure message (i.e. #1 PMS, #1 ECU MAINT, etc. caution messages, or #1 ECU FAIL warning message) is still present proceed as follows:

Non critical malfunction.

(Both engines remain in AUTO mode)

Proceed with flight and correct trouble before next flight.

Critical malfunction.

(The affected engine reverts to MANUAL mode)

Land as soon as practical flying with one engine in MANUAL mode.

NOTE

In case of critical malfunction the ECU reversion to MANUAL mode is independent from ENG GOV switch position.

Any small change in power will be compensated through the serviceable ECU engine.

Power lever movement (through ENG TRIM switch) of the engine in MANUAL is only required for large changes in power.

(See also the paragraph ″ENGINE OPERATION IN MANUAL MODE″).

AUTO MODE TO MANUAL MODE TRANSFER

Transfer from AUTO mode to MANUAL mode can occur as a result of:

- Pilot request by ENG GOV switch on collective control.
- ECU reversion to MANUAL due to AUTO mode control system critical fault.

NOTE

In presence of an ECU failure the engine control system reverts and operates in MANUAL mode independently from the ENG GOV switch position.

However it can be convenient to set the ENG GOV switch to MANUAL for congruence with the mode condition.

CAUTION

Be sure that PLA caution message is not displayed before selecting MANUAL mode.

NOTE

PLA caution message is active only when operating in AUTO mode.

The MANUAL mode condition is indicated to the pilot by the activation of MAN legend on EDU 1.

ENGINE OPERATION IN MANUAL MODE

Following a reversion to MANUAL mode, voluntary or automatic (MAN legend displayed on EDU 1), the FMM tracking system will fix the N1 speed to the last commanded value (fail fix mode of operation) maintaining the engine torque within ± 50 SHP.
The MANUAL mode control characteristic will vary depending on the engine condition at the mode transfer.

One engine in MANUAL mode and one in AUTO mode

It is recommended to set the ″MANUAL″ engine to a suitable fixed power related to the particular flight condition and let the other engine ECU maintain rotor speed. Monitor engine parameters to respect the limitations.
Any change of fuel flow and consequently of N1 speed may be accomplished through power lever movements.
Each power lever is motorized and it should be operated by the pilot through the relative ENG TRIM toggle switch placed on collective control.
Land as soon as practical.

Both engines in MANUAL mode

WARNING

Training with both engines in MANUAL mode should be avoided.

In case of critical malfunction of both ECUs (double failure), the pilot shall control both engines in MANUAL mode.
Each FMM tracking system will fix the N1 speed at the last commanded value.
Land as soon as practical and perform a running landing.

TO REACH THE LANDING SITE

ENG TRIM toggle switches	:	Operate to adjust the torque of one engine to about 50% of the total torque required for the desired airspeed. Adjust the other torque to maintain rotor speed between 95% and 102% N2/NR and desired flying parameters.

NOTE

To simplify the manual control of the engines the pilot is suggested to adjust only one engine torque by operating only one toggle switch.
Avoid torque settings too close to engine operating limits.

Airspeed	:	It is suggested to respect OEI V_{NE} limitations.
Manoeuvres	:	Avoid any manoeuvre requiring large and rapid changes of torque.

CAUTION

Take care of N2 response when lowering the collective in order to avoid engine overspeed.

APPROACH AND LANDING

Landing gear lever	:	DOWN.
Utility hydraulic system	:	Pressure within limits, check.
Nose wheel lock	:	ON (lever up).
Parking brake	:	OFF.
External light	:	As required.
Approach	:	Set one engine torque at 40% if at high gross weight and at 30% if at low gross weight. Use the other engine to establish a slightly steep approach (35 ±10 Kts IAS and 700 fpm maximum rate of descent at 102% NR). Control the landing point by changing the airspeed and minimize any further power adjustment.
Landing	:	Flare the helicopter to minimize the ground speed at landing. Use the collective to cushion the touchdown and, if necessary, leave the NR to drop down to 90%.

After the touchdown simultaneously lower the collective and reduce both engines to IDLE.

CAUTION

Take care of N2 response when lowering the collective in order to avoid engine overspeed.

Apply toe brakes.

Shutdown : Refer to paragraph ″ENGINE SHUTDOWN IN MANUAL MODE″.

MANUAL MODE TO AUTO MODE TRANSFER

Transfer to AUTO mode can only be accomplished by pilot demand.
The following procedure is recommended.
Transfer should be performed in steady state flight condition.

EDU 1 : Ensure ECU FAIL warning message is not displayed.

ENG GOV switch : Set AUTO.
MAN legend on EDU 1 suppressed.

Engine parameters : Check for correct engine operation in AUTO mode.

NOTE

The MANUAL/AUTO transfer may results in up to 10% torque transients.

Engine power lever : FLIGHT, check.
PLA caution message not displayed.

NOTE

If AUTO mode was selected when engine power lever was not in FLIGHT position, set the power lever correctly as soon as possible.

NOTE

PLA caution message is active only when operating in AUTO mode.

ENGINE STARTING IN MANUAL MODE (ON GROUND)

Following an ECU failure, flight may be initiated only in order to fly (VFR) without passengers on board to a repair facility.

NOTE

It is recommended to start the engine in manual mode only in case that starting in auto mode is not possible.

Before proceeding to start the engine in manual mode, perform an ECU power OFF-ON reset trying to clear all faults.
If critical faults are not cleared (ECU FAIL warning and MAN legend still displayed) proceed as follows.
Perform ENGINE PRE-START CHECK as per Section 2.

Collective control : Flat pitch, check.

Rotor brake (if installed) : Disengaged (lever full forward)

ENG GOV switch : MANUAL.

NOTE

In presence of an ECU failure the engine control system reverts and operates in MANUAL mode independently from the ENG GOV switch position.
However it can be convenient to set the ENG GOV switch to MANUAL for congruence with the mode condition.
The ENG GOV switch shall mandatorily be on MANUAL, only when the manual mode is selected voluntarily (i.e. for training).

EDU 1	:	Select START page.
ENG MODE switch (inoperative engine)	:	IDLE.
Engine power lever (inoperative engine)	:	FLIGHT, check.
Starting button	:	Push and hold. START and IGN legends vertically displayed beside N1 and TOT scales on the EDU.
Gas produce (N1)	:	Note increasing.
Engine temperature (TOT)	:	Note increasing at light-off, then keep it under control by moving the engine power lever as necessary to prevent exceeding the TOT transient.

NOTE

If engine hangs below idle (less than 54% N1), slowly move the power lever forward beyond FLIGHT position, until the engine accelerates, shutdown the engine by setting OFF the power lever and release the starting button.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit

shutdown the engine by setting OFF the pertinent power lever and release the starting button.

Starting button	:	Release when gas producer (N1) reaches 50%. START and IGN legends suppressed.
Engine power control	:	Set the power as required by using either the engine power lever or the ENG TRIM toggle switch on the collective control.
Engine oil pressure	:	Check.

Continue as per normal starting procedure.

ENGINE SHUTDOWN IN MANUAL MODE

This procedure will apply in case of ECU failure and consequent reversion to MANUAL mode (ECU FAIL warning and MAN legend displayed).

ENG GOV switch	:	MANUAL.

NOTE

In presence of an ECU failure the engine control system reverts and operates in MANUAL mode independently from the ENG GOV switch position.
However it can be convenient to set the ENG GOV switch to MANUAL for congruence with the mode condition.
The ENG GOV switch shall mandatorily be on MANUAL, only when the manual mode is selected voluntarily (i.e. for training).

ENG TRIM toggle switch	:	Operate to set N2 speed to 65% ± 1%.

Engine power lever : Set to OFF.

NOTE

If necessary the engine may be shutdown directly from FLT.

ENG MODE switch : OFF (full counterclockwise)

Fuel pump : OFF. FUEL PUMP caution message displayed.

CAUTION

During shutdown check that the N1 speed decelerates freely. Note any abnormal noise or rapid rundown and take corrective action as required per Maintenance Manual.

CAUTION

If there is any evidence of fire within the engine after shutdown, perform immediately a DRY MOTORING RUN as described in Section 2.

ENG FUEL switch : OFF. Fuel valve indicator horizontal.

CROSSFEED switch : CLOSED. Crossfeed indicator vertical.

RPM SWITCH MALFUNCTION

In case of RPM switch malfunction with consequent impossibility of changing NR/N2 selection (100% or 102%), the ECU will maintain the engine at the NR/N2 speed selected before the failure.
Proceed with flight taking care of the selected NR/N2 speed.

NOTE

In case of NR/N2 switch locked at 102% the pilot can complete the flight provided that the airspeed is limited to 130 KIAS.

Correct trouble before next flight.

DRIVE SYSTEM FAILURES

TAIL ROTOR FAILURE

A tail rotor drive failure results in a loss of yaw control with a consequent yaw to the right, which is the more rapid, the less the forward speed is and the more the torque level is.
In fact the vertical fin produces an antitorque component which is a function of the forward speed and which permits to control the helicopter in low torque conditions.
A tail rotor drive failure may be accompanied by noise, vibration or oscillation in the tail section.
The action to be taken is different whether the helicopter is in hover or in forward flight.
In both cases the landing should be made at the lowest possible power or even with both engines out.
While a tail rotor drive failure in hovering is immediately detected, the same failure may be less evident in cruise.
In case of an event making suspect of a possible trouble in the tail section when flying at cruise speed, proceed as follows:

Altitude	:	Maintain the cruise altitude.
Airspeed and pedal control	:	Reduce gradually to 60 KIAS and meanwhile check the helicopter response to pedal control displacement and the appearence of any anomalous vibrations and/or noise.

If the check confirms the tail rotor failure, proceed as per the paragraph ″COMPLETE LOSS OF TAIL ROTOR IN CRUISE″, otherwise carry out the following further check:

Airspeed	:	Maintain 60 KIAS.
Collective	:	Slowly raise, to increase the antitorque demand, as close as possible to maximum continuous power and let the helicopter climb.
Pedal control	:	Check the pedal effectiveness to control the yaw and any anomalous vibration and/or noise.

If the pedal effectiveness does not result sufficient to control the yaw, proceed as per the paragraph ″COMPLETE LOSS OF TAIL ROTOR IN CRUISE″.
If on the contrary nothing seems to confirm a tail rotor failure, continue flight.

COMPLETE LOSS OF TAIL ROTOR CONTROL IN HOVERING

PROCEDURE

Collective	:	Lower as necessary to reduce yaw rate and to land.

If height above ground permits:

Engine power levers	:	Slam to OFF.

NOTE

Operate engine power levers for a quick reaction either in AUTO mode or in MANUAL mode.

Collective : As necessary to cushion the touchdown.

NOTE

A slight rotation can be expected on touchdown.

COMPLETE LOSS OF TAIL ROTOR IN CRUISE

PROCEDURE

Collective : Lower as necessary to eliminate yaw to the right.

Airspeed/power : As necessary in order to reach a suitable landing site.

NOTE

Increased power will allow an extension of the flight path, however an increase in power necessitates an increase in speed to prevent the helicopter from turning.

Landing gear lever : DOWN.

Nose wheel lock : ON.

Parking brake : OFF.

On reaching the point of intended landing:

Collective and cyclic : As needed to control lateral-directional stability, touch down point, speed and attitude.

If the above procedure is not suitable to the landing site:

ENG MODE switches : OFF (full counterclockwise)

Engine power levers	:	OFF.
Fuel pumps	:	OFF.
ENG FUEL switches	:	OFF.
CROSS FEED switch	:	CLOSED.
GENerators 1 and 2	:	OFF.
BATTery	:	OFF (except as needed in night flight).

Perform an autorotative landing, into wind, maintaining forward speed if terrain permits.

SYSTEM FAILURES

SERVO HYDRAULIC SYSTEM MALFUNCTION

The helicopter is equipped with two independent servo hydraulic systems for cyclic and collective pitch control.
Either system can deliver adequate power to control the helicopter.
The tail rotor pedals are boosted only by N.1 system.
N.2 system delivers power to the utility hydraulic system.

PRESSURE LOSS IN N.1 MAIN HYDRAULIC SYSTEM

INDICATIONS:

EDU 1	:	SERVO 1 caution message and MASTER CAUTION light activated.
EDU 2	:	MAIN HYD 1 system pressure below minimum.

Pedals : They tend to move to tail rotor traction zero position (right pedal about three centimeters more forward than left pedal).

NOTE

The pedal movement is plus or minus evident depending from flight condition. It increases the quicker the pressure drops and more left pedal movement is required.

Control force on pedals increases to approximately 35 kg in hovering and less in forward flight.

Cyclic and collective : No change in control force.

PROCEDURE

Airspeed : Reduce gradually avoiding pull up manoeuvers; recommanded maximum 90 Kts and 25° bank angle in order to maintain acceptable control loads.

Flight controls : Avoid rapid movements.

Failed servo hydraulic system : OFF.

NOTE

The hydraulic system is designed in order to prevent the deactivation of the operative system when one of SERVO caution messages is activated.

Land as soon as practical.

WARNING

Following the pressure loss in N.1 main hydraulic system avoid landing and/or operating in conditions which require a high degree or maneuverability (i.e. avoid operating in enclosed areas, helicopter heading not windward, in particular with wind from the right).

PRESSURE LOSS IN N.2 MAIN HYDRAULIC SYSTEM

INDICATIONS:

EDU 1	: SERVO 2 caution message and MASTER CAUTION light activated.
EDU 2	: MAIN HYD 2 system pressure below minimum.
Cyclic, pedal and collective	: No change in control force.

Proceed as per failure of N.1 main hydraulic system.
Land as soon as practical.

NOTE

In the event of pressure loss in N.2 main hydraulic system there is sufficient pressure in the accumulator of the normal utility hydraulic system to operate the toe brakes.

JAMMING OF A SERVO VALVE

The helicopter is equipped with three hydraulic servo actuators, ″tandem″ type, on main rotor controls (cyclic and collective) and with one hydraulic servo actuator, single body type, on tail rotor control (pedal).

The jamming of a servo valve of main rotor servo actuators cannot be detected when both main hydraulic systems are functioning as the second hydraulic system will guarantee the full efficiency of the actuators.
Such kind of failure can be noticed only during the system check on ground before takeoff (refer to Section 2).
The jamming of the servo valve of tail rotor servo actuator will result in an increase of pedal control force of approximately 35 kg in hovering and less in forward flight.

PROCEDURE

Airspeed : Reduce gradually avoiding pull up manoeuvers; recommended maximum 90 Kts and 25° bank angle in order to maintain acceptable control loads.

Land as soon as practical.

WARNING

Following the loss of tail rotor servo actuator avoid landing and/or operating in conditions which require a high degree of maneuverability (i.e. avoid operating in enclosed areas, helicopter heading not windward, in particular with wind from the right).

LANDING GEAR MALFUNCTION

If the landing gear will not extend and/or will not lock:

PROCEDURE

EDU 2 (AUXILIARY) : Check hydraulic utility systems pressure within limits.

Landing gear lever lock : Set to EMERG (Turn clockwise breaking the lockwiring).

Landing gear lever	:	DOWN EMERG (full down) Check light sequence: - Red light on (extension) - Red light off - Green light on (extended and locked) LANDING GEAR caution message suppressed, if previously activated.

If the landing gear will not extend and/or lock also after the emergency procedure, proceed to a hovering at a height sufficient for ground personnel to check the gear.

ELECTRICAL POWER FAILURE

(Helicopters not equipped with emergency bus)

EDU message	Fault condition	Refer to para.
# 1 (# 2) DC GEN caution	d.c. generator failure	FAILURE OF A GENERATOR
BUS TIE caution	Bus tie malfunction. Check:	
	GEN BUS (1 or 2) switch OFF	FAILURE OF GENERATOR BUS # 1 OR # 2
	GEN BUS (1 and 2) switches OFF	FAILURE OF BOTH GENERATOR BUSSES
# 1 (# 2) DC GEN, BUS TIE cautions.	d.c. generator and bus tie malfunction. Check:	
	GEN BUS # 2 switch OFF	FAILURE OF GENERATOR 1 AND GENERATOR BUS # 2

EDU message	Fault condition	Refer to para.
	GEN BUS # 1 switch OFF	FAILURE OF # GENERATOR 2 AND GENERATOR BUS # 1
# 1 # 2 DC GEN, BUS TIE, INV1 (2), SAS1 (2) cautions.	d.c. gen and bus tie malfunction. Check:	
	GEN BUS # 2 switch OFF	FAILURE OF GENERATOR 1 GENERATOR BUS # 2
	GEN BUS # 1 switch OFF	FAILURE OF GENERATOR 2 AND GENERATOR BUS # 1.
# 1 and # 2 DC GEN cautions	Failure of both generators	FAILURE OF BOTH GENERATORS.
# 1 and # 2 DC GEN, BUS TIE, INV 1, INV 2, SAS 1 OFF, SAS 2 OFF cautions.	Failure of both generators and both generators busses.	FAILURE OF BOTH GENERATORS AND BOTH GENERATOR BUSSES
INV 1 or 2 cautions	Failure of an inverter.	FAILURE OF AN INVERTER
SAS 1 (2) OFF-VG1 (2) caution	Failure of an a.c. bus system N1 (N. 2)	FAILURE OF AN A. C. BUS SYSTEM (115V, 26V)

Failure of a generator

(Helicopters not equipped with emergency bus)

This failure is indicated by # 1 (#2) DC GEN caution message on EDU 1.

PROCEDURE

GEN 1 (2) : Reset, then ON. If generator does not remain ON, set switch to OFF. Power is supplied to all loads by the remaining generator.

GEN BUS 1 (2) : ON, check.

CAUTION

Check ammeter does not exceed limits.

Failure of generator bus # 1 or # 2

(Helicopters not equipped with emergency bus)

This failure is indicated by the BUS TIE caution message on EDU 1.

PROCEDURE

GEN BUS 1 or 2 : ON. If engagement of one switch trips the other, reengage the tripped switch and reset other switch to OFF.

If failure confirmed, no loads are lost.

Failure of both generator busses

(Helicopters not equipped with emergency bus)

This failure is indicated by the BUS TIE caution message on EDU 1.

PROCEDURE

GEN BUS 1 and 2 : ON, check.

If failure confirmed, no loads are lost.

NOTE

Following a failure of both Generator Busses the battery will continue to supply the following loads: DAU ch A, EDU Primary, ICS PLT.

CAUTION

In the event of subsequent double generators failure the flight can be continued on battery power for maximum of 25 minutes. Land as soon as possible.

Failure of #1 dc generator and generator bus #2

(Helicopters not equipped with emergency bus)

This failure is indicated by # 1 DC GEN, BUS TIE caution messages on EDU 1.

PROCEDURE

GEN 1 : Reset, then ON.

GEN BUS 2 : ON, check.

If failure is confirmed, proceed as follows:

GEN BUS 1 : OFF. All loads of Generator Bus #1 are lost. All loads of Essential Bus #1 are retained.

NOTE

In this event all loads on #2 Bus are maintained by Generator 2.

After switching off the GEN BUS 1 the following message will be indicated: INV 1, SAS 1 OFF.

NOTE

Following a failure of both Generator Busses the battery will comntinue to supply the following loads: DAU ch A, EDU Primary, ICS PLT.

CAUTION

In the event of subsequent Generator 2 failure the flight can be continued on battery power for maximum of 25 minutes. Land as soon as possible.

Failure of #2 dc generator and generator bus #1

(Helicopters not equipped with emergency bus)

This failure is indicated by # 2 DC GEN, BUS TIE caution messages on EDU 1.

PROCEDURE

GEN 2 : Reset, then ON.

GEN BUS 1 : ON, check.

If failure is confirmed, proceed as follows:

GEN BUS 2 : OFF. All loads of Generator Bus # 2 are lost. All loads of Essential Bus #2 are retained.

NOTE

In this event all loads on #1 Bus are maintained by generator 1.

After switching off the GEN BUS 2 the following message will be indicated: INV 2, SAS 2 OFF.

NOTE

Following a failure of both Generator Busses the battery will continue to supply the following loads: DAU ch A, EDU Primary, ICS PLT and VHF1 (if radio master switch incorporates the ground position GND).

CAUTION

In the event of subsequent Generator 1 failure the flight can be continued on battery power for maximum of 25 minutes. Land as soon as possible.

Failure of generator 1 or 2 and generator bus #1 or #2

(Helicopters not equipped with emergency bus)

The failure of Generator 1 and # 1 DC BUS is indicated by # 1 DC GEN, BUS TIE, INV 1, SAS 1 OFF caution messages on EDU 1.
All loads of Generator Bus #1 and both pilot and copilot EADI (if installed) are lost. All loads of Essential Bus #1 are retained.

NOTE

In this event all loads on #2 Bus are maintained by Generator 2.

The failure of Generator 2 and #2 DC BUS is indicated by # DC GEN, BUS TIE, INV 2, SAS2 OFF caution messages on EDU 1.
All loads of Generator Bus #2 and both pilot and copilot EHSI (if installed) are lost. All loads of Essential Bus #2 are retained.

NOTE

In this event all loads on #1 Bus are maintained by Generator 1.

PROCEDURE

GEN 1 (2) : Reset, then ON. - If #1 (#2) DC GEN caution light remain ON, select switch to OFF (loss of generator).

GEN BUS (1 and/or 2) : ON. If engagement of one switch trips the other, reengage the tripped switch and reset other switch to OFF (loss of affected bus).

NOTE

If both switches remaining engaged, power is supplied to all loads by remaining generator.

CAUTION

Check ammeter does not exceed limits.

Failure of both generators

(Helicopters not equipped with emergency bus)

This failure is indicated by #1 and #2 DC GEN caution messages on EDU 1.

PROCEDURE

GEN 1 and 2 : Reset, then ON.

If both generators are inoperative, continue flight on battery
Land as soon as possible.

GEN BUS 1 and 2 : ON, check.

WARNING

After both generators failure the battery will carry the electrical loads, for the basic configuration, for approximately 10 minutes. The battery operating time can be extended by selectively reducing system load.
Flight duration can be extended to 30 minutes provided that only the following electrical loads are selected:

- EDU1, EDU2, DAU channel A and B
- Inverter (1 or 2)
- Fuel pumps
- Fuel valves
- ICS Pilots and AWG
- ADI stand-by
- Force trim
- VHF (2 minutes Tx and 13 minutes Rx)
- Helipilot 1
- Landing light(s) or Searchlight
- Wander lights
- Trasponder

NOTE

The landing light(s) or the searchlight (if installed) can be turned ON for only one minute before landing.
Simultaneous operation of the landing light(s) and the searchlight is prohibited.

Failure of both generators and both generators busses

(Helicopter not equipped with emergency bus)

The failure of both Generators followed by either the simultaneous failure of both Generator Busses or their internal disconnection is indicated by #1 and #2 DC GEN, BUS TIE, INV 1, INV 2, SAS 1 OFF, SAS 2 OFF, caution messages on EDU 1.

All loads are lost except DAU ch A, EDU Primary, ICS PLT and VHF1 (if radio master switch incorporates the ground position).

PROCEDURE

GEN 1 and 2 : Reset, then ON.

GEN BUS 1 and 2 : ON, check generators still operative.

If both generators and generator busses are inoperative, continue flight on battery.
Land as soon as possible.

NOTE

After both generators and generator busses failure the following cautions will be displayed:
- IDS
- #1 and #2 FUEL LOW
- FUEL PUMP 1 and 2
- VG1 and 2
- #1 and #2 PLA
- #1 and #2 FIRE DET
- LANDING GEAR

Failure of an inverter

(Helicopter not equipped with emergency bus)

In the event of a main inverter failure (N° 1 or N° 2) the remaining inverter will automatically power the 115V and 26V busses of the failed inverter.
The failure is indicated by INV 1 (2) and SAS1 (2) OFF caution messages.
Attempt to reconnect the failed inverter.

If INV 1 (2) caution message does not extinguish (loss of affected inverter), proceed as follows:

Failed inverter : OFF.

SAS (1 or 2) switch : ON.

Failure of an a.c. bus system - 115V and 26V

(Helicopter not equipped with emergency bus)

The failure of an a.c. bus system (115V and 26V) is indicated by the simultaneous activation of:

SAS 1 OFF - VG 1 message: for a.c. bus system N° 1
SAS 2 OFF - VG 2 message: for a.c. bus system N° 2

WARNING

In the event of N° 1 a.c. bus system failure the following instruments and equipment will continue to operate:

- VOR/ILS N° 2
- ADF
- VG 2
- SAS 2

WARNING

In the event of N° 2 a.c. bus system failure the following instruments and equipment will continue to operate:

- VOR/ILS N° 1
- VG 1
- SAS 1
- GYRO COMPASS
- HSI

ELECTRICAL POWER FAILURE

(Helicopters equipped with emergency bus)

EDU message	Fault condition	Refer to para.
#1 (#2) DC GEN caution	d.c. generator failure	FAILURE OF A GENERATOR
BUS TIE caution	Bus tie malfunction. Check:	
	GEN BUS 1 or 2 switch OFF	FAILURE OF GENERATOR BUS #1 OR #2
	GEN BUS 1 and 2 switches OFF	FAILURE OF BOTH GENERATOR BUSSES
#1 (#2) DC GEN, BUS TIE cautions and BATT DISCH warning (if battery discharge control box is installed)	d.c. generator and bus tie malfunction. Check:	
	GEN BUS #2 switch OFF	FAILURE OF GENERATOR 1 AND GENERATOR BUS #2
	GEN BUS #1 switch OFF	FAILURE OF GENERATOR 2 AND GENERATOR BUS #1
#1 DC GEN, BUS TIE cautions.	d.c. generator and bus tie malfunction. Check:	FAILURE OF GENERATOR 1 OR 2 AND GENERATOR BUS #1 OR #2
#2 DC GEN, BUS TIE, INV 2 SAS 2 OFF cautions	GEN BUS #1 (#2) switch OFF	
ELECTRICAL and BATT DISCH warnings (if battery discharge control box is installed)	Failure of both generators.	FAILURE OF BOTH GENERATORS

EDU message	Fault condition	Refer to para.
BUS TIE, INV 2, SAS1 OFF, SAS 2 OFF caution messages and ELECTRICAL and BATT DISCH warnings (if battery discharge control box is installed).	Failure of both generators and both generators busses.	FAILURE OF BOTH GENERATORS AND BOTH GENERATORS BUSSES

NOTE

The BATT DISCH warning is only applicable to helicopter equipped with the battery discharge control box.

INV 1 or 2 caution	Failure of an inverter	FAILURE OF AN INVERTER
SAS1 (2) OFF - VG1 (2) cautions	Failure of an a.c. bus system N° 1 (N° 2)	FAILURE OF AN A.C. BUS SYSTEM (115V 26V)

Failure of a generator

(Helicopter equipped with emergency bus)

This failure is indicated by #1 (#2) DC GEN caution message on EDU 1.

PROCEDURE

GEN 1 (2) : Reset, then ON. If generator does not remain ON, set switch to OFF. Power is supplied to all loads by the remaining generator.

GEN BUS 1 (2) : ON, check.

CAUTION

Check ammeter does not exceed limit.

Failure of generator bus #1 or #2

(Helicopter equipped with emergency bus)

This failure is indicated by the BUS TIE caution message and after few seconds, if the battery discharge control box is installed, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN BUS (1 or 2) : ON. If engagement of one switch trips the other, reengage the tripped switch and reset other switch to OFF.

If failure is confirmed, no loads are lost.

:

If both switches trip, set BAT-Tery switch to OFF. (Loss of battery bus).

NOTE

In the event of switching off the battery the DAU ch A and EDU Primary are inoperative. All BUS #1 and #2 loads are supplied by N° 1 and N° 2 generators.

Failure of both generator busses

(Helicopter equipped with emergency bus)

This failure is indicated by the BUS TIE caution message and after few seconds, if the battery discharge control box is installed, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN BUS (1 and 2)	: ON, check.

If failure is confirmed, proceed as follows:

Pilot ICS panel switch	: Select FAIL position
BATTery	: OFF: This sheds the battery bus causing the loss of DAU ch A and EDU Primary. BATT OFF caution and ELECTRICAL warning messages will be indicated.
SAS 1 or 2 switch	: ON.

NOTE

In the event of switching off the battery the DAU ch A and EDU Primary are inoperative.
All Busses #1 #2 loads are supplied by N° 1 and N° 2 generators.

Failure of generator 1 and generator bus #2

(Helicopter equipped with emergency bus)

This failure is indicated by #1 DC GEN, BUS TIE caution messages and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN 1	: Reset, then ON.
GEN BUS 2	: ON, check.

If failure is confirmed, proceed as follows:

Pilot ICS panel switch	: Select FAIL position
BATTery switch	: OFF. All loads of Generator Bus #1 and Emergency Bus #1 are lost. All loads of Essential Bus #1 are retained.

NOTE

In this event all loads on #2 Bus are maintained by Generator 2.

After switching off the BATTery the following message will be indicated: INV 1, SAS 1 OFF, BATT OFF cautions, ELECTRICAL warning as well as DAU ch A and EDU Primary will be lost.

Failure of generator 2 and generator bus #1

(Helicopter equipped with emergency bus)

This failure is indicated by #2 DC GEN, BUS TIE caution messages and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN 2	: Reset, then ON.

GEN BUS 1 : ON, check.

If failure is confirmed, proceed as follows:

Pilot ICS panel switch : Select FAIL position

BATTery switch : OFF. All loads of Generator Bus #2 and Emergency Bus #2 are lost. All loads of Essential Bus #2 are retained.

NOTE

In this event all loads on #1 Bus are maintained by Generator 1.

After switching off the BATTery the following message will be indicated: INV 2, SAS 2 OFF, BATT OFF cautions, ELECTRICAL warning as well as DAU ch A and EDU Primary will be lost.

Failure of generator 1 or 2 and generator bus #1 or #2

(Helicopter equipped with emergency bus)

The failure of Generator 1 and # 1 DC BUS is indicated by #1 DC GEN, BUS TIE, caution messages on EDU 1.
All loads of Generator Bus #1 and both copilot EFIS (if installed) are lost. All loads of Essential Bus #1 and Emergency Bus #1 are retained.

NOTE

In this event all loads on #2 Bus are maintained by Generator 2.

NOTE

The landing light(s) or the searchlight (if installed) can be turned ON for only one minute before landing.
Simultaneous operation of the landing light(s) and the searchlight is prohibited.

The failure of Generator 2 and # 2 DC BUS is indicated by #2 DC GEN, BUS TIE, INV2, SAS2 OFF caution messages on EDU 1.
All loads of Generator Bus #2 and pilot EHSI (if installed) are lost. All loads of Essential Bus #2 and Emergency Bus #2 are retained.

NOTE

In this event all loads on #1 Bus are maintained by Generator 1.

NOTE

The landing light(s) or the searchlight (if installed) can be turned ON for only one minute before landing.
Simultaneous operation of the landing light(s) and the searchlight is prohibited.

PROCEDURE

GEN 1 (2)	: Reset, then ON. - If #1 (#2) DC GEN caution light remains ON, select switch to OFF (loss of generator).
GEN BUS (1 and/or 2)	: ON. If engagement of one switch trips the other, reengage the tripped switch and reset other switch to OFF (loss of affected bus).

NOTE

If both switches remain engaged, power is supplied to all loads by remaining generator.

CAUTION

Check ammeter does not exceed limits.

Failure of both generators

(Helicopter equipped with emergency bus)

This failure is indicated by ELECTRICAL warning message and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN 1 and 2 : Reset, then ON.

If both generators are inoperative, the flight can be continued on battery power for maximum of 30 minutes provided the following procedures must be accomplished:

NOTE

During the load shedding procedure all SAS will be lost until SAS 1 is reselected ON. It is recommended that prior to load shedding, speed is reduced to between 100 and 120 Kts in order to reduce pilot work load.

Cockpit wander light (pilot) : ON (at night).

GEN BUS 1 : OFF.

GEN BUS 2 : OFF.

SAS 1 switch : ON (SAS 1 OFF caution message suppressed).

NOTE

If the Generator Busses are not disconnected the battery is capable of supplying power for a further 10 minutes maximum, providing that VHF radio is not used in transmission for more than 3 minutes.

NOTE

Following the above selection sequence, the emergency busses (N° 1 and N° 2) will supply power to the following indications and systems:

EMERGENCY BUS N°1
- Fuel quantity N° 1
- Fuel pump N° 1
- Fuel shut-off valve N° 1
- Fuel crossfeed
- Engine 1 fire detector system
- N° 1 engine compartment fire extinguisher system
- Force trim
- Intercom system, copilot
- Attitude engage
- SAS 1 system
- ADI pilot (EHSI-74 version)
- HSI standby (EHSI-74 version)
- Landing gear position indicator
- Landing light(s)
- Emergency floats
- Inverter #1

EMERGENCY BUS N° 2
- Fuel quantity N° 2
- Fuel pump N° 2
- Fuel shut-off valve N° 2
- Engine 2 fire detector system
- N° 2 engine compartment fire extinguisher system
- Hydraulic system
- Eng. Gov. CTL
- EDU N°2
- DAU ch B
- ADI standby
- Cockpit wander light (pilot)
- Searchlight (if installed)
- EADI pilot (EFIS version)
- Emergency floats
- VHF N° 2
- VOR N° 2

NOTE

The pilot is allowed to operate the VHF N° 2 system for a period of 15 minutes reception and 3 minutes transmission during the next 25 minutes, commencing from the expiration of the first 5 minutes.

NOTE

The landing light(s) or the searchlight (if installed) can be turned ON for only one minute before landing.
Simultaneous operation of the landing light(s) and the searchlight is prohibited.

If conditions require and the Generator Busses have been intentionally disconnected, other system such as windshield wipers can be energized by switching related Generator Bus to ON. In this event, the battery operating time will be reduced due to the greater loads imposed.

Failure of both generators and both generators busses

(Helicopter equipped with emergency bus)

The failure of both Generators followed by either the simultaneous failure of both Generator Busses of their internal disconnection is indicated by BUS TIE, INV 2, SAS 1 OFF, SAS 2 OFF, caution messages and by ELECTRICAL warning messages and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE

GEN 1 and 2 : Reset, then ON.

GEN BUS 1 and 2 : ON, check if Generator reset.

If both generators remain inoperative, the flight can be continued on battery power for maximum of 30 minutes provided that the following procedures are accomplished:

NOTE

During the load shedding procedure all SAS will be lost until SAS 1 is reselected ON. It is recommended that prior to load shedding, speed is reduced to between 100 and 120 Kts in order to reduce pilot work load.

PROCEDURE

Cockpit wander light (pilot)	: ON (at night).
GEN BUS 1	: OFF.
GEN BUS 2	: OFF.
SAS 1 switch	: ON (SAS 1 OFF caution message extinguished).

NOTE

If the Generator Busses are not disconnected the battery is capable of supplying power for a further 10 minutes maximum, providing that VHF radio is not used in transmission for more than 3 minutes.

NOTE

Following the above selection sequence, the emergency busses (N° 1 and N° 2) will supply power to the following indications and systems:

EMERGENCY BUS N°1
- Fuel quantity N° 1
- Fuel pump N° 1
- Fuel shut-off valve N° 1
- Fuel crossfeed
- Engine 1 fire detector system
- N° 1 engine compartment fire extinguisher system
- Force trim
- Intercom system, copilot
- Attitude engage

- SAS 1 system
- ADI pilot (EHSI-74 version)
- HSI standby (EHSI-74 version)
- Landing gear position indicator
- Landing light(s)
- Emergency floats
- Inverter #1

EMERGENCY BUS N° 2
- Fuel quantity N° 2
- Fuel pump N° 2
- Fuel shut-off valve N° 2
- Engine 2 fire detector system
- N° 2 engine compartment fire extinguisher system
- Hydraulic system
- Eng. Gov. CTL
- EDU N°2
- DAU ch B
- ADI standby
- Cockpit wander light (pilot)
- Searchlight (if installed)
- EADI pilot (EFIS version)
- Emergency floats
- VHF N° 2
- VOR N° 2

NOTE

The pilot is allowed to operate the VHF N° 2 system for a period of 15 minutes reception and 3 minutes transmission during the next 25 minutes, commencing from the expiration of the first 5 minutes.

NOTE

The landing light(s) or the searchlight (if installed) can be turned ON for only one minute before landing.
Simultaneous operation of the landing light(s) and the searchlight is prohibited.

If conditions require and the Generator Busses have been intentionally disconnected, other system such as windshield wipers can be energized by switching related Generator Bus to ON. In this event, the battery operating time will be reduced due to the greater loads imposed.

Failure of an inverter

(Helicopter equipped with emergency bus)

In the event of a main inverter failure (N° 1 or N° 2) the remaining inverter will automatically power the 115V and 26V busses of the failed inverter.
The failure is indicated by INV 1 (2) and SAS1 (2) OFF caution messages.
Attempt to reset the failed inverter.

If INV 1 (2) caution message does not extinguish (loss of affected inverter), proceed as follows:

Failed inverter : OFF.

SAS (1 or 2) switch : ON.

Failure of an a.c. bus system - 115V and 26V

(Helicopter equipped with emergency bus)

The failure of an a.c. bus system (115V and 26v) is indicated by the simultaneous activation of:

SAS 1 OFF - VG 1 message: for a.c. bus system N° 1
SAS 2 OFF - VG 2 message: for a.c. bus system N° 2

WARNING

In the event of N° 1 a.c. bus system failure the following instruments and equipment will continue to operate:
- VOR/ILS N°2
- ADF
- VG 2
- SAS 2

WARNING

In the event of N°2 a.c. bus system failure the following instruments and equipment will continue to operate:
- VOR/ILS N°1
- VG 1
- SAS 1
- GYRO COMPASS
- HSI

RAD-MSTR (RADIO MASTER) SWITCH FAILURE

In case of RAD-MSTR (Radio Master) switch failure disengage the 1 and 2 RADIO MASTER OVRD circuit breakers then use the radio equipment as desired by means of their control panels.

HELIPILOT MALFUNCTION

During operation of the Helipilot System, malfunctions may occur which require pilot intervention.
These malfunctions are discussed in detail in the following paragraphs.

CAUTION

Following a single SAS failure, the AUTOTRIM function is inoperative. In this condition the pilot is recommended to fly continuously checking and re-centering the API position or to fly decoupled (in ATTD mode), due to the reduced system authority.

NOTE

HELIPILOT indicators normally refer to SAS 1. Turning SAS 1 off. HELIPILOT indicators will automatically switch to SAS 2.

NOTE

The attitude beep-trim on the cyclic is inoperative when flying coupled in the relative axis or when one of the two SAS is inoperative.

Repeated disturbances during pitch, roll, or yaw helipilot operation

Retrim the helicopter. Identify the affected axis and helipilot by observing HELIPILOT indicators (API), helicopter attitudes/rates and ADIs (normal and STBY).

NOTE

SAS 2 pitch and roll actuators position may be observed by pressing the monitor switch SAS 2 PUSH on HELIPILOT panel.

Disengage failed system and proceed as follows:

Cyclis and collective	: Hands on.
Maximum Airspeed	: Reduce below 120 KIAS.
Maximum ROC	: 500 ft/min.

Proceed with flight.

Repair failed system before next flight.

Oscillatory malfunction during pitch, roll or yaw helipilot operation

Airspeed	: Control. Reduce below 120 KIAS.
Power	: Reduce if practicable.
Rate of climb	: 500 ft/min maximum.

Identify the affected axis and helipilot by observing HELIPILOT indicators (API), helicopter attitudes/rates and ADIs (normal and STBY).
Switch off the affected SAS.
If SAS disengagement eliminates the oscillation, leave it off.
If oscillation persists, re-engage both SAS and switch off the other SAS.
If oscillation occurs on yaw, switch off SAS 1 leaving SAS 2 on.

NOTE

SAS 2 pitch and roll actuators position may be observed by pressing the monitor switch SAS 2 PUSH on HELIPILOT panel.

Proceed as follows:

Maximum airspeed	: Less than 120 KIAS.
Maximum ROC	: 500 ft/min.
Cyclic and collective	: Hands on.

Proceed with flight.

WARNING

Landing shall not be attempted while oscillatory malfunction exists.

AUTOTRIM MALFUNCTION

Autotrim malfunction is evidenced by observing an undesirable cyclic stick motion.
If malfunction occurs put AUTOTRIM switch to OFF.

INTERCOMMUNICATION SYSTEM FAILURE

In case of failure of either intercommunication system proceed as follows on the affected panel:

FAIL/NORM switch	: Set to FAIL.

Resume to operate normally.

INTEGRATED DISPLAY SYSTEM FAILURE

The internal IDS failures may differ according to the following cases:

EDU failure

INDICATIONS:

Affected EDU	: Blank or unusable
Healthy EDU	: Automatically set to REVERSIONARY mode. IDS caution message displayed.

PROCEDURE
Proceed with flight and correct trouble before next flight.

EDU display degradation

INDICATIONS:

Affected EDU	: Visible display degradation (i.e. graphical and/or lighting degradation)

PROCEDURE

ON/OFF switch on affected EDU	: OFF.
Healthy EDU	: REVERSIONARY mode displayed. IDS caution message activated.

Proceed with flight and correct trouble before next flight.

Failure of both EDUs

INDICATIONS:

EDU 1 and 2 : Blank or unusable.

PROCEDURE

EDU 1 and 2 : OFF.

CAUTION

Primary and secondary parameters, caution and warning messages no more available except:
- MASTER CAUTION and MASTER WARNING lights.
- Engine fire light on power lever grips.
- All vocal and acoustic alarms.

NOTE

Engine and rotor governing still maintained by ECUs.

Land as soon as practical taking care that fuel quantity and FUEL LOW caution indications are no more available.

Loss of ECU parameter

INDICATIONS:

In case of loss of N1 and/or torque:

EDU 1 : #1 (#2) ECU FAIL warning message and MAN legend displayed.

In case of loss of TOT:

EDU 1	: #1 (#2) TOT LIMITER caution message displayed.

PROCEDURE

EDU 1	: Check for correctness of backup indication.

If the affected ECU has reverted to manual mode, proceed as per paragraph ENGINE OPERATION in MANUAL MODE. Proceed with flight and correct trouble before next flight.

Complete loss of primary (main and backup) or secondary parameters

INDICATIONS:

EDU 1 (2)	: No indication of the affected parameter and red dashes in the digital box.

NOTE

For N2 indication red FAIL legend vertically displayed on the affected scale.

Proceed with flight and correct trouble before next flight.

Miscompare of DAU primary parameters

INDICATIONS:

EDU 1	: DAU MISCMP-P caution message displayed.

PROCEDURE

EDU 1 : Access MENU 2/3 page to verify DAU channels status. Select the DAU channel indicated by yellow legend and check for data discrepancy. Deselect the channel to return to normal mode operation.

Proceed with flight and correct trouble before next flight.

Failure of one DAU channel

INDICATIONS:

EDU 1 : IDS caution message displayed.

PROCEDURE

EDU 2 : Access MENU 2/3 page to verify DAU channels status. The failed DAU channel is indicated by yellow legend. Check for correctness of secondary parameters.

Proceed with flight ignoring the failed DAU channel.
Correct trouble before next flight.

Failure of both DAU channels

INDICATIONS:

EDU 1 : IDS caution message displayed.

EDU 2 : CRUISE mode displayed in the same format as in EDU 1.

CAUTION

Secondary parameters no more available as well as all caution and warning indications, except:
- ENG 1 (2) OUT message plus vocal alarm
- Engine fire light on power lever grips plus ENGINE ONE (TWO) FIRE vocal alarm.

MASTER CAUTION and MASTER WARNING lights inoperative.

NOTE

Engine and rotor governing still maintained by the ECUs.

Land as soon as practical taking care that fuel quantity and FUEL LOW caution indications are no more available.

FIRE

ENGINE FIRE

INDICATIONS:

EDU 1	: ENG 1 (2) FIRE warning message displayed and ENGINE ONE (TWO) FIRE aural message activated.
Engine power grip (affected engine)	: Illuminated.
MASTER WARNING light	: Illuminated.

ENGINE FIRE DURING START

PROCEDURE

Abort start of affected engine as follows:

ENG MODE switch	: OFF (full counterclockwise).
Power lever	: OFF.
ENG FUEL switch	: OFF.
CROSSFEED switch	: CLOSED.
Fuel pump	: OFF.

If the other engine was operating, complete engines shutdown.
Abandon the helicopter as soon as possible.

ENGINE FIRE IN FLIGHT

PROCEDURE

Initiate immediately an emergency descent .

Collective	: Decrease until torque is less than 50%.

Shutdown the affected engine as follows:

ENG MODE switch	: OFF (full counterclockwise).
Power lever	: OFF.
ENG FUEL switch	: OFF.
CROSSFEED switch	: CLOSED.
Fuel pump	: OFF.
GENerator	: OFF.

If the fire is extinguished:

EDU 1	: ENG 1 (2) FIRE warning message suppressed.
Engine power grip (affected engine)	: Light out.

Land as soon as practical.

If fire is not extinguished land as soon as possible.
Complete engines shutdown.
Abandon the helicopter as soon as possible.

CAUTION

Do not attempt to restart the engine.

SMOKE IN CABIN, TOXIC FUMES, ETC.

In case of smoke, toxic fumes, etc., proceed as follows:

VENT CKPT switch : HIGH.

VENT CAB switch (if installed) : HIGH.

Front ventilation ports : Open.

Sliding window (if installed) : Open.

If smoke persists:

Airspeed : Reduce to minimum.

Cabin doors : Jettison.

CAUTION

Jettison the cabin doors at low forward speed and low rate of descent to prevent possibility of doors to hit tail rotor or main rotor.

NOTE

If smoke is suspected to be of an electrical origin, attempt to isolate the source by switching OFF electrical circuits.

Land as soon as possible.

STATIC PORT OBSTRUCTION

When operating in adverse weather conditions (rain, snow, etc.), if erratic readings from the airspeed indicator and altimeter occur, with the STATIC source switch in NORMAL position, proceed as follows:

Sliding window (if installed) and vents	:	Closed.
ECS or Heater (if installed)	:	OFF.
STATIC source switch	:	Remove the guard and select ALTERNATE.

This procedure selects an alternate static source utilizing cabin air.
Proceed with flight.

CAUTION

When utilizing the alternate static source, decrease the altimeter readings by 300 ft.

NOTE

The airspeed indication obtained through the alternate static source is slightly higher than the actual value in all speed range.

FLIGHT IN THUNDERSTORM - LIGHTNING

When flying in thunderstorm, the helicopter may be struck by lightning.
If is suspected that the helicopter has been struck by lightning proceed as follows:

Airspeed	:	Reduce (V_{NE} 80 Kts)

CAUTION

Avoid to perform extreme manoeuvers.

Land as soon as practical.

EMERGENCY PROCEDURE FOR LIMIT OVERRIDE PUSHBUTTON OPERATION

The engine is protected from limits exceeding through the engine limit governing.
The engine limit governing is responsible for limiting measured engine parameters.
The limit governing is set not to overcome the following limits:

Torque OEI	142%
Torque AEO	110%
TOT	930°C
N1	102.4%

In emergency condition the pilot may operate the LIM OVRD red pushbutton located on the collective control to override the engine limit governing.

PROCEDURE

LIM OVRD pushbutton
(on collective control) : Press.

CAUTION

Limit override pushbutton operation is allowed for actual emergency only.

EDU 2 : LIMIT OVRD ON advisory message displayed.

EDU 1	:	OEI transient torque marking (red triangle) displayed, if in OEI mode.
Collective	:	Apply as necessary respecting following transients: Torque OEI 180% Torque AEO 166% TOT 980C° N1 103.4% paying attention to return within the operating limits in no more than 20 sec as the engine is not protected from exceeding the above transients.

The operation of LIM OVRD pushbutton must be noted on the helicopter log-book.
Refer to helicopter Maintenance Manual for the inspection to accomplish before next flight.

SECTION 4
PERFORMANCE DATA

TABLE OF CONTENTS

LIST OF ILLUSTRATIONS

List of illustrations (Contd.)

SECTION 4
PERFORMANCE DATA

INTRODUCTION

The performance data presented herein is derived from the engine manufacturer's specification power for the engine less installation losses. This data is applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag, or power available.

NOTE

Data shown at -30°C are provided to allow interpolation at -25°C.

POWER ASSURANCE CHECKS

The Power Assurance Check charts are provided in order to let the pilot determine if the engines can produce the installed specification power.
A power assurance check should be performed daily. Additional checks should be made if unusual operating conditions or indications arise.
The hover check is performed prior to takeoff and the in-flight check is provided for periodic in-flight monitoring of engine performance.
Either power assurance check method may be selected at the discretion of the pilot. It is pilot's responsibility to accomplish the procedure safely, considering passenger load, terrain being overflown and the qualification of persons on board to assist in watching for other air traffic and to record power check data.
If either one of the two engines does not meet the requirements of the hover or the in-flight check, the minimum performance requirements as per published data may not be obtained.
The cause of engine power loss, or excessive TOT or GAS PRODUCER RPM (N1) should be determined as soon as practical.
Refer to Engine Maintenance Manual.

POWER ASSURANCE CHECK HOVER

* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 102%.
* TEST ENGINE MODE SWITCH - FLIGHT.
* OTHER ENGINE MODE SWITCH - IDLE.

* INCREASE COLLECTIVE UNTIL LIGHT ON WHEELS OR HOVERING AT 3 FEET, DO NOT EXCEED 820∞C TOT OR 97.4% N1 OR 124% TORQUE.
* STABILIZE POWER ONE MINUTE, THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAXIMUM ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINATED AS SOON AS PRACTICAL.

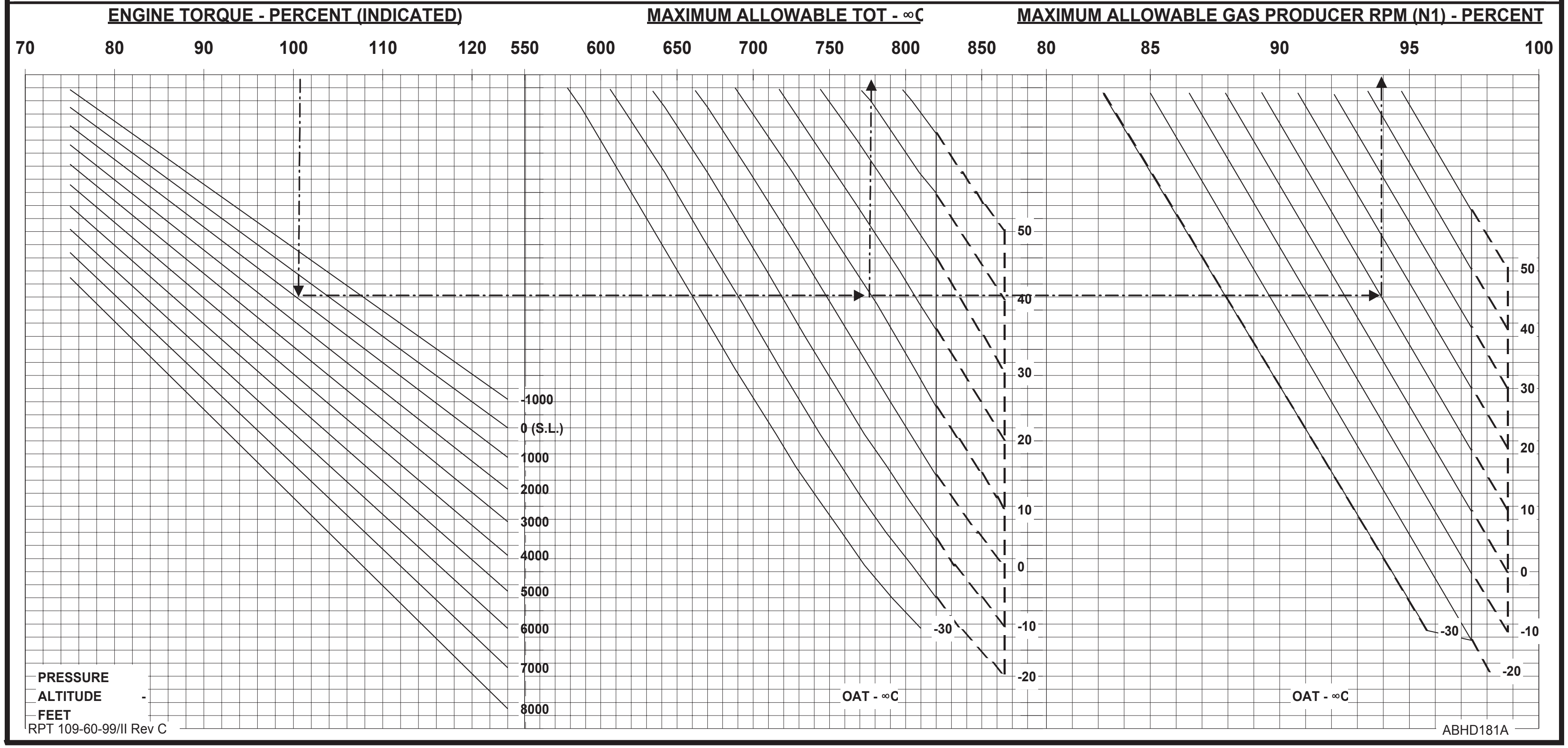


Figure 4-1. Power assurance check - hover.

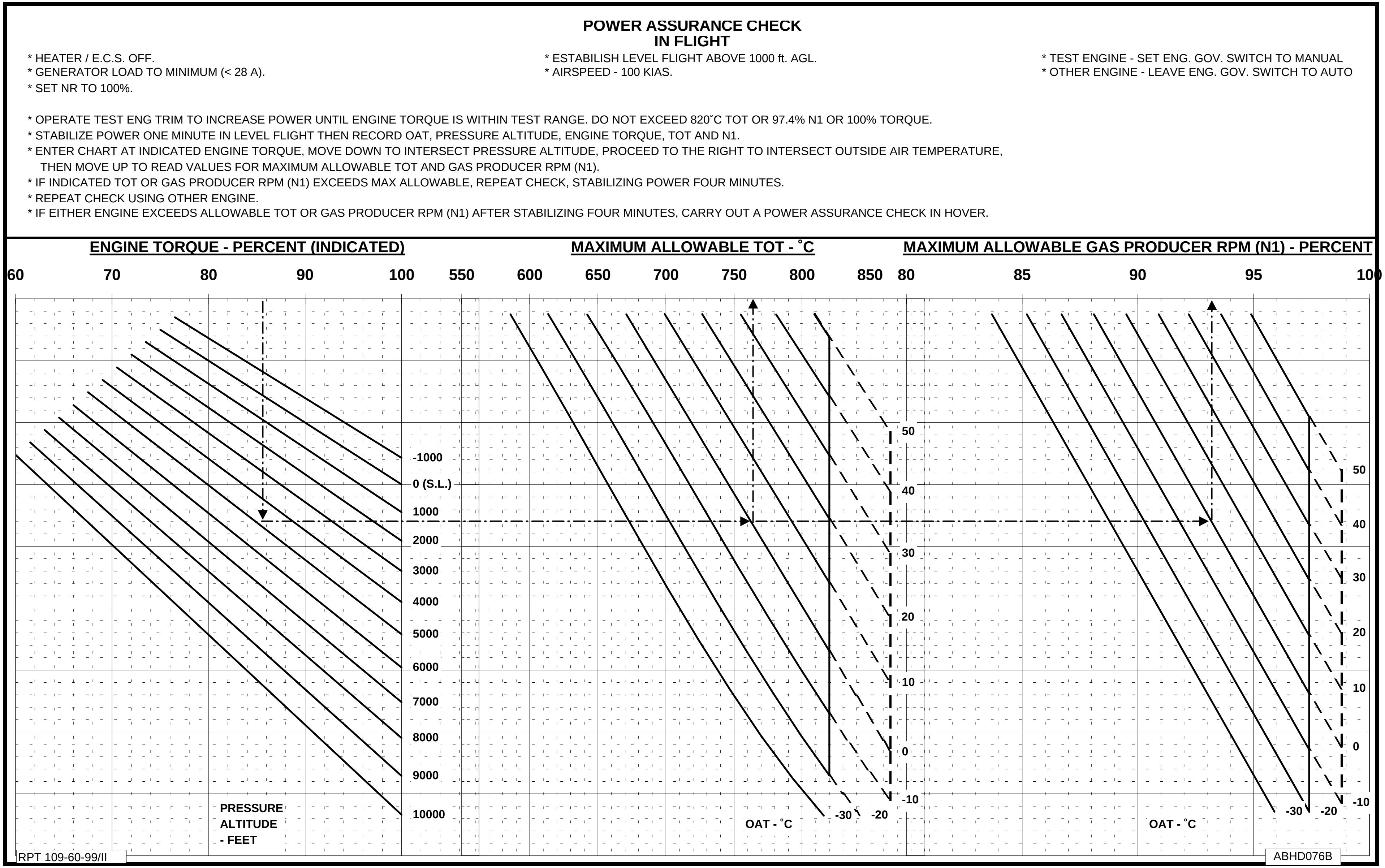


Figure 4-2. Power assurance check - in flight.

DENSITY ALTITUDE

A Density Altitude Chart is provided to aid in calculation of performance and limitations.

Density altitude (Hd) is an expression of the density of the air on terms of height above sea level; therefore, the less dense the air, the higher the density altitude.

For standard conditions of temperature and pressure, density altitude is the same as pressure altitude (Hp).

As temperature increases above standard for any altitude, the density altitude will also increase to values higher then pressure altitude.

The chart shows density altitude as a function of pressure altitude and temperature.

The chart also enable to obtain the inverse of the square root of the density ratio ($^1/_{\sqrt{\sigma}}$), which is used to calculate the True Airspeed (TAS) by the relation:

$$TAS = CAS \times {}^1/_{\sqrt{\sigma}}$$

where CAS is the Calibrated Airspeed.

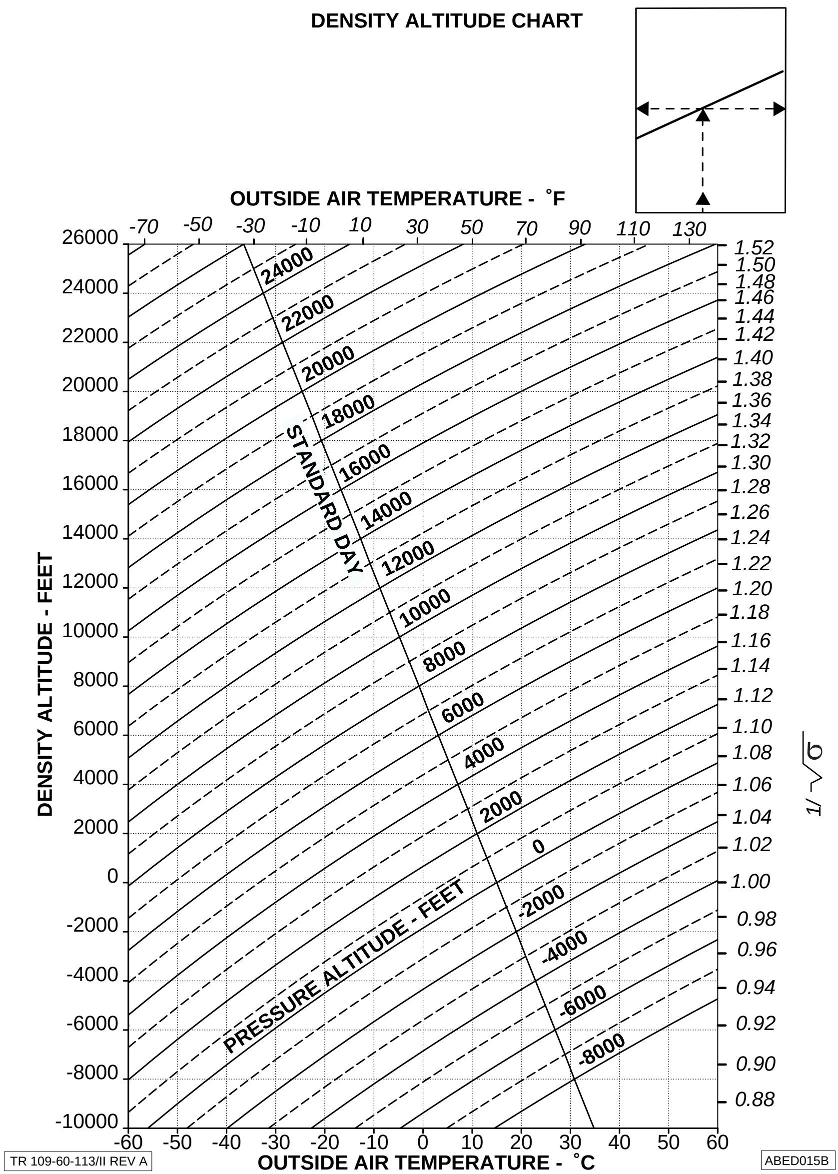


Figure 4-3. Density - altitude chart.

CONVERSION CHART

A Conversion Chart is also provided for altitude, weight, temperature, airspeed, rate of climb and capacity.

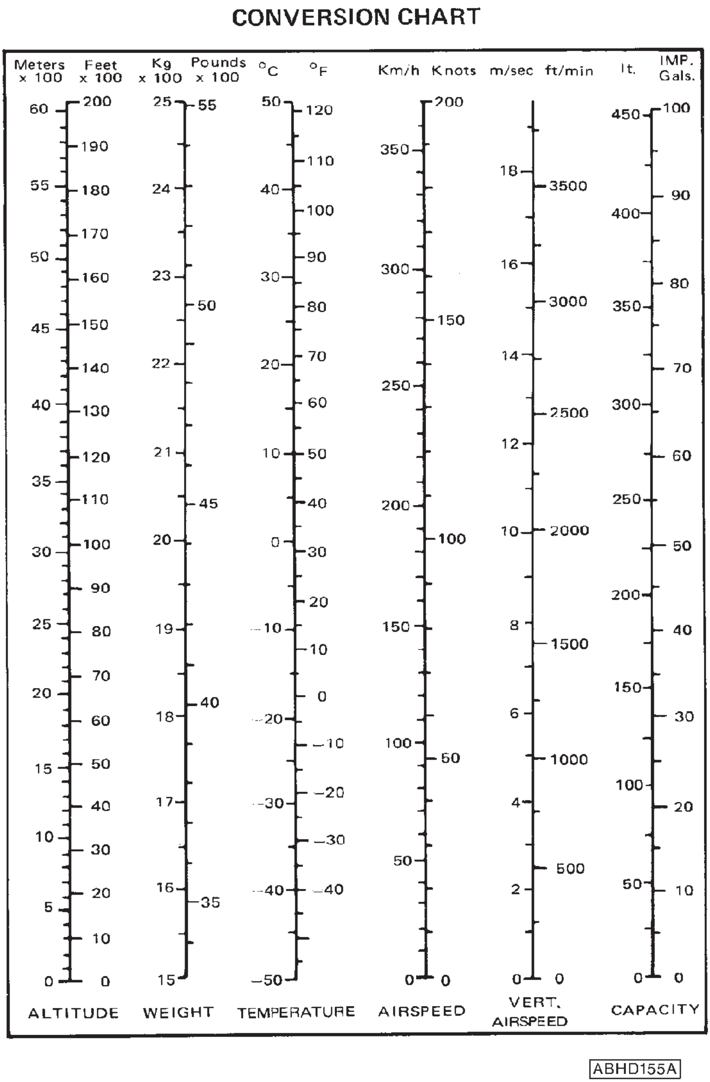


Figure 4-4. Conversion chart.

OPERATION VS ALLOWABLE WIND

Satisfactory stability and control in rearward and sideward flight has been demonstrated, at all loading conditions for hover in ground effect up to takeoff power, from -1000 ft to 8000 ft Hd, in the following wind/ground speed azimuth envelope:

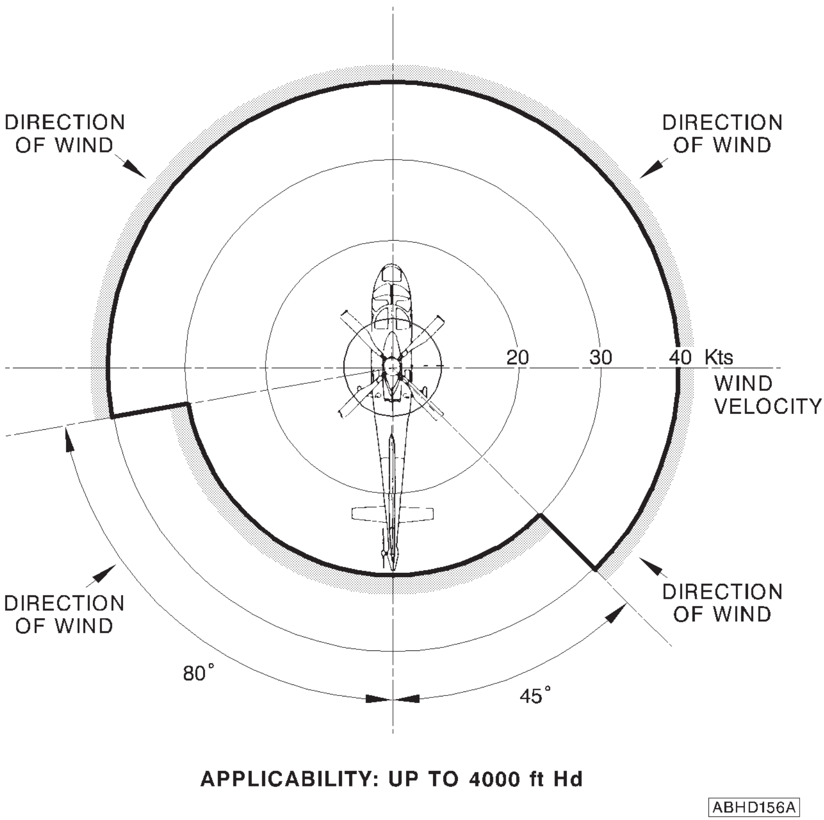


Figure 4-5 (Sheet 1 of 2). Wind/ground speed azimuth envelope.

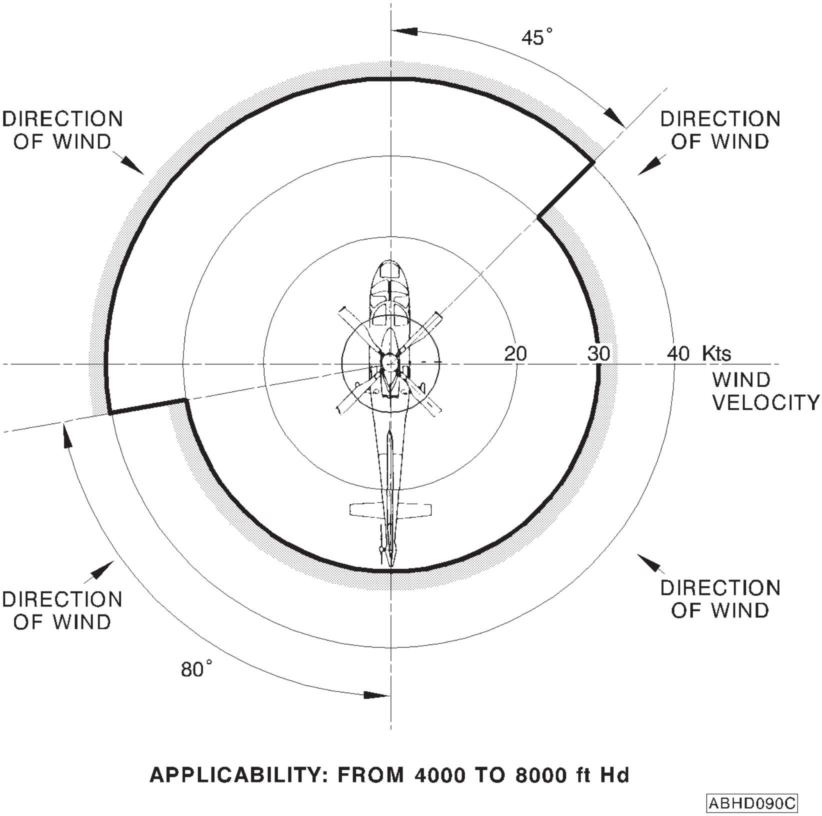


Figure 4-5 (Sheet 2 of 2). Wind/ground speed azimuth envelope.

HOVERING CEILING IGE

The Hovering Ceiling In Ground Effect charts provide the maximum gross weight for hovering IGE (3 ft above ground level) at all pressure altitudes and outside air temperatures, with main rotor speed at 102% and zero wind condition.
The charts are presented for takeoff power rating and for maximum continuous power rating.

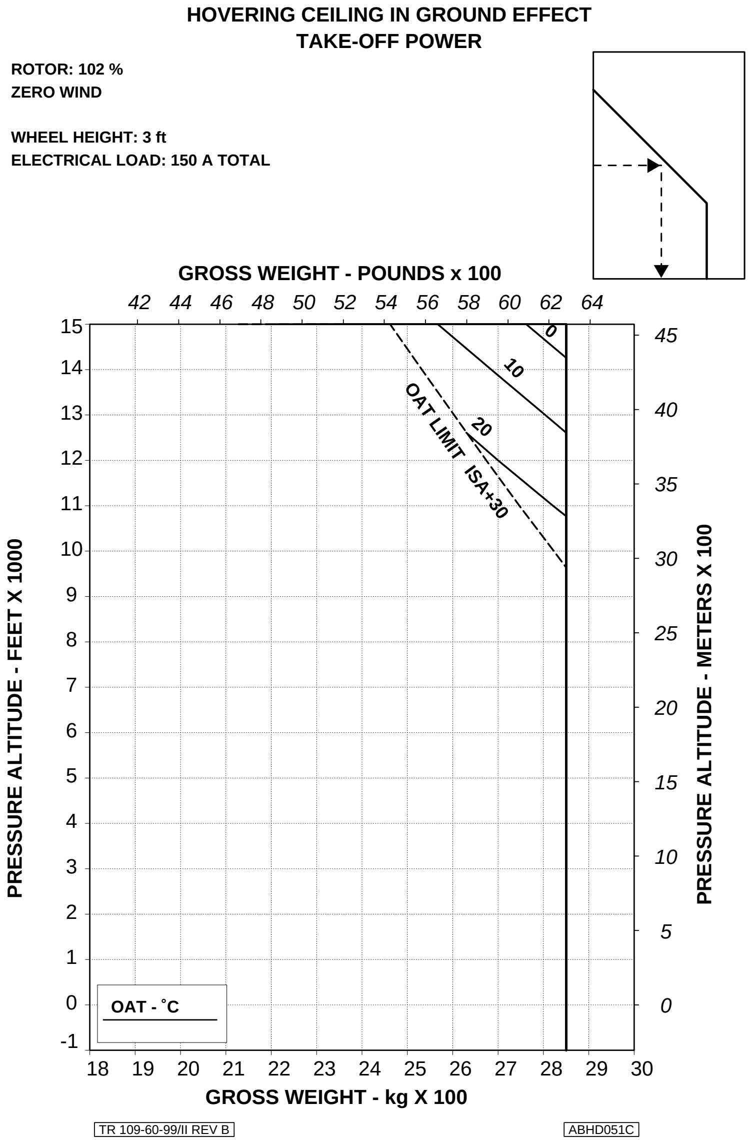


Figure 4-6. Hovering ceiling - in ground effect - take-off power.

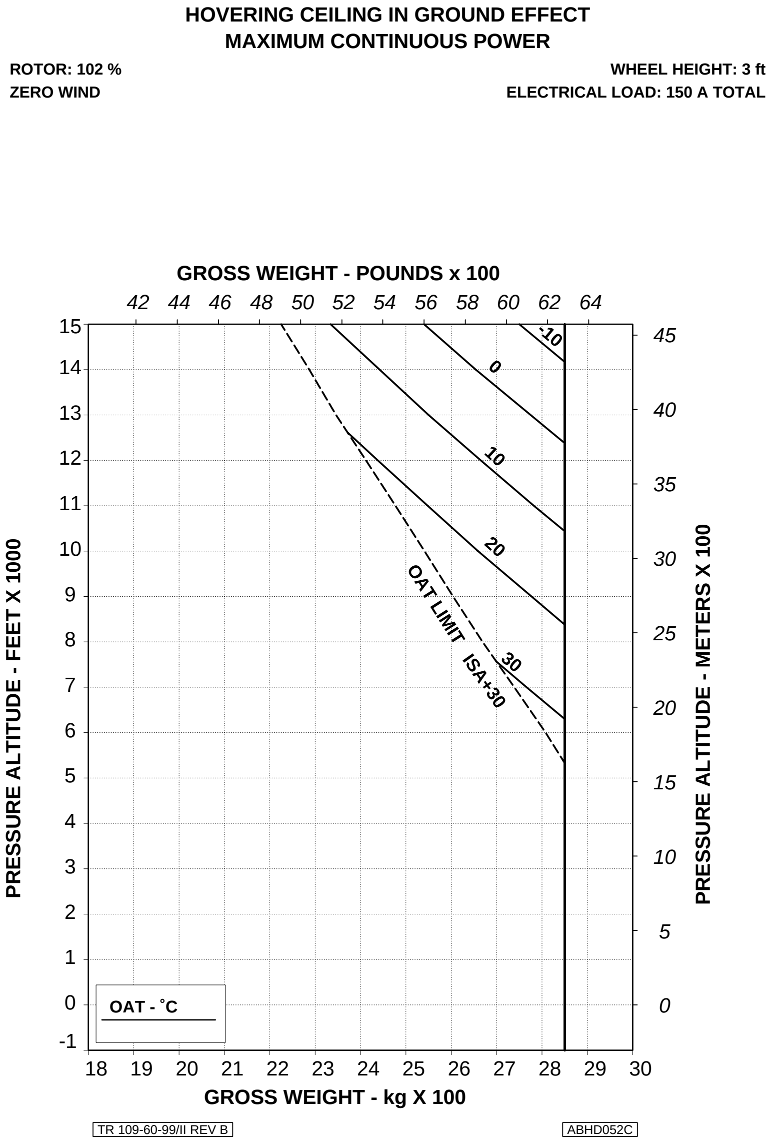


Figure 4-7. Hovering ceiling - in ground effect - maximum continuous power.

HOVERING CEILING OGE

The Hovering Ceiling Out of Ground Effect charts provide the maximum gross weight for hovering OGE (60 ft above ground level) at all pressure altitudes and outside air temperatures, with main rotor speed at 102% and zero wind condition.
The charts are presented for takeoff power rating and for maximum continuous power rating.

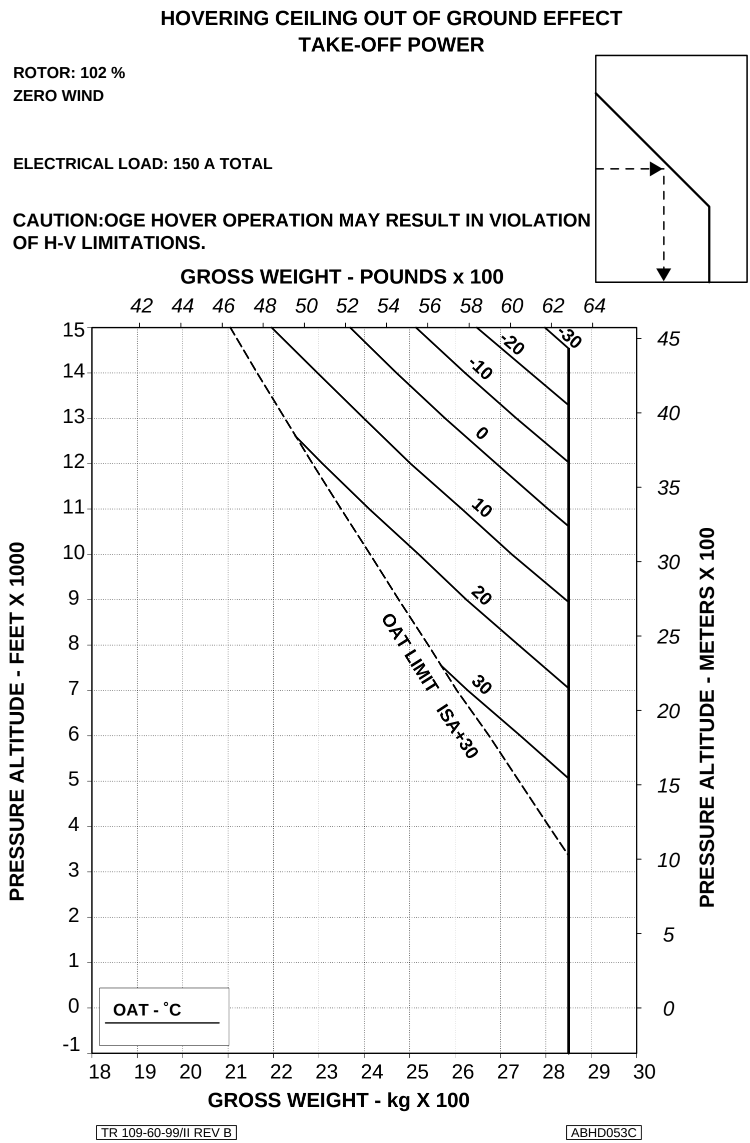


Figure 4-8. Hovering ceiling - out of ground effect - take-off power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

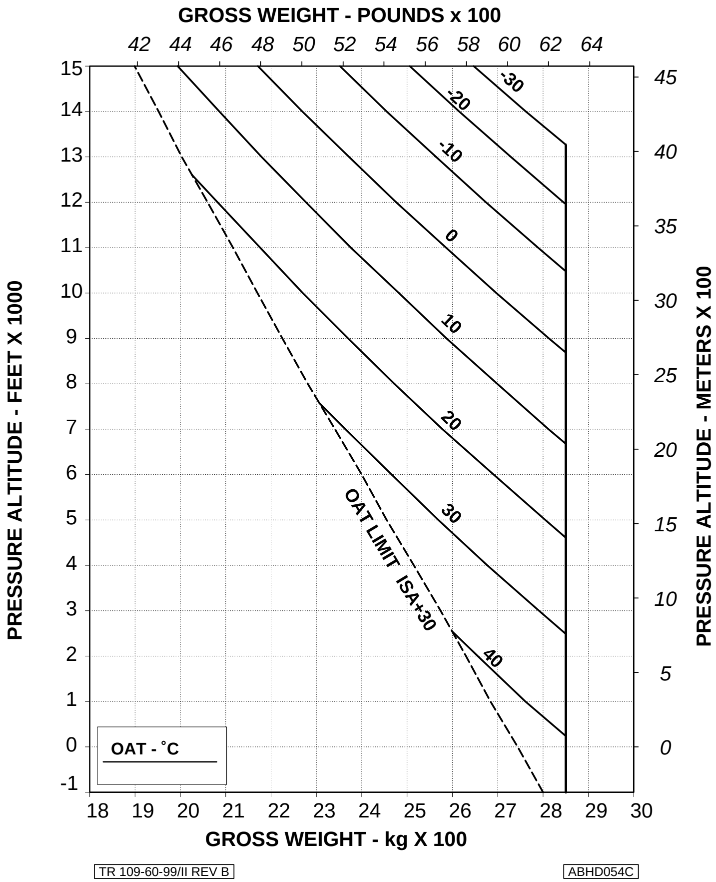


Figure 4-9. Hovering ceiling - out of ground effect - maximum continuous power.

HEIGHT - VELOCITY DIAGRAM

The Height - Velocity diagram enable to establish if, in the event of a single engine failure during takeoff, landing or other operation near the surface, a combination of airspeed and height above ground exists from which a safe single engine landing on a smooth, level and hard surface cannot be assured (dangerous zone).

The height - Velocity diagram is split in two charts.

Chart A shows the weight values, together with outside air temperature and altitude, at/below which the dangerous zone does not exist. For heavier weights refer to Chart B.

Chart B defines the combinations of height and airspeed to avoid for safe operations.

NOTE

The height - velocity diagram does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power off landing can be made.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

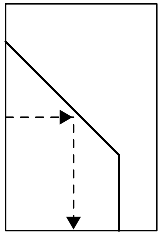


CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V LIMITATION FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

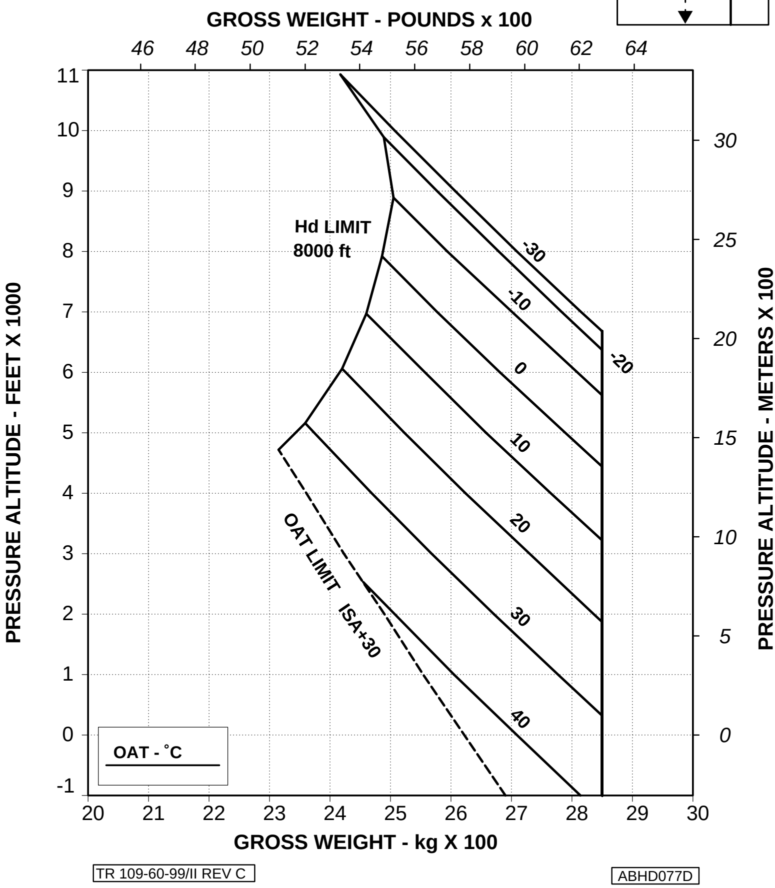


Figure 4-10 (sheet 1 of 2). Height - velocity diagram.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

CHART B

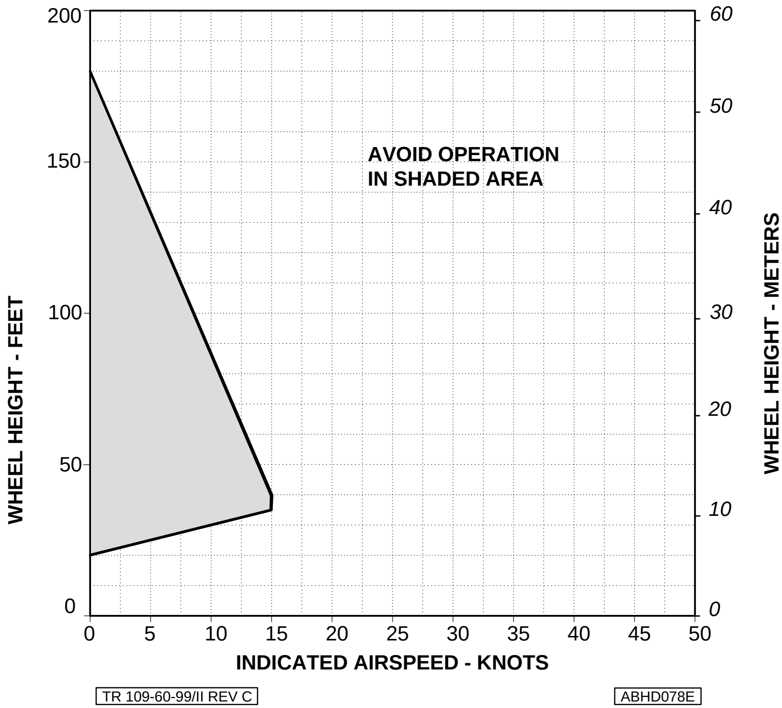


Figure 4-10 (sheet 2 of 2). Height - velocity diagram.

RATE OF CLIMB

The Rate of Climb charts are provided with all engines operative and with one engine inoperative. The rate of climb AEO charts are presented for takeoff power rating and for maximum continuous power rating, both with main rotor speed at 100%, and they refer to the best rate of climb speed of 60 Kts IAS.The rate of climb OEI charts are presented for 2.5 minutes power rating and for maximum continuous power rating, both with main rotor speed at 100%, and they refer to the best rate of climb speed of 60 Kts IAS.

NOTE

Single engine performance is intended for emergency use when one engine becomes inoperative due to an actual malfunction and for maintenance or training purpose.
For maintenance and training OEI operation shall be limited to the maximum continuos OEI power rating.

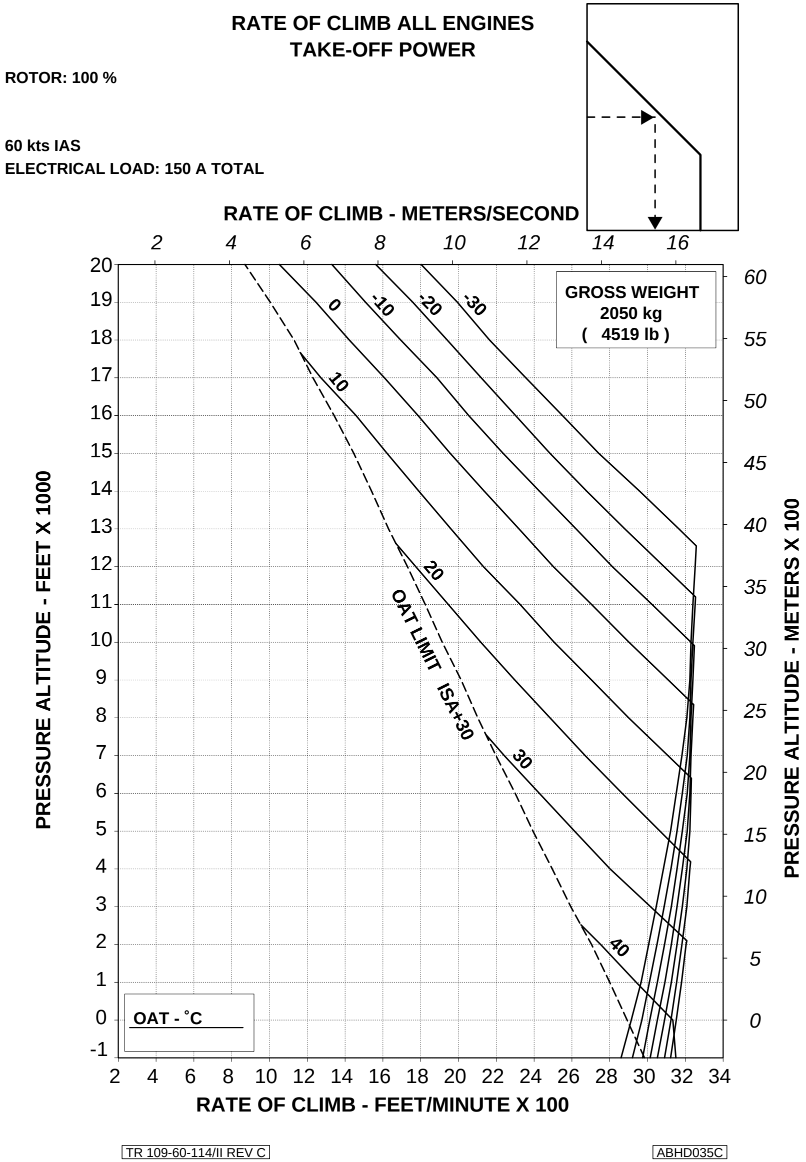


Figure 4-11. Rate of climb - all engines - take-off power - 2050 kg.

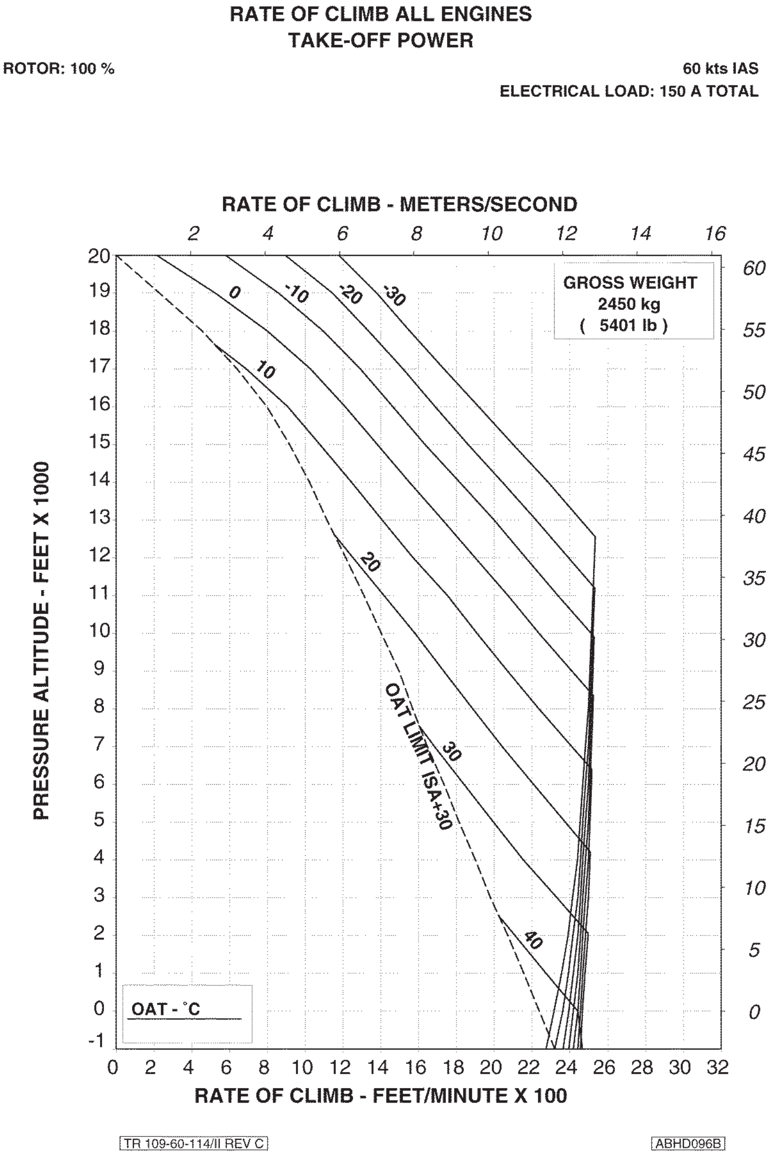


Figure 4-12. Rate of climb - all engines - take-off power - 2450 kg.

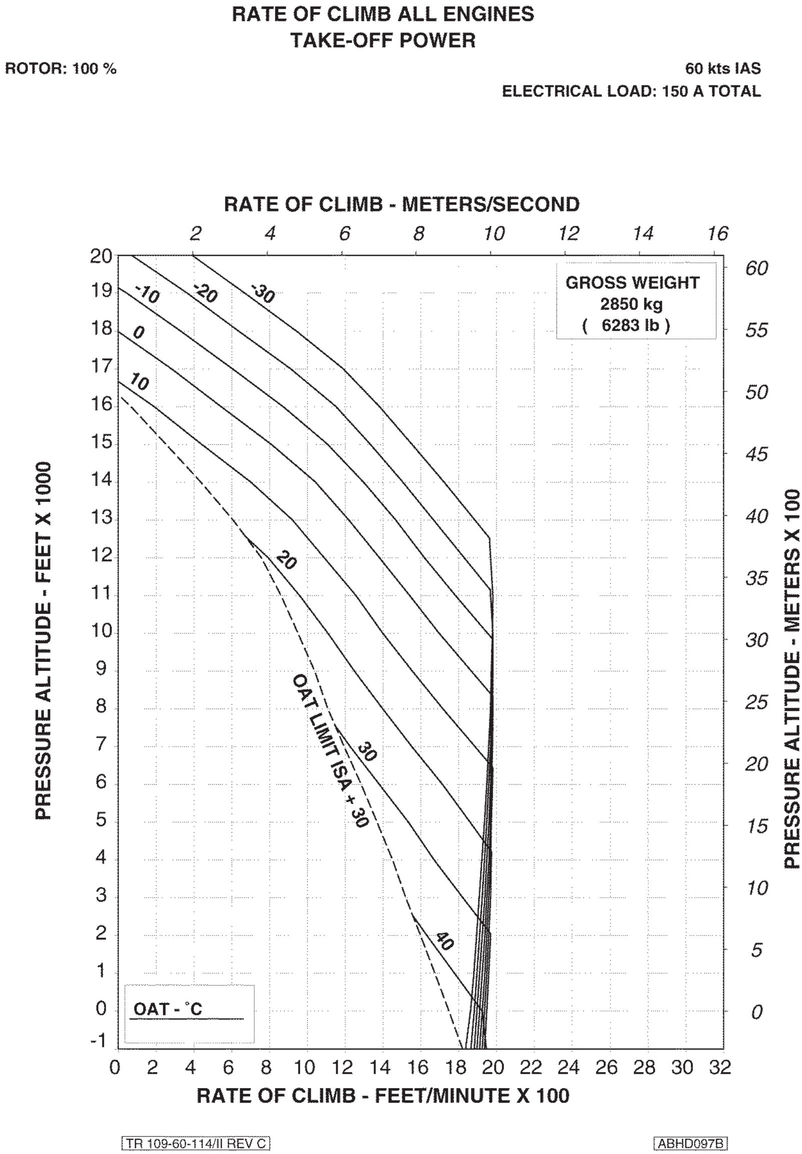


Figure 4-13. Rate of climb - all engines - take-off power - 2850 kg.

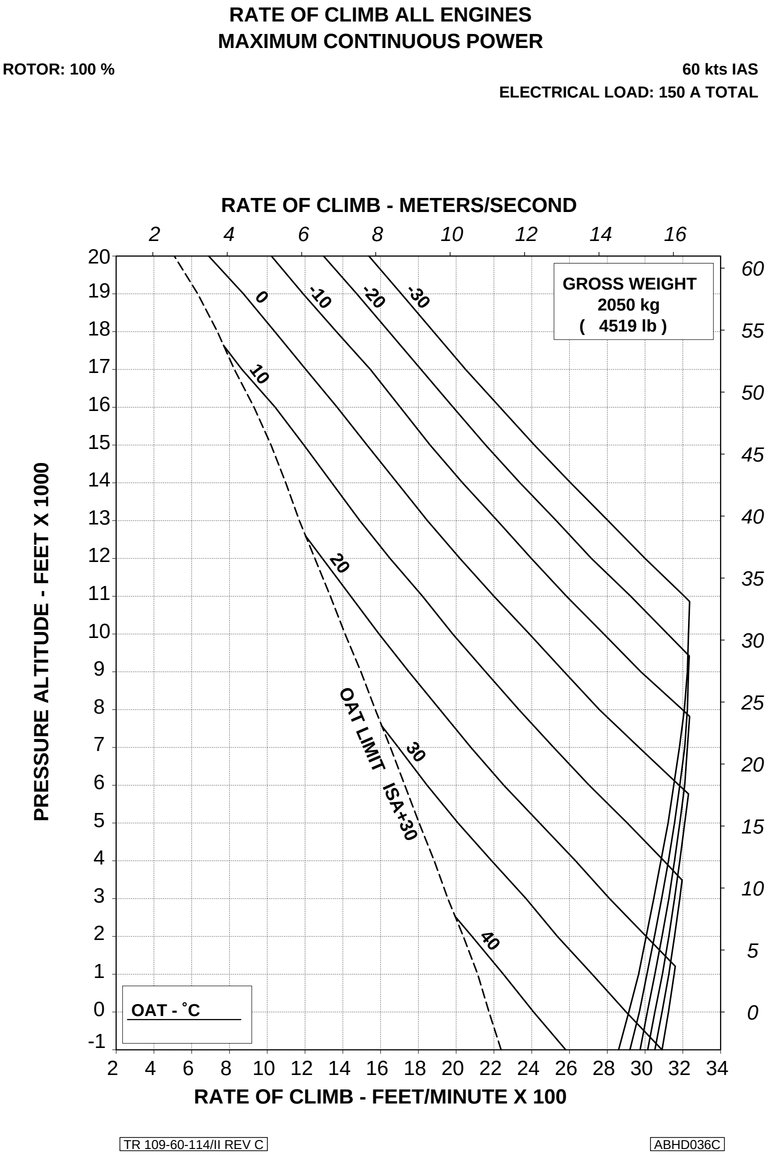


Figure 4-14. Rate of climb - all engines - maximum continuous power - 2050 kg.

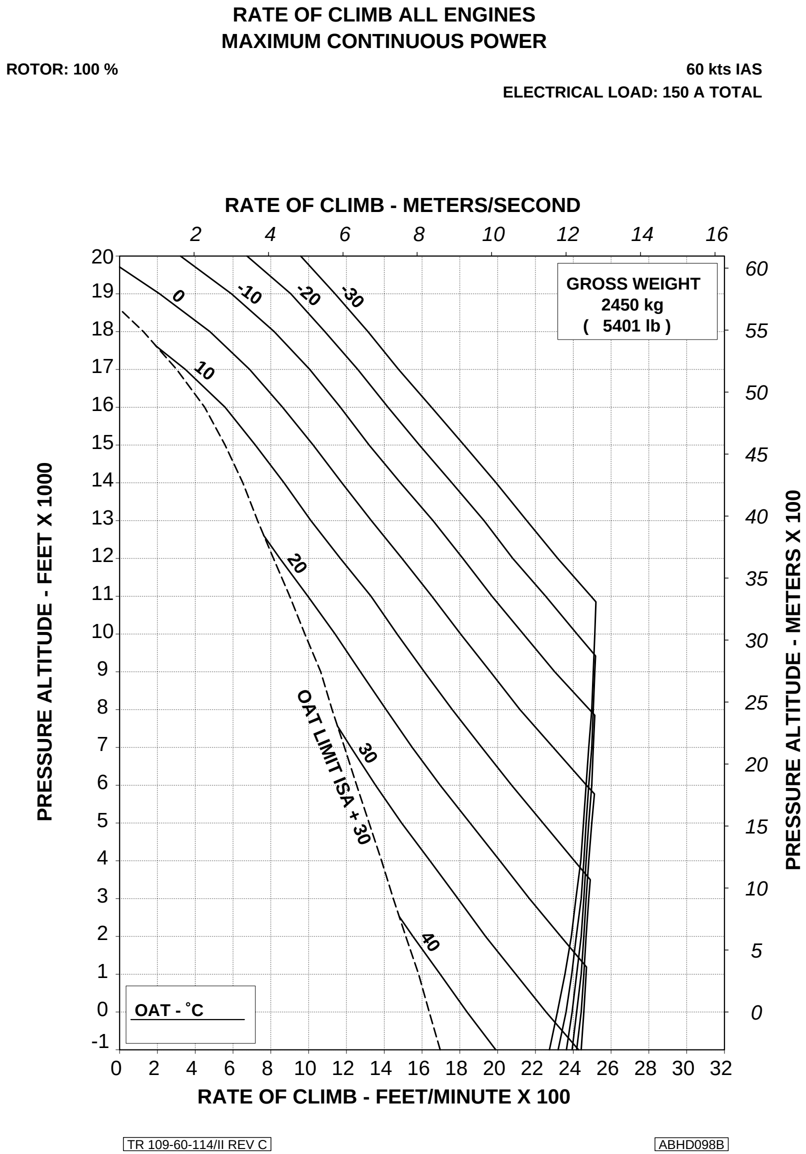


Figure 4-15. Rate of climb - all engines - maximum continuous power - 2450 kg.

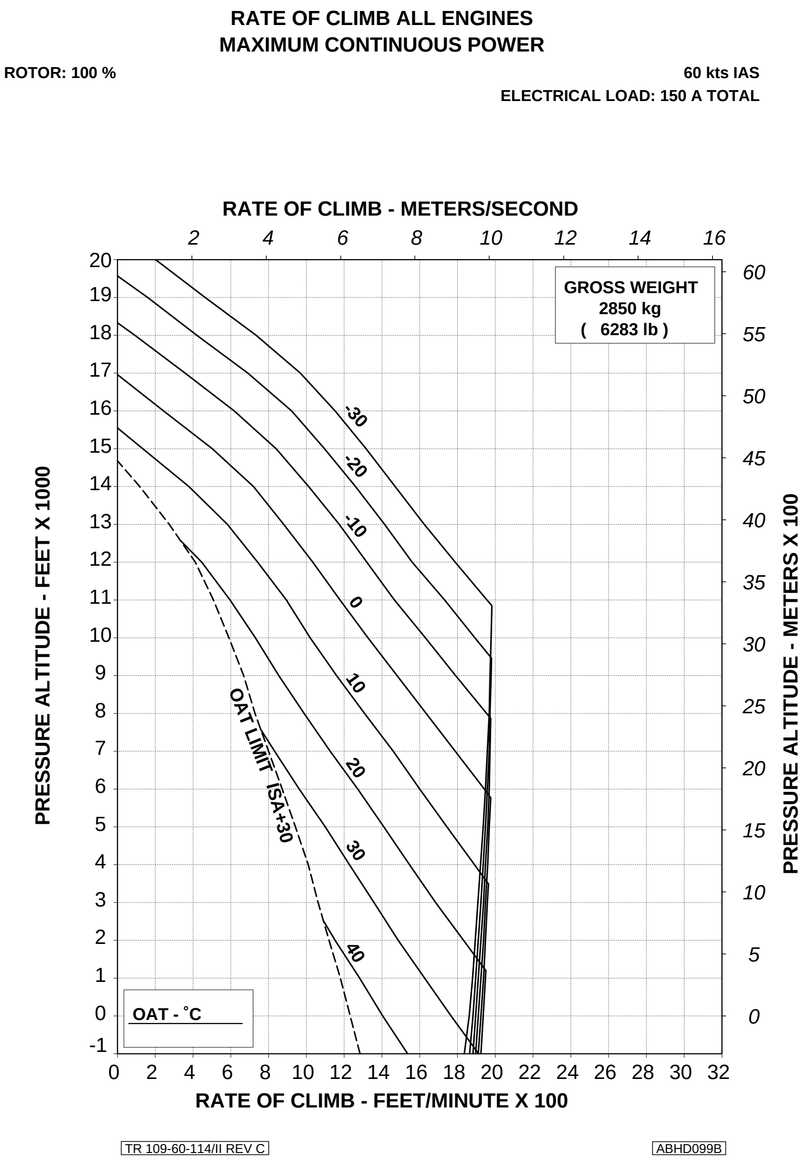


Figure 4-16. Rate of climb - all engines - maximum continuous power - 2850 kg.

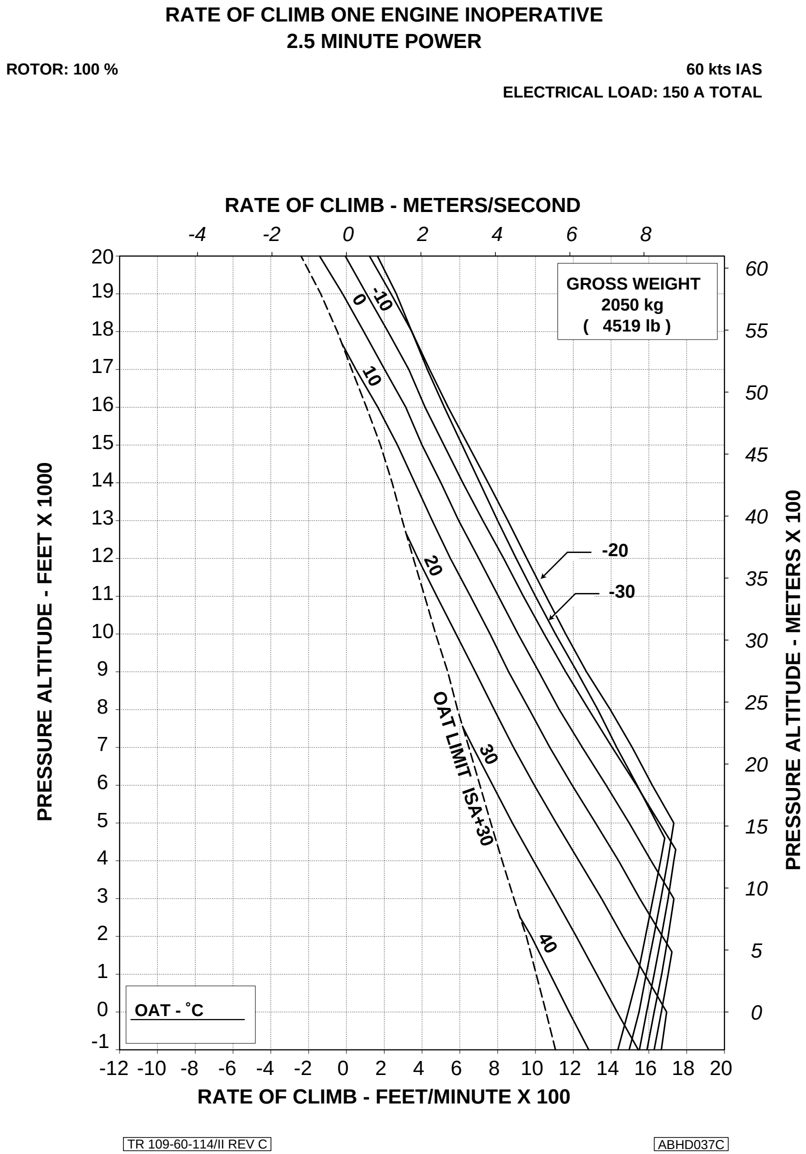


Figure 4-17. Rate of climb - OEI - 2.5 minutes power - 2050 kg.

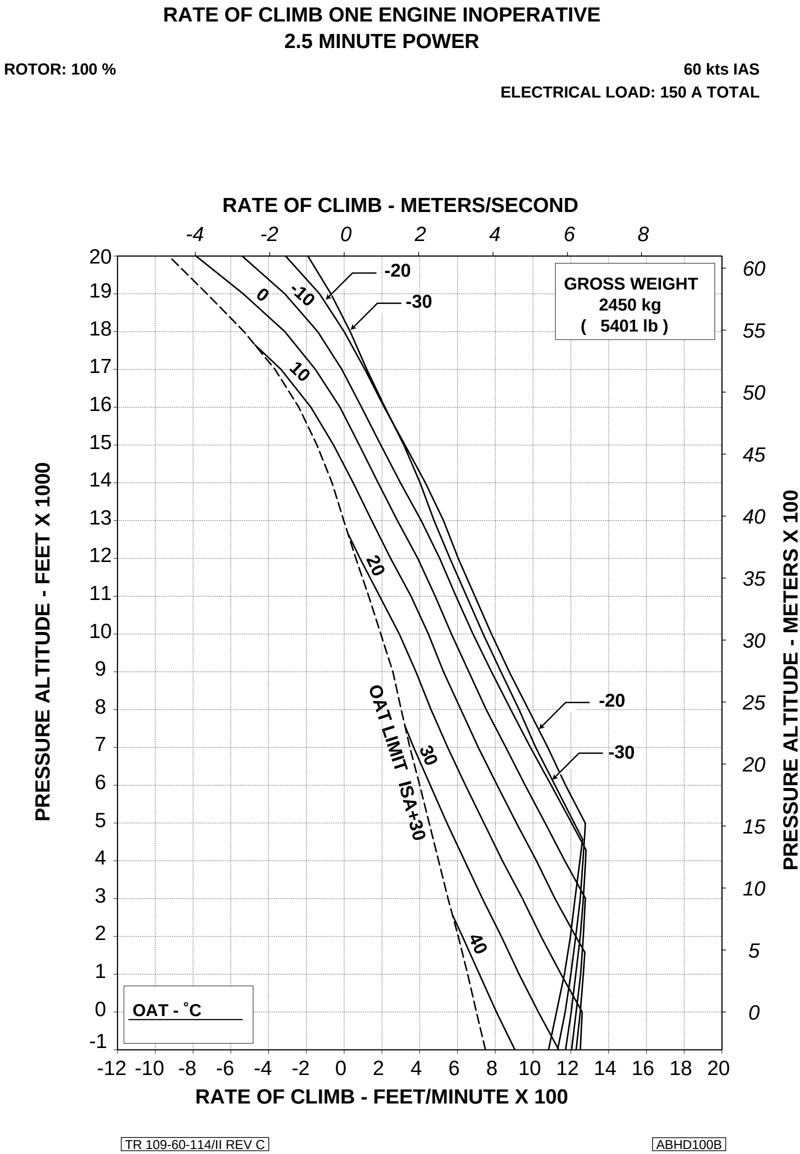


Figure 4-18. Rate of climb - OEI - 2.5 minutes power - 2450 kg.

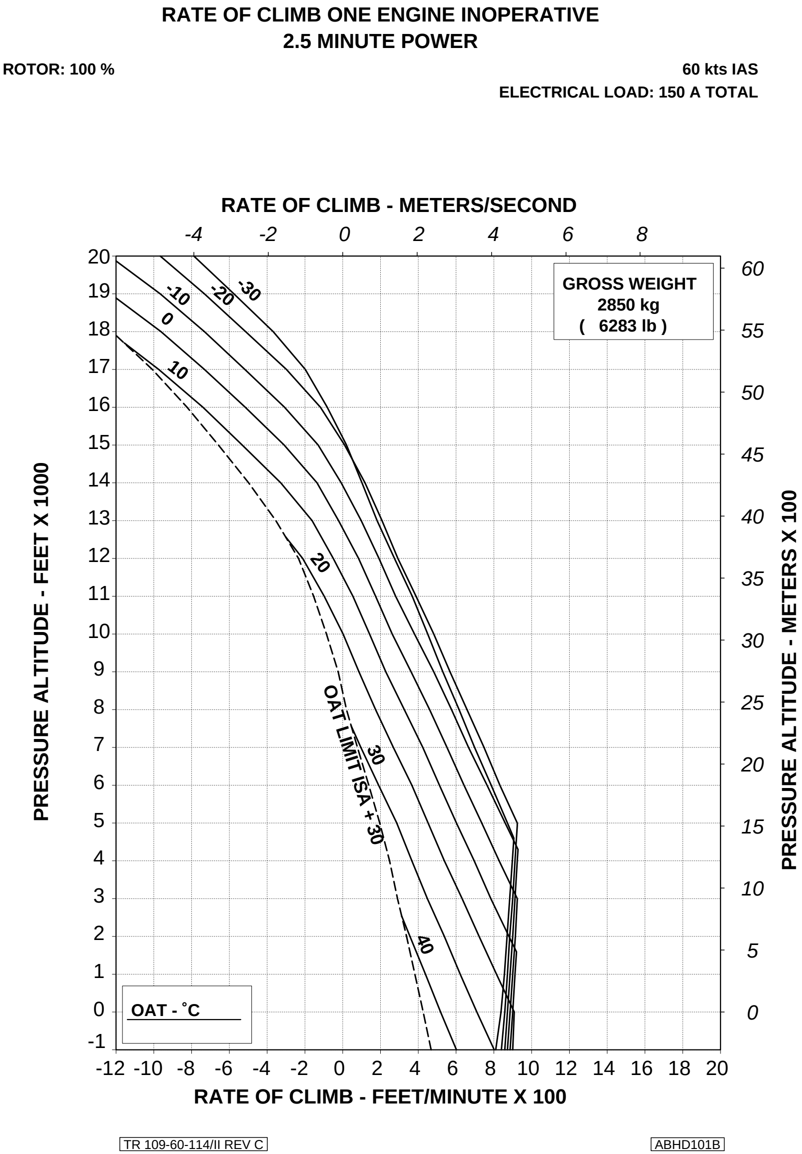


Figure 4-19. Rate of climb - OEI - 2.5 minutes power - 2850 kg.

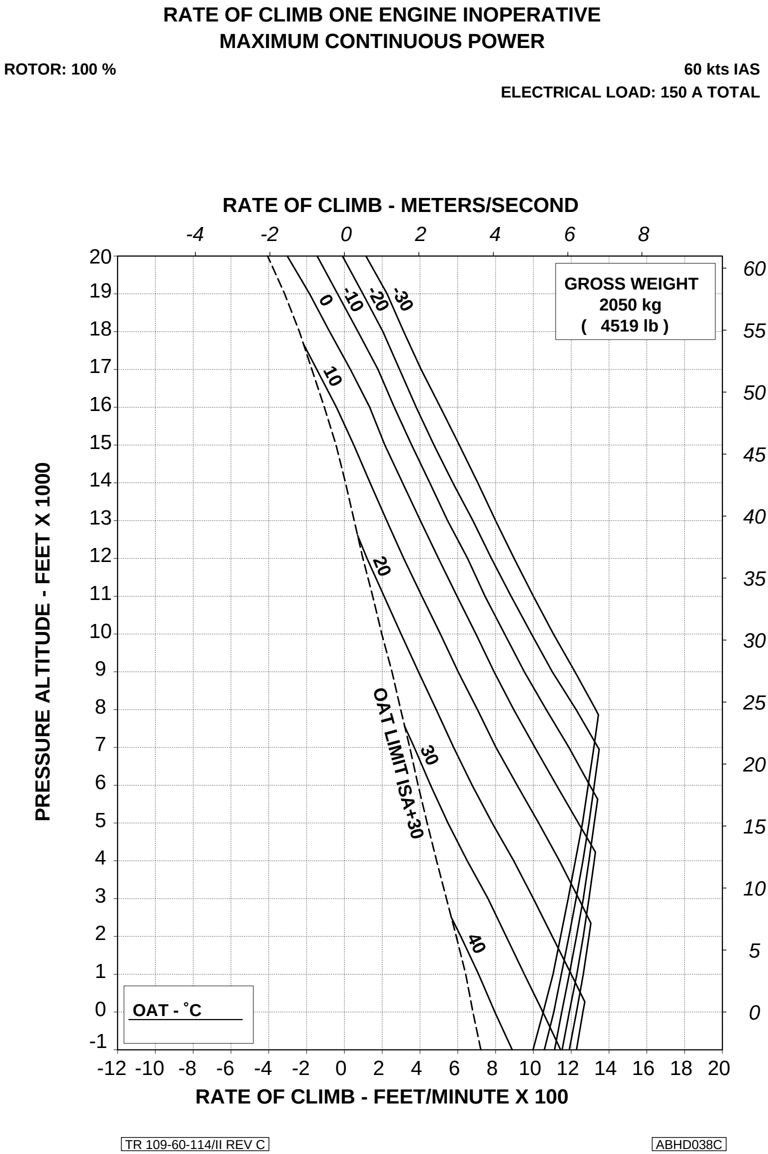


Figure 4-20. Rate of climb - OEI - maximum continuous power - 2050 kg.

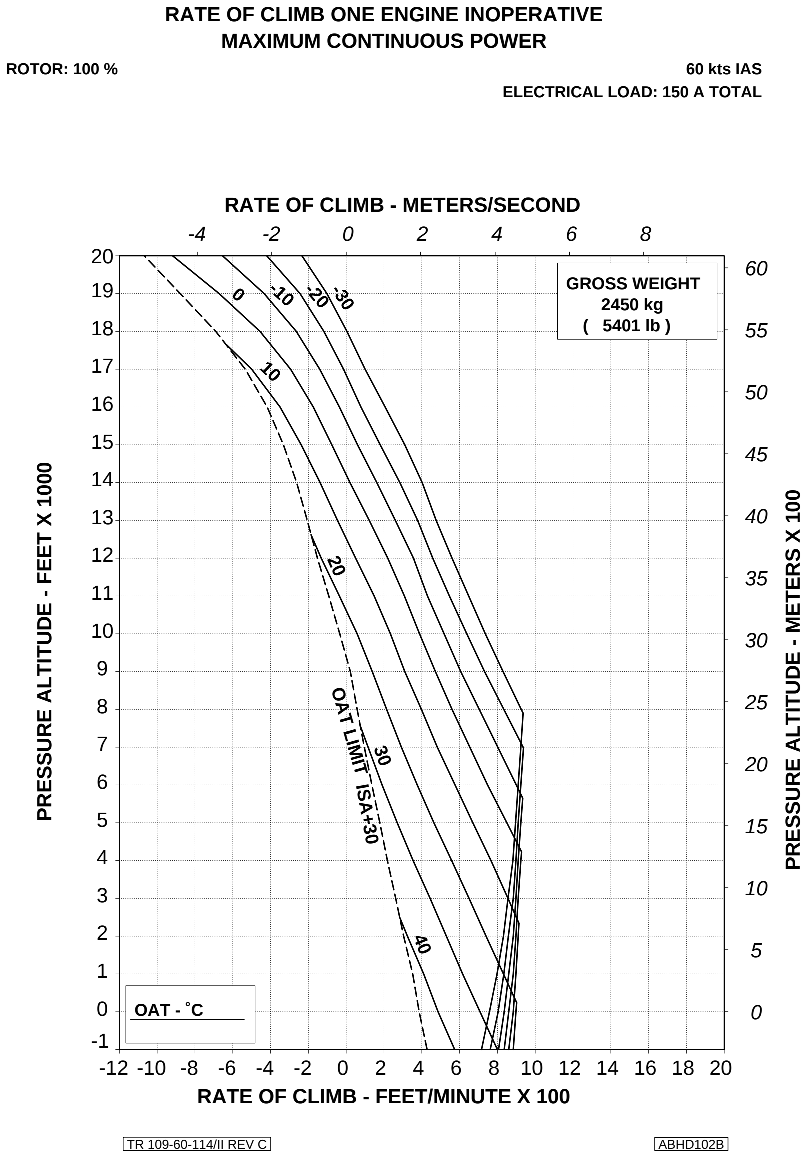


Figure 4-21. Rate of climb - OEI - maximum continuous power - 2450 kg.

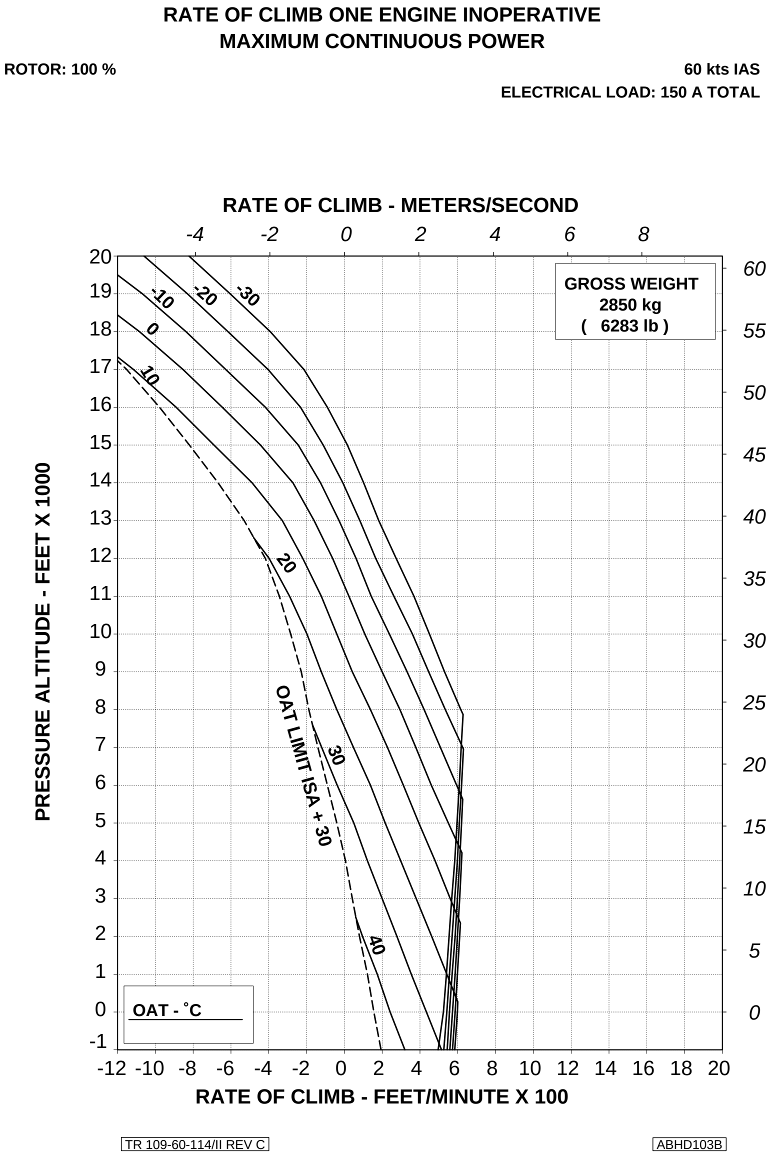
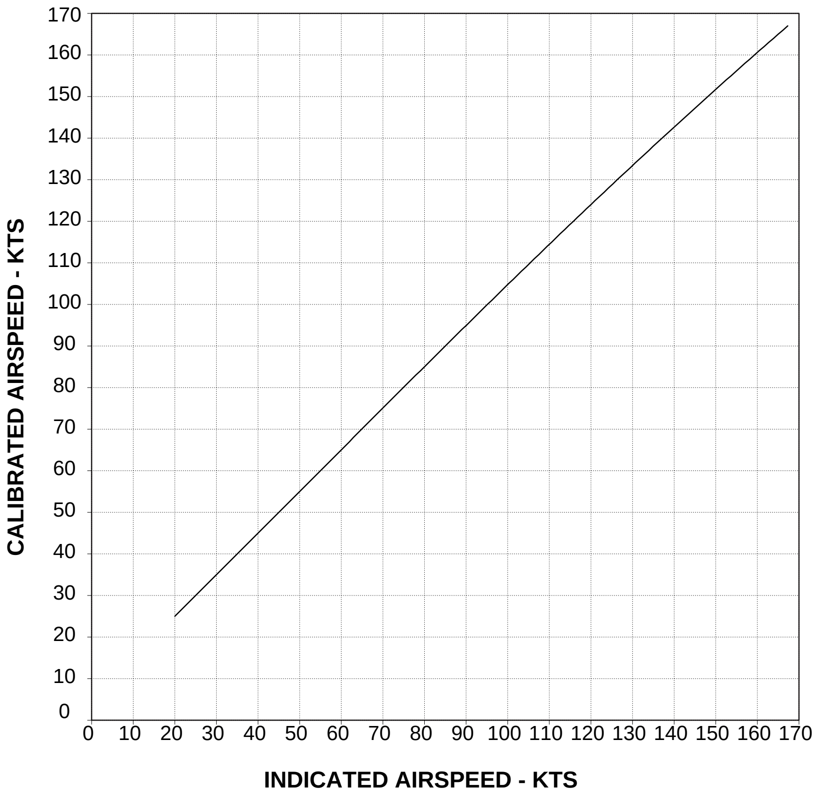


Figure 4-22. Rate of climb - OEI - maximum continuous power - 2850 kg.

AIRSPEED CALIBRATION

The Airspeed Calibration charts provide calibrated airspeed in forward flight for pilot and copilot airspeed systems.

AIRSPEED CALIBRATION CURVE
PILOT
(FORWARD FLIGHT)



RPT 109-60-99/II REV A

ABHD033A

Figure 4-23. Airspeed calibration curve pilot (forward flight).

**AIRSPEED CALIBRATION CURVE
COPILOT
(FORWARD FLIGHT)**

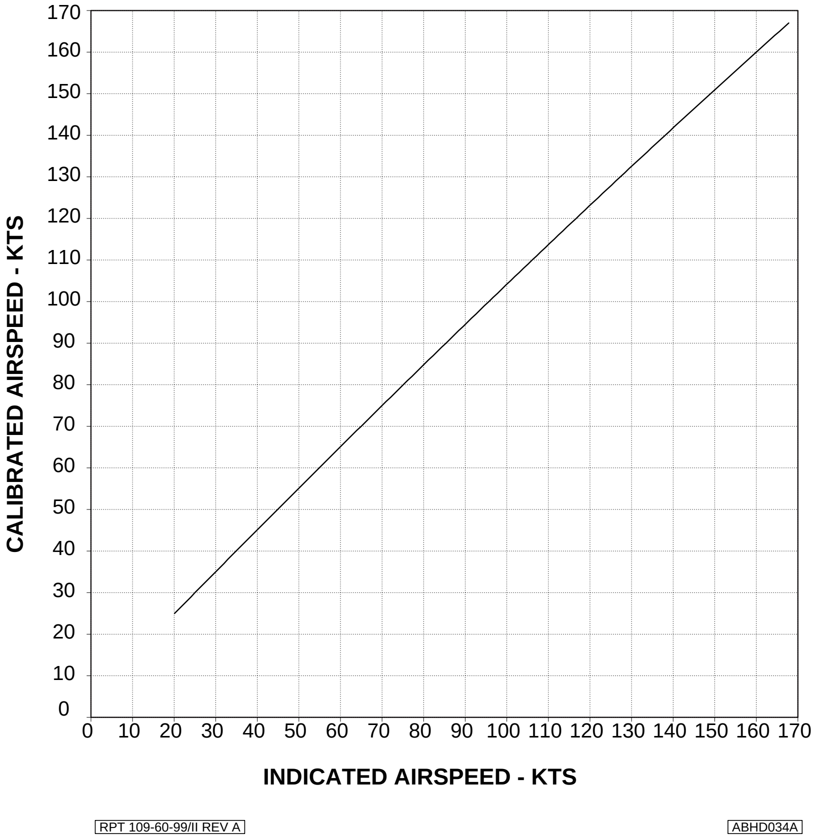


Figure 4-24. Airspeed calibration curve copilot (forward flight).

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SECTION 5
OPTIONAL EQUIPMENT

LIST OF APPENDICES

List of Appendices (Contd.)

Appendix No.	Name of equipment	No.
14	Battery 22 Ah	109-0812-04
15	EFIS (Electronic Flight Instrument System)	109-0900-57- -101/-103/-105 /-107
16	Global Position System Garmin 165	109-0811-53
17	Nightsun searchlight SX-16	109-0811-33 109-0813-46
18	Snow skid	109-0811-99
19	Global Position System Trimble 2101	109-0811-53 and 109-0822-91
20	Slump protection pads	109-0811-73
21	Emergency floats	109-0811-42
22	Wire strike protection system	109-0812-07
23	Equivalent category ″A″ operations training procedure	—
24	External hoist	109-0812-31
25	Skywatch Traffic Advisory System - SKY497	109-0812-39
26	Moving Map System -Skyforce Observer	109-0812-38
27	Satellite Telephone - AIRSAT1	109-0812-40
28	Particle Separator Engine Air Induction System	109-0811-55
29	Starter Generator Configuration APC P/N 160SG139Q1	—
30	External loudspeakers	109-0811-67

List of Appendices (Contd.)

Appendix No.	Name of equipment	No.
31	E.M.T. (Emergency Medical Transportation)	109-0812-64
32	HF System - Type HF950	109-0812-25
33	Cargo hook	109-0810-31
34	Cargo hook	109-0811-75
35	V_{LE} and V_{LO} extension up to 140 KIAS	—
36	Supplementary fuel tanks	109-0812-46
37	Multifunction dispaly KMD 550	109-0813-16
38	Main transmission gear box vibration isolator struts installation.	109-0822-99
39	Main transmission oil temperature limit extension	109-0823-24
40	Searchlight	109-0812-83
41	E.M.T. (Emergency Medical Transportation) Single and dual litter	109-0813-59
42	EFIS - Astronautics (Electronic Flight Instrument System)	109-0900-71-1A01
43	Nightsun searchlight SX-5	109-0813-72
44	Moving Map System Skyforce Observer	109-0812-38
45	Increase internal gross weight	109-0823-22
46	High temperature operation (ISA + 35°C)	109-0823-46
47	Searchlight	109-0813-76
48	HF System - Type KHF1050	109-0814-32
49	Wheater Radar PRIMUS 700	109-0813-23
50	Satcom Aircell	109-B811-14

OPTIONAL EQUIPMENT INCOMPATIBILITY

The following table shows the incompatibility of one or more optional equipment when installed on board the helicopter.

Table 5-1. Optional equipment incompatibility

Appendix No.	Name of equipment	Incompatibility (Appendix No.)
1	Rotor brake	None
2	Engine compartment fire extinguisher	None
3	Weather radar - RDR 2000	None
4	Pulsed chip detector	None
5	Environmental control system (air conditioning)	6
6	Bleed air heater	5
7	Searchlight	None
8	Supplementary fuel tanks	10 (dual litter version) 36 and 41
9	″Green aircraft configuration″	10, 31 and 41
10	EMS (emergency medical service)	8 (dual litter version), 9, 25, 26, 31, 36 and 41
11	Sliding doors	None
12	Equivalent category ″A″ operations	13 and 45
13	Integrated Display System configuration EDU 109-0900-42-101 DAU 109-0900-42-103	12, 23 and 39
14	Battery 22 Ah	None
15	EFIS (Electronic Flight Instrument System)	42
16	Global Position System Garmin 165	19
17	Nightsun searchlight SX-16	43
18	Snow skid	20
19	Global Position System Trimble 2101	16
20	Slump protection pads	18
21	Emergency floats	None
22	Wire strike protection system	None

Appendix No.	Name of equipment	Incompatibility (Appendix No.)
23	Equivalent category ″A″ operations training procedure	13 and 45
24	External hoist	27 and 38
25	Skywatch Traffic Advisory System SKY 497	10 (oxygen system 109-0811-76) and 32
26	Moving Map System Skyforce Observer	10 (oxygen system 109-0811-76), 32 and 44
27	Satellite telephone AIRSAT1	24 and 50
28	Particle separator Engine Air Induction System	None
29	Starter generator APC 160SG139Q1	None
30	External loudspeakers	40
31	EMT (Emergency Medical Transportation)	9, 10 and 41
32	HF System type HF950	25, 26 and 48
33	Cargo hook	38
34	Cargo hook	38
35	V_{LE} and V_{LO} extension up to 140 KIAS	—
36	Supplementary fuel tanks	8, 10 (dual litter version) and 41
37	Multifunction display KMD 550	None
38	Main transmission gear box vibration isolator struts	24, 33 and 34
39	Main transmission temperature limit extension	13
40	Searchlight	30 and 47
41	E.M.T. (Emergency Medical Transportation) Single and dual litter	8 (dual litter version), 9, 10, 31 and 36
42	EFIS - Astronautics (Electronic Flight Instrument System)	15
43	Nightsun searchlight SX-5	17
44	Moving Map System Skyforce Observer	26
45	Increased Internal Gross Weight	12 and 23
46	High temperature operation (ISA + 35°C)	None

Appendix No.	Name of equipment	Incompatibility (Appendix No.)
47	Searchlight	40
48	HF System type KHF1050	32
49	Weather Radar PRIMUS 700	None
50	Satcom Aircell	27

LIST OF SUPPLEMENTS

Supplement No.	Subject	Helicopter Applicability S/N
1	Electrical Provision 109-0742-74-103	11201
2	Observation System EOST-51	11201
3	Reserved	—
4	Reserved	—
5	Reserved	—
6	Reserved	—
7	FLIR LEO II A-3 System	11211/11212
8	Homing DF931-05	11211/11212
9	Rappelling Installation	11211/11212
10	Reserved	—
11	V/UHF AM-FM Flexcomm II	11211/11212
12	V/UHF AM-FM Flexcomm II	11202/11213/11226
13	Direction Finder DF931-05	11202/11213/11226
14	VHF/FM Radio V70/25-4	11202/11213/11226
15	Reserved	—
16	FLIR LEO II A-3 System	11223
17	TELIT SAT550 Telephone System	11223
18	Reserved	—
19	FLIR LEO II A-5	11224
20	Moving Map	11224
21	Radio V/UHF Flexcomm II	11224
22	Direction Finder 931-16	11224
23	NVG	11224
24	GSM Telephone Telit 550	11224
25	TELIT SAT550 Telephone System	11630
26	Moving Map	11630
27	NVG	11630
28	Medical Equipment	11630
29	Reserved	—
30	Reserved	—

Supplement No.	Subject	Helicopter Applicability S/N
31	NVG	11647/11649/ 11700/11704
32	Moving Map	11647/11649
33	SEA FLIR II	11647/11649
34	Deleted	—
35	TELIT GSM/SAT550 Telephone System	11654/11685
36	Reserved	—
37	Reserved	—
38	Reserved	—
39	Reserved	—
40	Reserved	—
41	Reserved	—
42	Reserved	—
43	Reserved	—
44	SEA FLIR III	11700/11704
45	UHF Motorola XTL 5000 Radio	11700/11704
46	HD Camera System	11724
47	Transmission Data Link	11724
48	Receiver Data Link	11724

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dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

ROTOR BRAKE

The Rotor Brake P/N 109-0810-63, consists of a disc type brake, operating lever, the tubing and fittings required for attachment.
Installation of the rotor brake permits rapid deceleration of the rotor after engine shut-down.
The rotor brake operation is pointed out by activation of ROTOR BRAKE ON caution message on EDU 1.

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SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After rotor brake installation the new empty weight and C.G. location must be determined.

PLACARDS

DO NOT APPLY ROTOR BRAKE ABOVE 40% ROTOR RPM

In clear view of the pilot

SECTION 2 - NORMAL PROCEDURES

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N.6 (Fuselage - lh side)

Transmission oil : Check correct level. Door secured.

NOTE

If rotor brake has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

AREA N.7 (Cabin Interior)

Rotor brake lever : Freedom of movement. Leave disengaged (full forward).

PILOT'S PREFLIGHT CHECK

(Every flight)

Rotor brake lever : Freedom of movement. Leave disengaged (full forward).

Transmision oil : Check correct oil level. Door secured.

NOTE

If rotor brake has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

ENGINE PRE-START CHECK

Rotor brake : Engage and check ROTOR BRK ON caution message on EDU 1 displayed; disengage and check caution message suppressed.

ENGINE START

CAUTION

The rotor brake must be disengaged before starting.

SHUTDOWN

CAUTION

The rotor brake must be disengaged whenever the engines are running.

Rotor brake : Apply below 40% rotor RPM.

POST FLIGHT CHECK

Rotor brake lever : Disengaged (full forward)

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
ROTOR BRK ON	Rotor brake lever not in OFF position.	Check rotor brake lever in OFF position (full forward). If message remains on, land as soon as practical.

SECTION 4 - PERFORMANCE DATA

No change.

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dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

ENGINE COMPARTMENT FIRE EXTINGUISHERS

The engine compartment fire extinguishers installation P/N 109-0811-39 consists of two systems, each of which can be used to discharge the extinguishing agent into the desired engine compartment.

Each system consists of a bottle, containing HALON 1301, located behind the engine bay above the baggage compartment, and the necessary tubing to carry the extinguishing agent into the engine compartments.

The installation also includes a control panel located on the overhead console, supports, electrical components and attaching hardware.

Each bottle is provided with an electronic pressure sensor connected with a pilot-lamp located into the avionic bay of baggage compartment to allow an easy check of the bottle pressure status (pilot-lamp illuminated with low pressure inside the bottle).

Moreover each bottle is connected to a disc-type discharge indicator, visible from the helicopter exterior, for an easy check of the bottle charge.

The control panel incorporates two pushbuttons, protected by guards and a three position selector switch.

The pushbuttons illuminate putting in evidence the wording FIRE to indicate the presence of fire in an engine compartment and permit to arm the fire extinguisher system and to operate the fuel shut-off valve to isolate the affected engine (S/OFF wording illuminated). The selector switch, which is spring-loaded to the central position, permits the discharge of one bottle when set to BTL 1 (2) and the discharge of the second bottle, if required, when set to BTL 2 (1).

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PART I - E.A.S.A. APPROVED

LIST OF ILLUSTRATIONS

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CENTER OF GRAVITY LIMITATIONS

After engine compartment fire extinguishers installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°2 (Fuselage - RH side)

Indicator disc	: Red, check.

NOTE

If the indicator disc is not in position, it means that the relative bottle has already been discharged and it needs to be replaced.

Extinguishing system nozzles (inside the engine bay)	: Free of obstructions.

AREA N° 6 (Fuselage - LH side)

Extinguishing bottles pressure status	: Pilot-lamps (in the avionic bay of baggage compartment) extinguished, check.

NOTE

Pilot-lamp lighted up means that the bottle pressure is low (bottle unserviceable) and that it needs to be replaced.

Indicator disc : Red, check.

NOTE

If the indicator disc is not in position, it means that the relative bottle has already been discharged and it needs to be replaced.

Extinguishing system nozzles (inside the engine bay) : Free of obstructions.

PILOT'S PREFLIGHT CHECK

(Every flight)

Extinguishing bottles pressure status : Pilot-lamps (in the avionic bay of baggage compartment) extinguished, check.

NOTE

Pilot-lamp lighted up means that the bottle pressure is low (bottle unserviceable) and that it needs to be replaced.

ENGINE PRE-START CHECK

FIRE EXTING circuit breakers : In.

EDU 1 : Select MENU, press TEST key and check (apart from the test sequence described in the basic Rotorcraft Flight Manual) the illumination of both FIRE pushbuttons.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

WARNING SYSTEM

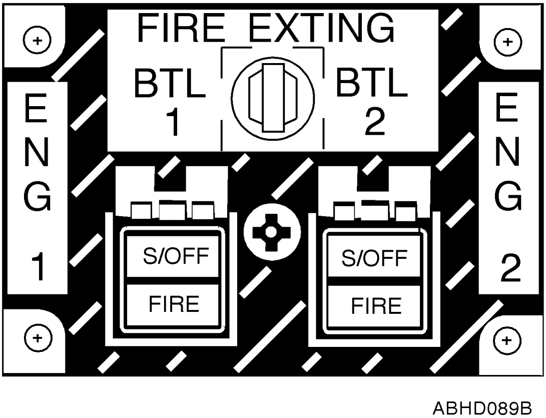

Figure 3-1A. Control panel on the overhead console (Applicable up to S/N 11674).

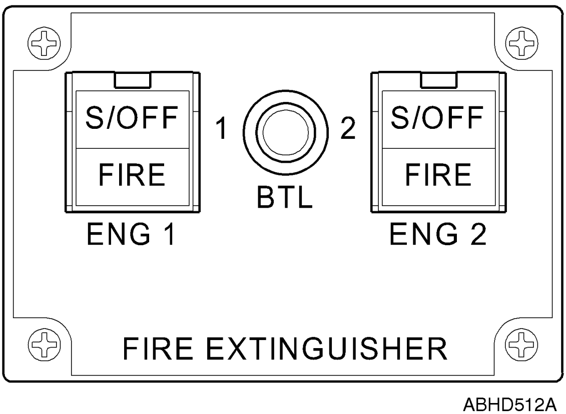

Figure 3-1B. Control panel on the overhead console (Applicable from S/N 11675 and subs).

WARNING MESSAGES (RED)

EDU Message	Fault condition	Corrective action
ENG 1 (2) FIRE	Fire in engine compartment.	Shut down the affected engine. Operate the engine fire extinguisher. Land as soon as possible. (See the pertinent paragraphs of this Section).

FIRE

ENGINE FIRE

INDICATIONS:

EDU 1 : ENG 1 (2) FIRE warning message displayed and ENGINE ONE (TWO) FIRE aural message activated.

Engine power grip (affected engine) : Illuminated.

MASTER WARNING light : Illuminated.

FIRE pushbutton (affected engine) : Illuminated.

ENGINE FIRE DURING START

PROCEDURE

To extinguish fire proceed as follows with the affected engine:

ENG MODE switch : OFF (full counterclockwise).

Power lever	:	OFF.
S/OFF FIRE pushbutton	:	Raise the guard and press. S/OFF wording illuminated.

WARNING

Press button once only to close the fuel shut-off valve and arm the fire extinguishers system. A further operation of the pushbutton re-opens the fuel shut-off valve and disables the system.

Fuel valve (affected engine)	:	Check OFF, fuel valve indicator horizontal.
FIRE EXTING selector switch	:	Set to BTL 1 to cause the discharge of bottle n°1 of the extinguishing agent.
ENG FUEL switch (affected engine)	:	OFF.
CROSS FEED switch	:	CLOSED.
Fuel pump (affected engine)	:	OFF.

If fire warning message does not extinguish:

FIRE EXTING selector switch	:	Set to BTL 2 to cause the discharge of bottle n°2 of the extinguishing agent.

If fire warning message is still present, complete engines shutdown. Abandon the helicopter as soon as possible.

ENGINE FIRE IN FLIGHT

PROCEDURE

Collective	: Decrease until torque is less than 50%.

To extinguish fire proceed as follows with the affected engine.

ENG MODE switch	: OFF (full counterclockwise).
Power lever	: OFF.
S/OFF FIRE pushbutton	: Raise the guard and press. S/OFF wording illuminated.

WARNING

Press button once only to close the fuel shut-off valve and arm the fire extinguishers system. A further operation of the pushbutton re-opens the fuel shut-off valve and disables the system.

Fuel valve (affected engine)	: Check OFF, fuel valve indicator horizontal.
FIRE EXTING selector switch	: Set to BTL 1 to cause the discharge of bottle n°1 of the extinguishing agent.
ENG FUEL switch (affected engine)	: OFF.
CROSS FEED switch	: CLOSED.
Fuel pump (affected engine)	: OFF.
GEN switch (affected engine)	: OFF.

If fire warning message does not extinguish:

FIRE EXTING selector switch : Set to BTL 2 to cause the discharge of bottle n°2 of the extinguishing agent.

If the fire is extinguished proceed to a suitable area and land.
If fire is not extinguished land as soon as possible.

NOTE
Do not attempt to restart the engine in flight.

SECTION 4 - PERFORMANCE DATA

No change.

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The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

WEATHER RADAR - RDR 2000 BENDIX KING

The Weather Radar RDR 2000 Bendix King P/N 109-0811-47 consists of a control panel, of an antenna and a transceiver on the forward part of the helicopter nose.
Basically the weather radar images are presented on a dedicated display on the instrument panel. When EFIS is installed the weather radar images are presented on EHSI of pilot and copilot in the modes ARC WX - ARC WX + MAP VERT PROFILE.

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SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After Weather Radar RDR 2000 installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

WARNING

The weather radar must not be used for obstacle or terrain avoidance.

WEATHER RADAR OPERATION

The Weather Radar cannot be operated when the helicopter is on ground: a microswitch connected to the main landing gear inhibits the radar radiation even if the selector switch is in ON position.

NOTE

When the EFIS is installed, the TX FLT (Transmission FauLT) message is shown in the lower right corner of EHSI (if relevant radar page is selected) if the helicopter is on ground with radar selector in ON position.

NOTE

When SBY or TEST mode is selected the radar does not radiate. As a precaution it is nevertheless convenient to operate as follows:

- Direct nose of helicopter such that antenna scan sector is free of large metallic objects such as hangars or other aircraft for a distance of 30 m (100 ft) and tilt antenna fully upwards.
- Avoid radar operation during refueling of helicopter or other refueling operations within 30 m (100 ft).
- Avoid radar operation if personnel is standing too close in the 180 degree forward sector of the helicopter.

WARNING

Avoid to set the radar ON when there is personnel within 1,5 m (5 ft) from the radar antenna.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

When EFIS is installed, the weather radar image is not available in Composite mode.

SECTION 4 - PERFORMANCE DATA

No change.

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The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

PULSED CHIP DETECTOR

The Pulsed Chip Detector system enables the pilot to burn the particles, collected by the chip detectors of the lubrication systems.
The pulsed Chip Detector is activated by means of the CHIP BURNER switch located on the overhead panel.
Two Pulsed Chip Detector configuration exist:
- P/N 109-0811-48-101 - enabling to burn particles collected by the chip detectors of the main transmission and tail rotor gearbox.
- P/N 109-0811-48-105 - enabling to burn particles collected by the chip detectors of the main transmission, tail rotor gearbox and engines.

The presence of particles on the chip detectors is pointed out by the activation of XMSN OIL CHIP, TGB OIL CHIP and # 1 (# 2) OIL CHIP caution messages on EDU 1.

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SECTION 1 - LIMITATIONS

No change.

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
XSMN OIL CHIP	Presence of metal particles in transmission oil.	Activate CHIP BURNER switch momentarily and release. If the message is not suppressed, reduce power, land as soon as possible.

CAUTION

It is prohibited to activate the switch within 30 minutes from the first activation and more than twice within a 50 hour period. On the third activation of concerned message, land as soon as possible. Maintenance action is required before next flight.
Make an appropriate log book entry for each message activation.

EDU message	Fault condition	Corrective action

NOTE

Terminated a cycle of 50 hours with not more than two message activations, the new 50 hour cycle begins from the last message activation.
Make an appropriate log book entry stating beginning of new 50 hour cycle.

EDU message	Fault condition	Corrective action
TGB OIL CHIP	Presence of metal particles in the tail rotor gearbox oil.	Activate CHIP BURNER switch momentarily and release. If the message is not suppressed, reduce power, land as soon as possible.

CAUTION

It is prohibited to activate the switch within 30 minutes from the first activation and more than twice within a 50 hour period. On the third activation of concerned message, land as soon as possible. Maintenance action is required before next flight.
Make an appropriate log book entry for each message activation.

NOTE

Terminated a cycle of 50 hours with not more than two message activations, the new 50 hour cycle begins from the last message activation.
Make an appropriate log book entry stating beginning of new 50 hour cycle.

EDU message	Fault condition	Corrective action
# 1 (# 2) OIL CHIP	Presence of metal particles in the engine oil.	Activate CHIP BURNER switch momentarily and release. If the message is not suppressed, reduce power, if possible, and shut down the affected engine as soon as practical to prevent damage.

CAUTION

A maximum of 3-fuzz burner activation is permitted per flight. Make an appropriate log book entry for each message activation.

If fuzz burner activation has been recorded, perform maintenance action as per Engine Maintenance Manual, before next flight.

NOTE

The fuzz burner shall not be used to eliminate an engine chip detector caution message when aircraft is on the ground or when the engine oil pressure is not within limits.

SECTION 4 - PERFORMANCE DATA

No change.

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The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

ENVIRONMENTAL CONTROL SYSTEM (AIR CONDITIONING)

The Environmental Control System (ECS) P/N 109-0811-43 consists of two bleed-air shut-off valves, one pressure and regulating shut-off valve, pressure and temperature switches, air conditioning unit, outside air intake and exhaust duct, pilot and passenger area air outlets, connecting ducts and tubing, a circuit switch and two control switches to operate the shut-off valves. The ECS consists also of a cabin air recirculating system. A selecting valve, to reduce the used bleed air, can be selected by an economy mode selector switch.

Bleed air and outside air are fed into the air conditioning unit where a temperature control sensor, set by a control knob (electrically connected to a temperature control system), determines the mixing ratio and the extent to which the air is then cooled to produce the desired temperature.
The system is provided with an overtemperature switch.

NOTE

When the economy mode selector switch is in ECON position the quantity of bleed air from the engines is less.

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ECS OPERATION

Above 2000 ft pressure altitude (Hp), the ECS shall be OFF during take off and landing and in flight below 200 ft AGL.
At any altitude the ECS shall be OFF during single engine operation and in all flight conditions requiring the maximum engine power available.

NOTE

If necessary, the ECS may be used in hovering in ground effect not above 3 ft wheel height and in hovering out of ground effect above 200 ft wheels height.

CENTER OF GRAVITY LIMITATIONS

After ECS installation the new empty weight and C.G. location must be determined.

MISCELLANEOUS LIMITATIONS

During take off and landing with ECS ON, the operations with cargo hook or external hoist are prohibited.

SECTION 2 - NORMAL PROCEDURES

ENGINE PRE-START CHECK

ECS switch	: OFF.
SHUT-OFF #1 switch	: OFF.
SHUT-OFF #2 switch	: OFF.

SYSTEMS CHECK

ENVIRONMENTAL CONTROL SYSTEM OPERATIONAL CHECK

The operational check may be accomplished at this time or at any time that environmental control system is desired.

NR/N2	:	100% stabilized, check.
ECS switch	:	ON, observe no air-flow from the outlets.
ECON/NORM selector switch	:	NORM.
SHUT-OFF #1 switch	:	ON, observe air-flow from outlets, then set to OFF.
SHUT-OFF #2 switch	:	ON, observe air-flow from outlets.
SHUT-OFF #1 switch	:	ON.
TEMP CONT knob	:	Set to MIN. Observe cool airflow from outlets.

NOTE

Condensation may be visible from the outlets when the humidity is high.

Engine temperatures (TOT)	:	Check.
ECON/NORM selector switch	:	ECON. Observe TOT decrease, then set to NORM again.
TEMP CONT knob	:	Set to MAX (by turning clockwise) and observe that the temperature of the airflow from the outlets increases.
ECS switch	:	OFF, observe that the airflow shuts off.
SHUT-OFF # 1 and # 2 switches	:	OFF.

CAUTION

Disengage ECS when:

1. The air flow does not shut-off when ECS switch is OFF. Operate the shut-off switches.
2. The ECS switch trips. Operate the shut-off switches.
3. The air flow does not shut-off when SHUT-OFF # 1, # 2 switches are OFF. Operate the ECS switch.

TAKE-OFF

(Below 2000 ft Hp)

If ECS operation is desired:

SHUT-OFF #1 switch	: ON.
SHUT-OFF #2 switch	: ON.
ECS switch	: ON.
TEMP CONT knob	: As desired.
ECON/NORM selector switch	: As desired.

IN FLIGHT

(Above 200 ft AGL)

If ECS operation is desired:

SHUT-OFF #1 switch	: ON.
SHUT-OFF #2 switch	: ON.
ECS switch	: ON.
TEMP CONT knob	: As desired.
ECON/NORM selector switch	: As desired.

APPROACH AND LANDING

(Above 2000 ft Hp and below 200 ft AGL)

Before descending below 200 ft AGL and landing:

ECS switch	:	OFF.
SHUT-OFF # 1 and # 2 switches	:	OFF.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

ENGINE FAILURES

FAILURE OF ONE ENGINE

Procedure

ECS switch	:	OFF.
SHUT-OFF # 1 and # 2 switches	:	OFF.

FAILURE OF TWO ENGINES

Procedure

ECS switch	:	OFF.
SHUT-OFF # 1 and # 2 switches	:	OFF.

ENGINE RESTART IN FLIGHT

Before attempting a restart:

ECS switch	:	OFF, check.
SHUT-OFF # 1 and # 2 switches	:	OFF, check.

FIRE

ENGINE FIRE IN FLIGHT

Procedure

ECS switch	:	OFF.
SHUT-OFF # 1 and # 2 switches	:	OFF.

SMOKE IN CABIN, TOXIC FUMES, ETC.

ECS switch : OFF.

SHUT-OFF # 1 and # 2 switches : OFF.

ENVIRONMENTAL CONTROL SYSTEM MALFUNCTION

If a malfunction in the ECS controls occurs, deactivate the system as follows:

ECS switch : OFF.

SHUT-OFF # 1 and # 2 switches : OFF.

Proceed with flight, correct trouble before next flight.

SECTION 4 - PERFORMANCE DATA

GENERAL

The maximum power available with the ECS operating is less than that available with the helicopter in basic configuration.
This power loss is caused by the compressor bleed-air used for the ECS.
Performance charts with the ECS switched ON are provided for all engine operative and for Height-Velocity diagram up to 2000 ft Hp.
During single engine operation the ECS must be switched OFF.
Performance with the ECS switched OFF is the same as that shown in the basic Rotorcraft Flight Manual.

HEIGHT - VELOCITY DIAGRAM

(Up to 2000 ft Hp)

The Height - Velocity diagram enable to establish if, in the event of a single engine failure during takeoff, landing or other operation near the surface, a combination of airspeed and height above ground exists from which a safe single engine landing on a smooth, level and hard surface cannot be assured (dangerous zone).

The Height - Velocity diagram is split in two charts.

Chart A shows the weight values, together with outside air temperature and altitude, at/below which the dangerous zone does not exist.

For heavier weights refer to Chart B.

Chart B defines the combinations of height and airspeed to avoid for safe operations.

NOTE

The Height - Velocity diagram does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power off landing can be made.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

ECS ON

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM FOR HEAVIER WEIGHTS, REFER TO CHART "B".

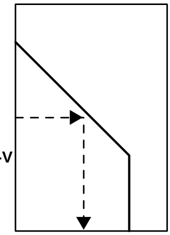

CHART A

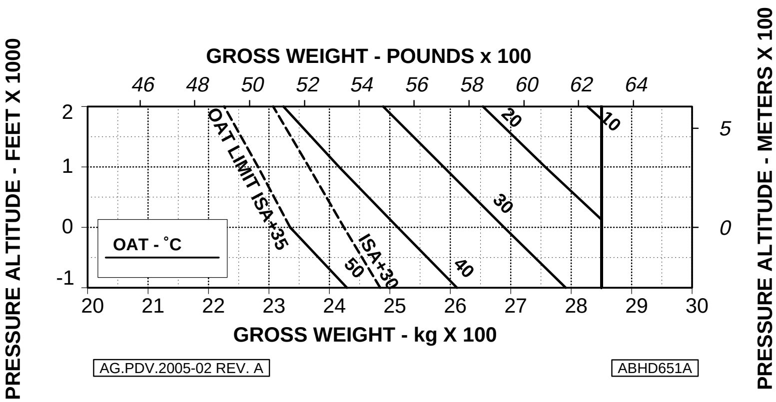

Figure 4-1. Height-Velocity diagram - basic with ECS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

EAPS OFF
ECS ON

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

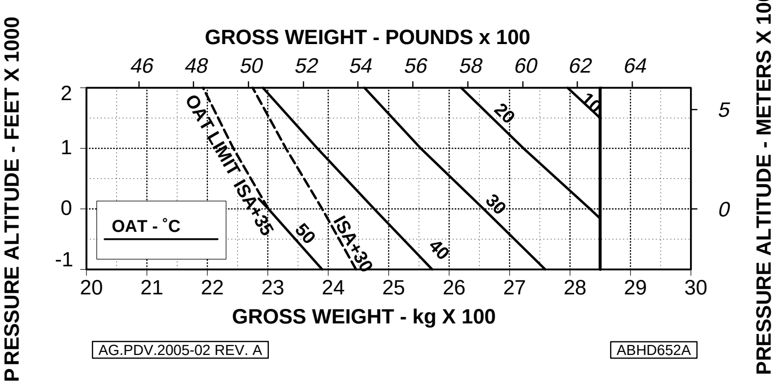

Figure 4-2. Height-Velocity diagram - EAPS OFF with ECS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

EAPS ON
ECS ON

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

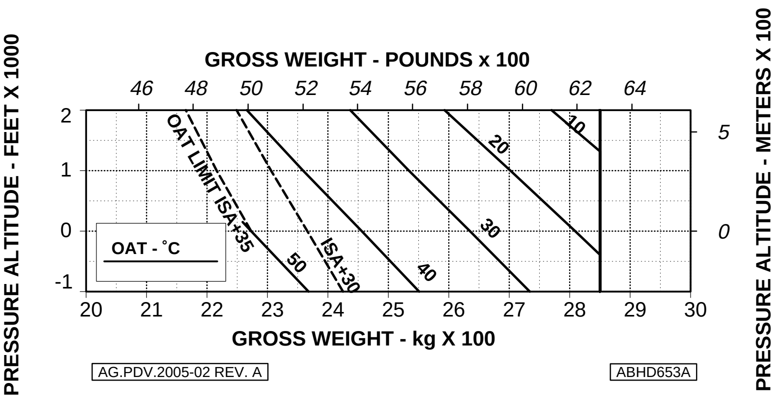

Figure 4-3. Height-Velocity diagram - EAPS ON with ECS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

GROSS WEIGHT: UP TO 3000 kg (6613 lb)

ECS ON (UP TO 2000 ft-HP)

WITHOUT EAPS AND WITH EAPS OFF/ON

CHART B

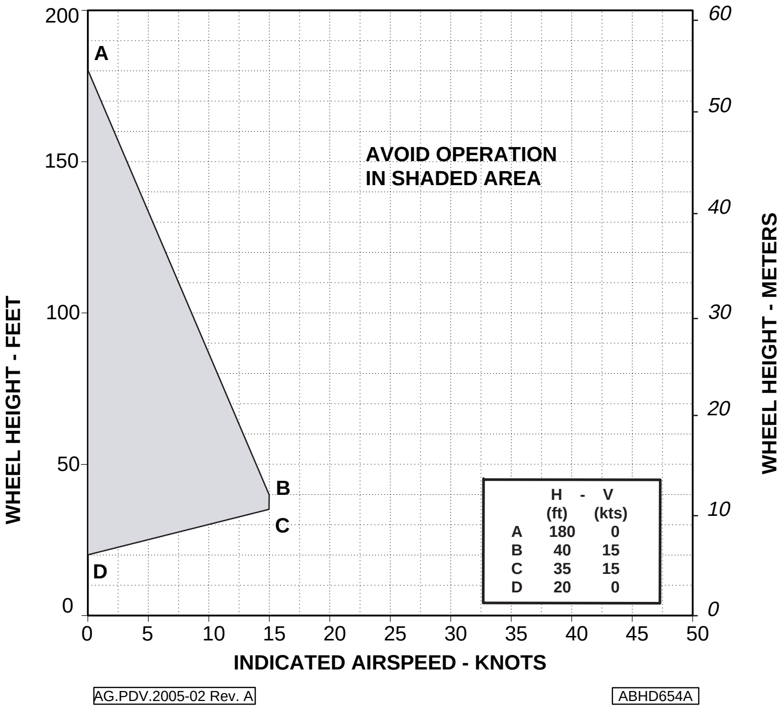

Figure 4-4. Height-Velocity diagram - Chart B.

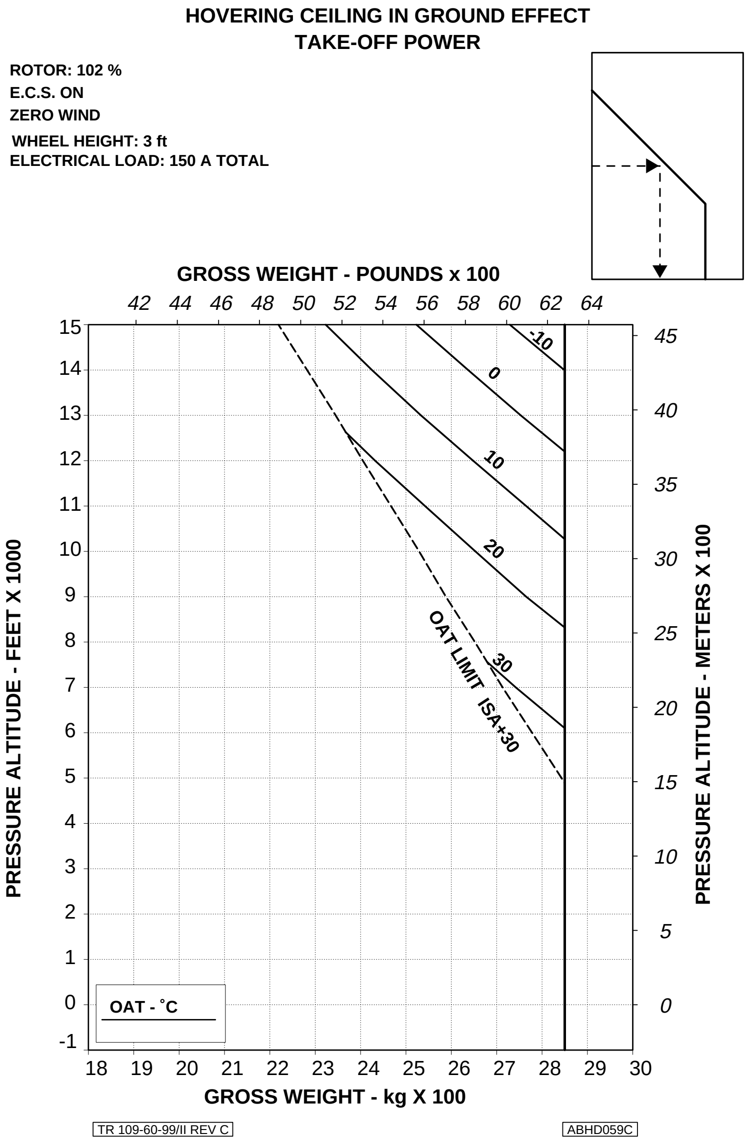

Figure 4-5. Hovering ceiling - IGE - take-off power.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
E.C.S. ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

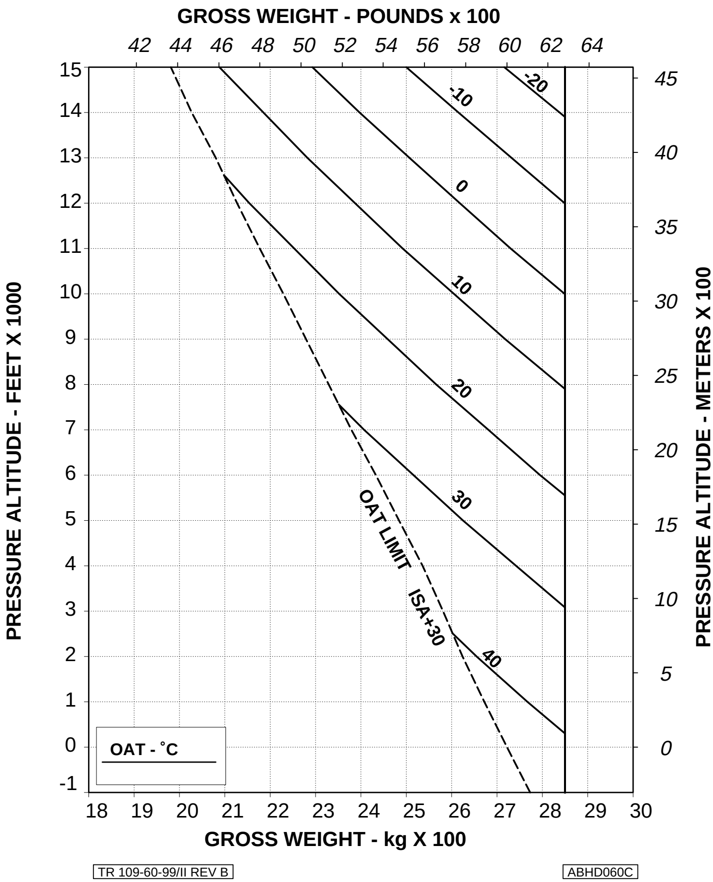

Figure 4-6. Hovering ceiling - IGE - maximum continuous power.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
E.C.S. ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

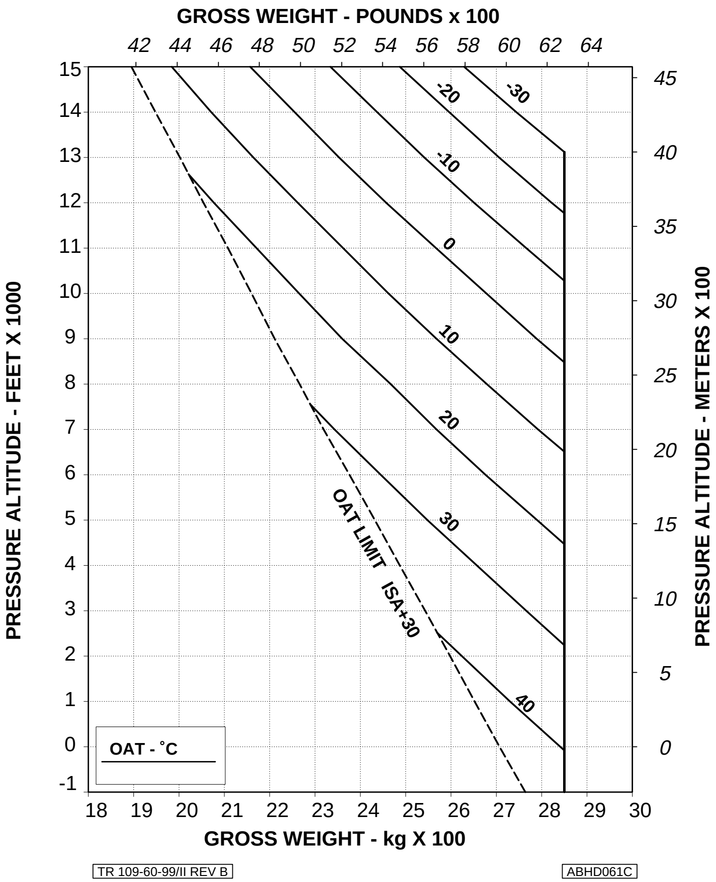

Figure 4-7. Hovering ceiling - OGE - take-off power.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
E.C.S. ON | ELECTRICAL LOAD: 150 A TOTAL

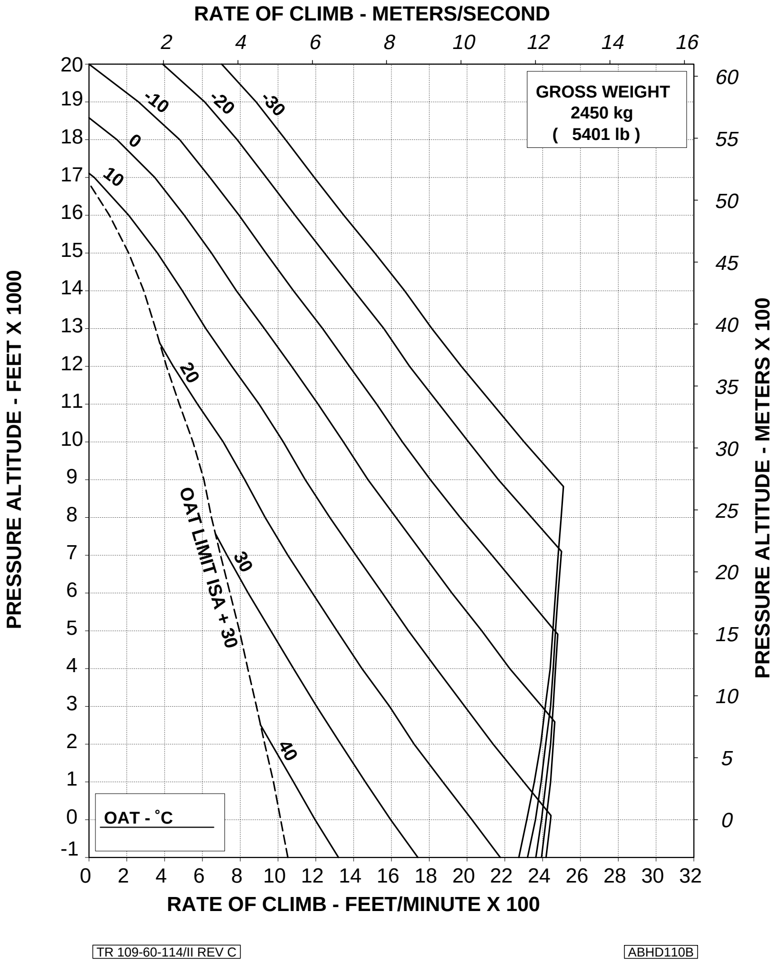

Figure 4-13. Rate of climb - all engines - maximum continuous power - 2450 kg.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
E.C.S. ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

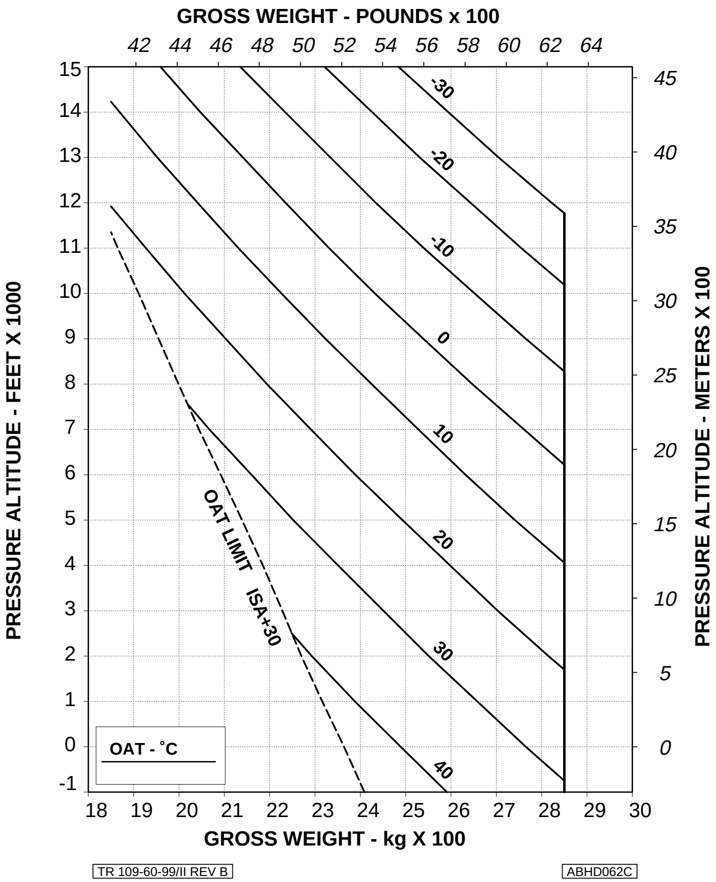

Figure 4-8. Hovering ceiling - OGE - maximum continuous power.

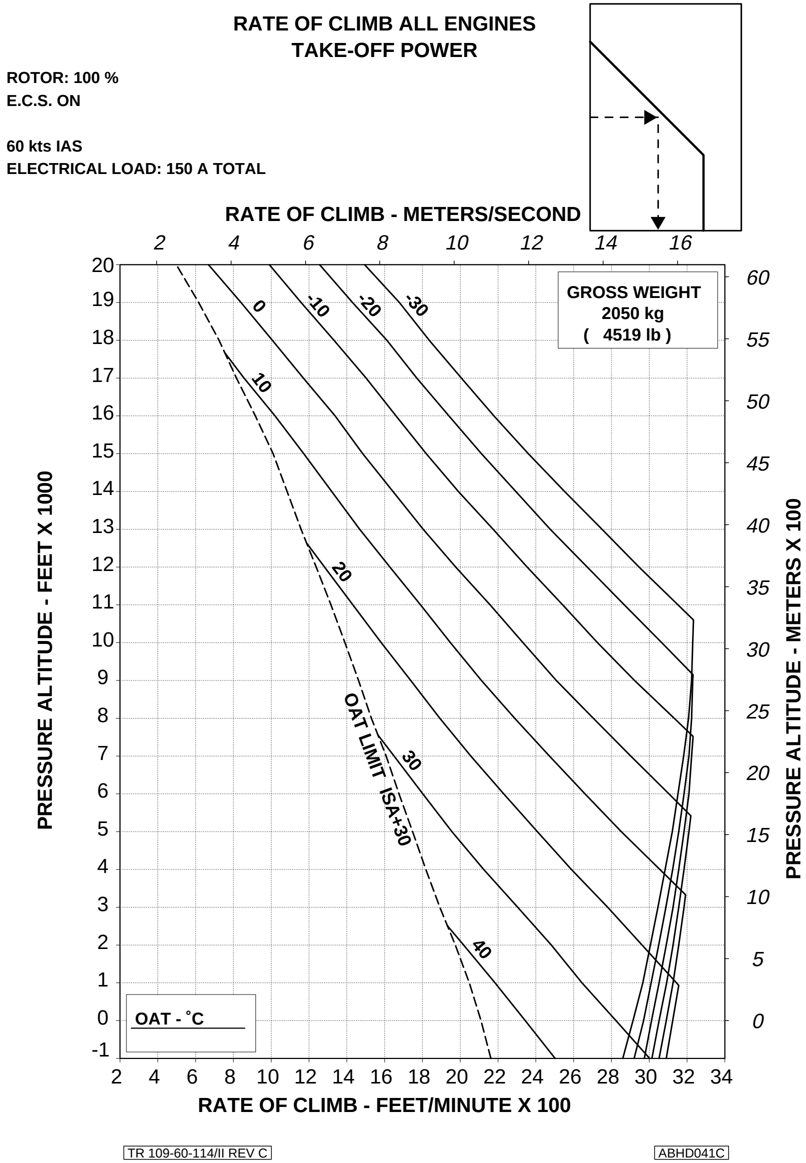

Figure 4-9. Rate of climb - all engines - take-off power - 2050 kg.

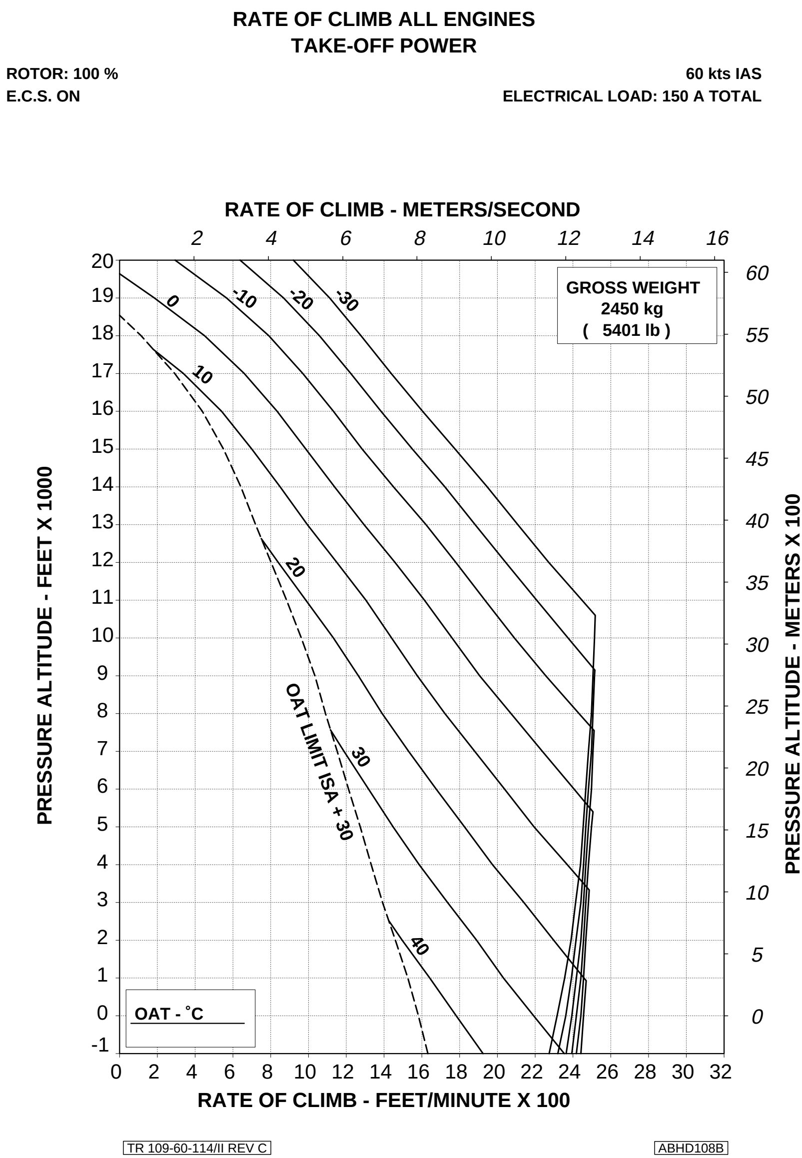

Figure 4-10. Rate of climb - all engines - take-off power - 2450 kg.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

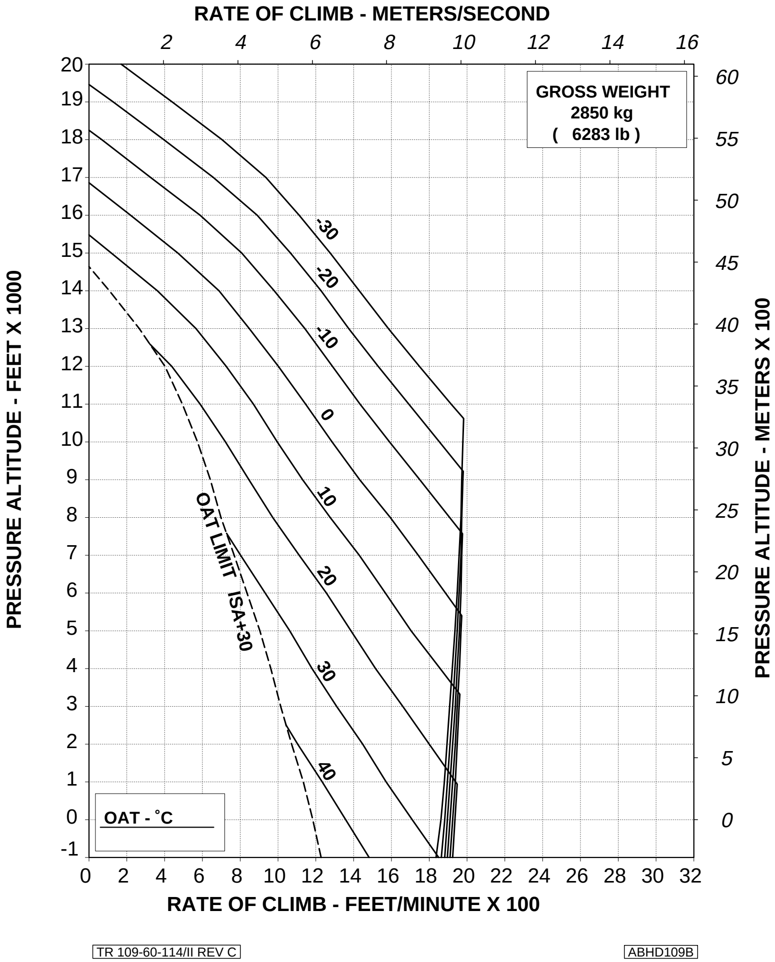

Figure 4-11. Rate of climb - all engines - take-off power - 2850 kg.

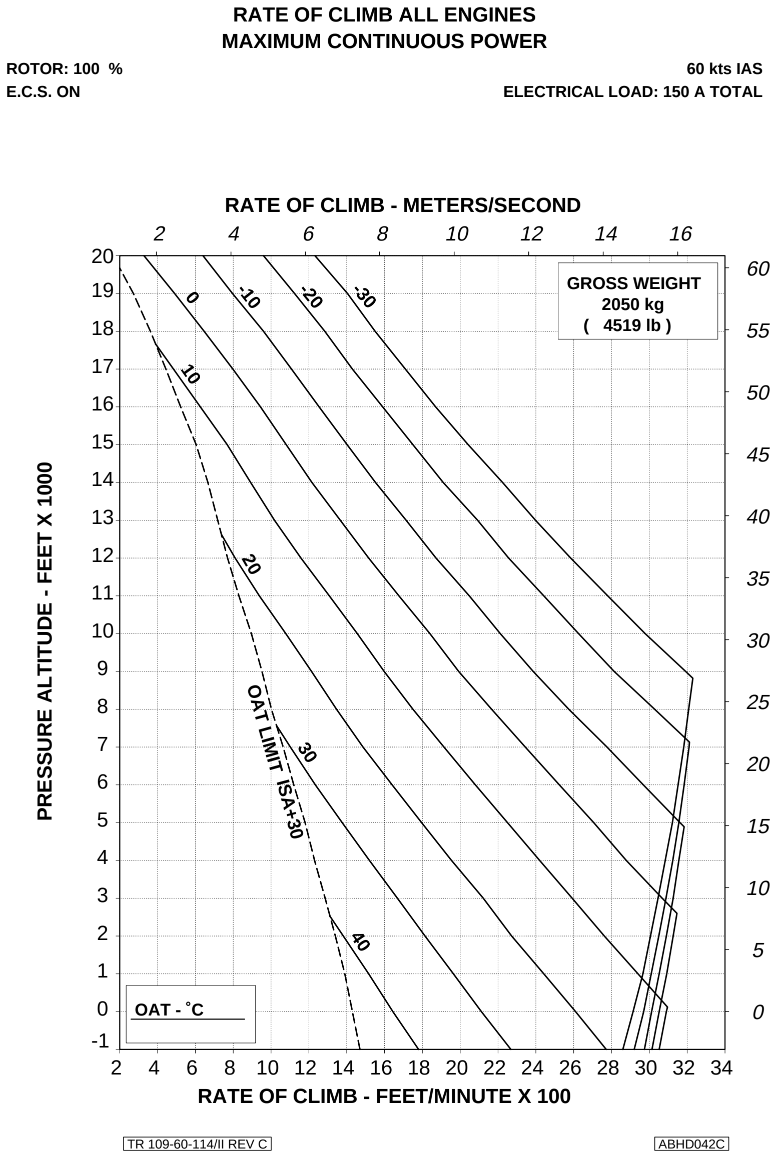

Figure 4-12. Rate of climb - all engines - maximum continuous power - 2050 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS

E.C.S. ON | ELECTRICAL LOAD: 150 A TOTAL

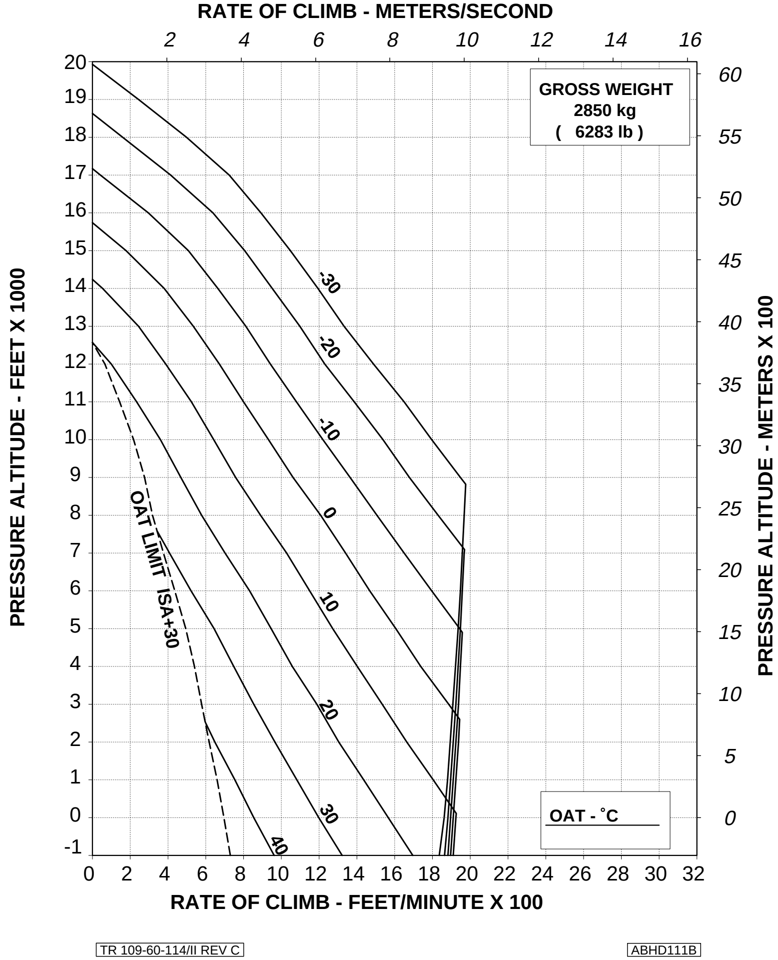

Figure 4-14. Rate of climb - all engines - maximum continuous power - 2850 kg.

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dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

BLEED-AIR HEATER

The Bleed-Air Heater P/N 109-0811-38 consists of two shut-off valves, one mixing valve with solenoid control valve, one outside air intake, a temperature sensor control for pilot and passenger area air outlets, connecting ducts and tubing, one circuit breaker switch and three control switches to operate shut-off valves and solenoid control valve.

Bleed air and outside air is fed into the mixing valve where a sensor determines the mixing ratio to produce the desired temperature. The temperature is regulated by a manual control knob and flexible cable connected to a variable remote sensor in the mixing valve.

The system is provided with an overtemperature switch.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

BLEED AIR OPERATION

The bleed-air heater shall be OFF during take off and landing, in flight below 200 ft AGL, during single engine operation and in all flight conditions requiring the maximum engine power available.

NOTE

If necessary the bleed air heater may be used in hovering in ground effect not above 3 ft wheel height and in hovering out of ground effect above 200 ft wheels height.

CENTER OF GRAVITY LIMITATIONS

After bleed-air heater installation the new empty weight and C.G. location must be determined.

PLACARDS

DO NOT OPERATE HEATER ABOVE 21°C OAT

SECTION 2 - NORMAL PROCEDURES

ENGINE PRE-START CHECK

HTR switch	: OFF.
SHUT-OFF #1 switch	: OFF.
SHUT-OFF #2 switch	: OFF
MIX switch	: OFF

SYSTEMS CHECK

BLEED-AIR HEATER OPERATIONAL CHECK

The operational check may be accomplished at this time or at any time the heater operation is desired.

NR/N2	: 100% stabilized, check.
TEMPerature CONTrol knob	: MIN.
HTR switch	: ON.
TEMPerature CONTrol knob	: Turn the knob clockwise (MAX) and observe no air-flow from outlets.
MIX switch	: ON and observe no air-flow.
SHUT-OFF # 1 switch	: ON and observe warm air-flow from outlets, then set to OFF.
SHUT-OFF # 2 switch	: ON and observe warm air-flow from the outlets.
SHUT-OFF # 1 switch	: ON.
TEMPerature CONTrol knob	: Turn the knob fully counter-clockwise (MIN), observe that the air-flow shut-off, then turn it again clockwise (MAX).
MIX switch	: OFF, observe that the airflow shut-off, then move to ON again.
HTR switch	: OFF, observe that the airflow shut-off.
TEMPerature CONTrol knob	: Turn fully counterclockwise (MIN).
MIX switch	: OFF.
SHUT-OFF # 1 and # 2 switches	: OFF.

CAUTION

Turn the heater OFF when:

1. The air flow does not shutoff when TEMP CONTrol knob is turned fully counterclockwise (MIN). Operate the shutoff switches.
2. The HTR switch trips. Operate the shutoff switches.
3. The air flow does not shutoff when SHUT-OFF #1 and #2 switches are OFF. Operate the HTR switch.

IN FLIGHT (ABOVE 200 FT AGL)

If heater operation is desired:

TEMP CONTrol knob	: Check fully counterclockwise (MIN).
SHUT-OFF # 1 switch	: ON.
SHUT-OFF # 2 switch	: ON.
MIX switch	: ON.
HTR switch	: ON.
TEMP CONTrol knob	: Turn clockwise (to increase temperature) and set to desired temperature.

CAUTION

Do not operate the heater above 21°C OAT.

APPROACH AND LANDING (BELOW 200 FT AGL)

Before descending below 200 ft AGL and landing:

HTR switch	: OFF

SHUT-OFF # 1 and # 2 switches : OFF.

MIX switch : OFF.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

ENGINE FAILURES

FAILURE OF ONE ENGINE

PROCEDURE

HTR switch : OFF.

SHUT-OFF # 1 and # 2 switches : OFF.

MIX switch : OFF.

FAILURE OF TWO ENGINES

PROCEDURE

HTR switch : OFF.

SHUT-OFF # 1 and # 2 switches : OFF.

MIX switch : OFF.

ENGINE RESTART IN FLIGHT

Before attemping a restart:

HTR switch : OFF, check.

SHUT-OFF # 1 and # 2 switches : OFF, check.

MIX switch : OFF, check.

FIRE

ENGINE FIRE IN FLIGHT

Procedure

HTR switch	: OFF.
SHUT-OFF # 1 and # 2 switches	: OFF.
MIX switch	: OFF:

SMOKE IN CABIN, TOXIC FUMES, ETC.

HTR switch	: OFF.
SHUT-OFF # 1 and # 2 switches	: OFF.
MIX switch	: OFF.

HEATER MALFUNCTION

If a malfunction in the bleed-air heater unit controls occurs disconnect the system as follows:

HTR switch	: OFF.
SHUT-OFF # 1 and # 2 switches	: OFF.
MIX switch	: OFF.

Proceed with flight, correct trouble prior to next flight.

SECTION 4 - PERFORMANCE DATA

GENERAL

The maximum power available with the Bleed-Air Heater system operating is less than that available with the helicopter in basic configuration.

This power loss is caused by the compressor bleed-air used for the heater system.

Performance charts with the bleed-air heater switched ON are provided for all engine operative.

During single engine operation the bleed-air heater system must be switched OFF.

Performance with the bleed-air heater system switched OFF is the same as that shown in the basic Rotorcraft Flight Manual.

List of Performance Data Charts:

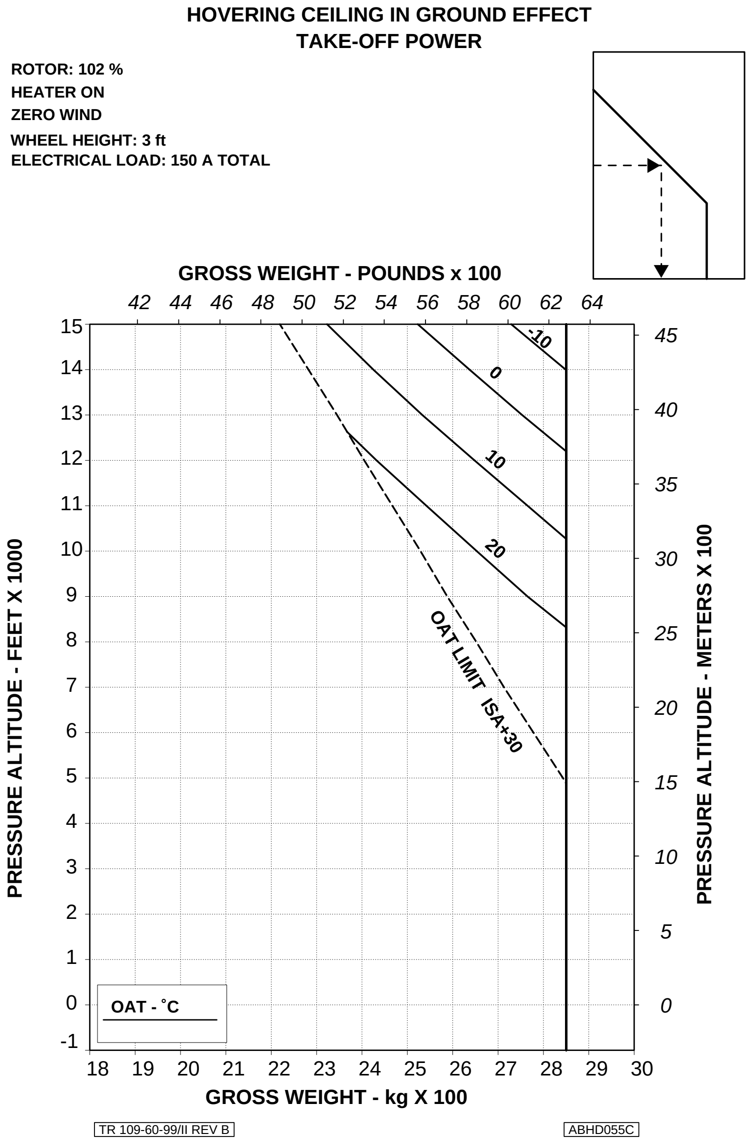

Figure 4-1. Hovering ceiling - IGE - take-off power.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

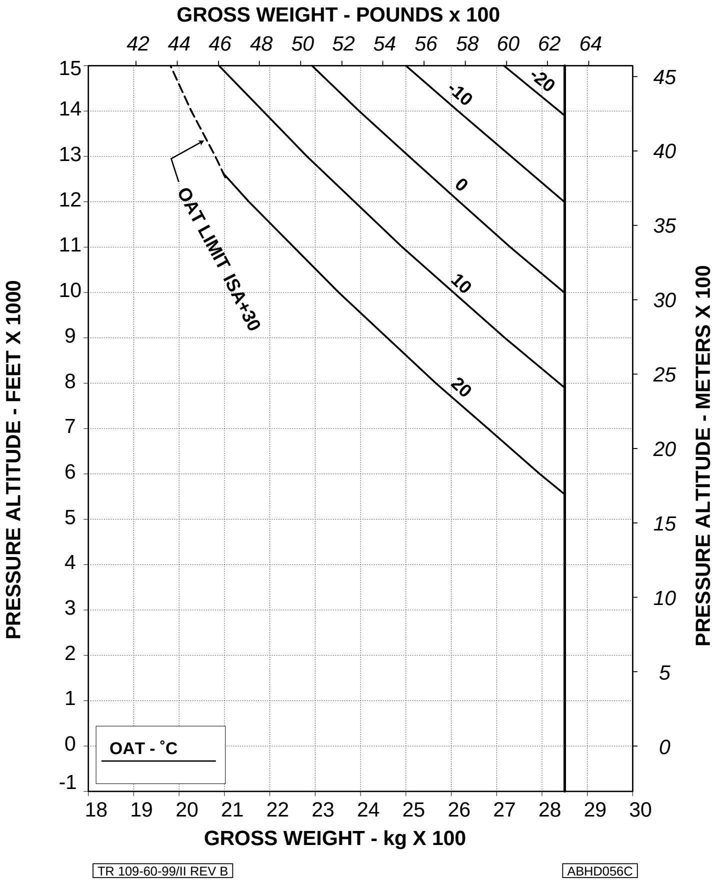

Figure 4-2. Hovering ceiling - IGE - maximum continuous power.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
HEATER ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

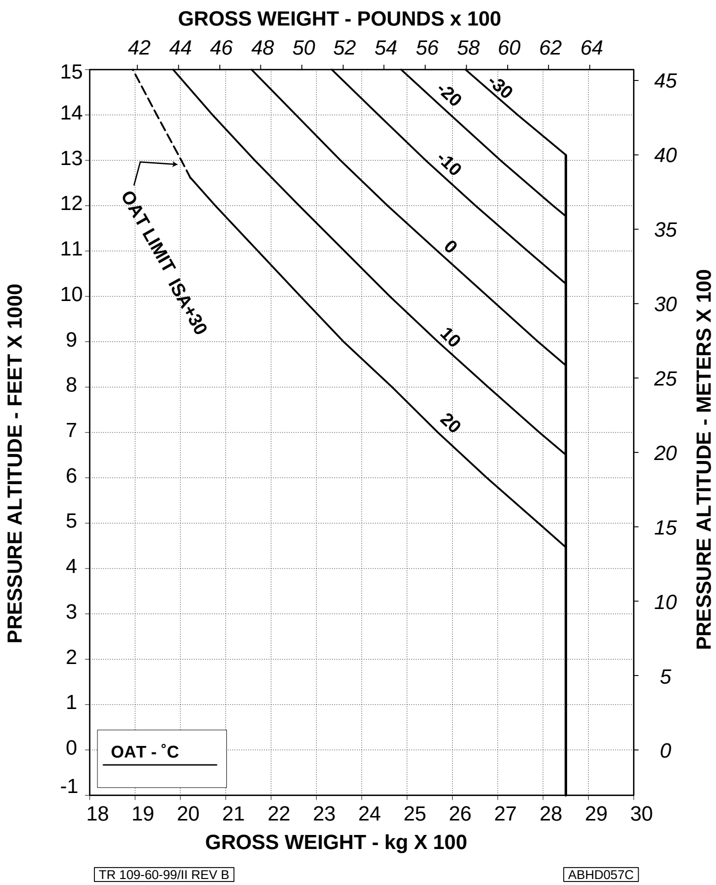

Figure 4-3. Hovering ceiling - OGE - take-off power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

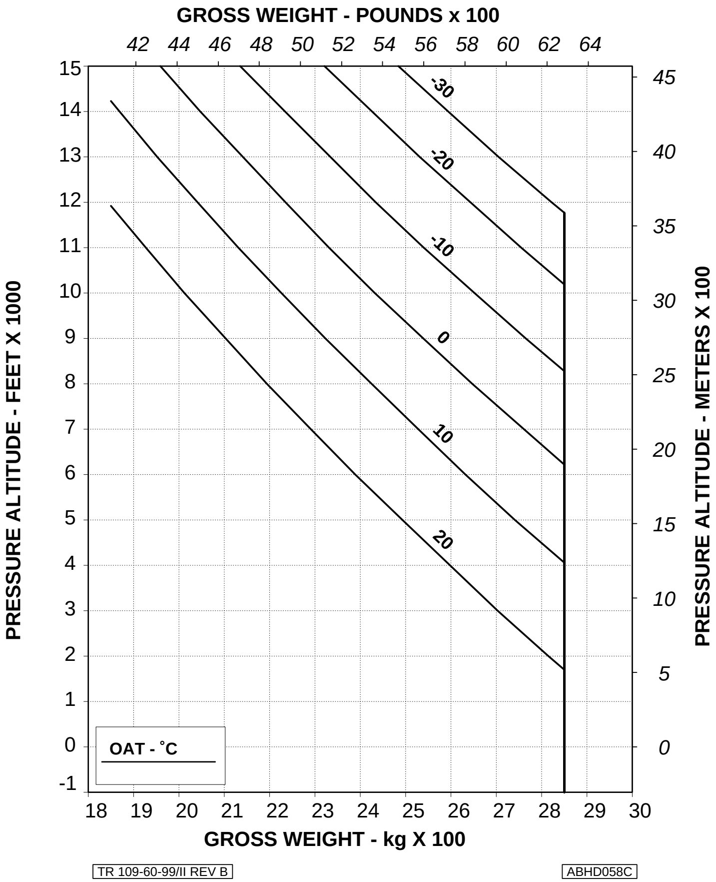

Figure 4-4. Hovering ceiling - OGE - maximum continuous power.

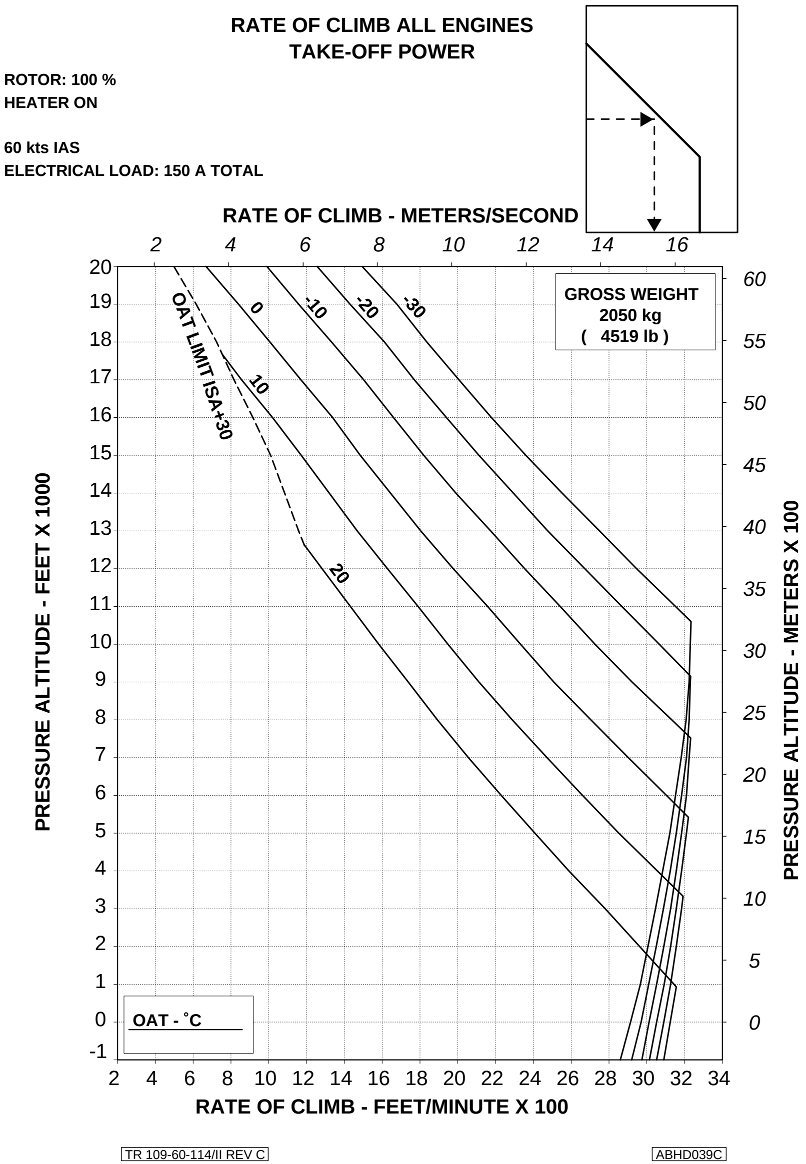

Figure 4-5. Rate of climb - all engines - take-off power - 2050 kg.

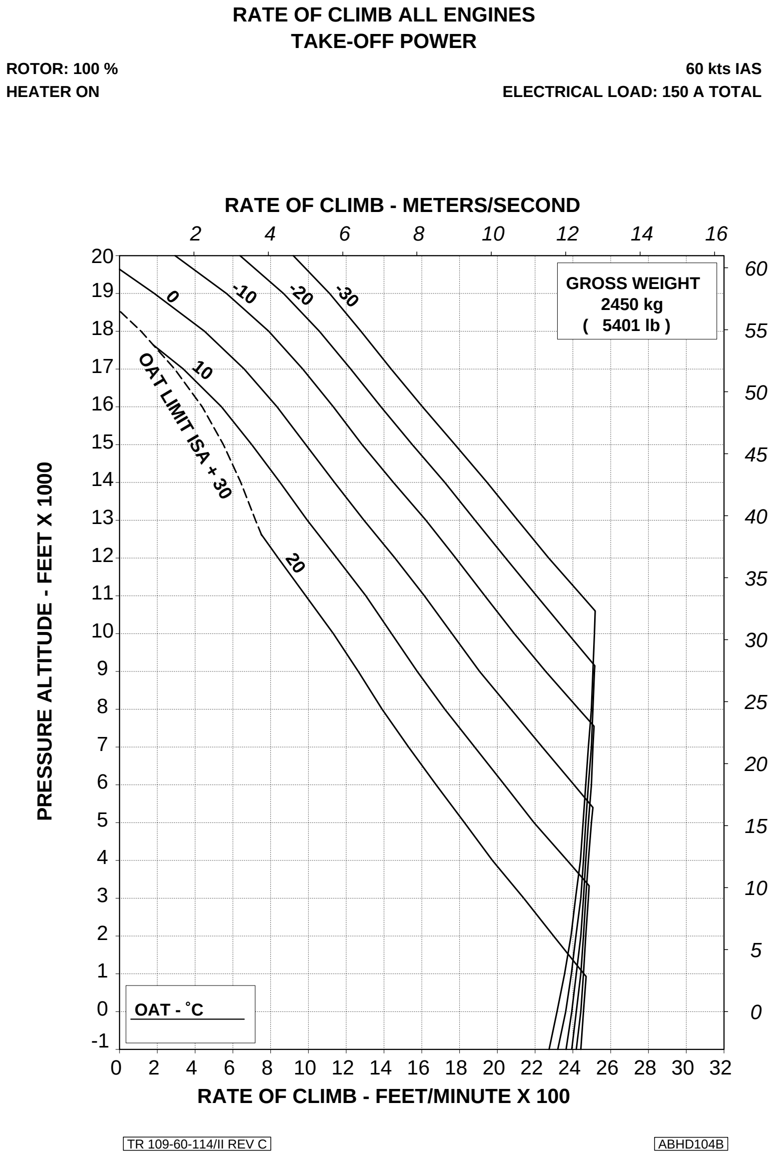

Figure 4-6. Rate of climb - all engines - take-off power - 2450 kg.

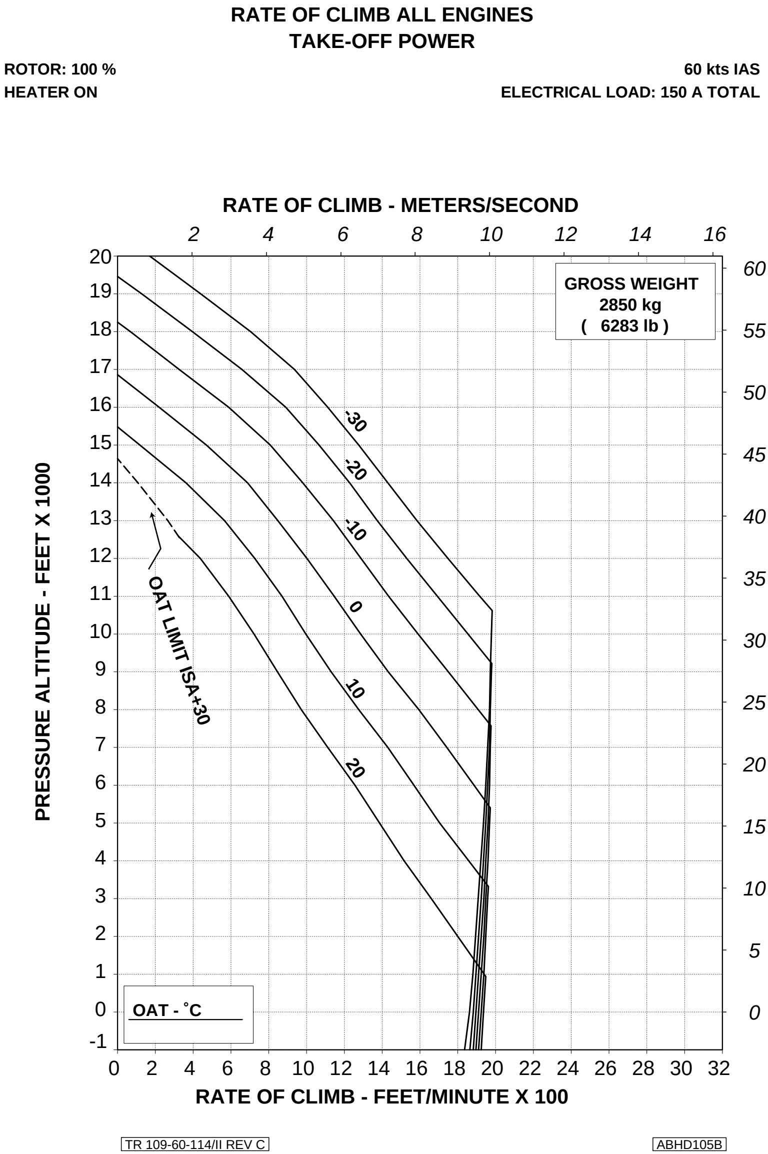

Figure 4-7. Rate of climb - all engines - take-off power - 2850 kg.

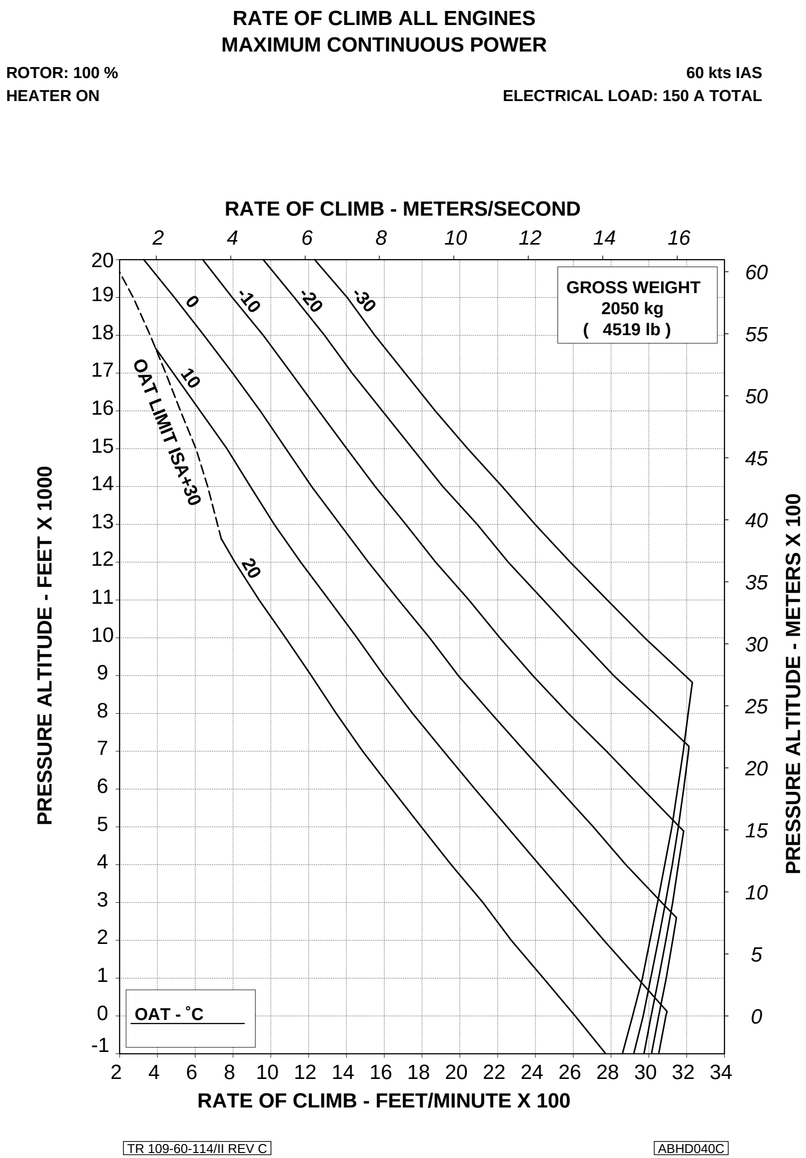

Figure 4-8. Rate of climb - all engines - maximum continuous power - 2050 kg.

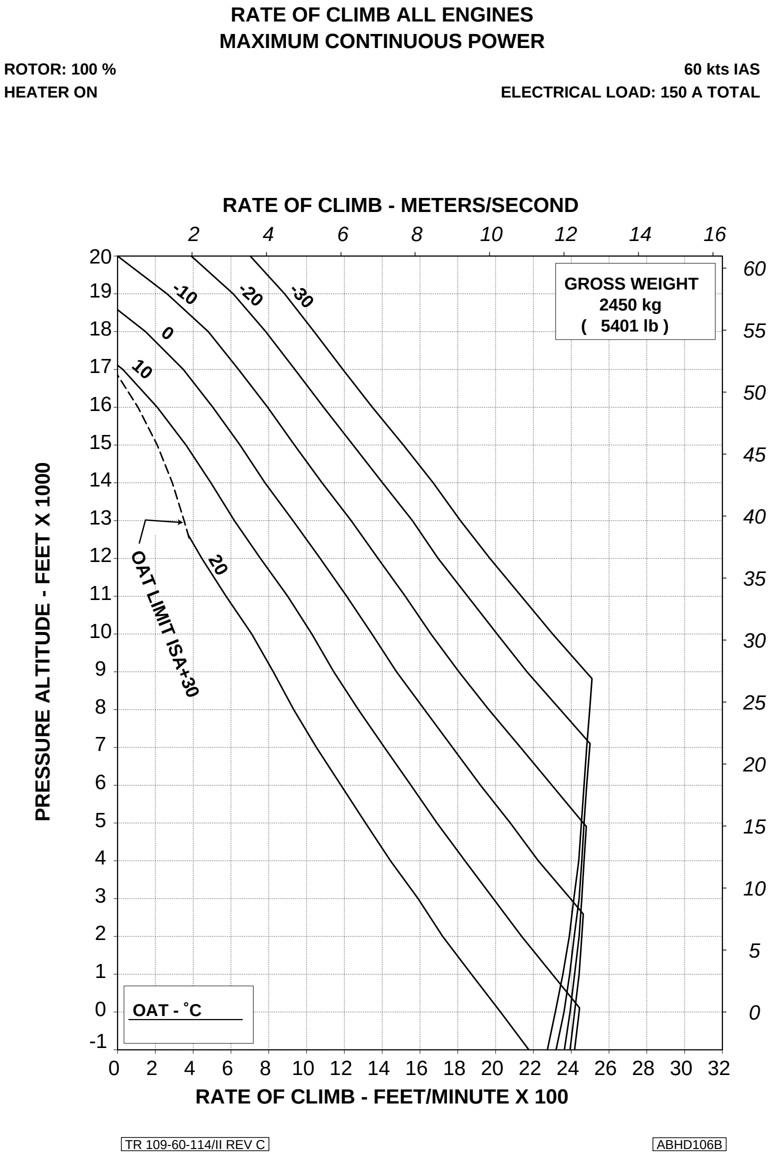

Figure 4-9. Rate of climb - all engines - maximum continuous power - 2450 kg.

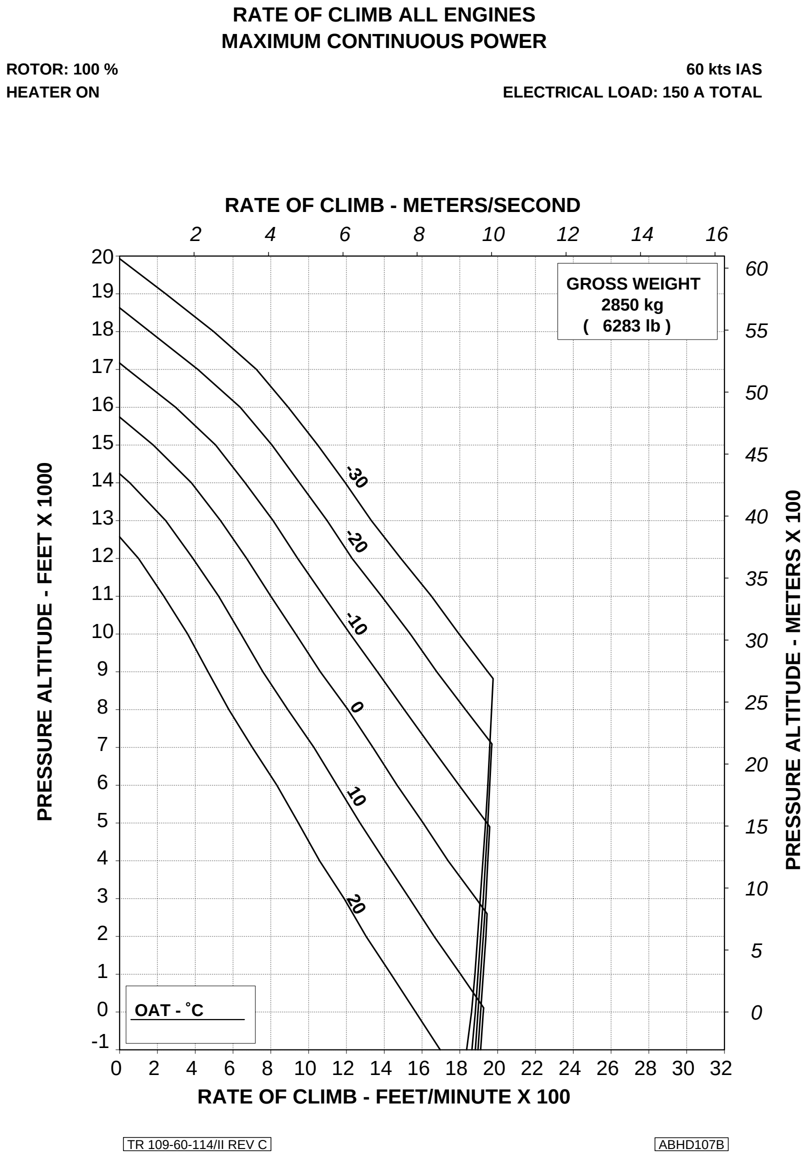

Figure 4-10. Rate of climb - all engines - maximum continuous power - 2850 kg.

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dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SEARCHLIGHT

The Searchlight installation P/N 109-0811-46 consists of a swinging light installed under the forward section of the helicopter fuselage. The light can be extended, stowed or swung as required by operating a switch on the collective lever.

NOTE

When the searchlight P/N 109-0811-46 is installed, one of the fixed landing lights is removed.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

AIRSPEED LIMITATIONS (KIAS)

SEARCHLIGHT INSTALLATION P/N 109-0811-46 (all dashes, other than -115)

Maximum speed for searchlight extension	: 120 KIAS
Maximum speed with searchlight extended, for orientation and retraction	: 150 KIAS

SEARCHLIGHT INSTALLATION P/N 109-0811-46-115 ONLY

Maximum speed for searchlight extension, for orientation and retraction	: 135 KIAS
Maximum speed with searchlight extended	: 150 KIAS

CENTER OF GRAVITY LIMITS

After searchlight installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT’S DAILY PREFLIGHT CHECK

(First flight of the day)

Area N°2 (Fuselage - rh side)

Searchlight	: Condition and cleanliness.

PILOT'S PREFLIGHT CHECK

(Every flight)

Searchlight : Condition and cleanliness.

ENGINE PRE-START CHECK

SRCH-CTL/SRCH-PWR circuit breakers : ON

IN FLIGHT

NOTE

When operating Searchlight, magnetic compass indication is not reliable.

SEARCHLIGHT OPERATING PROCEDURE

NOTE

The searchlight, consisting of a swinging light installed under the forward section of the helicopter fuselage, can be extended, stowed or swung as required by operating a switch on the collective lever.

EXTENSION

EXT/RETR/L/R switch on collective lever : EXT (to extend light in the desired position).

ON/OFF/STOW switch on collective lever : ON.

NOTE

With the switch in OFF position the light remains extinguished in the position where it has been left.

EXT/RETR/L/R switch on collective lever : Set as necessary.

NOTE

Moving switch to L or R position the searchlight rotates left or right. It is possible to adjust the light in an intermediate position, from stowed to extended, by temporarily moving the switch to EXT or RETR position.

RETRACTION

ON/OFF/STOW switch on collective lever : STOW then OFF.

NOTE

In STOW position the light is extinguished.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

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dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SUPPLEMENTARY FUEL TANKS

The supplementary fuel tanks installation P/N 109-0811-49 provides an additional 265 liters capacity. It consists of two tank cells (RH cell of 105 liters and LH cell of 160 liters) installed behind the passenger seat. However the installation can be arranged as follows:
- Both LH and RH tank cells installed.
- Only RH tank cell installed.

The fuel transfer from the supplementary fuel tank cells to the main fuel cell is by gravity. The two tank cells are separated by panels; each cell is provided with a fuel level probe.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After supplementary fuel tank installation, the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

ENGINE PRE-START CHECK

Fuel quantity : Check.

CAUTION

When only the RH tank cell is installed and fuel system is fully serviced, a difference of fuel quantity indication, equivalent to the fuel contained into the RH tank cell, is normal. Such difference decreases with the fuel consumption down to zero when about 110 kg of fuel is reached in each main tank.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

WEIGHTS - ARMS AND MOMENTS

LONGITUDINAL MOMENTS

USABLE FUEL - MAIN FUEL TANKS PLUS SUPPLEMENTARY TANKS			
WEIGHT (kg)	CAPACITY (l) (0.8 kg/l)	ARM (mm)	MOMENT (kgm)
20	25	3324	66.5
40	50	3327	133.1
60	75	3329	199.7
80	100	3331	266.5
100	125	3399	339.9
120	150	3461	415.3
140	175	3505	490.7
160	200	3539	566.2
180	225	3543	637.7
200	250	3551	710.2
220	275	3571	785.6
240	300	3614	867.4
260	325	3641	946.7
280	350	3668	1027.0
300	375	3691	1107.3
320	400	3711	1187.5
340	425	3729	1267.9
360	450	3746	1348.6
380	475	3760	1428.8
400	500	3773	1509.2
420	525	3785	1589.7
440	550	3796	1670.2
460	575	3806	1750.8
480	600	3815	1831.2
500	625	3823	1911.5

(Cont.)

USABLE FUEL - MAIN FUEL TANKS PLUS SUPPLEMENTARY TANKS			
WEIGHT (kg)	CAPACITY (l) (0.8 kg/l)	ARM (mm)	MOMENT (kgm)
520	650	3831	1992.1
540	675	3838	2072.5
560	700	3845	2153.2
580	725	3851	2233.6
600	750	3857	2314.2
620	775	3863	2395.1
640	800	3868	2475.5
660	825	3873	2556.2
680	850	3883	2640.4
688	860	3887	2674.3

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANK ONLY			
WEIGHT (kg)	CAPACITY (l) (0.8 kg/l)	ARM (mm)	MOMENT (kgm)
20	25	3324	66.5
40	50	3327	133.1
60	75	3329	199.7
80	100	3331	266.5
100	125	3399	339.9
120	150	3461	415.3
140	175	3505	490.7
160	200	3539	566.2
180	225	3543	637.7
200	250	3551	710.2
220	275	3571	785.6

(Cont.)

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANKS ONLY			
WEIGHT (kg)	CAPACITY (l) (0.8 kg/l)	ARM (mm)	MOMENT (kgm)
240	300	3614	867.4
260	325	3645	947.7
280	350	3671	1027.9
300	375	3695	1108.5
320	400	3715	1188.8
340	425	3733	1269.2
360	450	3750	1350.0
380	475	3764	1430.3
400	500	3778	1511,2
420	525	3796	1594.3
440	550	3816	1679.0
460	575	3834	1763.6
480	600	3851	1848.5
500	625	3866	1933.0
520	650	3880	2017.6
540	675	3893	2102.2
560	700	3906	2187.4

LATERAL MOMENTS

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANK ONLY			
WEIGHT (kg)	CAPACITY l (0.8 kg/l)	ARM (mm)	MOMENT (kgm)
20	20	0	0.0
40	50	0	0.0
60	75	0	0.0
80	100	0	0.0
100	125	0	0.0
120	150	0	0.0
140	175	0	0.0
160	200	0	0.0
180	225	0	0.0
200	250	0	0.0
220	275	0	0.0
240	300	0	0.0
260	325	15	3.9
280	350	28	7.8
300	375	40	12.0
320	400	49	15.7
340	425	58	19.7
360	450	66	23.8
380	475	73	27.7
400	500	79	31.6
420	525	79	33.2
440	550	75	33.0
460	575	72	33.1
480	600	69	33.1
500	625	66	33.0
520	650	64	33.3
540	675	61	32.9
560	700	59	33.0

SECTION 8 - HANDLING AND SERVICING

SERVICING

Supplementary fuel tanks

	RH fuel cell	LH fuel cell
Capacity	105 liters	160 liters
Usable	105 liters	160 liters

R.A.I. Approval Letter 97/3147/MAE
dated 30 July 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

″GREEN AIRCRAFT″ CONFIGURATION

The ″Green Aircraft″ configuration, as defined in the current Agusta Report N° 109-06-07, includes the following non-conformities with respect to the General Arrangement Drawing N° 109-9000-01-151:

— Interior arrangement (soundproofing, upholstery, ashtrays, etc.), seats and safety belts not installed except for pilot seat and relative safety belts and ashtray. The helicopter may not be painted.

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SECTION 1 - LIMITATIONS

FLIGHT CREW

The minimum flight crew consists of one pilot who shall operate the helicopter from the right crew seat.

NUMBER OF SEATS

One (pilot).

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 97/5288/MAE
dated 17 December 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

E.M.S.
EMERGENCY MEDICAL SERVICE

The Emergency Medical Service P/N 109-0811-70, for emergency, rescue and ambulance operation, may be equipped as follows:

— -113/-179 consists of a litter, a medical storage chest and three seats: one in the cabin forward position, swivelling type facing aft and two in the cabin aft position, fixed facing forward, panels at the upper side of lateral and aft walls for medical devices and lights. The litter, when installed, is equipped with three straps and is mounted longitudinally in the left hand side of the cabin.

— -119/-165 in addition to the components listed for -113/-179, it includes an oxygen system P/N 109-0811-76 that is supplied by a bottle of 2200 liters, installed in the baggage compartment, at a pressure of 70 psi and may be shut-off by means of a handle mounted on the left hand side of the pilot overhead aft panel.

— -127/-171 consists of a second litter installed on the right hand side of the cabin in conjunction with the E.M.S. P/N 109-0811-70-113/-179 or -119/-165.

— -131 consists of an internal arrangement without any litter, eight seats (including the pilot), panels at the upper side of lateral and aft walls for medical devices, lights and a provision for the oxygen system.

— -137/-173 in addition to the components listed for -119/-165 it includes different passengers belts P/N AGU-001A and a supplementary fuel tank P/N 109-0811-49, of 105 liters capacity (for version -137) and a supplementary fuel tank P/N 109-0812-46, of 94 liters capacity (for version -173), installed behind the aft passenger seat, RH side (refer to Appendix 8 for Supplementary Fuel Tanks P/N 109-0811-49 and Appendix 36 for Supplementary Fuel P/N 109-0812-46) of this manual.

— -143 in addition to the components listed for -119 it includes different passengers belts P/N AGU-001A.

— -145 in addition to the components listed for -127 it includes different passengers belts P/N AGU-001A.

— -163 as -183 but with passengers belts P/N AGU-001A.

— -183 as -173 but with oxygen system removed.

The E.M.S. comprises an intercommunication system for the patient attendants.

CAUTION

All medical equipment incorporating electronic devices must satisfy the electromagnetic compatibility requirements.

NOTE

For the E.M.S. P/N 109-0811-70-113/-119/-137/-143/-163/-165/-173/-179/-183, the electromagnetic compatibility has been verified with the following medical equipment:
- Lifepak 12 defibrillator plus monitor P/N VLP 12-02-000033 or
- Lifepak 12 defibrillator plus monitor P/N VLP 12-02-000602
- Oxylog 2000 pulmonary ventilator P/N 8413950
- Dialog 2000 monitor P/N 5706201
- Oxypac pulse oximeter P/N 2128578
- Syringe infusion pump IVAC P2000 P/N 2001FA2UGB0
- Medical suction unit 0B Minivac ASP 70170 P/N BF

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PART II - MANUFACTURER'S DATA

LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The helicopter in EMS configuration permits rescue and ambulance operation under day and night VFR and IFR non-icing conditions.

REQUIRED EQUIPMENT

Sliding doors kit installation is required for EMS installation and operation.

Whenever the E.M.S. supplementary cabin lights P/N 109-0812-51-107 are installed, the curtain installation P/N 109-0812-52 must be present on board.

VFR OPERATION

FLIGHT WITH PASSENGER CABIN DOORS OPEN OR REMOVED

Operation with passenger cabin doors open or removed is prohibited when patient is on board.

FLIGHT CREW

The minimum flight crew consists of one pilot and one attendant; both of whom shall be trained in and capable of assisting in litter patient emergency evacuation procedures.

NUMBER OF SEATS

(E.M.S. P/N 109-0811-70-113/-179, -119/-165, -137/-173, -143 and -163/-183)
Six (6) - including the pilot and litter patient for E.M.S.
However the number of seats depends on the interior arrangement.
(E.M.S. P/N 109-0811-70-127/-171 and -145)
Six (6) - including the pilot and litter patients for E.M.S.
However the number of seats depends on the interior arrangement.
(E.M.S. P/N 109-0811-70-131)
Eight (8) - including the pilot and up to seven seats.
However the number of seats depends on the interior arrangement.

BAGGAGE COMPARTMENT LIMITATIONS

(When oxygen bottle is installed)

Maximum load : 140 kg (309 lb)

CENTER OF GRAVITY LIMITATIONS

After E.M.S. installation the new empty weight and CG location must be determined.

PLACARDS

(When oxygen bottle is installed)

NO SMOKING WITH OXYGEN SYSTEM INSTALLED

In clear view of pilot/copilot and in passengers cabin, above passengers cabin doors

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°2 (Fuselage - RH side)

Utility access door to the oxygen bottle socket (if installed) : Secured.
UTIL DOOR caution message on EDU 1 out.

NOTE

The messages UTIL DOOR and OXYGEN CLOSED are respectively replaced by caution and advisory lights on the instrument panel when IDS configuration utilizes the EDU software version 06 or lower and DAU software version 05 or lower.

OXY H.P. RELIEF green indicator plug : Check in position.

NOTE

If the OXY H.P. RELIEF green indicator plug is not in position, the oxygen bottle is discharged.

AREA N°6 (Fuselage - LH side)

Baggage compartment : Check the oxygen bottle fairing is properly secured.

AREA N°7 (Cabin interior)

Cabin seats and litter(s) : Check for condition and fastened if unoccupied.

PILOT'S PREFLIGHT CHECK

(Every flight)

Utility access door to the oxygen bottle socket (if installed) : Secured.
UTIL DOOR caution message on EDU 1 out.

NOTE

The messages UTIL DOOR and OXYGEN CLOSED are respectively replaced by caution and advisory lights on the instrument panel when IDS configuration utilizes the EDU software version 06 or lower and DAU software version 05 or lower.

OXY H.P. RELIEF green indicator plug : Check in position.

NOTE

If the OXY H.P. RELIEF green indicator plug is not in position, the oxygen bottle is discharged.

Baggage compartment	:	Check the oxygen bottle fairing is properly secured.
Cabin seats and litter(s)	:	Check for condition and fastened if unoccupied.

ENGINE PRE-START CHECK

OXYGEN MANUAL SHUT-OFF handle	:	Check in closed position (in). OXYGEN CLOSED advisory message on EDU 2 displayed.

NOTE

The messages UTIL DOOR and OXYGEN CLOSED are respectively replaced by caution and advisory lights on the instrument panel when IDS configuration utilizes the EDU software version 06 or lower and DAU software version 05 or lower.

TAKE-OFF

Cabin curtain (if installed)	:	Open.

In case of night flight:

CAB switch	:	OFF.

IN FLIGHT

CAB switch	:	As required.

NOTE

Whenever the supplementary cabin lights are installed and switched on, cabin curtain must be closed.

APPROACH AND LANDING

Cabin curtain (if installed) : Open.

In case of night flight

CAB switch : OFF.

LITTER OPERATIONS

(All E.M.S. configurations except P/N 109-0811-70-131)

LITTER(S) LOADING

— Secure patient(s) to the litter using the three straps provided.
— Unlock and rise the central aft seat.
— Load left litter from LH door and right litter (if installed) from RH door.
— Secure litter(s) to the locks.

LITTER(S) UNLOADING

— Unlock and rise the central aft seat.
— Unlock the litter(s).
— Unload the right litter (if installed) from RH door and/or left litter from LH door.

OXYGEN SYSTEM OPERATIONS

OXYGEN MANUAL SHUT-OFF handle : Pull to open.
OXYGEN CLOSED advisory message on EDU 2 out.

NOTE

The messages UTIL DOOR and OXYGEN CLOSED are respectively replaced by caution and advisory lights on the instrument panel when IDS configuration utilizes the EDU software version 06 or lower and DAU software version 05 or lower.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

(E.M.S. P/N 109-0811-70-119/-165, -137/173 and -143)

WARNING SYSTEM

CAUTION MESSAGES (YELLOW)

NOTE

The messages UTIL DOOR and OXYGEN CLOSED are respectively replaced by caution and advisory lights on the instrument panel when IDS configuration utilizes the EDU software version 06 or lower and DAU software version 05 or lower.

EDU message	Fault condition	Corrective action
UTIL DOOR	The utility access door to the oxygen bottle socket is not correctly closed.	Close door correctly before flight. If the caution message is displayed during flight, land as soon as practicable.

In emergency, pilot(s) must manually close the shut-off valve of the oxygen supply system by pushing the OXYGEN MANUAL SHUT-OFF handle on the overhead panel (green OXYGEN CLOSED advisory message on EDU 2 displayed).

EVACUATION THROUGH EMERGENCY EXITS

Unstrap the patient(s) and evacuate either through the LH or RH emergency exit.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

PATIENT ARM ON LITTER (-113/-179, -119/-165, -137/-173, -143 AND -163/-183)

Longitudinal arm (STA)	: 2900 mm (114.2 inches) from STA 0.
Lateral arm (BL)	: 412 mm (16.2 inches) from the helicopter plane of simmetry.

PATIENTS ARM ON LITTER (-127/-171 AND -145)

Longitudinal arm (STA)	: 2900 mm (114.2 inches) from STA 0.
Lateral arm (BL) (left or right)	: 412 mm (16.2 inches) from the helicopter plane of simmetry.

FORWARD ATTENDANT ARM (-113/-179, -119/-165, -137/-173, -143 AND -163/-183)

Longitudinal arm (STA)	: 2455 mm (96.6 inches) from STA 0.
Lateral arm (BL)	: 0 to 412 mm (16.2 inches) from helicopter plane of simmetry.

FORWARD ATTENDANTS ARM (-131 ONLY)

Longitudinal arm (STA)	: 2455 mm (96.6 inches) from STA 0.

AFT ATTENDANT ARM (-113/-179, -119/-165, -137/-173, -143 AND -163/-183)

Longitudinal arm (STA)	: 3200 mm (126 inches) from STA 0.
Lateral, inboard attendant arm (BL)	: 0 mm.

Lateral, outboard attendant arm (BL) : 412 mm (16.2 inches) from the helicopter plane of simmetry.

AFT ATTENDANT ARM (-127/-171 AND -145)

Longitudinal arm (STA) : 3200 mm (126 inches) from STA 0.

AFT ATTENDANTS ARM (-131 ONLY)

Longitudinal arm (STA) : 3200 mm (126 inches) from STA 0.

SECTION 7 - SYSTEMS DESCRIPTION

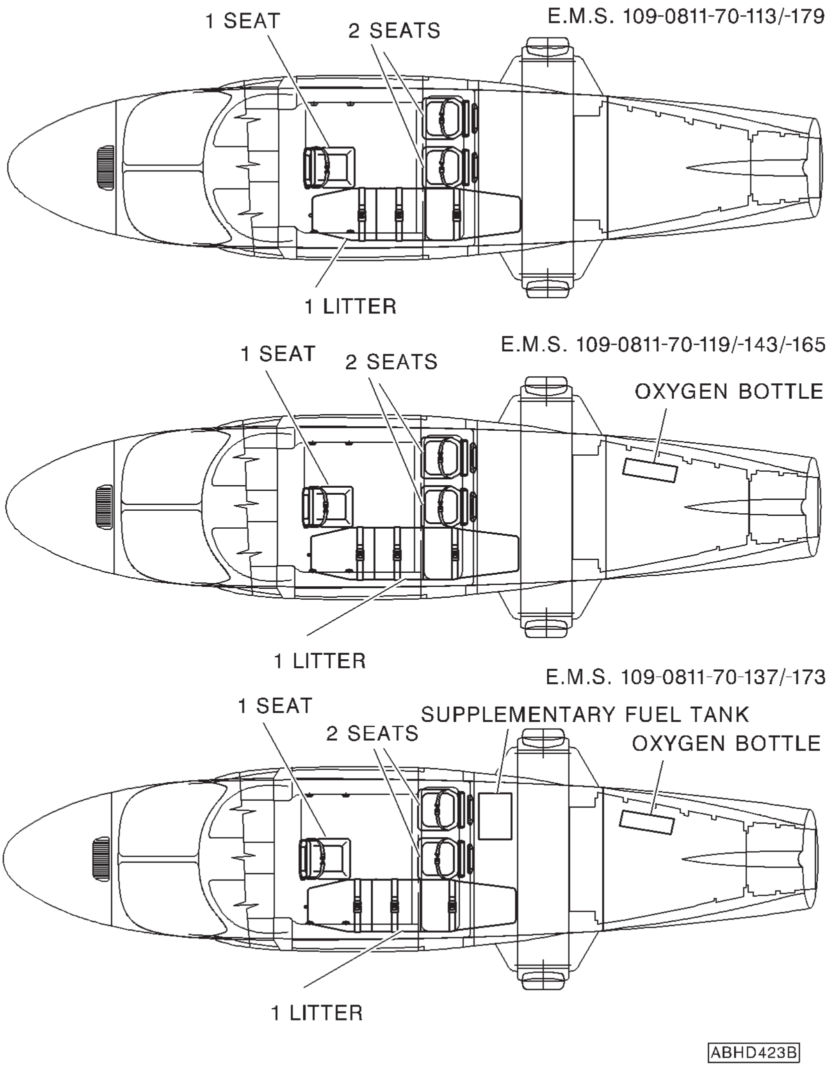


Figure 7-1. (sheet 1 of 2) E.M.S. Interior arrangement.

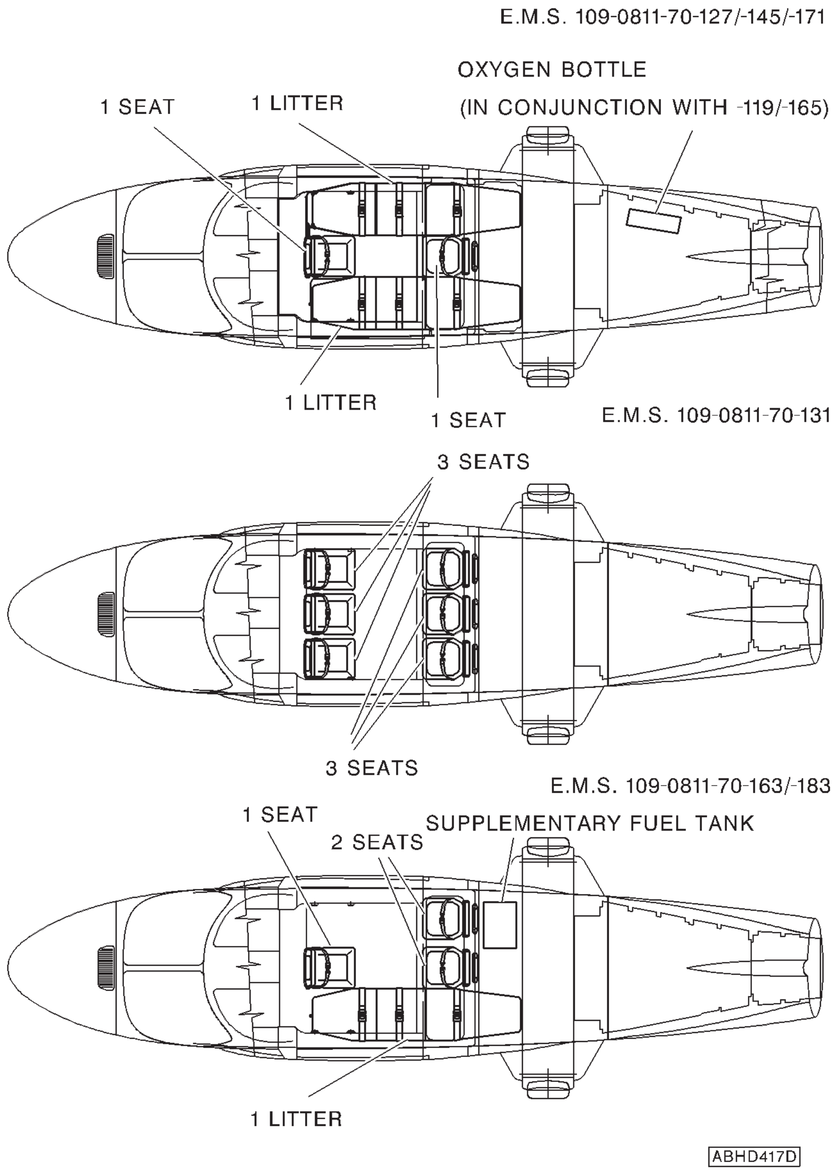
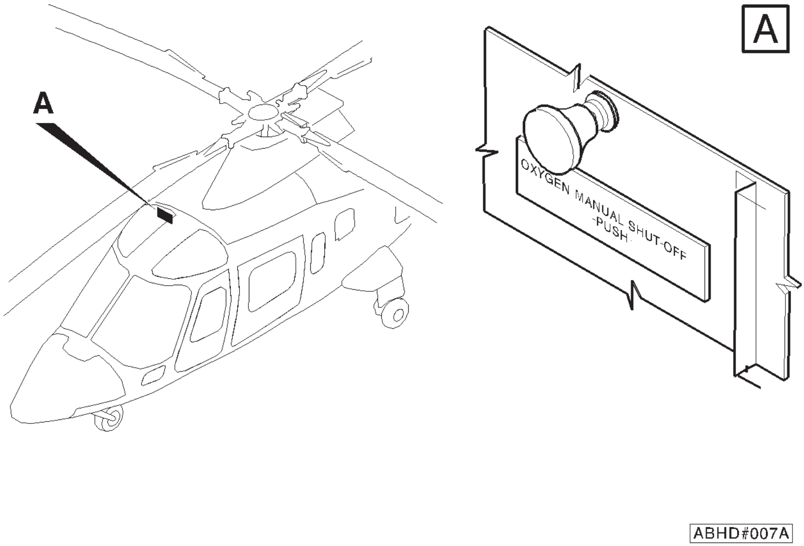


Figure 7-1. (sheet 2 of 2) E.M.S. Interior arrangement.



(WHEN OXYGEN BOTTLE IS INSTALLED)

Figure 7-2. Oxygen manual shut-off handle.

R.A.I. Approval Letter 97/5288/MAE
dated 17 December 1997

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SLIDING DOORS

The sliding doors P/N 109-0822-58 are installed in lieu of the cabin standard passenger doors. The doors, provided with an emergency jettison window, slide along airframe mounted tracks; the door aft travel is limited by a stop installed on the lower track.
Each sliding door is provided with internal and external operating handles, with a locking device, on the upper side, which permits to maintain the door secured in open position and with two pins, on the forward edge, that engage holes, provided in the fuselage, to maintain the door secured in closed position.

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SECTION 1 - LIMITATIONS

VFR OPERATION

VFR operation is approved with either one or both doors locked in open position or removed.

NOTE

Operating limitations are identical for both conditions either with doors locked in open position or removed.

IFR OPERATION

IFR operation is prohibited with either one or both doors locked in open position or removed.

AIRSPEED LIMITATIONS (KIAS)

V_{NE} with doors closed	: No change.
V_{NE} with one or both doors opened or removed	: 70 KIAS.
V_{NE} during doors opening and closing operations	: 70 KIAS.

CENTER OF GRAVITY LIMITATIONS

After sliding door installation the new empty weight and CG location must be determined.

CAUTION

Opening one or both sliding doors will result in a 13 mm maximum aft displacement change CG location.

PLACARDS

OPENING/CLOSING DOOR ABOVE 70 Kts IAS IS PROHIBITED

Below each sliding door handle

SECTION 2 - NORMAL PROCEDURES

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N° 2 (Fuselage - rh side)

Sliding door jettison window : Security of window and seal retainer.

AREA N° 6 (Fuselage - lh side)

Sliding door jettison window : Security of window and seal retainer.

AREA N° 7 (Cabin interior)

Sliding doors jettison windows : Security and check strap is secured.

Sliding doors : Security and correct operation of locking and mechanical stop devices.

Cabin interior	: Operation with passenger sliding doors open or removed requires removal or correct securing of all cabin equipment, installations and trim panels.

IN FLIGHT

CAUTION

Assure that doors are engaged correctly on stops during operation with doors open.

CAUTION

Only an operator secured to the provided belts can open/close the sliding doors in flight.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 98/401/MAE
dated 29-1-1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EQUIVALENT CATEGORY ″A″ OPERATIONS

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LIST OF ILLUSTRATIONS

List of Illustrations (Contd.)

INTRODUCTION

GENERAL INFORMATION

This Appendix provides all information allowing the aircraft to be operated in equivalent Category ″A″ as defined in JAR OPS 3.480.

DEFINITIONS

– CAT - A take-off	: Operation of the helicopter taking into account that, if one engine fails at any time after the start of the takeoff, the helicopter can: – at or prior to TDP, return to and safely stop on the takeoff area; or – at or after TDP, climb out from the point of failure and continue a single engine forward flight.
– CAT - A landing	: Operation of the helicopter taking into account that, if one engine fails at any time in the approach, the helicopter can: – at or after LDP, land and stop safely on the intended landing area; or – at or prior LDP, climb out from the point of failure and continue a single engine forward flight.

- Takeoff decision point (TDP) : It is the last point in the take off path at which, as result of power unit failure, a rejected takeoff can be assured, and the first point at which a completed takeoff can be assured.
- Landing decision point (LDP) : It is the last point on landing path that permits, if a single engine failure is experienced, either to land on a predeterminated area or to maintain a single engine forward flight.
- Takeoff distance : The horizontal distance from the start of the prescribed takeoff procedure to a point at least 35 ft (10.6 m) above the takeoff surface where V_2 and a positive rate of climb are attained following an engine failure occuring at or after TDP.
- Rejected takeoff distance : The horizontal distance from the start of takeoff procedure to the point where the helicopter lands and stops safely following an engine failure occuring at or prior to TDP.
- Takeoff flight path : The distance travelled from where the helicopter reaches V_2 at or above 35 ft (10.6 m) AGL to 1000 ft (305 m) AGL.
- Takeoff safety speed or balked landing safety speed (V_2) : The airspeed that assures the required climb performance in OEI condition.

– Landing distance	: The horizontal distance from a point where the helicopter is 50 ft (15.2 m) (25 ft for vertical landings) above the landing surface to the point where it is brought to stop on the predetermined area with one engine inoperative.
– Balked landing	: The interruption of a landing approach and the start of a climbout. Balked landing capability following an engine failure is assured at or before the LDP.
– Balked landing distance	: The horizontal distance from the LDP to the point in the balked landing profile at which a minimum of 35 ft (10.6 m) is attained at V_2 in a climbing posture.
– Backup distance	: The horizontal distance from the starting point of takeoff to the TDP during vertical take-off with backward flight procedure.
– Clear area	: Any takeoff and landing surface clear of obstacles where the landing field length is not a limiting factor.
– Short field	: Any takeoff and landing surface clear of obstacles where the landing field length is limited.
– Helipad	: Any identifiable surface on land or structure used for the landing and takeoff of helicopters.

– Congested area	:	A densely populated area substantially used for residential, commercial, or recreational purposes without adequate safe forced landing areas.

ABBREVIATIONS

– AGL	:	Above Ground Level.
– AHE	:	Above Helipad Elevation.
– GS	:	Ground Speed.
– LDP	:	Landing Decision Point.
– MPOG	:	Minimum Pitch On Ground.
– TDP	:	Takeoff Decision Point.
– OEI	:	One Engine Inoperative.
– V_1	:	Speed at TDP or at LDP.
– V_2	:	Takeoff Safety Speed or Balked Landing Safety Speed.
– V_Y	:	Best rate of climb speed.
– WAT	:	Weight - Altitude - Temperature.

SECTION 1 - LIMITATIONS

NOTE

The helicopter is certified for ″Equivalent Category A″ operations as defined in JAR OPS 3.480.

TYPE OF OPERATION

Takeoff and landing procedures from clear area, short field and from ground based or elevated helipads, herein described, are approved under day and night VFR, non-icing conditions.

REQUIRED EQUIPMENT

In addition to the basic required equipment the following items shall be installed and operative:

— engine fire extinguisher P/N 109-0811-39.

and moreover for night operation:

— swinging searchlight P/N 109-0811-46 or searchlight P/N 109-0813-76-109.

GROUND SPEED LIMITATIONS

Maximum touchdown speed with parking brake on : 5 Kts.

WEIGHT LIMITATIONS

The maximum takeoff and landing weight to operate from clear area, from short field and from helipad (ground based or elevated) are shown in figures 1-1, 1-2 and 1-3 respectively.

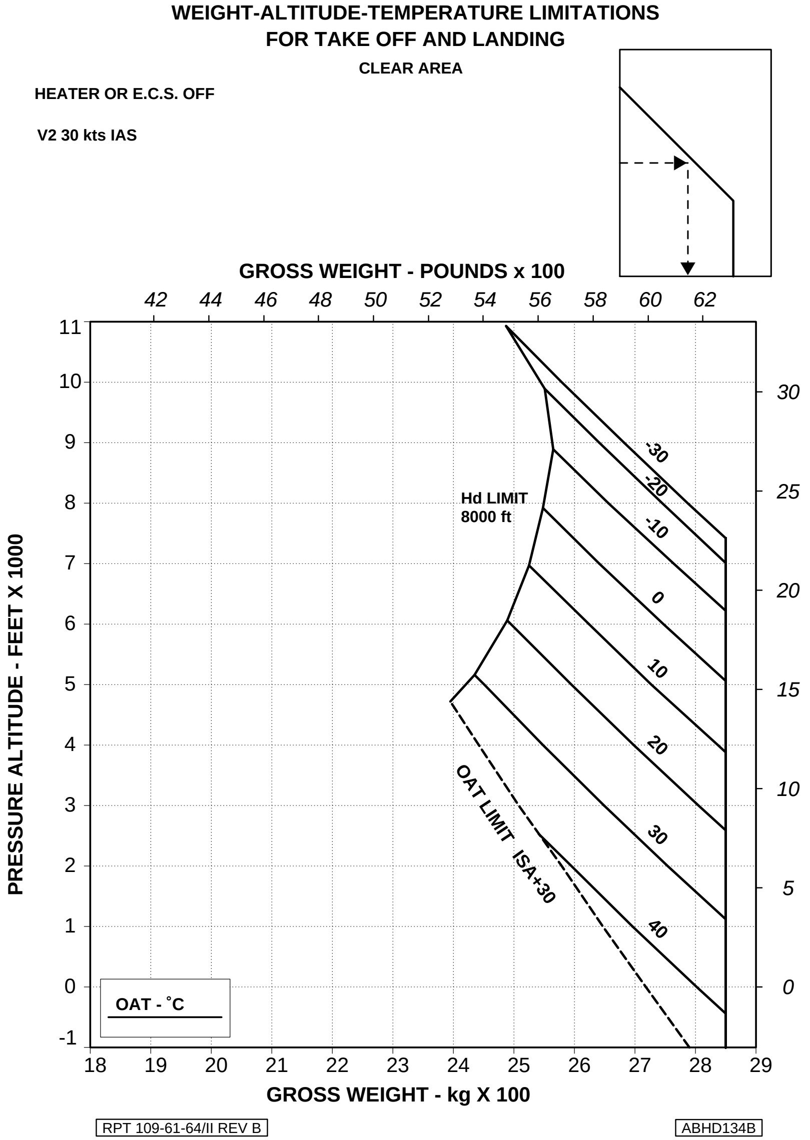


Figure 1-1. Weight - altitude - temperature limitations for takeoff and landing (clear area).

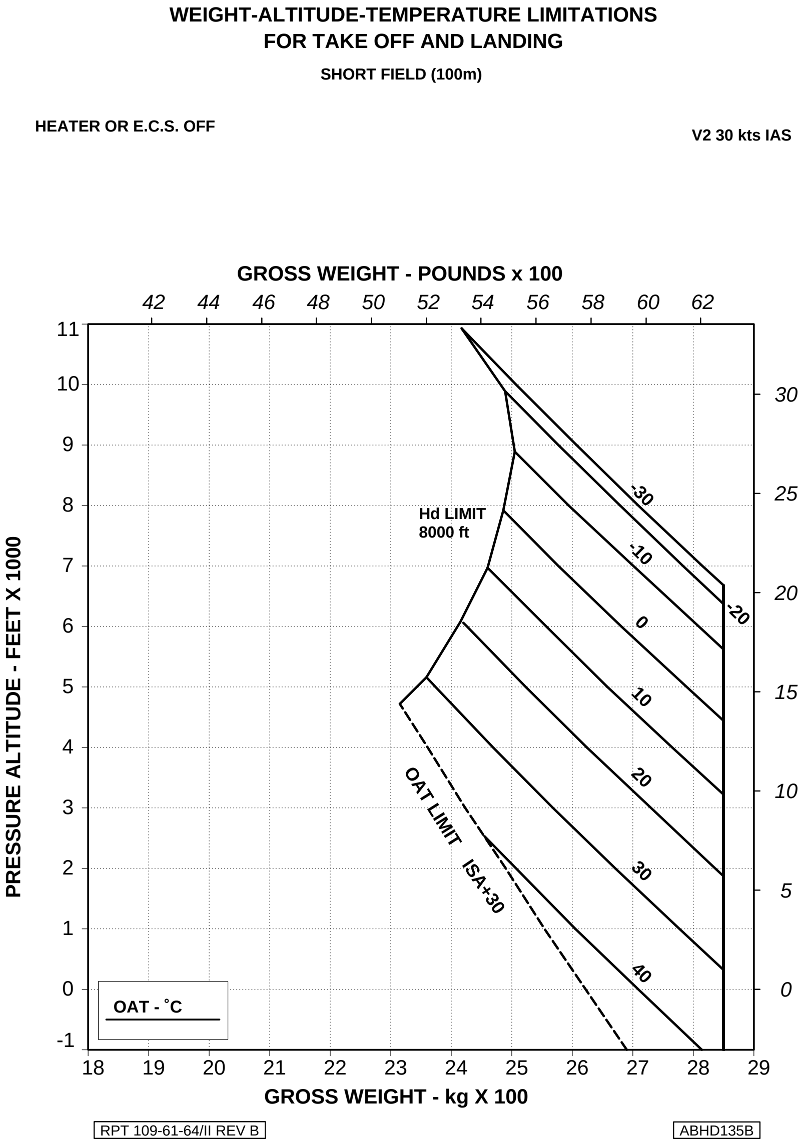


Figure 1-2. Weight - altitude - temperature limitations for takeoff and landing (short field).

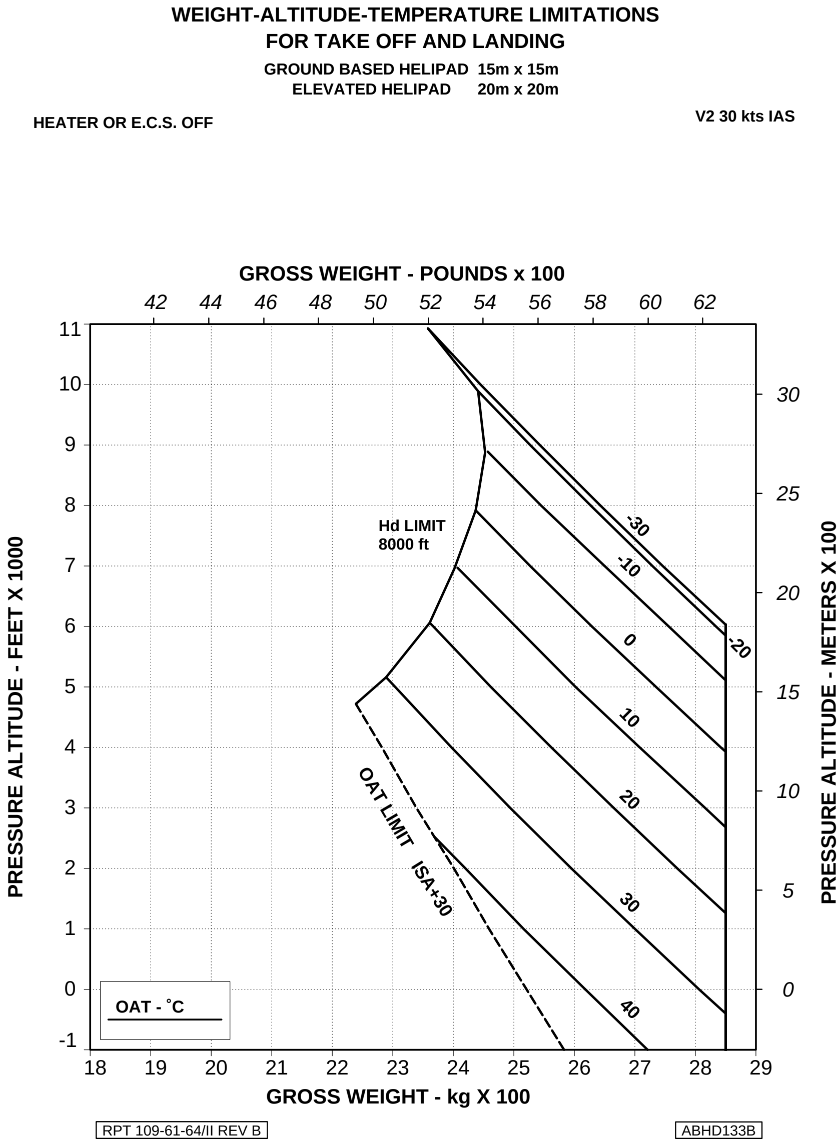


Figure 1-3. Weight - altitude - temperature limitations for takeoff and landing (ground based helipad/elevated helipad).

ALTITUDE LIMITATIONS

The maximum altitude for takeoff and landing is:

— operation from clear area 8000 ft (2438 m) Hd

NOTE

The ″soft″ takeoff manoeuvre is applicable up to 3000 ft (914 m) Hd only.

— operation from short field 8000 ft (2438 m) Hd

— operation from helipad (ground based or elevated) 8000 ft (2438 m) Hd

NOTE

The modified takeoff flight path from ground based helipad to clear high obstacles is applicable up to 3000 ft (914 m) Hd only.

CROSSWIND LIMITATIONS

Takeoff or landing downwind or with quartering tailwinds is prohibited.

SECTION 2 - NORMAL PROCEDURES

OPERATION FROM CLEAR AREA

GENERAL DATA

TAKEOFF DECISION POINT (TDP)

V_1	: 30 Kts IAS.
Wheels height	: 15 ft (4.6 m) AGL (Standard manoeuvre). 70 ft (21.3 m) AGL (″Soft″ manoeuvre).

NOTE

Radio altimeter heights are normally shown in all flight path profiles; barometric altitudes can also be used provided that a correct ″zero setting″ is performed in accordance with the takeoff procedure.

LANDING DECISION POINT (LDP)

V_1	: 25 Kts IAS.
Wheels height	: 80 ft (25.4 m) AGL.
Rate of descent	: 500 ±100 fpm.
V_2	: 30 Kts IAS.
V_Y	: 60 Kts IAS.

NOTE

Radio altimeter heights should be used when available.

TAKEOFF

There are two different takeoff procedures when operating from clear area: a dynamic takeoff manoeuvre **(standard takeoff manoeuvre)** which minimizes the takeoff and the rejected takeoff distances and a less dynamic takeoff manoeuvre **(″soft″ takeoff manoeuvre)** privileging the comfort of passengers against a longer rejected takeoff distance.
The ″soft″ takeoff manoeuvre is applicable when the takeoff site presents at least 400 m of runway clear of obstacles in front of the helicopter.
The ″soft″ takeoff manoeuvre is approved up to 3000 ft (914 m) Hd.

TAKEOFF FROM CLEAR AREA

CAUTION

The ECS or the bleed-air heater must be off during takeoff.

Collective	: MPOG.
Nose wheel lock	: ON (lever up).
Parking brake	: Off. Check PARK BRK ON caution message out.
Caution and warning messages	: Check none.
RPM switch	: Set 102%.
NR/N2 speed	: 102% stabilized, check.
Radio altimeter (if installed)	: Press the TEST pushbutton and verify for correct functioning.
ADI	: Verify for equivalent attitude indication between ADIs.
External lights	: As required.

NOTE

When use of search and landing light is required proceed as follows:

LDG LT switch	:	FWD.
SCHLT switch	:	ON, extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

Takeoff area	:	Check clear.
Collective	:	Slowly increase to hover at approximately 3 ft (1 m) wheels height and note torque.
Altimeter	:	Set zero pressure altitude.
ADI	:	Adjust (if applicable) the pitch bar to indicate zero.

NOTE

When taking off in windy conditions, set the ADI and the baro-altimeter on ground allowing for the anticipated attitude and altitude changes.

STANDARD TAKEOFF MANOEUVRE
(Fig 2-1)

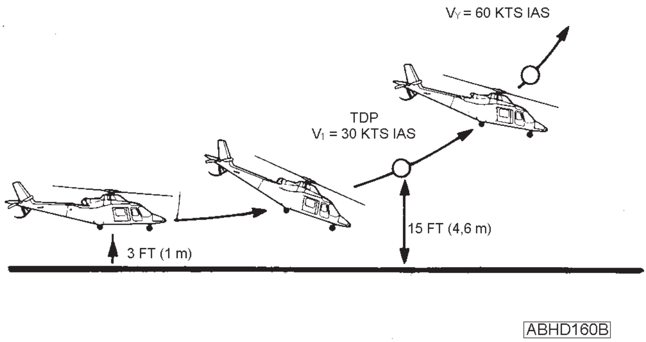


Figure 2-1. Takeoff flight path profile from clear area (Standard takeoff manoeuvre).

Collective and cyclic	: Apply as necessary to increase torque by 15% ±5% on each engine above that required to hover and set 25 degrees nose down attitude change.

NOTE
Do not exceed transmission torque, TOT or N1 limits.

Takeoff path	: Maintain pitch attitude and accelerate up to 30 Kts to achieve the TDP at 15 ft (4.6 m). After reaching TDP, select up to takeoff power, accelerate the helicopter to V_Y and reduce nose down attitude.
Landing gear	: UP (by 200 ft AGL).
RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.
Takeoff path	: Climb at V_Y to the desired altitude.

″SOFT″ TAKEOFF MANOEUVRE
(Fig 2-1A)

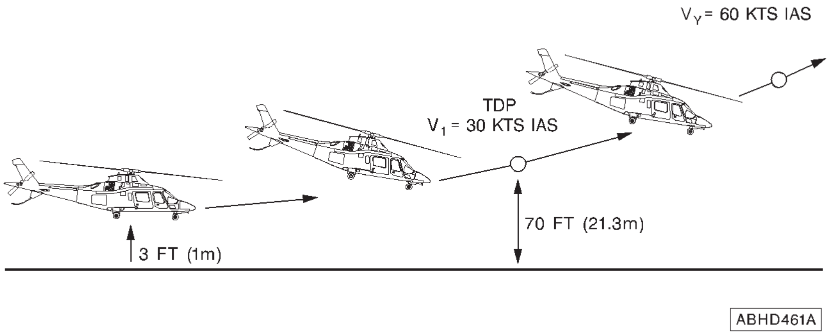


Figure 2-1A. Takeoff flight path profile from clear area (″Soft″ takeoff manoeuvre).

Collective and cyclic : Apply as necessary to increase torque by 15% ± 5% on each engine above that required to hover and set 15 degrees nose down attitude change.

NOTE

Do not exceed transmission torque, TOT or N1 limits.

Takeoff path : Maintain pitch attitude and accelerate up to 30 Kts to achieve the TDP at 70 ft (21.3 m). After reaching TDP, select up to takeoff power, accelerate the helicopter to V_y and reduce nose down attitude.

Landing gear	:	UP (by 200 ft AGL).
RPM switch	:	Set 100%.
NR/N2 speed	:	100% stabilized, check.
Takeoff path	:	Climb at V_y to the desired altitude.

APPROACH AND LANDING

LANDING ON CLEAR AREA
(Fig 2-2)

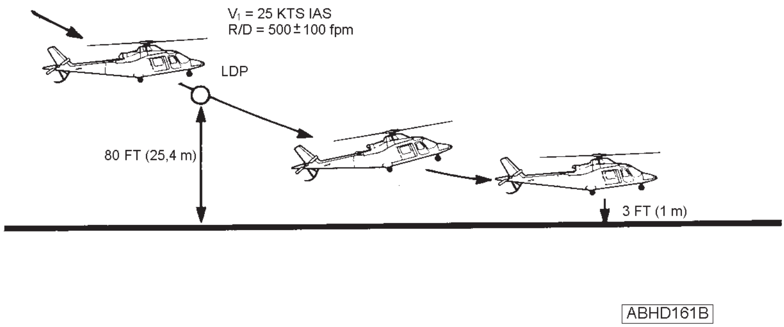


Figure 2-2. Landing flight path profile on clear area.

Altimeter	: Set the known QFE.
ADI	: Verifiy for equivalent attitude indication between ADIs.
RPM switch	: Set 102%.
NR/N2 speed	: 102% stabilized, check.
Landing gear	: DOWN.

CAUTION

Check the lever to be correctly engaged in the DOWN position to avoid inadvertent retraction of the landing gear.

Nose wheel lock	: ON (lever up).

Parking brake : Off. Check PARK BRK ON caution message out.

External lights : As required.

NOTE

When use of search and landing light is required proceed as follows:

LDG LT switch : FWD.

SCHLT switch : ON, extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

CAUTION

The ECS or the bleed-air heater must be off during landing.

Approach path : Initiate the approach to pass through the LDP, at 80 ft (25.4 m) above the landing site at 25 Kts IAS and with a rate of descent not higher than 500 ±100 fpm.
From the LDP, gradually slow down and decrease the rate of descent to achieve 3 ft (1 m) hover. Descend vertically to land.

NOTE

Attention should be paid to avoid touching the tail first.

OPERATION ON SHORT FIELD

THE SHORT FIELD PROCEDURE CAN BE USED IN CONGESTED AREA IN ACCORDANCE WITH THE APPROPRIATE OPERATING RULES.

GENERAL DATA

The minimum size for takeoff and landing area is 100 m (length) x 20 m (width) (328 ft x 66 ft) (figure 2-3).

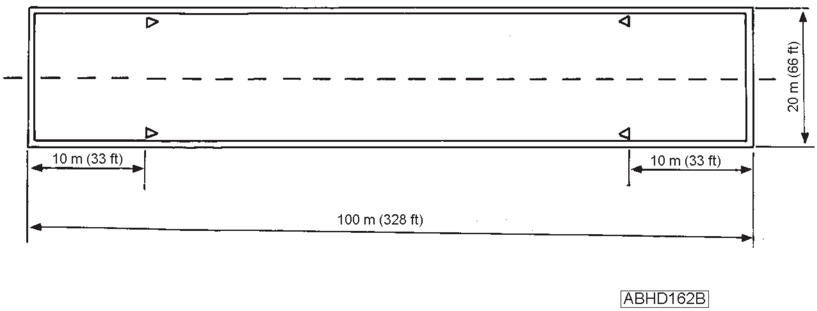


Figure 2-3. Minimum field size and example of marking.

NOTE

The takeoff has to be initiated with the helicopter positioned 10 m (33 ft) from the runway head as per the reference marks in figure 2-3. Refer to the same reference marks for the touchdown during landing.

TAKEOFF DECISION POINT (TDP)

V_1	: 0 Kts GS.
Wheel height	: 80 ft (25.4 m) AGL.

NOTE

Radio altimeter heights are normally shown in all flight path profiles; barometric altitudes can also be used provided that a correct ″zero setting″ is performed in accordance with the takeoff procedure.

LANDING DECISION POINT (LDP)

V_1	: 25 Kts IAS.
Wheel height	: 80 ft (25.4 m) AGL.
Rate of descent	: 500 ±100 fpm.
V_2	: 30 Kts IAS.
V_Y	: 60 Kts IAS.

NOTE

Radio altimeter heights should be used when available.

TAKEOFF

TAKEOFF FROM SHORT FIELD

(Fig 2-4)

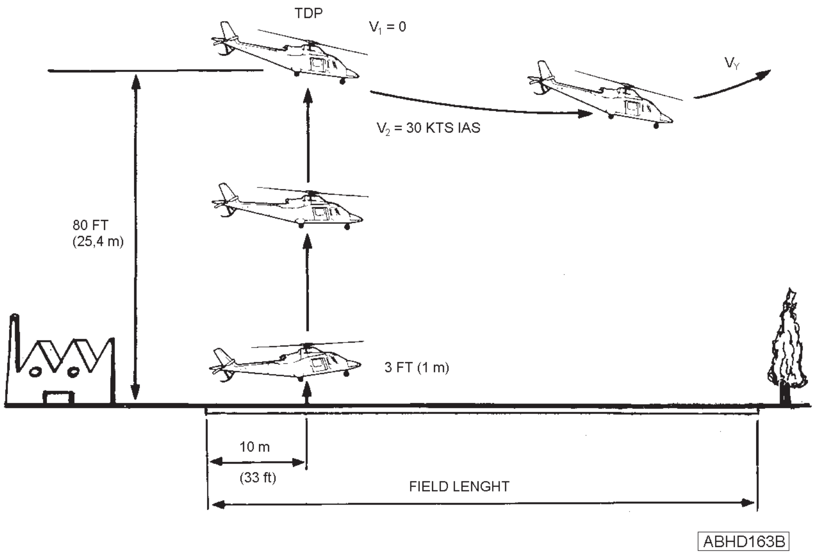


Figure 2-4. Takeoff profile from short field.

CAUTION

The ECS or the bleed-air heater must be off during takeoff.

NOTE

Takeoff will be initiated with the helicopter positioned, such that the takeoff reference marks are directly opposite the crew doors and the helicopter centered on the runway.
This will assure that the tail rotor is within the runway limits.

Collective	:	MPOG.
Nose wheel lock	:	ON (lever up).
Parking brake	:	Off. Check PARK BRK ON caution message out.
Caution and warning messages	:	Check none.
RPM switch	:	Set 102%.
NR/N2 speed	:	102% stabilized, check.
Radio altimeter (if installed)	:	Press the TEST pushbutton and verify for correct functioning.
ADI	:	Verify for equivalent attitude indication between ADIs.
External lights	:	As required.

NOTE

When use of search and landing light is required proceed as follows:

LDG LT switch	:	FWD.
SCHLT switch	:	ON, extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

Takeoff area	:	Check clear.
Collective	:	Slowly increase to hover at approximately 3 ft (1 m) wheels height and note torque.
Altimeter	:	Set zero pressure altitude.
ADI	:	Adjust (if applicable) the pitch bar to indicate zero.

NOTE

When taking off in windy conditions, set the ADI and the baro-altimeter on ground allowing for the anticipated attitude and altitude changes.

Collective	:	Apply as necessary to climb vertically with a rate of climb of 500 ±50 fpm.
Takeoff path	:	At TDP 80 ft (25.4 m) select up to takeoff power and rotate through 10 degrees nose down attitude simultaneously. When the speed reaches V_2 reduce nose down attitude and accelerate up to V_Y.
Landing gear	:	UP (by 200 ft AGL).
RPM switch	:	Set 100%.

NR/N2 speed : 100% stabilized, check.

Takeoff path : Climb at V_Y to the desired altitude.

APPROACH AND LANDING

LANDING ON SHORT FIELD
(Fig 2-5)

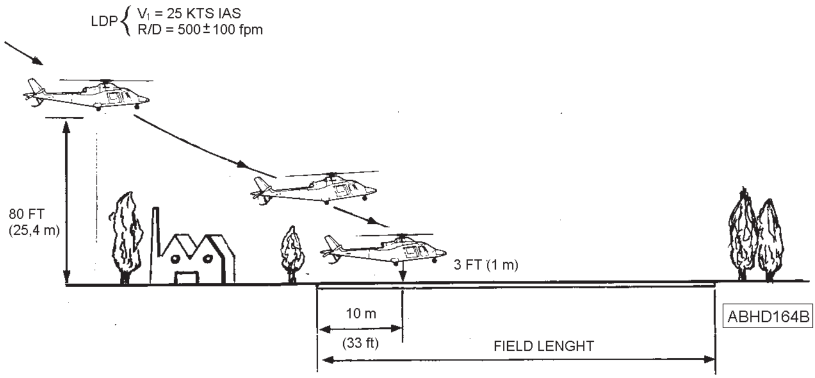


Figure 2-5. Landing profile on short field.

Altimeter : Set the known QFE.

ADI : Verify for equivalent attitude indication between ADIs.

RPM switch : Set 102%.

NR/N2 speed : 102% stabilized, check.

Landing gear : DOWN.

CAUTION

Check the lever to be correctly engaged in the DOWN position to avoid inadvertent retraction of the landing gear.

Nose wheel lock	:	ON (lever up).
Parking brake	:	Off. Check PARK BRK ON caution message out.
External lights	:	As required.

NOTE

When use of search and landing light is required proceed as follows:

LDG LT switch	:	FWD.
SCHLT switch	:	ON, extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

NOTE

The touchdown point shall be 10 m (33 ft) after the beginning of the runway head.

CAUTION

The ECS or the bleed-air heater must be off during landing.

Approach path	: Initiate the approach to pass through the LDP, at 80 ft (25.4 m) above the landing site, at 25 Kts IAS and with a rate of descent not higher than 500 ±100 fpm. From the LDP, gradually slow down and decrease the rate of descent to achieve 3 ft (1 m) hover. Descend vertically to land.

NOTE

Attention should be paid to avoid touching the tail first.

HELIPAD OPERATION (GROUND LEVEL OR ELEVATED)

GENERAL DATA

The minimum helipad sizes demonstrated for OEI landing procedure are:

— ground based helipad 15 m x 15m (49 ft x 49 ft)

— elevated helipad 20 m x 20 m (66 ft x 66 ft).

NOTE

The following data and procedures apply, unless differently specified, both from ground based helipad and from elevated helipad. All heights are referred to helipad level.

TAKEOFF DECISION POINT (TDP)

Wheels height : 80 ft (25.4 m) AHE.

NOTE

Radio altimeter heights are normally shown in all flight path profiles; barometric altitudes can also be used provided that a correct ″zero setting″ is performed in accordance with the takeoff procedure.

LANDING DECISION POINT (LDP)

V_1 : Up to 20 Kts IAS.

NOTE

Check for a positive forward ground speed.

NOTE

In windy conditions, V_1 could be greater than 20 Kts IAS according to the wind velocity to ensure a positive forward ground speed.

Wheels height : 80 ft (25.4 m) AHE.

Rate of descent	:	200 ±50 fpm.
V_2	:	30 Kts IAS.
V_Y	:	60 Kts IAS.

NOTE

Radio altimeter heights should be used when available if operating over a ground level helipad only.

TAKEOFF

TAKEOFF FROM HELIPAD (STANDARD TDP)

(Fig 2-6)

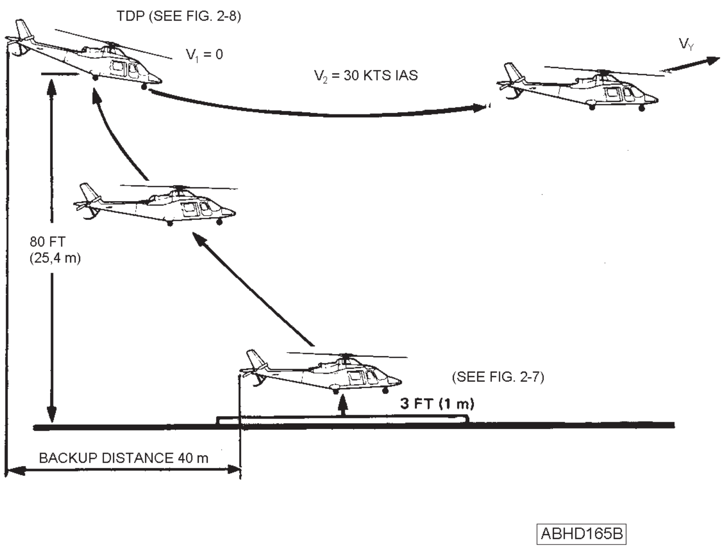


Figure 2-6. Takeoff profile from helipad.

CAUTION

The ECS or the bleed-air heater must be off during takeoff.

Collective	:	MPOG.
Nose wheel lock	:	ON (lever up).
Parking brake	:	ON. Check PARK BRK ON caution message displayed.
Caution and warning messages	:	Check none.
RPM switch	:	Set 102%.
NR/N2 speed	:	102% stabilized, check.
Radio altimeter (if installed)	:	Press the TEST pushbutton and verify for correct functioning.
ADI	:	Verify for equivalent attitude indication between ADIs.
External lights	:	As required.

NOTE

When the use of search and landing light is required proceed as follows:

LDG LT switch	:	FWD.
SCHLT switch	:	ON; extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

Takeoff area	:	Check clear.
Collective	:	Slowly increase to hover at approximately 3 ft (1 m) wheels height and note torque (see figure 2-7).

Altimeter	: Set zero pressure altitude.
ADI	: Adjust (if applicable) the pitch bar to indicate zero.

NOTE

When taking off in windy conditions, set the ADI and the baro-altimeter on ground allowing for the anticipated attitude and altitude changes.

Collective and cyclic	: Apply as necessary to obtain a vertical rate of climb of 500 ±100 fpm and simultaneously initiate a slow backward flight maintaining as visual reference the far right hand corner of the helipad.

NOTE

Use 5 - 10 degrees left yaw, if required, to maintain the visual reference.

Takeoff path	: At TDP 80 ft (25.4 m) (see figure 2-8) select up to takeoff power and rotate through 15 degrees nose down attitude change simultaneously. When the speed reaches V_2 reduce nose down attitude and accelerate up to V_Y.
Landing gear	: UP (by 200 ft AGL).
Parking brake	: OFF.
RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.

ABHD169B

Figure 2-7. Takeoff from helipad - view at 3 ft height.

ABHD187B

Figure 2-8. Takeoff from helipad - view at TDP (80 ft).

Takeoff path : Climb at V_Y to the desired altitude.

NOTE

For operations on elevated helipad, when obstacles and weather conditions permit, select a takeoff direction that allows the best use of available external cues, which may ease the possible reject manoeuvre. In case of wind consider the turbolence and down draft effects around the pad.

MODIFIED TAKEOFF FLIGHT PATH FROM GROUND BASED HELIPAD TO CLEAR HIGH OBSTACLES (INCREASED TDP) (Fig 2-8A)

The following procedure applies to all those circumstances in which obstacles in the close surroundings of the helipad require to increase the standard TDP height. The procedure does not depend on the location of the obstacle along the takeoff path, since it assures each point on the flight path, after the TDP, to stay at least 35 ft above the obstacle.

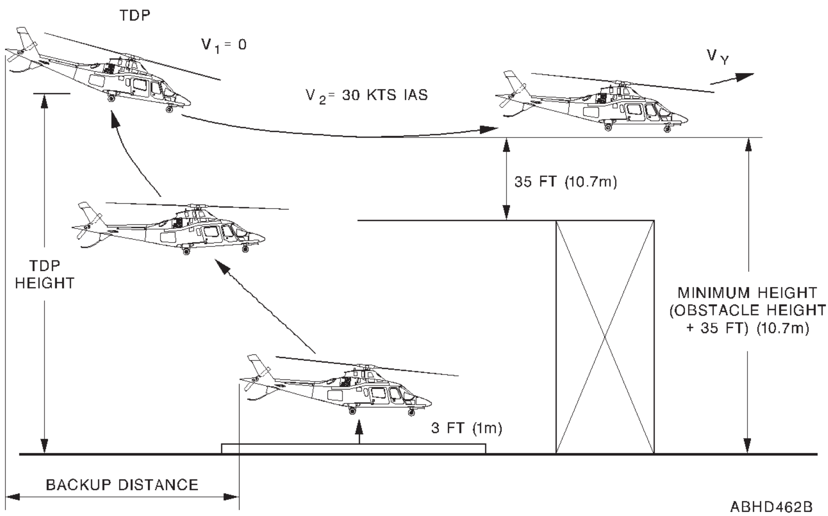


Figure 2-8A. Takeoff profile from helipad with an high obstacle on the path.

Procedure

Determine the TDP height such that the Minimum Height presented in Table 2-1 is not lower than the obstacle height plus 35 ft of clearance.
Perform the takeoff procedure from helipad utilizing the above TDP and taking into account of the increased Backup Distance.

TDP HEIGHT (ft)	BACKUP DISTANCE (m)	MINIMUM HEIGHT (ft)	HEIGHT AT THE END OF TAKEOFF DISTANCE[1] (ft)	TAKEOFF DISTANCE (m)
140	95	35	55	130
160	115	50	75	105
180	135	70	85	80
200	150	90	105	50
220	170	110	125	25
240	185	125	145	0
260	205	145	160	-30
280	225	165	180	-55
300	240	185	200	-80
320	260	200	215	-100
340	275	220	235	-115
360	295	240	255	-130
380	310	260	275	-145
400	330	270	295	-160
420	350	295	315	-175
440	365	315	340	-190
460	385	335	360	-210
480	400	350	380	-225
500	420	370	405	-240

1) Height at which V_2 and a positive rate of climb are achieved.

Table 2-1. Height and Distances versus the minimum height to clear.

Example

Determine the TDP height to takeoff from ground based helipad by clearing a 50 ft high obstacle.

Solution:

- Add 35 ft of minimum clearance to known obstacle height (50 ft) to obtain the Minimum Height (85 ft) on takeoff path.
- Enter table 2-1 with Minimum Height (85 ft) to obtain the corresponding TDP height: if the Minimum Height results to be between two values (in this example 70 ft and 90 ft) refer to the highest value (90 ft).
  In this case the TDP height is 200 ft.

APPROACH AND LANDING

LANDING ON HELIPAD

(Fig 2-9)

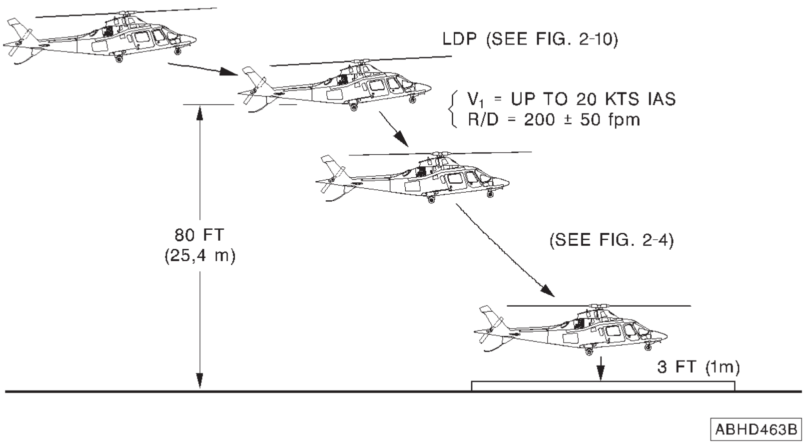


Figure 2-9. Landing profile on helipad.

Altimeter : Set the known QFE.

ADI	: Verify for equivalent attitude indication between ADIs.
RPM switch	: Set 102%.
NR/N2 speed	: 102% stabilized, check.
Landing gear	: DOWN.

CAUTION

Check the lever to be correctly engaged in DOWN position to avoid inadvertent retraction of the landing gear.

Nose wheel lock	: ON (lever up).
Parking brake	: ON. Check PARK BRK ON caution message displayed.
External lights	: As required.

NOTE

When the use of search and landing light is required proceed as follows:

LDG LT switch	: FWD.
SCHLT switch	: ON; extend and set as necessary.

CAUTION

Landing light operation shall be limited to the time strictly necessary to carry out takeoff and landing manoeuvres in order to avoid overheating.

CAUTION

The ECS or the bleed-air heater must be off during landing.

Approach path : Initiate the approach to pass through the LDP, at 80 ft (25.4 m) above the landing site, with an airspeed up to 20 Kts IAS (positive forward ground speed maintained) and with a rate of descent not higher than 200 ±50 fpm, keeping in sight the far right hand corner of the helipad (see figure 2-10). From LDP, gradually slow down and decrease the rate of descent to achieve 3 ft (1 m) hover.
Descend vertically to land.

NOTE

Attention should be paid to avoid touching the tail first.

ABHD032A

Figure 2-10. Landing on helipad - view at LDP (80 ft).

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

INTRODUCTION

The following tables list the EDU messages, the fault conditions, and corrective actions for emergencies and malfunctions that might occur during takeoff prior to TDP, during takeoff after TDP, during landing prior to LDP, and during landing after LDP.

All corrective action procedures herein listed assume the pilot to give first priority to aircraft control and to a safe flight path.

NOTE

The helicopter should not be operated following any emergency landing or shutdown until the cause of the malfunction has been determined and corrective maintenance action taken.

WARNING SYSTEM

WARNING MESSAGES (RED)

Takeoff (prior to TDP)

EDU message	Fault condition	Corrective action
ROTOR LOW	Rotor RPM low. Rotor RPM between 80% and 95.5% (power on) or 80% and 89.5% (power off).	Use collective to adjust RPM.

EDU message	Fault condition	Corrective action
ROTOR HIGH	Rotor RPM high. Rotor RPM above or equal to 105.5% (power on) or 110.5% (power off).	Use collective to adjust RPM.
ENG 1 (2) OUT	N1 abnormally low, below 35%, on engine 1 (2). Probable engine failure.	Land immediately. (See the pertinent paragraphs of this Section).
ENG 1 (2) FIRE	Fire in engine compartment.	Land immediately. Operate the engine fire extinguisher and shutdown the affected engine. (See the pertinent paragraph of Appendix 2).
XMSN OIL PRES	Transmission oil pressure below minimum limit.	Land immediately.
XMSN OIL HOT	Transmission oil temperature above maximum limit.	Land immediately.
BATT HOT	Battery temperature exceeding limits.	Land immediately. Switch battery off.
#1 (#2) ECU FAIL	Critical hardware failure of the engine control unit.	Automatic reversion to manual mode of the control of the affected engine. Land as soon as practical.

Takeoff (after TDP)

EDU message	Fault condition	Corrective action
ROTOR LOW	Rotor RPM low. Rotor RPM between 80% and 95.5% (power on) or 80% and 89.5% (power off).	Use collective to adjust RPM. If this message occurs during the continued takeoff manoeuvre, when rotor rpm is intentionally drooped, reset the audio warning and continue the manoeuvre respecting the minimum rotor rpm limit.
ROTOR HIGH	Rotor RPM high. Rotor RPM above or equal to 105.5% (power on) or 110.5% (power off).	Use collective to adjust RPM.
ENG 1 (2) OUT	N1 abnormally low, below 35%, on engine 1 (2). Probable engine failure.	Reach V_2. Shutdown the affected engine. Land as soon as possible. (See the pertinent paragraphs of this Section).
ENG 1 (2) FIRE	Fire in engine compartment.	Reach V_2 and operate the engine fire extinguisher. (See the pertinent paragraph of Appendix 2). Shutdown the affected engine. Land as soon as possible.

EDU message	Fault condition	Corrective action
XMSN OIL PRES	Transmission oil pressure below minimum limit.	Reach V_2, reduce power and land as soon as possible, however not later than 15 minutes.
XMSN OIL HOT	Transmission oil temperature above maximum limit.	Reach V_2, reduce power and land as soon as possible.
BATT HOT	Battery temperature exceeding limits.	Reach V_2 and switch battery off. Land as soon as possible.
#1 (#2) ECU FAIL	Critical hardware failure of the engine control unit.	Automatic reversion to manual mode of the control of the affected engine. Reach V_2. Land as soon as practical

Landing (prior to LDP)

NOTE

The decision to execute a balked landing when flying on one engine should be taken before the helicopter descends below LDP.

EDU message	Fault condition	Corrective action
ROTOR LOW	Rotor RPM low. Rotor RPM between 80% and 95.5% (power on) or 80% and 89.5% (power off).	Use collective to adjust RPM. If this message occurs during the balked landing manoeuvre, when rotor rpm is intentionally drooped, reset the audio warning and continue the manoeuvre respecting the minimum rotor rpm limit.
ROTOR HIGH	Rotor RPM high. Rotor RPM above or equal to 105.5% (power on) or 110.5% (power off).	Use collective to adjust RPM.
ENG 1 (2) OUT	N1 abnormally low, below 35%, on engine 1 (2). Probable engine failure.	Reach V_2. Shutdown the affected engine. Land as soon as possible. (See the pertinent paragraphs of this Section).

EDU message	Fault condition	Corrective action
ENG 1 (2) FIRE	Fire in engine compartment.	Reach V_2 and operate the engine fire extinguisher. (See the pertinent paragraph of Appendix 2). Shutdown the affected engine. Land as soon as possible.
XMSN OIL PRES	Transmission oil pressure below minimum limit.	Reach V_2, reduce power and land as soon as possible, however not later than 15 minutes.
XMSN OIL HOT	Transmission oil temperature above maximum limit.	Reach V_2, reduce power and land as soon as possible.
BATT HOT	Battery temperature exceeding limits.	Reach V_2 and switch battery off. Land as soon as possible.
#1 (#2) ECU FAIL	Critical hardware failure of the engine control unit.	Automatic reversion to manual mode of the control of the affected engine. Reach V_2. Land as soon as practical.

Landing (after LDP)

EDU message	Fault condition	Corrective action
ROTOR LOW	Rotor RPM low. Rotor RPM between 80% and 95.5% (power on) or 80% and 89.5% (power off).	Use collective to adjust RPM.
ROTOR HIGH	Rotor RPM high. Rotor RPM above or equal to 105.5% (power on) or 110.5% (power off).	Use collective to adjust RPM.
ENG 1 (2) OUT	N1 abnormally low, below 35%, on engine 1 (2). Probable engine failure.	Land immediately. (See the pertinent paragraph of this Section).
ENG 1 (2) FIRE	Fire in engine compartment.	Land immediately. Operate the engine fire extinguisher and shutdown the affected engine. (See the pertinent paragraph of Appendix 2).
XMSN OIL PRES	Transmission oil pressure below minimum limit.	Land immediately.

EDU message	Fault condition	Corrective action
XMSN OIL HOT	Transmission oil temperature above maximum limit.	Land immediately.
BATT HOT	Battery temperature exceeding limits.	Land immediately. Switch battery off.
#1 (#2) ECU FAIL	Critical hardware failure of the engine control unit.	Automatic reversion to manual mode of the control of the affected engine. Proceed to land.

EMERGENCY PROCEDURES FOR OPERATION ON CLEAR AREA

ENGINE FAILURE DURING TAKEOFF PRIOR TO/AT THE TAKE-OFF DECISION POINT (TDP)

(Fig 3-1)

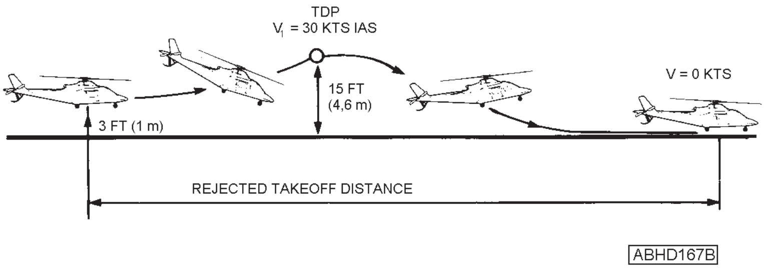


Figure 3-1. Rejected takeoff flight path profile (clear area).

NOTE

During the rejected takeoff manoeuvre, the pilot shall control the rotor speed, disregarding the other ECU controlled engine parameters.

Collective	:	As necessary to maintain rotor RPM around 100%.
Airspeed	:	Set a 30 degree nose up attitude change to reduce speed at constant height.

Collective : Level pitch attitude and as the helicopter starts to sink, gradually increase to cushion the touchdown.

NOTE

During a ″soft″ takeoff manoeuvre anticipate the recovery of normal attitude immediately after the pitch up manoeuvre, by slowly levelling the helicopter and by maintaining the level pitch attitude until touchdown.

Attitude : Maintain landing attitude before touchdown to avoid touching the tail first.

Touchdown : As the helicopter starts running on the ground reduce collective and apply toe brakes to stop.

Perform normal shutdown procedure.

ENGINE FAILURE DURING TAKEOFF AT/AFTER THE TAKEOFF DECISION POINT (TDP)

(Fig 3-2)

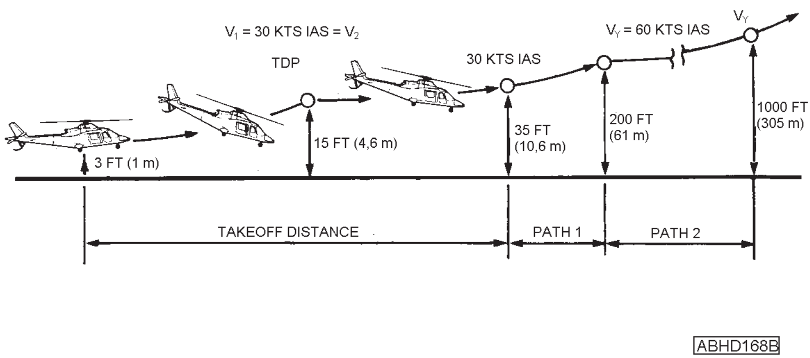


Figure 3-2. Continued takeoff flight path profile (clear area).

Collective	:	Adjust as necessary to drop NR not below 90%.
Airspeed	:	Increase to V_2 using up to 2.5 minute power and start to climb.
Continued takeoff manoeuvre	:	Recover rotor speed and climb at V_2 using 2.5 minute power to 200 ft (61 m) above the takeoff surface.
Landing gear	:	UP (by 200 ft AGL).
Airspeed	:	Accelerate to V_Y while climbing.
RPM switch	:	Set 100%.
NR/N2 speed	:	100% stabilized, check.
Takeoff path	:	Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).
Affected engine	:	Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING PRIOR TO/AT THE LANDING DECISION POINT (LDP)

(Fig 3-3)

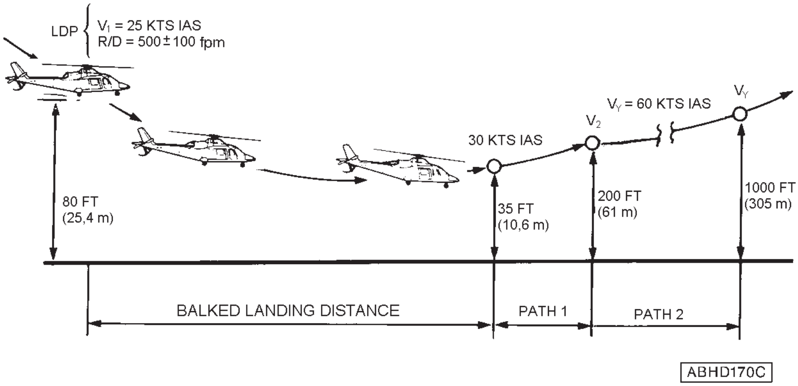


Figure 3-3. Balked landing flight path profile (clear area).

Balked landing manoeuvre	: Reach V_2 by pushing over 15 degrees attitude change and by setting the collective to obtain up to 2.5 minute power and to drop NR not below 90%. As the speed approaches V_2 reduce the nose down attitude. Recover rotor speed and climb at V_2 using 2.5 minute power to 200 ft (61 m) above the takeoff surface.
Landing gear	: UP (by 200 ft AGL).

Airspeed	: Accelerate to V_Y while climbing.
RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.
Climb path	: Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).
Affected engine	: Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING AT/AFTER THE LANDING DECISION POINT (LDP)
(Fig 3-4)

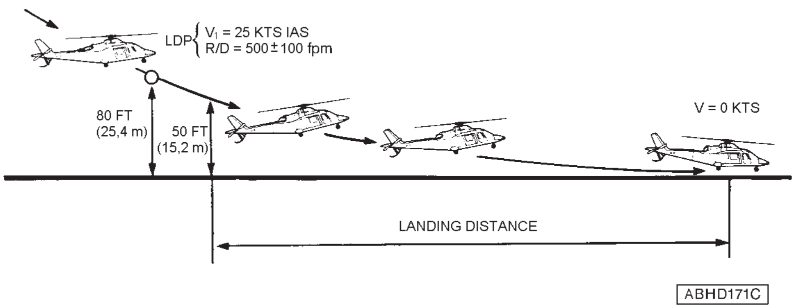


Figure 3-4. OEI landing flight path profile (clear area).

Collective	: Gradually increase to reduce the rate of descent maintaining rotor RPM around 100%.

Airspeed	:	Reduce by setting a 20 degrees nose up attitude change when approaching 15 ft.
Attitude	:	Assume landing attitude before touchdown to avoid touching the tail first.
Touchdown	:	As the helicopter starts running on the ground reduce collective and apply toe brakes to stop.

Perform normal shutdown procedure.

EMERGENCY PROCEDURES FOR OPERATION ON SHORT FIELD

ENGINE FAILURE DURING TAKEOFF PRIOR TO/AT THE TAKE-OFF DECISION POINT (TDP)

(Fig 3-5)

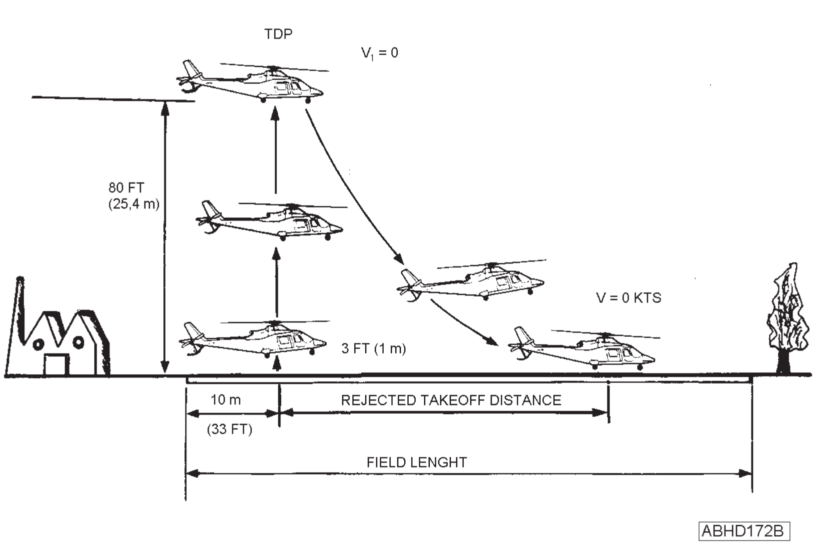


Figure 3-5. Rejected takeoff flight path profile (short field).

NOTE

During the rejected takeoff manoeuvre, the pilot shall control the rotor speed, disregarding the other ECU controlled engine parameters.

Collective	: As necessary to maintain rotor RPM around 100%.

Rejected takeoff manoeuvre	: If the failure occurs below 50 ft AGL push over slightly to bring the helicopter back to the ground. If the failure occurs at or above 50 ft AGL push over 20 degrees attitude change to obtain forward speed, and bring the helicopter back to the ground. Perform a flare and level the helicopter before accomplishing a running landing.

NOTE

Attention should be paid to avoid touching the tail first.

Touchdown	: As soon as the helicopter touches the ground reduce the collective and apply toe brakes to stop.

Perform normal shutdown procedure.

ENGINE FAILURE DURING TAKEOFF AT/AFTER THE TAKEOFF DECISION POINT (TDP)

(Fig 3-6)

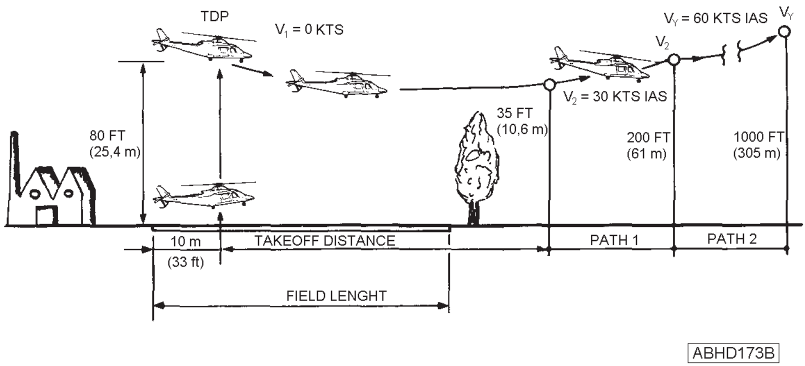


Figure 3-6. Continued takeoff flight path profile (short field).

Continued takeoff manoeuvre	: Push over 25 degrees attitude change to reach V_2. Adjust the collective to drop NR not below 90%. Recover rotor speed and climb at V_2 using 2.5 minute power up to 200 ft (61 m) above the takeoff surface.
Landing gear	: UP (by 200 ft AGL).
Airspeed	: Accelerate to V_Y while climbing.

RPM switch : Set 100%.

NR/N2 speed : 100% stabilized, check.

Takeoff path : Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).

Affected engine : Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING PRIOR TO/AT THE LANDING DECISION POINT (LDP)
(Fig 3-7)

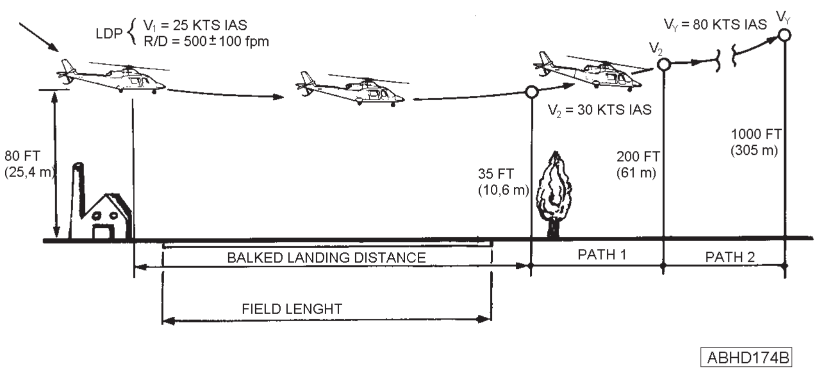


Figure 3-7. Balked landing flight path profile (short field).

Balked landing manoeuvre	: Reach V_2 by pushing over 20 degrees attitude change and by setting the collective to obtain up to 2.5 minute power and to drop NR not below 90%. As the speed approaches V_2 reduce the nose down attitude. Recover rotor speed and climb at V_2 using 2.5 minute power up to 200 ft (61 m) above the takeoff surface.
Landing gear	: UP (by 200 ft AGL).
Airspeed	: Accelerate to V_Y while climbing.
RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.
Climb path	: Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).
Affected engine	: Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING AT/AFTER THE LANDING DECISION POINT (LDP)

(Fig 3-8)

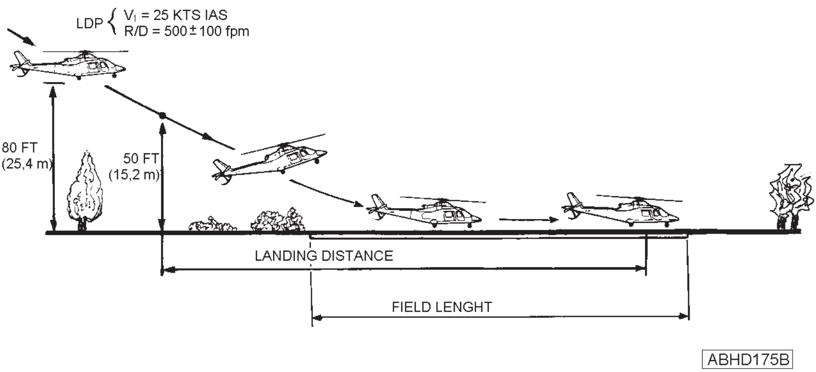


Figure 3-8. OEI landing flight path profile (short field).

Collective	: As necessary to maintain rotor RPM.
OEI landing manoeuvre	: If the failure occurs at or above 50 ft AGL push over 15 degrees attitude change to increase forward speed and approach the landing site. If the failure occurs below 50 ft AGL push over slightly to bring the helicopter to the ground.

Perform a flare and level the helicopter before accomplishing a running landing.

NOTE

Attention should be paid to avoid touching the tail first.

Touchdown	: As soon as the helicopter touches the ground reduce the collective and apply toe brakes to stop.

Perform normal shutdown procedure

EMERGENCY PROCEDURES FOR OPERATION ON HELIPAD (GROUND LEVEL OR ELEVATED)

ENGINE FAILURE DURING TAKEOFF PRIOR TO/AT THE TAKE-OFF DECISION POINT (TDP)

(Fig 3-9)

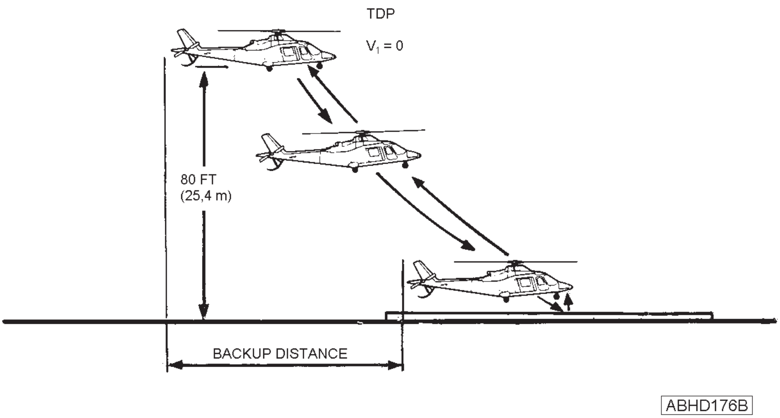


Figure 3-9. Rejected takeoff flight path profile (helipad).

Collective	: As necessary to maintain rotor RPM and OEI limits.
OEI landing manoeuvre	: If the failure occurs at or above 50 ft AGL push over 20 degrees attitude change to obtain forward speed. If the failure occurs below 50 ft AGL push over slightly to bring the helicopter back to the ground. In both cases carry out the manoeuvre maintaining the far right hand corner of the helipad in sight.

Perform a flare and level the helicopter to establish the minimum ground speed.

Collective : As necessary to cushion landing.

NOTE

Attention should be paid to avoid touching the tail first.

After landing perform normal shutdown procedure.

ENGINE FAILURE DURING TAKEOFF AT/AFTER THE TAKEOFF DECISION POINT (TDP)

(Fig 3-10)

Manoeuvre with standard TDP

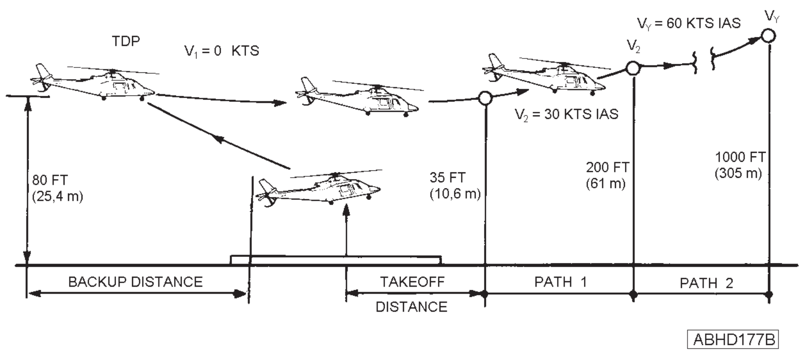


Figure 3-10. Continued takeoff flight path profile (helipad).

Continued takeoff manoeuvre	:	Set a nose down attitude of approximately 25 degrees to achieve V_2 by using up to the 2.5 minute power. Adjust collective to drop NR not below 90%. As the speed approaches V_2 reduce the nose down attitude. Recover rotor speed and climb at V_2 using 2.5 minute power up to 200 ft (61 m) above the helipad.
Landing gear	:	UP (by 200 ft AGL).
Airspeed	:	Accelerate to V_Y while climbing.
RPM switch	:	Set 100%.
NR/N2 speed	:	100% stabilized, check.
Takeoff path	:	Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).
Affected engine	:	Perform the complete shutdown procedure.

Land as soon as possible.

Manoeuvre from ground level helipad with increased TDP

Continued takeoff manoeuvre	: Set a nose down attitude change of approximately 25 degrees to achieve V_2 by using up to the 2.5 minute power. Adjust collective to drop NR not below 90%. As the speed approaches V_2 reduce the nose down attitude. Recover rotor speed and climb at V_2 using 2.5 minute power.

NOTE

If the minimum height in the continued flight path is 35 ft or higher, the takeoff distance is reached when V_2 and a positive rate of climb are achieved.

Landing gear	: UP (after having reached V_2 and a positive rate of climb).
Airspeed	: Accelerate to V_y after having reached V_2 and a positive rate of climb, but not below 200 ft AGL.

NOTE

When the TDP is 300 ft or higher the flight path 1 does not exist.

RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.

Takeoff path	: Select maximum continuous power OEI and keep on climbing at V_y up to 1000 ft (305 m).
Affected engine	: Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING PRIOR TO/AT THE LANDING DECISION POINT (LDP)
(Fig 3-11)

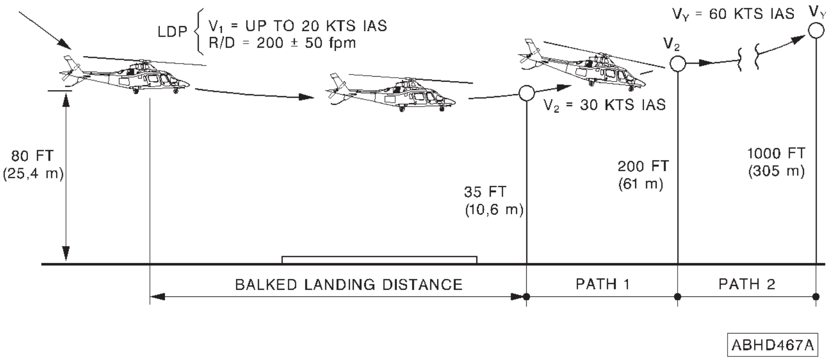


Figure 3-11. Balked landing flight path profile (helipad).

Balked landing manoeuvre	:	Reach V_2 by pushing over 15 degrees attitude change and by setting the collective to obtain up to 2.5 minute power and to drop NR not below 90%. As the speed approaches V_2 reduce the nose down attitude. Recover rotor speed and climb at V_2 using 2.5 minute power up to 200 ft (61 m) above the helipad.
Landing gear	:	UP (by 200 ft AGL).

Airspeed	: Accelerate to V_Y while climbing.
RPM switch	: Set 100%.
NR/N2 speed	: 100% stabilized, check.
Climb path	: Select maximum continuous power OEI and keep on climbing at V_Y up to 1000 ft (305 m).
Affected engine	: Perform the complete shutdown procedure.

Land as soon as possible.

ENGINE FAILURE DURING LANDING AT/AFTER THE LANDING DECISION POINT (LDP)
(Fig 3-12)

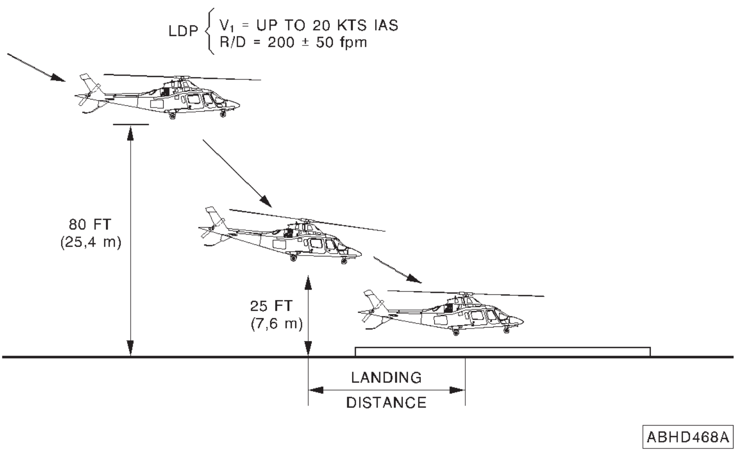


Figure 3-12. OEI landing flight path profile (helipad).

Collective	:	As necessary to maintain rotor RPM.
OEI landing manoeuvre	:	If the failure occurs at or above 50 ft above the helipad, push over 15 degrees attitude change to increase forward speed and approach the landing site. If the failure occurs below 50 ft above helipad push over slightly to bring the helicopter to land. In both cases always keep in sight the far right hand corner of the helipad. Perform a flare and level the helicopter before accomplishing a running landing.

NOTE

Attention should be paid to avoid touching the tail first.

After landing perform normal shutdown procedure.

SECTION 4 - PERFORMANCE DATA

POWER ASSURANCE CHECK

Refer to Section 4 of the basic Rotorcraft Flight Manual to determine if the engine can provide the installed specification power.

The hover check shall be performed daily prior to takeoff.

The in-flight check is provided for periodic in-flight monitoring of engine performance.

If one engine does not meet the requirements either of hover or in-flight power assurance check, the take-off and landing performance contained in this Appendix will not be achievable.

The cause of engine power loss shall be determined and engine maintenance action must be taken in accordance with the pertinent engine maintenance manual.

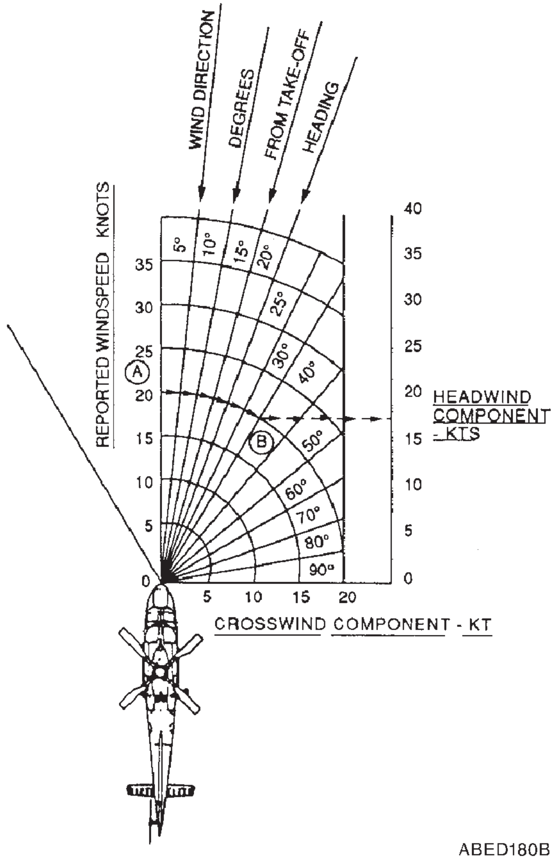
WIND EFFECT

Vertical takeoff and landing performance have been demonstrated up to 5 Kts crosswind.

For crosswind and headwind computation refer to Wind Component Chart (figure 4-1)

NOTE

Windspeed are unfactured. Unless otherwise authorized by operating regulations, the pilot is not authorized to credit more than 50 percent of the performance increase resulting from the actual favorable headwind component.



EXAMPLE

1. TAKE-OFF HEADING .... 170°
2. REPORTED WIND DIRECTION .... 200°
3. WIND DIRECTION, DEGREES FROM TAKE-OFF HEADING .... 30°
4. REPORTED WIND SPEED .... 20 KNOTS
5. ENTER CHART AT REPORTED WIND SPEED, POINT A
6. FOLLOW THE SHAPE OF THE CURVED LINES, TO WIND DIRECTION, DEGREES FROM TAKE-OFF HEADING, POINT B
7. PROCEED HORIZONTALLY TO THE HEADWIND COMPONENT SCALE AND READ HEADWIND COMPONENT .... 17 KNOTS
8. TAILWINDS HAVE NOT BEEN DEMONSTRATED

Figure 4-1. Wind component chart.

OPERATION FROM CLEAR AREA

NOTE

The distances are applicable to all weights, pressure altitudes and ambient temperatures, when operating at allowed gross weights (see WAT limitations).

TAKEOFF DISTANCE

The distance from takeoff starting to clear a 35 ft high obstacle at V_2 with a positive OEI rate of climb following an engine failure is:

− 370 m for a standard manoeuvre
− 130 m for a ″soft″ manoeuvre.

REJECTED TAKEOFF DISTANCE

The distance which defines the space necessary to accomplish a safe landing following an engine failure at or before the TDP, is:

− 240 m for a standard manoeuvre
− 400 m for a ″soft″ manoeuvre.

LANDING DISTANCE

The distance from a point at 50 ft (15.2 m) above the landing surface to the point at which the helicopter is brought to a stop on ground with one engine inoperative is 90 m.

BALKED LANDING DISTANCE

The balked landing distance never exceeds 370 m (takeoff distance obtained with standard manoeuvre).

OPERATION ON SHORT FIELD

NOTE

The distances are applicable to all weights, pressure altitudes and ambient temperatures, when operating at allowed gross weights (see WAT limitations).

TAKEOFF DISTANCE

The distance from takeoff starting to clear a 35 ft high obstacle at V_2 with a positive OEI rate of climb following an engine failure is 460 m.

REJECTED TAKEOFF DISTANCE

The distance which defines the space necessary to accomplish a safe landing following an engine failure at or before the TDP, is 80 m.

LANDING DISTANCE

The distance from a point at 50 ft (15.2 m) above the landing surface to the point at which the helicopter is brought to a stop on ground with one engine inoperative is 80 m.

BALKED LANDING DISTANCE

The horizontal distance from the LDP to the point at which a minimum of 35 ft (10.6 m) is attained at V_2 and with a positive OEI rate of climb is 250 m.

HELIPAD OPERATION (GROUND LEVEL OR ELEVATED)

NOTE

The distances are applicable to all weights, pressure altitudes and ambient temperatures, when operating at allowed gross weights (see WAT limitations).

TAKEOFF DISTANCE (MANOEUVRE WITH STANDARD TDP)

The horizontal distance from the start of takeoff to reach a point at least 35 ft (10.6 m) above the helipad at V_2 and with a positive OEI rate of climb is:
200 m up to 4000 ft Hd
400 m above 4000 ft Hd

BACKUP DISTANCE (MANOEUVRE WITH STANDARD TDP)

The horizontal distance from takeoff point to the TDP is 40 m.

TAKEOFF AND BACKUP DISTANCES (MANOEUVRE FROM GROUND LEVEL HELIPAD WITH INCREASED TDP)

In case of modified takeoff flight path from ground level helipad to clear high obstacles, the horizontal distance from the start of takeoff to reach V_2 with positive rate of climb is shown in table 4-1 together with the backup distance.

TDP HEIGHT (ft)	BACKUP DISTANCE (m)	TAKEOFF DISTANCE (m)
140	95	130
160	115	105
180	135	80
200	150	50
220	170	25
240	185	0
260	205	-30
280	225	-55

TDP HEIGHT (ft)	BACKUP DISTANCE (m)	TAKEOFF DISTANCE (m)
300	240	-80
320	260	-100
340	275	-115
360	295	-130
380	310	-145
400	330	-160
420	350	-175
440	365	-190
460	385	-210
480	400	-225
500	420	-240

Table 4-1. Distances versus the TDP.

LANDING DISTANCE

The horizontal distance from a point 25 ft (7.6 m) above the helipad to the landing point is 55 m.

BALKED LANDING DISTANCE

The horizontal distance from the LDP to the point at which a minimum of 35 ft (10.6 m) is attained at V_2 and with a positive OEI rate of climb is 300 m.

TAKEOFF FLIGHT PATH 1

MEAN HEIGHT GAINED IN 30 M HORIZONTAL DISTANCE

The mean height gained in 30 m of horizontal distance travelled during an OEI climb at V_2 after takeoff is shown in figures 4-2 thru 4-6 for various altitudes, temperatures, weights and headwind components.

NOTE

The headwind component should be obtained from the wind component chart shown in figure 4-1.

This chart applies from the end of takeoff distance to a height of 200 ft above the starting point.

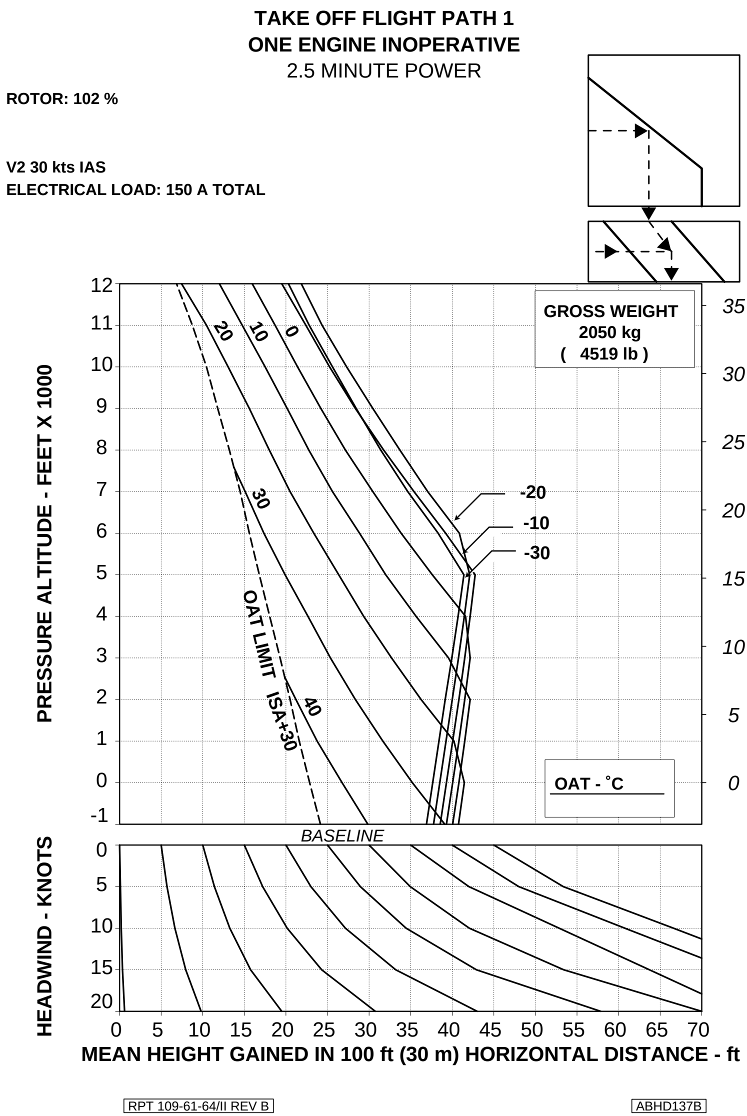


Figure 4-2. Takeoff flight path 1 - 2050 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
ELECTRICAL LOAD: 150 A TOTAL

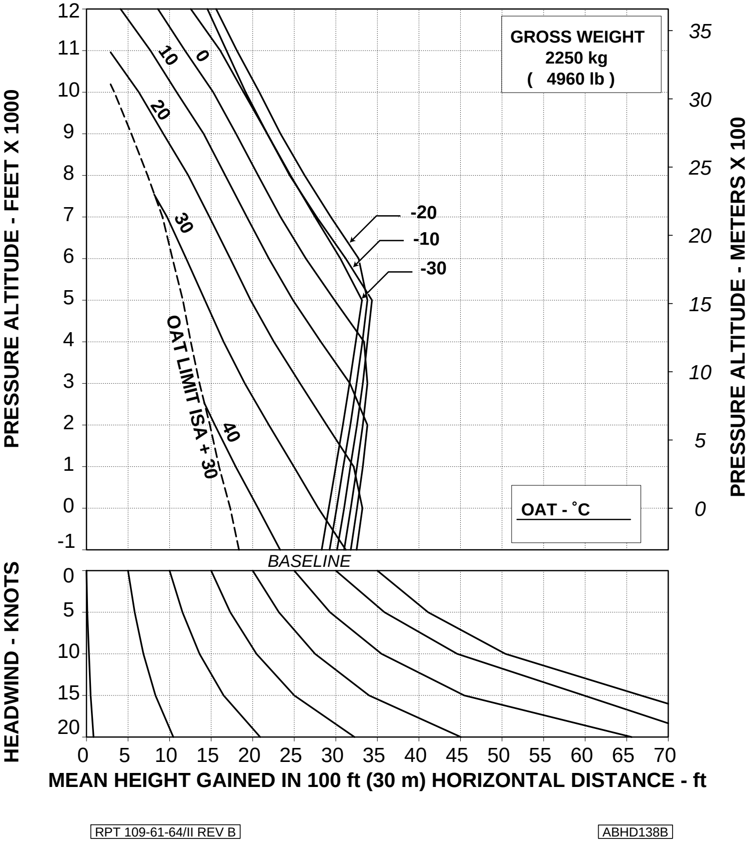


Figure 4-3. Takeoff flight path 1 - 2250 kg.

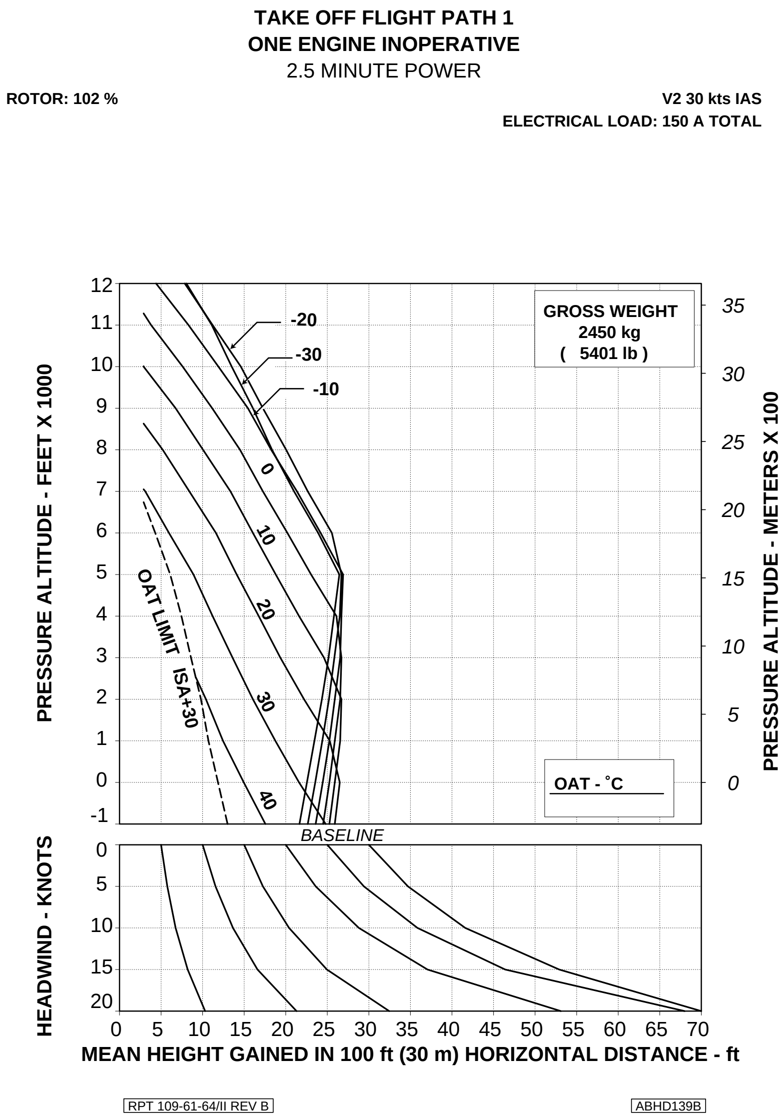


Figure 4-4. Takeoff flight path 1 - 2450 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

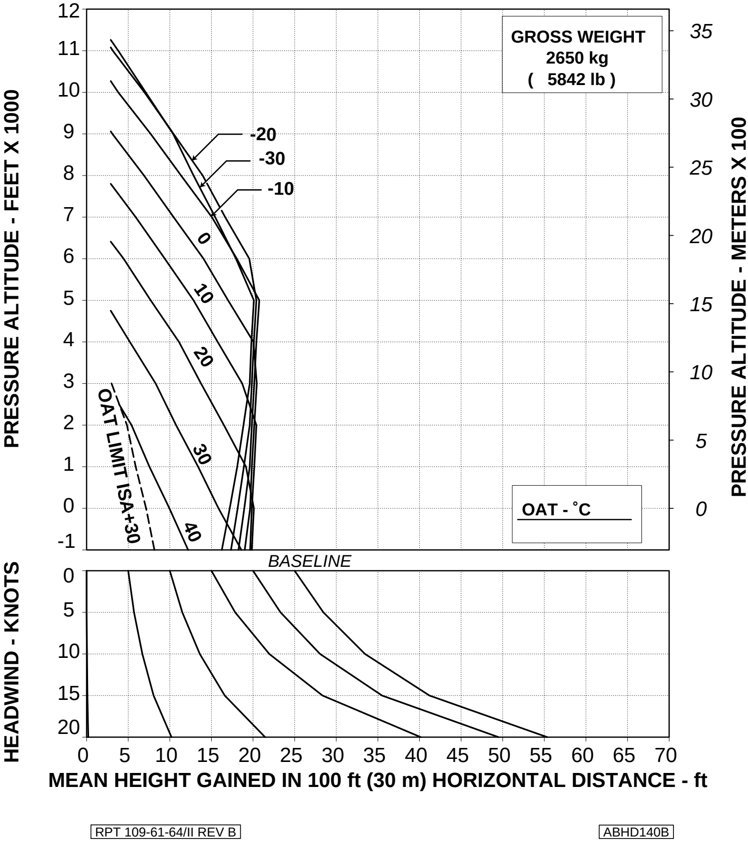


Figure 4-5. Takeoff flight path 1 - 2650 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

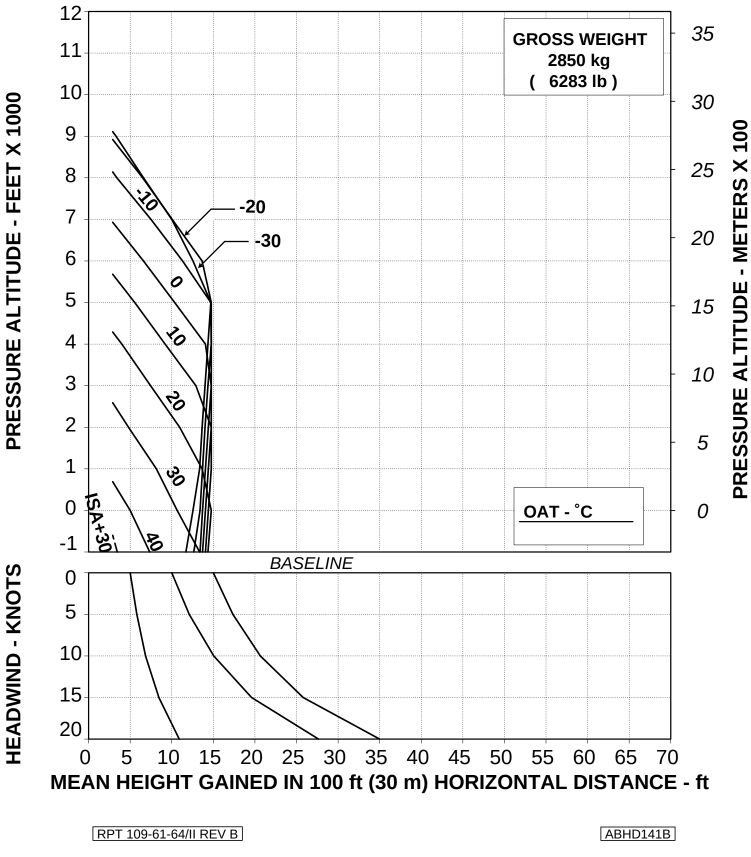


Figure 4-6. Takeoff flight path 1 - 2850 kg.

TAKEOFF FLIGHT PATH 2

MEAN HEIGHT GAINED IN 30 M HORIZONTAL DISTANCE

The mean height gained in 30 m of horizontal distance travelled during an OEI climb at V_Y after takeoff is shown in figures 4-7 thru 4-11 for various altitudes, temperatures, weights and headwind components.

NOTE

The headwind component should be obtained from the wind component chart shown in figure 4-1.

This chart applies from 200 ft to 1000 ft above the starting point.

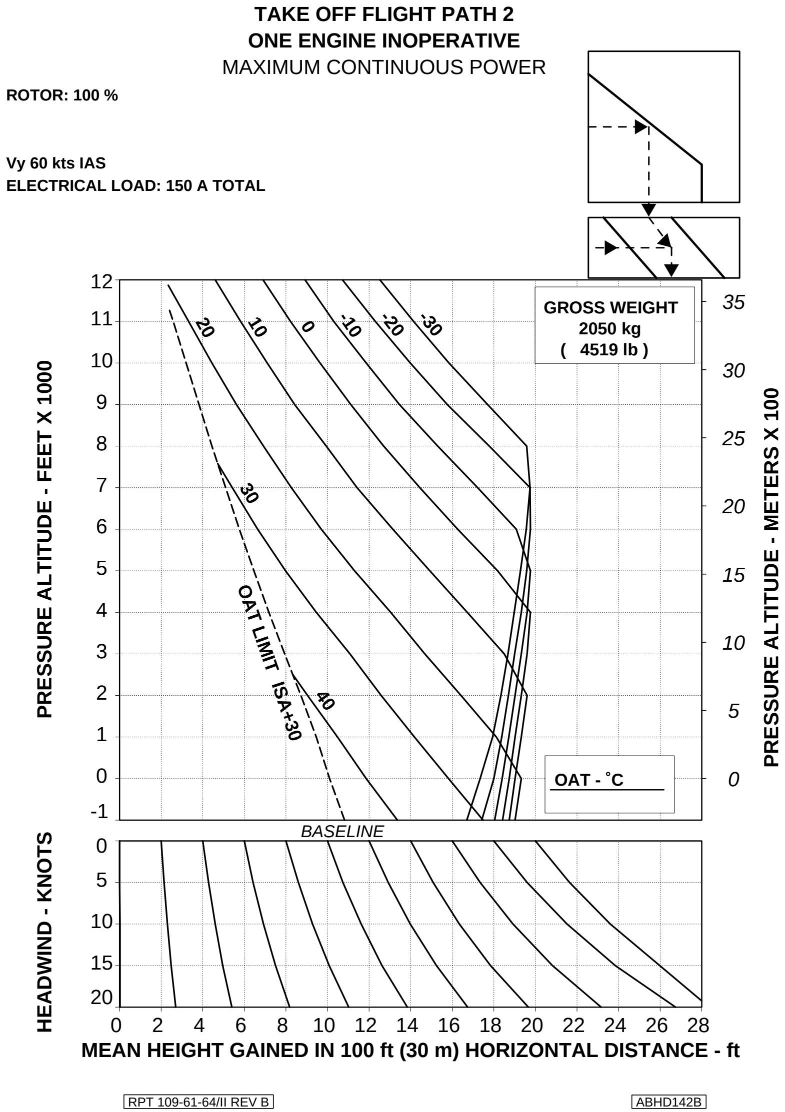


Figure 4-7. Takeoff flight path 2 - 2050 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

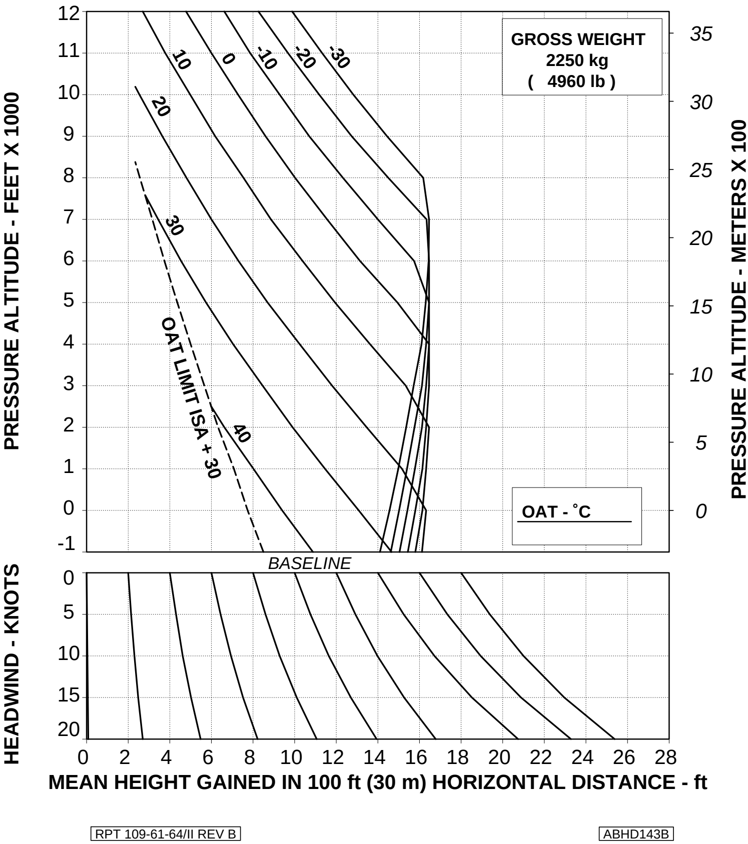


Figure 4-8. Takeoff flight path 2 - 2250 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

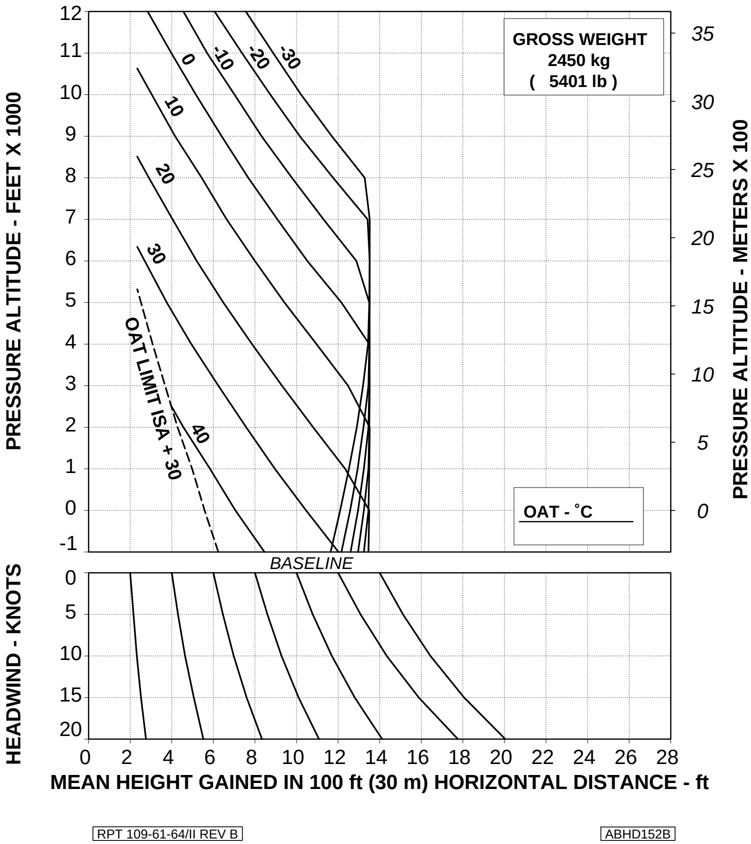


Figure 4-9. Takeoff flight path 2 - 2450 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

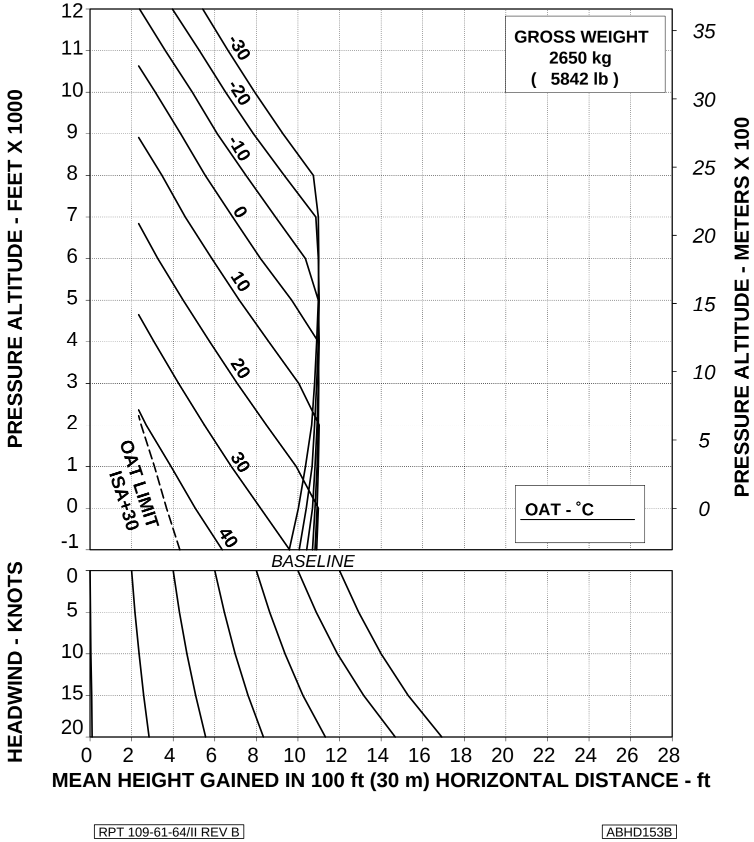


Figure 4-10. Takeoff flight path 2 - 2650 kg.

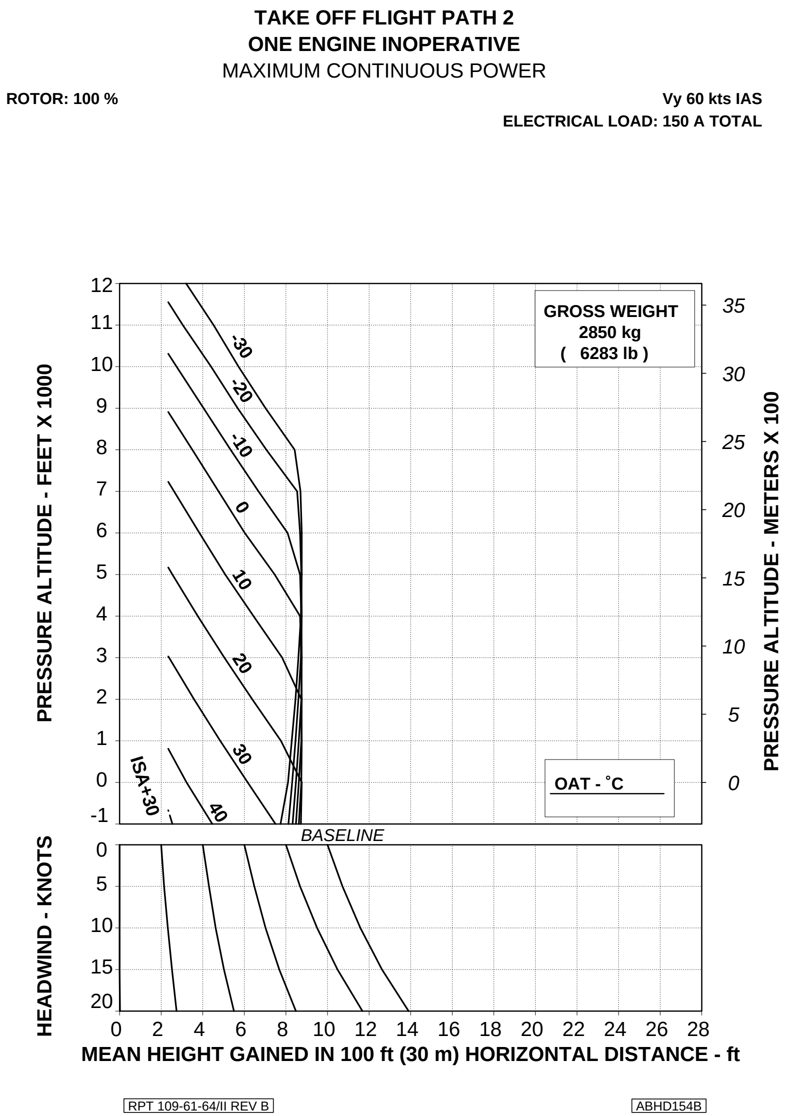


Figure 4-11. Takeoff flight path 2 - 2850 kg.

CLIMB PERFORMANCE

SINGLE ENGINE RATE OF CLIMB AT V_2 (30 KTS IAS)

The single engine rate of climb, at minimum V_2 and 2.5 minute power, is shown in figures 4-12 thru 4-16 for various altitudes, temperatures and weights.
These curves are for general information only, since the takeoff flight path charts are presented for flight path determination.

SINGLE ENGINE RATE OF CLIMB AT V_Y (60 KTS IAS)

The single engine rate of climb, at V_Y (60 Kts IAS) and maximum continuous power, is unchanged from the basic Rotorcraft Flight Manual.

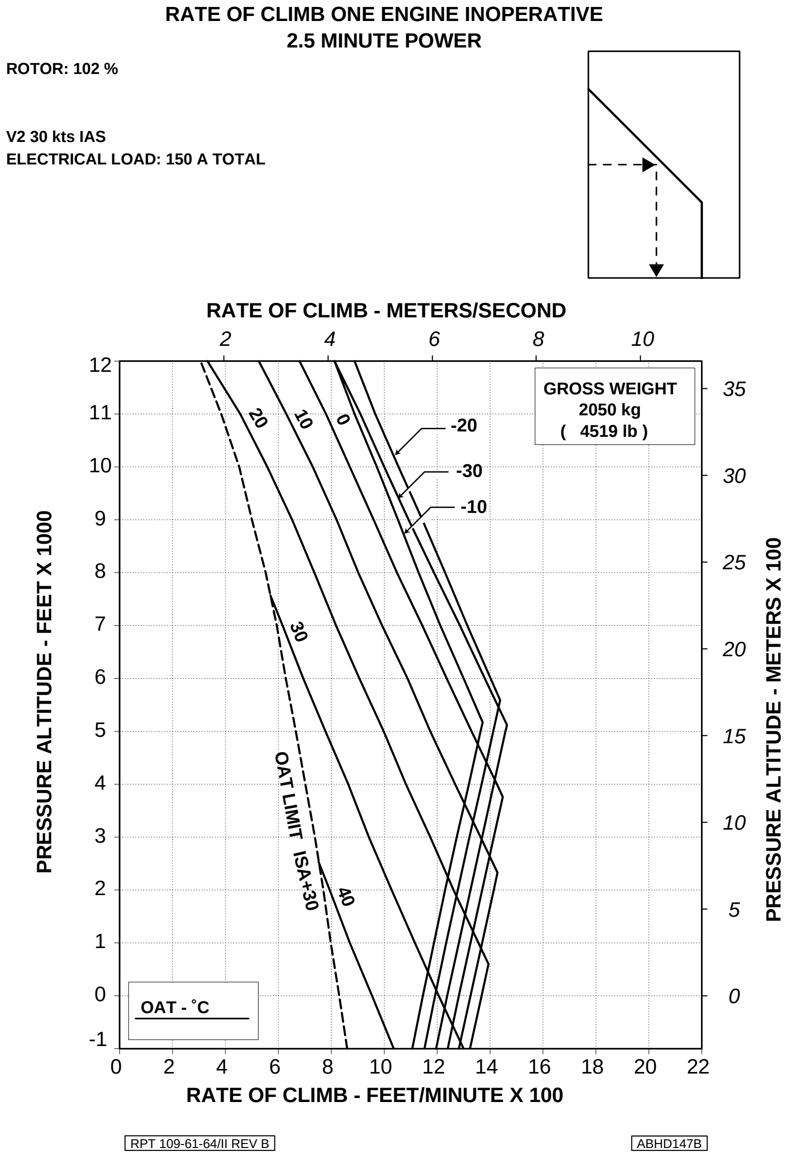


Figure 4-12. Rate of climb - OEI - 2.5 minutes power - 2050 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

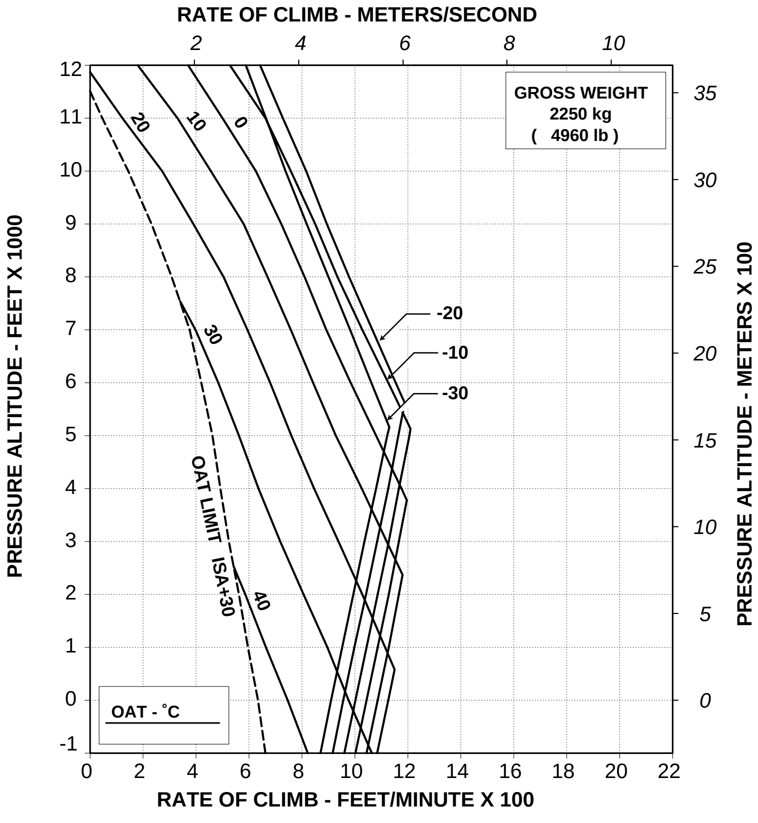


Figure 4-13. Rate of climb - OEI - 2.5 minutes power - 2250 kg.

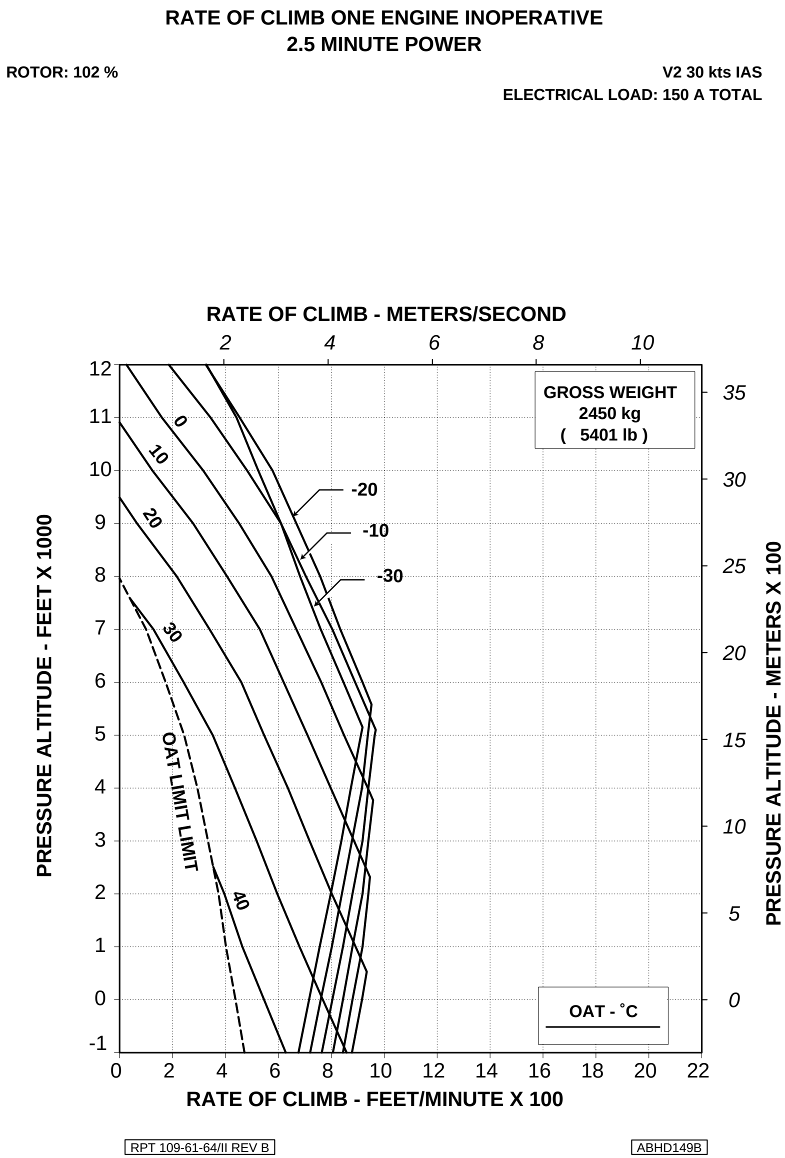
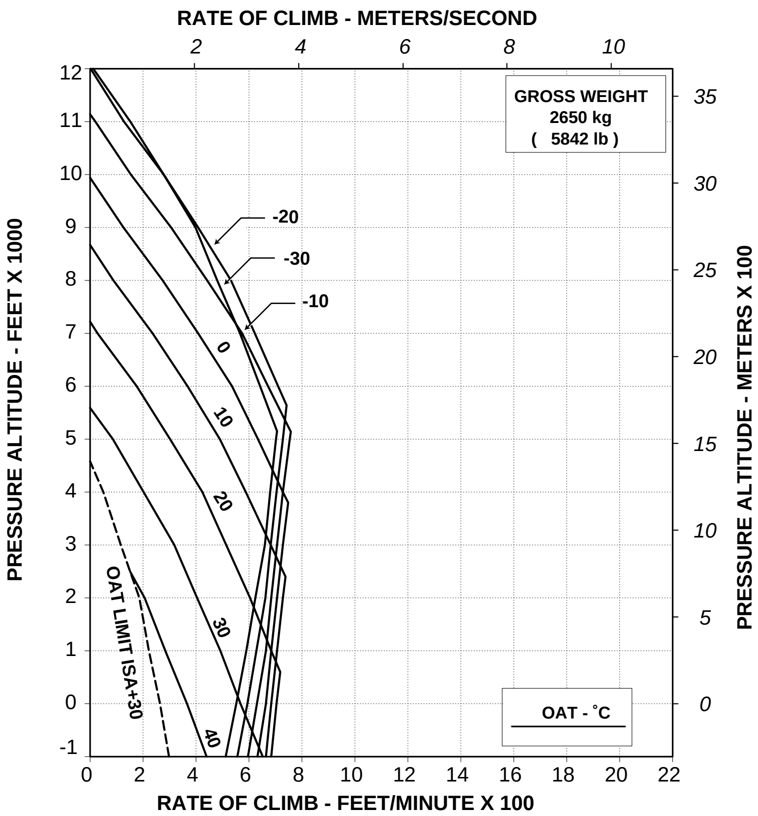


Figure 4-14. Rate of climb - OEI - 2.5 minutes power - 2450 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL



RPT 109-61-64/II REV B

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Figure 4-15 . Rate of climb - OEI - 2.5 minutes power - 2650 kg.

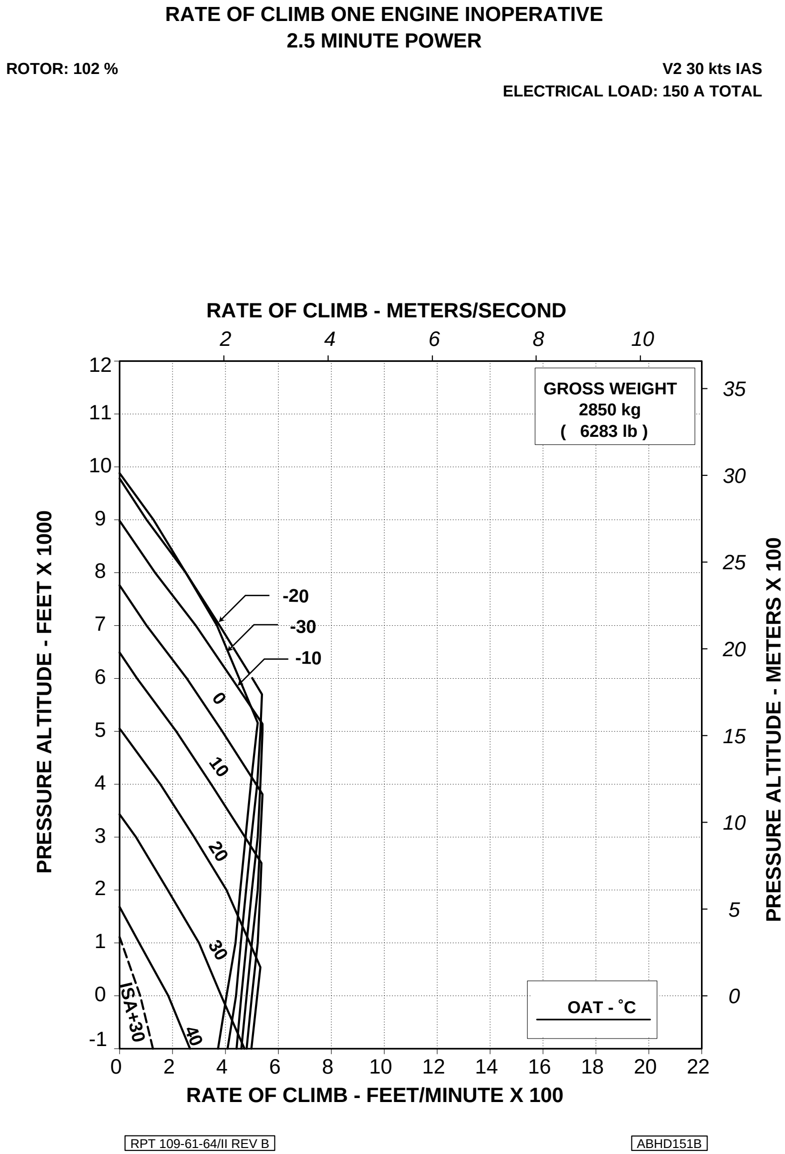

Figure 4-16. Rate of climb - OEI - 2.5 minutes power - 2850 kg.

R.A.I. Approval Letter 98/401/MAE
dated 29-1-1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

INTEGRATED DISPLAY SYSTEM CONFIGURATION EDU P/N 109-0900-42-101 DAU P/N 109-0900-42-103

This Appendix applies to the helicopters configurated with the Integrated Display System composed by EDU P/N 109-0900-42-101 and DAU P/N 109-0900-42-103.

Such IDS configuration utilizes the EDU software version 005 and the DAU software version 004.

TABLE OF CONTENTS

LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The equivalent Cat. A operations are prohibited.

POWER PLANT LIMITATIONS

POWER TURBINE (N2) RPM

One engine inoperative (OEI)

Transient	: 90%.
Minimum	: 99%.
Continuous operation	: 99 to 101%.
Takeoff and landing	: 101 to 102%.
Maximum	: 102%.
Transient (20 sec)	: 112.4%.

ROTOR LIMITATIONS

Power On (OEI)

Transient	: 90%.
Minimum	: 99%.
Continuous operation	: 99 to 101%.

Takeoff and landing (below 60 KIAS and 1000 ft AGL) : 101 to 102%.

NOTE

During an instrumental approach procedure the use of 102% NR is allowed up to 120 KIAS.

Maximum : 102%.

TRANSMISSION LIMITATIONS

TORQUE (TRQ%)

All engines operative (AEO)

NOTE

100% torque corresponds to 534 Nm (394 ft lb).

Maximum continuous : 100%.

Transient (6 sec) : 110%.

One engine inoperative (OEI)

NOTE

100% torque corresponds to 664 Nm (490 ft lb).

Maximum continuous : 100%.

2.5 min : 114%.

Transient (6 sec) : 125%.

NOTE

The One Engine Inoperative (OEI) rating use is intended for actual engine loss. For maintenance or training purposes OEI operation shall be limited to the maximum continuous power rating.

NOTE

Transient must not be used intentionally. Single engine transient may be used only in case of real emergency, when one engine becomes inoperative due to an actual malfunction.

INSTRUMENT MARKINGS

ELECTRONIC DISPLAY UNIT FORMATS

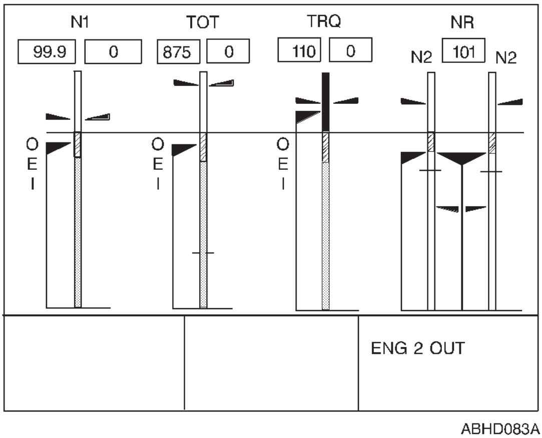

Figure 1-1. EDU 1 - OEI mode.

The OEI mode is a subset of CRUISE mode and is automatically selected if one engine becomes inoperative.
The OEI format is presented also in REVERSIONARY mode.

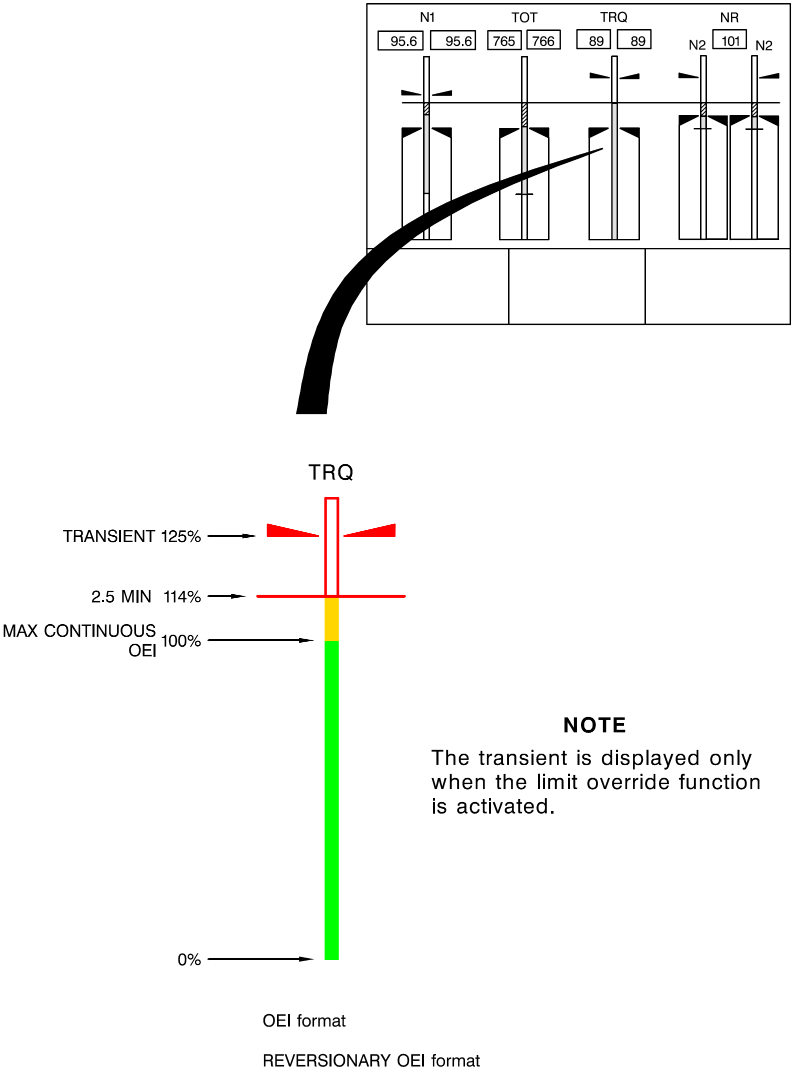

Figure 1-2. EDU 1 - TORQUE (OEI).

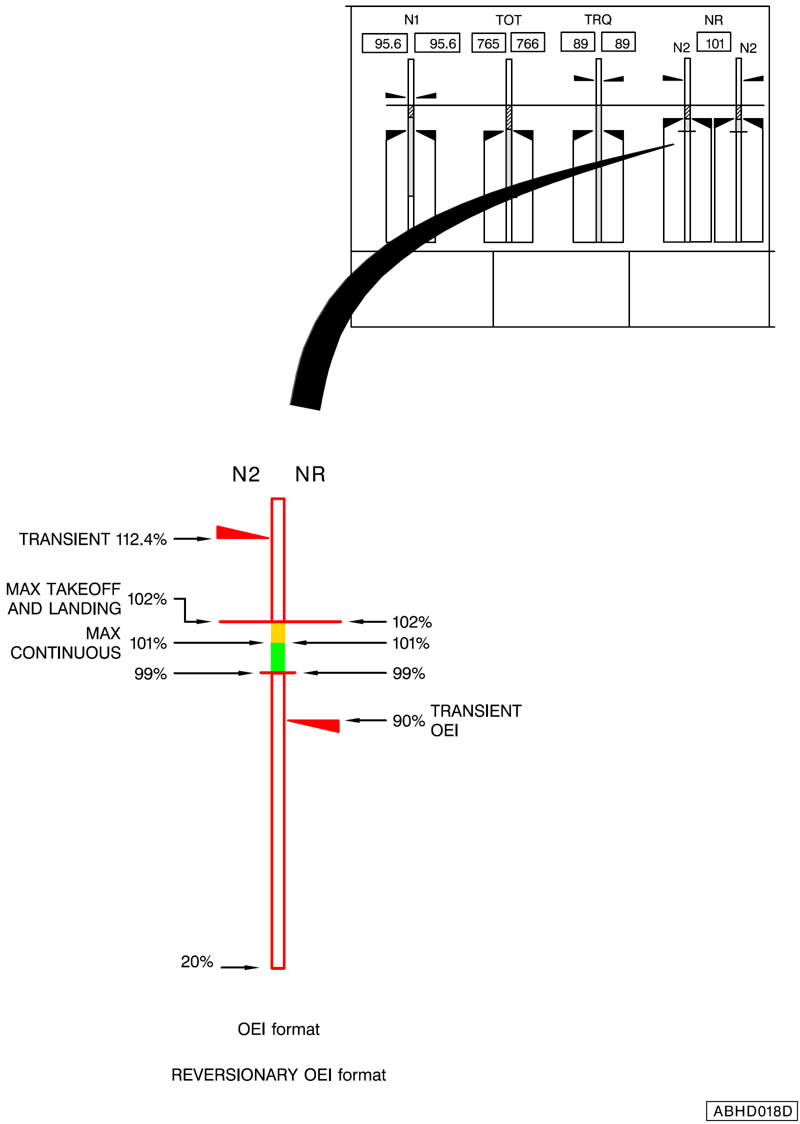

Figura 1-3. EDU 1 - ROTOR POWER TURBINE SPEED (OEI).

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

EMERGENCY PROCEDURE FOR LIMIT OVERRIDE PUSHBUTTON OPERATION

The engine is protected from limits exceeding through the engine limit governing.
The engine limit governing is responsible for limiting measured engine parameters through the control loop selection logic.
The limit governing is set not to overcome the following limits:

Torque OEI	114%
Torque AEO	110%
TOT	960°C
N1	102.4%

In emergency condition the pilot may operate the LIM OVRD red pushbutton located on the collective control to override the engine limit governing.

PROCEDURE

LIM OVRD pushbutton
(on collective control) : Press.

CAUTION

Limit override pushbutton operation is allowed for actual emergency only.

EDU 2 : LIMIT OVRD ON advisory message displayed.

EDU 1	:	OEI transient torque marking (red triangle) displayed, if in OEI mode.
Collective	:	Apply as necessary paying attention since the engine is no more protected from limits overcoming.

The operation of LIM OVRD pushbutton must be noted on the helicopter log-book.
Refer to helicopter Maintenance Manual for the inspection to accomplish before next flight.

SECTION 4 - PERFORMANCE DATA

POWER ASSURANCE CHECKS

The Power Assurance Check charts are provided in order to let the pilot determine if the engines can produce the installed specification power.

A power assurance check should be performed daily. Additional checks should be made if unusual operating conditions or indications arise.
The hover check is performed prior to takeoff and the in-flight check is provided for periodic in-flight monitoring of engine performance.
Either power assurance check method may be selected at the discretion of the pilot. It is pilot's responsibility to accomplish the procedure safely, considering passenger load, terrain being overflown and the qualification of persons on board to assist in watching for other air traffic and to record power check data.
If either one of the two engines does not meet the requirements of the hover or the in-flight check, the minimum performance requirements as per published data may not be obtained.
The cause of engine power loss, or excessive TOT or GAS PRODUCER RPM (N1) should be determined as soon as practical.
Refer to Engine Maintenance Manual.

POWER ASSURANCE CHECK
HOVER

* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 102%.
* TEST ENGINE MODE SWITCH - FLIGHT.
* OTHER ENGINE MODE SWITCH - IDLE.

* INCREASE COLLECTIVE UNTIL LIGHT ON WHEELS OR HOVERING AT 3 FEET, DO NOT EXCEED 820˚C TOT OR 97.4% N1 OR 100% TORQUE.
* STABILIZE POWER ONE MINUTE, THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAXIMUM ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINATED AS SOON AS PRACTICAL.

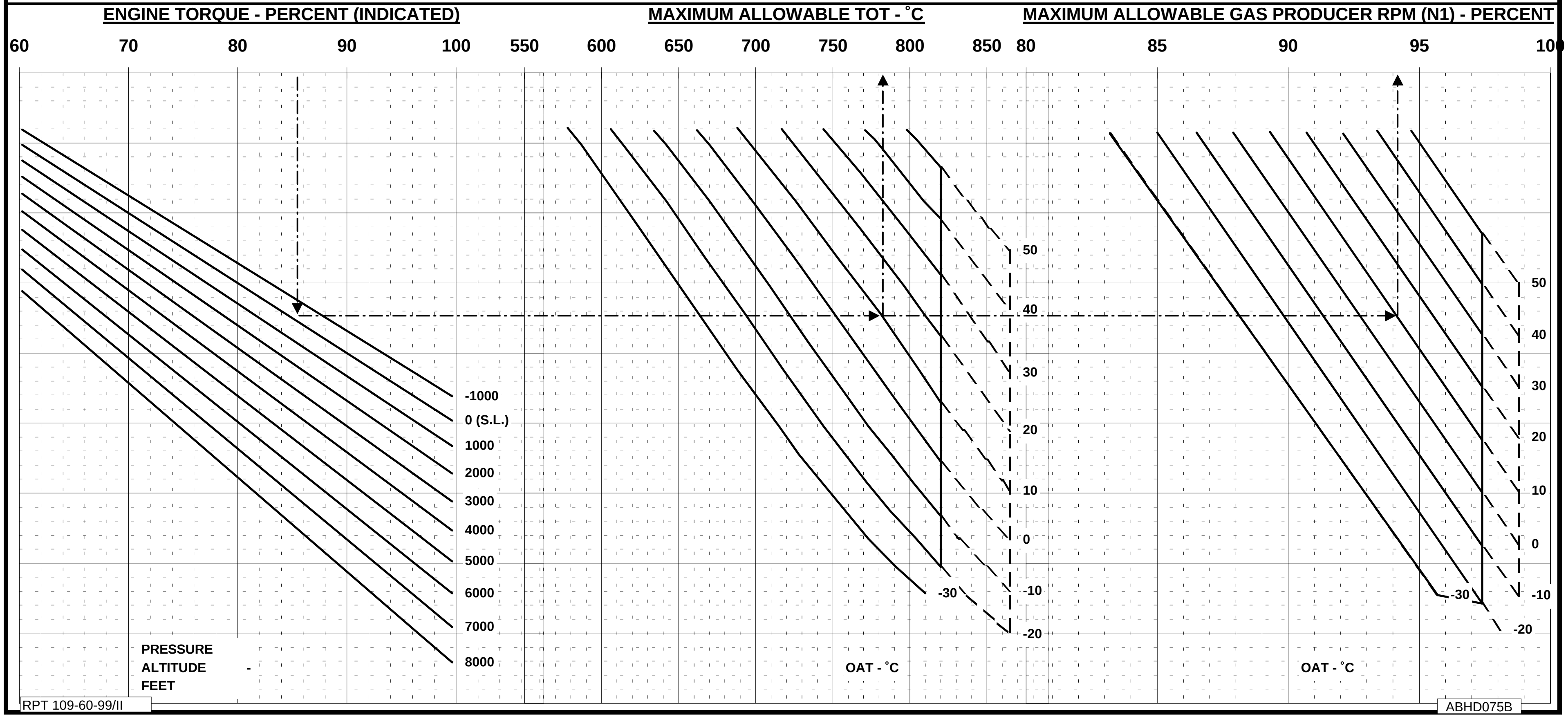

Figure 4-1. Power assurance check - hover.

POWER ASSURANCE CHECK IN FLIGHT

* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 100%.
* ESTABILISH LEVEL FLIGHT ABOVE 1000 ft. AGL.
* AIRSPEED - 100 KIAS.
* TEST ENGINE - SET ENG. GOV. SWITCH TO MANUAL
* OTHER ENGINE - LEAVE ENG. GOV. SWITCH TO AUTO

* OPERATE TEST ENG TRIM TO INCREASE POWER UNTIL ENGINE TORQUE IS WITHIN TEST RANGE. DO NOT EXCEED 820˚C TOT OR 97.4% N1 OR 100% TORQUE.
* STABILIZE POWER ONE MINUTE IN LEVEL FLIGHT THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAX ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, CARRY OUT A POWER ASSURANCE CHECK IN HOVER.

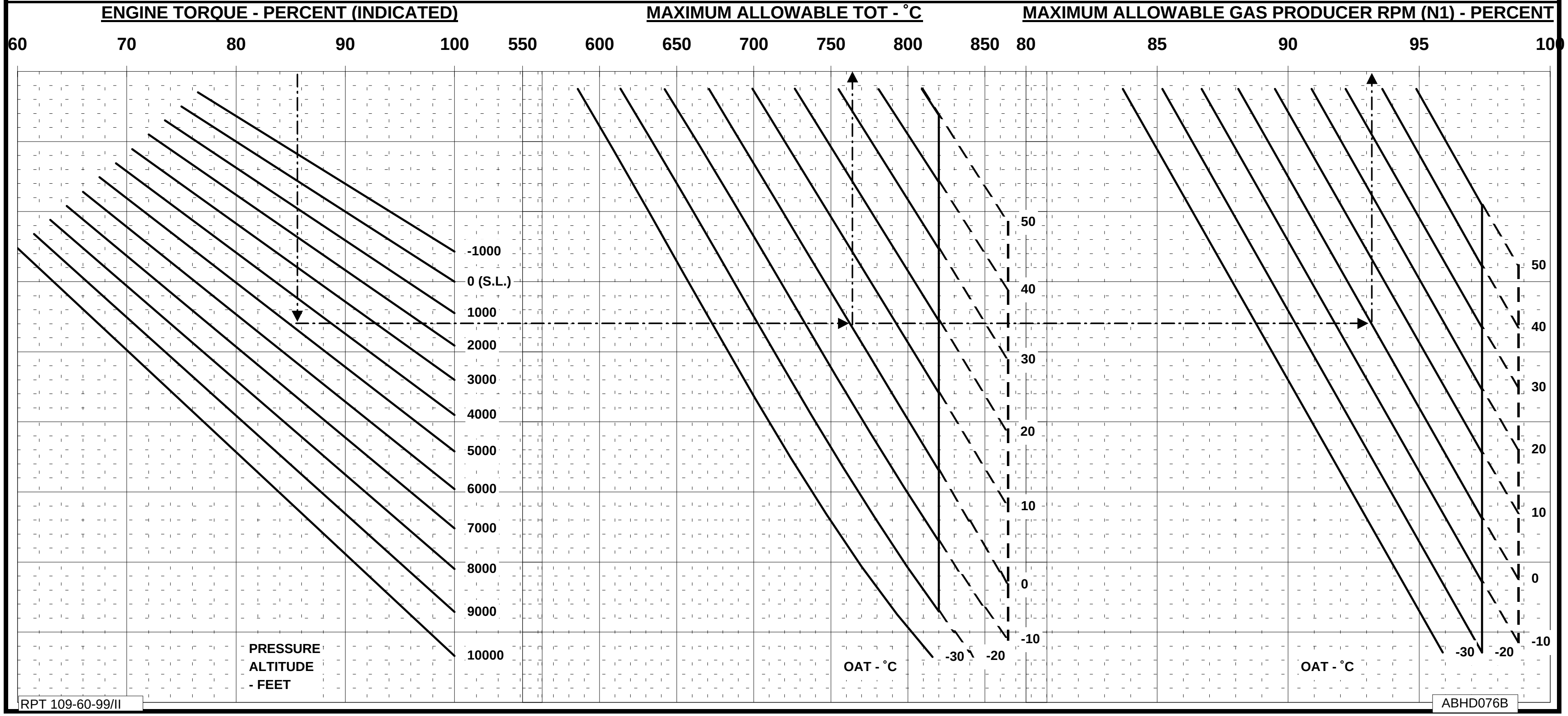

Figure 4-2. Power assurance check - in flight.

R.A.I. Approval Letter 98/809/MAE
dated 19-2-1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

BATTERY 22 AH

The battery installation P/N 109-0812-04 consists of a 22 Ah nickel-cadmium battery installed in lieu of the normal 27 Ah battery.
The installation includes vent tubes and attaching hardware.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

No change.

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

ELECTRICAL POWER FAILURE

Failure of both generators

Gen 1 and 2 : Reset, then ON. If both generators are inoperative continue flight on battery. Land as soon as possible.

WARNING

After both generators failure the battery will carry the electrical load for approximately 6 minutes. The battery operating time can be extended by selectively reducing system load.
Flight duration can be extended to 30 minutes provided that the following procedure is carried out.

GEN BUS 1 and 2 : OFF.

NOTE

Due to automatic load shedding, the following electrical loads will be supplied by the battery:

- EDU 1, EDU 2, DAU channel A and B
- Inverter 1
- Fuel pumps
- Fuel valves
- ICS Pilot and AWG
- ADI stand-by
- Force trim
- VHF 2 (2 minutes Tx and 13 minutes Rx)
- Helipilot 1
- Landing lights (2 minutes)
- Pilot spot light
- Fuel quantity 1 and 2
- Hydraulic system
- Pitot 2
- EADI pilot (if installed)
- Battery relay
- Emergency relay
- Landing gear indicator
- Attitude engage
- Fire detectors
- Fire extinguishers (if installed)
- Governor control

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 98/1385/MAE
dated 20-3-98

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EFIS

(ELECTRONIC FLIGHT INSTRUMENT SYSTEM)

The EFIS P/N 109-0900-57-101/-103/-105/-107 consists of four AMLCD (Active Matrix Liquid Crystal Display) displays installed in the instrument panel and performing as EADI and EHSI indicators.
The EFIS interfaces with the attitude and navigation systems and with the weather radar.
The pilot's EADI and EHSI and optionally also the copilot's ones are coupled also with the Flight Director computer (if installed).

AGUSTA

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SECTION 1 - LIMITATIONS

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PART II - MANUFACTURER'S DATA

LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION
(EFIS P/N 109-0900-57-101 only)

Operation is permitted under VFR flight only.

PLACARDS
(EFIS P/N 109-0900-57-101 only)

EFIS OPERATION IN VFR ONLY

In clear view of the pilot

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK
(First flight of the day)

Area N°7 (Cabin interior)

BATT RELAY circuit breaker : In. Door secured.

NOTE

The above mentioned circuit breaker is accessible from the pilot pedal bay through an inspection door and thus cannot be seen from the on-board seated position.

ENGINE PRE-START CHECKS

Stowable anti-glaze shield on the instrument panel : Adjust and fix as desired.

NORMAL ENGINE START

ENGINE 2 START

After RAD-MSTR switch activation:

EFIS displays	: Adjust brightness as necessary.

QUICK ENGINE START

After RAD-MSTR switch activation:

EFIS displays	: Adjust brightness as necessary.

SYSTEMS CHECK

EADI instrument (Pilot side):

ATT flag	: Out of view.
EADI	: Verify for consistency with the ADI stand-by.
ATT REV pushbutton	: Press - Check pushbutton illuminated - Check ADI reversion to opposite VG source - Check the consistency of the new ADI information with ADI stand-by - ATT 2 flag in view.
ATT REV pushbutton	: Press again.

Repeat the previous check with the copilot EADI and verify ATT 1 flag in view when ATT REV pushbutton is in.

Radio altimeter (EADI)	: Check zero altitude.

RA flag on EADI	:	Out of view.
DH	:	Set 200 ft.
RA TST button	:	Press and hold.
DH caption	:	In view.
Indication	:	100 ft (±5 ft).
RA TST button	:	Release.
DH	:	Set as desired.
F/D STBY button	:	Press and hold.
FD flag	:	Check, in view.
F/D STBY button	:	Release.

EHSI instrument (Pilot and copilot):

HDG flag	:	Out of view.
Compass heading	:	Consistent with magnetic compass reading. - Check yellow DG flag out of view (if in view set MAG/DG switch on compass control panel to MAG position).
HDG SEL	:	Select, as desired.
CRS SEL	:	Select, as desired.
BRG 1	:	As desired.
BRG 2	:	As desired.

IN FLIGHT

FLIGHT DIRECTOR OPERATION (if installed)

VOR (or GPS) coupled to NAV mode

CAUTION

When NAV mode on FD control panel is in CAP condition and a change of navigation source is carried out with the NAV key on the pilot (in command) EHSI, or the active navigation source goes to invalid (loss of signal), the relevant FD mode annunciation (NAV or GPS) changes from green to amber to alert the pilot of a possible degradation of the active FD mode.
Press the NAV pushbutton on FD mode selector to deactivate the mode.
Check the navigation source and select (if necessary) the new CRS and intercept HDG, then reselect the NAV pushbutton on FD control panel.

ILS mode

CAUTION

During ILS approach with FD engaged, the selection of the other navigation source (e.g. GPS) will be inhibited as soon as the LOC is captured.
Swapping from ILS 1 to ILS 2 is the only navigation source selection available.
If change between ILS 1 to ILS 2 carried out when FD is in LOC CAP status, the green LOC annunciation changes from green to amber to alert pilot of a possibke degradation of the active FD mode.
Press on FD control panel the NAV pushbutton to deselect ILS.
Check the navigation source and select the inbound course and intercept heading (if necessary) then reselect the NAV pushbutton on FD control panel.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

WARNING SYSTEM

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
VG 1	Gyro not erected, loss of power to gyro. ATT flag displayed on pilot's EADI.	Check breaker. Refer to stand-by attitude indicator. Pilot: revert to VG 2 through ATT REV switch. Check ATT 2 advisory message displayed on pilot's EADI.
VG 2	Gyro not erected, loss of power to gyro. ATT flag displayed on copilot's EADI.	Check breaker. Copilot: revert to VG 1 through ATT REV switch. Check ATT 1 advisory message displayed on copilot's EADI.

CAUTION LIGHT (YELLOW)

Panel Wording	Fault condition	Corrective action
EFIS FAN (if installed)	Failure of one or both EFIS fans.	Refer to para ″EFIS Fan failure″ in this Section.

SYSTEM FAILURES

EFIS FAILURES

The EFIS failures can affect separately or simultaneously each of four displays of the system and performing as EADI and EHSI indicators.
The cause can be an internal system failure or a connection system failure.
In the event of an internal system failure, the affected display (EADI or EHSI) becomes blank or an amber filled box with white FAIL is displayed on it, depending on the type of failure.

Failure of one EADI

In case of failure of pilot’s (or co-pilot’s) EADI, the pilot’s (or co-pilot’s) EHSI automatically reverts to composite mode and a red MON (″monitoring″) message appears at the same time on the displays showing the ADI to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of one EHSI

In case of failure of pilot’s (or co-copilot’s) EHSI, the pilot’s (or co-pilot’s) EADI automatically reverts to composite mode and a red MON (″monitoring″) message appears at the same time on the displays showing the ADI to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of pilot’s (or co-pilot’s) EADI and co-pilot’s (or pilot’s) EHSI

In case of a cross-failure of two EFIS displays, the remaining co-pilot’s (or pilot’s) EADI and pilot’s (or co-pilot’s) EHSI automatically revert to composite mode and a red MON (″monitoring″) message appears at the same time on both of them to point out the degradation of the comparison monitor function in the system.
The message ATT flag appears on pilot’s (or co-pilot’s) operating display to point out the loss of the primary attitude source (ATT 1 for pilot’s side and ATT 2 for co-pilot’s side).
Press the ATT REV pushbutton to switch the operating display over the alternative source.

On both displays the message ATT 2 (or ATT 1 if the remaining EADI is the co-pilot’s one) will appear.
Proceed with flight in the new system configuration.

Failure of both EADIs

In the event of a failure of both EADIs, the two EHSIs automatically revert to composite mode and a red MON (″monitoring″) message appears at the same time on both displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of both EHSIs

In the event of a failure of both EHSIs, the two EADIs automatically revert to composite mode and a red MON (″monitoring″) message appears at the same time on both displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of three out of four EFIS displays

In case of loss of three out of four displays constituting the EFIS, the screen of the remaining apparatus automatically reverts to composite mode and a red MON (″monitoring″) message appears to point out the failure of the monitoring intercommunication system inside the EFIS system.
Proceed with flight in the new system configuration.

EFIS Fan failure

In the event of failure of one or both EFIS cooling fans, the EFIS FAN light caption will illuminate.
If the failure occurs in flight and the Outside AIR Temperature (OAT) is greater than 45°C open vents and select VENT - CKPT switch to HIGH.
If ECS is installed, switch it ON.
Land as soon as practicable within 30 minutes.
With lower OATs proceed with flight and correct trouble before next flight.

EADI FAILURES

Radio altimeter failure

In the event of Radio Altimeter failure, a RA red flag replaces the numerical value; the rising runway and the DH indication (if present) disappear from the screen.
Aural messages ″ONE FIFTY FEET ″ and ″LANDING GEAR″ are inhibited while LANDING GEAR caution remains displayed on EDU.
Monitor barometric altitude during coupled ILS approach as the AUTOLEVEL function at 50 feet does not occur and the helicopter will continue to follow glide slope signal.

Flight Director failure (If installed)

In the event of Flight Director computer failure, a FD red flag is displayed. Moreover, FD command bars and mode indicators are removed from the screen.
Continue flight in ATT mode.

EHSI FAILURES

Weather radar failure

A red WX flag on EHSI indicates a failure within the WX Radar System.
Turn radar selector to OFF position.

EFIS FAULT MESSAGES

(Messages replacing the instrument indications).

EADI message	Fault condition	Corrective action
ATT	Gyro not erected, VG 1 (2) caution message displayed on EDU. Loss of power to gyro, loss of Roll or Pitch data on EADI.	Check breaker. Refer to stand-by attitude indicator. Revert to other VG through ATT REV pushbutton and check ATT 1 or ATT 2 displayed.

EHSI message	Fault condition	Corrective action
HDG	Aircraft heading data invalid or unavailable.	Refer to magnetic compass.

EFIS MISCOMPARE MESSAGES

(Messages displayed beside the instrument indications).

EADI message	Fault condition	Corrective action
PIT	Pitch attitude discrepancy between the gyro data.	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data.
ROL	Roll attitude discrepancy between the gyro data.	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data.

EADI message	Fault condition	Corrective action
ATT	Pitch and roll attitudes discrepancy between the gyro data.	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data.
LOC	Excessive discrepancy between the two localizer data.	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.
GS	Excessive discrepancy between the two glide slope data.	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.
ILS	Excessive discrepancy between both LOC and GS data.	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEMS DESCRIPTION

NAVIGATION SYSTEM

DME OPERATION

General

The following table shows the various combinations of possible DME data visualization on EHSI, depending on the selected navigation source.

NAV SELECTION		PLT IN CMD		CPLT IN CMD	
EHSI PLT	EHSI CPLT	PLT	CPLT	PLT	CPLT
VOR1	VOR1	DME1	DME1	DME1	DME1
VOR1	VOR2	DME1	—	—	DME2
VOR1	GPS	DME1	GPS	DME1	GPS
VOR2	VOR1	DME2	—	—	DME1
VOR2	VOR2	DME2	DME2	DME2	DME2
VOR2	GPS	DME2	GPS	—	GPS
GPS	VOR1	GPS	DME1	GPS	DME1
GPS	VOR2	GPS	—	GPS	DME2
GPS	GPS	GPS	GPS	GPS	GPS

Helicopters equipped with King Avionic suite

When on EHSI ″IN COMMAND″ the GPS 1 (or 2) is selected as navigator, the DME is automatically set on CH 1 (VOR 1).
If the pilot needs to use CH 2 (VOR 2), he must select VOR 2 as navigator, then select DME HOLD function and reselect the previous GPS sensor (1 or 2).

FLIGHT DIRECTOR OPERATION

When NAV, VOR APR, ILS or BC mode is active in CAP mode temporarily deselect the mode before performing any course change. Set the new course on the EHSI and intercept heading, then re-arm for capture.
Rotation of the course knob in NAV, VOR APR, ILS or BC modes while F/D system is in CAP condition, can falsely trigger the over station sensor and gives a degraded performance in the system.
During ILS (or BC) approach with F/D engaged, selection of other navigation sensors (e.g. GPS) will be inhibited as soon as the LOC is captured.
Swap from ILS 1 to ILS 2 is the only available navigation source selection.

NAV mode operation with GPS

On the GPS receiver, select the appropriate CDI scale before engaging the NAV mode of F/D.

The best CDI scale suggested for GPS coupled with F/D NAV mode is ±1 nm full scale.

EFIS COMPOSITE MODE

EFIS composite mode is used only in the event of an in-flight failure of any indicator (EADI or EHSI).

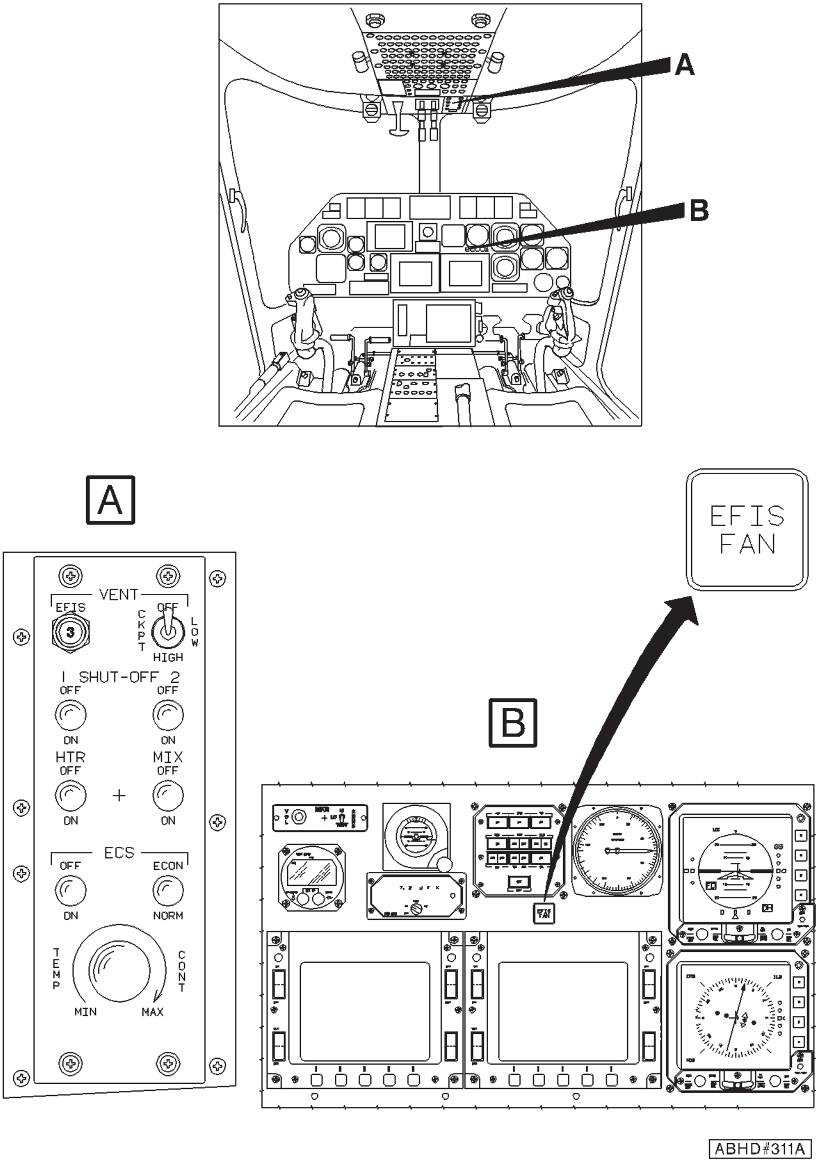

Figure 7-1. EFIS Fan (if installed) - Controls and indicators.

R.A.I. Approval Letter 98/1385/MAE
dated 20-03-98

The information contained herein supplements the information of the basic Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Flight Manual.

GLOBAL POSITION SYSTEM GARMIN 165

The GPS Garmin 165 system P/N 109-0811-53 consists of an antenna and of a receiver installed in the cockpit.

The function of the system is to acquire signals from the GPS system satellites and to process this information in real-time to obtain the user's position, velocity and time.

All data are shown on the receiver display.

The navigation data are also available on the EHSI (EFIS or EHSI 74 system) if the relative optional interface is mounted: in this configuration it is also possible to navigate with NAV mode of F/D coupled to GPS receiver.

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The GPS Garmin 165 system can be used in VFR conditions only, for use as a supplemental navigation system.

PLACARDS

GPS SYSTEM. USE ONLY DURING VFR OPERATION

In clear view of pilot

SECTION 2 - NORMAL PROCEDURES

NOTE

For correct and complete use of GPS Garmin 165, refer to Garmin GPS 165 Pilot's Guide, publication N° 190-00066-00 Rev. B or later.

SYSTEMS CHECK

GPS coupled with EFIS system (if installed)

Select the GPS on the EHSI through ″N″ button.

The GPS selection is indicated by a blue GPS legend on the left side of EHSI. All GPS data are also displayed in blue.

GPS coupled with EHSI 74B system (if installed)

Select the GPS on the EHSI through ″CRS″ button.

The GPS selection is indicated by a white LRN1 legend on the lower right corner of the EHSI.

All GPS data are also displayed in white.

NOTE

When GPS is selected on the EHSI 74B as navigation source, the navigation data coming from VOR 2 receiver are automatically available on pilot HSI (stand-by).

GPS operating procedure

— Select the GPS as navigation source on the EHSI (if available)
— Switch on the GPS system and wait for self-test procedure (automatically available for 5 seconds).

During self-test check that on EHSI:

— Distance datum = 10.0 NM.

— Desired track datum DTK = 150 deg.

— Course pointer shown at 150° on gyro card.

— Deviation bar positioned at 1 dot to the left.

Push the ENT key on GPS Garmin 165 receiver and verify, on the display, the beginning of the phase of satellite acquiring.

At the end of this phase (within 5 minutes), the system will show the page with the present position data.

Press the MSG key and check for the presence of the message ″READY FOR NAVIGATION″.

Press the SET key to select ″sensor status″ page to verify the receiver status (GPS-2D, GPS-3D).

IN FLIGHT

GPS coupled to EFIS or EHSI 74B system (if installed)

CAUTION

When one of F/D navigation modes (NAV-VOR APP-ILS-BC) is in CAP condition and a change of navigation source is carried out on the EHSI, press the relevant button on the F/D mode selector to deselect the mode.
Check or select the new course (or DTK), rearm the mode and carryout a normal capture (if necessary).

— Select GPS as navigation source on the EHSI.

— Select on GPS control panel the correct CDI scale (1 NM).

— Select D ⇒ function or create a route entering waypoints (max. 31)

— Activate the D ⇒ function or the Flight Plan.

— As soon as the course computed by GPS is displayed on the EHSI, select a heading to intercept the track manually or by using the NAV mode on the Flight Director.

To navigate with the NAV mode of Flight Director the following procedure should be followed:

— Select the heading bug on the EHSI to intercept heading and press the NAV key on the F/D control panel.

NOTE

With the NAV mode ”armed”, the helicopter will maintain the heading selected with HDG bug on EHSI, and will capture automatically the DTK when the deviation bar (on EHSI) is near the first dot (0.5 NM).

When the helicopter reaches the first waypoint in a flight plan sequence, steering to next waypoint is automatically provided.

CAUTION

Flying coupled with F/D can result in a 2 NM overshoot if DTK for next leg requires a heading change greater than 90 deg.
In order to prevent the overshoot, the pilot can deselect the NAV mode as soon as the GPS displays the message ″Next dtk xxx°″, fly manually to the new course and select again the NAV mode proceeding with a new intercepting.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

GPS power loss (stand alone system and EFIS system)

In the event of an electrical failure with consequent power loss, the GPS system will automatically revert to its internal battery. The pilot must confirm by pushing any switch within 30 seconds from failure, the up keeping of back-up status. The GPS system will continue to supply the route indications for about 2 hours.
During this time, the GPS system will continue to navigate, track satellites and drive the external CDI/HSI, but not display information on the unit's display until a key is pressed or a knob is turned. The pilot can press the SET key and choose in Battery saver page the desired selection (0 sec. to leave the display on at all time, without battery saver feature).

GPS power loss (EHSI 74B system)

In the event of an electrical failure with consequent power loss, the GPS system will automatically revert to its internal battery. The pilot must confirm by pushing any switch within 30 seconds from failure, the up keeping of back-up status.
The red NAV flag will appear on the EHSI 74B.
During operation coupled with F/D the lateral command bar will also disappear. The F/D automatic navigation pauses and at the same time the roll axis reverts to ATT/HOLD (wing level) mode even if the F/D NAV CAP light stays on.

Deselect NAV mode and proceed in HDG mode or in NAV mode with F/D coupled to a VOR receiver (if available). During this time, the GPS system will continue to navigate and track satellites but not display information on the unit's display until a key is pressed or a knob is turned. The pilot can press the SET key and choose in Battery saver page the desired selection (0 sec. to leave the display on at all time, without battery saver feature).

GPS failure (red ″GPS″ flag on EFIS or ″NAV″ flag on EHSI 74B)

In the event of a failure or total power loss of the GPS system, the red GPS flag will appear on the EFIS EHSI (NAV flag on the EHSI 74B).

During operation coupled with F/D the lateral command bar will also disappear. The F/D automatic navigation pauses and at the same time the roll axis reverts to ATT/HOLD (wing level) mode even if the F/D NAV CAP light (and GPS advisory on EFIS EADI) stays on.

Deselect NAV mode and proceed in HDG mode or in NAV mode with F/D coupled to a VOR receiver (if available).

GPS signal loss

In the event of signal loss, the message ″Poor GPS coverage″ is shown on the receiver display.

If the system is coupled to EHSI, the red GPS flag will appear on the EFIS EHSI (NAV flag on EHSI 74B).

During operation coupled with F/D the lateral command bar will also disappear. The F/D automatic navigation pauses and at the same time the roll axis reverts to ATT/HOLD (wing level) mode even if the F/D NAV CAP light (and GPS advisory on EFIS EADI) stays on.

Deselect NAV mode and proceed in HDG mode or in NAV mode with F/D coupled to a VOR receiver (if available).

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEMS DESCRIPTION

For GPS automatic navigation operation (Flight Plan mode) perform the following procedure:

— On the pilot EHSI select the GPS receiver as navigation source.

— Select on GPS receiver the correct CDI scale (1 NM full scale).

— Create a route by entering two or more waypoints (max. 31).

— Activate the route. As soon as the DTK relative to first leg is displayed on the EHSI, select heading which provides a track intercept and press NAV key on the F/D control panel.

The yellow ALRT (or WPT in the EFIS system) is displayed near the heading lubber line at reaching each enroute waypoint to advise the pilot the approaching of the waypoint.

After passing the waypoint check that the DTK, the CDI arrow and all the other information about the next leg are correctly displayed on the EHSI, and that the Flight Director automatically steers to the correct direction.

NOTE

During the leg change the helicopter is steered to the next leg by the turn anticipation feature. This function performs calculations, to execute the transition between the two legs with a nominal bank angle of 15 - 25 deg. (depending on airspeed).

With F/D in NAV CAP condition the helicopter bank angle is limited to 10 deg. An overshoot of 2NM can result during the leg change if the DTK for next leg requires a heading change greater than 90 deg.

In order to prevent the overshoot, the pilot can deselect the mode as soon as the GPS displays the message ″Next dtk xxx°″, fly manually to the new course and reselect again the NAV mode proceeding with a new intercepting.

Refer to Garmin 165 Pilot’s Guide for more detailed information relative to Turn anticipation feature.

R.A.I. Approval Letter 98/2201/MAE
dated 8 May 1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this Supplement, consult the basic Rotorcraft Flight Manual.

NIGHTSUN SEARCHLIGHT - SX16

The Nightsun Searchlight type SX16 is a high intensity light installed on a gimbal support located on the right hand side of the helicopter nose. Two possible versions are available:
- 109-0811-33 Standard version with white light
- 109-0813-46 IFCO version with NVG filter.

The installation consists of Nightsun Searchlight, NVG filter (for IFCO version only), gimbal support, remote control panel, hardware and cabling.
The xenon arc light may be switched on, focused and directed through the control panel.
If NVG filter is installed may be switched on using the IFCO switch on control panel.

NOTE

The operation with Night Vision Goggles (NVG) equipment has not been approved.

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LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

Activation of Nightsun Searchlight is prohibited in IFR condition.

AIRSPEED LIMITATIONS (KIAS)

Nightsun searchlight configured without NVG filter

Maximum airspeed (V_{NE})	: 140 KIAS

Nightsun searchlight configured with NVG filter

Maximum airspeed with NVG filter in operational position (V_{NE})	: 140 KIAS
Maximum airspeed for searchlight orientation with NVG filter stowed	: 80 KIAS
Maximun airspeed for moving NVG filter	: 60 KIAS

CENTER OF GRAVITY LIMITATIONS

After Nightsun Searchlight installation the new empty weight and C.G. location must be determined.

PLACARDS

Nightsun searchlight configured without NVG filter

CAUTION
DO NOT USE SX-16
SUN LT BELOW
50 FT AGL OR FOG
CONDITIONS.
MONITOR LOADMETER
WHEN USING SX-16
SUN LT.

MAXIMUM AIRSPEED
WITH SX-16 SUN LT
140 KTS IAS.

In clear view of the pilot

Nightsun searchlight configured with NVG filter

SX-16 SEARCHLIGHT

CAUTION
DO NOT USE
BELOW 50 FT AGL
OR FOG CONDITION.
MONITOR LOADMETER
WHEN IN USE.

MAXIMUM AIRSPEED
- NVG FILTER IN OPERATIONAL POSITION 140 KIAS
- NVG FILTER STOWED 80 KIAS
- MOVING NVG FILTER 60 KIAS

In clear view of the pilot

SECTION 2 - NORMAL PROCEDURES

PRE-FLIGHT CHECKS

PILOT'S DAILY PRE-FLIGHT CHECK

(First flight of the day)

AREA N°1 (Helicopter nose)

Nightsun searchlight	:	Security and wiring properly connected. Lens and NVG filter, if installed, for cleanliness.

AREA N°7 (Cabin Interior)

Nightsun searchlight	:	Check for correct operation of searchlight and NVG filter, if installed.

PILOT'S PREFLIGHT CHECK

(Every flight)

Nightsun searchlight	:	Security and wiring properly connected. Lens and NVG filter, if installed, for cleanliness.

ENGINE PRE–START CHECK

SUN LT circuit breaker	:	In.

IN FLIGHT

NOTE

When operating Nightsun Searchlight, magnetic compass indication is not reliable.

NIGHTSUN SEARCHLIGHT OPERATING PROCEDURE

CAUTION

Do not operate Nightsun Searchlight after a generator failure.

IFCO switch on searchlight remote control panel (if NVG filter is installed)	: As desired.
OFF-ON-START switch on searchlight remote control panel	: Hold in START position until the ignition has occured, approximately 5 seconds.

CAUTION

Holding the switch in START position after the ignition may damage the equipment.

Aim and focus	: As desired.

CAUTION

Do not direct the beam towards other aircraft or vehicles to prevent temporary blinding effect, if NVG filter is not installed or not utilized.

NVG FILTER OPERATION (if installed)

CAUTION

Do not move the filter when the searchlight is on to avoid glare to the pilot.

Proceed as follows to set the filter in operational position:

IFCO switch on searchlight remote control panel	: CLOSE. IR wording, on remote control panel display, in sight.

Proceed as follows to stow the filter:

IFCO switch on searchlight remote control panel	: OPEN. IR wording, on remote control panel display, out of sight.

APPROACH AND LANDING

OFF-ON-START switch on search-light remote control panel	: OFF.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

SYSTEM FAILURES

ELECTRICAL POWER FAILURE

Failure of a generator

OFF-ON-START switch on search-light remote control panel : OFF. Refer to paragraph ″FAILURE OF A GENERATOR″ of basic Rotorcraft Flight Manual.

Failure of generator 1 or 2 and generator bus #1 or #2

OFF-ON-START switch on search-light remote control panel : OFF. Refer to paragraph ″FAILURE OF GENERATOR1 OR 2 AND GENERATOR BUS #1 OR #2″ in the basic Rotorcraft Flight Manual.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEM DESCRIPTION

The Nightsun Searchlight SX-16 is provided for enhancing the searchlight capabilities during night or poor visibility conditions. The searchlight contains an air cooled, high intensity xenon arc lamp which start rapidly and can be operated continuosly or may be stopped and restarted as dictated by operational requirements.

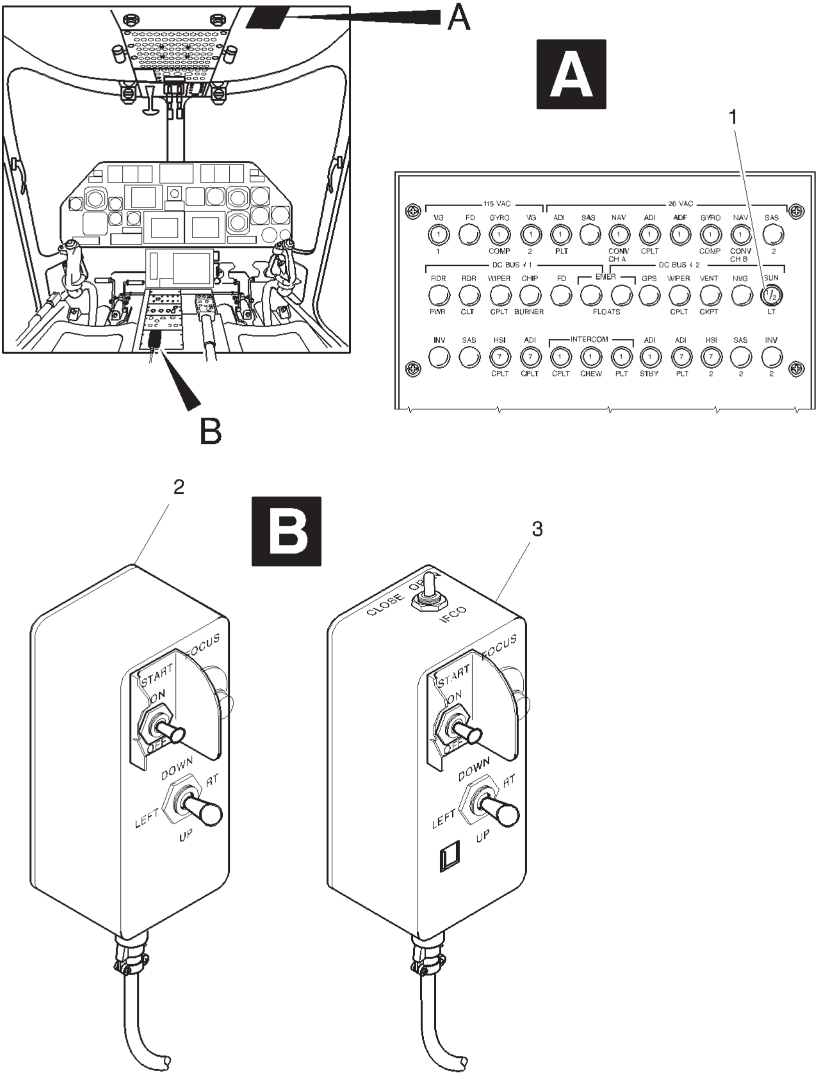
The SX-16 main features are:

- Lamp type	Xenon, 1600 Watts Short arc
- Peak Beam Intensity	80-100 Million Candelpower
- Typical range	2-4 km.

The helicopter is equipped with attachment points on the right side of the nose, at which the supporting frame is attached. The searchlight is mounted on a gimbal with motors and can be rotated in azimuth and elevation. Nightsun searchlight is powered at 28 VDC supplied through a dedicated circuit breaker located on the overhead panel.

The IFCO version of the searchlight is provided with the IR filter that converts the standard light to infrared providing complete compatibility with night vision goggles operations.

The searchlight is controlled by switches housed on a remote control panel placed on the lower side of the central console. If the IFCO version is installed, the switch that control the IR filter is also located on the remote control panel.



1. SUN LT circuit breaker
2. Remote control panel, standard version
3. Remote control panel, IFCO version

ABHD581A

Figure 7-1. Nightsun searchlight control panel.

R.A.I. Approval Letter 98/2201/MAE
dated 8 May 1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SNOW SKID

The snow skid installation P/N 109-0811-99 consists of three skids fixed to the axles of the landing gear wheels.
Each skid is held in position by two bungee cords which allow the skid to tilt during landing to adapt to ground surface.
The attachment of snow skid to main landing gear includes also a mechanical stop which reduces its rotation around the wheel axis.
The installation also includes a locking device to prevent operation of the landing gear control lever and a guard located over the nose wheel centering lock lever to prevent inadvertent operation

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After snow skid installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N.1 (Helicopter Nose)

Snow skid, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the skids, especially to any sheet metal layer separation in the cable attachment area. Check that the skid is not frozen and stuck to the ground.

AREA N°2 (Fuselage - rh side)

Snow skid, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the skids, especially to any sheet metal layer separation in the cable attachment area. Check that the skid is not frozen and stuck to the ground.

AREA N°6 (Fuselage - lh side)

Snow skid, steel cords, bungee cords
and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the skids, especially to any sheet metal layer separation in the cable attachment area. Check that the skid is not frozen and stuck to the ground.

PILOT'S PREFLIGHT CHECK

(Every flight)

Snow skids, steel cords, bungee cords
and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the skids, especially to any sheet metal layer separation in the cable attachment area. Check that the skid is not frozen and stuck to the ground.

CAUTION

Forward touchdown speed must be limited, taking into account the condition of the snow and ground.

OPERATION ON SNOW/ICE COVERED GROUND

Use caution when taxiing on soft and/or uneven snow.

Use caution when starting the engines with the helicopter on snow/ice covered ground due to the possibility of helicopter rotating before the tail rotor reaches effective RPM.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

If any malfunction of the skids occurs in flight, reduce the speed and land as soon as practical.

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 98/2201/MAE
dated 8-5-1998

The information contained herein supplements the information of the basic Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Flight Manual.

GLOBAL POSITION SYSTEM
TRIMBLE 2101

The GPS Trimble 2101 system is supplied in two configurations:

— One identified by P/N 109-0811-53, for use of GPS in VFR conditions as supplemental navigation system.

— One identified by P/N 109-0822-91, allows to operate the GPS as supplemental navigation system in IFR and Basic RNAV mode.

The GPS Trimble 2101 system 109-0811-53 consists of an antenna on the tail fin and of a receiver installed in the cockpit. The function of the system is to acquire signals from the GPS system satellites and to process this information in real-time to obtain the user’s position, velocity and time.
All data are shown on the receiver display.
The navigation data are also available on the EHSI (EFIS or EHSI 74 system) if the relative optional interface is mounted: in this configuration it is also possible to navigate with NAV mode of F/D coupled to GPS receiver.

The GPS Trimble 2101 system P/N 109-0822-91 consists of a GPS Trimble 2101 coupled with EFIS system and Flight Director plus a link Arinc 429 with Air Data System Penny & Gillets D60350MK211212.
The configuration permits IFR and Basic RNAV operation.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

STANDARD CONFIGURATION (P/N 109-0811-53)

The GPS Trimble 2101 system can be used in VFR conditions only, for use as a supplemental navigation system. The GPS system can not be used in the ″DR″ mode.

B-RNAV CONFIGURATION (P/N 109-0822-91)

NOTE

The GPS must not be used for approaches.

NOTE

Holding procedures with GPS must be executed only using HDG mode of the F/D.

This configuration can be used as a supplemental navigation system in IFR and for B-RNAV operation, only if the basic IFR equipment are on board and fully operative.
The GPS system can not be used in the ″DR″ mode (Dead Reckoning) and in any of the fault conditions listed in Chap. 9 (Message) of Trimble Pilot's Guide.

PLACARDS

With GPS Trimble 2101 system in standard configuration P/N 109-0811-53 only.

GPS SYSTEM. USE ONLY DURING VFR OPERATION

In clear view of pilot

With GPS Trimble 2101 system in B-RNAV configuration P/N 109-0822-91 only.

GPS SYSTEM MUST NOT BE USED FOR APPROACHES

In clear view of pilot

SECTION 2 - NORMAL PROCEDURES

CAUTION

The pertinent Trimble GPS 2101 Pilot's Guide must be available on board:

− for installation P/N 109-0811-53:

A. Publication N. 83162 rev. A, with current software release 243, or later approved;

B. Publication N. 82881 rev. E, with current software release 241c, or later approved.

− for installation P/N 109-0822-91:

Publication N. 82881 rev. E, with current software release 241c, or later approved.

SYSTEMS CHECK

Switch the system on and wait for the automatic built-in-test. At the end of the test the GPS displays the message ″Ready for Navigation″.
Press the AUX key twice to select ″sensor status″ page to verify the receiver status (GPS-2D, GPS-3D).
If the yellow ″MSG″ legend flashes on the GPS control panel, press the MSG function key on the GPS control panel, in order to display the message (a list of all the messages and their meaning are issued in Chapter 9 of GPS 2101 Pilot's Guide).
- Check on the GPS control panel the correct CDI scale (1 NM).

GPS coupled with EFIS system (if installed)

Select the GPS on the EHSI through ″N″ button.
The GPS selection is indicated by a blue GPS legend on the left side of EHSI.
All GPS data are also displayed in blue.

GPS coupled with EHSI 74B system (if installed)

Select the GPS on the EHSI through ″CRS″ button.

The GPS selection is indicated by a white LRN1 legend on the lower right corner of the EHSI.
All GPS data are also displayed in white.

NOTE

When GPS is selected on the EHSI 74B as navigation source, the navigation data coming from VOR 2 receiver are automatically available on pilot HSI (stand-by).

IN FLIGHT

If the yellow ″MSG″ legend flashes on the GPS control panel press the MSG function key on the GPS control panel, in order to display the message.
If the message is ″DEAD RECKONING ON″ refer to paragraph ″GPS signal loss″ in Section 3.

GPS system in IFR and B-RNAV configuration

In IFR and B-RNAV operation check the GPS mode, the RAIM availability and the RAIM mode in the sensor status page before activating any Flight Plan.

GPS coupled to EFIS or EHSI 74B system (if installed)

CAUTION

When one of F/D navigation modes (NAV-VOR APP-ILS-BC) is in CAP condition press the relevant button on the F/D mode selector to deselect the mode before any change of navigation source is carried out on the EHSI.
Check or select the new course (or DTK), rearm the mode and carryout a normal capture (if necessary).

— Select GPS as navigation source on the EHSI.

— Select a Direct To a waypoint or create a route entering waypoints (max. 40).

— Activate the D ⇒ function or the Flight Plan.

— As soon as the course computed by GPS is displayed on the EHSI, select a heading to intercept and hold the track manually or use the NAV mode on the Flight Director.

To use the NAV mode of the Flight Director the following procedure should be followed:

— Select the heading bug on the EHSI to intercept the track and press the NAV key on the F/D control panel to arm the NAV mode.

NOTE

With the NAV mode ”armed”, the helicopter will maintain the heading selected with HDG bug on the EHSI, and will capture automatically the DTK when the deviation bar (on EHSI) is near the first dot (0.5 NM).

When the helicopter reaches the first waypoint in a flight plan sequence, steering to the next waypoint is automatically provided.

CAUTION

Flying with the F/D ″coupled″ can result in a 2 NM overshoot if DTK for next leg requires a heading change greater than 90 deg. In order to prevent the overshoot, the pilot can deselect the NAV mode as soon as the GPS displays the message ″TURN TO INTERCEPT NEW COURSE XXX°″, fly manually to the new course and select again the NAV mode to follow the Flight plan.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

GPS failure (red flag ″GPS″ on EHSI or ″NAV″ flag on EHSI 74B)

In the event of a failure, or power loss of the GPS system, the red GPS flag will appear on the EFIS EHSI (NAV flag on EHSI 74B).

During operation coupled with F/D the lateral command bar will also disappear. The F/D automatic navigation pauses and at the same time the roll axis reverts to ATT/HOLD (wing level) mode even if the F/D NAV CAP light (and GPS advisory on EFIS EADI) stays on.

Deselect NAV mode and proceed in HDG mode or in NAV mode with F/D coupled to a VOR receiver (if available).

GPS signal loss

In the event of signal loss, the GPS system goes automatically in Dead Reckoning (DR) mode.

In this case the yellow ″MSG″ legend is displayed on the GPS control panel (and on the EFIS EHSI).

The message ″DEAD RECKONING ON″ appears on the GPS system display if the MSG key on the GPS control panel is pressed.

If the system is coupled to the EFIS or EHSI 74B system, the yellow DR advisory is displayed at right of the lubber line.

If the system is coupled also to the NAV mode function of Flight Director, deselect NAV mode and proceed in HDG mode or in NAV mode using VOR navaid (if available).

ADS failure (IFR and B-RNAV configuration)

In the event of ″PRESSURE ALTITUDE DATA INVALID″ message, press AUX key and check that GPS mode is at least 3D and the RAIM is available in GPS status page in the event of NO RAIM status, the use of GPS for enroute navigation is still approved only if a cross-check with VOR/DME systems is possible.
The use of system with NO RAIM is not allowed in terminal area and during approach; in these conditions the pilot must use the approved systems only (VOR/DME etc.).

RAIM message (IFR and B-RNAV configuration)

In the event of ″RAIM UNAVAILABLE″ message, the use of GPS for enroute navigation is still approved only if a cross-check with VOR/DME system is possible.
The use of system with this message is not allowed in terminal area and during approach; in these conditions the pilot must use the approved systems only (VOR/DME etc.).
The entire list of messages is included in Chapter 9 of Trimble Pilot’s Guide.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEMS DESCRIPTION

For GPS automatic navigation operation perform the following procedure:

— On the pilot EHSI select the GPS receiver as navigation source.

— Create a route by entering two or more waypoints (max. 40).

— Press twice the ″D″ function key to activate the route. As soon as the DTK relative to first leg is displayed on the EHSI, select heading which provides a track intercept and press NAV key on the F/D control panel.

The yellow ALRT (or WPT in the EFIS system) is displayed near the lubber line to get pilot attention when approaching each waypoint.
After passing the waypoint check that the DTK, the CDI arrow and all the other information about the next leg are correctly displayed on the EHSI, and that the Flight Director automatically steers to the correct direction.

NOTE

During the leg change the helicopter is steered to the next leg approximately 0.6 NM before reaching the active WPT.
With F/D in NAV CAP condition the helicopter bank angle is limited to 10 deg. An overshoot of 2NM can result during the leg change if the DTK for next leg requires a heading change greater than 90 deg.
In order to prevent the overshoot, the pilot can deselect the mode as soon as the GPS displays the message ″TURN TO INTERCEPT NEW COURSE XXX°″, fly manually to the new course and reselect again the NAV mode to follow the flight plan.

R.A.I. Approval Letter 98/5857/MAE
dated 26 November 1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SLUMP PROTECTION PADS

The slump protection pads installation P/N 109-0811-73 consists of three aluminium pads fixed to the axes of the landing gear wheels.
Each pad is held in position by two steel cords, one forward and one rearward, and two additional bungee cords which allow the pads to tilt during landing to adapt the ground surface.
The installation also includes a locking device to prevent operation of the landing gear control lever and a guard located over the nose wheel centering lock lever to prevent inadvertent operation.

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SECTION 1 - LIMITATIONS

AIRSPEED LIMITATIONS (IAS)

NOTE

Comply with airspeed limitations with landing gear extended.

CENTER OF GRAVITY LIMITATIONS

After slump protection pads installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°1 (Helicopter nose)

Pad, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the pad, especially to any sheet metal layer separation in the cable attachment area.

AREA N°2 (Fuselage - rh side)

Pad, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the pad, especially to any sheet metal layer separation in the cable attachment area.

AREA N°6 (Fuselage - lh side)

Pad, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the pad, especially to any sheet metal layer separation in the cable attachment area.

PILOT'S PREFLIGHT CHECK

(Every flight)

Pads, steel cords, bungee cords and relative supports : Condition and security.

NOTE

Special attention shall be paid to any permanent deformation to the cable supports (sheet metal parts) on the pad, especially to any sheet metal layer separation in the cable attachment area.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

If any malfunction of the pads or cords should occur in flight, reduce the speed and land as soon as practical.

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 98/6164/MAE
dated 17 December 1998

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EMERGENCY FLOATS

The emergency floats installation P/N 109-0811-42 consists of four float assemblies, two fitted on each side of the fuselage at crew door and baggage compartment stations and an electrically operated inflation system.
Each float assembly comprises a cylindrical float, an inflation system consisting of a cylinder charged with helium, a squib valve, rigid and flexible hoses and of a support assembly in light alloy provided with fittings to attach the assembly to the helicopter fuselage by means of quick release pins.
The electrical system for the inflation system consists of two circuits, each capable of inflating the four floats.
The emergency floats installation must include provision P/N 109-0703-92 for the installation of the prescribed survival equipment.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The emergency floats installation is approved for assistance during emergency ditching. The takeoff after ditching is prohibited.
In the event of unintended inflation, a landing must be accomplished at first suitable location (not on water) and system deflated and stowed or fully removed prior to further flight.
Ferry flight with floats inflated is prohibited.

REQUIRED EQUIPMENT

For over water operation the applicable operating regulations relating to emergency equipment (life jacket, rubber dinghy, signalling equipment etc.) must be complied with.
The following table shows the emergency equipments with reference to the applicable internal arrangement.

Emergency equipments	Survival equipment provision	Internal arrangement
109-0810-52-119	109-0703-92-129	109-0811-70-113
109-0810-52-119	109-0703-92-129	109-0811-70-119
109-0810-52-117	109-0703-92-127	109-0811-82-101
109-0810-52-121	109-0703-92-131	109-0811-70-127
109-0810-52-119	109-0703-92-129	109-0811-70-137
109-0810-52-119	109-0703-92-129	109-0811-70-143
109-0810-52-121	109-0703-92-131	109-0811-70-145
109-0810-52-125	109-0703-92-137	109-0811-82-113
109-0810-52-127	109-0703-92-141	109-0811-82-117/-119/-121
109-0810-52-129	109-0703-92-141	109-0811-59-101/-103
109-0810-52-131	109-0703-92-141	109-0813-59-105

NOTE

Survival equipment installation not compatible with drawer installation P/N 109-0329-21-101 or P/N 109-0709-61-101.

AIRSPEED LIMITATIONS (IAS)

Stowed floats V_{NE}	: No change.
Maximum speed for float inflation	: 80 Kts.
Inflated floats V_{NE}	: 80 Kts.

CENTER OF GRAVITY LIMITATIONS

After emergency floats installation the new empty weight and CG location must be determined.

ALTITUDE LIMITATIONS

Maximum operating altitude for floats inflation and flight with floats inflated	: 3000 ft (915 m) above intended ditching point.

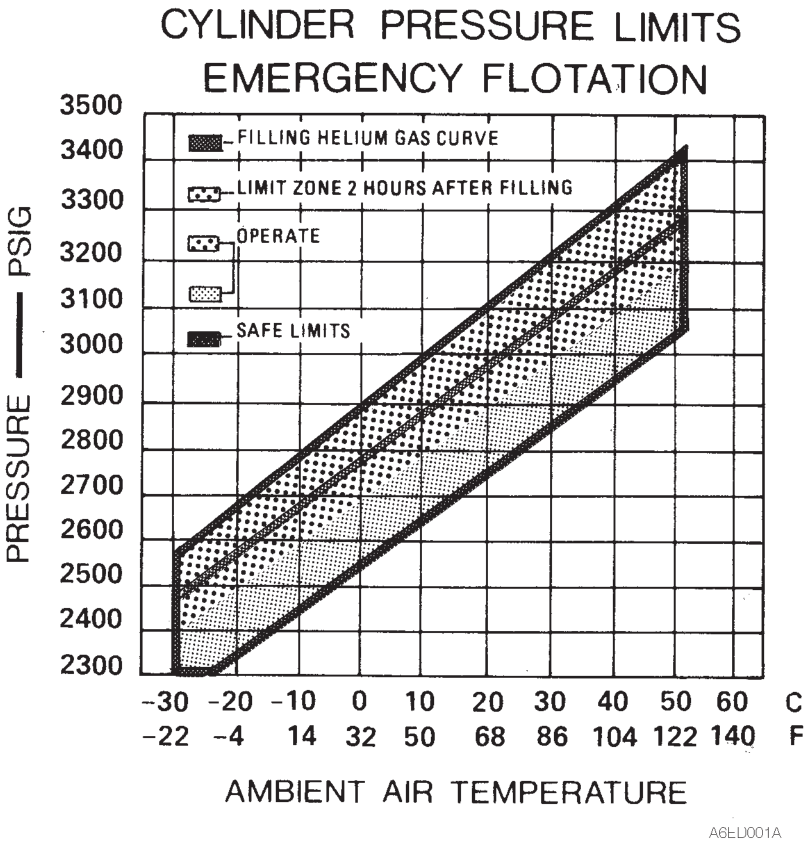
PLACARDS

EMERGENCY FLOATS INFLATED V_{NE}: 80 Kts
DO NOT INFLATE FLOATS ABOVE 3000 Ft GROUND
INFLATION ABOVE 80 Kts IS PROHIBITED

Within the V_{NE} placard

EMERGENCY FLOATS
ARMING ABOVE 80 KNOTS
AND 3000 FT GROUND
PROHIBITED

On the pedestal



Near each cylinder pressure gauge.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

Area N° 2 (Fuselage - rh side)

Cylinder pressure	: Check pressure gauge reading is in accordance with diagram on placard located near gauges.
Electric harness and connector	: Visually check for condition and security.
Floats	: Stowed. Check protective cover for cleanliness and condition.
Float packboards and fittings	: Visually check for security and condition.

Area N° 6 (Fuselage - lh side)

Cylinder pressure	: Check pressure gauge reading is in accordance with diagram on placard located near gauges.
Electric harness and connector	: Visually check for condition and security.
Floats	: Stowed. Check protective cover for cleanliness and condition.
Float packboards and fittings	: Visually check for security and condition.

PILOT'S PREFLIGHT CHECK

(Every flight)

Cylinder pressure	:	Check pressure gauge reading is in accordance with diagram on placard located near gauges.
Electric harness and connector	:	Visually check for condition and security.
Floats	:	Stowed. Check protective cover for cleanliness and condition.
Float packboards and fittings	:	Visually check for security and condition.

ENGINE PRE-START CHECK

NOTE

Pilot shall warn passengers on life jackets and life-raft position inside cabin.

EMER FLOATS circuit breakers	:	In.

SYSTEM CHECK

Before taking off

EMER FLOATS switch	:	Open the guard and set ARMED.
EDU 1	:	FLOATS ARMED caution message displayed.
EMER FLOATS switch	:	Set OFF and close guard, if not required.

NOTE

Do not arm the emergency float system when the takeoff is not carried out over water.

IN FLIGHT

It is allowed to arm the emergency float system only when flying over water below 3000 ft (915 m) AGL and at an airspeed less than 80 Kts IAS.
The emergency float system must be armed during all flight conditions over water at/below 200 ft AGL and at any airspeed less than 60 Kts IAS.
If the system has not been armed since the takeoff:

EMER FLOATS switch	: Open the guard and set ARMED.
EDU 1	: FLOATS ARMED caution message displayed.

APPROACH AND LANDING

Approaching to land by flying over water, if the system has not been armed before:

EMER FLOATS switch	: Open the guard and set ARMED .
EDU 1	: FLOATS ARMED caution message displayed.

SHUTDOWN

After engine shutdown.

EMER FLOATS switch	: OFF and guard closed.
EDU 1	: FLOATS ARMED caution message out.

EMER FLOATS circuit breakers : Out.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
FLOATS ARMED	Emergency float system armed.	Continue flight taking care that the emergency float system is operative.

FLOATS INFLATION

WARNING

Do not inflate floats at more than 3000 ft (915 m) above the intended ditching point and at airspeed higher than 80 Kts IAS.

EMER FLOATS switch : Open the guard and set ARMED (if the system is not yet armed).

EDU 1 : FLOATS ARMED caution message displayed.

FLOATS pushbutton : Depress.

NOTE

The inflation of floats is completed after approximately 5 seconds, however the floats are capable of sustaining the helicopter at the maximum gross weight after approximately the first 3 seconds.

CAUTION

The float inflation control system is entirely electrical. Thus, if the helicopter is operating with the battery as sole source of power due to generators failure, it is necessary to inflate the floats prior to complete battery discharge. (Refer also to paragraph ″Failure of both generators″ in basic Rotorcraft Flight Manual).

DITCHING

Power on procedure

If it is possible to hover, evacuate passengers and then fly to a distance of at least 100 m.
Head helicopter into wind (or against wave direction) when close to the water and proceed as follows:

Pilot's door	: Release.
Collective	: Lower to alight on water.
ENG 1 and 2 MODE switches	: OFF.
Engine power levers	: OFF (full aft).
ENG 1 and 2 FUEL switches	: OFF.
Fuel pumps 1 and 2	: OFF.
BATTery and GEN 1 and 2	: OFF.
INVerters 1 and 2	: OFF.

Evacuate helicopter as soon as rotor stops.

Power off procedure

Collective	: Reduce as required to establish autorotation at 70 to 75 Kts speed.

ENG 1 and 2 MODE switches	:	OFF.
Engine power levers	:	OFF (full aft).
ENG 1 and 2 FUEL switches	:	OFF.
Fuel pumps 1 and 2	:	OFF.
GEN 1 and 2 switches	:	OFF.

At approximately 100 ft to 70 ft above the water, depending on weight, initiate a flare to reduce the rate of descent and the forward speed to minimum; at approximately 10 ft bring the helicopter to a slightly nose-up attitude.
As the helicopter settles, apply collective to cushion the contact with the water.
When the helicopter is in the water, release doors.

NOTE

Jettison of pilot doors, after acting of release handle, is improved by a combined upwards/outwards handle pushing.

Evacuate helicopter as soon as the rotor stops.

CAUTION

People on board of the aircraft shall be trained for the proper handling and use of the emergency equipment associated with the aircraft configuration that are flying.

CAUTION

On calm sea, contact water with forward speed below 20 Kts, 5° - 10° nose-up attitude, zero yaw and minimum possible vertical speed. On rough sea, in addition to the above precautions, it is recommended to contact the water as close to the wave crest as possible or on the back of the wave.
If possible avoid ditching in shallow water areas as, for example, in the vicinity of sand banks.

NOTE

Representative model tests have indicated that sea states 4 condition with height/length ratios of less than 1:12.5 do not cause the helicopter to capsize.

LANDING AFTER AN UNINTENDED INFLATION

In event of an unintended inflation in flight a landing must be accomplished at the first suitable location, definitely not on water.
Do not exceed 80 Kts IAS when flying with the emergency floats inflated.
Approach the landing site, extend the landing gear and bring the helicopter to hover at 10 ft.
Slowly descend vertically to touchdown avoiding to crawl on ground.
Lower the collective and proceed to engine shutdown.

In case that, for particular reasons (i.e. strong wind), the landing manoeuvre with the floats inflated is not advisable, proceed to hover the helicopter close to the ground and ask ground personnel to deflate the floats by cutting them.

SECTION 4 - PERFORMANCE DATA

No change.

R.A.I. Approval Letter 99/1254/MAE
dated 1 April 1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

NOTE

The functional performance of the wire strike protection system has not been demonstrated.

WIRE STRIKE PROTECTION SYSTEM

The wire strike protection system P/N 109-0812-07 is designed to provide a measure of protection for the helicopter in the event of inadvertent flight into horizontally strung mechanical, electrical and communication wires and cables.
The wire strike protection system consists of a roof mounted wedge-type cutter, a bottom fuselage mounted cutter, a nose access door reinforcement, a window strut bumper and windshield mounted deflector to facilitate wire sliding.

NOTE

The lower cutter can be locked in two positions; in forward tilted position for flight and in backward tilted position for nose access door opening or for helicopter towing. The red-banded safety pin must be used to keep the lower cutter deflector in backward tilted position.

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SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After the wire strike protection system installation the new empty weight and CG location must be determined.

SECTION 2 - NORMAL PROCEDURES

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N° 1 (Helicopter nose)

NOTE

Before opening the access door of the nose compartment it is necessary: to remove the quick lock release pin from the lower cutter and tilt the deflector in backward position; to lock the lower cutter deflector in backward tilted position by means of the red banded safety pin; to remove the quick lock release pin which connects the nose access door reinforcement to the window strut bumper; to unlock the upper nose door fasteners and door latches.

Lower cutter : Condition and locked in forward position.

CAUTION

Damage may result from landing gear retraction if the lower cutter is not locked in forward position.

Nose access door reinforcement : Condition and connected to the window strut bumper.

AREA N° 2 (Fuselage - rh side)

Wiper deflectors and window strut bumper	: Condition and security.
Upper cutter	: Condition and security.

AREA N° 6 (Fuselage - lh side)

Upper cutter	: Condition and security.
Wiper deflectors and window strut bumper	: Condition and security.

PILOT'S PREFLIGHT CHECK

(Every flight)

Lower cutter	: Condition and locked in forward position.

CAUTION

Damage may result from landing gear retraction if the lower cutter is not locked in forward position.

Upper cutter, nose access door reinforcement, wiper deflectors and window strut bumper	: Condition.

TAKE-OFF

CAUTION

Excessive nose down attitude during transition to forward flight may cause the lower cutter deflector to hit the ground.

APPROACH AND LANDING

NOTE

Landing on even surfaces it is recommended when the wire strike protection system is installed.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

In the event of a wire strike, land as soon as possible and perform engine shutdown.

Inspect for damage and carry out the required maintenance actions before next flight.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEM DESCRIPTION

WIRE STRIKE PROTECTION SYSTEM

The wire strike protection system is designed to provide a measure of protection in the event of an inadvertent impact against horizontally strung wires.
The wire cutting is obtained by means of the converging upper and lower deflectors using the kinetic energy of the flying helicopter.
The safety envelope provided by the wire stike protection system is substantially reduced when the flight attitude departs from straight and level.

SECTION 8 - HANDLING AND SERVICING

TOWING

Before connecting the towing bar to the nose wheel, remove the quick lock release pin from the lower cutter deflector and tilt the deflector backward.
Lock the deflector in backward position by means of the red-banded safety pin.
When towing operation is completed and after towing bar removal, tilt the deflector in forward position and lock it by means of the quick lock release pin.

R.A.I. Approval Letter 99/1608/MAE
dated 23-4-1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EQUIVALENT CATEGORY ″A″ OPERATIONS TRAINING PROCEDURE

NOTE

This Appendix applies only to helicopters configured with the control panel P/N 109-0900-55 and with the IDS utilizing the EDU software version 007 or later, and DAU software version 006 or later.

Pilots should be properly trained to operate in equivalent Category A according to the AEO and OEI procedures contained in Appendix 12.

For such purpose this Appendix gives the pilot the possibility to train in a way as realistic as possible without causing damage to the helicopter or exceeding the operating limitations.

This has been obtained by reducing the maximum gross weight for training activity as well as the engine limit governing.

The engine limit governing in TRAINING mode has been set not to overcome the engine takeoff rating.

NOTE

The possibility to override the engine governing limits exists, in case of emergency, also in TRAINING mode.

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INTRODUCTION

DESCRIPTION

The TRAINING mode can be selected through the specific switch (figure I-1) placed on the engine control panel.

This is a magnetically latched three position switch.
In normal flight condition the switch must be set to neutral position (central).
When the pilot sets the switch to #1 (#2) position, the engine 1 (2) is set to TRAINING mode and at the same time the engine 2 (1) is governed at 90% N2.

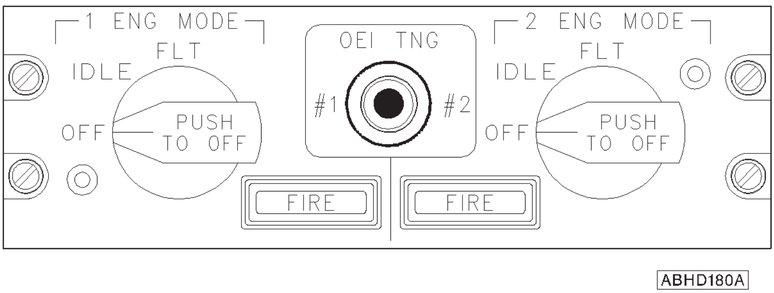

Figure I-1. OEI TRAINING switch.

SECTION 1 - LIMITATIONS

TRAINING PROCEDURE OPERATION

The training activity can be carried out, by day and by night, on clear area, on short field and on ground level helipad.

The night-exercise has to be practised only after having gained a good proficiency during the day.

The area around the helipad should be suitable in the event of an overrun of the helipad.

REQUIRED EQUIPMENT

In addition to the basic equipment for equivalent category A operations the helicopter must be equipped with dual controls P/N 109-0810-01 and with the control panel P/N 109-0900-55.

FLIGHT LIMITATIONS

Training in gusty wind conditions or with poor visibility is prohibited.

Takeoff or landing downwind or with quartering tailwinds is prohibited.

FLIGHT CREW

One instructor pilot and one trainee pilot.

WEIGHT LIMITATIONS

The maximum takeoff and landing weight for training is shown in figure 1-1 for clear area, in figure 1-2 for short field and in figure 1-3 for ground based helipad.

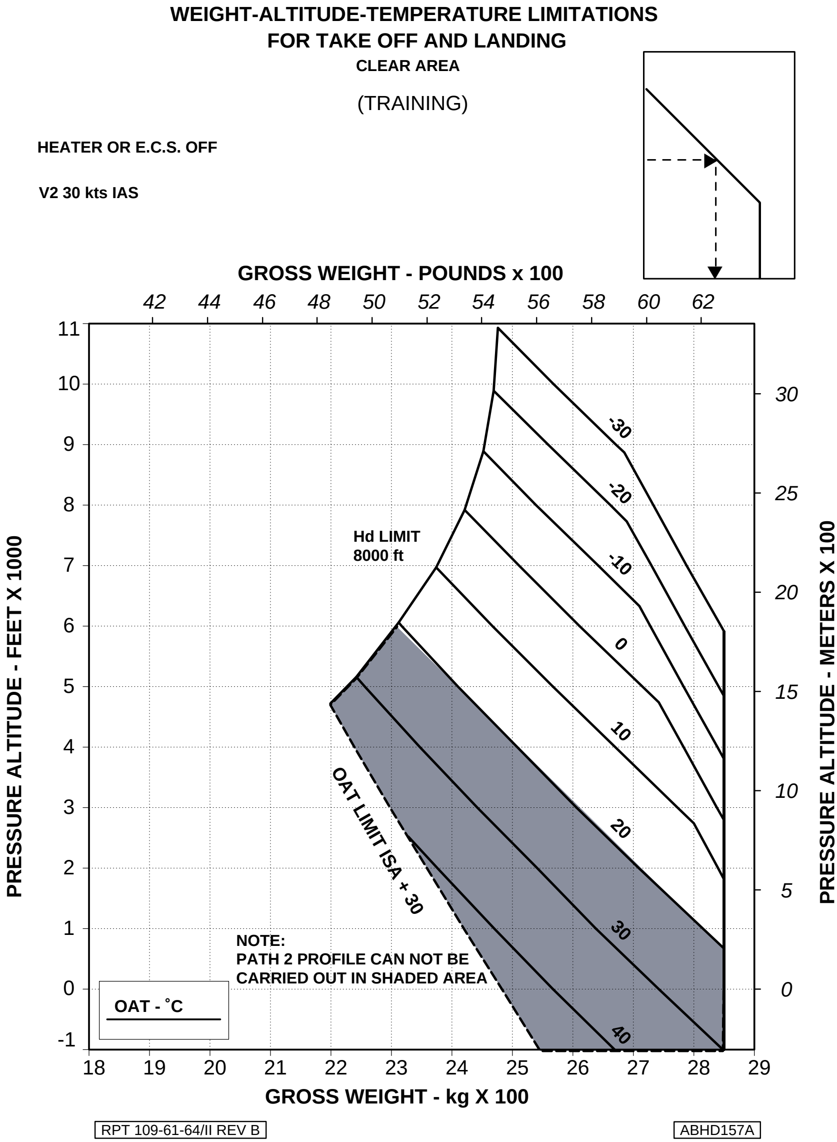

Figure 1-1. Maximum takeoff and landing weight for altitude and temperature (WAT for training on clear area).

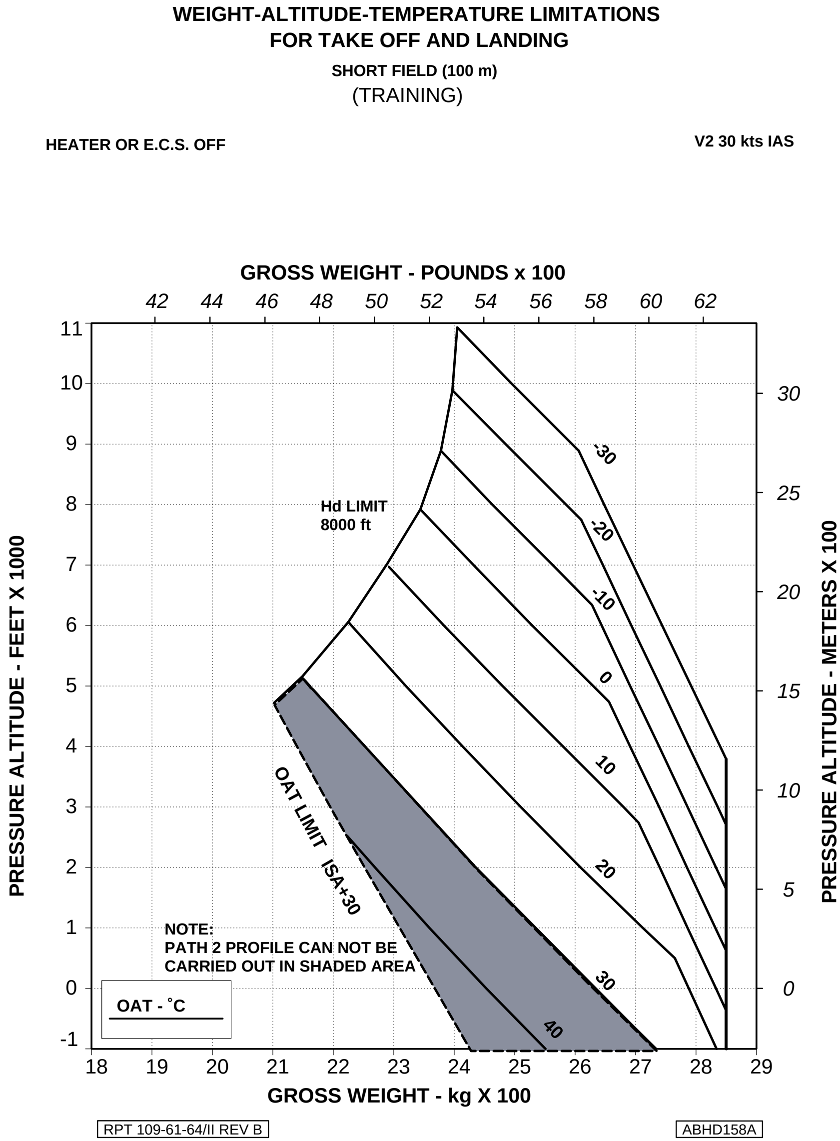

Figure 1-2. Maximum takeoff and landing weight for altitude and temperature (WAT for training on short field).

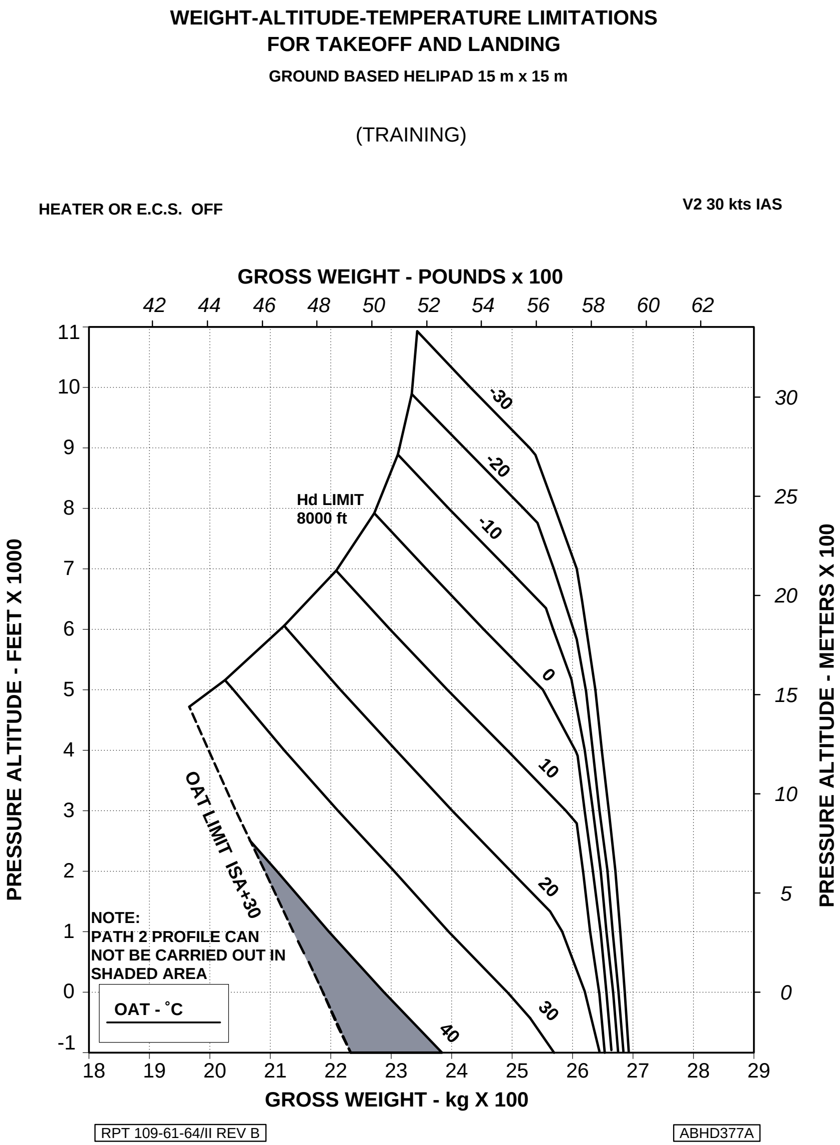

Figure 1-3. Maximum takeoff and landing weight for altitude and temperature (WAT for training on helipad).

POWER PLANT LIMITATIONS

GAS GENERATOR (N1) RPM

Training (OEI)

Continuous operation	: 50 to 97.4%.
Maximum continuous	: 97.4%.
2.5 min range	: 97.4 to 98.7%.
Maximum	: 98.7%.

POWER TURBINE (N2) RPM

Training (OEI)

Minimum	: 90%.
Takeoff and landing (real emergency or training, and/or equivalent cat. ″A″ operations only)	: 90 to 99%.
Continuous operation	: 99 to 101%.
Takeoff and landing	: 101 to 102%.
Maximum	: 102%.

TURBINE OUTLET TEMPERATURE (TOT)

Training (OEI)

Maximum continuous	: 820°C.
2.5 min range	: 820 to 863°C.
Maximum	: 863°C.

TRANSMISSION LIMITATIONS

TORQUE (TRQ%)

Training (OEI)

Maximum continuous : 102%.

2.5 min : 122%.

NOTE

The reduced OEI governing limits for training prevents engine and transmission system overstress and leaves the trainee a safety margin.

INSTRUMENT MARKINGS

ELECTRONIC DISPLAY UNIT FORMATS

The main display formats are:

EDU 1	EDU 2
- START	- MAIN
- CRUISE { AEO / OEI / AUTOROTATION	- AUXILIARY
- OEI TRAINING	

- REVERSIONARY { AEO / OEI / AUTOROTATION

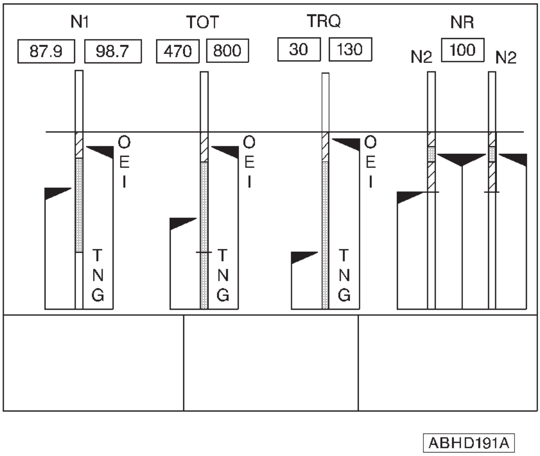

Figure 1-4. EDU 1 - OEI TRAINING mode.

NOTE

The OEI TRAINING mode can be selected by the pilot at any time but it is automatically reverted to CRUISE mode (either OEI or AEO, as applicable) if a single or dual engine out condition occurs or if the ECUs suppress the training authorization.

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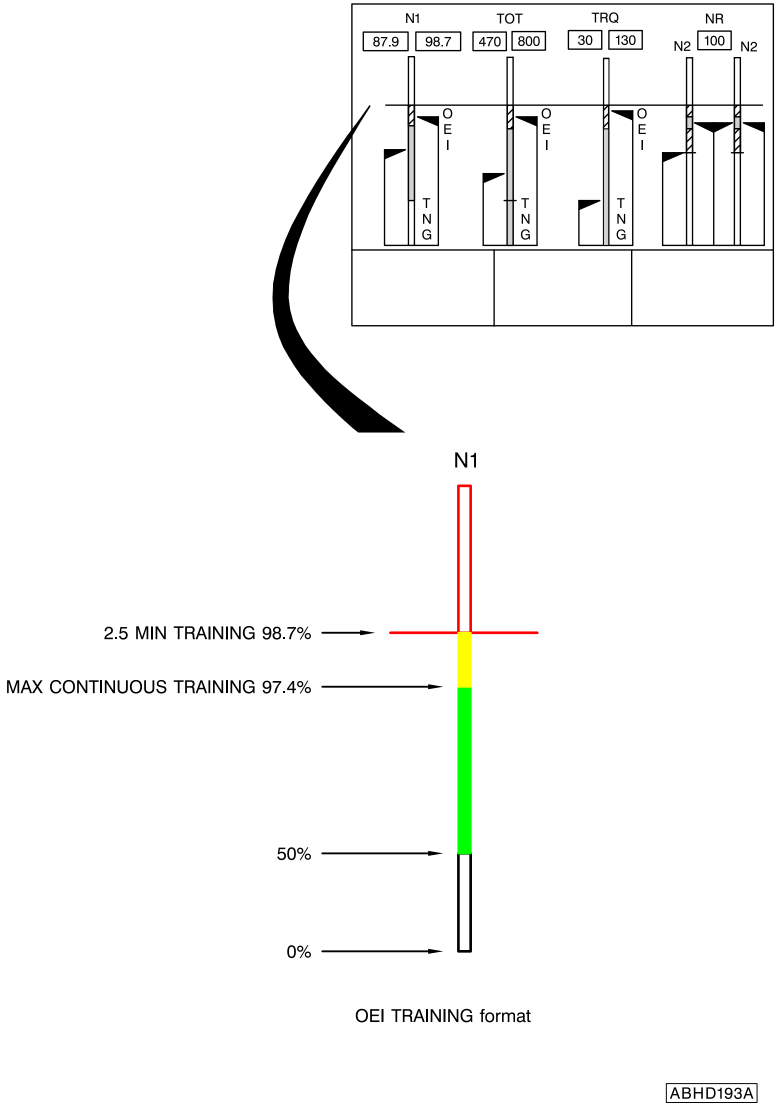

Figure 1-5. EDU 1 - GAS GENERATOR SPEED (OEI).

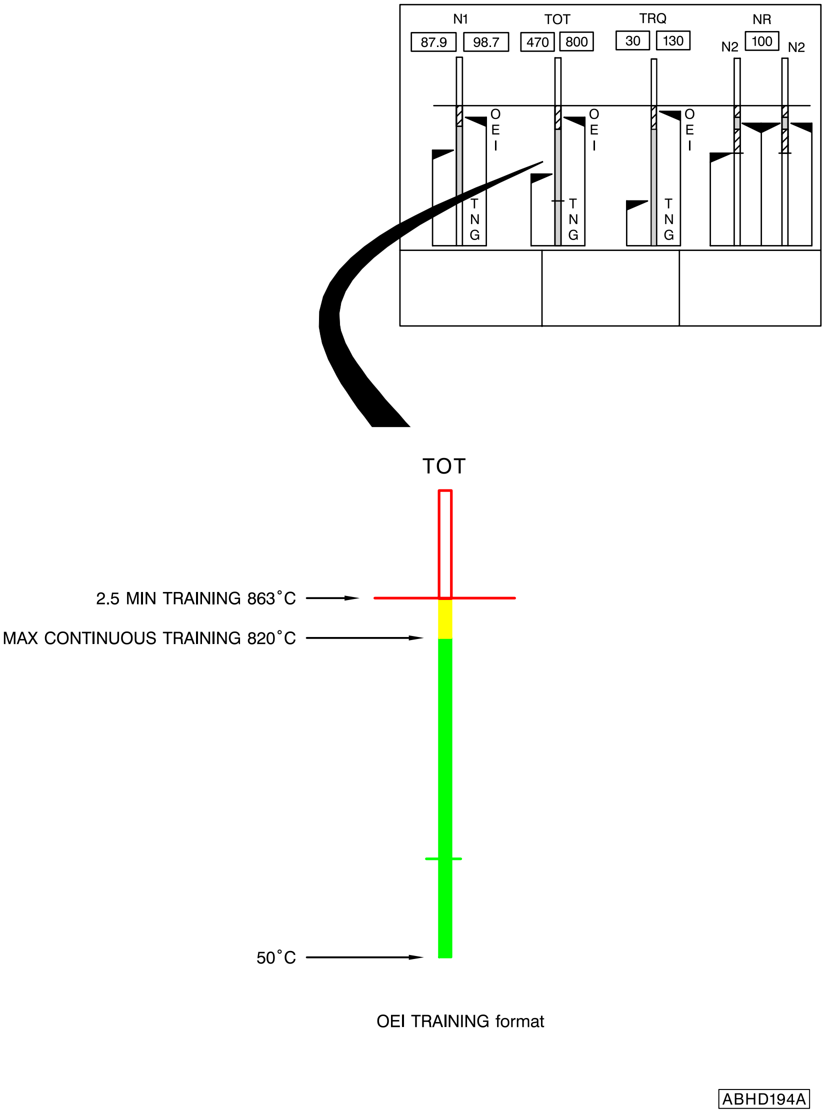

Figure 1-6. EDU 1 - TURBINE OUTLET TEMPERATURE (OEI).

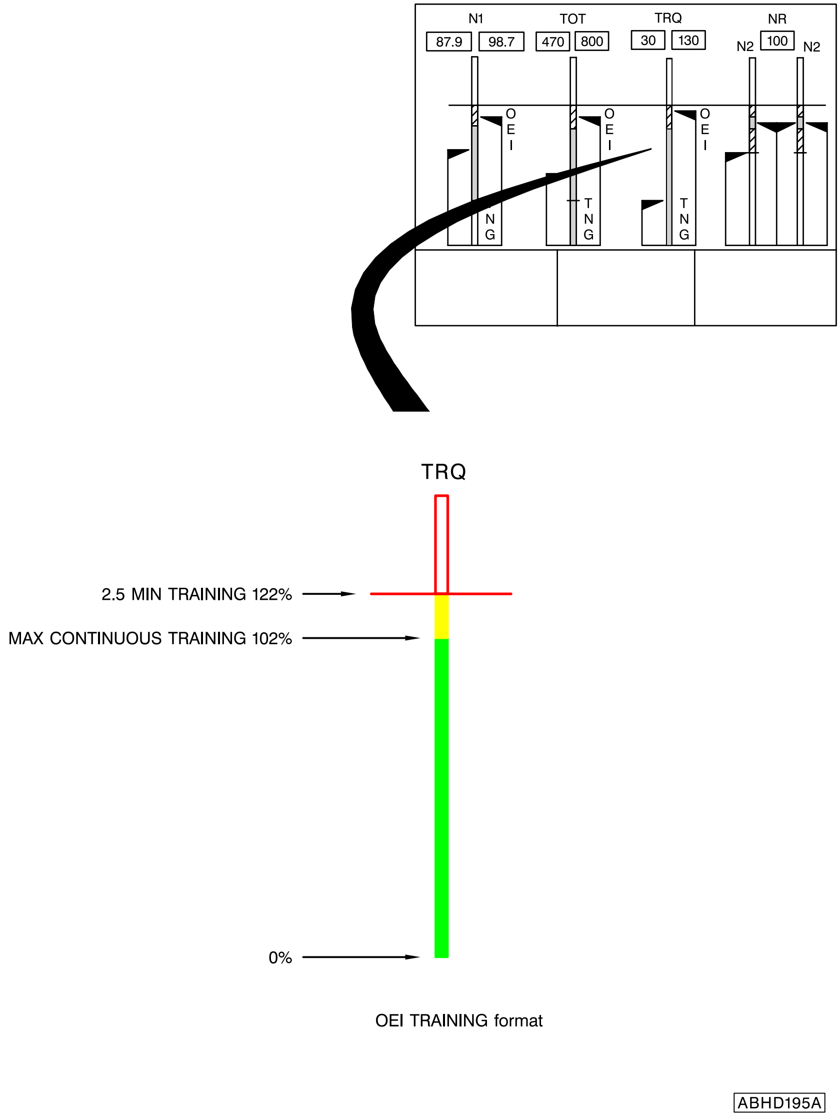

Figure 1-7. EDU 1 - TORQUE (OEI).

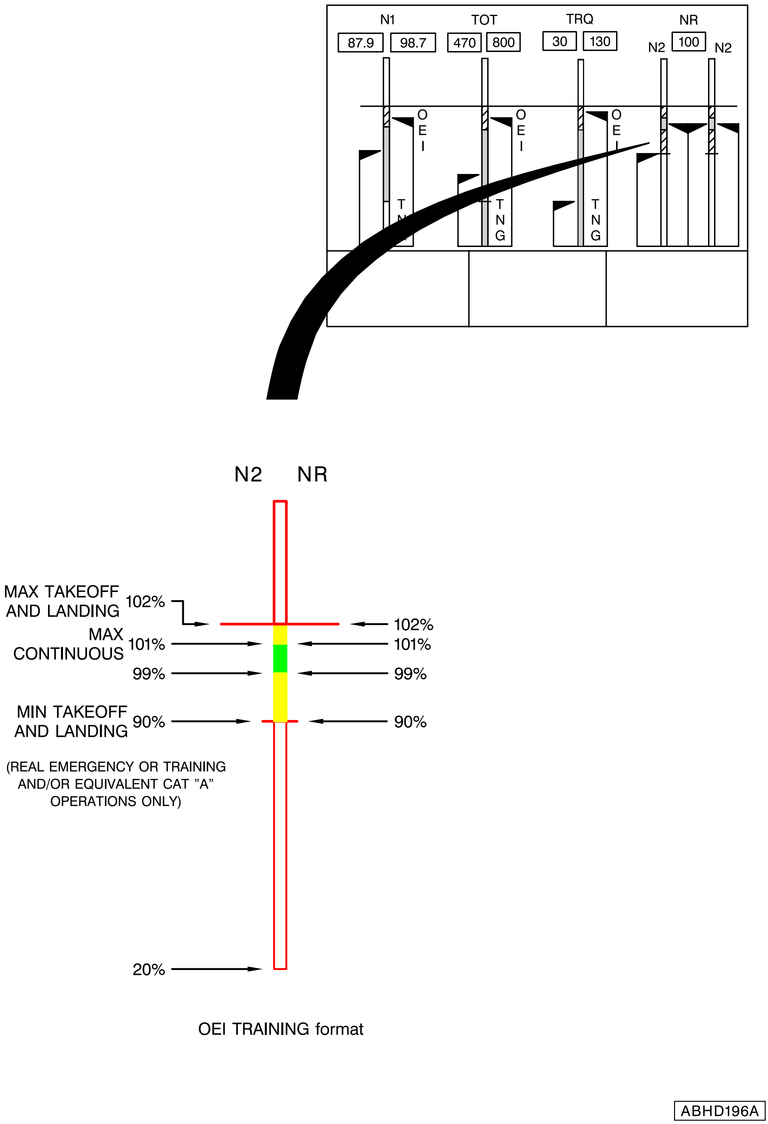

Figure 1-8. EDU 1 - ROTOR/POWER TURBINE SPEED (OEI).

SECTION 2 - NORMAL PROCEDURES

TRAINING PROCEDURES

Perform the training activity strictly following the same profiles presented in Section 2 and in Section 3 of Appendix 12.

Training procedure can be considered completed at the end of flight path 1 (upon reaching a height of 200 ft above the starting point).

Simulate one engine failure by setting the OEI TNG switch to #1 (#2). In this way the engine #1 (#2) will be set to TRAINING mode (see Section 1 for the specific limitations) and at the same time the engine #2 (#1) will be slowed down to 90% N2.

NOTE

It is prohibited to drop voluntarily the rotor speed (NR) below 90%.

Neverthelesss the ECU automatically aborts the TRAINING mode if the rotor speed (NR) accidentally drops at or below 87%.

NOTE

Every five (5) simulated engine failures utilize the other engine in order to consume the life of both engines equally.

Exit the TRAINING mode by setting the OEI TNG switch to the central position.

NOTE

The use of the limit override pushbutton to exit TRAINING mode is intended for emergency only.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

EMERGENCY DURING TRAINING

At any time during the drill the TRAINING mode can be aborted, automatically by the electronic engine control or voluntarily by the pilots in case of critical situations.
The electronic engine control automatically aborts or does not permit the TRAINING mode selection if any of the following occur:

— the rotor speed (NR) drops below 87% (i.e. as a consequence of a collective pull-up following a wrong manoeuvre);

— a critical failure (i.e. engine out) to either engine;

— an engine flameout or reduction at IDLE setting;

— an ECU failure or any discrete non-critical faults;

— a reversion to MANUAL mode;

— the loss of ARINC cross-talk between the two ECUs;

— any engine mode switch not at FLIGHT position;

— limit override pushbutton activated (Refer also to the pertinent paragraph in Section 3 of the basic Rotorcraft Flight Manual).

NOTE

When the TRAINING mode is aborted, the OEI TNG switch automatically trips to the neutral position (centered).

Both the instructor and the trainee pilot can, at any time, abort the TRAINING mode whether respectively by resetting the OEI TNG switch to the central position or, in an extreme case, by operating the LIM OVRD pushbutton; in the second case the action will also override the engine limit governing (refer also to Section 3 of the basic RFM).

In TRAINING mode the engine limit governing is set not to overcome the following limits:

Torque OEI	122%
TOT	863°C
N1	98.7%

Once the TRAINING mode is aborted, both engines resume running matched at the pilot requested power and the EDU 1 automatically reverts to CRUISE AEO mode.
If the LIM OVRD pushbutton has been operated, pay attention to manoeuvre the collective since the engines are no more protected from exceeding limits. The pilot shall press the LIM OVRD pushbutton a second time to restore the normal engine limit governing.

SECTION 4 - PERFORMANCE

The distance information in Appendix 12 is also valid for training.

TAKEOFF FLIGHT PATH 1

MEAN HEIGHT GAINED IN 30 M HORIZONTAL DISTANCE

The mean height gained in 30 m of horizontal distance travelled during an OEI climb at V_2 after takeoff in training is shown in figures 4-1 thru 4-5 for various altitudes, temperatures, weights and headwind components.

NOTE

The headwind component should be obtained from the wind component chart shown in figure 4-1 of Appendix 12.

These charts apply from the end of takeoff distance to a height of 200 ft above the starting point.

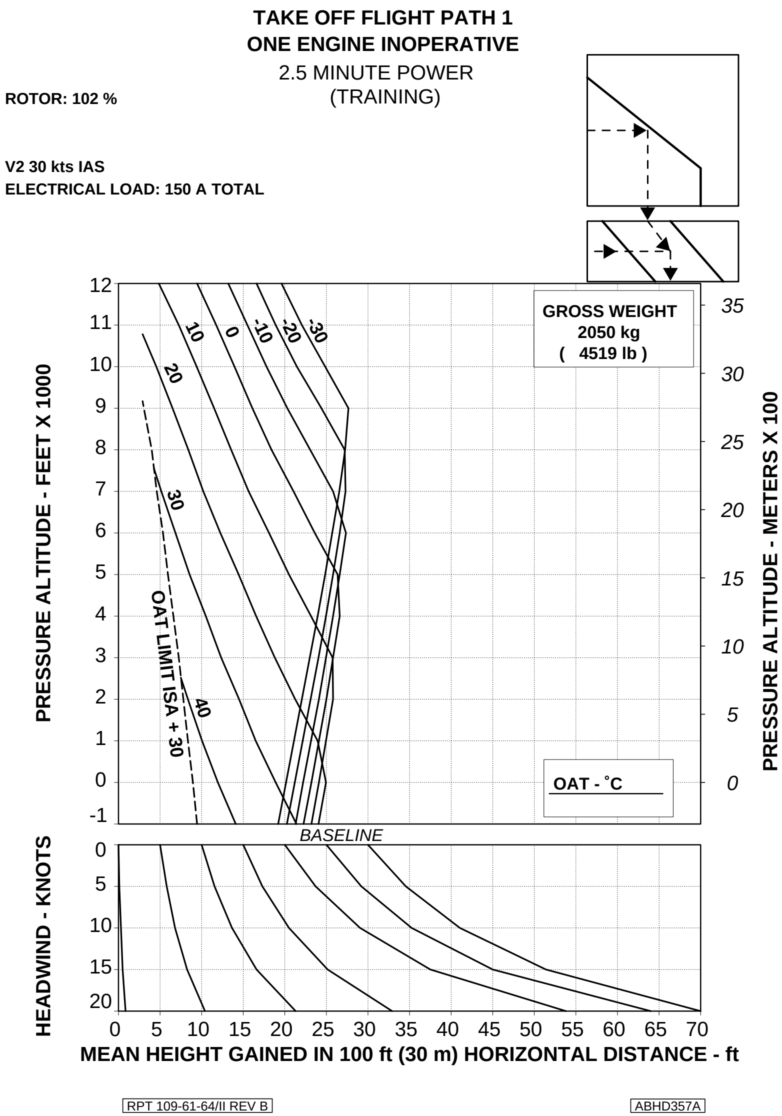

Figure 4-1. Takeoff flight path 1 - 2050 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

V2 30 kts IAS

ELECTRICAL LOAD: 150 A TOTAL

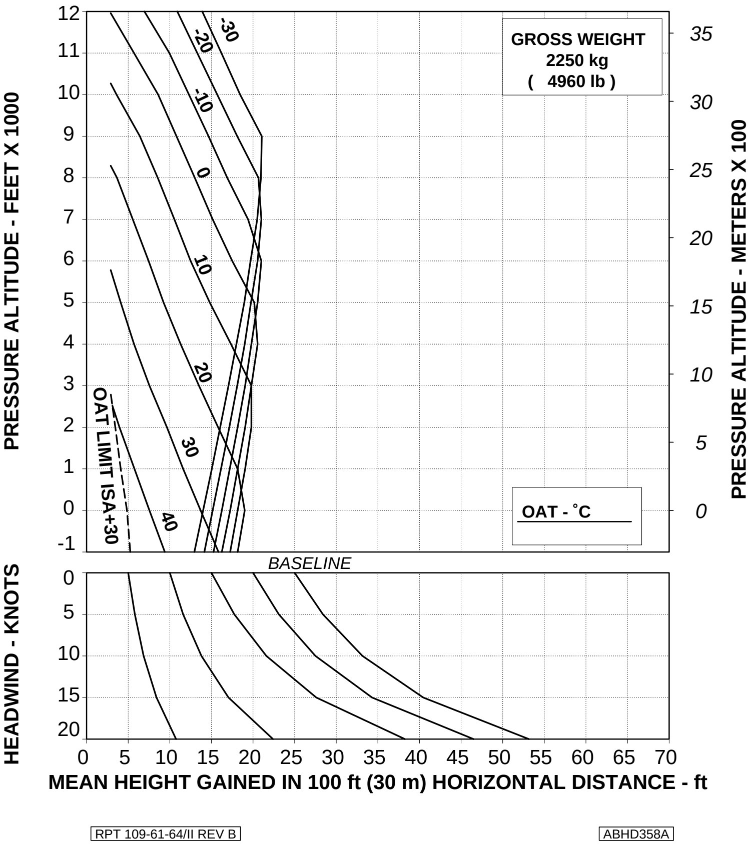

Figure 4-2. Takeoff flight path 1 - 2250 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

V2 30 kts IAS

ROTOR: 102 %

ELECTRICAL LOAD: 150 A TOTAL

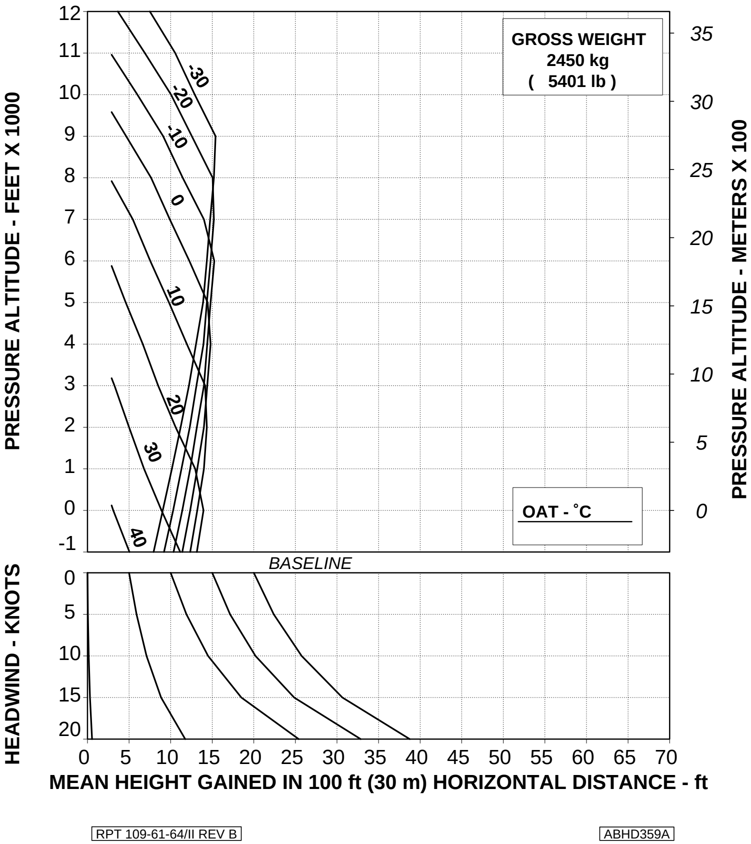

Figure 4-3. Takeoff flight path 1 - 2450 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

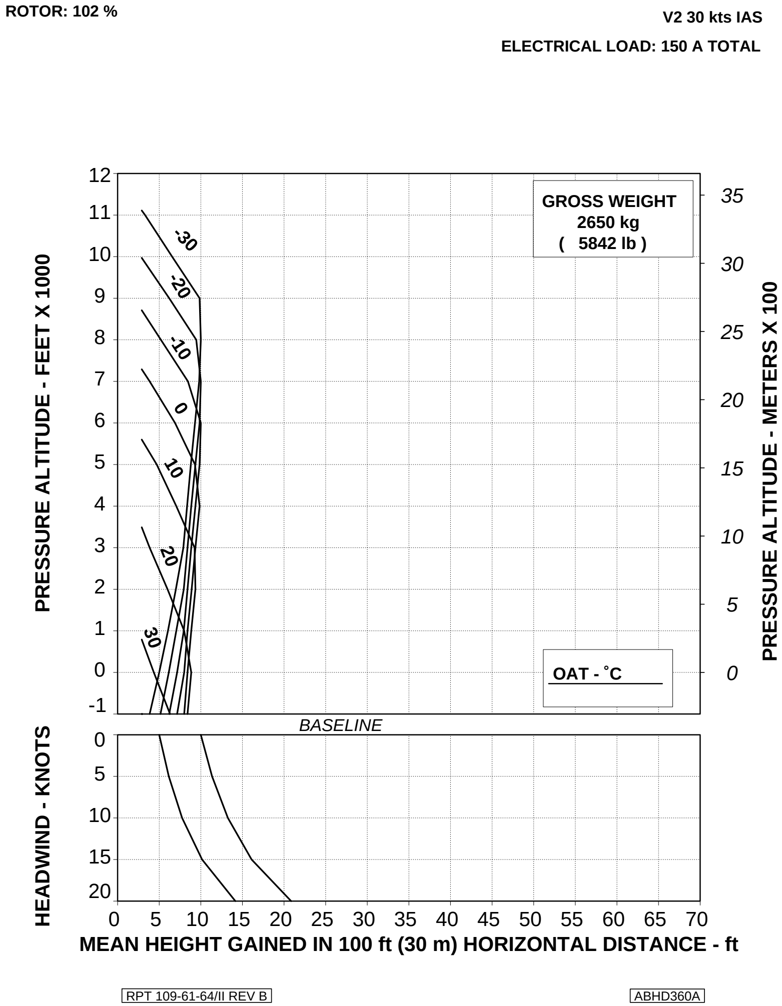

Figure 4-4. Takeoff flight path 1 - 2650 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

V2 30 kts IAS

ELECTRICAL LOAD: 150 A TOTAL

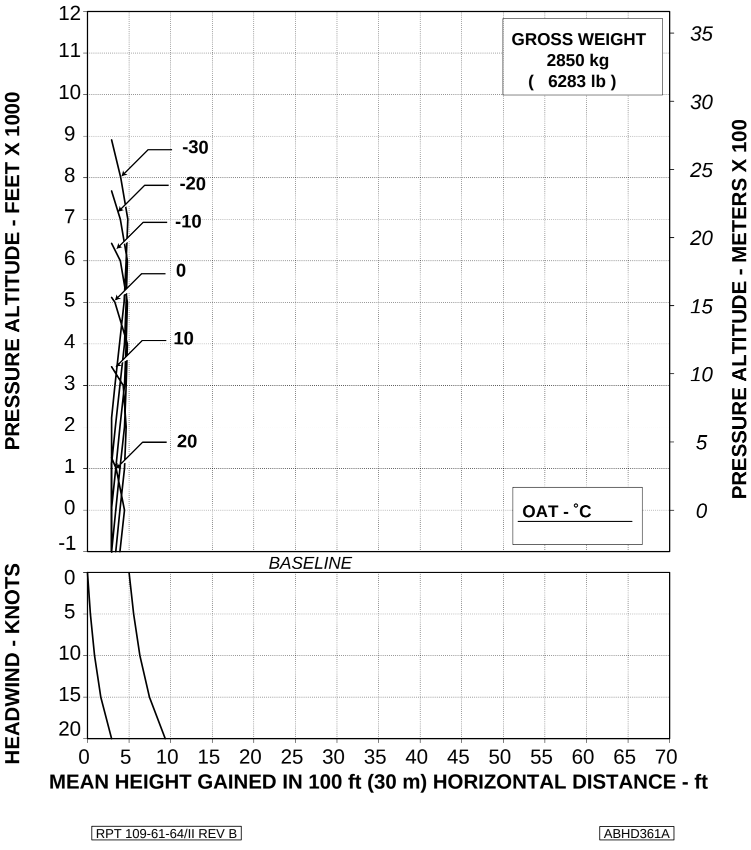

Figure 4-5. Takeoff flight path 1 - 2850 kg.

TAKEOFF FLIGHT PATH 2

MEAN HEIGHT GAINED IN 30 M HORIZONTAL DISTANCE

The mean height gained in 30 m of horizontal distance travelled during an OEI climb at V_Y after takeoff in training is shown in figures 4-6 thru 4-10 for various altitudes, temperatures, weights and headwind components.

NOTE

The headwind component should be obtained from the wind component chart shown in figure 4-1 of Appendix 12.

These charts apply from 200 ft to 1000 ft above the starting point.

WARNING

In some ambient conditions a climb in flight path 2 may not be achieved.
Refer to WAT limitations for training.

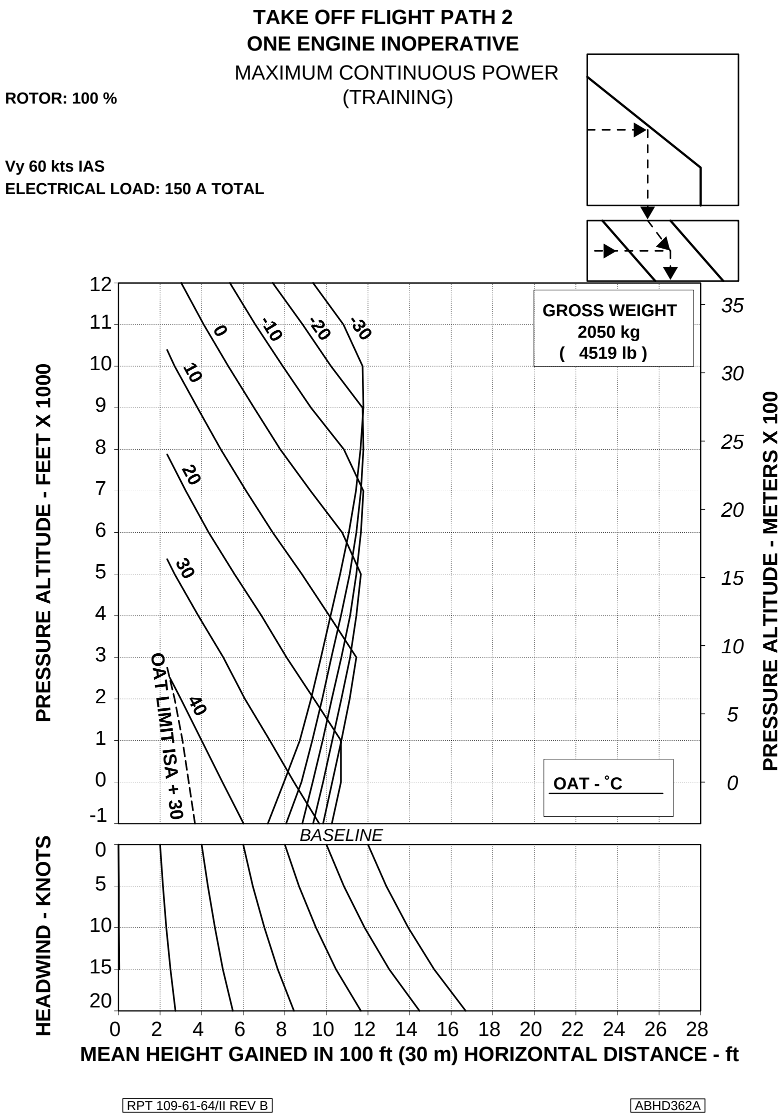

Figure 4-6. Takeoff flight path 2 - 2050 kg.

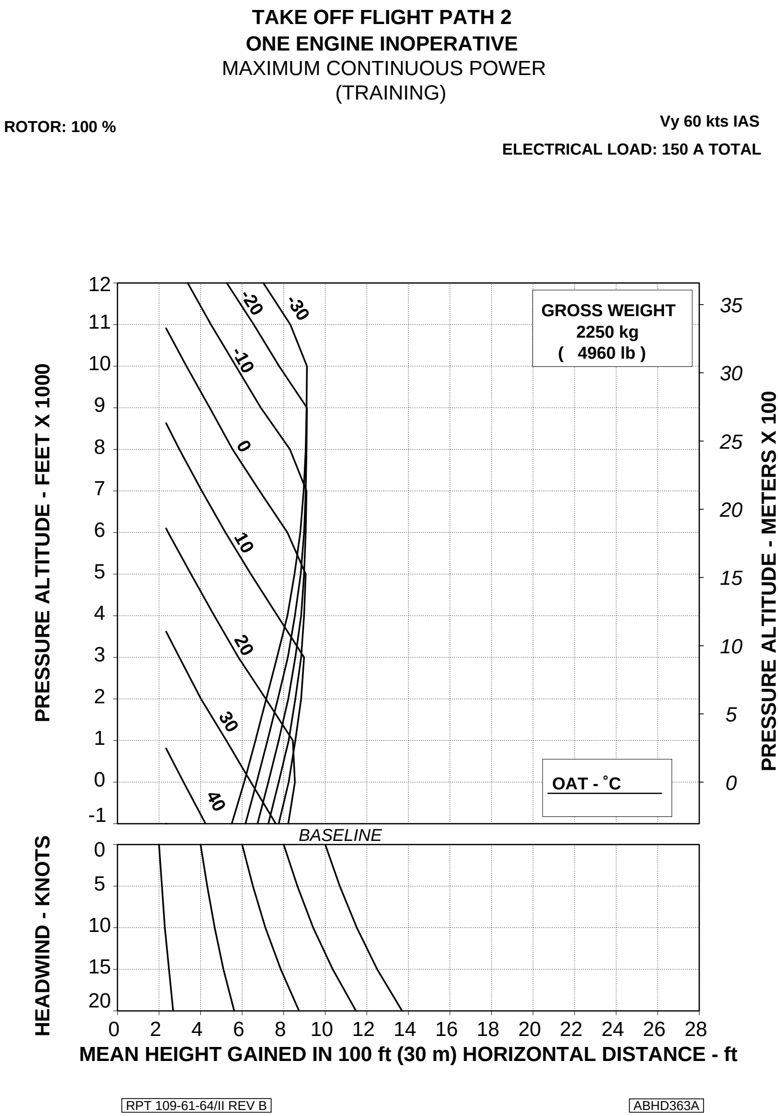

Figure 4-7. Takeoff flight path 2 - 2250 kg.

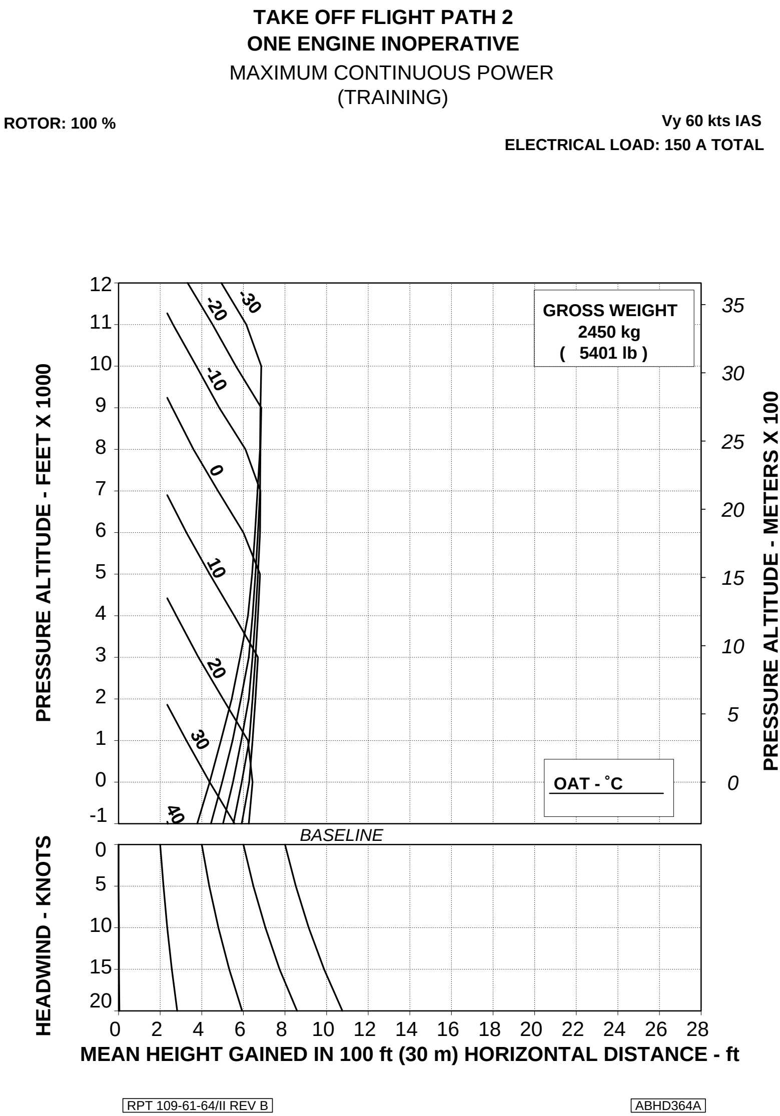

Figure 4-8. Takeoff flight path 2 - 2450 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**

ELECTRICAL LOAD: 150 A TOTAL

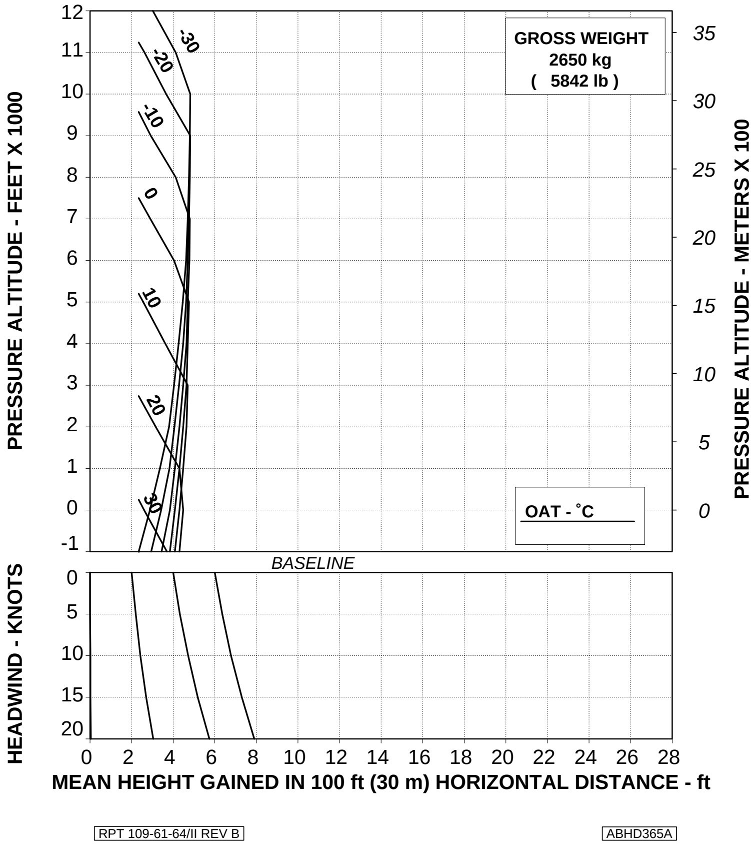

Figure 4-9. Takeoff flight path 2 - 2650 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 %

Vy 60 kts IAS

ELECTRICAL LOAD: 150 A TOTAL

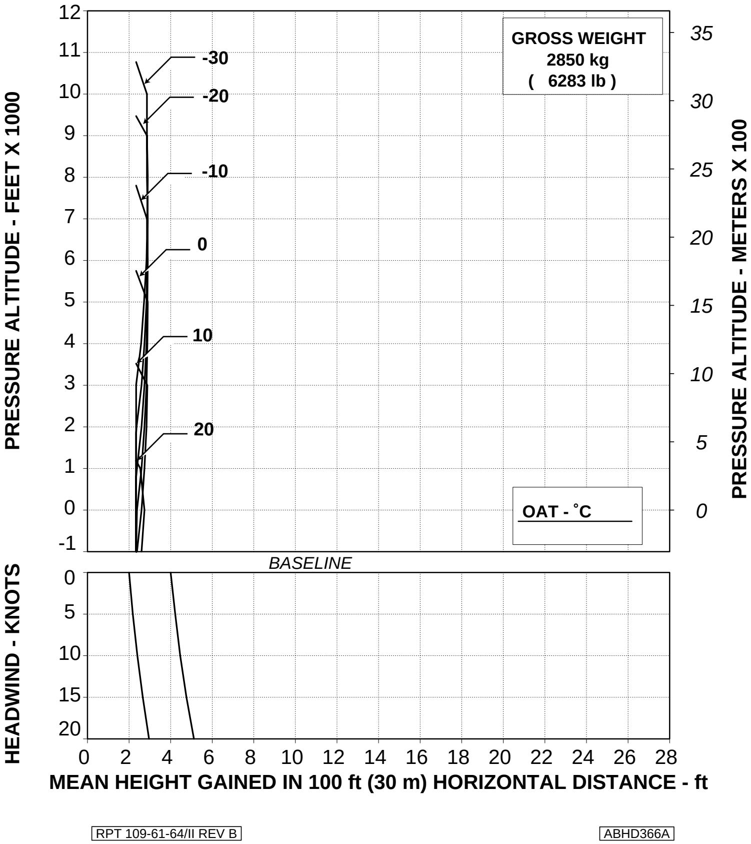

Figure 4-10. Takeoff flight path 2 - 2850 kg.

R.A.I. Approved

E.N.A.C. Approval Letter 99/2951/MAE
dated 30 July 1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EXTERNAL HOIST

The external hoist P/N 109-0812-31 enables cargo and emergency rescue operation in areas where landing cannot be accomplished. It consists of an electric hoist motor and winch assembly, mounting frame, an electronic control system that allows the pilot or the crew member to operate the hoist, a cable payout display, electrical components, wiring and relative hardware.

NOTE

For hoist operation the helicopter must be fitted with the step bar protection on the lower right side of the helicopter fuselage.

The winch unit contains 75 usable meters (245 ft) of hoist cable.
Cargo hoisting and lowering can be controlled by the crew member through the remote control thumbwheel providing variable cable speeds on command, or by the pilot through the hoist control switch on the cyclic stick at a fixed cable speed.
A load selection switch is located on the remote control on the right side of the direction/speed control.
Putting the load selection switch in 113 kg position, loads up to a maximum of 113 kg (250 lb) may be lifted or lowered at speeds up to 60 m per minute (200 ft/min). When the switch is placed in 272 kg position, loads up to 272 kg (600 lb) may be lifted or lowered at speeds up to 30 m per minute (100 ft/min).

The hoist system is provided with a cable foul protection system that stops the hoist if the cable is not properly stowed on the drum and activates a HOIST CABLE LKD caution message on EDU 1.

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TYPE OF OPERATION

Hoist operation is approved in VFR conditions with ground visual contact. Hoist operation is approved also with the right passenger door removed. Normal helicopter operation is approved, with the hoist installed, providing the hoist is not used and the hoist electrical system is deactivated.
IFR operation, with the external hoist installed and deactivated, is approved with doors closed.

REQUIRED EQUIPMENT

In addition to the basic required equipment the helicopter must be fitted with:
- electrical system with Emergency Bus #1 P/N 109-0812-04
- step bar protection P/N 109-0715-46
- IDS EDU software release 009 and DAU software release 008.

Operations at gross weight greater than 2850 kg up to 3000 kg are permitted provided that the following assembly is installed:

— Tail rotor installation P/N 109-0136-02-103.

For operations and performance at gross weight greater than 2850 kg up to 3000 kg refer to Appendix 45.

FLIGHT CREW

One pilot and one operator.
The operator must be restrained by a safety and shoulder harness equipped with manual cable cutter during all phases of hoist operation. The operator shall wear protective gloves for guiding cable during operation. The hoist operator shall be familiar with hoist operating procedures and limitations.

AIRSPEED LIMITATIONS (IAS)

MAXIMUM OPERATING LIMIT SPEED (V_{MO})

Hoist operation (load raising or lowering) is permitted with aircraft in stationary hover only.
Aircraft horizontal translation with external hoisted load outside the aircraft cabin is approved in any azimuth direction limited to 20 knots relative airspeed up to 8000 ft Hd.

CAUTION

Airspeed with external load is limited by controllability. Caution should be exercised when carrying an external load as the handling characteristics may be affected by the size, weight and shape of the load.

The vertical speed shall be limited to ± 200 fpm.
At altitudes higher than 8000 ft, flight with external hoisted load outside the aircraft cabin is prohibited.

NOTE

Flight with external hoisted load shall be limited to reach a place suitable for load recovering.

WARNING

Hoisting or lowering an empty litter in open position is prohibited.

HOIST LIMITATIONS

Maximum hoist load : 272 kg (600 lb).

Maximum available hoist cable length : 75 m (245 ft).

Hoist allowable load is variable with helicopter gross weight in order not to exceed lateral CG limits. Refer to figure 1-2 for an example of hoist load configuration.

NOTE

The hoist-loading chart is based on the most adverse loading combinations of pilot, hoist operator and attendant. For further cargo load configurations, refer to figures 1-1 to 1-6 for CG computation.

CENTER OF GRAVITY LIMITATIONS

After hoist installation the new empty weight and CG location must be determined.
See figure 1-3 for longitudinal CG limits and figure 1-4 for lateral CG limits.

For gross weight greater than 2850 kg (6283 lb), see figure 1-5 for longitudinal CG limits and 1-6 for lateral CG limits.

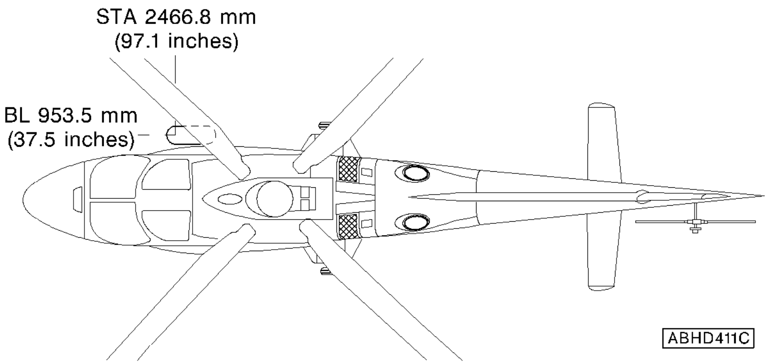
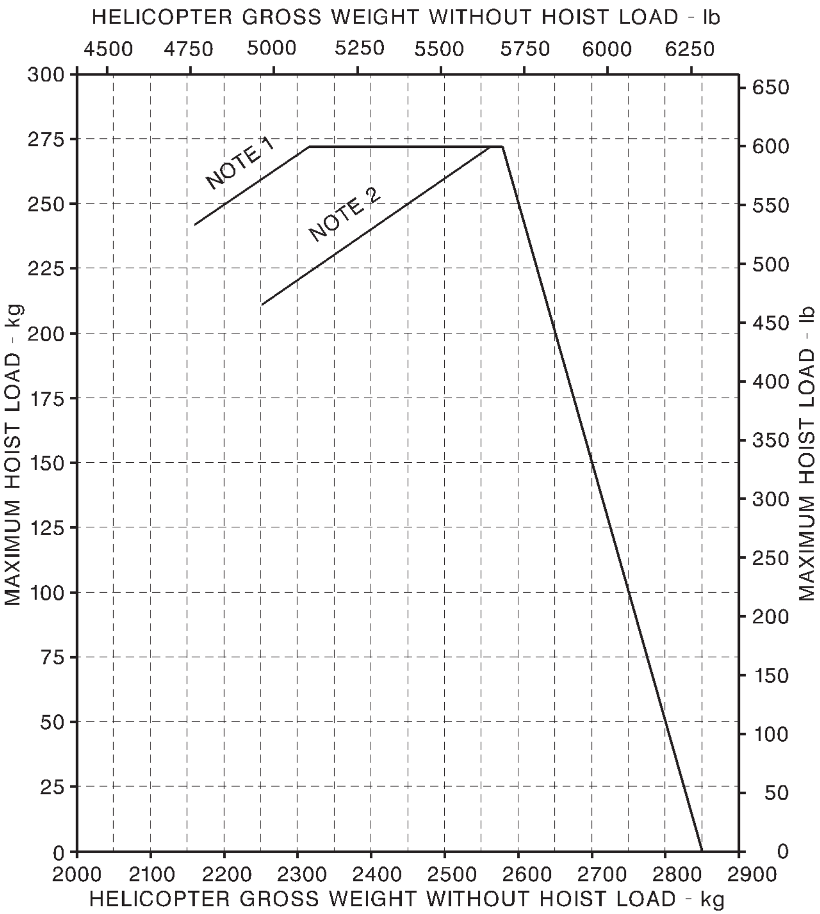


Figure 1-1. Hoist station and butt line.



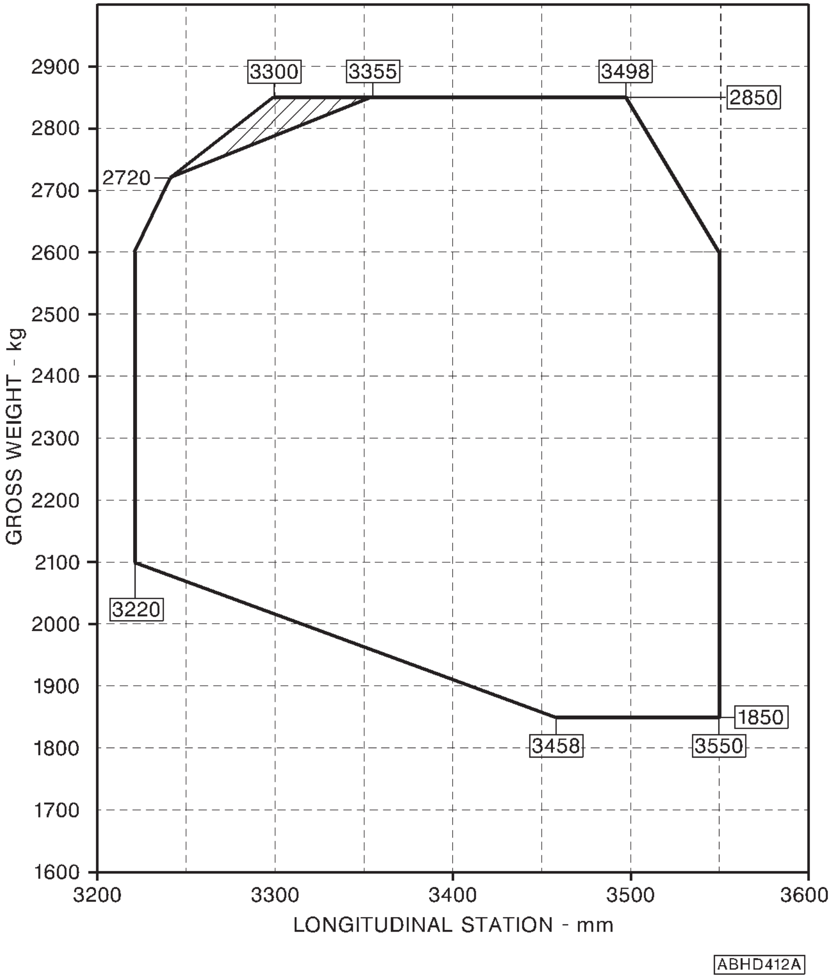
KEY :

NOTE 1 Pilot (90 kg) plus an operator at BL 625 mm

NOTE 2 Pilot (90 kg) plus an operator at BL 625 mm and one at BL 430 mm

ABED225A

Figure 1-2. Hoist loading chart (examples of weight layout).



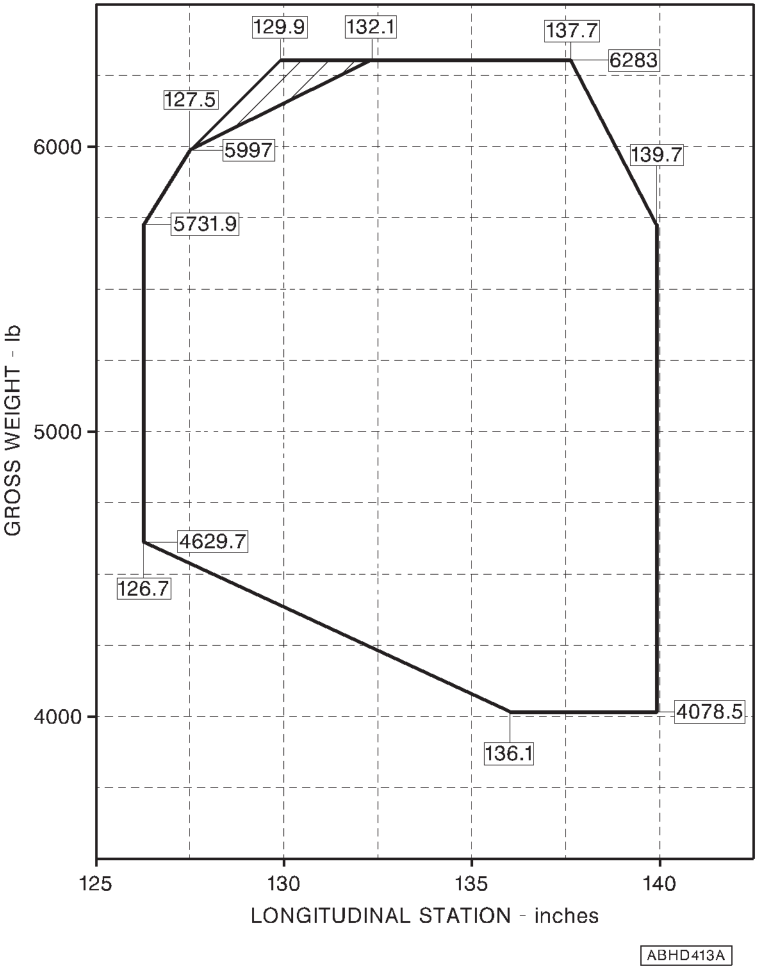
NOTE

Longitudinal station ″0″ is 1835 mm forward of the front jack point.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-3 (sheet 1 of 2). Longitudinal CG limits (Metric).



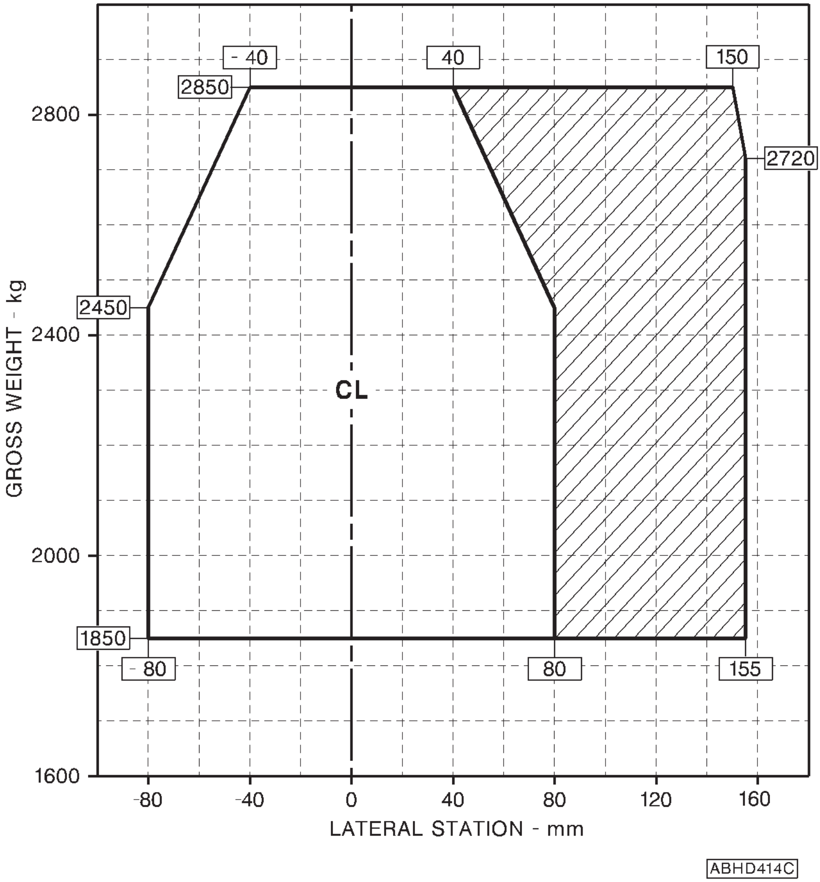
NOTE

Longitudinal station ″0″ is 72.2 in. forward of the front jack point.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-3 (sheet 2 of 2). Longitudinal CG limits (English).



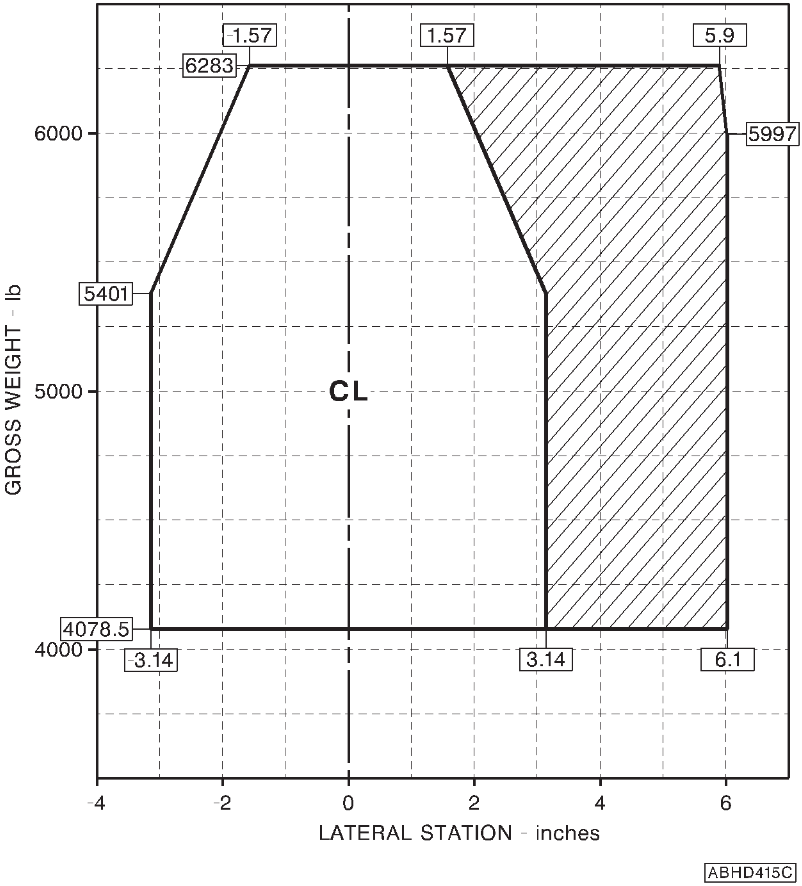
NOTE

The lateral station ″0″ is 450 mm inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-4 (sheet 1 of 2). Lateral CG limits (Metric).



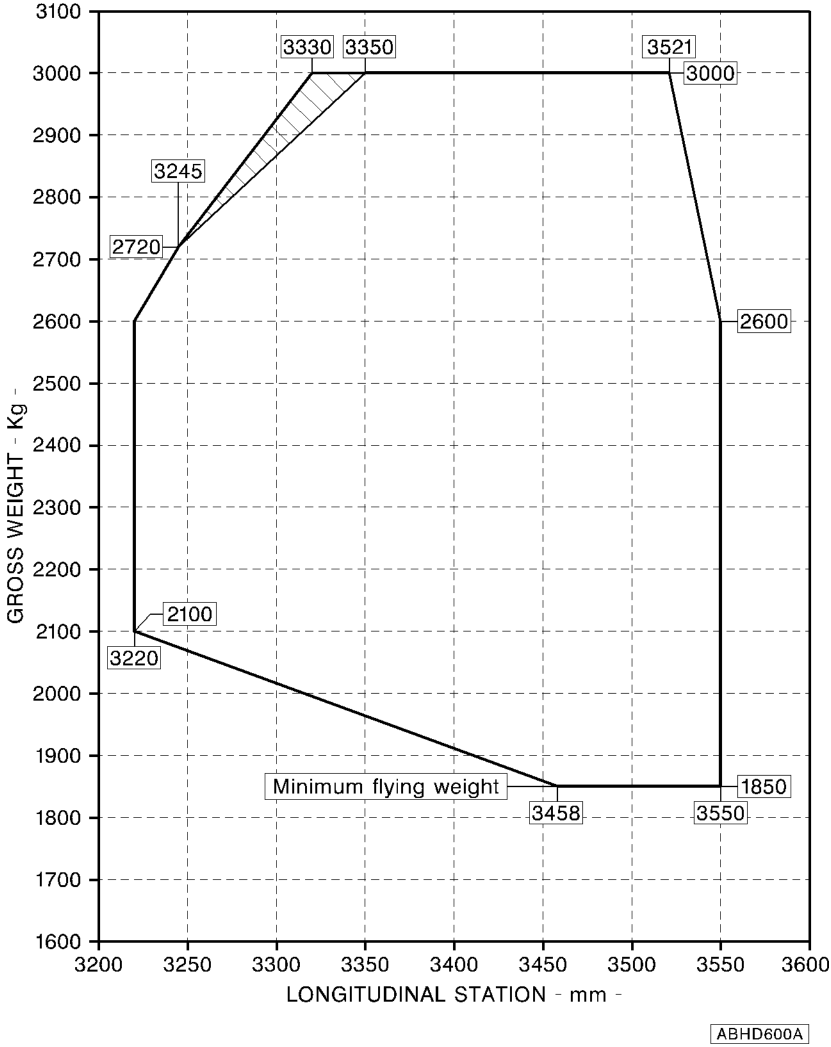
NOTE

The lateral station ″0″ is 17.7 in. inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

NOTE

The diagram shaded area is applicable only during hoist operation.

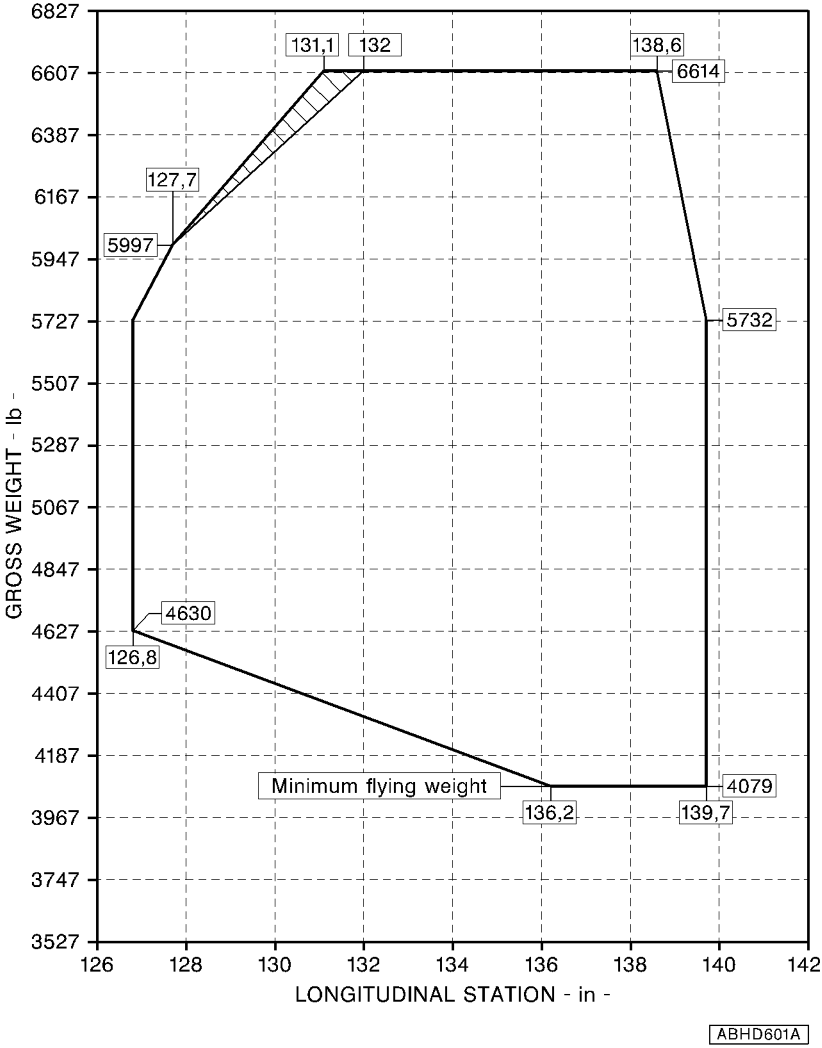
Figure 1-4 (sheet 2 of 2). Lateral CG limits (English).



NOTE
Longitudinal station ″0″ is 1835 mm forward of the front jack point.

NOTE
The diagram shaded area is applicable only during hoist operation.

Figure 1-5 (sheet 1 of 2). Longitudinal CG limit (metric).



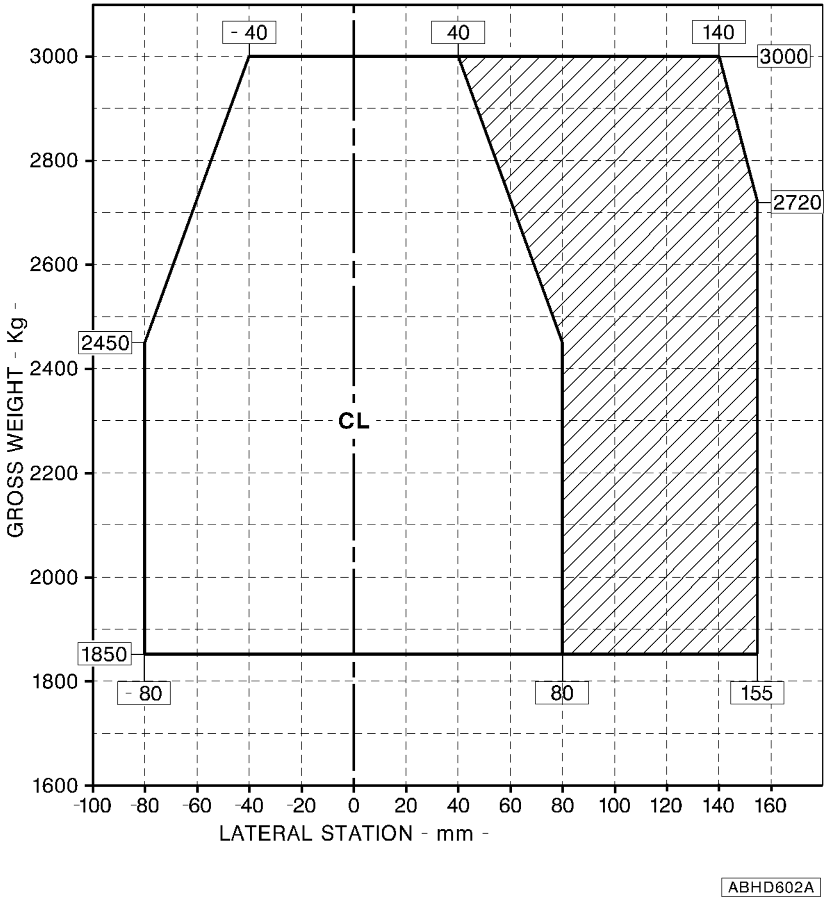
NOTE

Longitudinal station ″0″ is 72.2 in. forward of the front jack point.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-5 (sheet 2 of 2). Longitudinal CG limit (english).



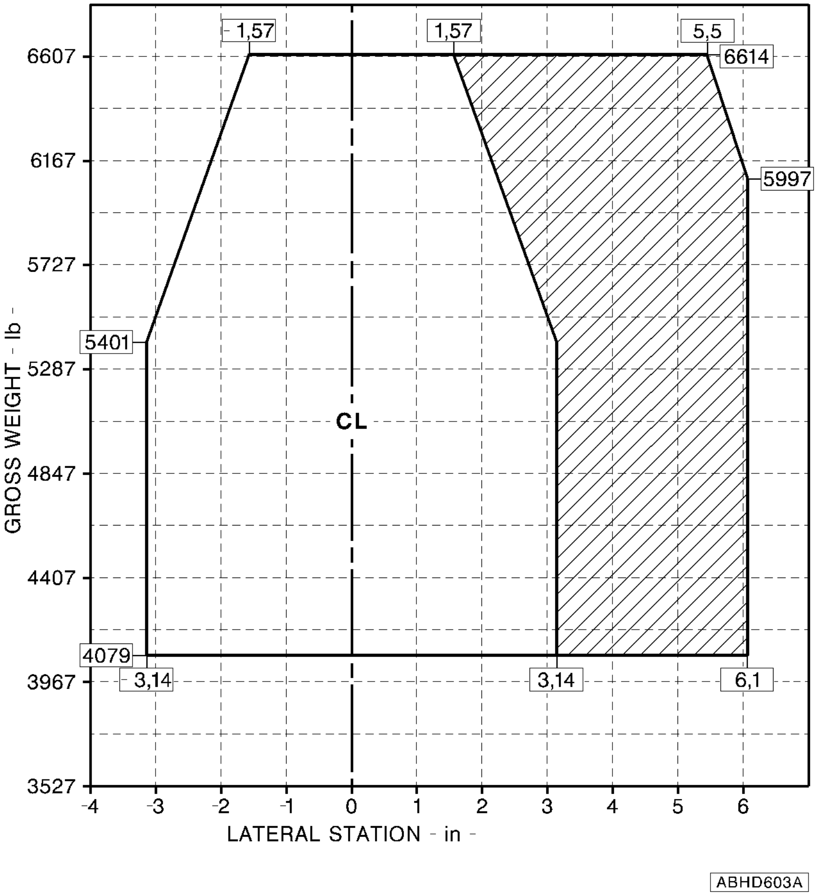
NOTE

The lateral station ″0″ is 450 mm inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-6 (sheet 1 of 2). Lateral CG limit (metric).



NOTE

The lateral station ″0″ is 17.7 in. inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

NOTE

The diagram shaded area is applicable only during hoist operation.

Figure 1-6 (sheet 2 of 2). Lateral CG limit (english).

PLACARDS

CAUTION If for any reason the hoist stops, during rising or lowering operation, do not attempt to free it by actuating the controller. Continued operation may cause total cable failure. Land as soon as possible.

In clear view of hoist operator and pilot

CAUTION The operator's hand must be on the cable at any time during hoist operation with load on the hook.

In clear view of hoist operator

V_{MO} with hoist suspended: 20 kts relative airspeed up to 8000 ft.

In clear view of the pilot

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N° 2 (Fuselage - rh side)

Hoist : Condition, security, oil leaks and wiring connected.

CAUTION

Crushable bumper shall be inspected to insure the bumper is not being crushed. Cause of any bumper crush must be corrected prior subsequent hoist operation, damaged bumper may lead to cable failure.

Hoist cowling : Secured.

Hoist mount : Condition and security.

AREA N° 7 (Cabin interior)

Operator strap : Correct installation.

Manual cable cutter : Verify presence of manual cable cutter.

CABLE CUT switch (on pilot's cyclic stick) : OFF, guard closed.

HOIST PWR circuit breaker : In.

For the following checks connect the d.c. electrical supply:

Main HOIST switch	: ON. HOIST ON advisory message on EDU 2 displayed.
Hoist cable payout display (both pilot and remote control)	: Check built-in-test reading is: 88.8.

WARNING

Protective gloves should always be used whenever handling the hoist cable to prevent injury from any possible broken cable strands.

CAUTION

Care should be taken to prevent cable damage caused by kinking when handling it on the ground. The cable should lie onto a clean surface whenever possible.

Hoist operation	: Check by reeling-out and reeling-in approximately three (3) meters of cable.

NOTE

When reeling-in the cable with no load, apply tension with gloved hands to ensure smooth and even wrapping.

Cable and terminal	: Condition and security; check cable for abrasion, kinking and corrosion.

NOTE

Ensure that setcrew locking hook assembly and bumper assembly is properly seated in body of assembly and setscrew is locked in place with cotter pin.

Full-in limit switch	:	Check for correct operation.
Main HOIST switch	:	OFF. HOIST ON advisory message on EDU 2 suppressed.

Disconnect the d.c. electrical supply.

HOIST PWR circuit breaker	:	Out.

ENGINE PRE-START CHECK

CABLE CUT switch (on pilot's cyclic stick)	:	OFF, guard closed. HOIST CUT ARMD caution message on EDU 1 not displayed.
HOIST circuit breakers	:	In.

SYSTEMS CHECK

Main HOIST switch	:	ON. HOIST ON advisory message on EDU 2 displayed.
HOIST CABLE LKD caution message (on EDU 1)	:	Not displayed, check.

Hoist operator:

Safety vests and straps	:	On and secured to helicopter.
Gloves	:	On.
HOIST control switch (on pilot's cyclic stick)	:	DOWN.

NOTE

When reeling-in the cable with no load, apply tension with gloved hands to ensure smooth and even wrapping. Approximately 9 kg tension is recommended.

Hoist hook	:	Check lowering.
Cable payout display	:	Check for meter counting.
HOIST control switch (on pilot's cyclic stick)	:	UP.
Hoist hook	:	Check raising.
HOIST thumbwheel (on remote control)	:	Rotate to DN and to UP.

NOTE

The hoist check, lowering and raising, shall be accomplished with the load selection switch in both positions: 113 kg and 272 kg.

Hoist hook	:	Check lowering or raising according to thumbwheel selection.
HOIST control switch (on pilot's cyclic stick)	:	DOWN or UP, checking that it overrides the thumbwheel selection.
HOIST thumb-wheel (on remote control)	:	OFF.
Main HOIST switch	:	OFF. HOIST ON advisory message on EDU 2 suppressed.

IN FLIGHT

HOIST OPERATING PROCEDURE

Main HOIST switch : ON. HOIST ON advisory message on EDU 2 displayed.

WARNING

Hoist operator shall be secured to helicopter with an approved safety harness during hoist operations.

Establish hover over hoist operation area.

CAUTION

Avoid, whenever possible, operating the hoist with crosswind or rear wind.

Right passenger door (if installed) : Open and locked.

Load selection switch (on remote control) : Set to 113 kg or 272 kg as required.

WARNING

During hoist operation the operator must always maintain his hand on the cable and verify the correctness of cable unwinding and rewinding.

HOIST control switch (pilot) or
HOIST thumbwheel (operator) : DOWN.

Cable speed (operator only) : As desired by means of the HOIST thumbwheel on remote control.

NOTE

As hook nears the up or down limits, hoist speed automatically slows.

CAUTION

During hoist operation the cable trail angle must be kept to minimum. If for any reason the trail angle exceeds 15 degrees (30 deg. cone angle), refer to helicopter Maintenance Manual for pertinent actions.

CAUTION

Do not allow cable to drag on the ground or any other surface which could contaminate or damage the cable.

CAUTION

Static electricity should be dissipated by suitable means before ground personnel touch the hook or cable.

HOIST control switch (pilot)
or HOIST thumbwheel (operator) : UP.

Cable speed (operator only) : As desired by means of the HOIST thumbwheel on remote control.

NOTE

With load selection switch set to 113 kg mode, operation of the hoist with loads exceeding 113 kg (250 lb) will cause the MOTOR WRN light located on the remote control to flash. The hoist control circuit will automatically shift and latch the hoist system to operate in the 272 kg mode. The hoist system will stay

″latched″ in the 272 kg mode and the MOTOR WRN light will continue to flash until the system is reset by switching the load select switch to the 272 kg mode. The high speed mode (113 kg) should not be selected until the present load is removed from the cable.

CAUTION

If the MOTOR WRN light remains illuminated (steady light), complete the hoist cycle and let the motor cool down (light extinguished) before resuming the hoist operations.
Prolonged operation of equipment with MOTOR WRN light illuminated (steady) will result in damaged or ″burned out″ hoist motor.

Maintain hover until load is into cabin.
If the place is not suitable to complete the load recovery on board, make transition to forward flight into wind, if possible, allowing adequate hoist load clearance over obstacles.

Airspeed : As required for adequate controllability, not to exceed 20 KIAS.

Once a suitable place has been reached, establish hover and proceed to recover load on board or to lay it on ground.

CAUTION

Hoist operation (load raising or lowering) is permitted with aircraft in stationary hover only.

Maintain hover until hoist operations are completed and the cargo is completely on board.

Right passenger door (if installed) : Close.

Main hoist switch : OFF. HOIST ON advisory message on EDU 2 suppressed.

NOTE

During hoist operation, the operator shall record any shock load to the cable; in this event, the cable must be replaced prior to the next flight.

LITTER HOISTING

When emergency transportation with the litter is essential, every effort should be made to land the helicopter for litter loading. Litter hoisting can be hazardous and should be accomplished only when a landing is not feasible. In addition to all other procedures contained herein, the following shall apply to litter hoisting operations.

LITTER

WARNING

Hoisting or lowering an empty litter in open position is prohibited. An empty litter suspended from hoist in open position can oscillate uncontrollably in rotor wash and can fly upward, striking fuselage or tail rotor.

Prior to hoisting or lowering an empty litter, litter shall be closed and secured with straps. Litter should be suspended in a near-vertical position and sling straps should be drawn tight.

WARNING

Hoist hook catch shall be secured with safety pin prior to hoisting litter.

Litter sling straps should be adjusted so that litter is 610 to 710 mm (24 to 28 in.) below hoist hook.

NOTE

If litter is suspended too far below hook. litter can not be loaded into helicopter with hoist hook at up limit.

CAUTION

A loaded litter can rotate around cable during hoisting. Hoist operator may have to grasp litter sling straps to control rotation as litter approaches landing gear.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

HOIST LOAD JETTISON

The external hoist installation is provided with an electrical cable cut system operated by the pilot. If an emergency condition should require the release of hoisted load proceed as follows:

CABLE CUT switch (on pilot's cyclic stick) : Lift the guard to arm the system: HOIST CUT ARMD caution message on EDU 1 displayed. Operate the switch.

In the event of failure of the electrical cable cut system, cut the cable with the manual cable cutter accessible to the hoist operator.
Cut the cable as close to the hoist as possible.

HOIST MOTOR WARNING LIGHT

The hoist MOTOR WRN light provides two features:

— Hoist system overload: detected by a flashing MOTOR WRN light.
— Motor overtemperature: detected by a continuously illuminated MOTOR WRN light.

In the event of a hoist system overload when the load selection switch is set to 113 kg mode, the hoist control circuitry automatically shifts and latches the hoist system to operate in the 272 kg mode and consequently to limit cable speed to 30 m per minute.
The high speed mode (113 kg) should not be selected until the present load is removed from the cable.
In the event of a motor overtemperature, complete the hoist cycle and let the motor cool down (light extinguished) before resuming the hoist operations.

CAUTION

Prolonged operation of hoist with MOTOR WRN light illuminated (steady) will result in damaged or ″burned out″ hoist motor.

UNEVEN CABLE WINDING

The external hoist installation is provided with a cable foul indicator system which protects the cable from the effects of continued operating with a fouled cable.
When a cable foul develops on the hoist drum the actuator of the cable foul indicator system interrupts the electrical circuit of the hoist motor in either the up or down direction.

INDICATIONS:

EDU 1	:	HOIST CABLE LKD caution message illuminated.
HOIST control switch (pilot) and HOIST thumbwheel (operator)	:	Inoperative.

PROCEDURE:

Proceed in forward flight with load suspended to a suitable site, allowing adequate hoist load clearance over obstacles.

Airspeed	:	As required for adequate controllability, not to exceed 20 KIAS.

Once a suitable site has been reached, establish hover, then slowly descend to lay the load on ground.
Recover manually the cable on board then proceed to land.
Correct the problem of cable foul incident before restoring hoist to operational conditions.

SYSTEM FAILURES

ELECTRICAL POWER FAILURE

Failure of both generators
In case of failure of both generators during hoist operation the pilot is allowed to complete only the outstanding recovery cycle.
Refer to Section 3 of the basic Rotorcraft Flight Manual for procedures.

NOTE

One hoist recovery cycle reduces the flight time on battery power to 20 minutes.

Failure of generator bus # 1 or # 2

If the failure occurs during hoist operation, proceed as follows:

GEN BUS 1 and/or 2 switch	:	ON.

If engagement of one switch trips the other, reengage the tripped switch and reset the other switch to OFF.
If both switches trip, complete the outstanding hoist recovery cycle then set BATT switch to OFF.
Refer also to Section 3 of the basic Rotorcraft Flight Manual.

Failure of both generator busses

This falure is indicated by BUS TIE caution message and after few seconds, if the battery discharge control box is installed, by BATT DISCH warning message on EDU 1. No loads are lost.

PROCEDURE:

GEN BUS 1 and 2 switches	: ON, check.

If failure is confirmed, complete the outstanding hoist recovery cycle then proceed as follows:

BATT switch	: OFF. ELECTRICAL warning and BATT OFF caution messages displayed.

Loss of DAU ch A and EDU Primary data

SAS 1 or 2, as applicable	: ON.

Failure of # 1 DC generator and generator bus # 2

This failure is indicated by #1 DC GEN, BUS TIE caution messages and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE:

GEN 1 switch	: Reset, then ON.
GEN BUS 2 switch	: ON, check.

If failure is confirmed, complete the outstanding hoist recovery cycle then proceed as follows:

BATT switch	:	OFF. ELECTRICAL warning, INV 1, SAS 1 OFF, BATT OFF caution messages displayed.

All loads of GENerator BUS # 1 and Emergency Bus # 1 are lost.
All loads on Essential Bus # 1 are retained.
DAU ch A and EDU Primary data are lost.

NOTE

In this event all loads on # 2 Bus are maintained.

Failure of # 2 DC generator and generator bus # 1

This failure is indicated by # 2 DC GEN, BUS TIE caution messages and, if the helicopter is equipped with battery discharge control box, by BATT DISCH warning message on EDU 1.

PROCEDURE:

GEN 2 switch	:	Reset, then ON.
GEN BUS 1 switch	:	ON, check.

If failure is confirmed, complete the outstanding hoist recovery cycle then proceed as follows:

BATT switch	:	OFF. ELECTRICAL warning, INV 2, SAS 2 OFF, BATT OFF caution messages displayed.

All loads of GENerator BUS # 2 and Emergency Bus # 2 are lost.
All loads on Essential Bus # 2 are retained.
DAU ch A and EDU Primary data are lost.

Pilot ICS panel switch	:	Select FAIL position.

NOTE

In this event all loads on # 1 Bus are maintained.

Failure of both generators and generators busses

The failure of both generators followed by either the simultaneous failure of both Generator Busses or their intentional disconnection is indicated by BUS TIE, INV 2, SAS 1 OFF, SAS 2 OFF caution messages by ELECTRICAL warning messsage and, if battery discharge control box is installed, by BATT DISCH warning message on EDU 1.

PROCEDURE:

GEN 1 and 2 switches	: Reset, then ON.
GEN BUS 1 and 2 switches	: ON, check if Generator reset.

If both generators remain inoperative, complete the outstanding hoist recovery cycle on battery power then proceed as per Section 3 of basic Rotorcraft Flight Manual.

NOTE

One hoist recovery cycle reduces the flight time on battery power to 20 minutes.

SECTION 4 - PERFORMANCE DATA

OPERATION VS ALLOWABLE WIND

Satisfactory stability and control in rearward and sideward flight has been demonstrated, at all loading conditions for hover out of ground effect up to takeoff power, from -1000 ft to 8000 ft Hd, in the following wind/ground speed azimuth envelope:

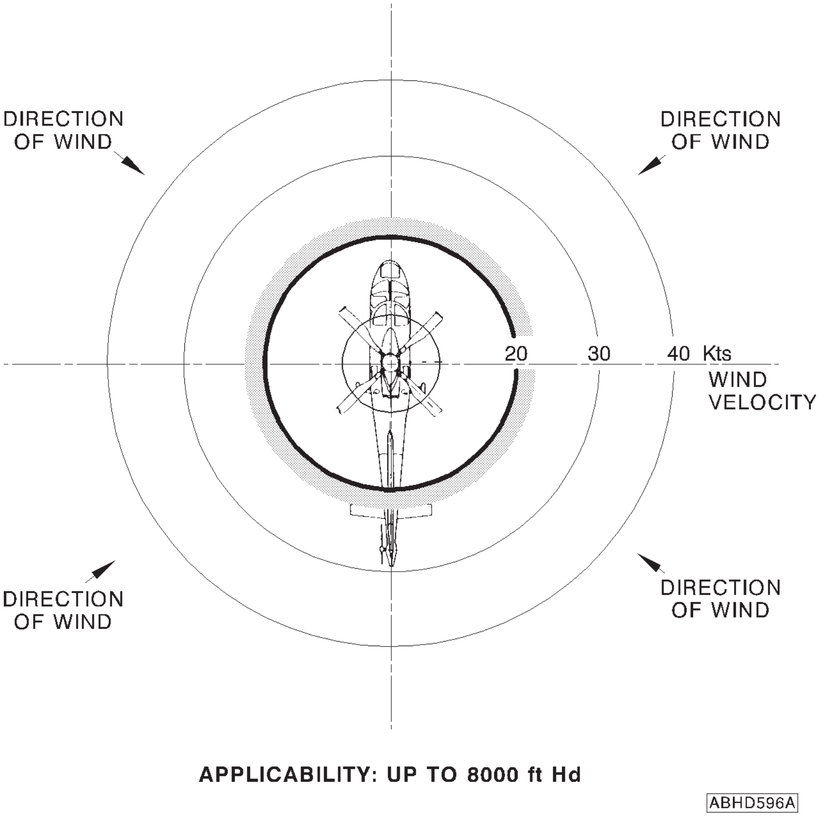
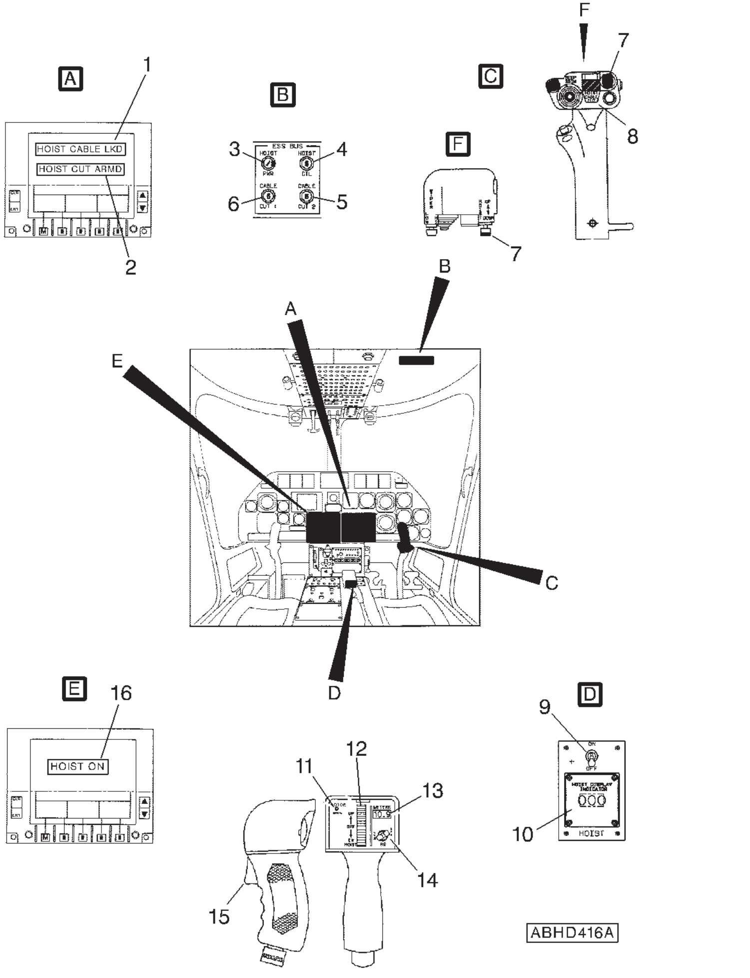


Figure 4-1. Wind/ground speed azimuth envelope.

SECTION 7 - SYSTEMS DESCRIPTION



1. HOIST CABLE LKD caution message
2. HOIST CUT ARMD caution message
3. HOIST-PWR circuit breaker
4. HOIST-CTL circuit breaker
5. CABLE-CUT 2 circuit breaker
6. CABLE-CUT 1 circuit breaker
7. HOIST control switch
8. HOIST CABLE CUT switch
9. HOIST switch
10. Pilot display indicator
11. MOTOR WRN light
12. HOIST CONTROL switch (thumbwheel)
13. Cable payout display
14. Load selection switch
15. ICS switch
16. HOIST ON advisory message

Figure 7-1. Hoist controls.

E.N.A.C. Approval Letter 99/4800/MAE
dated 28 December 1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SKYWATCH TRAFFIC ADVISORY SYSTEM SKY497

The SKYWATCH TAS SKY497 P/N 109-0812-39 is an onboard traffic advisory system which monitors a radius of nominally 6 NM around the helicopter by interrogating any intruding aircraft transponder, determinates if a potential conflict with other aircraft exists.
This is done by computing the relative range, altitude, bearing and closure rate of other transponder-equipped aircraft respect the helicopter.
The SKYWATCH system consists of the following items: TRC497 T/R computer, TA SKYWATCH display and NY156 directional antenna.
The Skywatch system is interfaced with enconding altimeter, directional gyro, weight-on-wheel switch and landing gear selector.
As an option the display can be shared with a STORMSCOPE weather mapping system (model WX-1000) using a remote TAS/WX MAP mode switch.
As an option TA SKYWATCH display can be replaced by a KMD 550 multifunction display (only for traffic alert function).
Refer also to ″MULTIFUNCTION DISPLAY KMD 550″ (Appendix 37).

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The TAS SKY497 can be used in VFR and IFR conditions as an aid to visually acquiring traffic.

NOTE

The pilot should maneuver the aircraft based only on ATC guidance or positive visual acquisition of the conflicting traffic.
Maneuver should be consistent with ATC instructions.
No maneuvers should be based only on a Traffic Advisory.
ATC should be contacted for resolution of the traffic conflict.

REQUIRED EQUIPMENT

In addition to the basic required equipment the helicopter must be fitted with:
- encoding altimeter.
- directional gyro.

COMPATIBILITY

The installation of the SKYWATCH TAS SKY 497 is not physically compatible with the oxygen system P/N 109-0811-76.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S PREFLIGHT CHECK

(Every flight)

SKYWATCH TAS SKY497 Pilot's Guide : Check on board.

SKY497 INTEGRATED WITH KMD 550

KMD 550/850 Multifunction Display Pilot's Guide : Check on board

KMD 550/850 Multifunction Display (TCAS/TAS) Pilot's Guide Addendum : Check on board

ENGINE START (NORMAL OR QUICK)

CAUTION

Do not turn ON the Radio Master switch before engine(s) starting to avoid damaging the TAS SKY497 system.

SYSTEMS CHECK

SKY497 IN STAND-ALONE CONFIGURATION

TAS/WX MAP pushbutton (if installed) : Press to select TAS mode.

TAS SKY497 OFF/BRT knob : Turn clockwise to the desired display brightness. After the completion of power on self test, press TEST button and verify that test screen appears while self-test is in process and that upon successful completion of the self-test, the audio message ″TRAFFIC ADVISORY SYSTEM TEST PASSED″ is heard and the display reverts to the standby screen.

NOTE

If the audio message ″TRAFFIC ADVISORY SYSTEM TEST FAILED″ is heard and a SKY497 FAILED wording appears on the screen, push the test button again. If the system continues to fail and for any other malfunction refer to SKYWATCH Pilot's Guide.

NOTE

The self-test is inhibited when the helicopter is in flight.

SKY497 INTEGRATED WITH KMD 550

KMD 550 : Switch ON. At the end of the self-test, press TRFC key and select 5 nm range. Check for TAS SBY operating mode.

TAS TEST push button : Press and verify that test screen appears while self test is in proces and that upon successful completion of the self-test, the audio message ″TRAFFIC ADVISORY SYSTEM TEST PASSED″ is heard and the display reverts to the standby mode.

NOTE

If the audio message ″TRAFFIC ADVISORY SYSTEM TEST FAILED″ is heard and the TAS FAIL legend appears in the screen, push the TEST button again. If the failure aural and caution message remain, check for the REPORTED TRAFFIC SYSTEM FAILURES in the TRFC page and refer to SKYWATCH pilot’s guide.

TAKE OFF

TAS SKY 497 : Check OPR mode is operating within 10 seconds from take off.

IN FLIGHT

NOTE

When the SKYWATCH system is used and a Traffic Alert is issued, scan outside for the intruder aircraft, call ATC for guidance and, if the traffic is visually acquired, use normal procedures to maintain separation.

WARNING

Do not attempt maneuvers based solely on traffic information shown on SKY 497 display.
Information on the display is provided to the flight crew as an aid in visually acquiring traffic; it is not a replacement for ATC and See & Avoid techniques.

NOTE

When SKY 497 shares the display with STORMSCOPE model WX1000 and the display is selected WX MAP mode, if SKY 497 generates a Traffic Alert (TA), the display automatically reverts to SKYWATCH mode until the TA is cleared.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE

No change.

SECTION 7 - SYSTEMS DESCRIPTION

The Traffic Advisory (TA) display indicates the relative position of an intruder when it is approximately 30 seconds from the Closest Point of Approch (CPA) or inside a range of 0.55 NM and ± 800 ft from own helicopter.
If these conditions are detected a voice message ″TRAFFIC TRAFFIC″ is generated in pilot and copilot intercommunication system to alert of the hazardous condition.
In approach or departure areas (with landing gear extended) the above criteria are reduced to 15/20 seconds from CPA or inside a range of 0.2 NM and ± 600 ft, but no audio alert is generated.
The system is intended to assist the pilot in achieving visual acquisition of the threat aircraft, and it must be considered a backup system to the ″SEE AND AVOID″ concept and the ATC radar environment.
The system does not work if the encoding altimeter is inoperative.
When a STORMSCOPE WX1000 is interfaced with the SKY 497, it is possible to select SKYWATCH or STORMSCOPE display mode using the remote TAS/WX MAP pushbutton. Once in STORMSCOPE display mode, the buttons on the display bezel allow to control STORMSCOPE function.
The SKY 497 does not superimpose SKYWATCH data on top of STORMSCOPE data or viceversa; however if the SKY 497 is in STORMSCOPE mode and the SKY 497 detects a Trafic Alert, the display automatically reverts to SKYWATCH mode until the TA disappears. At the same way if the display is in STORMSCOPE mode and the SKY 497 detects a failure, the SKY 497 Failed Screen appears with a message to ″Press Any Key to Ack″. Pressing any key switches the SKY 497 back to STORMSCOPE mode.

E.N.A.C. Approval Letter 99/4800/MAE
dated 28-12-1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

MOVING MAP SYSTEM SKYFORCE OBSERVER

The SKYFORCE OBSERVER P/N 109-0812-38 is a GPS driven detailed mapping system.
It is designed to provide the pilot with a sophisticated mapping and navigation function, which is capable of displaying highly detailed digital and scanned area maps.
The SKYFORCE OBSERVER system consists of a 200 MMX Pentium processing unit, a control unit and a TFT colour display.

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The navigation can not be based on the SKYFORCE OBSERVER System.

COMPATIBILITY

The installation of the SKYFORCE OBSERVER System is not physically compatible with the oxygen system P/N 109-0811-76.

SECTION 2 - NORMAL PROCEDURES

The pertinent SKYFORCE OBSERVER Operating & Installation manual must be available on board (publication N.0B1001 rev.4).

ENGINE START (NORMAL OR QUICK)

CAUTION

Do not turn on the Skyforce Observer before engine starting to avoid damaging the system.

After both engines starting:

SKYFORCE OBSERVER Control Unit : Select ON. Verify that ″SKYFORCE OBSERVER″ header screen appears.
Check that the ON and HDD lamps illuminate and that the TEMP lamp is off.

NOTE

The TEMP lamp is red to indicate ″Too Hot″ and amber to indicate ″Heater in Operation″.

Verify that, after approximately 100 seconds, the WARNING screen appears.
Check the correct software standard.

SYSTEMS CHECK

SKYFORCE OBSERVER display head : Select CONTINUE on the warning screen to display the MAIN MENU.
Check that initially the Map Mode comes up as a Static System with NO ENTRY icon in the centre of the screen.
Check that Map Mode switches from a NO ENTRY to helicopter icon once GPS has FIX.
Check that in Map Mode the functions ZOOM IN, ZOOM OUT, SEARCH and MAIN MENU are active.
Then return to MAIN MENU.

IN FLIGHT

SKYFORCE OBSERVER display head : As required.

SHUTDOWN

(After engines shutdown)

SKYFORCE OBSERVER display head : Go to MAIN MENU and select SHUT DOWN.
Press the YES key.

NOTE

The SHUT DOWN key must be used to shut the system software down; it ensures that the operating system and all data files are stored and secured prior to the power down of the Observer hardware.

At the end of shutdown sequence verify that the ″IT IS SAFE TO TURN OFF YOUR COMPUTER″ message is displayed before to proceed to hardware turn off.

SKYFORCE OBSERVER Control Unit : Select OFF.
Check that LEDs extinguish.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.N.A.C. Approval Letter 99/4800/MAE
dated 28-12-1999

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SATELLITE TELEPHONE AIRSAT 1

The AIRSAT 1 P/N 109-0812-40 is a satellite telephone system offering the possibility of worldwide telephone communications through the 66-satellite IRIDIUM network.
The AIRSAT 1 system consists of an Iridium transceiver unit, a blade antenna, two backlit digital handsets (one in pilot cabin on the pedestal and the second in passenger cabin in the cocktail cabinet) and a control panel at pilot's disposal.

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The use in flight of AIRSAT 1 satellite telephone must be authorized by the pilot.
The use of AIRSAT 1 satellite telephone during takeoff and landing is prohibited.

COMPATIBILITY

The installation of the AIRSAT 1 System is not physically compatible with the External Hoist installation P/N 109-0812-31.

SECTION 2 - NORMAL PROCEDURES

IN FLIGHT

SATELLITE TELEPHONE OPERATION

PILOT/CREW pushbutton	: As required.
Phone handset	: Turn the phone on and digit the PIN. Operate the phone as required

NOTE

The phone call on arrival are announced by the blinking of CALL light at the selected phone station (pilot or crew).

To switch the phone operation from pilot to crew station or viceversa, operate as follows:

Phone handset (in use)	: Turn the phone off.

PILOT/CREW pushbutton	:	Push to select
Phone handset (to use)	:	Turn the phone on and digit the PIN. Operate the phone as required.

CAUTION

Do not try to switch the phone control from pilot to crew station or viceversa without turning off the telephone

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.N.A.C. Approval Letter N. 00/504/MAE
dated 16 February 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

PARTICLE SEPARATOR ENGINE AIR INDUCTION SYSTEM

The particle separator P/N 109-0811-55 protects the engine against ingestion of sand, dust and other particles.
The system consists of two particle separators located in front of each engine air intake, tubing, hoses, electrical cables and controls for the engine bleed air shut-off valves and the hardware required to complete the installation. Foreign particles are separated by the swirling action of the air passing through the vortex generators contained into the particle separator. The clean air enters the engine air intake, while the foreign particles are directed by the scavenge flow into a lower chamber. From this chamber the foreign particles are discharged outboard through an ejector activated by the engine bleed air.
The particle separator may be activated by the pilot through a control switch on the overhead panel which permits air bleeding from the engine for particle separator ejector operation. The particle separator can anyhow operate with no bleed air, with a separation efficiency of 50%. In this case, particles will be accumulated in the chamber of the ejector up to its complete filling, after which external particles can no further be separated thus flowing directly into the engine.

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

Flight in falling and/or blowing snow is prohibited.

REQUIRED EQUIPMENT

In addition to the basic required equipment the helicopter must be fitted with:

— the Integrated Display System composed by EDU P/N 109-0900-42-2A01 and DAU P/N 109-0900-42-6A01.

— EDU software release 006 or later, and DAU software release 005 or later.

WEIGHT LIMITATIONS

The maximum takeoff and landing weight to operate in:

— Equivalent Category ″A″ from clear area, from short field and from helipad (ground based or elevated) are shown in figures 1-1 thru 1-6;

— Equivalent Category ″A″ - Training procedures from clear area, from short field and from helipad (ground based) are shown in figures 1-7 thru 1-9.

CENTER OF GRAVITY LIMITATIONS

After the particle separator installation the new empty weight and C.G. location must be determined.

ALTITUDE LIMITATIONS

Maximum operating pressure altitude : 15000 ft (4572 m)

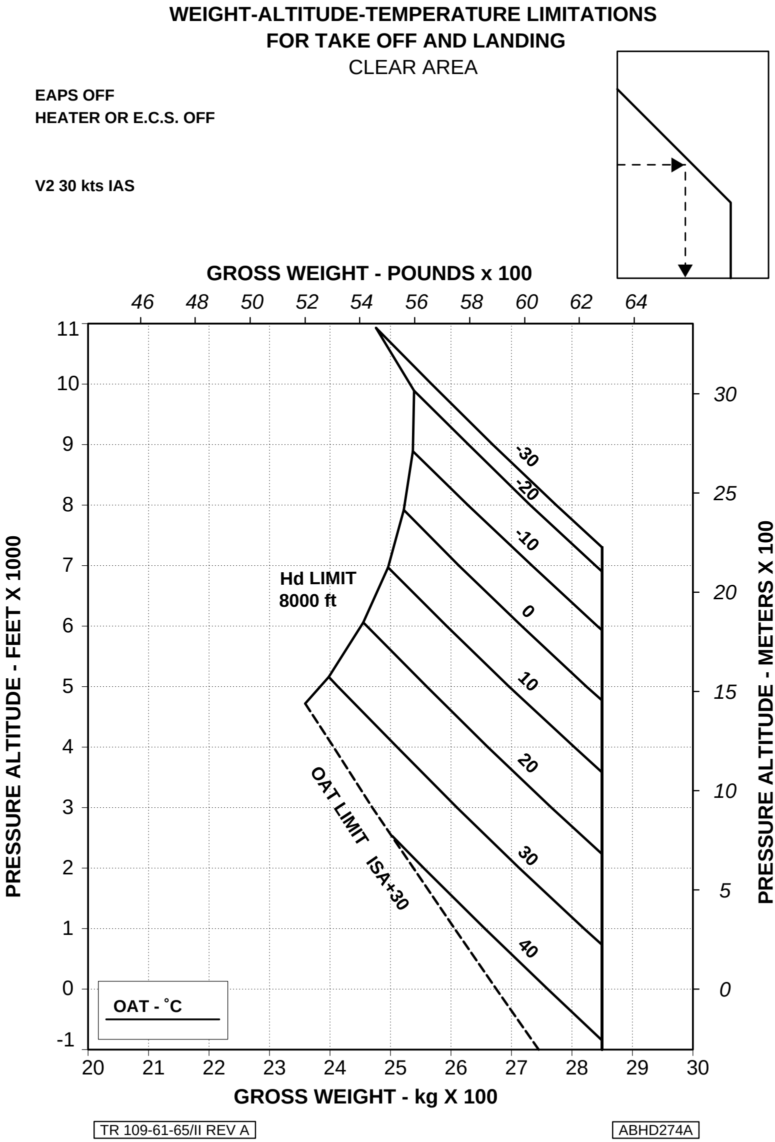

Figure 1-1. Weight - altitude - temperature limitations for take off and landing (clear area) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
CLEAR AREA

EAPS ON
HEATER OR E.C.S. OFF

V2 30 kts IAS

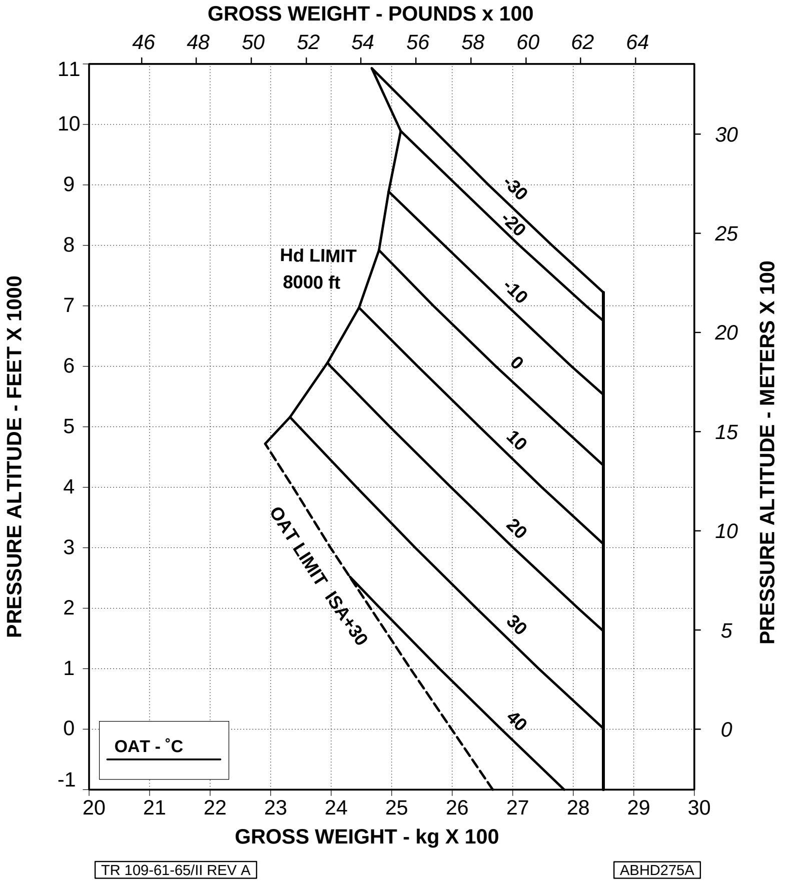

Figure 1-2. Weight - altitude - temperature limitations for take off and landing (clear area) - EAPS ON.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

SHORT FIELD (100 m)

EAPS OFF **V2 30 kts IAS**

HEATER OR E.C.S. OFF

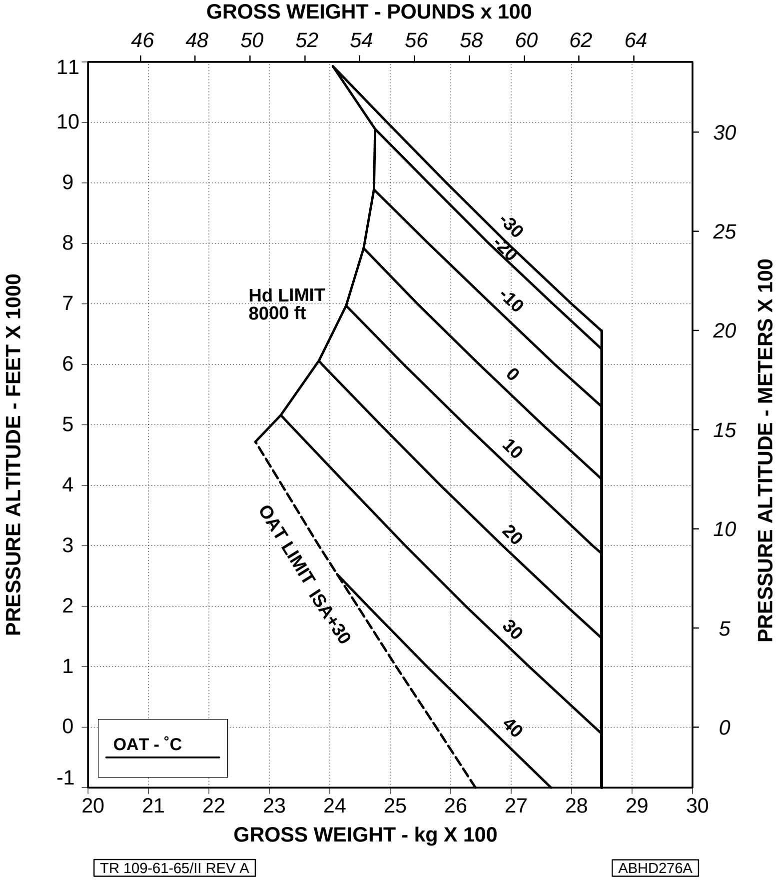

Figure 1-3. Weight - altitude - temperature limitations for take off and landing (short field) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
SHORT FIELD (100 m)

EAPS ON
HEATER OR E.C.S. OFF

V2 30 kts IAS

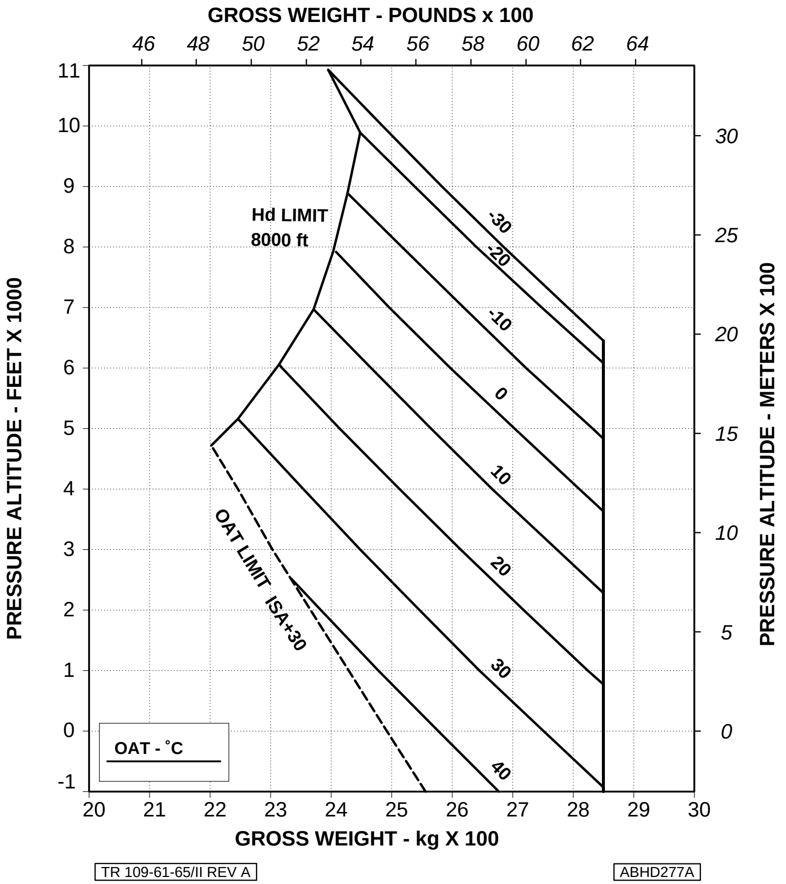

Figure 1-4. Weight - altitude - temperature limitations for take off and landing (short field) - EAPS ON.

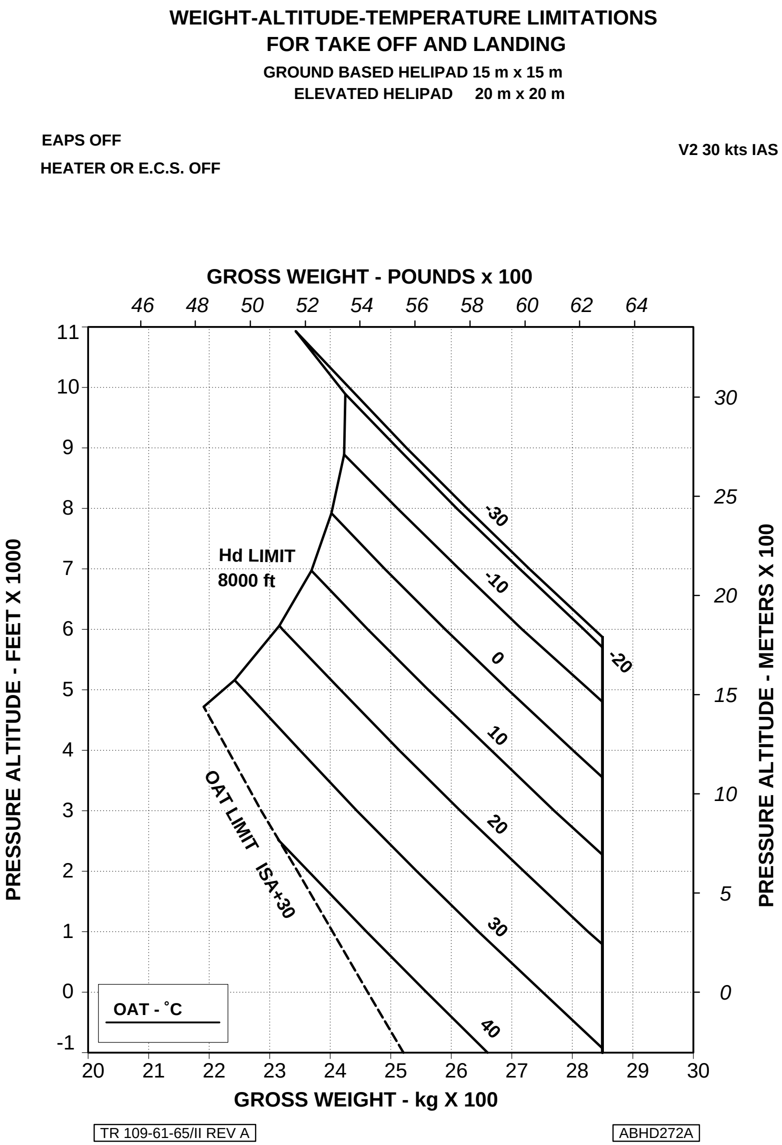

Figure 1-5. Weight - altitude - temperature limitations for take off and landing (ground based helipad/elevated helipad) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
GROUND BASED HELIPAD 15 m x 15 m
ELEVATED HELIPAD 20 m X 20 m

EAPS ON

HEATER OR E.C.S. OFF

V2 30 kts IAS

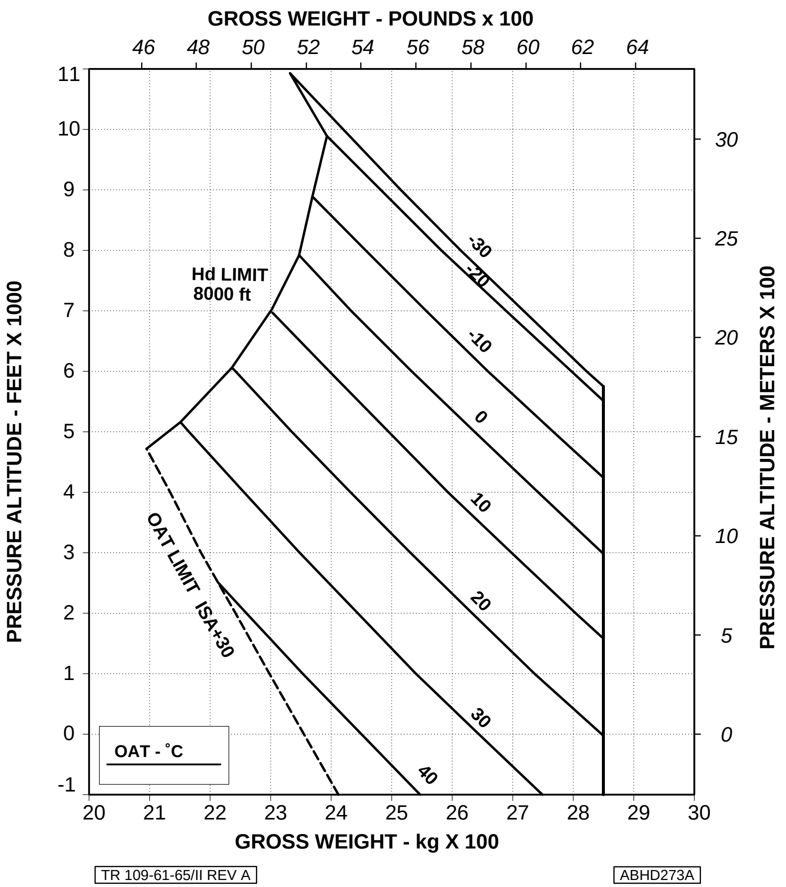

Figure 1-6. Weight - altitude - temperature limitations for take off and landing (ground based helipad/elevated helipad) - EAPS ON.

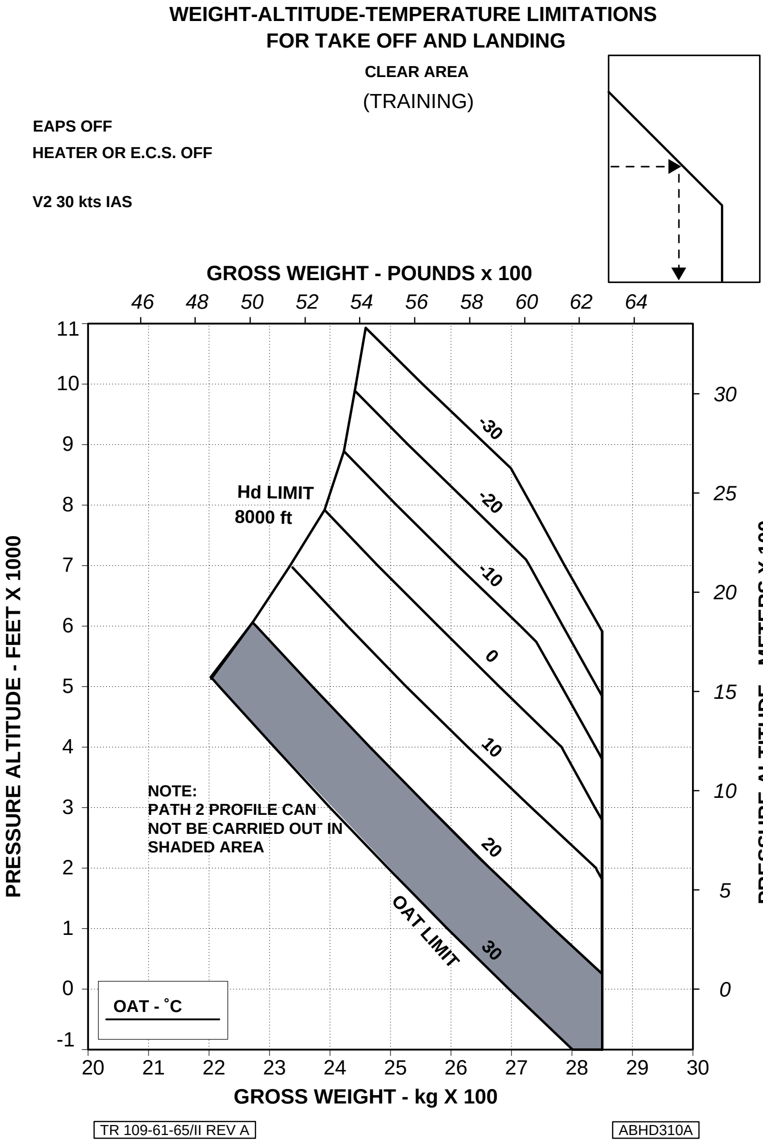

Figure 1-7. Weight - altitude - temperature limitations for take off and landing (clear area) - training - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

SHORT FIELD (100 m)

(TRAINING)

EAPS OFF

HEATER OR E.C.S. OFF

V2 30 kts IAS

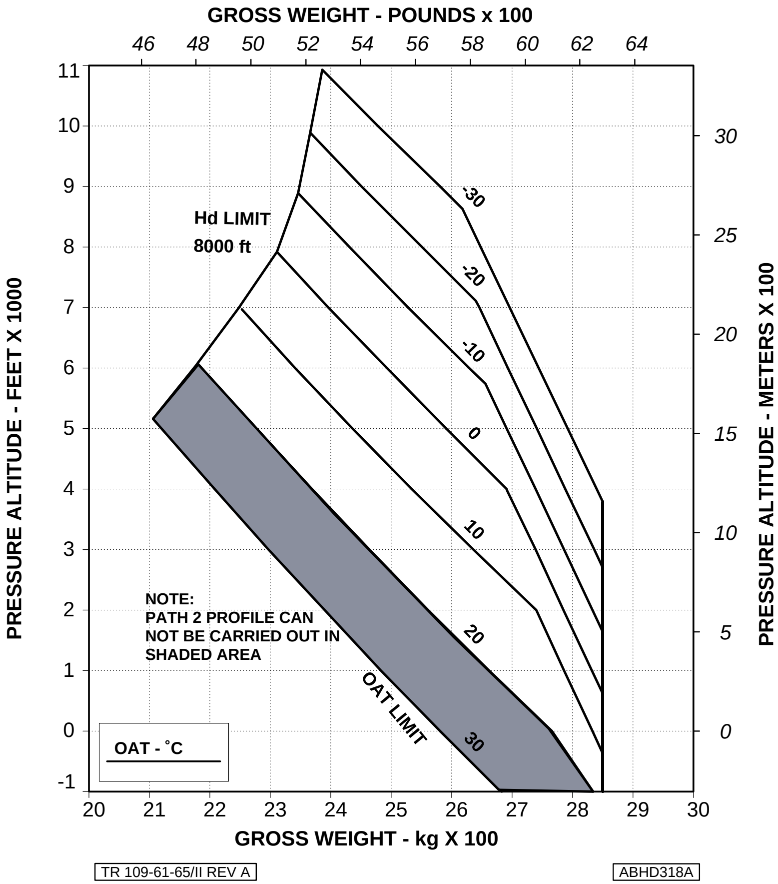

Figure 1-8.Weight - altitude - temperature limitations for take off and landing (short field) - training - EAPS OFF.

 E.N.A.C. Approved

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
GROUND BASED HELIPAD 15 m x 15 m

(TRAINING)

EAPS OFF **V2 30 kts IAS**

HEATER OR E.C.S. OFF

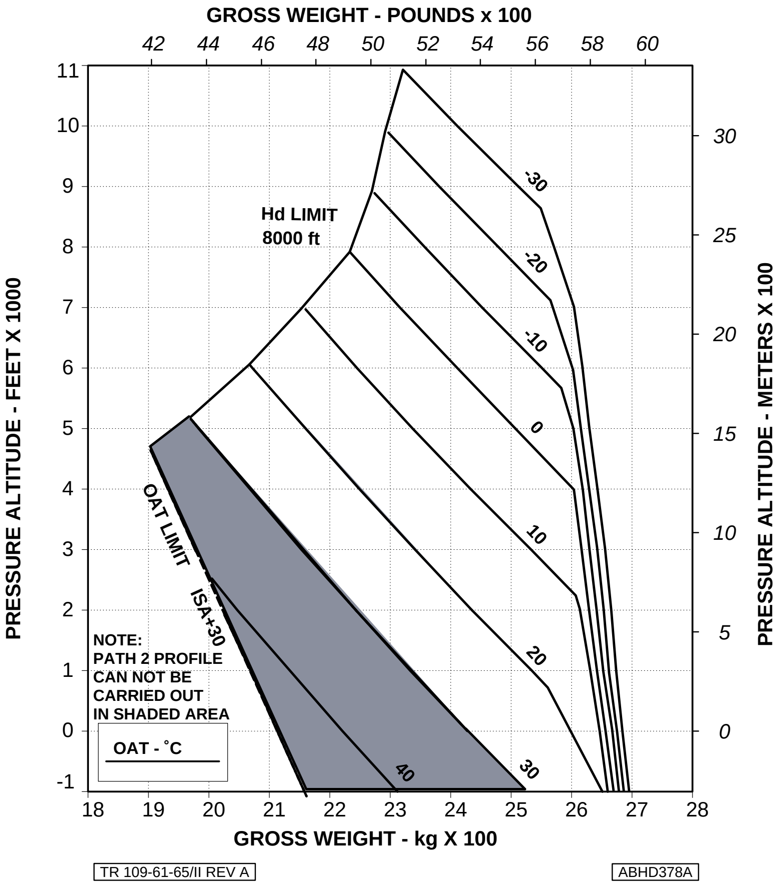

Figure 1-9. Weight - altitude - temperature limitations for take off and landing (ground based helipad) - training - EAPS OFF.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

Area N° 2 (Fuselage - rh side)

Engine air intake and particle separator screen and ejector	: Cover removed; foreign matter, integrity and free of obstructions.

Area N° 6 (Fuselage - lh side)

Engine air intake and particle separator screen and ejector	: Cover removed; foreign matter, integrity and free of obstructions.

PILOT'S PREFLIGHT CHECK

(Every flight)

Engine air intakes and particle separator screens and ejectors	: Covers removed; foreign matter, integrity and free of obstructions.

ENGINE PRE-START CHECK

(at the end of ENGINE PRE-START CHECK sequence)

EAPS 1 and 2 switches	:	ON. Check #1 and #2 EAPS ON advisory messages displayed on EDU 2, and #1 and #2 EAPS PRES caution messages displayed on EDU 1.
EAPS 1 and 2 switches	:	OFF.

SYSTEMS CHECK

EAPS 1 and 2 switches	:	ON. #1 and #2 EAPS ON advisory messages on EDU 2 displayed.
Engine temperature (TOT)	:	Note increasing by about 15°C.

On completion of systems check:

EAPS 1 and 2 switches	:	As required.

IN FLIGHT

EAPS 1 and 2 switches	:	As required.

APPROACH AND LANDING

EAPS 1 and 2 switches	:	As required.

SHUTDOWN

(Before ENG 1 and 2 MODE switches to OFF)

EAPS 1 and 2 switches : OFF (if utilized) and check #1 and #2 EAPS ON advisory messages on EDU 2 out.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
#1 (#2) EAPS PRES	Malfunction of engine #1 (#2) bleed air shut-off valve: non-opening of the valve with EAPS switch in ON position, or non-closure of the valve with EAPS switch in OFF position.	Proceed with flight. Correct trouble prior to next flight.

ENGINE FAILURES

FAILURE OF ONE ENGINE

EAPS 1 and 2 switches : OFF (if utilized).
#1 and #2 EAPS ON advisory messages on EDU 2 out.

ENGINE RESTART IN FLIGHT

Before attempting a restart either in AUTO mode or in MANUAL mode:

EAPS switch (affected engine) : OFF, check.

ENGINE STARTING IN MANUAL MODE (ON GROUND)

EAPS switch (affected engine) : OFF.

ENGINE SHUTDOWN IN MANUAL MODE

(Before engine power lever to OFF)

EAPS switch (affected engine) : OFF.

FIRE

ENGINE FIRE IN FLIGHT

EAPS 1 and 2 switches : OFF (if utilized).
#1 and #2 EAPS ON advisory messages on EDU 2 out.

FLIGHT IN THUNDERSTORM - LIGHTNING

Induced current overloads caused by lightnings could trip the EAPS switches. Reset EAPS switches to ON, if required.

SECTION 4 - PERFORMANCE DATA

PERFORMANCE CHARTS

This Section includes the performance data for A109E helicopters equipped with particle separator and related to the following configurations:

— helicopters in basic configuration;
— helicopters with Bleed Air Heater or E.C.S.;

Moreover it presents the performance data with the particle separator for:

— Equivalent Category ″A″ operations;
— Equivalent Category ″A″ Training Procedures.

The presence of the particle separator, even with bleed air shut-off valve closed (EAPS OFF), causes an increase of power losses due to a pressure reduction at the engine intake.
The increase of power losses is more evident when the bleed air shut-off valve is open (EAPS ON).
In both cases the helicopter performance shows, therefore, a reduction if compared with a helicopter not equipped with particle separator.

Helicopters in basic configuration

POWER ASSURANCE CHECK
HOVER

* EAPS OFF.
* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 102%.

* TEST ENGINE MODE SWITCH - FLIGHT.
* OTHER ENGINE MODE SWITCH - IDLE.

* INCREASE COLLECTIVE UNTIL LIGHT ON WHEELS OR HOVERING AT 3 FEET, DO NOT EXCEED 820˚C TOT OR 97.4% N1 OR 124% TORQUE.
* STABILIZE POWER ONE MINUTE, THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAXIMUM ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINATED AS SOON AS PRACTICAL.

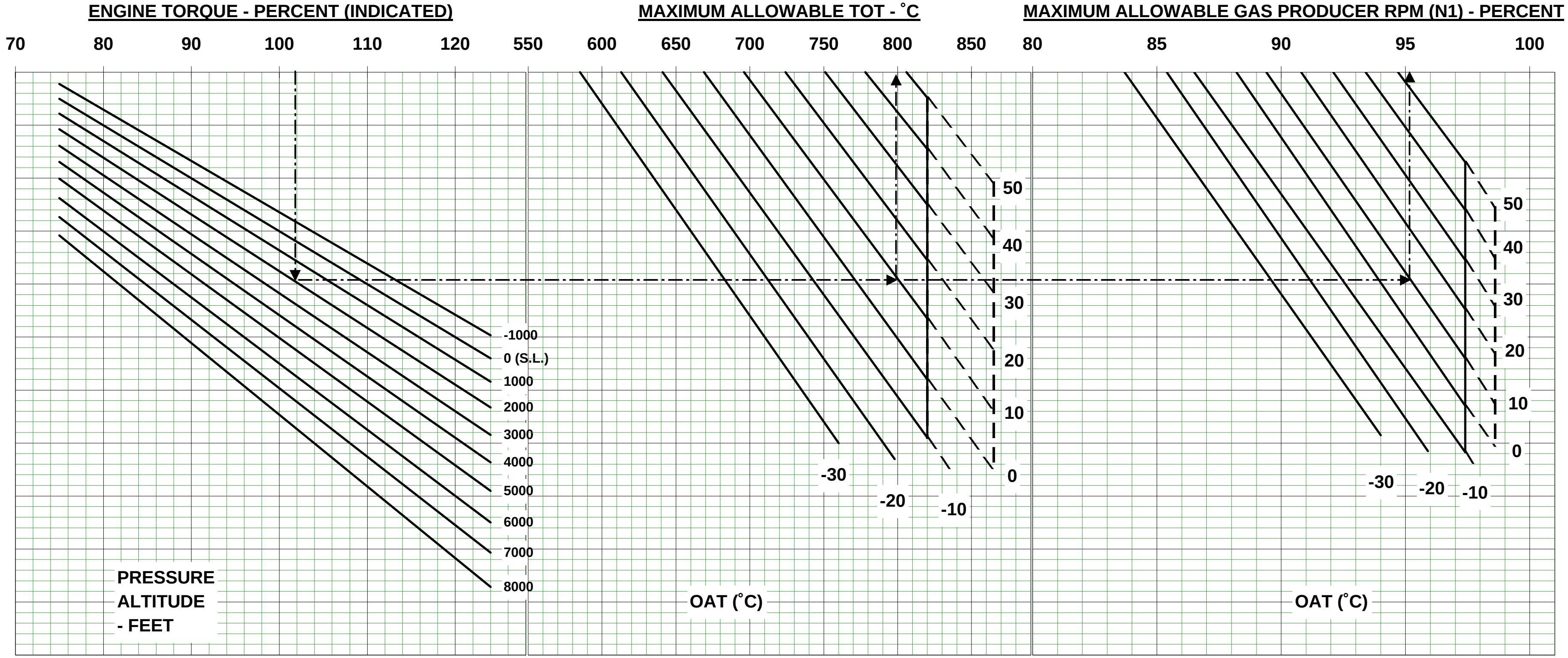

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Figure 4-1. Power assurance check - hover - EAPS OFF.

POWER ASSURANCE CHECK
HOVER

* EAPS ON.
* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 102%.

* TEST ENGINE MODE SWITCH - FLIGHT.
* OTHER ENGINE MODE SWITCH - IDLE.

* INCREASE COLLECTIVE UNTIL LIGHT ON WHEELS OR HOVERING AT 3 FEET, DO NOT EXCEED 820˚C TOT OR 97.4% N1 OR 124% TORQUE.
* STABILIZE POWER ONE MINUTE, THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAXIMUM ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINATED AS SOON AS PRACTICAL.

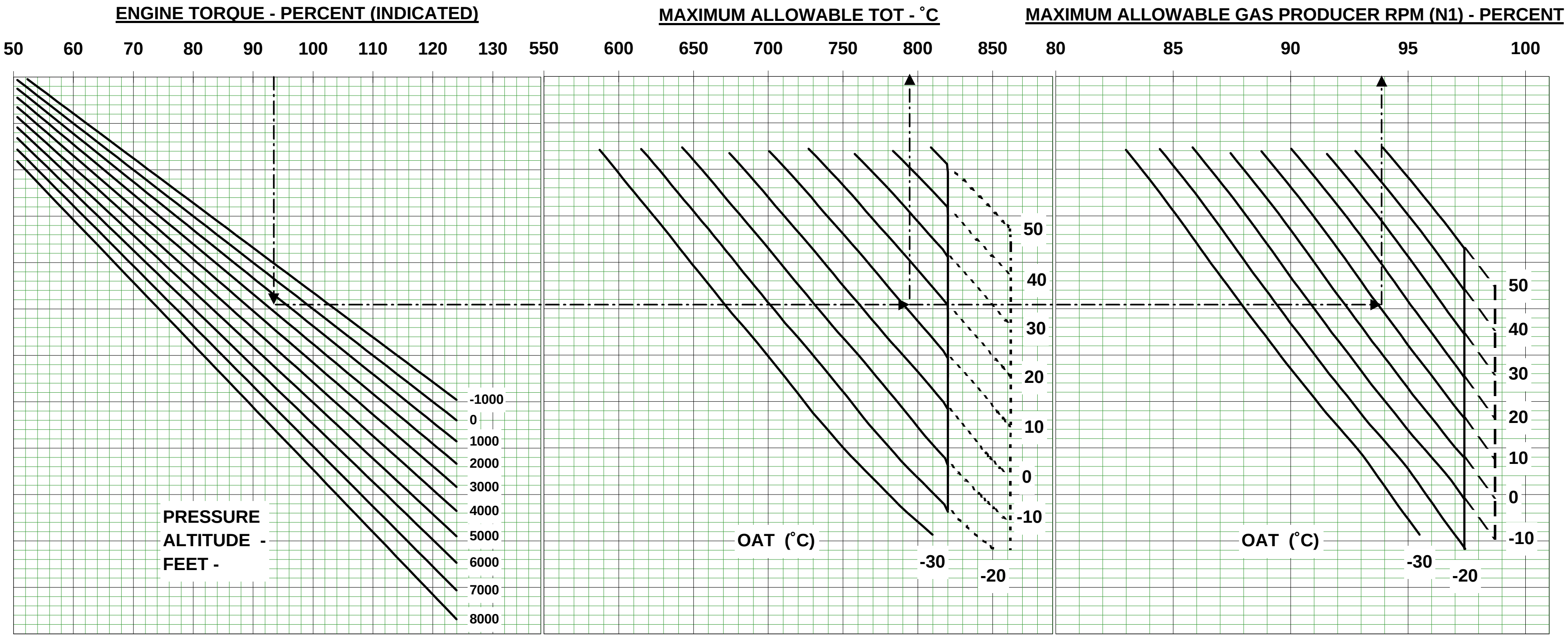

Figure 4-2. Power assurance check - hover - EAPS ON.

POWER ASSURANCE CHECK
IN FLIGHT

* EAPS OFF
* HEATER / E.C.S. OFF.
* GENERATOR LOAD TO MINIMUM (< 28 A).
* SET NR TO 100%.

* ESTABLISH LEVEL FLIGHT ABOVE 1000 ft. AGL.
* AIRSPEED - 100 KIAS.

* TEST ENGINE - SET ENG. GOV. SWITCH TO MANUAL
* OTHER ENGINE - LEAVE ENG. GOV. SWITCH TO AUTO

* OPERATE TEST ENG TRIM TO INCREASE POWER UNTIL ENGINE TORQUE IS WITHIN TEST RANGE. DO NOT EXCEED 820˚C TOT OR 97.4% N1 OR 100% TORQUE.
* STABILIZE POWER ONE MINUTE IN LEVEL FLIGHT THEN RECORD OAT, PRESSURE ALTITUDE, ENGINE TORQUE, TOT AND N1.
* ENTER CHART AT INDICATED ENGINE TORQUE, MOVE DOWN TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE TOT AND GAS PRODUCER RPM (N1).
* IF INDICATED TOT OR GAS PRODUCER RPM (N1) EXCEEDS MAX ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.
* REPEAT CHECK USING OTHER ENGINE.
* IF EITHER ENGINE EXCEEDS ALLOWABLE TOT OR GAS PRODUCER RPM (N1) AFTER STABILIZING FOUR MINUTES, CARRY OUT A POWER ASSURANCE CHECK IN HOVER.

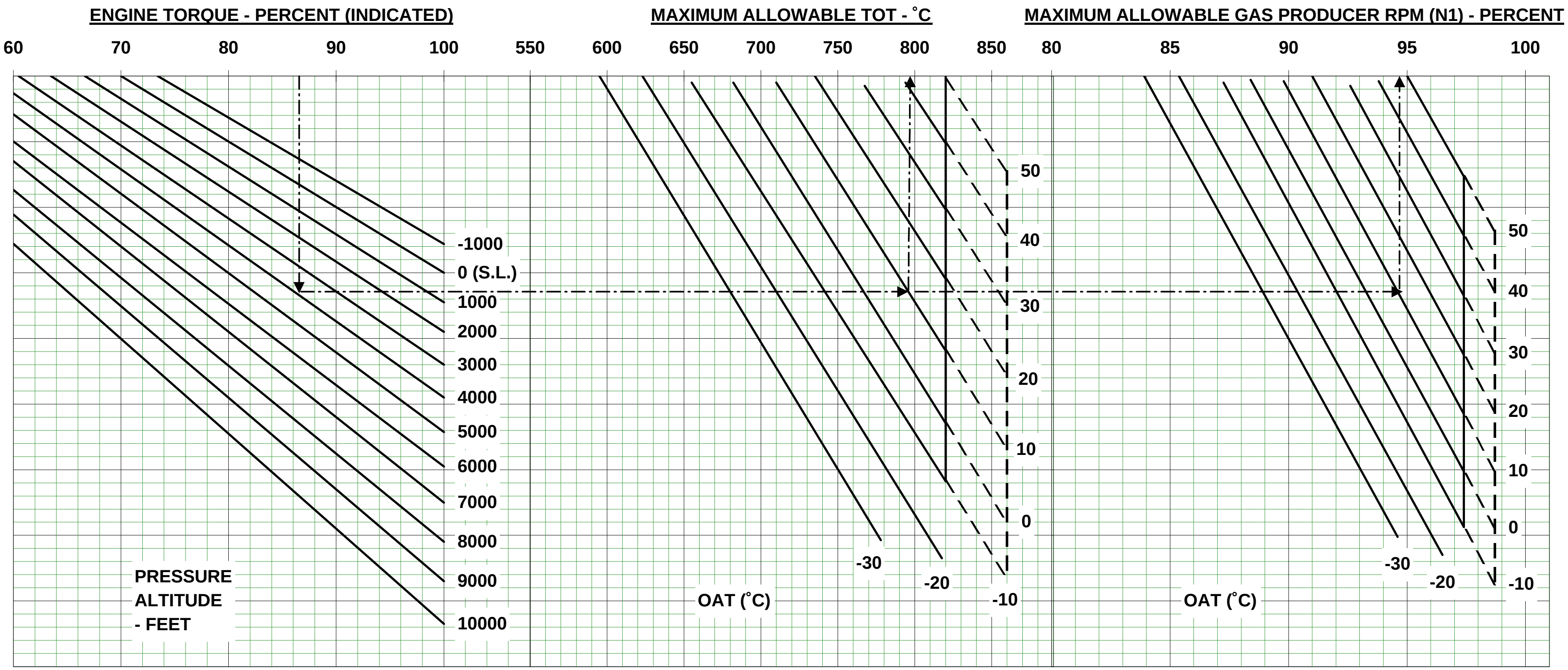

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Figure 4-3. Power assurance check - in flight - EAPS OFF.

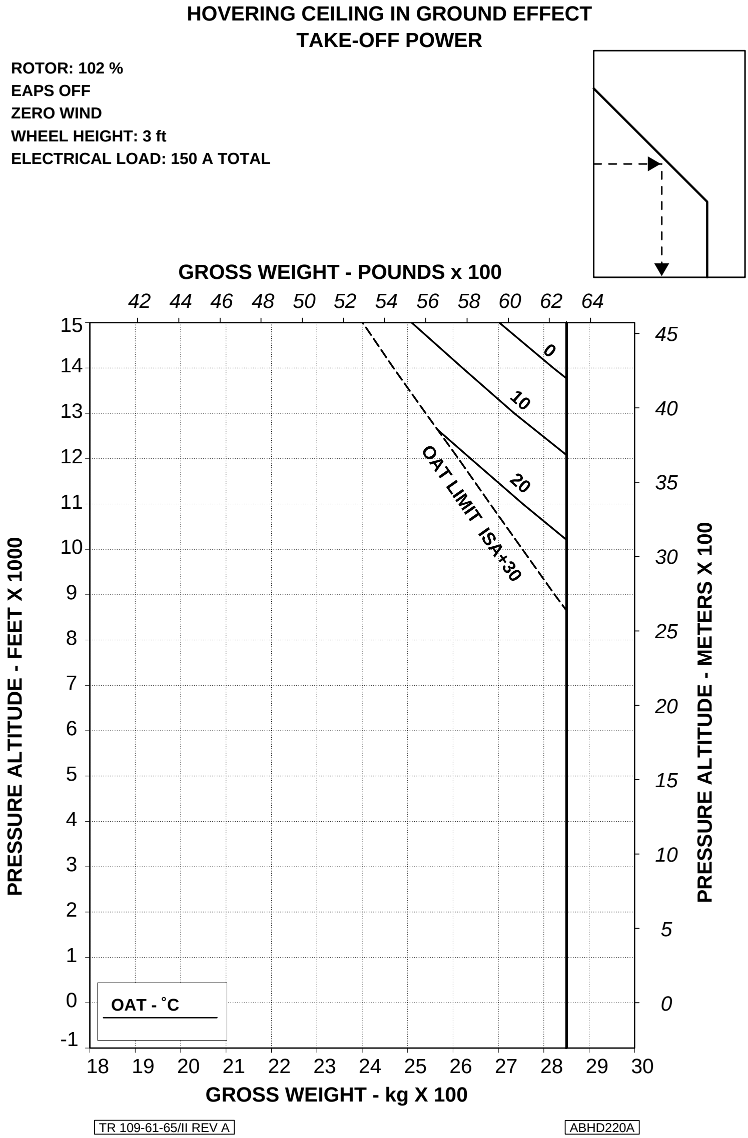


Figure 4-4. Hovering ceiling - IGE - take off power - EAPS OFF.

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

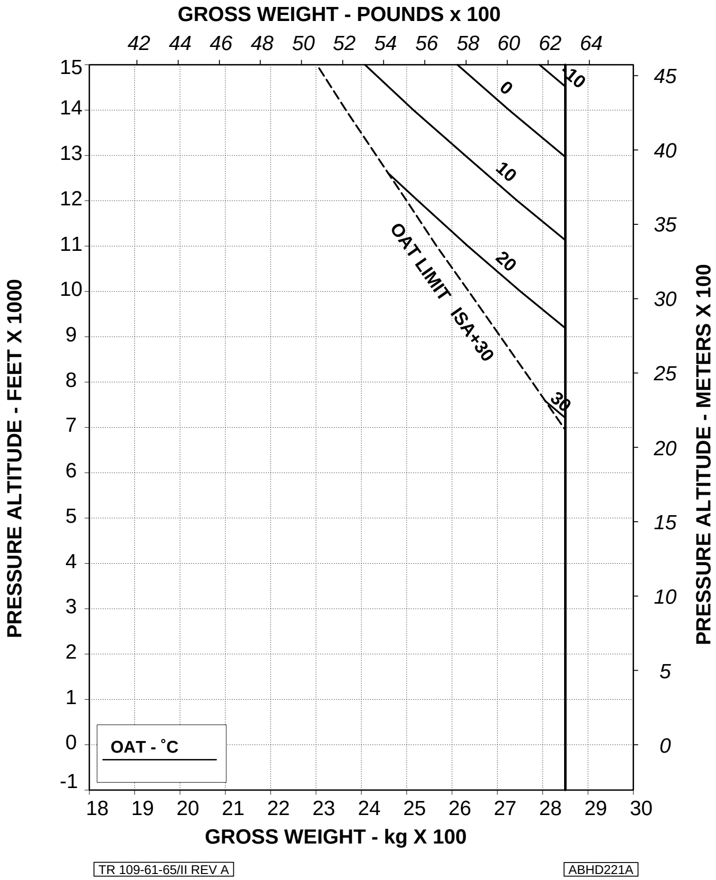


Figure 4-5. Hovering ceiling - IGE - take off power - EAPS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

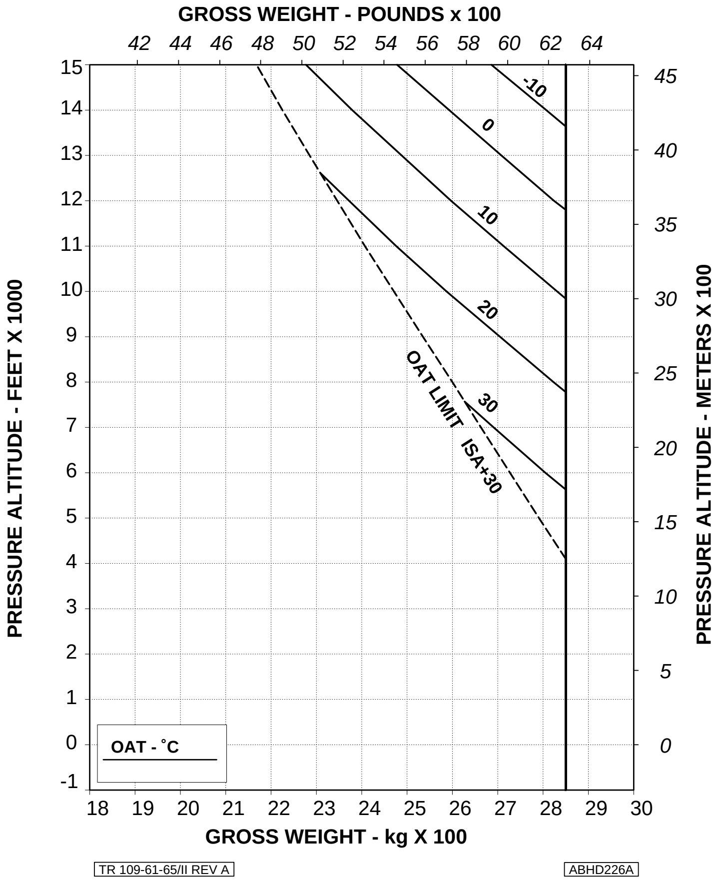


Figure 4-6. Hovering ceiling - IGE - maximum continuous power - EAPS OFF.

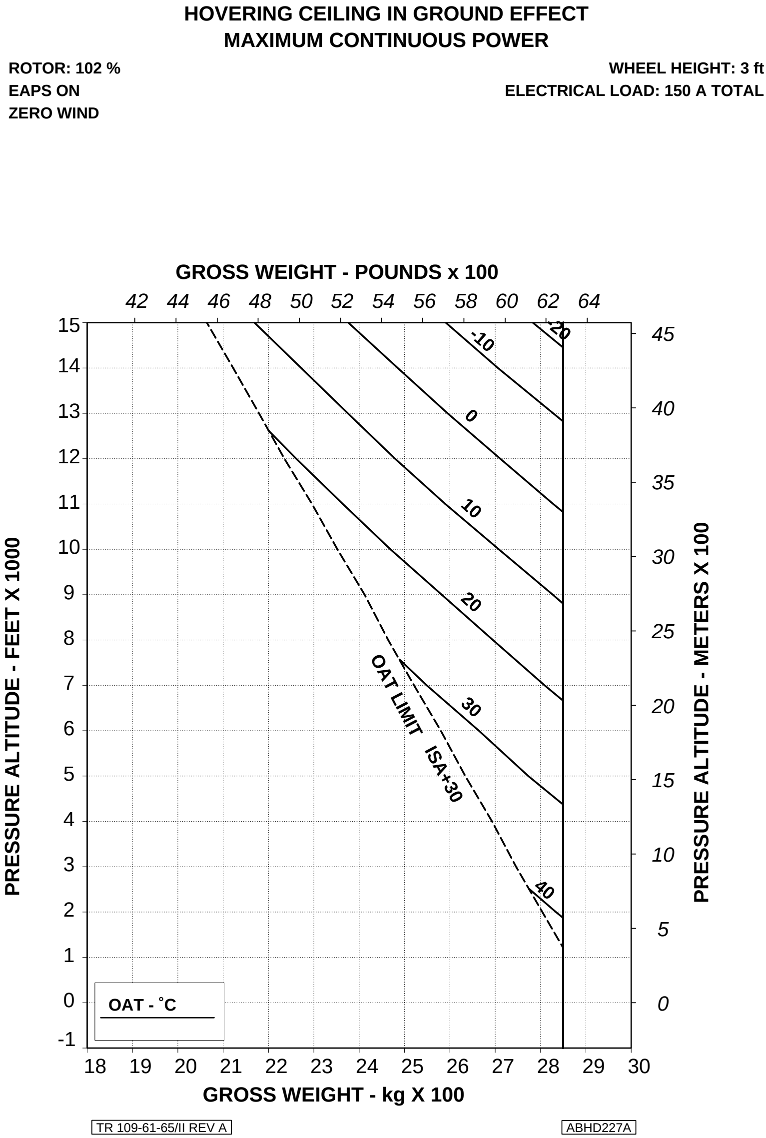


Figure 4-7. Hovering ceiling - IGE - maximum continuous power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

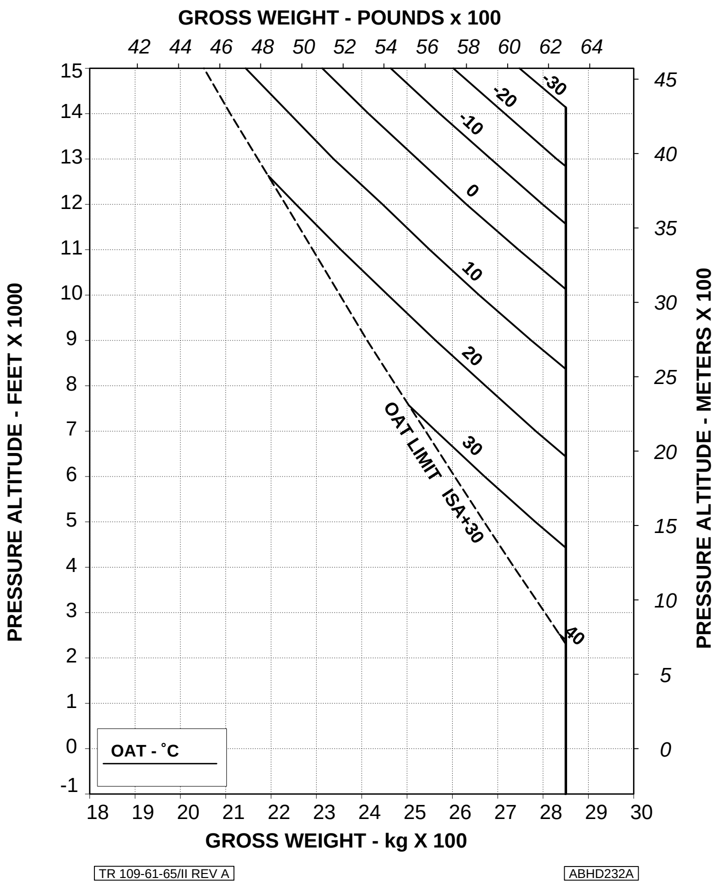


Figure 4-8. Hovering ceiling - OGE - take off power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

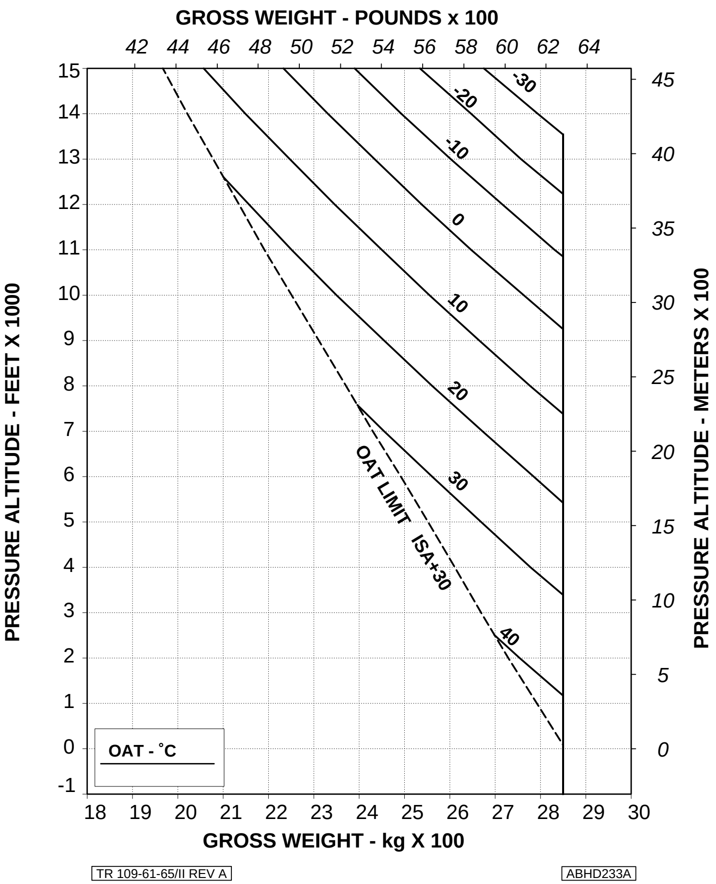


Figure 4-9. Hovering ceiling - OGE - take off power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

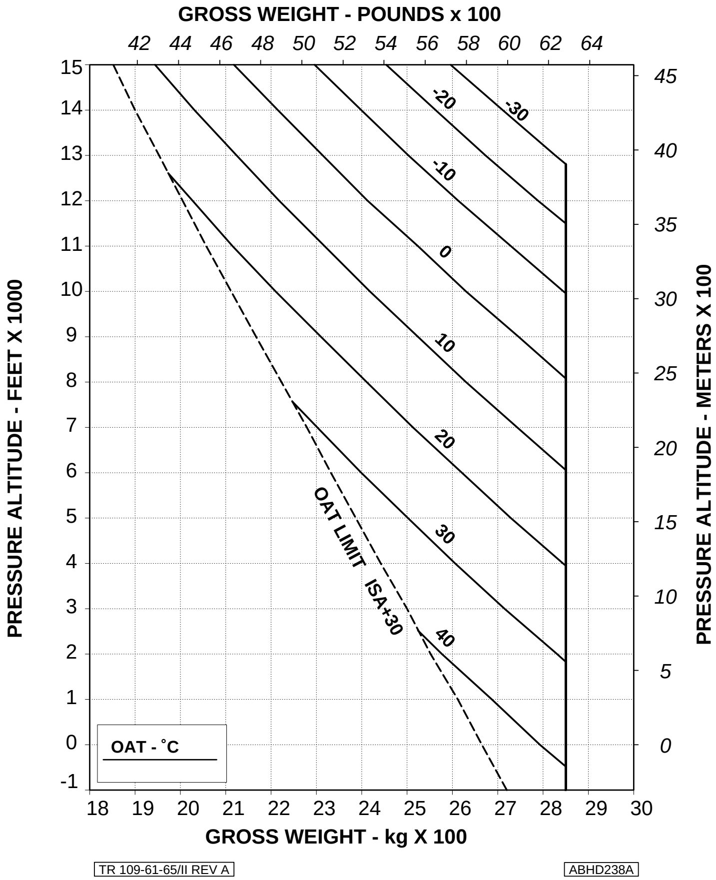


Figure 4-10. Hovering ceiling - OGE - maximum continuous power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

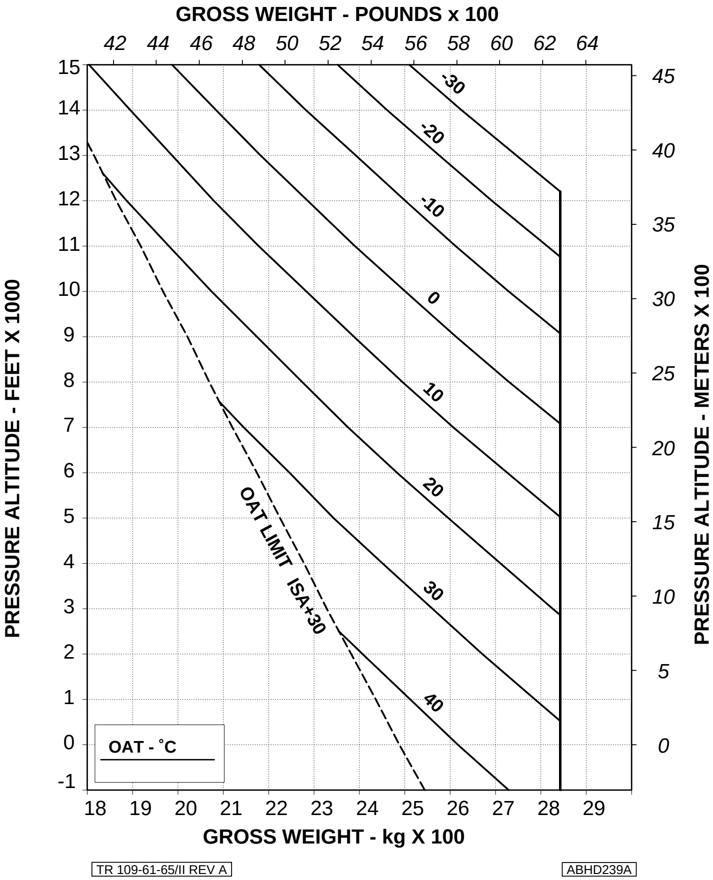


Figure 4-11. Hovering ceiling - OGE - maximum continuous power - EAPS ON.

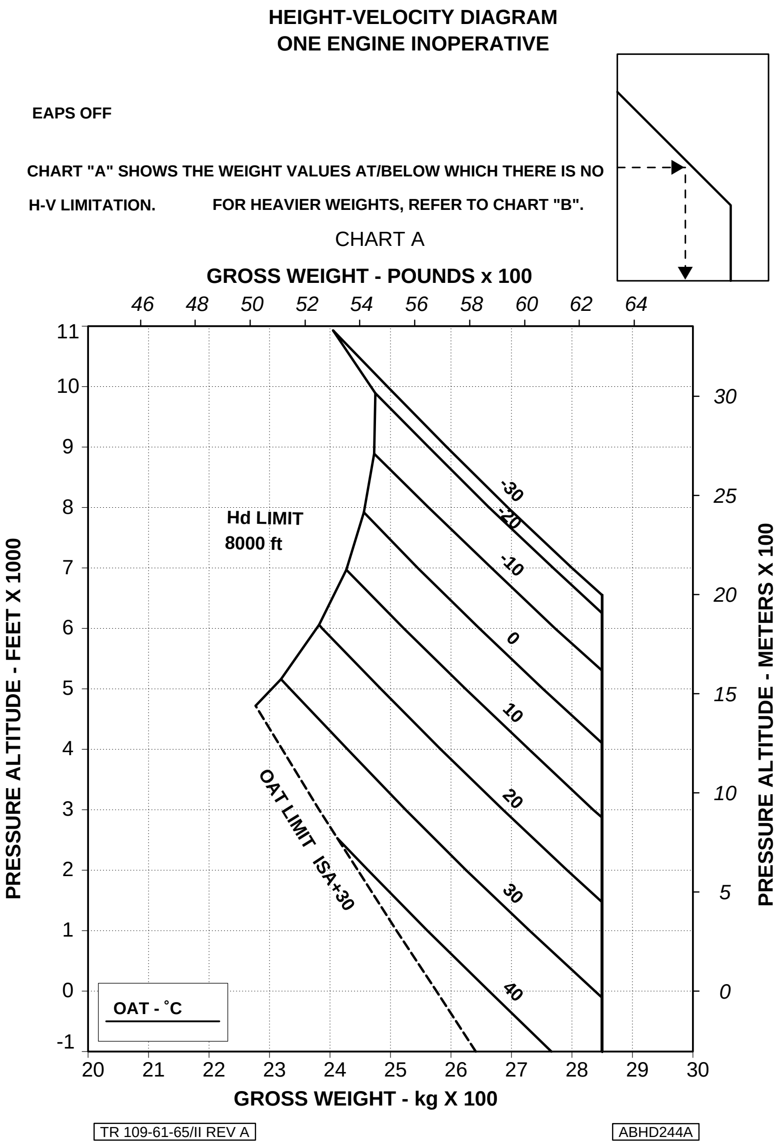


Figure 4-12 (sheet 1 of 2). Height - velocity diagram - EAPS OFF.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

(FOR SMOOTH, LEVEL AND HARD SURFACE)

EAPS OFF

CHART B

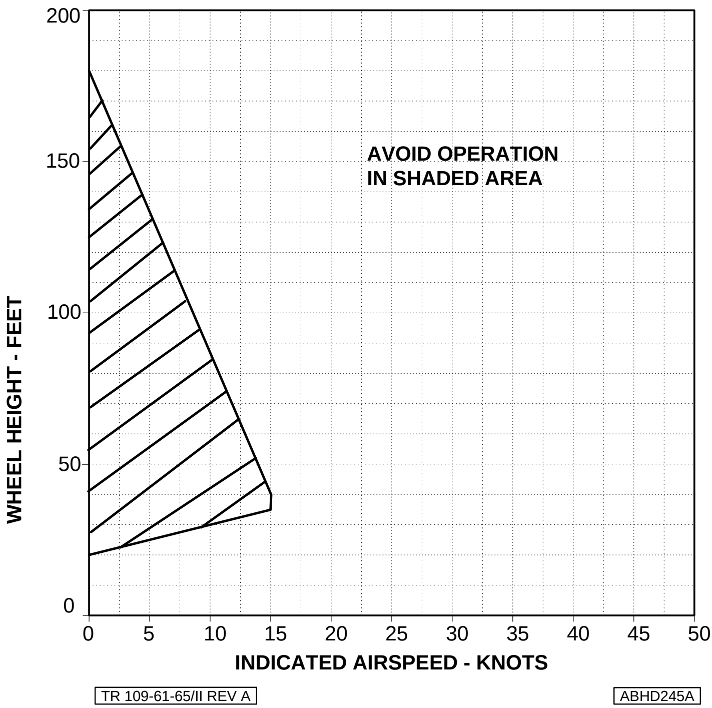


Figure 4-12 (sheet 2 of 2). Height - velocity diagram - EAPS OFF.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

EAPS ON

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V LIMITATION.
FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

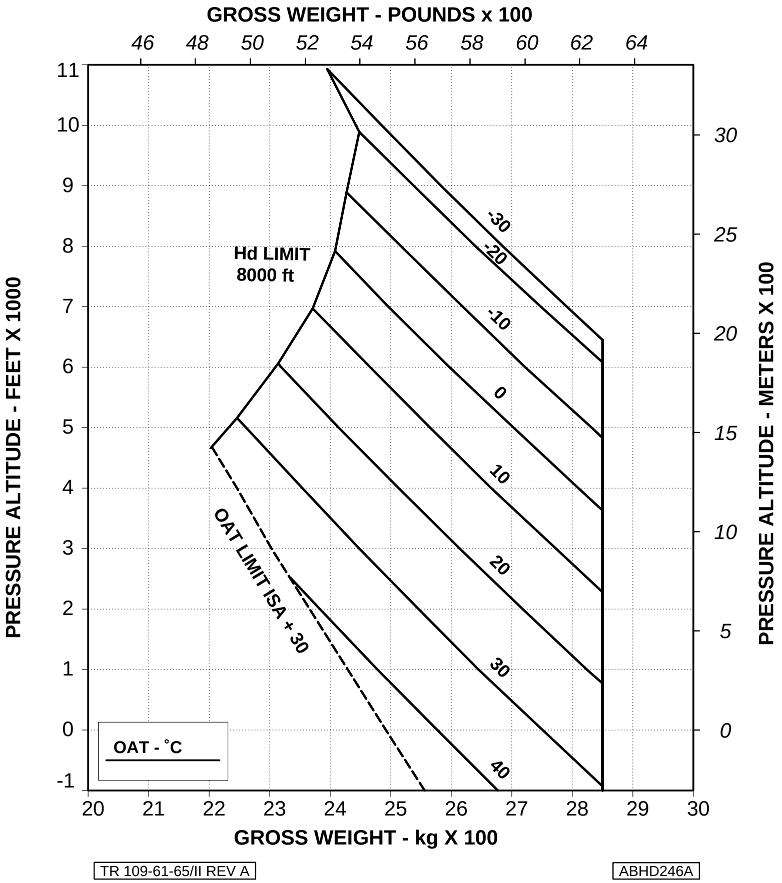


Figure 4-13 (sheet 1 of 2). Height - velocity diagram - EAPS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

(FOR SMOOTH, LEVEL AND HARD SURFACE)

EAPS ON

CHART B

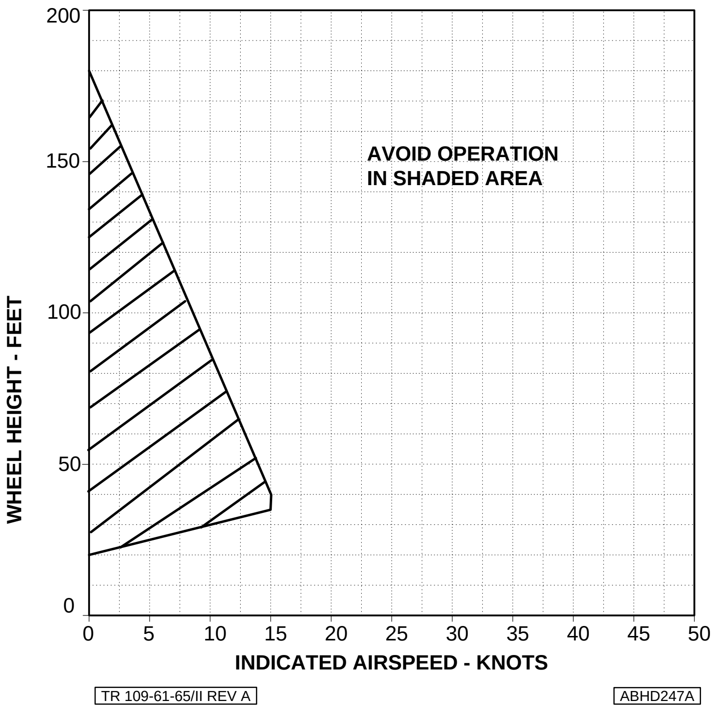


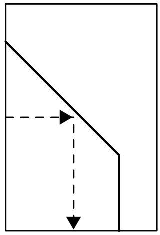
Figure 4-13 (sheet 2 of 2). Height - velocity diagram - EAPS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 350 ft/min (1.78 m/s)



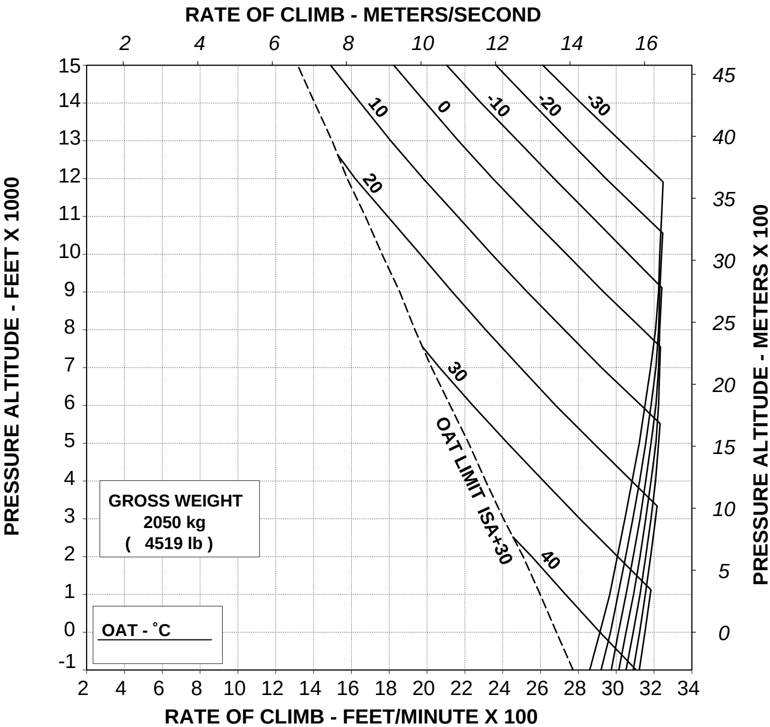


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Figure 4-14. Rate of climb - all engines - take off power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 300 ft/min (1.52 m/s)

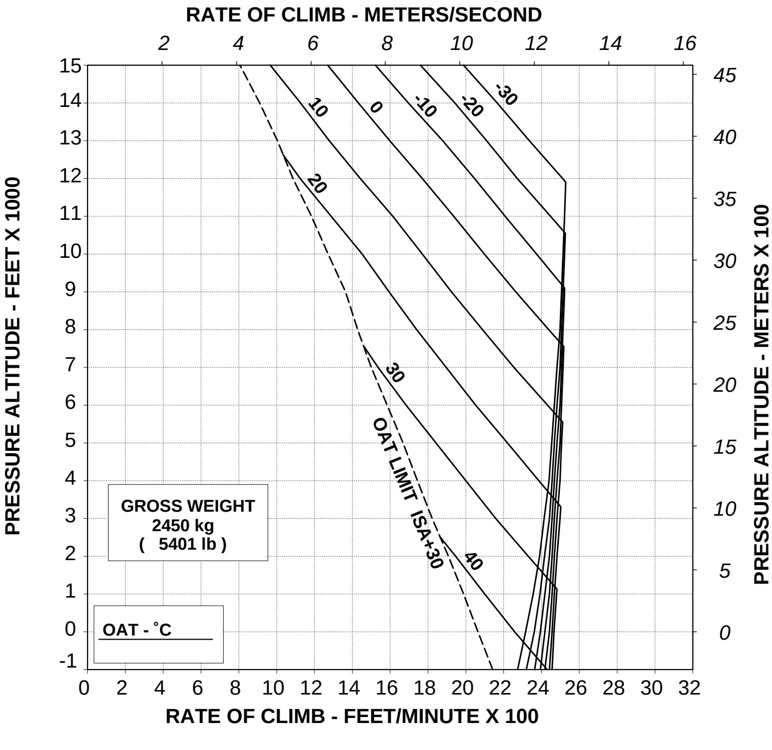

Figure 4-15. Rate of climb - all engines - take off power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS

EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 250 ft/min (1.27 m/s)

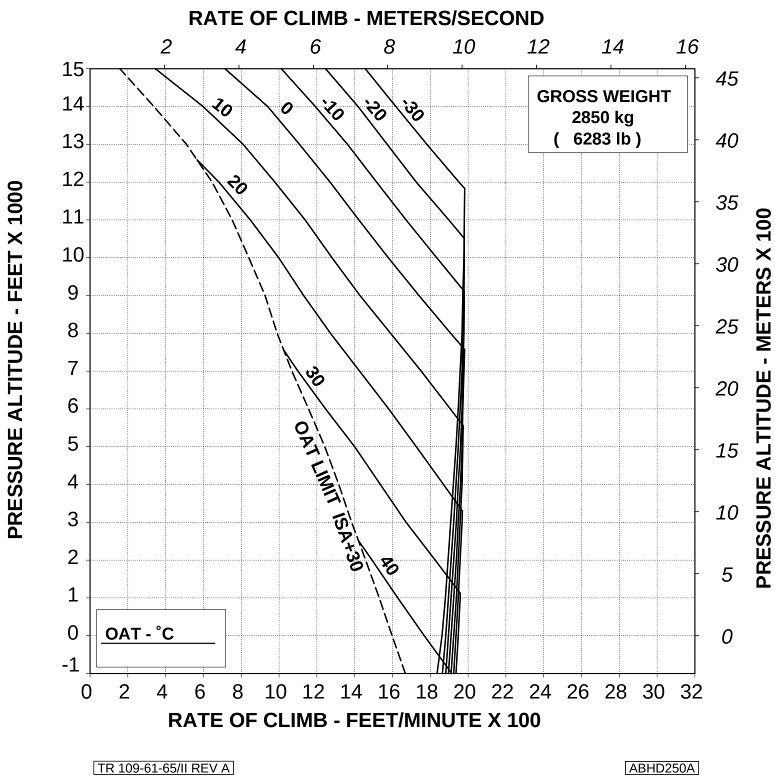

Figure 4-16. Rate of climb - all engines - take off power - 2850 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | **60 kts IAS**
EAPS OFF | **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 300 ft/min (1.52 m/s)

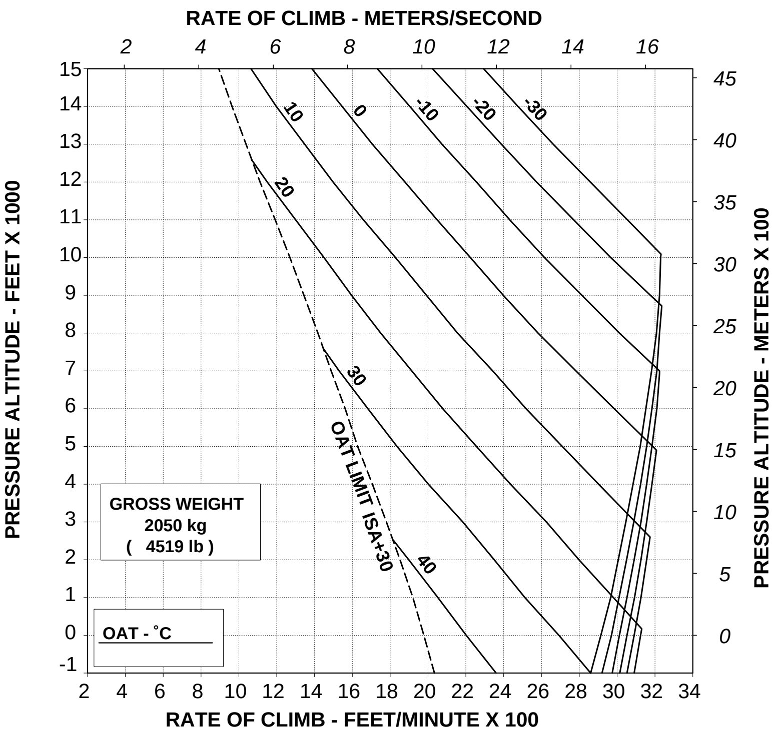

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Figure 4-17. Rate of climb - all engines - maximum continuous power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 250 ft/min (1.27 m/s)

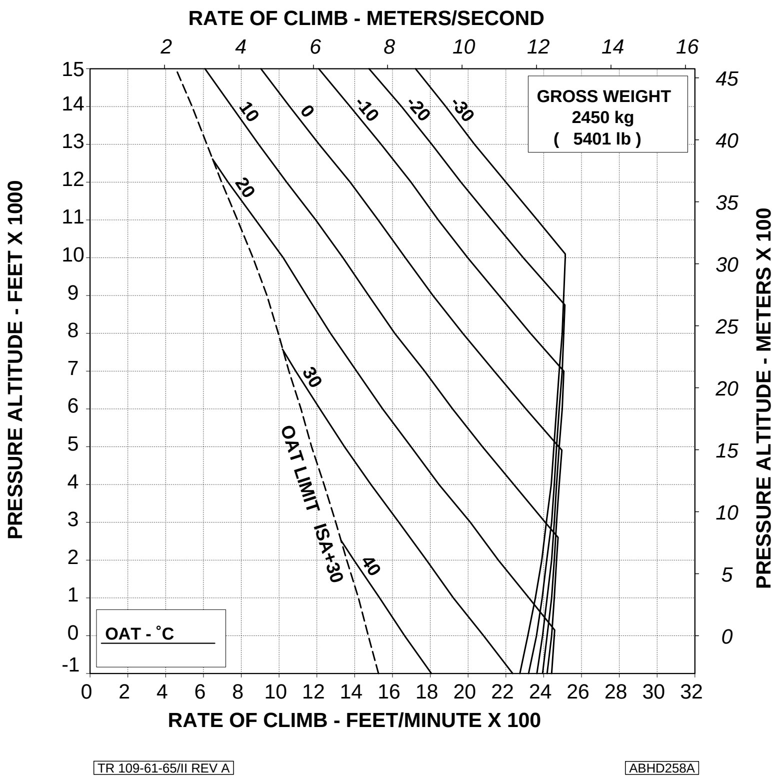


Figure 4-18. Rate of climb - all engines - maximum continuous power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 250 ft/min (1.27 m/s)

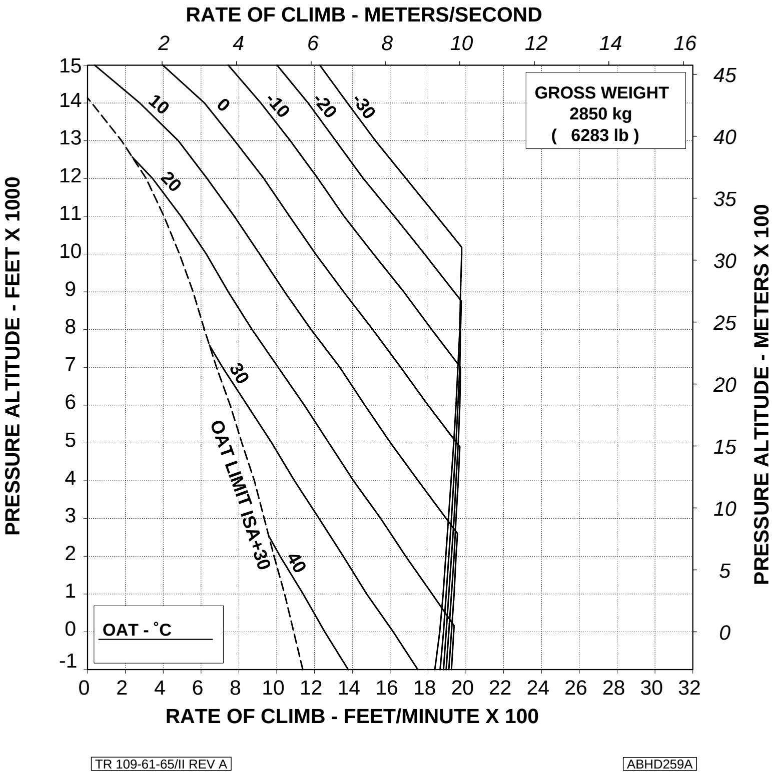
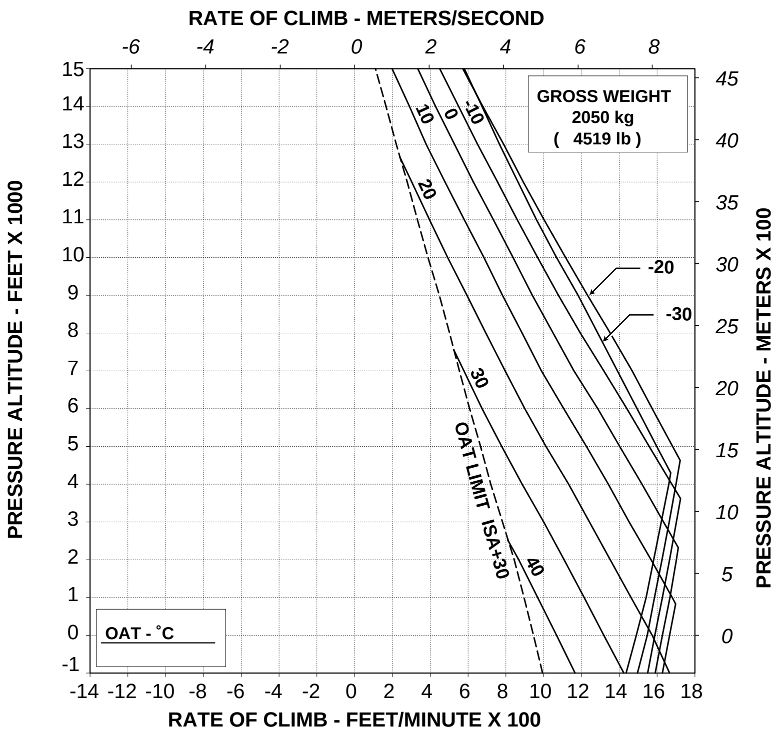


Figure 4-19. Rate of climb - all engines - maximum continuous power - 2850 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 175 ft/min (0.89 m/s)



TR 109-61-65/II REV A ABHD266A

Figure 4-20. Rate of climb - OEI - 2.5 minutes power - 2050 kg - EAPS OFF/ON.

AGUSTA

RATE OF CLIMB ONE ENGINE INOPERATIVE 2.5 MINUTE POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 150 ft/min (0.76 m/s)

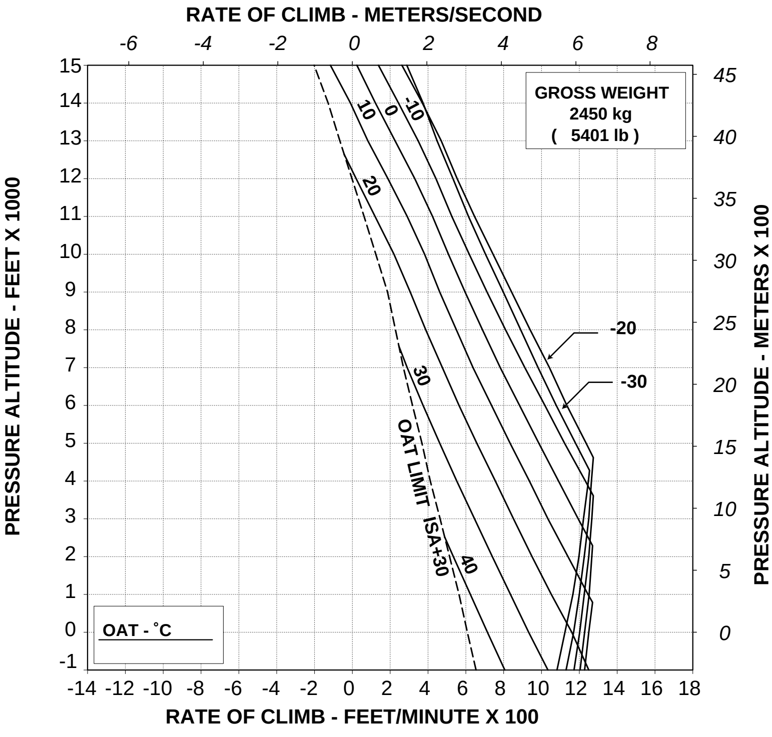


Figure 4-21. Rate of climb - OEI - 2.5 minutes power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % | **60 kts IAS**
EAPS OFF | **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 125 ft/min (0.64 m/s)

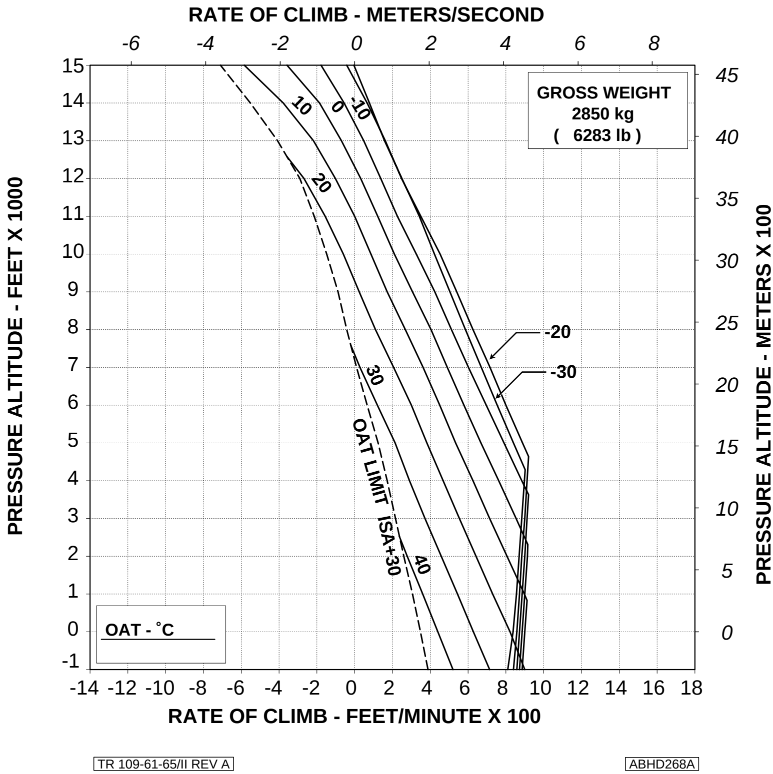


Figure 4-22. Rate of climb - OEI - 2.5 minutes power - 2850 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 150 ft/min (0.76 m/s)

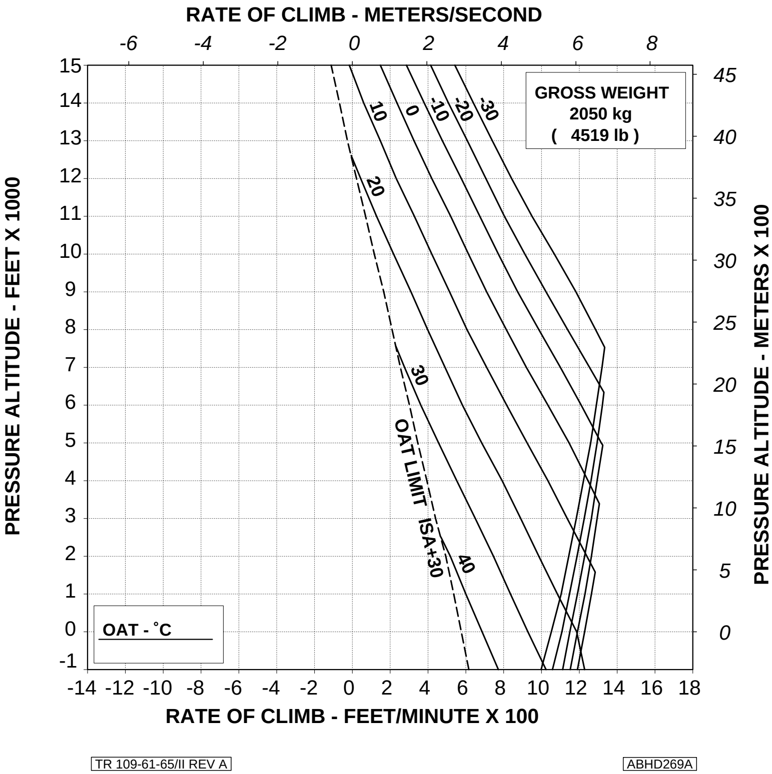


Figure 4-23. Rate of climb - OEI - maximum continuous power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON : DECREASE RATE OF CLIMB BY 150 ft/min (0.76 m/s)

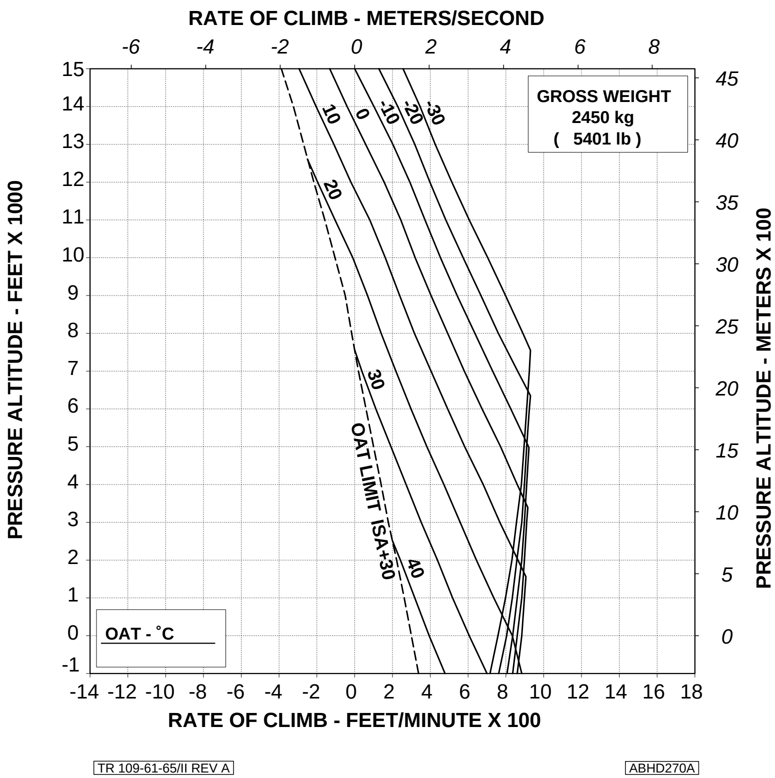


Figure 4-24. Rate of climb - OEI - maximum continuous power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % — 60 kts IAS

EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 125 ft/min (0.64 m/s)

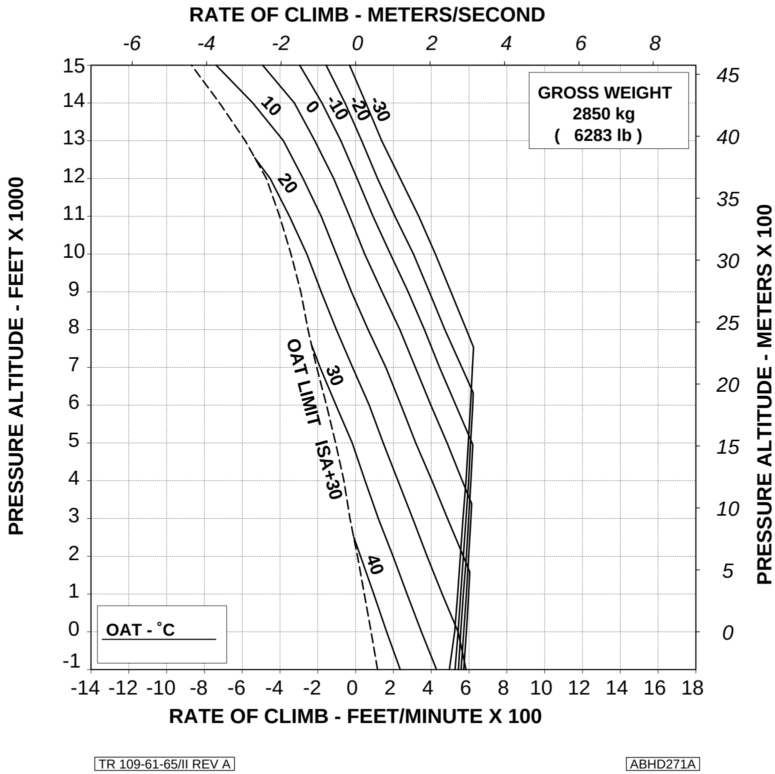


Figure 4-25. Rate of climb - OEI - maximum continuous power - 2850 kg - EAPS OFF/ON.

Helicopters in basic configuration with bleed air heater or ECS (refer also to Appendix 5 or 6)

— Figure 4-26. Hovering ceiling - IGE - take off power - EAPS OFF/ON - heater or ECS ON.
— Figure 4-27. Hovering ceiling - IGE - maximum continuous power - EAPS OFF/ON - heater or ECS ON.
— Figure 4-28. Hovering ceiling - OGE - take off power - EAPS OFF/ON - heater or ECS ON.
— Figure 4-29. Hovering ceiling - OGE - maximum continuous power - EAPS OFF/ON - heater or ECS ON.
— Figure 4-30. Rate of climb - all engines - take off power - 2050 kg - EAPS OFF/ON - heater or ECS ON.
— Figure 4-31. Rate of climb - all engines - take off power - 2450 kg - EAPS OFF/ON - heater or ECS ON.
— Figure 4-32. Rate of climb - all engines - take off power - 2850 kg - EAPS OFF/ON - heater or ECS ON.
— Figure 4-33. Rate of climb - all engines - maximum continuous power - 2050 kg - EAPS OFF/ON - heater or ECS ON.
— Figure 4-34. Rate of climb - all engines - maximum continuous power - 2450 kg - EAPS OFF/ON - heater or ECS ON.
— Figure 4-35. Rate of climb - all engines - maximum continuous power - 2850 kg - EAPS OFF/ON - heater or ECS ON.

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
HEATER OR E.C.S. ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE GROSS WEIGHT BY 40 kg (88 lb)

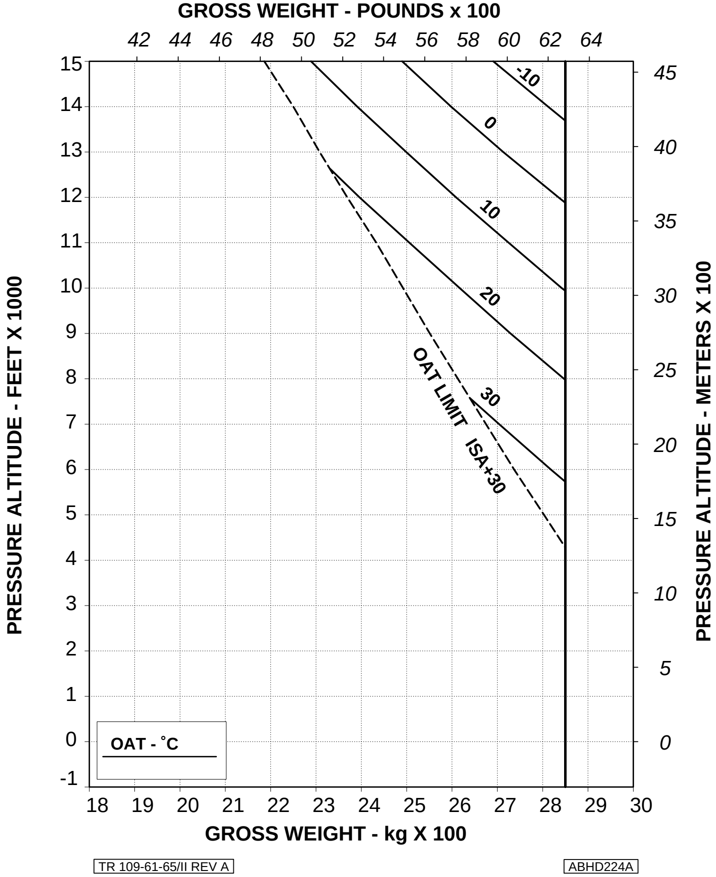


Figure 4-26. Hovering ceiling - IGE - take off power - EAPS OFF/ON - heater or ECS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
HEATER OR E.C.S. ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE GROSS WEIGHT BY 50 kg (110 lb)

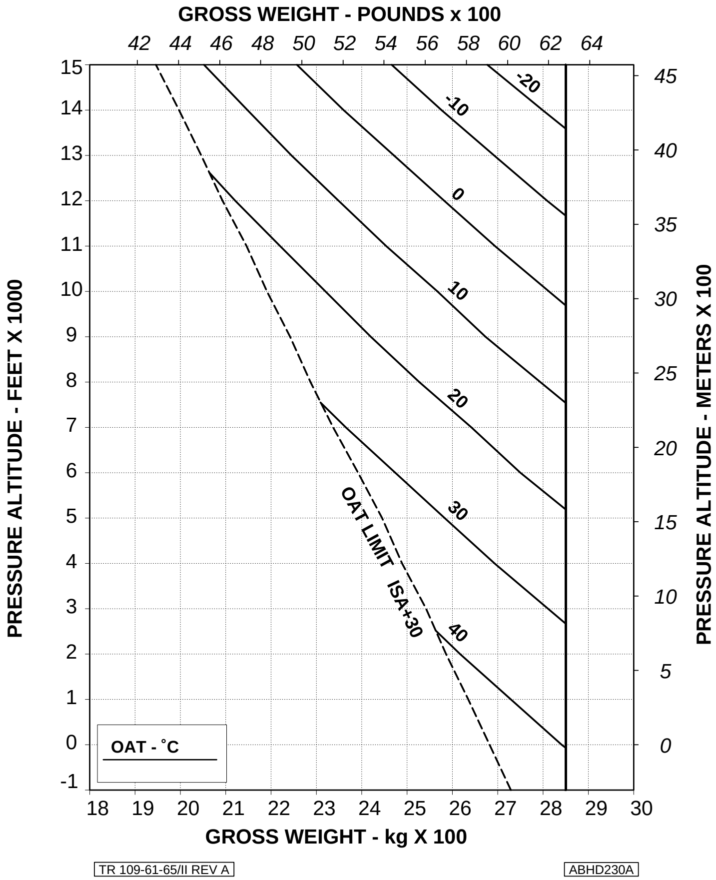


Figure 4-27. Hovering ceiling - IGE - maximum continuous power - EAPS OFF/ON - heater or ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
HEATER OR E.C.S. ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION:OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

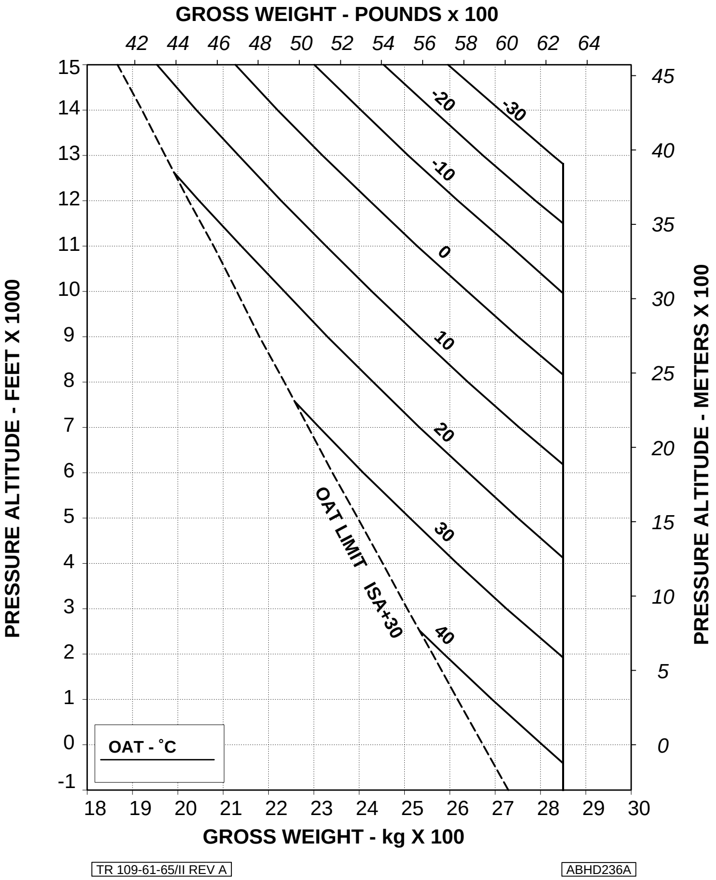


Figure 4-28. Hovering ceiling - OGE - take off power - EAPS OFF/ON - heater or ECS ON.

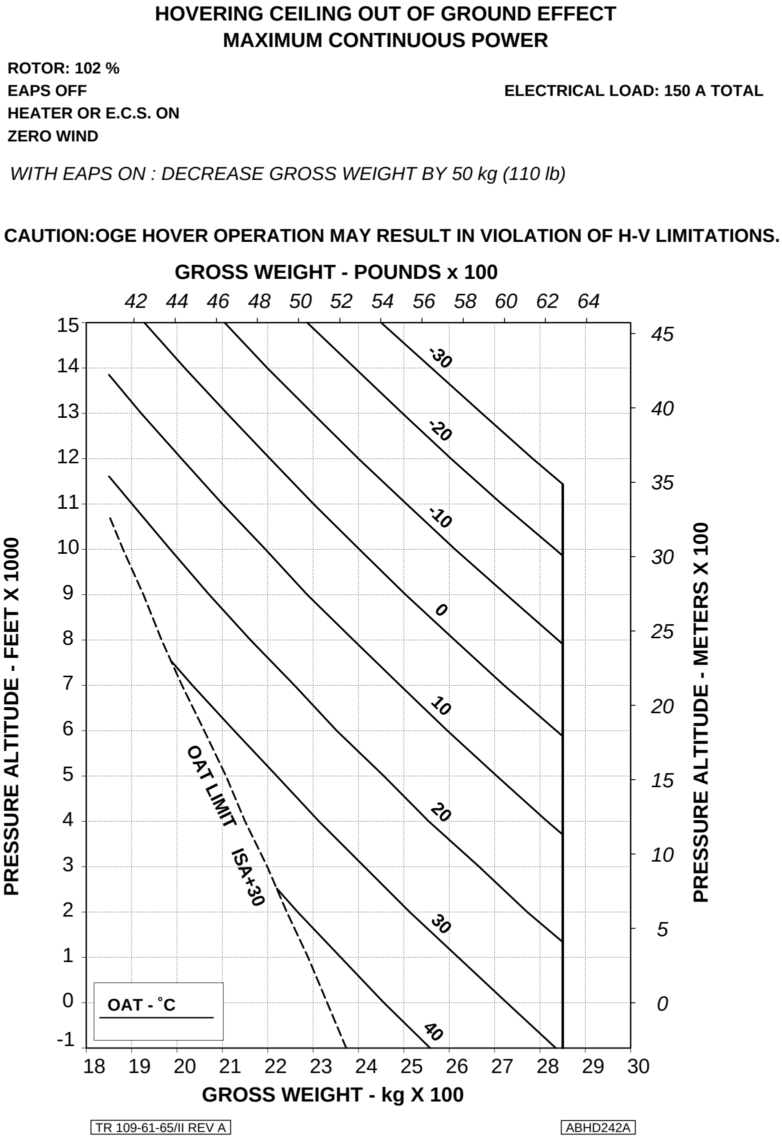


Figure 4-29. Hovering ceiling - OGE - maximum continuous power - EAPS OFF/ON - heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**
HEATER OR E.C.S. ON

WITH EAPS ON : DECREASE RATE OF CLIMB BY 100 ft/min (0.51 m/s)

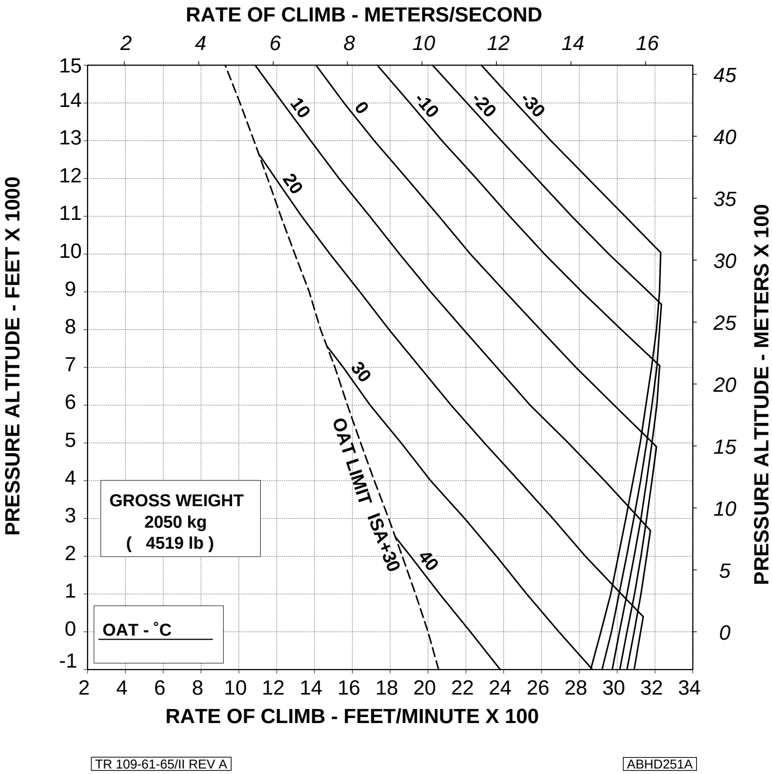


Figure 4-30. Rate of climb - all engines - take off power - 2050 kg - EAPS OFF/ON - heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON : DECREASE RATE OF CLIMB BY 50 ft/min (0.25 m/s)

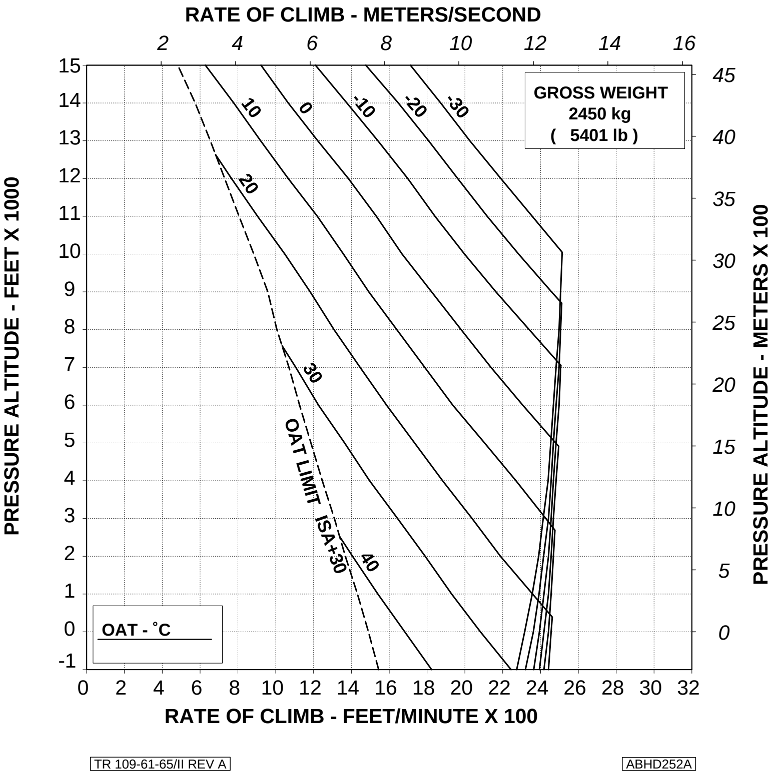


Figure 4-31. Rate of climb - all engines - take off power - 2450 kg - EAPS OFF/ON - heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — **60 kts IAS**
EAPS OFF — **ELECTRICAL LOAD: 150 A TOTAL**
HEATER OR E.C.S. ON

WITH EAPS ON : DECREASE RATE OF CLIMB BY 50 ft/min (0.25 m/s)

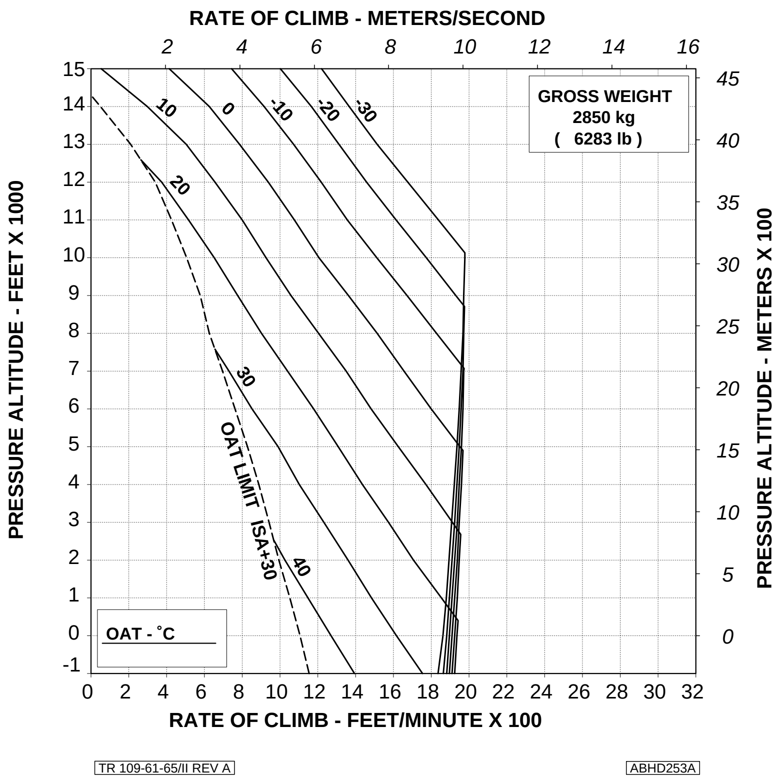


Figure 4-32. Rate of climb - all engines - take off power - 2850 kg - EAPS OFF/ON - heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**
HEATER OR E.C.S. ON

WITH EAPS ON : DECREASE RATE OF CLIMB BY 100 ft/min (0.51 m/s)

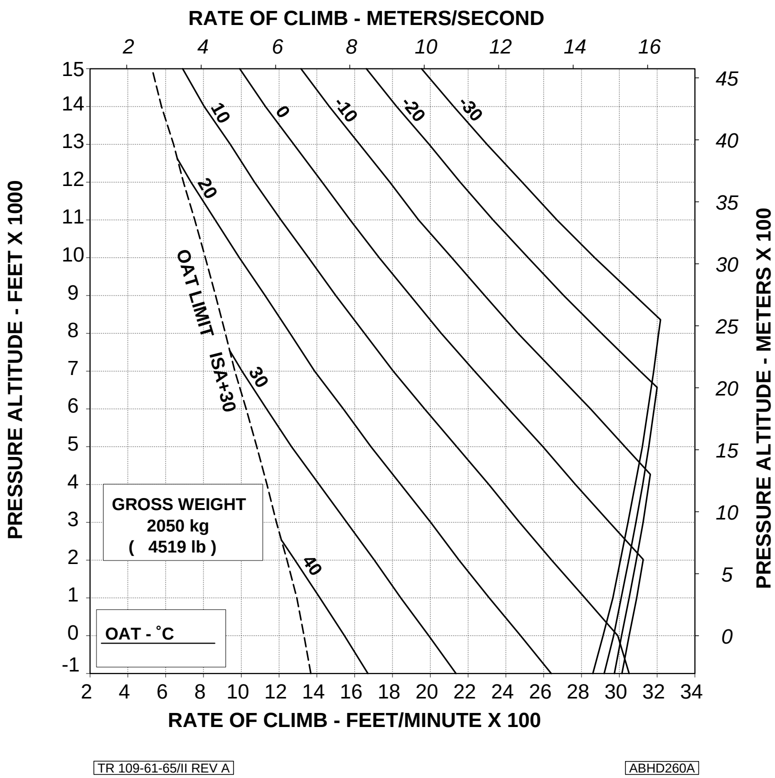


Figure 4-33. Rate of climb - all engines - maximum continuous power - 2050 kg - EAPS OFF/ON - heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | **60 kts IAS**
EAPS OFF | **ELECTRICAL LOAD: 150 A TOTAL**
HEATER OR E.C.S. ON

WITH EAPS ON : DECREASE RATE OF CLIMB BY 100 ft/min (0.51 m/s)

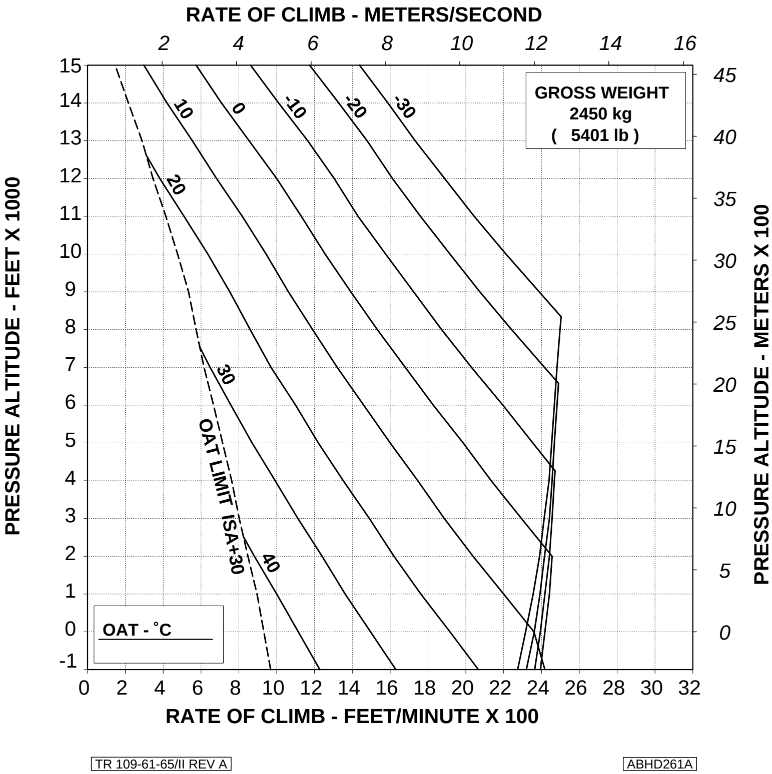


Figure 4-34. Rate of climb - all engines - maximum continuous power - 2450 kg - EAPS OFF/ON - heater or ECS ON.

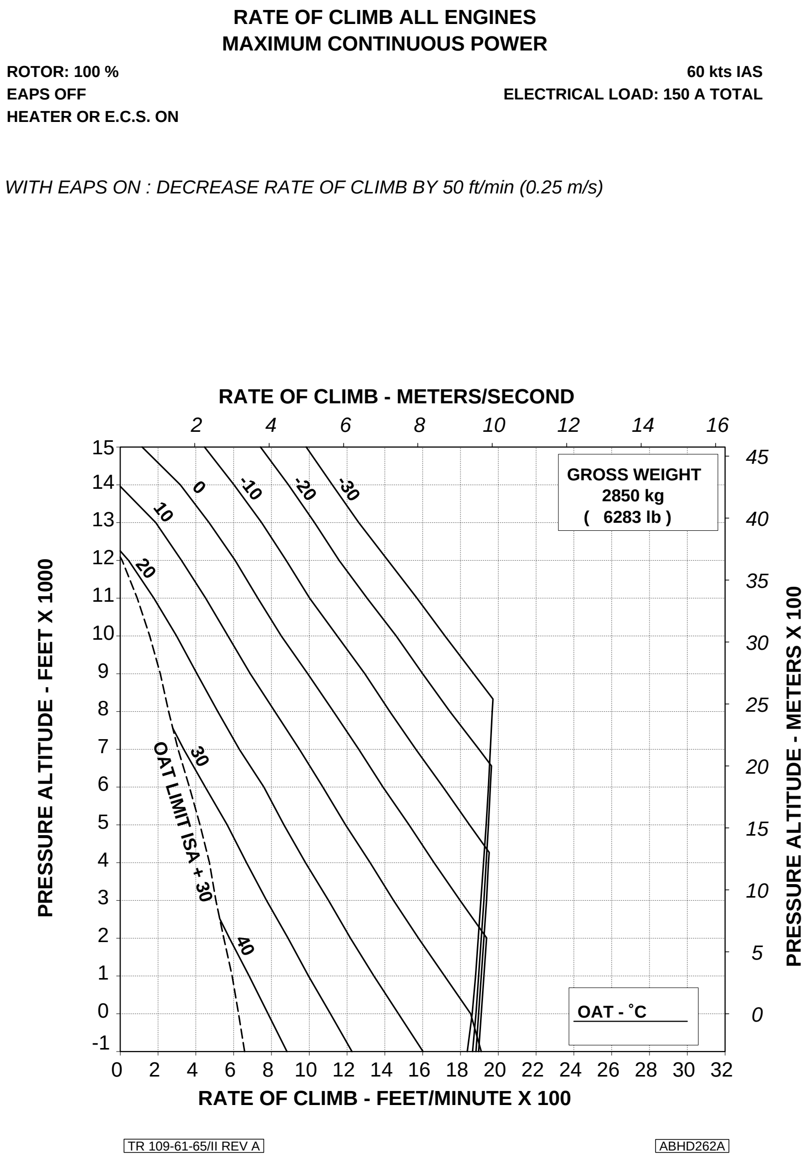


Figure 4-35. Rate of climb - all engines - maximum continuous power - 2850 kg - EAPS OFF/ON - heater or ECS ON.

Helicopters in basic configuration for Equivalent category ″A″ operations (refer also to Appendix 12)

— Figure 4-63. Rate of climb - OEI - 2.5 minutes power - 2650 kg - EAPS ON.
— Figure 4-64. Rate of climb - OEI - 2.5 minutes power - 2850 kg - EAPS OFF.
— Figure 4-65. Rate of climb - OEI - 2.5 minutes power - 2850 kg - EAPS ON.

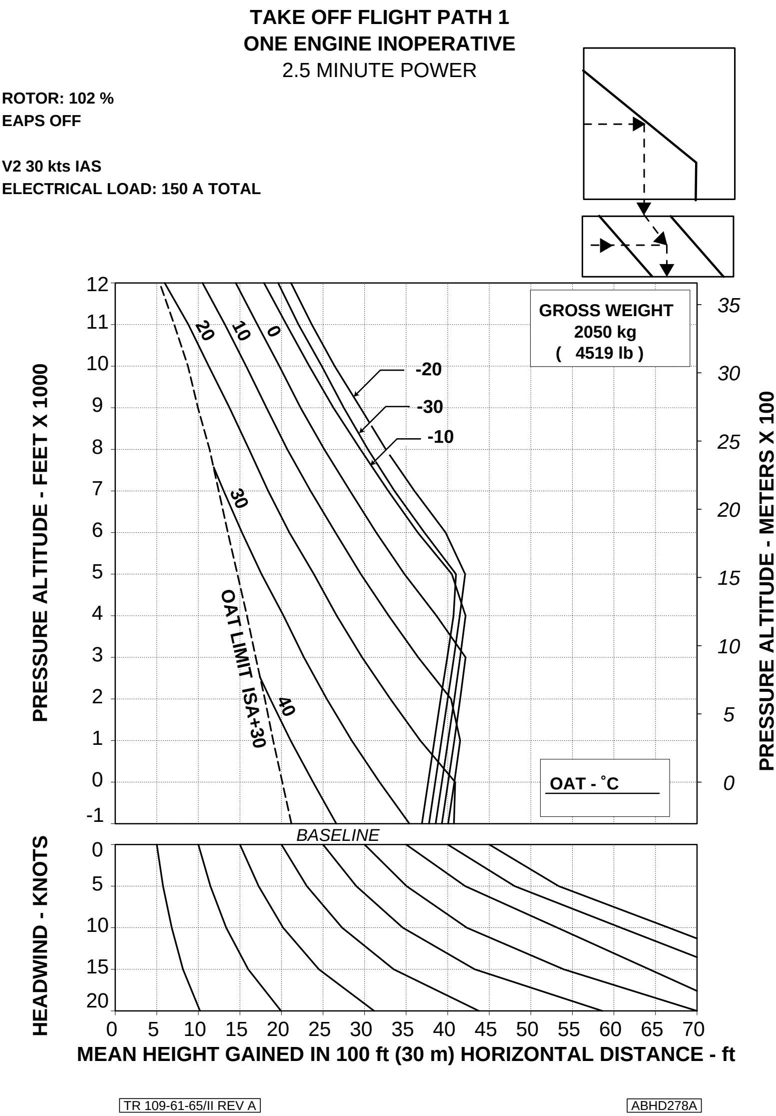


Figure 4-36. Takeoff flight path 1 - 2050 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

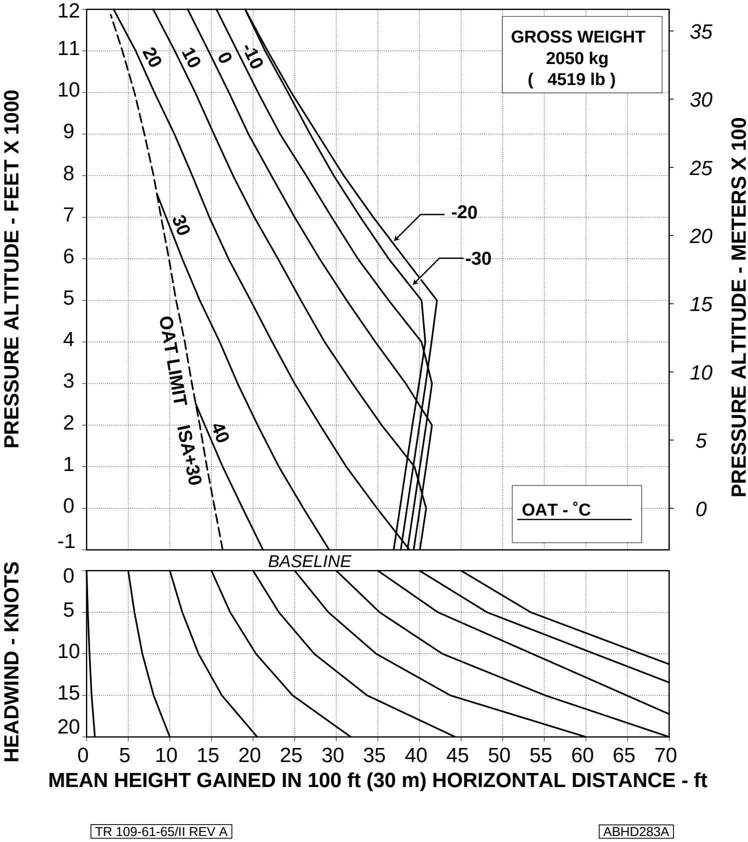


Figure 4-37. Takeoff flight path 1 - 2050 kg - EAPS ON.

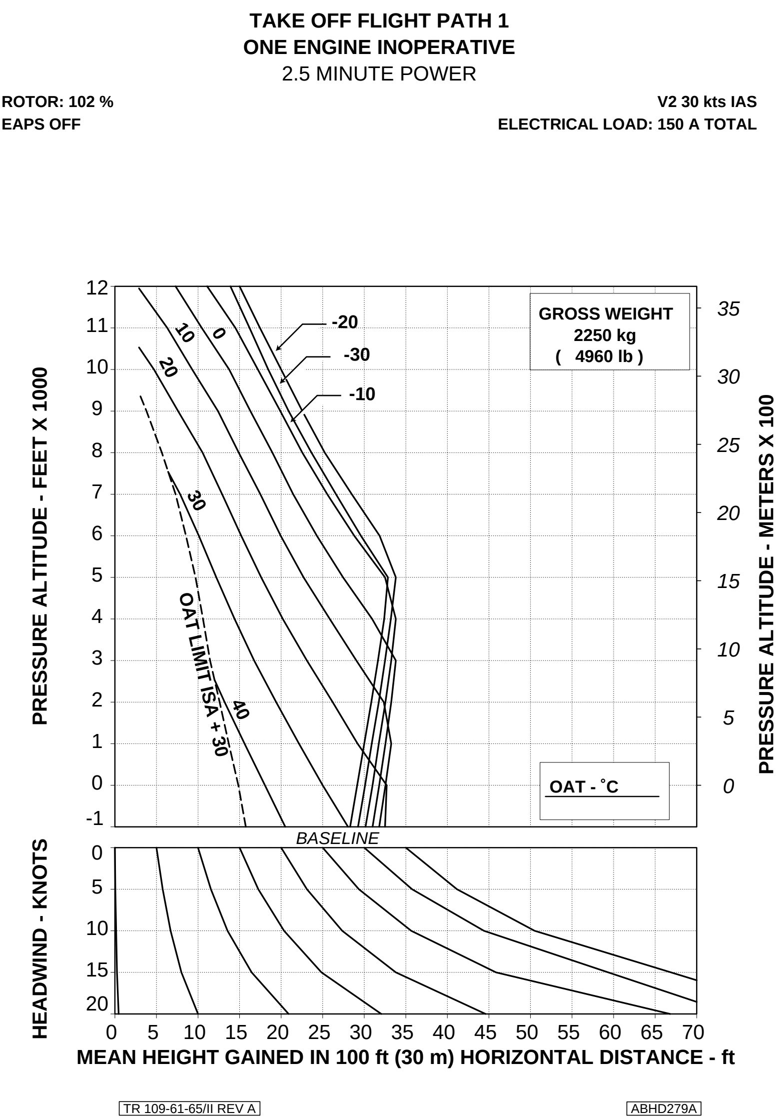


Figure 4-38. Takeoff flight path 1 - 2250 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS ON

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

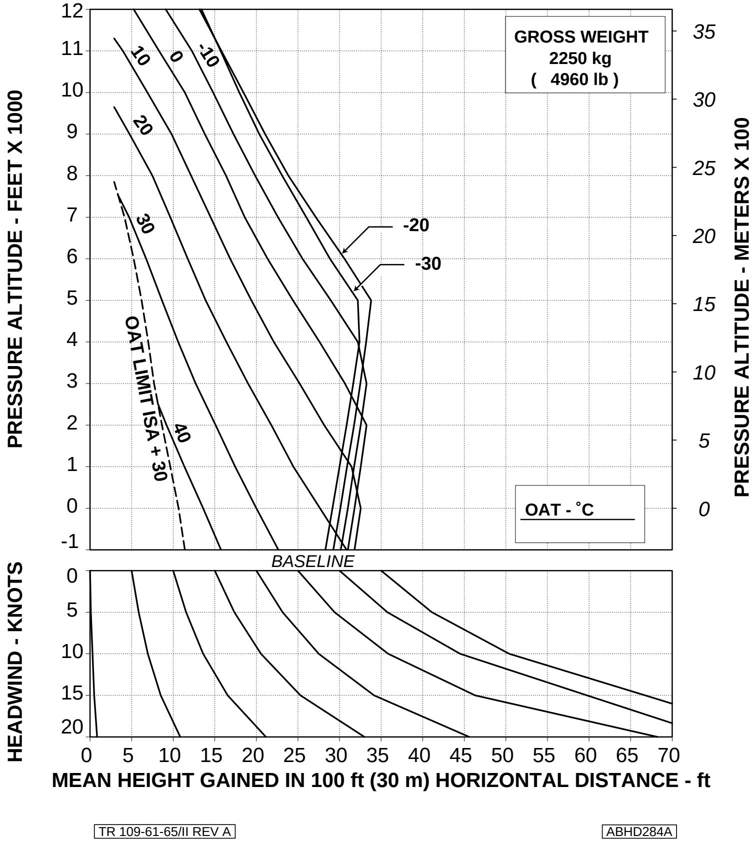


Figure 4-39. Takeoff flight path 1 - 2250 kg - EAPS ON.

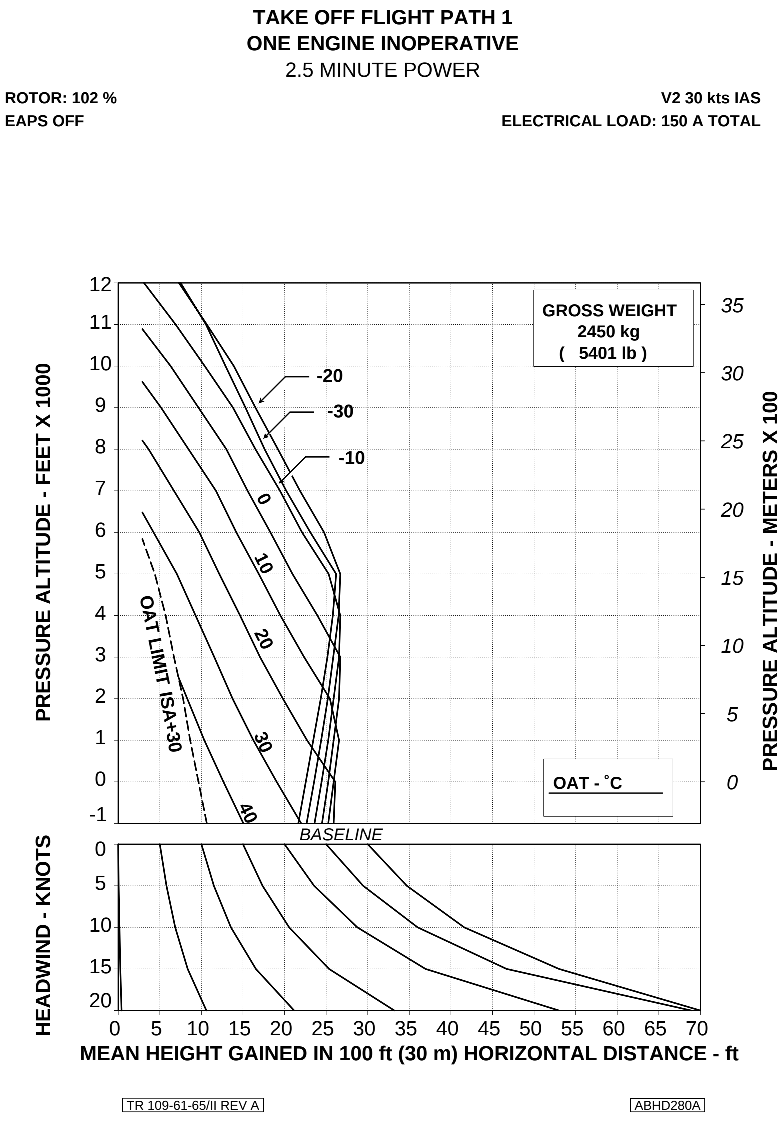


Figure 4-40. Takeoff flight path 1 - 2450 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

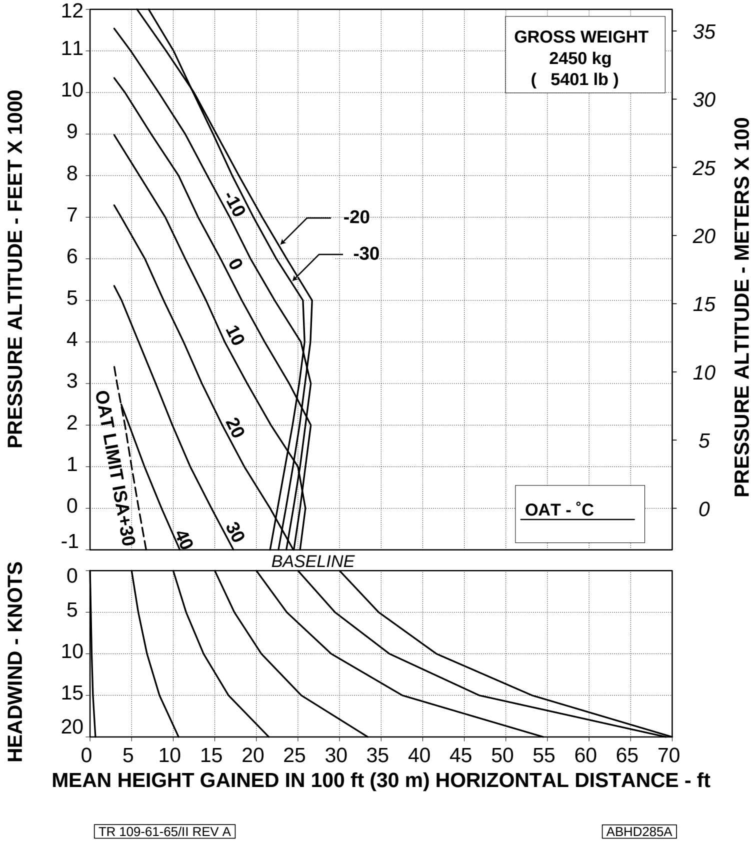


Figure 4-41. Takeoff flight path 1 - 2450 kg - EAPS ON.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

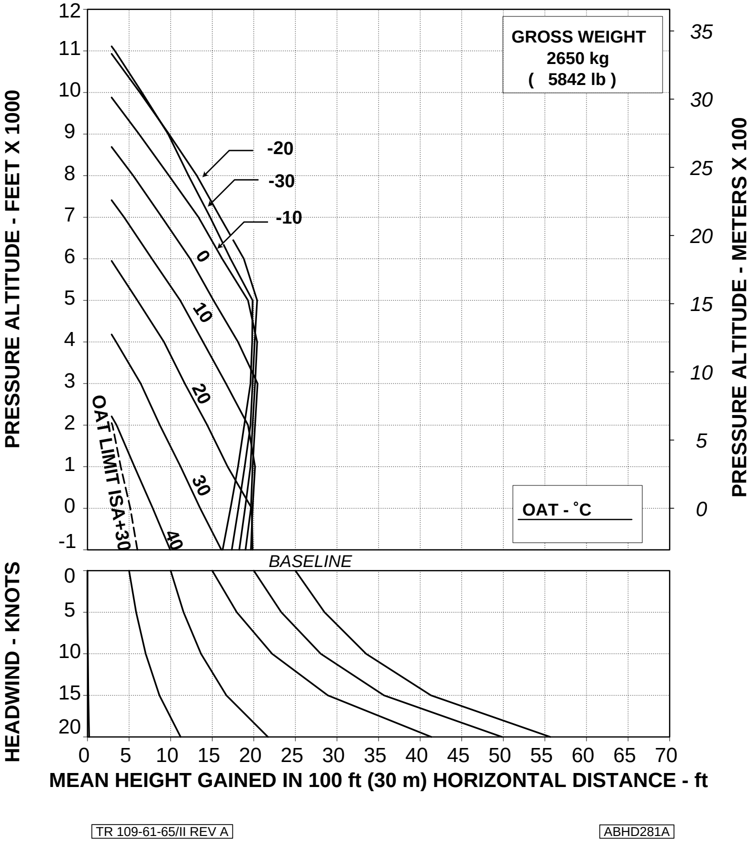


Figure 4-42. Takeoff flight path 1 - 2650 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

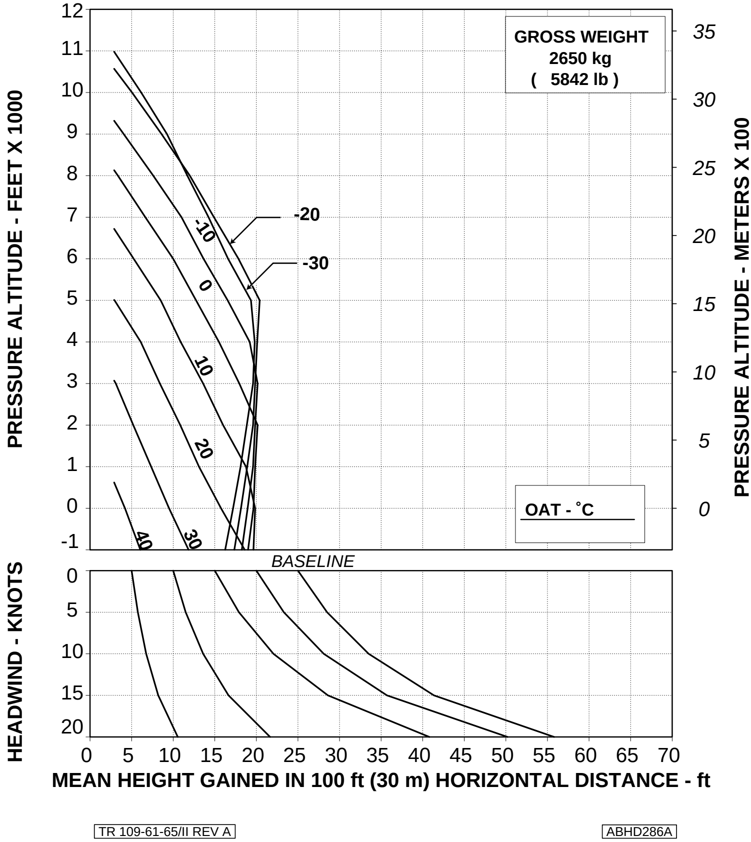


Figure 4-43. Takeoff flight path 1 - 2650 kg - EAPS ON.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

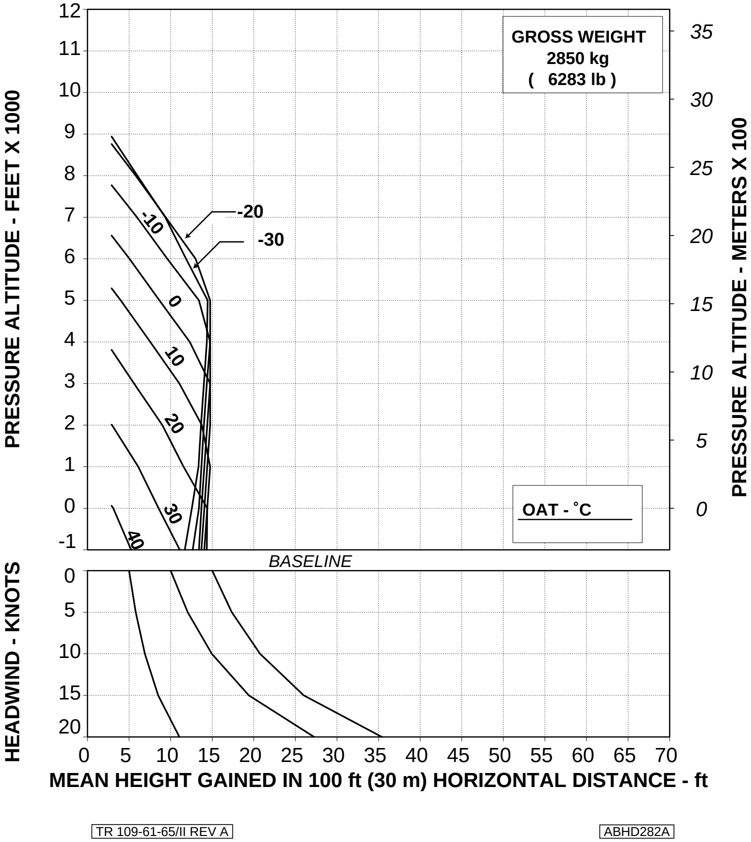


Figure 4-44. Takeoff flight path 1 - 2850 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS ON

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

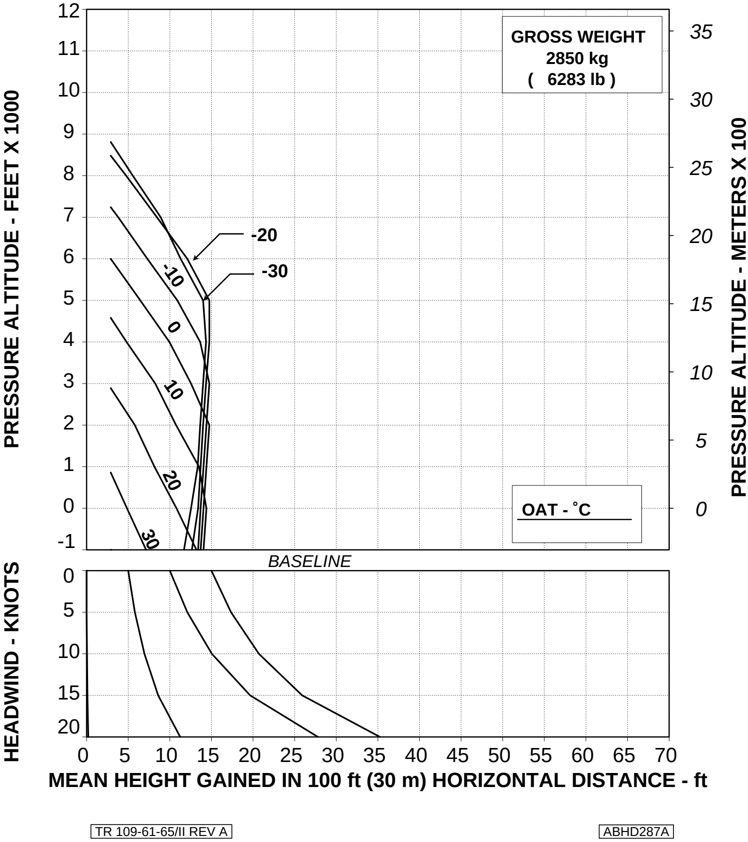


Figure 4-45. Takeoff flight path 1 - 2850 kg - EAPS ON.

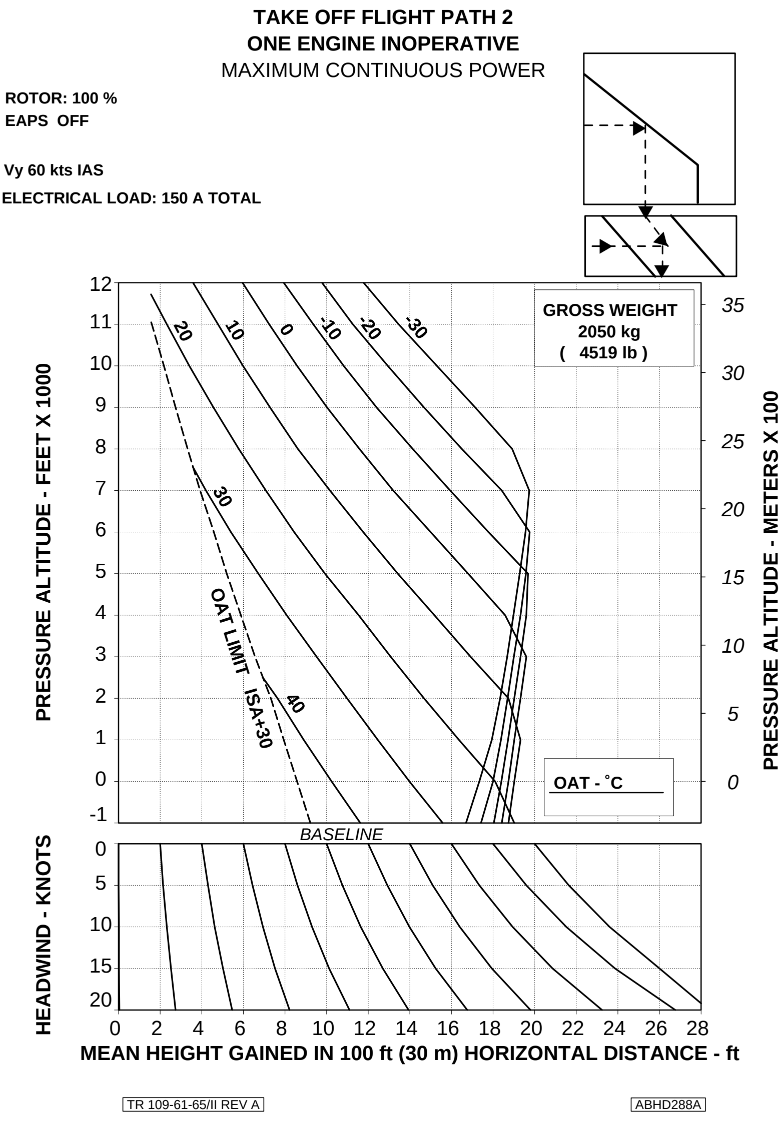


Figure 4-46. Takeoff flight path 2 - 2050 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

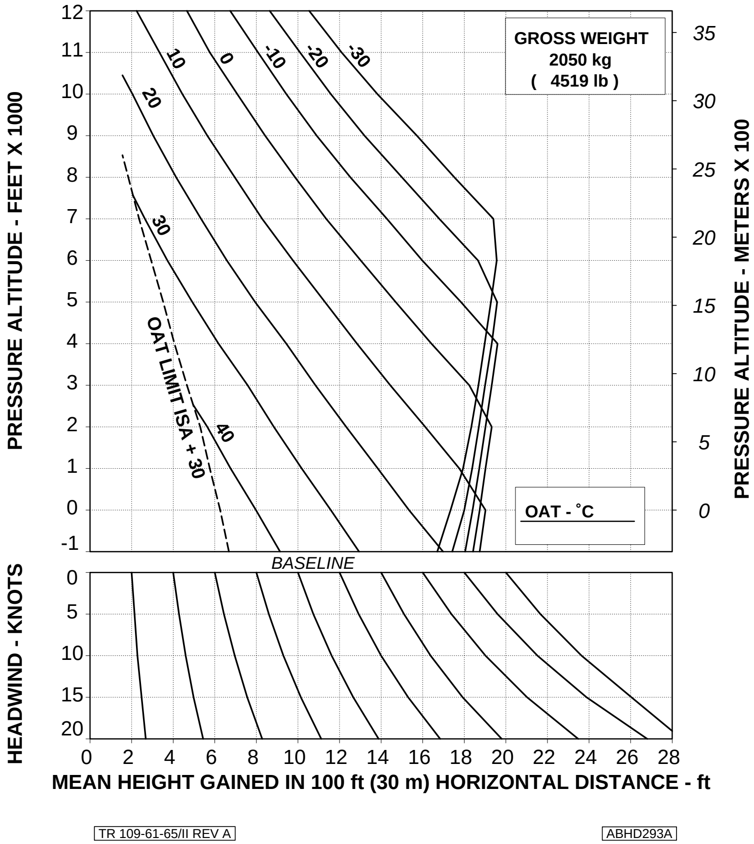


Figure 4-47. Takeoff flight path 2 - 2050 kg - EAPS ON.

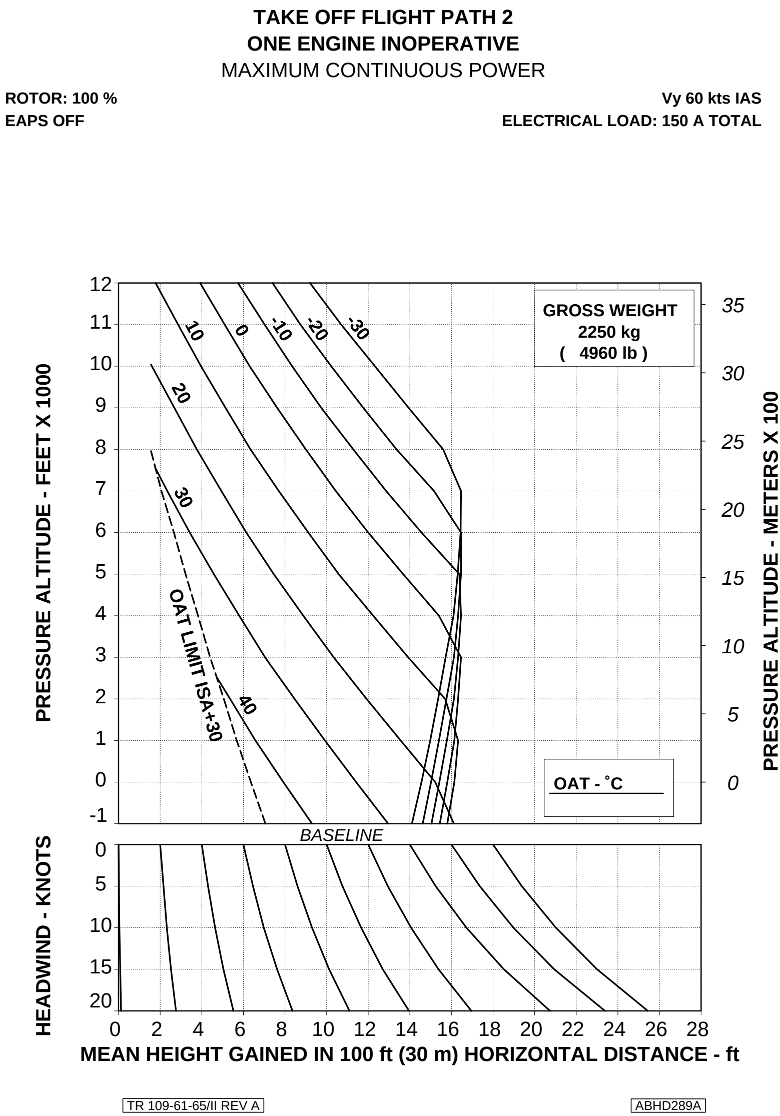


Figure 4-48. Takeoff flight path 2 - 2250 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

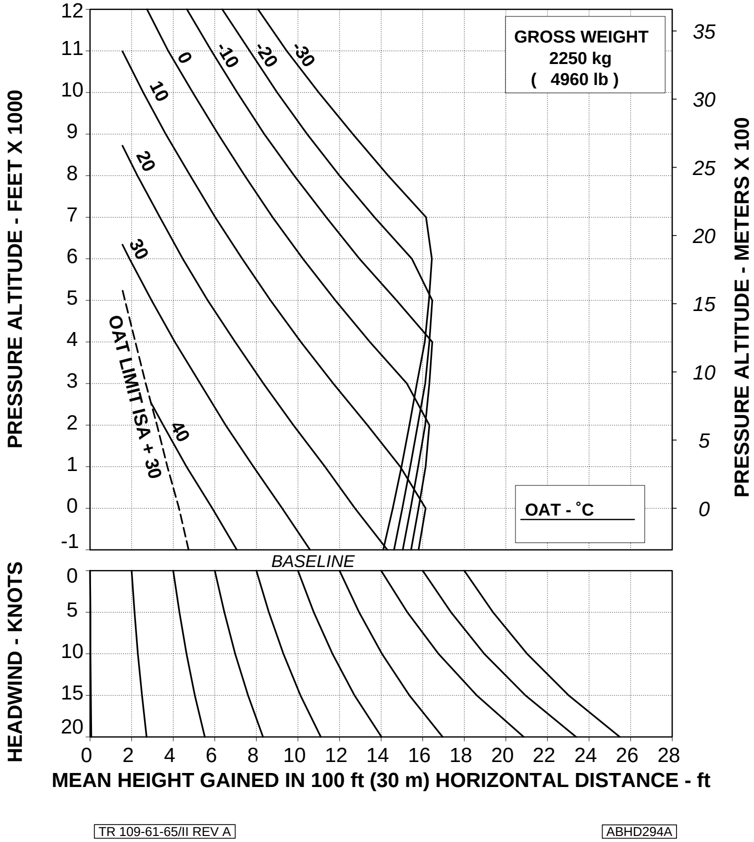


Figure 4-49. Takeoff flight path 2 - 2250 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

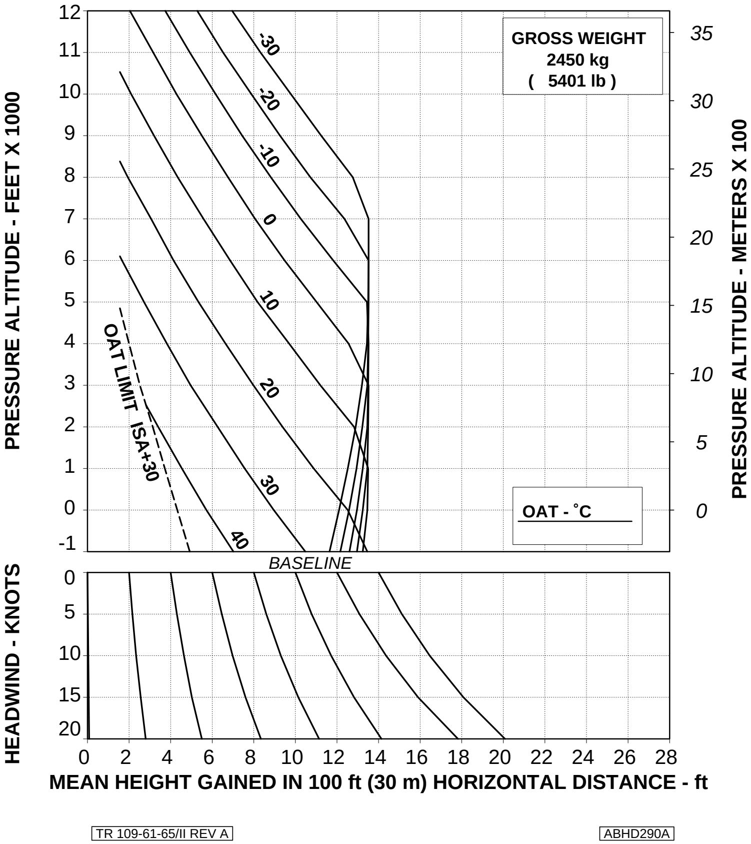


Figure 4-50. Takeoff flight path 2 - 2450 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

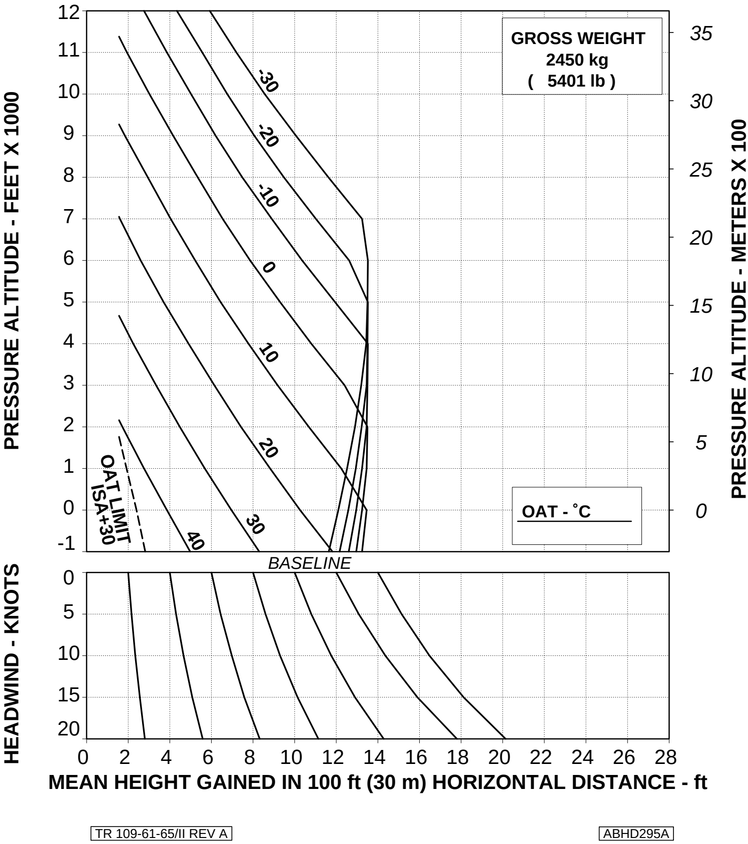


Figure 4-51. Takeoff flight path 2 - 2450 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

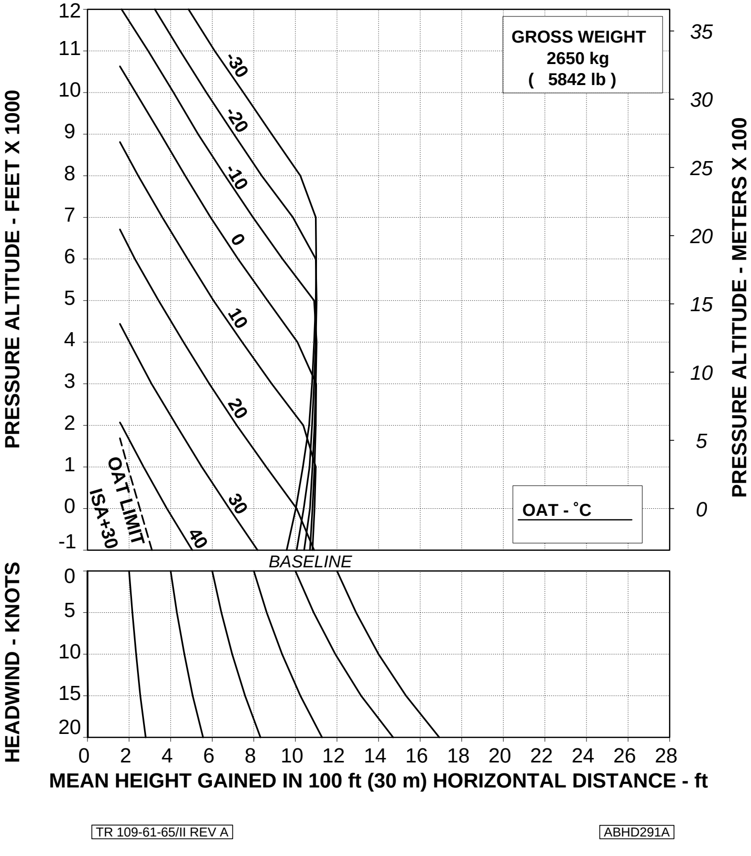


Figure 4-52. Takeoff flight path 2 - 2650 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

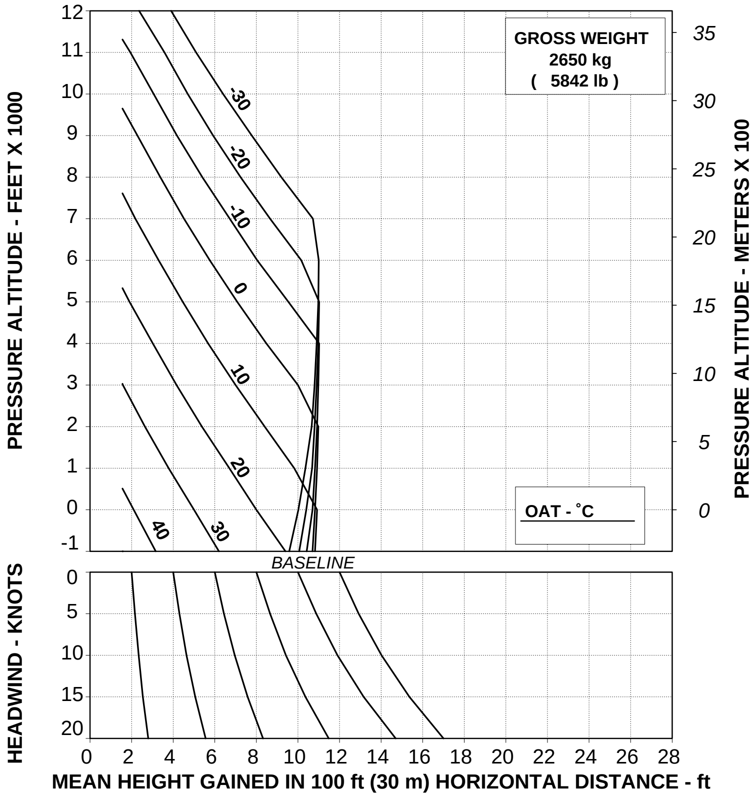


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Figure 4-53. Takeoff flight path 2 - 2650 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

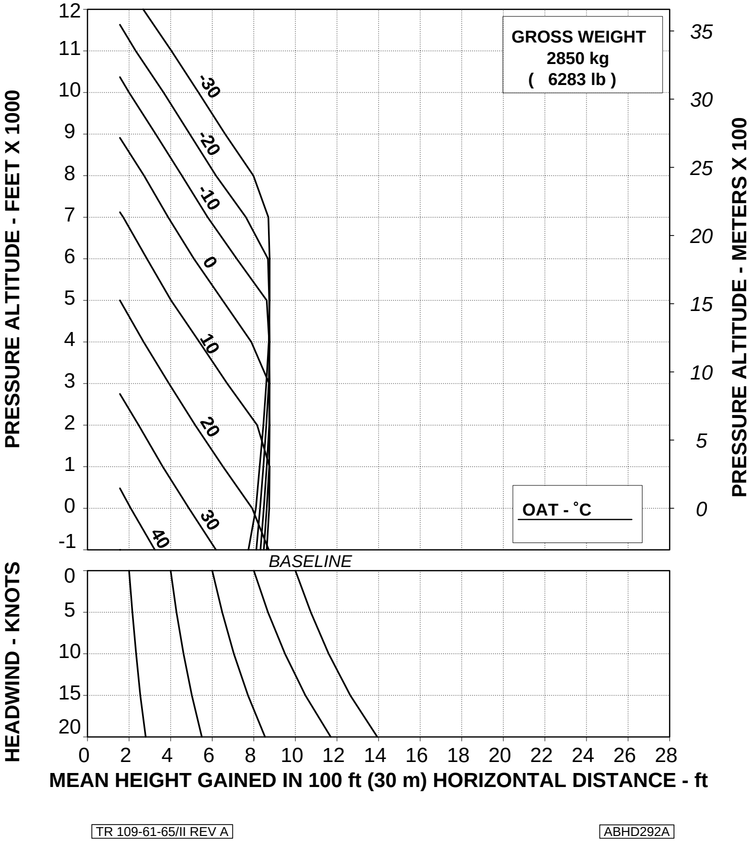

Figure 4-54. Takeoff flight path 2 - 2850 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

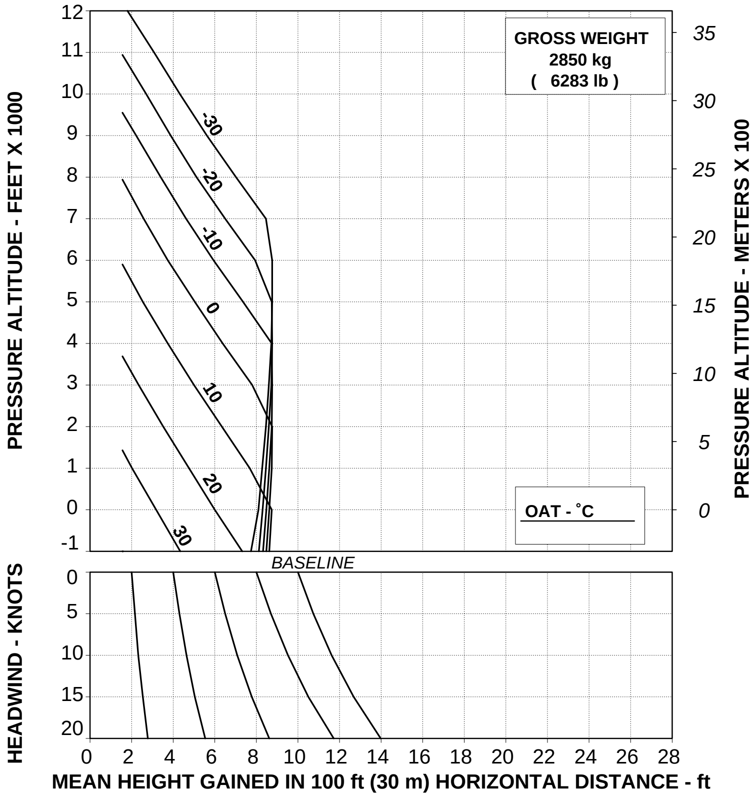

Figure 4-55. Takeoff flight path 2 - 2850 kg - EAPS ON.

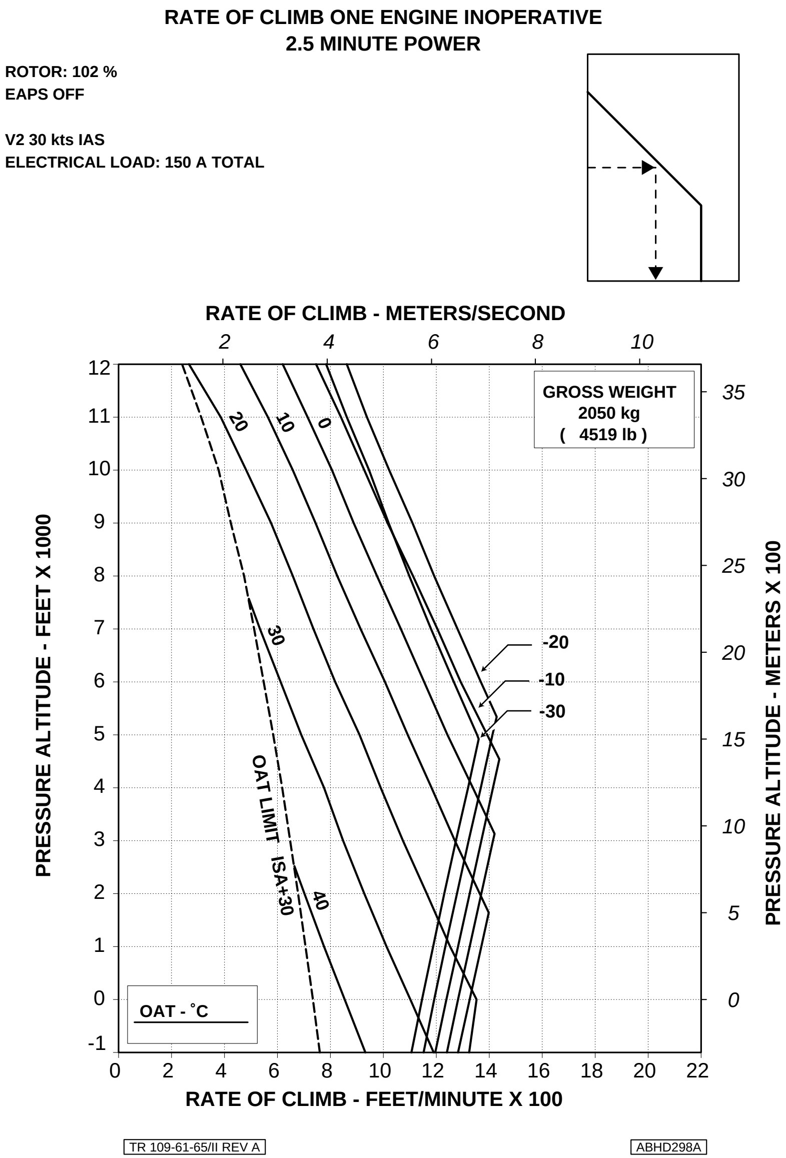

Figure 4-56. Rate of climb - OEI - 2.5 minutes power - 2050 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

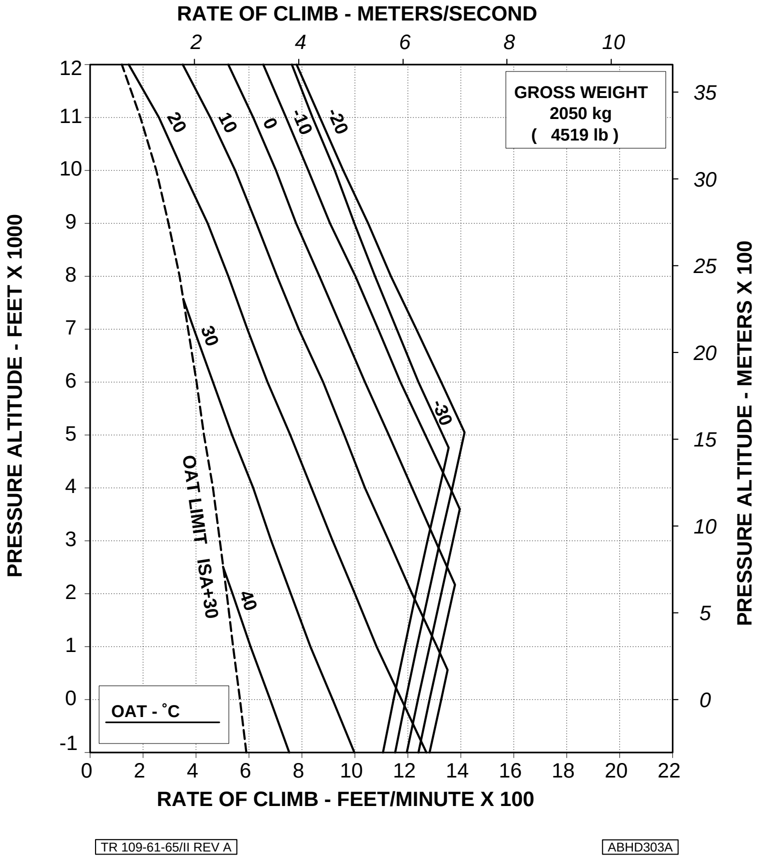

Figure 4-57. Rate of climb - OEI - 2.5 minutes power - 2050 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

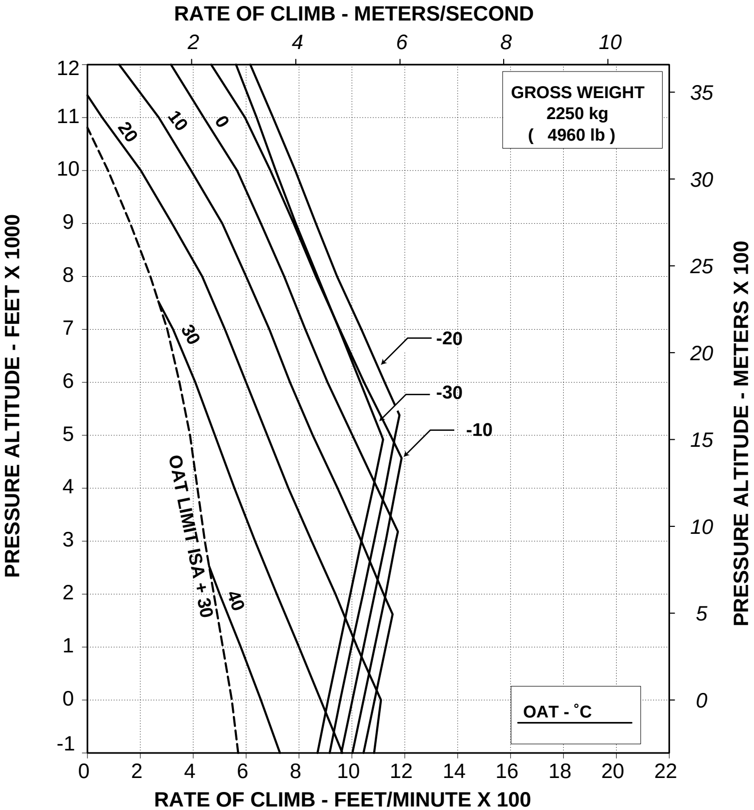

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Figure 4-58. Rate of climb - OEI - 2.5 minutes power - 2250 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % — V2 30 kts IAS

EAPS ON — ELECTRICAL LOAD: 150 A TOTAL

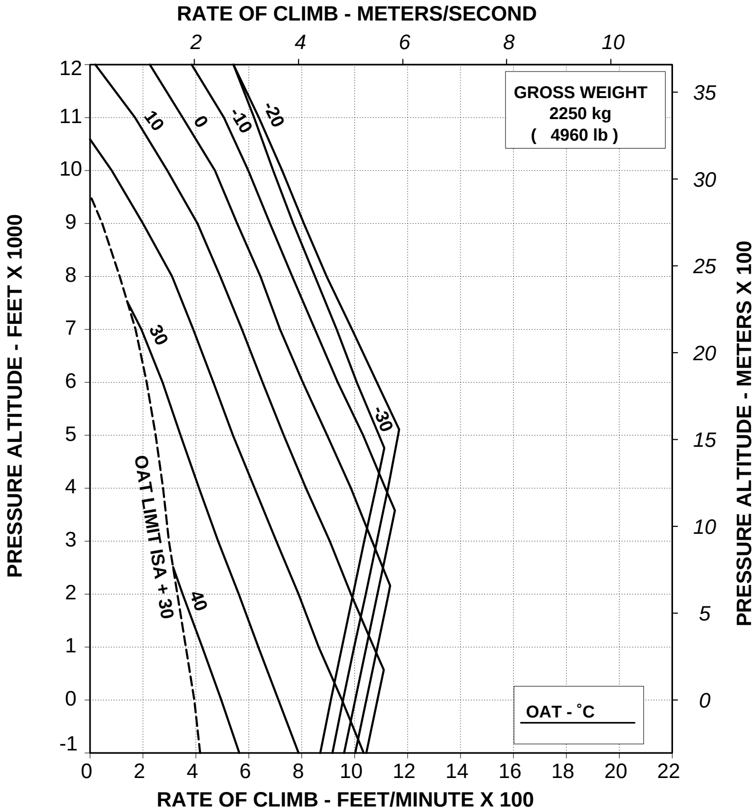


Figure 4-59. Rate of climb - OEI - 2.5 minutes power - 2250 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

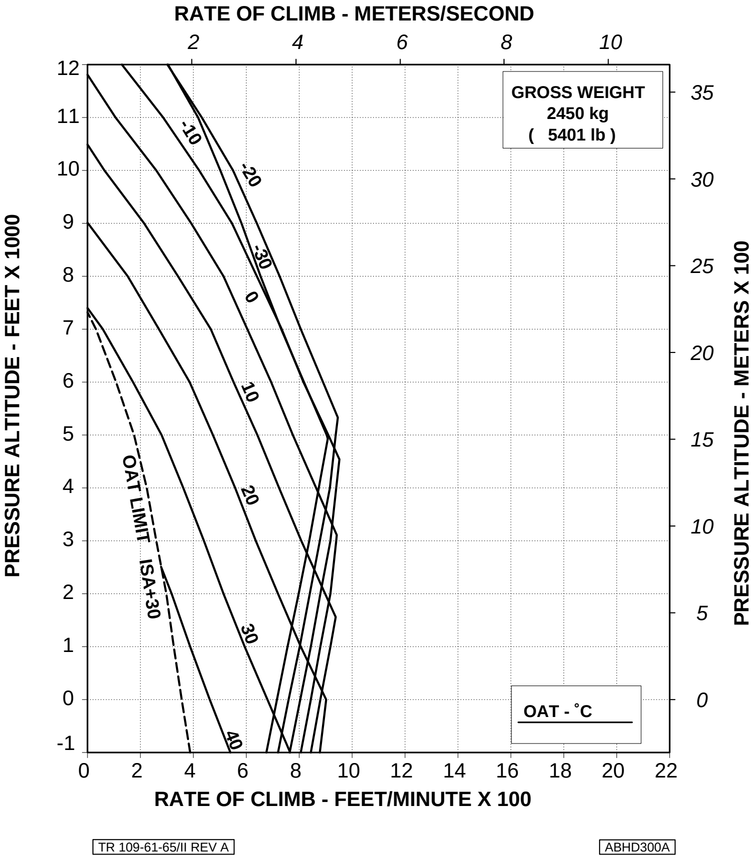


Figure 4-60. Rate of climb - OEI - 2.5 minutes power - 2450 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % V2 30 kts IAS
EAPS ON ELECTRICAL LOAD: 150 A TOTAL

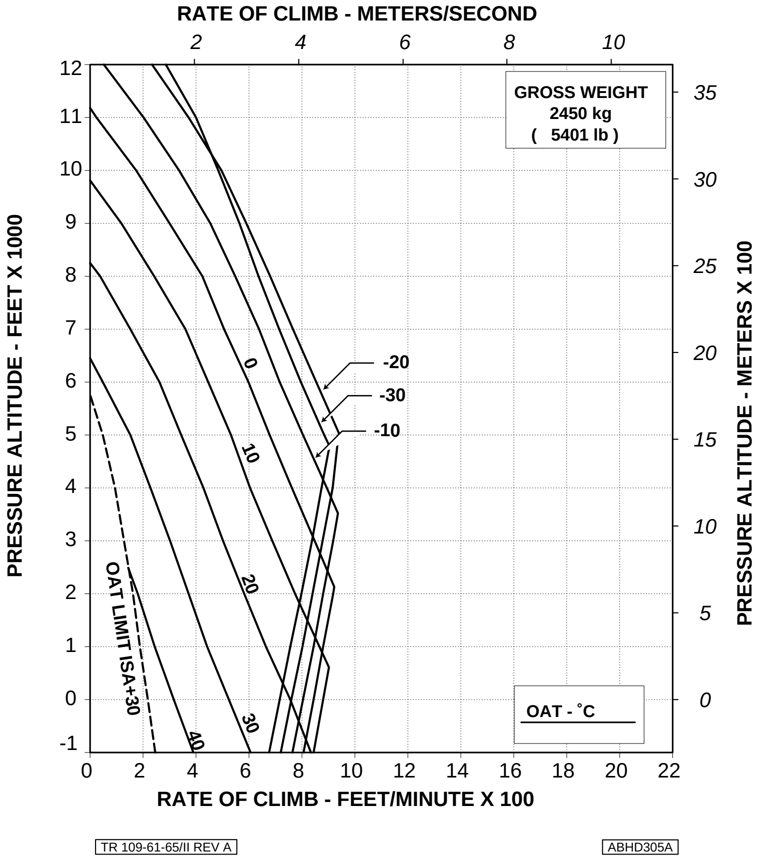


Figure 4-61. Rate of climb - OEI - 2.5 minutes power - 2450 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

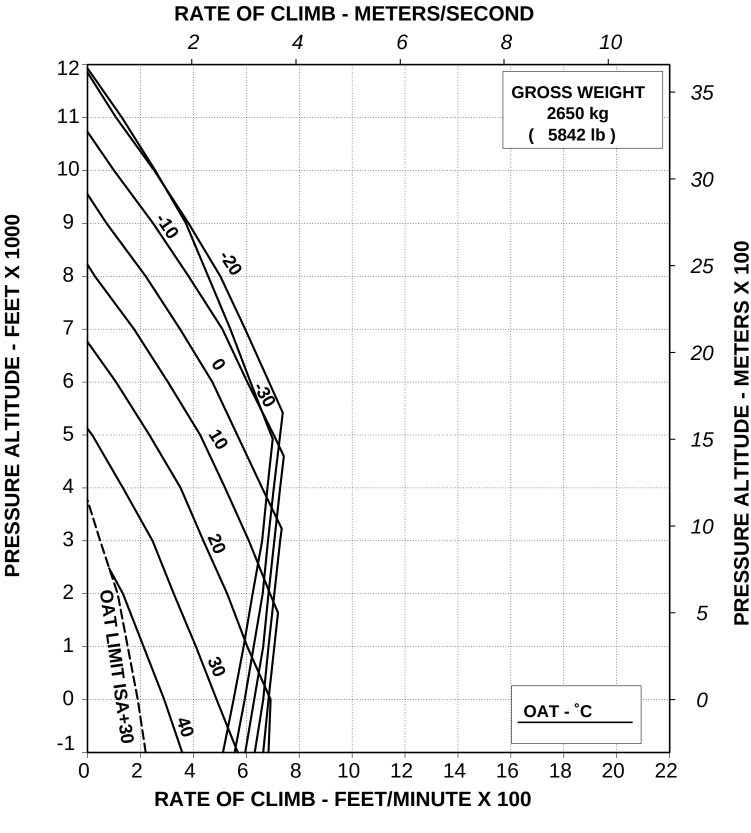


Figure 4-62. Rate of climb - OEI - 2.5 minutes power - 2650 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % | V2 30 kts IAS

EAPS ON | ELECTRICAL LOAD: 150 A TOTAL

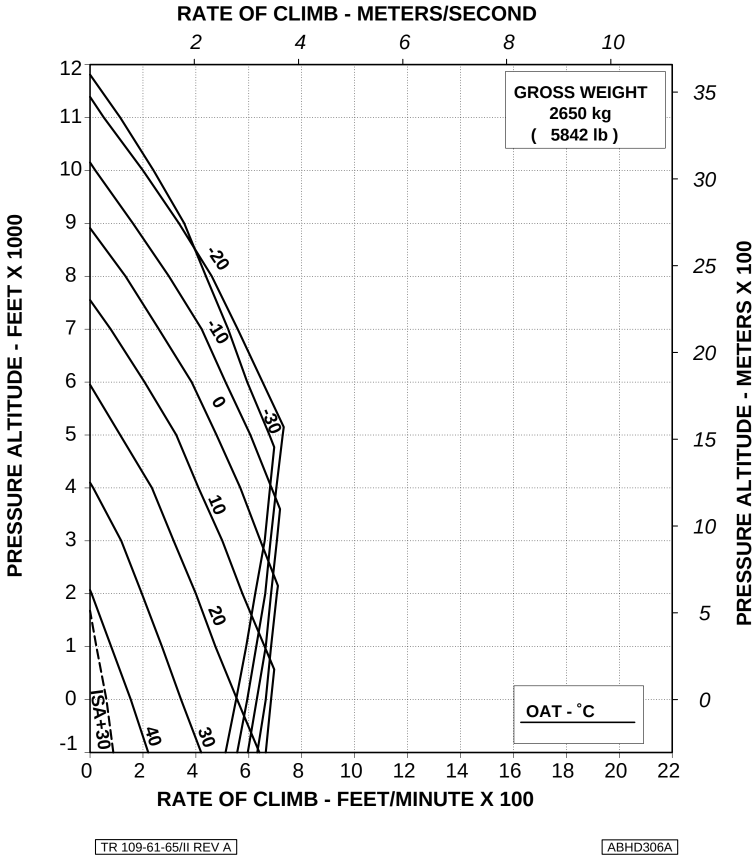


Figure 4-63. Rate of climb - OEI - 2.5 minutes power - 2650 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % — V2 30 kts IAS

EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL

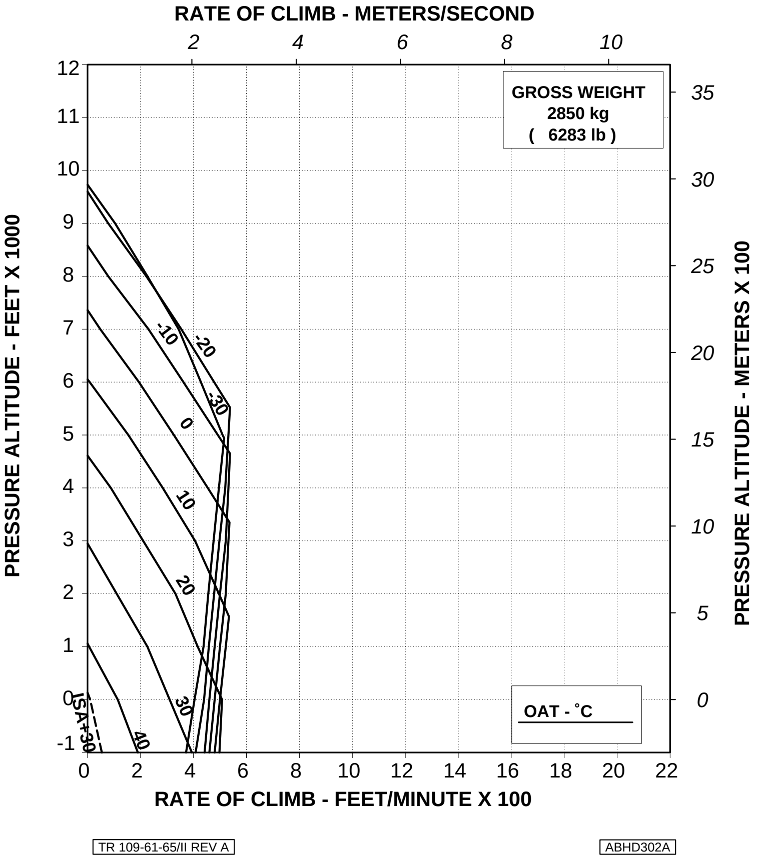


Figure 4-64. Rate of climb - OEI - 2.5 minutes power - 2850 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **30 KNOTS IAS**
EAPS ON **ELECTRICAL LOAD: 150 A TOTAL**

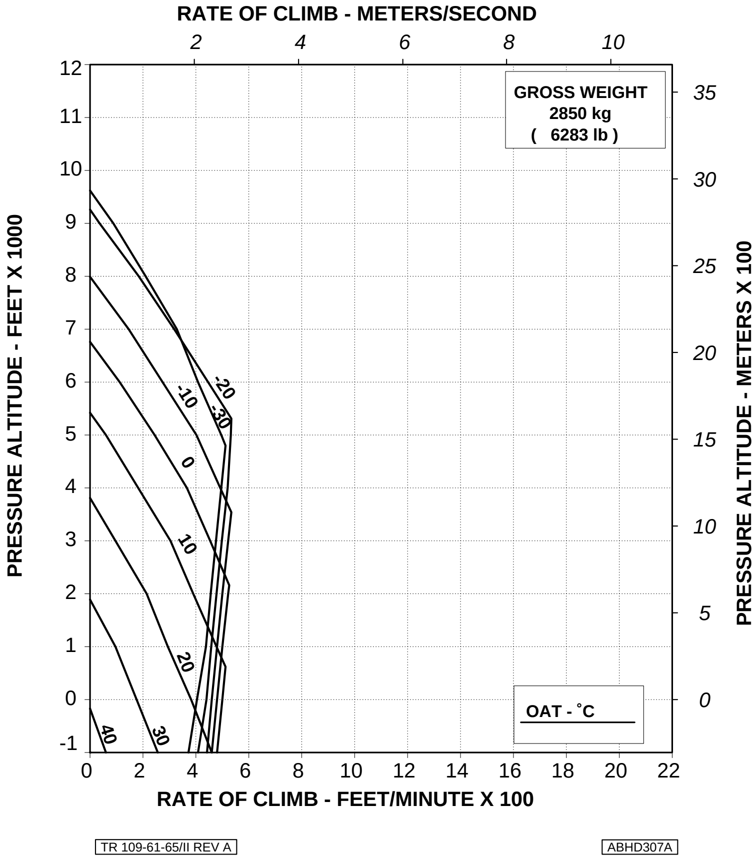


Figure 4-65. Rate of climb - OEI - 2.5 minutes power - 2850 kg - EAPS ON.

Helicopters in basic configuration for Equivalent category ″A″ operations - Training procedures (refer also to Appendix 23)

— Figure 4-66. Takeoff flight path 1 - 2050 kg - training - EAPS OFF.
— Figure 4-67. Takeoff flight path 1 - 2250 kg - training - EAPS OFF.
— Figure 4-68. Takeoff flight path 1 - 2450 kg - training - EAPS OFF.
— Figure 4-69. Takeoff flight path 1 - 2650 kg - training - EAPS OFF.
— Figure 4-70. Takeoff flight path 1 - 2850 kg - training - EAPS OFF.
— Figure 4-71. Takeoff flight path 2 - 2050 kg - training - EAPS OFF.
— Figure 4-72. Takeoff flight path 2 - 2250 kg - training - EAPS OFF.
— Figure 4-73. Takeoff flight path 2 - 2450 kg - training - EAPS OFF.
— Figure 4-74. Takeoff flight path 2 - 2650 kg - training - EAPS OFF.
— Figure 4-75. Takeoff flight path 2 - 2850 kg - training - EAPS OFF.

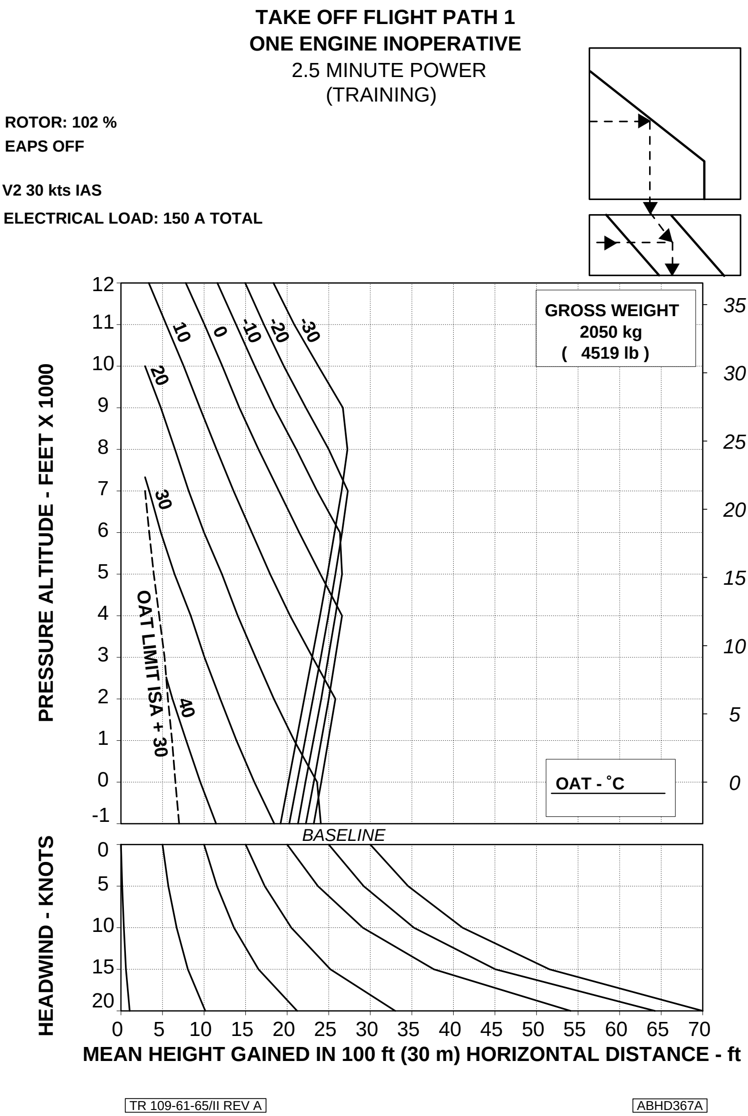


Figure 4-66. Takeoff flight path 1 - 2050 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

EAPS OFF

V2 30 kts IAS

ELECTRICAL LOAD: 150 A TOTAL

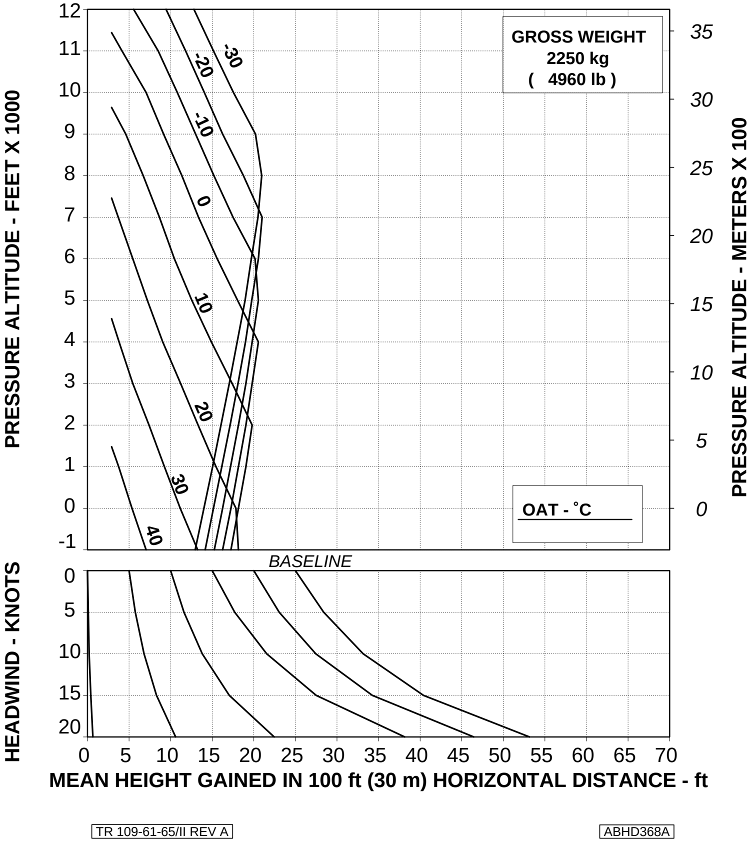


Figure 4-67. Takeoff flight path 1 - 2250 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

EAPS OFF

V2 30 kts IAS

ELECTRICAL LOAD: 150 A TOTAL

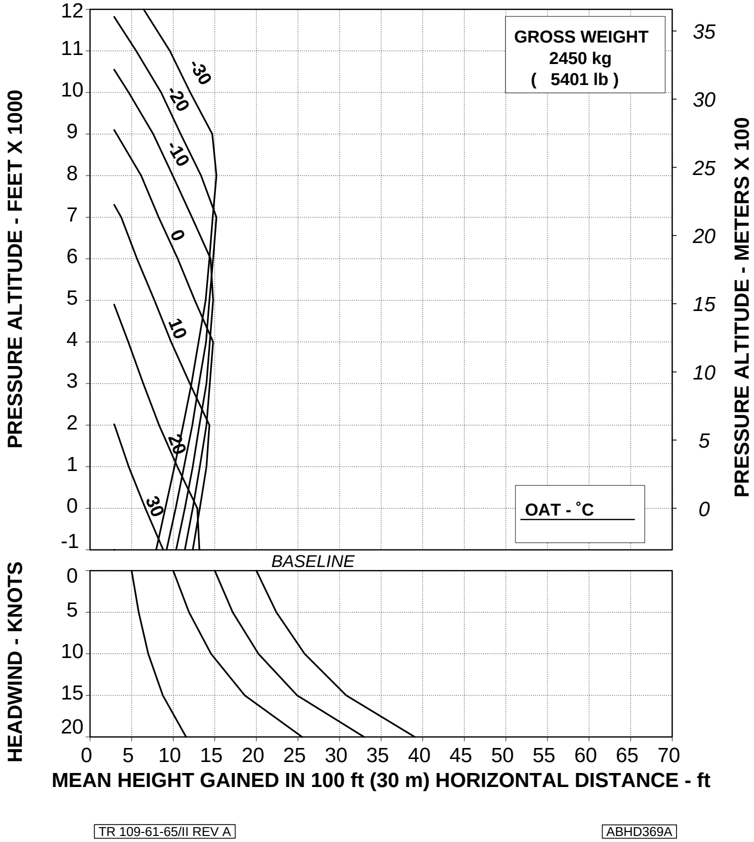


Figure 4-68. Takeoff flight path 1 - 2450 kg - training - EAPS OFF.

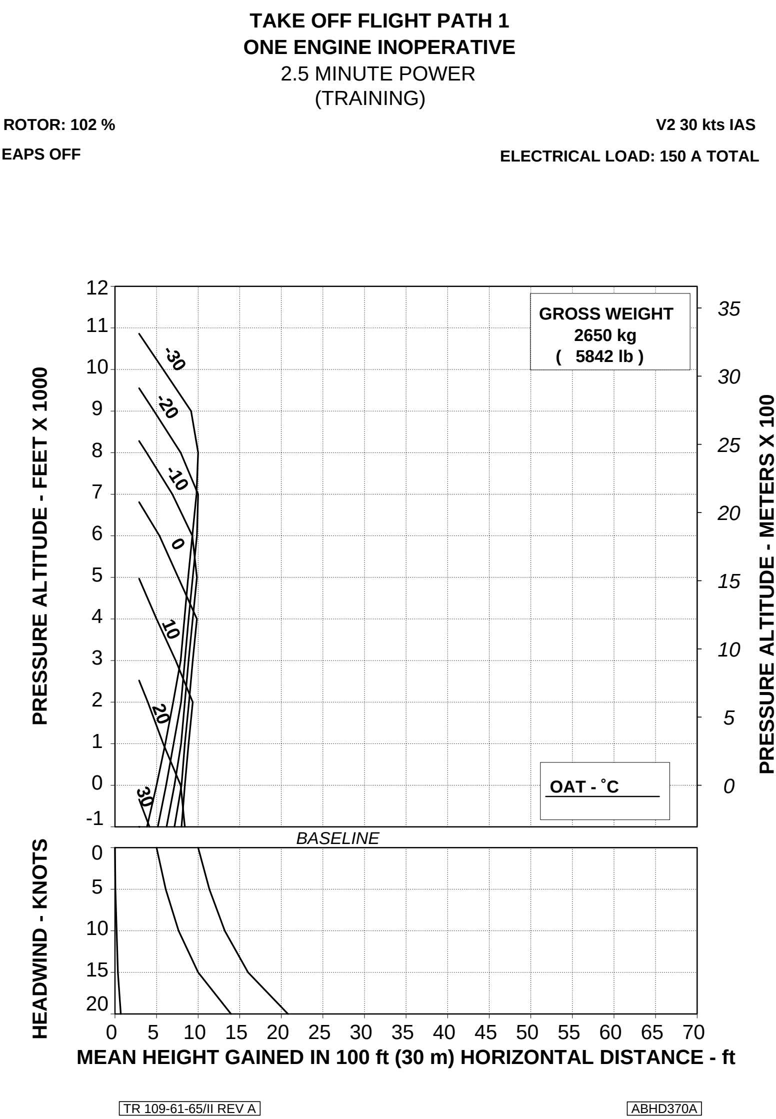
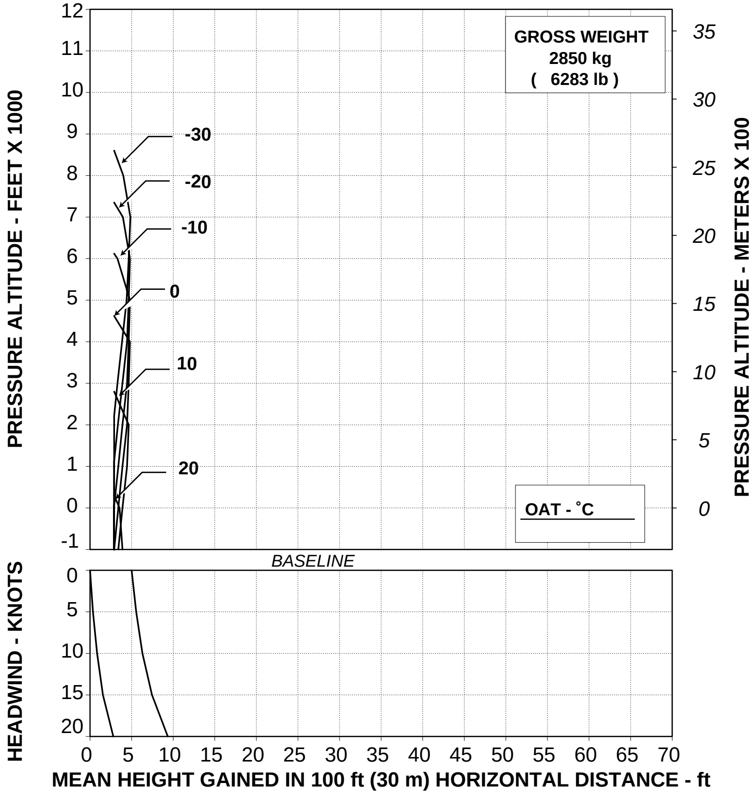


Figure 4-69. Takeoff flight path 1 - 2650 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**



TR 109-61-65/II REV A ABHD371A

Figure 4-70. Takeoff flight path 1 - 2850 kg - training - EAPS OFF.

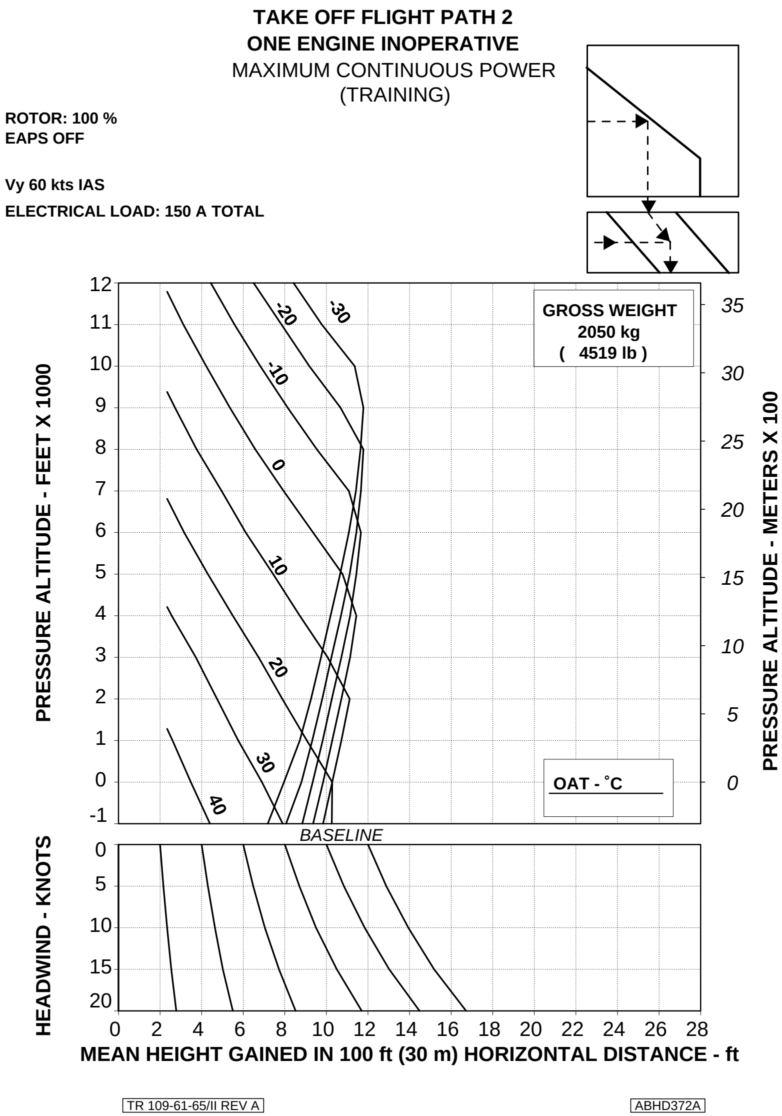


Figure 4-71. Takeoff flight path 2 - 2050 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**

EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

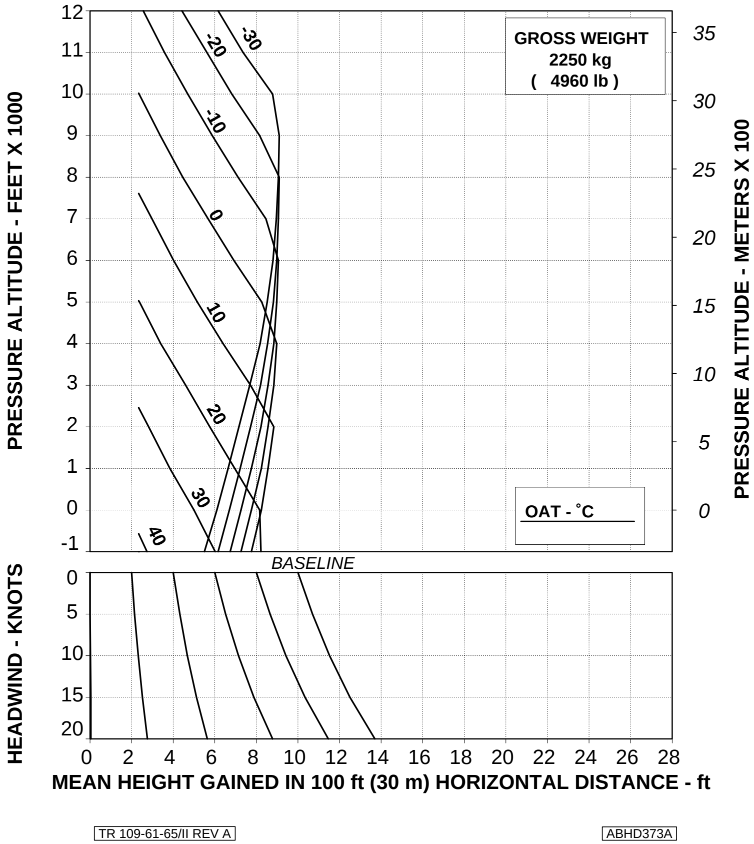
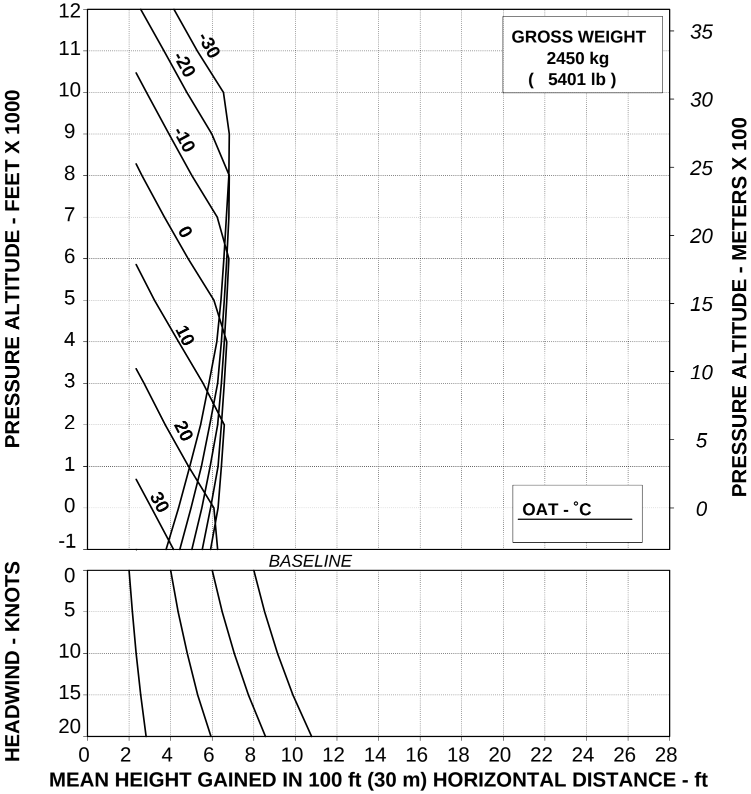


Figure 4-72. Takeoff flight path 2 - 2250 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % | Vy 60 kts IAS

EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL



TR 109-61-65/II REV A

ABHD374A

Figure 4-73. Takeoff flight path 2 - 2450 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 %
EAPS OFF

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

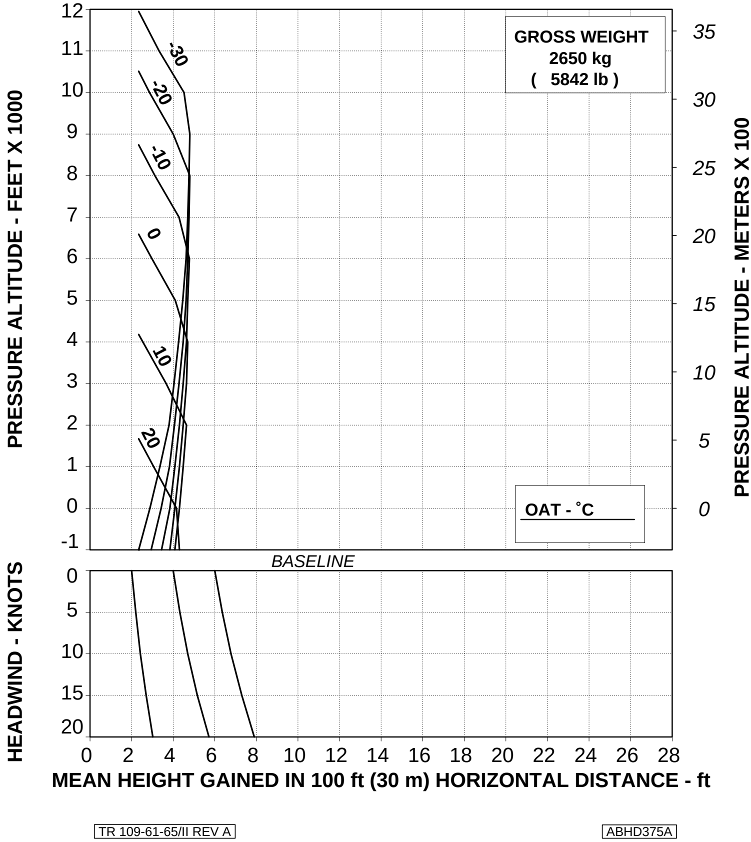


Figure 4-74. Takeoff flight path 2 - 2650 kg - training - EAPS OFF.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**

EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

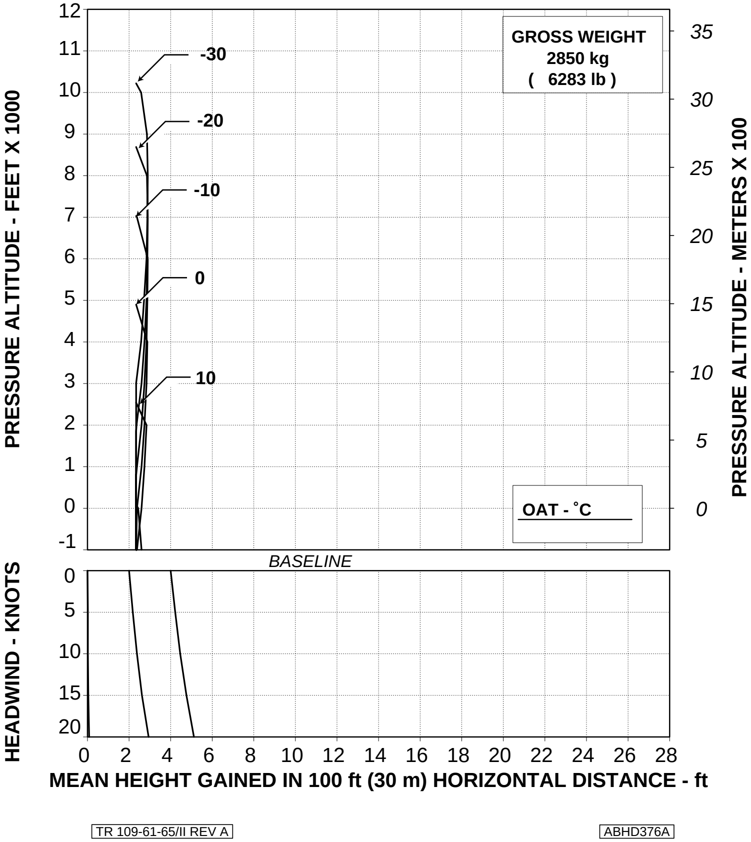
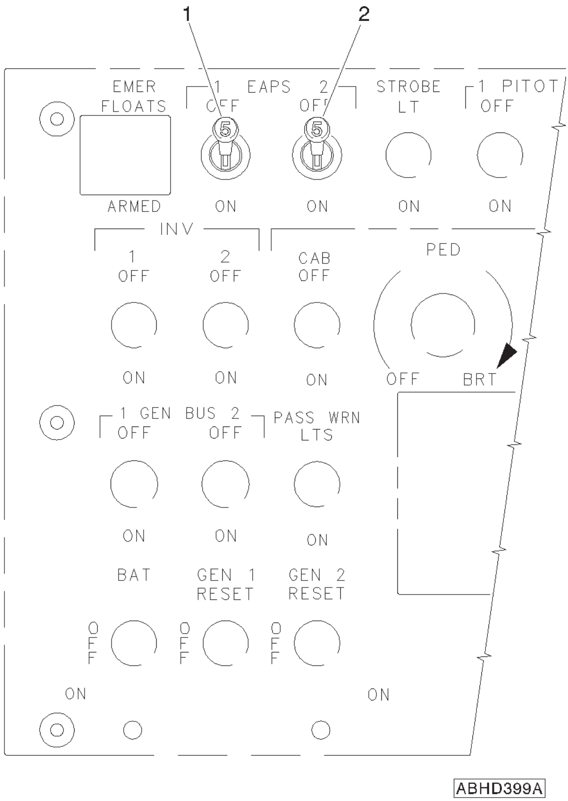


Figure 4-75. Takeoff flight path 2 - 2850 kg - training - EAPS OFF.

SECTION 7 - SYSTEM DESCRIPTION

INSTRUMENT PANEL AND CONSOLE



1. EAPS 1 switch
2. EAPS 2 switch

Figure 7-1. Overhead panel (partial).

INTEGRATED DISPLAY SYSTEM (IDS)

DATA DISPLAY

Display of Advisory/Status Messages

Advisory Message	Description
#1 (#2) EAPS ON	Engine Air Particle Separator on

E.N.A.C. Approval Letter N. 00/505/MAE
dated 16 February 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

STARTER GENERATOR CONFIGURATION APC P/N 160SG139Q1

This Appendix applies to the helicopter installing the starter generators APC P/N 160SG139Q1. This model of starter does not allow any generator-assisted start of the second engine (cross start operation).

TABLE OF CONTENTS

PART I

E.N.A.C. APPROVED

SECTION 1 - LIMITATIONS

CROSS START OPERATION

Any generator-assisted start of the second engine (cross start operation) is prohibited.

PLACARDS

ENGINE CROSS START IS PROHIBITED

In clear view of the pilot.

SECTION 2 - NORMAL PROCEDURES

ENGINE PRE-START CHECK

BATTery	: ON
GEN BUS 1 and 2	: ON
GEN 1 and 2	: OFF, check.

NORMAL ENGINE START

Either engine may be started first.
Either engine may be started using either the auto or the manual mode.

NOTE

It is recommended the normal engine starts be made using the auto mode. For starting procedure in manual mode refer to Section 3.

Collective control	: Flat pitch, check.

Rotor brake (if installed) : Disengaged (lever full forward)

EDU 1 : Select START page.

ENGINE 1 START

ENG 1 MODE switch : IDLE

NOTE

It is recommended to start the engine to IDLE, nevertheless, if necessary, it is possible to start to FLIGHT by setting the ENG MODE switch directly to FLT.

Gas Producer (N1) : Note increasing and START legend vertically displayed.

Engine temperature (TOT) : Note increasing and IGN legend vertically displayed.

CAUTION

Monitor engine start and if any of the following occurs:
− lightup is not obtained within 10 seconds of fuel on
− abnormal noises are heard
−TOT increases beyond start limit
− NI or N2 increases beyond engine limits
− engine hangs (stagnation in N1 below 54%)
shutdown the engine by setting OFF the pertinent ENG MODE switch.

CAUTION

Following aborted start shutdown perform the following procedure before restarting:
− Allow 30 seconds fuel drain period.
− Perform a 30 seconds DRY MOTORING RUN.
Refer to Section 1 for engine starter limitations and to DRY MOTORING RUN procedure in this Section.

Engine oil pressure : Check.

NOTE

During cold starting conditions the engine oil pressure can rise up to a transient of 200 PSI. The pressure decreases as oil temperature rises. The oil pressure limits in Sectioj1, Fig. 1-5 must be respected when main oil temperature is 71°C - 120°C.

CAUTION

Do not apply power until engine oil temperature reaches 10°C.

WARNING

If the main rotor has not begun to rotate when gas producer (N1) reaches 25%, abort the start by setting the ENG MODE switch to OFF.

Engine N°1 starter : Automatically deactivated when N1 is 50%.
START and IGN legends automatically suppressed.

Main hydraulic system : When the main rotor begins to rotate, check rise in main hydraulic pressure.

Hydraulic utility system (normal and emergency) : When accumulators are discharged, as main rotor begins to rotate, check pressure rise in both systems and note the activation of MAIN UTIL CHRG and EMER UTIL CHRG caution messages. Note both caution messages are suppressed when systems are charged.

CAUTION

If MAIN UTIL CHRG and/or EMER UTIL CHRG caution messages are not displayed during systems charging, abort and correct trouble.

N°1 engine power turbine speed (N2) : Check stabilized to IDLE speed of 65% ± 1%.

NOTE

If the engine has been started directly to FLT (flight) the N2 will stabilize to 100%.

NOTE

Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

Engine and transmission oil : Check pressure and temperature.

NOTE

On ground, in IDLE condition, the transmission oil pressure indication can be below the green lower limit.

ENG 1 MODE switch : FLT.

NOTE

In the starting phase it is suggested to select FLIGHT mode as soon as possible in order to speed up the engine oil heating.

BEFORE ENGINE 2 START (ON BATTERY)

During starting on battery, when battery is not fully charged, before proceeding to start the second engine it is convenient to recharge the battery as follows:

GEN 1	:	ON.
N°1 engine power turbine speed (N2)	:	100% stabilized, check.
Ammeter (Auxiliary page EDU 2)	:	Check until current within limits.
GEN1	:	OFF.

CAUTION

Cross starting using battery and starter-generator to start the opposite engine can cause starter-generator shaft shearing.

ENGINE 2 START

Repeat above procedure to start engine N°2.

CAUTION

Ensure that the second engine engages as the N2 increases ti IDLE. A nonengaged engine shows positive N2 value and near zero torque. If a nonengagement occurs, shut down the nonengaged engine first. When the nonengaged engine has stopped, shut down the engaged engine. If a sudden (hard) engagement occurs, shut down both engines. Maintenance action is required.

Engine parameters	:	Check within limits.
External power	:	Disconnect (if used)
GEN 1 and 2	:	ON.
INV 1 and 2 switches	:	ON.
RAD MSTR switch	:	ON.
Clock	:	Set.
Rotor speed (NR)	:	100% check.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

ENGINE RESTART IN FLIGHT

Each engine may be started using either the auto or manual mode It is recommended that normal engine starts be made using the auto mode.

CAUTION

When the cause of an engine flameout is suspected to be mechanical, do not attempt a restart.

Engine restart in flight with ECU operative (AUTO mode)

Airspeed	: 70 Kts.
ENG MODE switch (inoperative engine)	: OFF, check.
Engine power lever (inoperative engine)	: FLIGHT.
ENG GOV switch (inoperative engine)	: AUTO, check.
ENG FUEL switch (inoperative engine)	: ON.
FUEL PUMP switch (inoperative engine)	: ON.
CROSS FEED switch	: CLOSED, check.

GEN switches : Both OFF.

CAUTION

Cross starting using battery and starter-generator to start the opposite engine can cause starter-generator shaft shearing.

CAUTION

The pilot shall engage the starter and the ignition only when N1 is below 20%.

ENG MODE switch (inoperative engine) : IDLE.

NOTE

It is recommended to start the engine to IDLE, neverthless, if necessary, it is possible to start to FLIGHT by setting the ENG MODE switch directly to FLT.

Gas producer (N1) : Note increasing and START legend vertically displayed.

Engine temperature (TOT) : Note increasing at light-off and IGN legend vertically displayed.

CAUTION

Until N1 is below 35% be ready to abort the starting sequence in case that the TOT overcomes the starting limit (half red dot).

NOTE

The engine governing logic becomes operative and sets the TOT limit to 650°C during start when N1 is in the range from 35% to 54%.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

Engine oil pressure : Check.

Engine starter : Automatically deactivated when N1 is 50%. START and IGN legends automatically suppressed.

Engine power turbine speed (N2) : Check stabilized to idle speed of 65% ± 1%.

NOTE

If the engine has been started directly to FLIGHT the N2 will stabilize to 100%.

GEN switches : Both ON.

CROSS FEED switch : NORM (bar vertical).

Engine restart in flight with ECU inoperative (MANUAL mode)

Airspeed : 70 Kts.

ENG MODE switch (inoperative engine) : OFF, check.

Engine power lever (inoperative engine) : OFF, check.

ENG GOV switch : MANUAL.

NOTE

In presence of an ECU failure the engine control system reverts and operates in manual mode independently from the ENG GOV switch position.
However it can be convenient to set the ENG GOV switch to MANUAL for congruence with the mode condition.
The ENG GOV switch shall mandatorily be on MANUAL, only when the manual mode is selected voluntarily (i.e. for training).

ENG FUEL switch (inoperative engine) : ON.

FUEL PUMP switch (inoperative engine) : ON.

CROSS FEED switch : CLOSED, check.

GEN switches : Both OFF.

CAUTION

Cross starting using battery and starter-generator to start the opposite engine can cause starter-generator shaft shearing.

CAUTION

The pilot shall engage the starter and the ignition only when N1 is below 20%.

ENG MODE switch (inoperative engine) : IDLE.

Engine power lever (inoperative engine)	:	Move to FLIGHT.
Starting button	:	Push and hold. START and IGN legends vertically displayed beside N1 and TOT scales on the EDU panel.
Gas producer (N1)	:	Note increasing.
Engine temperature (TOT)	:	Note increasing at light-off, then keep it under control by moving the engine power lever as necessary to prevent exceeding the TOT transient.

NOTE

If engine hangs below idle (less than 54% N1), slowly move power lever forward beyond FLIGHT position, until the engine accelerates, monitoring TOT, N1 and NR. If the engine does not accelerate shutdown the engine by setting OFF the power lever and release the starting button.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limits

shutdown the engine by setting OFF the pertinent power lever and release the starting button.

Starting button	:	Release when gas producer (N1) reaches 50%. START and IGN legends suppressed.

Engine power control	:	Set the power as required by using either the engine power lever or the ENG TRIM toggle switch on the collective control.
GEN switches	:	Both ON.
Engine oil pressure	:	Check.
CROSS FEED switch	:	NORM (bar vertical).

E.N.A.C. Approval Letter N. 00/851/MAE
dated 16 March 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EXTERNAL LOUDSPEAKERS

The external loudspeakers P/N 109-0811-67 permit helicopter crew to communicate to ground personnel while remaining in flight.
The system consists of two loudspeakers, a control panel, an amplifier, switches and hardware.
The external loudspeakers are located underneath the fuselage, at the baggage station, and they are mounted in a fixed position.
A control panel, on the central console, permits to operate the system by means of two switches, identified POWER ON and SIREN OFF/WAIL/YELP, and a control knob for the volume (operative olly for PA function).

NOTE

The switch identified PA/RADIO is active only in PA position (Public Address).

When the system is operating the light POWER ON on the control panel is illuminated.

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CENTER OF GRAVITY LIMITATIONS

After external loudspeakers installaion the new empty weight and CG location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°2 (Fuselage - rh side)

External loudspeakers	:	Check condition and security. Wiring connections for security.

AREA N°6 (Fuselage - lh side)

External loudspeakers	:	Check condition and security. Wiring connections for security.

IN FLIGHT

WARNING

Avoid loudspeakers operation when on ground as this may cause injury to personnel. Ground support personnel in vicinity of helicopter should wear protective hearing devices.

PUBLIC ADDRESS OPERATION

POWER switch	: ON. Check POWER ON light illuminated.
PA/RADIO selector switch	: Check in PA position.
Transmit selector (on ICS control panel)	: PA
PTT trigger switch (on cyclic stick)	: Push at the second detent.

Any message spoken to the headset microphone will be spread through the horn speakers.

VOL knob (on loudspeakers control panel)	: as required.

When system operation is no longer required:

POWER switch	: OFF. Check POWER ON light extinguished.

SIREN OPERATION

POWER switch	: ON. Check POWER ON light illuminated.
SIREN switch	: WAIL or YELP, as required.

When system operation is no longer required:

SIREN switch : OFF

POWER switch : OFF

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change

SECTION 7 - SYSTEM DESCRIPTION

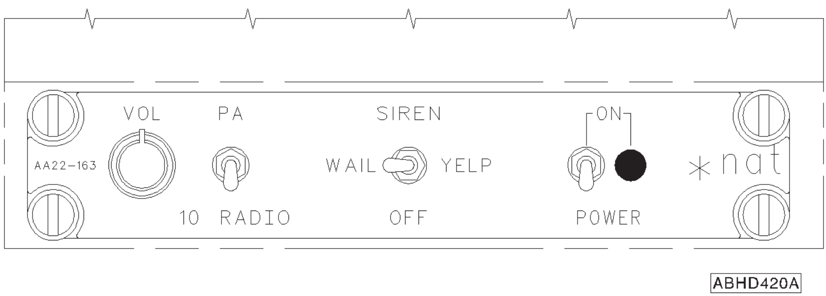

Figure 7-1. External loudspeakers control panel.

E.N.A.C. Approval Letter No. 00-5638-TMI/C
dated 31 March 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

E.M.T.
EMERGENCY MEDICAL TRANSPORTATION

The Emergency Medical Transportation P/N 109-0812-64, for emergency, rescue and ambulance operation, may be equipped as follows:

— -101 consists of a litter, a forward, swivelling type, medical seat and an aft single seat, on the right side, P/N 109-0718-64-101 (confort type).
— -105 is similar to -101 with the exception of the aft single seat, on the right side, P/N 109-0702-89-175 (standard type).

This installation allows, in emergency situation, injured people to be taken on a standard litter, positioned longitudinally on the right side of passenger compartment.

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PART II - MANUFACTURER'S DATA

LIST OF ILLUSTRATIONS

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The helicopter in EMT configuration permits rescue and transportation of injured people under day and night VFR and IFR non-icing conditions.

REQUIRED EQUIPMENT

Sliding doors kit installation is required for E.M.T. installation and operation.

VFR OPERATION

FLIGHT WITH PASSENGER CABIN DOORS OPEN OR REMOVED

Operation with passenger cabin doors open or removed is prohibited when patient is on board.

FLIGHT CREW

The minimum flight crew consists of one pilot and one attendant; both of whom shall be trained in and capable of assisting in litter patient emergency evacuation procedures.

NUMBER OF SEATS

Five (5) - including the pilot and the litter patient.

CENTER OF GRAVITY LIMITATIONS

After E.M.T. interior installation the new empty weight and CG location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°7 (Cabin interior)

E.M.T. interior	:	Check for condition.
Cabin seats and litter	:	Check for condition and fastened if unoccupied.

PILOT'S PREFLIGHT CHECK

(Every flight)

E.M.T. interior	:	Check for condition.
Cabin seats and litter	:	Check for condition and fastened if unoccupied.

LITTER OPERATIONS

LITTER LOADING

— Secure patient to the litter using the three straps provided.

— Secure litter to the locks.

LITTER UNLOADING

— Unlock the litter from the locks.

— Unload the litter from LH door.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

EVACUATION THROUGH EMERGENCY EXITS

Unstrap the patient and evacuate either through the LH or RH emergency exits.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

PATIENT ARM ON LITTER

Longitudinal arm (STA)	: 2900 mm (114.2 inches) from STA 0.
Lateral arm (BL) (left)	: -412 mm (16.2 inches) from the helicopter plane of symmetry.

FORWARD ATTENDANT ARM

Longitudinal arm (STA)	: 2455 mm (96.6 inches) from STA 0.
Lateral arm (BL) (right)	: 0 to 412 mm (16.2 inches) from helicopter plane of symmetry.

AFT ATTENDANT ARM

Longitudinal arm (STA)	: 3200 mm (126 inches) from STA 0.
Lateral arm (BL) (right)	: 412 mm (16.2 inches) from the helicopter plane of symmetry.

SECTION 7 - SYSTEM DESCRIPTION

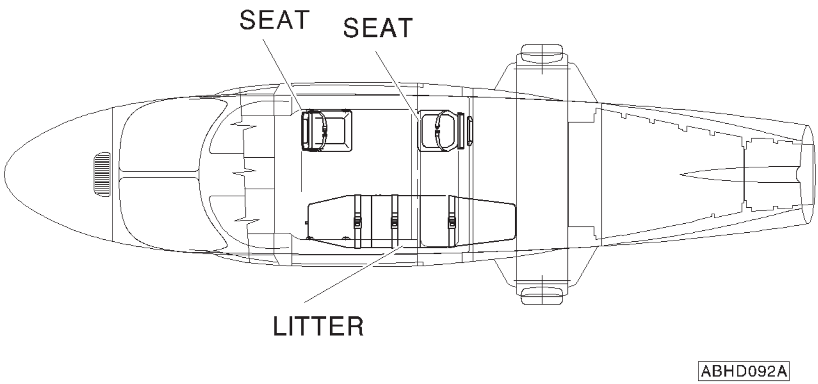

Figure 7-1. E.M.T. Interior arrangment

R.A.I. Approval Letter N. 00-6654 TMI/C
dated 24 May 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

HF SYSTEM - TYPE HF950

The HF system, type HF950 P/N 109-0812-25, is used for long distance communication, in the frequency range of 2 to 29.999 MHz in 100 Hz increments.
The system, connected to the I.C.S., provides two-way voice radio communication in AM and USB mode.
The system consists of:

- a control panel located on the pedestal
- a transceiver and an antenna coupler located in the baggage compartment.
- an antenna installed on the tail boom, right hand side.

As an optional device, a SECAL decoder receiver (CDS-10-101) may be coupled to HF system; in this configuration a light, located on the instrument panel, and an audio signal, in the I.C.S. headset, warn the pilot of an HF incoming call.

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PART I - E.N.A.C. APPROVED

SECTION 1 - LIMITATIONS

No change.

SECTION 2 - NORMAL PROCEDURES

IN FLIGHT

CAUTION

While transmitting with the HF950, the bearing indications of ADF system are not reliable.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE

No change.

E.N.A.C. Approval Letter 00/2775/MAE
dated 27 September 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

CARGO HOOK

The cargo hook P/N 109-0810-31 consists of a support frame, a hook, a rearview mirror, P/N 109-0858-68 an electrical and manual (emergency) release system, attaching hardware, an electronic hook load measuring system and a digital readout indicator. The cargo hook installation includes a suitable locking device for the landing gear control lever to prevent the landing gear retraction. A spring system is attached to the cargo hook which provides the stowing when the hook is not in use.

NOTE

The swiveling link is not supplied with the cargo hook; however, it is recommended to use it between the suspension cable and the cargo hook.

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LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

CARGO HOOK OPERATION

The cargo hook is authorized only as ″Class B rotorcraft-load combination″ as follows:
″Class B rotorcraft-load combination means one in which the external load is jettisonable and is lifted free of land or water during the rotorcraft operation″.

NOTE

Cat. ″A″ operations are permitted only with the cargo hook in the stowed position.

FLIGHT CREW

NOTE

When operating with cargo on the hook, only the personnel involved in the mission are allowed on board.

AIRSPEED LIMITATIONS (IAS)

Flight operations are only permitted with landing gear extended when the helicopter is equipped with cargo hook.
V_{NE} with external loads attached to cargo hook:

From sea level to 5000 ft	: 100 Kts
Above 5000 ft	: Decrease 3 Kts every 1000 ft from 100 Kts.

NOTE

The V_{NE} has been demonstrated with a ballasted cargo sling. The airspeed with external cargo is limited by controllability. Caution should be exercised, when carrying external cargo, as the handling characteristics may be affected by the size, weight and shape of the cargo load, as well as by the distance between the load and the hook.

WEIGHT LIMITATIONS

Maximum Gross Weight with external loads attached to cargo hook	: 3000 kg (6613 lb)

NOTE

For maximum takeoff and landing weight refer to Section 1 of the basic Rotorcraft Flight Manual.

CARGO HOOK LIMITATIONS

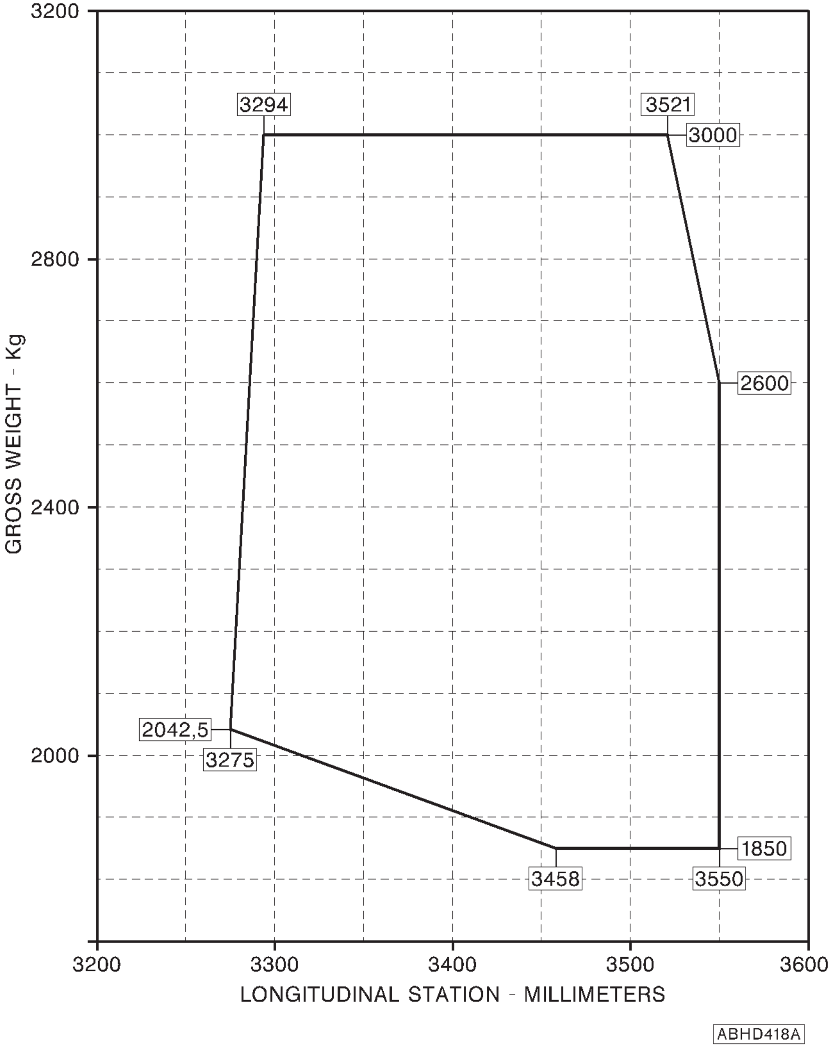
Cargo hook loading limit	: 1000 kg (2205 lb)

WARNING

Flight with unballasted sling as an external load is prohibited.

CENTER OF GRAVITY LIMITATIONS

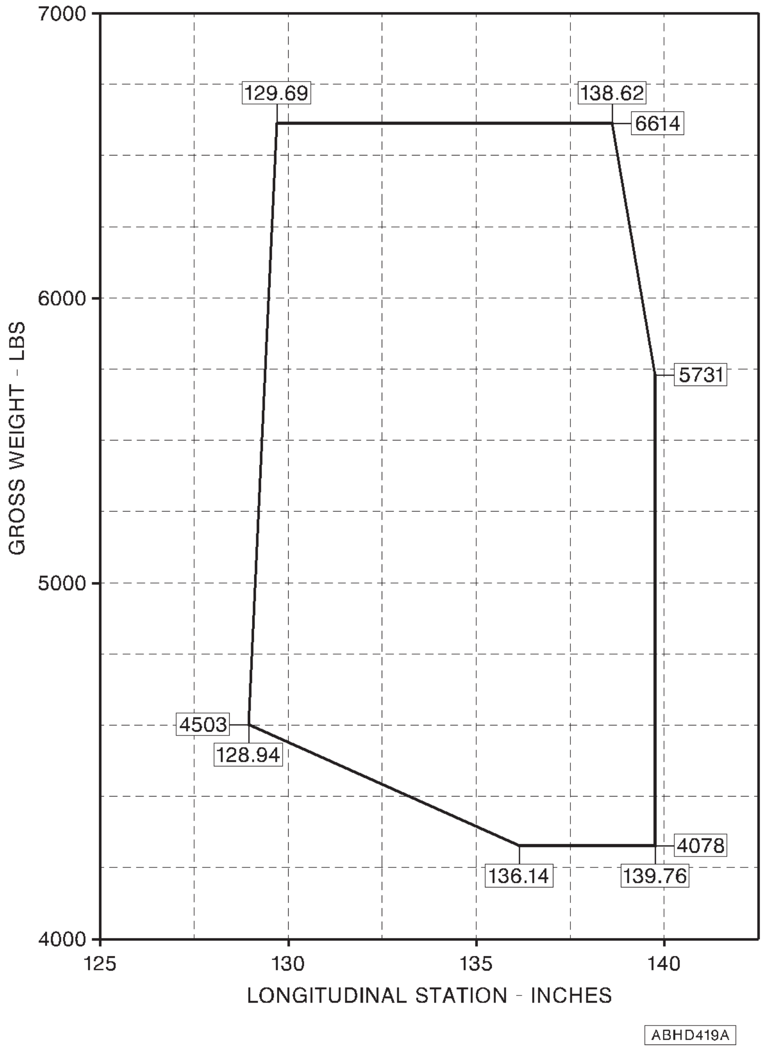
After cargo hook installation the new empty weight and C.G. location must be determined.
Refer to figure 1-1 for Longitudinal CG limits and figure 1-2 for Lateral CG limits when flying with external load attached to cargo hook.



NOTE

Longitudinal station ″0″ is 1835 mm forward of the front jack point.

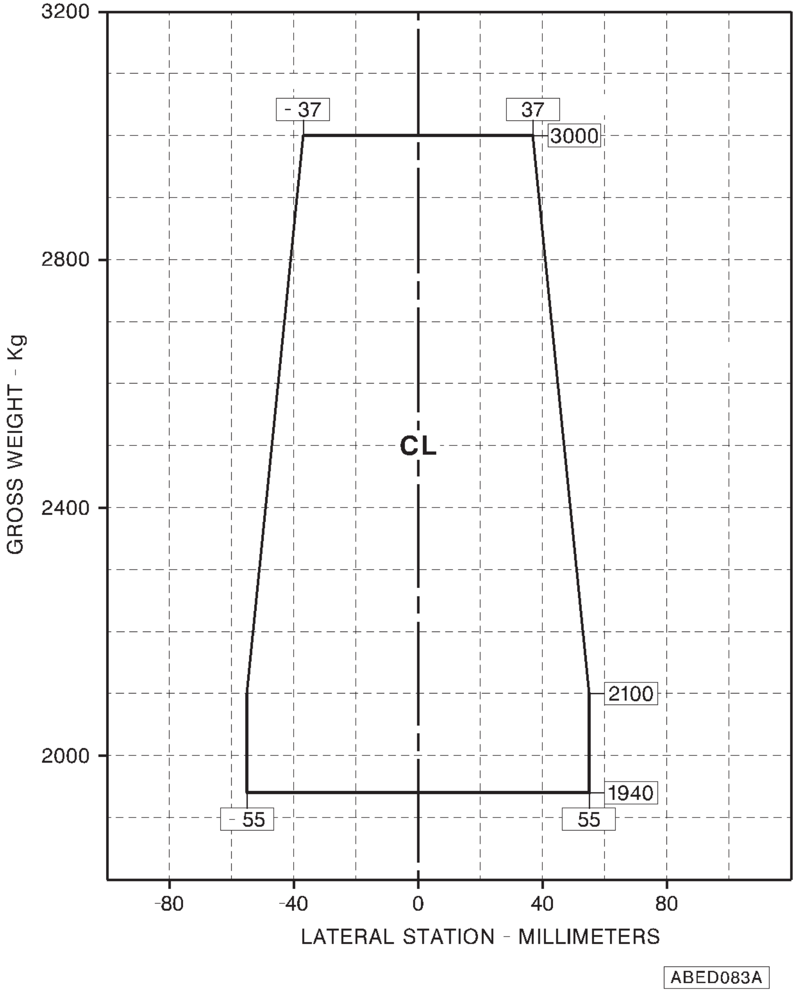
Figure 1-1 (sheet 1 of 2). Longitudinal CG limits (metric).



NOTE

Longitudinal station ″0″ is 72.2 in. forward of the front jack point.

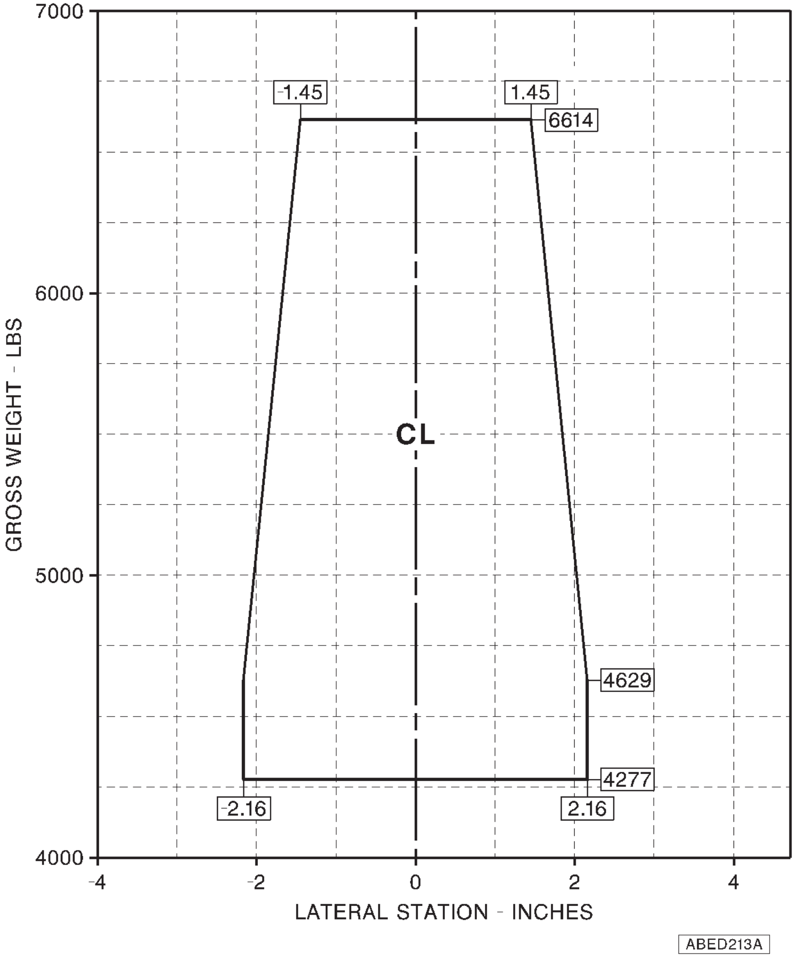
Figure 1-1 (sheet 2 of 2). Longitudinal CG limits (english).



NOTE

The lateral station ″0″ is 450 mm inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

Figure 1-2 (sheet 1 of 2). Lateral CG limits (metric).



NOTE

The lateral station ″0″ is 17.7 in. inboard from each main jack point and coincides with the helicopter longitudinal plane of symmetry.

Figure 1-2 (sheet 2 of 2). Lateral CG limits (english).

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°1 (Helicopter nose)

Rearview mirror : Condition, cleanliness and security.

AREA N°2 (Fuselage - rh side)

Cargo hook : Condition and security.

AREA N°7 (Cabin interior)

For the following checks connect the d.c. supply.

NOTE

Ground personnel shall assist the pilot during the cargo hook checks.

EDU 1 : HOOK ARMED caution message out.

CARGO HOOK pushbutton on pilot cyclic stick : Lift the guard to arm the release system.

EDU 1 : HOOK ARMED caution message displayed.

CARGO HOOK pushbutton on pilot cyclic stick : Push. Verify the opening of the cargo hook and HOOK OPEN caution message on EDU 1 displayed. Release the switch and verify that the cargo hook returns to closed position and that HOOK OPEN caution message is suppressed.

NOTE

The cargo hook is provided with a spring which keeps it permanently in closed position even when the opening system releases the lock-device.
A force of approximately 5 kg must be applied to the cargo hook to overcome the spring-force and to verify the hook opening.

CARGO HOOK pushbutton on pilot cyclic stick : Lower the guard to protect the release pushbutton.

EDU 1 : HOOK ARMED caution message out

Repeat cargo hook checks by using CARGO HOOK pushbutton on copilot cyclic stick.

EMER CARGO RELEASE PULL handle (emergency) : PULL. Verify the opening of the cargo hook and HOOK OPEN caution message on EDU 1 displayed. Release the handle and verify that the cargo hook returns to closed position and that HOOK OPEN caution message is suppressed.

PILOT’S PREFLIGHT CHECK

(Every flight)

Rearview mirror	: Condition, cleanliness and security.
Cargo hook	: Condition and security.

SYSTEMS CHECK

Hook load indicator	: Set to zero.

NOTE

Adjust the hook load indicator after a 5 minutes warm up with no load on the hook.

TAKE-OFF

CARGO ATTACHMENT

Take off and stabilize in hovering at sufficient height to allow crew member to discharge helicopter static electricity and to attach cargo sling to the cargo hook.

NOTE

The distance between load and hook shall be kept as short as possible. For cargo attachment to the hook refer to the instructions contained in the ″Owner Manual″ n. 120-071-00 dated January 10th 1998.

NOTE

Better directional control may be obtained by avoiding relative winds from critical azimuth area while performing external cargo operations. See ″Operation vs Allowable Wind″ in Section 4 of basic Rotorcraft Flight Manual.

WARNING

Discharge helicopter static electricity, before attaching cargo, by touching the airframe with a ground wire or if a metal sling is used, the hook-up ring can be struck against the cargo hook.
If contact has been lost after initial grounding of the helicopter, it should be electrically regrounded and, if possible, contact maintained until hook-up is completed.

NOTE

Attachment of cargo sling to the hook can be observed by means of the rearview mirror.

After cargo attachment slowly increase the collective pitch and ascend vertically to take-up the slack of cargo sling.
Lift vertically cargo from surface and read the hook load indicator to verify the cargo weight to be within the hook loading limitations.
Hover to check for satisfactory controllability and power within limits.

IN FLIGHT

Enter into slow forward speed and verify that uncontrollable or hazardous flight conditions do not exist. Allow adequate sling load clearance over obstacles. Increase forward speed and select an operational airspeed at which no hazardous oscillation is encountered.

APPROACH AND LANDING

CARGO RELEASE

CARGO HOOK pushbutton on cyclic stick : Lift the guard to arm the release system.

EDU 1 : HOOK ARMED caution message displayed.

Perform the approach to the cargo release area with care and at low speed. Stabilize hover above release point, then slowly descend until cargo lies down on ground.

NOTE

Prolongated hover OGE operations in hot day conditions at maximum gross weight may result in an increase of main transmission oil temperature.

CARGO HOOK pushbutton on cyclic stick : Push to release cargo. Verify HOOK OPEN caution message on EDU 1 is displayed.

NOTE

The load is released only when its weight overcomes the spring-force of the hook.

Rearview mirror : Check load released.

NOTE

In case of non-release of cargo, the pilot should slowly increase the collective pitch to ascend, as much necessary to strain the cable, before operating again the CARGO HOOK pushbutton.

NOTE

In the event of an electrical failure pull mechanical manual release control handle (EMER CARGO RELEASE PULL) to drop cargo.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

WARNING SYSTEM

CAUTION MESSAGES (AMBER)

EDU message	Fault condition	Corrective action
HOOK ARMED	Cargo release system armed.	No corrective action is required. Take care to avoid accidental release of cargo.
HOOK OPEN	Hook in open position.	Verify if hook release pushbutton or emergency handle were operated.

EMERGENCY CARGO RELEASE

In case that an emergency situation requires to release the cargo, operate the electrical release system through the pushbutton on cyclic stick.
In case this fails to operate, pull the handle of the mechanical release system.

SECTION 4 - PERFORMANCE DATA

List of Performance Data Charts:

NOTE

Forward flight performance will be degraded due to the load aerodynamic drag. The level of degradation will be dependent on load size and shape.

There are no significant load effects on hover performance.

OPERATION VS ALLOWABLE WIND

Satisfactory stability and control in rearward and sideward flight has been demonstrated, at all loading conditions for hover out of ground effect up to takeoff power, from -1000 ft to 8000 ft Hd, in the following wind/ground speed azimuth envelope:

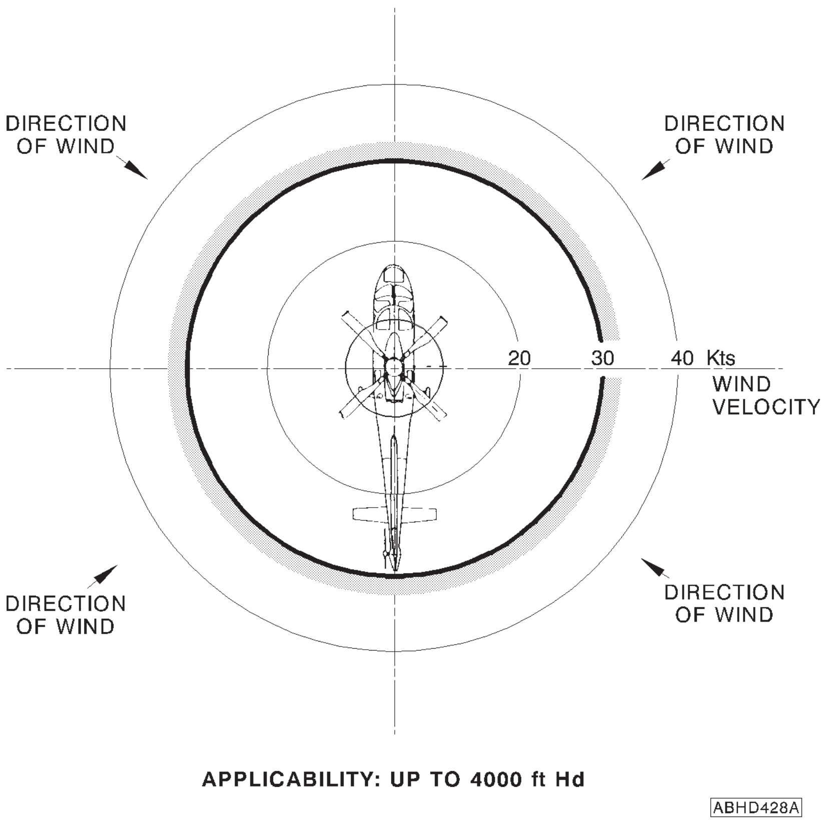


Figure 4-1 (sheet 1 of 2). Wind/ground speed azimuth envelope.

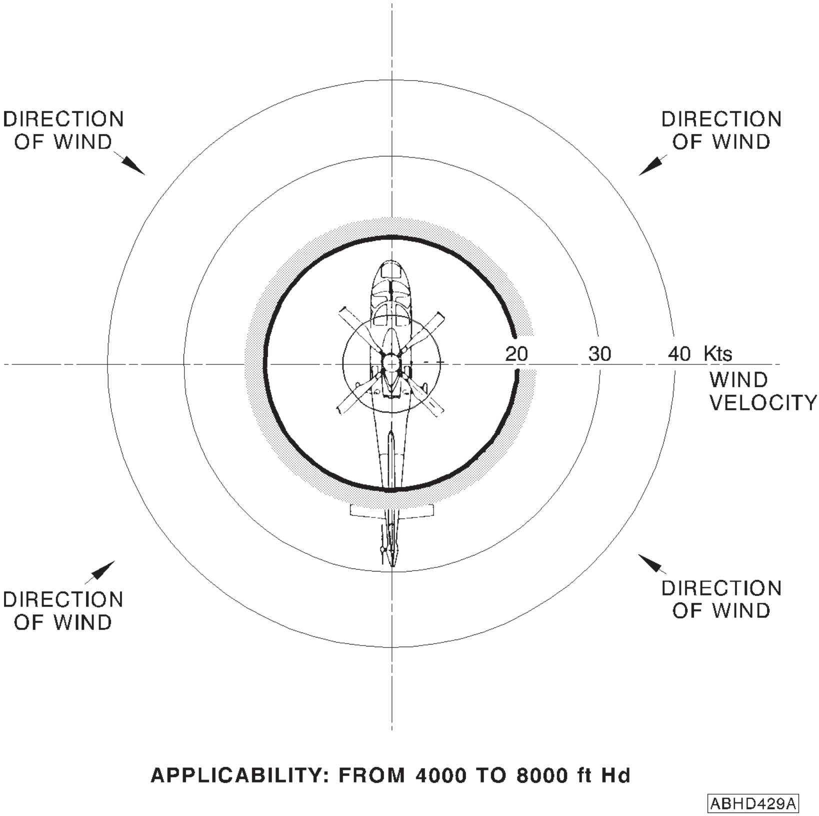


Figure 4-1 (sheet 2 of 2). Wind/ground speed azimuth envelope.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

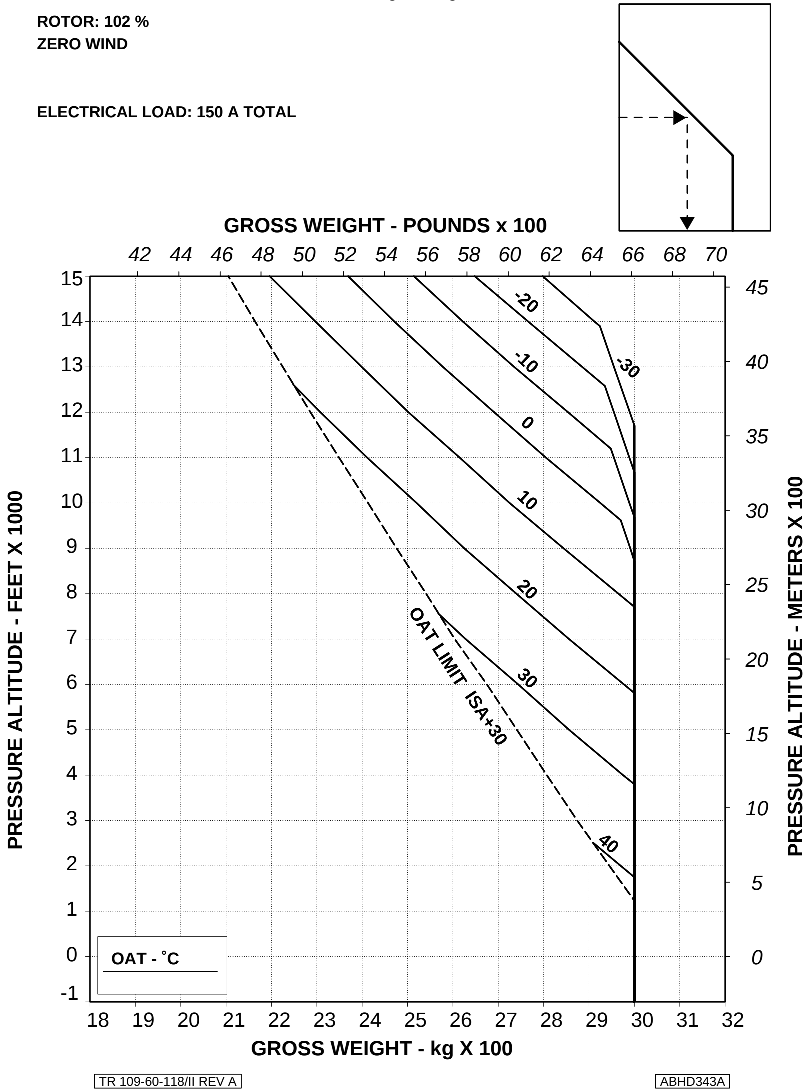


Figure 4-2. Hovering ceiling - OGE - take-off power.

AGUSTA

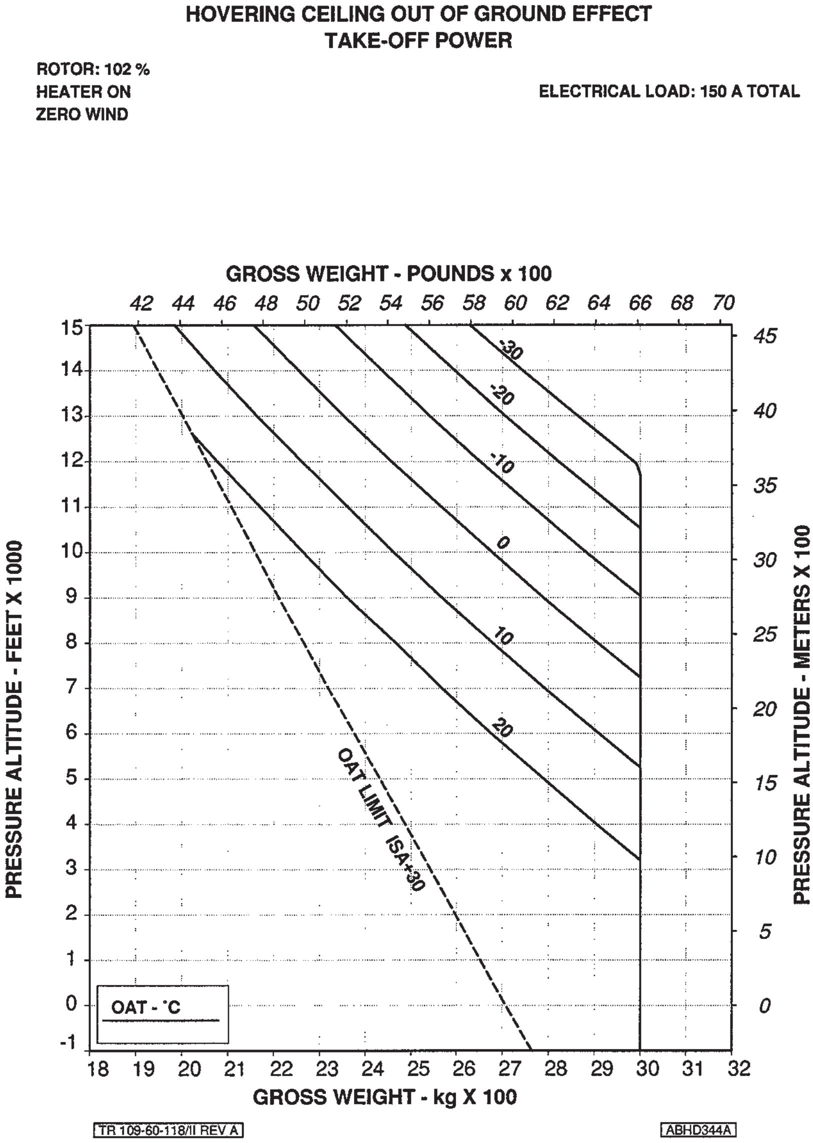


Figure 4-3. Hovering ceiling - OGE - take-off power - heater ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
E.C.S. ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

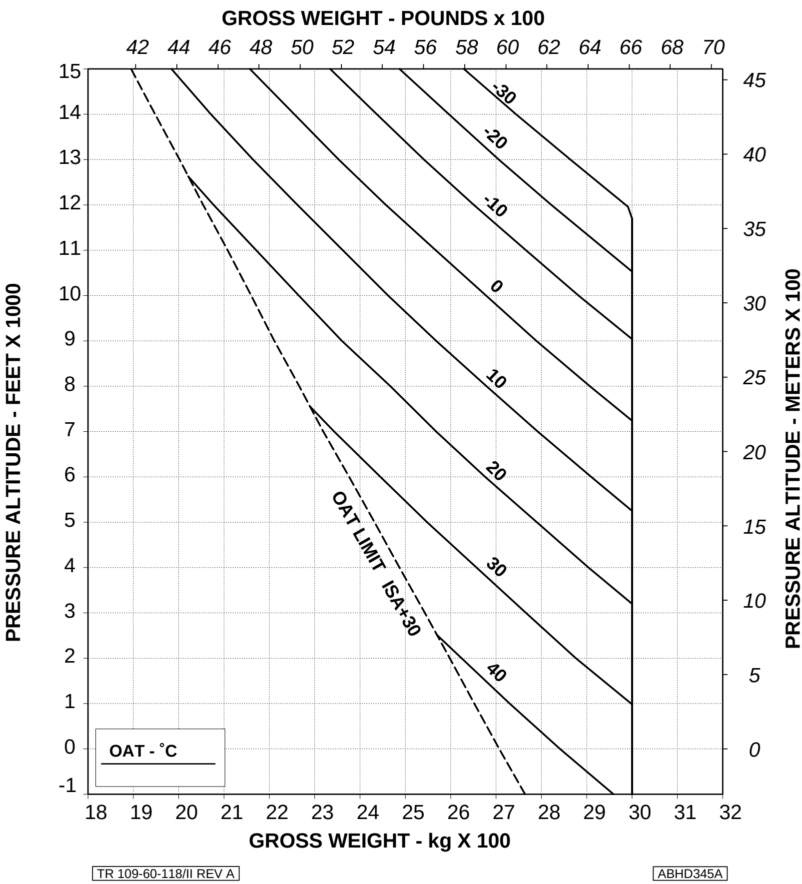


Figure 4-4. Hovering ceiling - OGE - take-off power - ECS ON.

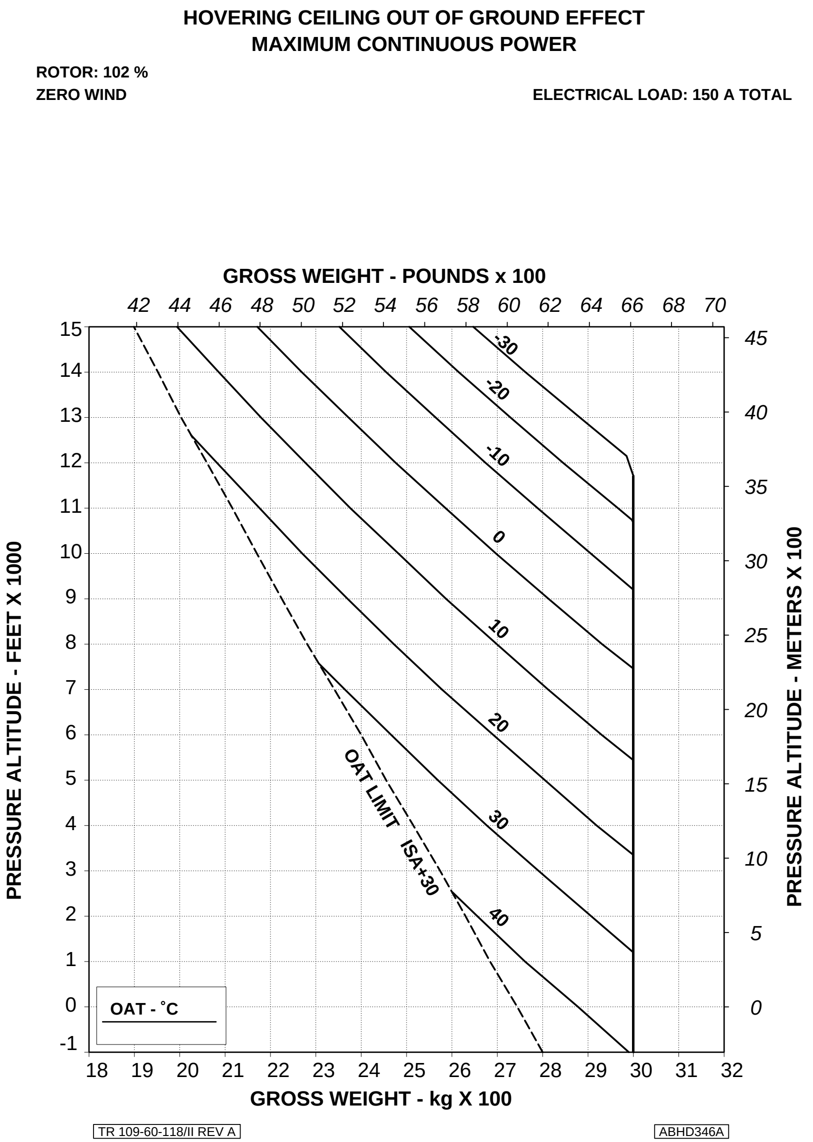


Figure 4-5. Hovering ceiling - OGE - maximum continuous power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

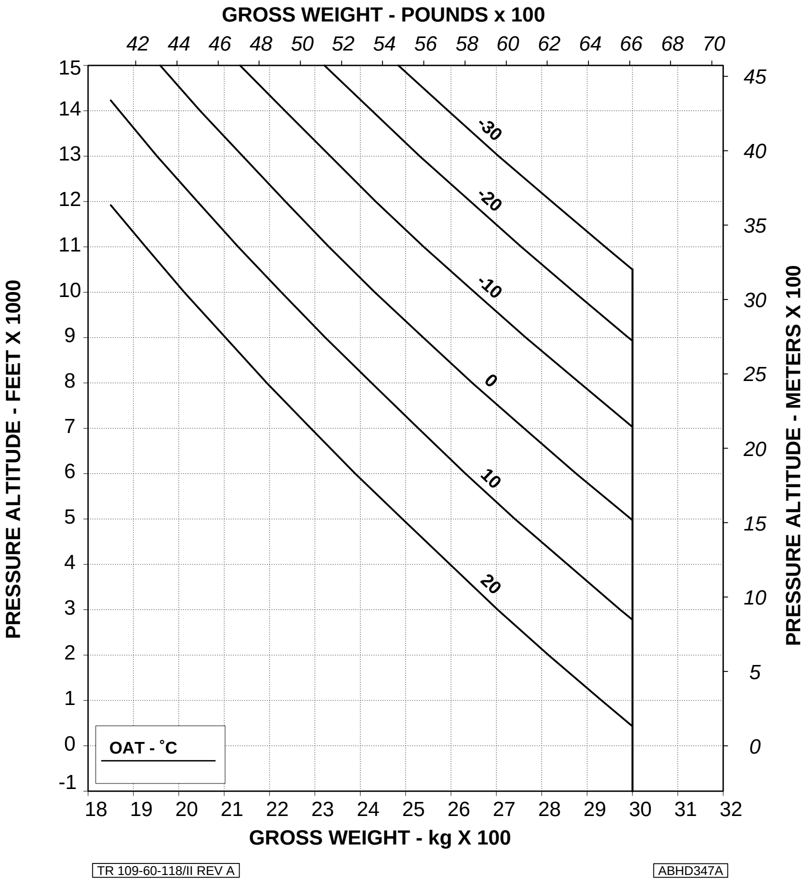


Figure 4-6. Hovering ceiling - OGE - maximum continuous power - heater ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
E.C.S. ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

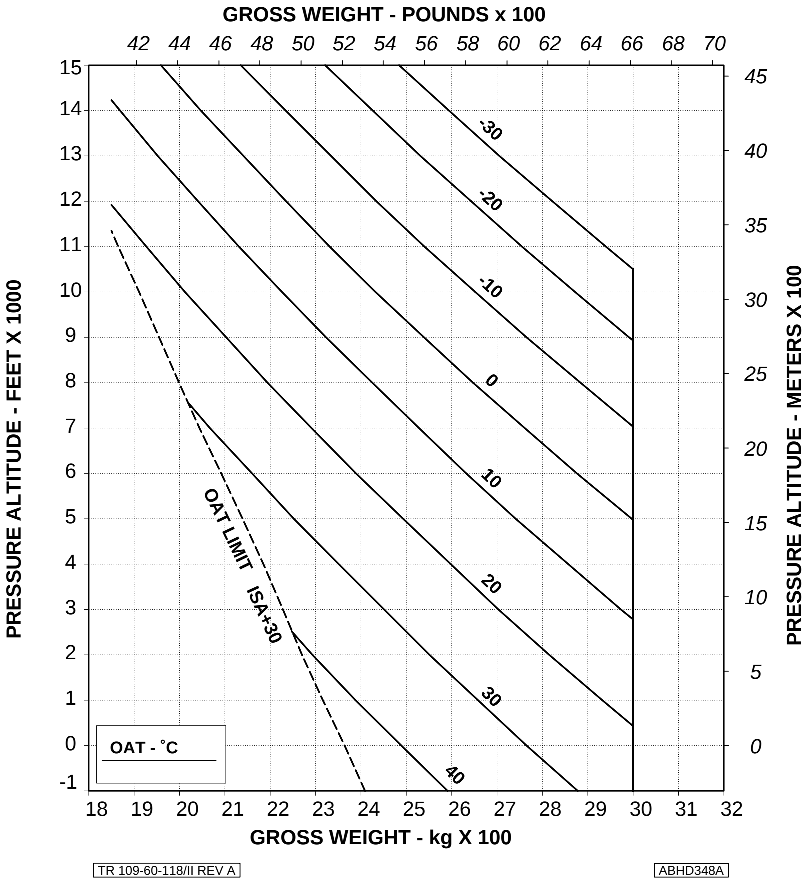


Figure 4-7. Hovering ceiling - OGE - maximum continuous power - ECS ON.

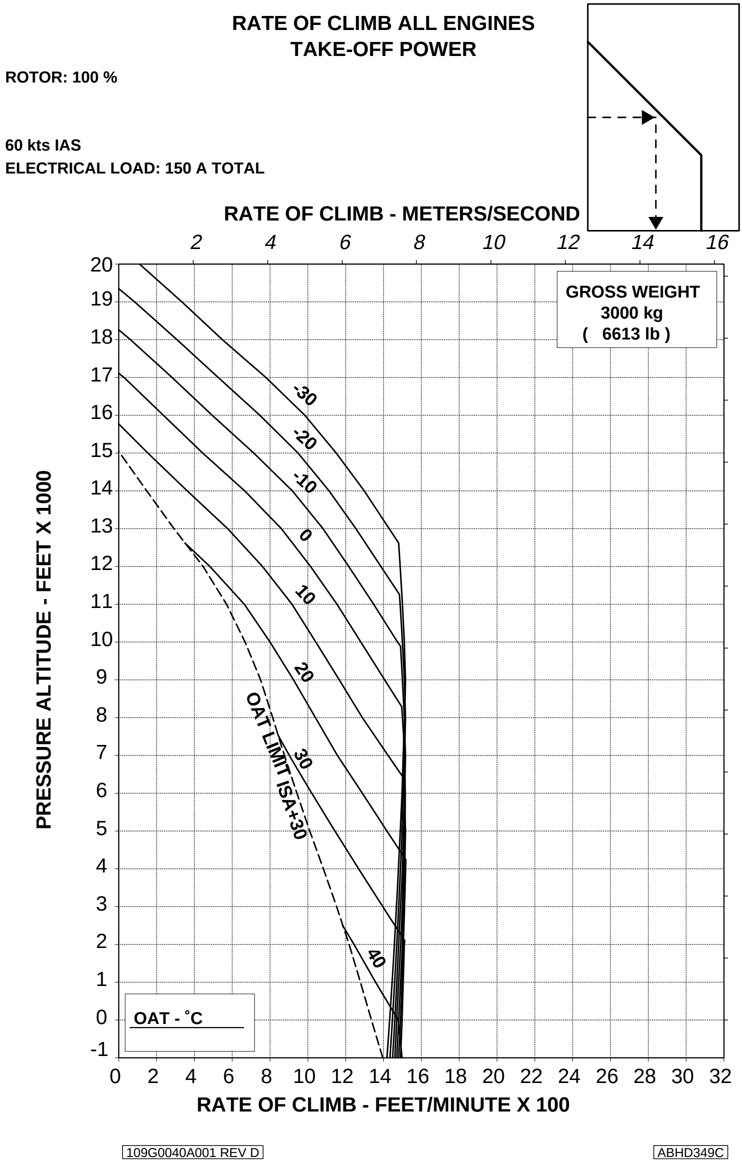


Figure 4-8. Rate of climb - all engines - take-off power.

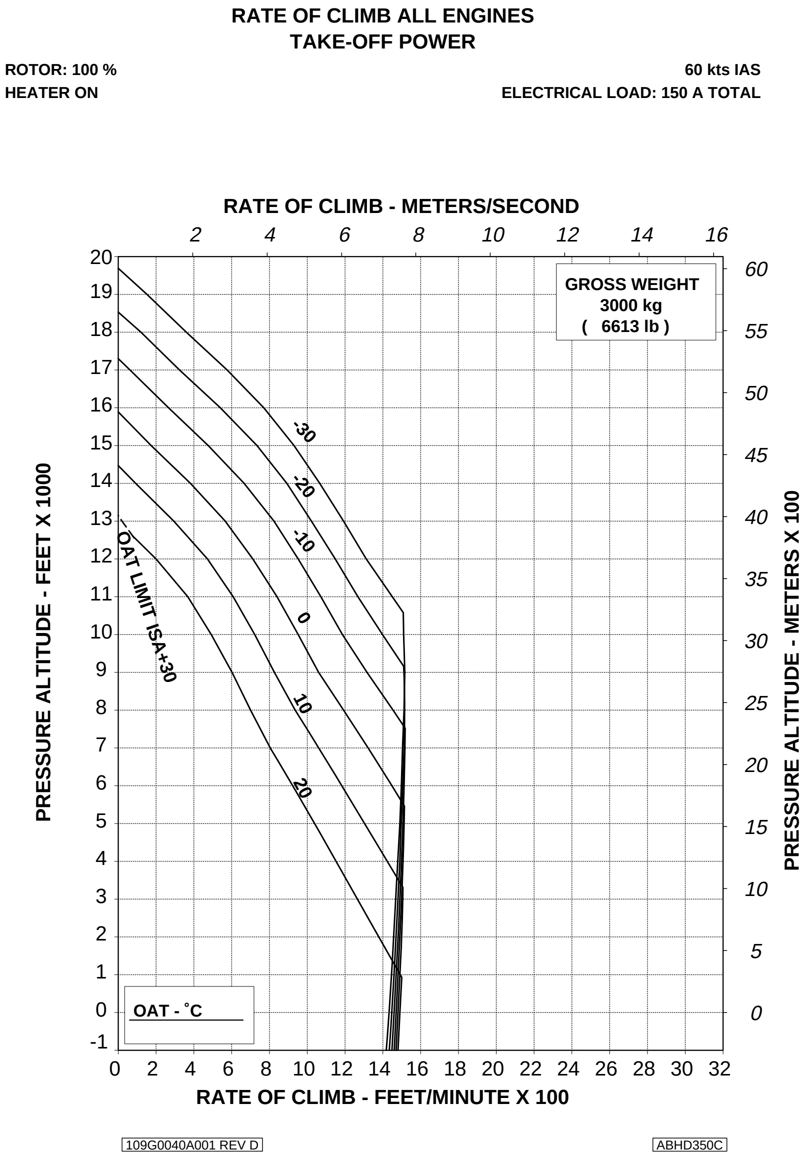


Figure 4-9. Rate of climb - all engines - take-off power - heater ON.

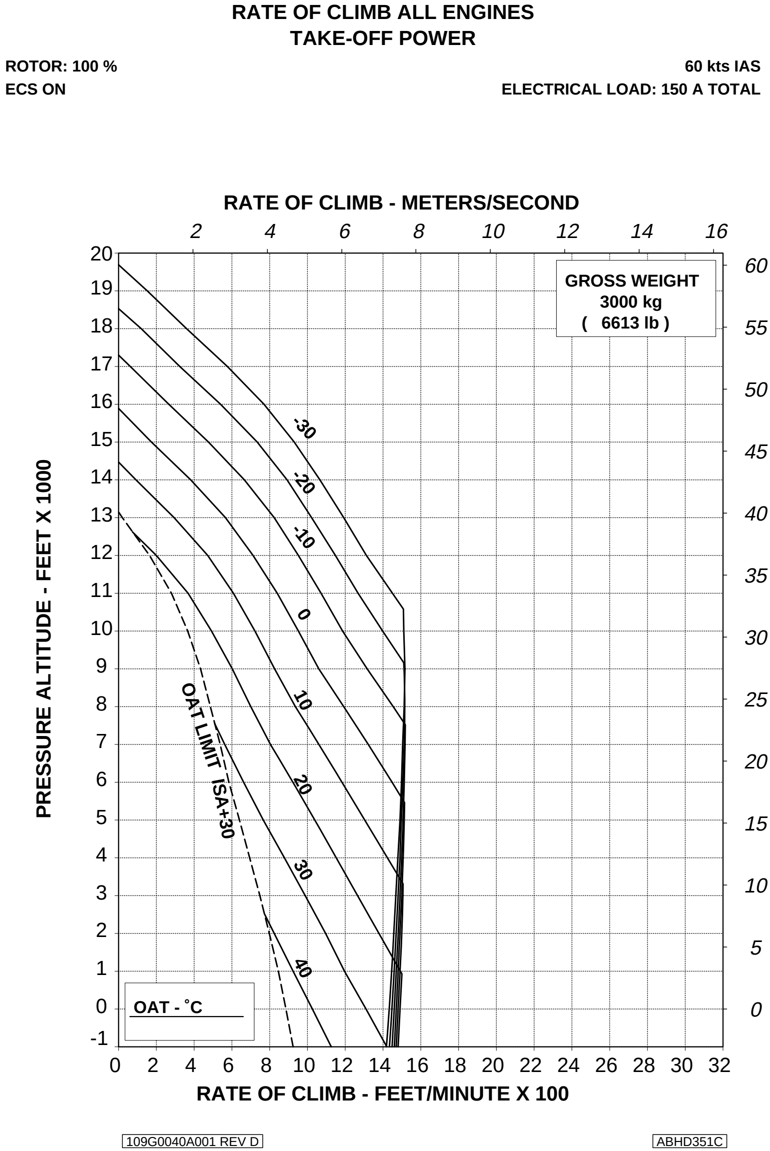


Figure 4-10. Rate of climb - all engines - take-off power - ECS ON.

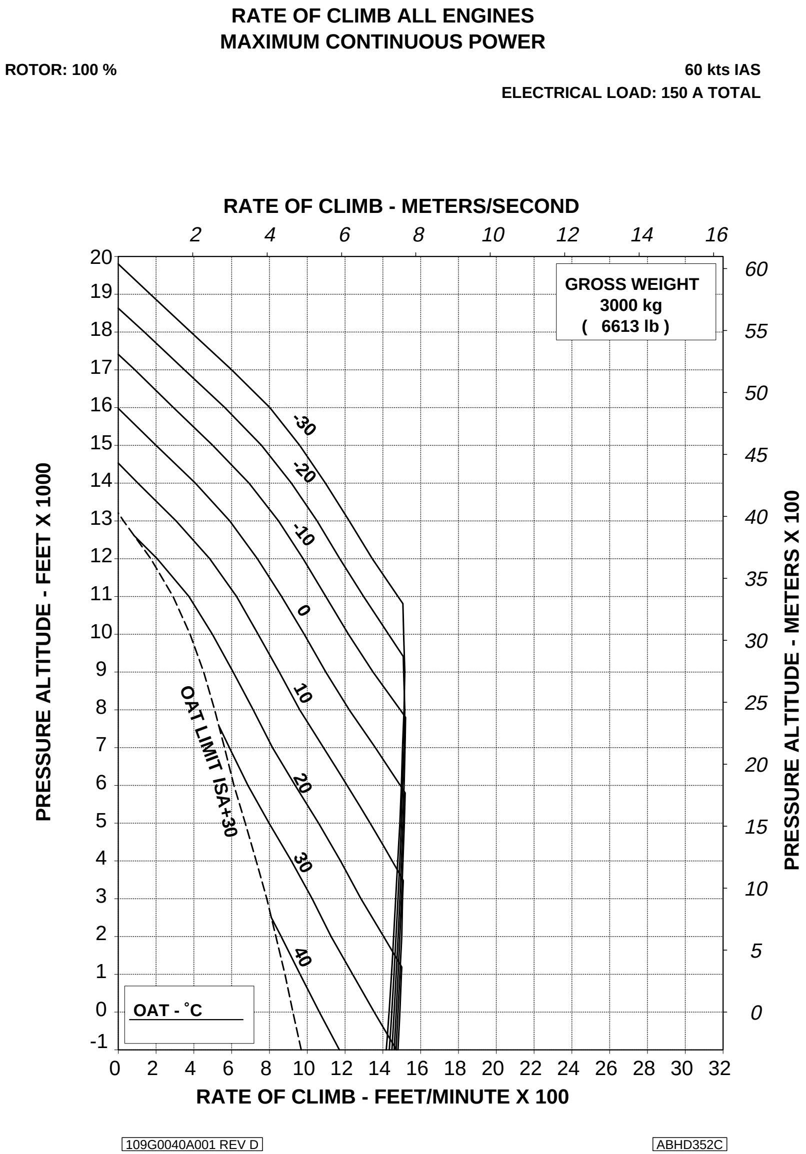


Figure 4-11. Rate of climb - all engines - maximum continuous power.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
HEATER ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

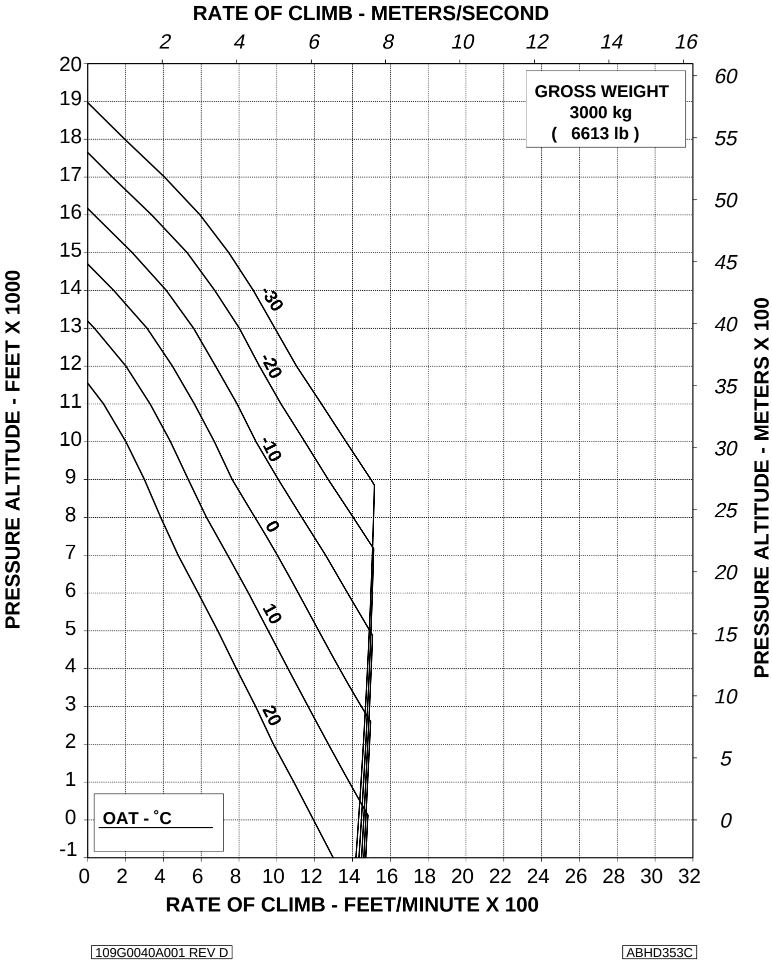


Figure 4-12. Rate of climb - all engines - maximum continuous power - heater ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
ECS ON **ELECTRICAL LOAD: 150 A TOTAL**

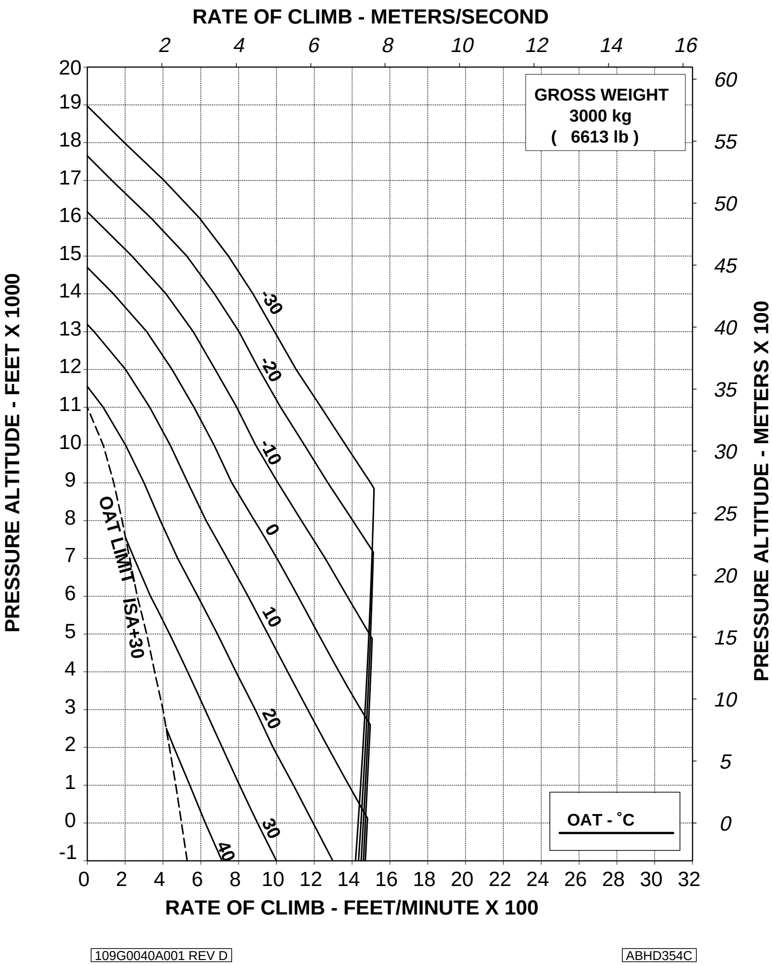


Figure 4-13. Rate of climb - all engines - maximum continuous power - ECS ON.

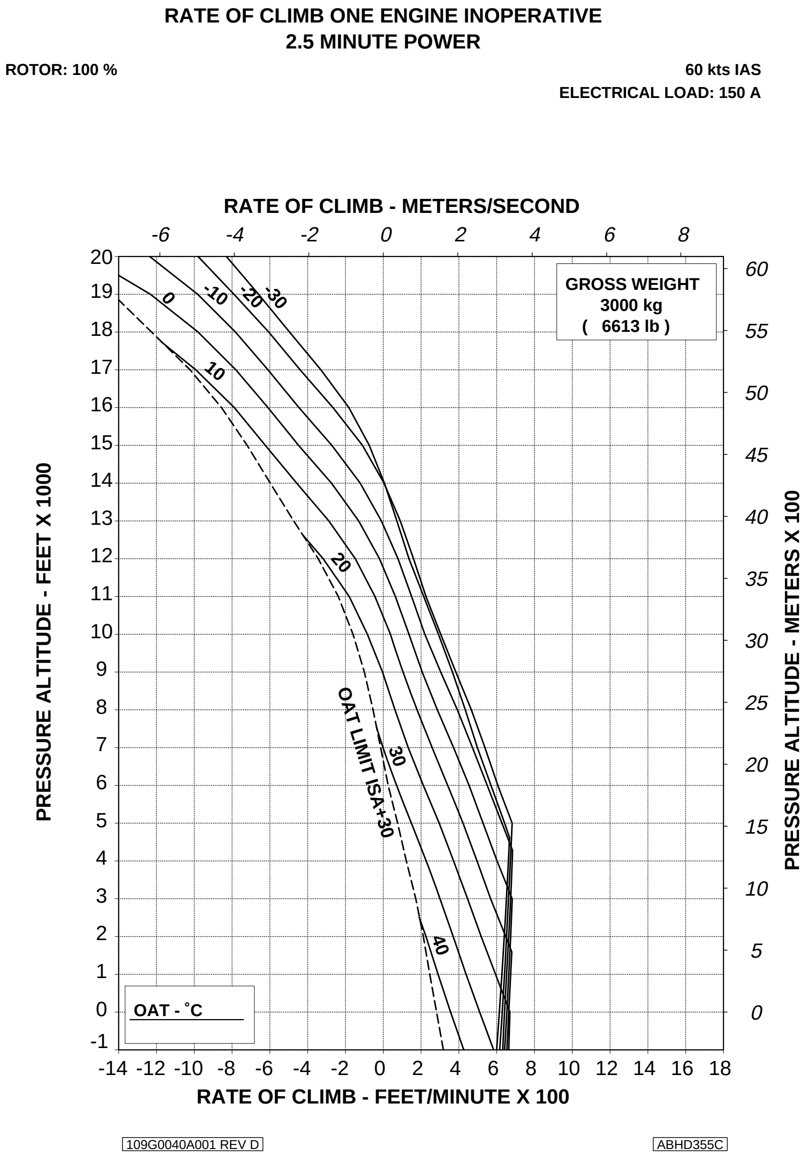


Figure 4-14. Rate of climb - OEI - 2.5 minutes power.

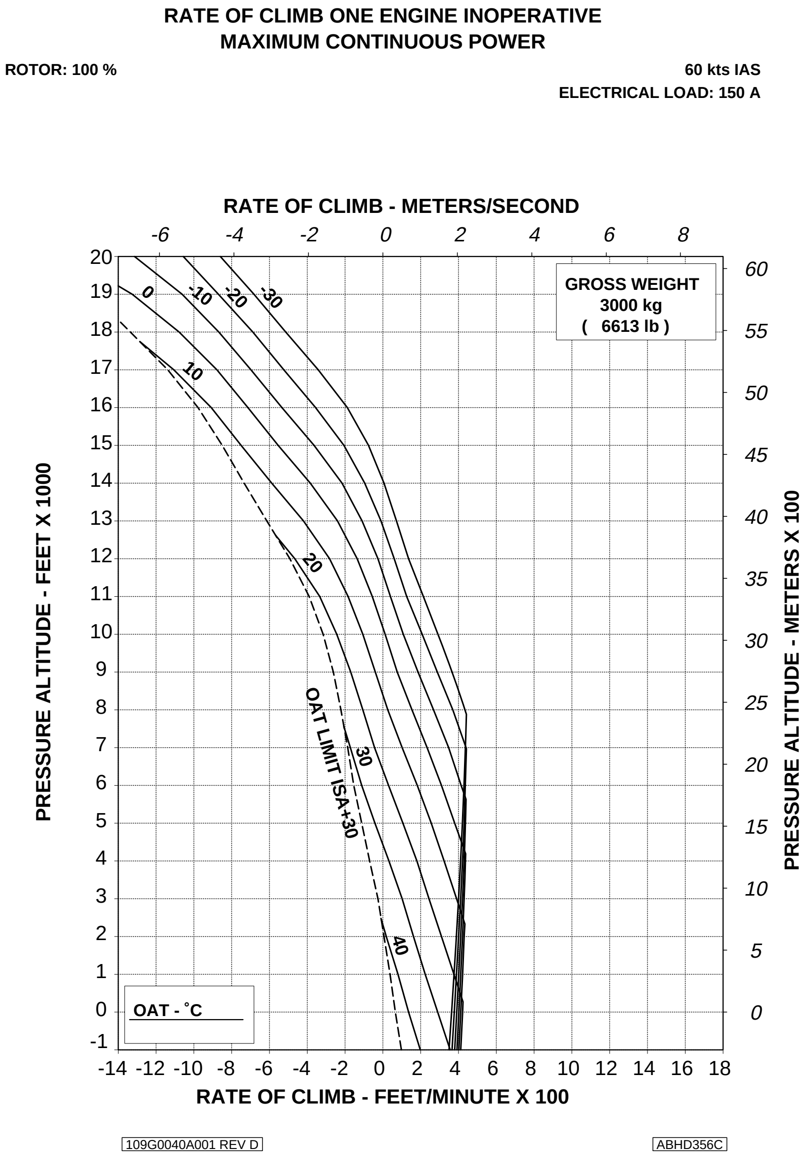


Figure 4-15. Rate of climb - OEI - maximum continuous power.

SECTION 6 - WEIGHT AND BALANCE

DATUM LINE LOCATIONS

Figure 6-1 presents the cargo hook station data to aid in weight and balance computations.

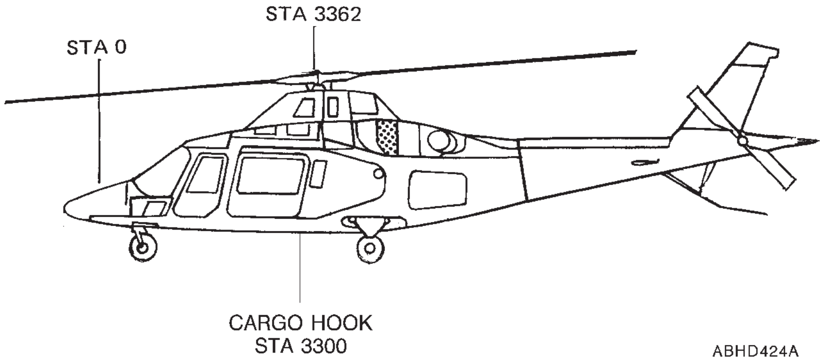
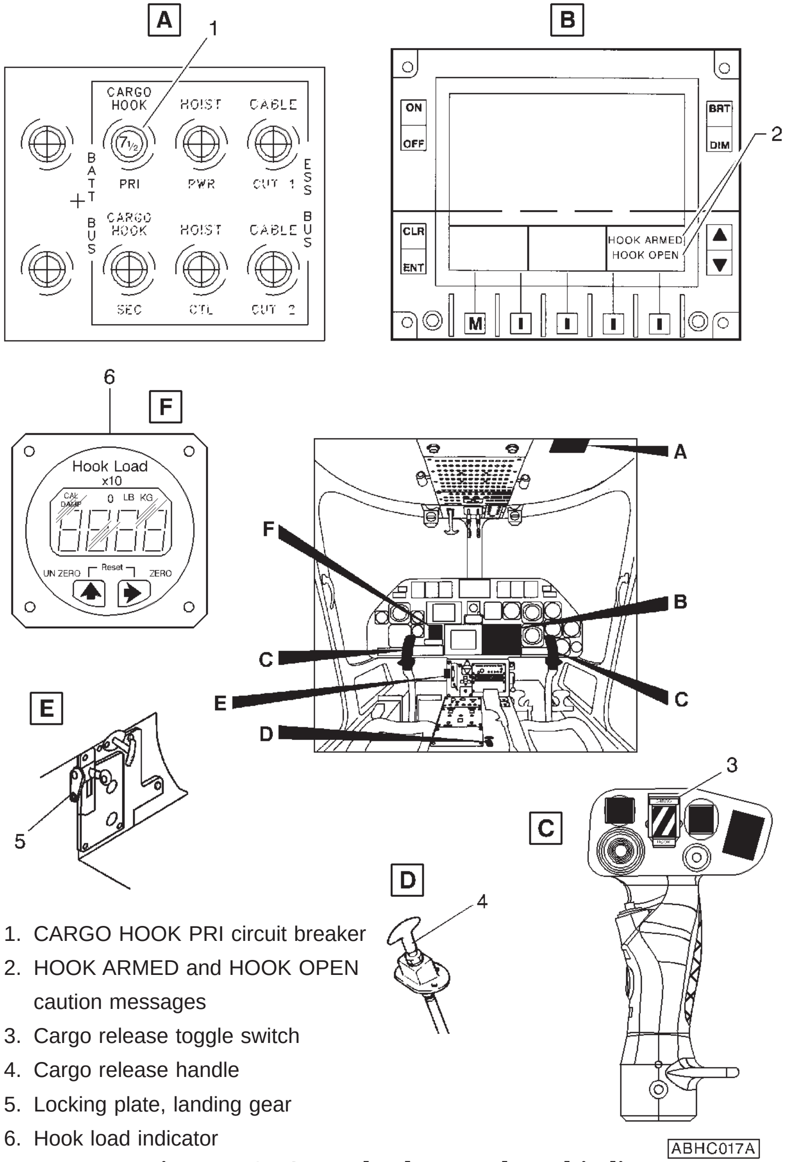


Figure 6-1. Cargo hook station diagram.

SECTION 7 - SYSTEMS DESCRIPTION



1. CARGO HOOK PRI circuit breaker
2. HOOK ARMED and HOOK OPEN caution messages
3. Cargo release toggle switch
4. Cargo release handle
5. Locking plate, landing gear
6. Hook load indicator

Figure 7-1. Cargo hook controls and indicators.

APPENDIX 34

E.N.A.C. Approval Letter 00/2775/MAE
dated 27 September 2000

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

CARGO HOOK

NOTE

This Appendix may be only used as a complement to the Appendix 33.

The cargo hook P/N 109-0811-75 consists of a support frame, a hook, an electrical and manual (emergency) release system and attaching hardware.
The cargo hook P/N 109-0811-75 (secondary) is compatible and can be installed only together with the cargo hook P/N 109-0810-31 (primary).
When both hooks are installed simultaneously, the secondary hook P/N 109-0811-75 may be used only as a safety device of the other hook.

NOTE

The swiveling link is not supplied with the cargo hook; however, it is recommended to use it between the suspension cable and the cargo.

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LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

CARGO HOOKS OPERATION

The cargo hooks may be used only as ″Class B rotorcraft-load combination″ as follows:

″Class B rotorcraft-load combination means one in which the external load is jettisonable and is lifted free of land or water during the rotorcraft operation″.
Trasporting of passengers suspended from the external cargo hook is restricted to emergency situations only. The applicable national regulations shall be complied with. All cargo must be secured to the primary cargo hook. The function of the Secondary cargo hook is to serve as a safety-of-flight back-up in the event of failure of the primary cargo hook.

NOTE

Cat. ″A″ operations are permitted only with the cargo hook P/N 109-0810-31 in the stowed position and the cargo hook P/N 109-0810-75 removed.

CAUTION

Simultaneous operation with individual loads, attached separately to the two hooks, is prohibited.

NOTE

When cargo hook P/N 109-0811-75 is utilized it is necessary the use of cable P/N 109-0811-86-101 to connect this hook with the hook P/N 109-0810-31. The cable installation P/N 109-0811-86-101 also comprises the cable P/N 109-0811-86-149 to which the cargo must be secured.

WEIGHT LIMITATIONS

Maximum Gross Weight with external
loads attached to cargo hook : 3000 kg (6613 lb)

NOTE

For maximum takeoff and landing weight refer to Section 1 of the basic Rotorcraft Flight Manual.

CARGO HOOK LOADING LIMITATIONS

Cargo hook loading limit : 500 kg (1102.5 lb)

CAUTION

When cargo hook P/N 109-0811-75 is used the load must be attached to cargo hook P/N 109-0810-31 which is provided with a load indicator, and however must not exceed 500 Kg (1102.5 lb).

WARNING

Flight with unballasted sling as an external load is prohibited.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

NOTE

The following preflight checks complete those scheduled for primary cargo hook P/N 109-0810-31.

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

Suspension safety cable and harnesses : Condition and security.

AREA N°2 (Fuselage - rh side)

Cargo hooks : Condition and security.

AREA N°7 (Cabin interior)

For the following checks connect the d.c. supply.

NOTE

Ground personnel shall assist the pilot during the cargo hook checks.

EDU 1 : HOOK ARMED caution message suppressed.

CARGO HOOK pushbutton on pilot cyclic stick : Lift the guard to arm the release system.

EDU 1 : HOOK ARMED caution message displayed.

CARGO HOOK pushbutton on pilot cyclic stick : Push. Verify the opening of both cargo hooks and HOOK OPEN caution message on EDU 1 displayed. Release the switch and verify that both cargo hooks return to closed position and that HOOK OPEN caution message is suppressed.

NOTE

Both cargo hooks are provided with a spring which keeps them permanently in closed position even when the opening system releases the lock-device.
A force of approximately 5 kg must be applied to each cargo hook to overcome the spring-force and to verify the hook opening.

CARGO HOOK pushbutton on pilot cyclic stick : Lower the guard to protect the release pushbutton.

EDU 1 : HOOK ARMED caution message suppressed.

Repeat cargo hooks checks by using CARGO HOOK pushbutton on copilot cyclic stick.

EMER CARGO RELEASE PULL handles (emergency) : Pull one at a time and verify opening of the respective cargo hook and HOOK OPEN caution message on EDU 1 displayed. Release the handles and verify that both cargo hooks return to closed position and that HOOK OPEN caution message is suppressed.

PILOT'S PREFLIGHT CHECK

(Every flight)

Cargo hooks : Condition and security.

SYSTEMS CHECK

Hook load indicator
(primary hook only) : Set to zero.

NOTE

Adjust the look load indicator after a 5 minutes warm up with no load on the hook.

TAKE-OFF

CARGO ATTACHMENT

Take off and stabilize in hovering at sufficient height to allow crew member to discharge helicopter static electricity and to attach cargo sling to the cargo hook.

NOTE

The distance between load and hook shall be kept as short as possible.

CAUTION

The cargo must be secured to primary hook P/N 109-0810-31, while the cargo hook P/N 109-0811-75 must be used to attach the security cable.

NOTE

Better directional control may be obtained by avoiding relative winds from critical azimuth area while performing external cargo operations. See ″Operation vs Allowable Wind″ in Section 4 of basic Rotorcraft Flight Manual.

WARNING

Discharge helicopter static electricity, before attaching cargo, by touching the airframe with a ground wire or if a metal sling is used, the hook-up ring can be struck against the cargo hook.
If contact has been lost after initial grounding of the helicopter, it should be electrically regrounded and, if possible, contact maintained until hook-up is completed.

NOTE

Attachment of cargo sling to the hook can be observed by means of the rearview mirror.

After cargo attachment slowly increase the collective pitch and ascend vertically to take-up the slack of cargo sling.
Lift vertically cargo from surface and read the hook load indicator to verify the cargo weight to be within the hook loading limitations.
Hover to check for satisfactory controllability and power within limits.

IN FLIGHT

Enter into slow forward speed and verify that uncontrollable or hazardous flight conditions do not exist. Allow adequate sling load clearance over obstacles. Increase forward speed and select an operational airspeed at which no hazardous oscillation is encountered.

APPROACH AND LANDING

CARGO RELEASE

CARGO HOOK pushbutton on cyclic stick	: Lift the guard to arm the release system.

EDU 1 : HOOK ARMED caution message displayed.

Perform the approach to the cargo release area with care and at low speed. Stabilize hover above release point, then slowly descend until cargo lies down on ground.

NOTE

Prolongated hover OGE operations in hot day conditions at maximum gross weight may result in an increase of main transmission oil temperature.

CARGO HOOK pushbutton on cyclic stick : Push to release cargo. Verify the HOOK OPEN caution message on EDU 1 is displayed.

NOTE

The load is released only when its weight overcomes the spring-force of the hooks.

Rearview mirror : Check load released.

NOTE

In case of non-release of cargo, the pilot should slowly increase the collective pitch to ascend, as much necessary to strain the cable, before operating again the CARGO HOOK pushbutton.

NOTE

In the event of an electrical failure of one or both cargo hooks, pull mechanical manual release control handle EMER CARGO RELEASE PULL of secondary hook (P/N 109-0811-75, see SECONDARY CARGO HOOK placard on pedestal) then mechanical manual release control handle EMER CARGO RELEASE PULL of the primary hook (P/N 109-0810-31, see PRIMARY CARGO HOOK placard on pedestal) to release cargo.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

WARNING SYSTEM

CAUTION MESSAGES (AMBER)

EDU message	Fault condition	Corrective action
HOOK ARMED	Cargo release system armed.	No corrective action is required. Take care to avoid accidental release of cargo.
HOOK OPEN	One and/or both hooks in open position	Verify if hook release pushbutton or emergency handles were operated.

EMERGENCY CARGO RELEASE

In case that an emergency situation requires to release the cargo, operate the electrical release system through the pushbutton on cyclic stick.
In case this fails to operate, pull mechanical manual release control handle EMER CARGO RELEASE PULL of secondary hook then mechanical manual release control handle EMER CARGO RELEASE PULL of the primary hook to release cargo.

SECTION 4 - PERFORMANCE DATA

Refer to Appendix 33.

SECTION 6 - WEIGHT AND BALANCE

DATUM LINE LOCATIONS

Figure 6-1 presents the primary and secondary cargo hooks stations data to aid in weight and balance computations.

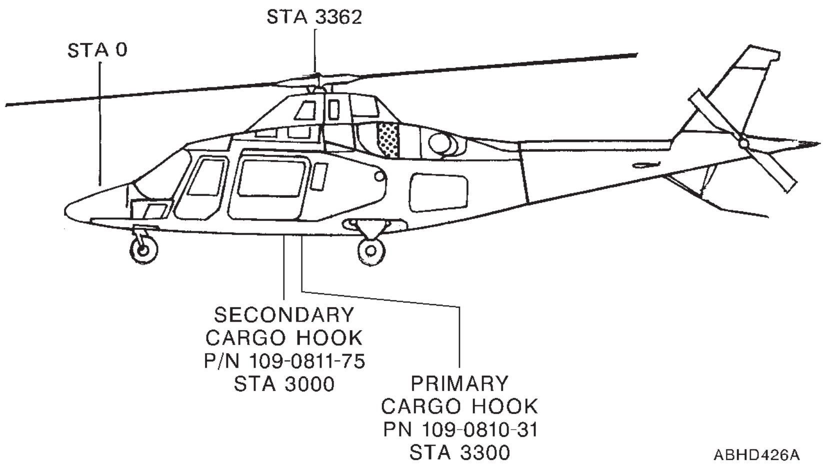


Figure 6-1. Primary and secondary cargo hooks stations diagram.

SECTION 7 - SYSTEMS DESCRIPTION

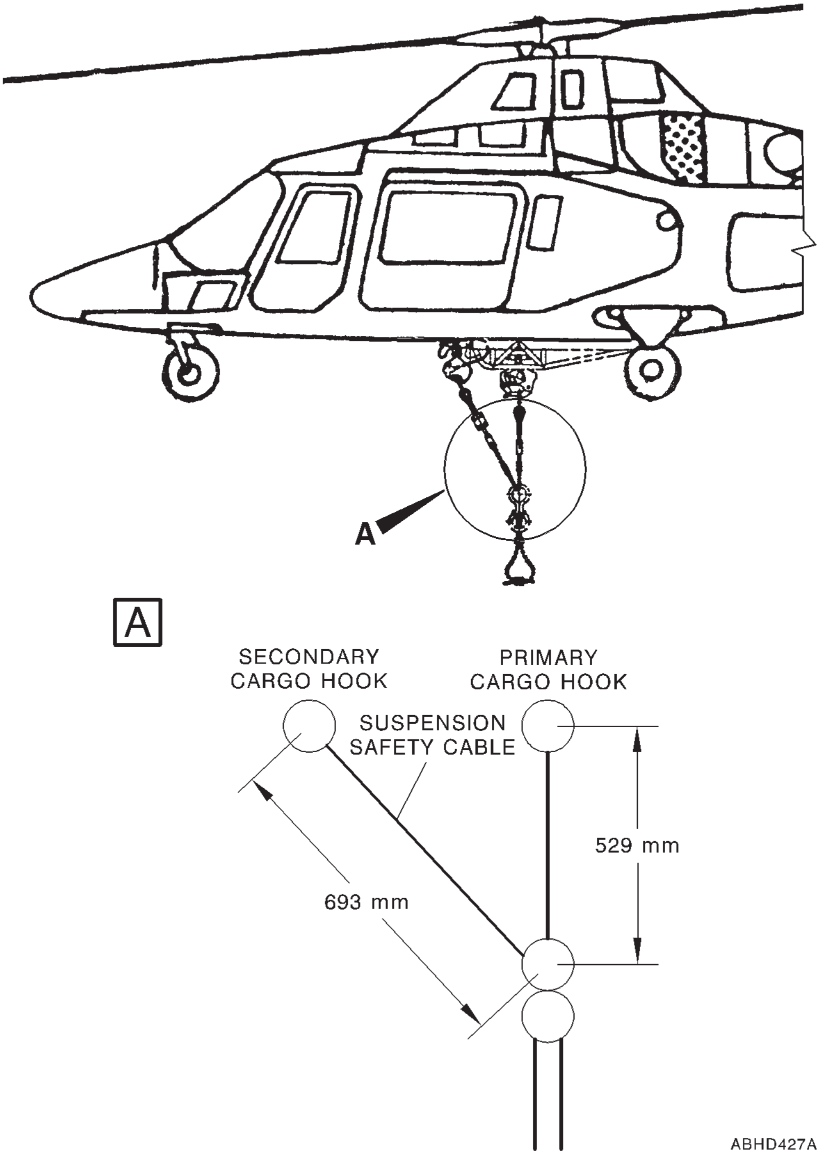
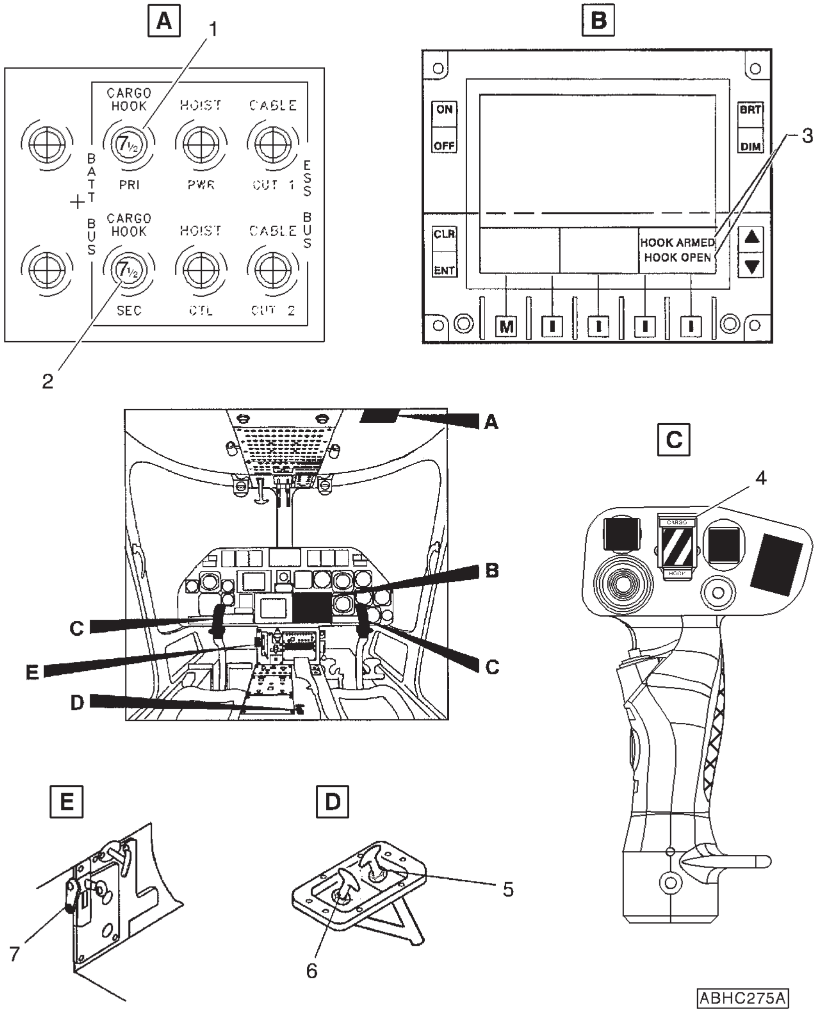


Figure 7-1. External load rigging.



1. CARGO HOOK PRI circuit breaker
2. CARGO HOOK SEC circuit breaker
3. HOOK ARMED and HOOK OPEN caution messages
4. Cargo release toggle switch
5. Secondary hook, cargo release handle
6. Primary hook, cargo release handle
7. Locking plate, landing gear

Figure 7-2. Secondary cargo hook controls and indicators.

E.N.A.C. Approval Letter 171035/SPA
dated 11 July 2001

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

V_{LO} AND V_{LE} EXTENSION UP TO 140 KIAS

The extension of V_{LO} and V_{LE} airspeed limitations to 140 KIAS aims to improve the helicopter performance with landing gear extended and in particular to let speed up the instrumental landing procedure. The V_{LO} and V_{LE} limit extension compels the modification P/N 109-0813-12 consisting of new composite doors for the forward landing gear doors with improved hinges.

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PART I - E.N.A.C. APPROVED

SECTION 1 - LIMITATIONS

REQUIRED EQUIPMENT

In addition to the basic required equipment the helicopter must be fitted with forward landing gear door P/N 109-0313-88-101.

AIRSPEED LIMITATIONS (KIAS)

Maximum landing gear operating speed V_{LO} 140 KIAS.

Maximum landing gear extended speed V_{LE} 140 KIAS.

PLACARDS

AIRSPEED LIMITATIONS V_{NE} - KIAS

Hp-ft / OAT°C	-1000 TO SL	3000	6000	9000	12000	15000	18000	20000
45	168	165	–	–	–	–	–	–
35	168	168	158	–	–	–	–	–
25	168	168	161	152	143	–	–	–
15	168	168	164	155	146	136	111	–
5	168	168	167	157	148	139	114	97
-5	168	168	168	160	151	142	117	100
-15	168	168	168	164	154	145	120	103
-25	168	168	168	167	157	148	123	106

AUTOROTATION OR OEI:
REDUCE V_{NE} BY 40 KIAS

MAX FWD TOUCHDOWN SPEED: 40 Kts
V_{LO} AND V_{LE} : 140 KIAS

ABHD144D

In clear view of the pilot

SECTION 2 - NORMAL PROCEDURES

TAKE-OFF

HOVER TAKE-OFF

CAUTION

Do not operate landing gear at speeds above 140 KIAS.
Do not fly with landing gear extended at speeds above 140 KIAS.

ROLLING TAKE-OFF

CAUTION

Do not operate landing gear at speeds above 140 KIAS.
Do not fly with landing gear extended at speeds above 140 KIAS.

IN FLIGHT

CAUTION

Do not operate landing gear at speeds above 140 KIAS.
Do not fly with landing gear extended at speeds above 140 KIAS.

APPROACH AND LANDING

CAUTION

Do not operate landing gear at speeds above 140 KIAS.
Do not fly with landing gear extended at speeds above 140 KIAS.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE

No change.

E.N.A.C. Approval Letter 171300/SPA
dated 30 November 2001

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SUPPLEMENTARY FUEL TANKS

The supplementary fuel tanks installation P/N 109-0812-46 provides an additional 230 liters capacity. It consists of two tank cells (RH cell of 94 liters and LH cell of 136 liters) installed behind the passenger seat. However the installation can be arranged as follows:
- both LH and RH tank cells installed;
- only RH tank cell installed.

The fuel transfer from the supplementary fuel tank cells to the main fuel cell is by gravity. The two tank cells are separated by panels; each cell is provided with a fuel level probe.

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SECTION 1 - LIMITATIONS

CENTER OF GRAVITY LIMITATIONS

After supplementary fuel tank installation, the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

ENGINE PRE-START CHECK

Fuel quantity : Check.

CAUTION

When only the RH tank cell is installed and fuel system is fully serviced, a difference of fuel quantity indication, equivalent to the fuel contained into the RH tank cell, is normal. Such difference decreases with the fuel consumption down to zero when about 110 kg of fuel is reached in each main tank.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

WEIGHTS - ARMS AND MOMENTS

LONGITUDINAL MOMENTS

USABLE FUEL - MAIN FUEL TANKS PLUS SUPPLEMENTARY TANKS			
WEIGHT (kg)	**CAPACITY (l) (0.8 kg/l)**	**ARM (mm)**	**MOMENT (kgm)**
20	25	3324	66.5
40	50	3327	133.1
60	75	3329	199.7
80	100	3331	266.5
100	125	3399	339.9
120	150	3461	415.3
140	175	3505	490.7
160	200	3539	566.2
180	225	3543	637.7
200	250	3551	710.2
220	275	3571	785.6
240	300	3614	867.4
260	325	3650	949.0
280	350	3675	1029.0
300	375	3695	1108.5
320	400	3715	1188.8
340	425	3735	1269.9
360	450	3755	1351.8
380	475	3775	1434.5
400	500	3785	1514.0
420	525	3798	1595.2
440	550	3806	1674.6
460	575	3818	1756.3
480	600	3830	1838.4

(Cont.d)

USABLE FUEL - MAIN FUEL TANKS PLUS SUPPLEMENTARY TANKS			
WEIGHT (kg)	**CAPACITY (l) (0.8 kg/l)**	**ARM (mm)**	**MOMENT (kgm)**
500	625	3840	1920.0
520	650	3847	2000.4
540	675	3856	2082.2
560	700	3863	2163.3
580	725	3872	2245.8
600	750	3876	2325.6
620	775	3883	2407.5
640	800	3890	2489.6
660	825	3897	2572.0

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANK ONLY			
WEIGHT (kg)	**CAPACITY (l) (0.8 kg/l)**	**ARM (mm)**	**MOMENT (kgm)**
20	25	3324	66.5
40	50	3327	133.1
60	75	3329	199.7
80	100	3331	266.5
100	125	3399	339.9
120	150	3461	415.3
140	175	3505	490.7
160	200	3539	566.2
180	225	3543	637.7
200	250	3551	710.2
220	275	3571	785.6

(Cont.d)

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANK ONLY			
WEIGHT (kg)	**CAPACITY (l) (0.8 kg/l)**	**ARM (mm)**	**MOMENT (kgm)**
240	300	3614	867.4
260	325	3653	949.8
280	350	3687	1032.4
300	375	3716	1114.8
320	400	3741	1197.1
340	425	3764	1279.8
360	450	3784	1362.2
380	475	3802	1444.8
400	500	3819	1527.6
420	525	3834	1610.3
440	550	3847	1692.7
460	575	3860	1775.6
480	600	3871	1858.1
500	625	3883	1941.5
520	650	3893	2024.4
540	675	3901	2106.5
551	689	3910	2155.2

LATERAL MOMENTS

USABLE FUEL - MAIN FUEL TANKS PLUS RH SUPPLEMENTARY TANK ONLY			
WEIGHT (kg)	**CAPACITY l (0.8 kg/l)**	**ARM (mm)**	**MOMENT (kgm)**
20	20	0	0.0
40	50	0	0.0
60	75	0	0.0
80	100	0	0.0
100	125	0	0.0
120	150	0	0.0
140	175	0	0.0
160	200	0	0.0
180	225	0	0.0
200	250	0	0.0
220	275	0	0.0
240	300	0	0.0
260	325	8	2.1
280	350	15	4.2
300	375	20	6.0
320	400	25	8.0
340	425	30	10.2
360	450	34	12.2
380	475	38	14.4
400	500	42	16.8
420	525	45	18.9
440	550	48	21.1
460	575	52	23.9
480	600	55	26.4
500	625	56	28.0
520	650	57	29.6
540	675	58	31.3
551	689	58	32.0

SECTION 8 - HANDLING AND SERVICING

SERVICING

Supplementary fuel tanks

	RH fuel cell	LH fuel cell
Capacity	94 liters	136 liters
Usable	94 liters	136 liters

E.N.A.C. Approval Letter 171350/SPA
dated 19 December 2001

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

MULTIFUNCTION DISPLAY KMD 550

The KMD 550 P/N 109-0813-16 is a multifunction display system . Basically the system is interfaced with an external GPS receiver for moving map operation. The system is capable of displaying highly detailed and scanned area maps.
As an option the system may be interfaced with an external TAS SKY497.
Refer to ″SKYWATCH TRAFFIC ADVISORY SYSTEM SKY 497″ (Appendix 25).

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The KMD 550 system can be used in VFR conditions only as a supplemental navigation system.
The navigation must not be based on the KMD 550 system only.
The functions associated with key TRFC are available only if system is interfaced with TAS SKY 497. The function associated with keys WX and TERR are not available.

SECTION 2 - NORMAL PROCEDURES

NOTE

For correct and complete use of KMD 550, refer to Honeywell KMD 550/850 Pilot's Guide. If the system is interfaced with TAS SKY 497, refer also to Honeywell ″Traffic Avoidance Function (TCAS/TAS) pilot's guide addendum″.

SYSTEMS CHECK

NOTE

Before switching on the system check functionality of the relative GPS system (Trimble 2101).

KMD 550 : Turn the OFF/ON knob to ON position and adjust brightness. At the end of self-test, select MAP function key and check for valid GPS fix data. NO EXTERNAL GPS DATA advisory should not be shown in the middle of the display.

IN FLIGHT

KMD 550 : As required.

NOTE

If the GPS is lost at any time during normal operation of the unit, the ″NO EXTERNAL GPS DATA″ is overlaid on the map.

NOTE

In map mode the system, when interfaced with SKY497 TAS , is not able to display traffic information on the screen if heading data is invalid.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE

No change.

E.N.A.C. Approval Letter 171301/SPA
dated 30 November 2001

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

MAIN TRANSMISSION GEAR BOX VIBRATION ISOLATOR STRUTS

The main transmission gear box vibration isolator struts installation P/N 109-0822-99 consists of two vibration isolator struts connected to the main transmission gear box and the cabin roof in lieu of the standard forward fixed struts. Their purpose is to provide forward structural support to the main transmission gear box and reduce the vibratory forces transmitted to the aircraft fuselage. Each strut consists of an elastomeric component containing fluid, connected to a tubolar metallic strut. The struts are attached to the main transmission gear box and to the cabin roof as for the standard rigid struts.

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SECTION 1 - LIMITATIONS

REQUIRED EQUIPMENT

Operations at gross weight greater than 2850 kg up to 3000 kg are permitted provided that the following assembly is installed:

— Tail rotor installation P/N 109-0136-02-103.

CENTER GRAVITY LIMITATIONS

After installation of main transmission gear box vibration isolator struts, the new empty weight and CG location must be determined.

COMPATIBILITY

The main transmission gear box vibration isolator struts P/N 109-0822-99 installation is not compatible with the following installations:

—	109-0812-31	External hoist
—	109-0810-31	Cargo hook
—	109-0811-75	Cargo hook

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.N.A.C. Approval Letter No. 02/171602/SPA
dated 12 November 2002

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

MAIN TRANSMISSION OIL TEMPERATURE LIMIT EXTENSION

The modification P/N 109-0823-24-101 allows to extend the main transmission oil temperature limit up to 120° Centigrade.
The modification consists of a new transmission oil filter provided with a temperature switch activating the XMSN OIL HOT warning message at 120° Centigrade, a new EDU P/N 109-0900-42-2A08 (standard configuration) and a new DAU P/N 109-0900-42-6A08.

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LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

MAIN TRANSMISSION LUBRICATION SYSTEM LIMITATIONS

Oil pressure

No change.

Oil temperature

Maximum	: 120°C

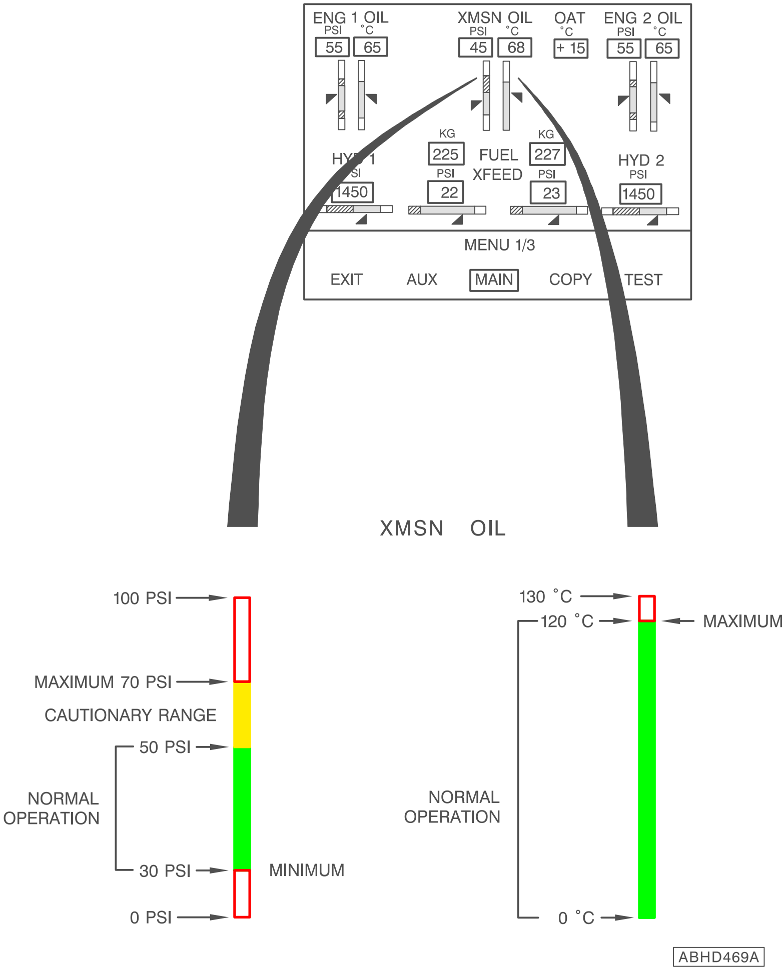

Fig. 1-1. EDU 2 (MAIN) - TRANSMISSION OIL (PRESSURE AND TEMPERATURE)

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.N.A.C. Approval Letter 03/171104/SPA
dated 11 March 2003

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SEARCHLIGHT

The Searchlight installation P/N 109-0812-83 consists of a swinging light installed under the rear section of the helicopter fuselage. The light can be extended, stowed or swung as required by operating the switches on the central console.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

SEARCHLIGHT LIMITATIONS

The operation of searchlight P/N 109-0812-83 by the pilot during takeoff and landing is prohibited.
During takeoff and landing the searchlight P/N 109-0812-83 may be operated only by the co-pilot.

AIRSPEED LIMITATIONS (KIAS)

Maximum speed for searchlight extension	: 120 KIAS
Maximum speed with searchlight extended, for orientation and retraction	: 150 KIAS

CENTER OF GRAVITY LIMITATIONS

After searchlight installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

Area N°2 (Fuselage - RH side)

Searchlight	: Condition and cleanliness.

PILOT'S PREFLIGHT CHECK

(Every flight)

Searchlight : Condition and cleanliness.

ENGINE PRE-START CHECK

SCHLT 2 CLT/SCHLT 2 PWR circuit breakers : ON

SYSTEMS CHECK

Searchlight ON/OFF/STOW switch on central console : OFF, check.

IN FLIGHT

NOTE

When operating Searchlight, magnetic compass indication is not reliable.
Be aware of Searchlight deactivation before proceeding to an IFR Flight.

SEARCHLIGHT OPERATING PROCEDURE

NOTE

The searchlight, consisting of a swinging light installed under the rear section of the helicopter fuselage, can be extended, stowed or swung as required by operating the switches on the central console.

EXTENSION

EXT/RETR/L/R switch on central console : EXT (to extend light in the desired position).

ON/OFF/STOW switch on central console : ON.

NOTE

With the switch in OFF position the light remains extinguished in the position where it has been left.

EXT/RETR/L/R switch on central console : Set as necessary.

NOTE

Moving switch to L or R position the searchlight rotates left or right. It is possible to adjust the light in an intermediate position, from stowed to extended, by temporarily moving the switch to EXT or RETR position.

RETRACTION

ON/OFF/STOW switch on central console : STOW then OFF.

NOTE

In STOW position the light is extinguished.

Check for Searchlight deactivation after use by copilot or crew member.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.N.A.C. Approval Letter 03/171177/SPA
dated 2 May 2003

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

E.M.T. (EMERGENCY MEDICAL TRANSPORTATION) SINGLE AND DUAL LITTER

The Emergency Medical Transportation P/N 109-0813-59 provides the possibility to convert the VIP Utility Interior Arrangement P/N 109-0811-82 configured helicopters into an E.M.T. interior.
The configuration changes are realized by removing the seat cushions and by folding the seat back panels. This provides the necessary supporting and locking structure for a standard litter, allowing the transportation in an emergency situation of an injured person. The E.M.T. kit can be provided for a single or dual litter configuration.
In the single litter configuration (-101/-103/-107/-109) the litter is positioned on the left hand side of the cabin.
The dual litter (-105/-111) E.M.T. kit provides the capability of a second litter transportation on the right hand side of the cabin.
The single litter E.M.T. provides also seats for up to four (4) passengers into the cabin compartment, while the dual litter can provide two (2) passenger seats in between the litters. Single litter kit (-101/-107) can also be provided with the right hand side auxiliary fuel tank installation.

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SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The helicopter in E.M.T. configuration permits transportation of injured people under day and night VFR and IFR not icing conditions.

REQUIRED EQUIPMENT

The E.M.T. is a development of a VIP Utility configuration. The helicopter arrangement shall include :

For single litter configuration P/N 109-0813-59-101

— the installation P/N 109-0811-82-119

For single litter configuration P/N 109-0813-59-103

— the installation P/N 109-0811-82-121

For single litter configuration P/N 109-0813-59-107

— the installation P/N 109-0811-82-133

For single litter configuration P/N 109-0813-59-109

— the installation P/N 109-0811-82-135

For dual litter configuration P/N 109-0813-59-105

— the installation P/N 109-0811-82-121

For dual litter configuration P/N 109-0813-59-111

— the installation P/N 109-0811-82-135

The sliding doors P/N 109-0822-58.

Whenever the E.M.S. supplementary cabin lights P/N 109-0812-51-107 are installed, the curtain installation P/N 109-0812-52 must be present on board.

VFR OPERATION

FLIGHT WITH PASSENGER CABIN DOORS OPEN OR REMOVED

Operation with passenger cabin doors open or removed is prohibited when patient is on board.

FLIGHT CREW

The minimum flight crew consists of one pilot and an attendant; both of whom shall be trained in and capable of assisting in litter patient emergency evacuation procedures.

NUMBER OF SEATS

Seven (7) including the pilot and the litter patient, for single litter configuration Six (6) including the pilot and the litter patients, for dual litter configuration.

CENTER OF GRAVITY LIMITATIONS

After E.M.T., single or dual litter, installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N°7 (Cabin interior)

E.M.T. interior	:	Check for condition.
Cabin seats and litter	:	Check for condition and straps fastened if unoccupied.

PILOT'S PREFLIGHT CHECK

(Every flight)

E.M.T. interior	: Check for condition.
Cabin seats and litter	: Check for condition and straps fastened if unoccupied.

TAKE-OFF

Cabin curtain (if installed) : Open.

In case of night flight:

CAB switch : OFF.

IN FLIGHT

CAB switch : As required.

NOTE

Whenever the supplementary cabin lights are installed and switched on, cabin curtain must be closed.

APPROACH AND LANDING

Cabin curtain (if installed) : Open.

In case of night flight

CAB switch : OFF.

LITTER OPERATIONS

LITTER(S) LOADING

— Secure patient(s) to the litter(s) using the three straps provided;

— remove seat cushion of the central aft seat;

— unlock and lower the back of the central aft seat;

— load left litter from the left door and right litter (if installed) from the right door;

— secure litter(s) to the inboard locks;

— raise the back of the central aft seat and lock;

— Install seat cushion of the central aft seat.

LITTER(S) UNLOADING

— Remove seat cushion of the central aft seat;

— unlock and lower the back of central aft seat;

— unlock the litter(s) from the locks;

— unload the left litter from the left door and unload the right litter (if installed) from the right door;

— raise the back of the central aft seat and lock;

— install seat cushion of the central aft seat.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

EVACUATION THROUGH EMERGENCY EXITS

Unstrap the patient(s) and evacuate either through the LH or RH emergency exits.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

PATIENT ARM ON LITTER

Longitudinal arm (STA)	:	2900 mm (114.2 inches) from STA 0.
Lateral arm (BL) (left and/or right)	:	412 mm (16.2 inches) from the helicopter plane of symmetry.

FORWARD ATTENDANTS ARM

Longitudinal arm (STA)	:	2455 mm (96.6 inches) from STA 0.
Lateral, inboard attendant arm (BL)	:	BL 0.
Lateral, outboard attendant arm (BL)	:	412 mm (16.2 inches) from the helicopter plane of symmetry.

AFT ATTENDANTS ARM

Longitudinal arm (STA)	:	3200 mm (126 inches) from STA 0.
Lateral, inboard attendant arm (BL)	:	BL 0.
Lateral, outboard attendant arm (BL)	:	412 mm (16.2 inches) from the helicopter plane of symmetry.

SECTION 7 - SYSTEM DESCRIPTION

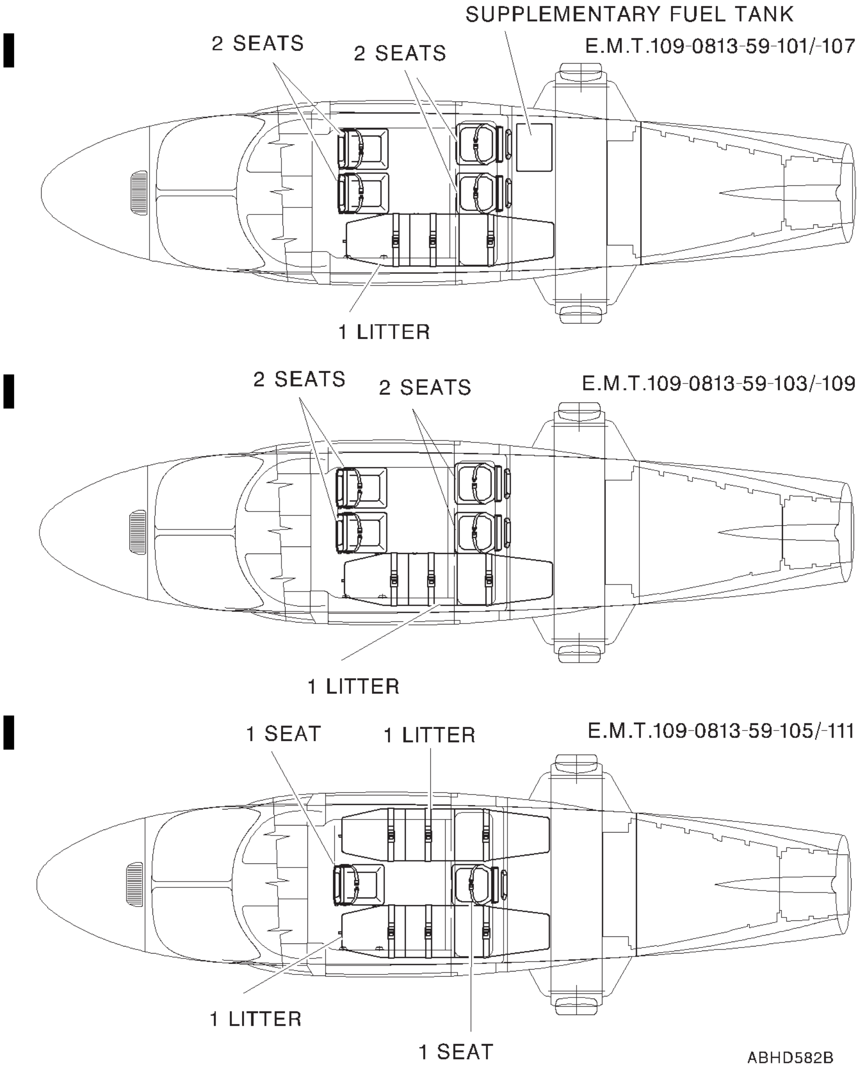

Figure 7-1. E.M.T. kit.

E.N.A.C. Approval Letter 03/171177/SPA
dated 2 May 2003

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

EFIS - ASTRONAUTICS

(ELECTRONIC FLIGHT INSTRUMENT SYSTEM)

The EFIS P/N 109-0900-71-1A01 or 2A01 (NVG version) consists of four AMLCD (Active Matrix Liquid Crystal Display) displays installed in the instrument panel and performing as EADI and EHSI indicators.
The EFIS interfaces with the attitude and navigation systems and with the weather radar and/or Stormscope WX 1000 (if installed).
The EADI and EHSI, in front of each pilot, are coupled also with the Flight Director computer (if installed).

AGUSTA

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SECTION 1 - LIMITATIONS

No change.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

Pilot's Daily preflight check (First flight of the day)

Area N.7 (Cabin interior)

BATT RELAY circuit breaker	: In. Door secured.

NOTE

The above mentioned circuit breaker is accessible from the pilot pedal bay through an inspection door and thus cannot be seen from the on-board seated position.

ENGINE PRE-START CHECKS

Stowable anti-glaze shield on the instrument panel	: Adjust and fix as desired.

NORMAL ENGINE START

ENGINE 2 START

After RAD-MSTR switch activation:

EFIS displays	: Adjust brightness as necessary.

QUICK ENGINE START

After RAD-MSTR switch activation:

EFIS displays	: Adjust brightness as necessary.

SYSTEMS CHECK

EADI instrument:

ATT FAIL flag	:	Out of view.
ADI	:	Verify for consistency with the ADI stand-by
ATT REV pushbutton	:	Press - Check pushbutton illuminated - Check ADI revertion to opposite VG source - Check the consistency of the new ADI information with ADI stand-by - ATT 2 flag in view.
ATT REV pushbutton	:	Press again

Repeat the previous check with copilot instruments and verify ATT 1 flag in view when ATT REV pushbutton is in.

Radio altimeter	:	Check zero altitude (± 5 ft)
RA 1 FAIL flag	:	Out of view.
DH	:	Set 50 ft and check DH flag in view
RA button	:	Press and hold - Check for TEST100RA1 indication (± 5 ft) - DH flag out of view.
RA button	:	Release
DH	:	Set as desired

F/D STBY button (on FD control panel)	:	Press and hold
FD FAIL flag	:	Check, in view
F/D STBY button	:	Release

EHSI instrument:

Compass HDG FAIL message	:	Out of view
Compass heading	:	Consistent with magnetic compass reading. - Check white DG flag out of view (if in view set MAG/DG switch on compass control panel to MAG position)
HDG bug selector	:	As desired
NAV source selector	:	As desired
CRS selector	:	As desired
BRG 1 selector	:	As desired
BRG 2 selector	:	As desired

IN FLIGHT

FLIGHT DIRECTOR OPERATION (if installed)

VOR (or GPS) coupled to NAV (or LNAV) mode

CAUTION

When NAV mode on FD control panel is in CAP condition and a change of navigation source is carried out with the NAV key on the pilot (in command) EHSI, or the active navigation source goes to invalid (loss of signal), the relevant FD mode annunciation (NAV or LNAV) changes from green to amber to alert the pilot of a possible degradation of the active FD mode.
Press the NAV pushbutton on FD mode selector to deactivate the mode.
Check the navigation source and select (if necessary) the new CRS and intercept HDG, then reselect the NAV pushbutton on FD control panel.

ILS mode

CAUTION

During ILS approach with FD engaged, the selection of the other navigation source (e.g. LNAV) will be inhibited as soon as the LOC is captured.
Swapping from ILS 1 to ILS 2 is the only navigation source selection available.
If change between ILS 1 to ILS 2 is carried out when FD is in LOC CAP status, the green LOC annunciation changes from green to amber to alert pilot of a possible degradation of the active FD mode.
Press on FD control panel the NAV push button to deselect ILS.
Check the navigation source and select the inbound course and intercept heading (if necessary) then reselect the NAV pushbutton on FD control panel.

STORMSCOPE WX 1000 OPERATION (if installed)

After Stormscope activation (pushbutton LX set to ON), data can be displayed on EHSI in MAP mode (both center or arc format) by pressing the WX/LX pushbutton.
The selection of STORMSCOPE overlay page is annunciated by ″LX″ blue legend in the bottom-right side of display.
If data are processed from system and EHSI is not selected to STORMSCOPE overlay page, an amber caution STORM is displayed in the upper right corner to indicate to the pilot the presence of storm cell.
Press CLKR key to clear data overlayed on the display.

RADAR METEO OPERATION (if installed)

CAUTION

Do not select radar function by using EFIS soft key to avoid affecting the operational logic of the radar system. Only select radar functions from the relevant radar control panel.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

WARNING MESSAGES (RED)

EADI message	Fault condition	Corrective action
ATT FAIL	Gyro not erected, loss of power to gyro, loss of Roll or Pitch data on EADI	Check breaker. Refer to stand-by attitude indicator. Revert to other VG through ATT REV push-button and check ATT 1 or ATT 2 displayed

EHSI message	Fault condition	Corrective action
HDG FAIL	Aircraft heading data invalid or unavailable.	Refer to magnetic compass.

CAUTION MESSAGES (YELLOW)

EDU message	Fault condition	Corrective action
VG 1	Gyro not erected, loss of power to gyro	Check breaker. Refer to stand-by attitude indicator. Pilot: revert to VG 2 through ATT REV push-button

EDU message	Fault condition	Corrective action
VG 2	Gyro not erected, loss of power to gyro	Check breaker. Copilot: revert to VG 1 through ATT REV push-button.

EFIS MISCOMPARE MESSAGES

(Messages displayed beside the instrument indications).

EADI message	Fault condition	Corrective action
PIT	Pitch attitude discrep-ancy between the gyro data	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data
ROL	Roll attitude discrep-ancy between the gyro data	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data
ATT	Pitch and roll attitudes discrepancy between the gyro data	Compare the EADIs data with the ADI stand-by. Select the ATT REV pushbutton on the EADI providing misleading data

EADI message	Fault condition	Corrective action
LOC	Excessive discrepancy between the two localizer data	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.
GS	Excessive discrepancy between the two glide slope data	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.
ILS	Excessive discrepancy between both LOC and GS data	Verify that both receivers are tuned on the same frequency. If this is confirmed abort the ILS approach.

CAUTION LIGHT (YELLOW)

Panel Wording	Fault condition	Corrective action
EFIS FAN (if installed)	Failure of one or both EFIS fans.	Refer to para. ″EFIS Fan failure″ in this Section.

SYSTEM FAILURES

EFIS FAILURES

The EFIS failures can affect separately or simultaneously each of four displays of the system and performing as EADI and EHSI indicators.
The cause can be an internal system failure or a connection system failure.
In the event of an internal system failure, the affected display (EADI or EHSI) becomes blank or an amber EFI FAIL caution is displayed on it, depending on the type of failure.

Failure of one EADI.

In case of failure of pilot’s (or co-pilot’s) EADI, the pilot’s (or co-pilot’s) EHSI automatically reverts to composite mode and a yellow MON DGRD (″monitoring degraded″) message appears at the same time on all three displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of one EHSI.

In case of failure of pilot’s (or co-copilot’s) EHSI, the pilot’s (or co-pilot’s) EADI automatically reverts to composite mode and a yellow MON DGRD (″monitoring degraded″) message appears at the same time on all three displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of pilot’s (or co-pilot’s) EADI and co-pilot’s (or pilot’s) EHSI.

In case of a cross-failure of two EFIS displays, the remaining co-pilot’s (or pilot’s) EADI and pilot’s (or co-pilot’s) EHSI automatically revert to composite mode and a yellow MON DGRD (″monitoring degraded″) message appears at the same time on both of them to point out the degradation of the comparison monitor function in the system.
The message ATT FAIL appers on pilot’s (or co-pilot’s) EHSI to point out the loss of the primary attitude source (ATT 1 for pilot’s EHSI and ATT 2 for co-pilot’s EHSI).
Press the ATT REV pushbutton to switch the operating EHSI over the alternative source.
On both displays the message ATT 2 (or ATT 1 if the remaining EHSI is the co-pilot’s one) will appear.
Proceed with flight in the new system configuration.

Failure of both EADIs.

In the event of a failure of both EADIs, the two EHSIs automatically revert to composite mode and a yellow MON DGRD (″monitoring degraded″) message appears at the same time on both displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of both EHSIs.

In the event of a failure of both EHSIs, the two EADIs automatically revert to composite mode and a yellow MON DGRD (″monitoring degraded″) message appears at the same time on both displays to point out the degradation of the comparison monitor function in the system.
Proceed with flight in the new system configuration.

Failure of three out of four EFIS displays.

In case of loss of three out of four displays constituting th EFIS, the screen of the remaining apparatus automatically reverts to composite mode and a yellow MON FAIL (″monitoring failure″) message appears to point out the failure of the monitoring intercommunication system inside the EFIS system.
Proceed with flight in the new system configuration.

EFIS Fan failure

In the event of failure of one or both EFIS cooling fans, the EFIS FAN light caption will illuminate.
If the failure occurs in flight and the Outside Air Temperature (OAT) is greater than 45°C open vents and select VENT - CKPT switch to HIGH.
If ECS is installed, switch it ON.
Land as soon as practicable within 30 minutes.
With lower OATs proceed with flight and correct trouble before next flight.

EADI FAILURES

Radio altimeter failure

In the event of Radio Altimeter failure, an amber caution RA FAIL replaces the numerical value; the rising runway and the DH indication (if present) disappear from the screen.
Aural message ″150 feet″ and ″Landing Gear″ and relative cautions on EDU are inhibited if Radio Altimeter is invalid.
Monitor barometric altitude during coupled ILS approach as the AUTO LEVEL function at 50 feet does not occur and the helicopter will continue to follow glide slope signal.

Flight Director failure

In the event of FD computer failure, an amber caution FD FAIL is displayed. Moreover, FD command bars and mode indicators are removed from the screen.
Continue flight in ATT mode.

EHSI FAILURES

Weather radar failure

An amber caution WX FAIL on EHSI indicates a failure within the WX Radar System.
Turn radar selector to OFF position.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEMS DESCRIPTION

DME OPERATION

When on EHSI ″IN COMMAND″ the LNAV 1 (or 2) is selected as navigator, the DME is automatically set on CH 2 (VOR 2). If the pilot needs to use CH 1 (VOR 1), he must select VOR 1 as navigator then select DME HOLD function and reselect the previous LNAV sensor (1 at 2).
The following table shows the various combinations of possible DME data visualization on EHSI, depending on the selected navigation source.

NAV SELECTION		PLT IN CMD		CPLT IN CMD	
EHSI PLT	EHSI CPLT	PLT	CPLT	PLT	CPLT
VOR1	VOR1	DME1	DME1	DME1	DME1
VOR1	VOR2	DME1	—	—	DME2
VOR1	GPS	DME1	GPS	DME1	GPS
VOR2	VOR1	DME2	—	—	DME1
VOR2	VOR2	DME2	DME2	DME2	DME2
VOR2	GPS	DME2	GPS	—	GPS
GPS	VOR1	GPS	DME1	GPS	DME1
GPS	VOR2	GPS	—	GPS	DME2
GPS	GPS	GPS	GPS	GPS	GPS

For helicopter equipped with King Avionic Suite, the letter H (DME HOLD Mode) can falsely appear below the not related bearing selected indication. Check for correct DME 1 (or 2) Hold selection and indication on dedicated DME indicator.

FLIGHT DIRECTOR OPERATION

When NAV, VOR APR, ILS or BC mode is active in CAP mode temporarily deselect the mode before performing any course change. Set new course on the EHSI and intercept heading, then re-arm for capture.

Rotation of the course knob in NAV, VOR APR, ILS or BC modes while FD system is in CAP condition, can falsely trigger the over station sensor and gives a degraded performance in the system.
During ILS (or BC) approach with FD engaged, selection of the other navigation sensors (e.g. LNAV) will be inhibited as soon as the LOC is captured.
Swap from ILS 1 to ILS 2 is the only available navigation source selection.

NAV mode operation with GPS

On the GPS receiver, select the appropriate CDI scale before engaging the NAV mode of FD.

The best CDI scale suggested for GPS coupled with F/D NAV mode is ±1 nm full scale.

EFIS COMPOSITE MODE

EFIS composite mode is used only in the event of an in-flight failure of any display (EADI or EHSI).

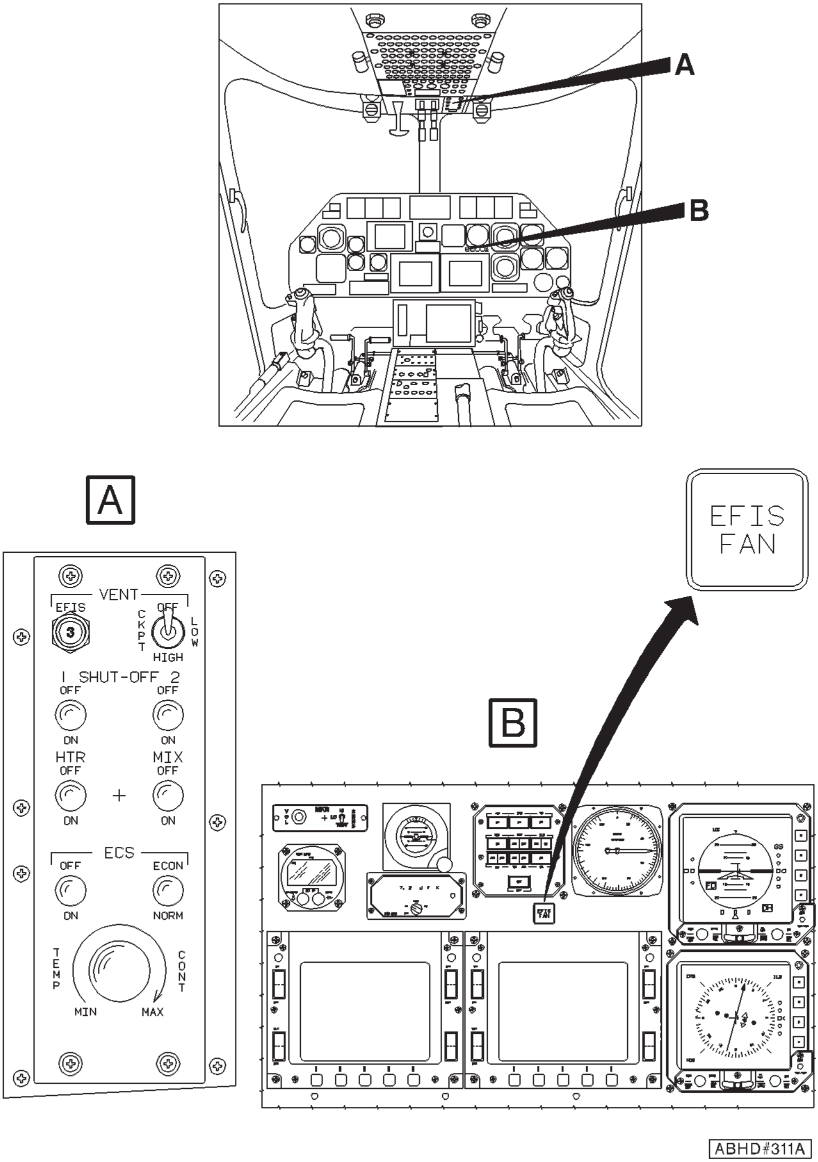

Figure 7-1. EFIS Fan (if installed) - Controls and Indicators.

E.A.S.A. Approval N° 2004-1960
dated 3 March 2004

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

NIGHTSUN SEARCHLIGHT SX-5

The Nightsun Searchlight, type SX-5, P/N 109-0813-72-101 is a high intensity light installed on a gimbal support located on the right hand side of the helicopter nose.
The xenon arc light may be switched on, focused and directed through the control panel.
The installation consists of a Nightsun Searchlight, gimbal support, control panel, hardware and cabling.

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LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

Activation of Nightsun Searchlight SX-5 in IFR condition is prohibited.

AIRSPEED LIMITATIONS (KIAS)

Maximum airspeed (V_{NE}) : 140 KIAS.

CENTER OF GRAVITY LIMITATIONS

After Nightsun Searchlight SX-5 installation the new empty weight and C.G. location must be determined.

ALTITUDE LIMITATIONS

Maximum pressure altitude for operation with searchlight SX-5 : 15000 ft (4572 m).

Maximum pressure altitude with searchlight SX-5 non operative : 20000 ft (6096 m).

MISCELLANEOUS LIMITATIONS

SEARCHLIGHT USE

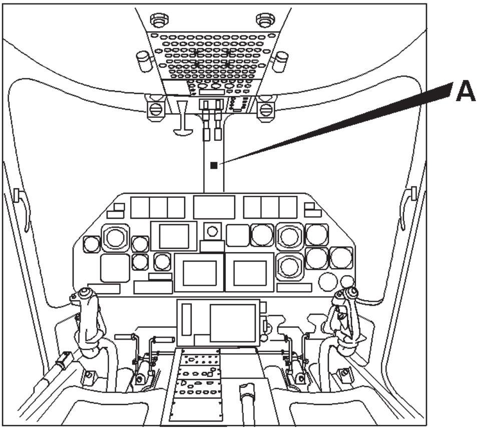
The searchlight SX-5 must not be switched on:

— Below 50 ft (15 m) AGL

— In cloud

— In fog conditions.

PLACARDS



A

CAUTION

DO NOT USE SX-5 SUN
LT BELOW 50 FT AGL
OR IN CLOUD OR IN
FOG CONDITIONS.
MONITOR LOADMETER
WHEN USING SX-5
SUN LT.

MAXIMUM AIRSPEED
WITH SX-5 SUN LT
140 KTS IAS.

ABHD592A

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

AREA N° 1 (Helicopter nose)

Nightsun Searchlight SX-5	: Condition and security. Wiring properly connected. Lens for cleanliness.

PILOT'S PREFLIGHT CHECK

(Every flight)

Nightsun Searchlight SX-5	: Condition and security. Wiring properly connected. Lens for cleanliness.

ENGINE PRE-START CHECK

SUN LT circuit breaker	: In.

IN FLIGHT

NOTE

When operating the Nightsun Searchlight SX-5, magnetic compass indication is not realiable.

NIGHTSUN SEARCHLIGHT OPERATING PROCEDURE

CAUTION

Do not operate the Nightsun Searchlight SX-5 after a generator failure.

SUN LT switch : Hold in START position approximately 5 seconds or until the ignition has occured.

CAUTION

Holding switch in START position after ignition may damage equipment.

Aim and focus : As desired.

CAUTION

Do not direct the beam towards other aircraft or vehicles to prevent temporary blinding effect.

APPROACH AND LANDING

SUN LT switch : OFF.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

SYSTEM FAILURES

ELECTRICAL POWER FAILURE
(Helicopters not equipped with emergency bus)

Failure of a generator
(Helicopters not equipped with emergency bus)

SUN LT switch : OFF. Refer to paragraph ″**Failure of a generator**″ of basic Rotorcraft Flight Manual.

Failure of generator 1 or 2 and generator bus #1 or #2
(Helicopters not equipped with emergency bus)

SUN LT switch : OFF. Refer to paragraph ″**Failure of generator 1 or 2 and generator bus #1 or #2**″ of basic Rotorcraft Flight Manual.

ELECTRICAL POWER FAILURE
(Helicopters equipped with emergency bus)

Failure of a generator
(Helicopters equipped with emergency bus)

SUN LT switch : OFF. Refer to paragraph ″**Failure of a generator**″ of basic Rotorcraft Flight Manual.

Failure of generator 1 or 2 and generator bus #1 or #2
(Helicopters equipped with emergency bus)

SUN LT switch : OFF. Refer to paragraph ″**Failure of generator 1 or 2 and generator bus #1 or #2**″ of basic Rotorcraft Flight Manual.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 7 - SYSTEMS DESCRIPTION

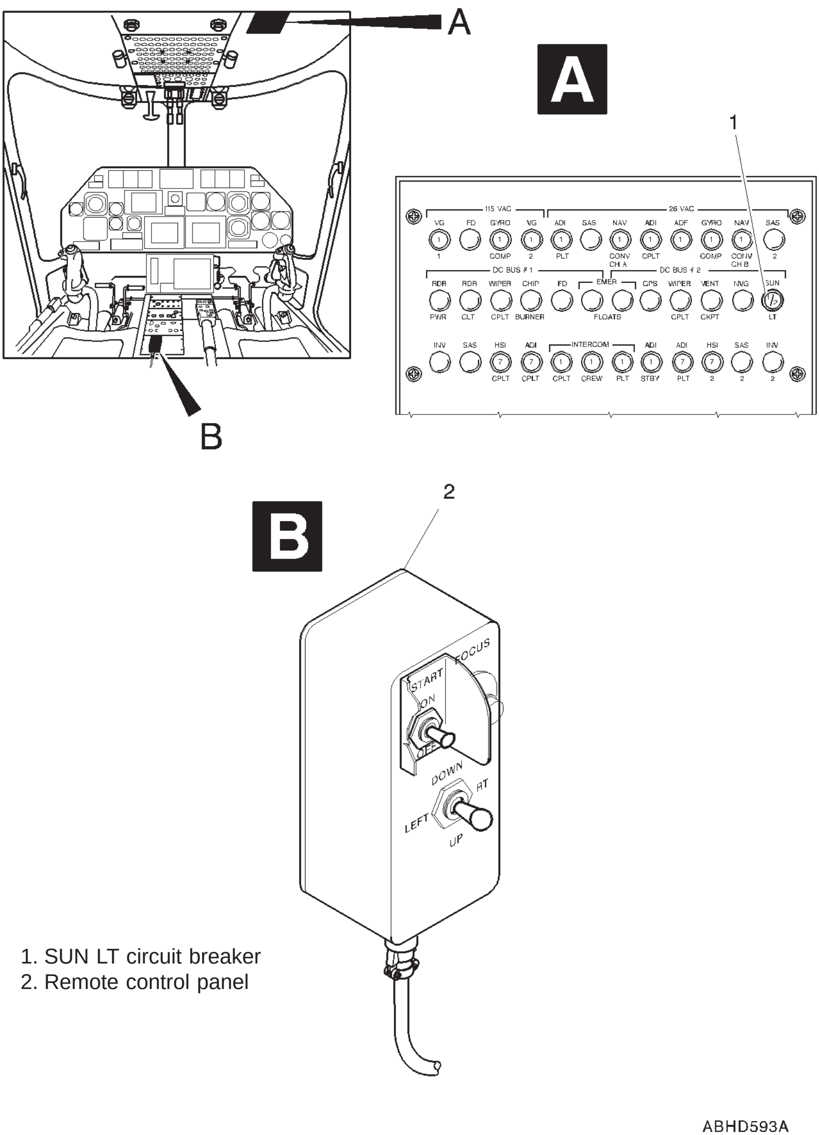
The Nightsun Searchlight SX-5 is provided for enhancing the searchlight capabilities during night or poor visibility conditions. The searchlight SX-5 contains an air cooled, high intensity xenon arc lamp which start rapidly and can be operated continuously or may be stopped and restarted as dictated by operational requirements.

The SX-5 main features are:

- Lamp type	Xenon, 800 Watts Short arc
- Peak Beam Intensity	15 Million Candelpower
- Typical range	70 - 300 m.

The helicopter is equipped with attachment points on the right side of the nose, at which the supporting frame is attached. The searchlight SX-5 is mounted on a gimbal with motors and can be rotated in azimuth and elevation. Nightsun searchlight SX-5 is powered at 28 VDC supplied through a dedicated circuit breaker located on the overhead panel.

The searchlight SX-5 is controlled by switches housed on a remote control panel placed on the lower side of the central console.



1. SUN LT circuit breaker
2. Remote control panel

Figure 7-1. Nightsun Searchlight SX-5 control panel.

E.A.S.A. Approval N° 2004-1960
dated 3 March 2004

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

MOVING MAP SYSTEM
SKYFORCE OBSERVER

The Skyforce Observer P/N 109-0812-38-109 is a GPS driven detailed mapping system.
It provides the pilot, by means of a display, with digital moving map images and navigation data aids.
The Skyforce Observer system consists of:

— a processing unit;
— a control unit;
— a colour display;
— a dedicated GPS antenna.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

Use of the Skyforce Observer System is permitted only during VFR operations with visual contact of the ground.
The navigation can not be based on the Skyforce Observer System.

PLACARDS

DIGITAL MAP GENERATOR. USE ONLY DURING VFR OPERATION WITH VISUAL CONTACT OF THE GROUND.

In clear view of pilot

SECTION 2 - NORMAL PROCEDURES

SYSTEMS CHECK

After both engine starting:

Skyforce Observer Control Unit : ON.
Check that the POWER and HDD lamps illuminate, and that the TEMP lamp is off.

NOTE

The TEMP lamp is red to indicate ″Too Hot″ and amber to indicate ″Heater in Operation″.
When the TEMP lamp is amber not all maps are available.
When the TEMP lamp is red the system shall be shut down.

Skyforce Observer display : Carry out the the power-up sequence until the MAIN MENU page displays.

Select MAP key.

NOTE

The message NO FIX POSSIBLE appears if the GPS has no fix valid data.

Check the helicopter icon appears in the center of the map in the correct geographic position.

IN FLIGHT

Skyforce Observer display : As required.

SHUTDOWN

After engines shutdown:

Skyforce Observer display : Go to MAIN MENU and carry out the SHUT DOWN sequence. Press the YES key.

NOTE

The SHUT DOWN key must be used to shut the system software down; it ensures that the operating system and all data files are stored and secured prior to the power down of the Skyforce Observer hardware.

The system display will then automatically switch to a black screen indicating that it is now safe to turn the system off.

Skyforce Observer Control Unit	:	Select OFF. Check that LEDs extinguish.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

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dated 14 March 2005

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

INCREASED INTERNAL GROSS WEIGHT

This Appendix contains the limitations and performance applicable when the modification P/N 109-0823-22 is installed, and when the helicopter is operated at an internal gross weight greater than 2850 kg up to a maximum of 3000 kg.

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LIST OF ILLUSTRATIONS

List of Illustrations (Cont.d)

SECTION 1 - LIMITATIONS

REQUIRED EQUIPMENT

Operations at gross weight greater than 2850 kg up to 3000 kg are permitted provided that the following assembly is installed:

— Tail rotor hub and blade assy P/N 109-8131-02-157
  or
— Tail rotor installation P/N 109-0136-02-103.

VFR OPERATION

FLIGHT WITH ONE OF THE TWO HELIPILOTS INOPERATIVE

Following one helipilot failure, cruise flight above 120 KIAS requires the pilot to maintain hands on flight controls.

AIRSPEED LIMITATIONS (KIAS)

V_{NE}(Power on)	:	See Figure 1-1
Minimum IFR airspeed	:	60 KIAS
Minimum speed during IFR approach	:	60 KIAS
Best rate of climb speed in IFR	:	65 KIAS

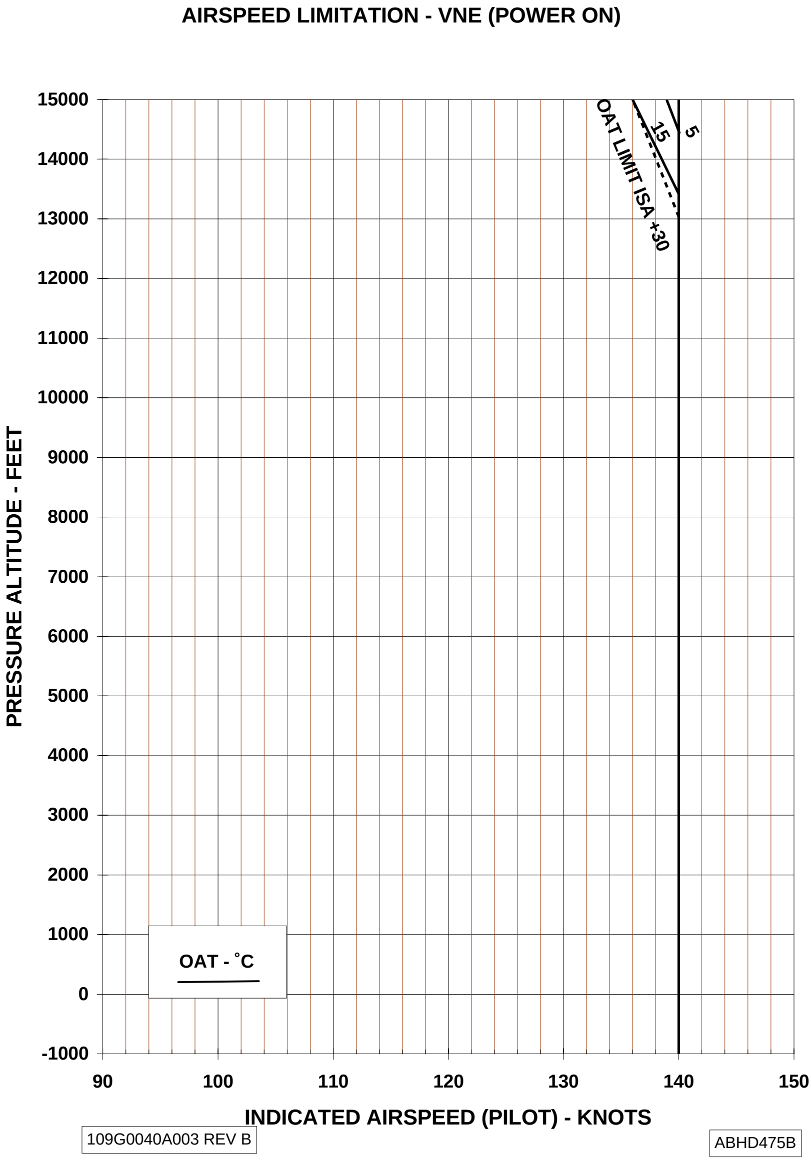


Figure 1-1. Airspeed Limitations - V_{NE}(Power on).

WEIGHT LIMITATIONS

Maximum gross weight
for takeoff and landing : 3000 kg (6613 lbs)

NOTE

The maximum takeoff and landing weight may be limited by performance data contained in Section 4.

CENTER OF GRAVITY LIMITATIONS.

Refer to Figure 1-2 for longitudinal CG limits and Figure 1-3 for lateral CG limits.

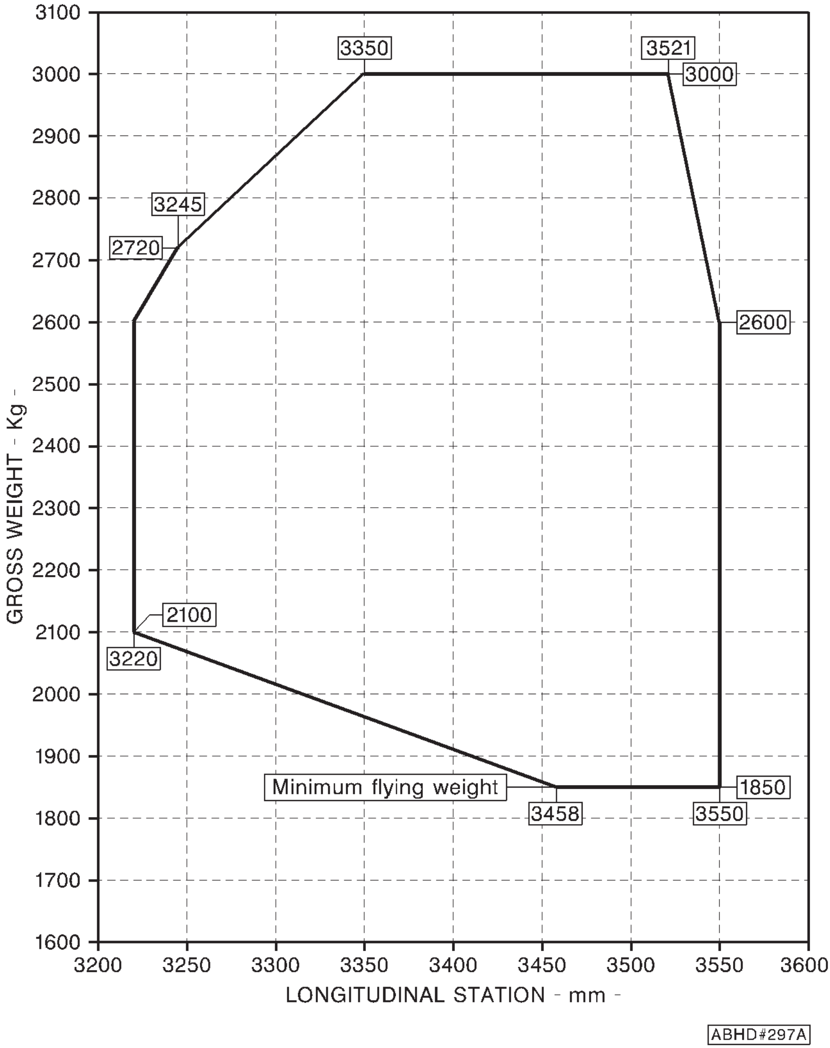


Figure 1-2 (sheet 1 of 2). Longitudinal CG limit (metric).

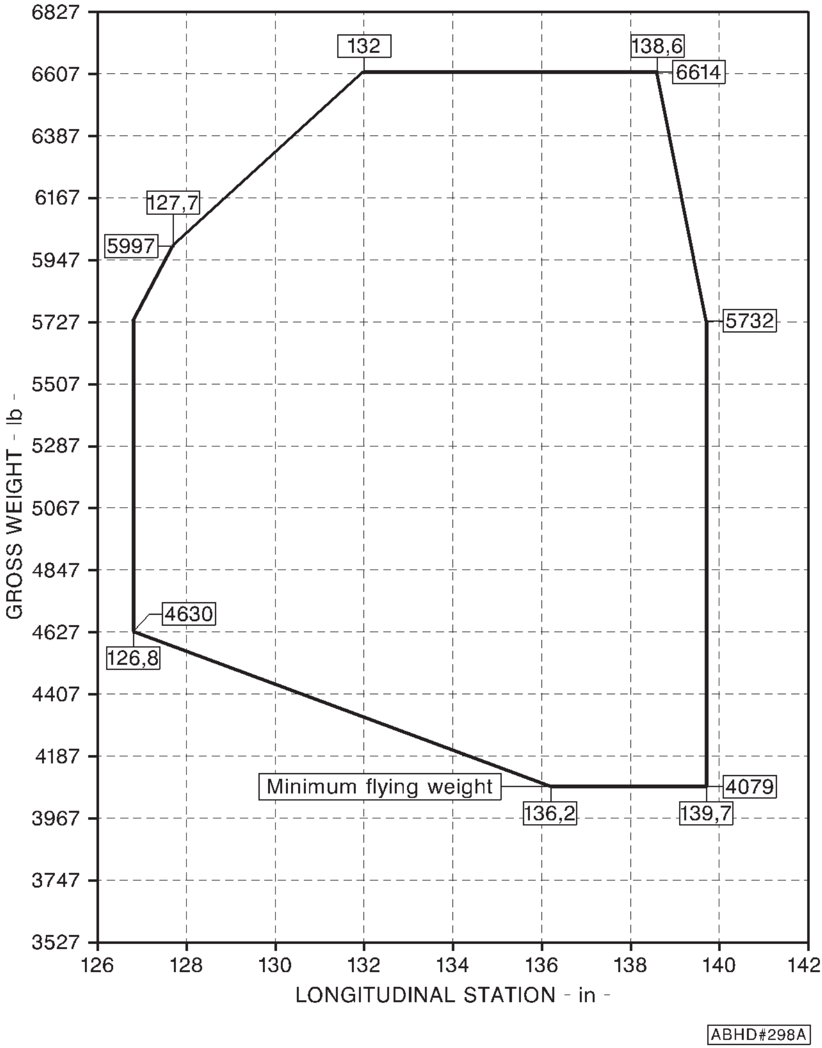


Figure 1-2 (sheet 2 of 2). Longitudinal CG limit (english).

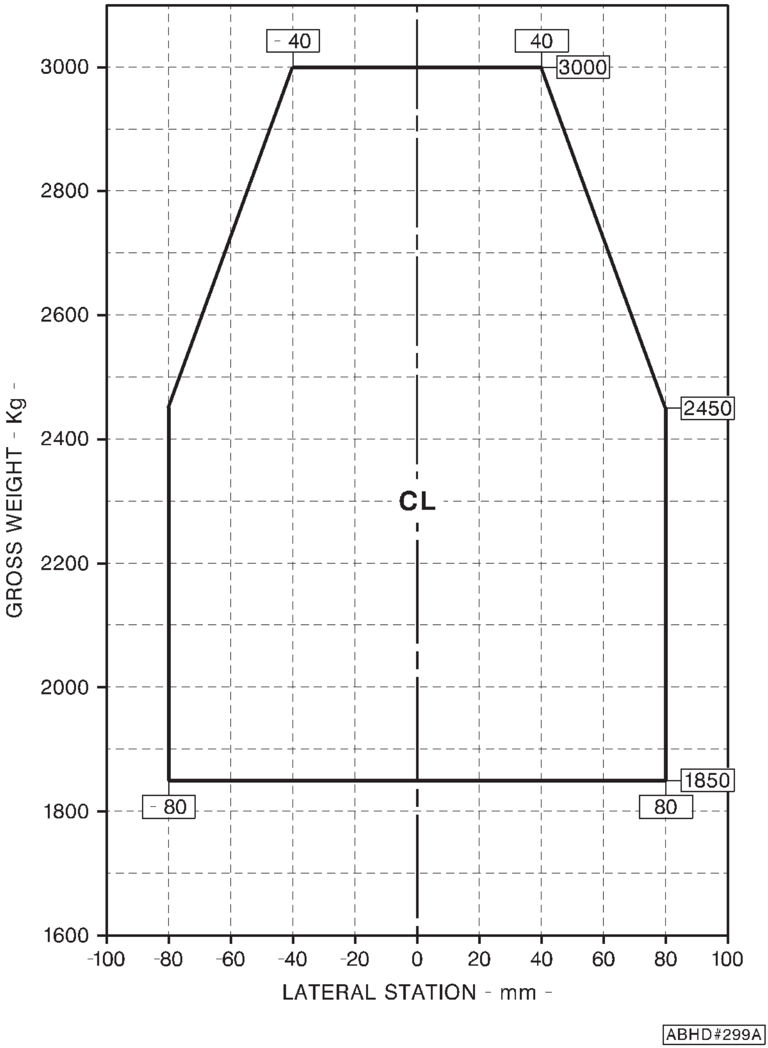


Figure 1-3 (sheet 1 of 2). Lateral CG limit (metric).

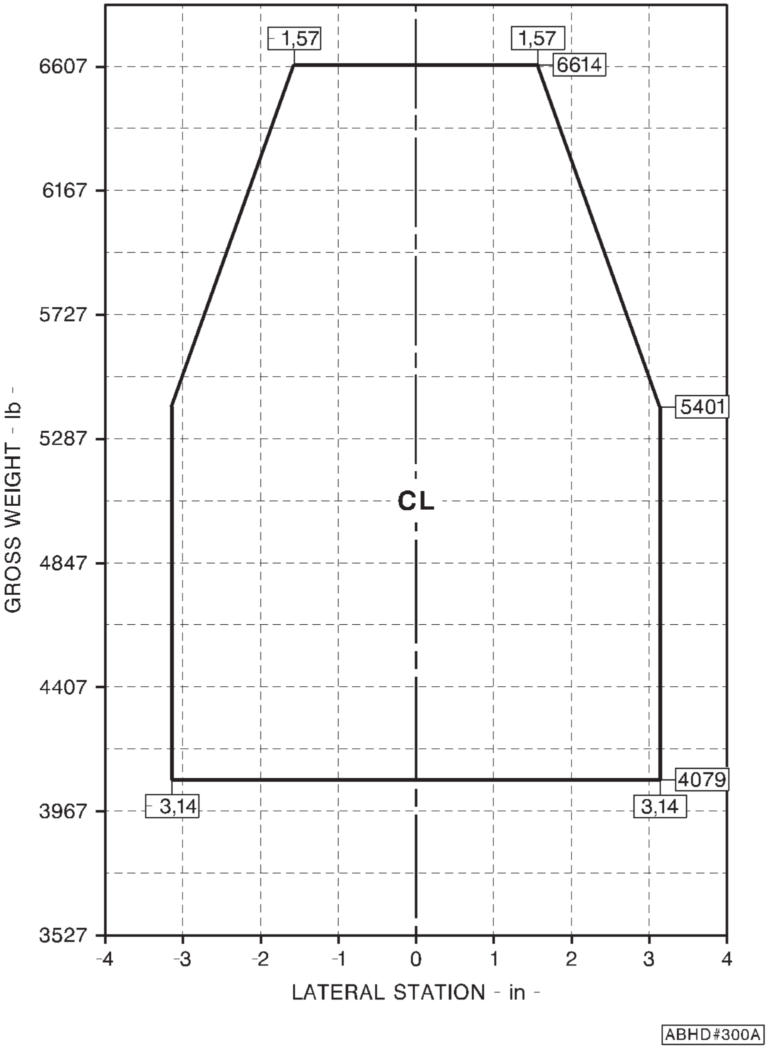


Figure 1-3 (sheet 2 of 2). Lateral CG limit (english).

ALTITUDE LIMITATIONS

Maximum pressure altitude for takeoff and landing	: 10000 ft (3048m)
Maximum operating pressure altitude	: 15000 ft (4572 m)

AMBIENT AIR TEMPERATURE LIMITATIONS

The maximum sea level ambient air temperature is +45 °C (+113 degree °F) and decreases with pressure altitude at the standard lapse rate of 2 °C (3.6 °F) every 1000 ft (305 m) up to 15000 ft (4572 m).

PLACARDS

FOR GROSS WEIGHT IN EXCESS OF 2850 kg UP TO 3000 kg. **− MAXIMUM AIRSPEED AEO: THE LESSER OF 140 KIAS AND Vne** **− AUTOROTATION OR OEI Vne: SAME AS UP TO 2850 kg** **− MAXIMUM OPERATING ALTITUDE: 15000 ft hp**

In clear view of the pilot
(grouped with Vne placard)

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

OPERATION VS ALLOWABLE WIND

Satisfactory stability and control in rearward and sideward flight has been demonstrated, at all loading conditions for hover in ground effect up to take-off power, from -1000 ft to 7000 ft Hd, in the following wind/ground speed azimuth envelope:

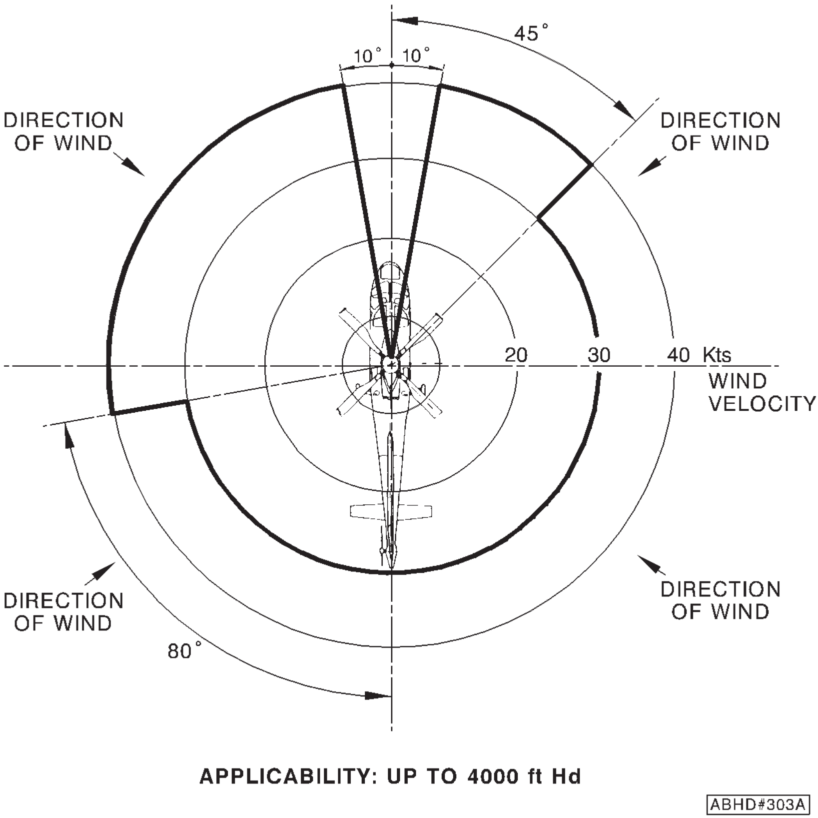


Figure 4-1 (Sheet 1 of 2). Wind/ground speed azimuth envelope.

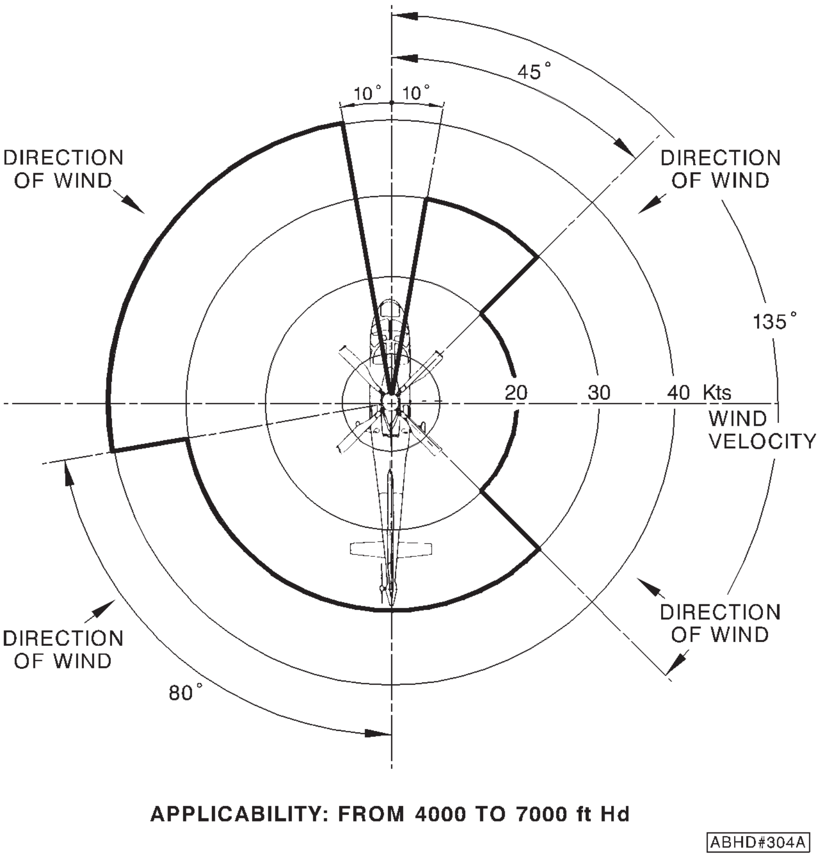


Figure 4-1 (Sheet 2 of 2). Wind/ground speed azimuth envelope.

PERFORMANCE CHARTS

This Section includes the performance data for A109E in the following configuration:

- helicopter in basic configuration;
- helicopter with Heater;
- helicopter with Environmental Control System (ECS);
- helicopter with Engine Air Particle Separator (EAPS) ON or OFF;
- helicopter with Engine Air Particle Separator (EAPS) and Heater or Environmental Control System (ECS).

NOTE

For operations at best rate of climb speed in IFR (65 KIAS), rate of climb charts for 60 KIAS are applicable.

For operations below 10000 ft Hd reduce climb performance by 40 fpm.

For operations above 10000 ft Hd reduce climb performance by 100 fpm.

RATE OF CLIMB

For operation with Snow Skid or Slump Protection Pads installed, reduce by 150 ft/min the Rate of Climb (AEO and OEI) given in this Appendix when one or more of the following Optional Equipments are installed:

- Nightsun searchglight SX-16 or SX-5;
- External hoist.

Helicopter in basic configuration

— Figure 4-2. Hovering ceiling - in ground effect - take-off power.
— Figure 4-3. Hovering ceiling - in ground effect - maximum continuous power.
— Figure 4-4. Hovering ceiling - out ground effect - take-off power.
— Figure 4-5. Hovering ceiling - out ground effect - maximum continuous power.
— Figure 4-6. Height-Velocity diagram - one engine inoperative.
— Figure 4-7. Rate of climb - all engines - take-off power.
— Figure 4-8. Rate of climb - all engines - maximum continuous power.
— Figure 4-9. Rate of climb - OEI - 2.5 minute power
— Figure 4-10. Rate of climb - OEI - maximum continuos power.

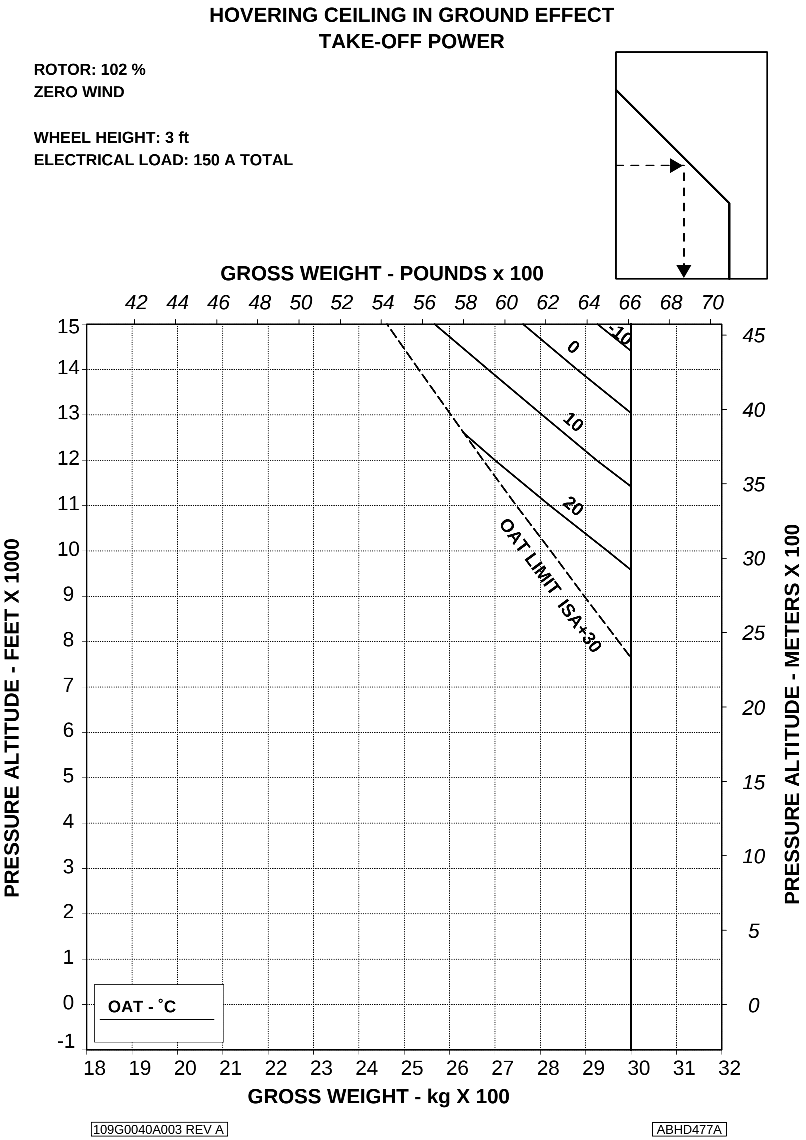


Figure 4-2. Hovering ceiling - in ground effect - take-off power.

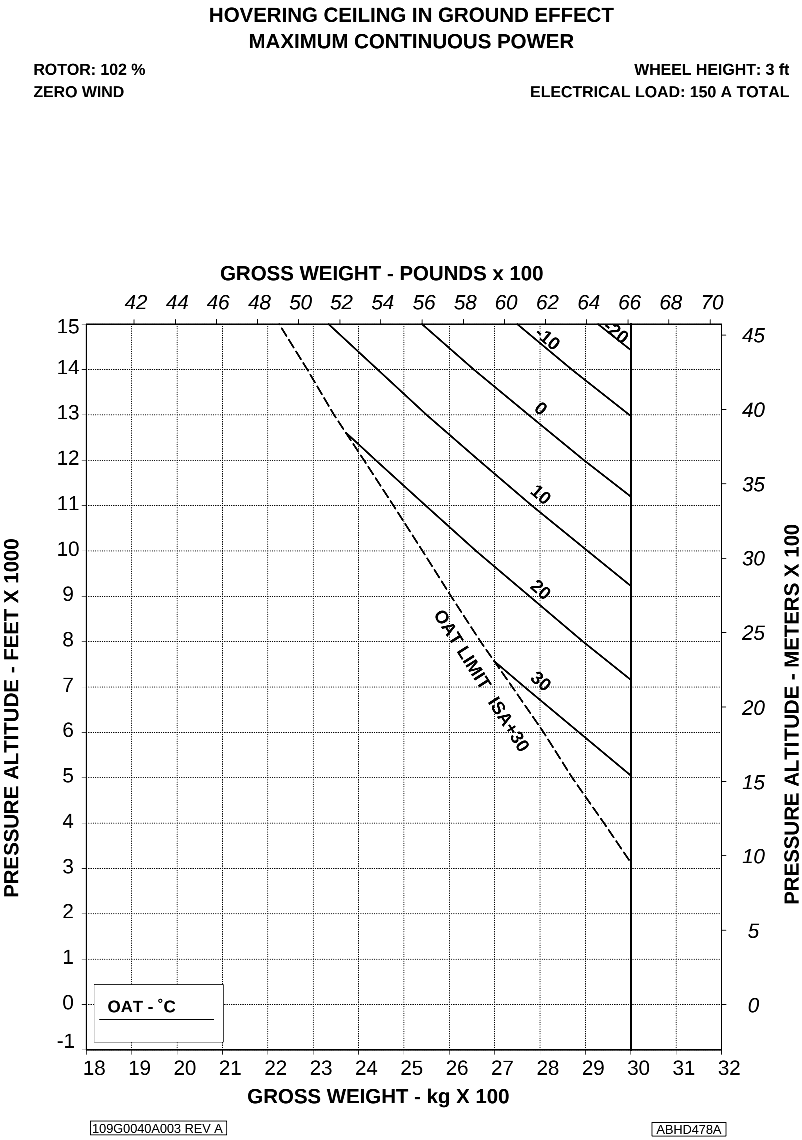
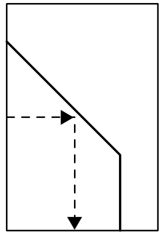


Figure 4-3. Hovering ceiling - in ground effect - maximum continuous power.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER



ROTOR: 102 %
ZERO WIND
ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

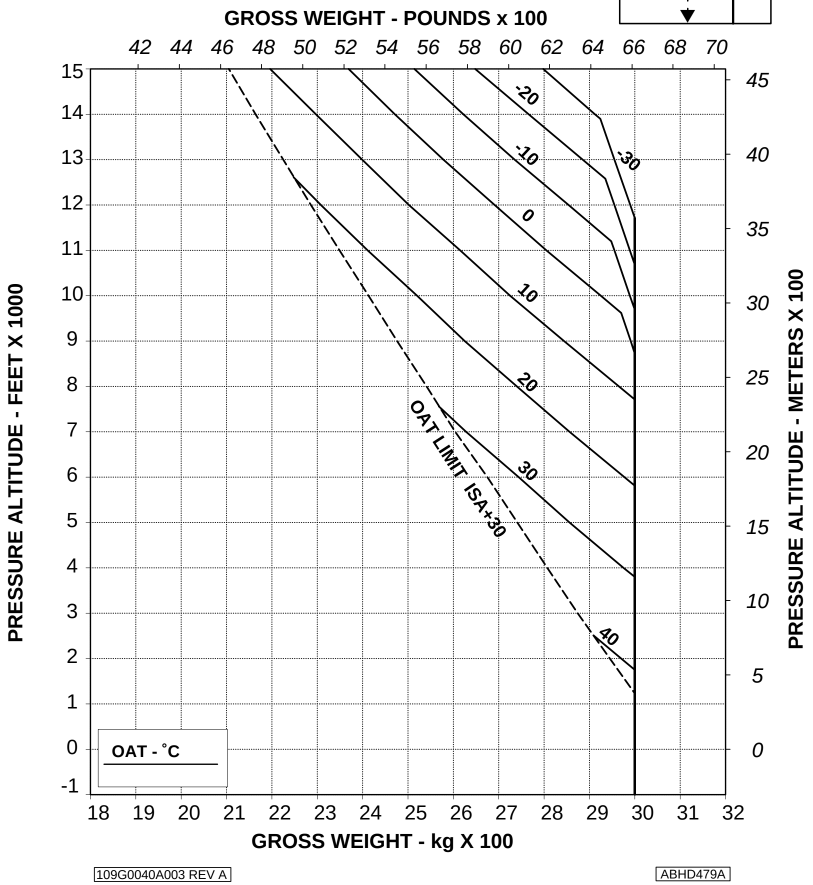


Figure 4-4. Hovering ceiling - out of ground effect - take-off power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

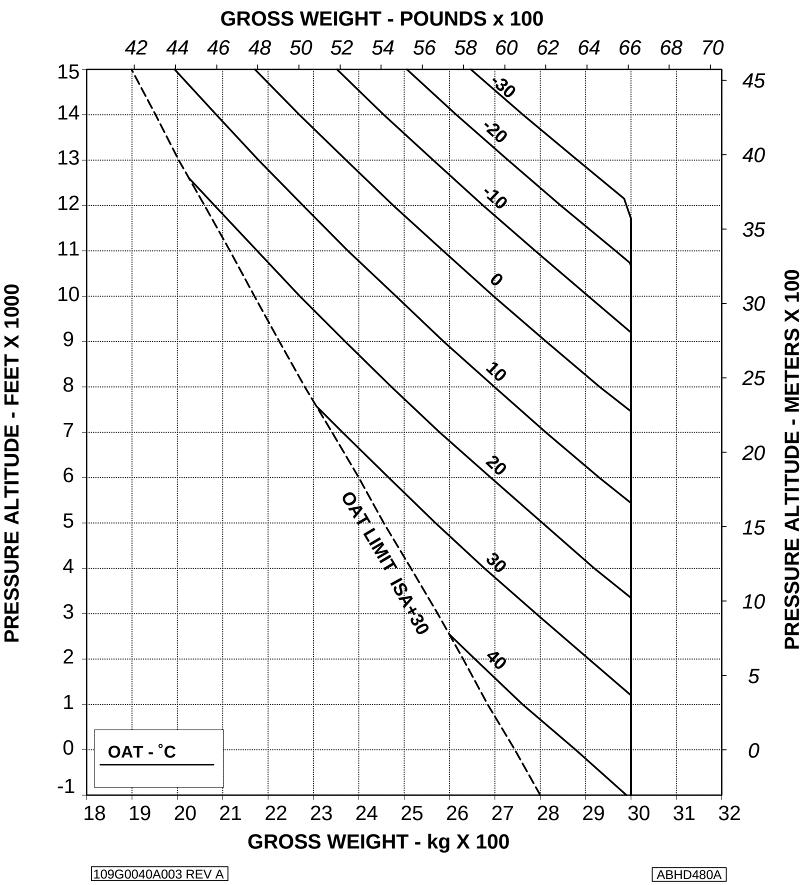


Figure 4-5. Hovering ceiling - out of ground effect - maximum continuous power.

HEIGHT-VELOCITY DIAGRAM

The Height - Velocity diagram is to identify the region where, in the event of a single engine failure during take-off, landing or operation near the ground, a combination of airspeed and height above ground exists from which a safe single engine landing on a smooth, level and hard surface cannot be assured (dangerous zone).

The following Height-Velocity diagram applies to all gross weights between 2850 and 3000 kg and to all configurations (without EAPS, with EAPS installed and OFF or with EAPS installed and ON).

Refer to the basic rotorcraft flight manual or to Appendix 28 for gross weight less than 2850 kg.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

GROSS WEIGHT: ABOVE 2850 kg (6283 lb) UP TO 3000 kg (6613 lb)

DENSITY ALTITUDE: UP TO 7000 ft

WITHOUT EAPS AND WITH EAPS OFF/ON

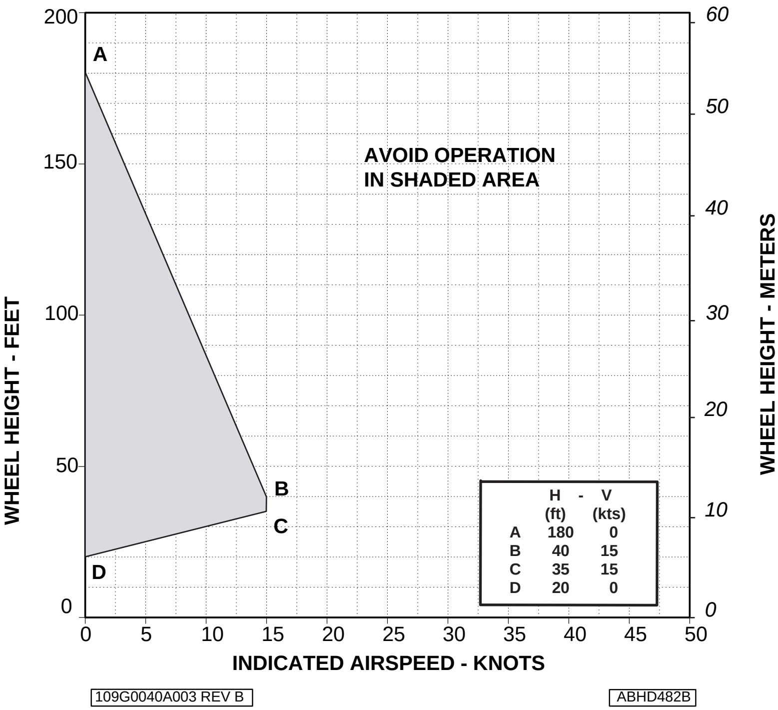


Figure 4-6. Height-velocity diagram - one engine inoperative.

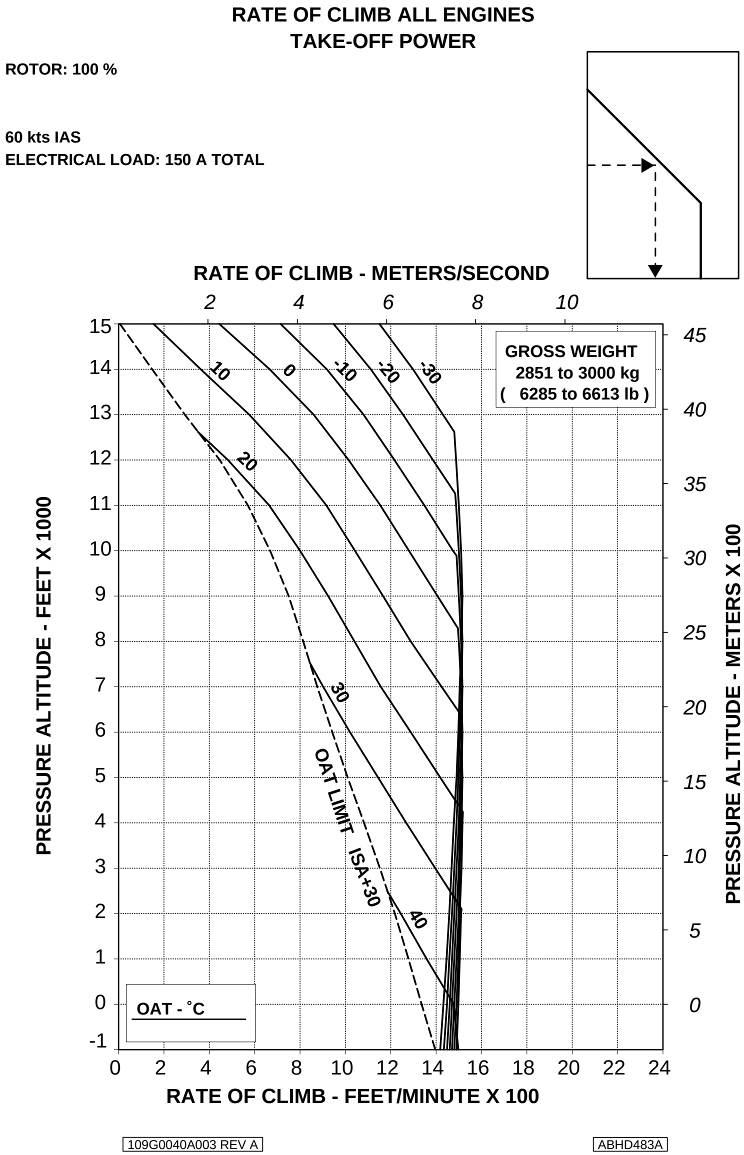


Figure 4-7. Rate of climb - all engines - take-off power.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

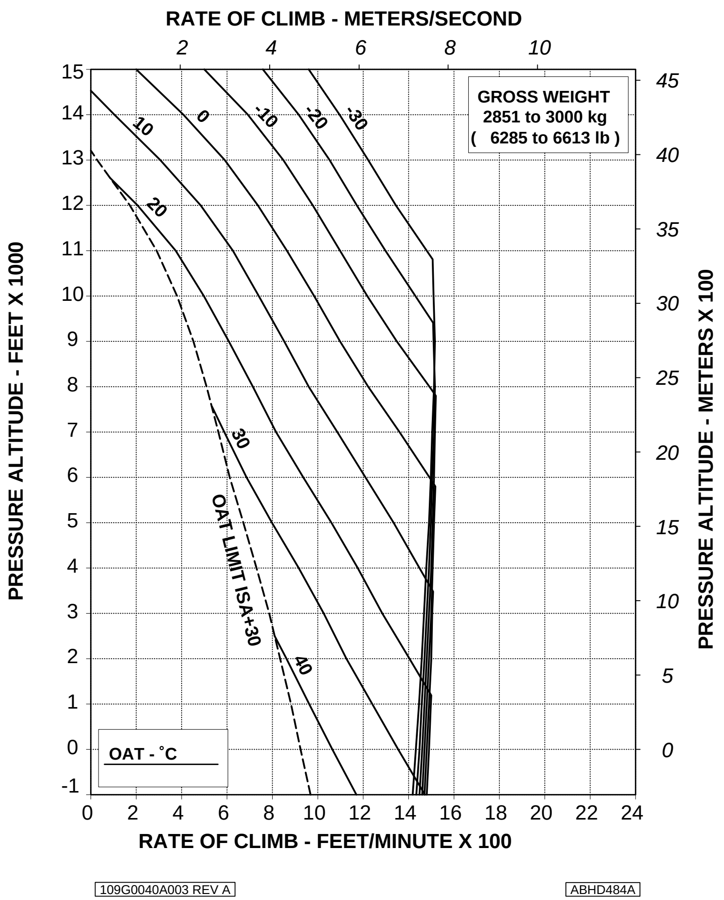


Figure 4-8. Rate of climb - all engines - maximum continuous power.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % **60 kts IAS**
ELECTRICAL LOAD: 150 A TOTAL

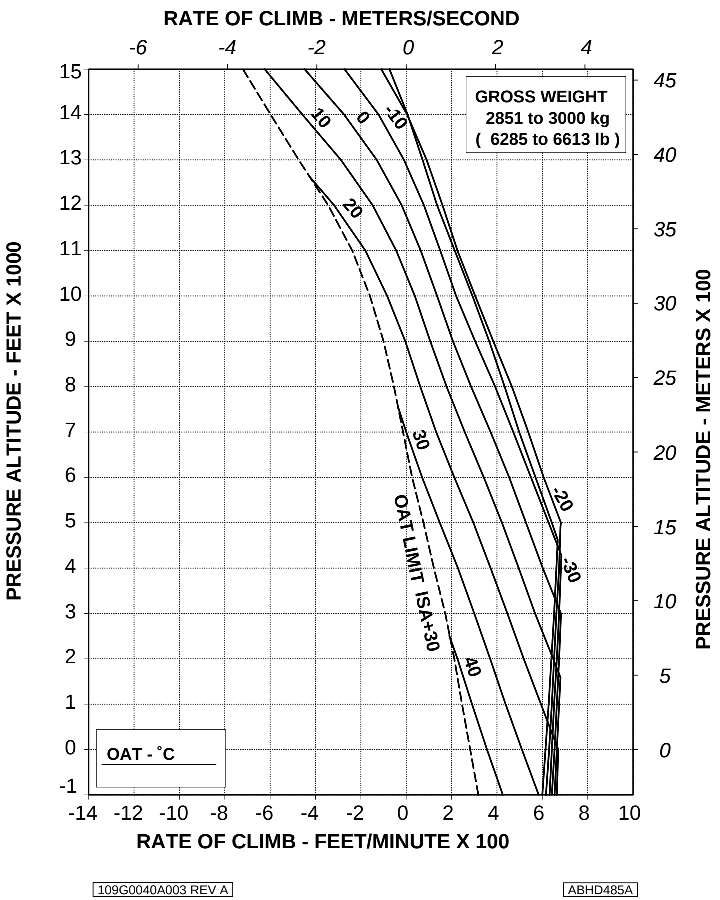


Figure 4-9. Rate of climb - OEI - 2,5 minute power.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

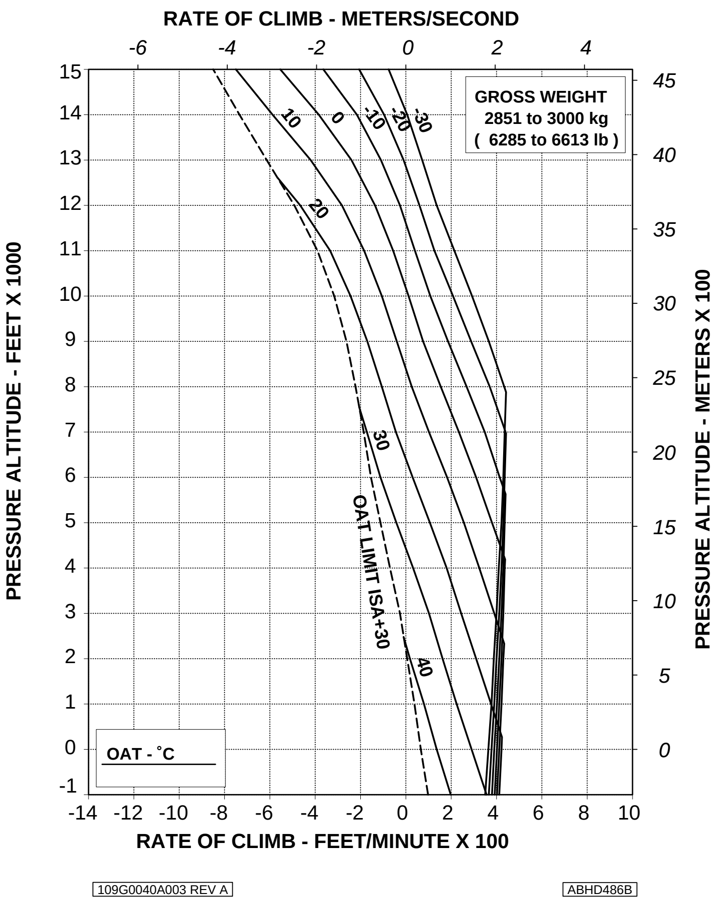


Figure 4-10. Rate of climb - OEI - maximum continuous power.

Helicopter with heater

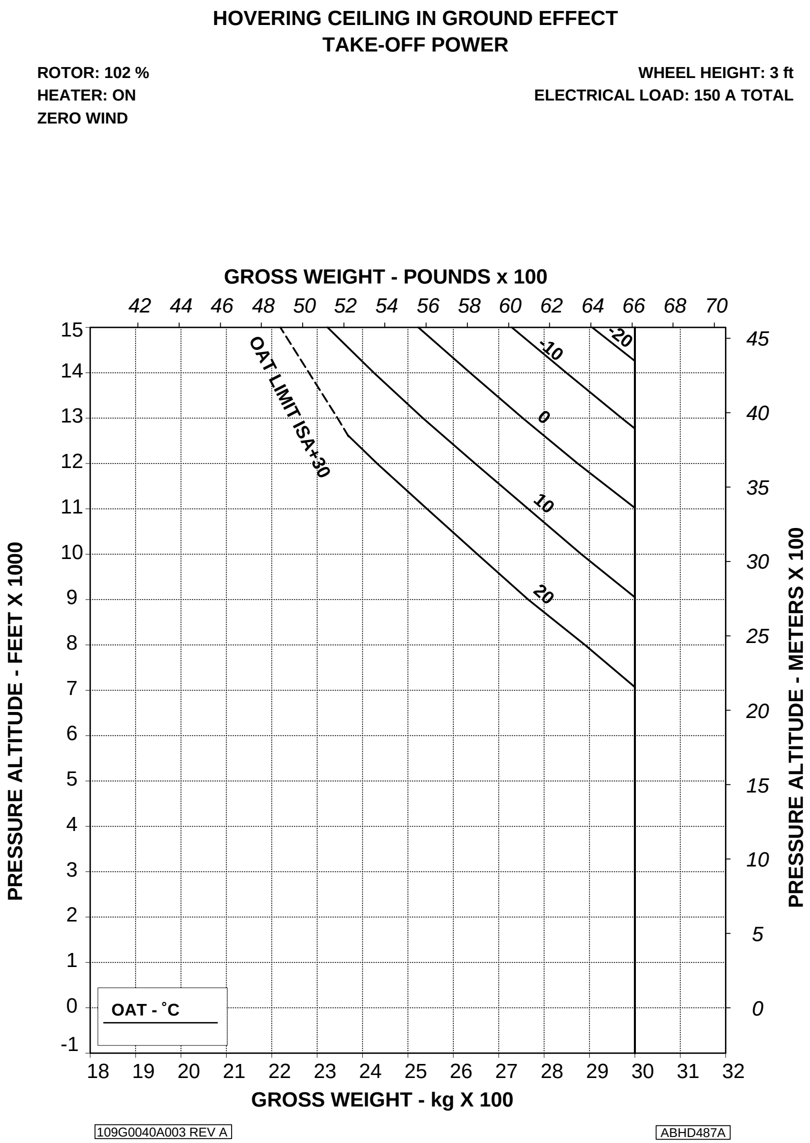


Figure 4-11. Hovering ceiling - in ground effect - take-off power - heater on.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

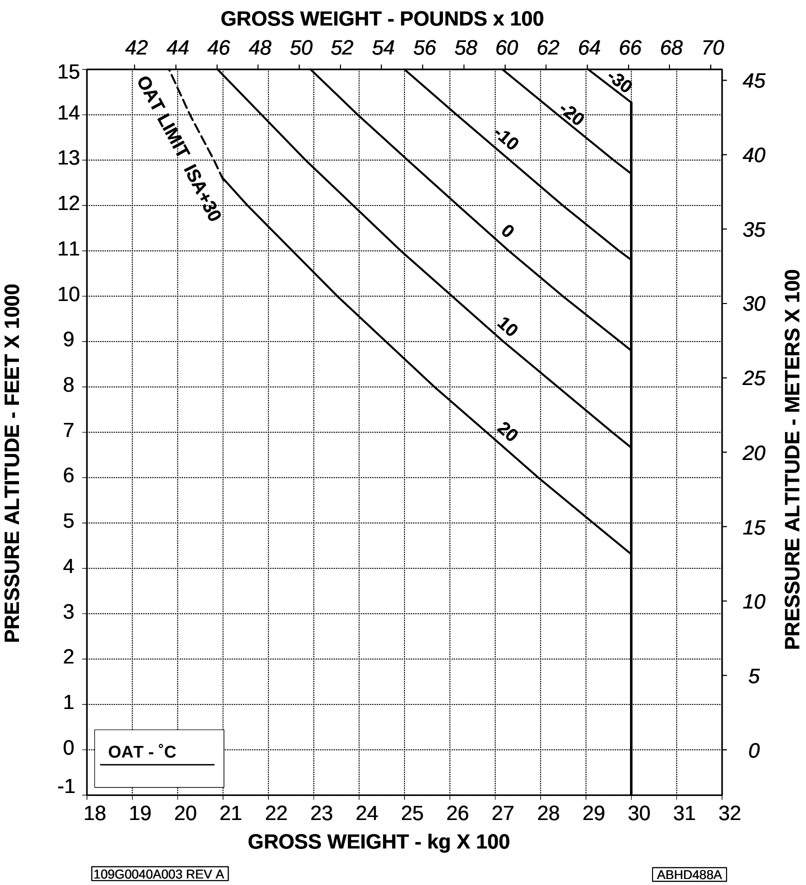


Figure 4-12. Hovering ceiling - in ground effect - maximum continuous power - heater on.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
HEATER: ON **ELECTRICAL LOAD: 150 A TOTAL**
ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

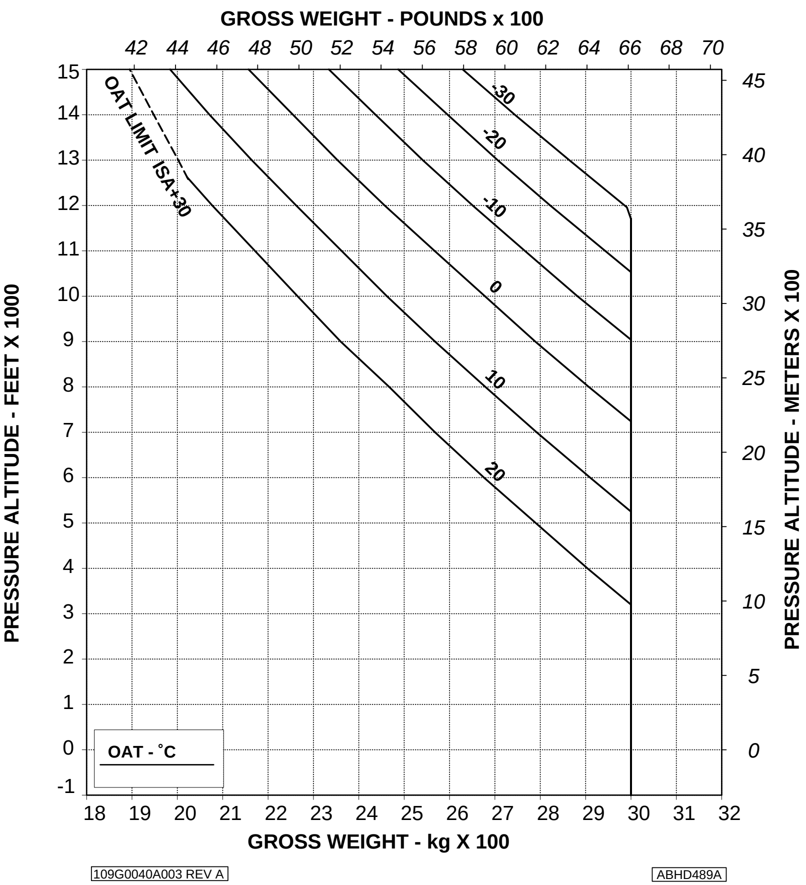


Figure 4-13. Hovering ceiling - out of ground effect - take-off power - heater on.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

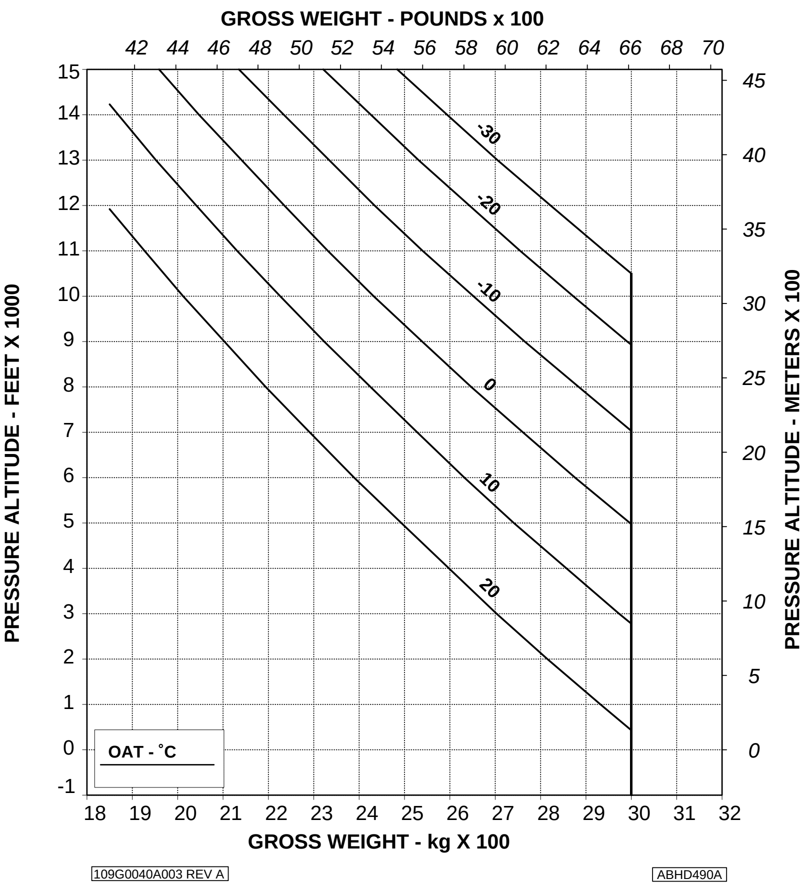


Figure 4-14. Hovering ceiling - out of ground effect - maximum continuous power - heater on.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS

HEATER: ON — ELECTRICAL LOAD: 150 A TOTAL

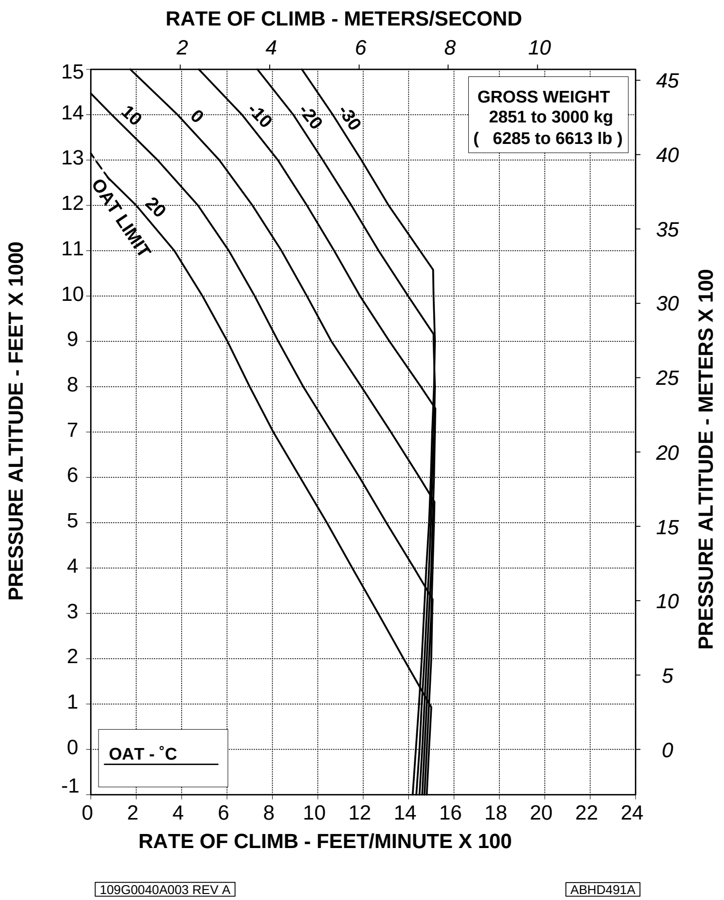


Figure 4-15. Rate of climb - all engines - take-off power - heater on.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
HEATER: ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

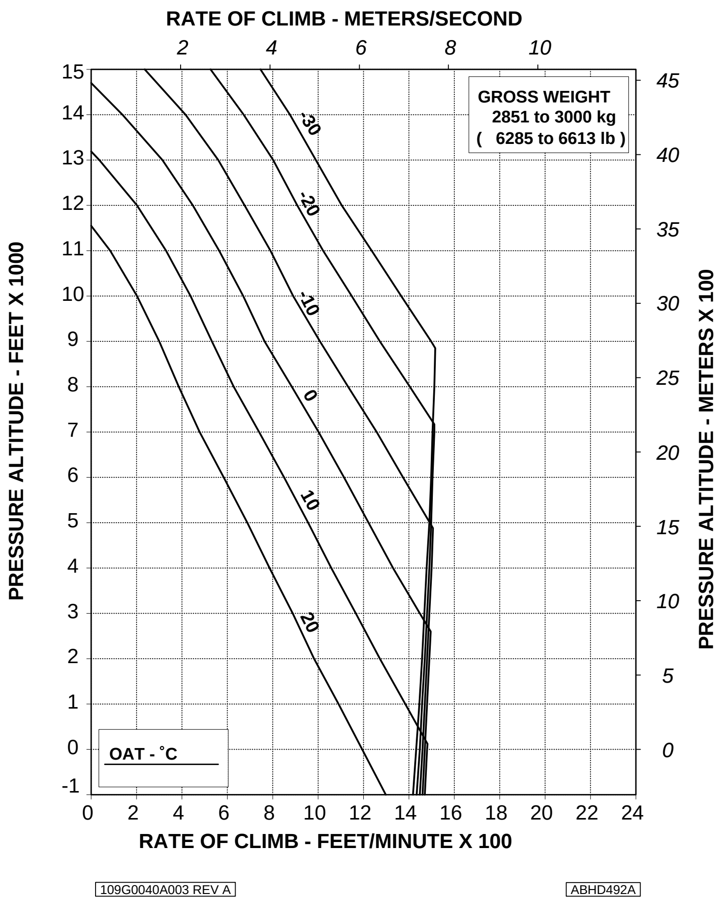


Figure 4-16. Rate of climb - all engines - maximum continuous power - heater on.

Helicopter with Environmental Control System (ECS)

- — Figure 4-17. Hovering ceiling - in ground effect - take-off power - ECS on.
- — Figure 4-18. Hovering ceiling - in ground effect - maximum continuous power - ECS on.
- — Figure 4-19. Hovering ceiling - out of ground effect - take-off power - ECS on.
- — Figure 4-20. Hovering ceiling - out of ground effect - maximum continuous power - ECS on.
- — Figure 4-21. Rate of climb - all engines - take-off power - ECS on.
- — Figure 4-22. Rate of climb - all engines - maximum continuous power - ECS on.

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
E.C.S.: ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

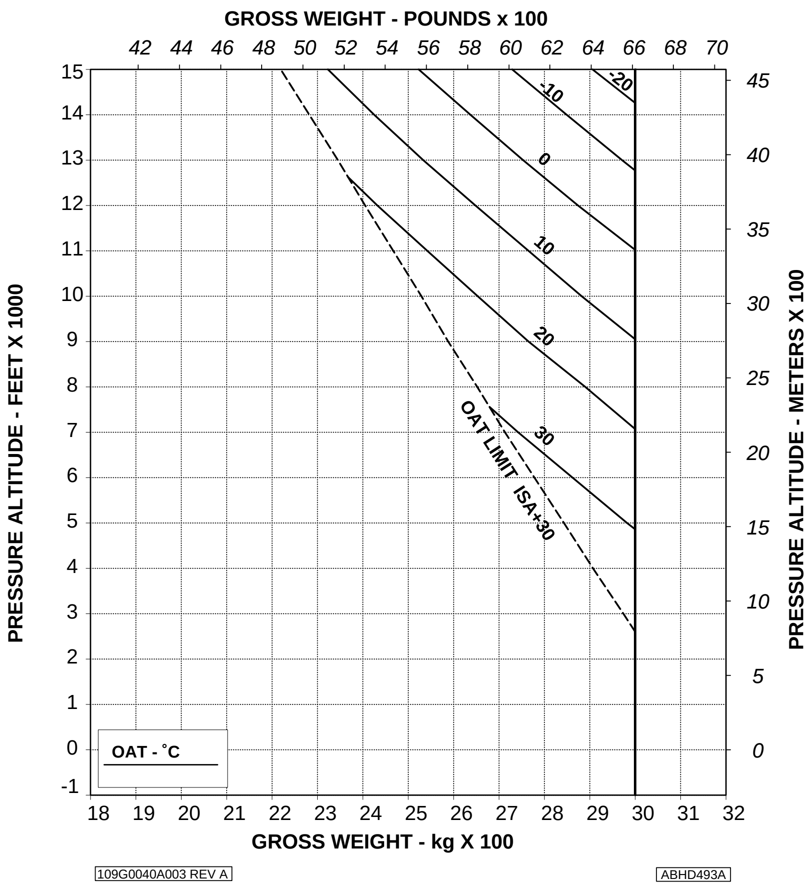


Figure 4-17. Hovering ceiling - in ground effect - take-off power - ECS on.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
E.C.S.: ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

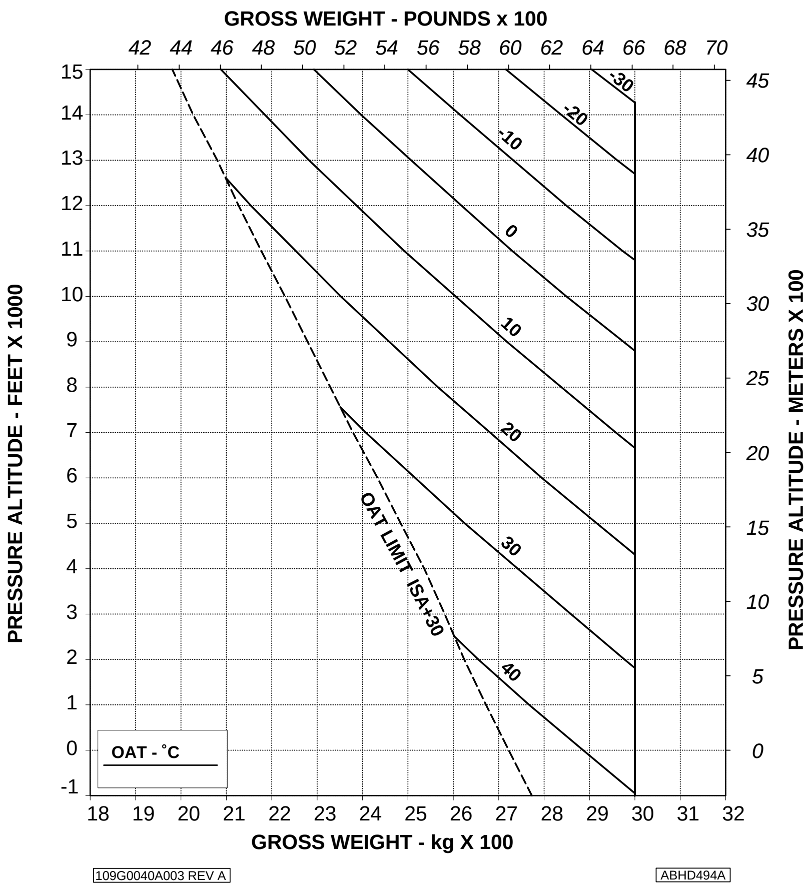


Figure 4-18. Hovering ceiling - in ground effect - maximum continuous power - ECS on.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
E.C.S.: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

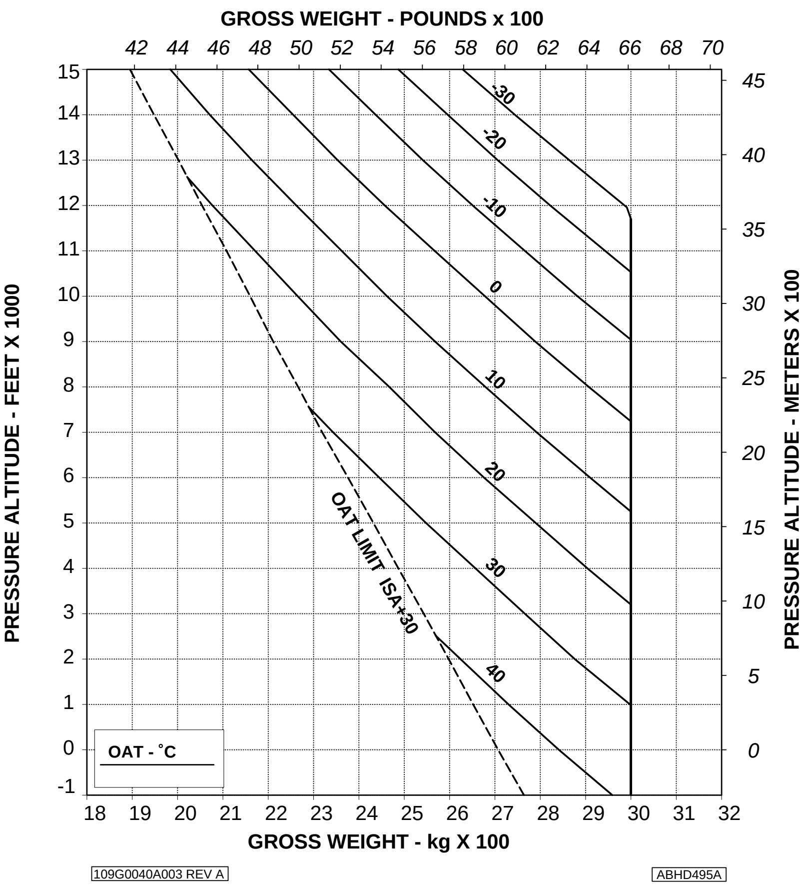


Figure 4-19. Hovering ceiling - out of ground effect - take-off power - ECS on.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
E.C.S.: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

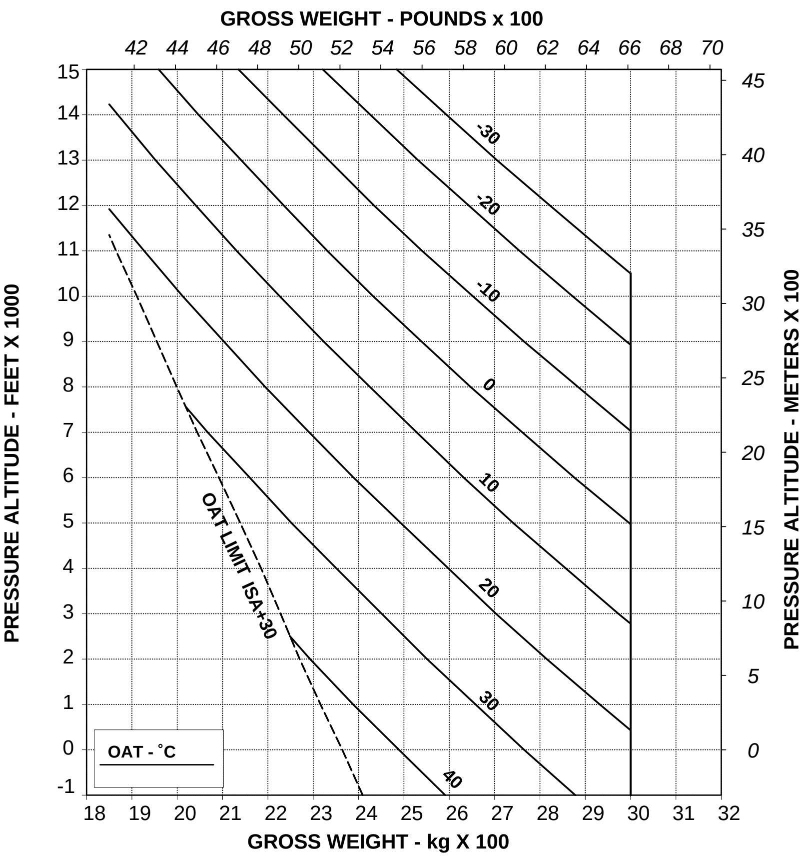


Figure 4-20. Hovering ceiling - out of ground effect - maximum continuous power - ECS on.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS
E.C.S.: ON — ELECTRICAL LOAD: 150 A TOTAL

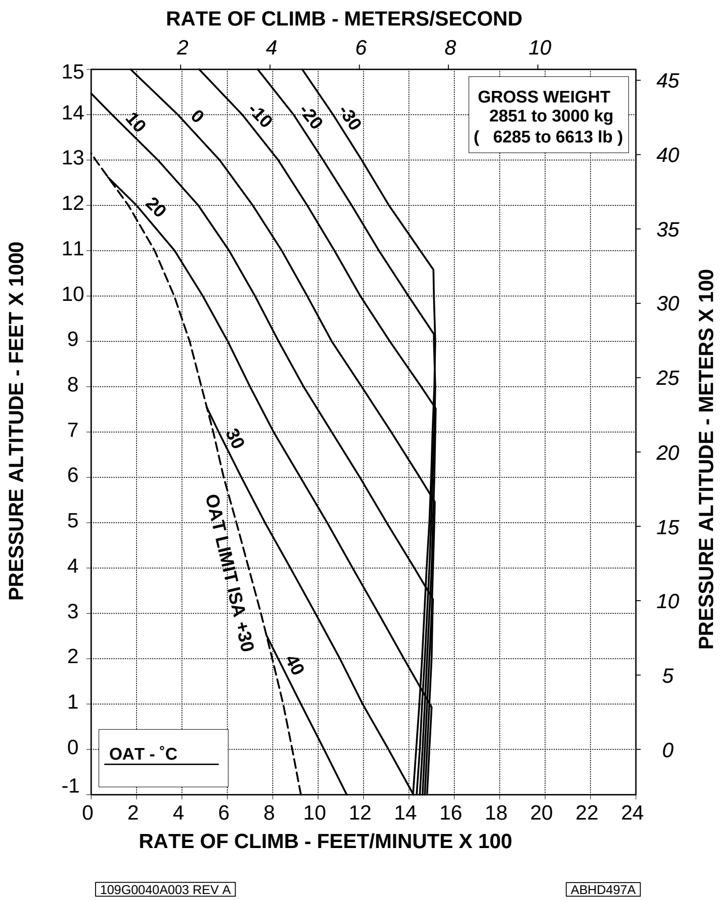


Figure 4-21. Rate of climb - all engines - take-off power - ECS on.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % — 60 kts IAS
E.C.S.: ON — ELECTRICAL LOAD: 150 A TOTAL

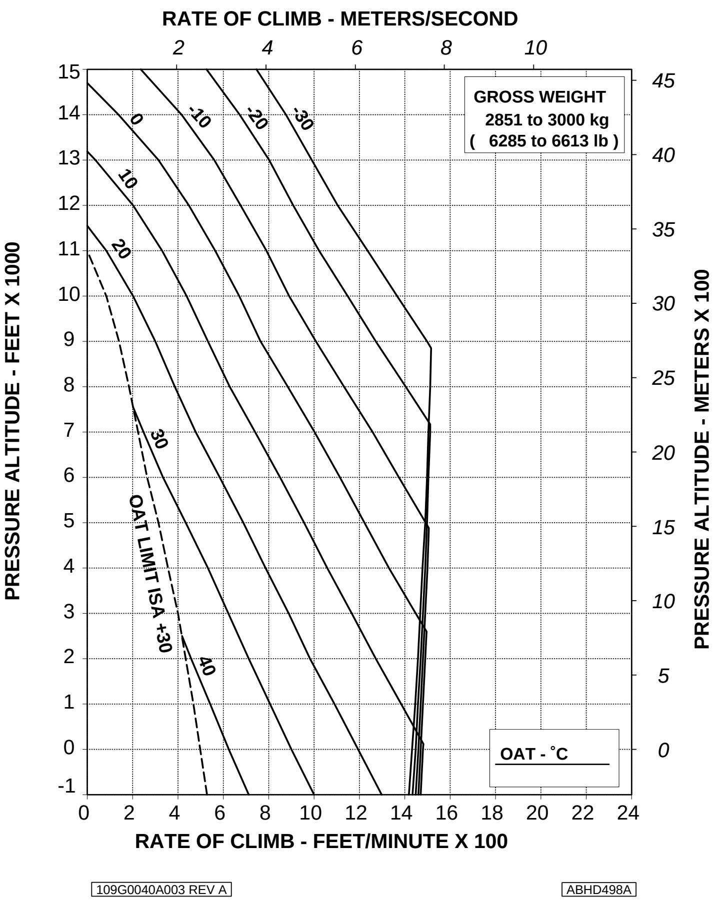


Figure 4-22. Rate of climb - all engines - maximum continuous power - ECS on.

Helicopter with Engine Air Particle Separator (EAPS) OFF or ON

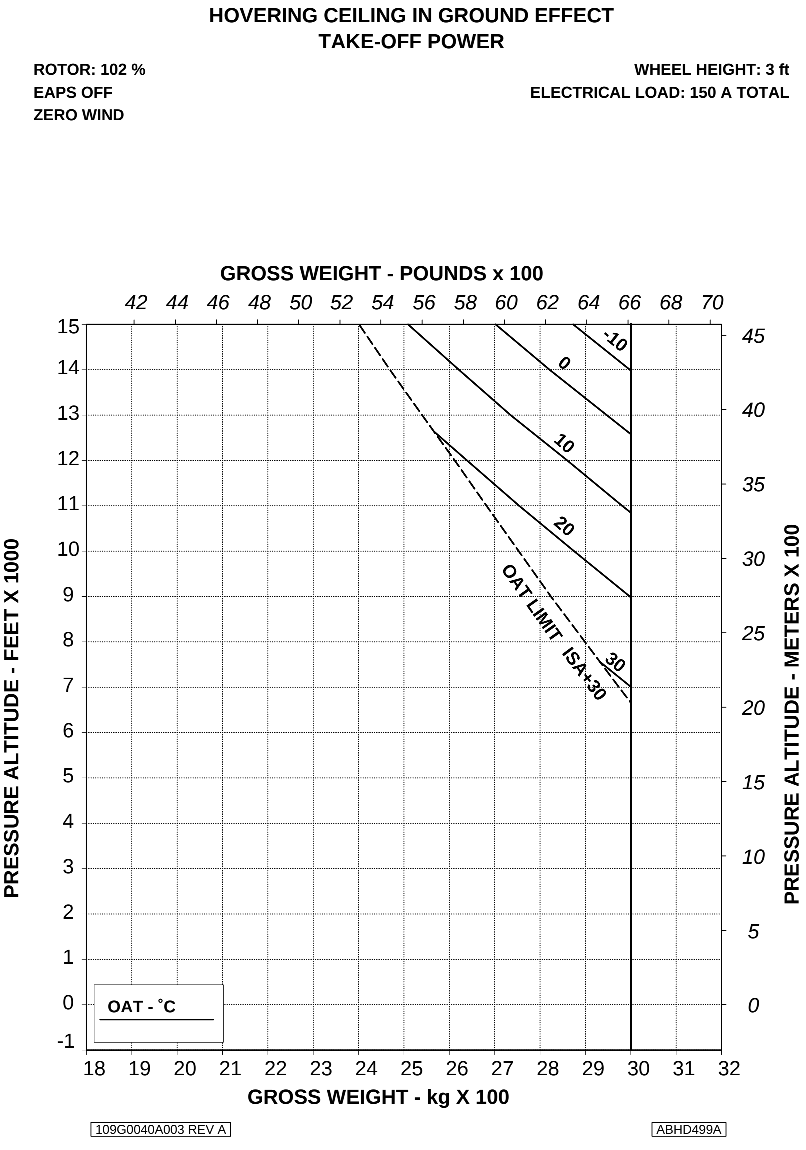


Figure 4-23. Hovering ceiling - in ground effect - take-off power - EAPS off.

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

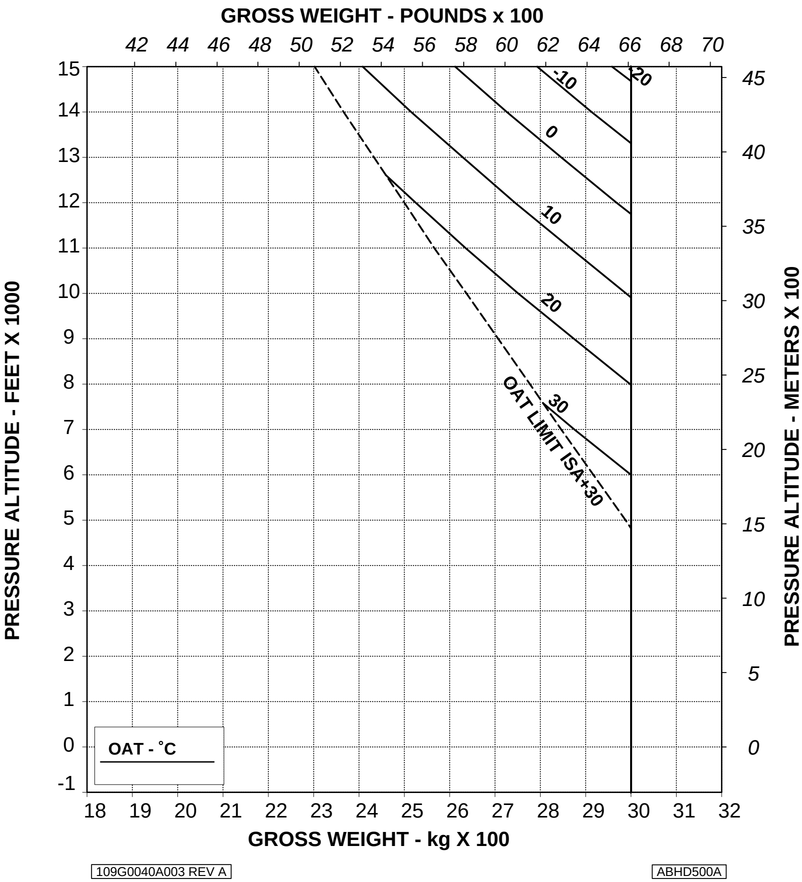


Figure 4-24. Hovering ceiling - in ground effect - take-off power - EAPS on.

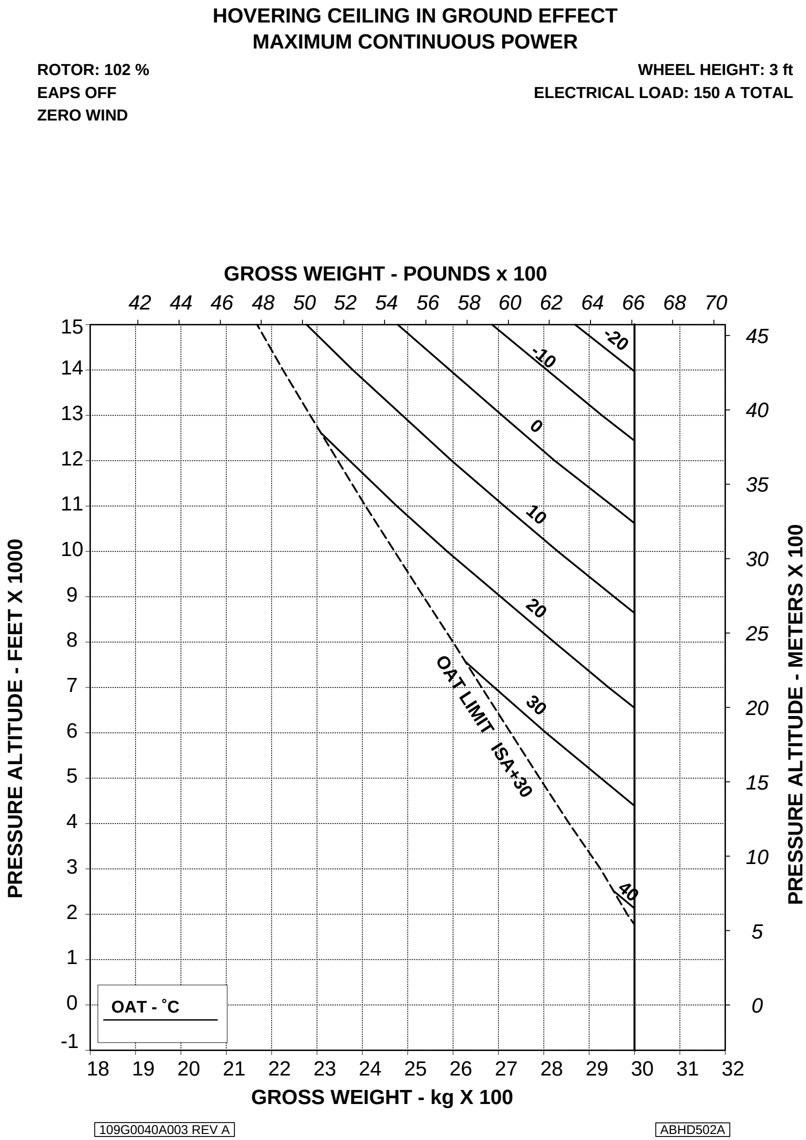


Figure 4-25. Hovering ceiling - in ground effect - maximum continuous power - EAPS off.

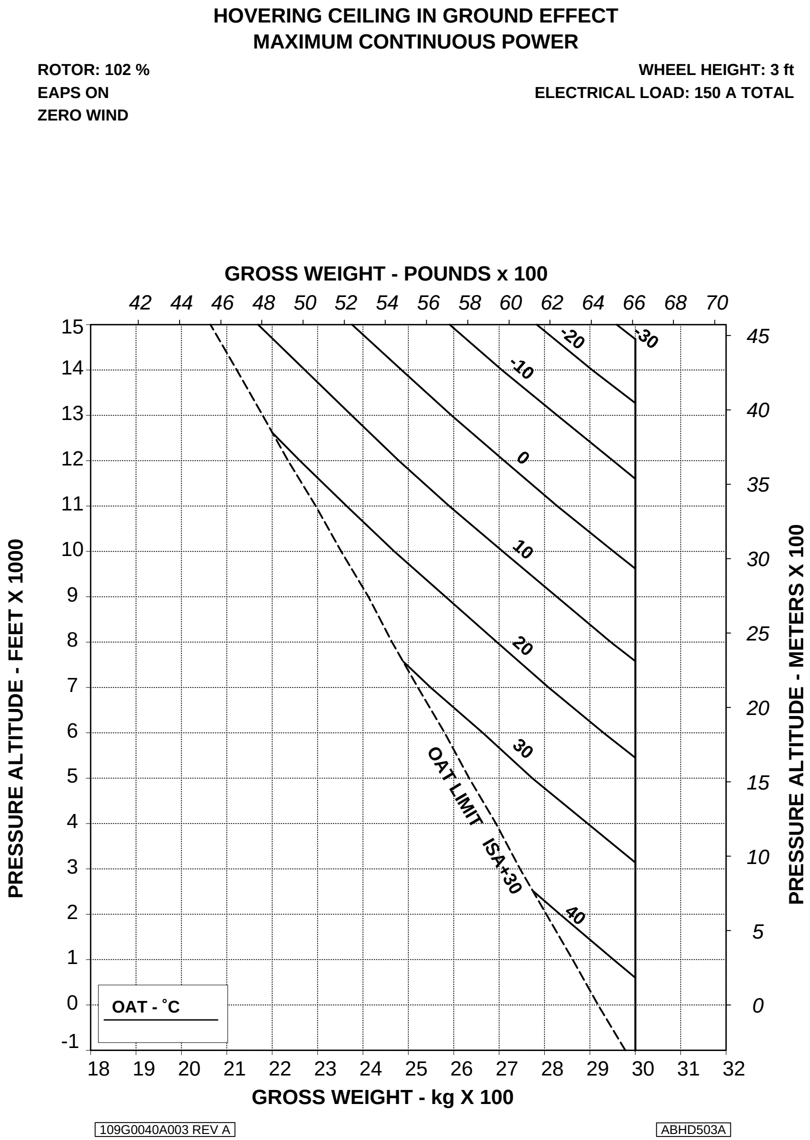


Figure 4-26. Hovering ceiling - in ground effect - maximum continuous power - EAPS on.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

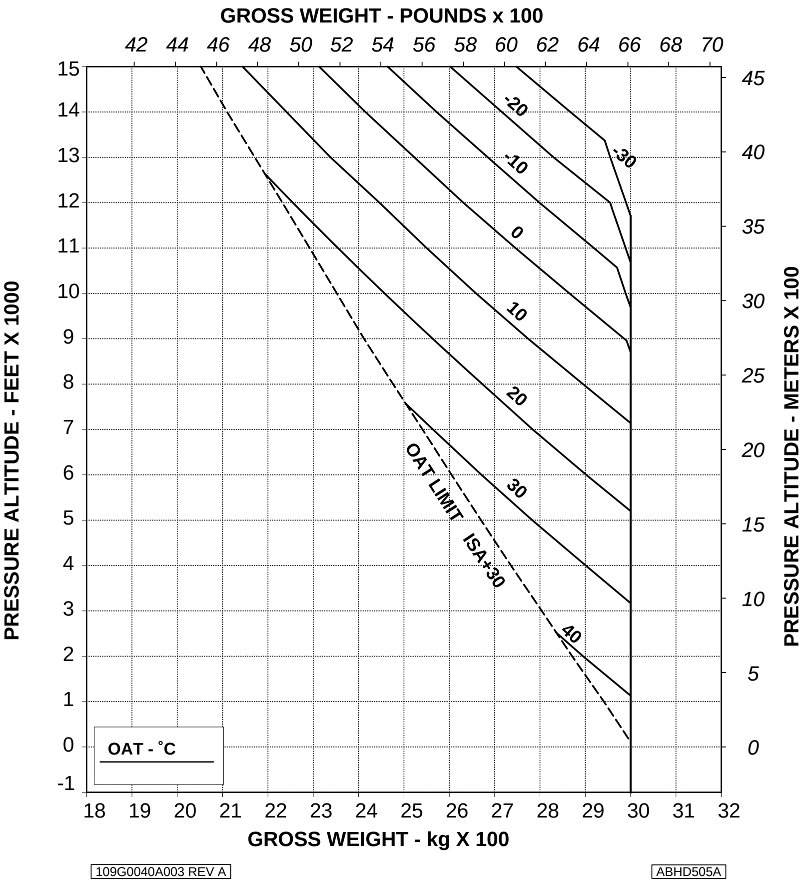


Figure 4-27. Hovering ceiling - out of ground effect - take-off power - EAPS off.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

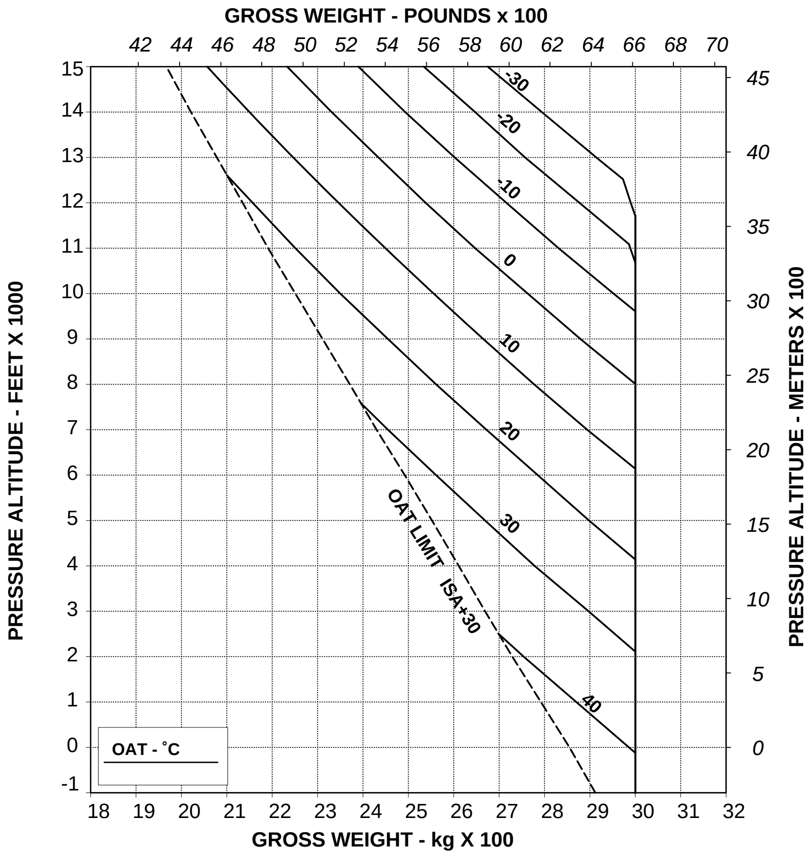


Figure 4-28. Hovering ceiling - out of ground effect - take-off power - EAPS on.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

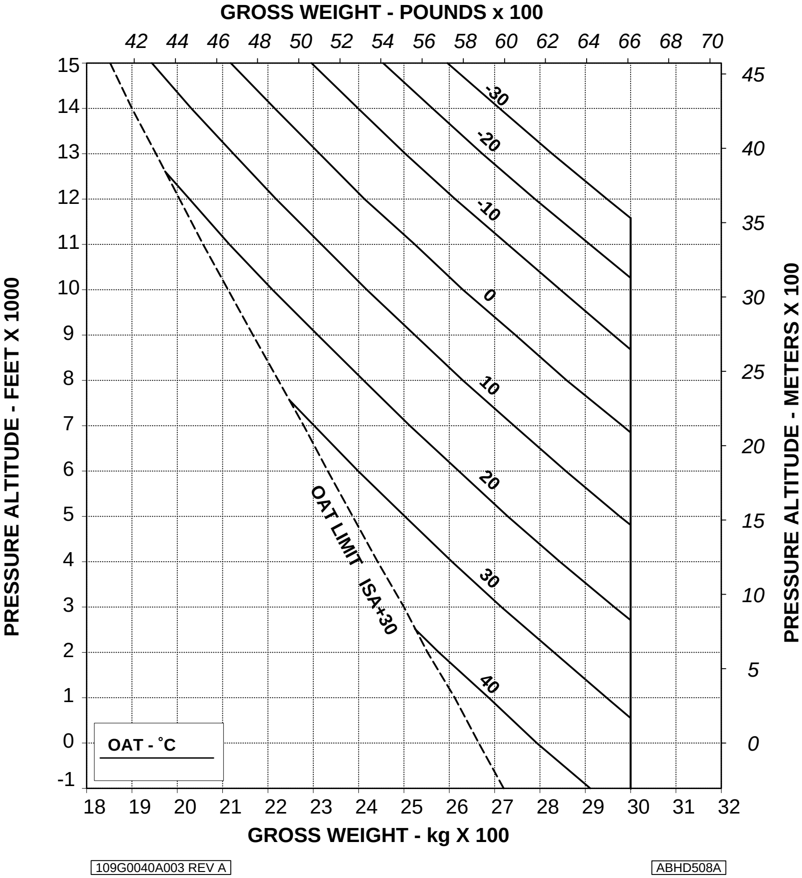


Figure 4-29. Hovering ceiling - out of ground effect - maximum continuous power - EAPS off.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

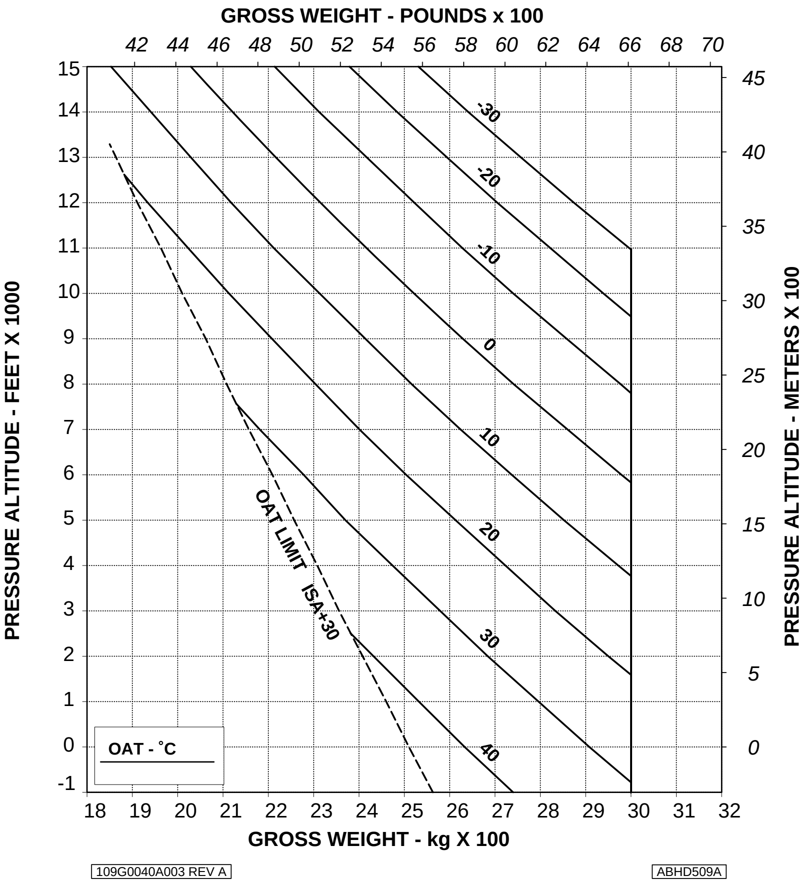


Figure 4-30. Hovering ceiling - out of ground effect - maximum continuous power - EAPS on.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS
EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27m/s)

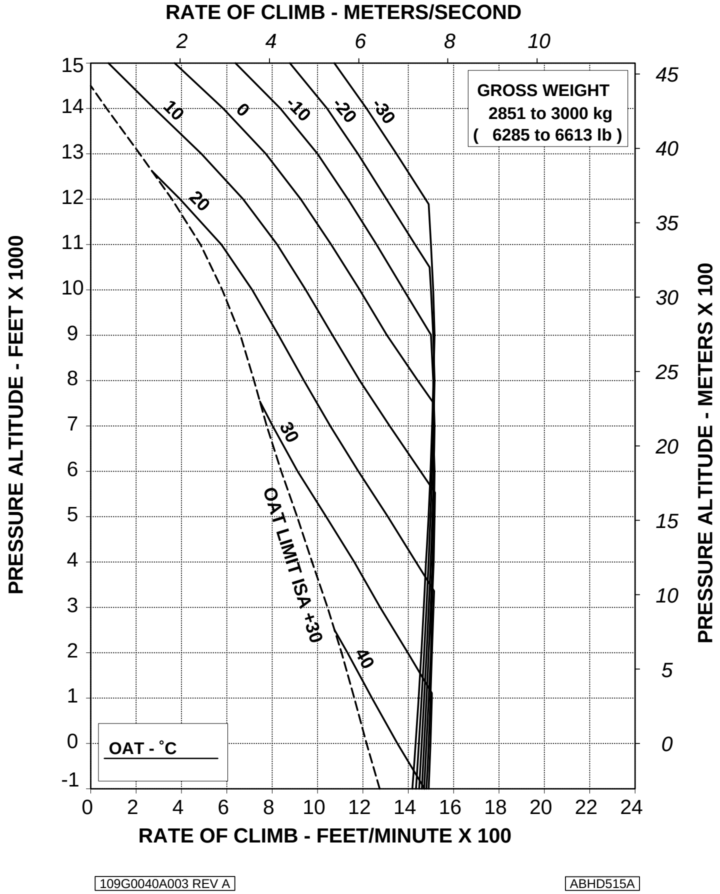


Figure 4-31. Rate of climb - all engines - take-off power - EAPS off/on.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON: DECREASE RATE OF CLIMB BY 200 ft/min (1,02 m/s)

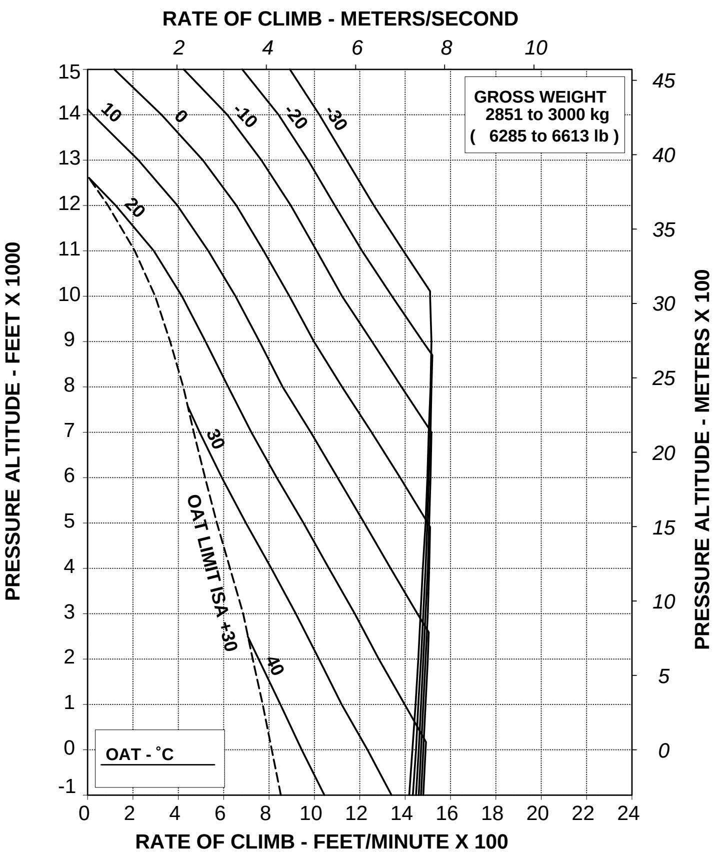


Figure 4-32. Rate of climb - all engines - maximum continuous power - EAPS off/on.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % | 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

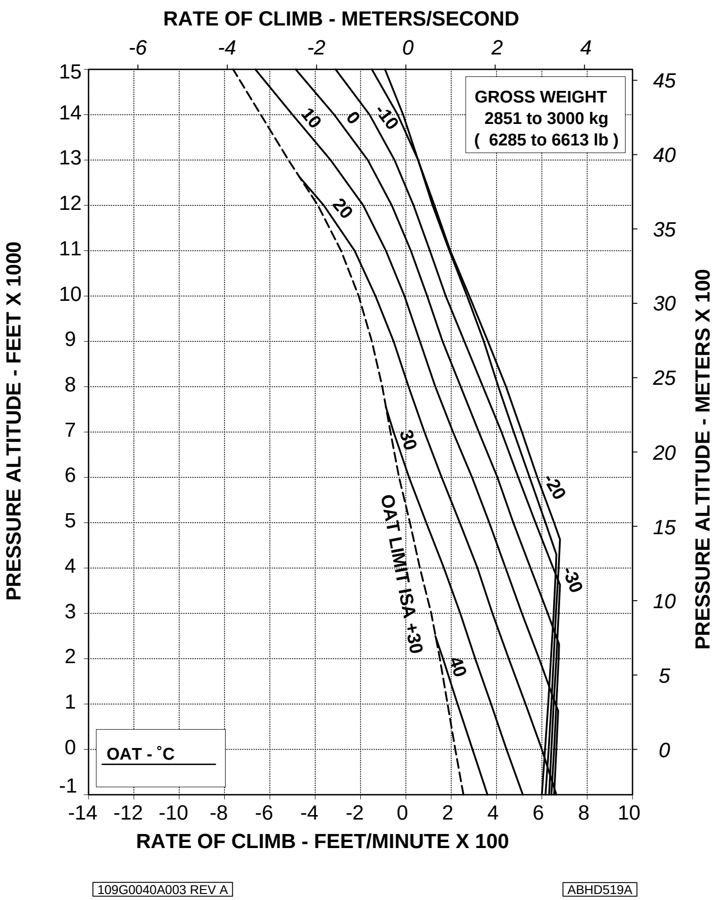


Figure 4-33. Rate of climb - OEI -2,5 minute power - EAPS off/on.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % — 60 kts IAS

EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

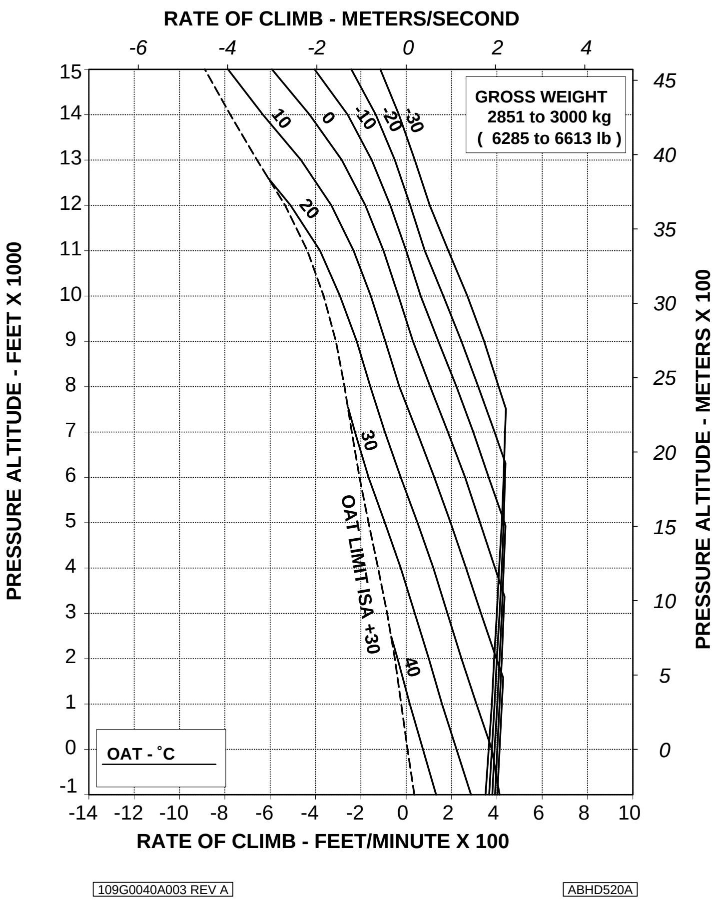


Figure 4-34. Rate of climb - OEI - maximum continuous power - EAPS off/on.

Helicopter with Engine Air Particle Separator (EAPS) and Heater or Environmental Control System (ECS).

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %	WHEEL HEIGHT: 3 ft
EAPS OFF	ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND	HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 40 kg (88 lb)

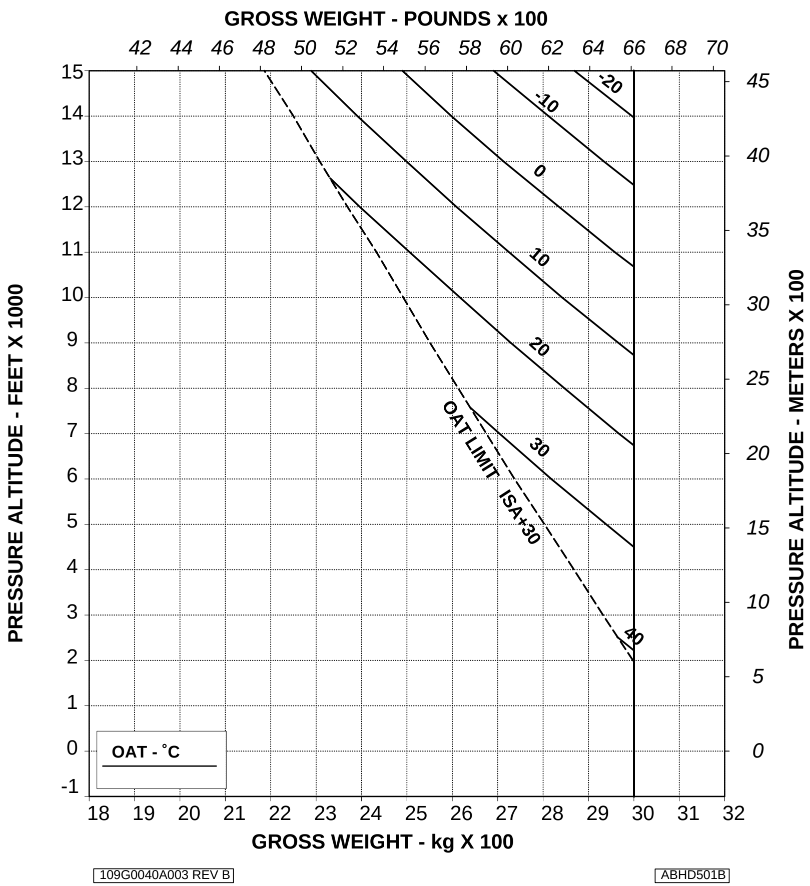


Figure 4-35. Hovering ceiling - in ground effect - take-off power - EAPS off/on - Heater or ECS on.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 % | WHEEL HEIGHT: 3 ft
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND | HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

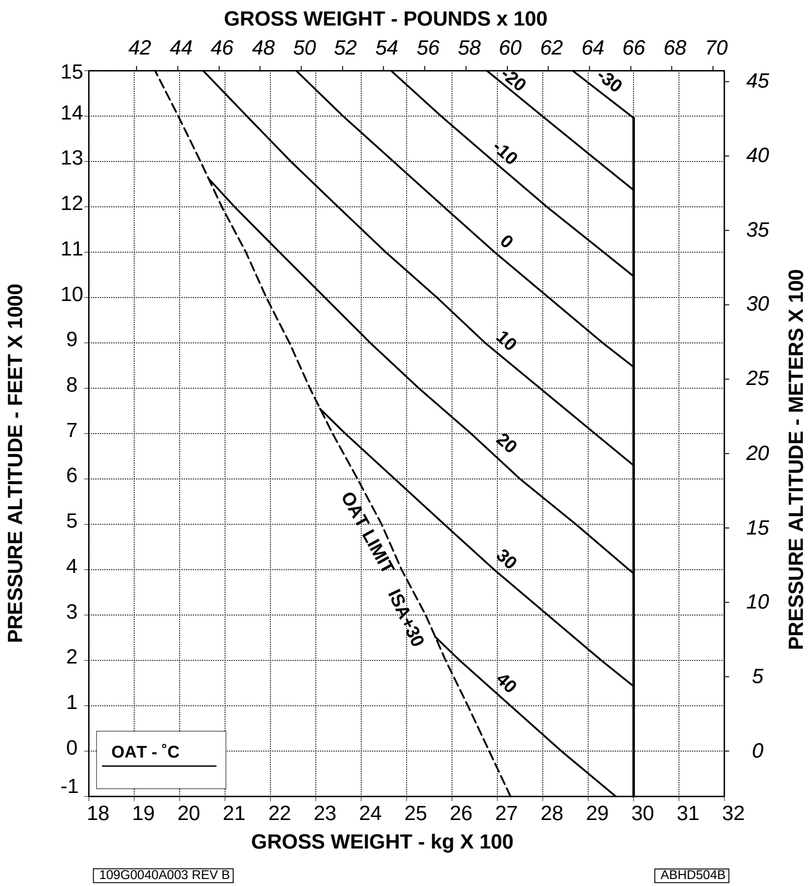


Figure 4-36. Hovering ceiling - in ground effect - maximum continuous power - EAPS off/on - Heater or ECS on.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

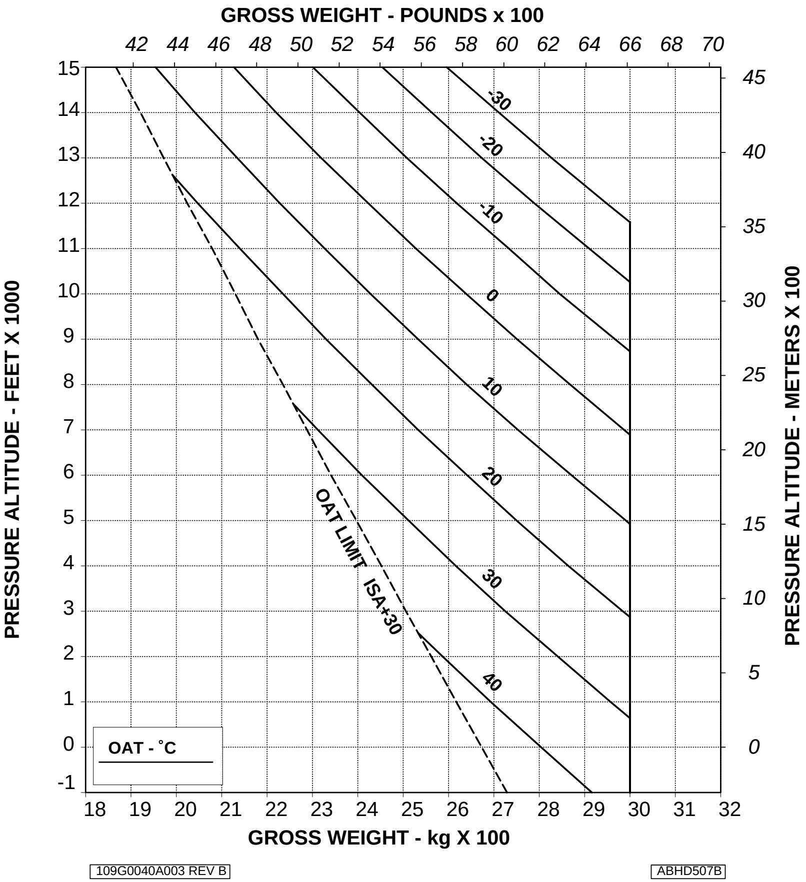


Figure 4-37. Hovering ceiling - out of ground effect - take-off power - EAPS off/on - Heater or ECS on.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

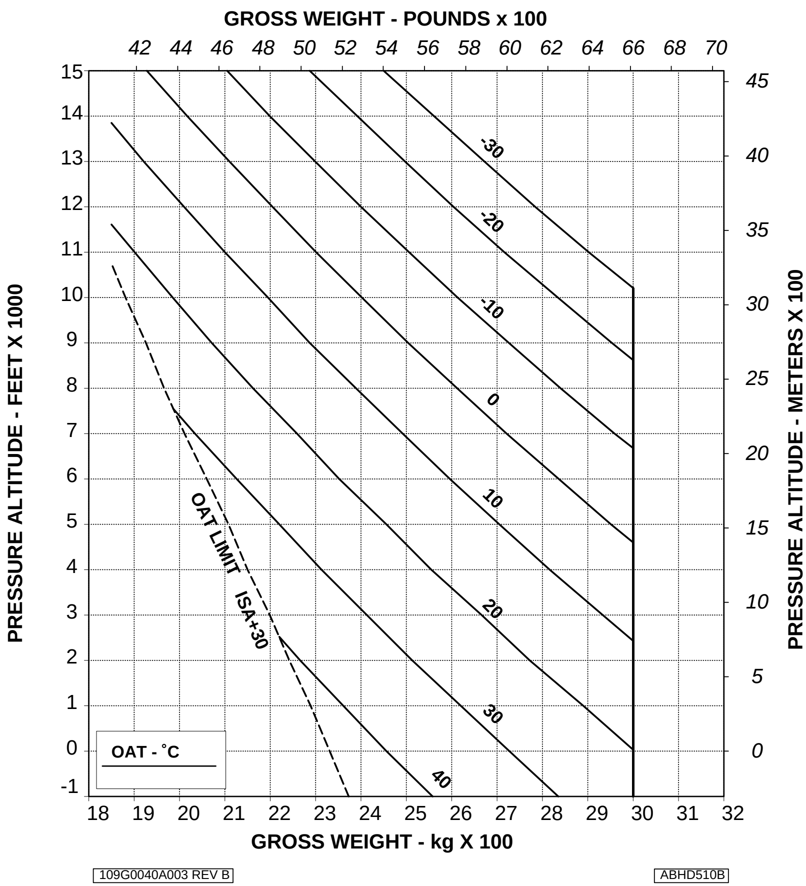


Figure 4-38. Hovering ceiling - out of ground effect - maximum continuous power - EAPS off/on - Heater or ECS on.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS
EAPS OFF — ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

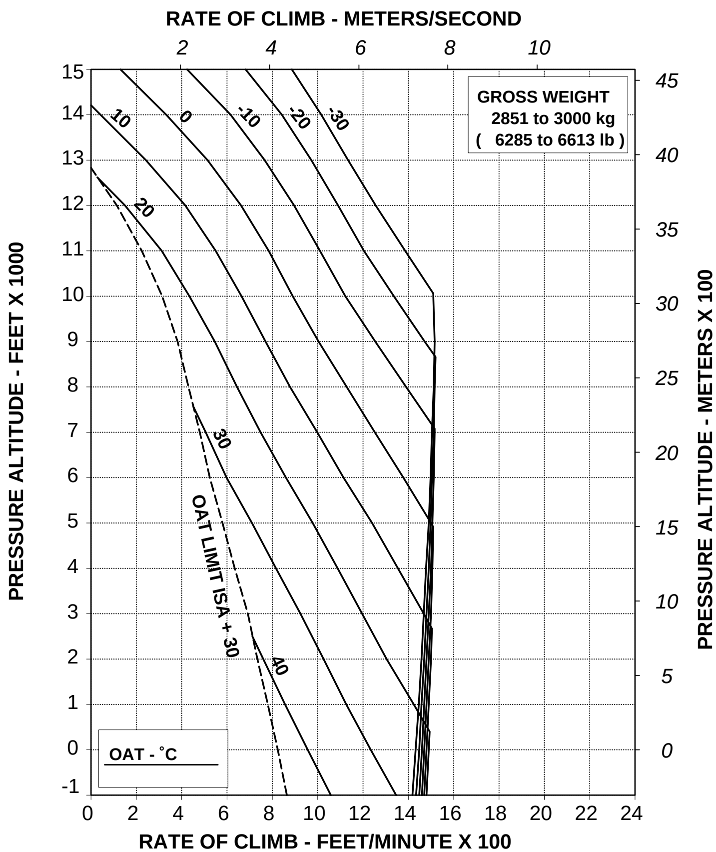


Figure 4-39. Rate of climb - all engines - take-off power - EAPS off/on - Heater or ECS on.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A TOTAL**
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

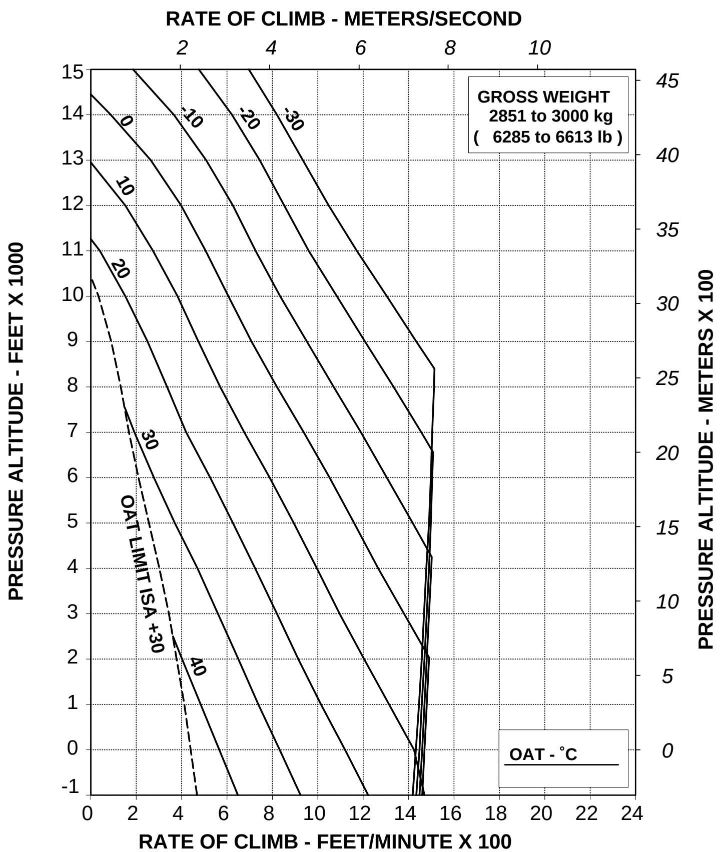


Figure 4-40. Rate of climb - all engines - maximum continuous power - EAPS off/on - Heater or ECS on.

NOISE CHARACTERISTICS

The following noise levels comply with ICAO Annex 16, Chapter 8 ″Noise requirements″ and FAR part 36 Appendix H.

Model:A109E	**Engine: Pratt & Whitney 206C**		**Gross Weight: 3000 Kg**			
Configuration	Level flyover EPNL (EPNdB)		Takeoff EPNL (EPNdB)		Approach EPNL (EPNdB)	
Clean aircraft No external kits installed	FAA 91.2	ICAO 91.3	FAA 91.8	ICAO 91.7	FAA 91.6	ICAO 91.7

SECTION 6 - WEIGHT AND BALANCE

No changes.

SECTION 7 - SYSTEM DESCRIPTION

No changes.

SECTION 8 - HANDLING AND SERVICING

No changes.

SECTION 9 - SUPPLEMENTAL PERFORMANCE INFORMATION

GENERAL INFORMATION

The Supplemental Performance Information contained in this section is provided for use in conjunction with Section 4 and optional equipment Appendices, as applicable.

This section contains cruise charts, to determine max endurance and recommended cruise, and hovering ceiling charts with headwind effect for one engine inoperative

HELICOPTER CONFIGURATION.

Clean configuration.

CRUISE CHARTS

Fuel flows are based on calculations and limited flight test data.

These data do not include the effects of ECS or heater on fuel consumption.

Data are applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag or power available.

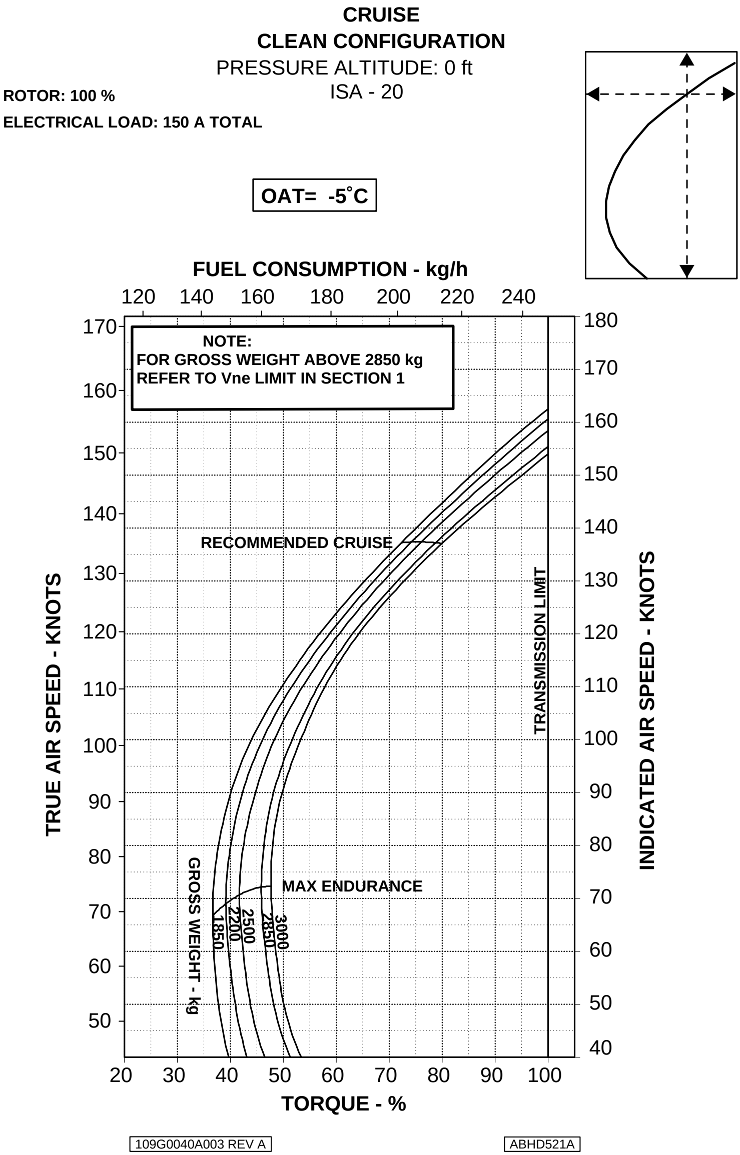


Figure 9-1. Cruise - Pressure Altitude 0 ft (Sea Level) - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 0 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

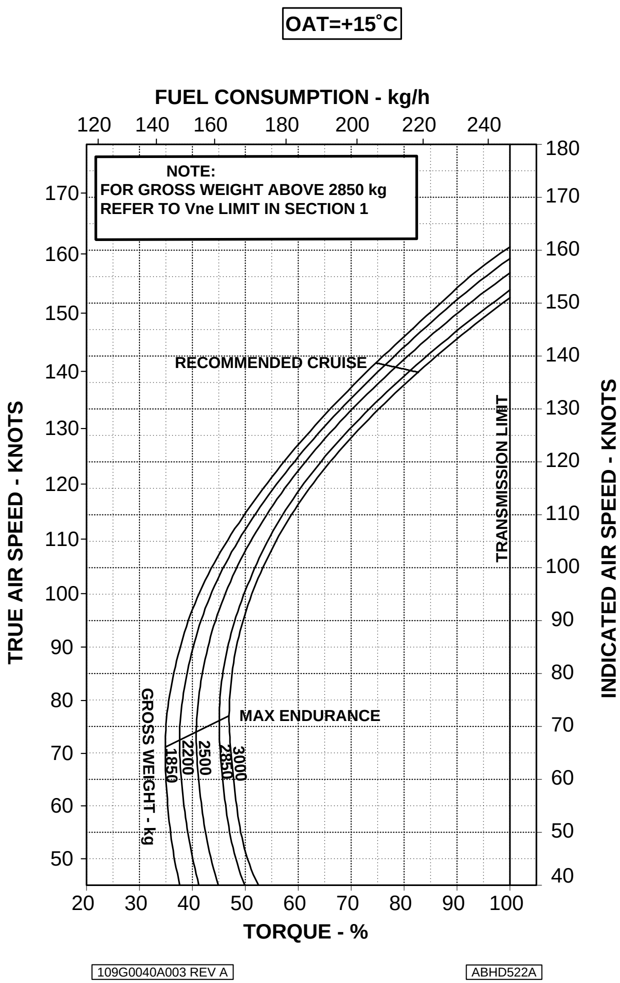


Figure 9-2. Cruise - Pressure Altitude 0 ft (Sea Level) - ISA.

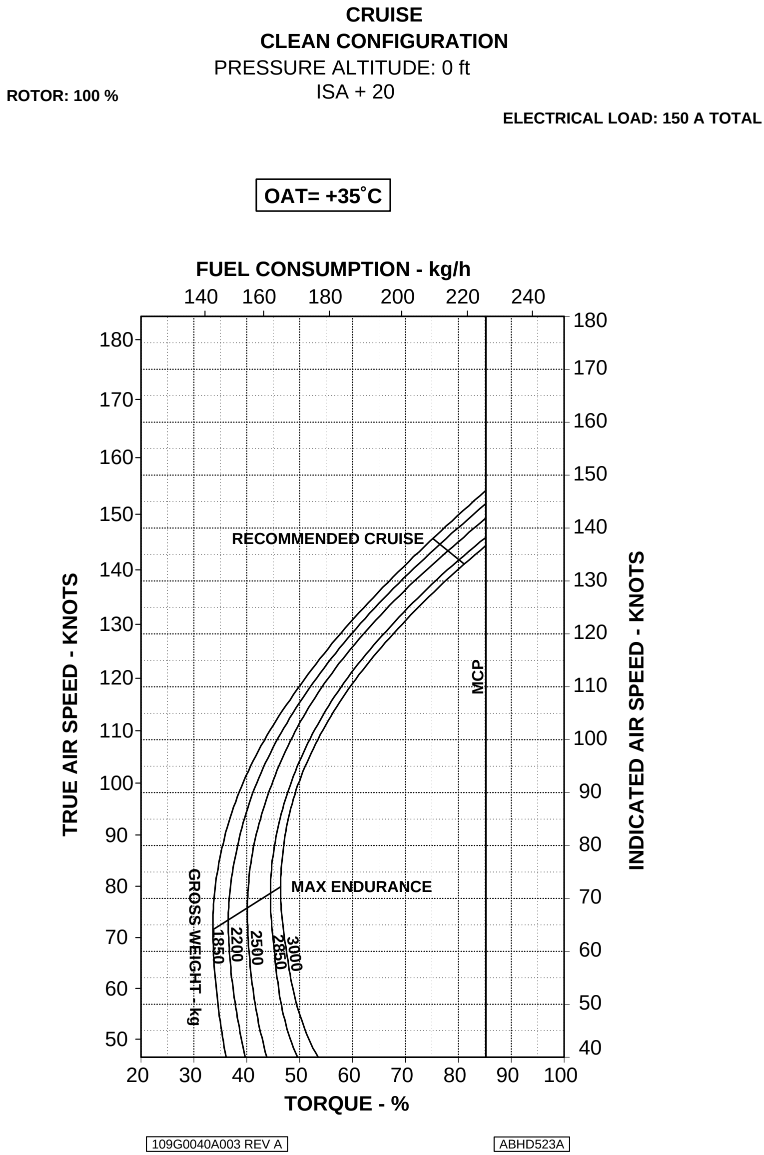


Figure 9-3. Cruise - Pressure Altitude 0 ft (Sea Level) - ISA +20°C.

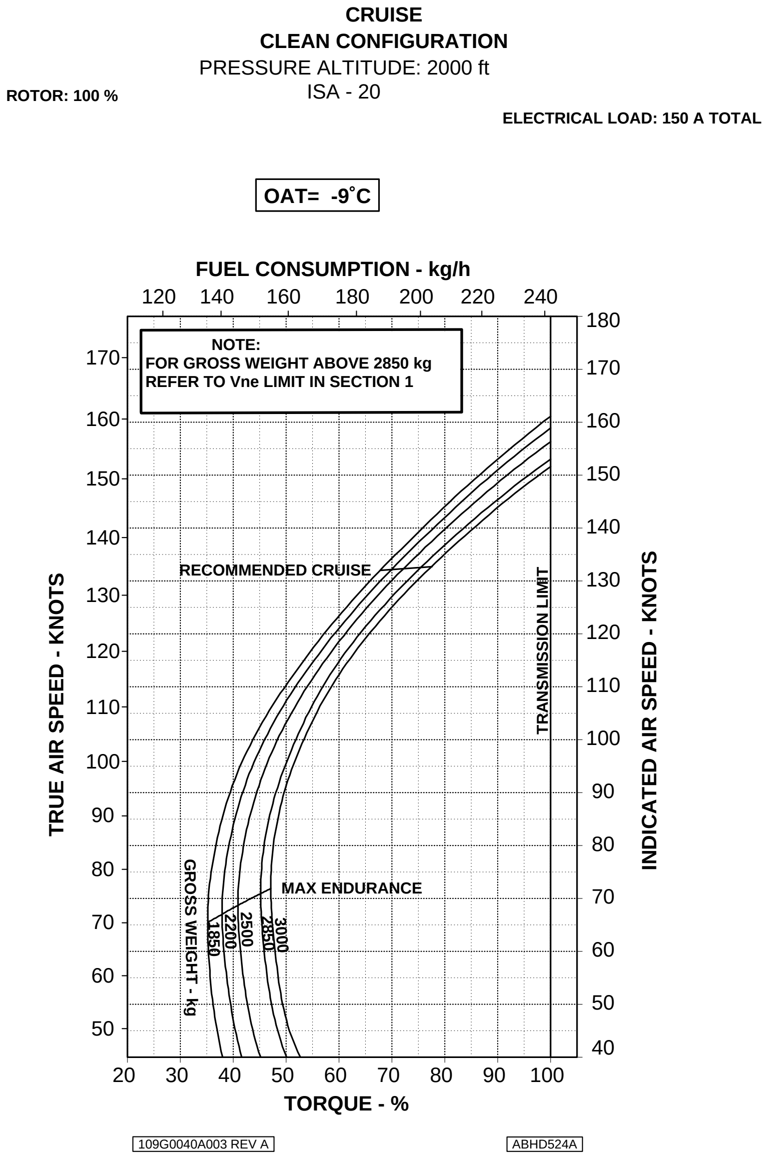


Figure 9-4. Cruise - Pressure Altitude 2000 ft - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 2000 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

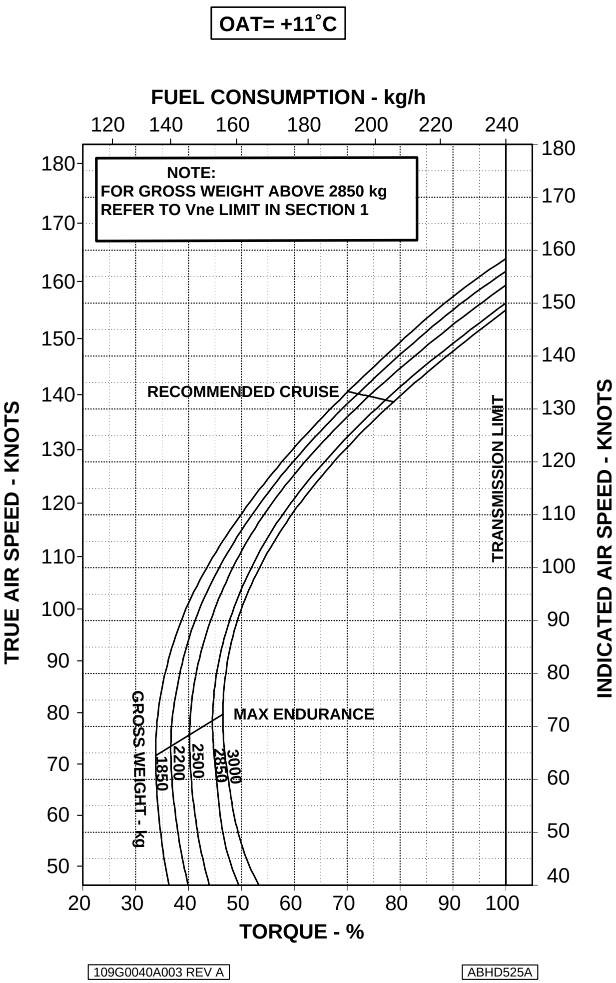


Figure 9-5. Cruise - Pressure Altitude 2000 ft - ISA

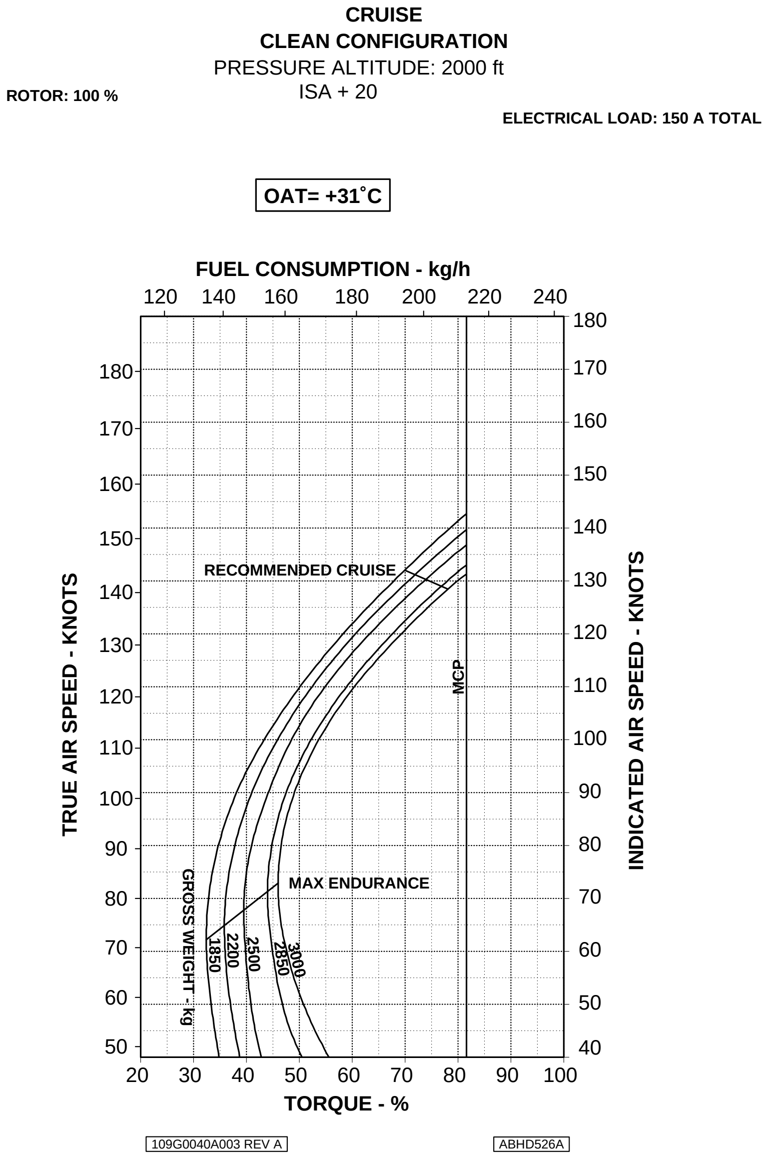


Figure 9-6. Cruise - Pressure Altitude 2000 ft - ISA +20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE: 4000 ft
ISA - 20

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= -13˚C

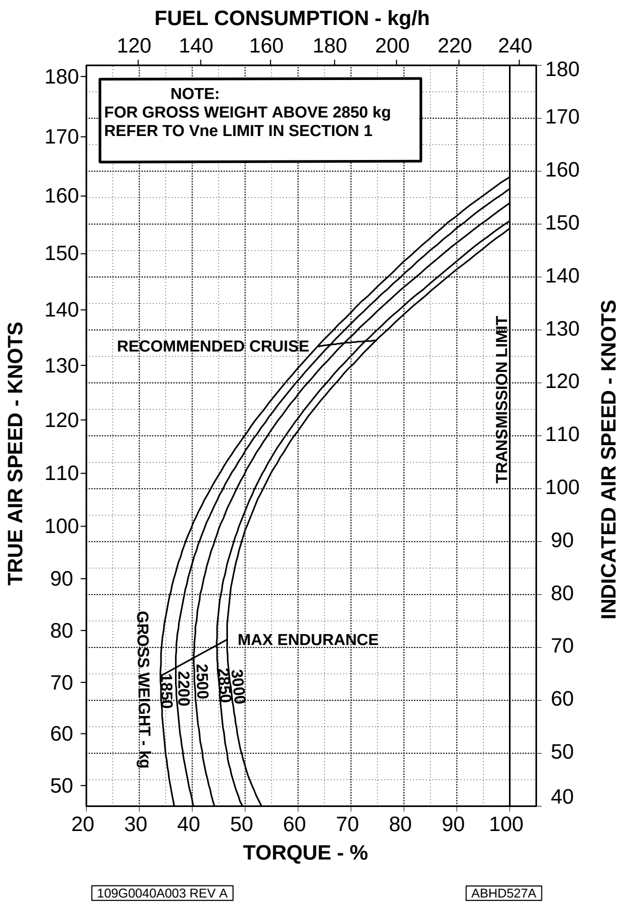


Figure 9-7. Cruise - Pressure Altitude 4000 ft - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 4000 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= + 7°C

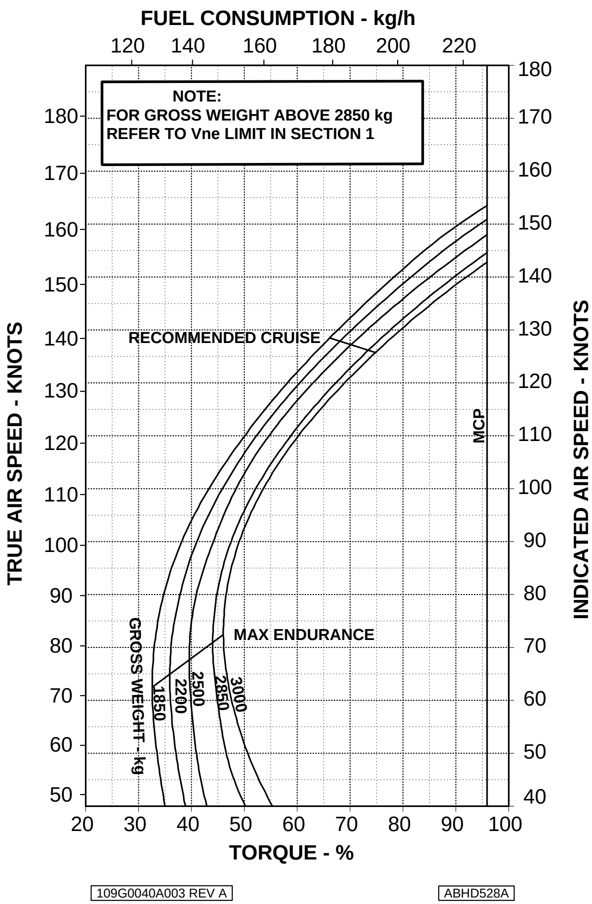


Figure 9-8. Cruise - Pressure Altitude 4000 ft - ISA.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE: 4000 ft
ISA + 20

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= +27°C

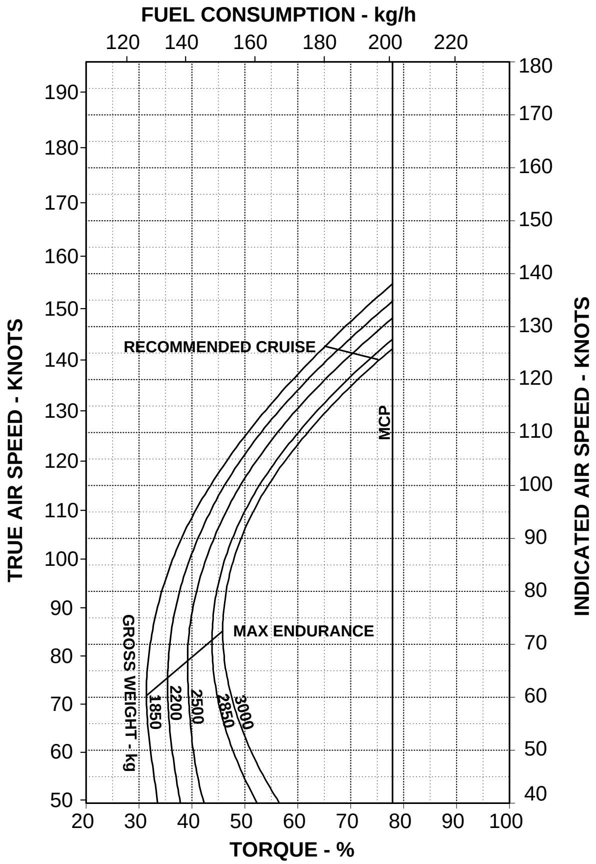


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Figure 9-9. Cruise - Pressure Altitude 4000 ft - ISA +20°C.

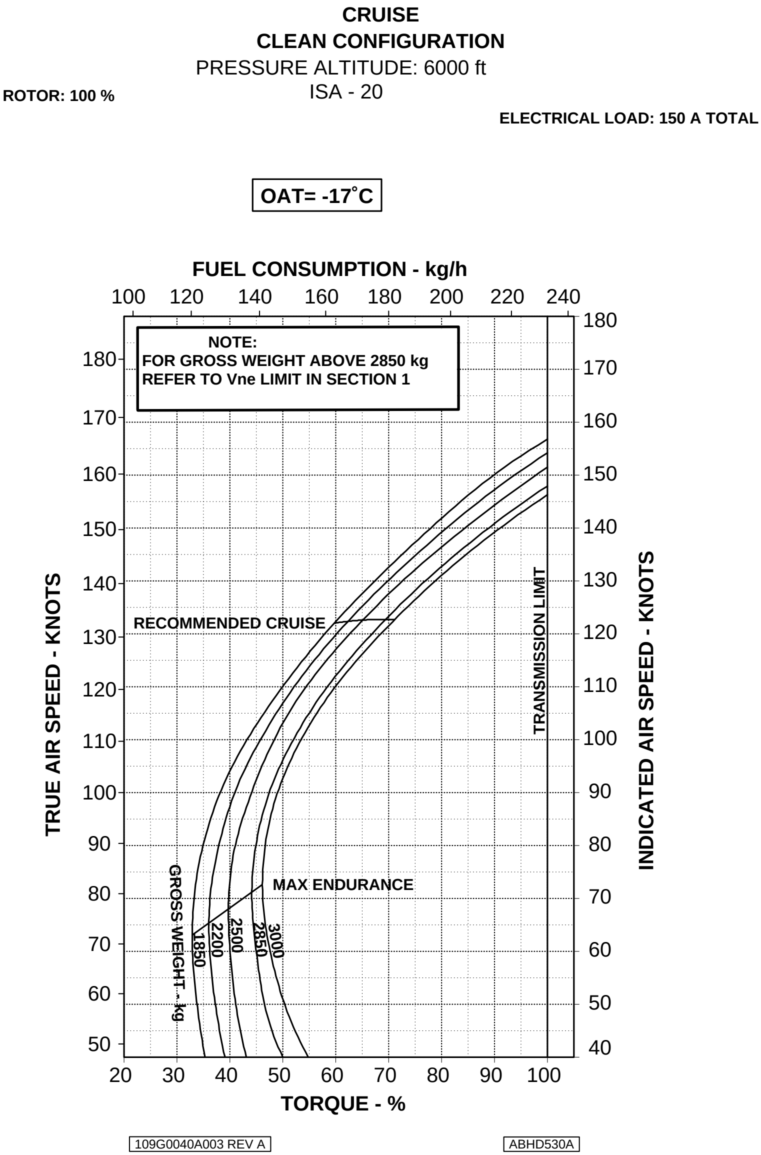

Figure 9-10. Cruise - Pressure Altitude 6000 ft - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 6000 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= + 3˚C

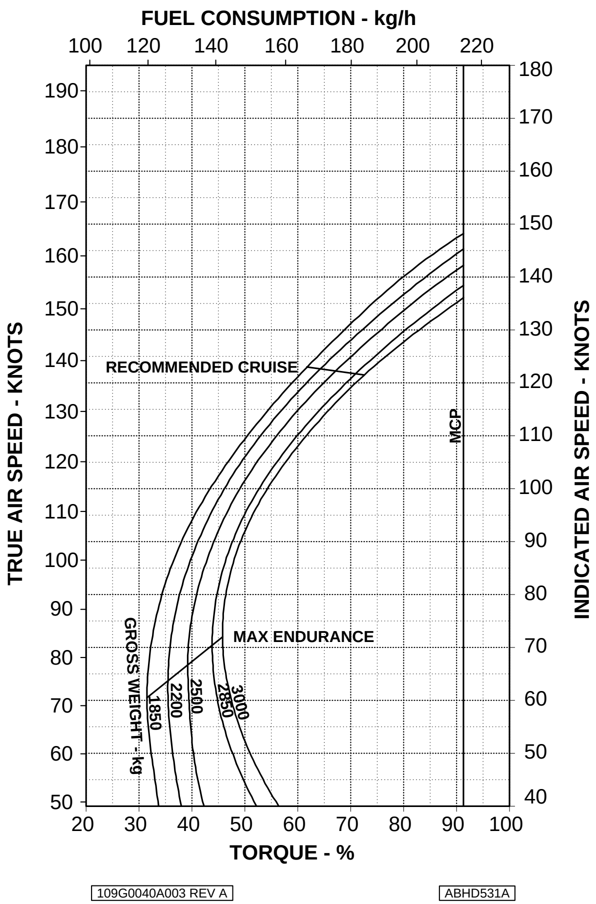

Figure 9-11. Cruise - Pressure Altitude 6000 ft - ISA.

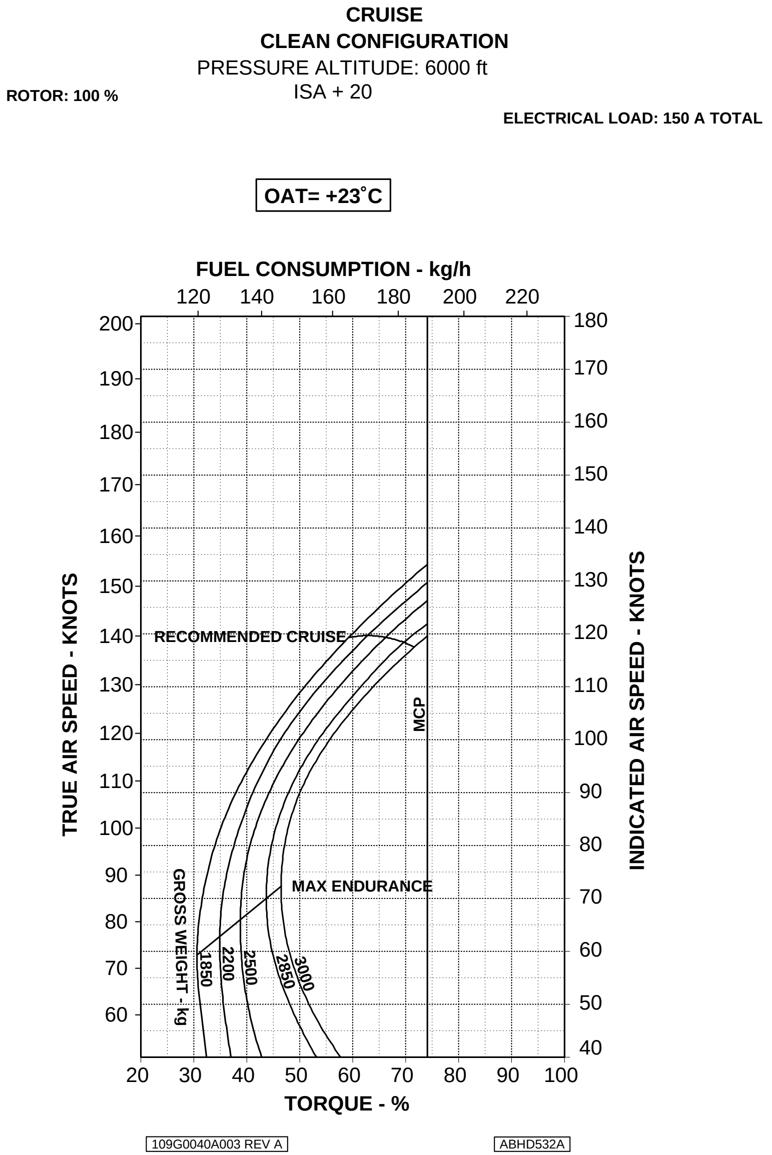

Figure 9-12. Cruise - Pressure Altitude 6000 ft - ISA +20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE: 8000 ft
ISA - 20

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= -21˚C

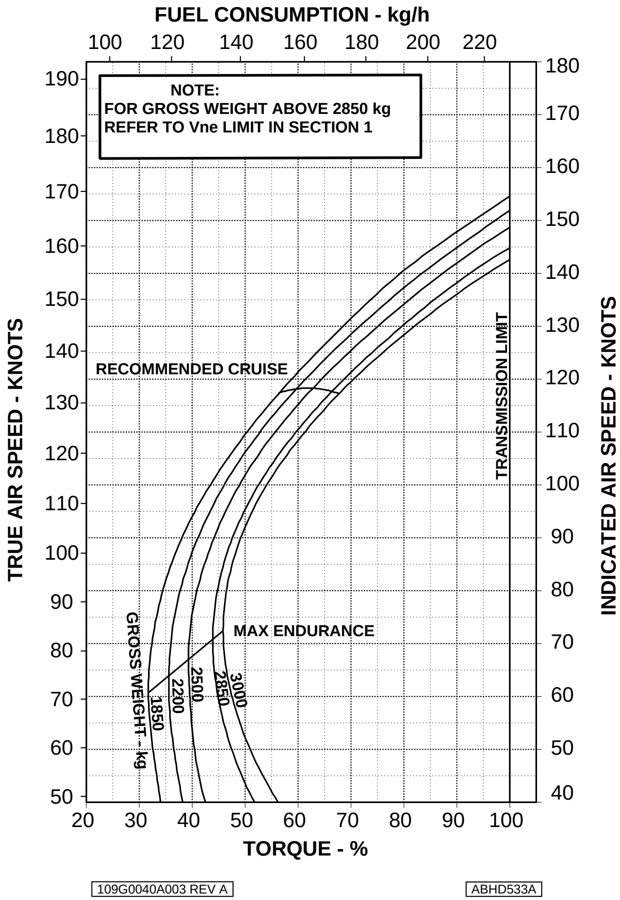

Figure 9-13. Cruise - Pressure Altitude 8000 ft - ISA -20°C.

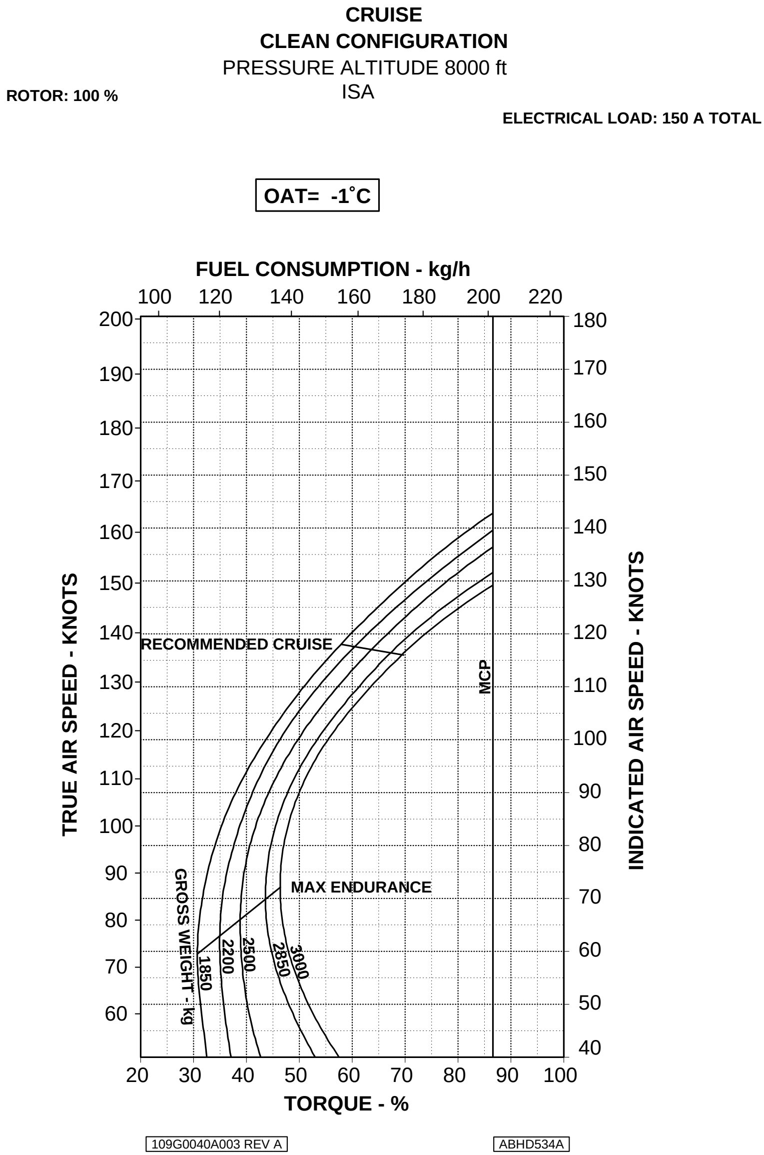

Figure 9-14. Cruise - Pressure Altitude 8000 ft - ISA.

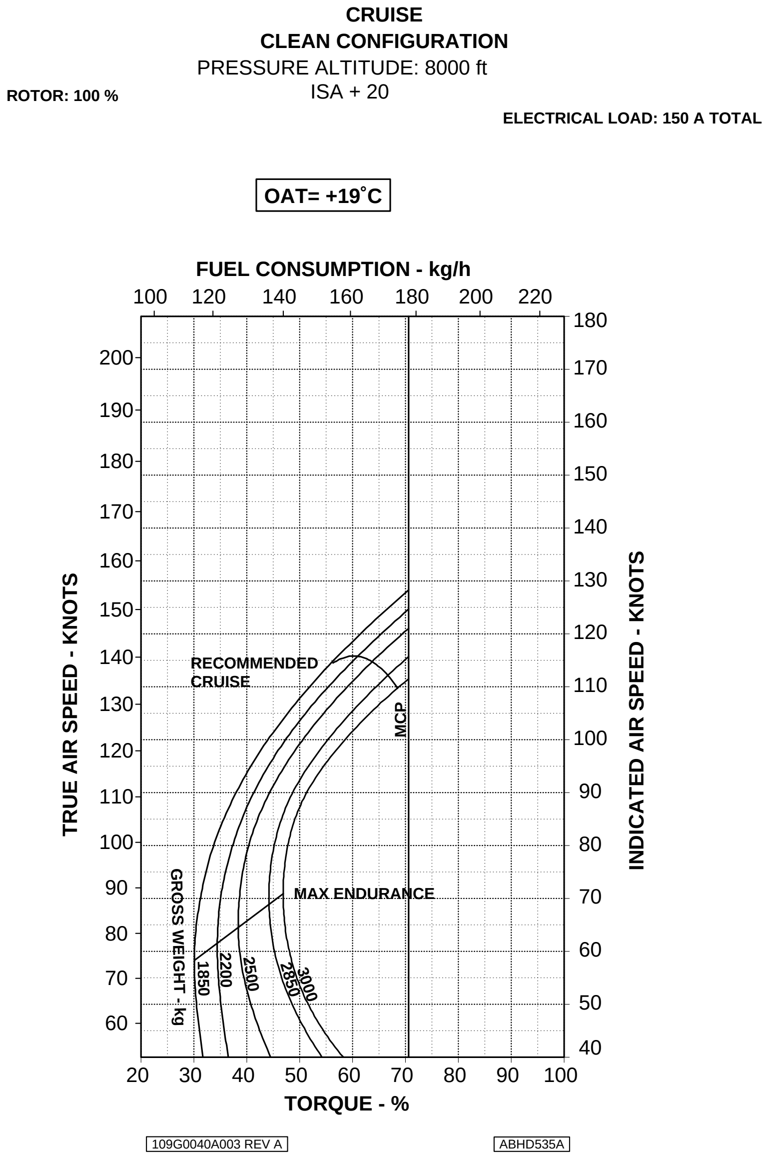

Figure 9-15. Cruise - Pressure Altitude 8000 ft - ISA +20°C.

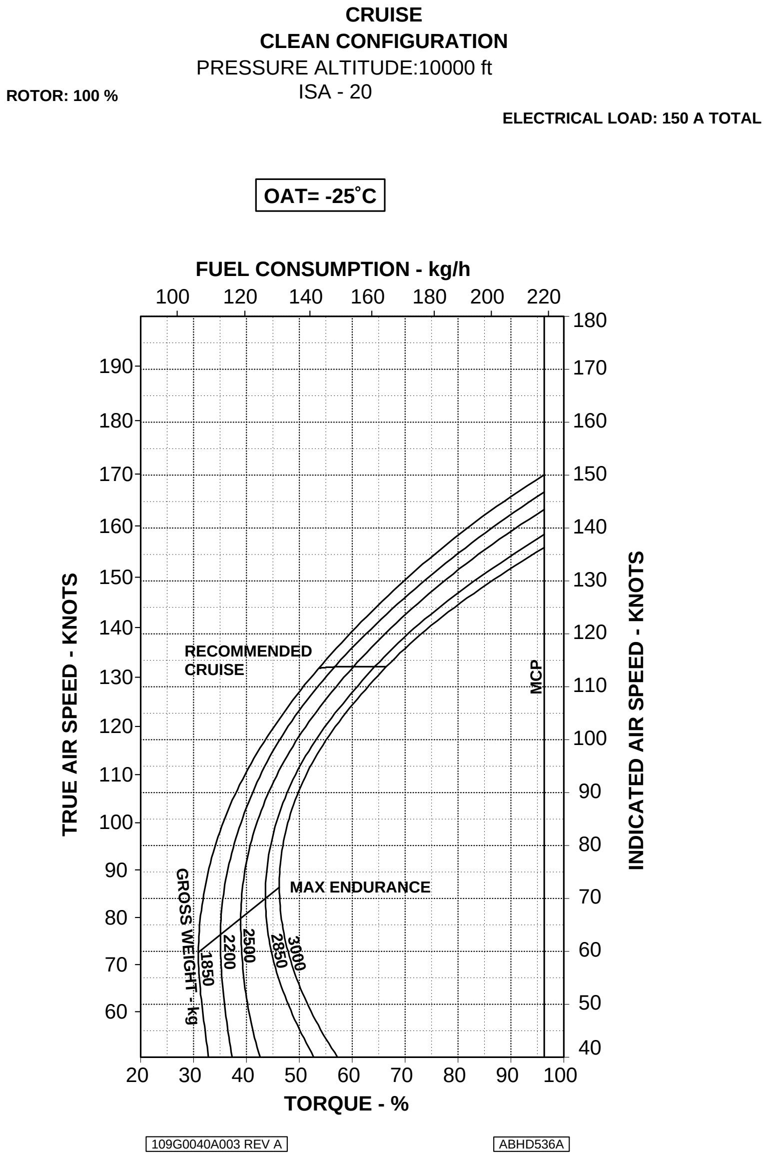

Figure 9-16. Cruise - Pressure Altitude 10000 ft - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 10000 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= -5°C

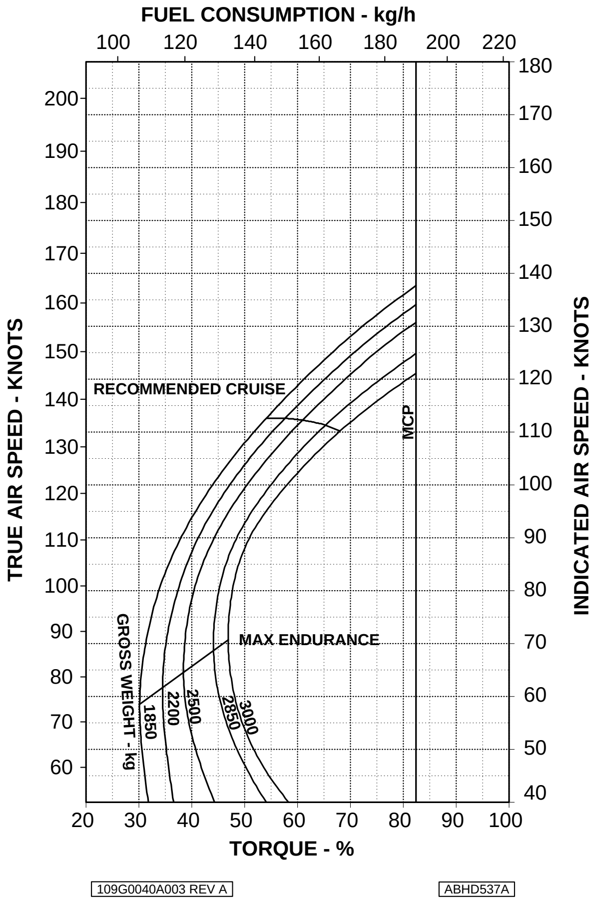

Figure 9-17. Cruise - Pressure Altitude 10000 ft - ISA.

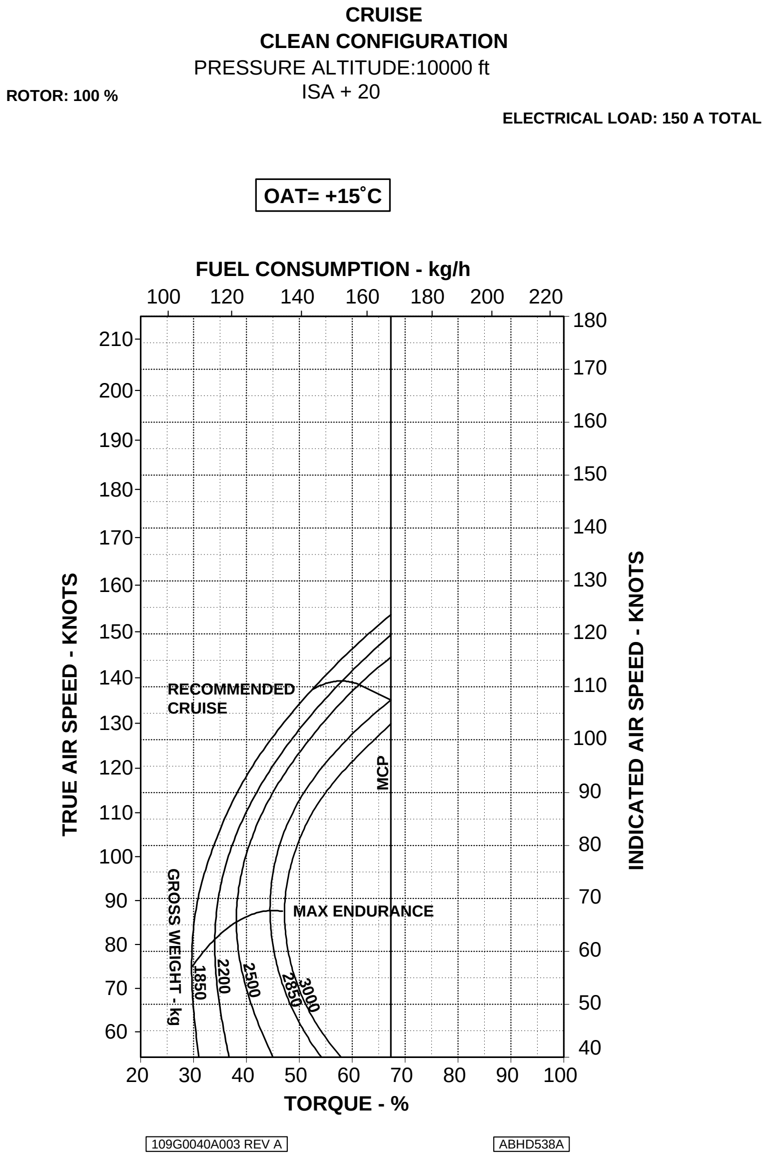

Figure 9-18. Cruise - Pressure Altitude 10000 ft - ISA +20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE:15000 ft
ISA - 20

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= -25˚C

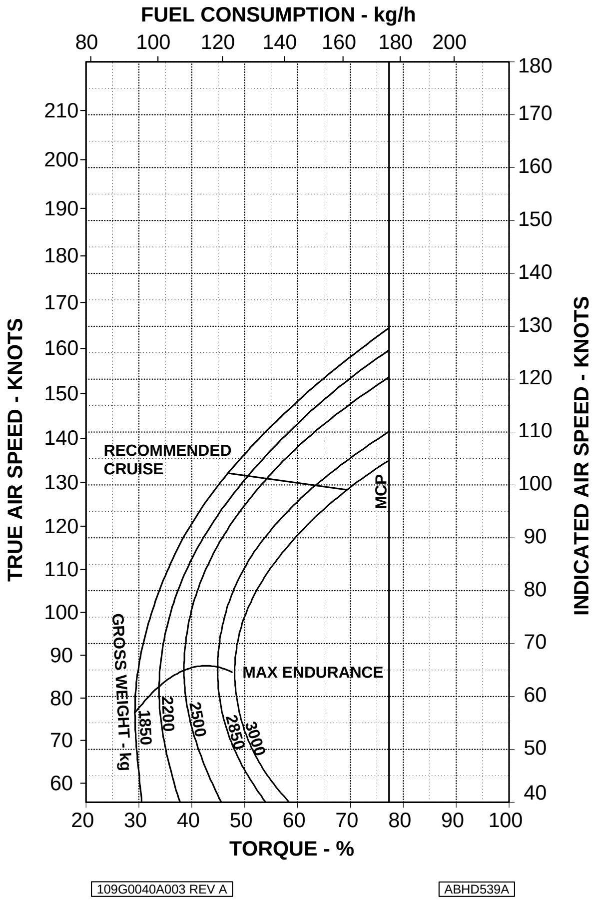

Figure 9-19. Cruise - Pressure Altitude 15000 ft - ISA -20°C.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE 15000 ft
ISA

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= -15˚C

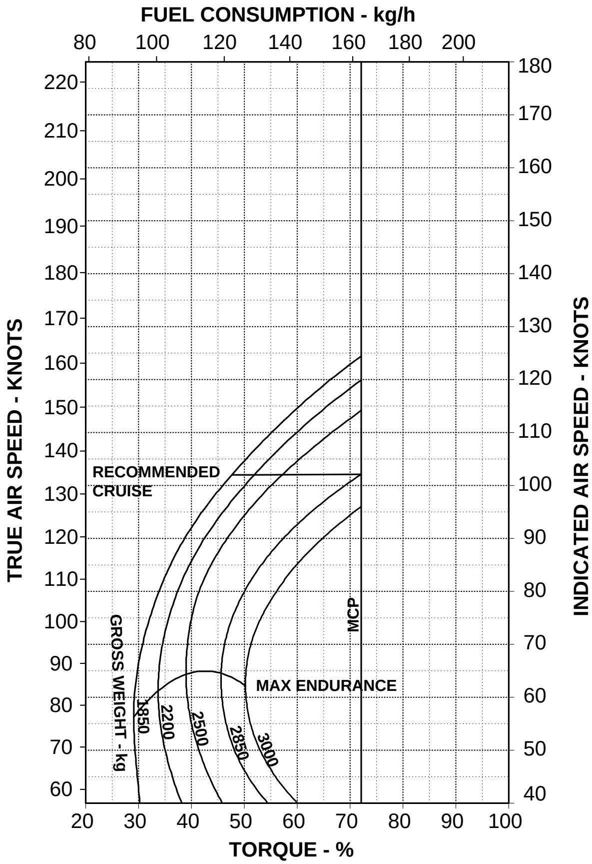

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Figure 9-20. Cruise - Pressure Altitude 15000 ft - ISA.

CRUISE
CLEAN CONFIGURATION
PRESSURE ALTITUDE:15000 ft
ISA + 20

ROTOR: 100 %

ELECTRICAL LOAD: 150 A TOTAL

OAT= + 5˚C

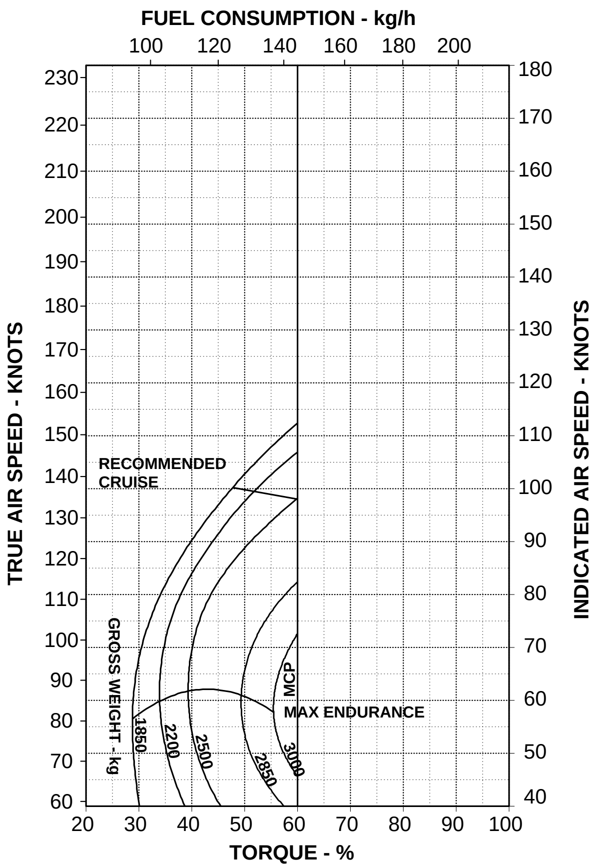

Figure 9-21. Cruise - Pressure Altitude 15000 ft - ISA +20°C.

HOVERING CEILING - ONE ENGINE INOPERATIVE CHARTS

The hovering ceiling charts are presented for the OEI 2.5 minute power rating In and Out Ground Effect with wind effect.

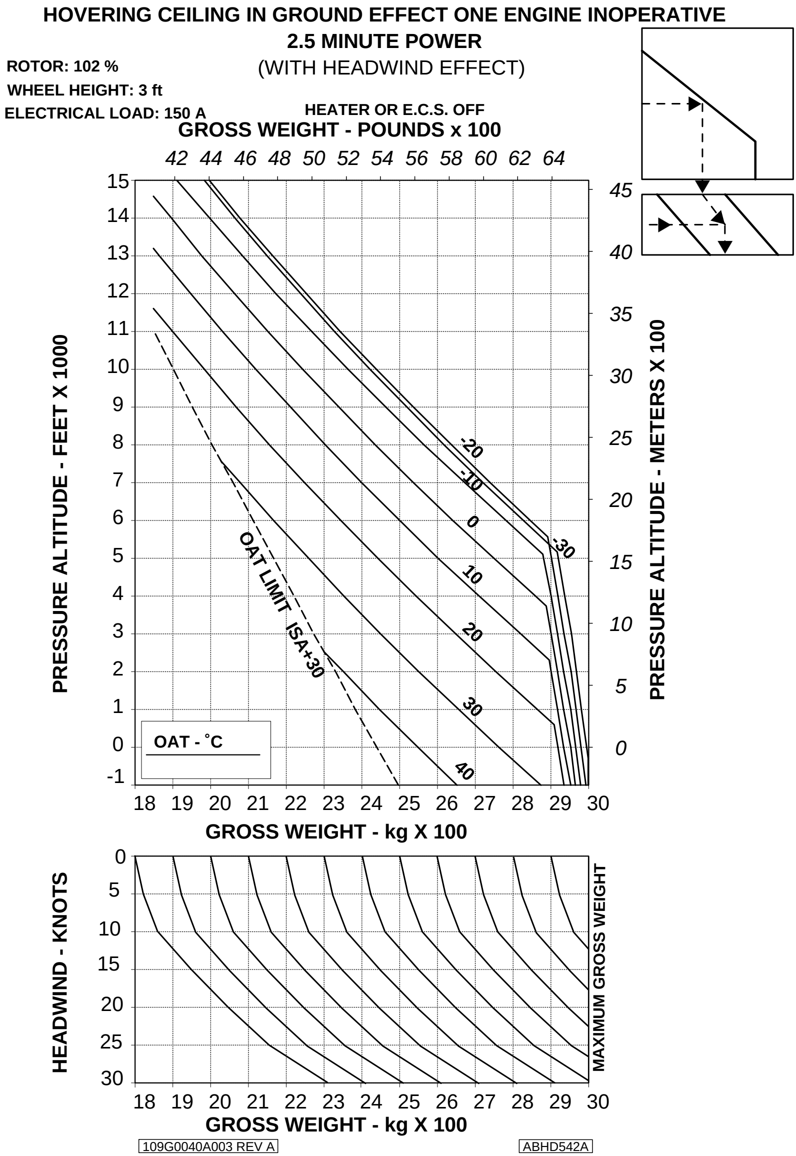

Figure 9-22. Hovering ceiling - IGE - OEI - 2.5 minute power.

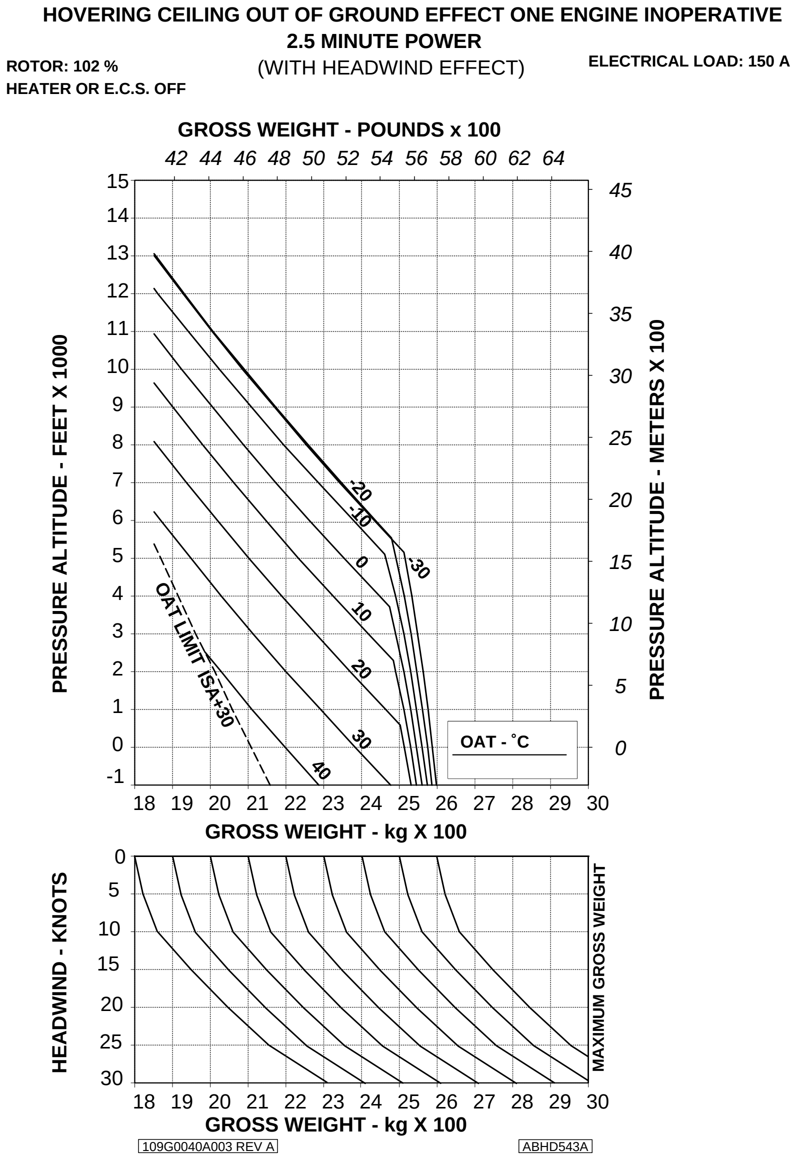

Figure 9-23. Hovering ceiling - OGE - OEI - 2.5 minute power.

E.A.S.A. Approval EASA. R.C.01098
dated 3 August 2005

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

HIGH TEMPERATURE OPERATION (ISA + 35°C)

The modification P/N 109-0823-46 allows to increase the maximum sea level ambient air temperature for operation up to +50°C (ISA +35°C).

Operation at maximum sea level ambient air temperature from +45°C up to +50°C, are permitted provided that the following assembly is installed (as applicable):

— Main Transmission Oil Temperature Limit Extension P/N 109-0823-24;

— Engine ejectors P/N 109-0601-51;

— V_{NE} placard P/N 109-0740L36;

— Flight instrument protection P/N 109-0732-02.

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PART I - E.A.S.A. APPROVED

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List of Illustrations (Cont.d)

SECTION 1 - LIMITATIONS

REQUIRED EQUIPMENT

Operation at maximum sea level ambient air temperature from +45°C up to +50°C, are permitted provided that the following assembly is installed (as applicable):

— Main Transmission Oil Temperature Limit Extension P/N 109-0823-24;

— Engine ejectors P/N 109-0601-51;

— V_{NE} placard P/N 109-0740L36;

— Flight instrument protection P/N 109-0732-02.

AIRSPEED LIMITATIONS (KIAS)

V_{NE} (Power on) .... See Figure 1-1

V_{NE} (Power off).. See Figure 1-2

V_{NE} (Power on at Gross Weight from 2850 up to 3000 kg).. See Figure 1-3

WEIGHT LIMITATIONS

The maximum weight for take-offs and landings to operate in:

— Equivalent Category ″A″ from clear area, from short field and from helipad (ground based or elevated) are shown in figures 1-4 thru 1-6;

— Equivalent Category ″A″ - Training procedures from clear area, from short field and from helipad (ground based) are shown in figures 1-7 thru 1-9.

— Equivalent Category ″A″ with particle separator from clear area, from short field and from helipad (ground based or elevated) are shown in figures 1-10 thru 1-15;

— Equivalent Category ″A″ - Training procedures with particle separator from clear area, from short field and from helipad (ground based) are shown in figures 1-16 thru 1-18.

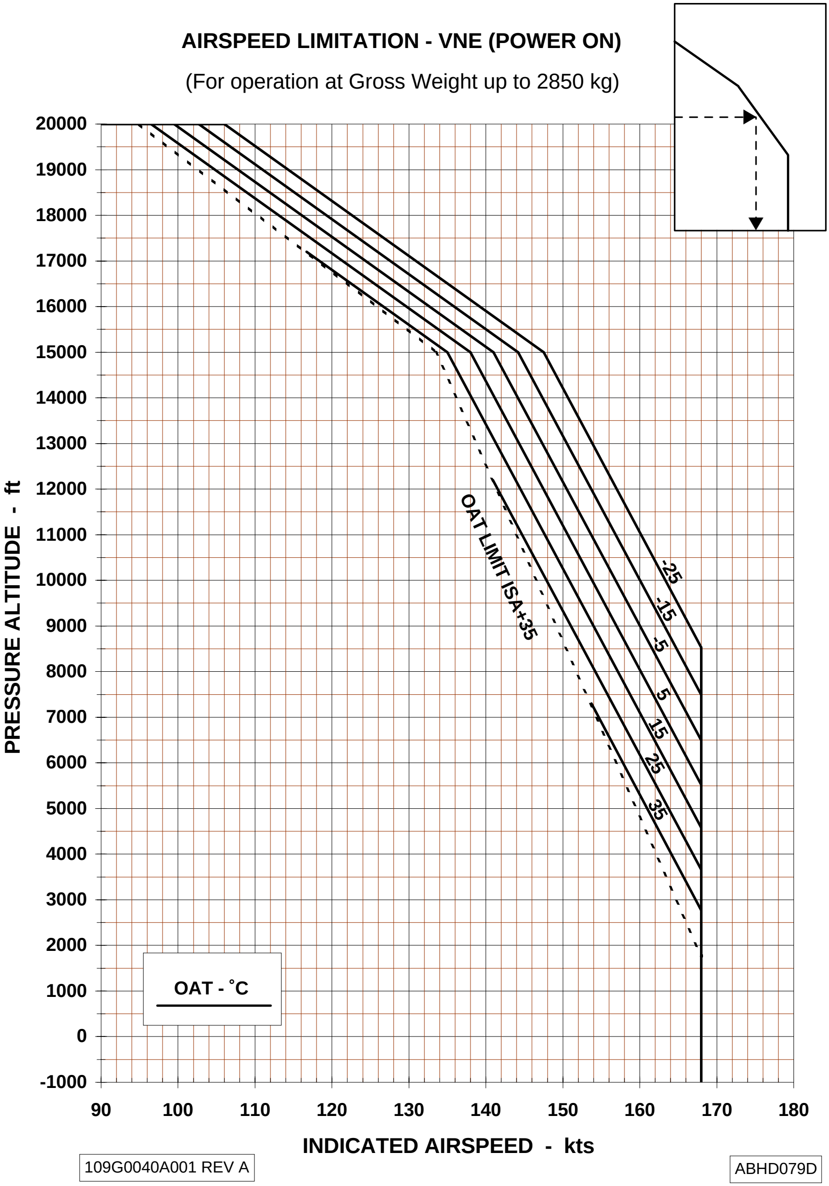


Figure 1-1. Airspeed Limitations - V_{NE} (Power on)

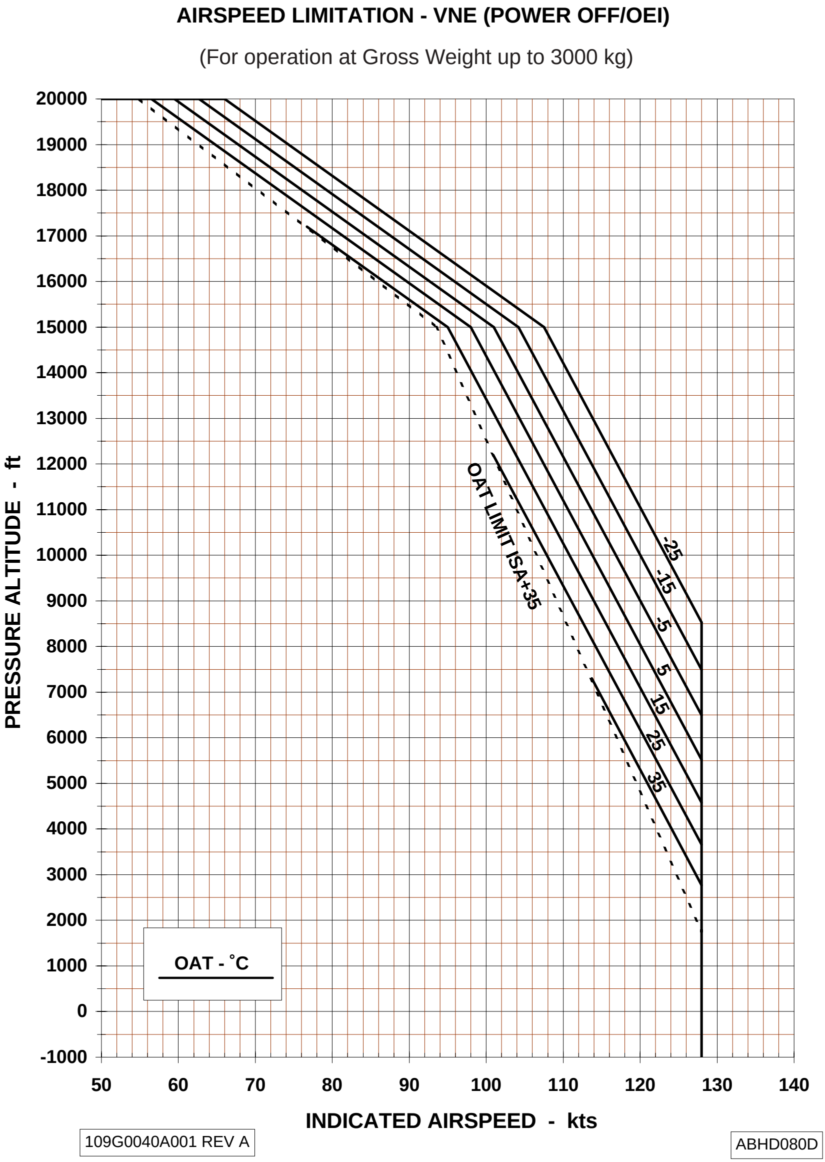


Figure 1-2. Airspeed Limitations - V_{NE} (Power off / OEI)

AGUSTA

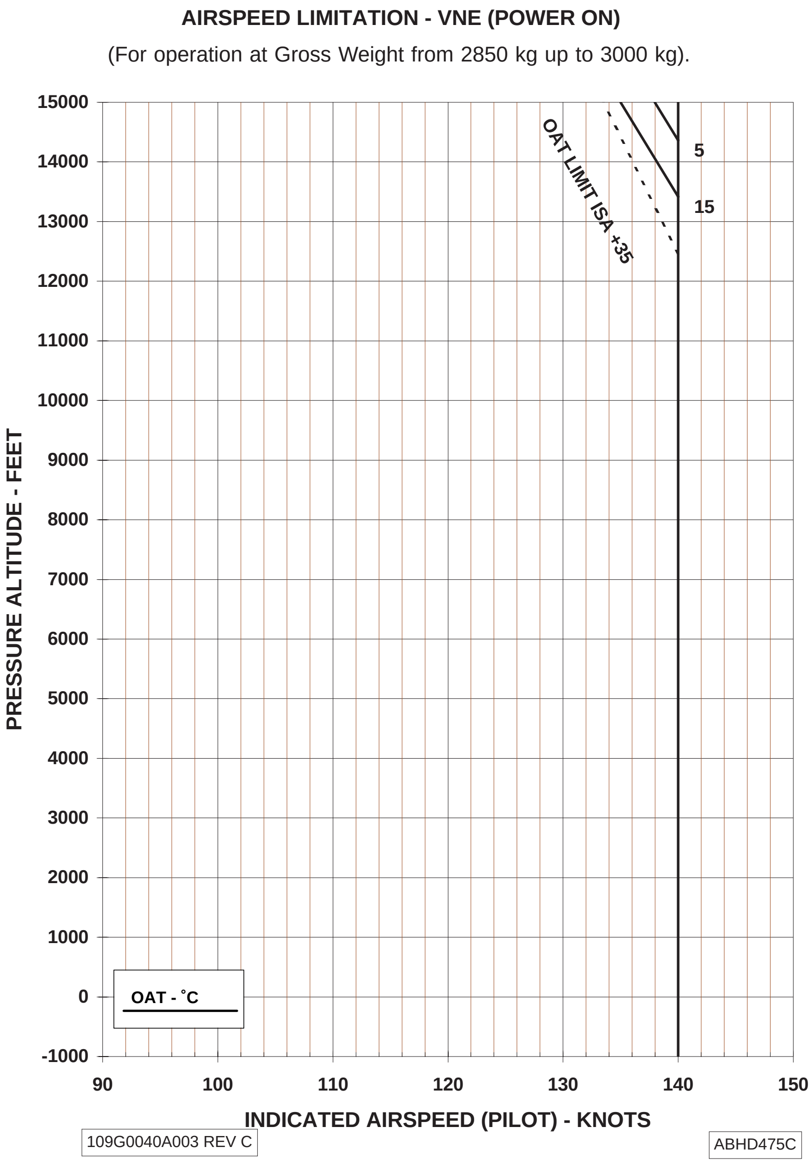


Figure 1-3. Airspeed Limitations - V_{NE}(Power on).

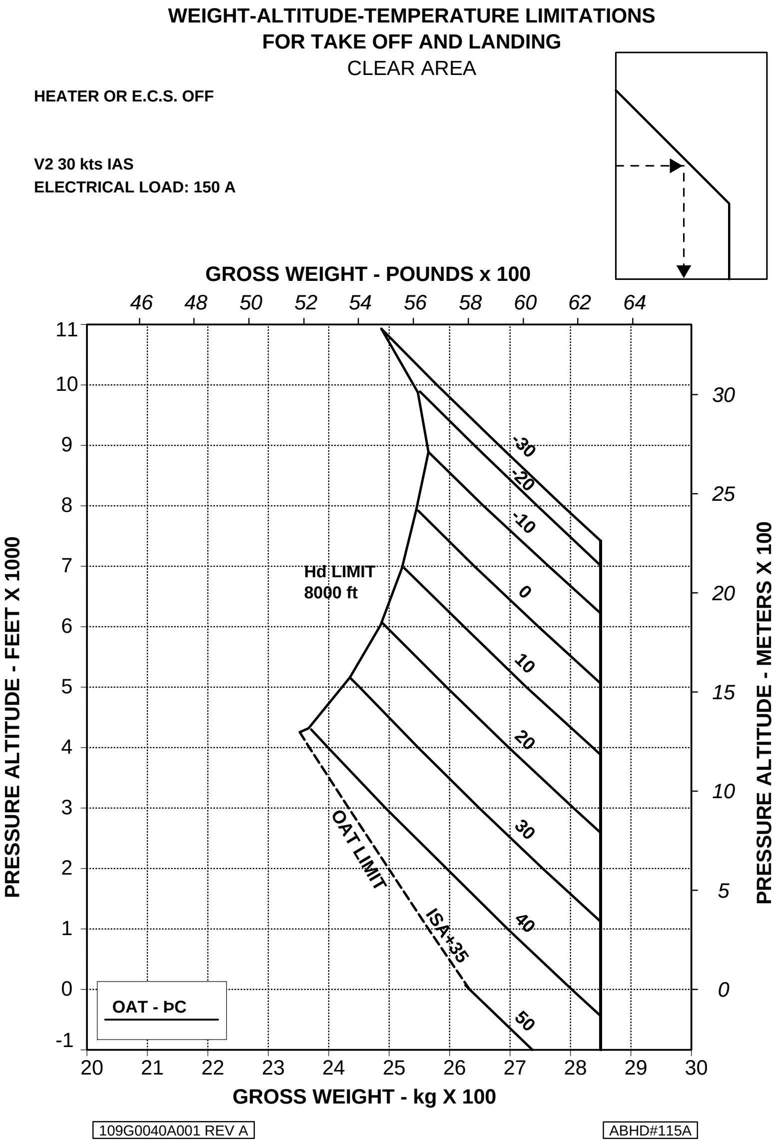


Figure 1-4. Weight-Altitude-Temperature limitations for take-off and landing (clear area).

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

SHORT FIELD (100 m)

HEATER OR E.C.S. OFF

V2 30 kts IAS

ELECTRICAL LOAD: 150 A

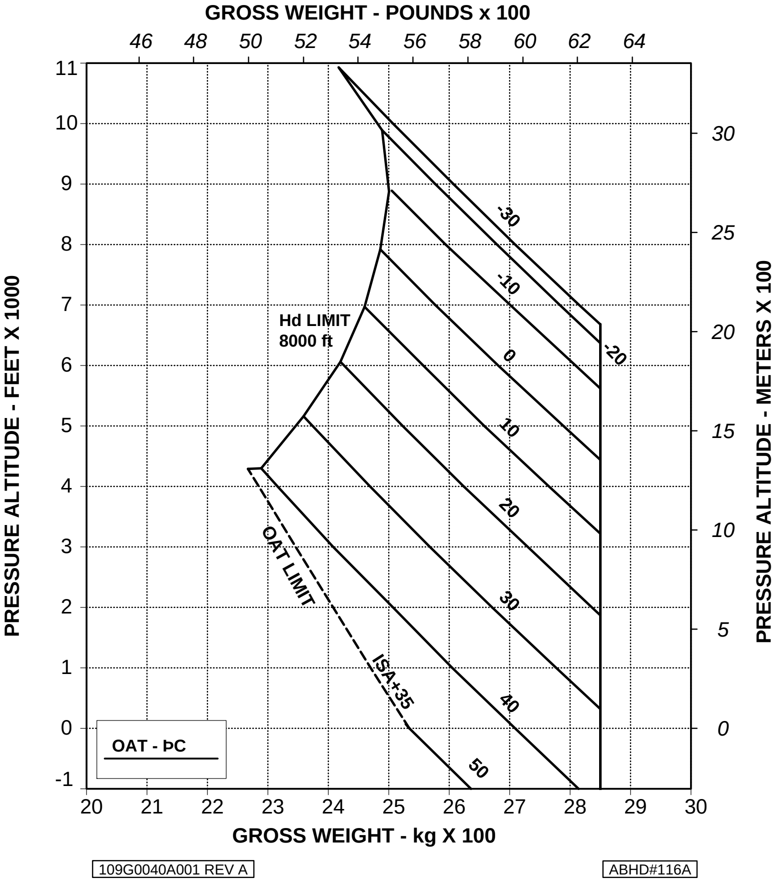


Figure 1-5. Weight-Altitude-Temperature limitations for take-off and landing (short field).

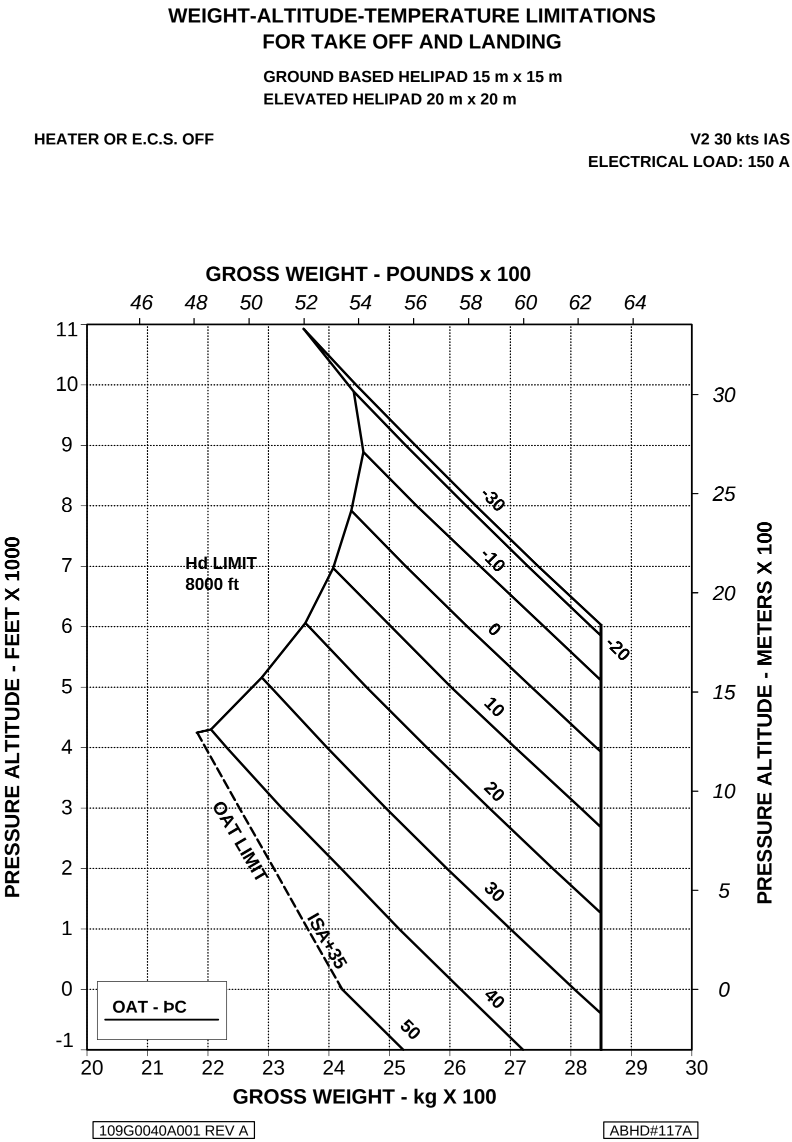


Figure 1-6. Weight-Altitude-Temperature limitations for take-off and landing (ground based helipad/elevated helipad).

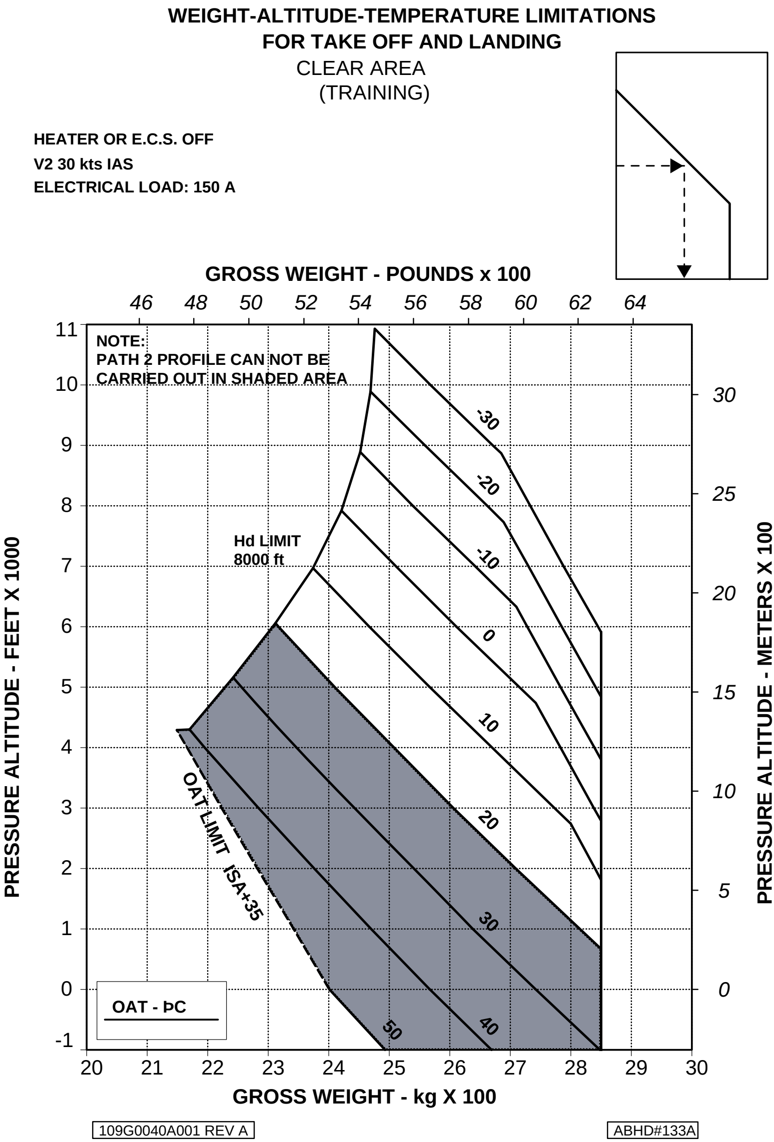


Figure 1-7. Weight-Altitude-Temperature limitations for take-off and landing (clear area) - Training.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
SHORT FIELD (100 m)
(TRAINING)

HEATER OR E.C.S. OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

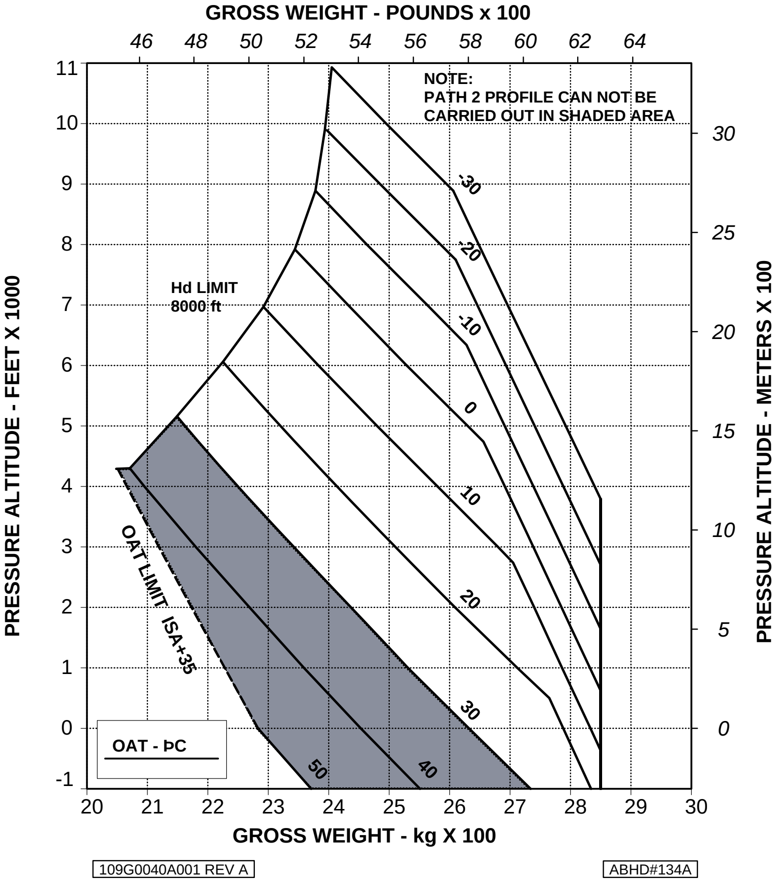


Figure 1-8. Weight-Altitude-Temperature limitations for take-off and landing (short field) - Training.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

GROUND BASED HELIPAD 15 m x 15 m

(TRAINING)

HEATER OR E.C.S. OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

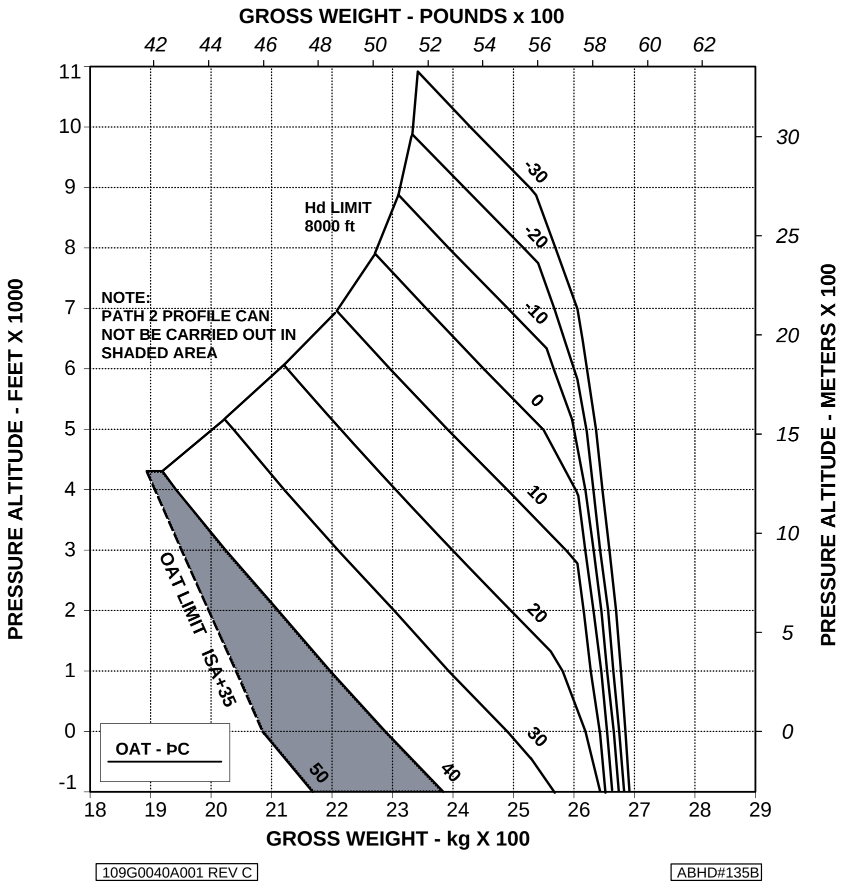


Figure 1-9. Weight-Altitude-Temperature limitations for take-off and landing (ground based helipad/elevated helipad) - Training.

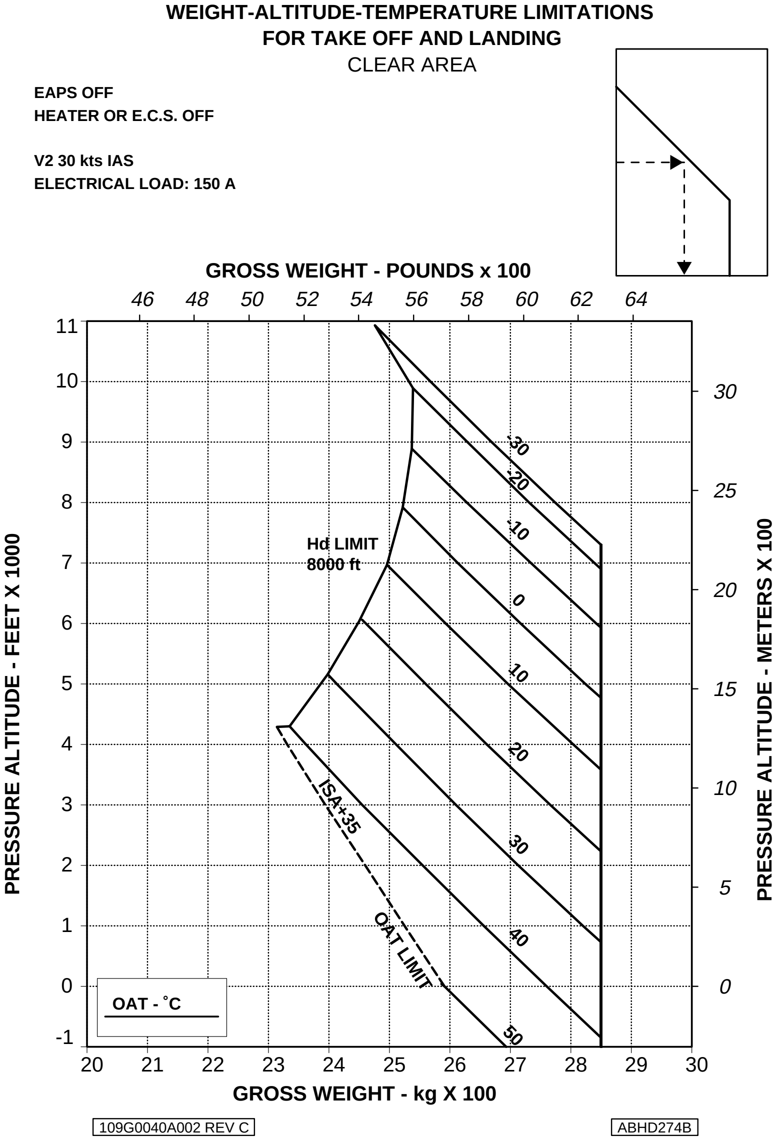


Figure 1-10. Weight-Altitude-Temperature limitations for take-off and landing (clear area) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
CLEAR AREA

EAPS ON **V2 30 kts IAS**
HEATER OR E.C.S. OFF **ELECTRICAL LOAD: 150 A**

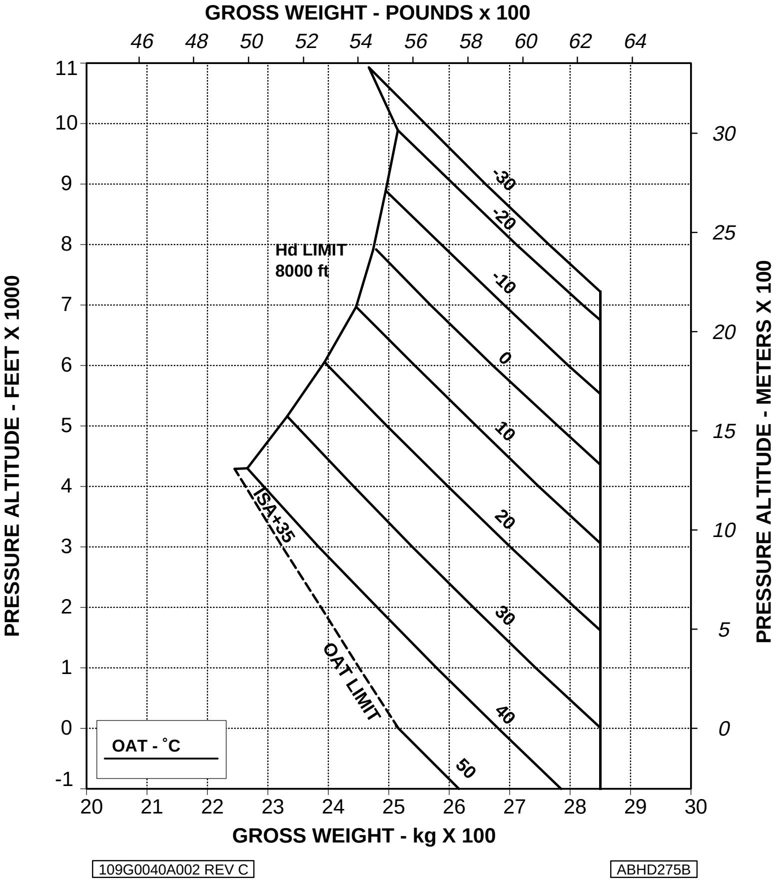


Figure 1-11. Weight-Altitude-Temperature limitations for take-off and landing (clear area) - EAPS ON.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

SHORT FIELD (100 m)

EAPS OFF
ELECTRICAL LOAD: 150 A
HEATER OR E.C.S. OFF

V2 30 kts IAS

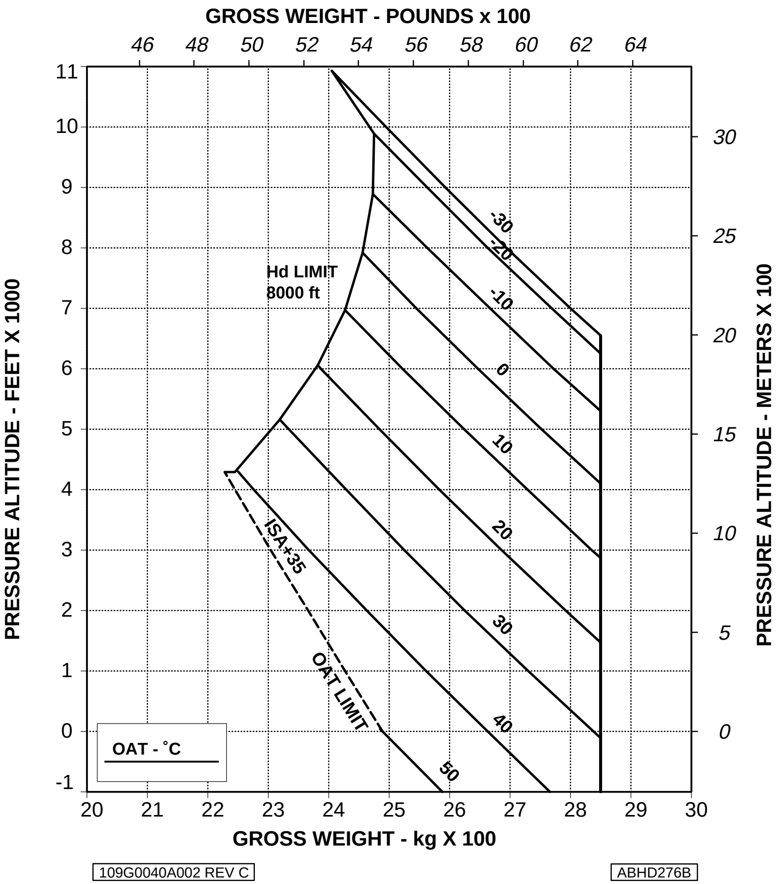


Figure 1-12. Weight-Altitude-Temperature limitations for take-off and landing (short field) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
SHORT FIELD (100 m)

EAPS ON
ELECTRICAL LOAD: 150 A
HEATER OR E.C.S. OFF

V2 30 kts IAS

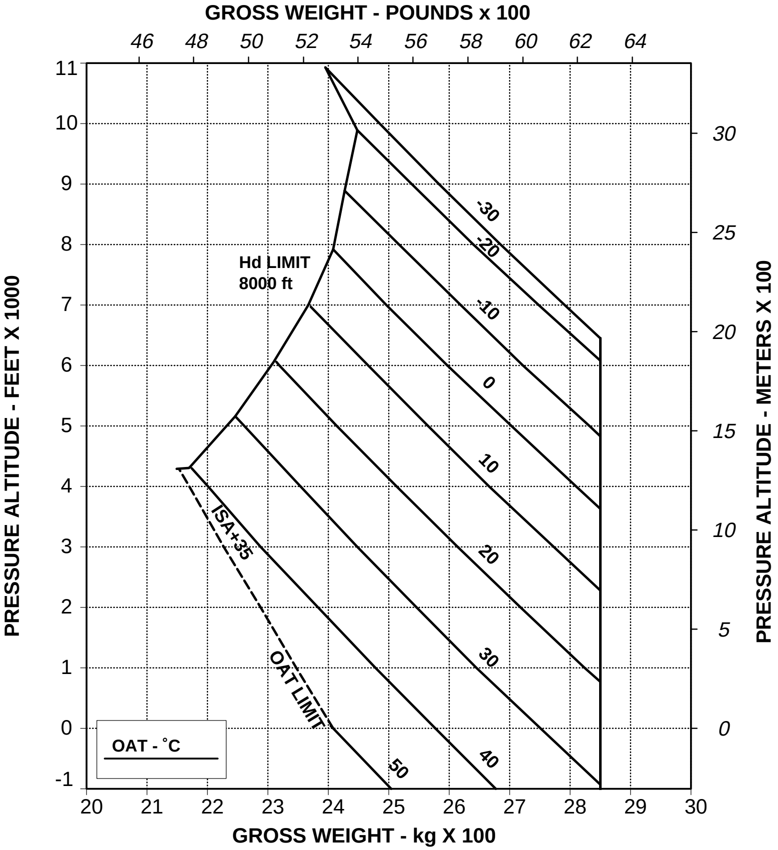


Figure 1-13. Weight-Altitude-Temperature limitations for take-off and landing (short field) - EAPS ON.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
GROUND BASED HELIPAD 15 m x 15 m
ELEVATED HELIPAD 20 m x 20 m

EAPS OFF
HEATER OR E.C.S. OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

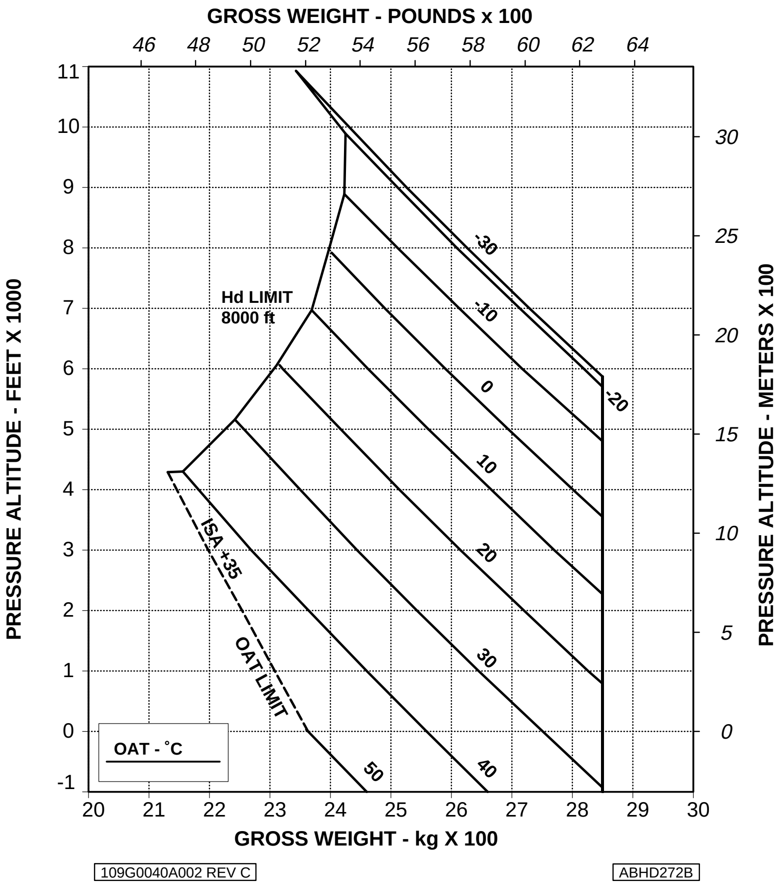


Figure 1-14. Weight-Altitude-Temperature limitations for take-off and landing (ground based helipad/elevated helipad) - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING

GROUND BASED HELIPAD 15 m x 15 m
ELEVATED HELIPAD 20 m x 20 m

EAPS ON
HEATER OR E.C.S. OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

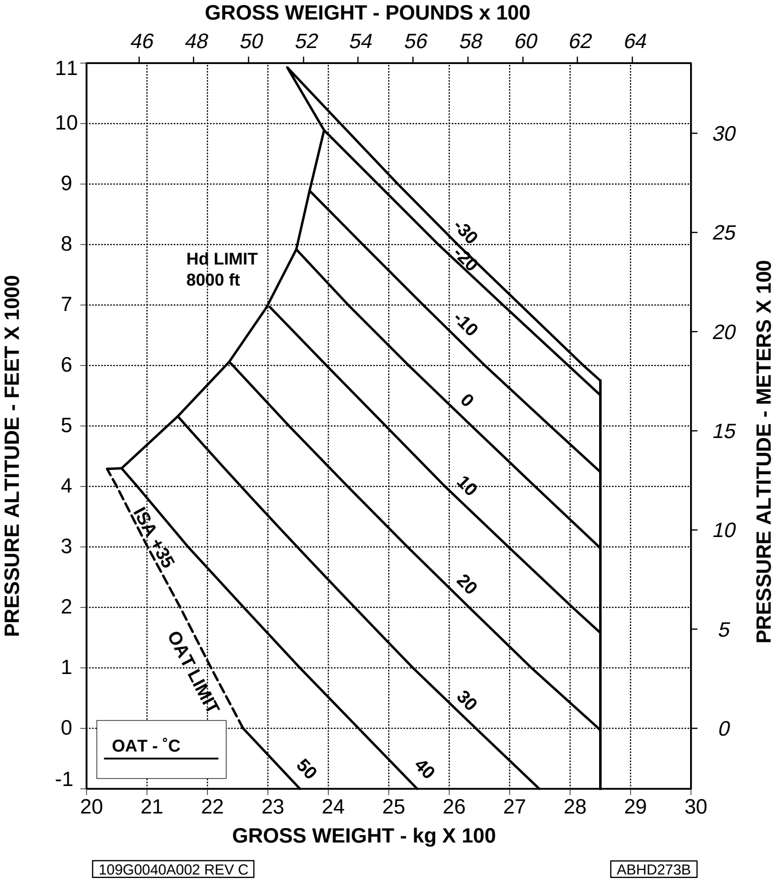


Figure 1-15. Weight-Altitude-Temperature limitations for take-off and landing (ground based helipad/elevated helipad) - EAPS ON.

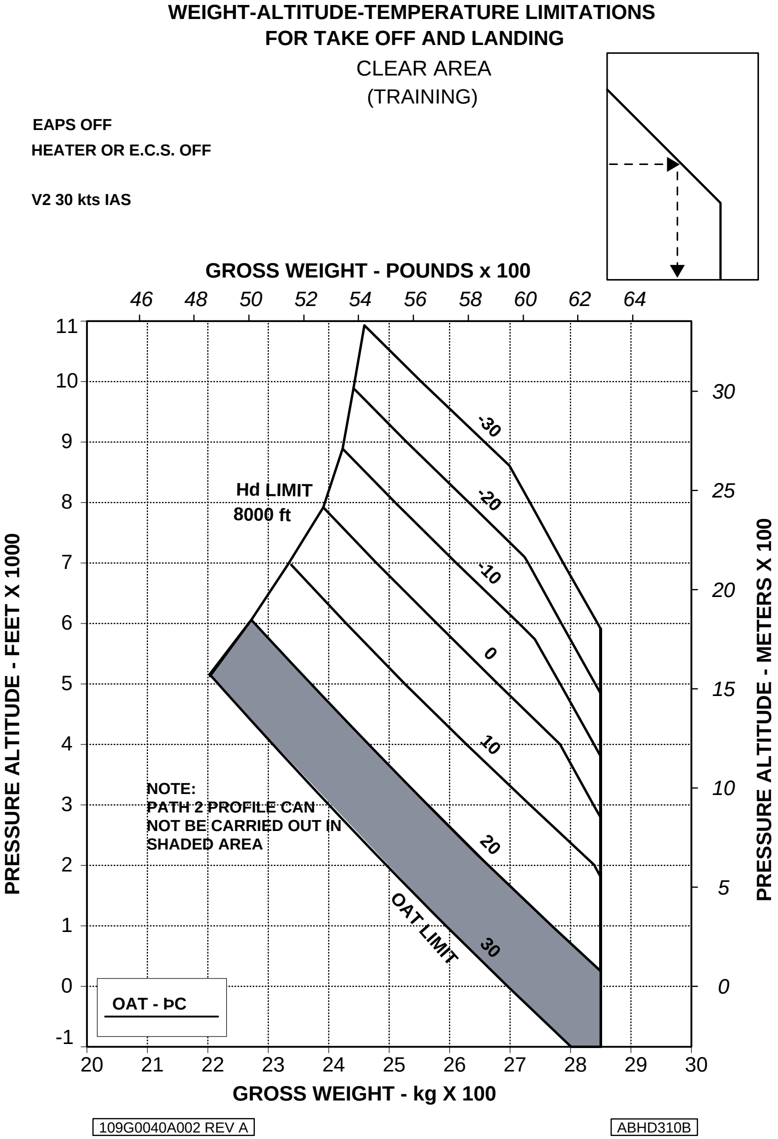


Figure 1-16. Weight-Altitude-Temperature limitations for take-off and landing (clear area) - Training - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS FOR TAKE OFF AND LANDING

SHORT FIELD (100 m)

(TRAINING)

EAPS OFF

HEATER OR E.C.S. OFF

V2 30 kts IAS

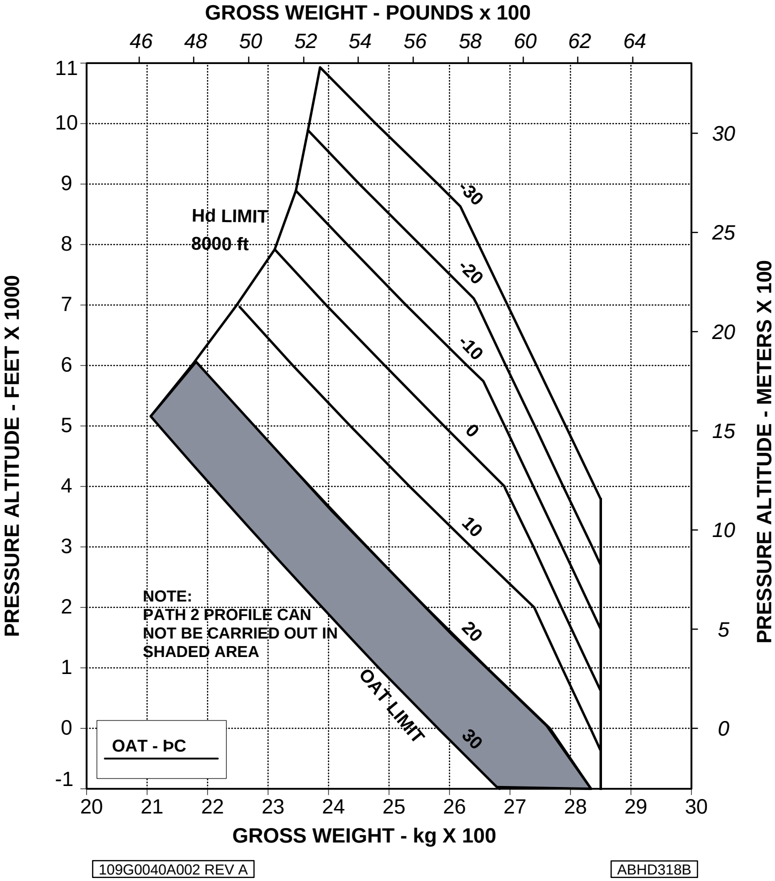


Figure 1-17. Weight-Altitude-Temperature limitations for take-off and landing (short field) - Training - EAPS OFF.

WEIGHT-ALTITUDE-TEMPERATURE LIMITATIONS
FOR TAKE OFF AND LANDING
GROUND BASED HELIPAD 15 m x 15 m
ELEVATED HELIPAD 20 m x 20 m
(TRAINING)

EAPS OFF
HEATER OR E.C.S. OFF

V2 30 kts IAS

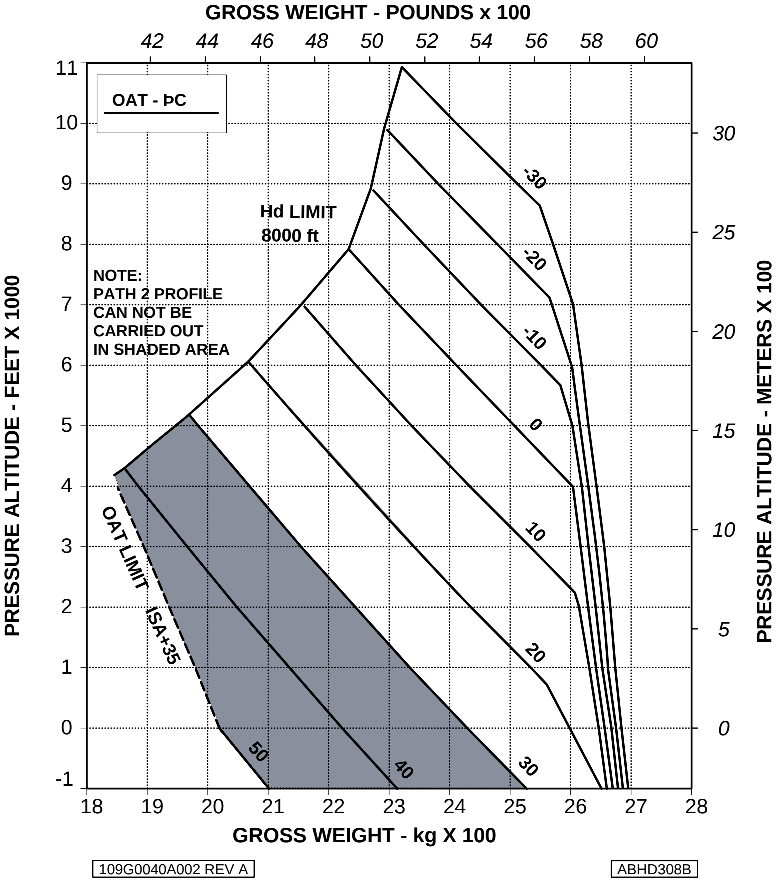
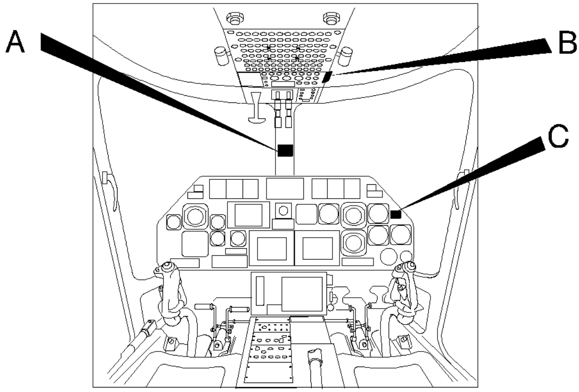


Figure 1-18. Weight-Altitude-Temperature limitations for take-off and landing (ground based helipad/elevated helipad) - Training - EAPS OFF.

AMBIENT AIR TEMPERATURE LIMITATIONS

The minimum ambient air temperature for operation is -25°C (-13°F).
The maximum sea level ambient air temperature for operation is +50°C (122°F) and decreases with pressure altitude at the standard lapse rate of 2°C (3.6°F) every 1000 ft (305 m) up to 20000 ft (6096 m).

PLACARDS



A

AIRSPEED LIMITATIONS V_{NE} - KIAS

Hp (ft) / OAT (°C)	-1000 to SL	3000	6000	9000	12000	15000	18000	20000
50	168	–	–	–	–	–	–	–
40	168	166	–	–	–	–	–	–
30	168	168	159	150	–	–	–	–
20	168	168	162	153	143	134	–	–
10	168	168	165	155	146	136	112	95
0	168	168	168	159	149	140	115	98
-10	168	168	168	162	152	143	118	101
-20	168	168	168	165	155	146	121	104
-25	168	168	168	167	157	148	123	106

FOR GROSS WEIGHT ABOVE 2850 Kg UP TO 3000 Kg:
- V_{NE} AEO: THE LESSER OF 140 KIAS AND ABOVE V_{NE};
- MAXIMUM OPERATING ALTITUDE: 15000 ft Hp.

V_{NE} OEI OR POWER OFF:
REDUCE ABOVE V_{NE} BY 40 KIAS

EMERGENCY FLOATS INFLATED: V_{NE} 80 KIAS;
ARMING ABOVE 80 KIAS OR 3000 ft AGL IS PROHIBITED.

V_{LO} AND V_{LE} : 120 KIAS
MAX FWD TOUCHDOWN SPEED: 40 Kts

ABHD511B

Figure 1-19. Airspeed (Vne) placard

SECTION 2 - NORMAL PROCEDURES

No change.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

This Section includes the performance data affected by the extension to +50°C of the ambient air temperature limitations and related to the following configurations:

— Helicopter in basic configuration
— Helicopter with E.C.S.
— Helicopter with Bleed Air Heater
— Helicopter with Particle Separator
— Helicopter with Cargo Hook

The performance data also include:

— Equivalent Category ″A″ operation
— Equivalent Category ″A″ operation - Training Procedures.
— Helicopter with Increased Internal Gross Weight.

Refer to the basic Rotorcraft Flight manual for all other performance data.

HELICOPTER IN BASIC CONFIGURATION

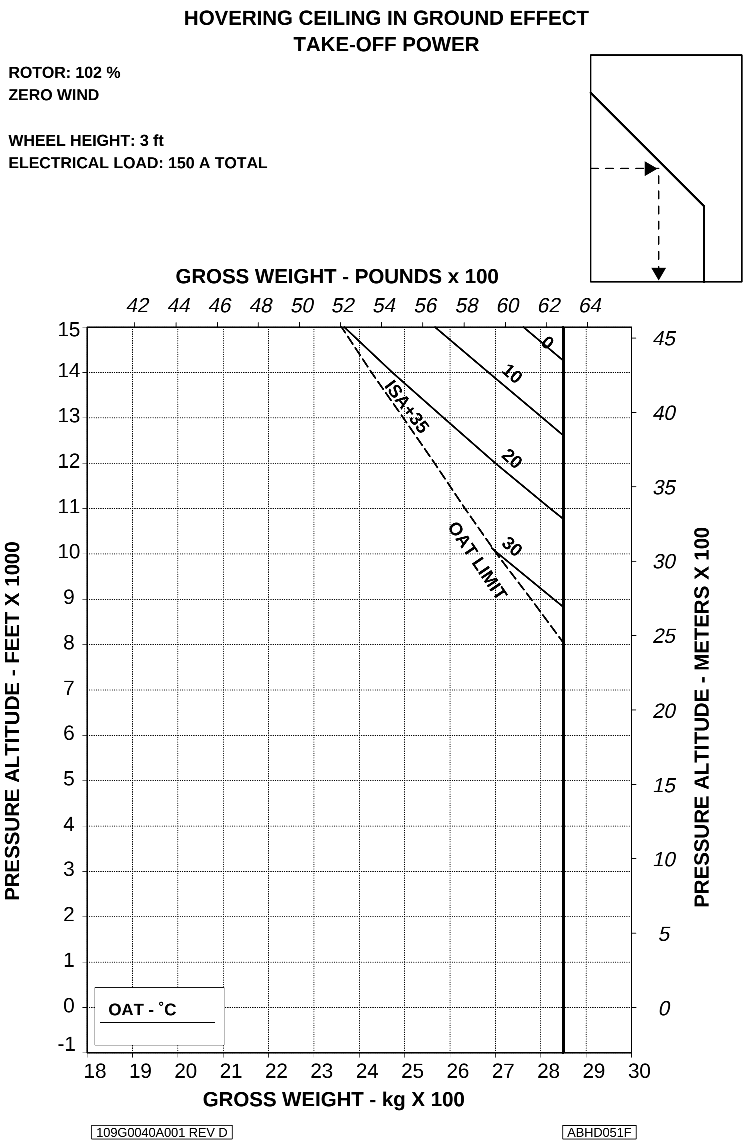


Figure 4-1. Hovering ceiling - IGE - Take-off power.

AGUSTA

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

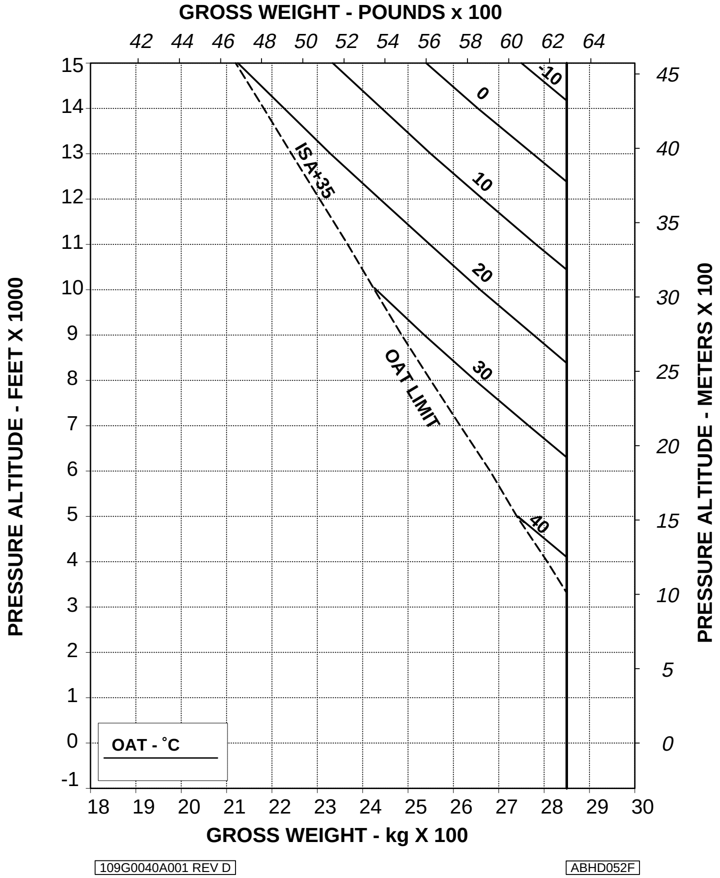


Figure 4-2. Hovering ceiling - IGE - Maximum continuous power.

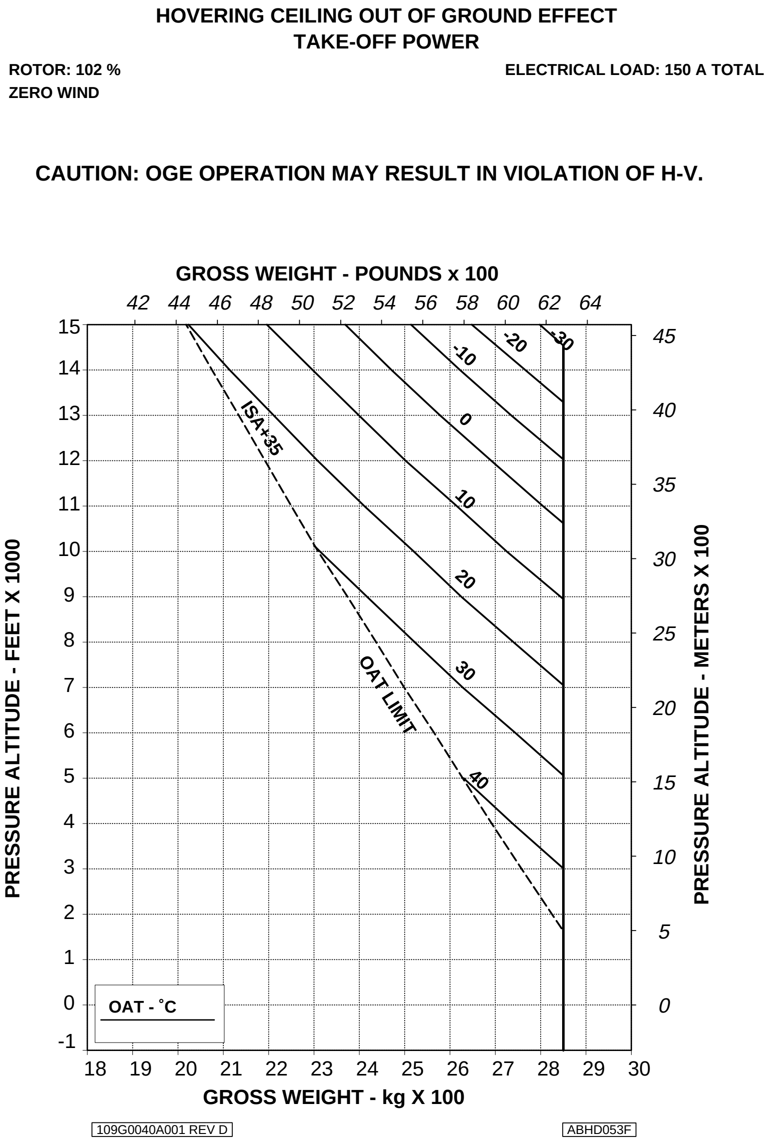


Figure 4-3. Hovering ceiling - OGE - Take-off power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 % ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

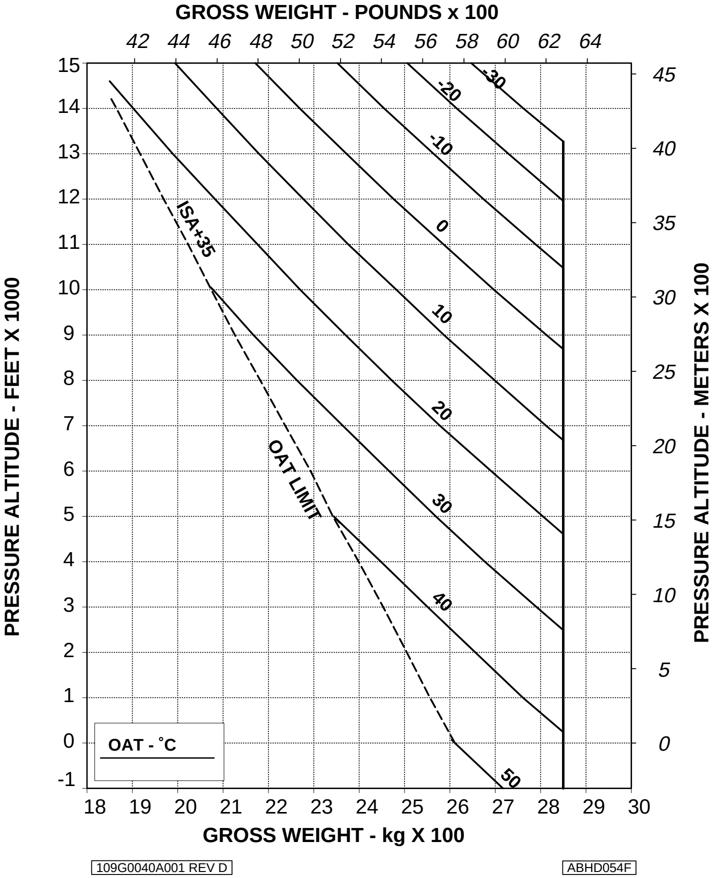


Figure 4-4. Hovering ceiling - OGE - Maximum continuous power.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM. FOR HEAVIER WEIGHTS, REFER TO CHART "B".

ELECTRICAL LOAD: 150 A

CHART A

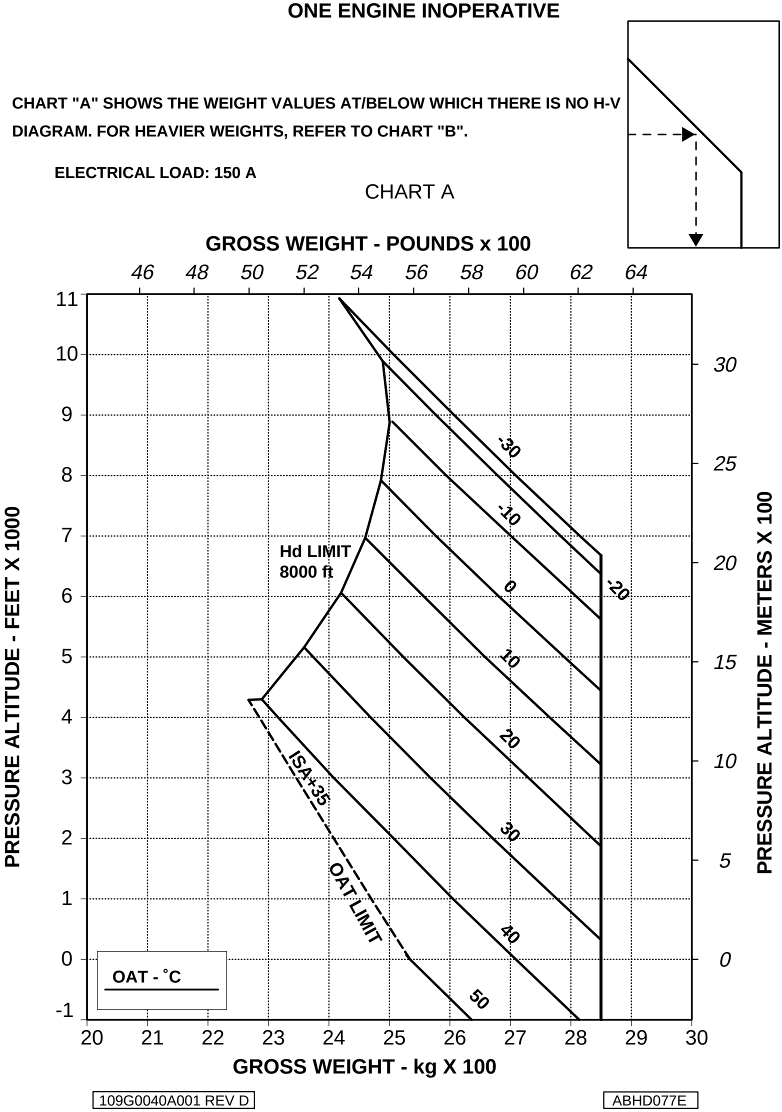


Figure 4-5. Height-Velocity diagram - OEI - Chart A.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

CHART B

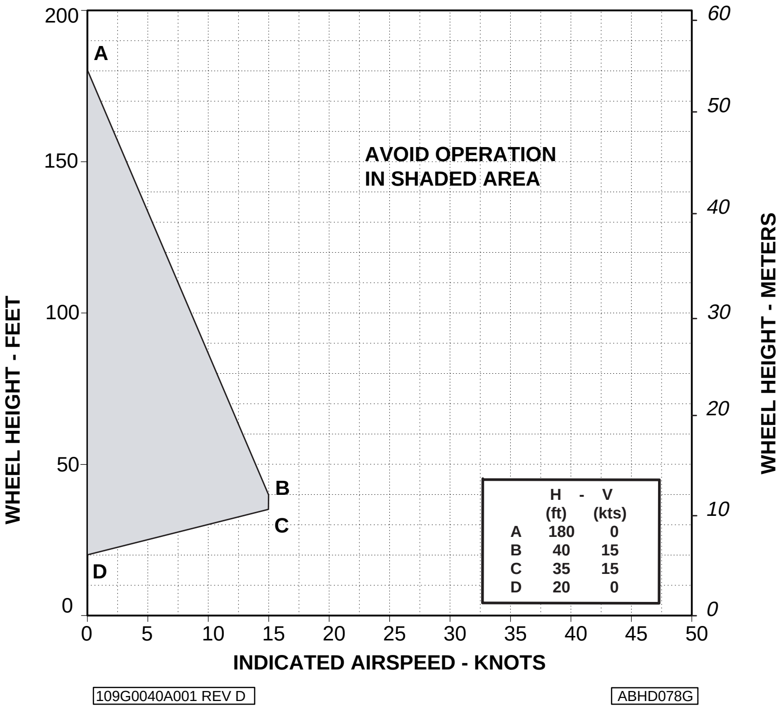


Figure 4-6. Height-Velocity diagram - OEI - Chart B.

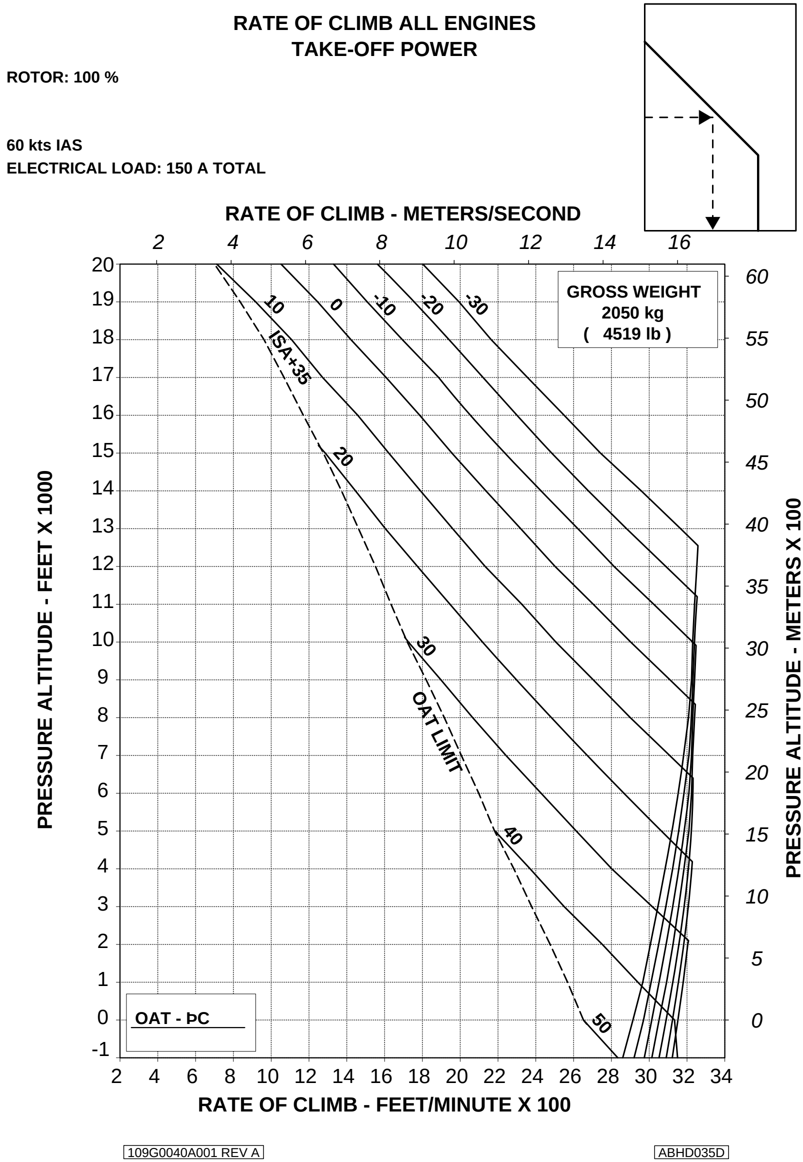


Figure 4-7. Rate of climb - All engines - Take-off power - 2050 kg.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

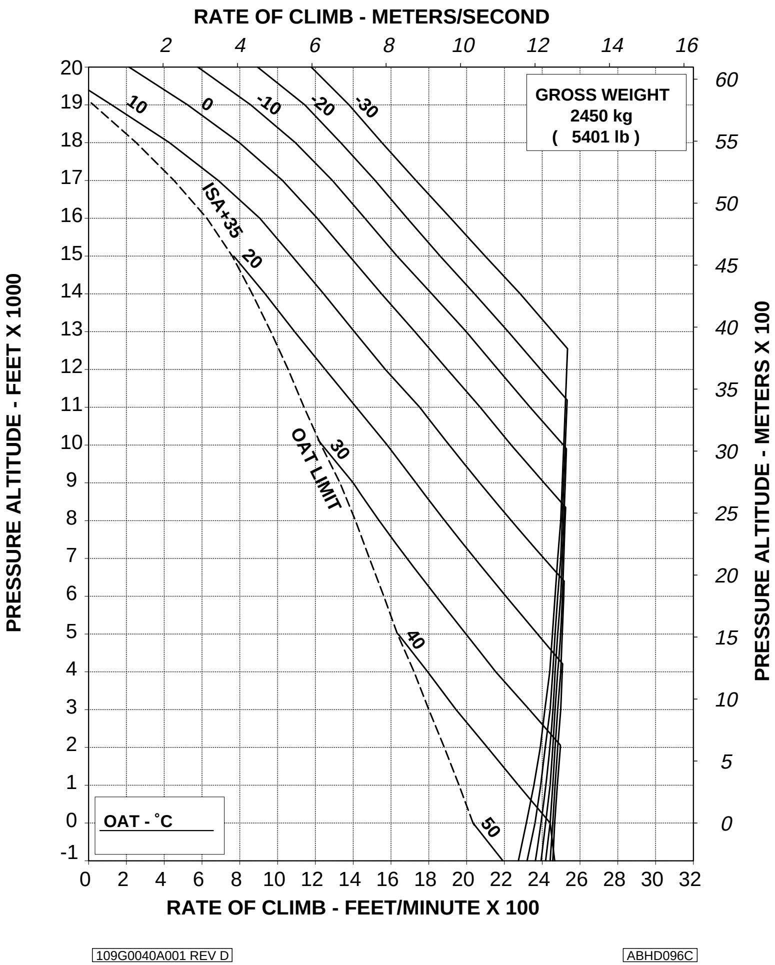


Figure 4-8. Rate of climb - All engines - Take-off power - 2450 kg.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

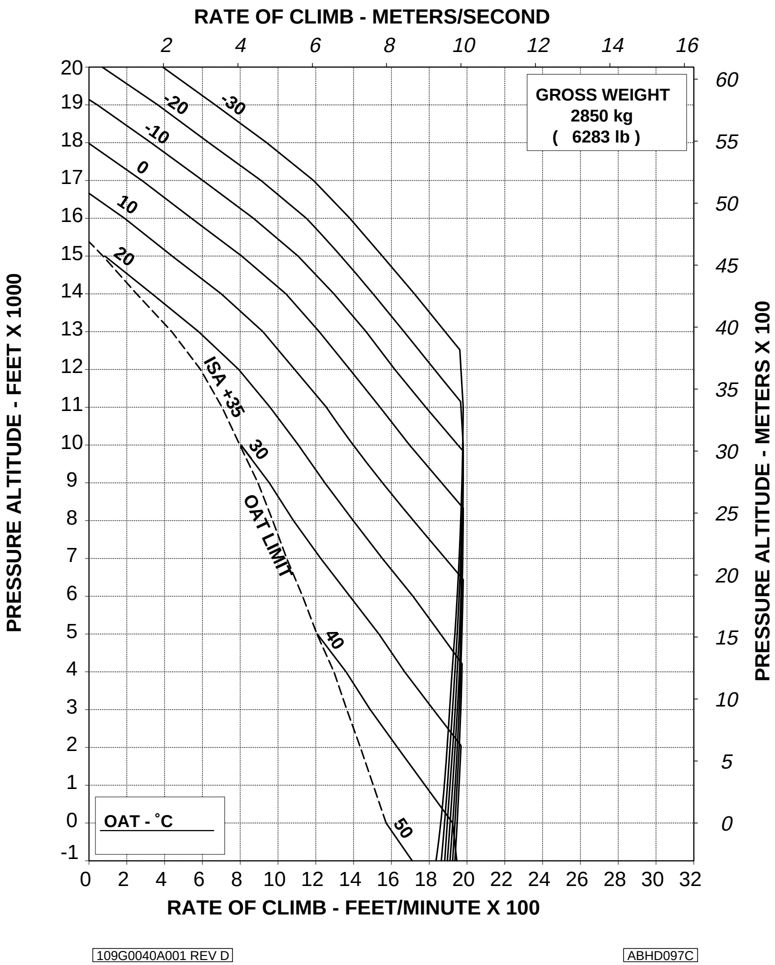


Figure 4-9. Rate of climb - All engines - Take-off power - 2850 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

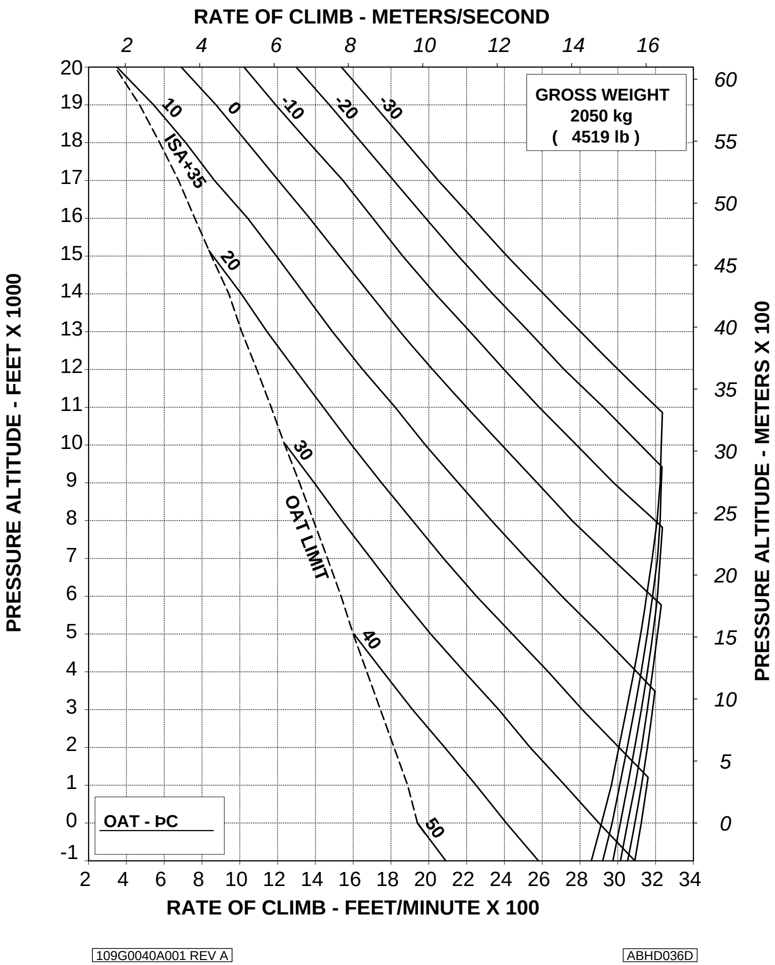


Figure 4-10. Rate of climb - All engines - Maximum continuous power - 2050 kg.

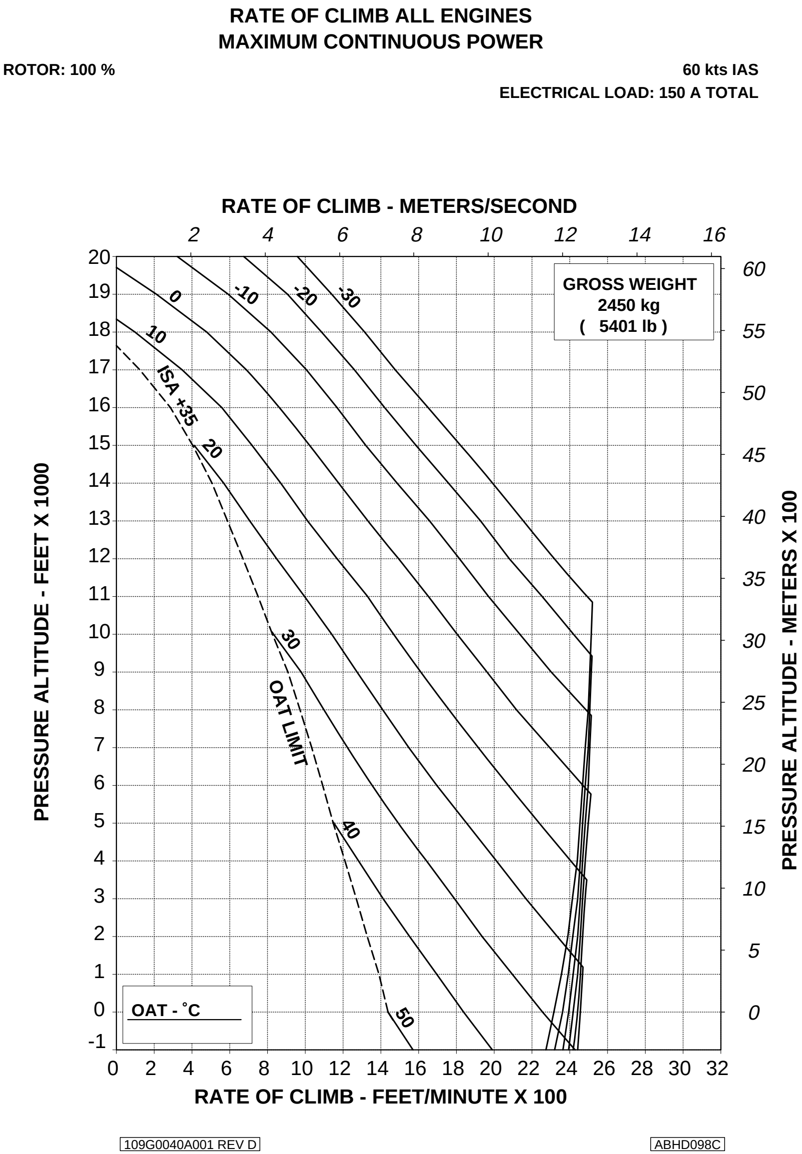


Figure 4-11. Rate of climb - All engines - Maximum continuous power - 2450 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

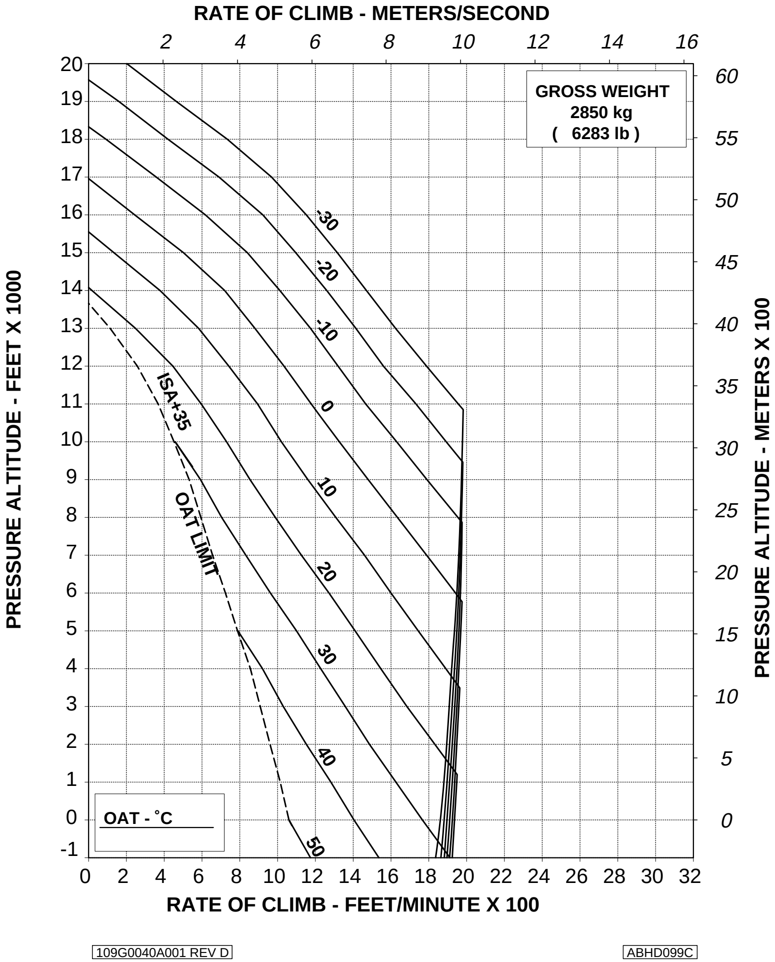


Figure 4-12. Rate of climb - All engines - Maximum continuous power - 2850 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 %

60 kts IAS

ELECTRICAL LOAD: 150 A

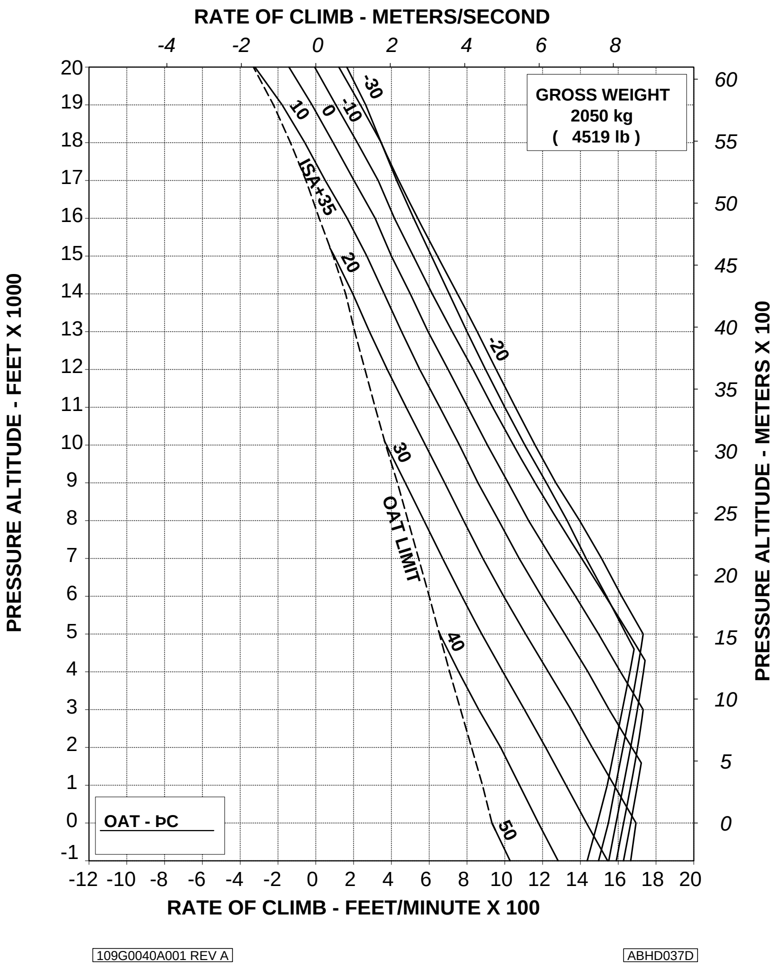


Figure 4-13. Rate of climb - OEI - 2,5 minute power - 2050 kg.

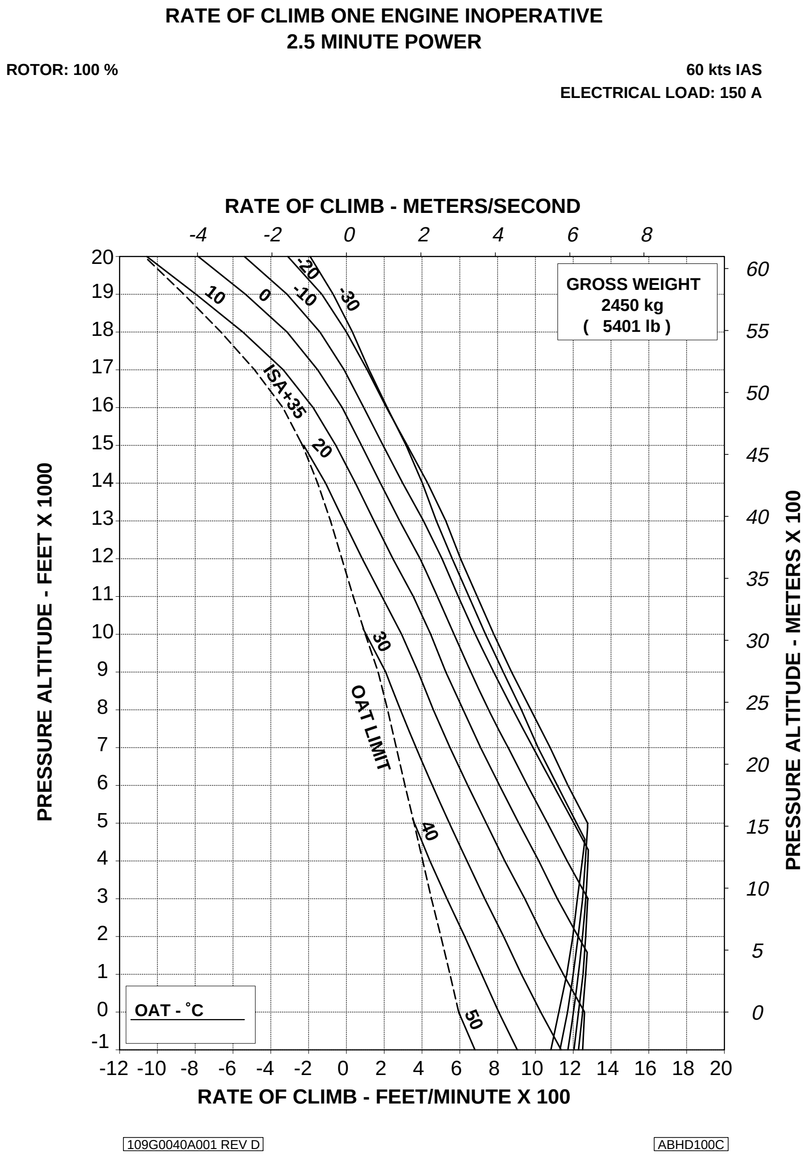


Figure 4-14. Rate of climb - OEI - 2,5 minute power - 2450 kg.

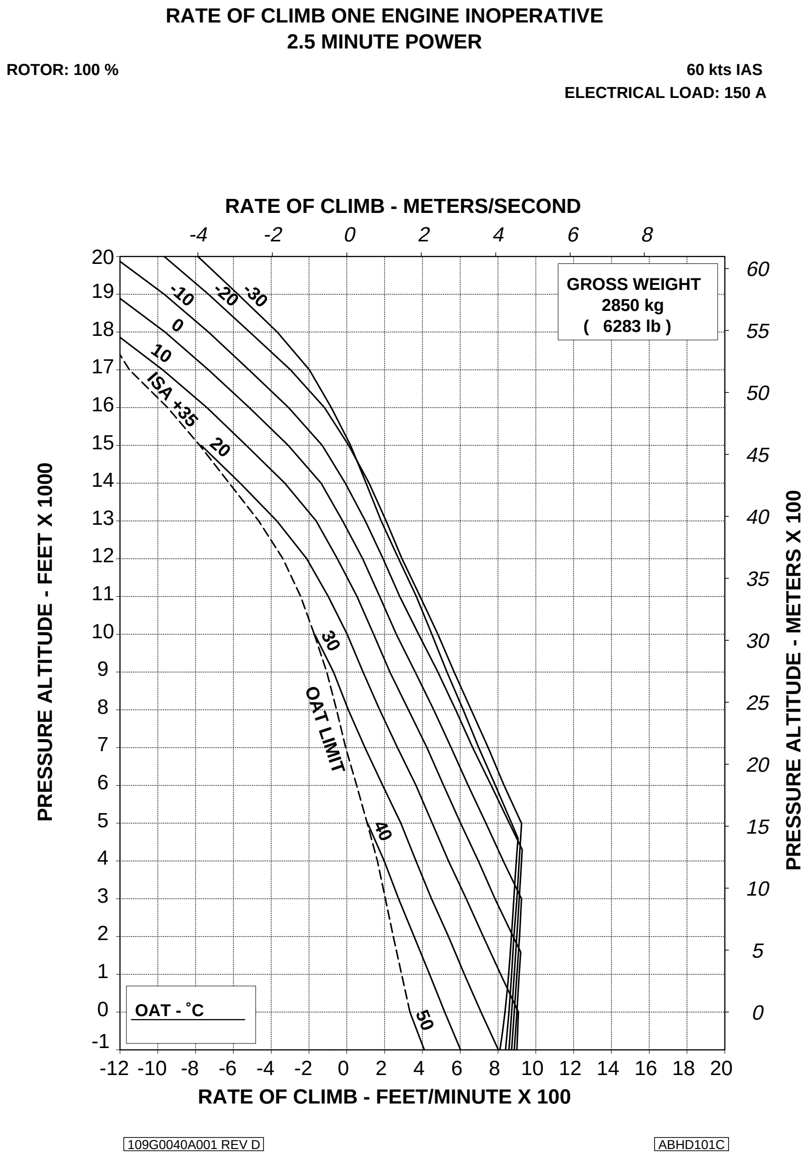


Figure 4-15. Rate of climb - OEI - 2,5 minute power - 2850 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS

ELECTRICAL LOAD: 150 A

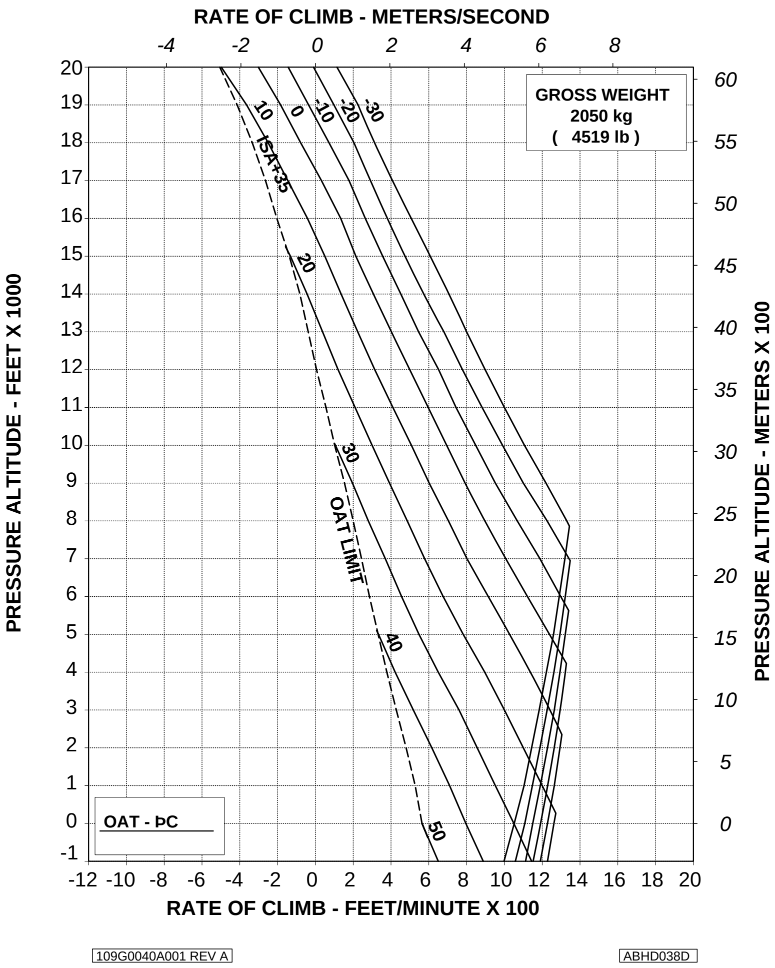


Figure 4-16. Rate of climb - OEI - Maximum continuous power - 2050 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A

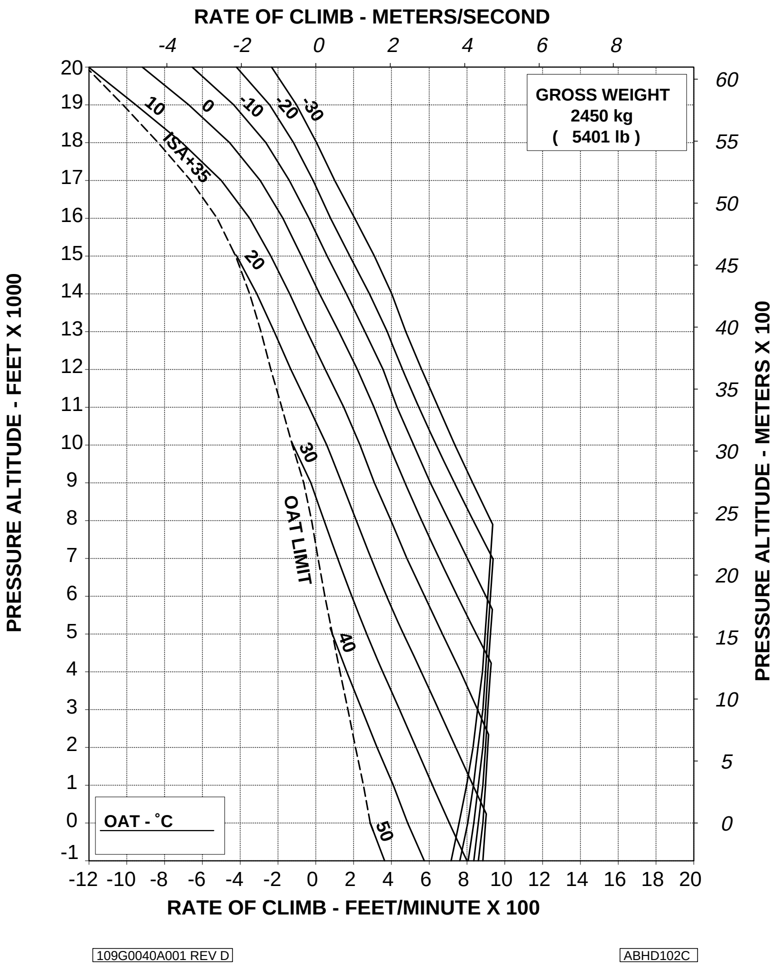


Figure 4-17. Rate of climb - OEI - Maximum continuous power - 2450 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A

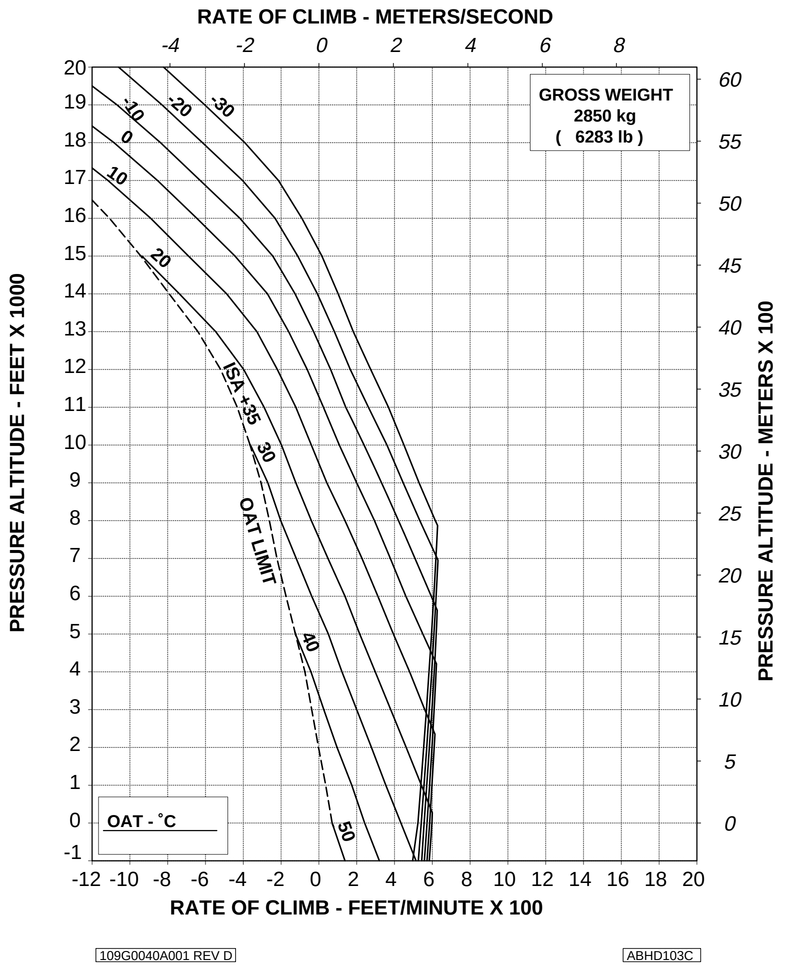


Figure 4-18. Rate of climb - OEI - Maximum continuous power - 2850 kg.

HELICOPTER WITH ENVIRONMENTAL CONTROL SYSTEM (ECS)

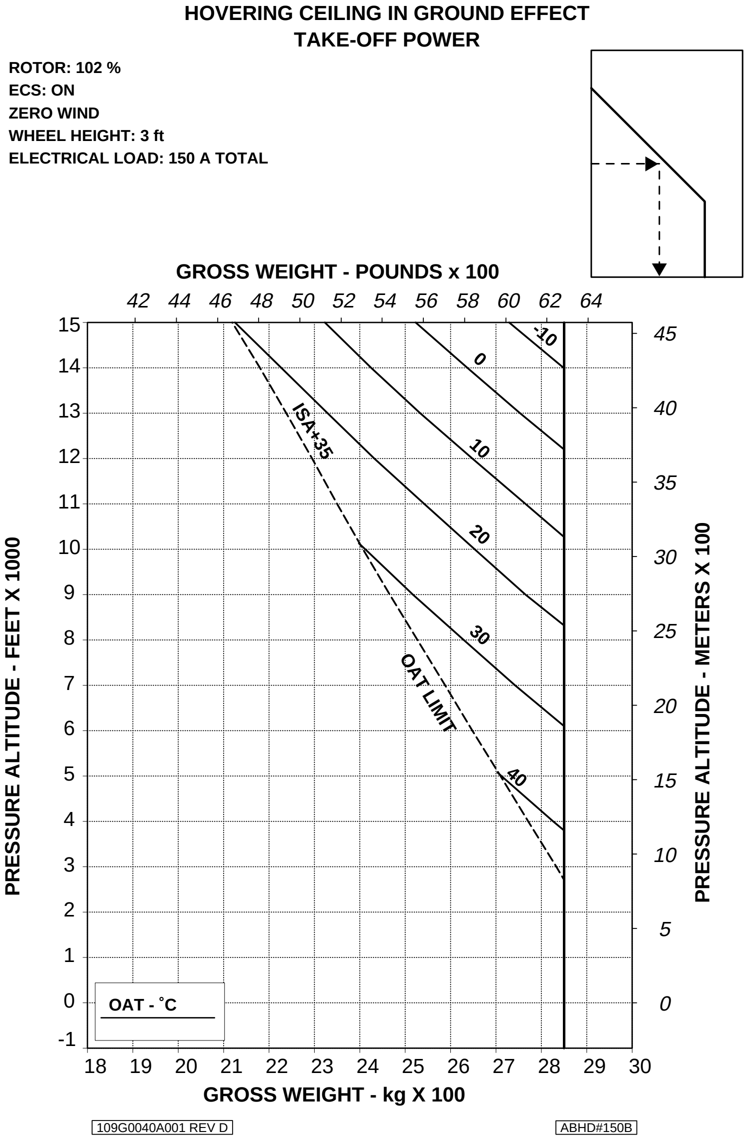


Figure 4-19. Hovering ceiling - IGE - Take-off power - ECS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

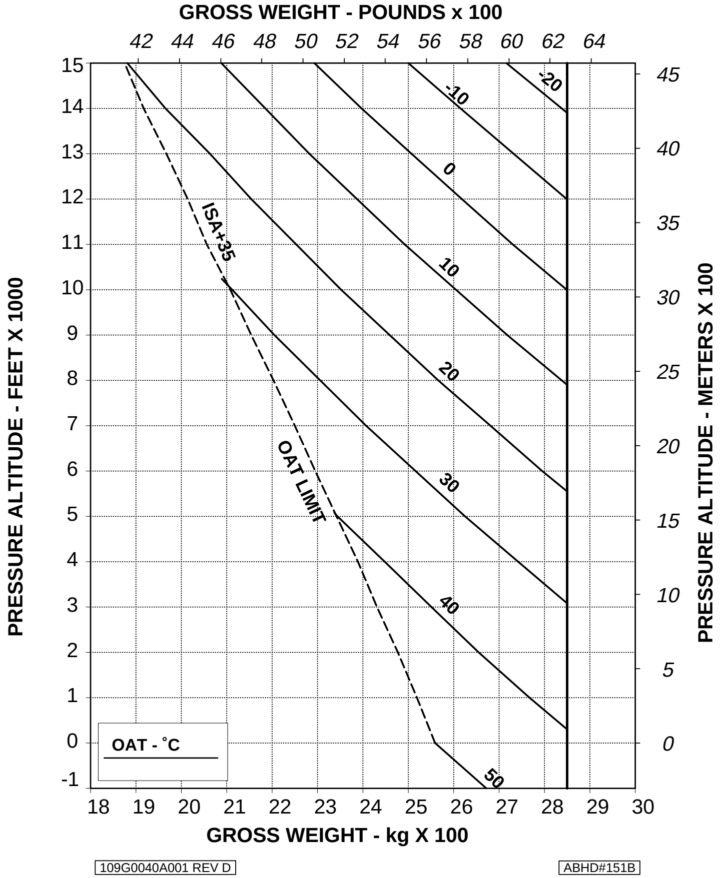


Figure 4-20. Hovering ceiling - IGE - Maximum continuous power - ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

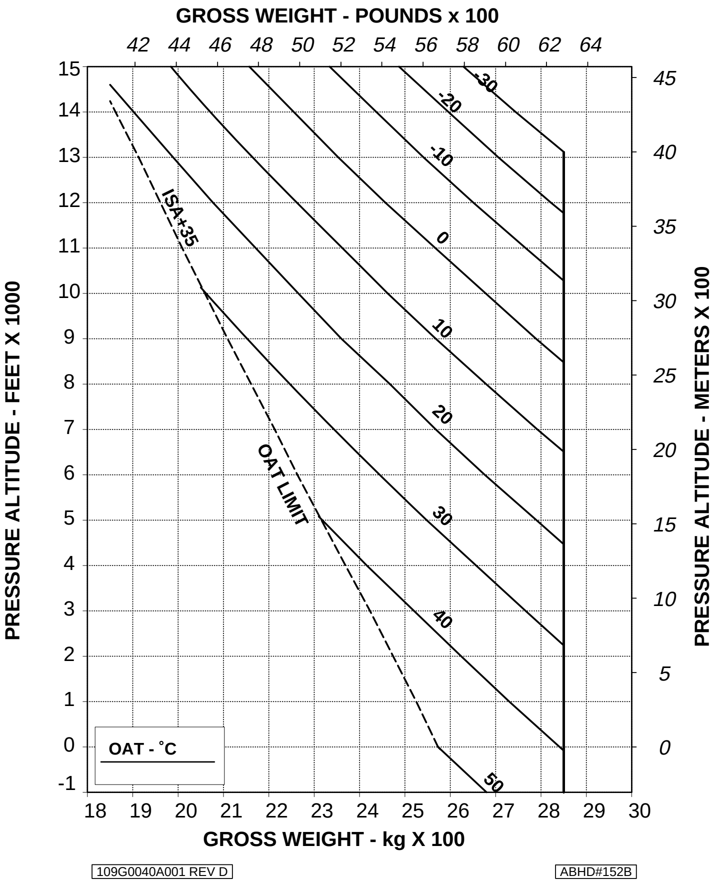


Figure 4-21. Hovering ceiling - OGE - Take-off power - ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

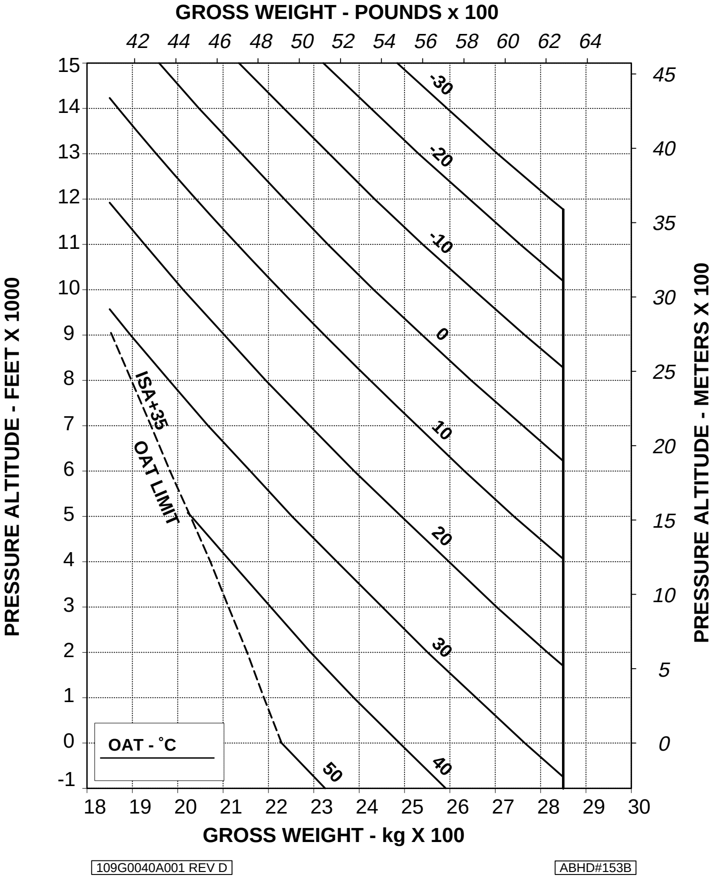


Figure 4-22. Hovering ceiling - OGE - Maximum continuous power - ECS ON.

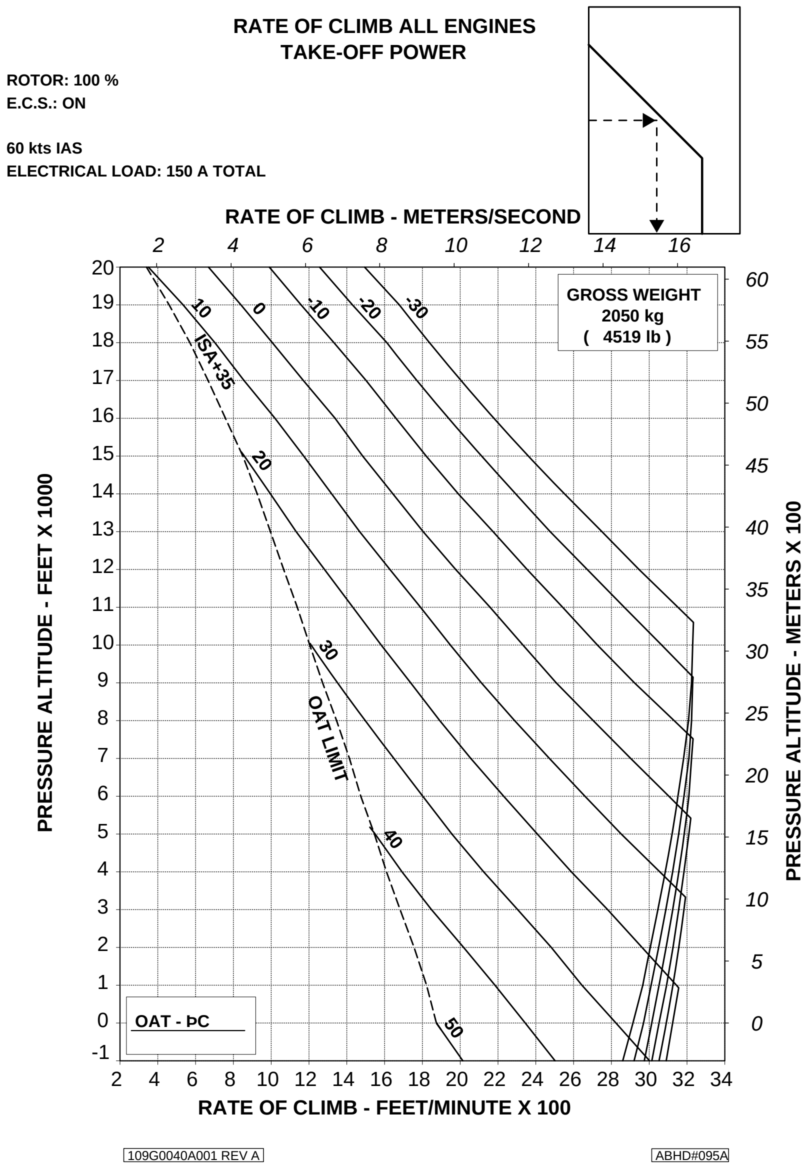


Figure 4-23. Rate of climb - All engines - Take-off power - ECS ON - 2050 kg.

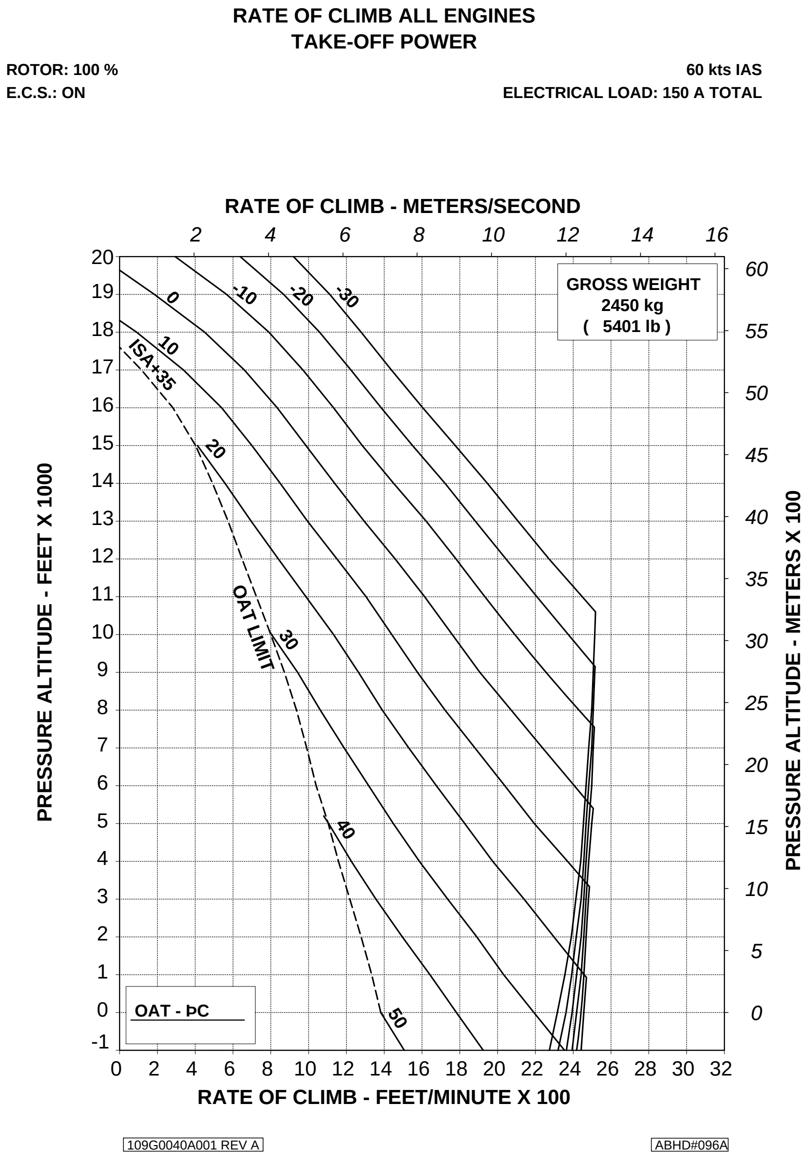


Figure 4-24. Rate of climb - All engines - Take-off power - ECS ON - 2450 kg.

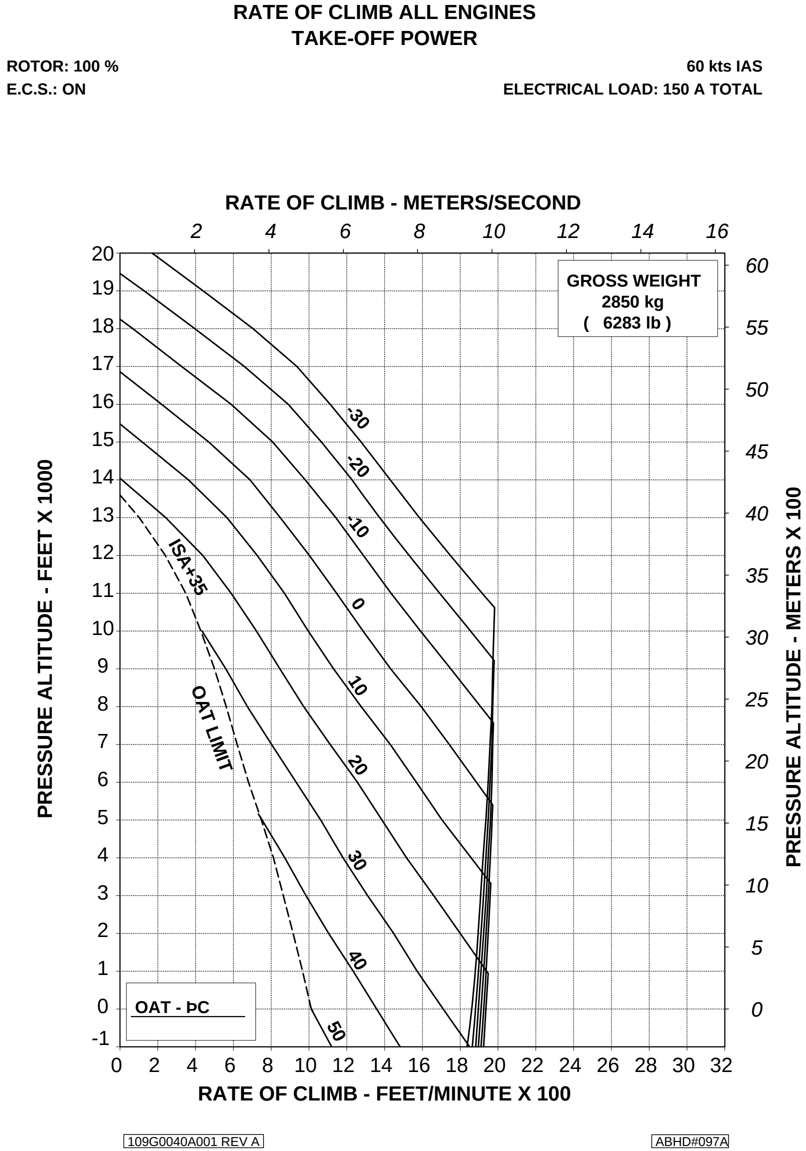


Figure 4-25. Rate of climb - All engines - Take-off power - ECS ON - 2850 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
E.C.S.: ON | ELECTRICAL LOAD: 150 A TOTAL

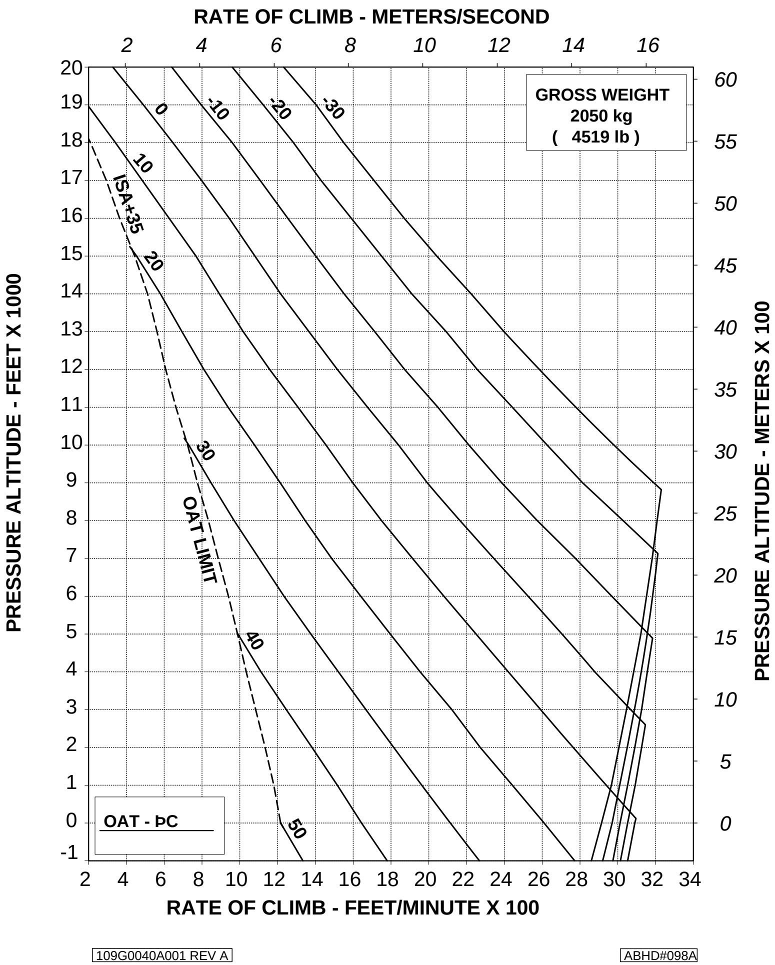


Figure 4-26. Rate of climb - All engines - Maximum continuous power - ECS ON - 2050 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % 60 kts IAS
E.C.S.: ON ELECTRICAL LOAD: 150 A TOTAL

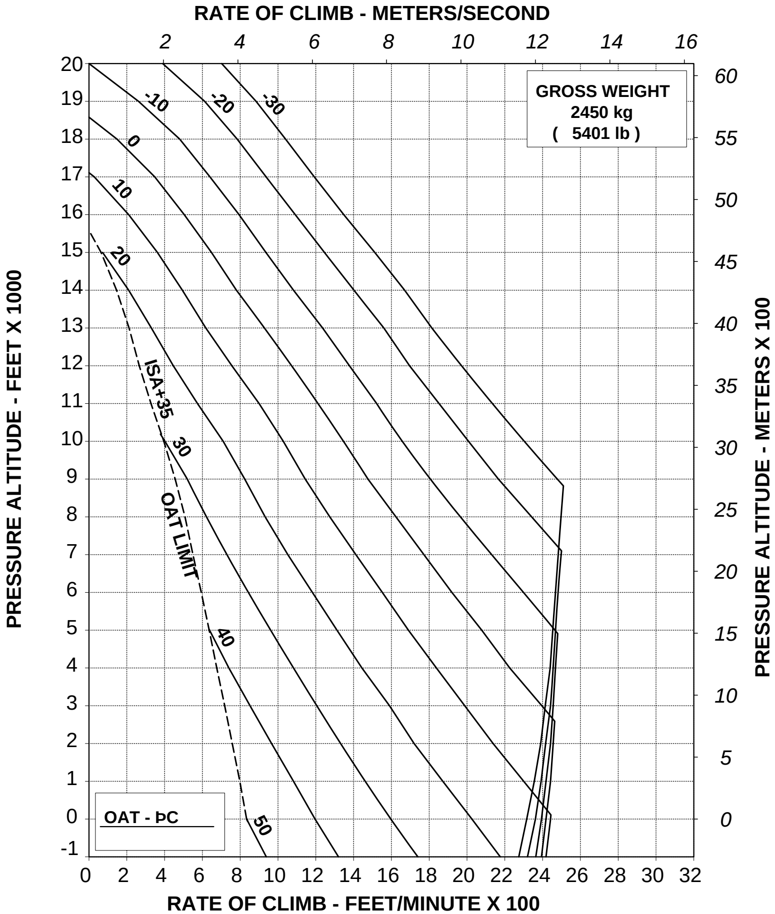


Figure 4-27. Rate of climb - All engines - Maximum continuous power - ECS ON - 2450 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % 60 kts IAS
E.C.S.: ON ELECTRICAL LOAD: 150 A TOTAL

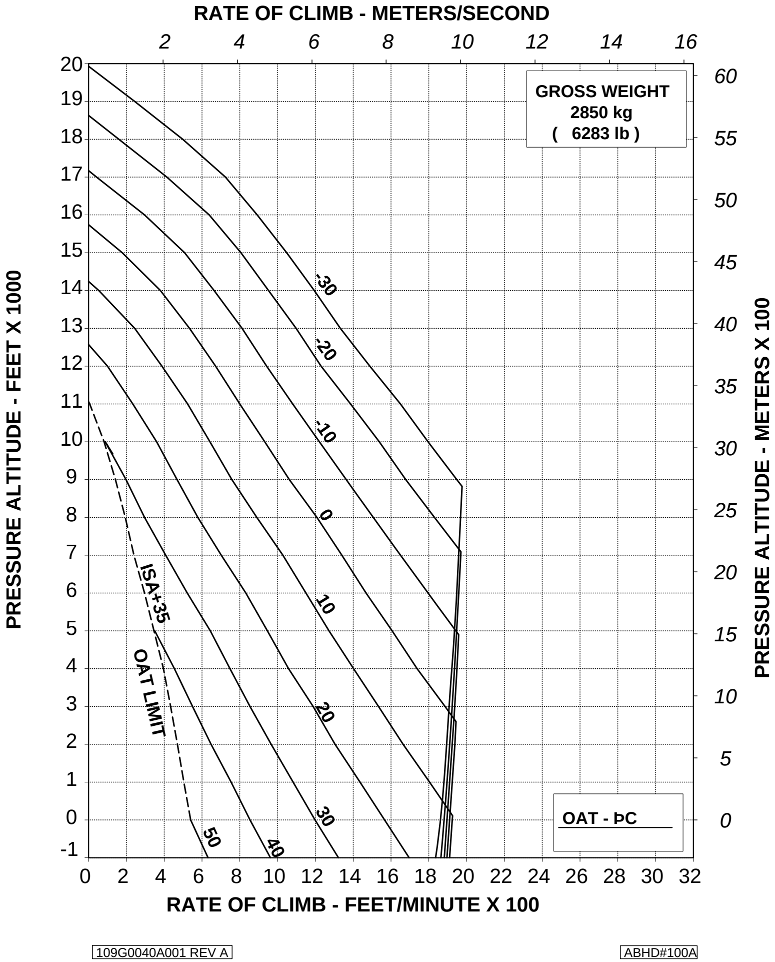


Figure 4-28. Rate of climb - All engines - Maximum continuous power - ECS ON - 2850 kg.

HELICOPTER WITH BLEED-AIR HEATER

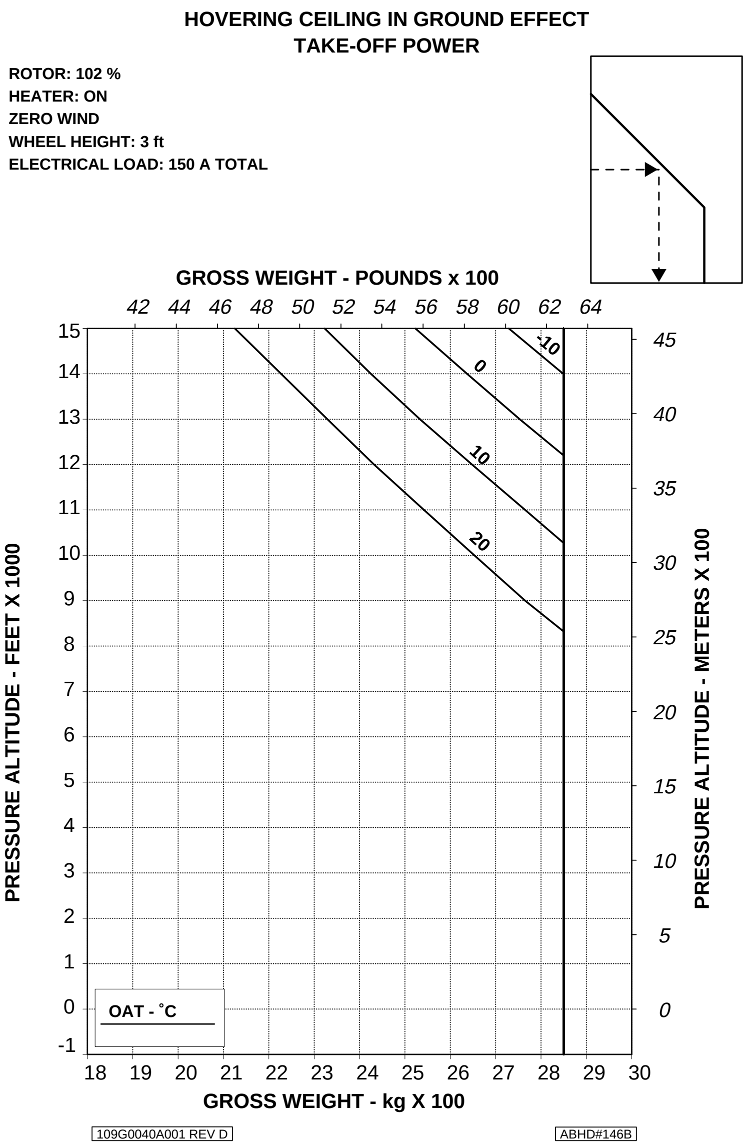


Figure 4-29. Hovering ceiling - IGE - Take-off power - Heater ON.

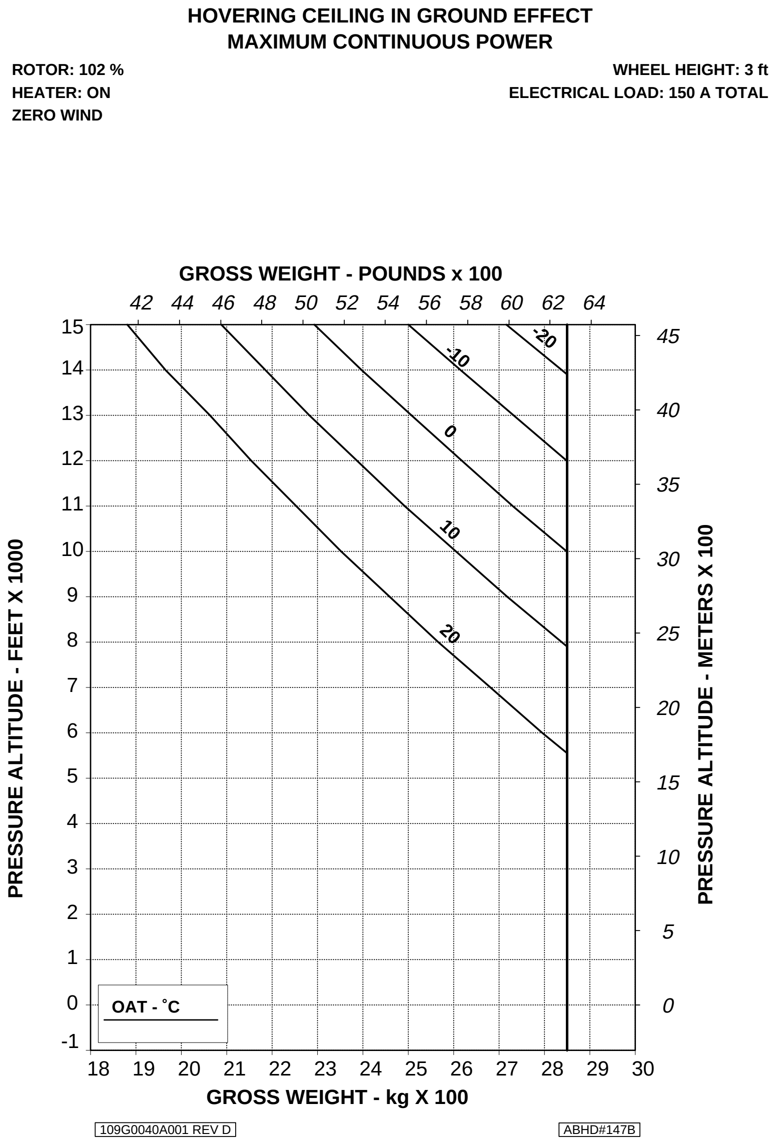


Figure 4-30. Hovering ceiling - IGE - Maximum continuous power - Heater ON .

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

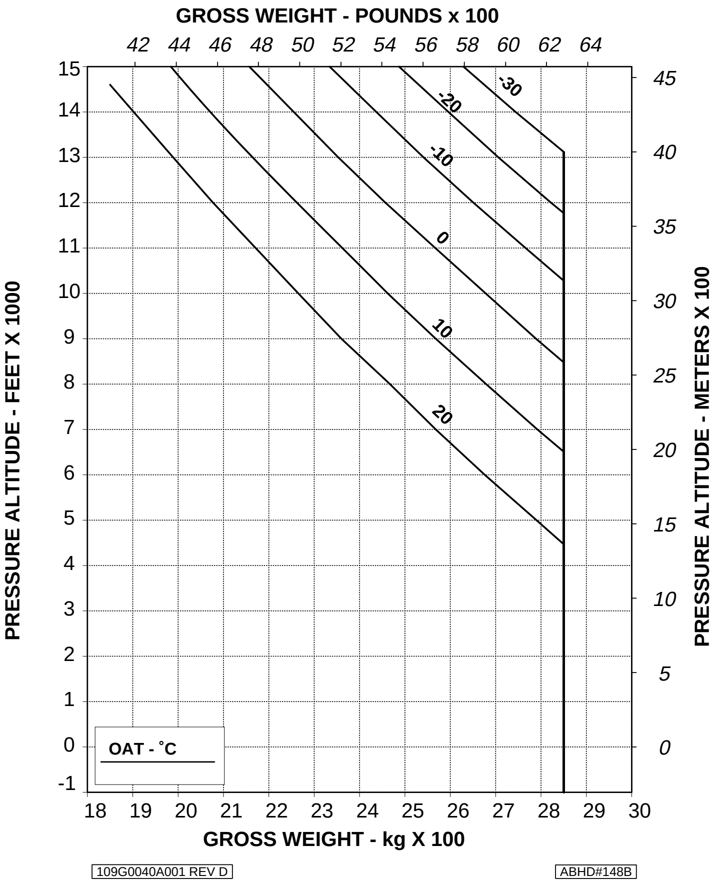


Figure 4-31. Hovering ceiling - OGE - Take-off power - Heater ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

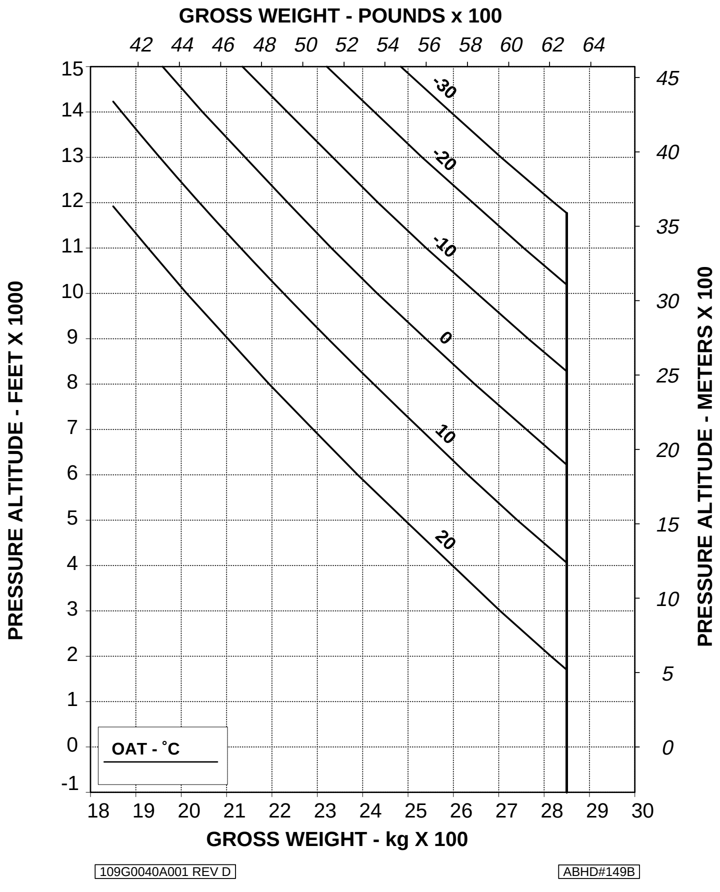


Figure 4-32. Hovering ceiling - OGE - Maximum continuous power - Heater ON.

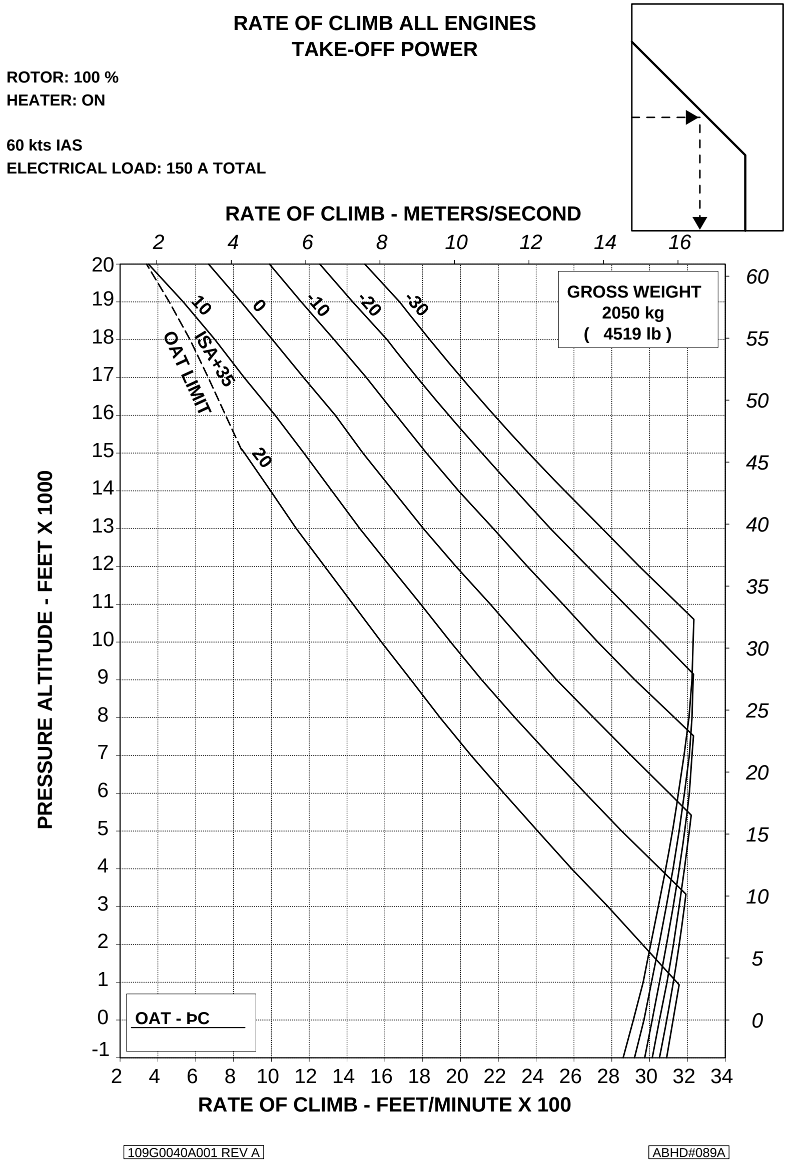


Figure 4-33. Rate of climb - All engines - Take-off power - Heater ON - 2050 kg.

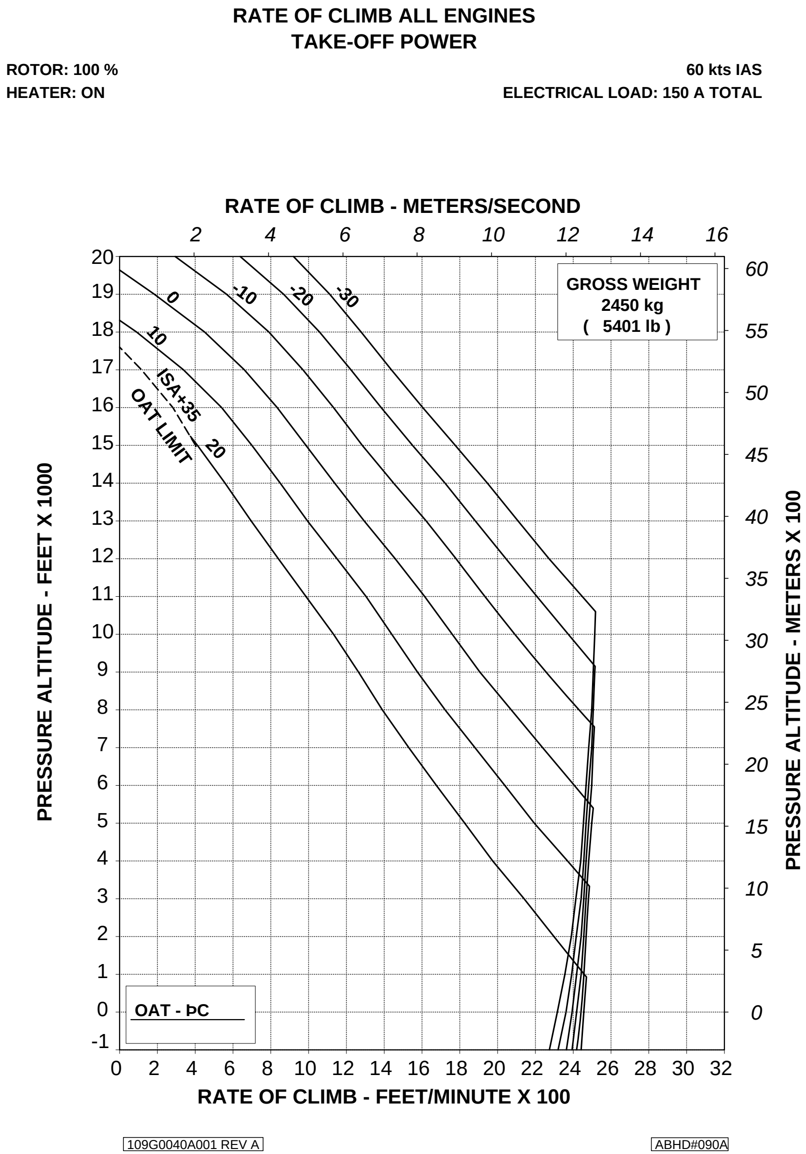


Figure 4-34. Rate of climb - All engines - Take-off power - Heater ON - 2450 kg.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % | 60 kts IAS
HEATER: ON | ELECTRICAL LOAD: 150 A TOTAL

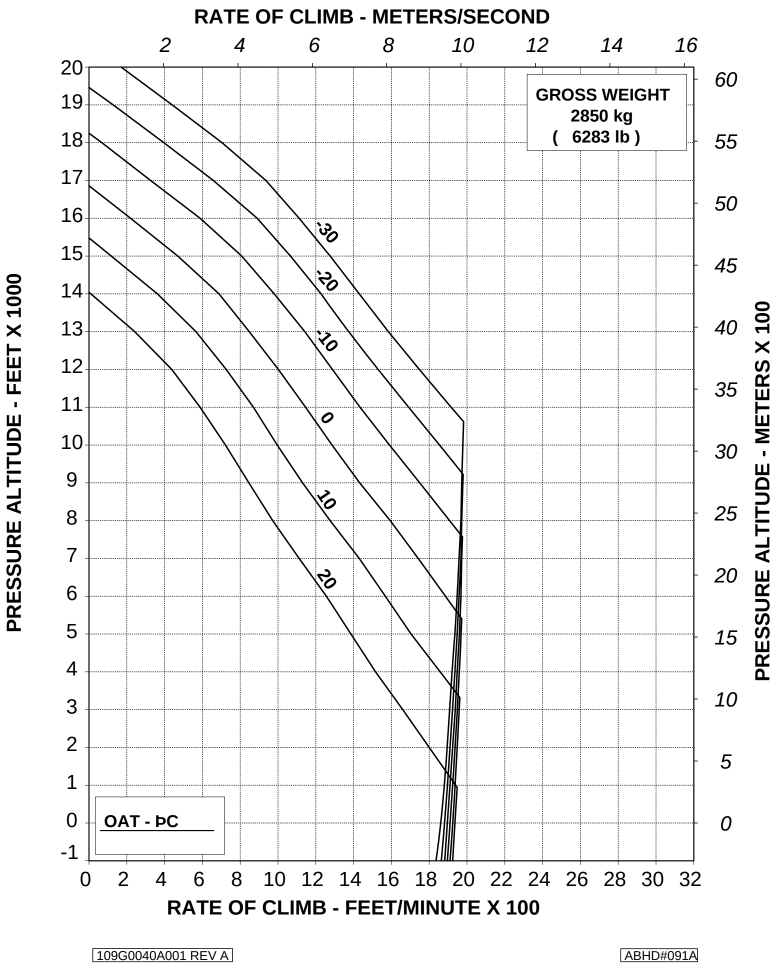


Figure 4-35. Rate of climb - All engines - Take-off power - Heater ON - 2850 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
HEATER: ON **ELECTRICAL LOAD: 150 A TOTAL**

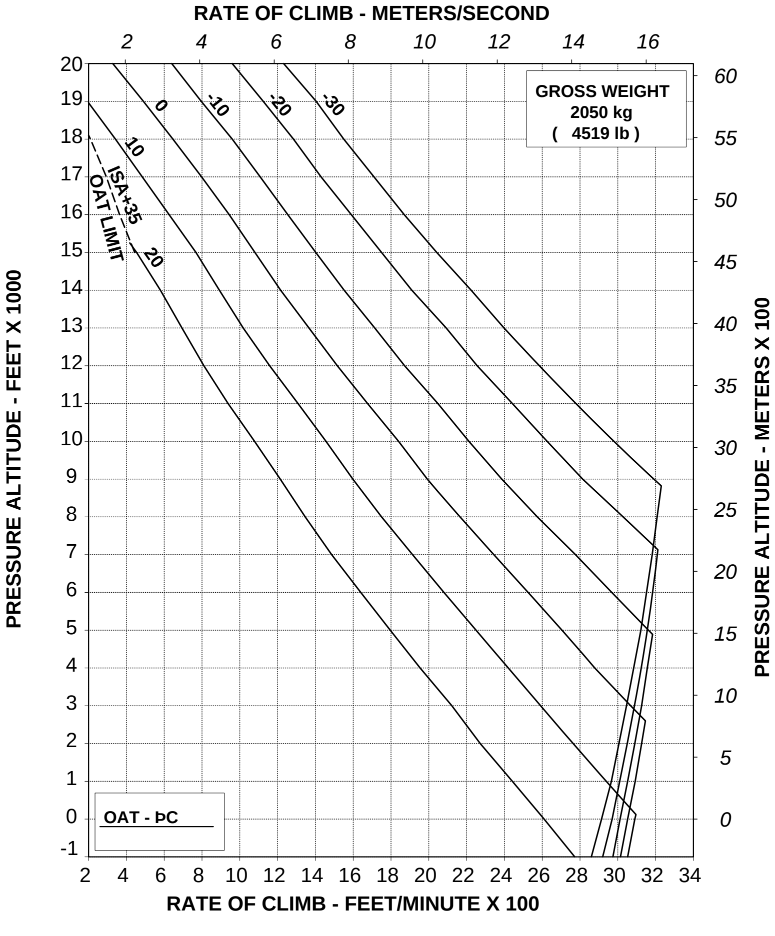


Figure 4-36. Rate of climb - All engines - Maximum continuous power - Heater ON - 2050 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
HEATER: ON | ELECTRICAL LOAD: 150 A TOTAL

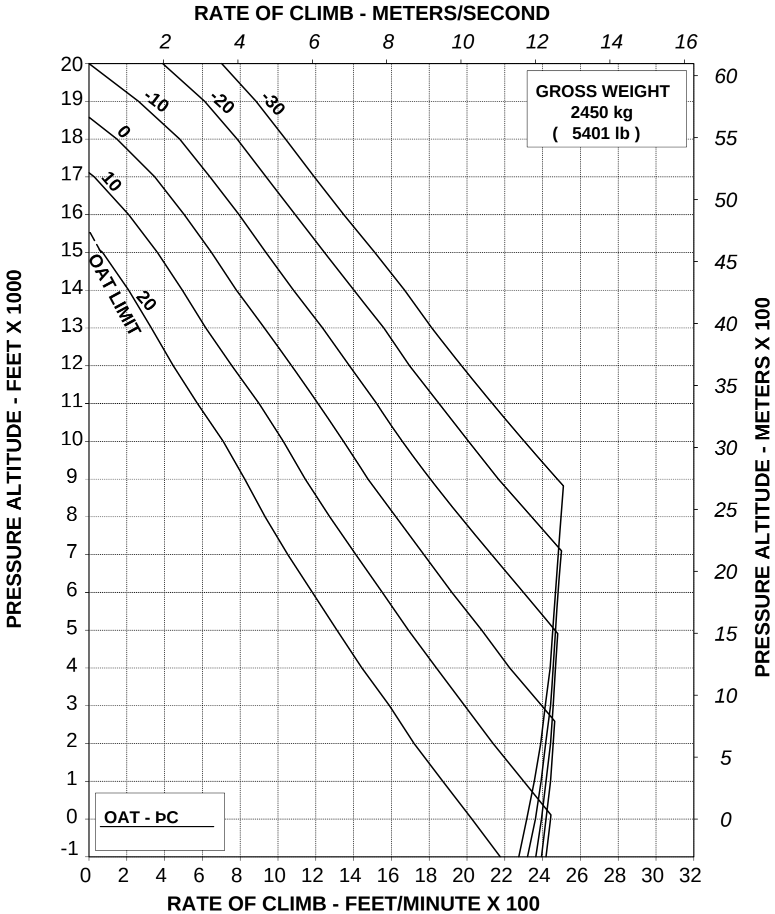


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Figure 4-37. Rate of climb - All engines - Maximum continuous power - Heater ON - 2450 kg.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
HEATER: ON **ELECTRICAL LOAD: 150 A TOTAL**

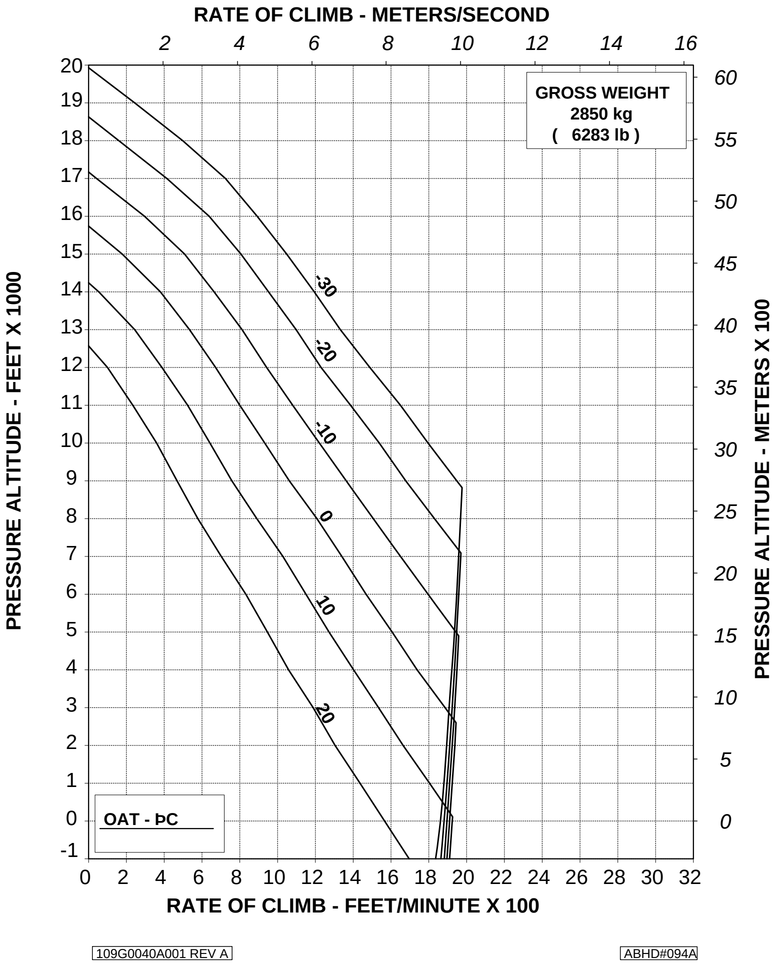

Figure 4-38. Rate of climb - All engines - Maximum continuous power - Heater ON - 2850 kg.

HELICOPTER WITH ENGINE AIR PARTICLE SEPARATOR (EAPS)

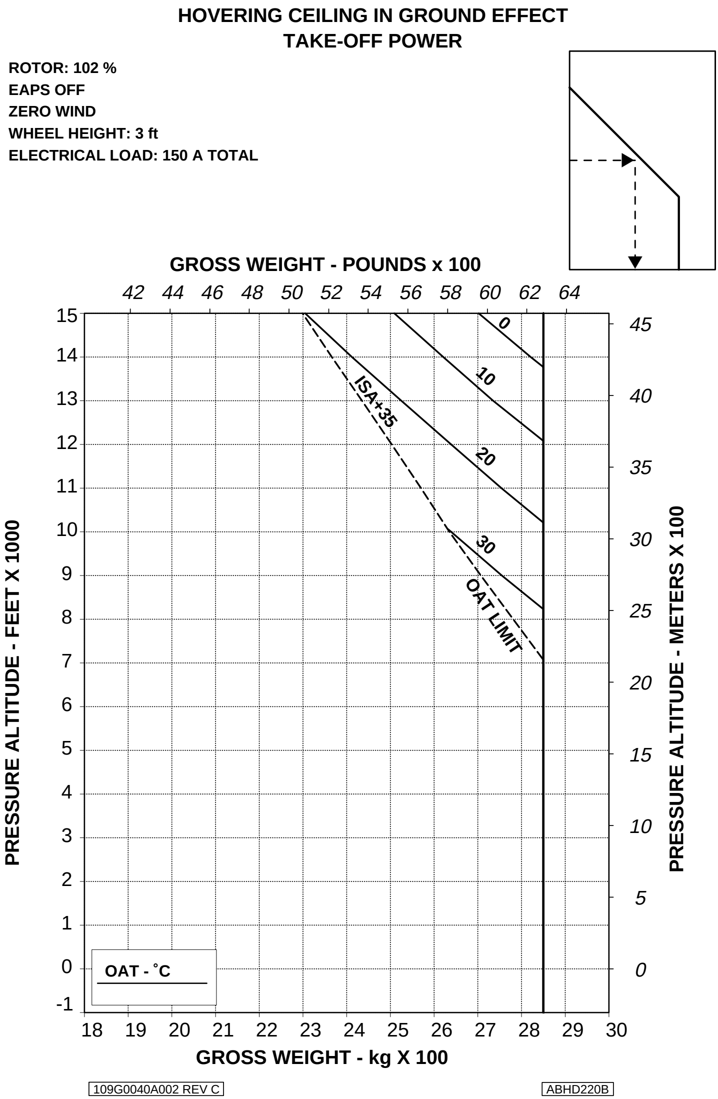

Figure 4-39. Hovering ceiling - IGE - Take-off power - EAPS OFF.

AGUSTA

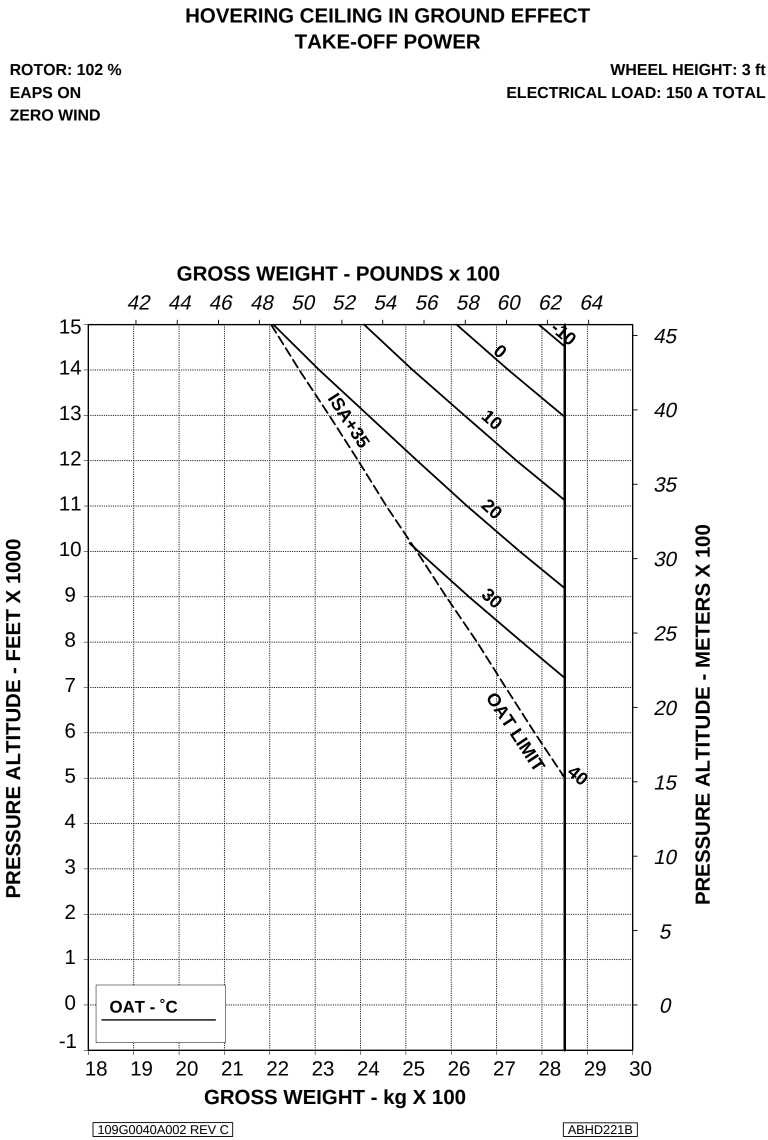

Figure 4-40. Hovering ceiling - IGE - Take-off power - EAPS ON.

 E.A.S.A. Approved

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 40 kg (88 lb)

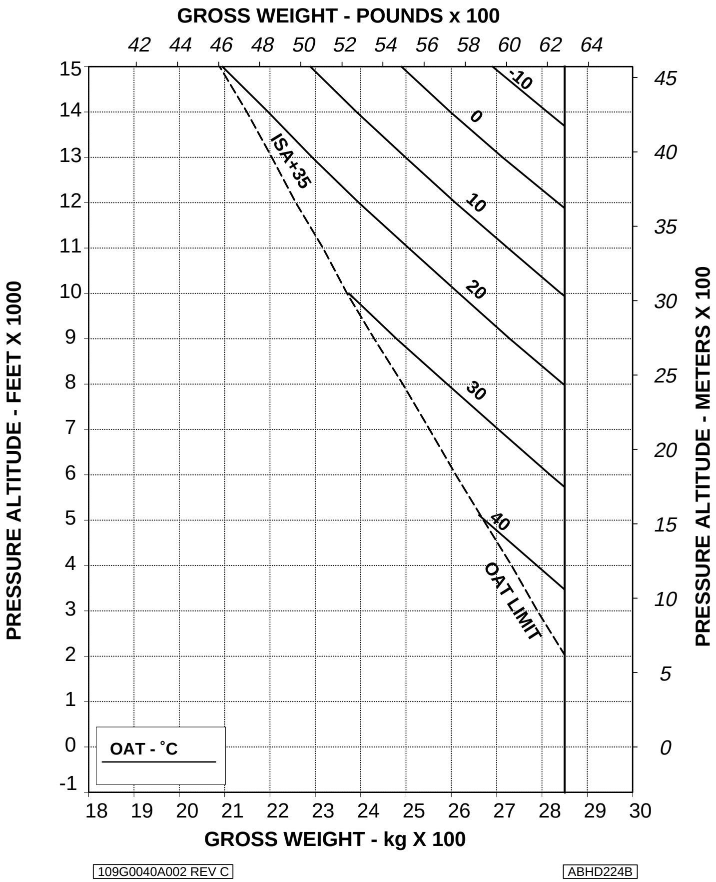

Figure 4-41. Hovering ceiling - IGE - Take-off power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

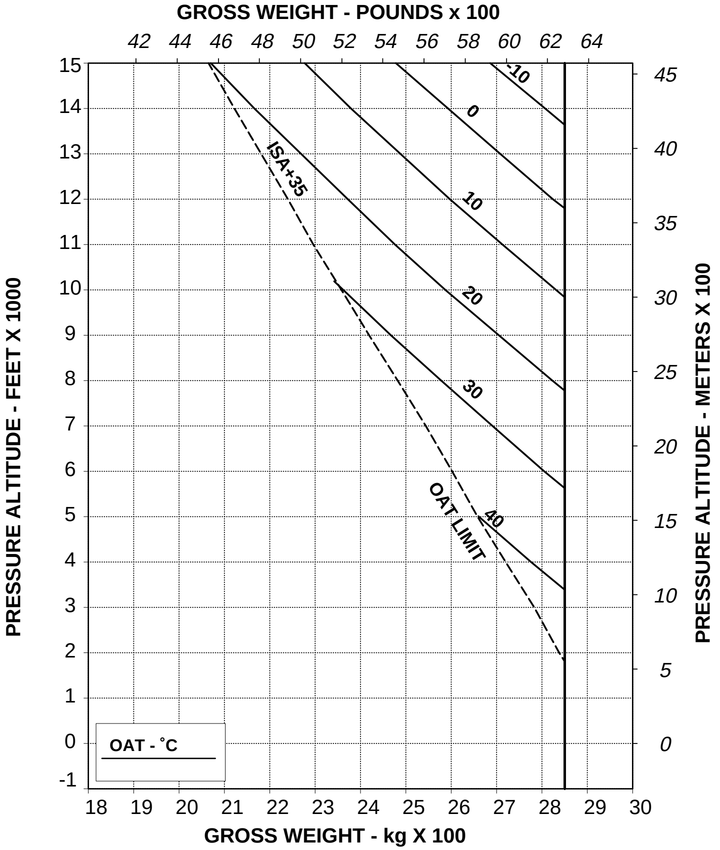

Figure 4-42. Hovering ceiling - IGE - Maximum continuous power - EAPS OFF.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

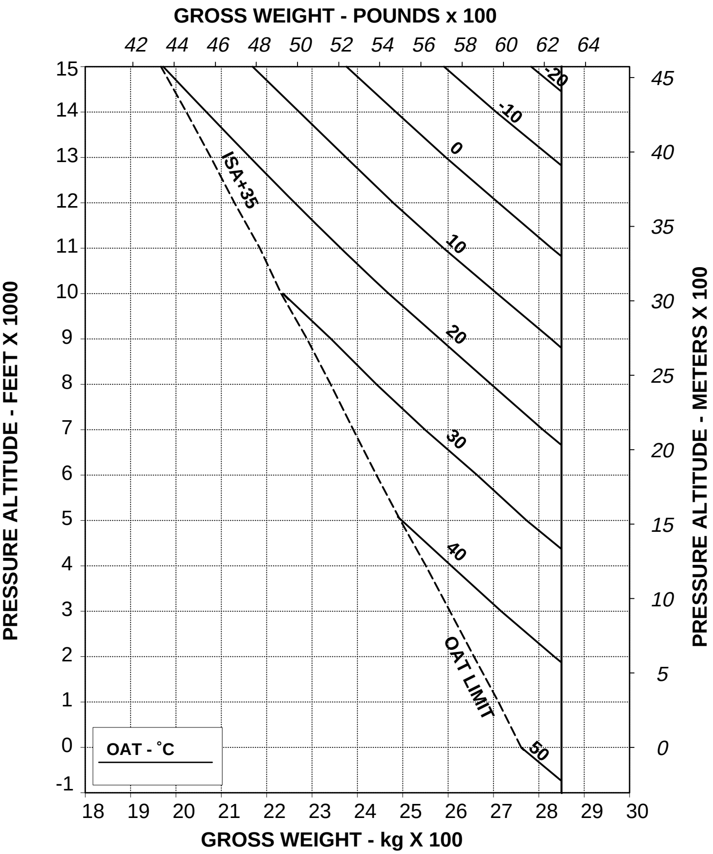

Figure 4-43. Hovering ceiling - IGE - Maximum continuous power - EAPS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

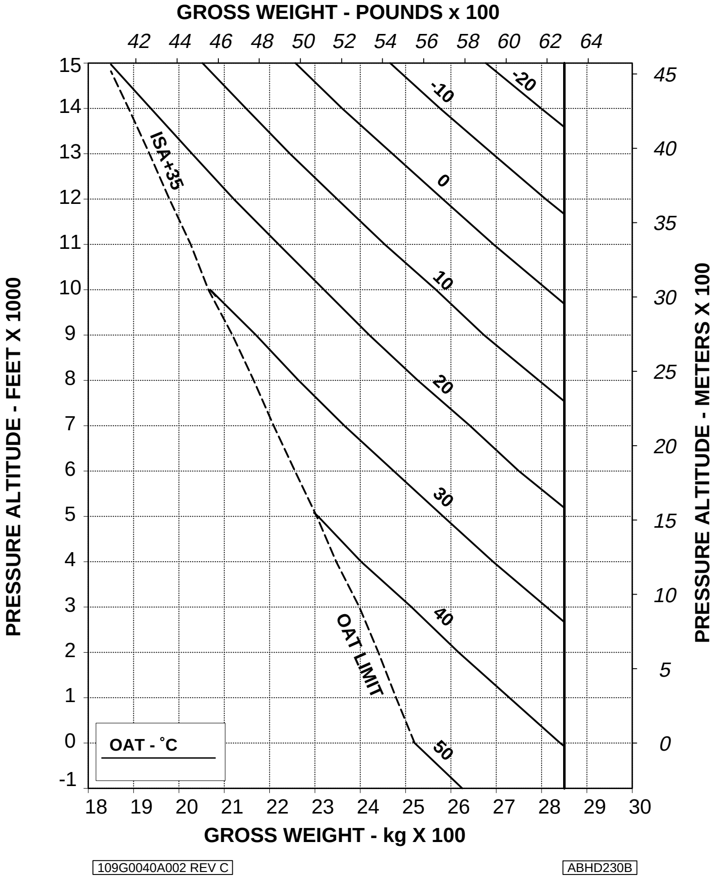

Figure 4-44. Hovering ceiling - IGE - Maximum continuous power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

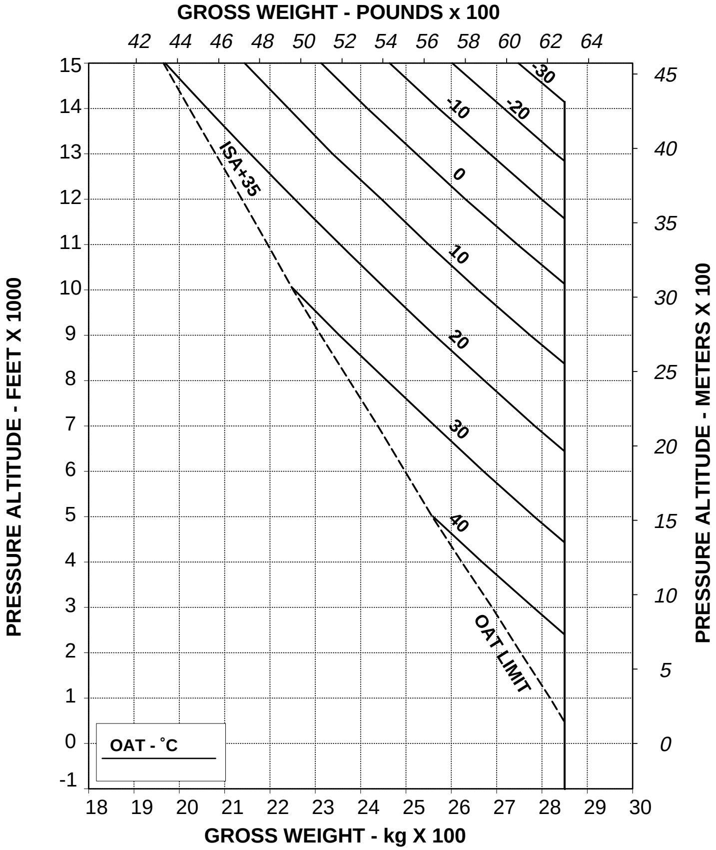

Figure 4-45. Hovering ceiling - OGE - Take-off power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

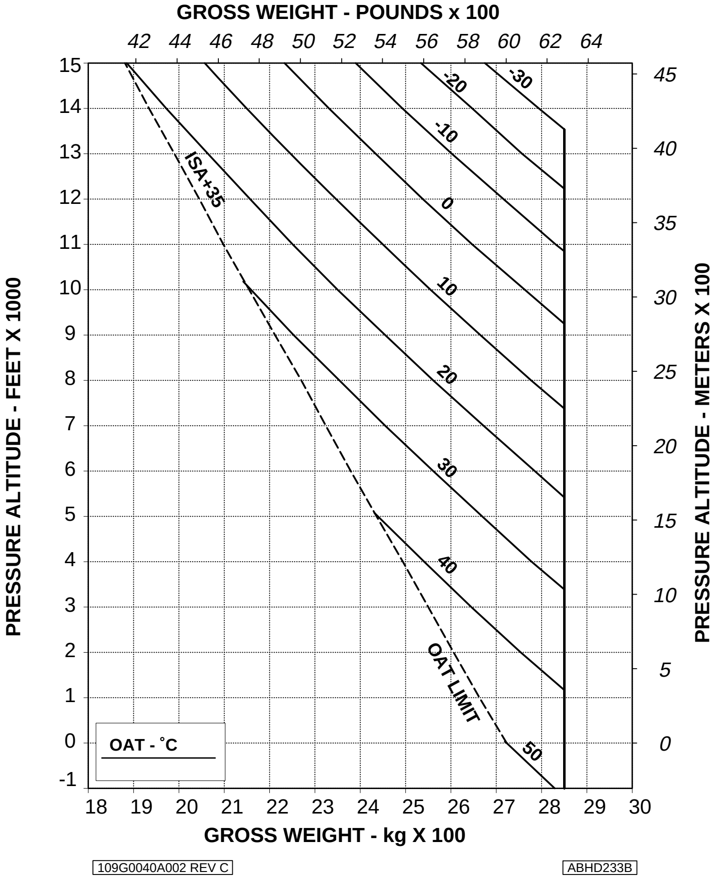

Figure 4-46. Hovering ceiling - OGE - Take-off power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

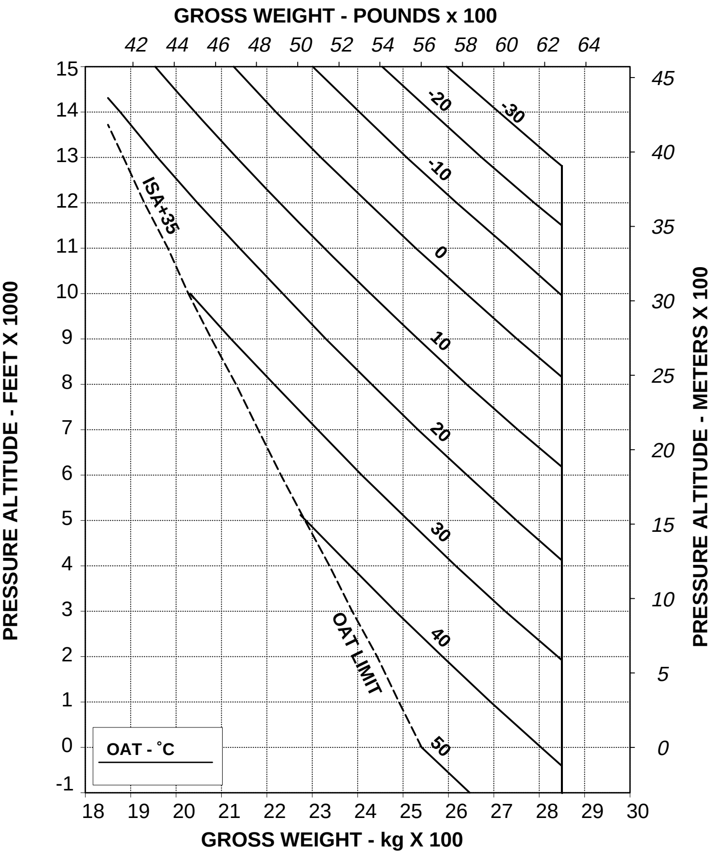

Figure 4-47. Hovering ceiling - OGE - Take-off power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

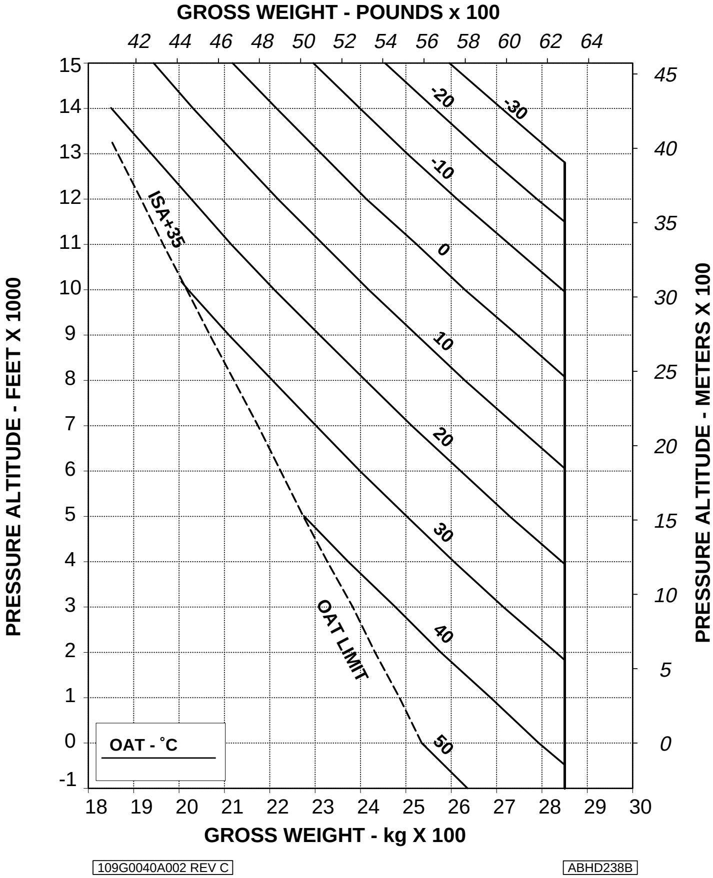

Figure 4-48. Hovering ceiling - OGE - Maximum continuous power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

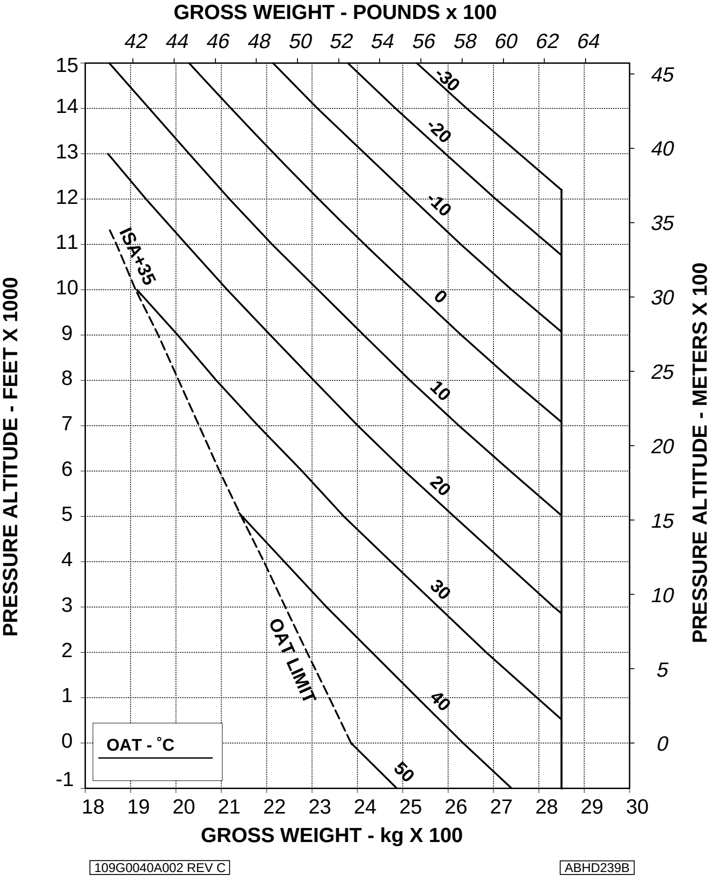

Figure 4-49. Hovering ceiling - OGE - Maximum continuous power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL
HEATER OR E.C.S. ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

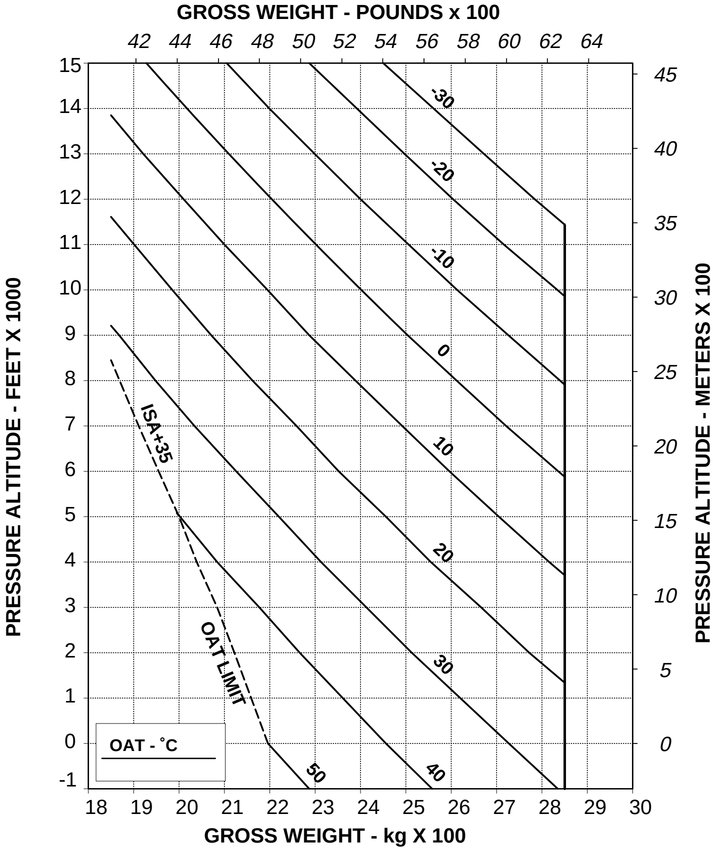

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Figure 4-50. Hovering ceiling - OGE - Maximum continuous power - EAPS OFF/ON - Heater or ECS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

EAPS OFF
ELECTRICAL LOAD: 150 A

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM. FOR HEAVIER WEIGHTS, REFER TO CHART "B".

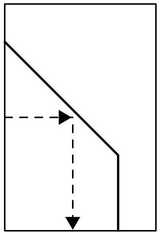

CHART A

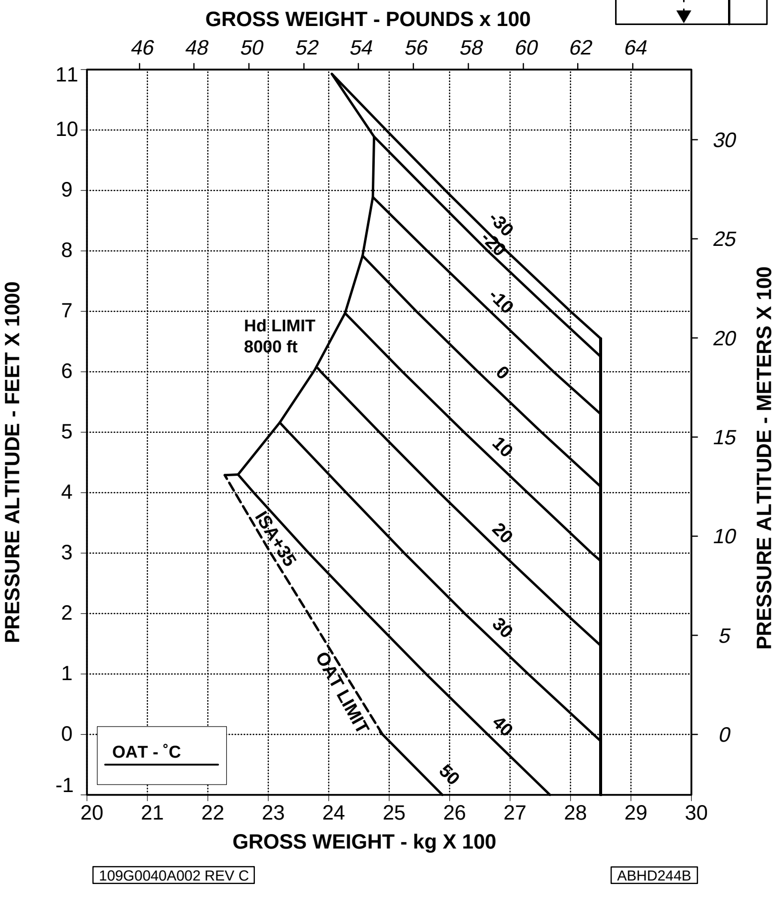

Figure 4-51. Height-Velocity diagram - OEI - Chart A - EAPS OFF.

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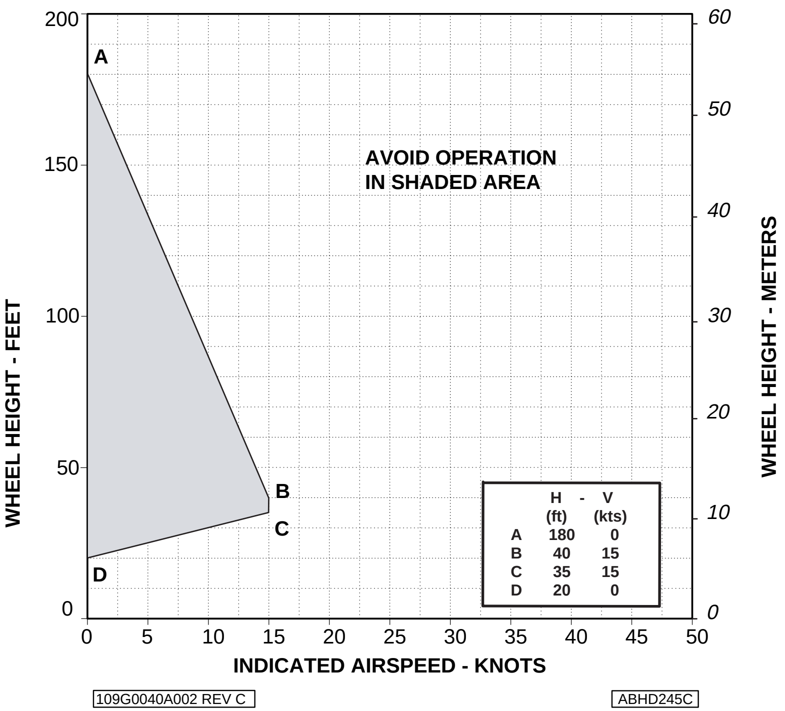
HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

EAPS OFF

CHART B



	H (ft)	V (kts)
A	180	0
B	40	15
C	35	15
D	20	0

Figure 4-52. Height-Velocity diagram - OEI - Chart B - EAPS OFF.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

EAPS ON
ELECTRICAL LOAD: 150 A

CHART "A" SHOWS THE WEIGHT VALUES AT/BELOW WHICH THERE IS NO H-V DIAGRAM. FOR HEAVIER WEIGHTS, REFER TO CHART "B".

CHART A

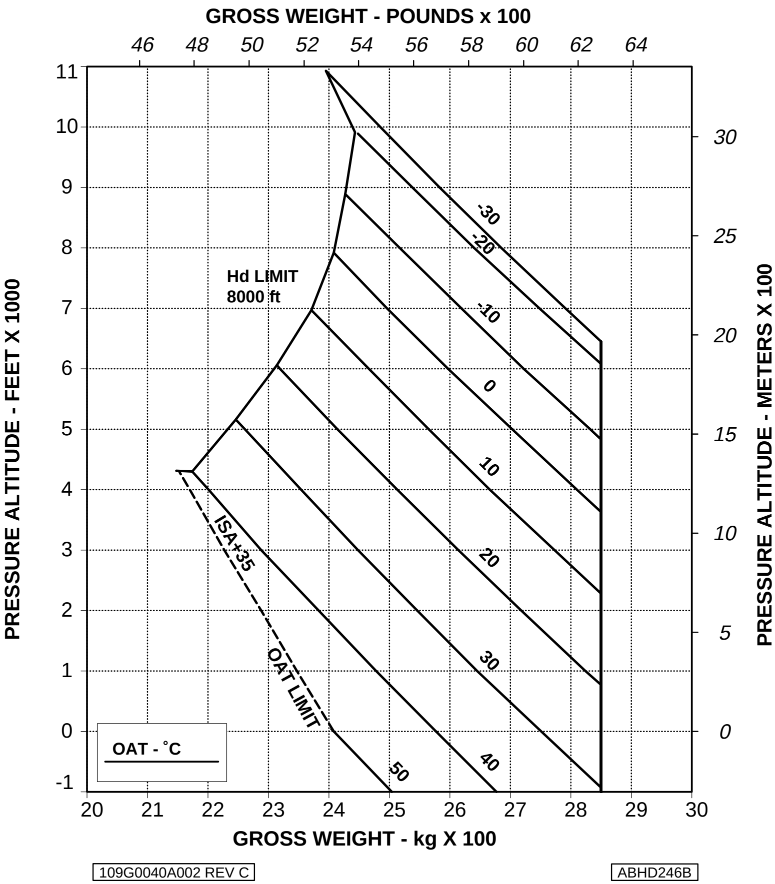


Figure 4-53. Height-Velocity diagram - OEI - Chart A - EAPS ON.

HEIGHT-VELOCITY DIAGRAM
ONE ENGINE INOPERATIVE

FOR SMOOTH, LEVEL, HARD SURFACES

EAPS ON

CHART B

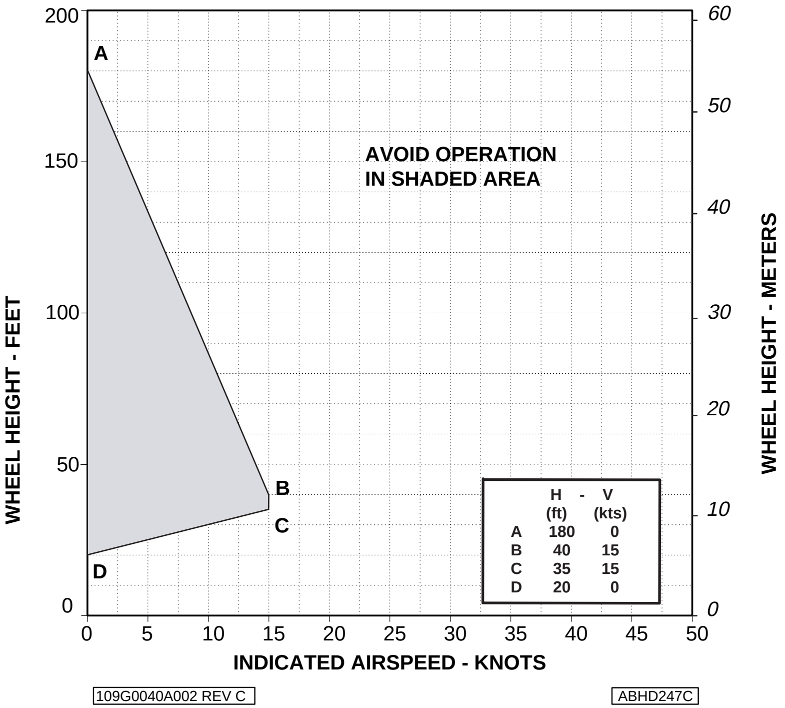
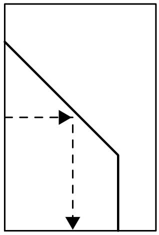


Figure 4-54. Height-Velocity diagram - OEI - Chart B - EAPS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL



WITH EAPS ON: DECREASE RATE OF CLIMB BY 350 ft/min (1,78 m/s)

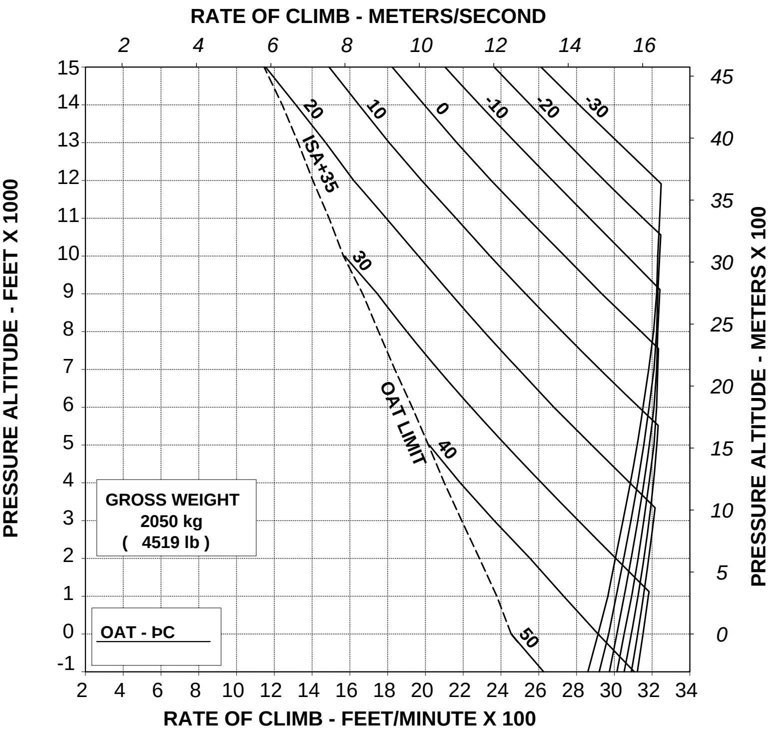


Figure 4-55. Rate of climb - All engines - Take-off power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 300 ft/min (1,52 m/s)

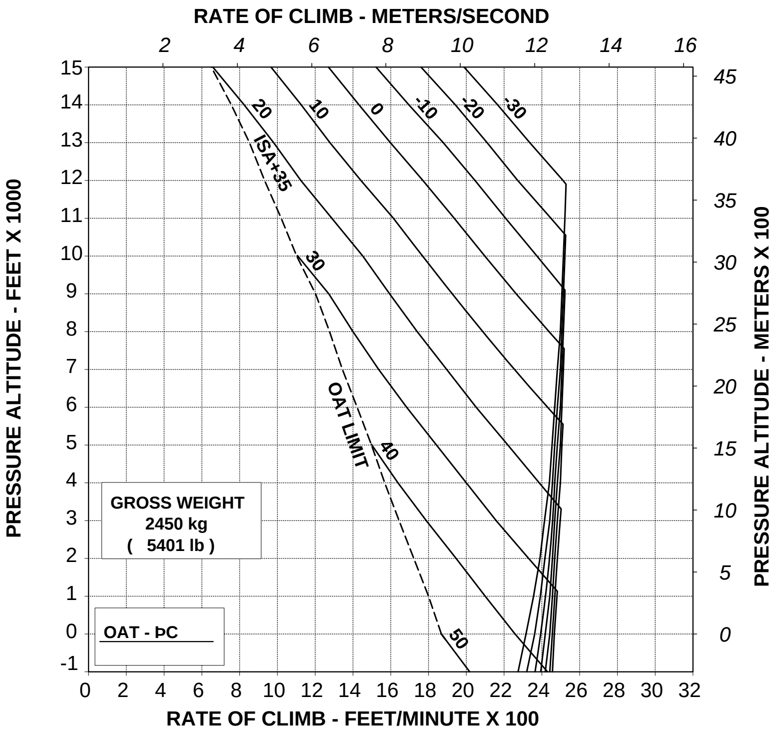


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Figure 4-56. Rate of climb - All engines - Take-off power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27 m/s)

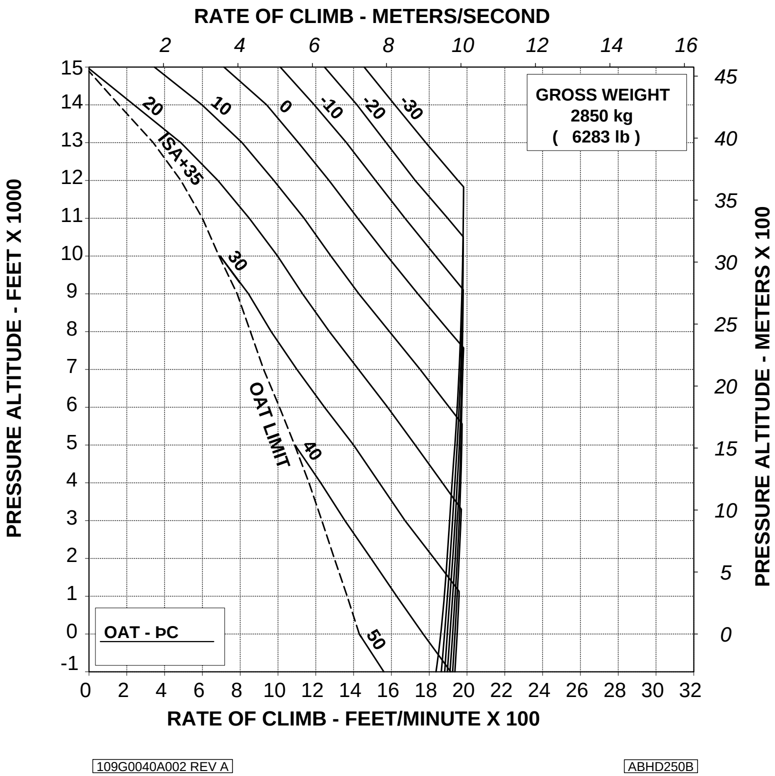

Figure 4-57. Rate of climb - All engines - Take-off power - 2850 kg - EAPS OFF/ON.

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RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

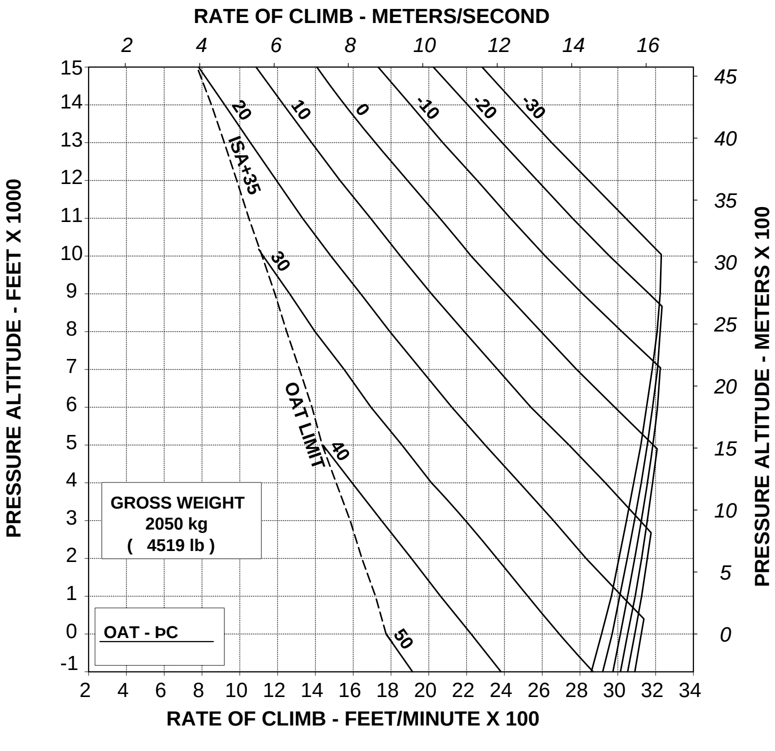

Figure 4-58. Rate of climb - All engines - Take-off power - 2050 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

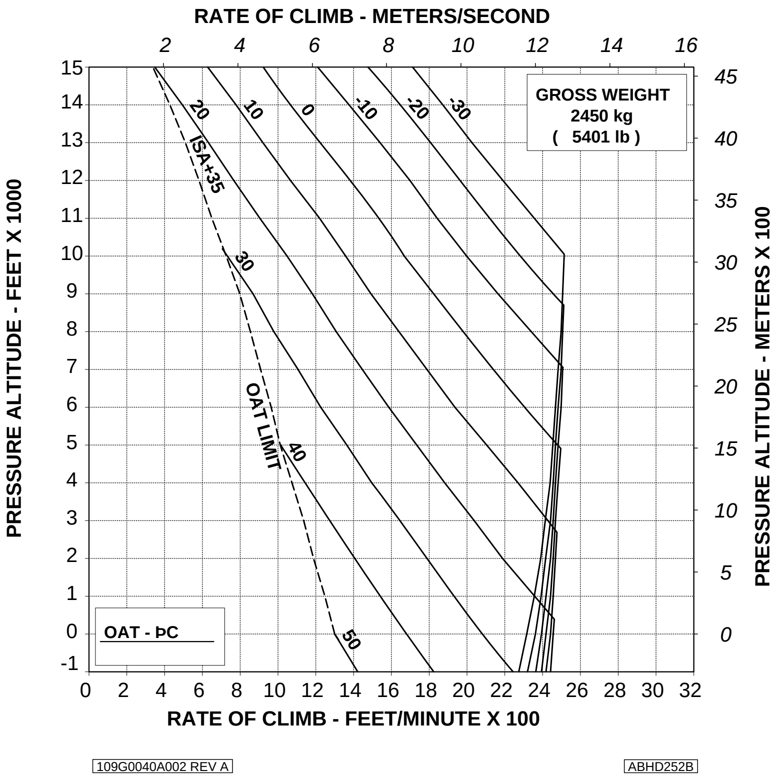

Figure 4-59. Rate of climb - All engines - Take-off power - 2450 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

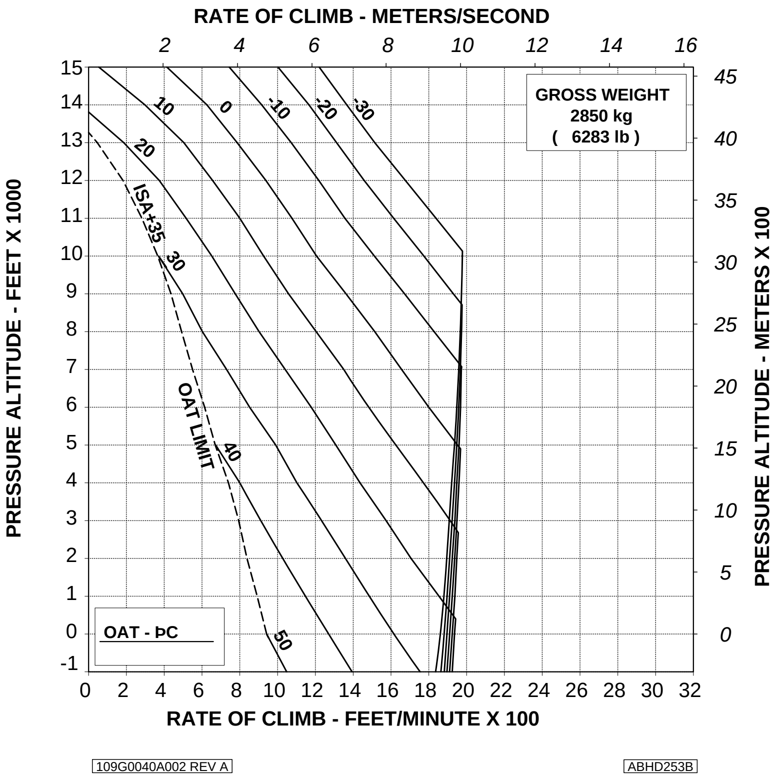

Figure 4-60. Rate of climb - All engines - Take-off power - 2850 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 300ft/min (1,52 m/s)

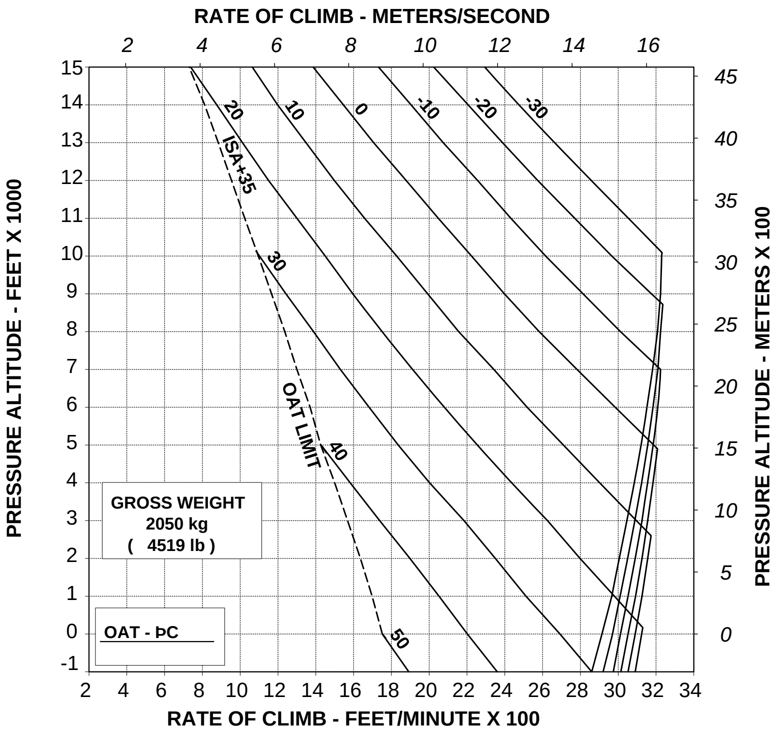

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Figure 4-61. Rate of climb - All engines - Maximum continuous power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | **60 kts IAS**
EAPS OFF | **ELECTRICAL LOAD: 150 A TOTAL**

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27 m/s)

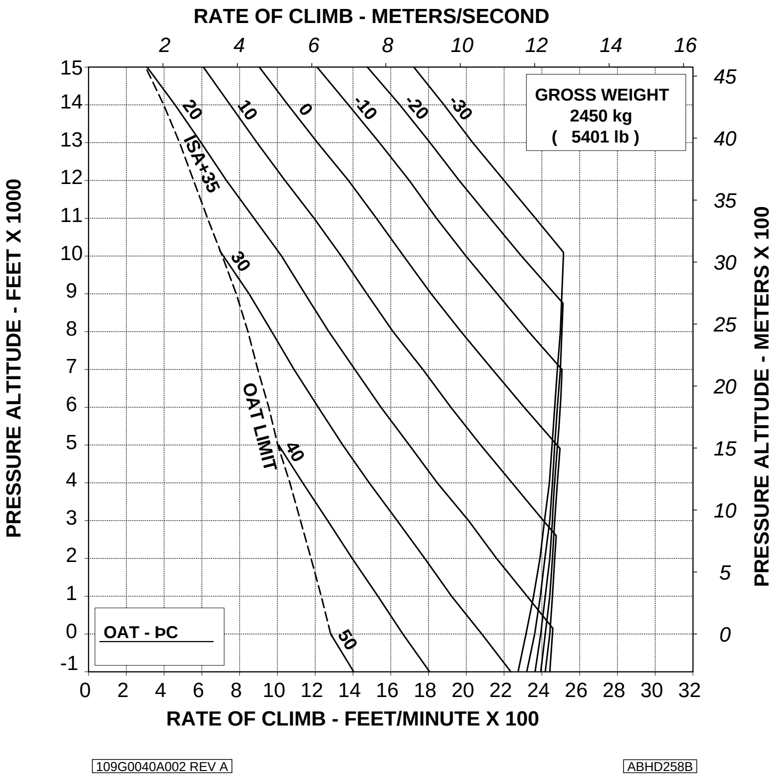

Figure 4-62. Rate of climb - All engines - Maximum continuous power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27 m/s)

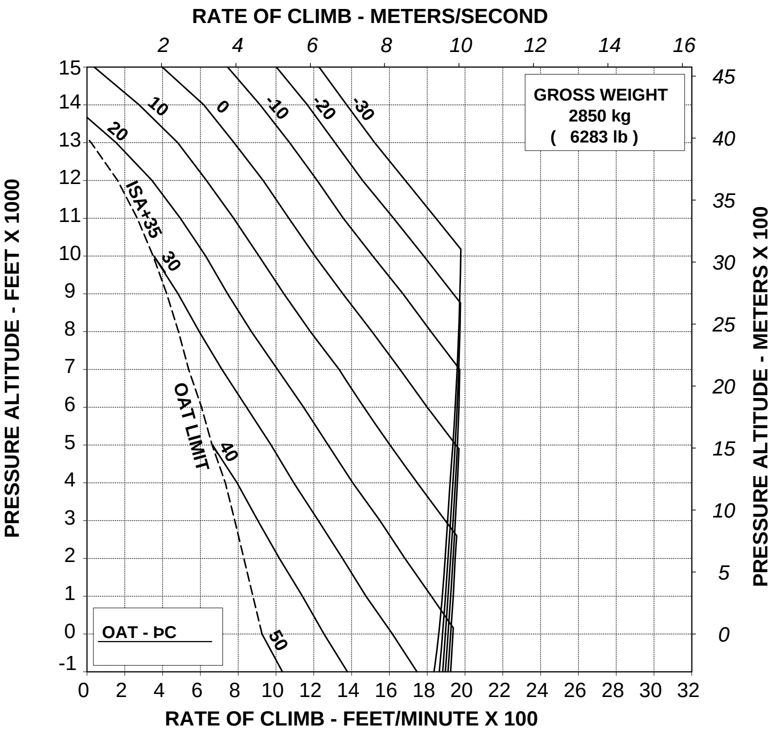

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Figure 4-63. Rate of climb - All engines - Maximum continuous power - 2850 kg - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

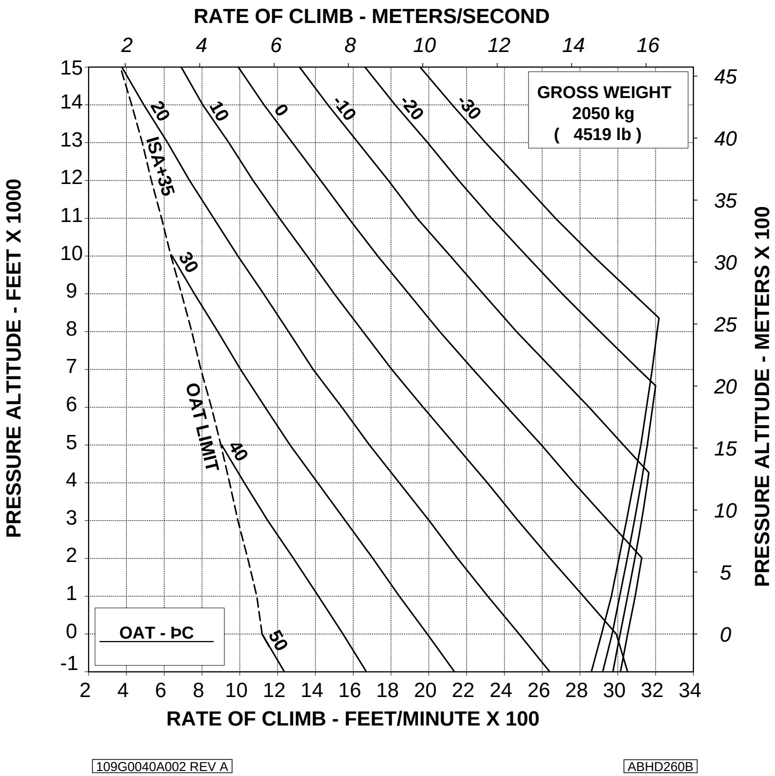

Figure 4-64. Rate of climb - All engines - Maximum continuous power - 2050 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

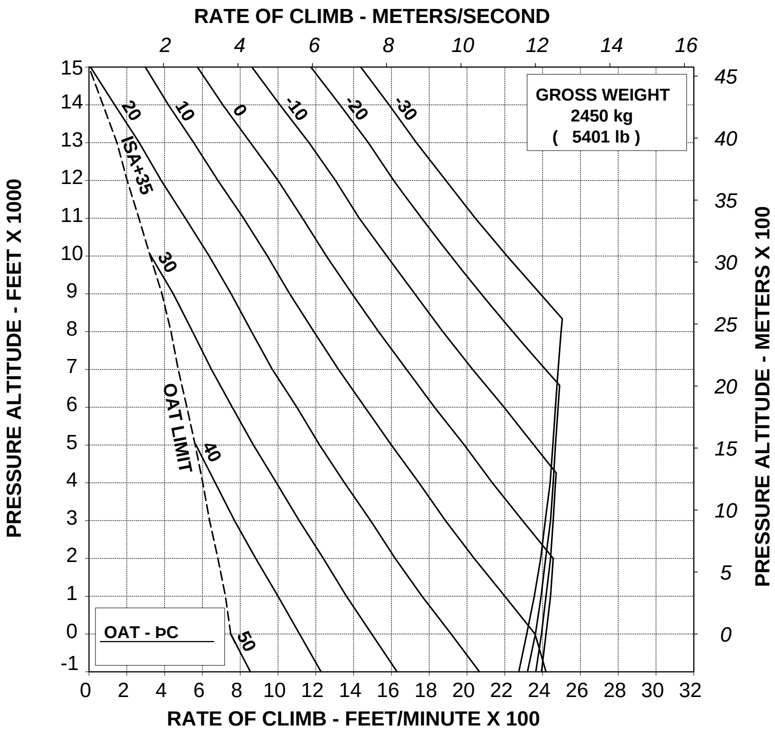

Figure 4-65. Rate of climb - All engines - Maximum continuous power - 2450 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
HEATER OR E.C.S. ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

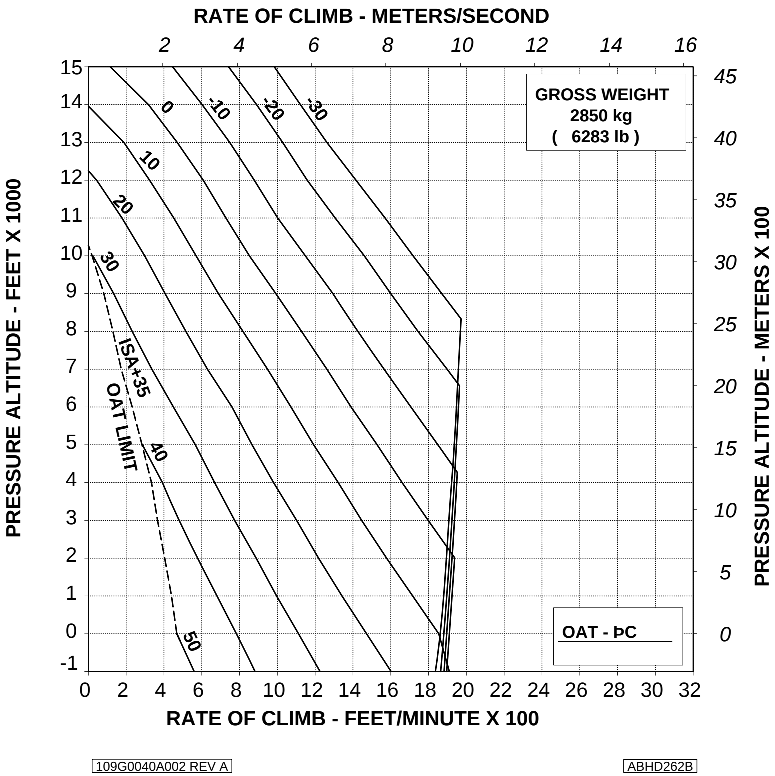

Figure 4-66. Rate of climb - All engines - Maximum continuous power - 2850 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

WITH EAPS ON: DECREASE RATE OF CLIMB BY 175 ft/min (0,89 m/s)

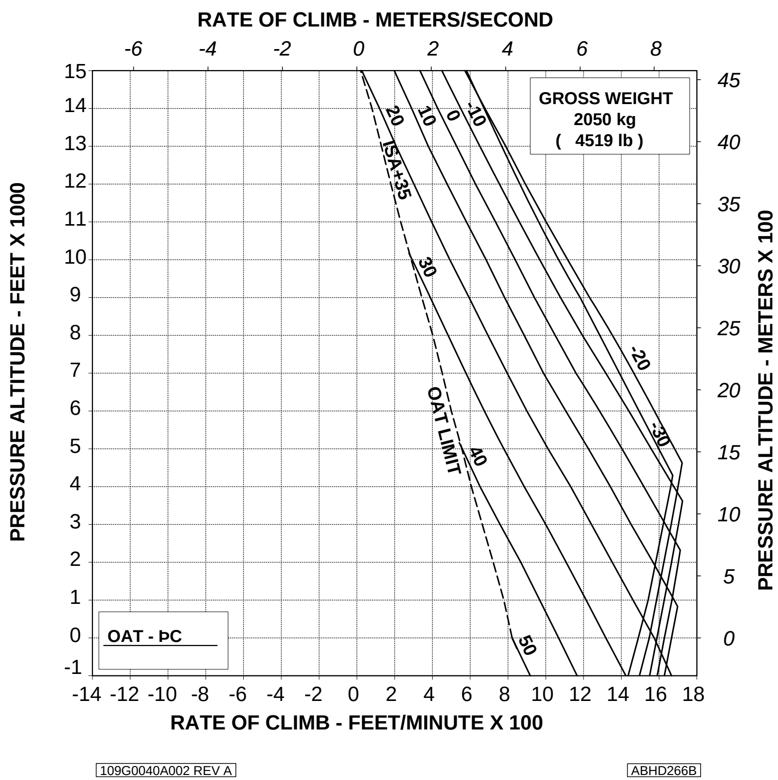

Figure 4-67. Rate of climb - OEI - 2.5 minute power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

WITH EAPS ON: DECREASE RATE OF CLIMB BY 150 ft/min (0,76 m/s)

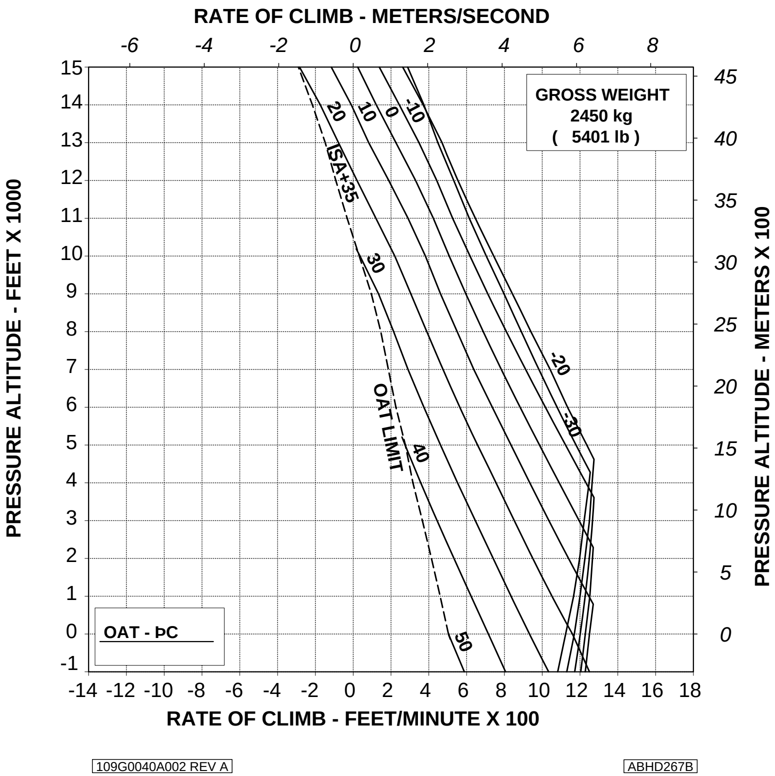

Figure 4-68. Rate of climb - OEI - 2.5 minute power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 125 ft/min (0,64 m/s)

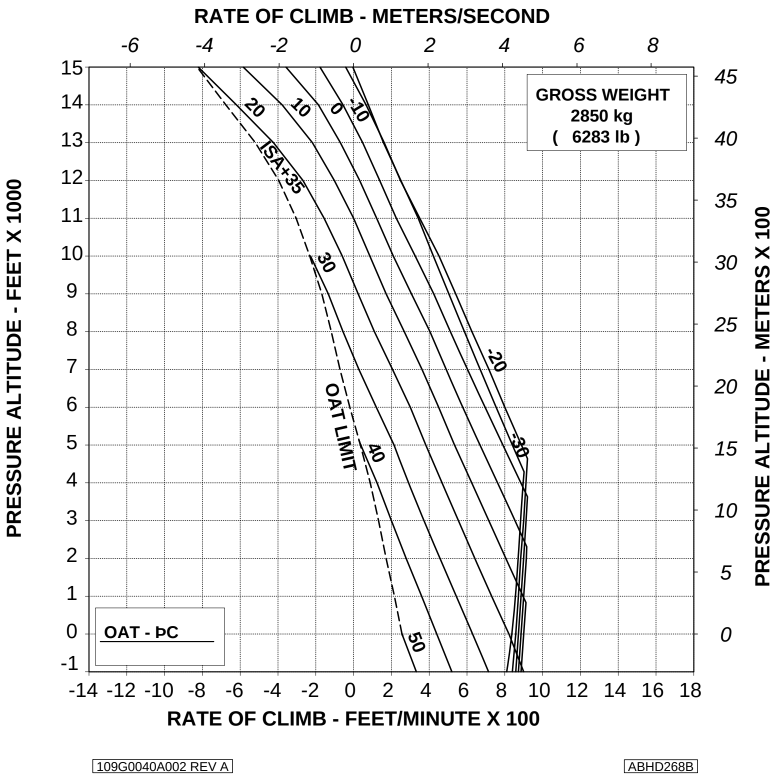

Figure 4-69. Rate of climb - OEI - 2.5 minute power - 2850 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 150 ft/min (0,76 m/s)

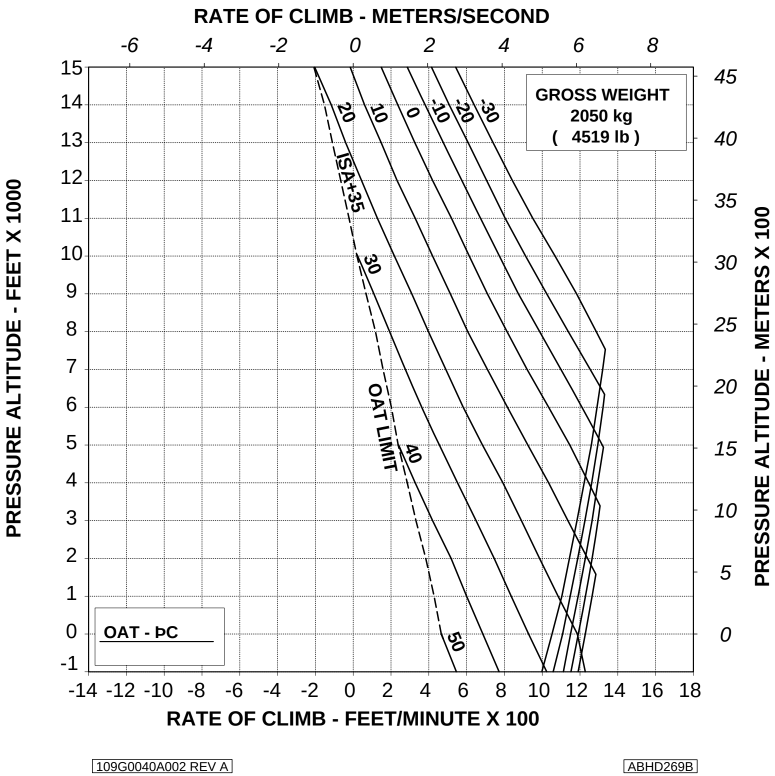

Figure 4-70. Rate of climb - OEI - Maximum continuous power - 2050 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % — 60 kts IAS
EAPS OFF — ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 150 ft/min (0,76 m/s)

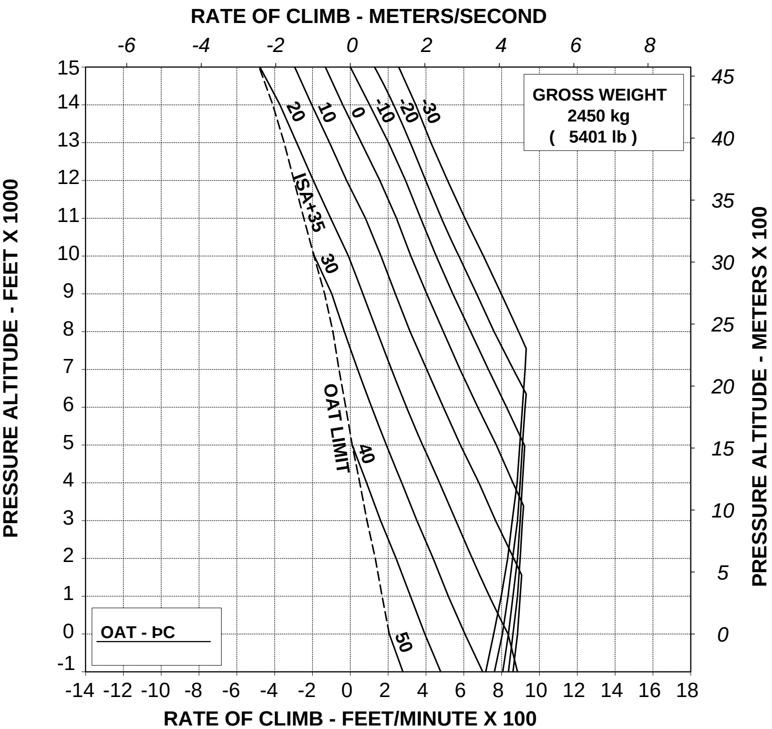

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Figure 4-71. Rate of climb - OEI - Maximum continuous power - 2450 kg - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

WITH EAPS ON: DECREASE RATE OF CLIMB BY 125 ft/min (0,64 m/s)

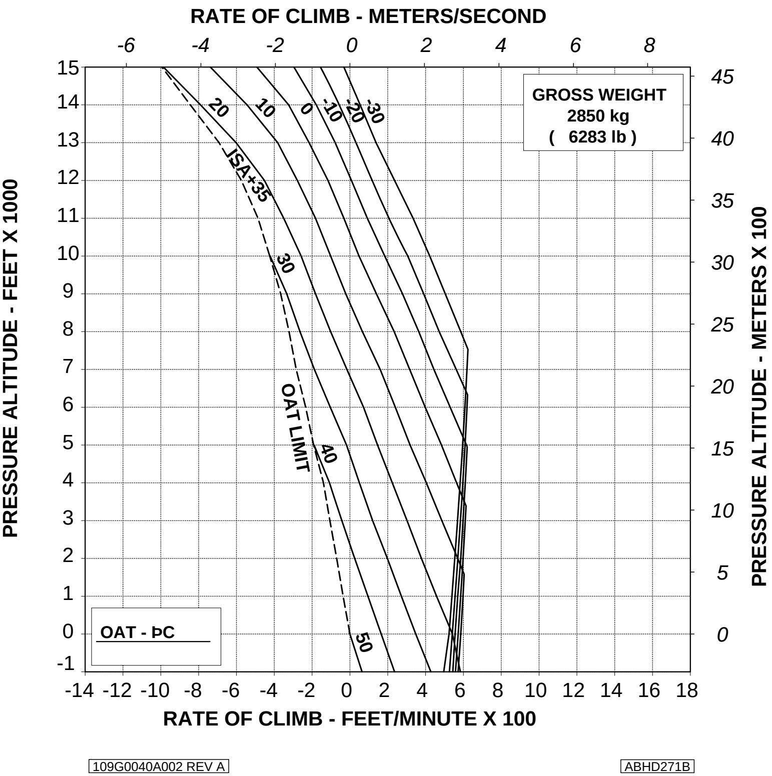

Figure 4-72. Rate of climb - OEI - Maximum continuous power - 2850 kg - EAPS OFF/ON.

CATEGORY ″A″ OPERATIONS (WITH EAPS)

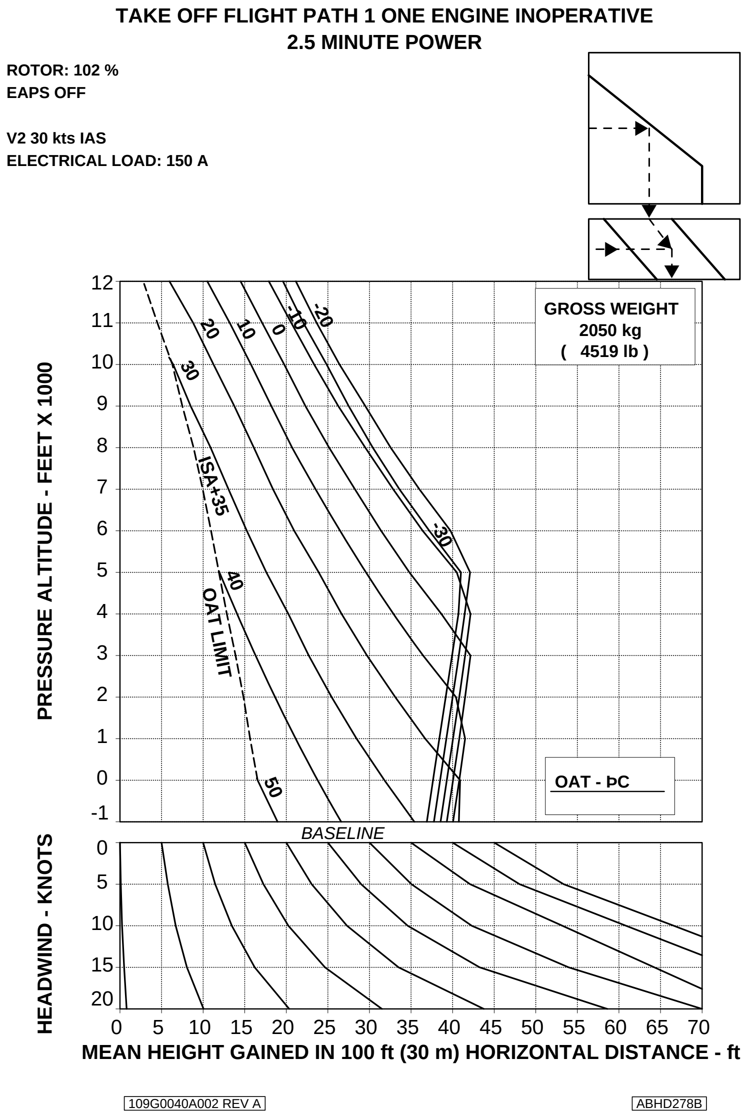

Figure 4-73. Take-off flight path 1 - 2050 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

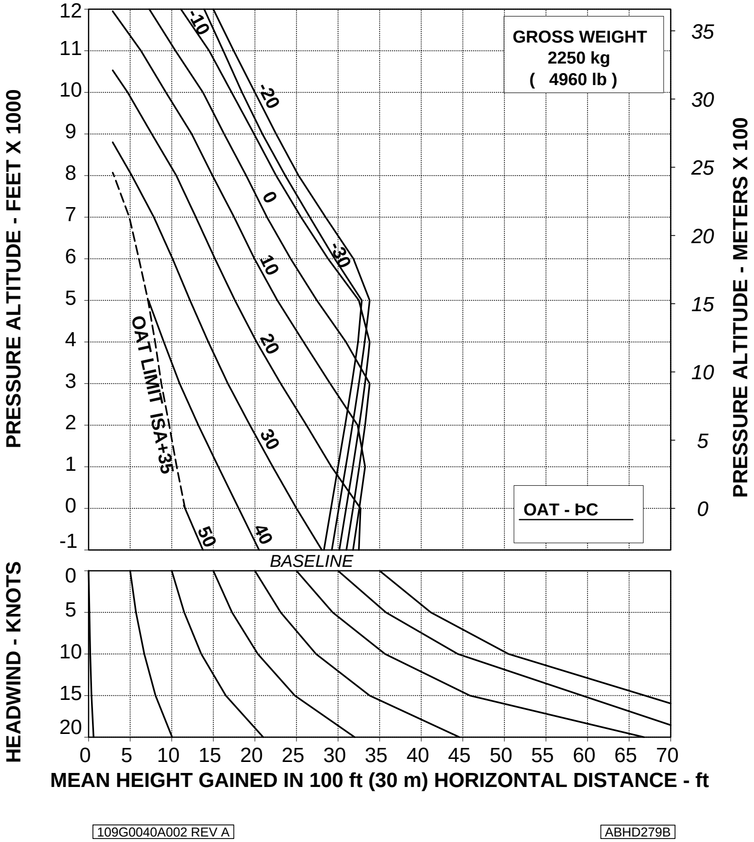

Figure 4-74. Take-off flight path 1 - 2250 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

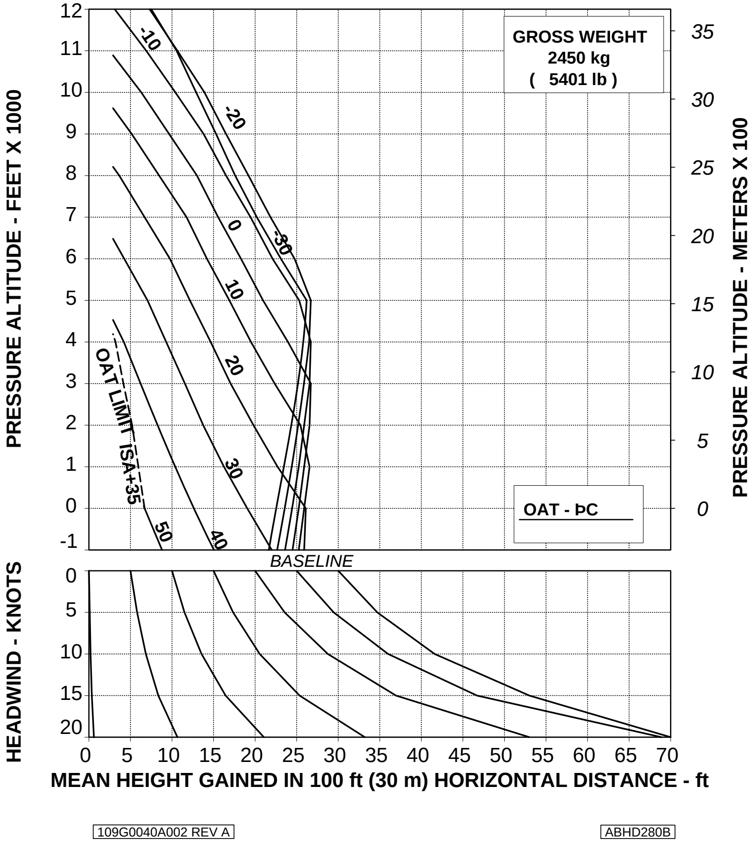

Figure 4-75. Take-off flight path 1 - 2450 kg - EAPS OFF.

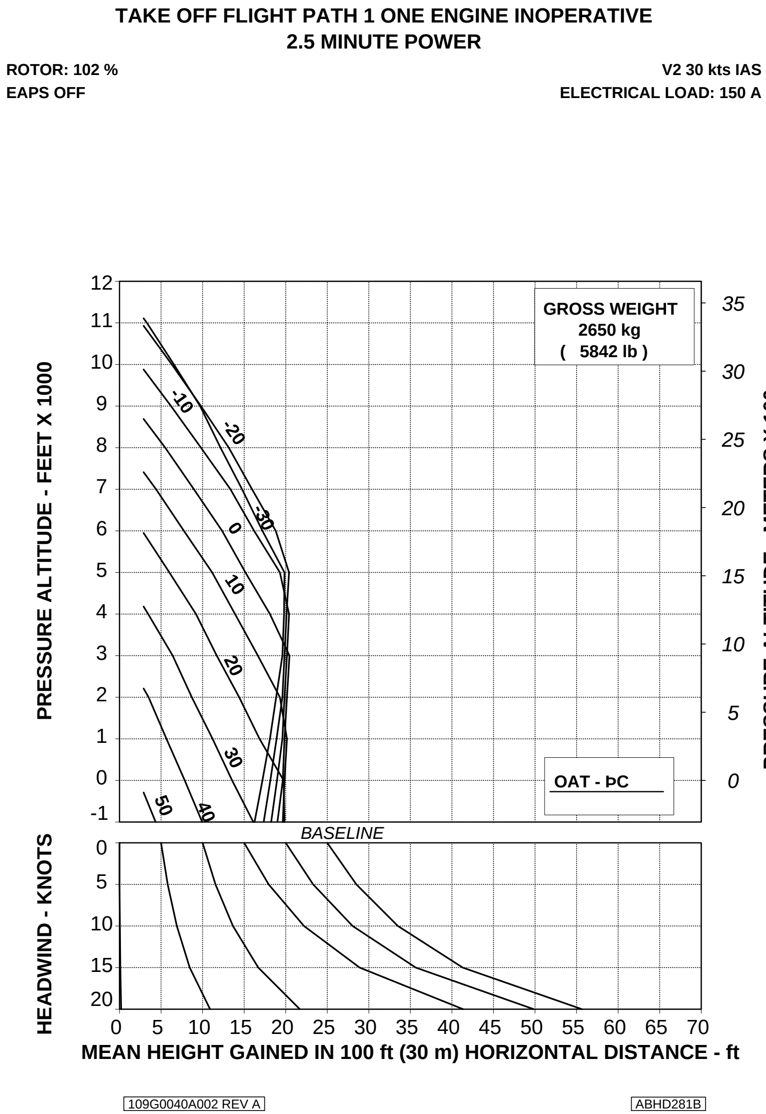

Figure 4-76. Take-off flight path 1 - 2650 kg - EAPS OFF.

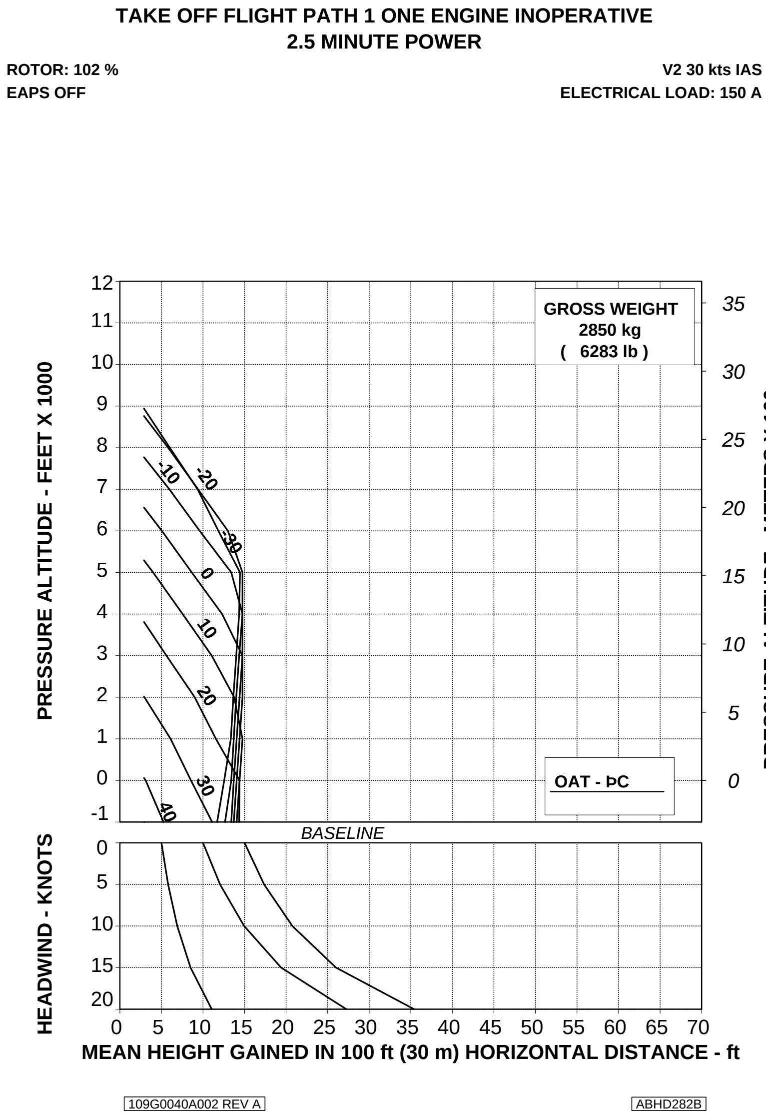

Figure 4-77. Take-off flight path 1 - 2850 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A**

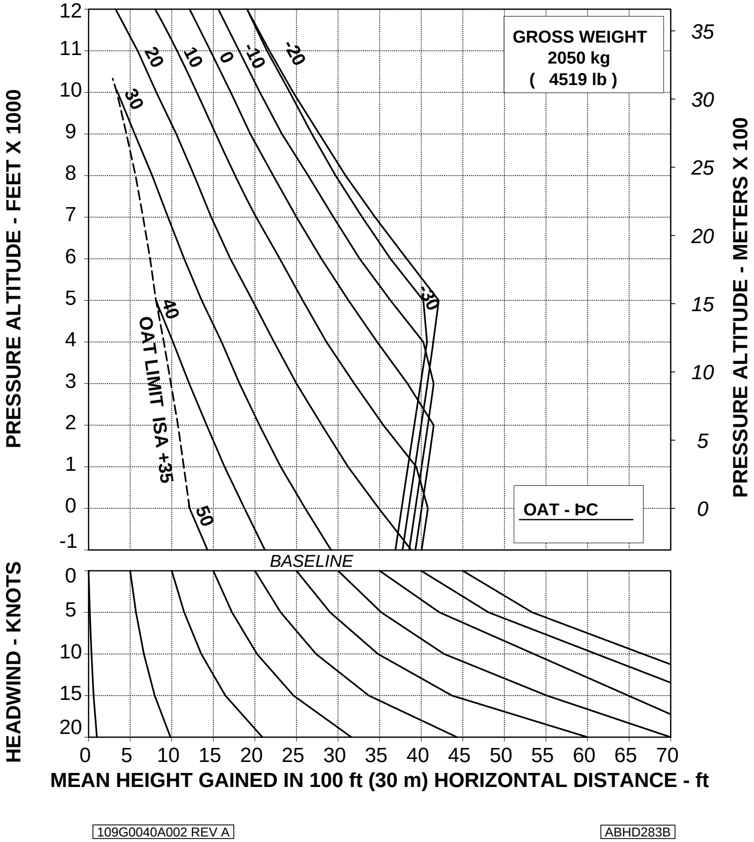

Figure 4-78. Take-off flight path 1 - 2050 kg - EAPS ON.

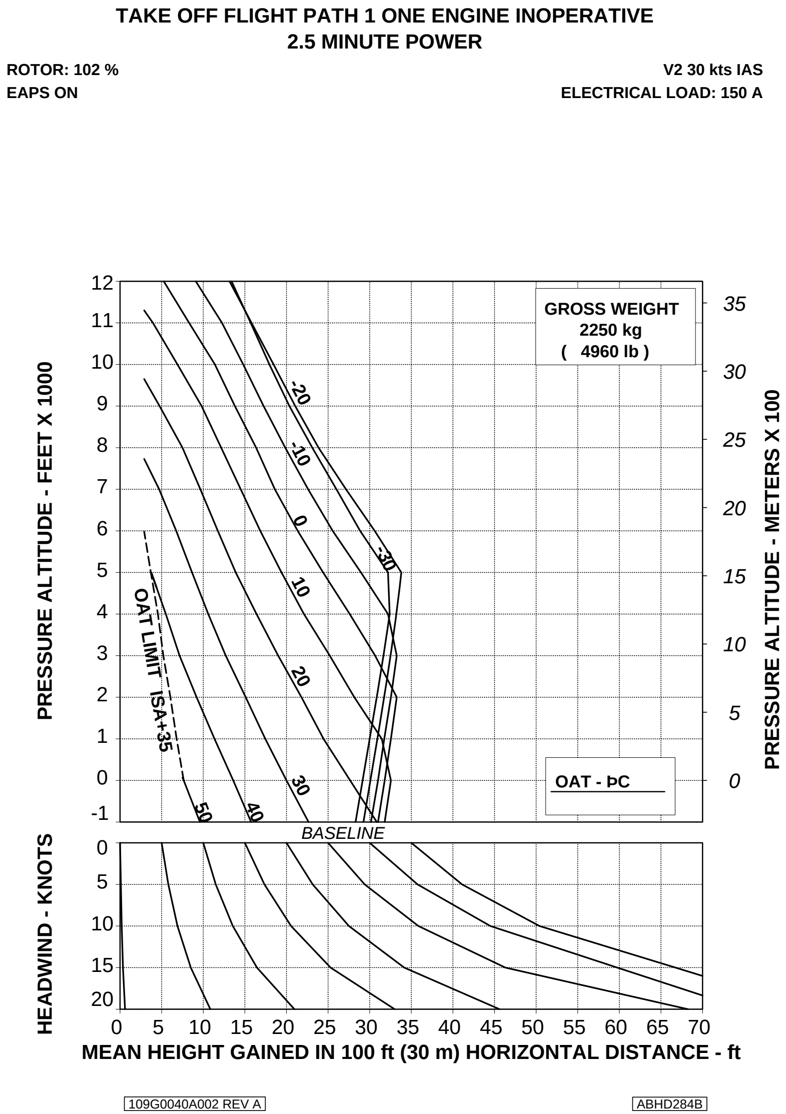

Figure 4-79. Take-off flight path 1 - 2250 kg - EAPS ON.

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TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % V2 30 kts IAS

EAPS ON ELECTRICAL LOAD: 150 A

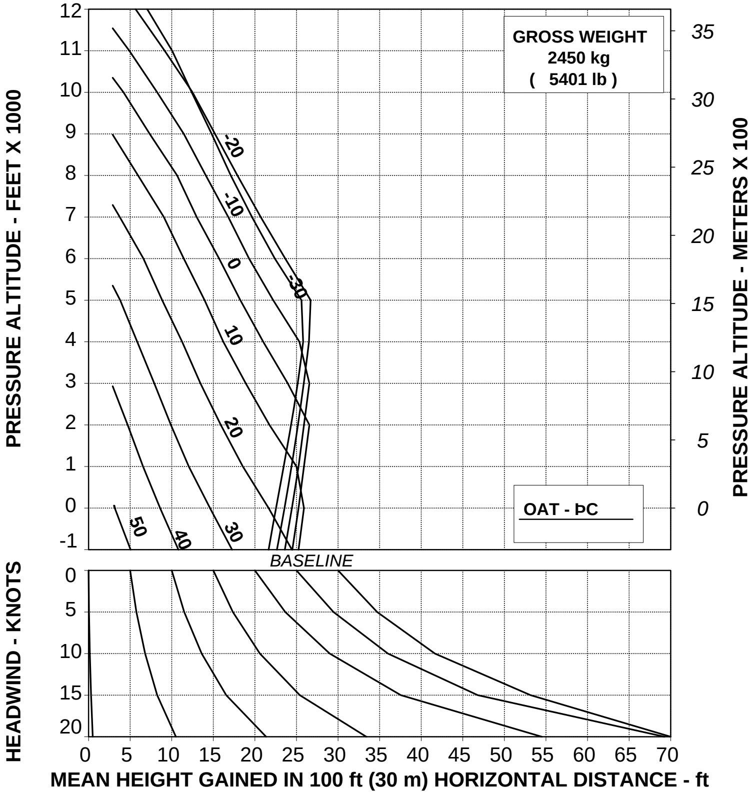

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Figure 4-80. Take-off flight path 1 - 2450 kg - EAPS ON.

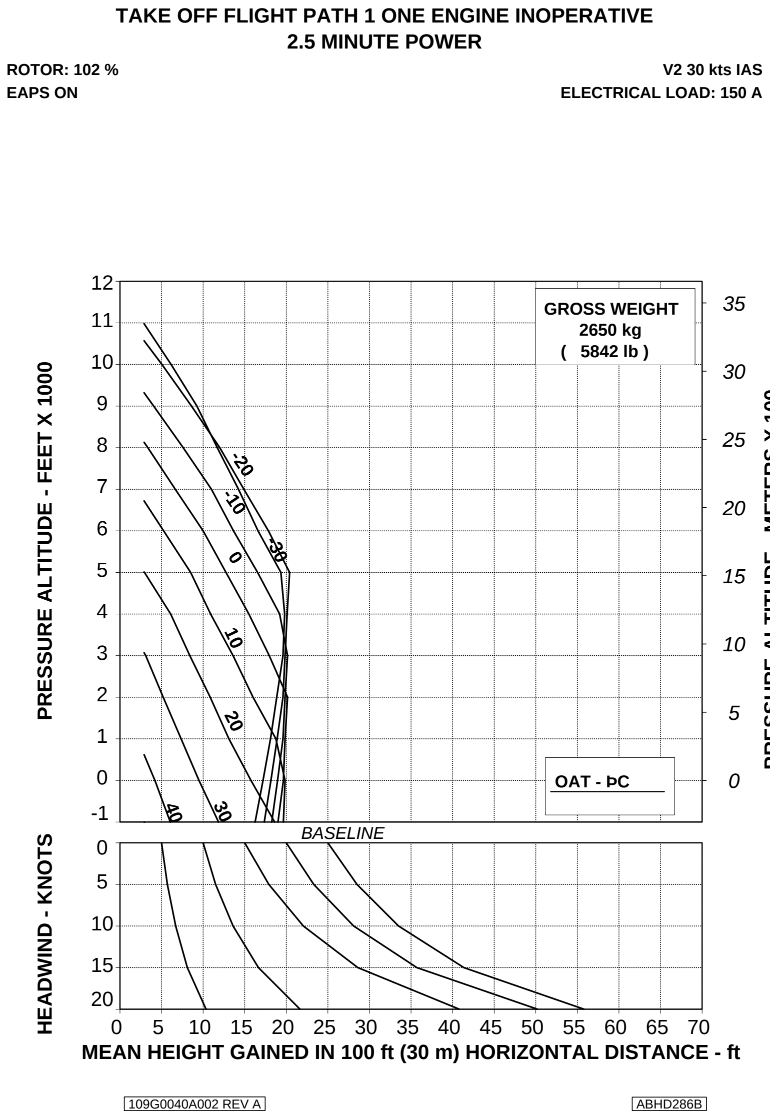

Figure 4-81. Take-off flight path 1 - 2650 kg - EAPS ON.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS ON

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

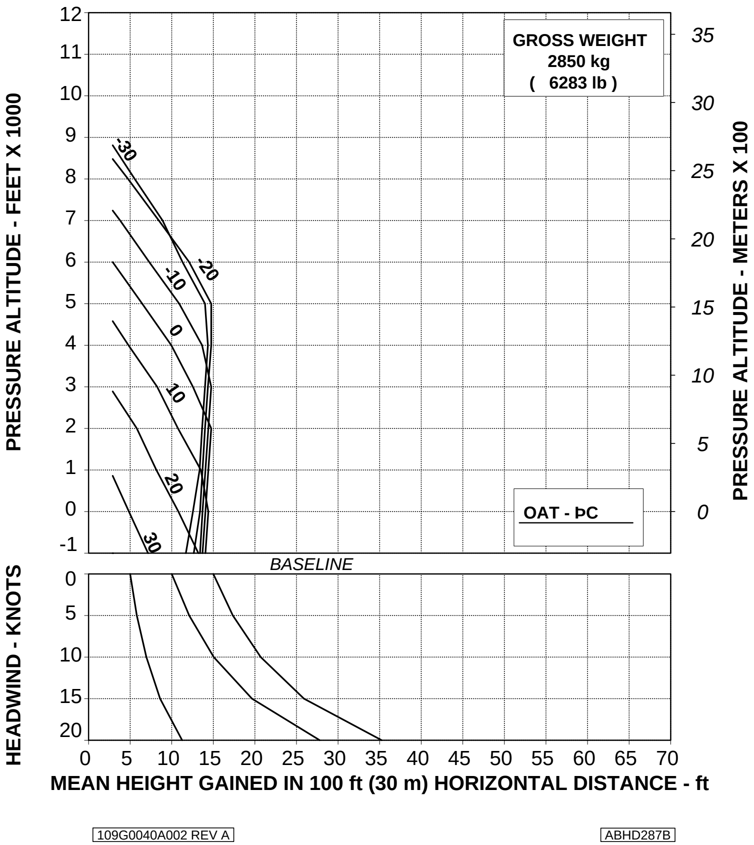

Figure 4-82. Take-off flight path 1 - 2850 kg - EAPS ON.

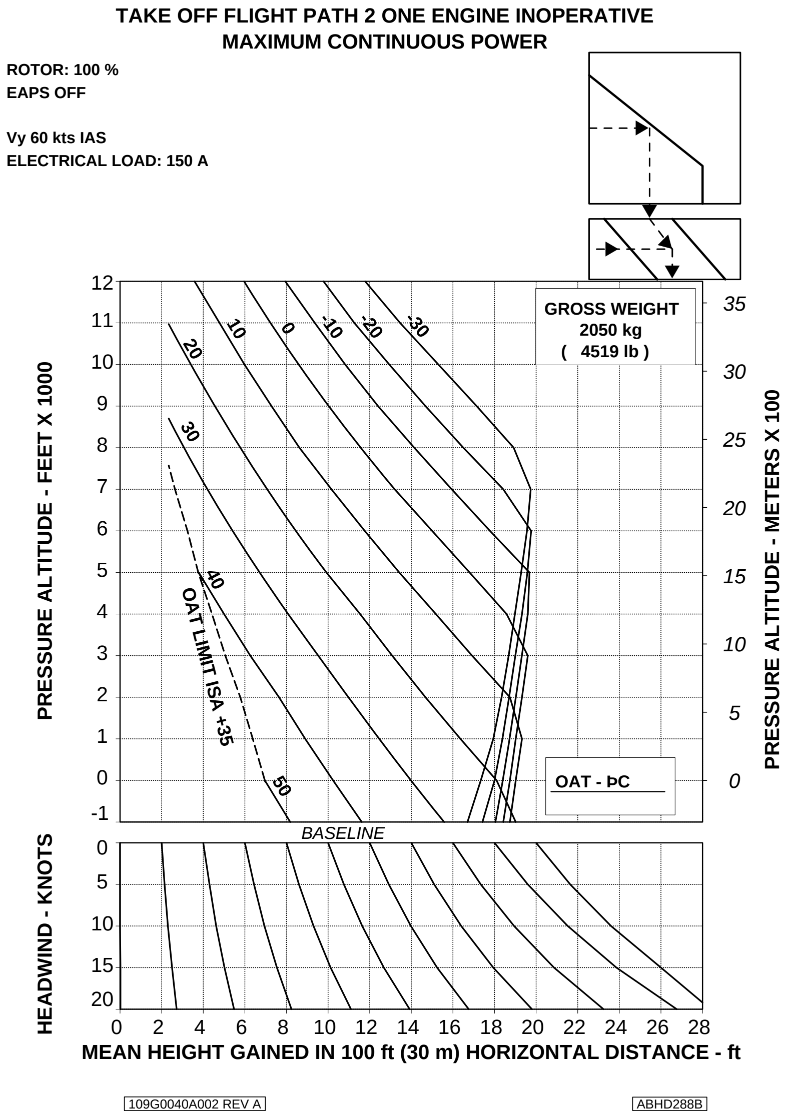

Figure 4-83. Take-off flight path 2 - 2050 kg - EAPS OFF.

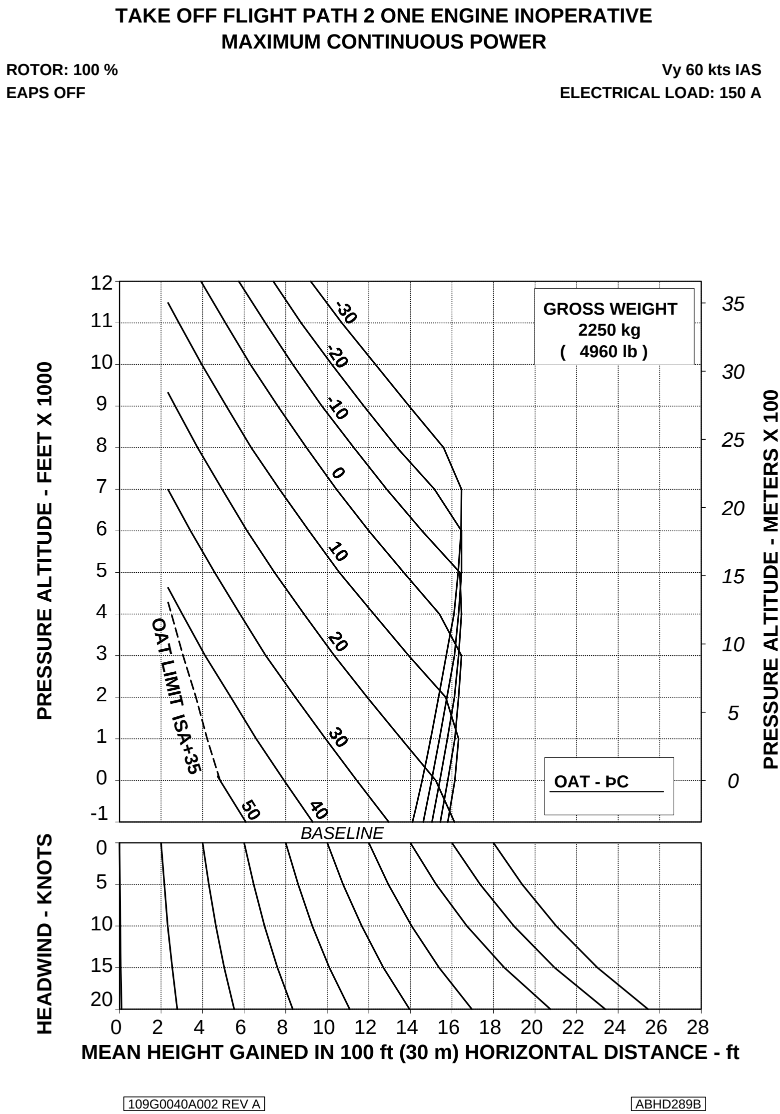

Figure 4-84. Take-off flight path 2 - 2250 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % Vy 60 kts IAS
EAPS OFF ELECTRICAL LOAD: 150 A

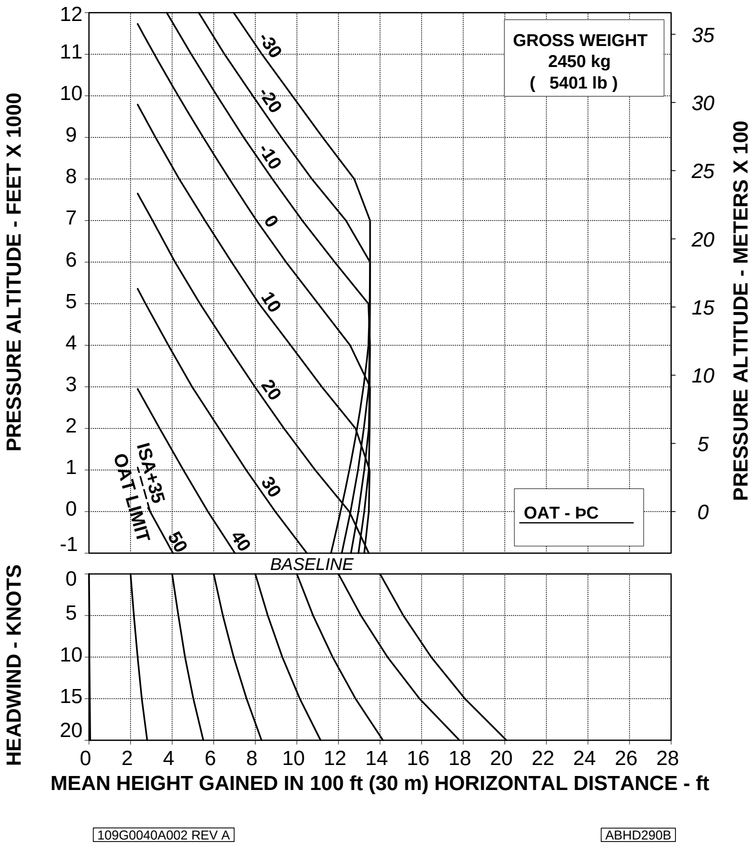

Figure 4-85. Take-off flight path 2 - 2450 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

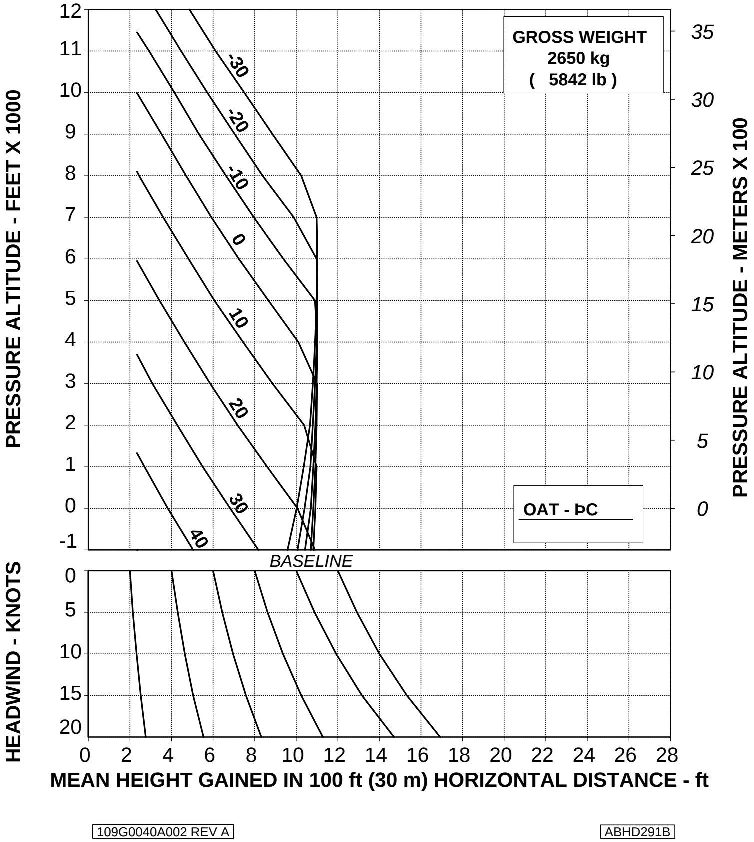

Figure 4-86. Take-off flight path 2 - 2650 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | Vy 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A

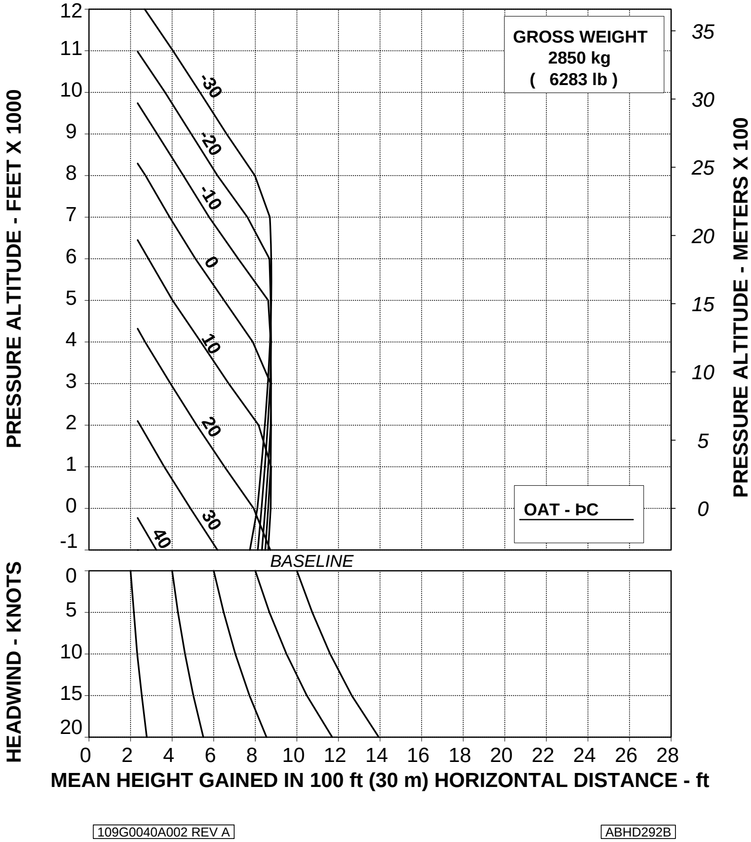

Figure 4-87. Take-off flight path 2 - 2850 kg - EAPS OFF.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A**

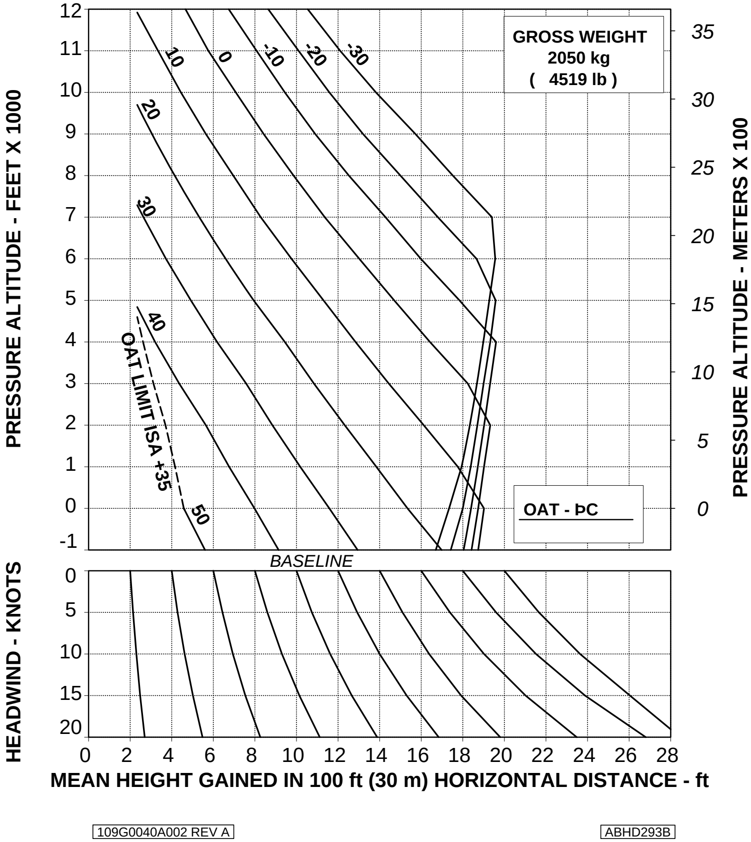

Figure 4-88. Take-off flight path 2 - 2050 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | Vy 60 kts IAS
EAPS ON | ELECTRICAL LOAD: 150 A

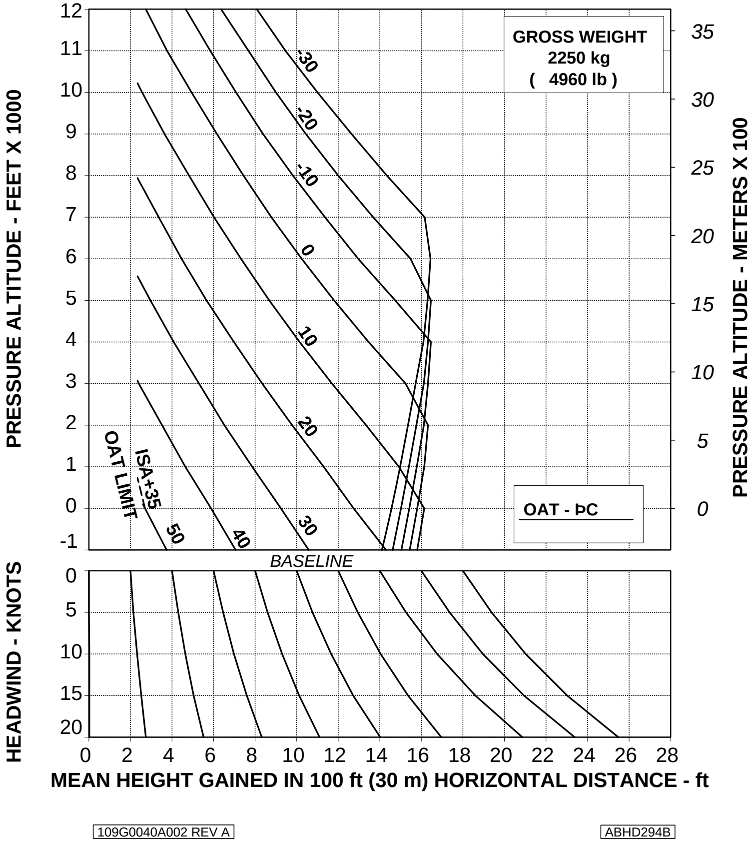

Figure 4-89. Take-off flight path 2 - 2250 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **Vy 60 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A**

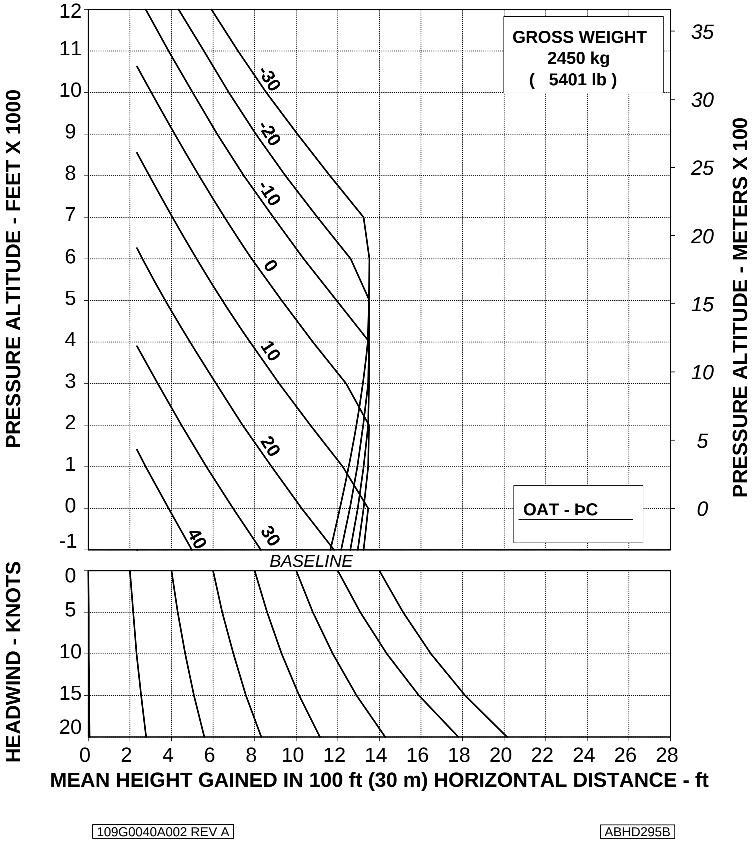

Figure 4-90. Take-off flight path 2 - 2450 kg - EAPS ON.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % Vy 60 kts IAS
EAPS ON ELECTRICAL LOAD: 150 A

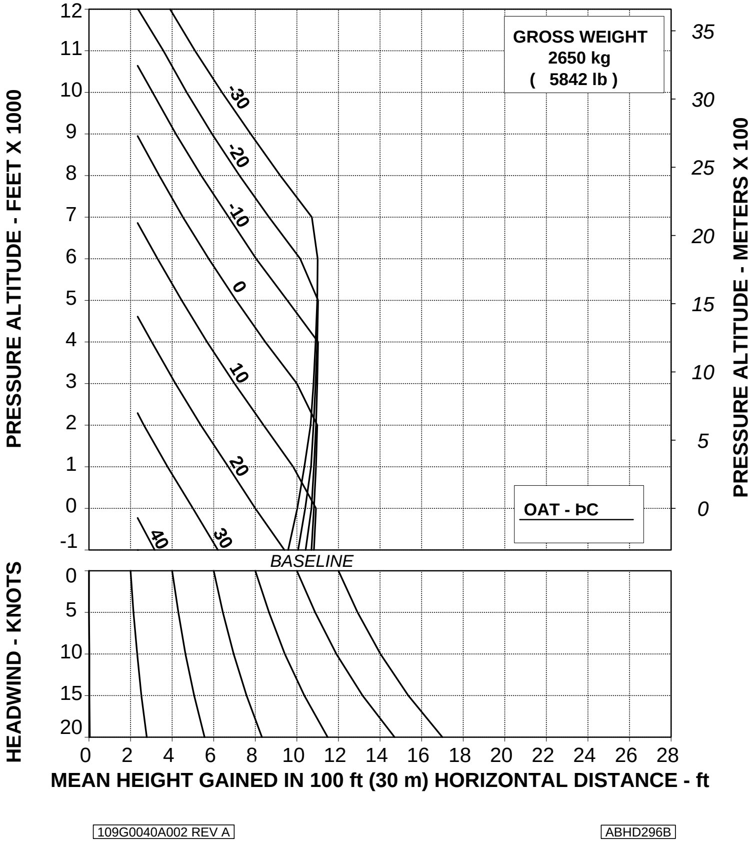

Figure 4-91. Take-off flight path 2 - 2650 kg - EAPS ON.

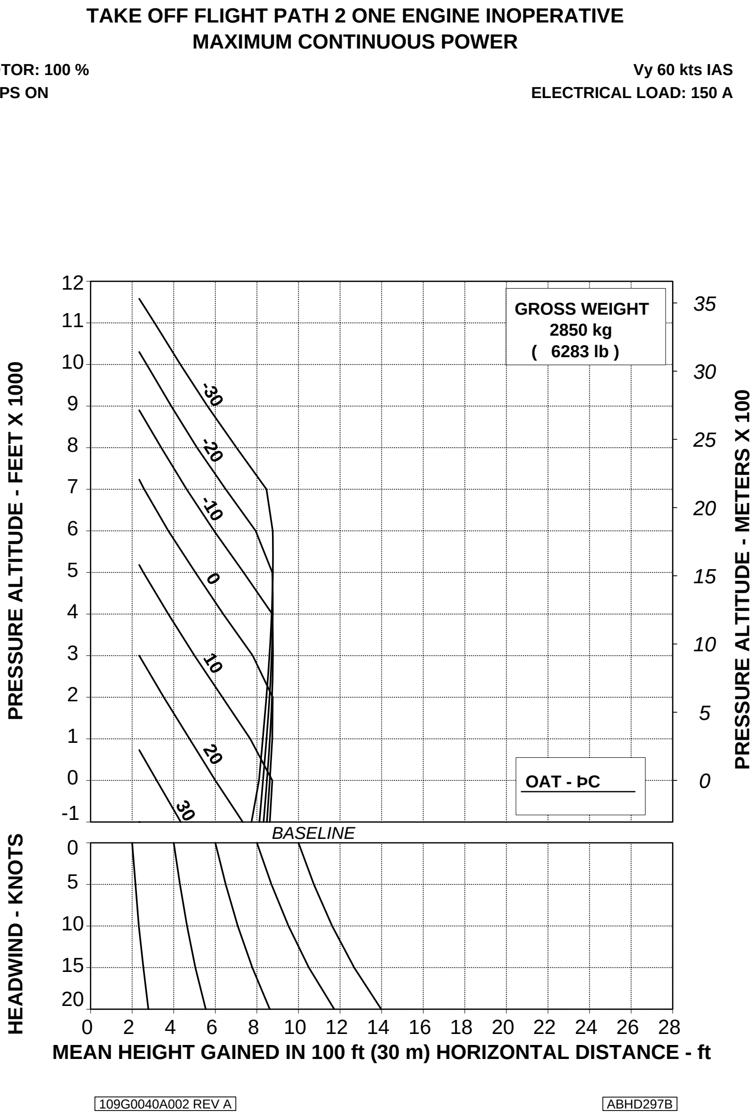

Figure 4-92. Take-off flight path 2 - 2850 kg - EAPS ON.

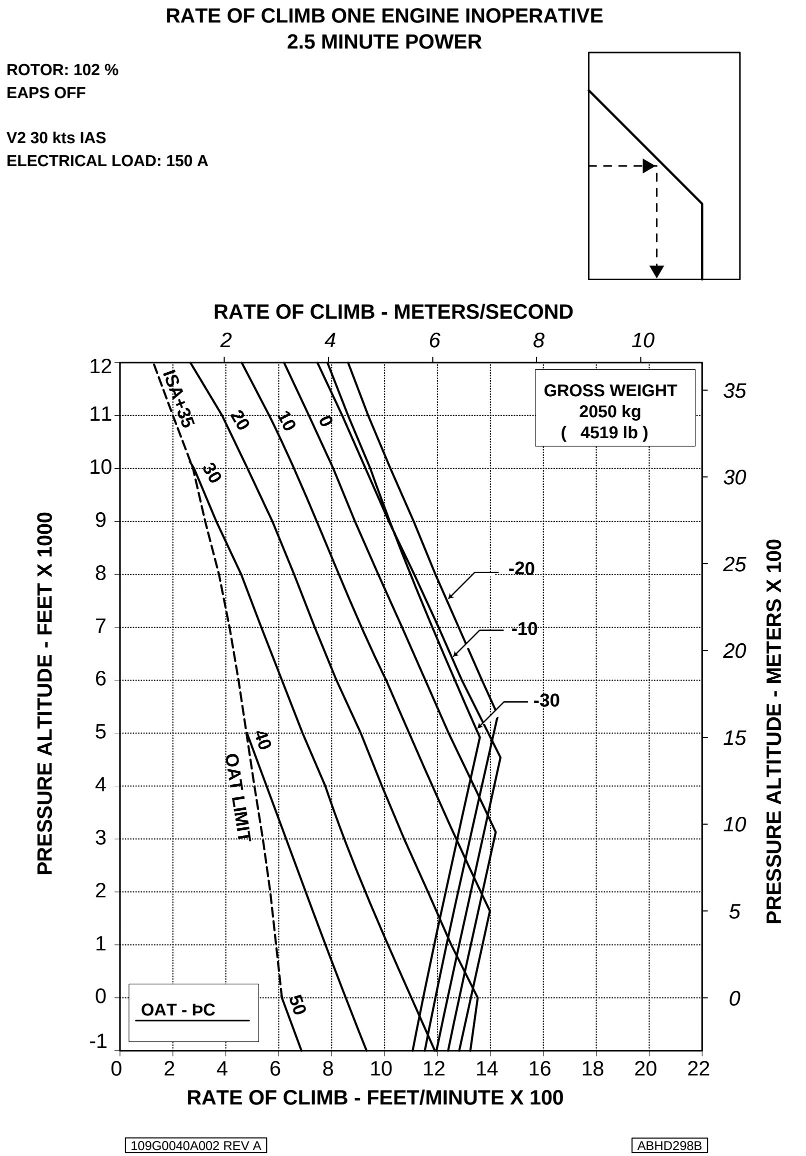

Figure 4-93. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2050 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % | V2 30 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A

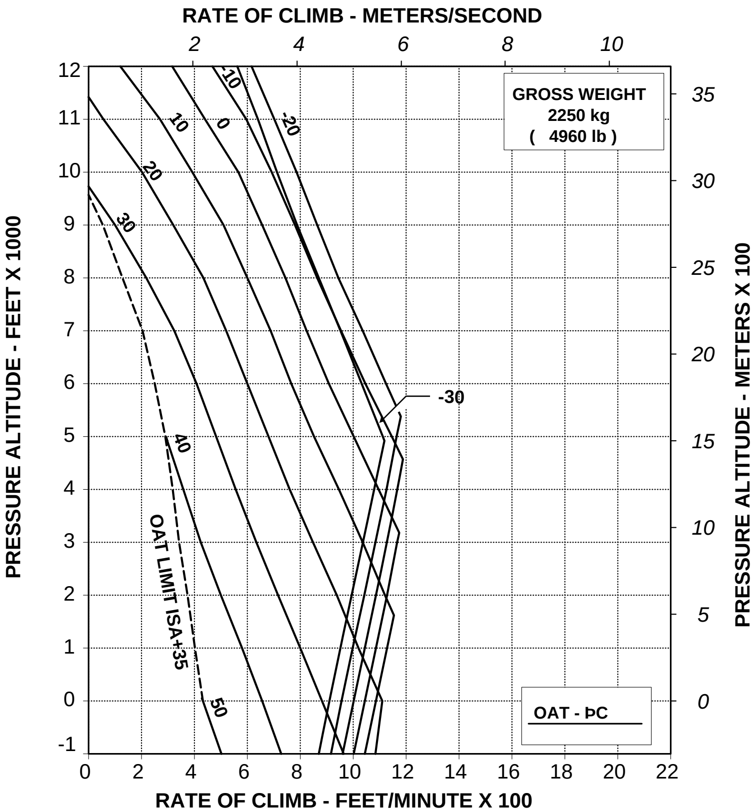

Figure 4-94. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2250 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

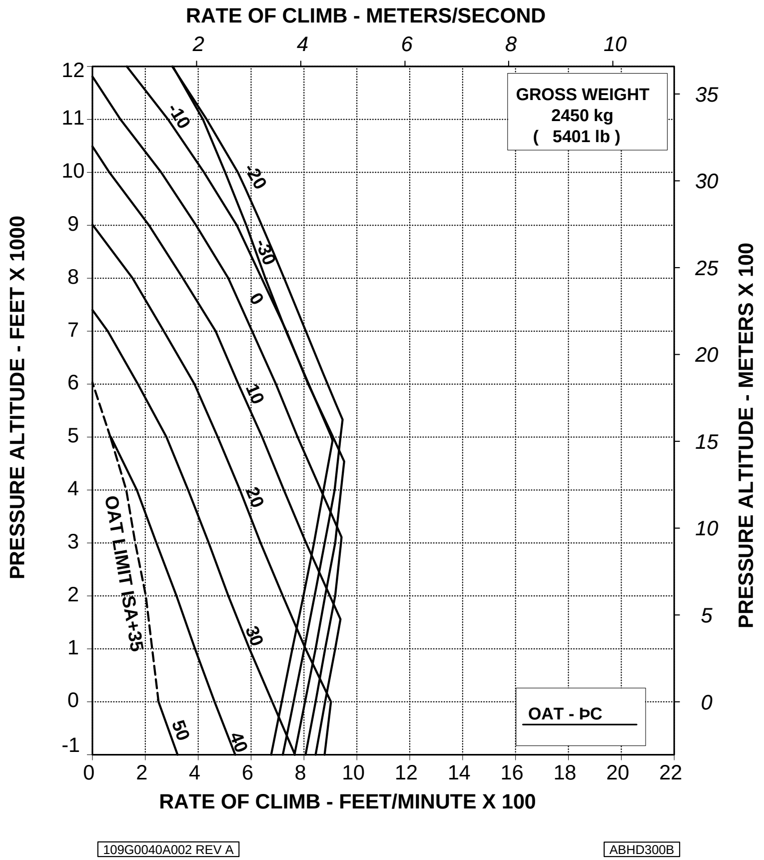

Figure 4-95. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2450 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

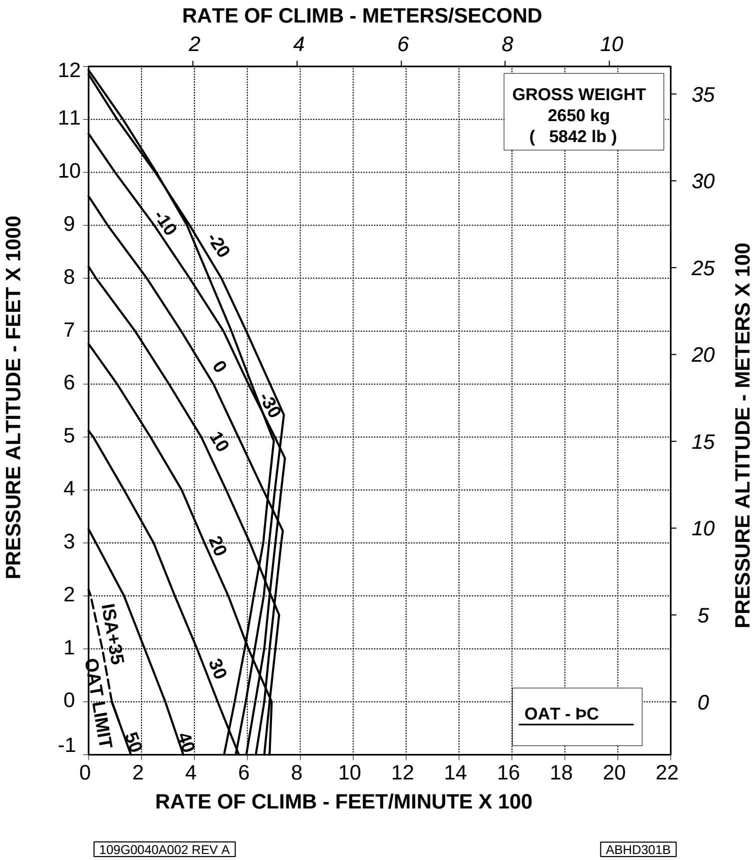

Figure 4-96. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2650 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % | V2 30 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A

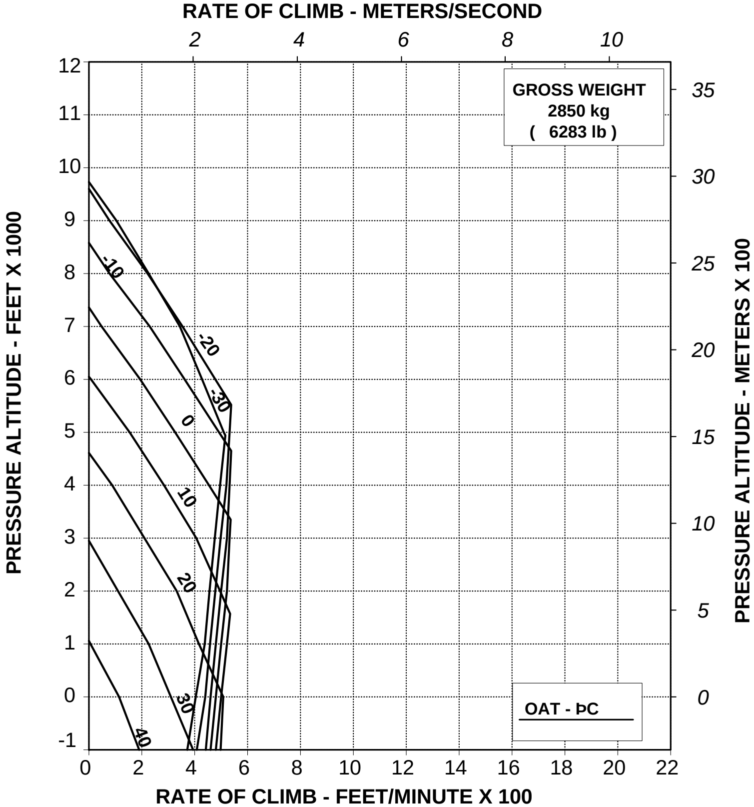

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Figure 4-97. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2850 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**

EAPS ON **ELECTRICAL LOAD: 150 A**

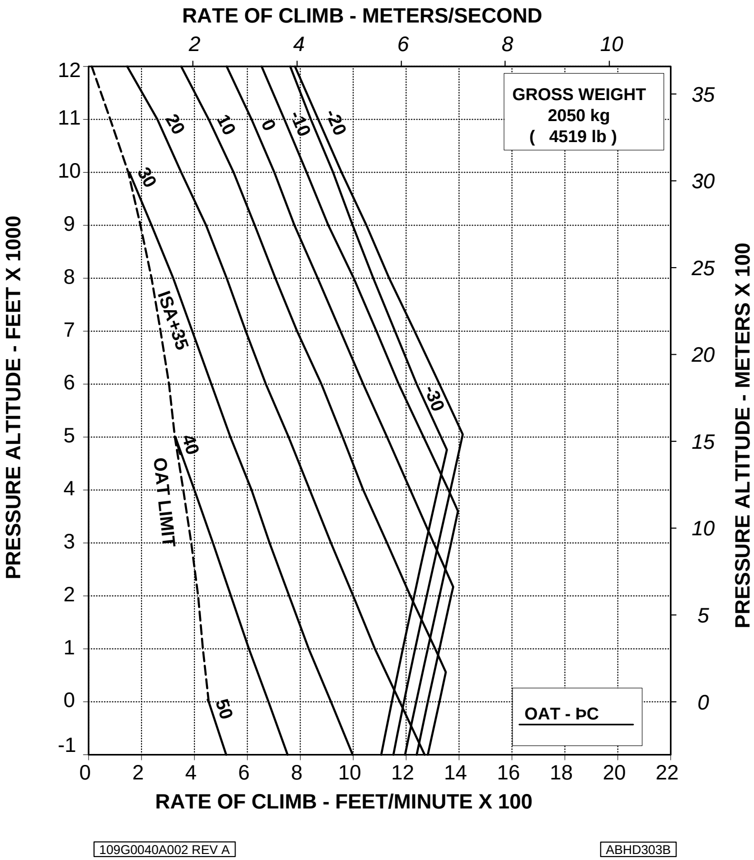

Figure 4-98. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2050 kg - EAPS ON.

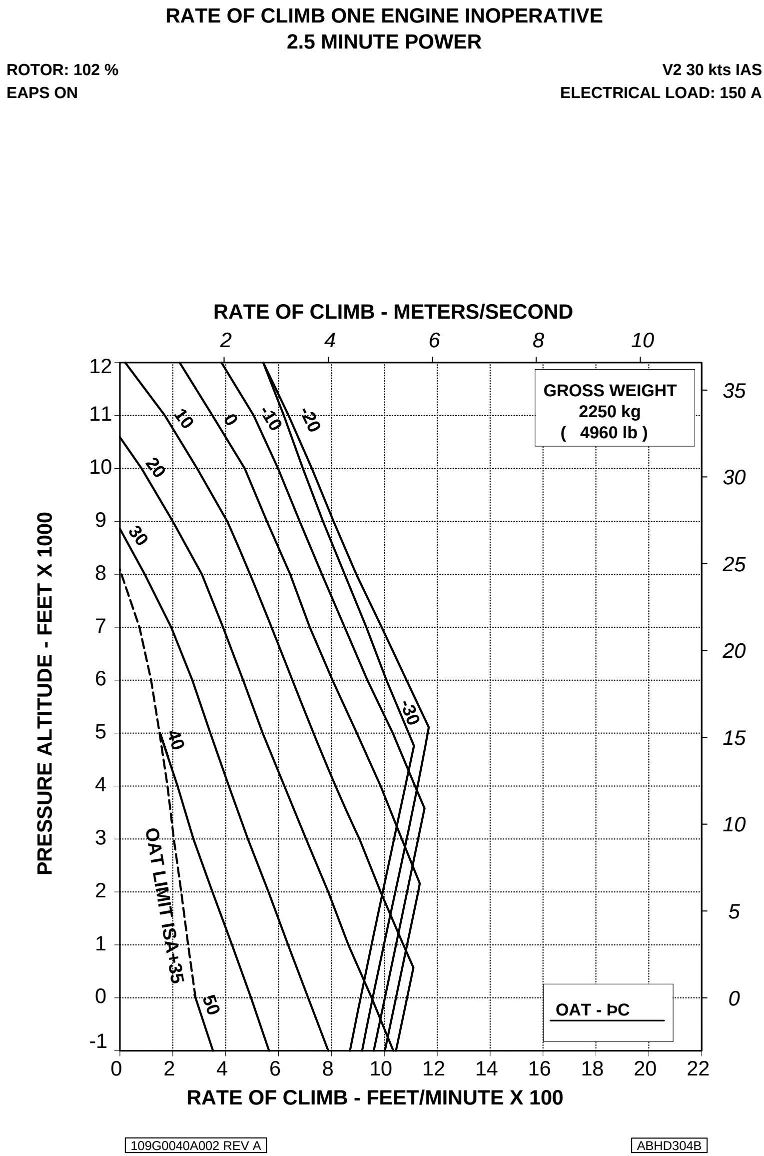

Figure 4-99. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2250 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
EAPS ON **ELECTRICAL LOAD: 150 A**

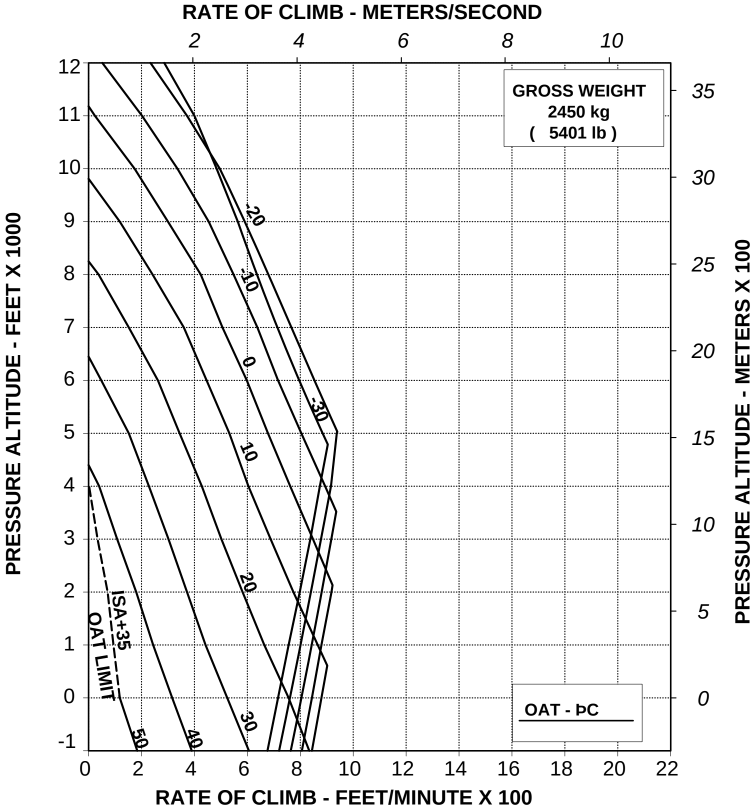

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Figure 4-100. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2450 kg - EAPS ON.

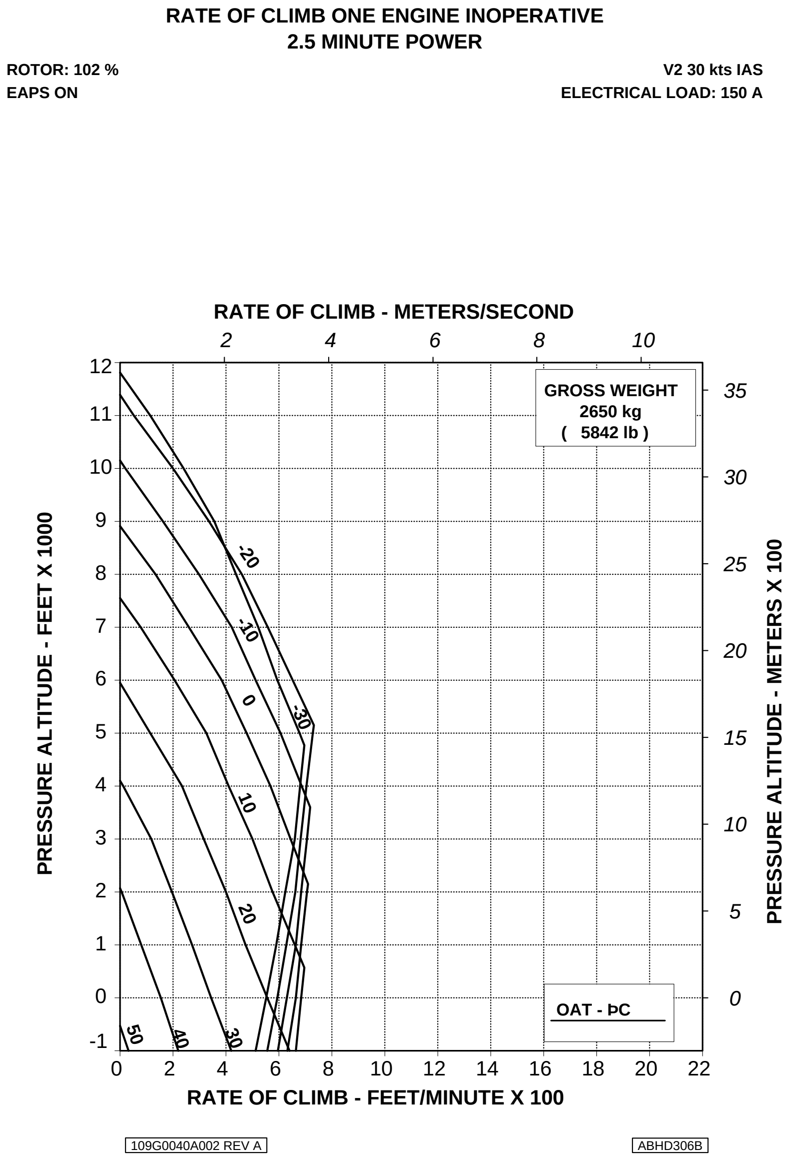

Figure 4-101. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2650 kg - EAPS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %
EAPS ON

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

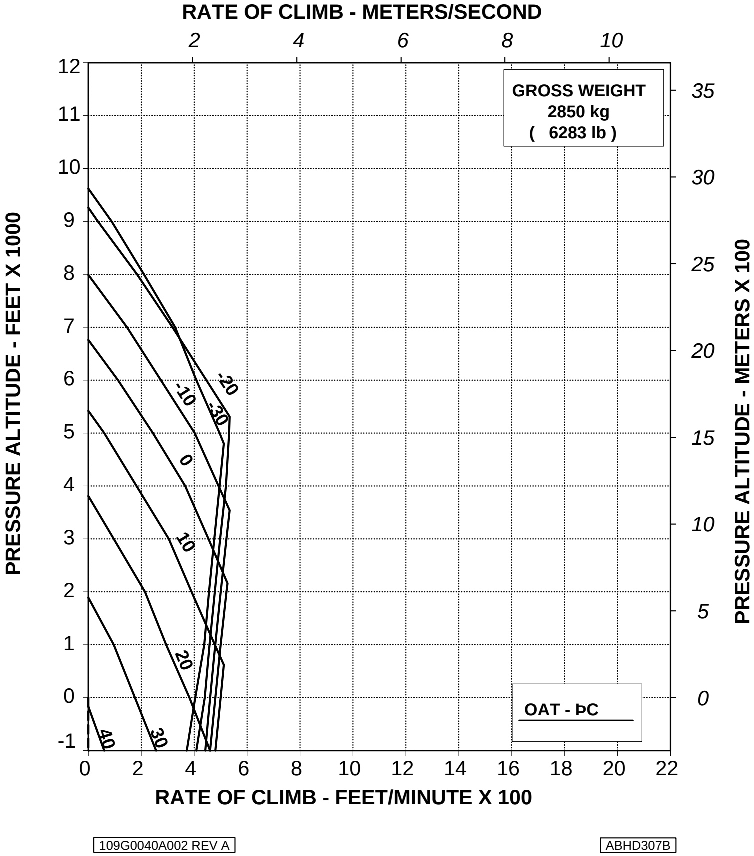

Figure 4-102. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2850 kg - EAPS ON.

CATEGORY ″A″ OPERATIONS - TRAINING PROCEDURES (WITH EAPS)

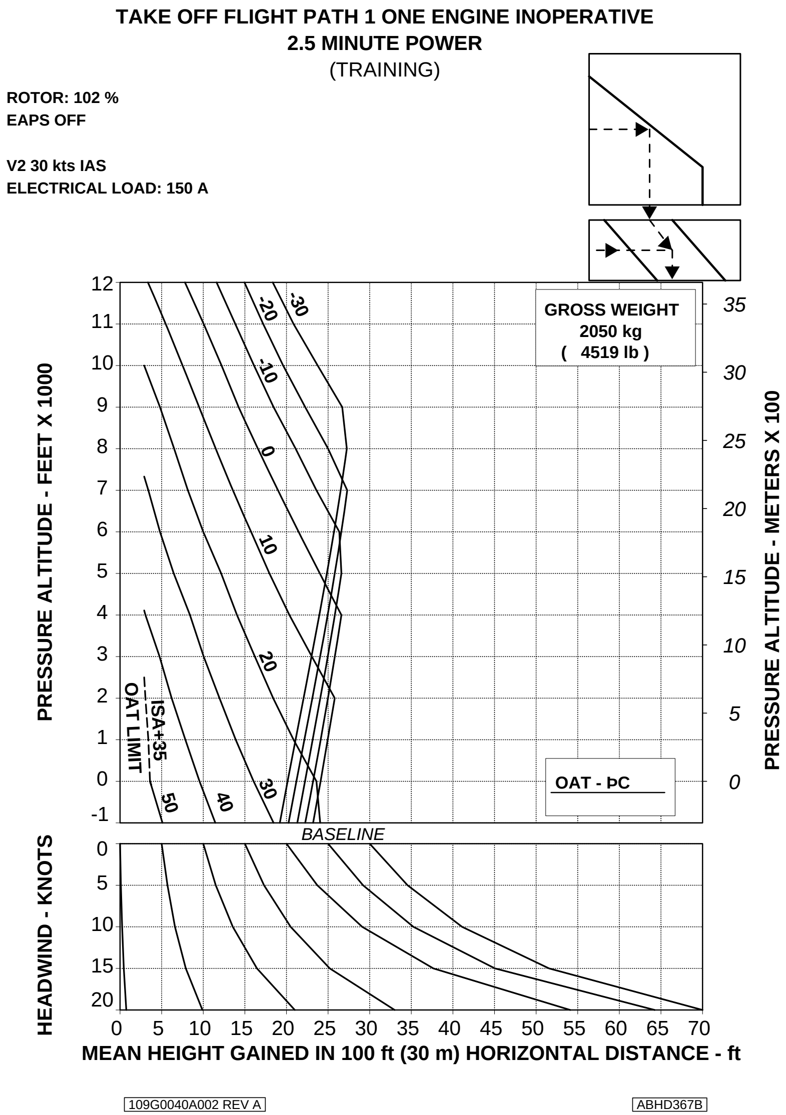

Figure 4-103. Take-off flight path 1 - 2050 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

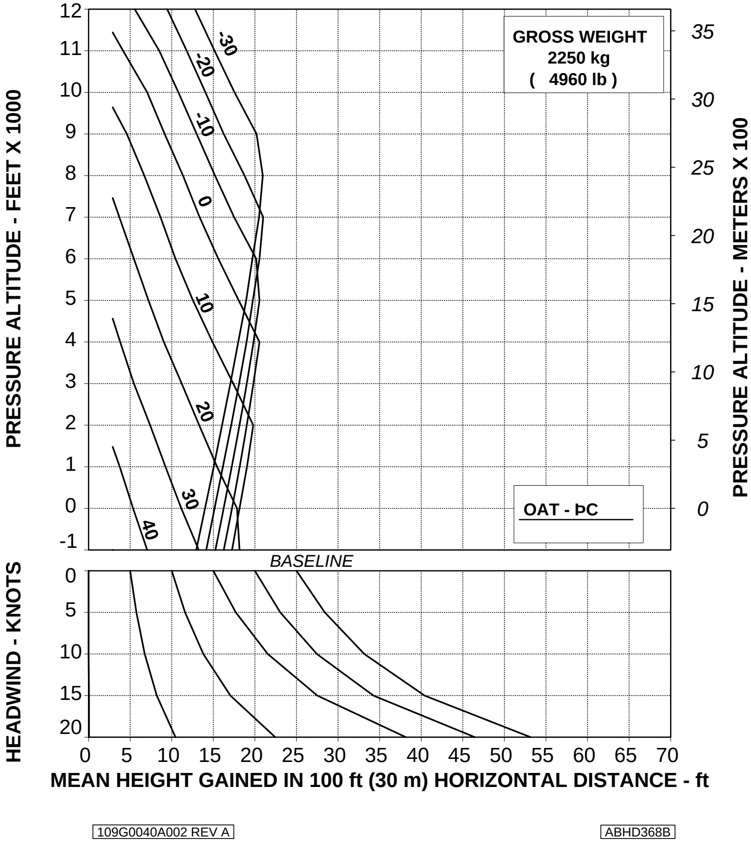

Figure 4-104. Take-off flight path 1 - 2250 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 % | V2 30 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A

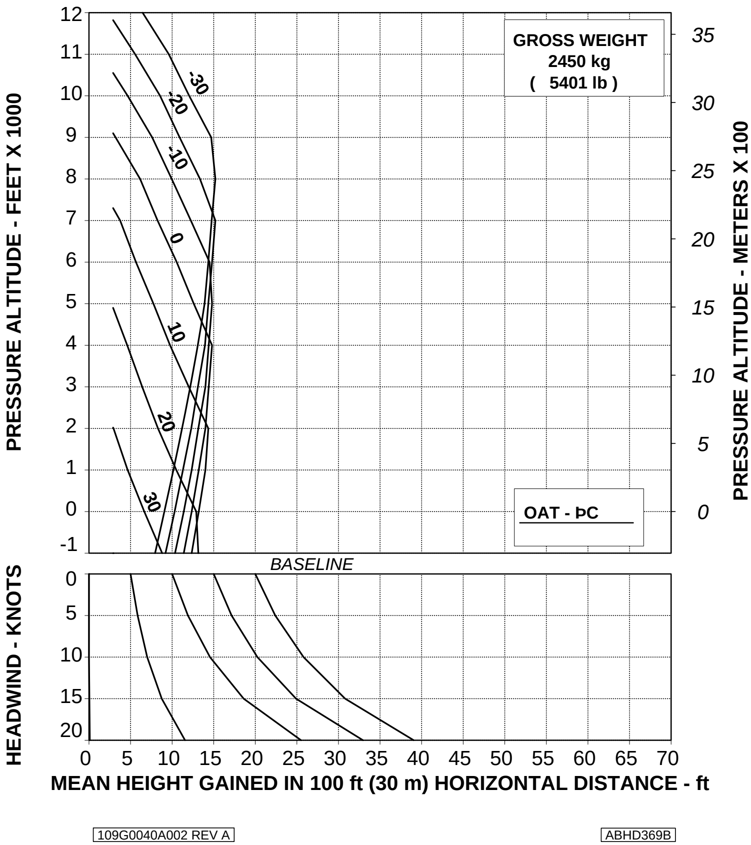

Figure 4-105. Take-off flight path 1 - 2450 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 % **V2 30 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

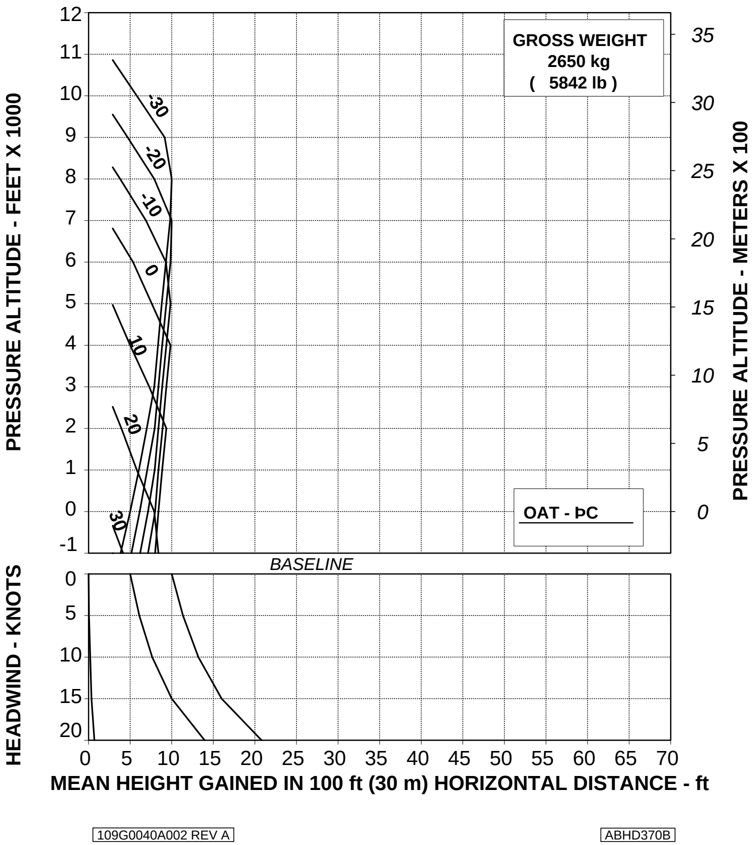

Figure 4-106. Take-off flight path 1 - 2650 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 1 ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %
EAPS OFF

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

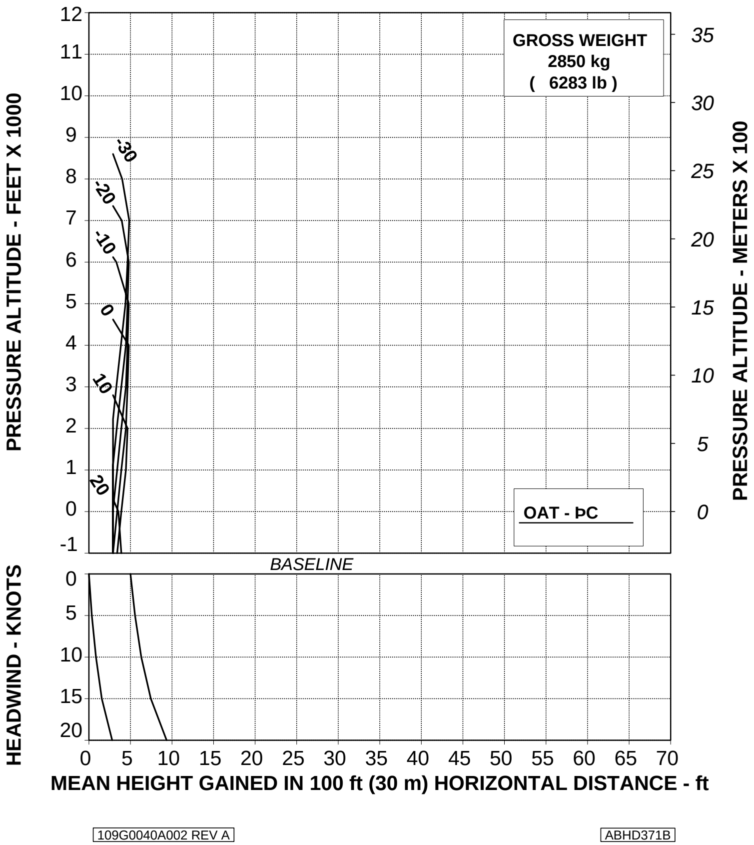

Figure 4-107. Take-off flight path 1 - 2850 kg - EAPS OFF - Training.

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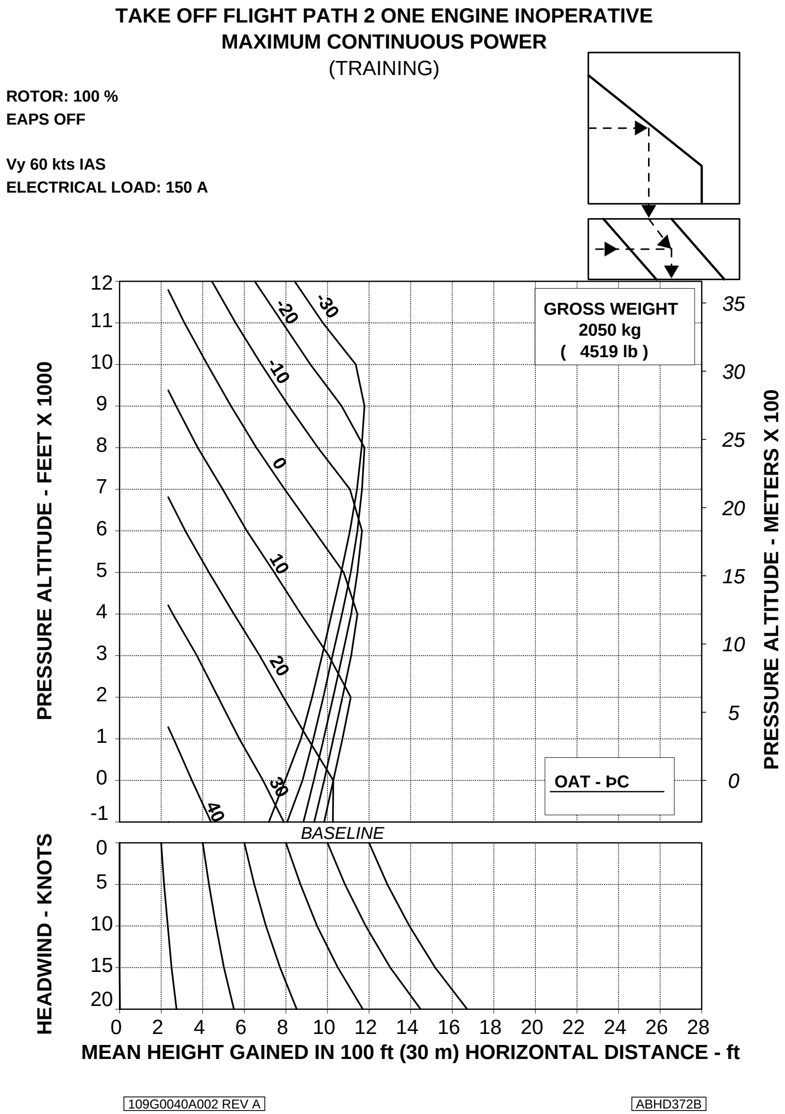

Figure 4-108. Take-off flight path 2 - 2050 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

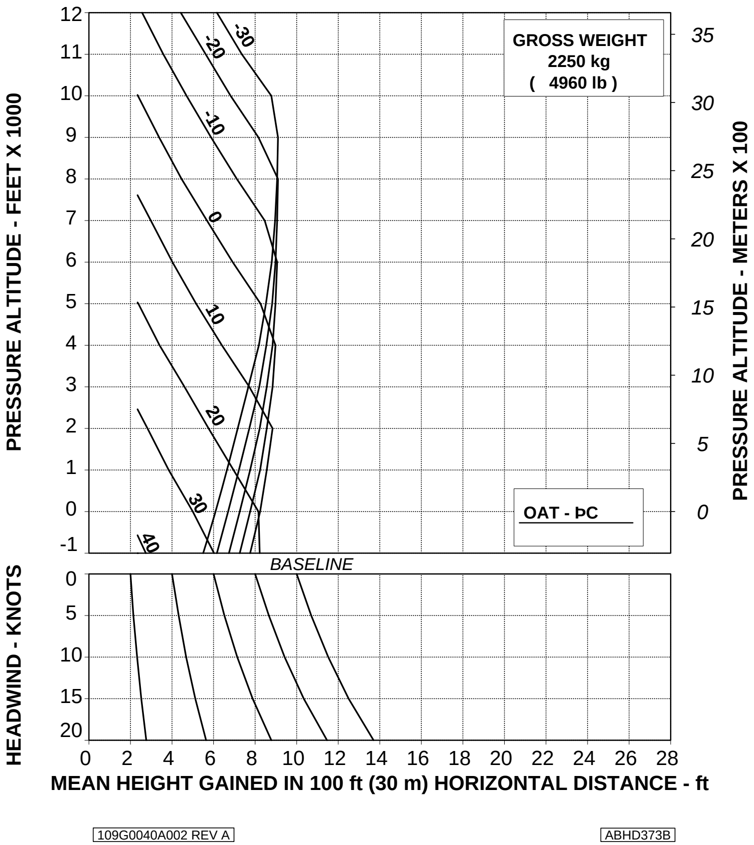

Figure 4-109. Take-off flight path 2 - 2250 kg - EAPS OFF - Training.

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TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

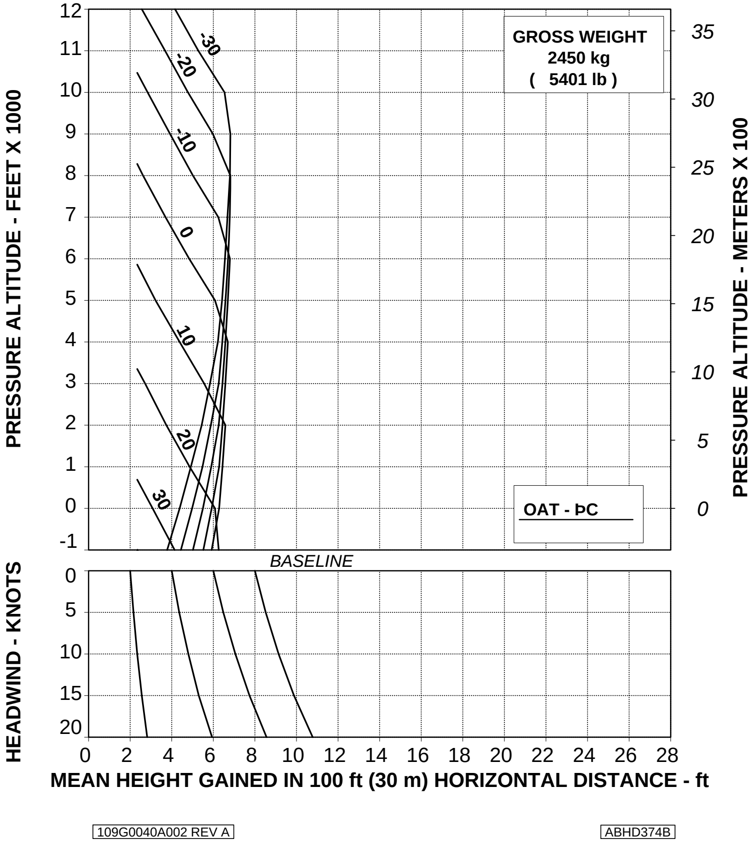

Figure 4-110. Take-off flight path 2 - 2450 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

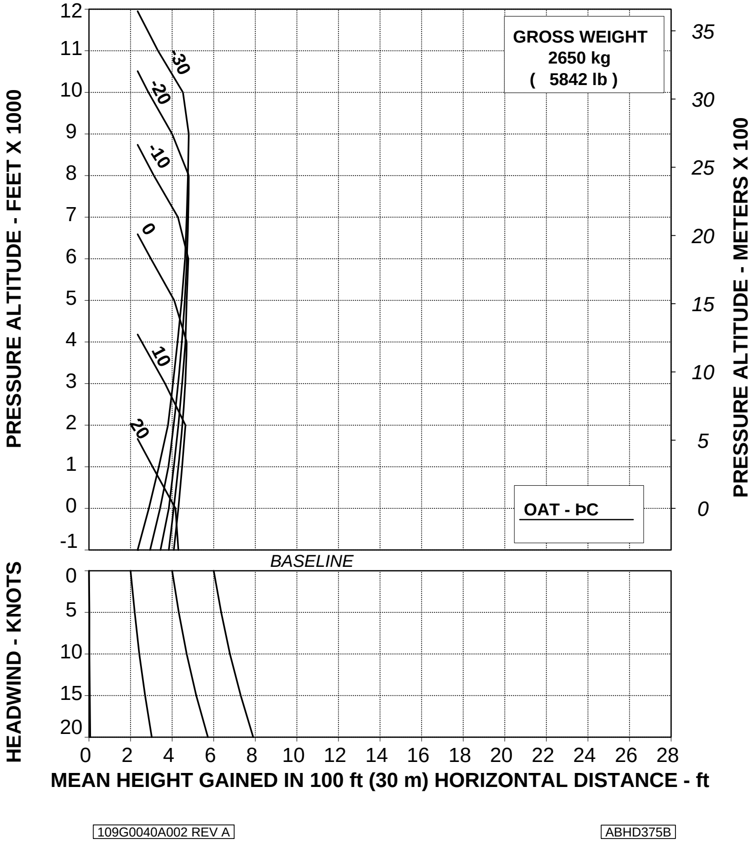

Figure 4-111. Take-off flight path 2 - 2650 kg - EAPS OFF - Training.

TAKE OFF FLIGHT PATH 2 ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 % **Vy 60 kts IAS**
EAPS OFF **ELECTRICAL LOAD: 150 A**

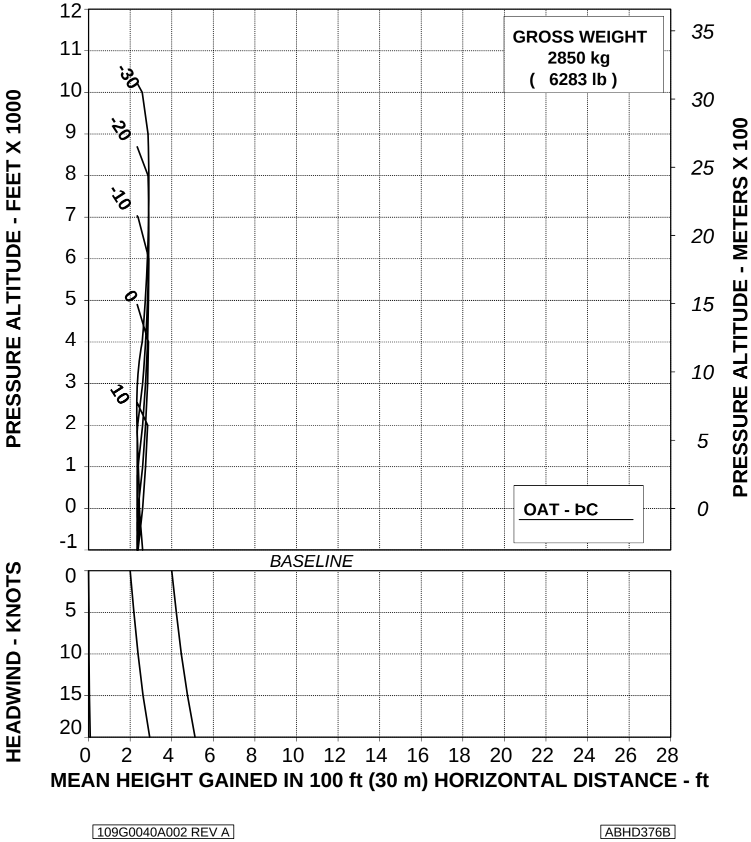

Figure 4-112. Take-off flight path 2 - 2850 kg - EAPS OFF - Training.

CARGO HOOK OPERATIONS (WITH EAPS)

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
CARGO HOOK OPERATION

ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

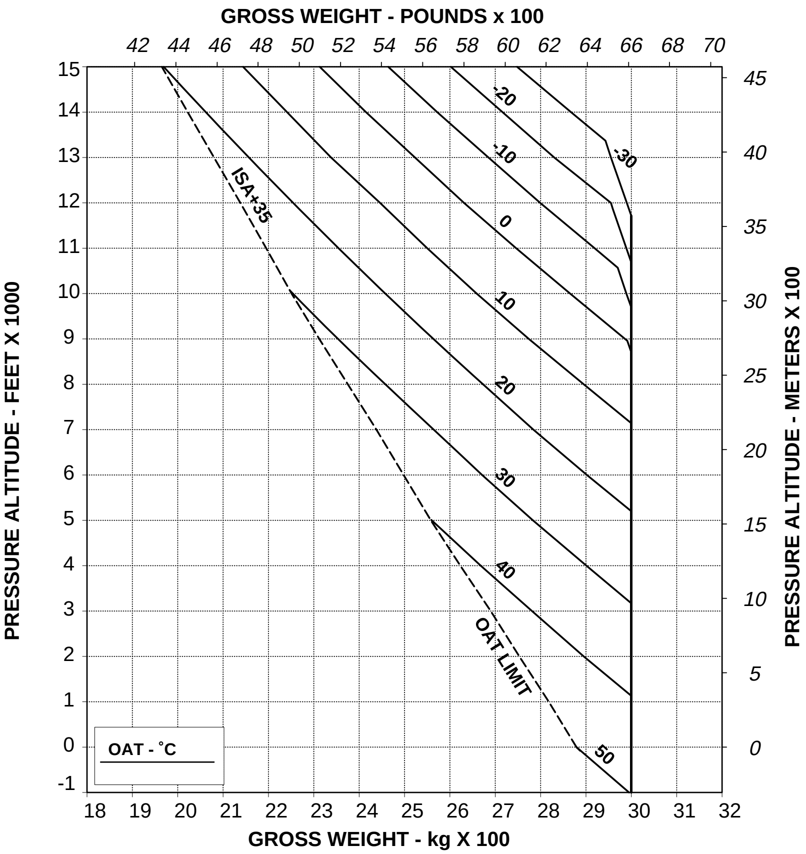

Figure 4-113. Hovering ceiling - OGE - Take-off power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
CARGO HOOK OPERATION

ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

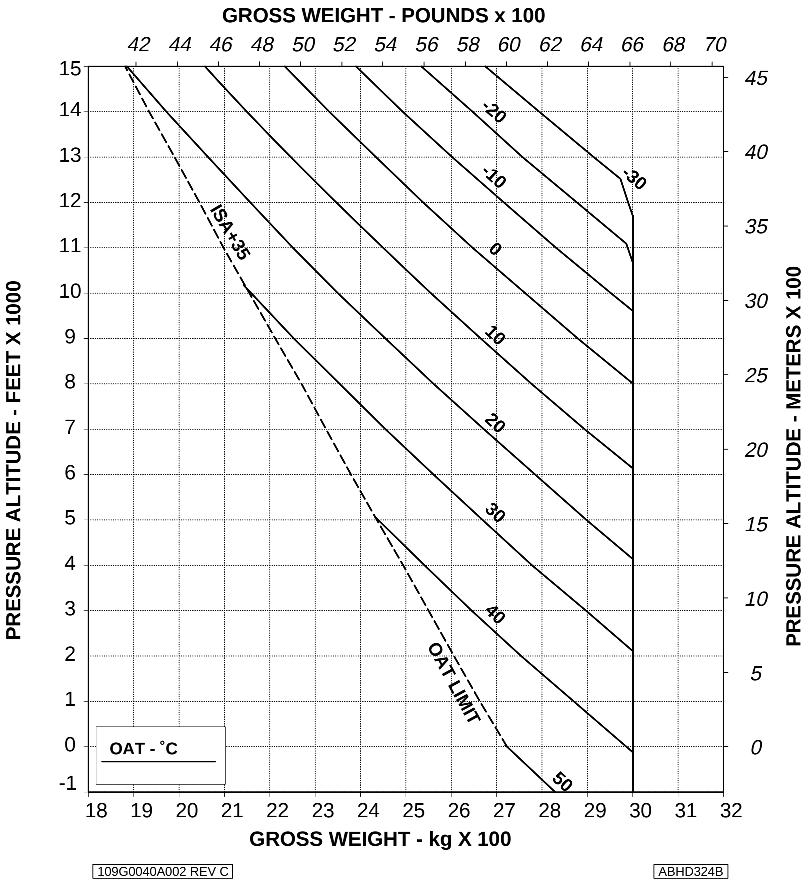

Figure 4-114. Hovering ceiling - OGE - Take-off power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
CARGO HOOK OPERATION

HEATER OR E.C.S. ON
ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

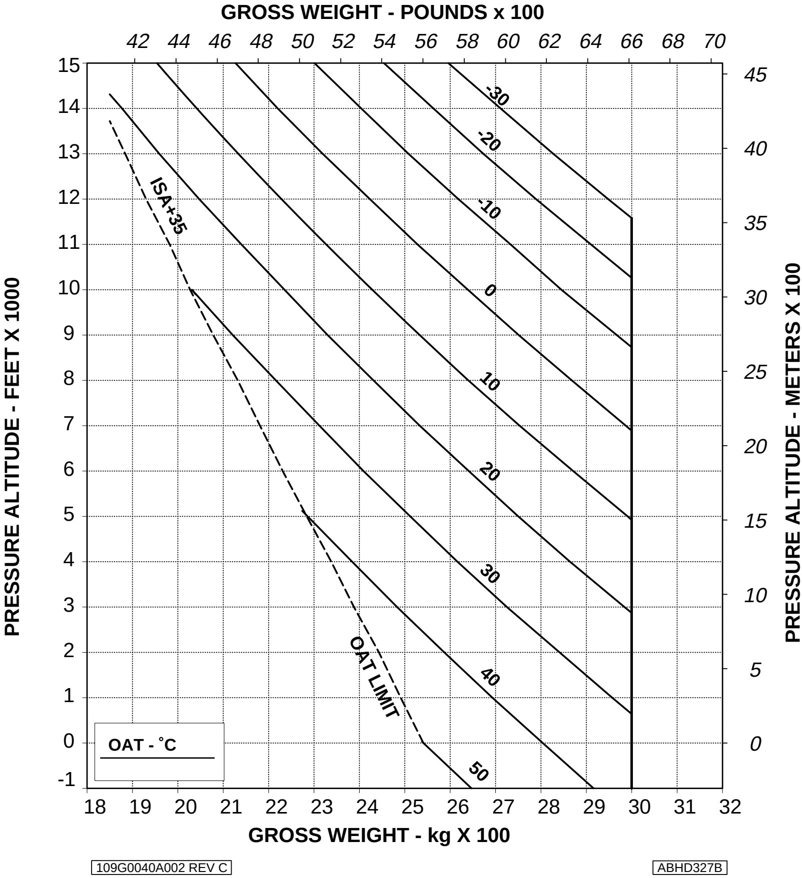

Figure 4-115. Hovering ceiling - OGE - Take-off power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
CARGO HOOK OPERATION

ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

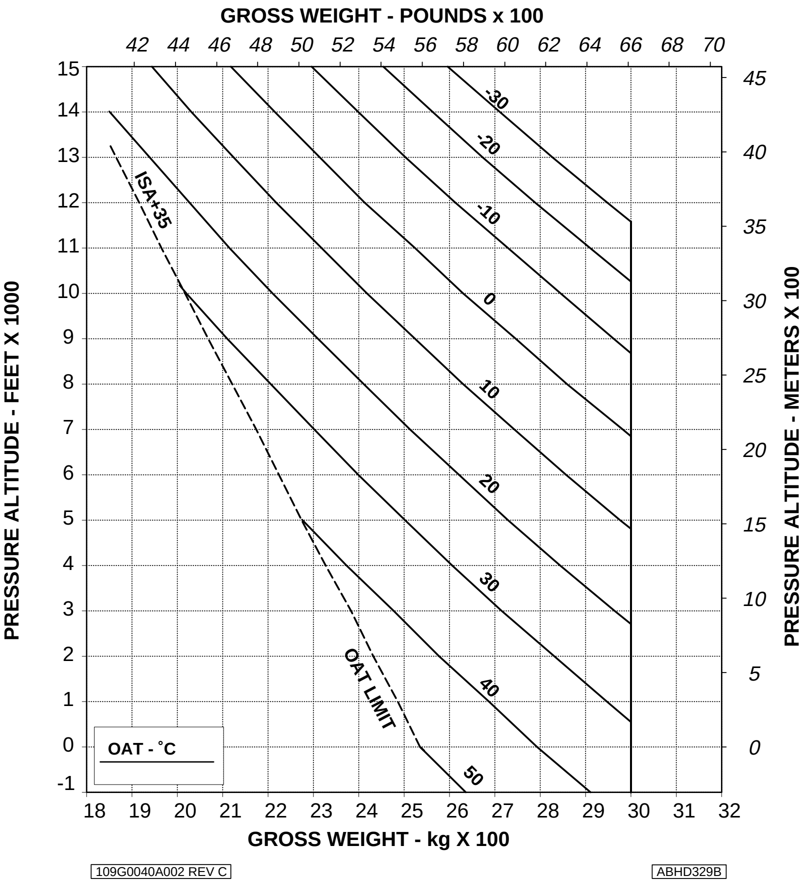

Figure 4-116. Hovering ceiling - OGE - Maximum continuous power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
CARGO HOOK OPERATION

ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

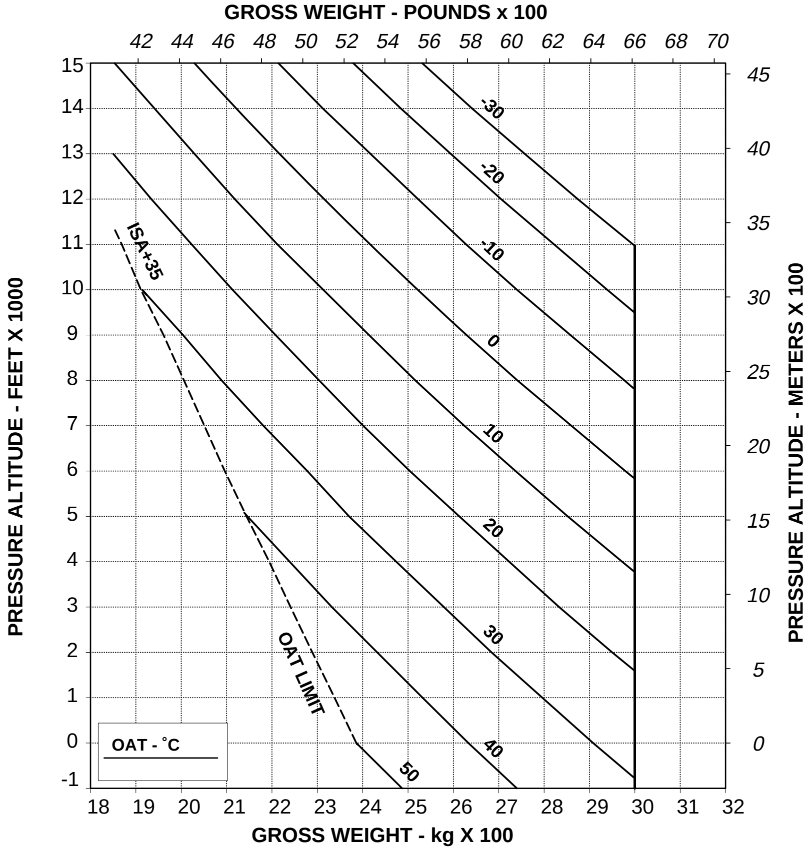

Figure 4-117. Hovering ceiling - OGE - Maximum continuous power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
CARGO HOOK OPERATION

HEATER OR E.C.S. ON
ELECTRICAL LOAD: 150 A TOTAL
ZERO WIND

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

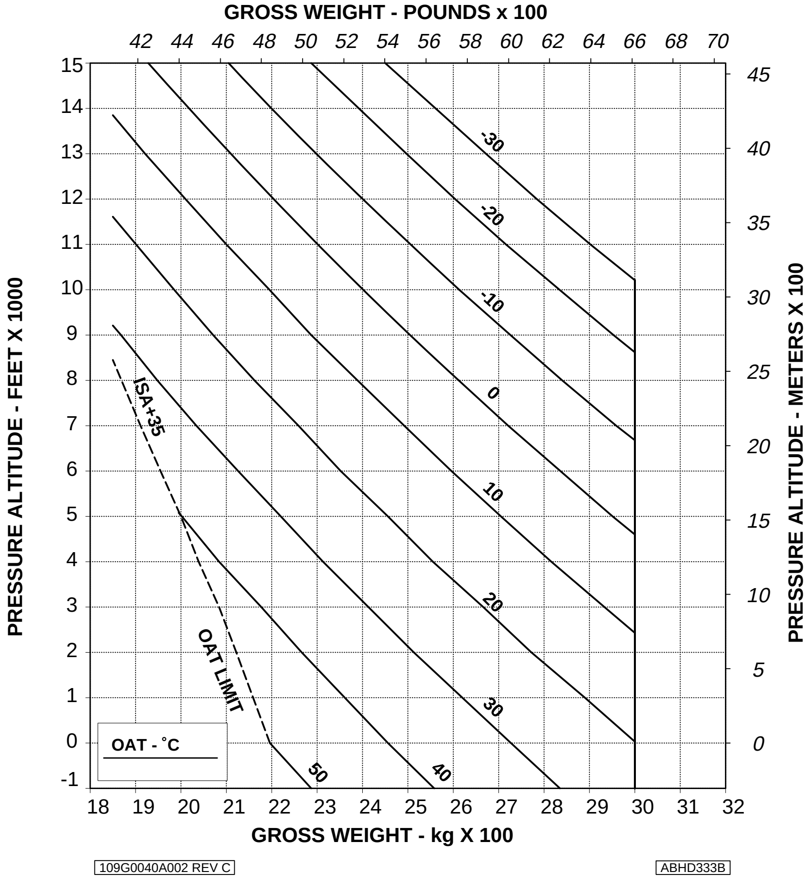

Figure 4-118. Hovering ceiling - OGE - Maximum continuous power - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % | 60 kts IAS
EAPS OFF | ELECTRICAL LOAD: 150 A TOTAL
CARGO HOOK OPERATION

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27 m/s)

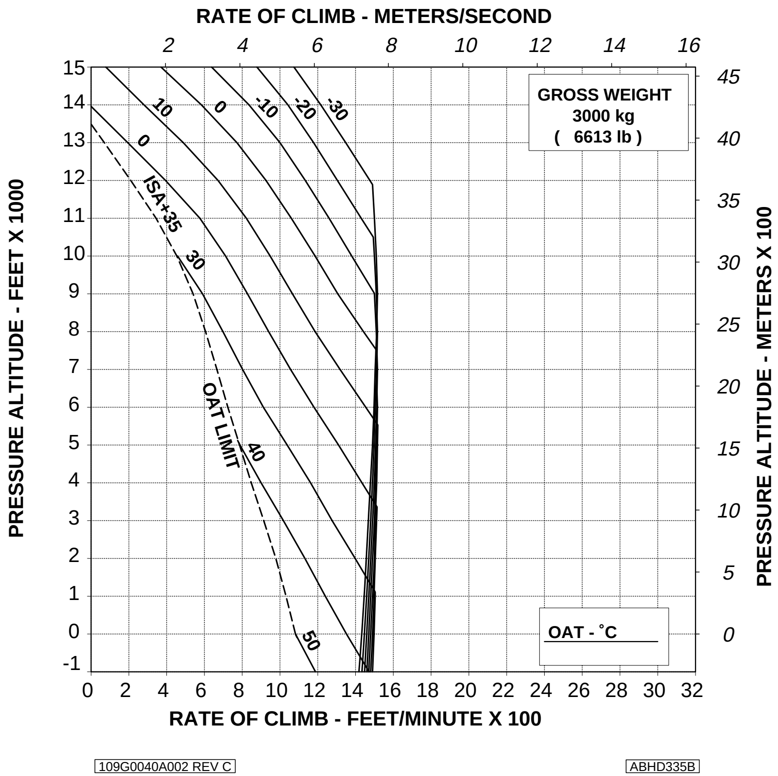

Figure 4-119. Rate of climb - All engines - Take-off power - 3000 kg - EAPS OFF.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100.00 %
EAPS OFF
CARGO HOOK OPERATION

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

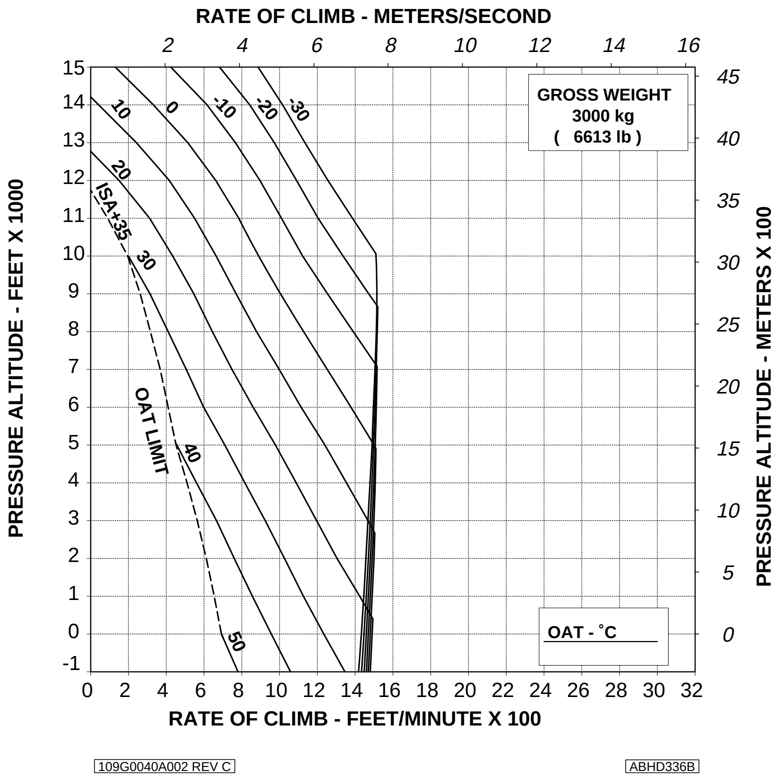

Figure 4-120. Rate of climb - All engines - Take-off power - 3000 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
CARGO HOOK OPERATION

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

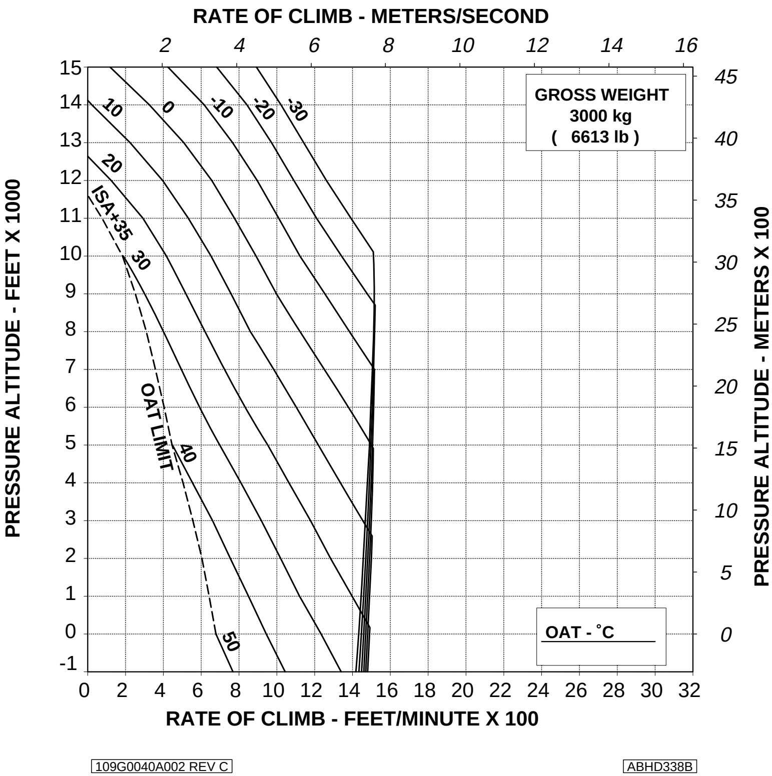

Figure 4-121. Rate of climb - All engines - Maximum continuous power - 3000 kg - EAPS OFF.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
CARGO HOOK OPERATION

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)

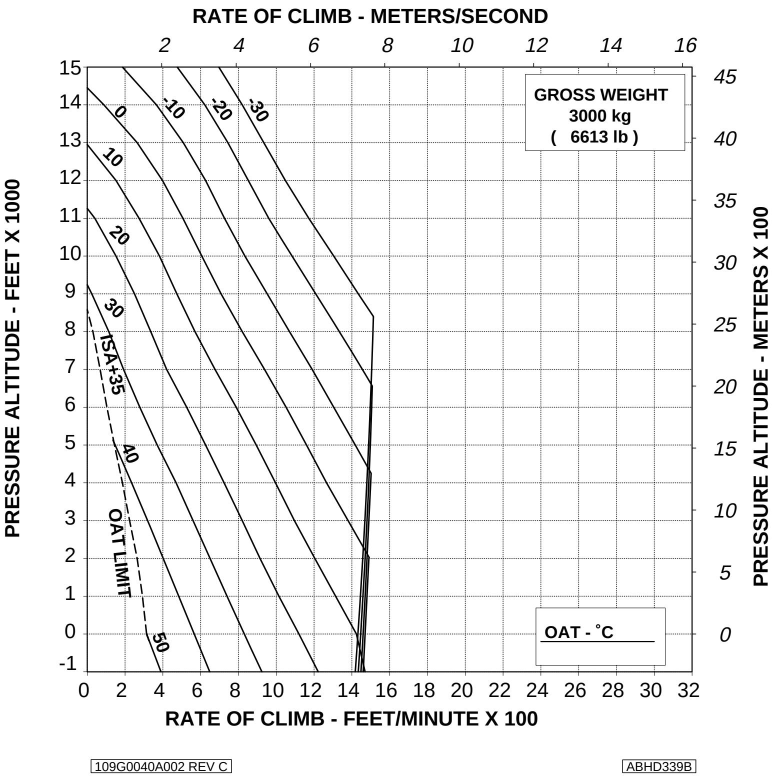

Figure 4-122. Rate of climb - All engines - Maximum continuous power - 3000 kg - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 % **60 kts IAS**

ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

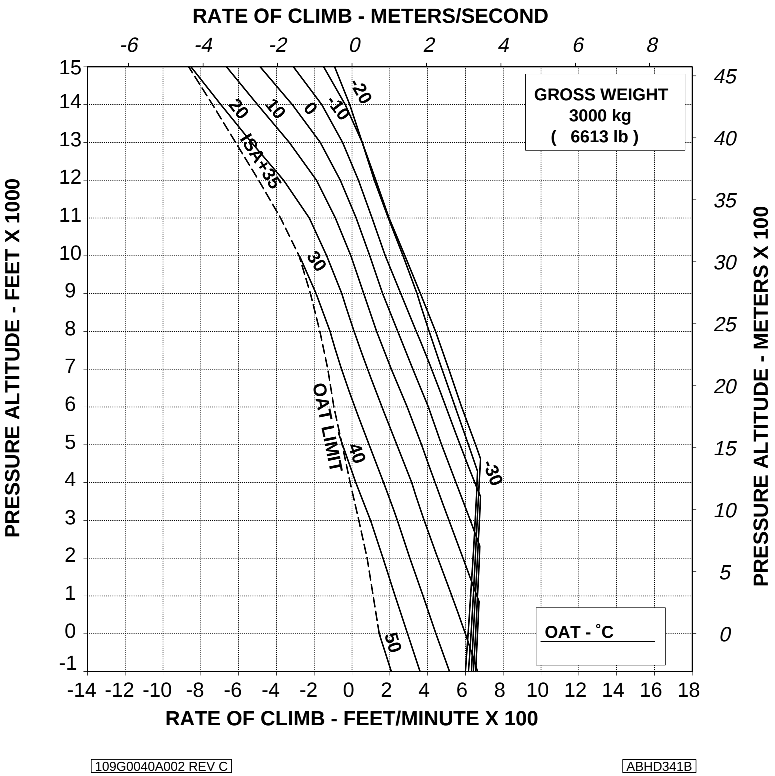

Figure 4-123. Rate of climb - OEI - 2.5 minute power - 3000 kg - EAPS OFF.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

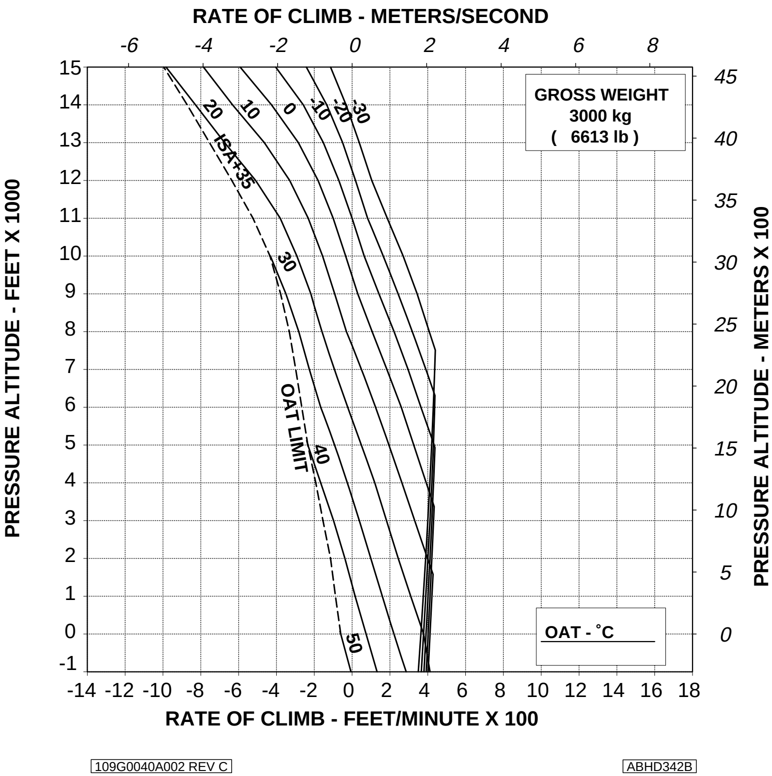

Figure 4-124. Rate of climb - OEI - Maximum continuous power - 3000 kg - EAPS OFF.

HELICOPTER WITH CARGO HOOK

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
ZERO WIND
ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

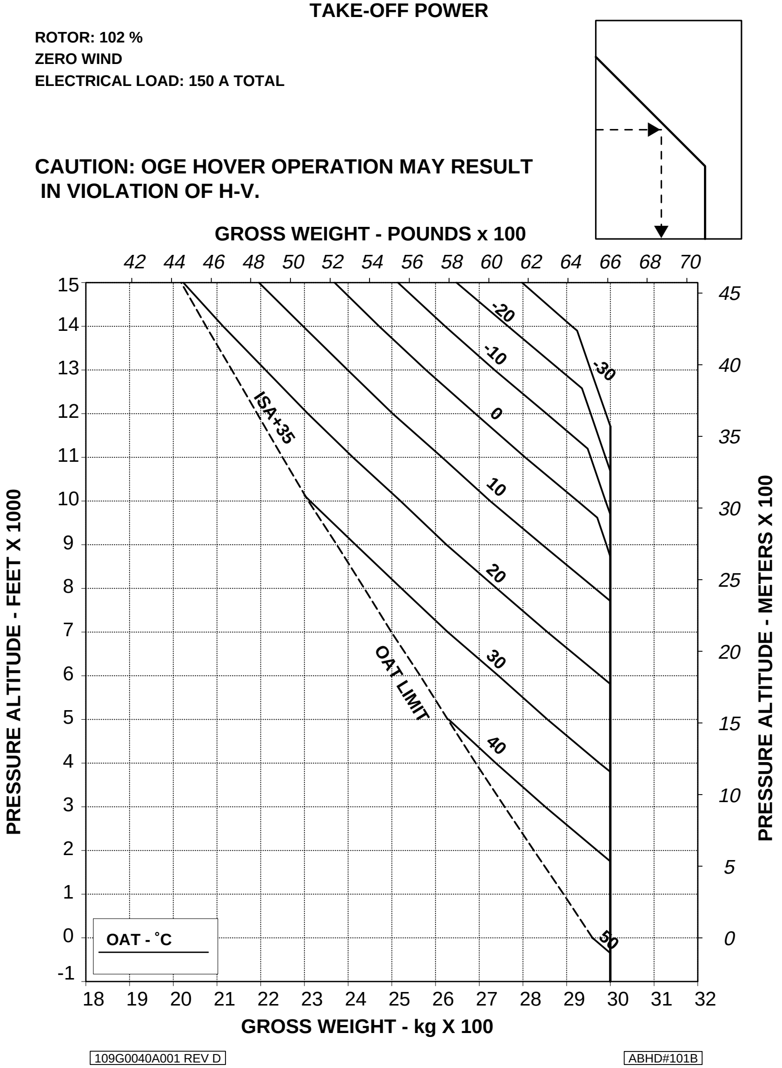

Figure 4-125. Hovering ceiling - OGE - Take-off power - 3000 kg (Cargo hook operations).

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
HEATER ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

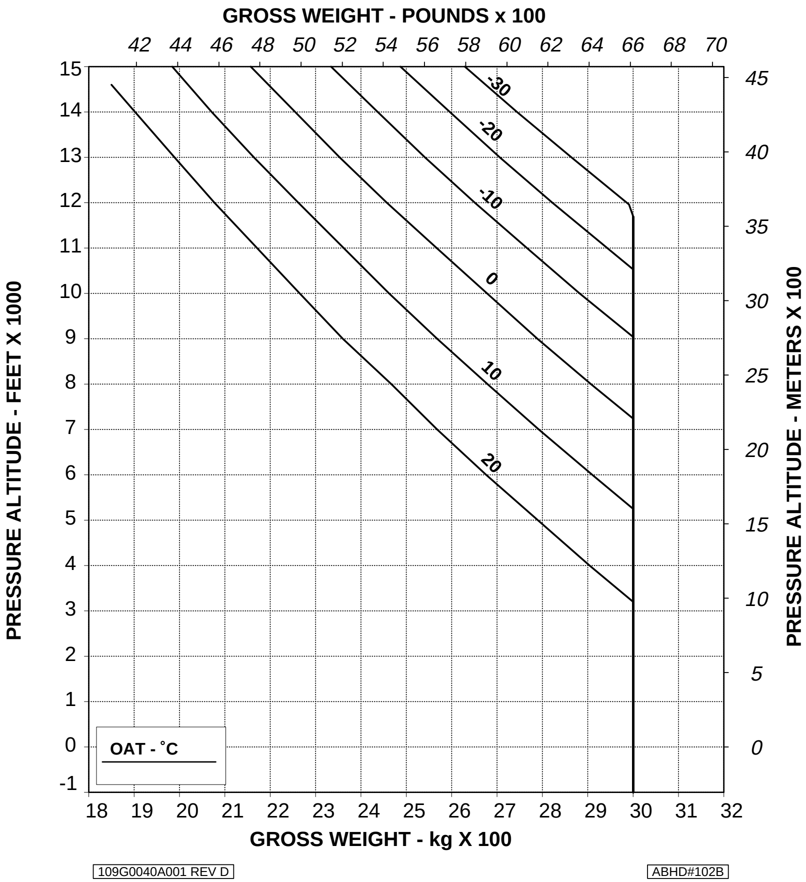

Figure 4-126. Hovering ceiling - OGE - Take-off power - Heater ON - 3000 kg (Cargo hook operations).

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
ECS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

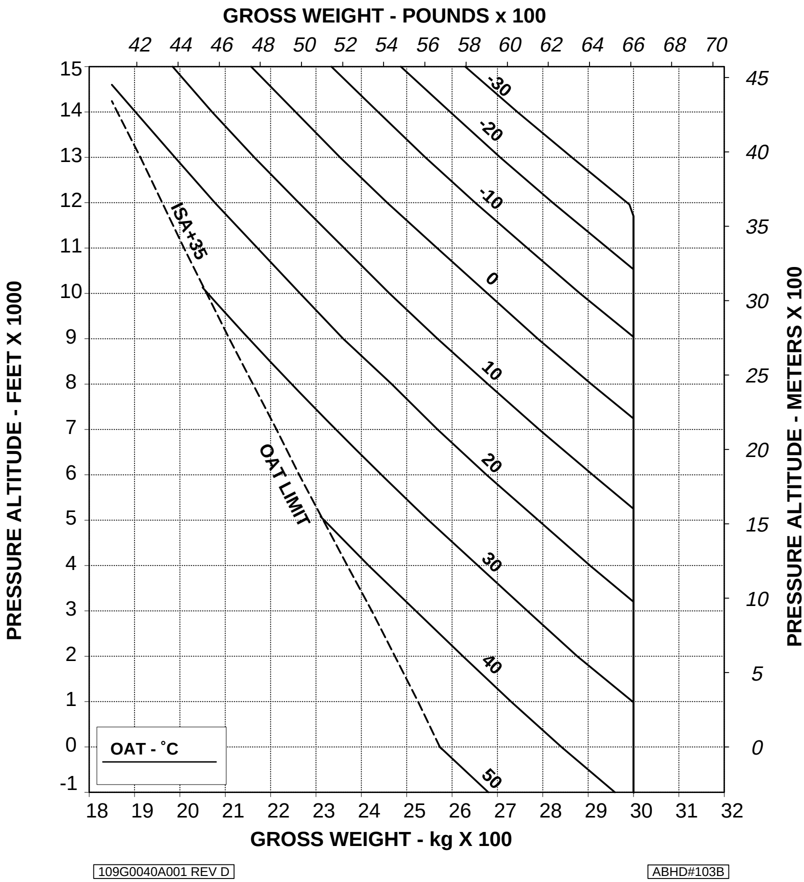

Figure 4-127. Hovering ceiling - OGE - Take-off power - ECS ON - 3000 kg (Cargo hook operations).

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 % **ELECTRICAL LOAD: 150 A TOTAL**

ZERO WIND

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

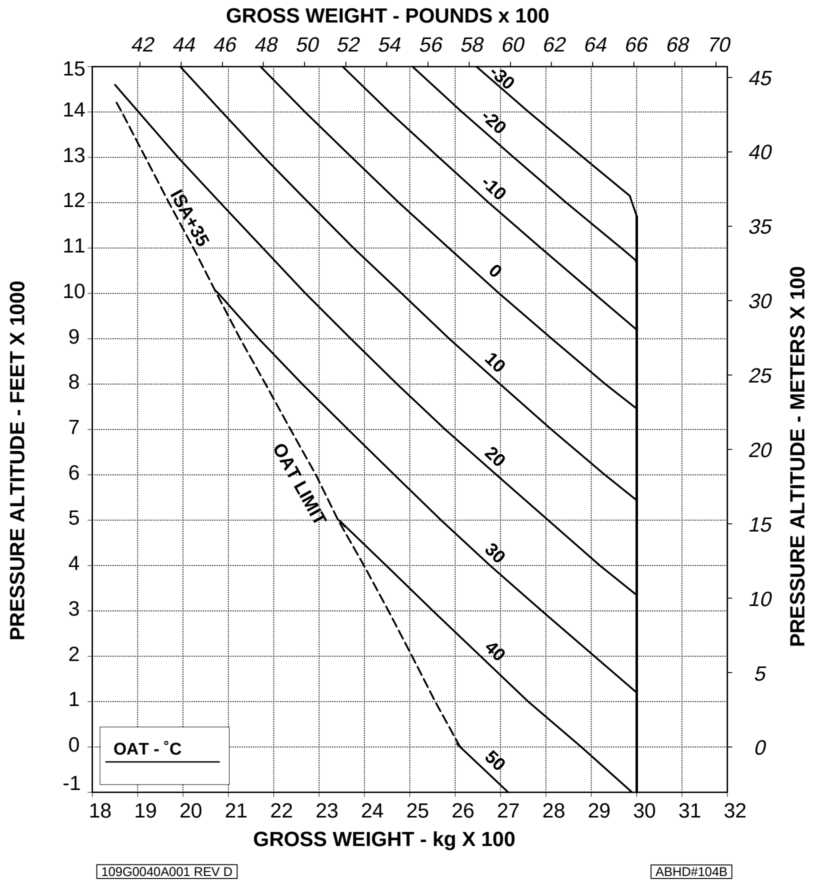

Figure 4-128. Hovering ceiling - OGE - Maximum continuous power - 3000 kg (Cargo hook operations).

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE OPERATION MAY RESULT IN VIOLATION OF H-V.

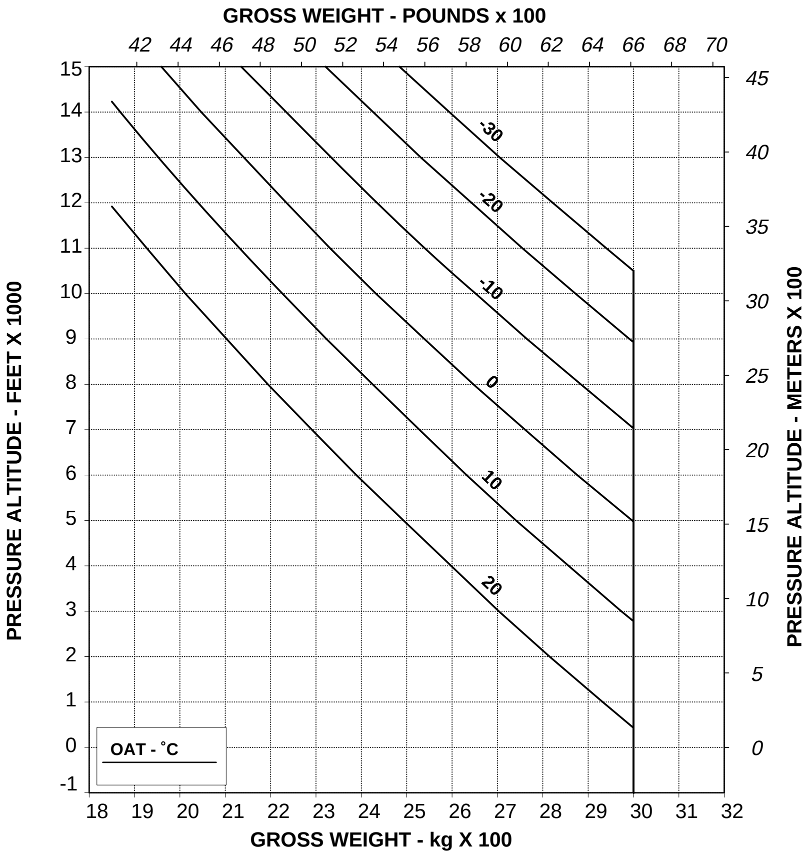

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Figure 4-129. Hovering ceiling - OGE - Maximum continuous power - Heater ON - 3000 kg (Cargo hook operations).

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ECS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

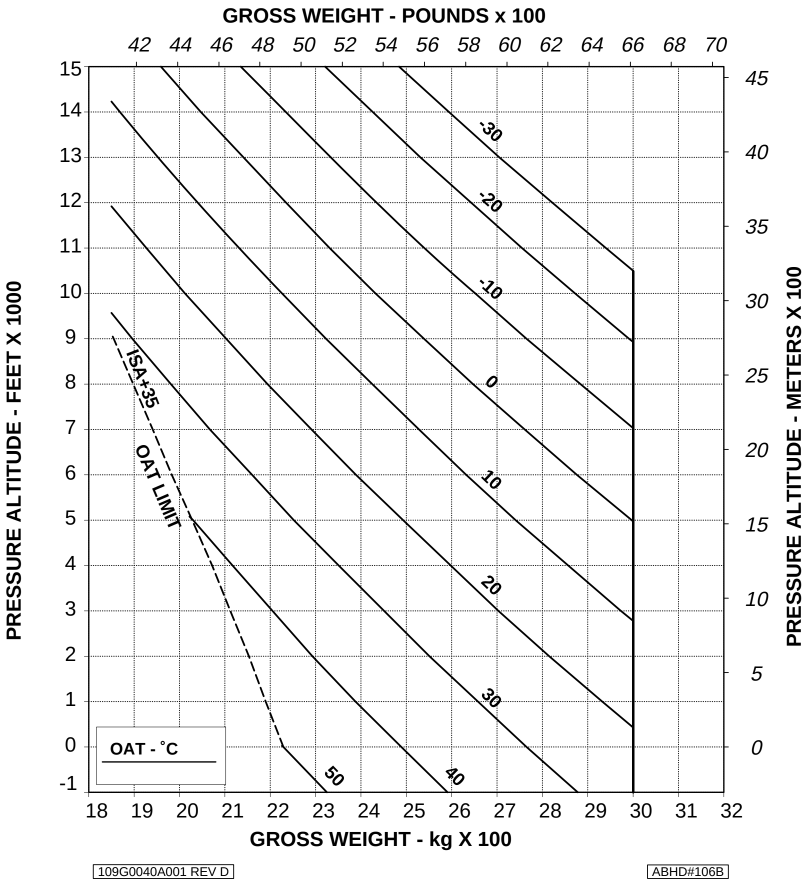

Figure 4-130. Hovering ceiling - OGE - Maximum continuous power - ECS ON - 3000 kg (Cargo hook operations).

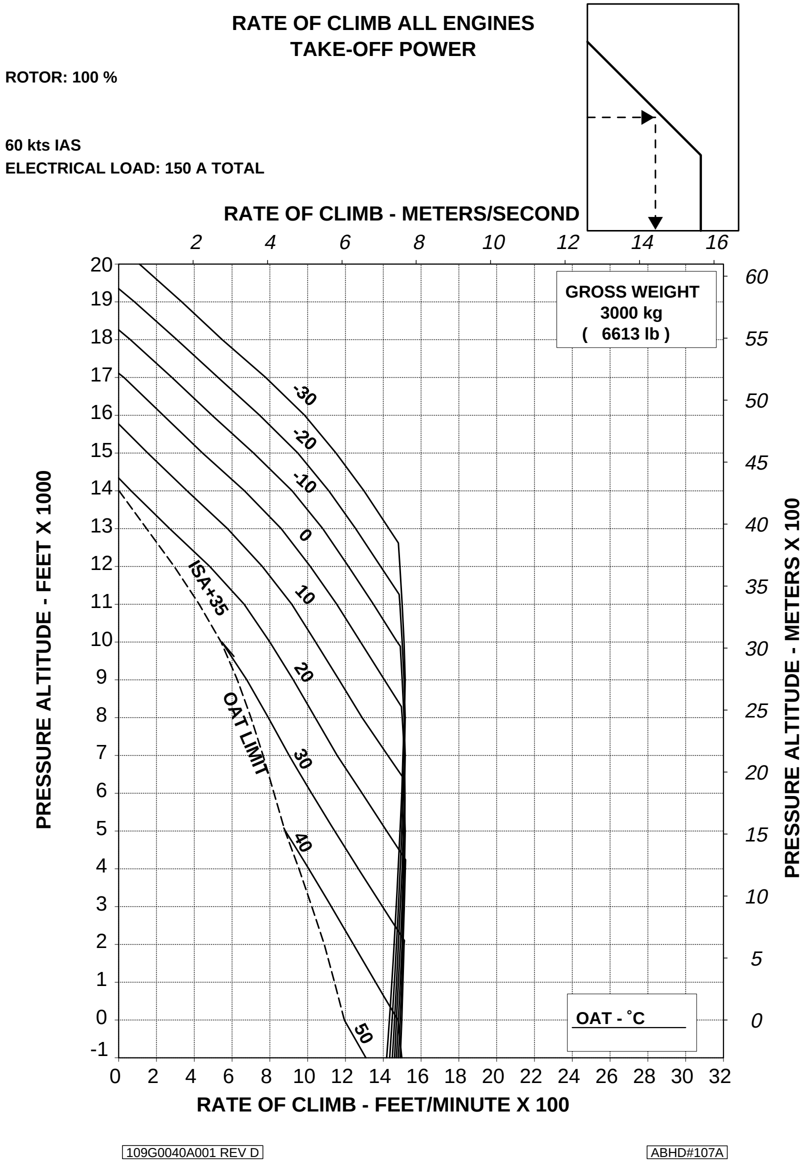

Figure 4-131. Rate of climb - All engines - Take-off power - 3000 kg (Cargo hook operations).

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % | 60 kts IAS
HEATER: ON | ELECTRICAL LOAD: 150 A TOTAL

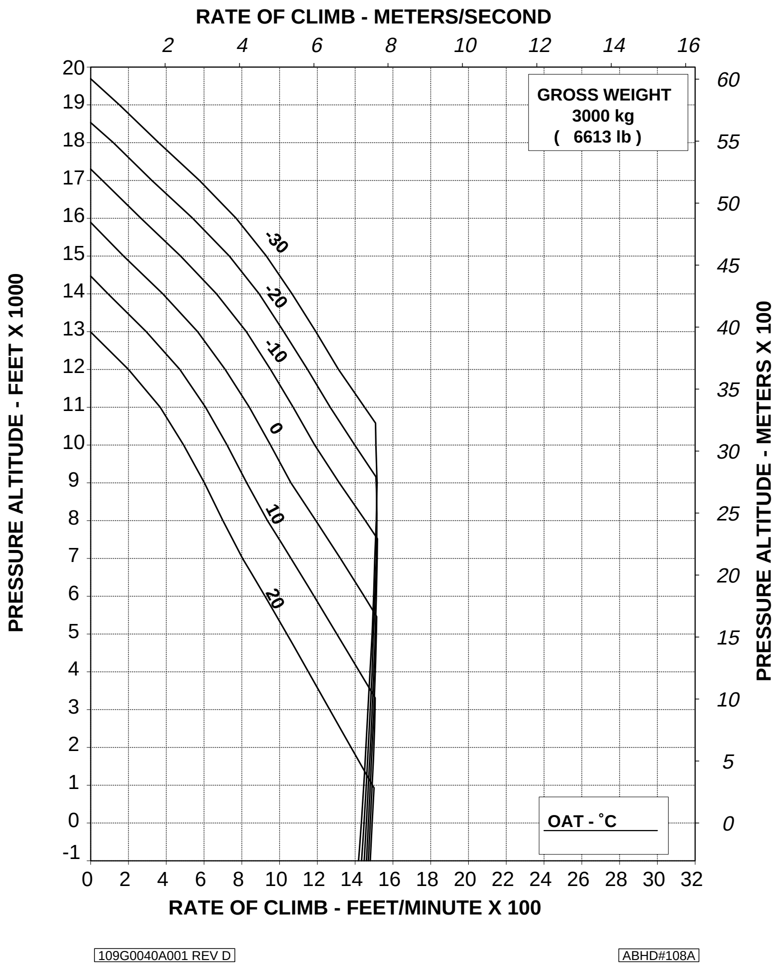

Figure 4-132. Rate of climb - All engines - Take-off power - Heater ON - 3000 kg (Cargo hook operations).

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % — 60 kts IAS
ECS: ON — ELECTRICAL LOAD: 150 A TOTAL

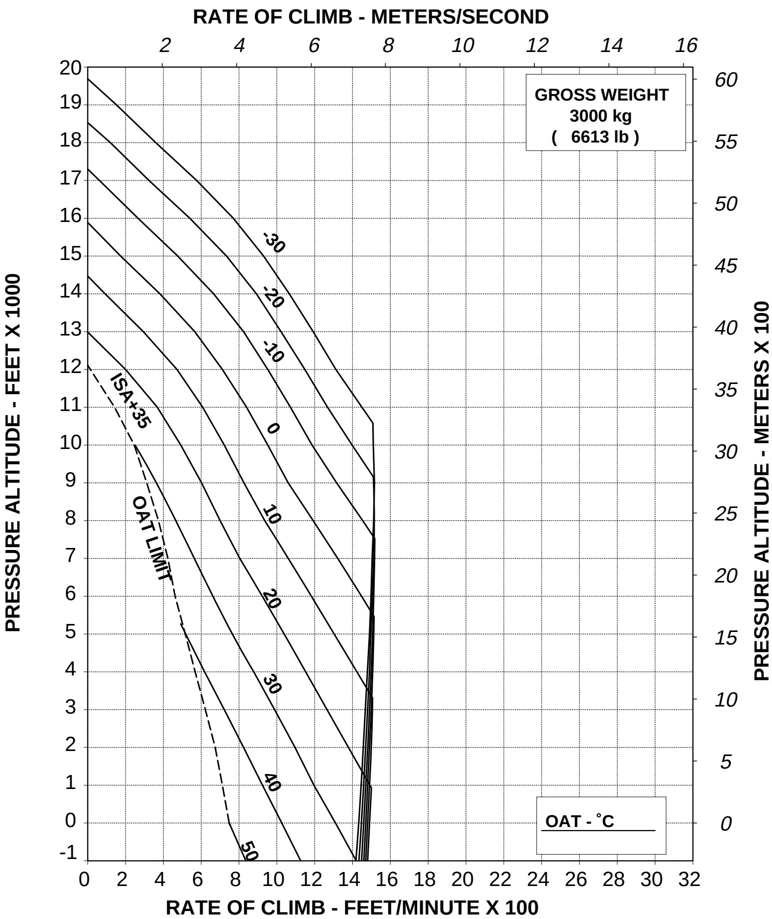

Figure 4-133. Rate of climb - All engines - Take-off power - ECS ON - 3000 kg (Cargo hook operations).

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % **60 kts IAS**
ELECTRICAL LOAD: 150 A TOTAL

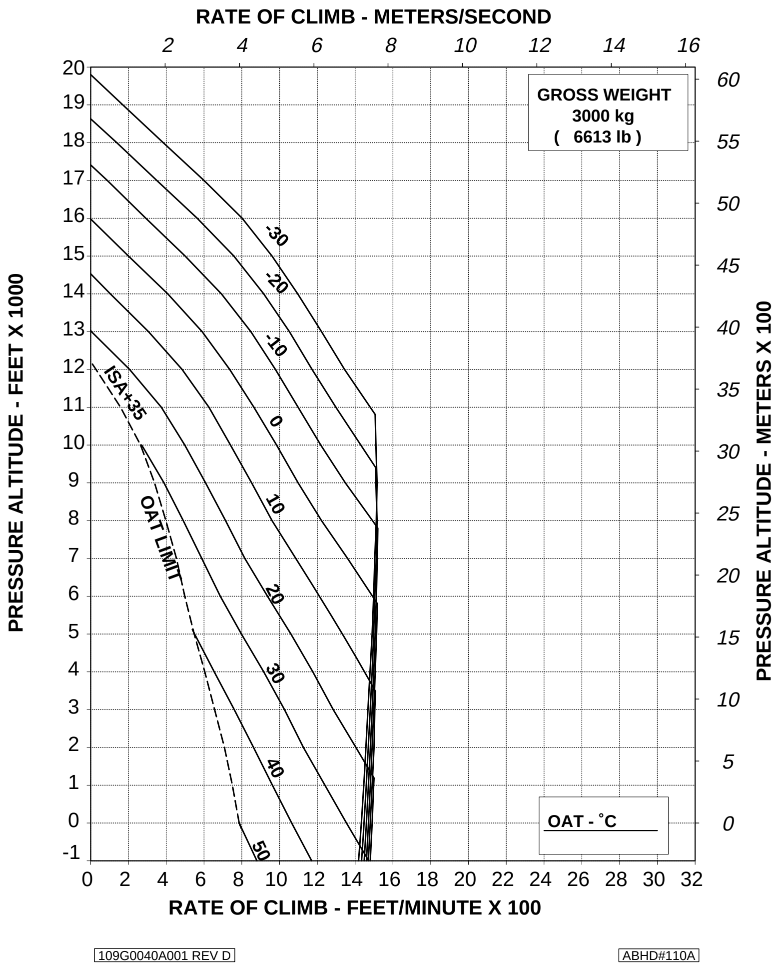

Figure 4-134. Rate of climb - All engines - Maximum continuous power - 3000 kg (Cargo hook operations).

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % 60 kts IAS
HEATER: ON ELECTRICAL LOAD: 150 A TOTAL

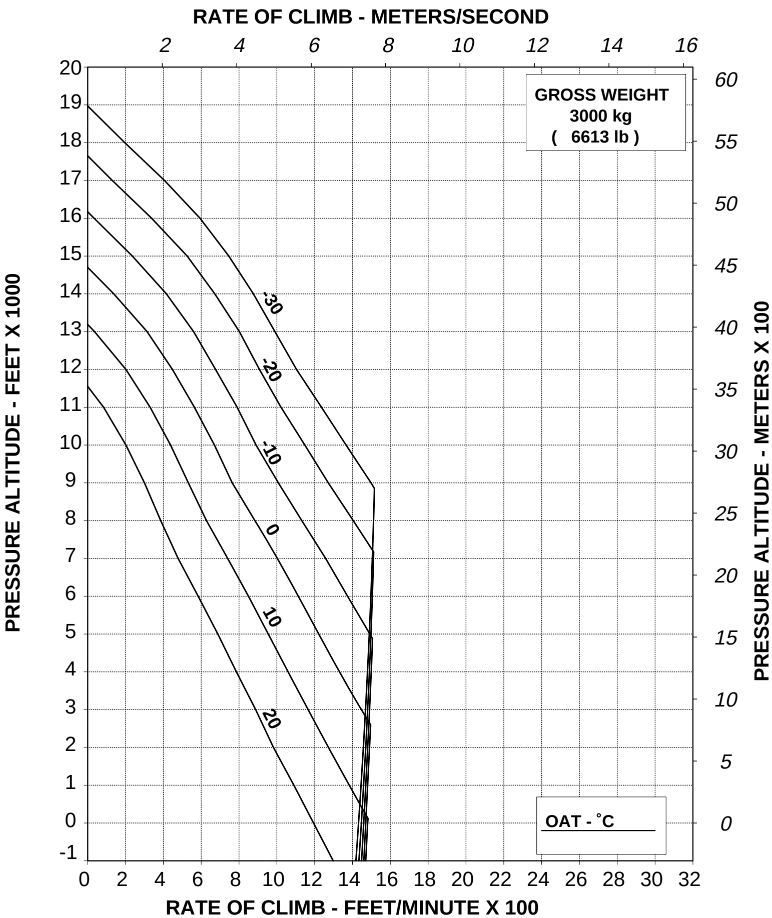

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Figure 4-135. Rate of climb - All engines - Maximum continuous power - Heater ON - 3000 kg (Cargo hook operations).

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 % | 60 kts IAS
ECS: ON | ELECTRICAL LOAD: 150 A TOTAL

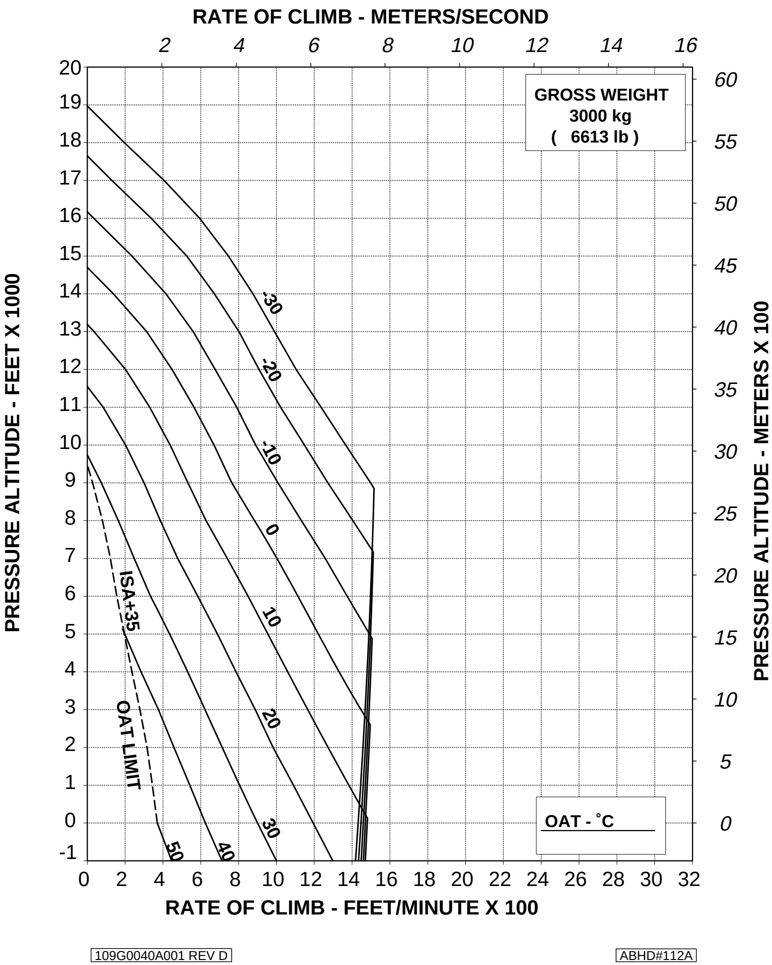

Figure 4-136. Rate of climb - All engines - Maximum continuous power - ECS ON - 3000 kg (Cargo hook operations).

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A

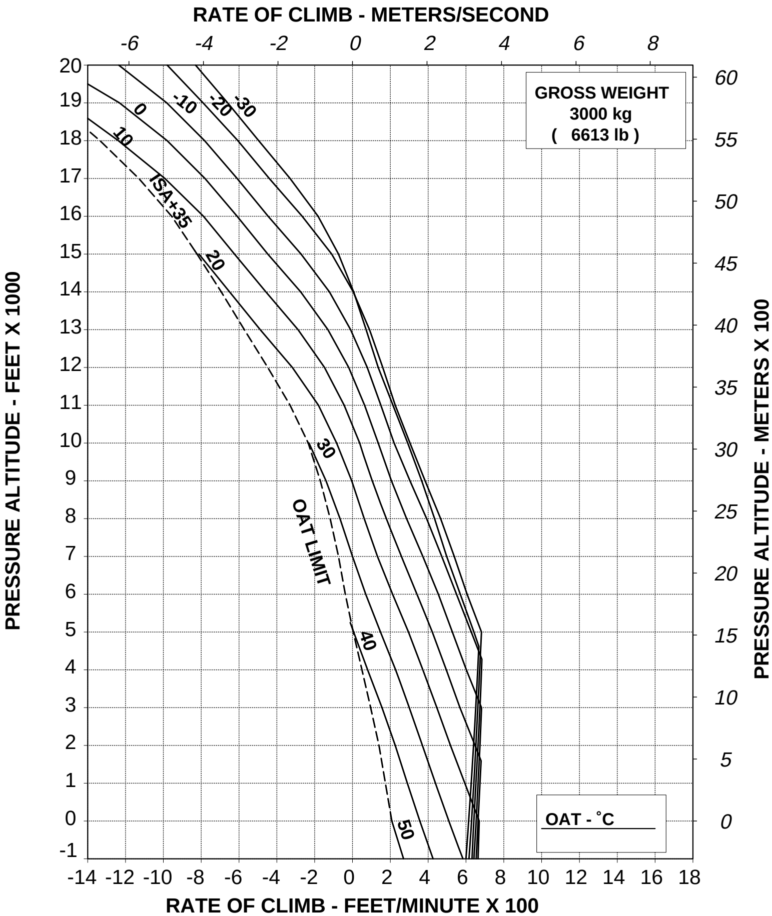

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Figure 4-137. Rate of climb - OEI - 2.5 minute power - 3000 kg (Cargo hook operations).

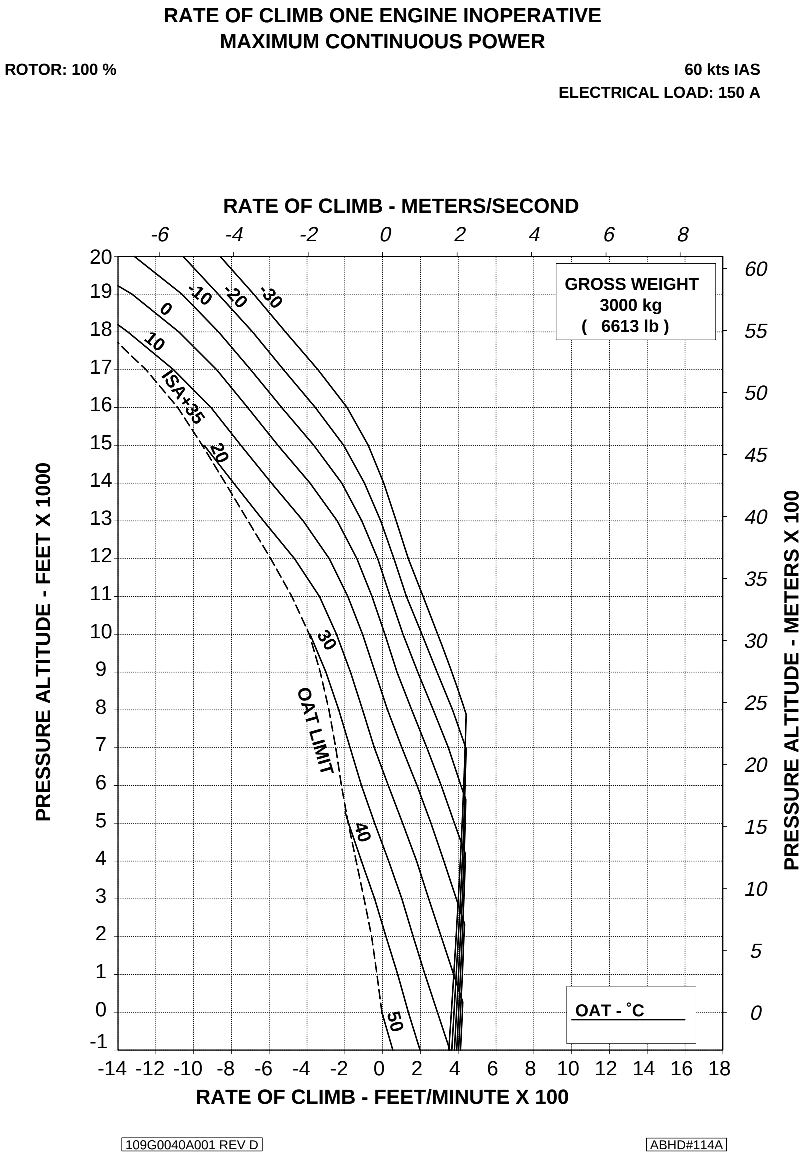

Figure 4-138. Rate of climb - OEI - Maximum continuous power - 3000 kg (Cargo hook operations).

EQUIVALENT CATEGORY ″A″ OPERATIONS

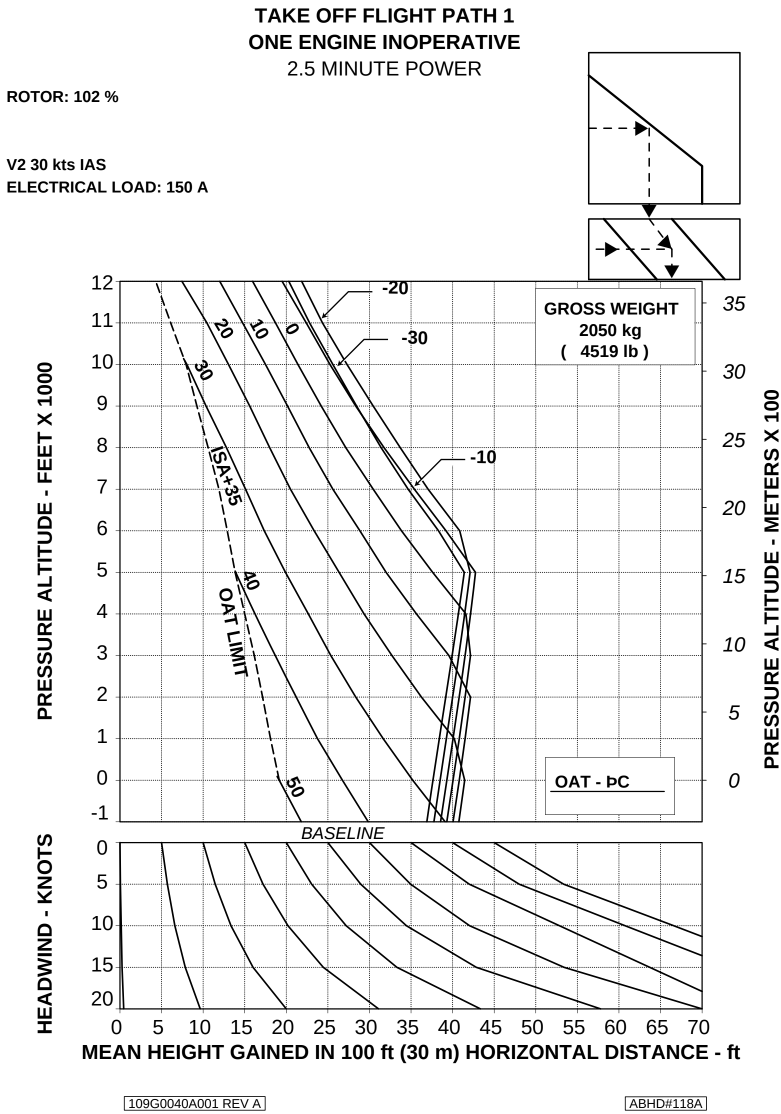

Figure 4-139. Take-off flight path 1 - 2050 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

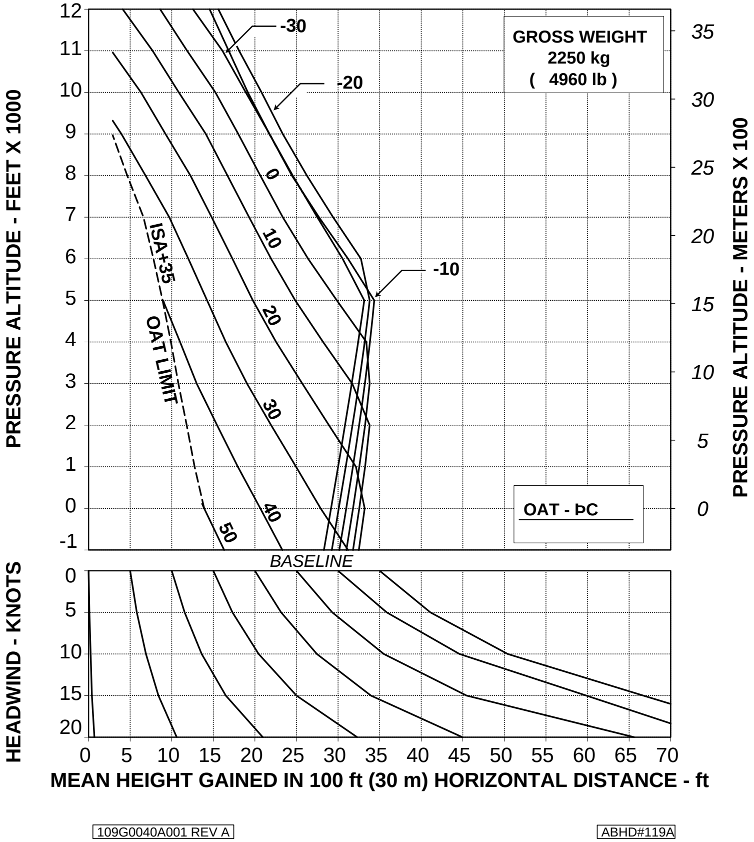

Figure 4-140. Take-off flight path 1 - 2250 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

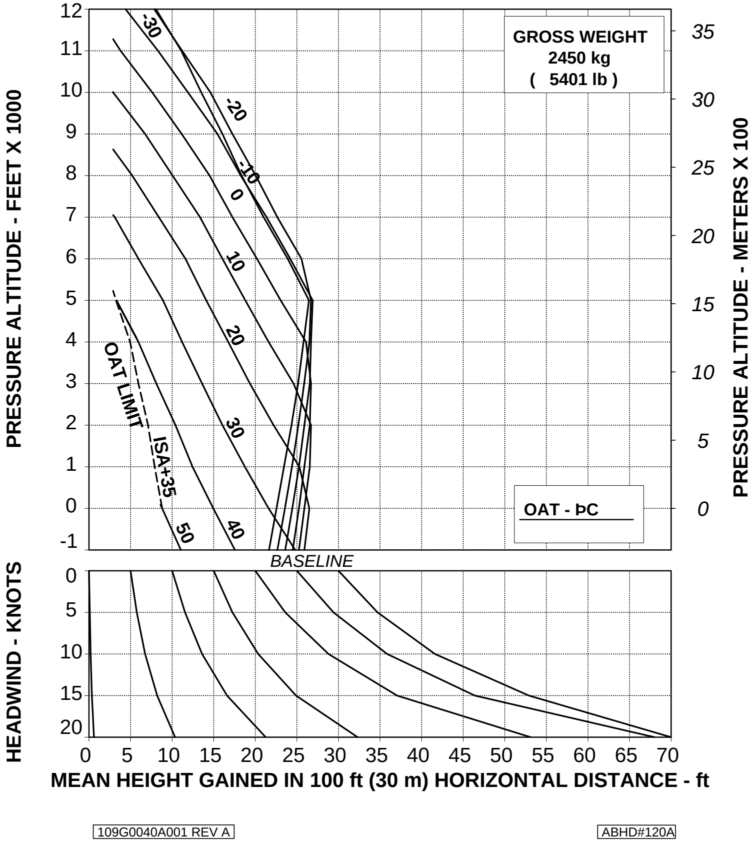

Figure 4-141. Take-off flight path 1 - 2450 kg.

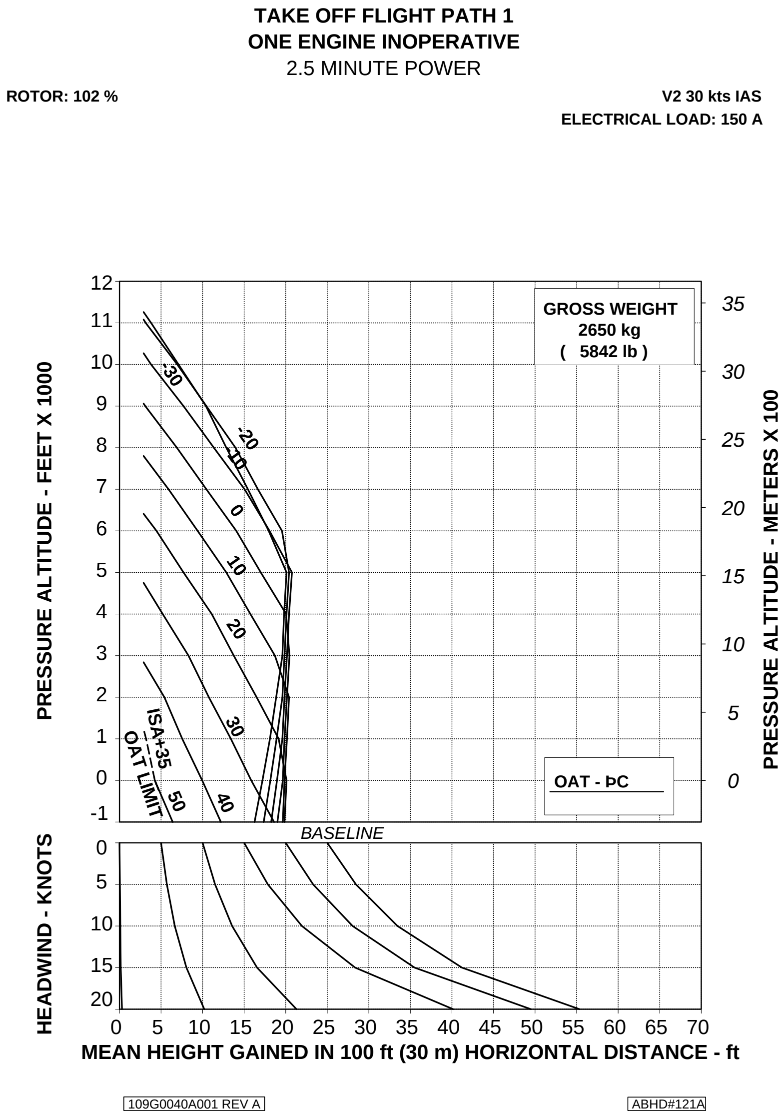

Figure 4-142. Take-off flight path 1 - 2650 kg.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
ELECTRICAL LOAD: 150 A

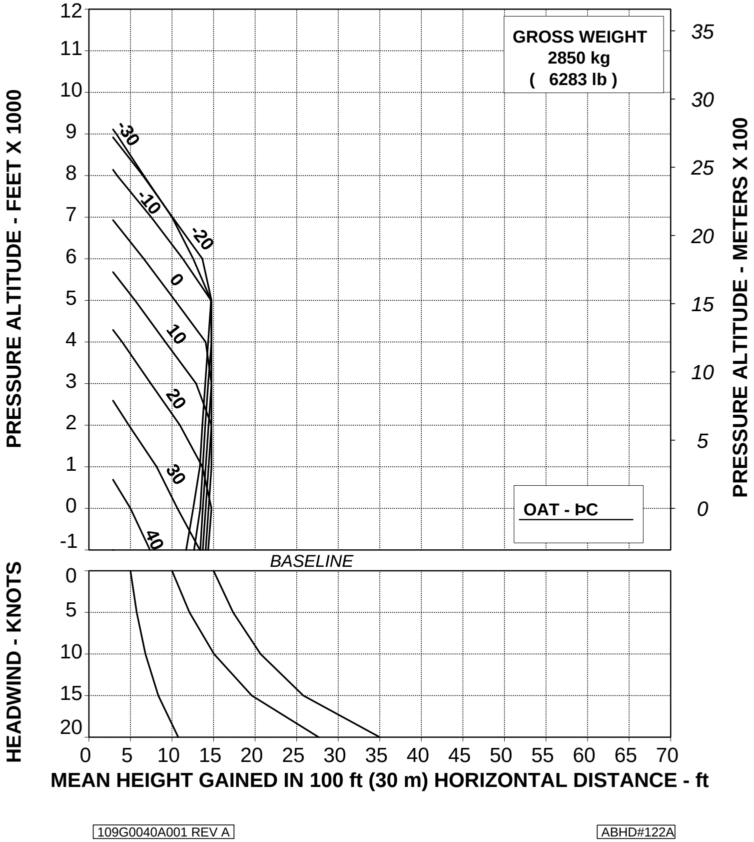

Figure 4-143. Take-off flight path 1 - 2850 kg.

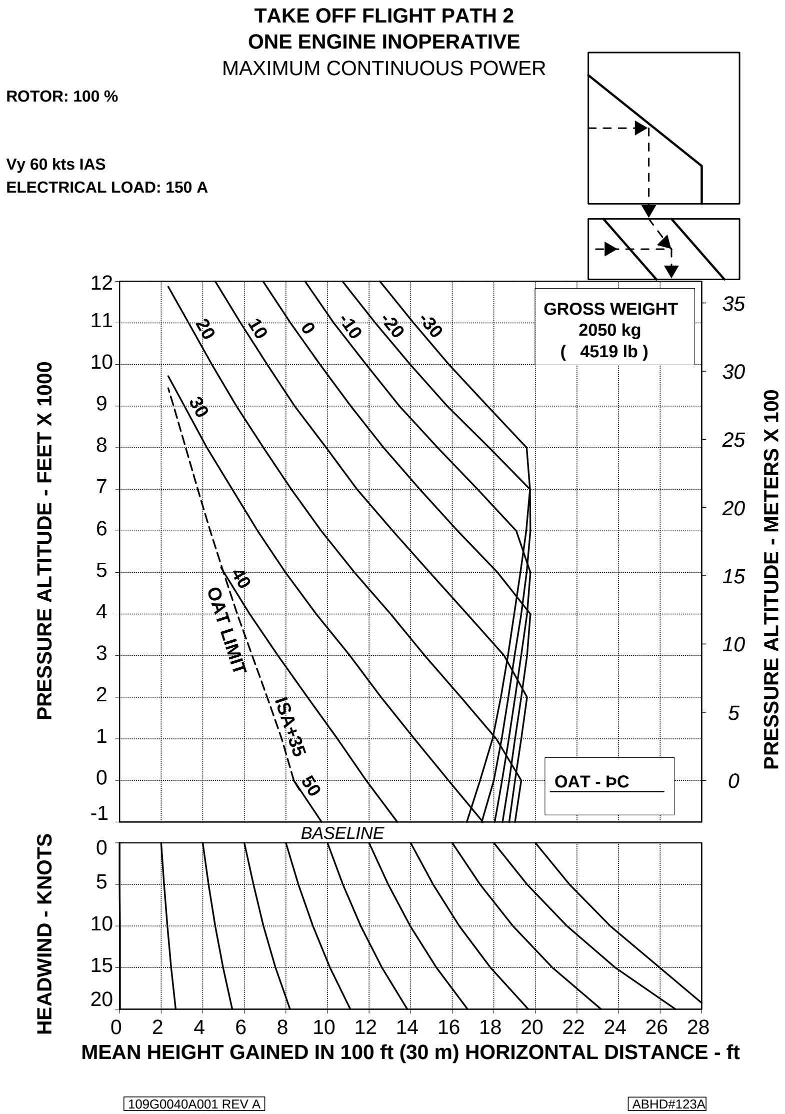

Figure 4-144. Take-off flight path 2 - 2050 kg.

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TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A

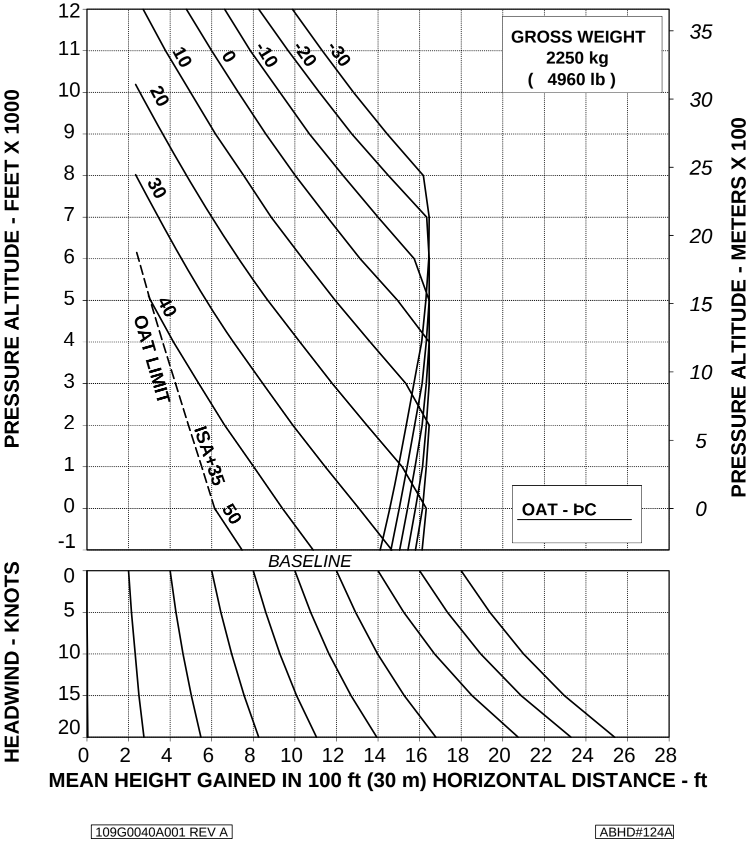

Figure 4-145. Take-off flight path 2 - 2250 kg.

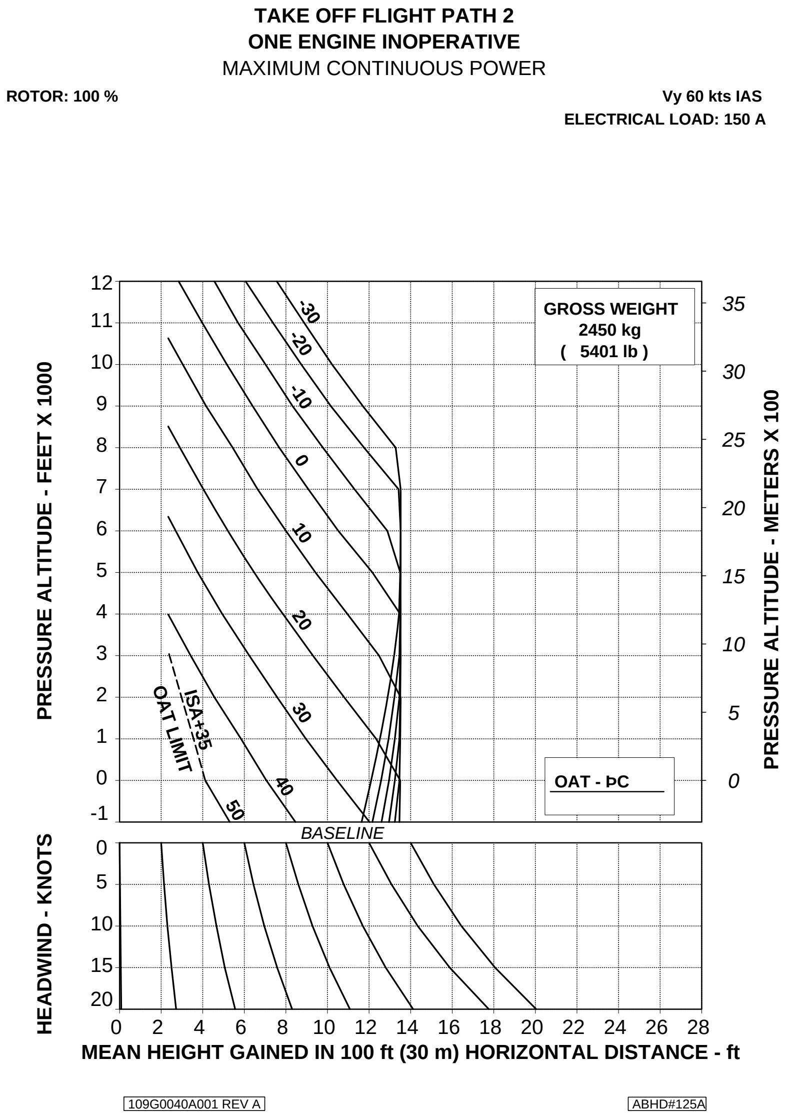

Figure 4-146. Take-off flight path 2 - 2450 kg.

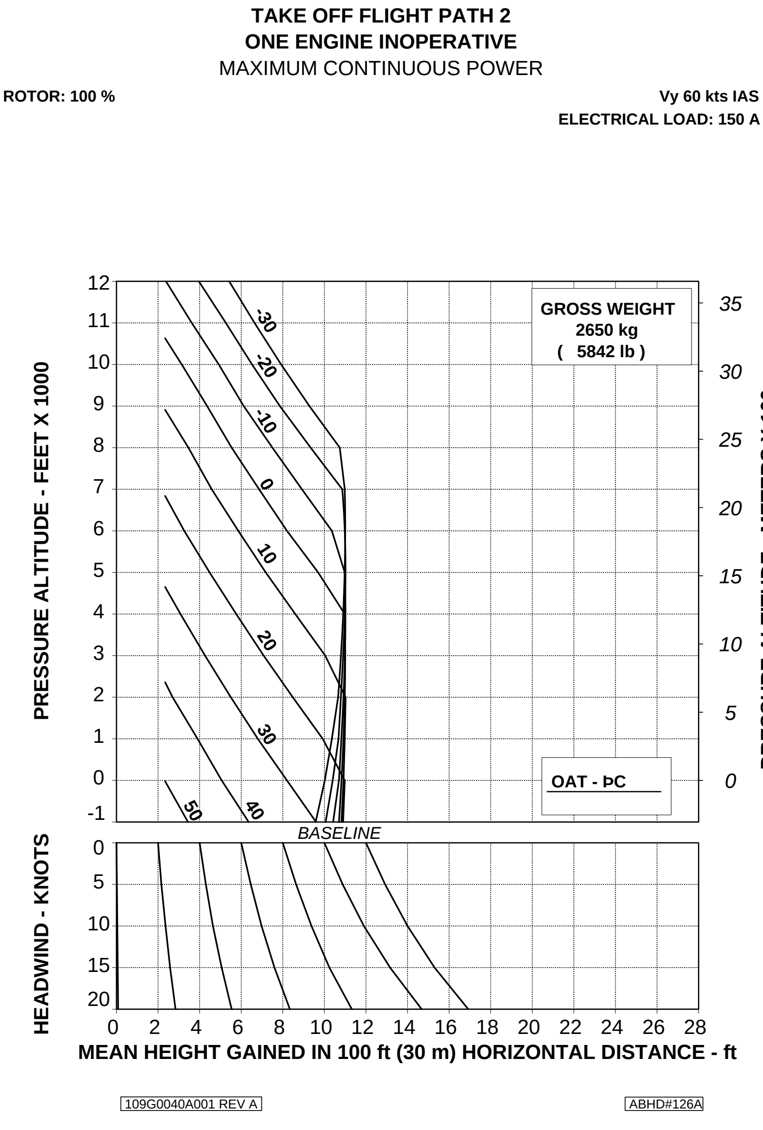

Figure 4-147. Take-off flight path 2 - 2650 kg.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A

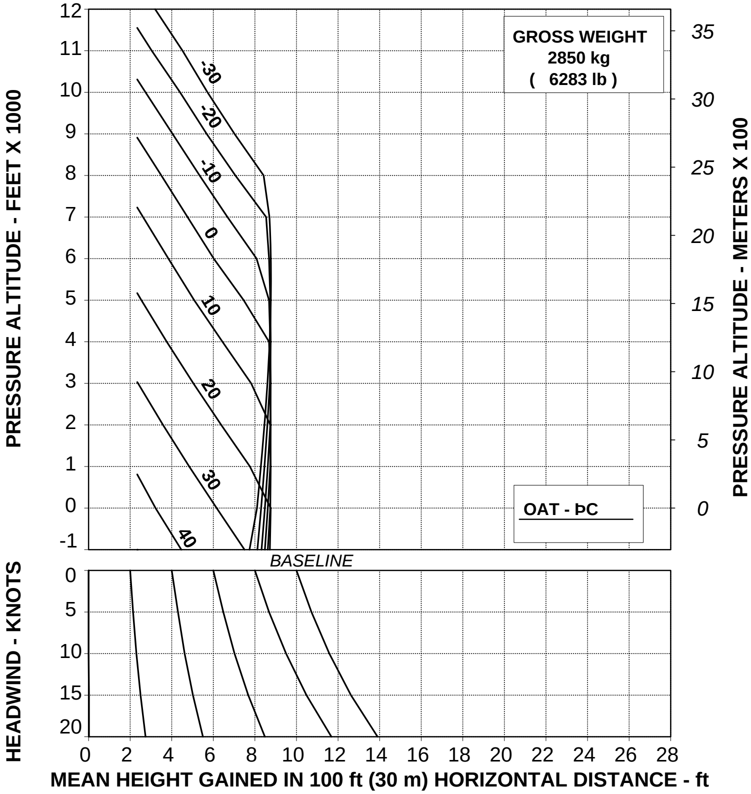

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Figure 4-148. Take-off flight path 2 - 2850 kg.

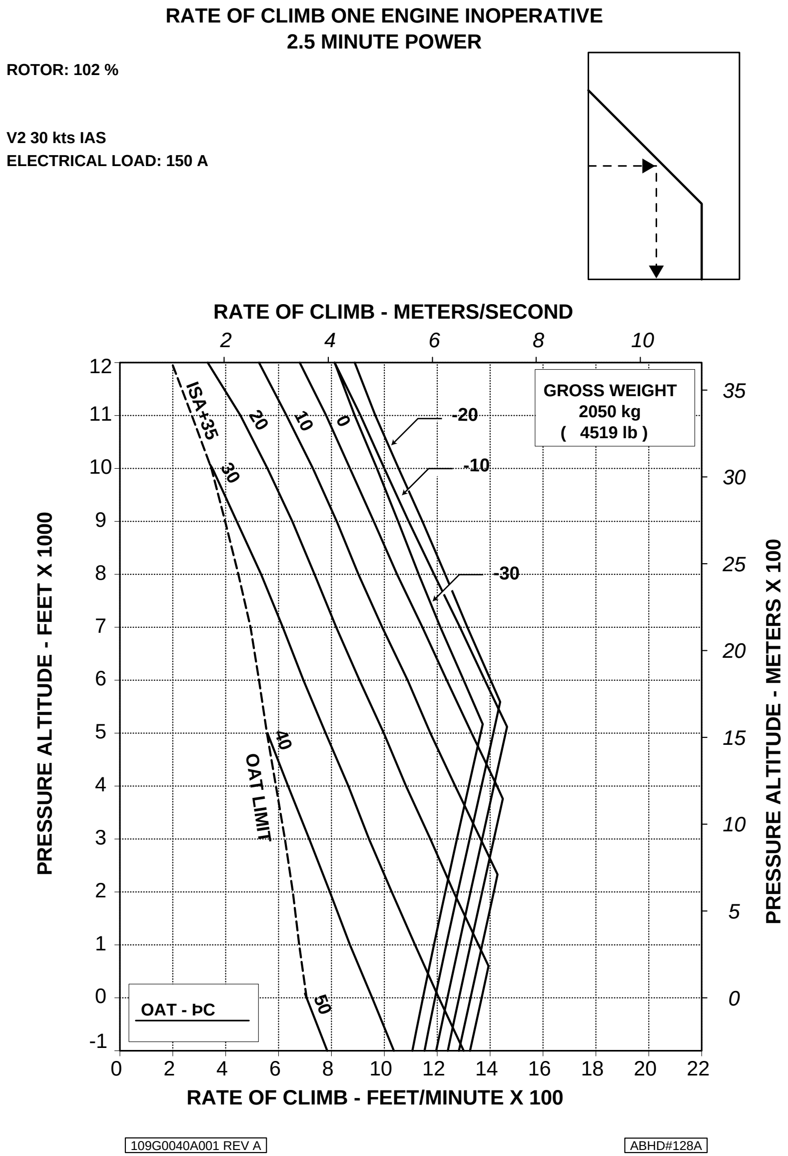

Figure 4-149. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2050 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS

ELECTRICAL LOAD: 150 A

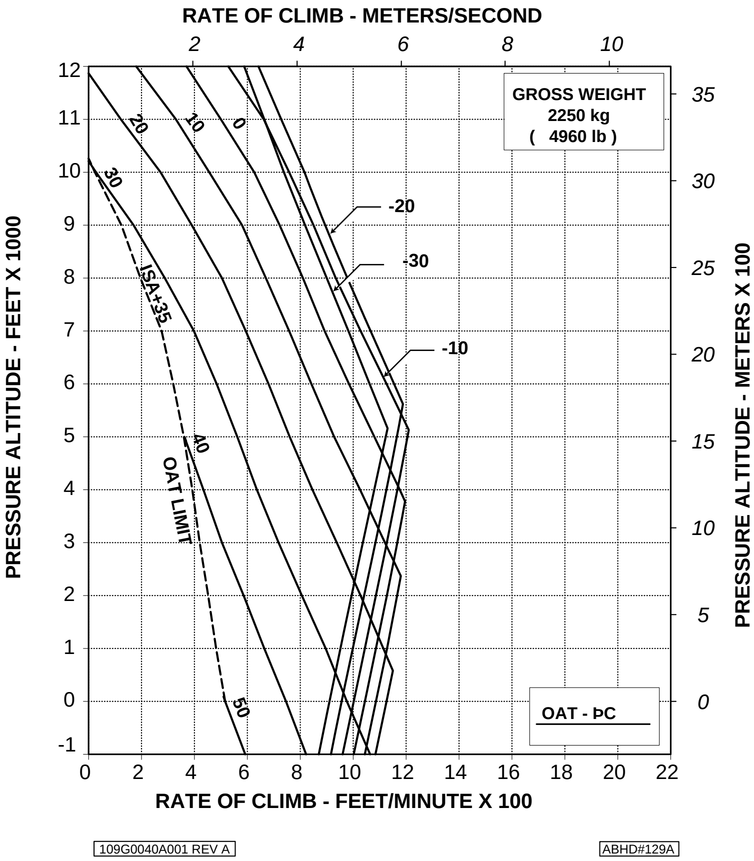

Figure 4-150. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2250 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

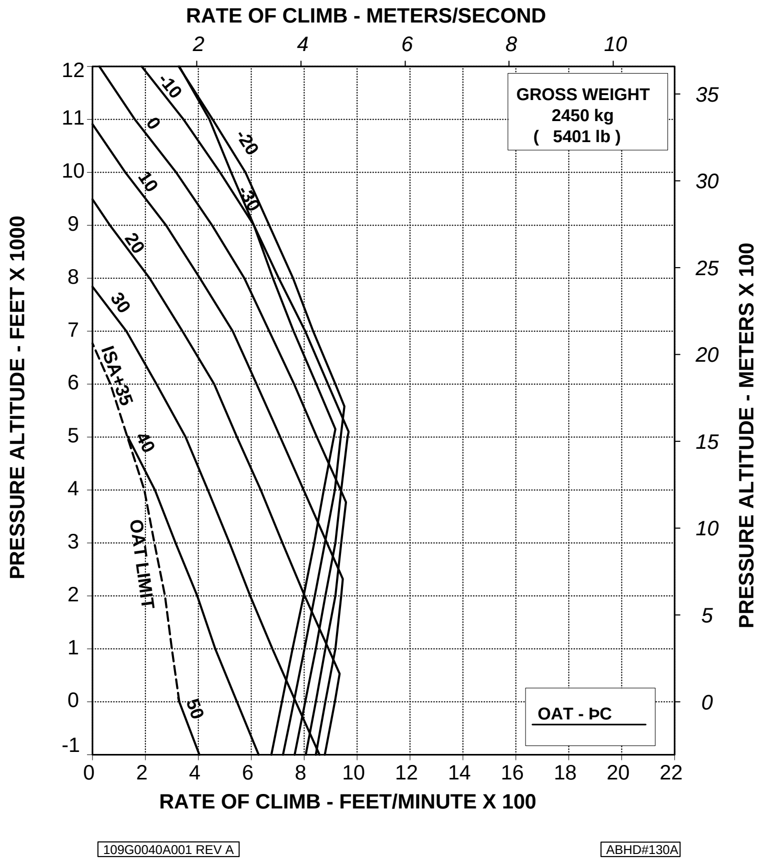

Figure 4-151. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2450 kg.

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RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 % **V2 30 kts IAS**
ELECTRICAL LOAD: 150 A

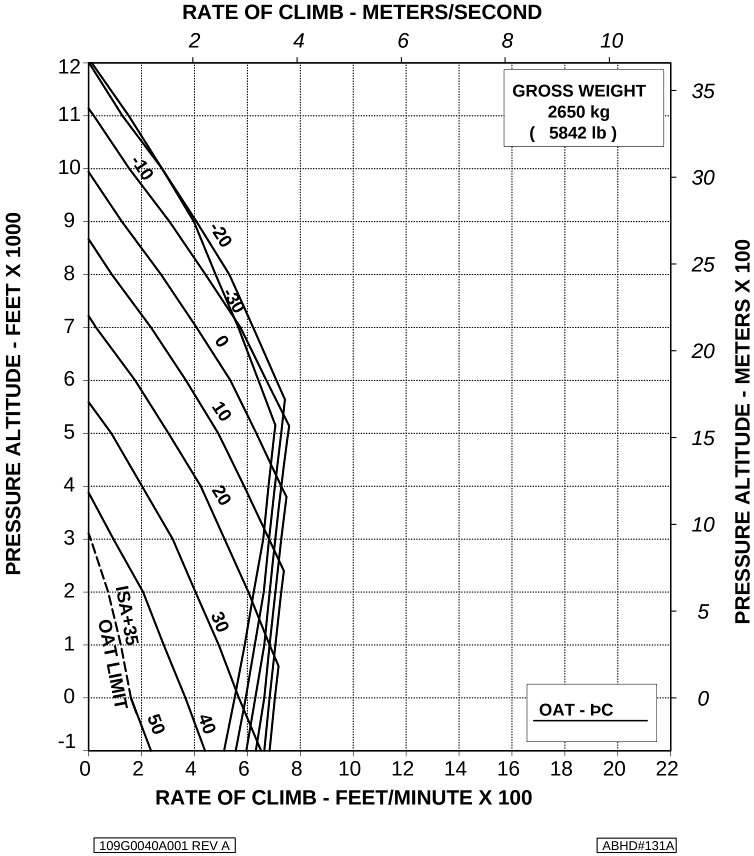


Figure 4-152. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2650 kg.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

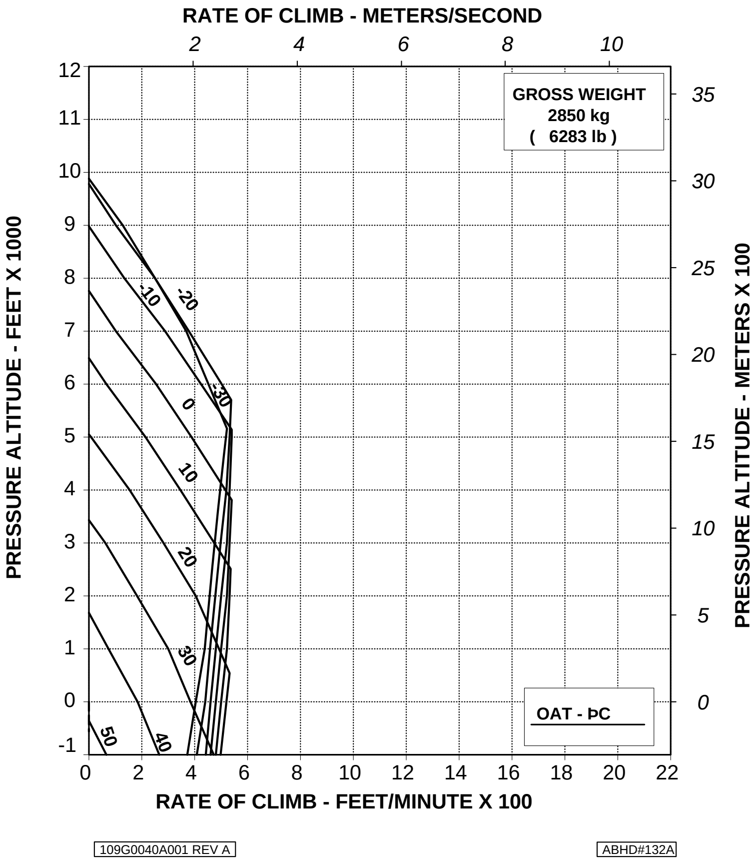


Figure 4-153. Rate of climb - OEI - 2.5 minute power - V_2 30 KIAS - 2850 kg.

EQUIVALENT CATEGORY "A" OPERATIONS - TRAINING PROCEDURES

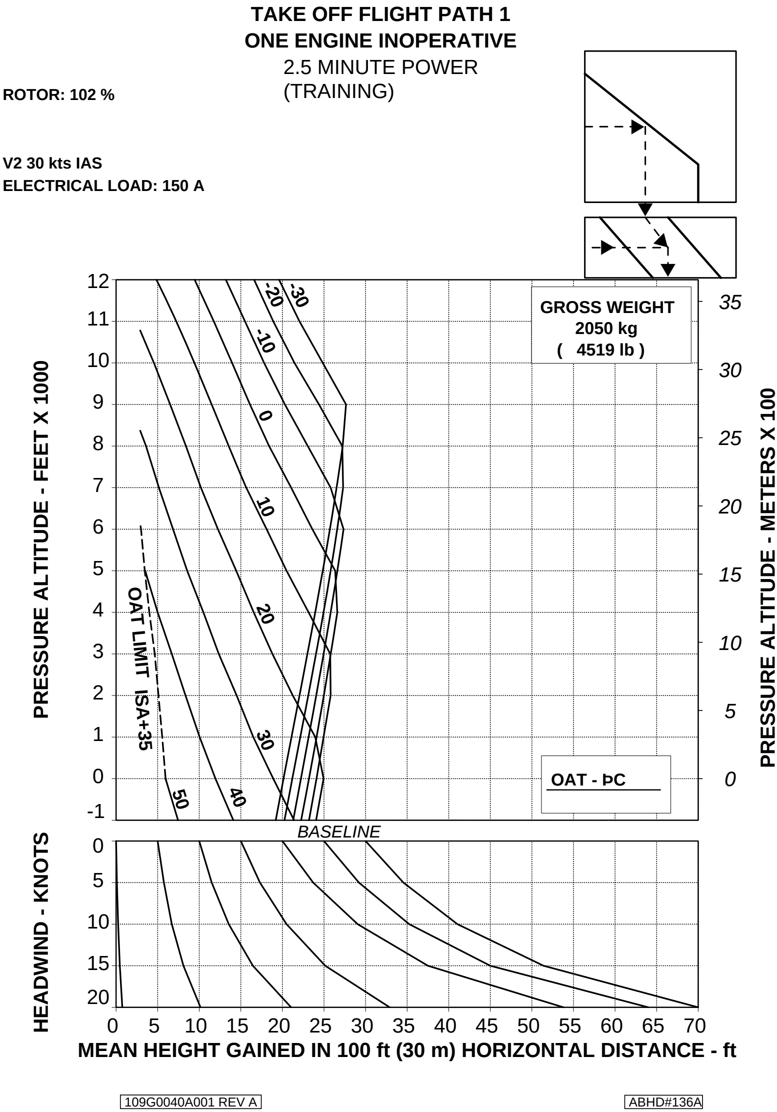


Figure 4-154. Take-off flight path 1 - 2050 kg - Training.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

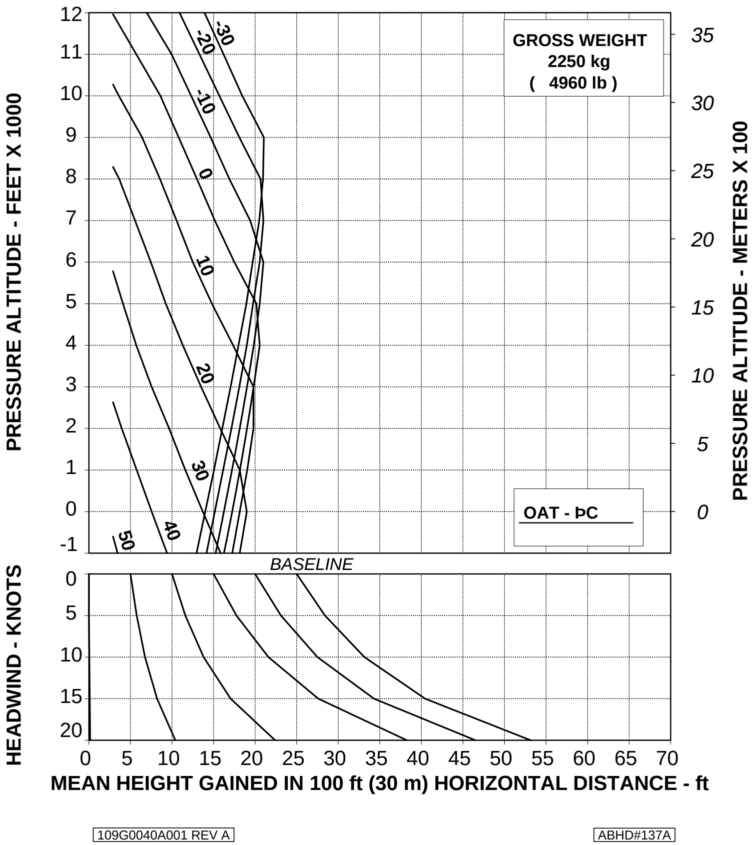


Figure 4-155. Take-off flight path 1 - 2250 kg - Training.

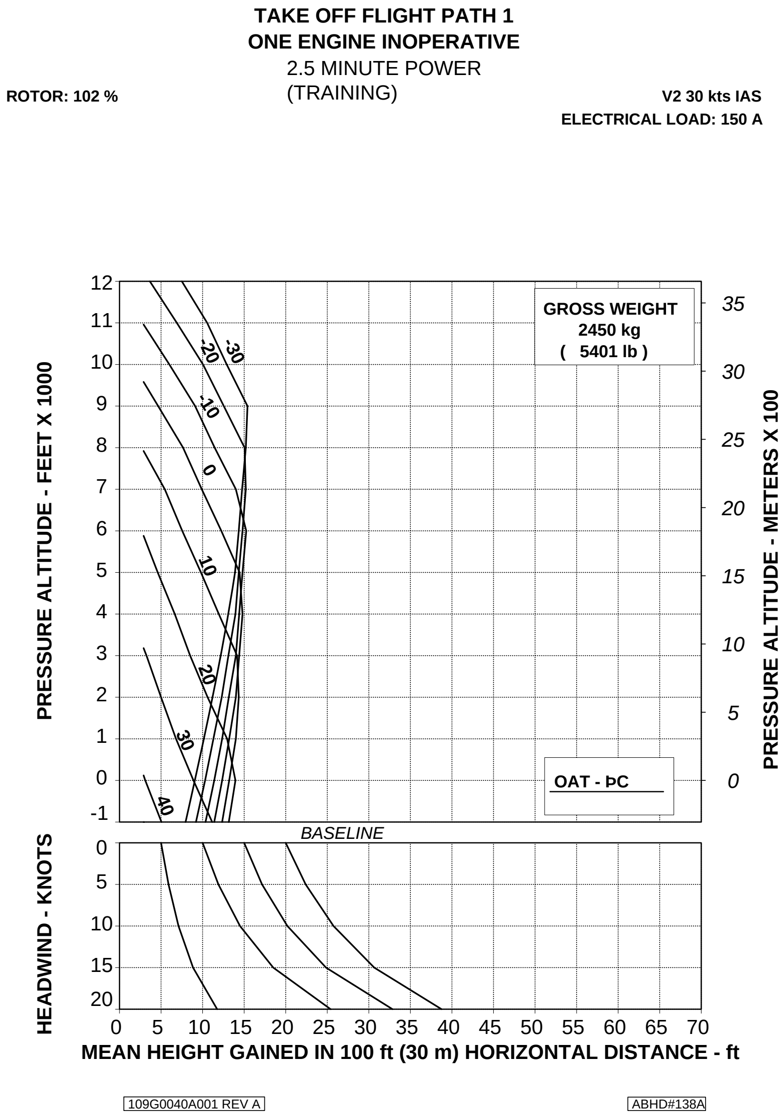


Figure 4-156. Take-off flight path 1 - 2450 kg - Training.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

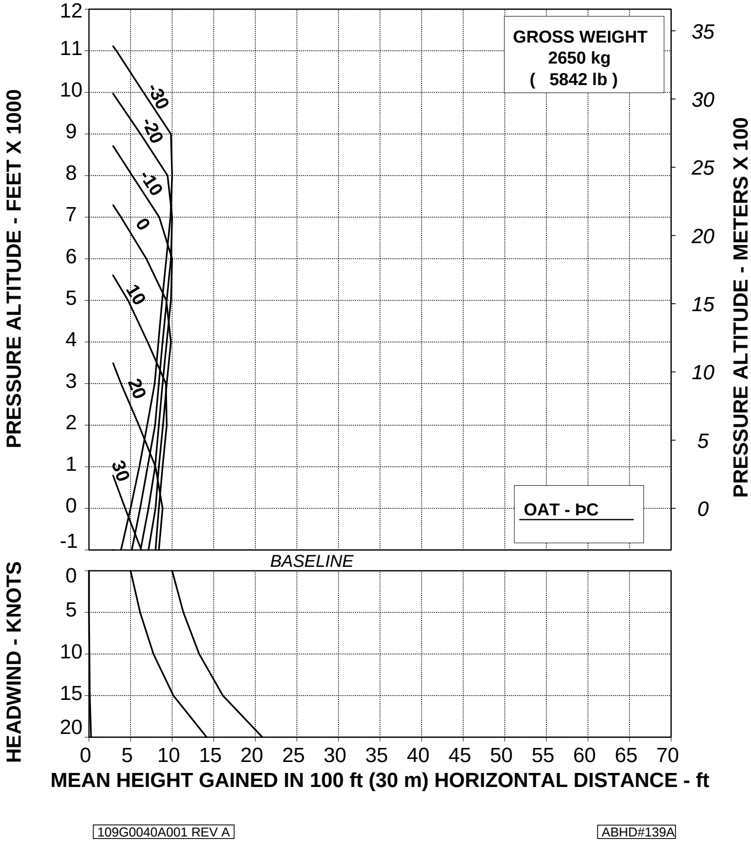


Figure 4-157. Take-off flight path 1 - 2650 kg - Training.

TAKE OFF FLIGHT PATH 1
ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(TRAINING)

ROTOR: 102 %

V2 30 kts IAS
ELECTRICAL LOAD: 150 A

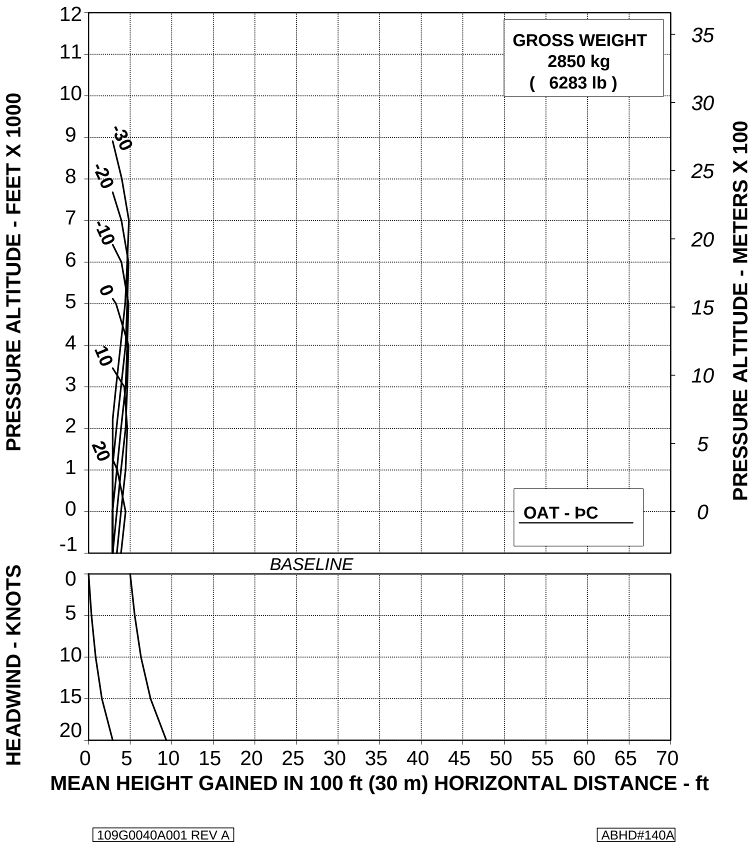


Figure 4-158. Take-off flight path 1 - 2850 kg - Training.

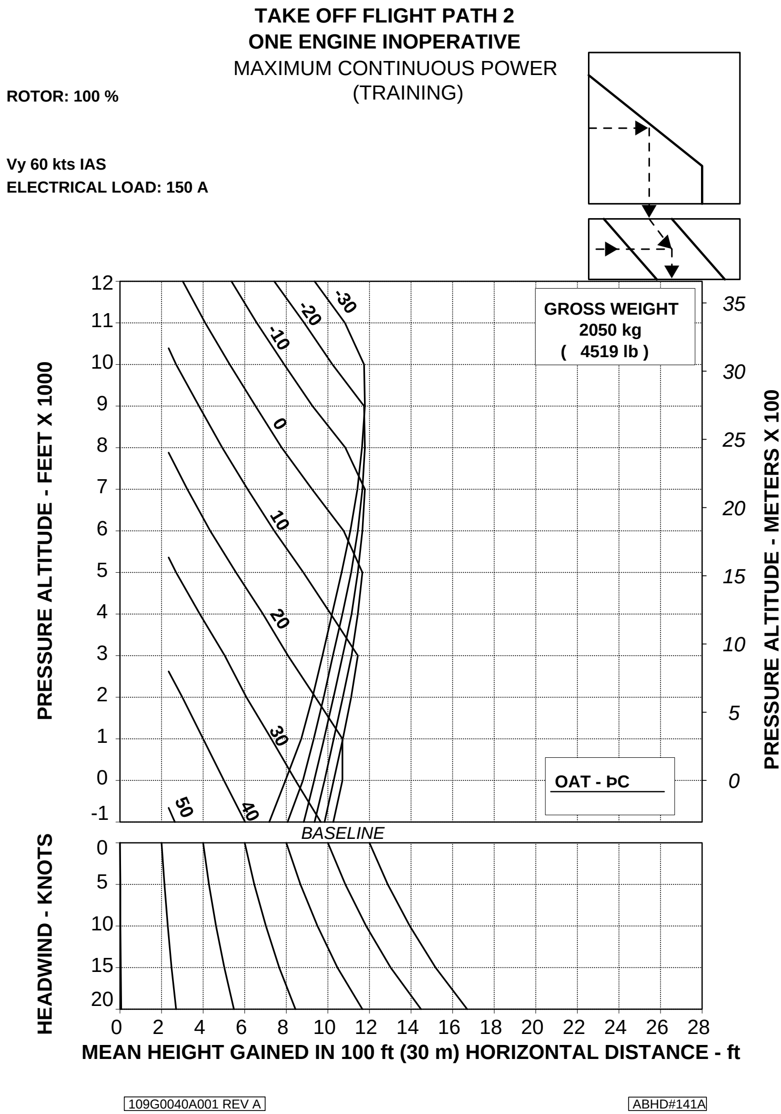


Figure 4-159. Take-off flight path 2 - 2050 kg - Training.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A

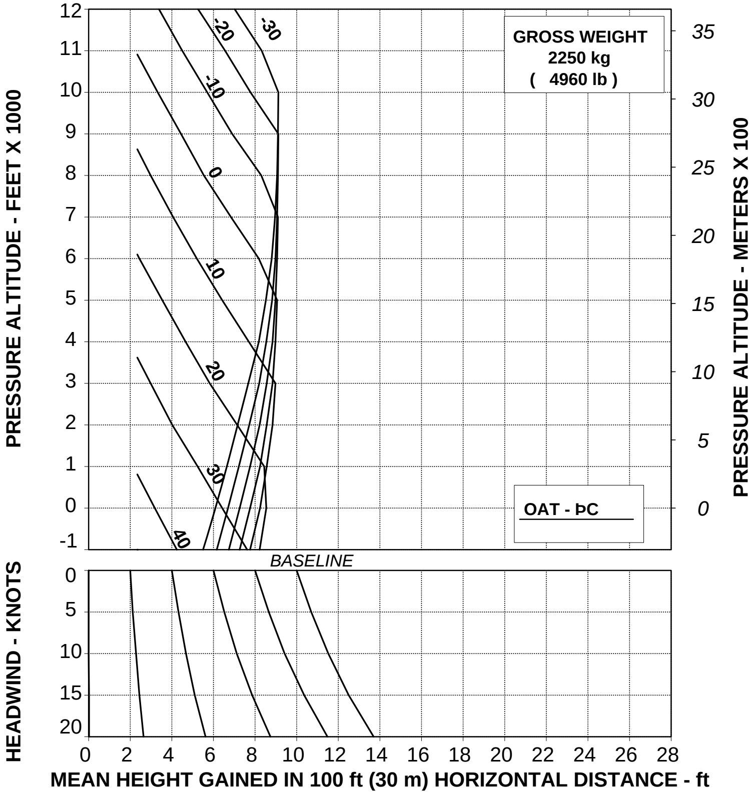


Figure 4-160. Take-off flight path 2 - 2250 kg - Training.

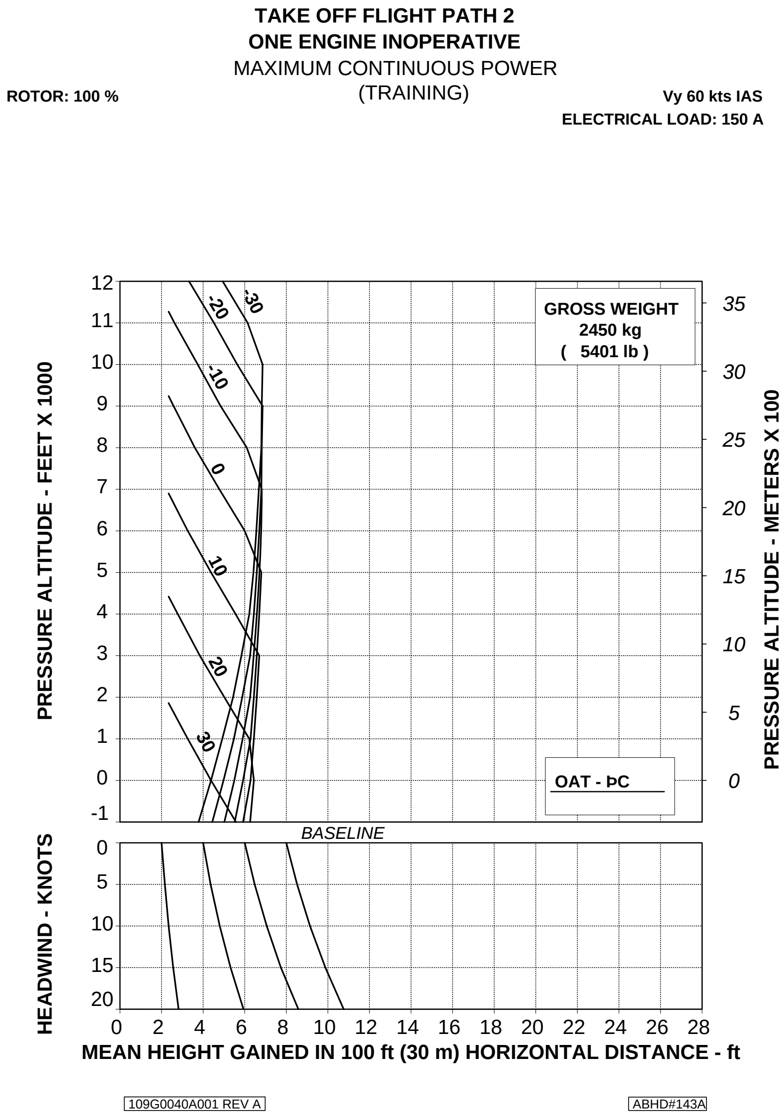


Figure 4-161. Take-off flight path 2 - 2450 kg - Training.

TAKE OFF FLIGHT PATH 2
ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER
(TRAINING)

ROTOR: 100 %

Vy 60 kts IAS
ELECTRICAL LOAD: 150 A

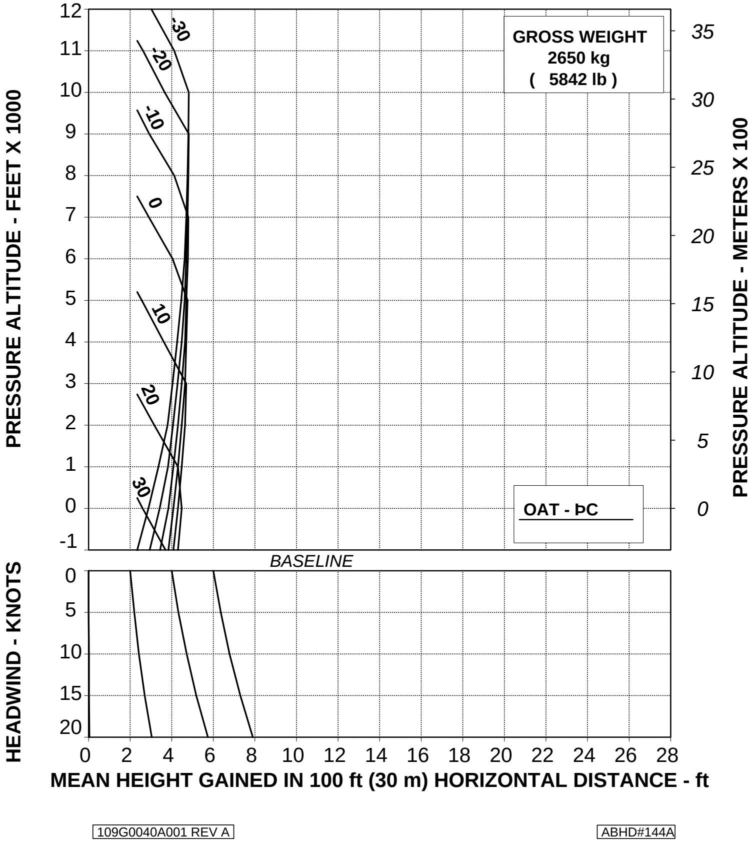


Figure 4-162. Take-off flight path 2 - 2650 kg - Training.

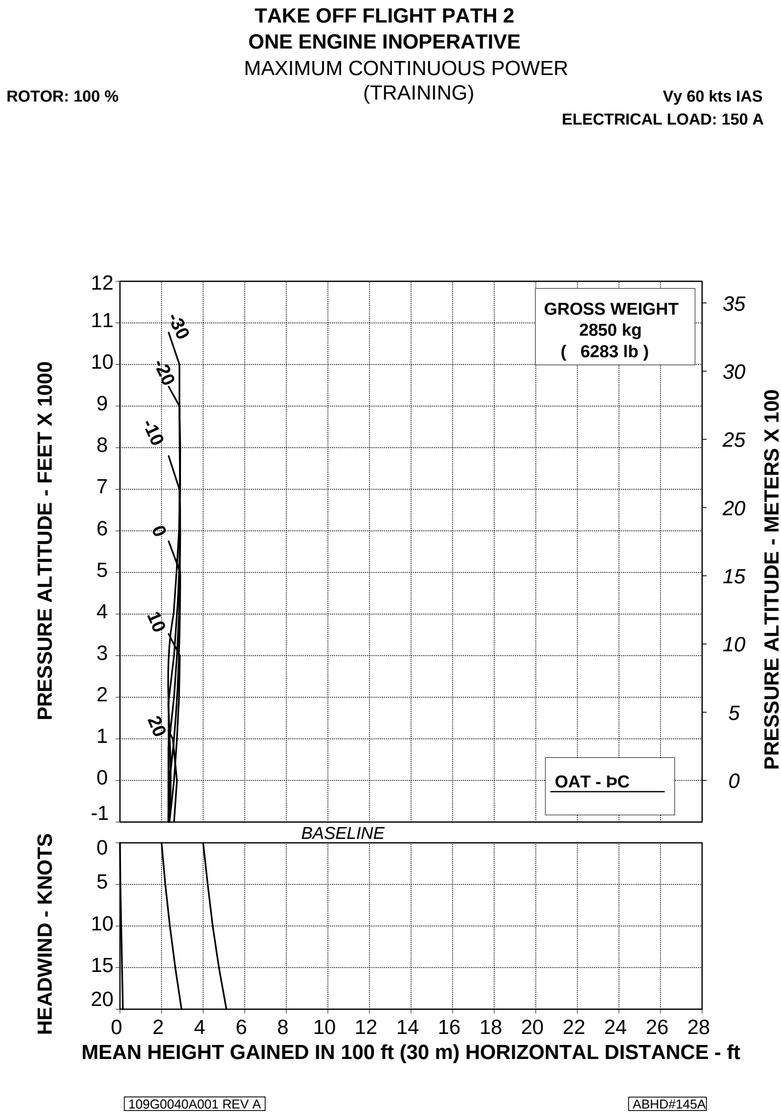


Figure 4-163. Take-off flight path 2 - 2850 kg - Training.

INCREASED INTERNAL GROSS WEIGHT

(For operation at Gross Weight from 2850 kg up to 3000 kg)

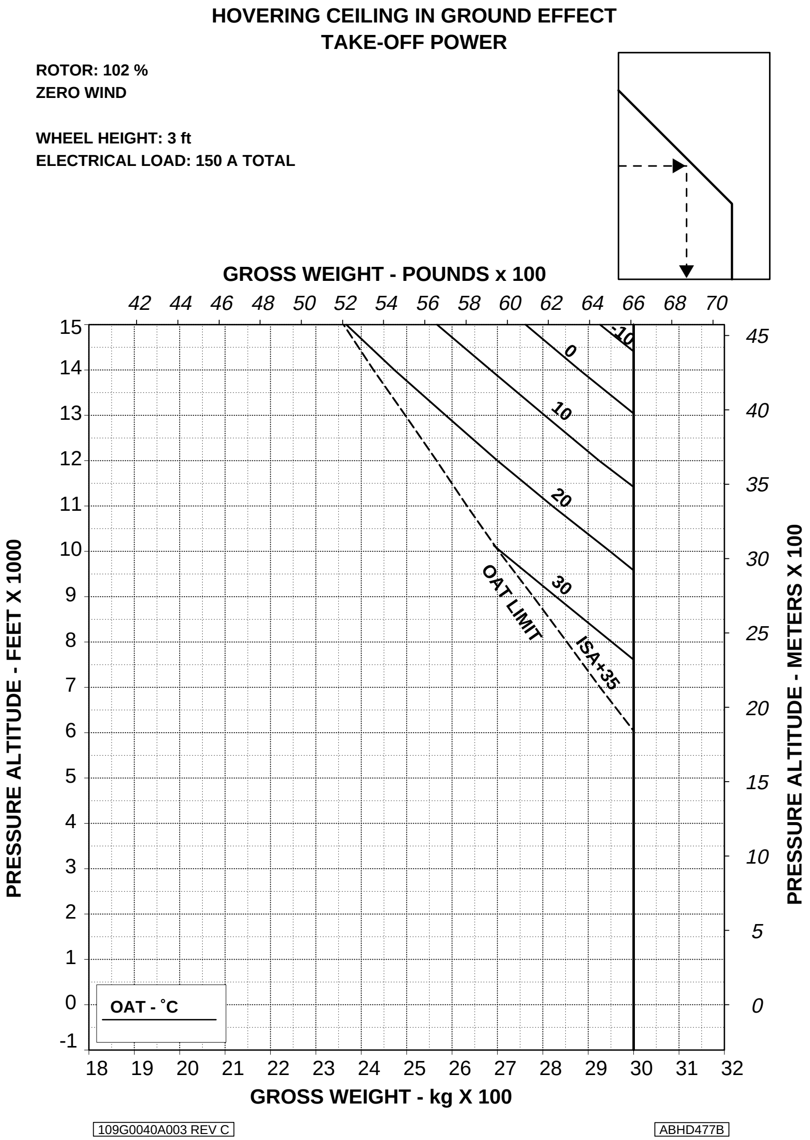


Figure 4-164. Hovering ceiling - IGE - Take-off power.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

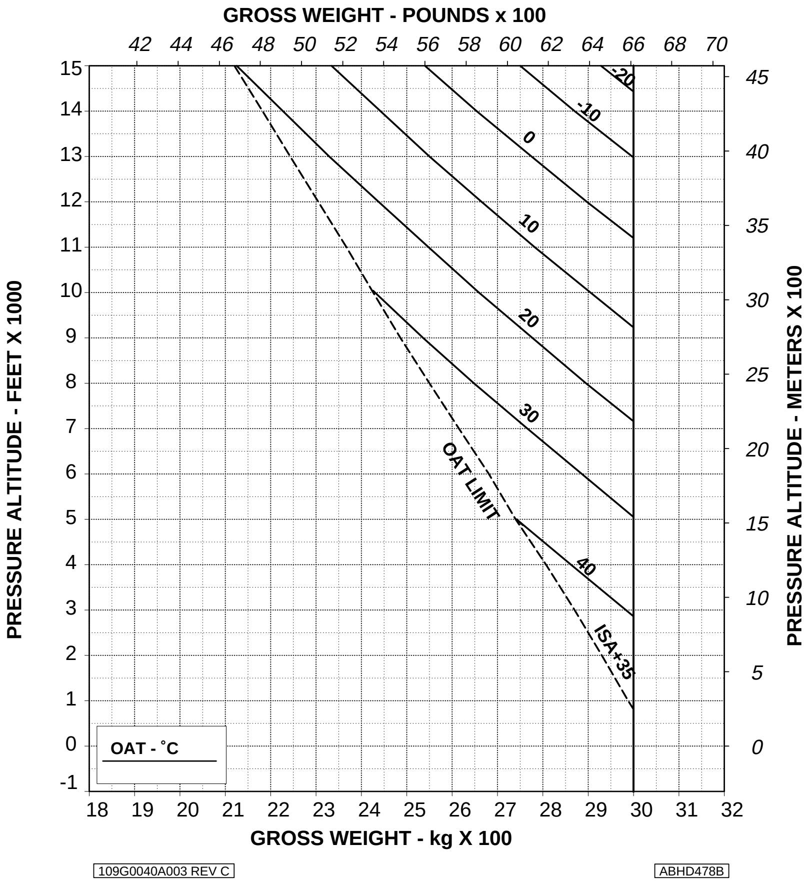
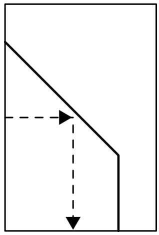


Figure 4-165. Hovering ceiling - IGE - Maximum continuous power.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
ZERO WIND
ELECTRICAL LOAD: 150 A TOTAL



CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

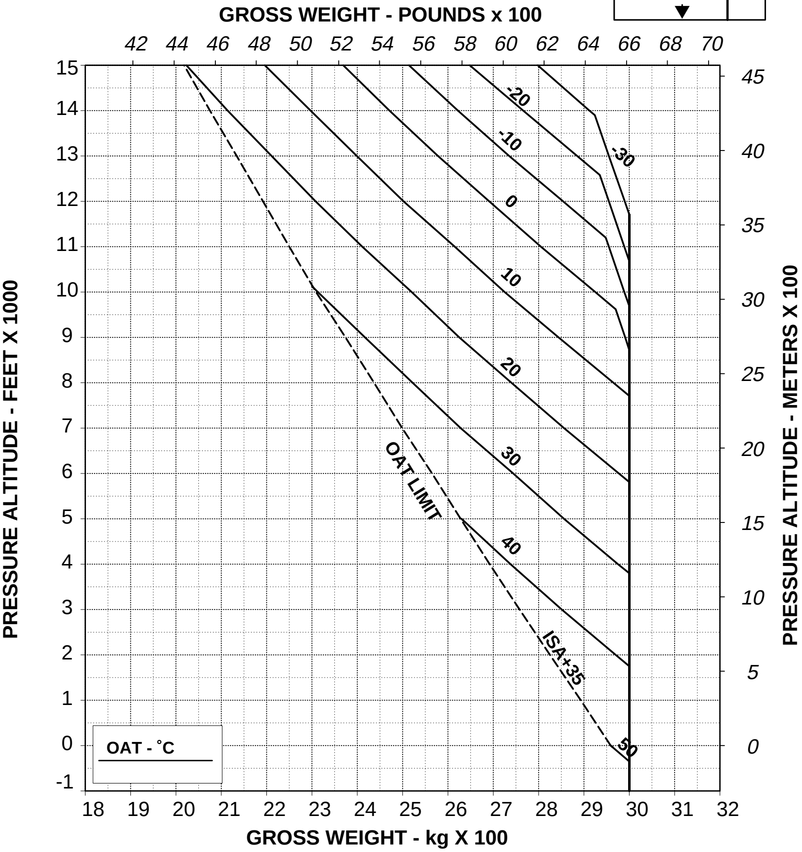


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Figure 4-166. Hovering ceiling - OGE - Take-off power.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

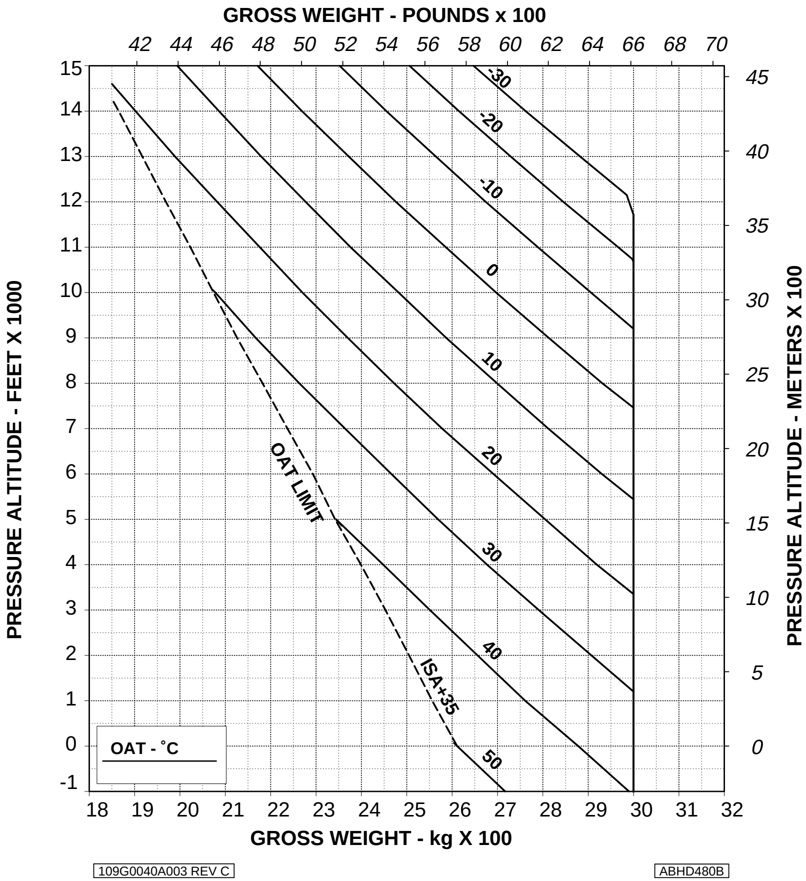

Figure 4-167. Hovering ceiling - OGE - Maximum continuous power.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

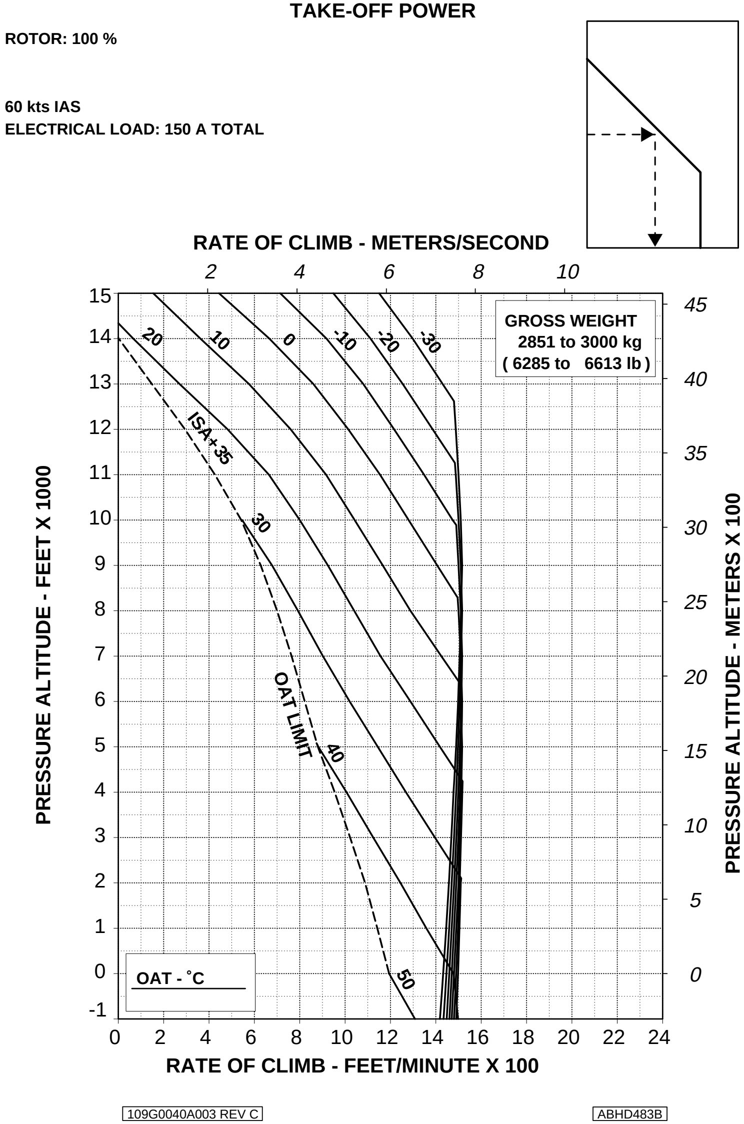

Figure 4-168. Rate of climb - All engines - Take-off power.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

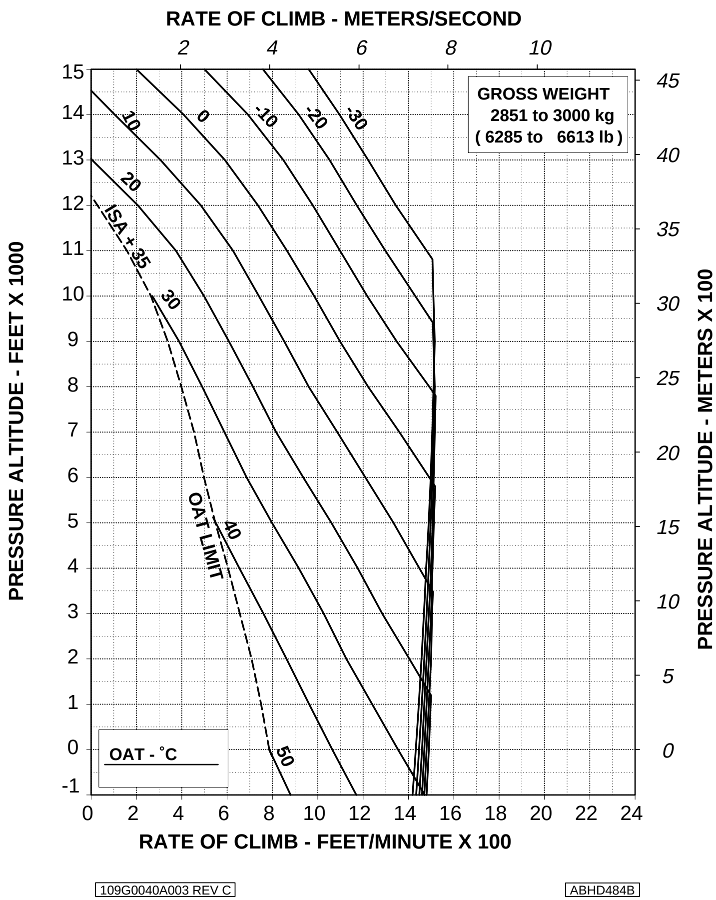

Figure 4-169. Rate of climb - All engines - Maximum continuous power.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 %

60 kts IAS
ELECTRICAL LOAD: 150 A

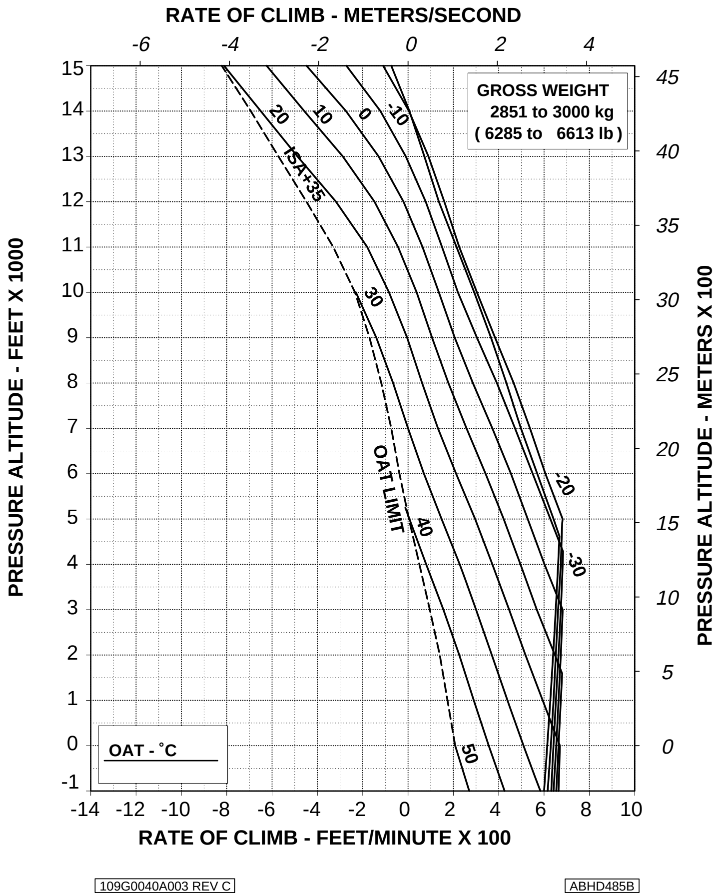

Figure 4-170. Rate of climb - OEI - 2,5 minute power.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %

60 kts IAS

ELECTRICAL LOAD: 150 A

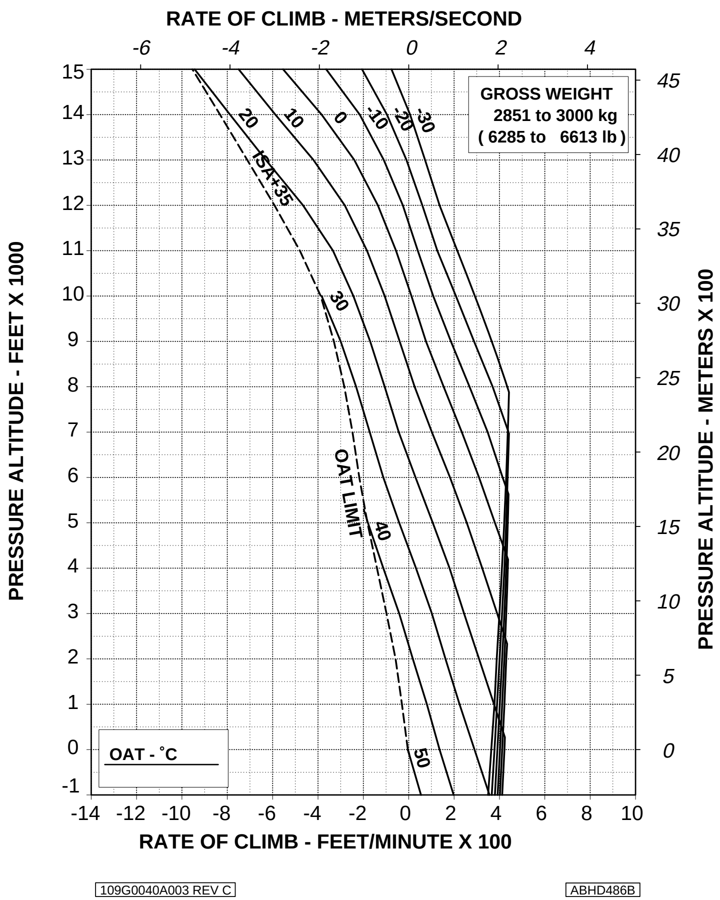

Figure 4-171. Rate of climb - OEI - Maximum continuous power.

HELICOPTER WITH BLEED-AIR HEATER

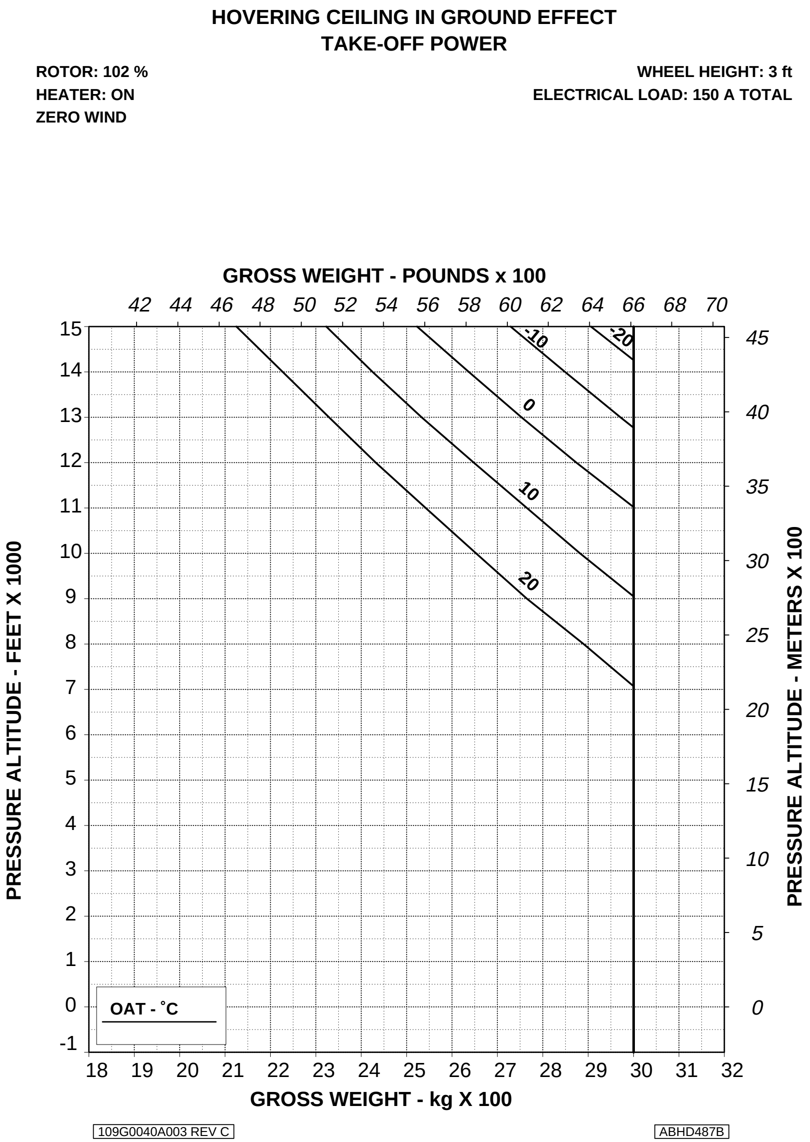

Figure 4-172. Hovering ceiling - IGE - Take-off power - Heater ON.

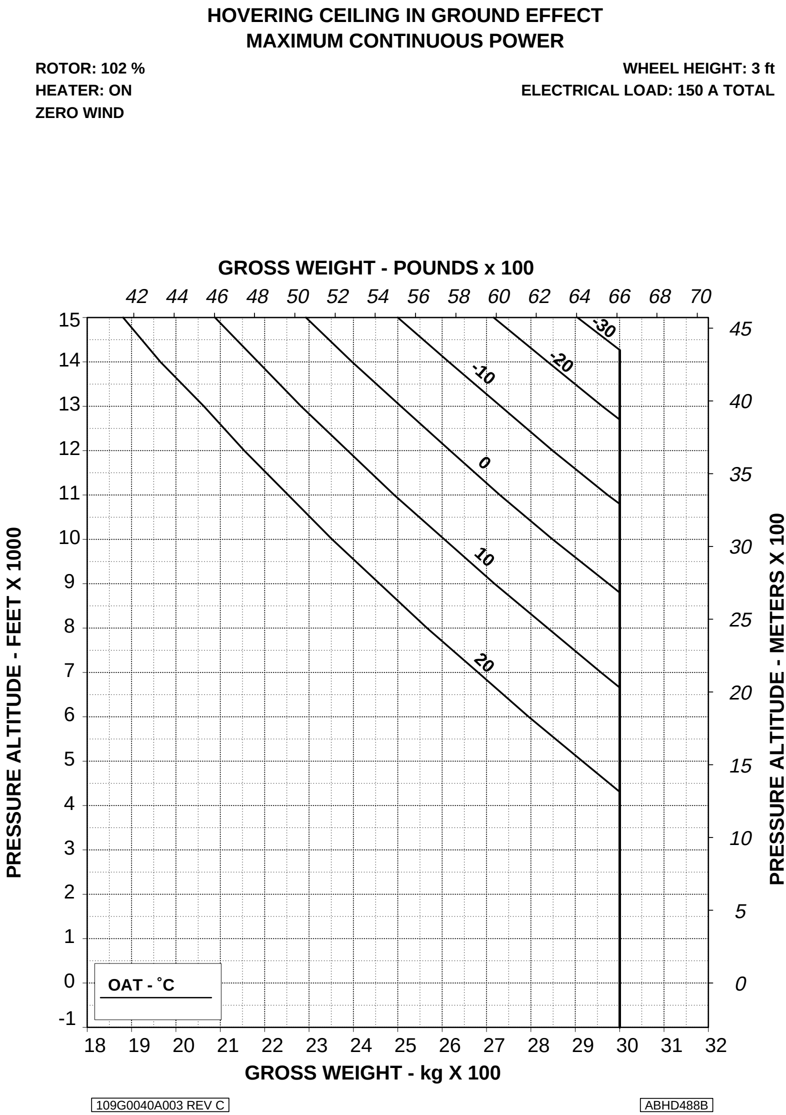

Figure 4-173. Hovering ceiling - IGE - Maximum continuous power - heater on.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

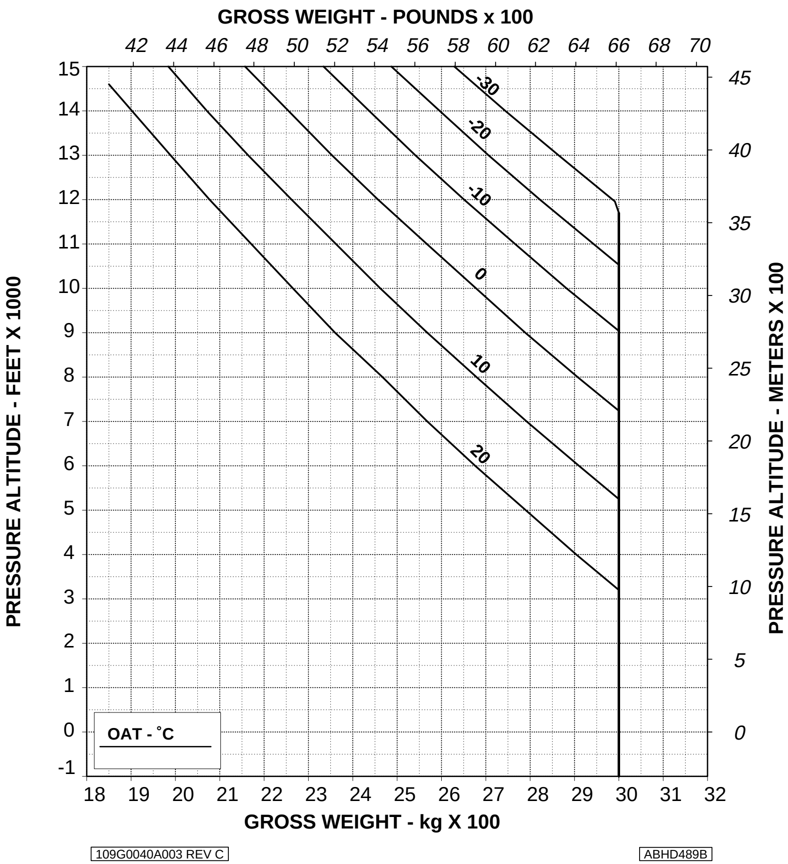

Figure 4-174. Hovering ceiling - OGE - Take-off power - Heater ON.

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HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
HEATER: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

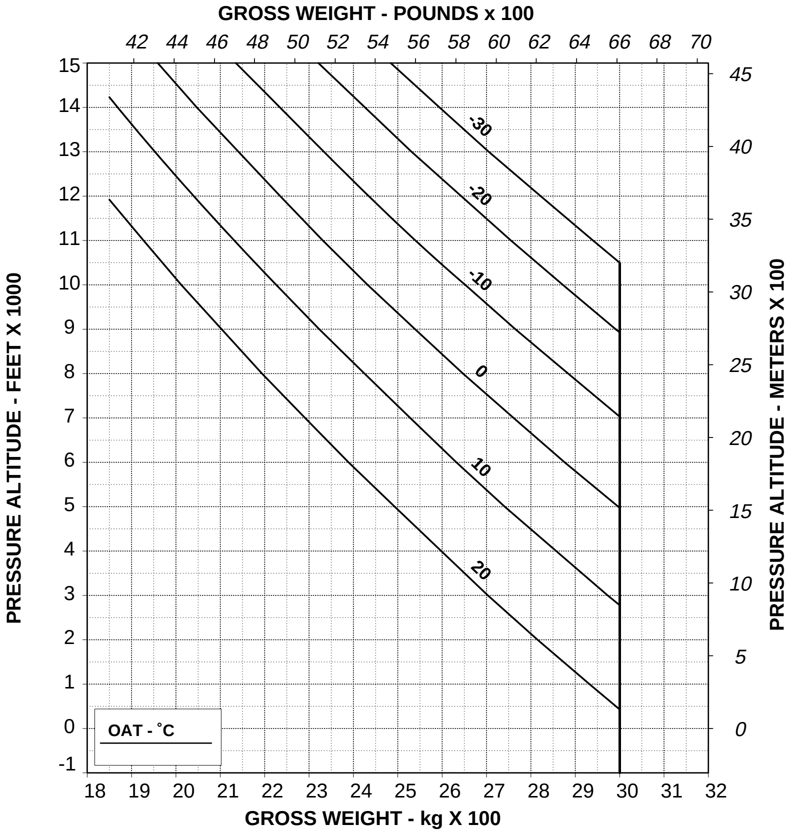

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Figure 4-175. Hovering ceiling - OGE - Maximum continuous power - Heater ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
HEATER: ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

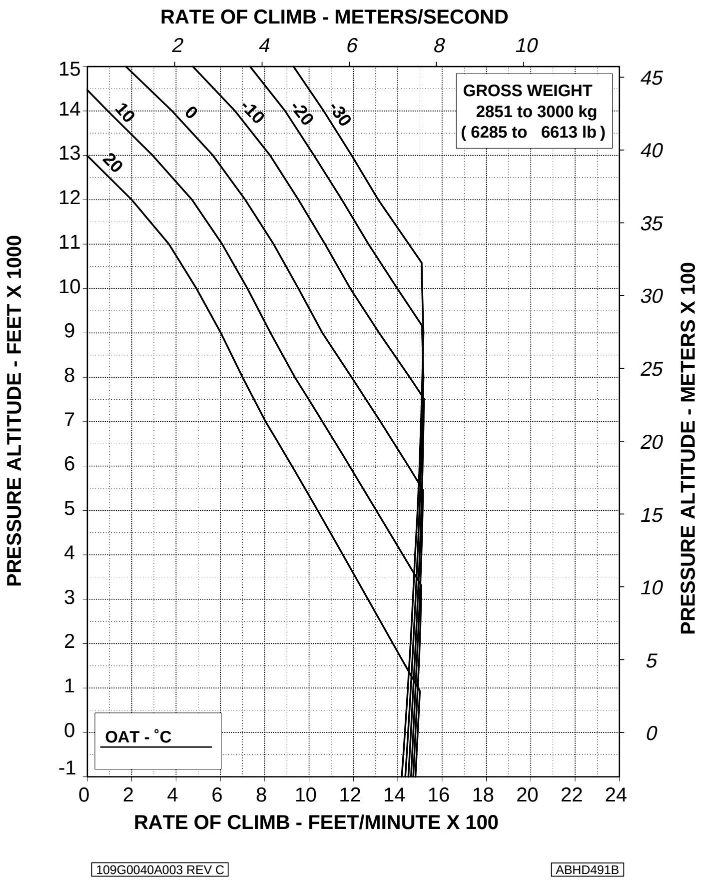


Figure 4-176. Rate of climb - All engines - Take-off power - Heater ON.

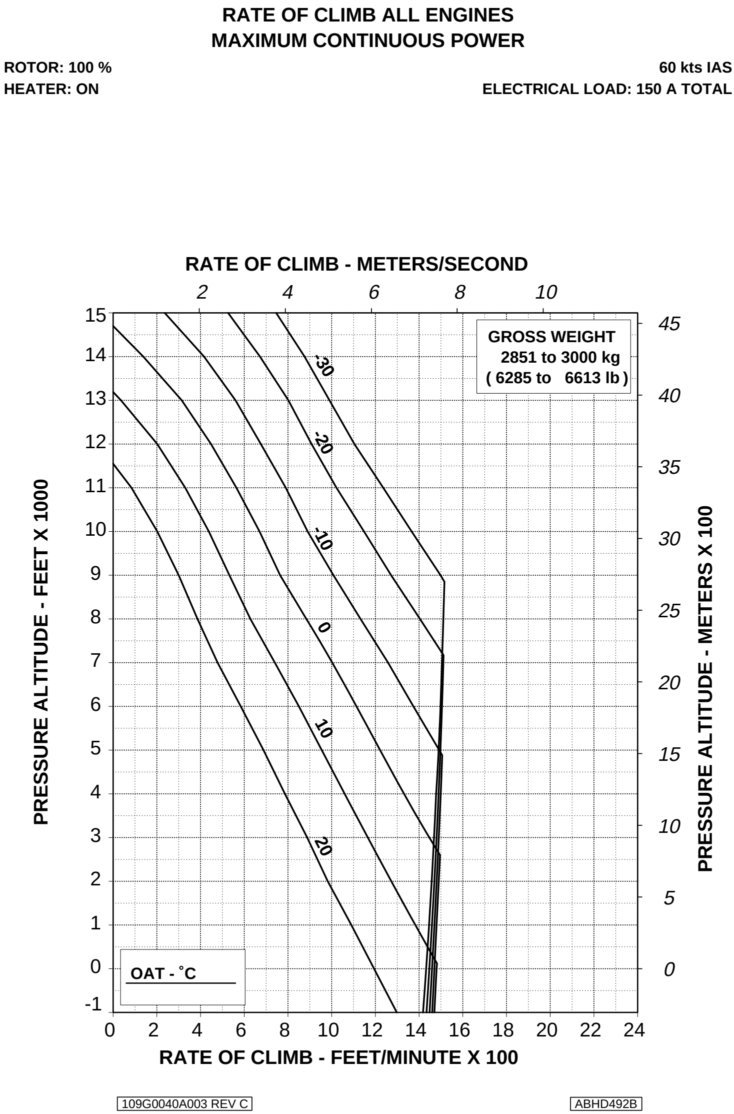


Figure 4-177. Rate of climb - All engines - Maximum continuous power - Heater ON.

HELICOPTER WITH ENVIRONMENTAL CONTROL SYSTEM (ECS)

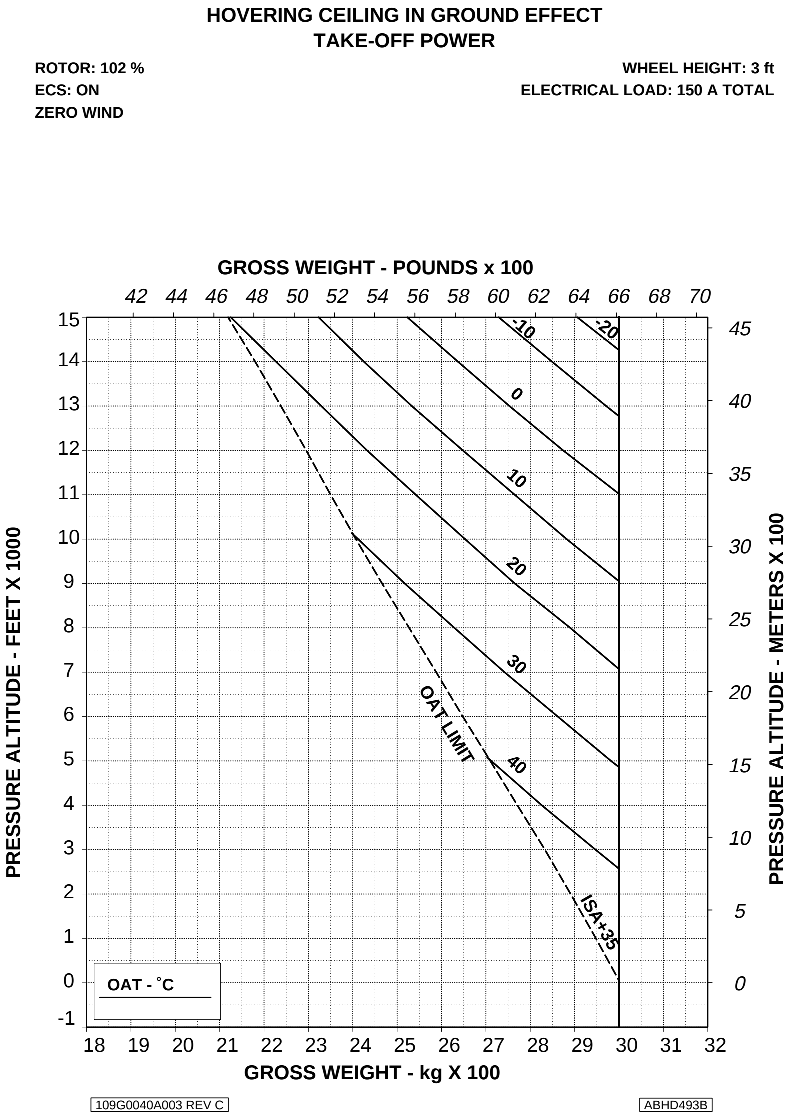


Figure 4-178. Hovering ceiling - IGE - Take-off power - ECS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL

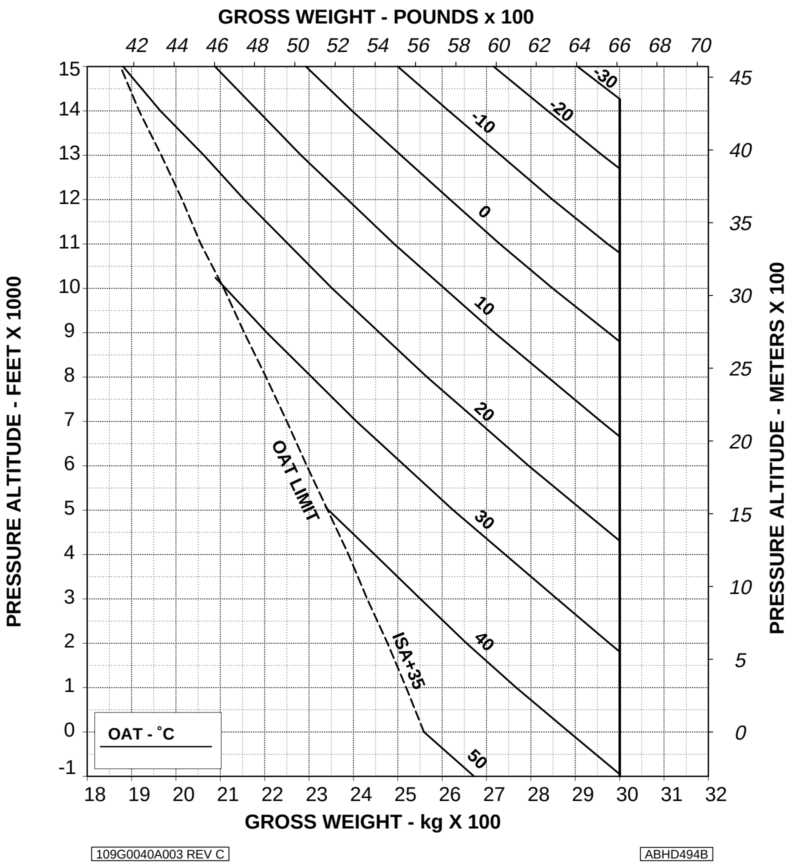


Figure 4-179. Hovering ceiling - IGE - Maximum continuous power - ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

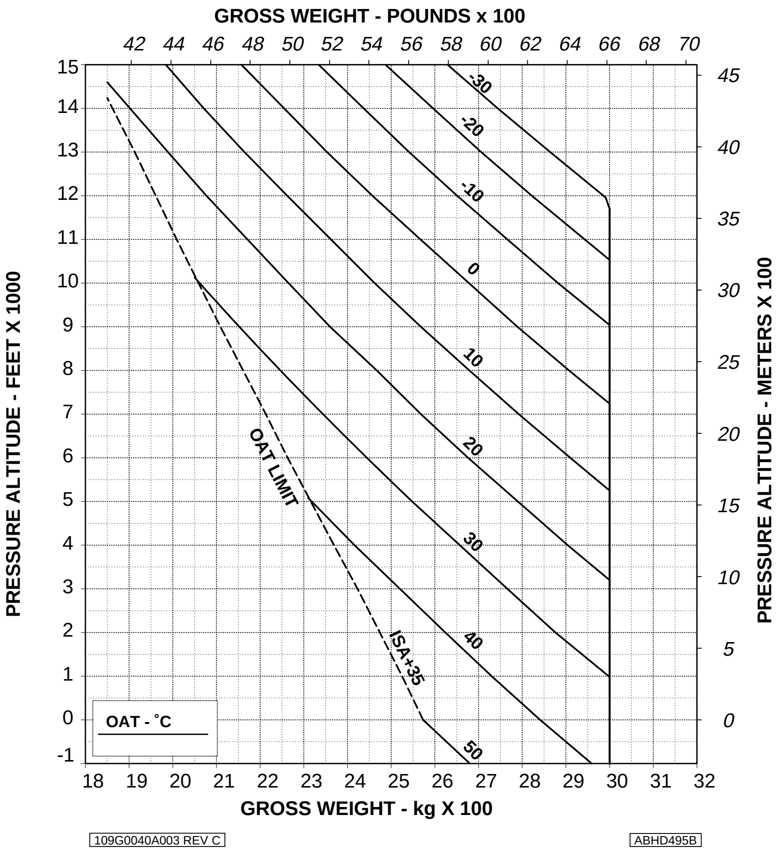


Figure 4-180. Hovering ceiling - OGE - Take-off power - ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
ECS: ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

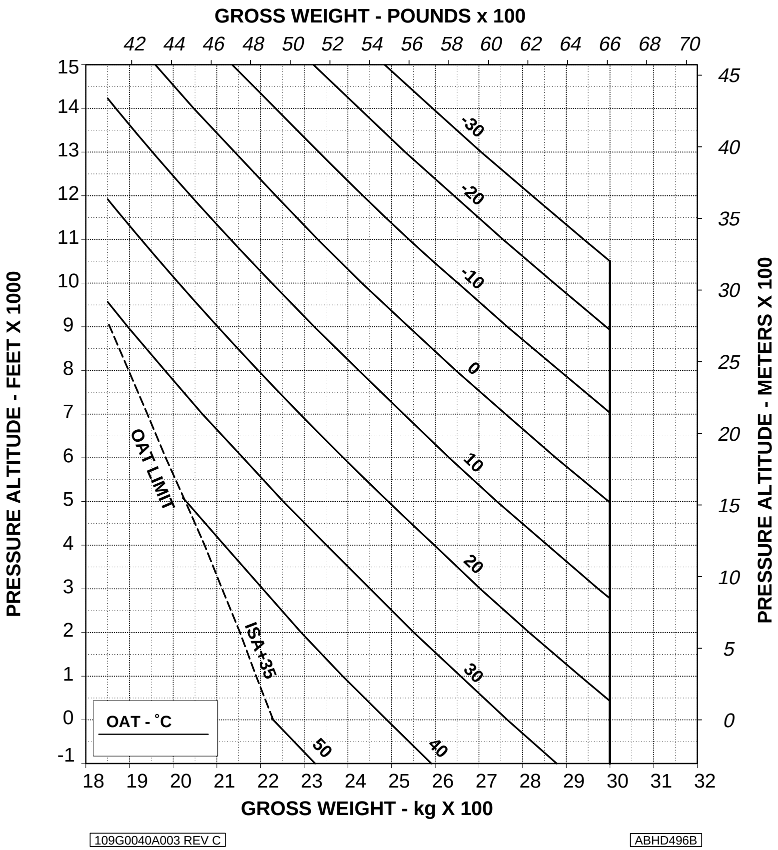


Figure 4-181. Hovering ceiling - OGE - Maximum continuous power.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 % 60 kts IAS
ECS: ON ELECTRICAL LOAD: 150 A TOTAL

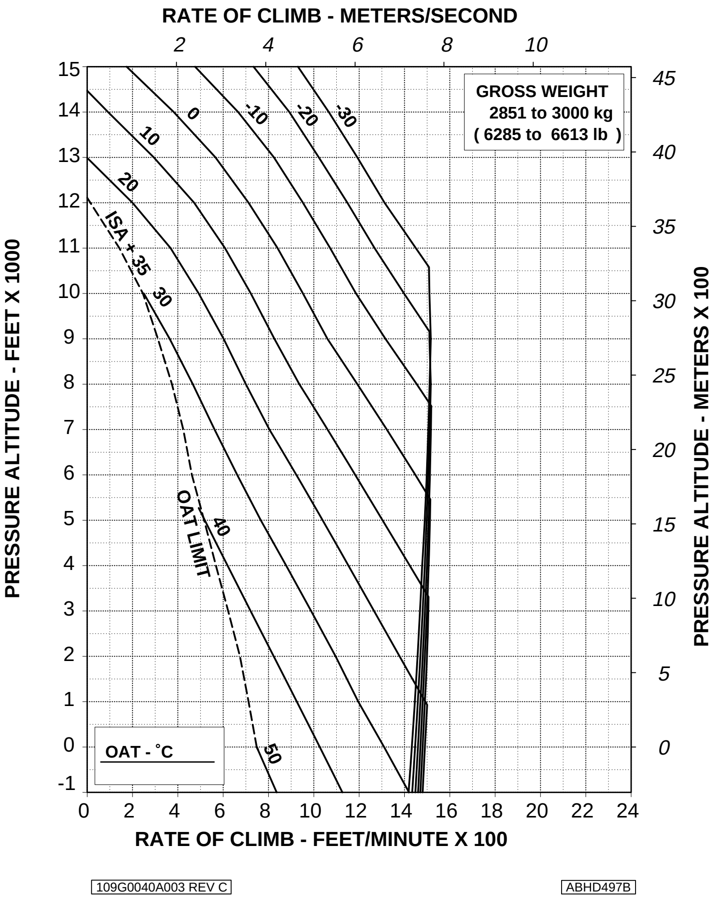


Figure 4-182. Rate of climb - All engines - Take-off power - ECS ON.

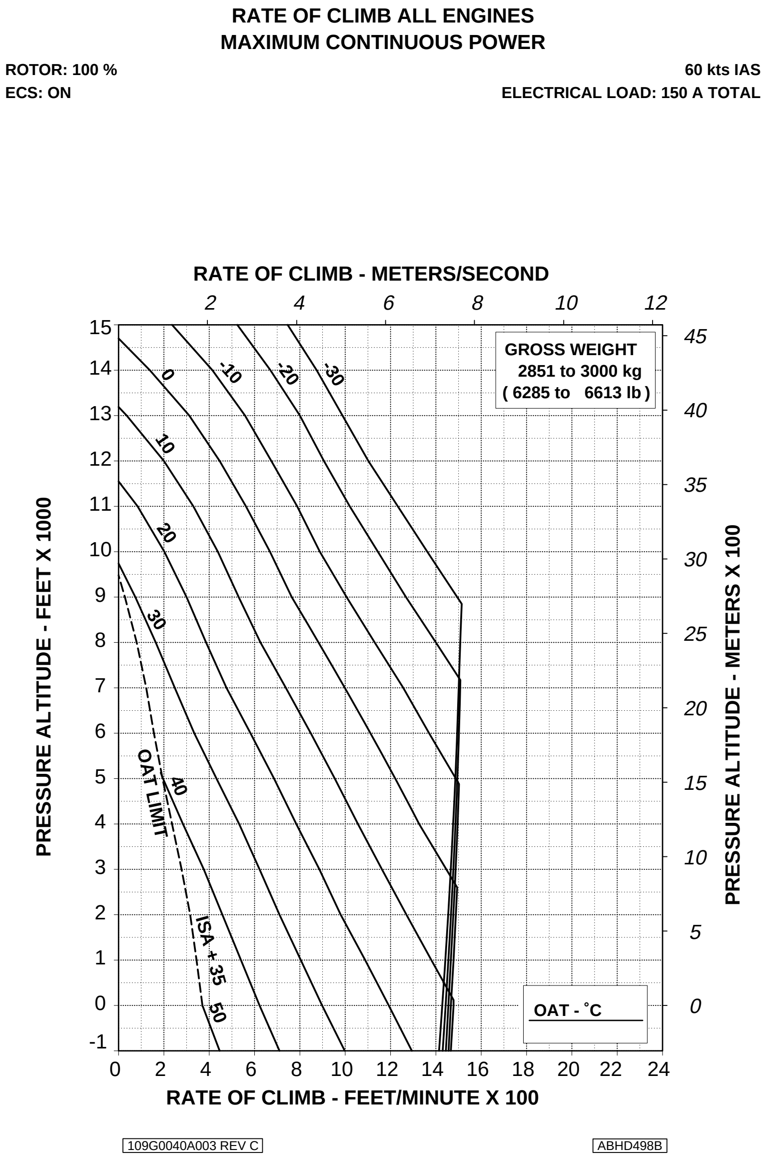


Figure 4-183. Rate of climb - All engines - Maximum continuous power - ECS ON.

HELICOPTER WITH ENGINE AIR PARTICLE SEPARATOR (EAPS) OFF or ON

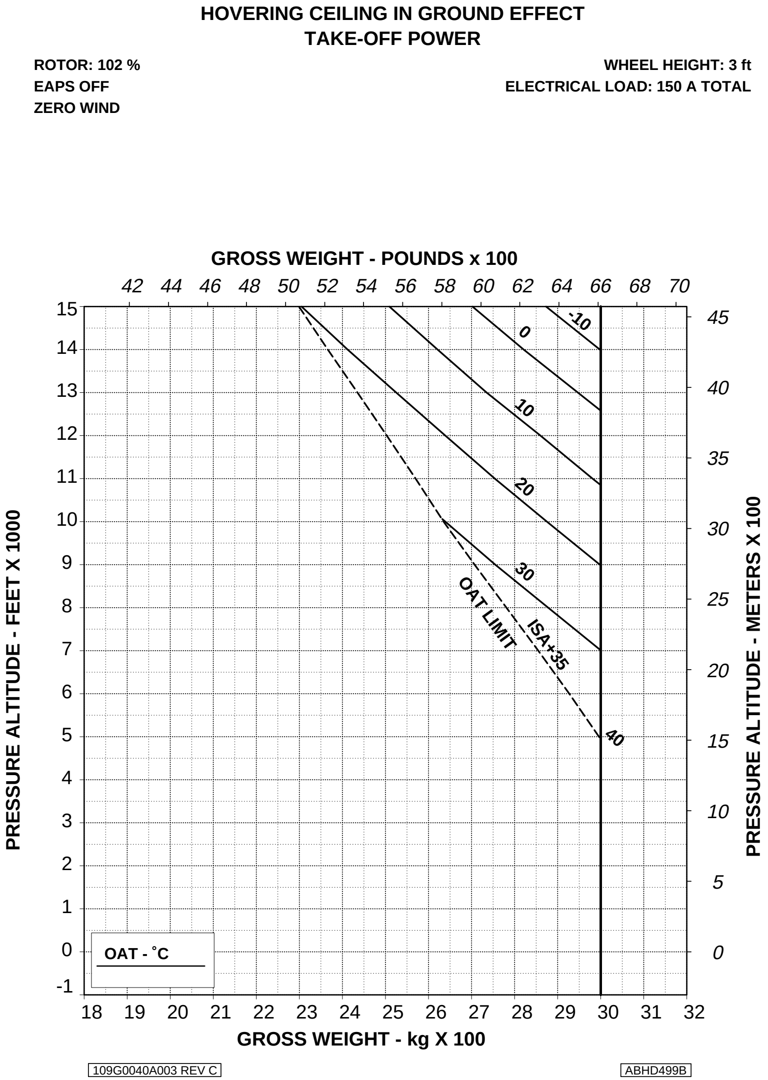


Figure 4-184. Hovering ceiling - IGE - Take-off power - EAPS OFF.

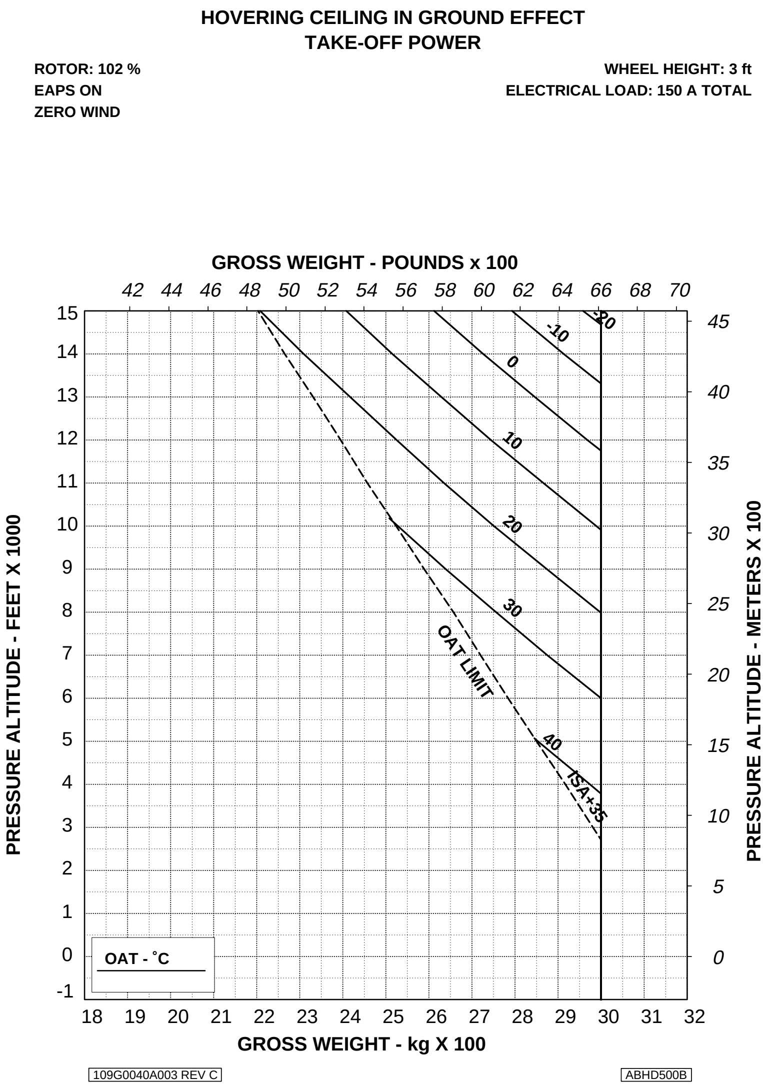


Figure 4-185. Hovering ceiling - IGE - Take-off power - EAPS ON.

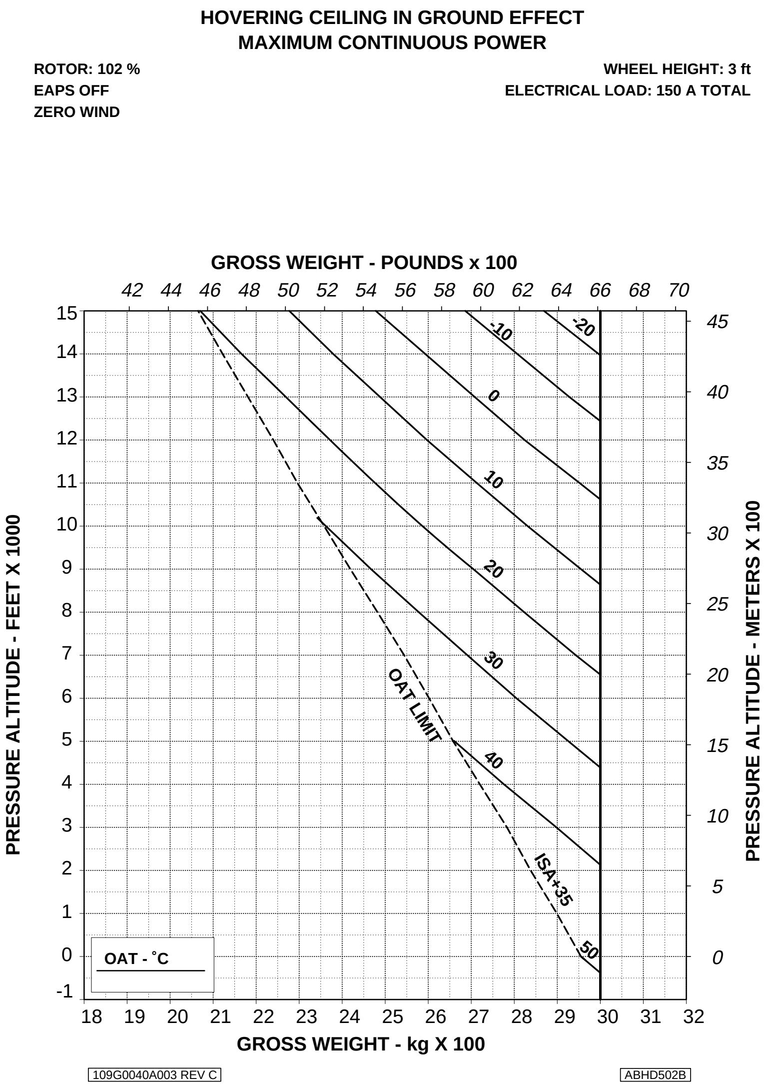


Figure 4-186. Hovering ceiling - IGE - maximum continuous power - EAPS OFF.

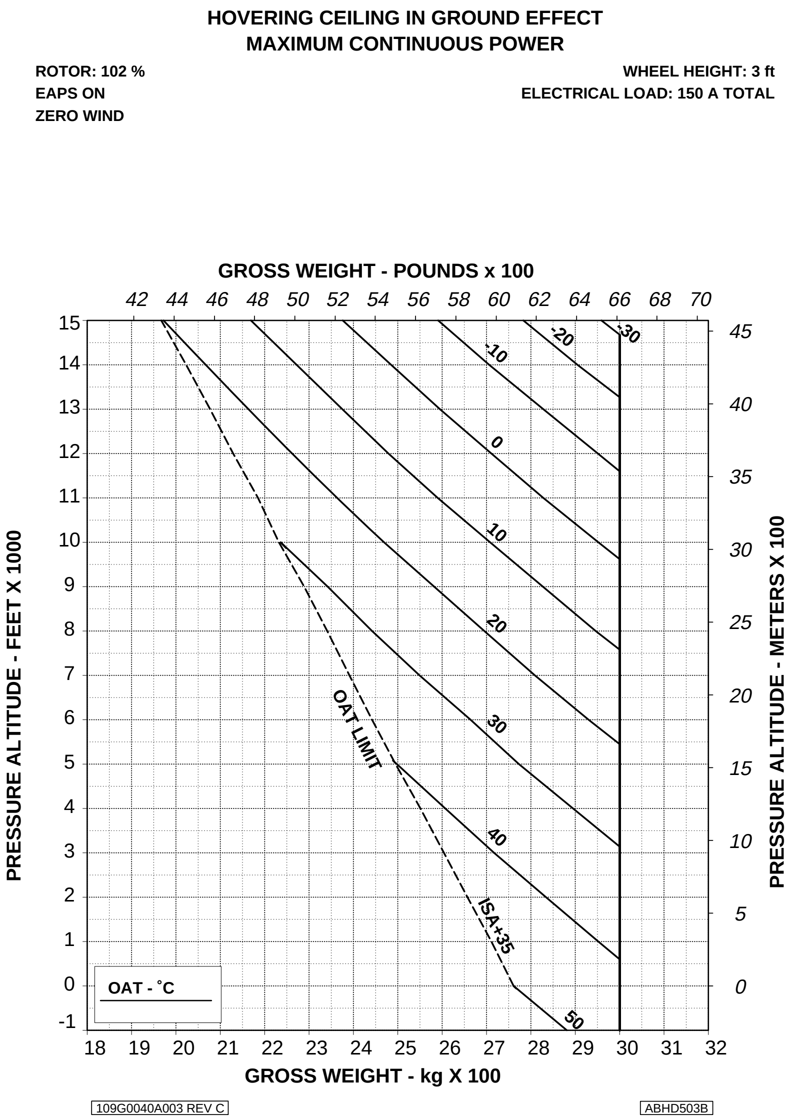


Figure 4-187. Hovering ceiling - IGE - maximum continuous power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

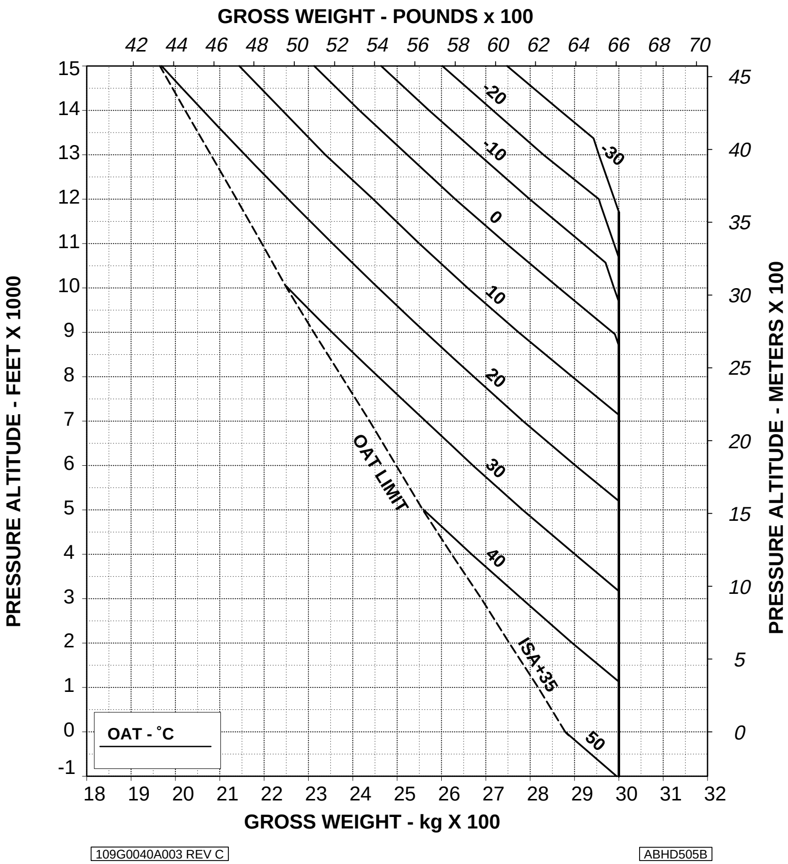


Figure 4-188. Hovering ceiling - OGE - Take-off power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

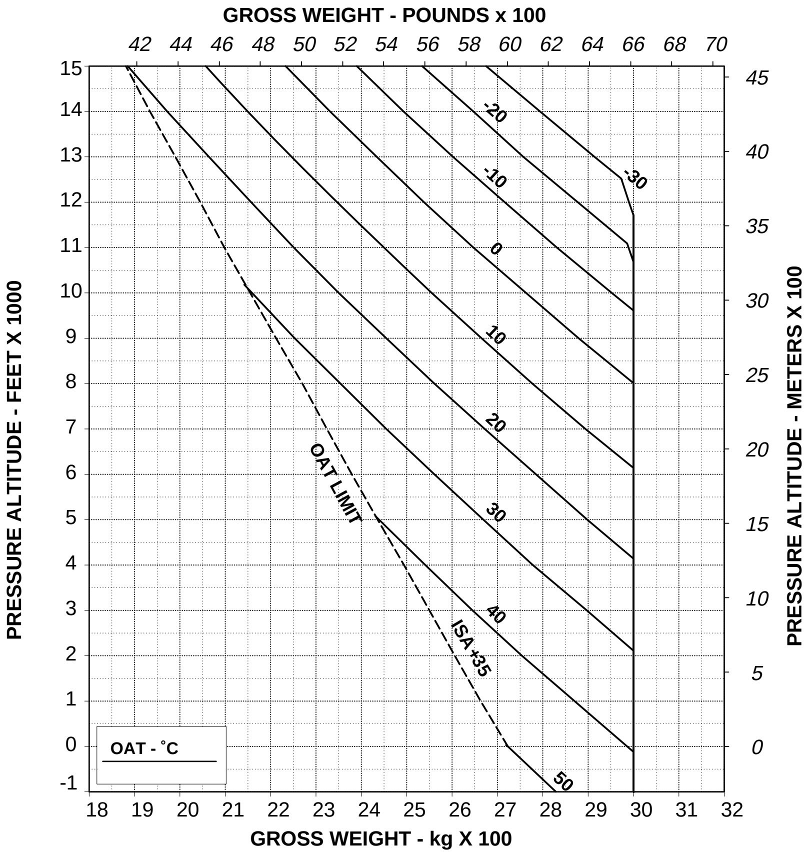


Figure 4-189. Hovering ceiling - OGE - Take-off power - EAPS ON.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

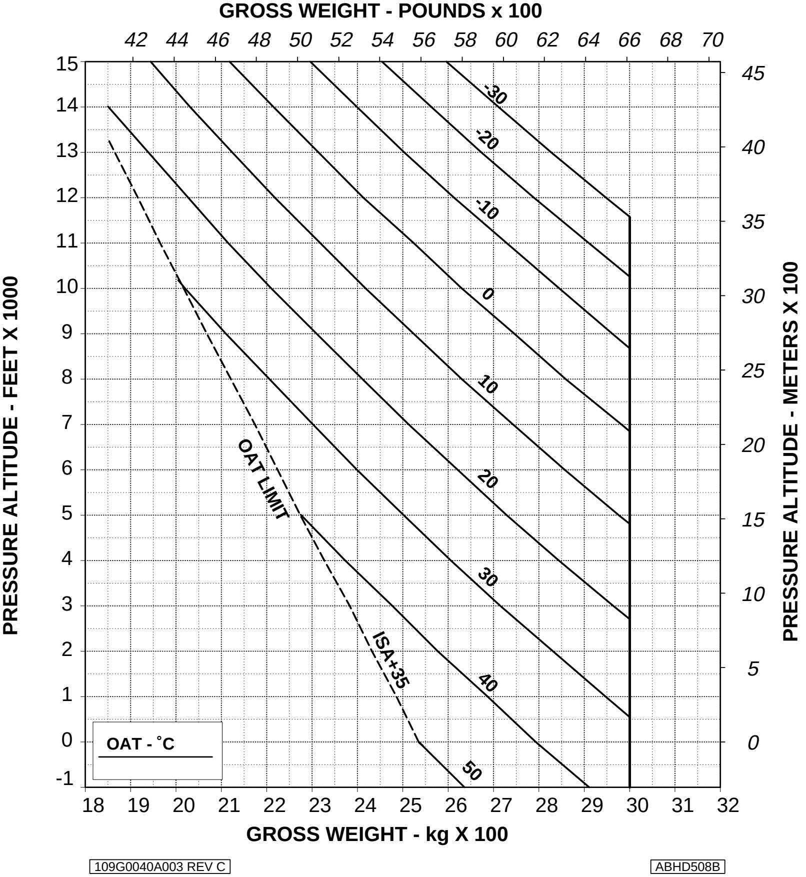


Figure 4-190. Hovering ceiling - OGE - maximum continuous power - EAPS OFF.

HOVERING CEILING OUT OF GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS ON
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

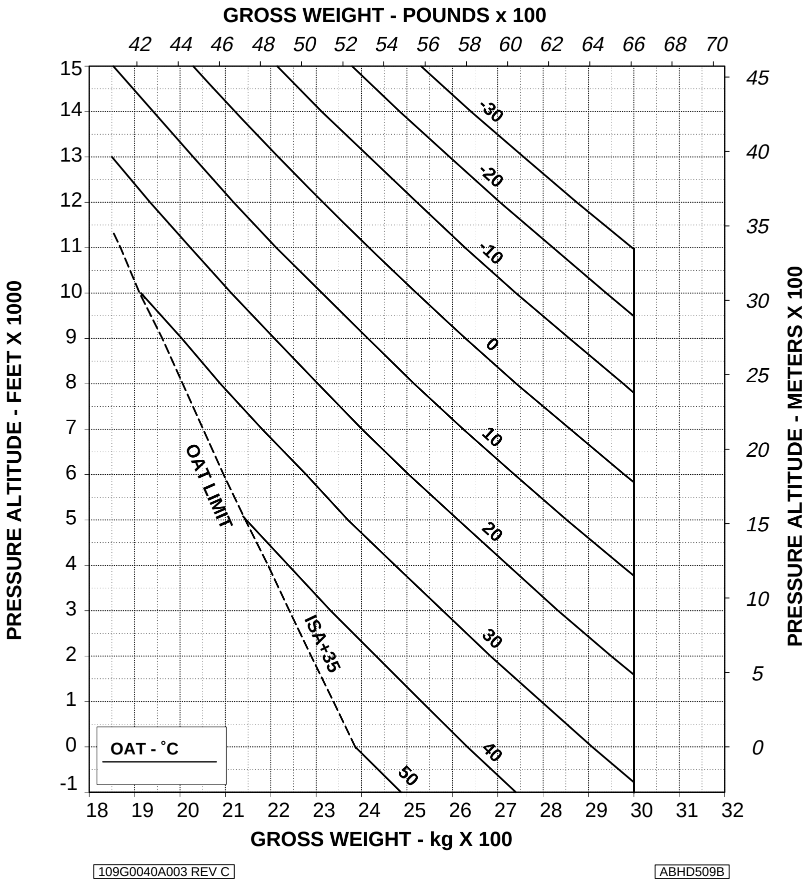


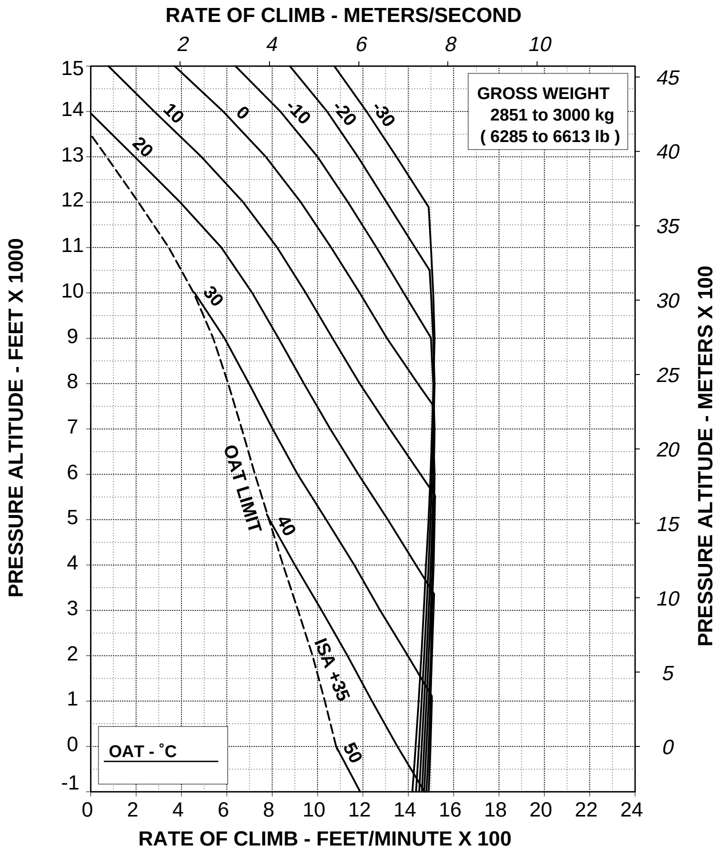
Figure 4-191. Hovering ceiling - OGE - maximum continuous power - EAPS ON.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 250 ft/min (1,27 m/s)



109G0040A003 REV C

ABHD515B

Figure 4-192. Rate of climb - All engines - Take-off power - EAPS OFF/ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 200 ft/min (1,02 m/s)

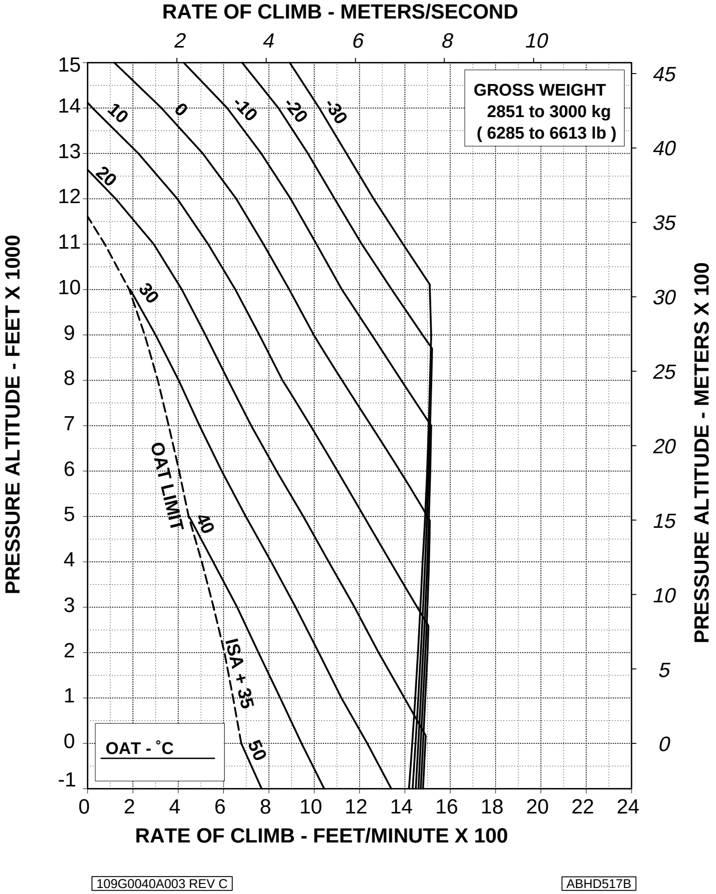


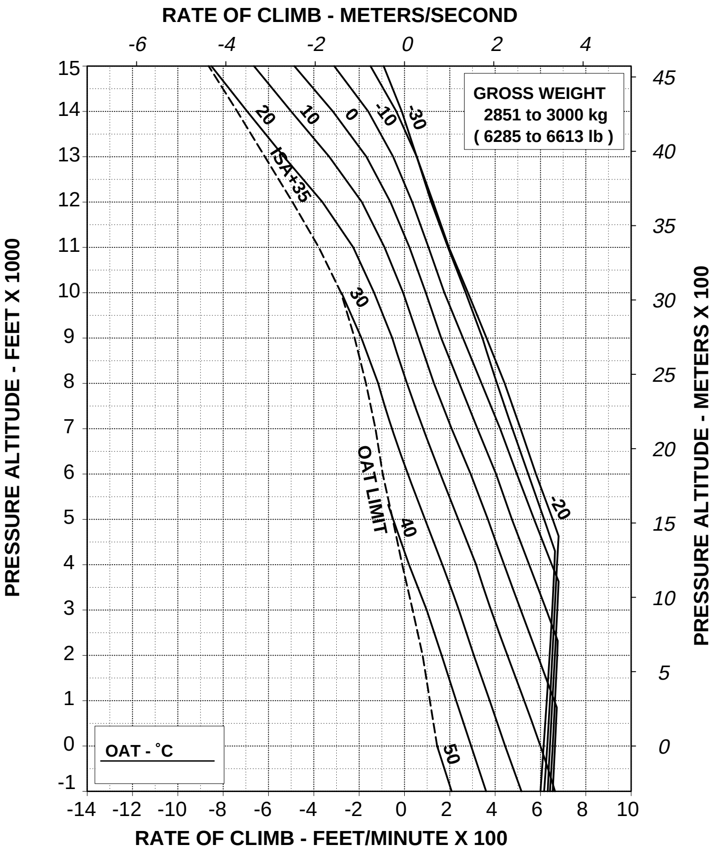
Figure 4-193. Rate of climb - All engines - Maximum continuous power - EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
2.5 MINUTE POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)



109G0040A003 REV C

ABHD519B

Figure 4-194. Rate of climb - OEI - 2,5 minute power - EAPS OFF/ON.
- EAPS OFF/ON.

RATE OF CLIMB ONE ENGINE INOPERATIVE
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF

60 kts IAS
ELECTRICAL LOAD: 150 A

WITH EAPS ON: DECREASE RATE OF CLIMB BY 100 ft/min (0,51 m/s)

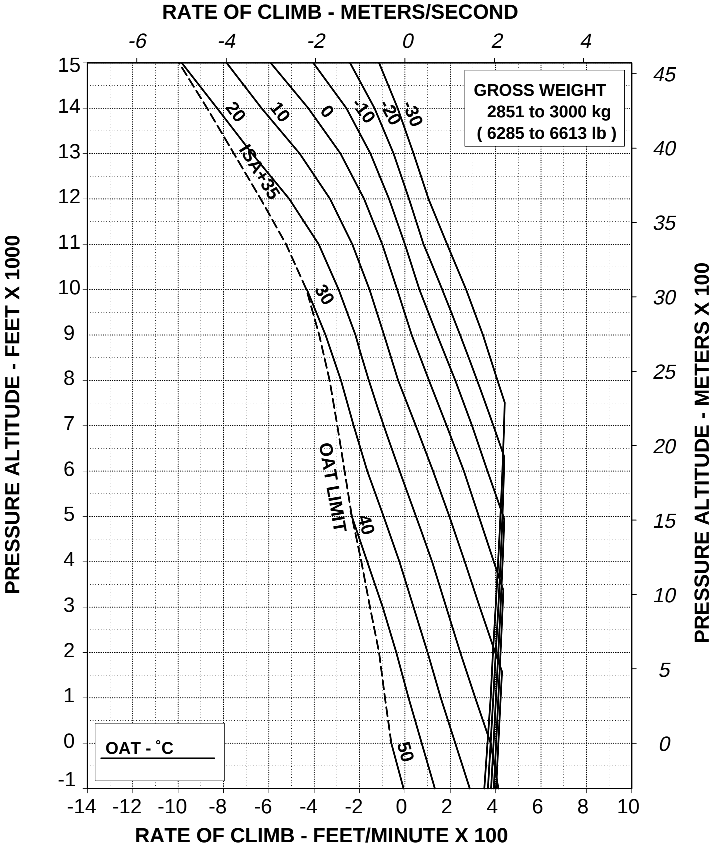


Figure 4-195. Rate of climb - OEI - Maximum continuous power - EAPS OFF/ON.

HELICOPTER WITH ENGINE AIR PARTICLE SEPARATOR (EAPS) AND HEATER OR ENVIRONMENTAL CONTROL SYSTEM (ECS)

HOVERING CEILING IN GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 40 kg (88 lb)

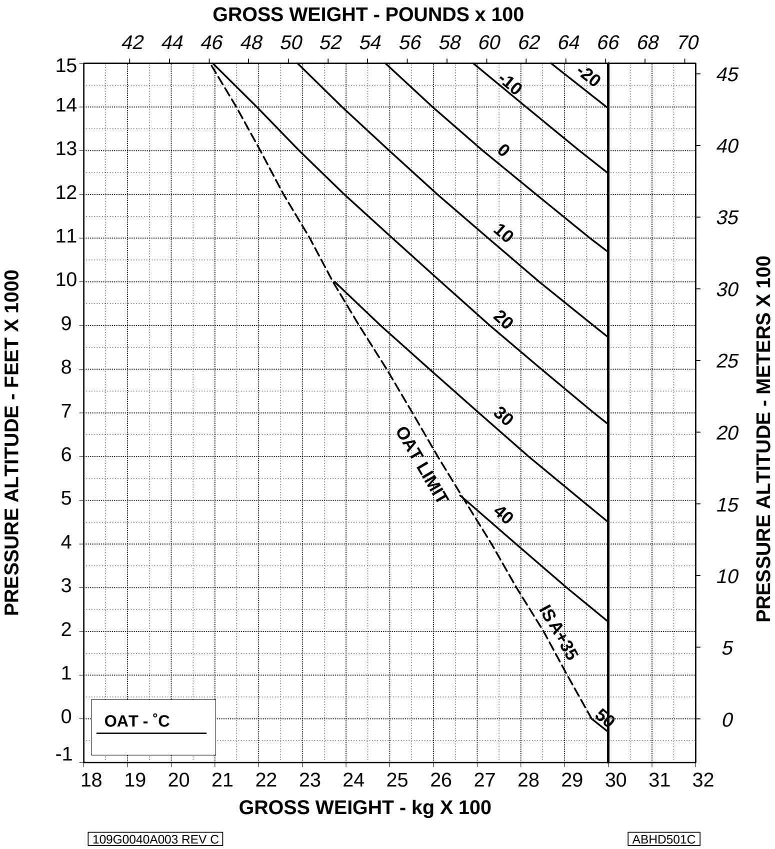


Figure 4-196. Hovering ceiling - IGE - Take-off power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING IN GROUND EFFECT
MAXIMUM CONTINUOUS POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

WHEEL HEIGHT: 3 ft
ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

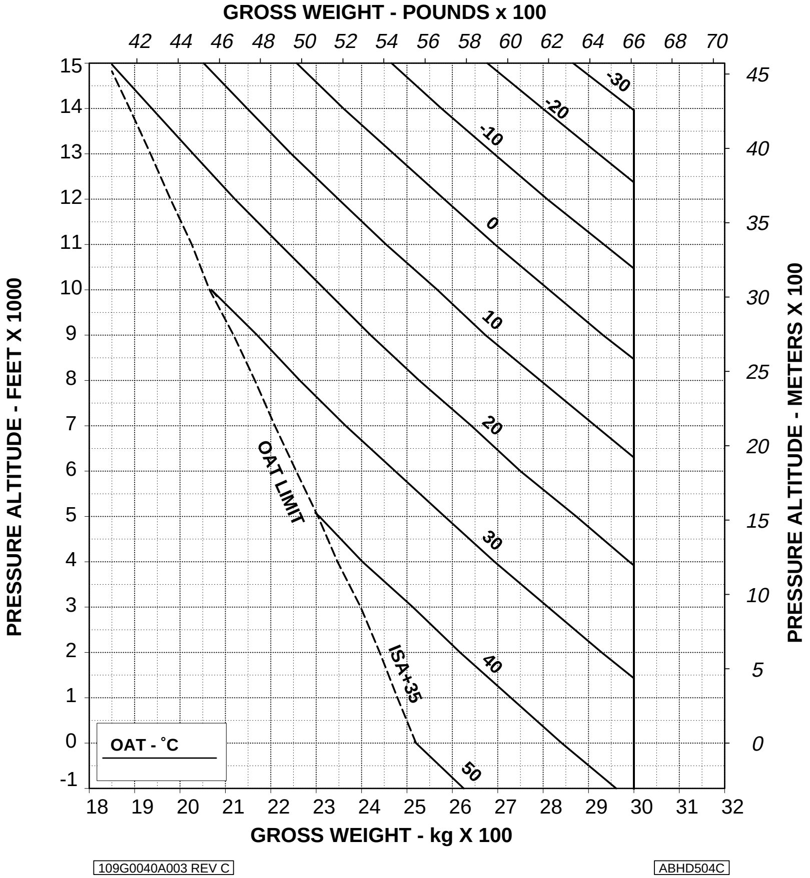


Figure 4-197. Hovering ceiling - IGE - Maximum continuous power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING OUT OF GROUND EFFECT
TAKE-OFF POWER

ROTOR: 102 %
EAPS OFF
ZERO WIND

ELECTRICAL LOAD: 150 A TOTAL
HEATER OR ECS ON

WITH EAPS ON: DECREASE GROSS WEIGHT BY 50 kg (110 lb)

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V.

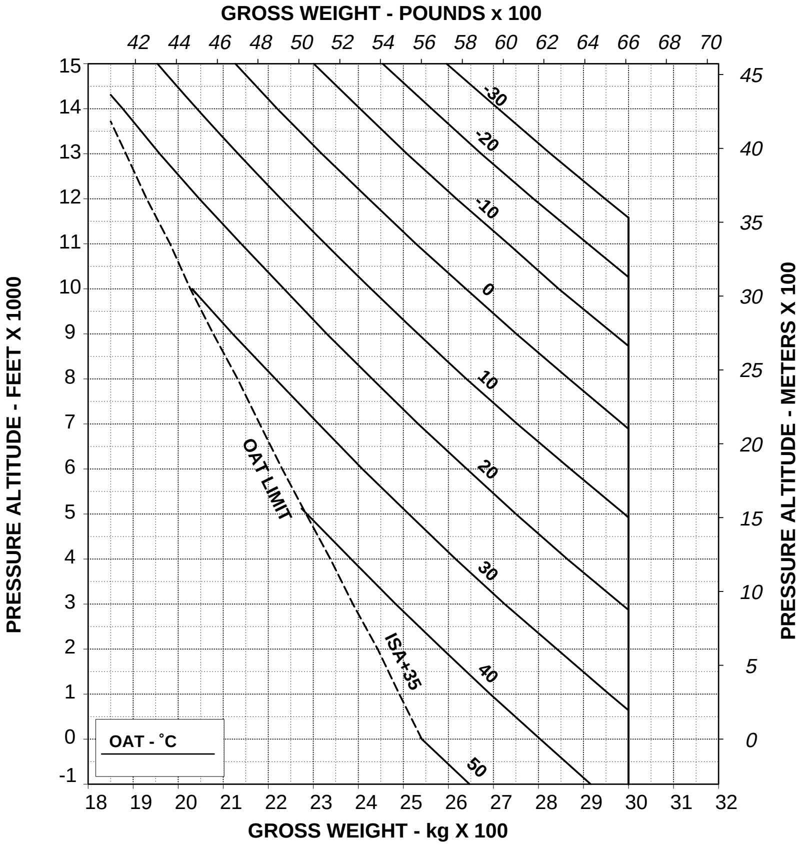


Figure 4-198. Hovering ceiling - OGE - Take-off power - EAPS OFF/ON - Heater or ECS ON.

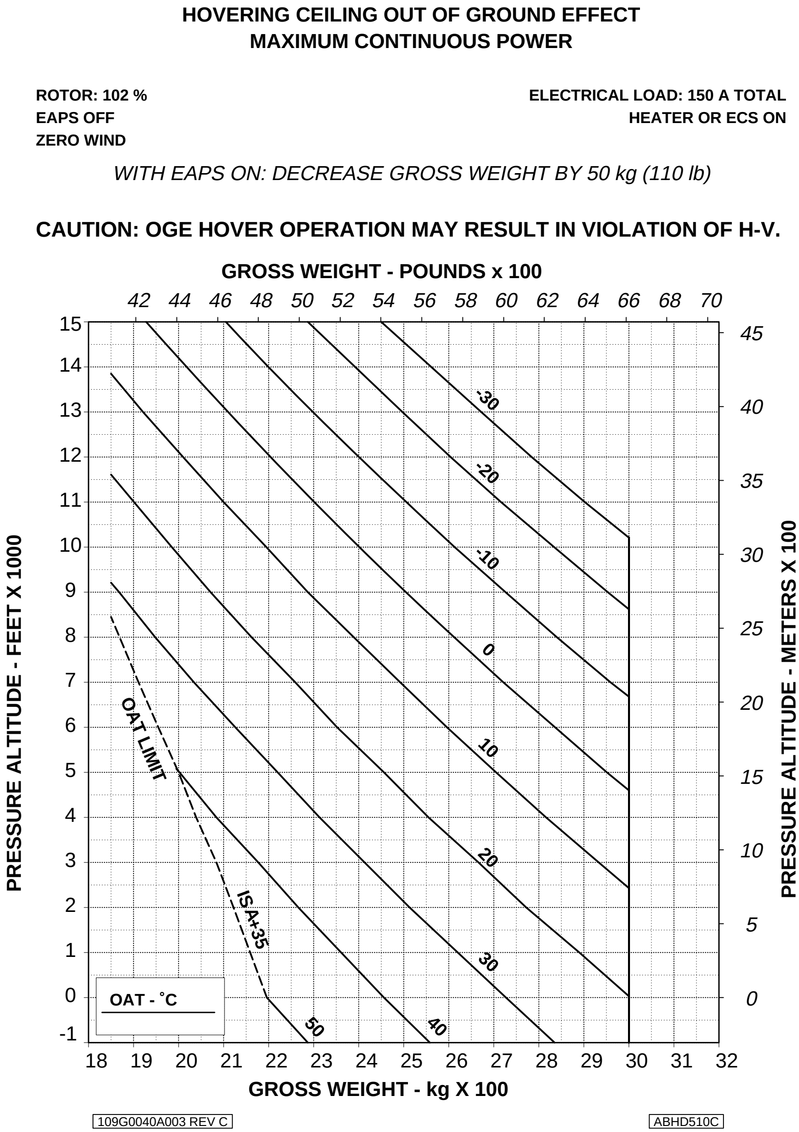


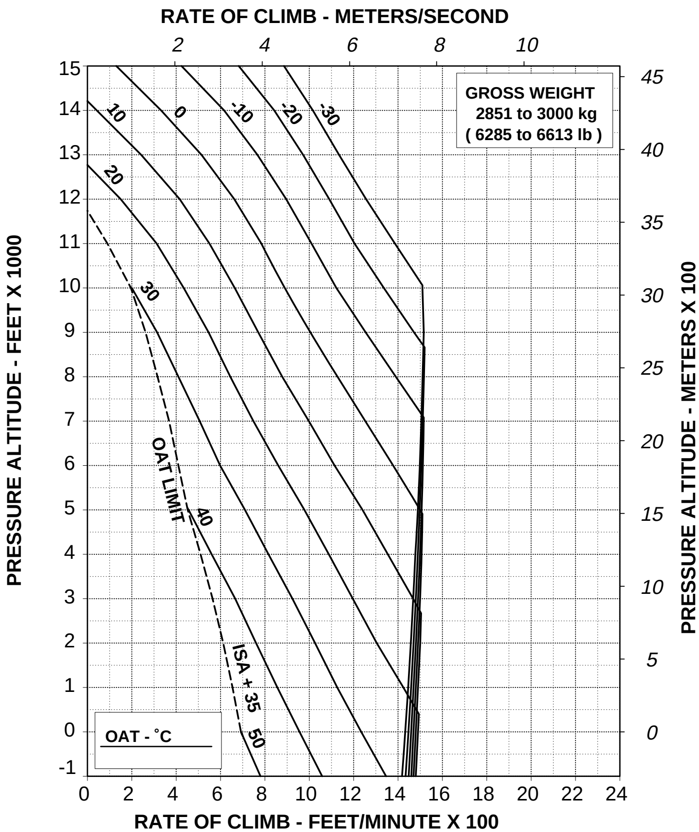
Figure 4-199. Hovering ceiling - OGE - Maximum continuous power - EAPS OFF/ON - Heater or ECS on.

RATE OF CLIMB ALL ENGINES
TAKE-OFF POWER

ROTOR: 100 %
EAPS OFF
HEATER OR ECS ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)



109G0040A003 REV C

ABHD516B

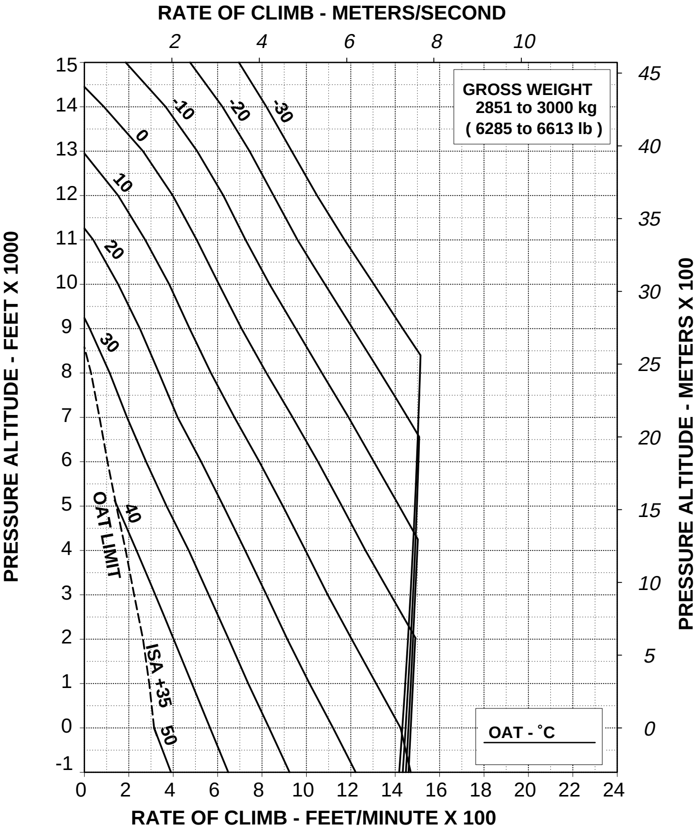
Figure 4-200. Rate of climb - All engines - Take-off power - EAPS OFF/ON - Heater or ECS ON.

RATE OF CLIMB ALL ENGINES
MAXIMUM CONTINUOUS POWER

ROTOR: 100 %
EAPS OFF
HEATER OR ECS ON

60 kts IAS
ELECTRICAL LOAD: 150 A TOTAL

WITH EAPS ON: DECREASE RATE OF CLIMB BY 50 ft/min (0,25 m/s)



109G0040A003 REV C

ABHD518B

Figure 4-201. Rate of climb - All engines - maximum continuous power - EAPS OFF/ON - Heater or ECS ON.

HOVERING CEILING - ONE ENGINE INOPERATIVE CHARTS

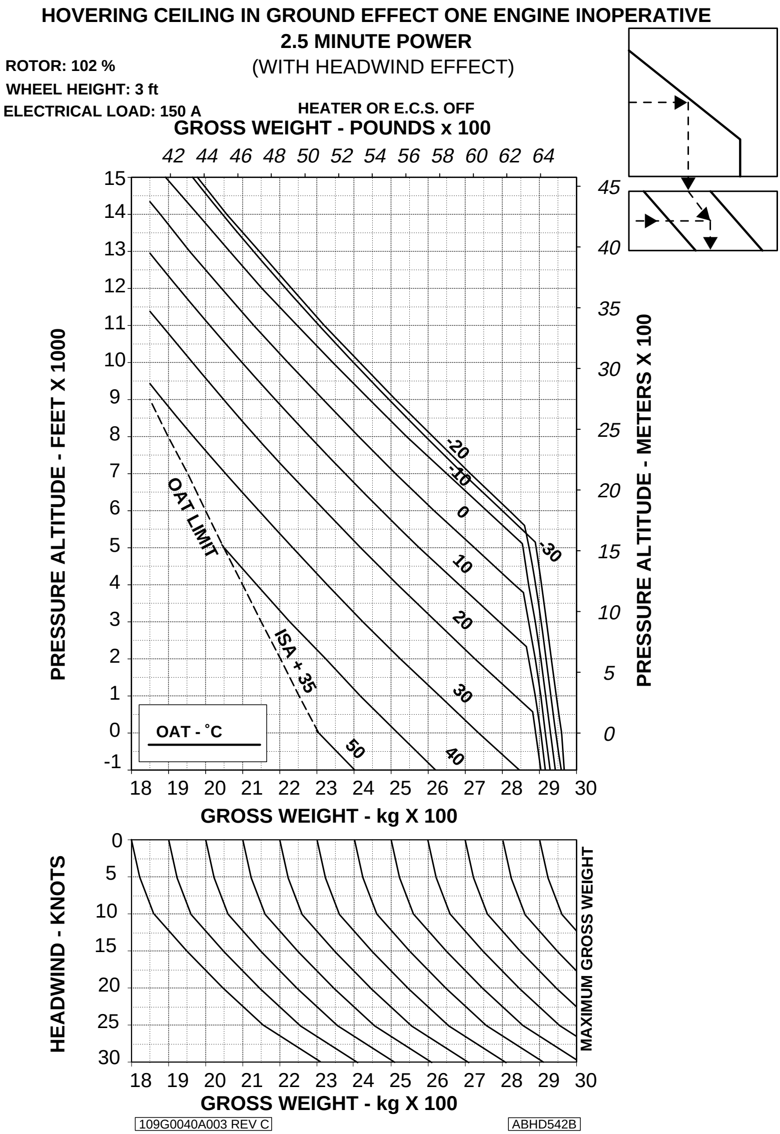


Figure 9-1. Hovering ceiling - IGE - OEI - 2.5 minute power.

HOVERING CEILING OUT OF GROUND EFFECT ONE ENGINE INOPERATIVE
2.5 MINUTE POWER
(WITH HEADWIND EFFECT)

ROTOR: 102 %
HEATER OR E.C.S. OFF
ELECTRICAL LOAD: 150 A

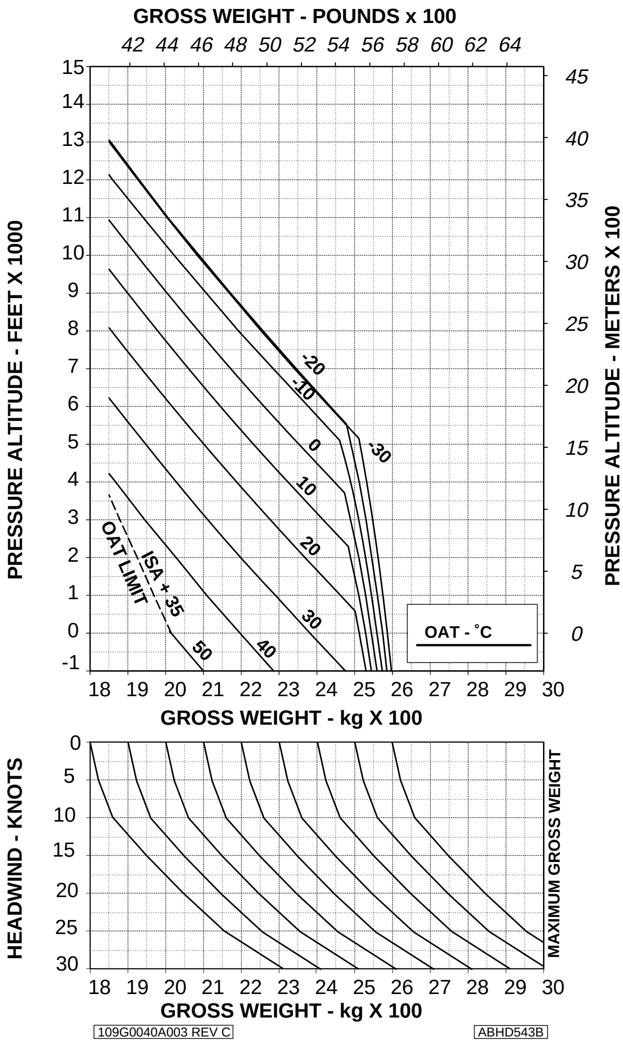


Figure 9-2. Hovering ceiling - OGE - OEI - 2.5 minute power.

**E.A.S.A. Approval EASA. R.C.01252
dated 27 September 2005**

*The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.*

SEARCHLIGHT

The Searchlight installation P/N 109-0813-76 consists of a swinging light installed under the forward section of the helicopter fuselage. The light can be extended, stowed or swung as required by operating a switch on pilot and copilot collective lever.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

AIRSPEED LIMITATIONS (KIAS)

Maximum speed for searchlight extension, for orientation and retraction	: 135 KIAS.

CENTER OF GRAVITY LIMITS

After searchlight installation the new empty weight and C.G. location must be determined.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

Area N°2 (Fuselage - rh side)

Searchlight	: Condition and cleanliness.

PILOT'S PREFLIGHT CHECK

(Every flight)

Searchlight	: Condition and cleanliness.

ENGINE PRE-START CHECK

SRCH-CTL/SRCH-PWR circuit breakers	: ON.

SYSTEM CHECK

Searchlight ON/OFF/STOW switch on pilot collective lever : OFF, check.

Searchlight ON/OFF/STOW switch on copilot collective lever : OFF, check.

IN FLIGHT

SEARCHLIGHT OPERATING PROCEDURE

NOTE

The searchlight, consisting of a swinging light installed under the forward section of the helicopter fuselage, can be extended, stowed or swung as required by operating a switch on the collective lever.

EXTENSION

EXT/RETR/L/R switch on pilot or copilot collective lever : EXT (to extend light in the desired position).

ON/OFF/STOW switch on pilot or copilot collective lever : ON.

NOTE

With the switch in OFF position the light remains extinguished in the position where it has been left.

EXT/RETR/L/R switch on pilot or copilot collective lever : Set as necessary.

NOTE

Moving switch to L or R position the searchlight rotates left or right. It is possible to adjust the light in an intermediate position, from stowed to extended, by temporarily moving the switch to EXT or RETR position.

RETRACTION

ON/OFF/STOW switch on pilot or co-pilot collective lever : STOW then OFF.

NOTE

In STOW position the light is extinguished.

Check for Searchlight deactivation after use by pilot and copilot.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.A.S.A. Approval EASA. R.C.01251
dated 28 September 2005

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

HF SYSTEM - TYPE KHF1050

The HF system, type KHF1050 P/N 109-0814-32, is used for long distance communication, in the frequency range of 2 to 29.999 MHz in 1 KHz increments.
The system, connected to the I.C.S., provides two-way voice radio communication in USB (voice and data) and AM mode.

The system consists of:

— a control panel;
— a transceiver;
— a power amplifier;
— an ″antenna coupler″;
— an antenna installed on the tail boom.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

No change.

SECTION 2 - NORMAL PROCEDURES

IN FLIGHT

CAUTION

While transmitting with KHF1050, the magnetic direction indicator can have a deviation up to 10 degrees on any heading.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

E.A.S.A. Approval EASA. R.C.02435
dated 25 September 2007

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

WEATHER RADAR PRIMUS 700

The Weather Radar PRIMUS 700 P/N 109-0813-23 is a lightweight, X-band digital radar that provides the crew with a visual information of the weather activity in a 120°/60° sector.
The Weather Radar system is a Multy-mode radar, that shows weather or mapping information on the radar display.

TABLE OF CONTENTS

LIST OF ILLUSTRATIONS

SECTION 1 - LIMITATIONS

TYPE OF OPERATION

The Weather Radar PRIMUS 700 must not be used for terrain avoidance.

SECTION 2 - NORMAL PROCEDURES

PREFLIGHT CHECKS

PILOT'S PREFLIGHT CHECK

(Every flight)

Weather Radar control panel : OFF or SBY.

SYSTEM CHECK

NOTE

The weather radar can not be operated when the helicopter is on the ground since a microswitch, connected to the main landing gear, inhibits the radar radiation.
As a precaution, it is nevertheless advisable to operate the weather radar as follows:

— The airborne radar should not be operated when the helicopter is in a Hangar or other enclosure, unless the radar transmitter is not operating or the energy is directed toward an absorption shield which dissipates the radio frequency energy.

— Avoid radar operation if personnel are standing in the 270 degree forward sector within 10 m (33 ft) of the helicopter.

— The airborne weather radar should not be operated while an aircraft is being refueled or defueled to prevent possible fuel ignition.

WARNING

Do not turn ON the Weather Radar if there are personnel within 6 m (20 ft) from the antenna.

IN FLIGHT

Weather Radar control panel : As required.

APPROACH AND LANDING

Weather Radar control panel : OFF or SBY.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

PERFORMANCE DATA

No change.

SECTION 7 - SYSTEM DESCRIPTION

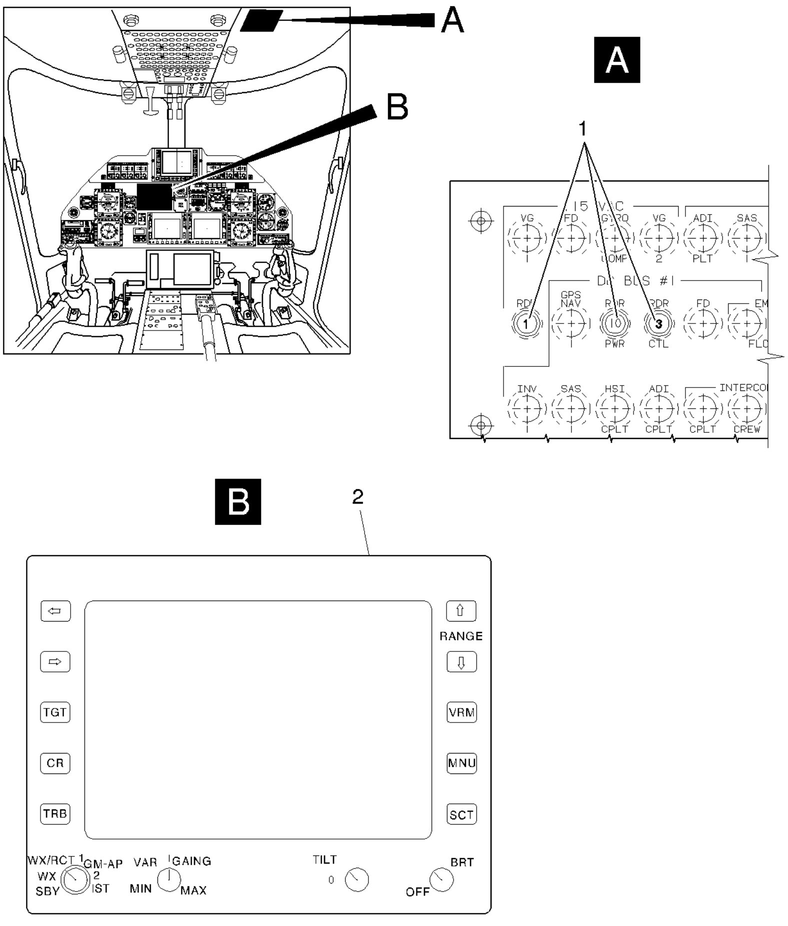
The PRIMUS 700 system provides ground/sea mapping and weather detection features. In ground/sea mapping mode, video levels of increasing reflectivity are displayed as cyan, yellow and magenta, corresponding to levels 1, 2 and 3 respectively. The system detects storms along the flight path of the aircraft and gives the pilot a visual indication of storm location and intensity in color.
For the weather detection mode, weather target returns are displayed in one of five colors (video levels 0, 1, 2, 3 or 4) with 0-level represented by a black screen indication of weak or no return, and levels 1, 2, 3 or 4 represented by green , yellow, red and magenta to show progressively stronger returns.

The Weather Radar PRIMUS 700 system consists of the following equipments:

— One Receiver/Transmitter located in the nose avionic bay .

— One antenna and its support located in the nose avionic bay.

— One Radar control panel and display (2) located on the instrument panel that controls the system (Refer to Figure 7-1).



1. Radar circuit breakers
2. Radar control panel

ABHD#315A

Figure 7-1. Weather Radar PRIMUS 700 - Controls and Indicators.

E.A.S.A. Approval EASA.R.C.03167
dated 05 December 2008

The information contained herein supplements the information of the basic Rotorcraft Flight Manual.
For limitations, procedures and performance data not contained in this appendix, consult the basic Rotorcraft Flight Manual.

SATCOM AIRCELL

The Satcom Aircell P/N 109-B811-14 is a satellite telephone system offering the possibility of worldwide telephone communications through the 66 satellite IRIDIUM network.
The Satcom Aircell system consists of an Iridium transceiver unit, an antenna and a telephone control panel at pilot's disposal.
The control panel allows the telephone system to interface intercommunication system (ICS) and allows hands free operation in pilot cabin.

TABLE OF CONTENTS

SECTION 1 - LIMITATIONS

MISCELLANEOUS LIMITATIONS

SATCOM AIRCELL

The use of the Satcom Aircell telephone system is prohibited for the pilot flying, both in transmission and receiving modes while the aircraft is airborne.

The use of the telephone in flight must be authorized by the pilot in command.

It is responsability of the operator to obtain the operational approval of the Satcom Aircell telephone from the Competent Authority.

PLACARD

IT IS PROHIBITED FOR THE PILOT FLYING TO USE THE TELEPHONE WHILE AIRBORNE

Near pilot audio panel

SECTION 2 - NORMAL PROCEDURES

EXTERNAL PREFLIGHT CHECKS

PILOT'S PREFLIGHT CHECK

Area N°2 (Fuselage - RH side)

Antenna : Check condition.

Area N°6 (Fuselage - LH side)

TEL circuit breaker
(baggage compartment) : In.

Antenna : Check condition.

IN FLIGHT

SATCOM AIRCELL OPERATION

(Not be used by the pilot flying)

To make a telephone call:

ICS selector	:	Set to TEL.
Telephone control panel	:	Press HOOK pushbutton and dial the number to make a telephone call.
PTT trigger switch (on cyclic stick)	:	Push the 2^{nd} detent to talk.
Telephone control panel	:	Press HOOK pushbutton to close communication.
ICS selector	:	Set as desired.

To answer a telephone call:

NOTE

Incoming telephone calls are announced by the CALL flashing light on the istrument panel and the light on the telephone control panel.

ICS selector	:	Set to TEL.
Telephone control panel	:	Press HOOK pushbutton.
PTT trigger switch (on cyclic stick)	:	Push the 2^{nd} detent to talk.
Telephone control panel	:	Press HOOK pushbutton to close communication.
ICS selector	:	Set as desired.

SECTION 3 - EMERGENCY AND MALFUNCTION PROCEDURES

No change.

SECTION 4 - PERFORMANCE DATA

No change.

SECTION 6 - WEIGHT AND BALANCE

No change.

SECTION 7 - SYSTEM DESCRIPTION

The Satcom Aircell system provides access to a range of two-ways communication options from the helicopter, including voice and data applications. These services are available when the aircraft is airborne or on the ground. The Satcom Aircell system accesses the Iridium Satellite Network of 66 Low-Earth Orbit (LEO) satellites and operates in the frequency range of 1616 MHz to 1625.5 MHz.
The primary components of the Satcom Aircell system are:

— The antenna;

— The transceiver;

— The telephone adapter;

— The CALL flashing lights.

The antenna is installed on the top of the cabin compartment. A radome protects it against rain, ice and lightning strikes. The antenna is connected to the transceiver with a coaxial cable.
The transceiver is installed in the nose compartment.
The telephone adapter is installed on the interseat console.
Function buttons, an alphnumeric keyboard, a display and two LED’s provide controls and indications for all the system functions.
One of the CALL flashing lights is installed in the passenger cabin on the right lining panel above the cabin door and the other in the cockpit on the intrument panel. These lights flashing when the telephone call is coming.

To make a telephone call from the passenger cabin, the HOOK pushbutton must be pressed and the number must be dialed from the telephone control panel.
The ICS of the passenger cabin must be selected in the telephone position. The PTT pushbutton on the relative Passeger Service Unit (PSU) must be pressed to talk.

Incoming telephone calls are announced by the CALL flashing light on the pilot instrument panel and in the passenger cabin.

To answer a telephone call from the passenger cabin, the HOOK pushbutton on the telephone control panel must be pressed.
The ICS of the passenger cabin must be selected in the telephone position. The PTT pushbutton on the relative PSU must be pressed to talk.

When the telephone call is finished, the HOOK pushbutton on the telephone control panel must be pressed to close the communication.

PART II

MANUFACTURER'S DATA

LIST OF REVISED PAGES

Revision No.	Subject	Date
—	Issue	31-5-1996
—	Issue (Supersedes issue dated 31-5-1996)	25-7-1996
—	Issue (Supersedes issue dated 25-7-1996)	30-7-1997
1		Not affected
2	Revised pages C-1/C-2, 7-21 and 7-24.	29-1-1998
3		Not affected
4		Not affected
5	Revised pages C-1/C-2, Section 7 and Section 8.	8-5-1998
6	Revised pages C-1/C-2 and 7-2	29-5-1998
7	Revised pages C-1/C-2, 6-4, 6-11, 7-ii, 7-45 and 7-46. Added page 7-47/7-48.	26-11-1998
8		Not affected
9		Not affected
10		Not affected
11	Revised pages C-1/C-2, 6-2, 6-5, 6-6, 7-39, 7-40 and 7-41. Added page 6-6A and 6-6B.	26-7-1999
12		Not affected
13		Not affected
14		Not affected
15		Not affected
16		Not affected
17		Not affected

LIST OF REVISED PAGES (CONT.)

Revision No.	Subject	Date
18		Not affected
19	Revised pages C-1, 7-21, 7-22, 8-9, 8-10. Added page C-2.	16-3-2000
20		Not affected
21	Revised pages C-2 and 7-44.	24-5-2000
22		Not affected
23		Not affected
24		Not affected
25		Not affected
26		Not affected
27		Not affected
28		Not affected
29		Not affected
30		Not affected
31		Not affected
32		Not affected
33		Not affected
34		Not affected
35	Revised pages C-2, 6-i/6-ii blank), 6-11, 6-12, 7-45, 7-46 and S-1/(S-2 blank).	03-03-2004
36		Not affected
37	Revised pages C-2, 7-45 and 7-46.	14-07-2004
38		Not affected
39		Not affected
40		Not affected
41	Revised pages C-2, 7-ii, 7-47 and 7-48. Added pages 7-iii, 7-iv, 7-49, 7-50.	08-06-2005

SECTION 6
WEIGHT AND BALANCE

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LIST OF ILLUSTRATIONS

SECTION 6
WEIGHT AND BALANCE

NOTE

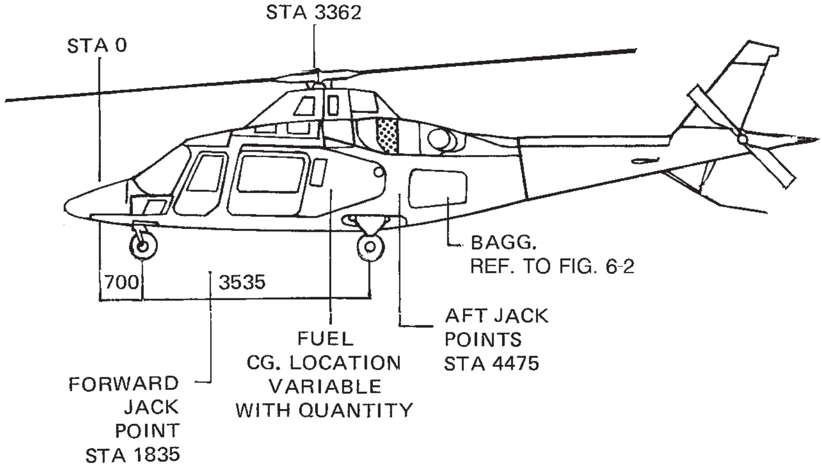
In accordance with R.A.I. procedures, the detail weight and balance data of this Section are not subject to R.A.I. approval. The loading instructions of this Section, however, have been accepted by R.A.I. as satisfying all requirements for instructions on loading of the rotorcraft within approved limits of weight and center of gravity, and on maintaining the loading within such limits.

GENERAL

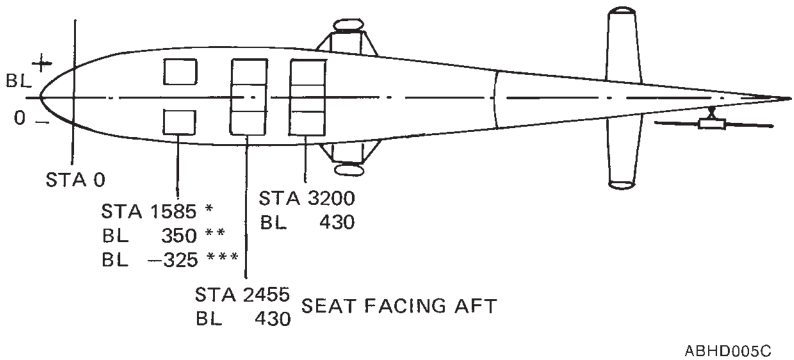
This Section provides information for the weight and balance computation of A109E helicopter.
It is the pilot's responsibility to ensure that the helicopter is properly loaded to maintain all flight long the center of gravity within the limitations defined in Section 1 of the Rotorcraft Flight Manual.
Figures, charts and examples are provided to assist the pilot in computing the proper loading conditions.

DATUM LINE LOCATIONS

Figure 6-1 presents fuselage station and buttock line data to aid in weight and balance computations.



STANDARD VERSION
PILOTS AND PASSENGERS STATION



* ADJUSTABLE FROM 1565 TO 1630

** PILOT

*** COPILOT/PASSENGER

Figure 6-1. Helicopter Stations Diagram.

WEIGHTS - ARMS AND MOMENTS

LONGITUDINAL MOMENTS

Pilots and Passengers			
Weight (kg)	Pilot (*) Copilot or Passenger (Arm 1585) Moment (kgm)	Passengers 3 places central seat aft facing (Arm 2455) Moment (kgm)	Passengers 3 places aft seat (Arm 3200) Moment (kgm)
60	95	147	192
65	103	160	208
70	111	172	224
75	119	184	240
80	127	196	256
85	135	209	272
90	143	221	288
95	151	233	304
100	158	245	320
120	190	295	384
140	222	344	448
160	254	393	512
180	285	442	576
200	317	491	640
220		540	704
240		589	768
260		638	832
280		687	896
300		736	960
320		786	1024

(*) Adjustable from 1565 to 1630

USABLE FUEL (Arm - see below)			
Weight (kg)	(l) (0.8 kg/l)	Arm (mm)	Moment (kgm)
20	25	3324	66.5
40	50	3327	133.1
60	75	3329	199.7
80	100	3331	266.5
100	125	3399	339.9
120	150	3461	415.3
140	175	3505	490.7
160	200	3539	566.2
180	225	3543	637.7
200	250	3551	710.2
220	275	3571	785.6
240	300	3614	867.4
260	325	3662	952.1
280	350	3703	1036.8
300	375	3739	1121.7
320	400	3770	1206.4
340	425	3797	1291.0
360	450	3821	1375.6
380	475	3843	1460.3
400	500	3863	1545.2
420	525	3880	1629.6
440	550	3897	1714.7
460	575	3911	1799.1
476	595	3921	1866.4

UNUSABLE FUEL (Arm - see below)			
Weight (kg)	(l) (0.8 kg/l)	Arm (mm)	Moment (kgm)
8	10	3320	26.6

ENGINE OIL (Arm 4280)		
Weight (kg)	**(l)**	**Moment (kgm)**
10.5	10.24	44.9

UNDRAINABLE ENGINE OIL (Arm 4280)		
Weight (kg)	**(l)**	**Moment (kgm)**
2.09	2.15	8.9

MAIN TRANSMISSION OIL (Arm 3441)		
Weight (kg)	**(l)**	**Moment (kgm)**
10.7	11	36.8

ALLOWABLE BAGGAGE LOAD
(MAXIMUM TOTAL LOAD = 150 kg)

	ZONE 1 c.g. 4880 Note 1	ZONE 2 c.g. 5240 Note 1	ZONE 3 c.g. 5560 Note 1	ZONE 4 c.g. 5960 Note 1	ZONE 5 c.g. 6430 Note 1
Baggage load (KG)	**Baggage moment (KGM)**				
10	49	52	56	60	64
20	98	105	111	119	129
30	147	157	167	179	193
40	195	209	222	238	257
50	244	262	278	298	321
60	293	314	333	358	386
70	342	367	389	417	450
80	391	419	444	477	514
90	440	471	500	536	578
100	489	524	556	596	643
105	513	550	583	626	675
108	528	565	600	644	694
110	537	576	611	656	*
115	562	602	639	685	*
120	586	628	667	715	*
123	601	644	683	733	*
130	635	681	722	*	*
132	645	691	733	*	*
135	660	707	*	*	*
140	684	733	*	*	*
150	733	*	*	*	*

* No loading area
NOTE 1: Refer to figure 6-2.

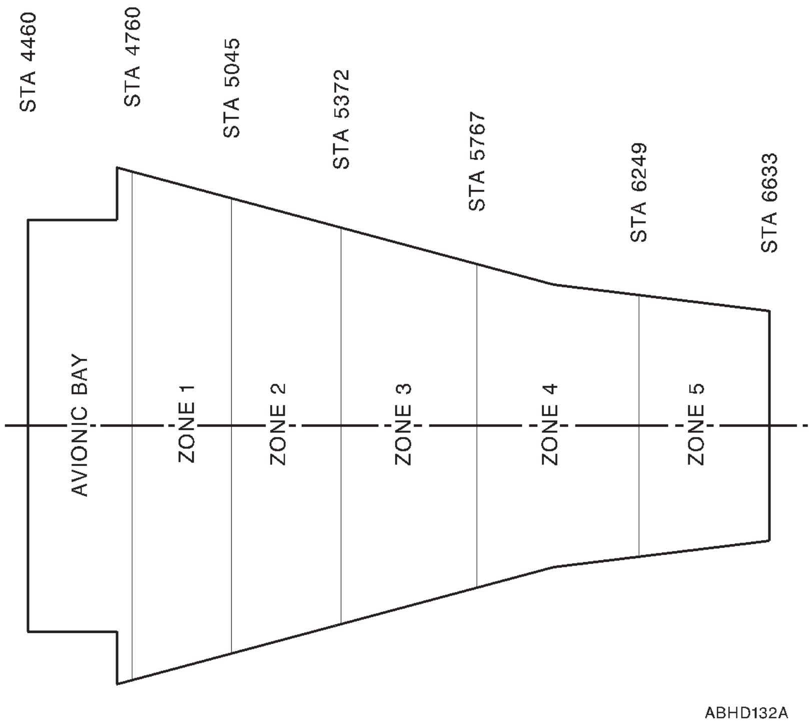


Figure 6-2. Baggage loading zones.

LATERAL ARMS

Item	Buttock line (mm)
Pilot	+350
Copilot	- 325
Passengers	See figure 6-1
Baggage	0

COMPUTATION OF LOADING

LONGITUDINAL LOADING SAMPLES

The empty weight and moment of the A109E can be found in the helicopter Weighing Record (Chart B). The values in Chart B were obtained by weighing the aircraft and computing the empty weight, moment and CG therefrom.

Subsequently as items of equipment are added or removed, entries should be made in the Basic Weight and Balance Record (Chart C) and the new empty weight and moment computed.

The empty weight includes fixed ballast, hydraulic fluid, unusable fuel, undrainable engine oil and transmission oil.

Three sample loadings are shown below (in the sample loading the empty weight and CG arm are assumed to be 1600 kg and 3550 mm).

a) Item	Weight (kg)	CG Arm (mm)	Moment (kgm)
Empty Weight	1600	3550	5680
Engine oil	10	4280	43
Pilots (2)	160	1585	254
Passengers (3) Front	240	2455	589
Baggage	50	5300	265
Total weight	2060	3316	6831
Fuel at Take-off	100	3399	340
Take-off Weight	2160	3320	7171
Fuel at Landing	70	3330	233
Landing Weight	2130	3316	7064

b) Item	Weight (kg)	CG Arm (mm)	Moment (kgm)
Empty Weight	1600	3550	5680
Engine oil	10	4280	43
Pilot	70	1585	111
Passengers (3) Front	250	2455	614
Passengers (3) Rear	220	3200	704
Baggage	90	5300	477
Total weight	2240	3406	7629
Fuel	370	3833	1418
Take-off Weight	2610	3466	9047
Fuel at Landing	70	3330	233
Landing Weight	2310	3403	7862

c) Item	Weight (kg)	CG Arm (mm)	Moment (kgm)
Empty Weight	1600	3550	5680
Engine oil	10	4280	43
Pilots (2)	140	1585	222
Passengers (3) Front	250	2455	614
Passengers (3) Rear	224	3200	714
Baggage	150	5300	795
Total weight	2374	3398	8068
Fuel	476	3912	1862
Take-off Weight	2850	3484	9930
Fuel at Landing	70	3330	233
Landing Weight	2444	3396	8301

The Weight - CG combination in the three examples above fall within the approved limits.

LATERAL LOADING SAMPLE

The empty weight CG is assumed to be at station ″0″ unless a different entry has been made in Chart C. With the empty CG at station ″0″ the approved lateral loading limits will not be exceeded except under extreme lateral loadings. An example of an asymmetric loading computation is given below.

Item	Weight (kg)	CG Arm (mm)	Moment (kgm)
Empty Weight	1600	0	0
Engine oil (N°1)	5	- 300	- 1.5
Engine oil (N°2)	5	+ 300	+ 1.5
Pilot	80	+ 350	28
Pass (1) right	80	+ 430	34.40
Pass (1) center	80	0	0
Fuel (right lower cell)	80	+ 340	27.20
Fuel (left lower cell)	40	- 340	- 13.60
Baggage	50	0	0
Take-off Weight	2020	37.6	76

The weight-CG combination computed above falls within the approved limits.

WEIGHT AND BALANCE DETERMINATION

Instructions for weight and balance determination are herewith enclosed with instructions for use of charts to enable the operator to obtain all necessary data as to basic helicopter configuration, empty weight and center of gravity. These charts will also provide for continuous control of weight and balance of the helicopter.

This system of weight and balance computation requires the use of charts and forms. They are identified as follows:

a. Chart A - Equipment List.
b. Chart B - Helicopter Weighing Record.
c. Chart C - Basic Weight and Balance Record.

WEIGHT AND BALANCE DATA RESPONSIBILITY

The aircraft manufacturer inserts all helicopter identifying data on the various charts. This record constitutes the basic weight and balance data of the helicopter, to which the Rotorcraft Flight Manual was assigned, for the condition shown on chart A. The operator shall keep this data up-to-date by recording all changes made to the configuration of the helicopter.

HELICOPTER WEIGHING

The helicopter must be weighed:

a. When major modifications or repairs are made, or kits are installed/removed.
b. When the basic weight data is suspected to be an error.
c. At time of major overhaul.
d. In accordance with R.A.I instructions.

USE OF CHARTS AND FORMS

USE OF CHART A

The Chart A gives the weight, arm and moment of all the standard and optional equipment. The manufacturer of the helicopter places check marks in the ″Basic Configuration″ column to identify the items of equipment in the helicopter for the weighing condition. A check (V) in the columns headed ″In Helicopter″ indicates the presence of the item in the helicopter, and a zero (0) indicates its absence. The next columns of chart A will permit inspection of the helicopter for equipment actually installed. When making an inventory, note whether any items of equipment have been installed or removed and if so enter corresponding weight and moment change on Chart C.

Subsequent check list inventories shall be carried out in the following cases:

a. When the helicopter undergoes modification, major repair or overhaul.
b. When changes in equipment are made for a different type of operation.
c. When the helicopter is reweighed.

WEIGHING INSTRUCTIONS

Preliminary instructions

a. Remove dirt, grease, moisture, etc. from the helicopter.

NOTE

The helicopter must be weighed in a closed hangar.

b. Drain fuel from tanks.

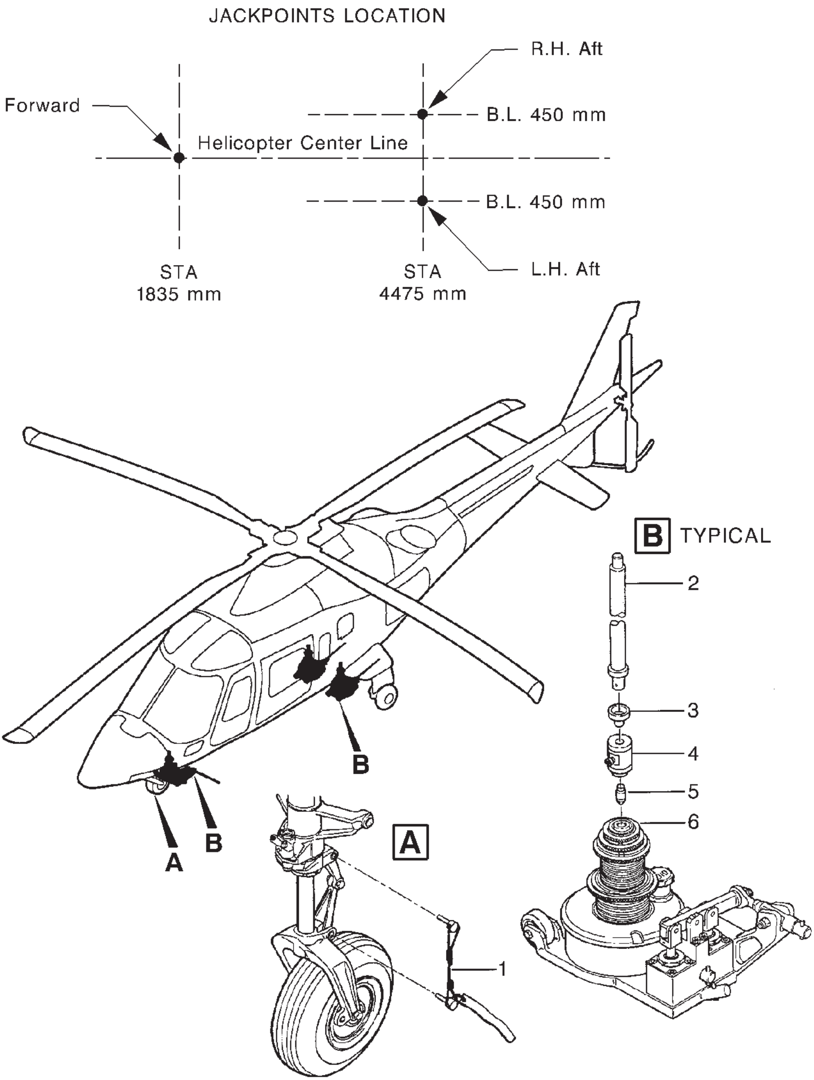
NOTE

If it is impossible to drain fuel due to fire hazard or local regulations, fill fuel tanks to capacity. Since the weight of fuel varies with temperature, determine actual specific gravity by use of a hydrometer. Multiply by tank capacity to obtain total fuel weight. Never weigh with partially filled tanks.

c. Drain engine oil or fill to normal capacity.
d. Fill main transmission to normal level.
e. Fill hydraulic fluid reservoirs to normal level.
f. Conduct an inventory of the operating equipment actually installed in the helicopter by means of Chart A.

<u>Weighing procedure</u>
Fig. 6-2

a. Install levelling equipment on helicopter as outlined in Section 08-20 of Maintenance Manual.
b. Make sure that weighing equipment is zeroed.
c. Install nose gear shock locking cable, see Fig. 6-3.
d. Connect to each of the three weighing equipment cell (4), adapter (3) and centering pin (5).
e. Position jacks (6), weighing equipment cells (4) and appropriate adapters (2) under respective jacking points.
f. Raise helicopter until wheels are clear of ground and level it as outlined in Section 08-20 of Maintenance Manual.
g. Read and note scale readings and enter data in appropriate Chart B.
h. Lower helicopter on the ground and remove all equipment used for weighing.



1. Nose gear shock strut locking cable
2. Jacking adapter (typ)
3. Adapter
4. Weighing equipment cell
5. Centering pin
6. Jack

Figure 6-3. Weighing of helicopter.

USE OF CHART B

a. Enter the actual scale readings in the first column of sheet 1. Subtract tare, if any, from the scale readings to obtain the net weight.
b. Multiply the net weights by their respective arms.
c. Add the net weight and moments.
d. Divide the total moment by the net weight to obtain ″as weighed″ CG position. Transfer the ″TOTAL″ (as weighed) weight arm and moment to the sheet 2 of Chart B.
e. Subtract the weight and moment of oil (if reservoir is full) from the ″as weighed″ total.
f. Subtract the total weight and moment of equipment weighed but not part of the basic helicopter (list these items in column one).
g. Add the weight and moment of unusable fuel.
h. Add the total weight and moment of the basic items not in helicopter when weighed (list these in column two). Added items shall be marked on Chart A.
i. Enter the new basic weight and moment on Chart C.

USE OF CHART C

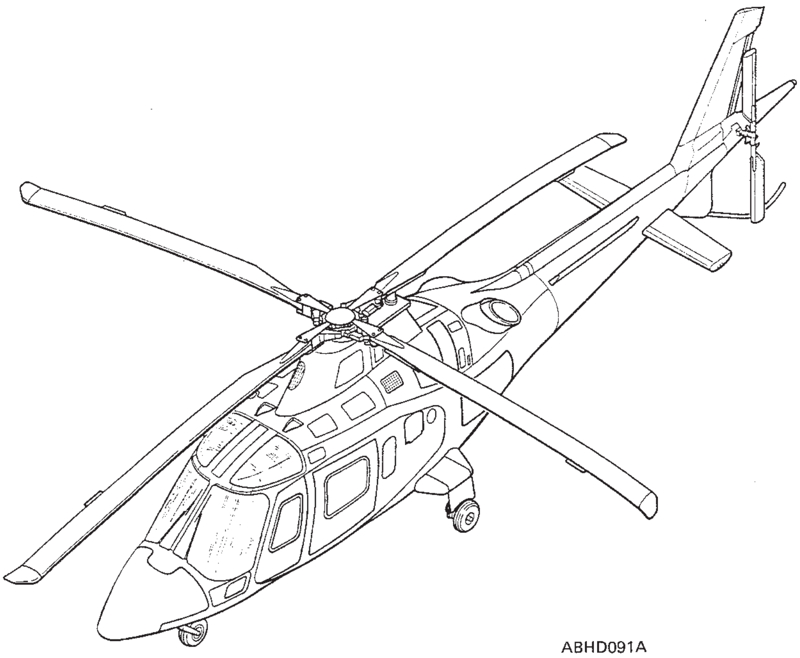
Chart C is a continuous history of the basic weight and moment resulting from modifications and equipment is considered the current weight and balance status of the basic helicopter. Make additions or subtractions to the basic weight and moment in Chart C as follows:

a. When the helicopter undergoes modification, major repair or overhaul.
b. When changes in equipment are made for a different type of operation.
c. When the helicopter is reweighed.

NOTE

If any equipment is not listed on Chart A, determine its weight and arm, and list corresponding data on Charts A and C.

WEIGHT AND BALANCE DATA



HELICOPTER A109E

SERIAL NUMBER ....

REGISTRATION MARKS ....

Page of

RECORD OF CHECKING
(Date and Signature)

CHART A – EQUIPMENT LIST
Report

MODEL: | **REGISTRATION MARKS** | **S/N**

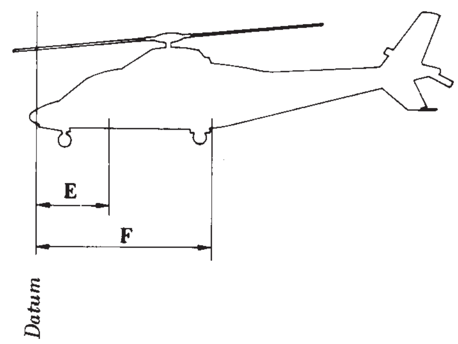
ITEM NUMBER	AUXILIARY EQUIPMENT P/N	AUXILIARY EQUIPMENT DENOMINATION	Q.TY	WEIGHT (kg)	ARM (mm)	MOMENT (kgmm)	Basic Configuration	In helicopter	Chart C entry	In helicopter	Chart C entry	In helicopter	Chart C entry

CHART B – HELICOPTER WEIGHING RECORD		Sheet 1 of 2
DATE	MODEL	S/N
PLACE	SIGNATURE	REGISTRATION MARKS

JACKPOINTS	SCALE READING	TARE	NET WEIGHT	ARM	MOMENT
L.H. FORWARD					
R.H. FORWARD					
TOTAL FORWARD					
L.H. AFTWARD					
R.H. AFTWARD					
TOTAL AFTWARD					
TOTAL (As weighed)					

E = Distance from the reference datum (STA.0) to the forward jackpoints Station (See Chart D).

F = Distance from the reference datum (STA.0) to the aft jackpoints Station (See Chart D).



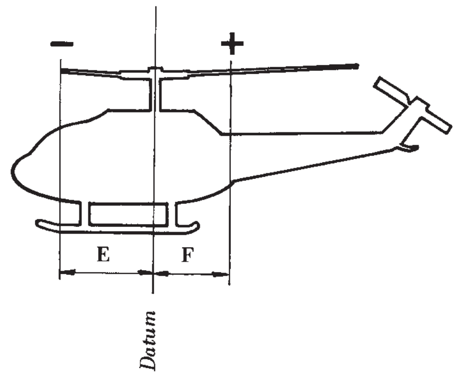


CHART B – HELICOPTER WEIGHING RECORD			Sheet 2 of 2
DESCRIPTION	NET WEIGHT	ARM	MOMENT
TOTAL (As weighed)			
SUBTRACT (See column 1 below)			
ADD (See column 2 below)			
TOTAL (Post to Chart C)			

COLUMN 1				COLUMN 2			
ITEMS WEIGHED BUT NOT PART OF BASIC WEIGHT	Weight	ARM	MOMENT	BASIC ITEMS NOT INSTALLED WHEN WEIGHED	Weight	ARM	MOMENT
TOTAL				TOTAL			

Reason for Weighing	Type Scales

REMARKS :

CHART C – BASIC WEIGHT AND BALANCE RECORD

Page No.

MODEL | S/N | REGISTRATION MARKS

DATE	Item No. See Chart A		DESCRIPTION	WEIGHT CHANGE ADDED (+) REMOVED (–)			TOTAL			SIGNATURE and/or VISAS
	IN	OUT		Weight	Arm	Moment	Weight	Moment	C.G.	

C.G. = Helicopter C.G. location. *IN = Installed.* *OUT = Removed.*

SECTION 7

SYSTEMS DESCRIPTION

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LIST OF ILLUSTRATIONS

LIST OF TABLES

SECTION 7
SYSTEMS DESCRIPTION

INTRODUCTION

This section provides a general description of systems of A109E helicopter .

HELICOPTER DECRIPTION

The A109E is a high-speed, high-performance, multi-purpose helicopter powered by two Pratt & Whitney Canada PW206C engines, with four-bladed fully articulated main rotor, two-bladed tail rotor and a retractable tricycle-type landing gear.
The airframe consists of two major assemblies: the forward fuselage and the aft fuselage (tail boom).
The forward fuselage comprises the nose section, the cabin and the rear section. The nose section includes an upper compartment for the electrical and electronic equipment, and a lower compartment which accommodates the hydraulic system accumulators, the nose landing gear and the other hydraulic components.
The cabin includes the crew compartment (cockpit) and the passenger compartment. Seating is provided for the pilot (right side) and a passenger (or copilot) in the cockpit, and up to six passengers in the relevant compartment.
The rear section accommodates the fuel tanks, the main landing gear compartments, the baggage compartment and the electrical and electronic equipment compartment.
The upper deck, located above and aft of the cabin area, accommodates the hydraulic system reservoirs and filter goups, the main transmission, the oil coolers and the engines.
The tail boom is bolted to the forward fuselage and supports the tail rotor and the relevant drive system. The tail boom includes the elevators, the vertical upper and lower fins, the tail skid and the tail cone.

PRINCIPAL DIMENSIONS

Refer to figure 7-1 for the airframe principal dimensions and to figure 7-2 for main internal dimensions and volumes.

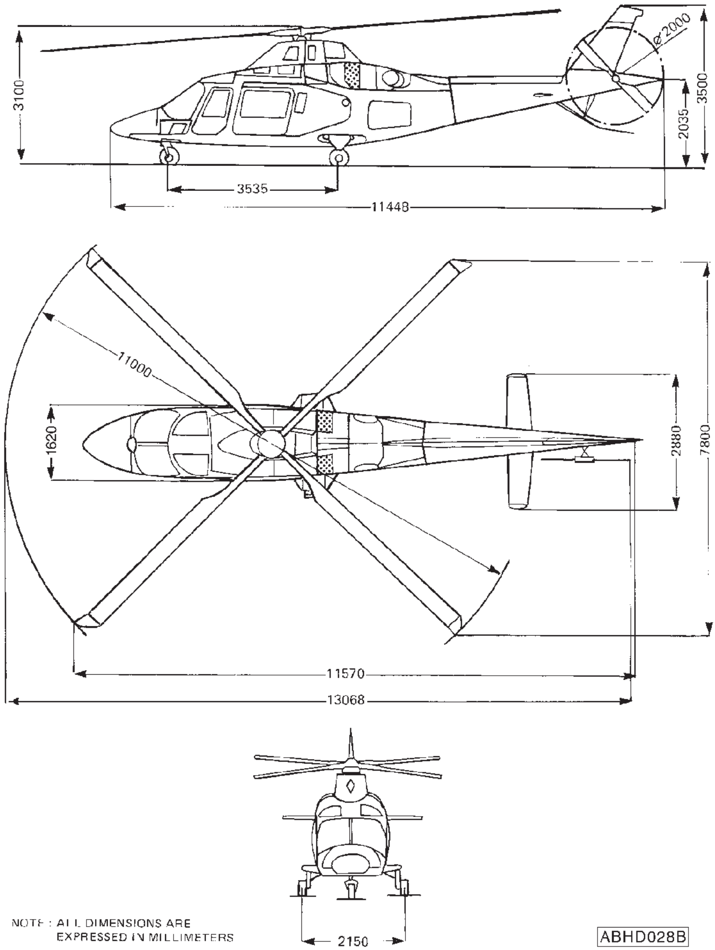


Figure 7-1. Airframe principal dimensions

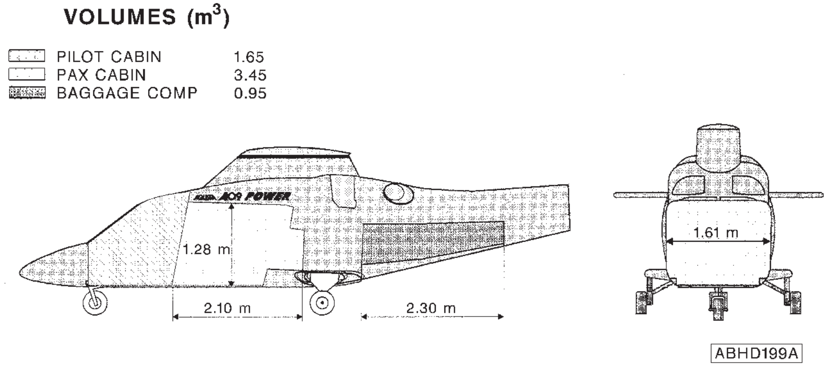
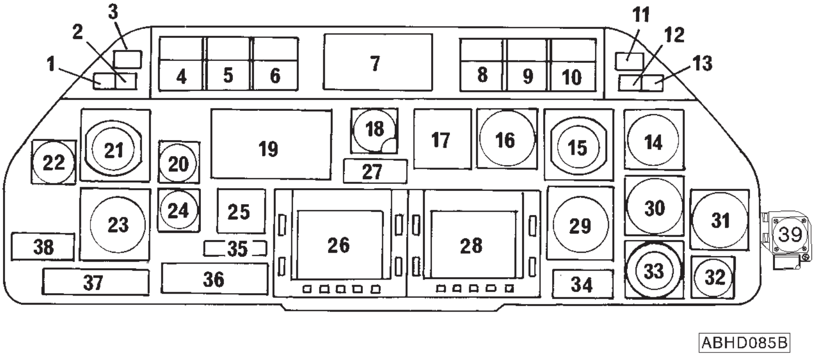


Figure 7-2. Main internal dimensions and volumes.

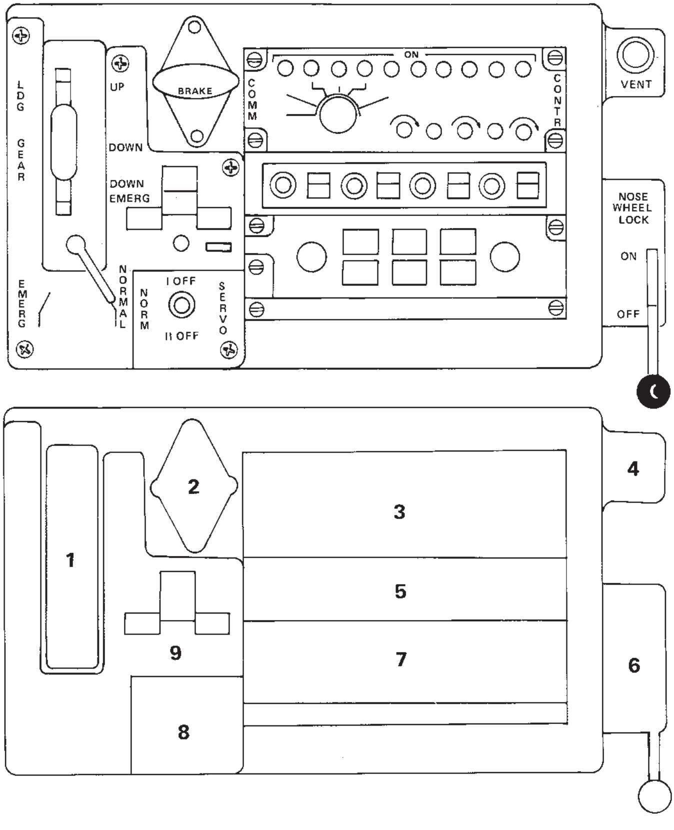
INSTRUMENT PANEL AND CONSOLE

Refer to figure 7-3 for instrument panel lay-out, to figure 7-4 for front console, to figure 7-5 for central console and to figure 7-6 for overhead panel.



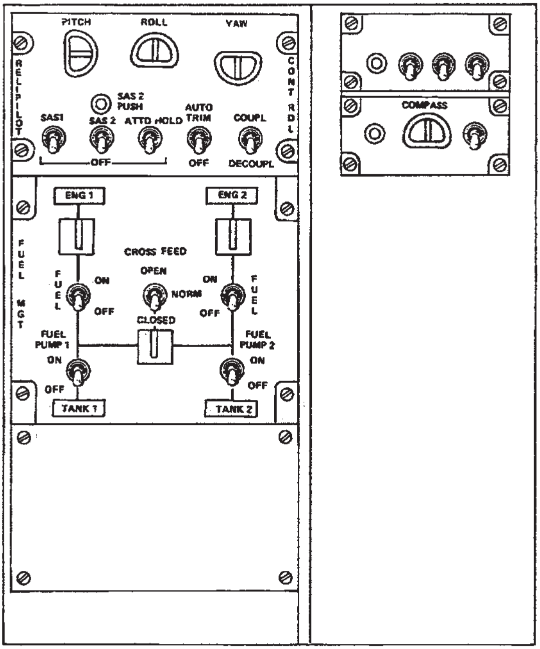
1. MASTER CAUTION lighted push-button
2. MASTER WARNING lighted push-button
3. Marker Beacon indicator
4. A.D.F. control panel
5. NAV2 control panel (VOR2)
6. NAV1 control panel (VOR1)
7. Empty
8. Transponder control panel
9. VHF/AM2 control panel
10. VHF/AM1 control panel
11. Marker Beacon indicator
12. MASTER CAUTION lighted push--button
13. MASTER WARNING lighted push--button
14. Altimeter, encoder
15. Attitude Director Indicator
16. Airspeed indicator
17. Flight Director mode selector
18. Attitude Director Indicator, stand-by
19. Empty
20. Altimeter
21. Attitude Director Indicator
22. Airspeed indicator
23. E.H.S.I.
24. Instantaneous vertical speed (I.V.S.I.)
25. Empty
26. E.C.U.2
27. D.M.E. indicator
28. E.C.U.1
29. E.H.S.I.
30. Instantaneous vertical speed (I.V.S.I.)
31. Radio altimeter indicator
32. Clock
33. Horizontal Situation Indicator
34. Empty
35. Marker Beacon control panel
36. Intercommunication control panel
37. E.H.S.I. control panel
38. Empty
39. Magnetic compass

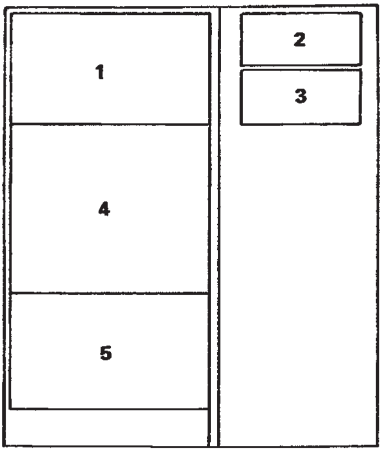
Figure 7-3. Instrument panel lay-out (typical).



1. Landing gear selector
2. Parking brake handle
3. Intercommunication control panel
4. Cabin ventilation knob
5. ICS selector control panel
6. Nose wheel lock lever
7. EHSI control panel
8. Hydraulic system switch
9. Landing gear position indicator

Figure 7-4. Front console (typical)





1. Helipilot control panel
2. Miscellaneous panel
3. Gyrocompass control panel
4. Fuel management control panel
5. Empty

ABHD087A

Figure 7-5. Central console (typical)

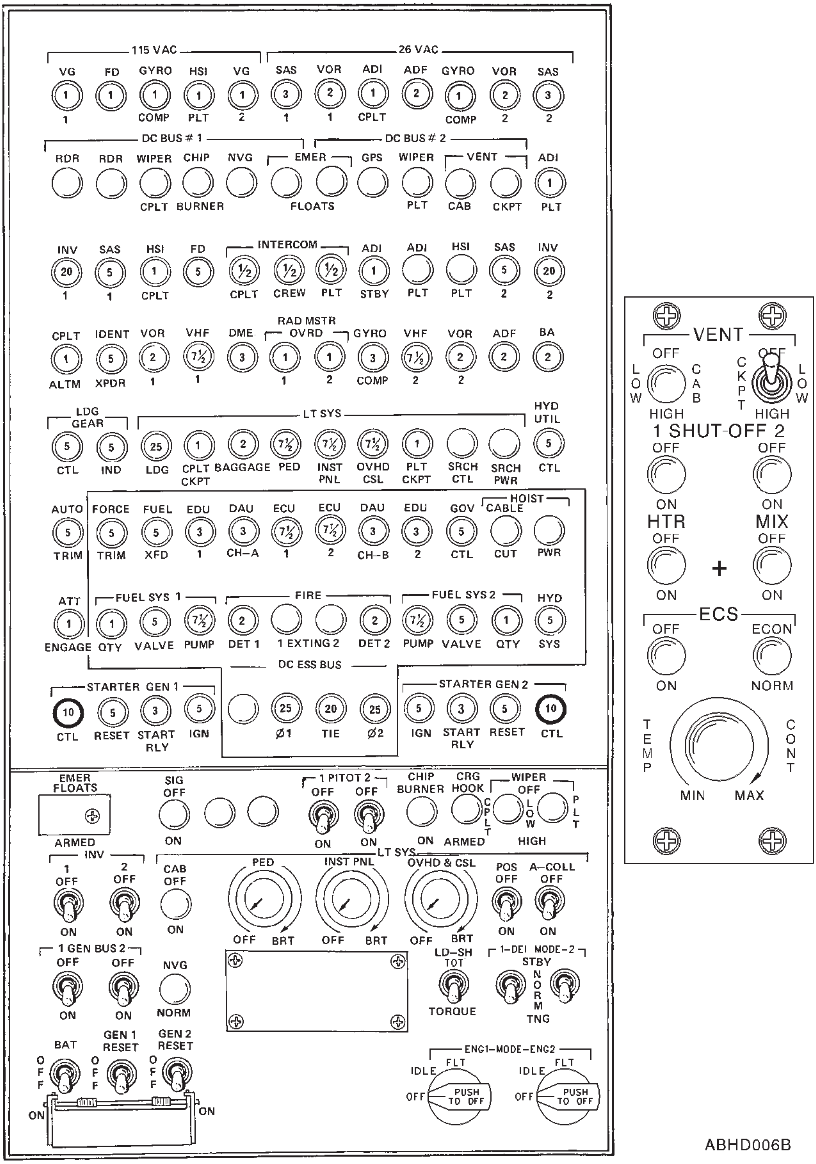
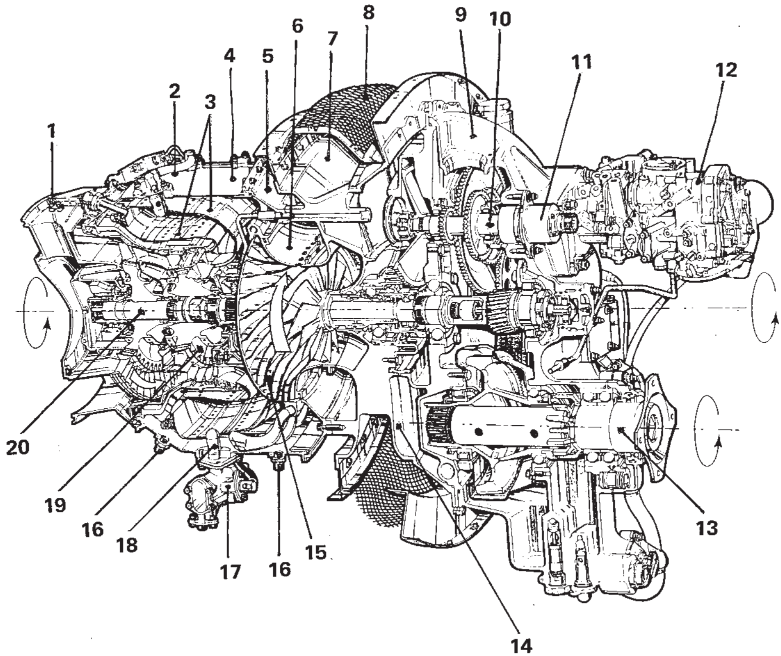


Figure 7-6. Overhead console (typical).

POWER PLANT

The A109E is powered by two Pratt & Whithey Canada PW206C engines (figure 7-7).
The PW206C engine is a lightweight, free turbine, turboshaft engine incorporating a single-stage centrifugal compressor driven by a single stage compressor turbine. Metered fuel is sprayed into e reservoir flow annular combustion chamber through twelve (12) individual fuel nozzles mounted around the gas generator case. A single channel, Fully Authority Digital Electronic Control (FADEC) with a mechanical backup (FMM ″FUEL MANAGEMENT MODULE″) ensures accurate control of the engine output speed and fast response changes in power demand.
The PW206C cosists of two modules:
- Turbomachinery module
- Reduction gearbox module
The turbomachinery module comprises the cold section and the hot section, while the reduction gearbox provides to reduce power turbine speed to one suitable for rotorcraft transmission operation.



1. Exhaust case
2. Fuel manifold
3. Combustion chambers
4. Gas generator case
5. Diffuser case
6. Impeller housing
7. Compressor inlet case
8. Air inlet screen
9. Reduction gearbox module
10. Input shaft
11. Permanent magnet alternator
12. Fuel Management Module
13. Output shaft
14. Oil tank cavity
15. Centrifugal impeller
16. Fuel drain valve
17. Fuel flow divider
18. Fuel nozzle
19. Compressor turbine
20. Power turbine shaft

Figure 7-7. Pratt & Whitney Canada 206C engine

ENGINE OIL SYSTEM

The engine oil system is subdivided into two independent circuits, each connected to the respective engine.
The engine oil system is of the dry-sump type and is supplied from an engine internal tank (one for each engine). Lubrication of different engine components is assured by a pump assembly, composed of a pressure pump and two scavenge pumps, driven by the accessory gearbox, and a pressure circuit filter.
An electrical signal supplies by N.1 or N.2 FADEC (ECC) to DAU which sends signal to EDU1 that displays # 1 OIL PRES and # 2 OIL PRES caution legends.
Two magnetic chip detectors, one located in the bottom of the accessory gearbox, the other in the scavenge pumps outlet, are electrically connected with EEC and DAU which sends signal to EDU1 that displays # 1 OIL CHIPS and # 2 OIL CHIPS caution legends.
The operation of the oil system is fully automatic and therefore no action is required from the pilot except for monitoring the system through EDUs display.
The oil system schematic is illustrated in figure 7-8.

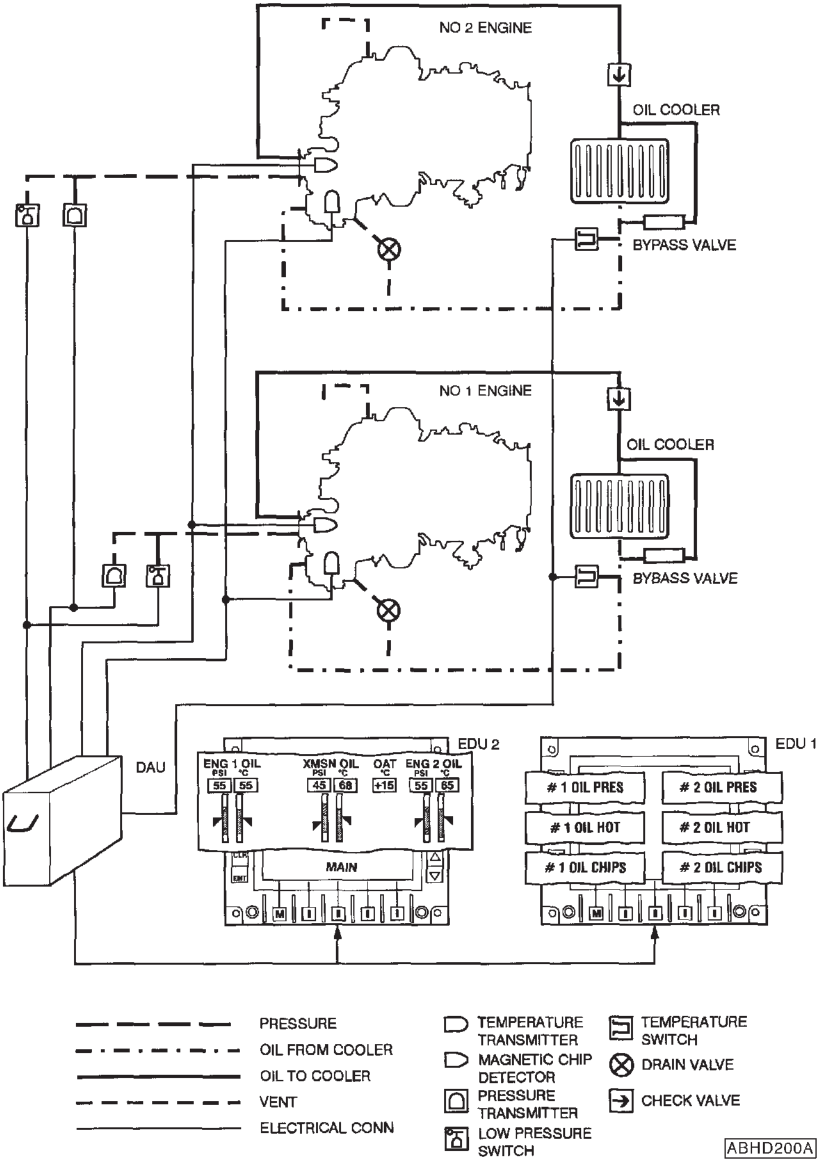


Figure 7-8. Engine oil supply system schematic.

FUEL SYSTEM

The fuel system (figure 7-9) consists of the following sub-systems:
- the storage system
- the distribution system
- the indication system

The storage system consists of two main tanks and one main rear tank. Each forward tank supplies fuel to the associated engine.
The distribution system consists of two identical independent circuits, each connected to associated engine. Each circuit comprises one fuel pump, a filter assembly, a shut off valve, a pressure transmitter and a differential pressure switch. A crossfeed valve allows the fuel to supply both engines from only one tank. The indicating system comprises the pressure indicating and fuel quantity indicating system and the fuel caution circuit.

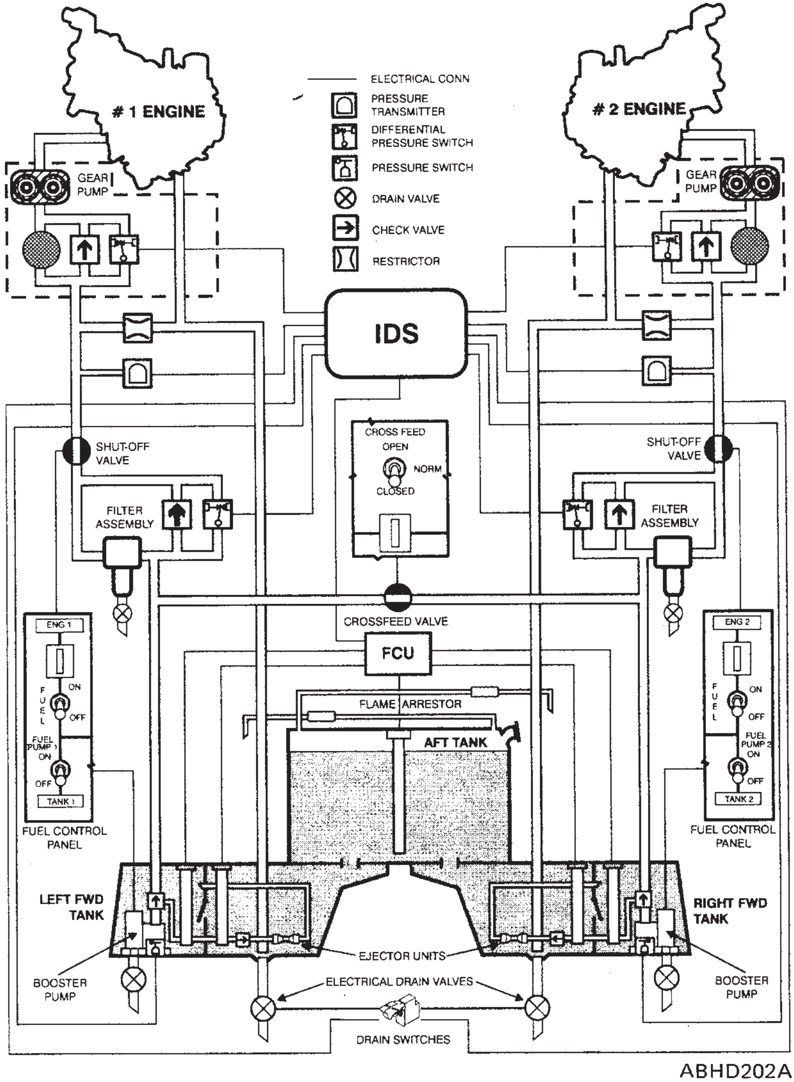


Figure 7-9. Airframe fuel system schematic

FLIGHT CONTROL SYSTEMS

The flight control systems (figure 7-10) provide the the correct control responses when the pilot makes control selections, giving him positive control of the attitude, speed and altitude of the helicopter. The A109E incorporates conventional helicopter flight controls: collective, cyclic and antitorque operated by cockpit controls: collective pitch lever, cyclic stick and tail rotor pedals. A mixing unit in the collective/cyclic control systems integrates the control inputs from both systems and provides a common output to the collective/cyclic actuators.
Refer to figure 7-11 for the collective grip assembly and to figure 7-12 for cyclic grip assembly.

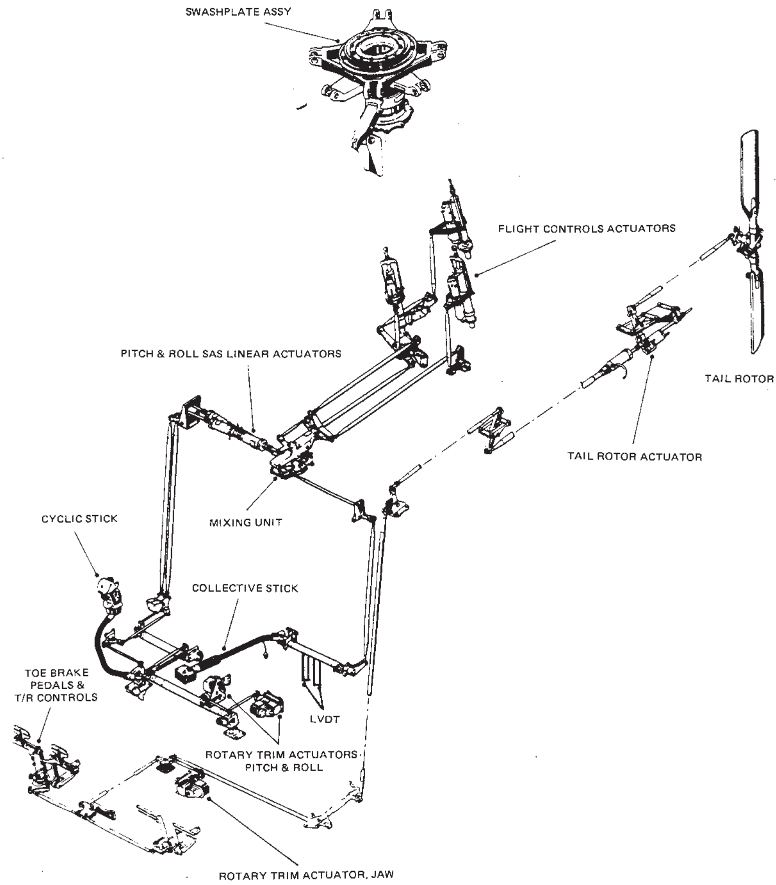
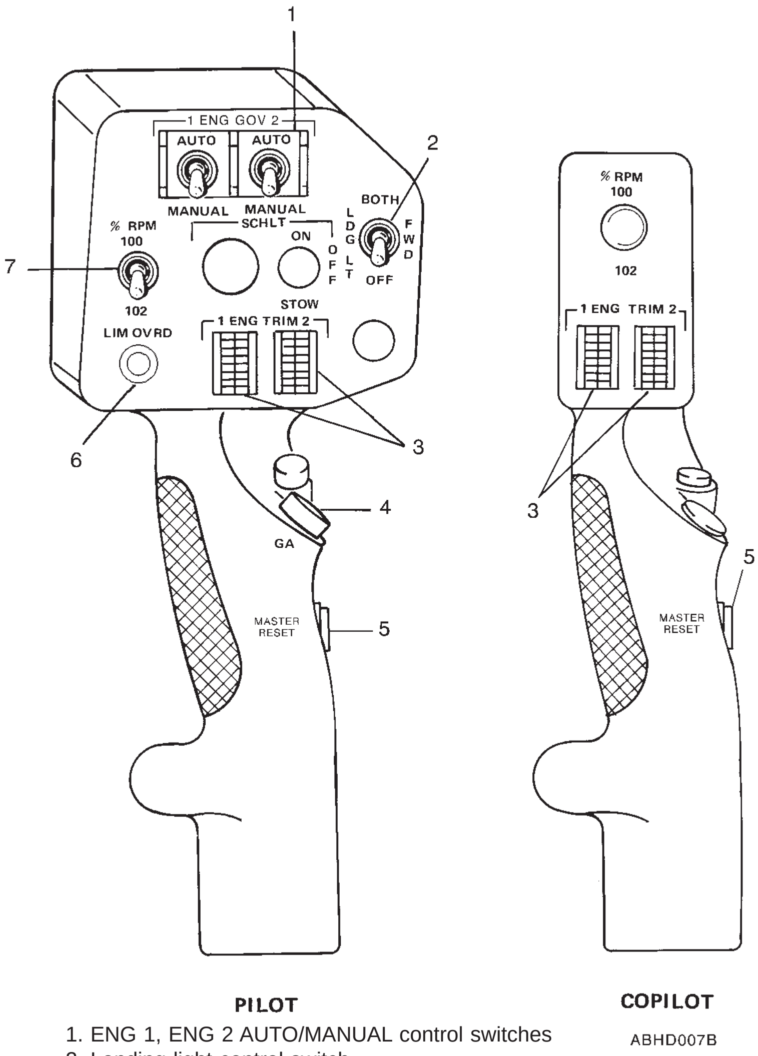
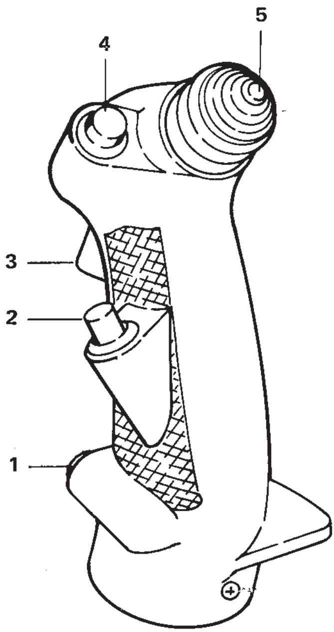


Figure 7-10. Flight control systems.



1. ENG 1, ENG 2 AUTO/MANUAL control switches
2. Landing light control switch
3. ENG 1, ENG 2 trim switch
4. Go around switch
5. Master caution/warning light reset pushbutton switch
6. LIM OVRD (limit override) switch pushbutton
7. RPM selector switch

Figure 7-11. Grip assembly, collective (typical).



1. Cargo release pushbutton switch
2. Force trim pushbutton switch
3. Microphone/Intercommunication trigger switch
4. Flight Director remove/standby pushbutton switch
5. Beeper trim switch

Figure 7-12. Grip assembly, cyclic (typical).

HYDRAULIC SYSTEM

Two independent systems (figure 7-13) supply the power to operate the flight control system and are used to provide the hydraulic power for operation of the main rotor servo actuators (both systems) and the tail rotor servo actuator (N.1 system only). In addition the N.2 system is used to provide the hydraulic power to the utility hydraulic system, necessary for operation of the landing gear and brakes.
The hydraulic power consists of the following sub-systems:
- N.1 hydraulic system
- N.2 hydraulic system
- Utility hydraulic system.

N.1 HYDRAULIC SYSTEM

This system, which operates at a maximum pressure of 1550 psi, consists of a suction circuit, a pressure circuit, a return circuit and a bypass circuit. The hydraulic fluid is contained in the reservoir located on the right side of the cabin roof. The fluid is sucked by a pump, driven by main transmission, and is supplied to the servo actuators through the filter group and the accumulator. The system is controlled by the hydraulic control panel, located on the front console, operated by the pilot. A ground test fitting is provided, pressure monitoring circuit and a low pressure monitoring circuit. The N.1 hydraulic system supplies the main rotor servo actuators and the tail rotor servo actuators.

N.2 HYDRAULIC SYSTEM

This system is similar to N.1 system. It supplies the main rotor servo actuators and the utility hydraulic system.

UTILITY HYDRAULIC SYSTEM

This system receives the power pressure from N.2 hydraulic system and supplies the pressure to operate the landing gear, wheel brakes, rotor brake and nose gear centering lock.
There are two sources of pressure energy supplied by this system:

MAIN (NORMAL): to provide energy to operate landing gear actuators, landing gear uplocks, nose gear centering lock, wheel brakes (toe pedals) and wheel brakes (park selector).
EMERGENCY: to provide energy to operate landing gear uplocks (unlocks), landing gear actuators (lower and lock), wheel brakes (park selector operation) and nose gear centering lock.

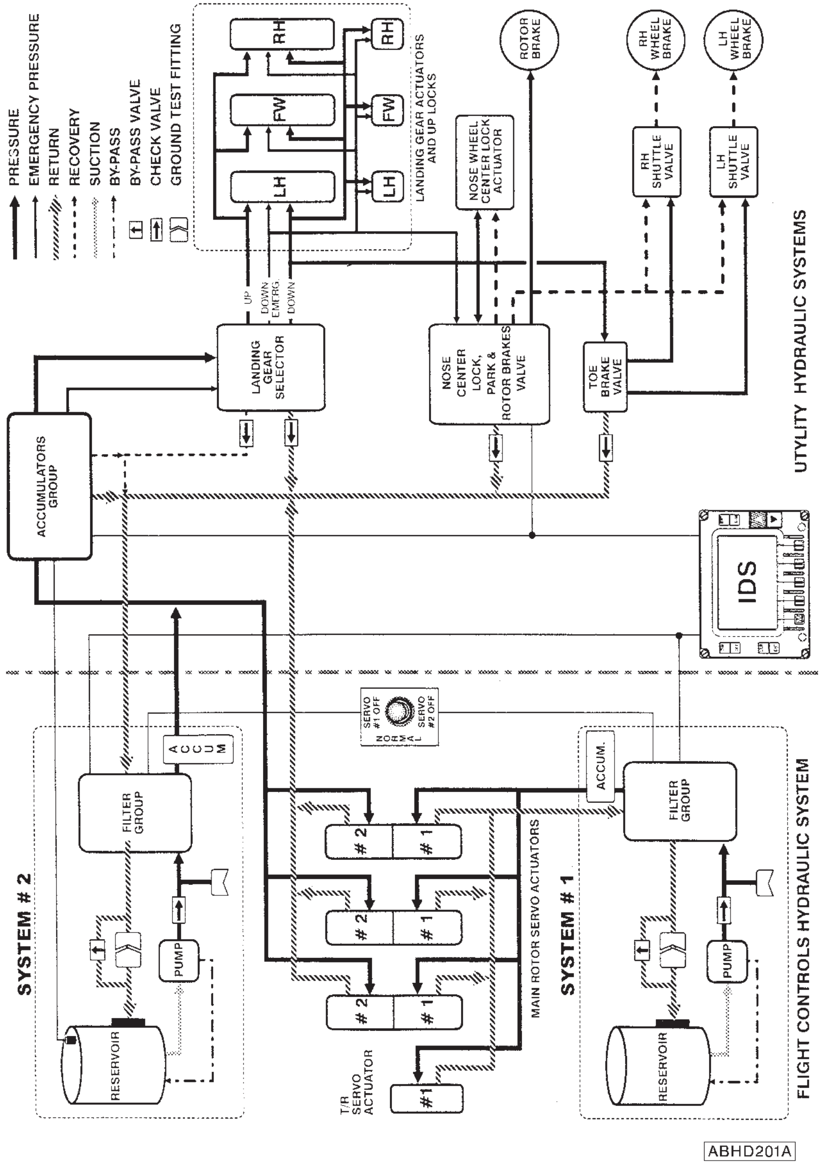


Figure 7-13. Hydraulic systems.

ELECTRICAL SYSTEMS

The electrical and electronic systems (figure 7-14) are powered by single wire circuit with common ground return through the helicopter structure.
The helicopter power supplies are:
- 28 V dc
- 115 V ac 400 hz single phase
- 26 V ac 400 Hz single phase

Two generators, a battery and, for ground handling, an external power receptacle, are the dc power main sources.
Two static inverters, powered by dc voltage, are the ac sources.
Both dc and ac powers are distributed through a bus bar system and operated by control switches located on the overhead console.
The electrical system is interfaced with the IDS for voltage, current, advisory, caution and warning indications.

DC ELECTRICAL SYSTEM

The dc electrical system (figure 7-15) is a 28 V direct current single conductor system, using the helicopter structure as a negative ground.
The main components of the systems are:
- Two starter generators
- Two dc control boxes
- Battery
- An external power receptacle
- One dc relay box

ESSENTIAL ELECTRICAL DC EQUIPMENT (TYPICAL)

The following equipment are supplied by the essential busses

Essential bus #1
ECU #1, Fuel cross feed, Force trim, Fire extinguisher, Fire det., Fuel pump #1, Fuel valve #1, Fuel q.ty #1.

Essential bus #2
Hydraulic system, Fuel q.ty #2, Fuel valve #2, Fuel pump #2, Fire det., Fire extinguisher, Hoist pwr, Hoist cable cut, Eng. gov. CTL, EDU #2, DAU CH-B, ECU #2.

Battery

The helicopter is equipped with a 24 V, 27 or 22 Ah or with a 25.2 V, 28 Ah nickel-cadmium battery located in the nose compartment.
A temperature switch, inside of the battery and connected to the IDS, detects the internal temperature of the battery, giving a BATT HOT warning message on the EDU 1 in case of battery overtemperature.

External power

The helicopter is provided with an external power receptacle on the rear right side of the fuselage. A microswitch, activated by the receptacle door, gives the EXT PWR ON advisory message on the EDU 1 when the door is in the open condition.

Starter-generator

Two starter-generators, installed each on the proper engine reduction gear-box, provide engine start when operated as an electric starter motor; after the engine start, the started generator, driven by the engine, reverts into a dc generator providing the necessary 28 V dc power.

AC ELECTRICAL POWER

The alternate power (figure 7-16) is supplied by two 250 VA single phase static inverters via two sensor relays. The inverters require 28 V dc power input suppied by:
- dc main bus N.1, through the INV 1 circuit breaker and INV 1 ON/OFF switch, and
- dc main bus N.2, through the INV 2 circuit braker and INV 2 ON/OFF switch.

The circuit breakers and control switches are located on the overhead console.
Each inverter supplies 115 V ac and 26 V ac to its own 115 V ac and 26 V ac distribution busses to which the helicopter ac loads are connected. In the event of an inverter failure, the relative sensing relay deenergizes, connecting the failed inverter busses to the other operating inverter.

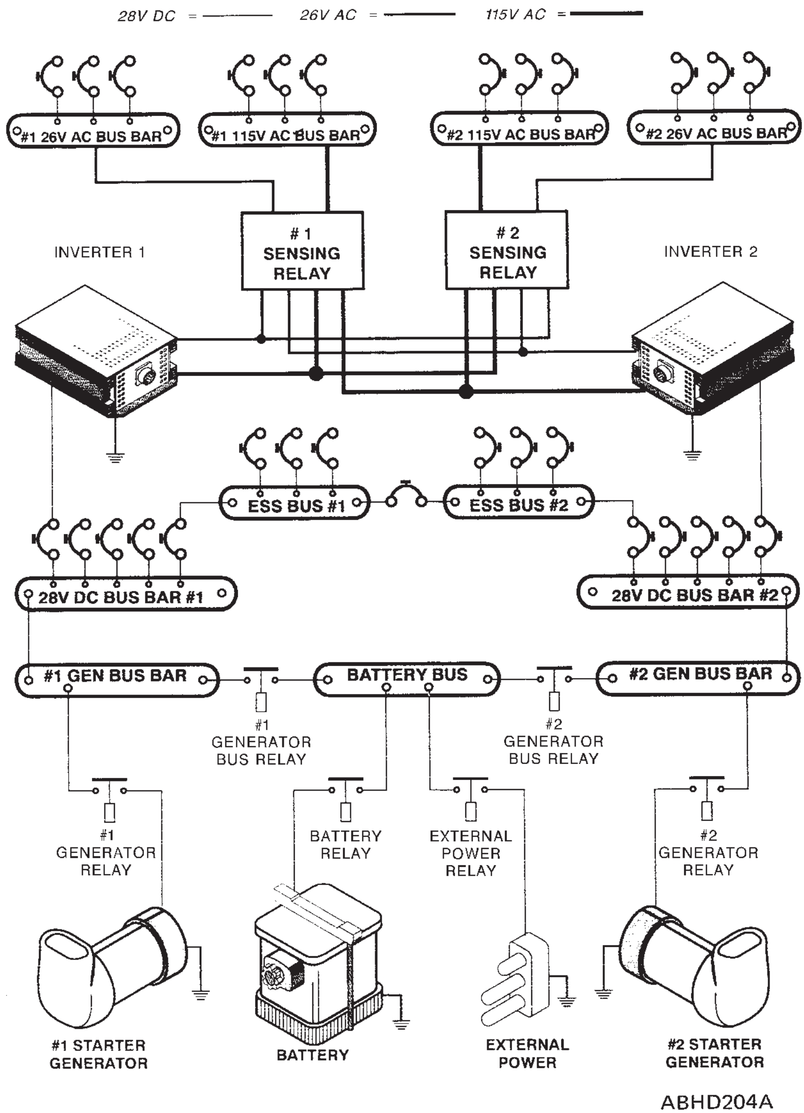


Figure 7-14. Electrical power systems.

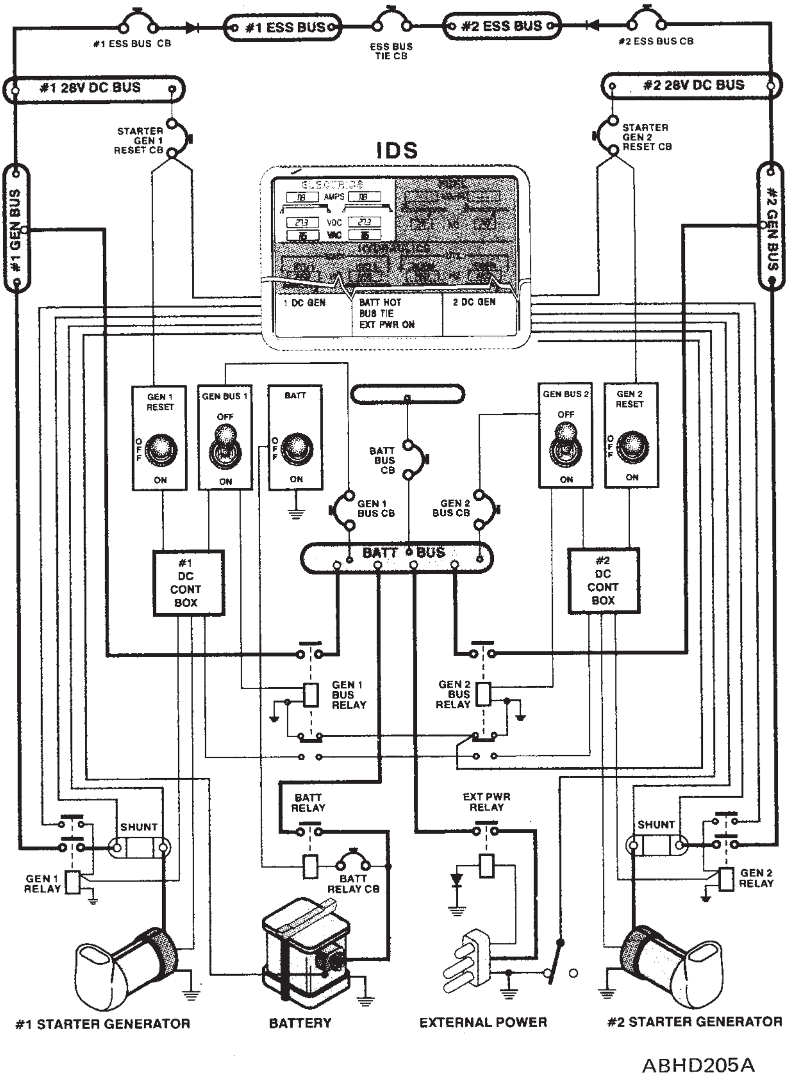


Figure 7-15. DC electrical system block diagram.

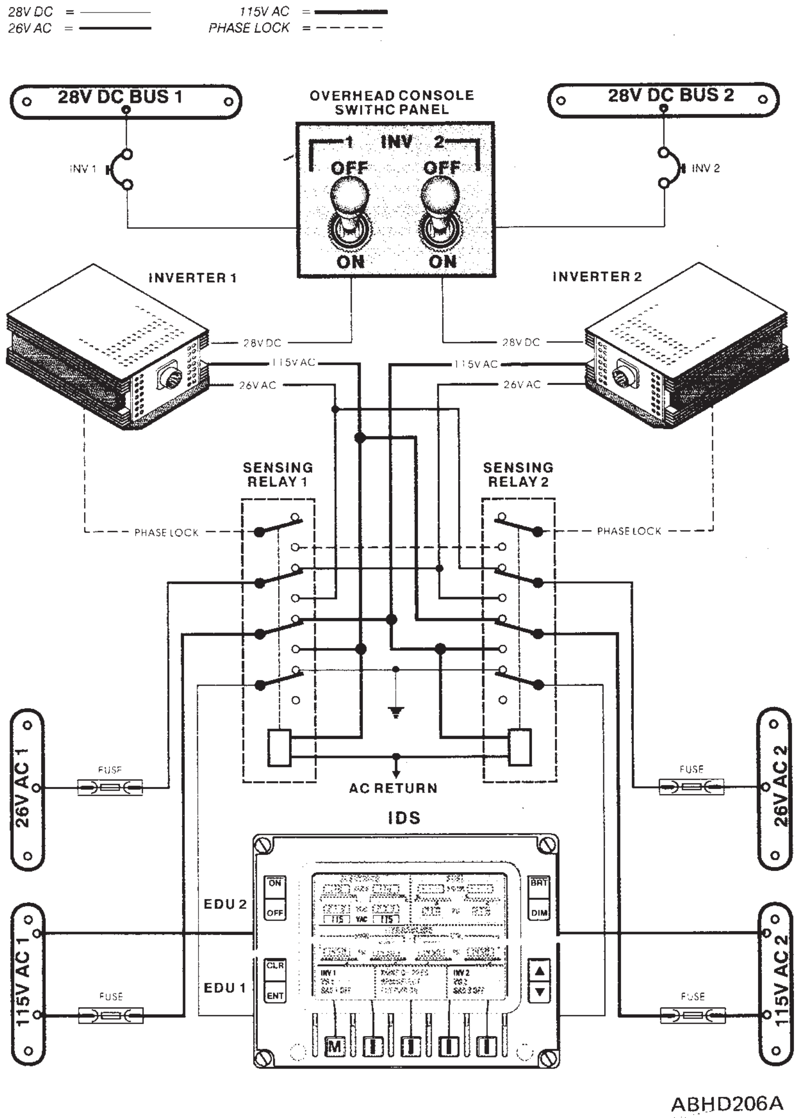


Figure 7-16. AC electrical system block diagram.

HELIPILOT/FLIGHT DIRECTOR

OPERATING PROCEDURES

HELIPILOT IFR OPERATION

The Helipilot Computer will perform all electronic computation and processing necessary for stabilization and automatic flight path control of the helicopter when coupled with the Helcis Flight Director Computer.

The Helipilot System is operated from cockpit located switches.

Solenoid held switches allow engagement of the Helipilot System, SAS 1, and ATTD HOLD, and they also supply excitation voltages to the servo amplifiers.

SAS 1 supplies pitch, roll and yaw axes stabilizing signals through control actuators and SAS 2 contains redundant pitch and roll electronics and actuators.

IFR operation is with SAS 1 and SAS 2 and ATTD HOLD engaged. SAS 1 actuator activity can be monitored by the pilot observing actuator trim meters located on the instrument panel.

By pressing the monitor switch SAS 2 PUSH located below the trim meters, SAS 2 actuators activity can be monitored.

When the pilot desires to disengage SAS 1 or SAS 2 he may move the appropriate switch to OFF position.

The IFR Mode operates in either of two possible conditions: automatic path control or attitude hold.

The automatic path control is dependent on the Flight Director Computer and is automatically achieved upon selection of a valid flight director mode on the FD controller. If a valid flight director mode is not selected in either the pitch or roll axis, that axis reverts to attitude hold.

When flying in attitude hold, the pilot may command attitude changes in the normal manner through the cyclic control.

Permanent attitude changes are achieved by operation of the aircraft FTR button located on the cyclic grip.

When the Helipilot System is coupled to the FD Computer and a valid flight director mode is selected, the aircraft flight path is automatically maintained by series actuators; however, any long term trimming operations are accomplished by the pilot.

The requirement to re-trim the controls is noted by the pilot by monitoring the actuator trim meters or will be noted by observing large deviations on the flight director indicator.
Retrimming is accomplished in the normal manner, i.e. by operating the trim release button on the cyclic and repositioning the cyclic as necessary.
Retrimming is necessary only in the yaw axis when Autotrim is engaged.
SFENA motorized trim provides both lateral and longitudinal cyclic trim through the activation of the AUTOTRIM switch and Flight Director modes during COUPLED operations.
AUTOTRIM is activated once the SAS actuator(s) are displaced from their midpoint by 30% of their travel.
If this or greater displacement is maintained for 2.5 seconds, the trim motor(s) are keyed and run until SAS actuator is recentered.
The autotrim function is cancelled by either the FTR button on the pilot's cyclic stick (or copilot's if installed) or when the Flight Director is in STANDBY (SBY) mode.
The coupled FD modes may be disengaged by pressing the FD-SBY button located on the cyclic, or moving the COUPLED/DECOUPLED switch to DECOUPLED position located on the console.
In this case the FD command bars are still present on ADI.
The attitude retention SAS 1, SAS 2 and ATTD HOLD shall be maintained throughout IFR Flight.

HELIPILOT VFR OPERATION

The engagement of SAS 1 and/or SAS 2, provides short term stabilizing signals to the helipilot actuators.
Normal operation is with SAS 1, SAS 2 and ATTD HOLD engaged and the pilot flies the helicopter in the normal manner.
The short term stabilization provided by the helipilots improves basic aircraft handling qualities and reduces the pilot work load.
To change from SAS only to the normal IFR Attitude Retention engage ATTD HOLD.
Flight with SAS only is reserved for VFR operation only.

FLIGHT DIRECTOR COMPUTER

The Flight Director Computer will perform all electronic computation and processing necessary for displaying flight director commands on the attitude indicator and for automatic path control of the helicopter when coupled with the Helipilot Computer.

Lateral modes available for either flight director or automatic coupling include heading select, NAV/VOR-VOR APP and localizer (both normal and back course).

In each case the navigation errors are processed by the flight director computer, providing roll attitude command which results in interception and tracking of the desired lateral flight path.

The pilot selects a desired mode (HDG or VOR APP or BC).

The vertical command bar on the ADI is then automatically followed by the Helipilot Computer when coupled.

The pilot when the system is uncoupled, flies the command bars and cross checks raw data in order to keep the aircraft on course. Longitudinal/vertical modes of operation include barometric altitude hold (ALT), airspeed hold (IAS) vertical speed select (VS) and glideslope capture and track (GS).

A vertical path can be flown IAS or ALT or VS or G/S.

Airspeed errors will then be processed to provide pitch attitude commands (displayed on the horizontal command bar on the ADI), when the ALT mode will be engaged, and altitude errors provide pitch attitude command.

Similarly, vertical speed and glideslope may be flown via pitch.

In the airspeed mode, changes to the existing speed may be commanded by operation of BEEP TRIM switch.

Altitude changes are normally made by engaging VS, setting the desired climb or descent rate on the IVSI and re-engaging ALT when the desired altitude is reached.

Glideslope mode is armed by pushing the ILS button.

Glideslope Engagement is automatic at the proper capture point. In addition to the above modes, an additional feature is provided. For Flight Director approaches, either on glideslope or open loop via selected vertical speed, a command to level out is made automatically as radar altitude decreases to 50 feet (if radar altimeter is installed).

At this point, the pitch command cue is tripped from G/S or VS automatically to maintain constant radar altitude. This mode is intended only as a safety

back-up in case the pilot has not taken over the aircraft manually at decision height or otherwise performed a go-around.
The go-around mode is engaged by the operation of the push-button located on the collective stick.
Engagement of this mode commands pitch attitude to achieve and maintain 750 FPM for helicopters with Flight Director wiring 109-0771-92-103 (1000 FPM for helicopters with Flight Director wiring 109-0771-92-107) climb if the airspeed is greater than 55 kias and the roll axis commands level attitude.
A pre-selected go-around heading may be engaged at any time by re-engaging the HDG mode, while the longitudinal/vertical axes remain in go-around.

IN FLIGHT OPERATION

Climb

Climb may be accomplished using either heading hold (HDG) or navigation (NAV-VOR APP) for roll control and either airspeed hold IAS or vertical speed hold (VS) for pitch control. The procedure for initiating a climb immediately following take off is:

SAS 1, SAS 2 and ATTD HOLD	:	On before take off
Collective	:	Climb power
FD	:	Appropriate pitch and roll modes selected after translation.
TRIM	:	As required.

Cruise

Level-off with the Helipilots engaged, or in Flight Director flight, may be made by selecting ″ALT″ at the desired altitude.
This reference altitude may be changed at any time by pressing off ALT and flying to a new altitude and re-selecting ″ALT″.
If cruising on ″IAS″ a new airspeed may be selected by activating the airspeed increase/decrease switch on the pilot’s collective, which alters the reference airspeed at approximately 2 knots per second.

VOR

VOR interception and tracking

To intercept a preselected VOR course, the flight director features an automatic capture capability which is armed prior to approaching an on-course condition.
Automatic capture is armed by pushing the NAV button, which will cause both HDG and NAV Arm buttons to illuminate if the course deviation indicator (CDI) is deflected off course.
During this armed mode, the autopilot responds only to the heading ″bug″.
As the helicopter approaches a point of interception from the selected VOR course, the HDG light goes out and NAV CAP comes on, and the autopilot now responds to the NAV information. This mode is designed for cross country navigation and any radial capture should be effectuated more than 15 NM from the station and at proper altitude. For capture less than 15 NM use VOR APP mode. When approaching a VOR station, the radio beam width becomes too narrow for the helicopter to follow. Once the CDI moves off-scale rapidly, the over-station sensor is activated. Once activated, over station sensing will be maintained for a fixed period of time from the last major deflection of the CDI to allow for station passage (NAV: approx. 45 secs., VOR APP: approx. 5 secs.).
During this time the autopilot will respond to the ″course arrow″ in the same manner as it does to the ″heading bug″ when operating in the heading hold mode.
After the over-station sensor is deactivated the helicopter will turn to make a normal course interception in NAV mode and in VOR APP mode.

NOTE

> Since over-station sensing is activated by rapid off scale movement of the CDI, a false over-station sensing can be induced by making rapid course changes with the course selector knob while tracking on VOR.

The helicopter is on the localizer when it approaches the glideslope. Glideslope capture is identified by the illumination of the ″ILS CAP″ button on the mode controller.
If the helicopter is not on the localizer when the helicopter intercepts the glideslope, automatic glideslope capture will not occur.

Manual glideslope capture may be initiated by pressing the ILS button at the appropriate point.

BC (BACK COURSE) LOC

A loc back course approach is made by pressing the BC button located on the FD control panel. The CAPture will occur at two dots HSI deviation.
The HSI is operated per normal front course instrument procedures.

NOTE

The BC mode renders the Helcis G/S mode inoperative, G/S raw data may still be present on the ADI and HSI due to spurious front course signal reflections.
Do not attempt to utilize this information.

AUTO LEVEL (IF RADAR ALTIMETER IS INSTALLED)

As an additional feature, the flight director has programmed a minimum altitude feature which operates during approach and commands an automatic leveling of the helicopter with the pitch cue, if the flight director is still in use at 50 feet radar altimeter.
This 50 feet altitude will be maintained indefinitely unless the pilot returns the FD mode control to stand-by or go-around.
The autopilot should be disengaged and the flight director on stand-by once the helicopter reaches decision height (DH) and the airport in sight.
If the airport is not in sight at DH, a go-around may be commanded by activating the ″GO-AROUND″ button on the collective.

G/A - GO-AROUND

Go-Around, for helicopters with Flight Director wiring 109-0771-92-103, is programmed to command the following:

— ″Wings level″.
— Climb at 750 FPM.
— Necessary power to achieve the above parameters.

Go-around, for helicopters with Flight Director wiring 109-0771-92-107, is programmed to command the following:

— ″Wing level″.

— Climb at 1000 FPM and approx. 80 Kts.

— Necessary power to achieve the above parameters.

Activation of the ″Go-around″ button will over-ride all other flight director modes and may be used at any time a climb is desired.

NOTE

Do not select IAS while in go-around, as the mode controller button will illuminate but will have no function.

NOTE

(for helicopters with Flight Director wiring 109-0771-92-107 only).
The suggested airspeed to perform the ″Go-around″ procedure is 100 to 140 Kts.
Airspeed lower than 100 Kts may determine a significant power demand to centre the collective cue. This power change will produce an helicopter pitch up variation to acquire the desired climb rate, that may reduce the initial speed to 50 Kts or below (depending on the helicopter maximum gross weight and on the rate of power change).
The helicopter will then accelerate to normal go-around speed ($\leqslant$ 80 Kts).
In order to minimize this airspeed reduction, it is suggested (below 100 Kts) a gradual application of the power required to centre the collective cue.

INTEGRATED DISPLAY SYSTEM (IDS)

The IDS is composed of two identical Electronic Display Units (EDU) and a dual redundant Data Acquisition Unit (DAU). The DAU and EDUs communicate over ARINC 429 data busses.
The IDS performs the following functions:

DATA DISPLAY

The IDS displays Primary and Secondary aircraft data, as well as Warning, Caution and Advisory messages. In normal operating mode, EDU 1 displays Primary parameters (RPMs, TOT, Torque), and EDU 2 displays Secondary parameters (temperatures, pressures, electrical quantities, etc.). Various display modes are possible, depending on the aircraft status (i.e. in flight or on ground) and the System status (i.e. both EDUs operating or one EDU operating). Primary and Secondary data are described below.
Further definition of the Data Display function includes:

Display of Engine Control Unit (ECU) Primary Parameters

The IDS does the following:

— DAU and both EDUs receive primary engine parameters from both ECUs. Also receive additional parameters, such as discretes, status and fault words from both ECUs.

— The DAU repeats ECU data to the EDUs over the ARINC 429 data busses. Primary data includes N1, TOT, Torque and NR/N2.

Display of DAU Primary Parameters

When selected either automatically or manually, the DAU provides the Primary data to EDUs.

Display of Warning and Caution Messages

Under normal conditions, Warning and Caution messages are displayed on EDU 1. EDU 1 message display area is fixed; if message conditions exist

which overrun the allowed display area, an indication of ″more pages″ is given by means of broken vertical lines and the pilot must use the up/down rocker switch on EDU bezel to scroll through messages.
At initial power-up, the Warning and Caution messages are displayed in a pre-defined priority, in order to provide a visual cue to the pilot during the engine start. At the end of pre-start check routine, the pilot may have to reestablish the priority order by pressing the CLR (Clear) switch on EDU bezel.
The IDS suppresses the active messages according to the following matrix:

Helicopter on ground	
Failure or Status Condition	**Message Suppressed**
Engine Out	PLA Caution
Engine Out and N 1 < 50%	ENG OUT Warning (enabled if ENG MODE switch not OFF and N 1 > 50%)
ECU in Manual Mode	PLA Caution
ECU Failure	PLA Caution
ENG MODE switch not OFF	ECU MAINT Caution

Helicopter in flight	
Failure or Status Condition	**Message Suppressed**
Engine Out	PLA, DC GEN, OIL PRES Cautions
N 2 Sensor Failure (FAIL legend)	IDLE Legend
ECU in Manual Mode	PLA Caution
ECU Failure	PLA Caution

Display of Secondary Parameters

Secondary parameters are processed by the DAU and displayed on EDU 2 as follows:

Indication	MAIN page	AUX page
Engine Oil Temperature and Pressure	X	
Transmission Oil Temperature and Pressure	X	
Fuel Pressure	X	
Fuel Quantity	X	X
Main Hydraulic Oil Pressure	X	X
Utility Hydraulic Oil Pressure		X
Outside Air Temperature (OAT)	X	
Voltmeters and Ammeters		X

Display of Advisory/Status Messages

Under normal operating conditions, Advisory and Status messages are displayed on EDU 2 in green and cyan, respectively. The pilot may have to use the up/down rocker switch on the EDU bezel to scroll through messages. Following is the list of Advisory (green) messages:

Advisory Message	Description
PITOT 1 (2) HEAT	Pitot 1 (2) heater on
ECS ON	Environmental Conditioning System on
VENT ON	Ventilation system on
LIMIT OVRD ON	LIM OVRD switch activated and ECU power limits (N1, TOT and Torque) overridden
FD DECPLED	Flight Director in decoupled mode
LANDING LT ON	Landing lights on
AUTOTRIM OFF	Autotrim off
FT OFF	Force trim off

Following is the list of Status (cyan) messages:

Status Message	Description
CHECK DATA	Some pressure sensors out-of-tolerance in DATA page. With both engines off, check for possible off-set errors on pressure indications on EDU 2. (See also the A109E Maintenance Manual).

The following green messages are displayed on EDU 1:

— START on N1 scale
— IGN on TOT scale
— IDLE on N2 scale

while the green XFEED message is displayed on EDU 2 between the Fuel Quantity readouts.

Color Coding

Under normal operating conditions, the scale pointers are presented in white and the associated digital boxes are presented in green. When any parameter exceeds the normal range of operation (green band), the color of associated pointer and digital box turns to yellow or red, as applicable, in order to highlight that particular condition. Furthermore, if that parameter enters in the red band, this latter turns to full red area to enhance the emergency condition. During engine start in manual mode, the digital boxes and pointers of N 1, TOT and Torque parameters turn to magenta until N 1 > 50% or the Engine Out discrete becomes inactive.

Display of IDS Malfunctions

In case of an internal IDS failure, the IDS Caution message is displayed on EDU 1. If either EDU fails, the healthy EDU automatically displays the REVERSIONARY mode. The REVERSIONARY mode can also be commanded via the ON/OFF switch on the EDU bezel. In case of complete loss of DAU, both EDUs display the CRUISE mode, as Secondary data are no longer available.

Display of Sensors Malfunctions

In case of loss of any analog input signal, red dashes in a white box are displayed on the affected EDU. In case of loss on any N 2 input signal, a red ″FAIL″ legend is vertically displayed adjacent to the affected N 2 scale.

Selection of Data Source

The DAU and the two ECUs interface with various analog and discrete sensors. The ECU data, when″healthy″, is always the source of Primary data for display by the IDS system. When an ECU informs the IDS that one or more pieces of its data are not valid, the IDS automatically reverts to the DAU as the source for the Primary data. However, the DAU can also be forced to be the source of Primary data for display via EDU bezel keys.

Selection of Data Display Formats

The EDUs determine, based on the data they receive, the format that the data is displayed in. Display formats which can be manually selectable are accessible via the ″M″ (Menu) key on EDU bezel. Display formats include:

— **EDU 1**

START Mode. This mode can be selected via the START key in MENU page 1 of EDU 1 when the helicopter is on ground.

CRUISE Mode. This mode is automatically presented at initial power up or when selected via the CRUISE key in MENU page 1 of EDU 1. This display format is also automatically presented on both EDUs in case of complete loss of DAU, as Secondary data are no longer available.

One Engine Inoperative (OEI) Mode. This mode is automatically presented when an Engine Out condition exists or when the ENG MODE switch is set to IDLE or OFF. An OEI legend is vertically displayed adjacent to N 1, TOT and Torque scales and color coded as follows:

- **White**: Indicates that the OEI format is being used.
- **Yellow**: Indicates that the 2.5 minute power rating is authorized by the ECU.
- **Red flashing**: Indicates that the 2.5 minute power rating is expired, as commanded by the ECU.

If an in-flight engine re-start is performed, the red dot indicating the engine temperature start limit is only presented on the side of TOT scale related to the engine being started. The display format automatically reverts to the AEO limits if the start has been successful.

AUTOROTATION Mode. This mode is automatically presented when both Engine Out signals are active, both ENG MODE switches are set to IDLE or OFF, an RPM split condition exists or both Torque signals are < 2%.

The IDS activates the corresponding AWG vocal messages based on the following power-off limits:

Rotor High when NR > 110%

Rotor Low when NR < 90%

REVERSIONARY Mode.This mode is automatically presented in case of either EDU failure or when manually commanded via the ON/OFF switch on EDU bezel.

The following display formats are also available:

REVERSIONARY - AEO

REVERSIONARY - OEI

REVERSIONARY - AUTOROTATION

As Secondary parameters are presented only in digital format without the box, a yellow or red dot (virtual light) is displayed adjacent to the affected title to indicate the operational band in which the relevant parameter is.

The warning, caution and advisory messages are displayed in priority order at the bottom of EDU 1 screen. If ″more pages″ of messages exist, the pilot must use the up/down rocker switch to scroll through them.

— **EDU 2**

MAIN Mode. This mode is automatically presented at initial power-up or when selected via the MAIN key in MENU page 1 of EDU 2.

Auxiliary (AUX) Mode. This mode is presented when selected via the AUX key in MENU page 1 of EDU 2. If a specific Warning or Caution message is set, the display automatically reverts to the MAIN page.

REVERSIONARY Mode. Same as in EDU 1.

Other Display Formats/Functions Selectable (EDU 1 and EDU 2)

TEST Mode (MENU page 1). Pre-flight test mode used to initiate the BIT function. This mode can only be accessed when the helicopter is on ground.

DATA Mode (MENU page 2). The IDS can, upon command via the CAL key in DATA page, calibrate all the pressure analog signal inputs, accounting for offsets in the sensors outputs with zero pressure applied. This format can only be accessed when the helicopter is on ground and is to be used at first DAU installation or when any pressure sensor is replaced.

DAU CH-A Selection (MENU page 2). This key is used to manually select the DAU Channel A as primary source for display. At initial power-up, the legend CH-A is displayed in green to indicate that this is the active channel. Once selected, a white box surrounds the green legend. The channel failure is indicated in yellow.

DAU CH-B Selection (MENU page 2). This key is used to manually select the DAU Channel B as primary source for display. At initial power-up, the legend CH-B is displayed in white to indicate that this is the hot-standby channel. Once selected, a white box surrounds the green legend. The channel failure is indicated in yellow.

MAIN TEST Mode (MENU page 3). Maintenance test mode used to verify the EDU screen integrity, as well as to present the system diagnostics. This mode can only be accessed when the helicopter is on ground.

DISPLAY DIMMING

The IDS does the following:

— Adjust EDU backlighting automatically based on input from Ambient Light Sensor (ALS) located on EDU bezel.

— Adjust EDU backlighting modifying ALS control based on manual input from the BRT/DIM (Bright/Dim) rocker switch located on EDU bezel.

— Adjust EDU backlighting based on status of Day/Night discrete input.

— Adjust intensity of Master Warning amd Master Caution Lights, if they are on, based on status of Day/Night discrete input.

COMPARISION MONITORING OF DAU DATA

The IDS displays ″DAU MISCMP-P″ (DAU Miscompare - Primary data) if primary data from each DAU channel disagree with one another.

MASTER WARNING AND MASTER CAUTION LIGHTS INTERFACE

One MWL and one MCL are installed at each pilot's station. The IDS does the following:

— Activate the MWLs when a Warning condition occurs. MWLs remain activated until the condition no longer exists, the Warning reset switch or the collective mounted reset switch is momentarily pressed.

— Activates the MCLs when a Caution condition occurs. MCLs remain activated until the condition no longer exists, the Caution reset switch or the collective mounted reset switch is momentarily pressed.

AURAL WARNING GENERATOR (AWG) INTERFACE.

The AWG sets a tone or vocal message in the pilots headsets when aircraft conditions warrant. The IDS is not the only system which interfaces with the AWG. The IDS does the following:

— Activates the AWG under Rotor High, Rotor Low, Transmission Overtorque or other warning conditions. AWG activation remains until the condition no longer exists, the Warning reset switch, the Caution reset switch or the collective mounted reset switch is momentarily pressed, as applicable.

— Activates the Rotor Low horn to provide a separate tone under low rotor speed conditions.

FUEL SYSTEM INTERFACE.

The IDS does the following:

— Provide excitation for each fuel pressure sensor and monitor its output signal.

— Monitor the ″Fuel Low″ discrete input for each tank and display the corresponding Caution message, if set.

— Test the low level sensor for correct operation when commanded through the IDS TEST key and display the ″F LOW FAIL″ Caution message, if set.

— Monitor each fuel quantity signal provided by the Fuel Computer Unit (FCU).

— Test FCU for correct operation when commanded via the IDS TEST key.

ENGINE FIRE DETECTOR INTERFACE.

The IDS does the following:

— Monitor the ″Engine Fire″ discrete input for each engine and display the corresponding Warning message, if set.

— Monitor the ″Engine Fire Detector″ discete input from each engine and display the ″FIRE DET″ Caution message, if set.

— Test the ″Engine Fire Detector″ for correct operation when commanded via the IDS TEST key.

WEIGHT ON WHEELS (WOW) INTERFACE.

The IDS interfaces with WOW discrete input to determine that the aircraft is on the ground.

ENGINE BLEED AIR VALVE CONTROL.

The IDS interfaces with each engine Bleed Valve and does the following:

— Activate the relay driver which closes the corresponding Bleed Valve under Engine Out Condition or both Bleed Valves under Engine Fire condition, until the condition no longer exists.

— Deactivate the relay driver (Valve open) under all other conditions.

BUILT-IN TEST (BIT)

Both the EDUs and the DAU are able to conduct self-tests and fault isolation. Testing is performed at power-up, as well as continuously during operation. Self-test of IDS can also be commanded via EDU TEST key. The following messages are displayed on EDU 2 if the pilot initiated test fails:

EDU 1 SW FAIL:	Red message flashing to indicate that the EDU 1 BIT checksum failure is detected.
EDU 2 SW FAIL:	Red message flashing to indicate that the EDU 2 BIT checksum failure is detected.
DAU-A SW FAIL:	Red message flashing to indicate that the DAU CH-A BIT checksum failure is detected.
DAU-B SW FAIL:	Red message flashing to indicate that the DAU CH-B BIT checksum failure is detected.
EDU 1 BIT BUSY:	Yellow message flashing to indicate that the EDU 1 has not yet passed the first pass through BIT.
EDU 2 BIT BUSY:	Yellow message flashing to indicate that the EDU 2 has not yet passed the first pass through BIT.
DAU-A BIT BUSY:	Yellow message flashing to indicate that the DAU CH-A has not yet passed the first pass through BIT.
DAU-B BIT BUSY:	Yellow message flashing to indicate that the DAU CH-B has not yet passed the first pass through BIT.
EDU 1 BIT FAIL:	Red message flashing to indicate that at least one BIT routine has failed.
EDU 2 BIT FAIL:	Red message flashing to indicate that at least one BIT routine has failed.
DAU-A BIT FAIL:	Red message flashing to indicate that at least one BIT routine has failed.

EDU 1 SW FAIL: Red message flashing to indicate that the EDU 1 BIT checksum failure is detected.

DAU-B BIT FAIL: Red message flashing to indicate that at least one BIT routine has failed.

IFR INSTALLATION

The IFR installation is offered in a single pilot P/N 109-0810-22-139/-143 and in a dual control P/N 109-0810-22-141/-145 configuration (see Table 7-1).

Part Number	Denomination
DUAL CONTROL CONFIGURATION	
109-0771-89-137/141	ICS
109-0771-89-117	ADF-60A
109-0771-89-109	VOR/ILS VIR 32 N°1
109-0771-89-113	VOR/ILS VIR 32 N°2.
109-0771-89-101	VHF/AM VHF 22A N°1
109-0771-89-105	VHF/AM VHF 22A N°2
109-0771-89-159	PILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-89-165	PILOT INSTRUMENTS (NAV.) (with EFIS)
109-0771-91-105	SPERRY AFCS
109-0771-89-125/225	TRANSPONDER TDR 90 / TDR94 Mod. S
109-0771-89-121	DME 42
109-0810-01-117	DUAL CONTROLS
109-0771-89-163	COPILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-89-167	COPILOT INSTRUMENTS (NAV.) (with EFIS)
109-0771-89-129	GYROCOMPASS INSTALLATION
109-0771-89-151	VERTICAL GYRO 1
109-0771-89-155	VERTICAL GYRO 2
SINGLE PILOT CONFIGURATION	
109-0771-89-137	ICS
109-0771-89-117	ADF-60A
109-0771-89-109	VOR/ILS VIR 32 N°1
109-0771-89-113	VOR/ILS VIR 32 N°2.
109-0771-89-101	VHF/AM VHF 22A N°1
109-0771-89-105	VHF/AM VHF 22A N°2
109-0771-89-159	PILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-89-165	PILOT INSTRUMENTS (NAV.) (with EFIS)
109-0731-89-107	ADDITIONAL INSTRUMENTS (with EFIS)
109-0731-96-107	ADDITIONAL INSTRUMENTS (with EHSI)
109-0771-91-105	SPERRY AFCS
109-0771-89-125/225	TRANSPONDER TDR 90 / TDR94 Mod. S
109-0771-89-121	DME 42
109-0771-89-129	GYROCOMPASS INSTALLATION
109-0771-89-151	VERTICAL GYRO 1
109-0771-89-155	VERTICAL GYRO 2

Table 7-1 (sheet 1 of 2). IFR installation components ″COLLINS configuration″.

Part Number	Denomination
DUAL CONTROL CONFIGURATION	
109-0771-88-137/133	ICS
109-0771-88-117	ADF KR87
109-0771-88-121	MARKER BEACON
109-0731-96-109	FLIGHT INSTRUMENTS (COPILOT) (with EHSI)
109-0731-89-109	FLIGHT INSTRUMENTS (COPILOT) (with EFIS)
109-0771-88-101	NAV/COMM 1 KX165
109-0771-88-105	NAV/COMM 2 KX165
109-0771-88-157	PILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-88-163	PILOT INSTRUMENTS (NAV.) (with EFIS)
109-0771-91-105	SPERRY AFCS
109-0771-88-113/203	TRANSPONDER KT71 / KT70 Mod. S
109-0771-88-109	DME KDM706A
109-0810-01-117	DUAL CONTROLS
109-0771-88-161	COPILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-88-165	COPILOT INSTRUMENTS (NAV.) (with EFIS)
109-0771-88-125	GYROCOMPASS INSTALLATION
109-0771-88-149	VERTICAL GYRO 1
109-0771-88-153	VERTICAL GYRO 2
SINGLE PILOT CONFIGURATION	
109-0771-88-133	ICS
109-0771-88-117	ADF KR87
109-0771-88-121	MARKER BEACON
109-0731-96-107	ADDITIONAL INSTRUMENTS (with EHSI)
109-0731-89-107	ADDITIONAL INSTRUMENTS (with EFIS)
109-0771-88-101	NAV/COMM 1 KX165
109-0771-88-105	NAV/COMM 2 KX165
109-0771-88-157	PILOT INSTRUMENTS (NAV.) (with EHSI)
109-0771-88-163	PILOT INSTRUMENTS (NAV.) (with EFIS)
109-0771-91-105	SPERRY AFCS
109-0771-88-113/203	TRANSPONDER KT71 / KT70 Mod. S
109-0771-88-109	DME KDM706A
109-0771-88-125	GYROCOMPASS INSTALLATION
109-0771-88-149	VERTICAL GYRO 1
109-0771-88-153	VERTICAL GYRO 2

Table 7-1 (sheet 2 of 2). IFR installation components ″KING configuration″.

LIGHTING SYSTEMS

The lighting system include all the lights utilized for the helicopter interior and exterior illumination.
The exterior lights include three position lights, two anti-collision lights and two landing lights.
The interior lights include instrument lights, utility lights, baggage light, cabin lights and advisory lights.

EMERGENCY EQUIPMENT

PORTABLE FIRE EXTINGUISHER

The helicopter is equipped with a portable, manually operated fire extinguisher installed on the central console, between the pilot's and copilot's seats. The mounting bracket is of the quick release type, for rapid removal of the fire extinguisher.

FIRST AID KIT

The first aid kit consists of a medical bag secured with strips of velcro to the vertical panel below the rear passenger seat.

COCKPIT VOICE DATA RECORDER

The Cockpit Voice Data Recorder (CVDR) System model FA23XX MADRAS consists of:

— a Recorder Unit;

— a Control Unit;

— an Area Microphone;

— a Mounting Tray.

The CVDR is housed in ARINC 404A, 1/2-ATR short case (refer to Figure 7-17).

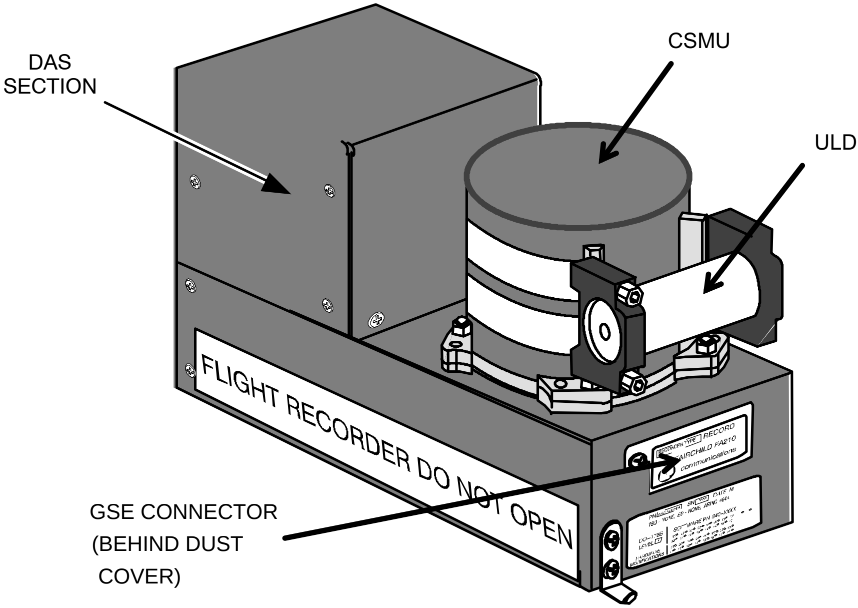


Figure 7-17. Cockpit Voice Data Recorder Model FA23XX MADRAS.

The chassis and Crash Survivable Memory Unit (CSMU) are painted international orange. Two reflective stripes are located on the CSMU.
The CSMU contains the solid state flash memory used as the recording medium. The micro Data Acquisition Section (DAS) input is self-contained in the rear chassis assembly.
An Underwater Locator Device (ULD) is mounted horizontally on the front of the CSMU and is also used as the recorder’s carrying handle.
The Ground Support Equipment (GSE) connector is located on the front of the MADRAS. This connector provides the interface from the recorder to GSE for checkout of the recorder, or to transfer data to a readout devices.
The CVDR is a ″On Condition Line Replaceble Unit″ (LRU) that simultaneously records both cockpit voice and flight data.

The CVDR-MADRAS records a minimum of 120 minutes of high quality audio from the following four cockpit audio inputs:

Channel 1: Pilot’s Audio
Channel 2: Co-pilot’s Audio
Channel 3: Operator’s Audio
Channel 4: Cockpit Area Microphone (CAM) Audio

The Flight Data Recorder (FDR) function receives flight data at 256 words-per-second (wps). The flight data is stored in flash memory segregated from the cockpit voice data. The CVDR is capable of storing a minimum of 25 hours of flight data. Flight data stored in 25-hour configuration CVDRs can be downloaded in approximately five minutes.

Previously recorded voice information cannot be read from the CVDR while it is installed in the aircraft. However, the flight data may be monitored real time, or copied out to the Ground Station Equipment.

Operator’s interface with CVDR unit is obtained by means of a control panel located in cockpit roof, in the vicinity of overhead panel. The control panel is shown in Figure 7-18.

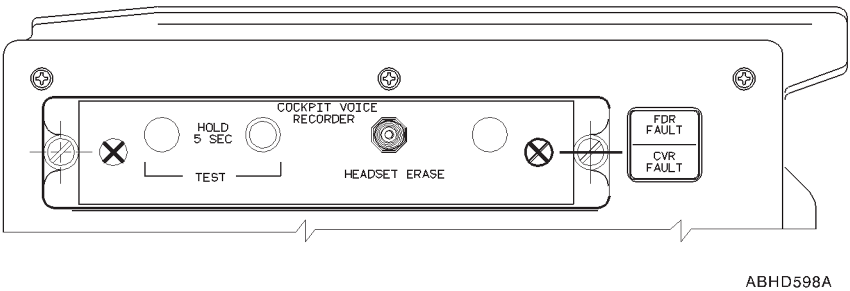


Figure 7-18. Cockpit Voice Data Recorder Control Panel.

The following information and data are provided to CVDR unit:

— A collection of Engines' and Helicopter's parameters, WARNING/CAUTION/ADVISORY messages by mean of serial connection between the Data Acquisition Unit (DAU) of the Integrated Display System (IDS) and the CVDR unit;

— Flight controls position by means of synchro-transmitters located on flight controls rods underneath pilot's seats;

— Helicopter's center of gravity accelerations by means of accelerometer located in the rear avionic bay;

— Airspeed and altitude by means of serial connection between the Air Data Unit (ADU) and the CVDR;

— Pilot/co-pilot's and operator's ICS communications, cockpit and passengers cabins audio signals by means of an area microphone located in the cockpit's front window central post (left side).

SECTION 8

HANDLING AND SERVICING

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HANDLING AND SERVICING

TOWING
(Figure 8-1)

The helicopter can be manoeuvered on the ground manually, or by a suitable vehicle, using the towing bar which can be secured to the nose wheel bar. The wheel can be steered 40° each side from the center.

TAXIING
(Figure 8-2)

The taxiing can only be performed by qualified pilots, but assistance from ground crew may be necessary when taxiing in obstructed areas.

PARKING
(Figure 8-3)

Park the helicopter in desired parking area on a level surface, when possible. Attach static ground wire and check that all switches are in OFF or neutral position. Install the approved straps and covers, as required.
For extended parking, disconnect the battery, engage rotor brake, lock main landing gear wheels, center and lock nose wheel, close all access doors and panels and install protective covers as shown on figure 8-4.

MOORING
(Figure 8-5)

The helicopter can be moored on a paved ramp, if available, with suitably spaced tie-down rings, and headed in direction from which forecast wind is expected.

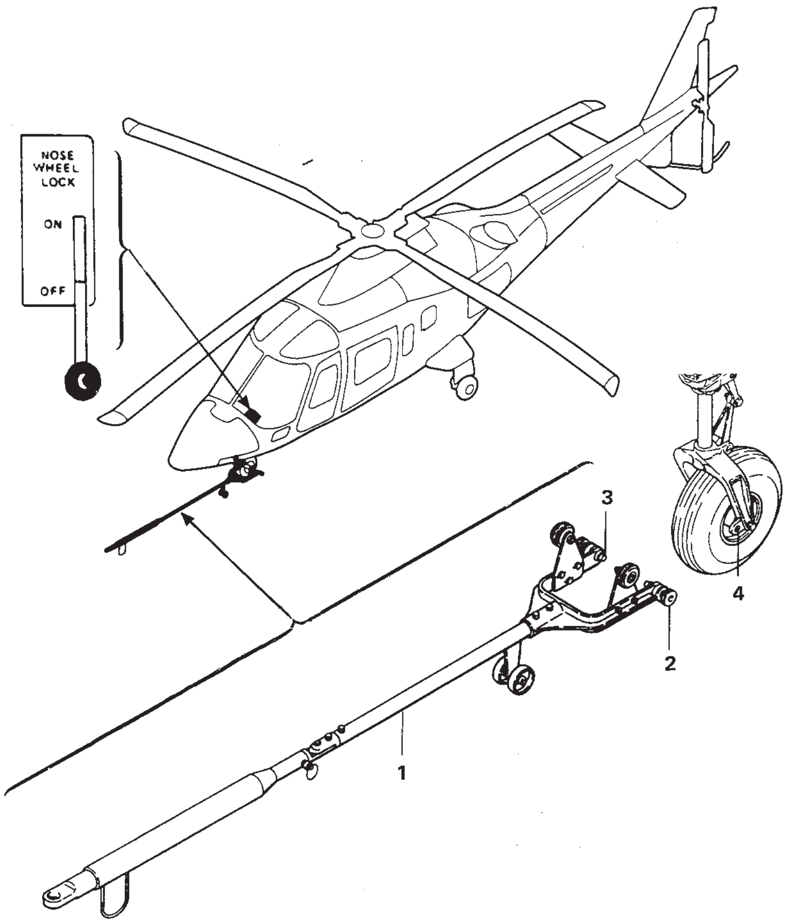
CAUTION

If forecast wind velocity exceeds 60 Kts, moor helicopter in a sheltered area, or place it in a hangar.

If suitable paved ramp and tie-down rings are not available, park helicopter on an appropriate parking area, headed into wind and use appropriate mooring anchors or make ″dead man″ anchors.
Engage rotor brake, lock main landing gear wheels, center and lock nose wheel, close all access doors and panels. Remove from parking area all loose equipment that can be lifted by wind.

SERVICING

Refer to Figure 8-6.



1. Tow bar
2. Pin grip (two)
3. Pin (two)
4. Pin hole (two)

Figure 8-1. Towing.

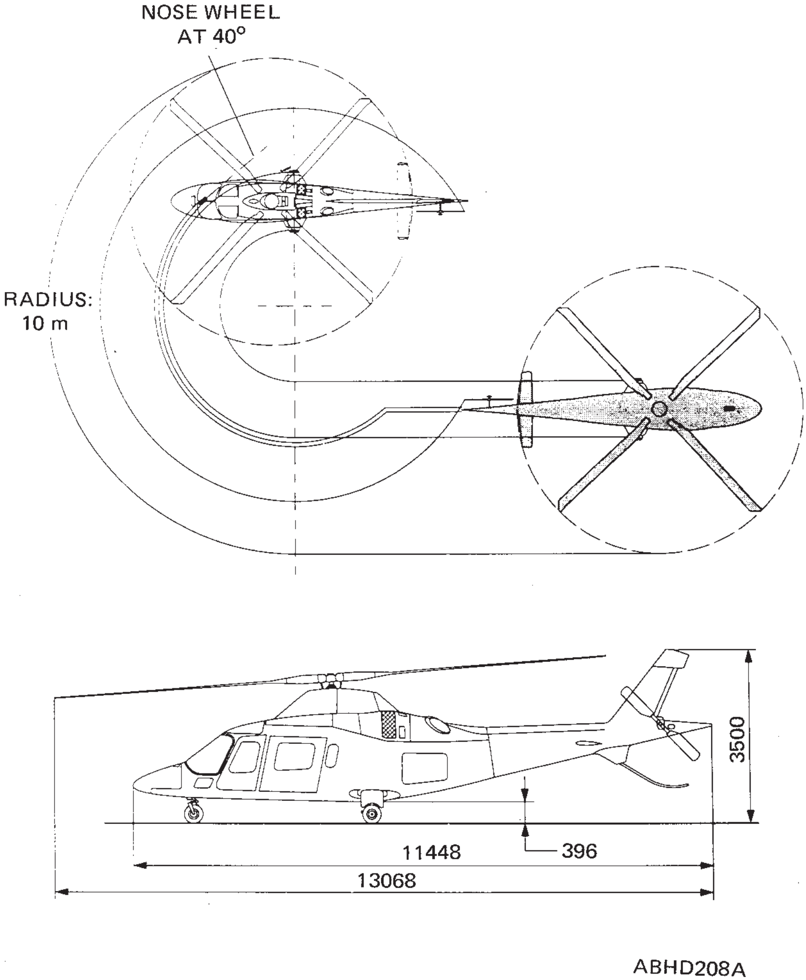
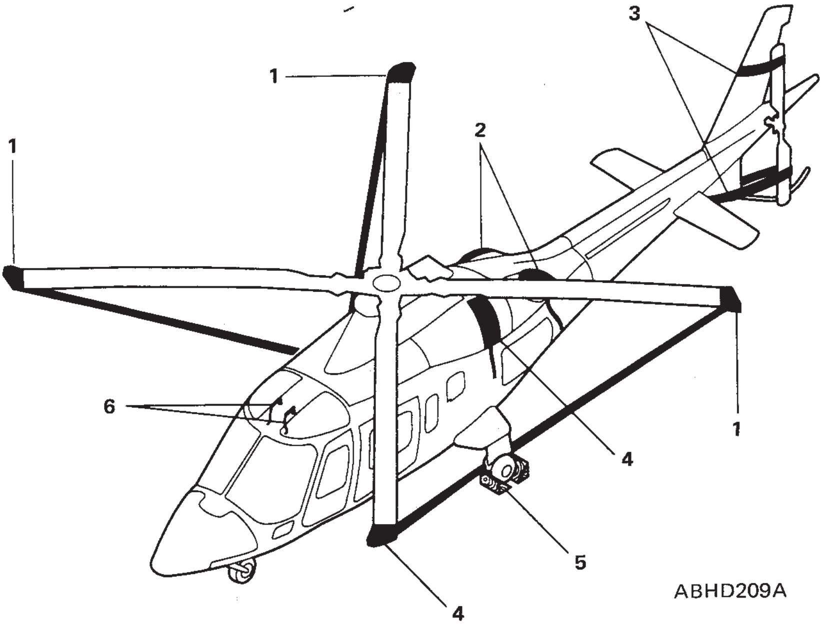
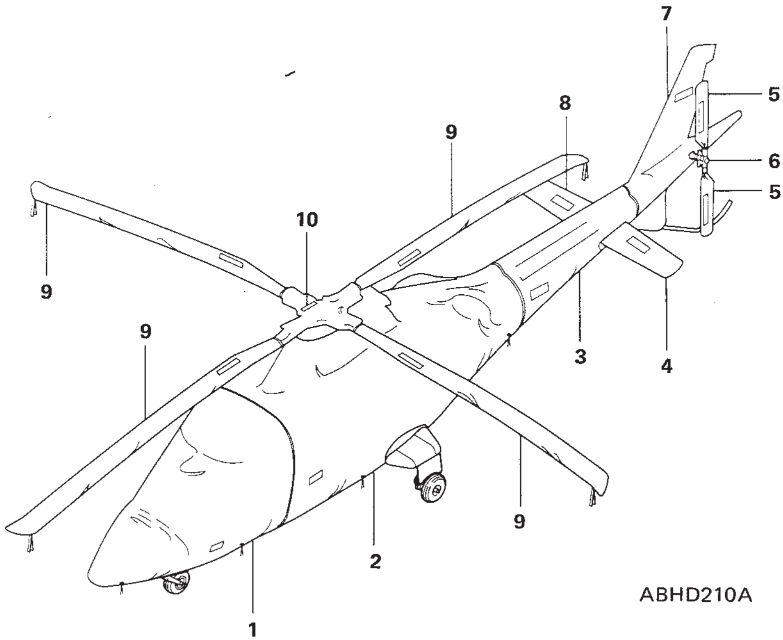


Figure 8-2. Taxiing.



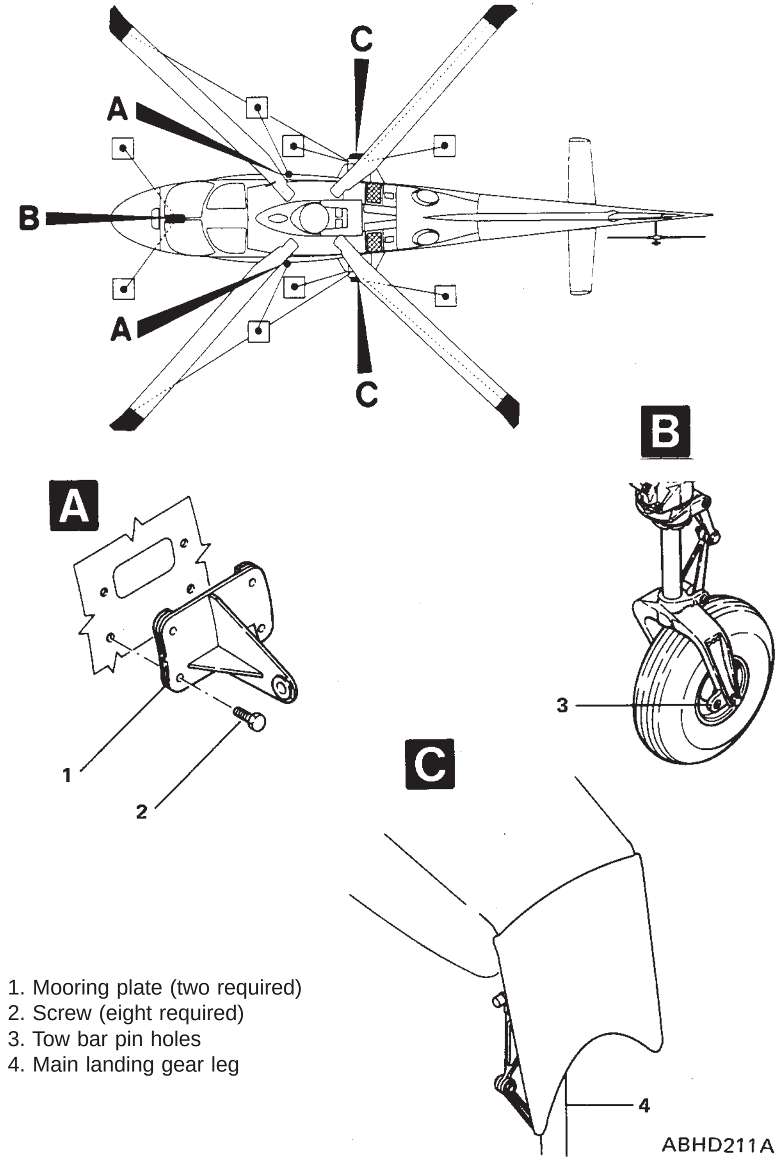
1. Main rotor blade straps
2. Engine exhaust duct covers
3. Tail rotor blade straps
4. Left engine air intake cover
   (Typical for right engine air intake cover)
5. Wheel chocks
6. Pitot tube covers

Figure 8-3. Parking.



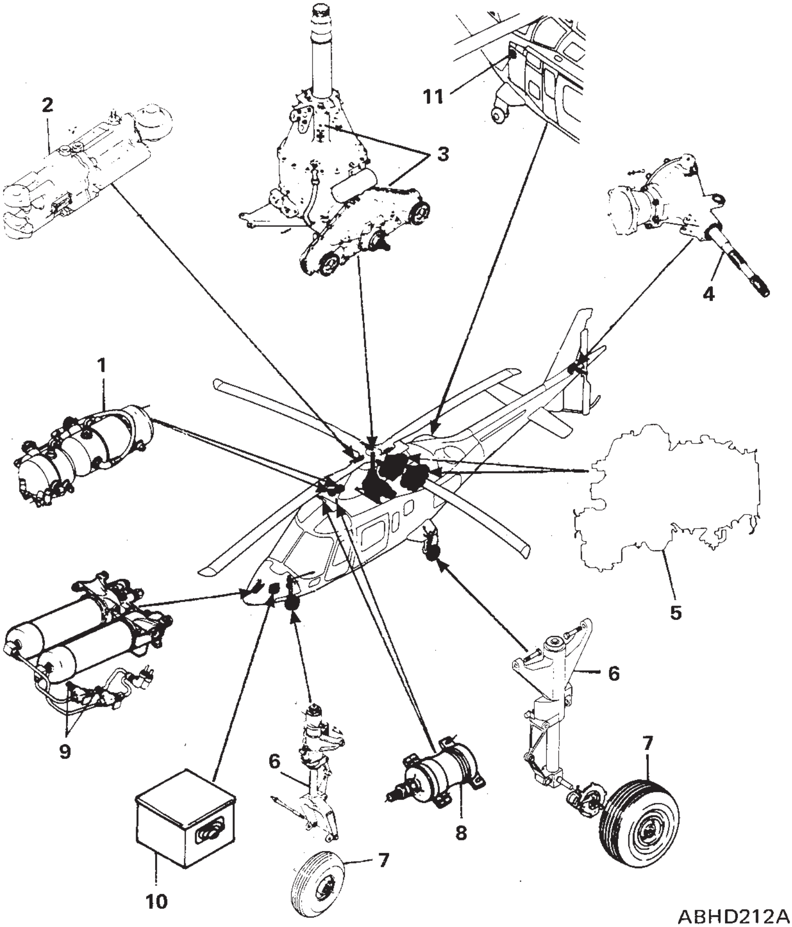
1. Fuselage nose section cover
2. Fuselage rear section cover
3. Tail boom cover
4. Left elevator cover
5. Tail rotor blade cover (two)
6. Tail rotor hub cover
7. Vertical empennage cover
8. Right elevator cover
9. Main rotor blade cover (four)
10. Main rotor hub cover

Figure 8-4. Protective covers.



1. Mooring plate (two required)
2. Screw (eight required)
3. Tow bar pin holes
4. Main landing gear leg

Figure 8-5. Mooring.



1. Flight control hydraulic system reservor (two)
2. Main rotor damper (four)
3. Main transmission
4. 90-degree gearbox
5. Engine (two)
6. Landing gear shock strut (three)
7. Landing gear wheel tires (three)
8. Flight control hydraulic system accumulators (two)
9. Utility hydraulic system accumulators (two)
10. Battery
11. Fuel filler point

Figure 8-6 (sheet 1 of 2). Servicing.

N°	ITEM	CAPACITY (liter)	NOTE
1	Hydraulic system	1.6	**N° 1 system**
1	Hydraulic system	3	N° 2 system
2	Dampers	0.05	Each damper
3	Main transmission	11	
4	90° gearbox	0.24	
5	Engine oil system	5.12	Each system
6	Landing gear strut		Nitrogen. Main LG: 188.5 psi Nose LG: 100 psi
7	Landing gear tires		
8	Flight controls accumulators		Nitrogen
9	Accumulators (utility hydraulic system)		Nitrogen. Charge to 427 psi
10	Battery		Distilled water. Add as necessary.
11	Fuel system	605	Usable: 595 liters

For fuel and oil Specification refer to Section 1 of this manual.

Figure 8-6 (sheet 2 of 2). Servicing.

SECTION 9

SUPPLEMENTAL PERFORMANCE INFORMATION

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SUPPLEMENTAL PERFORMANCE INFORMATION

GENERAL INFORMATION

The Supplemental Performance Information contained in this section is provided for use in conjunction with Section 4 and optional equipment Appendices, as applicable.
This section contains useful charts of cruise to determine max endurance and cruise recommended and hovering ceiling-single engine charts.

HELICOPTER CONFIGURATION

Standard configuration with engine air intakes.

CRUISE CHARTS

The cruise charts are based on estimates and limited flight test data. Fuel consumption may vary between engines. It is recommended that the operator conduct measurements to be used to adjust the presented data as required. These data do not include the effects of bleed air on fuel consumption. Fuel flow data are applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag, or power available.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 0 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -5°C

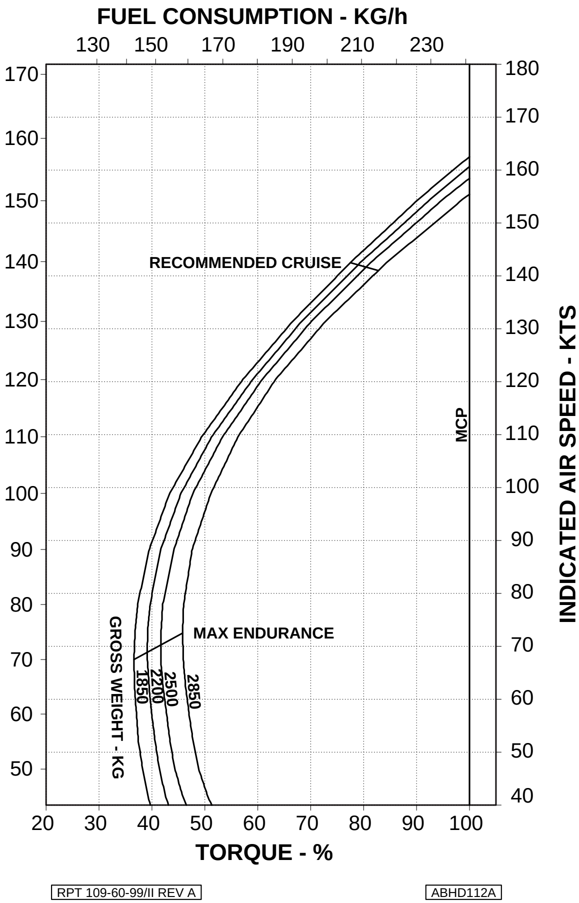


Figure 9-1. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 0 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 15°C

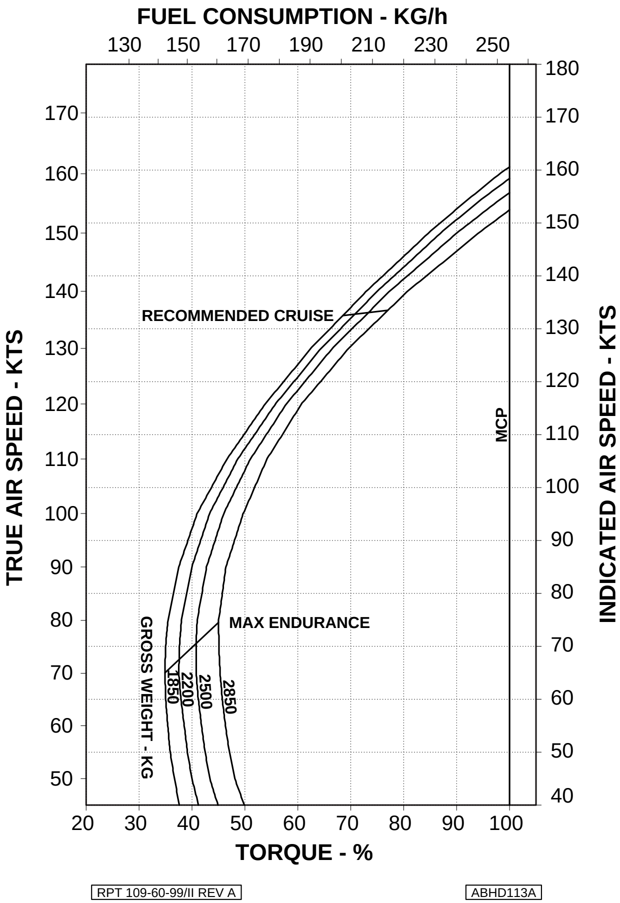


Figure 9-2. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 0 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 35˚C

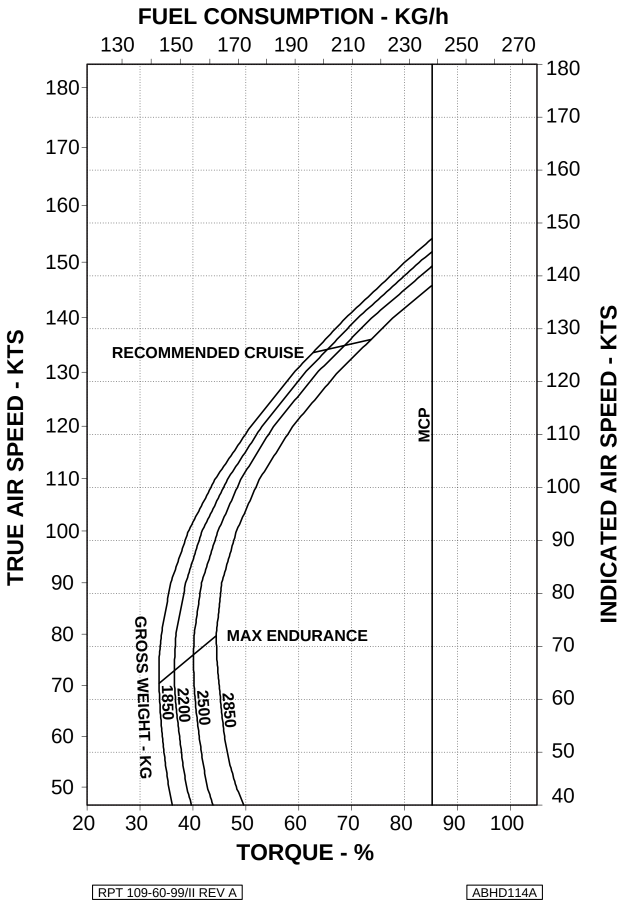


Figure 9-3. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 2000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -9˚C

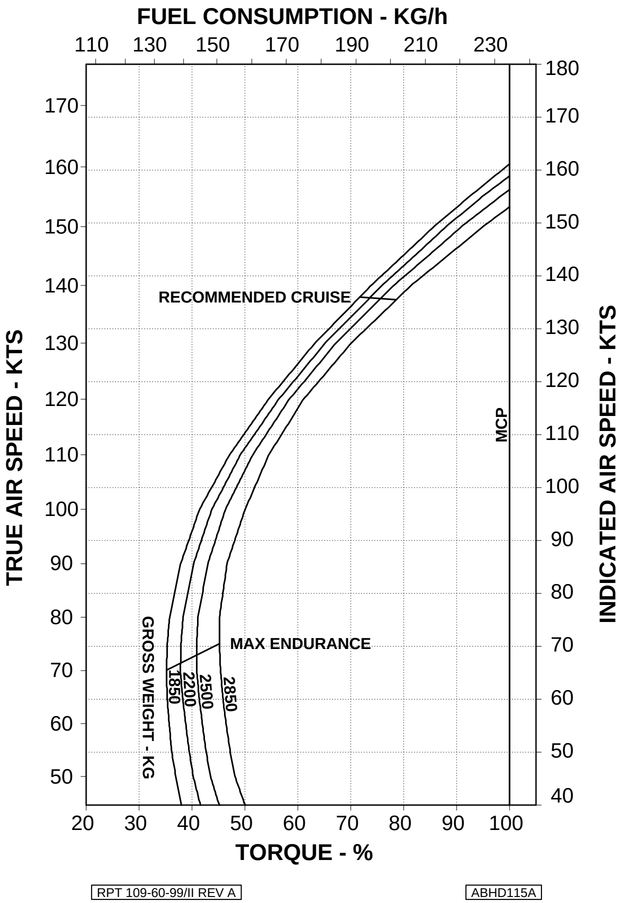


Figure 9-4. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 2000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 11°C

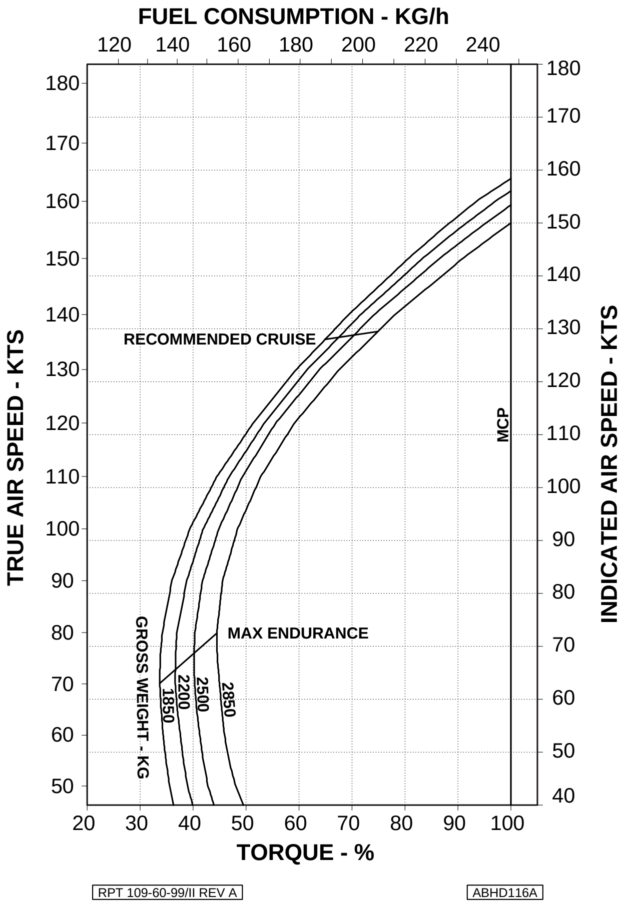


Figure 9-5. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 2000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 31°C

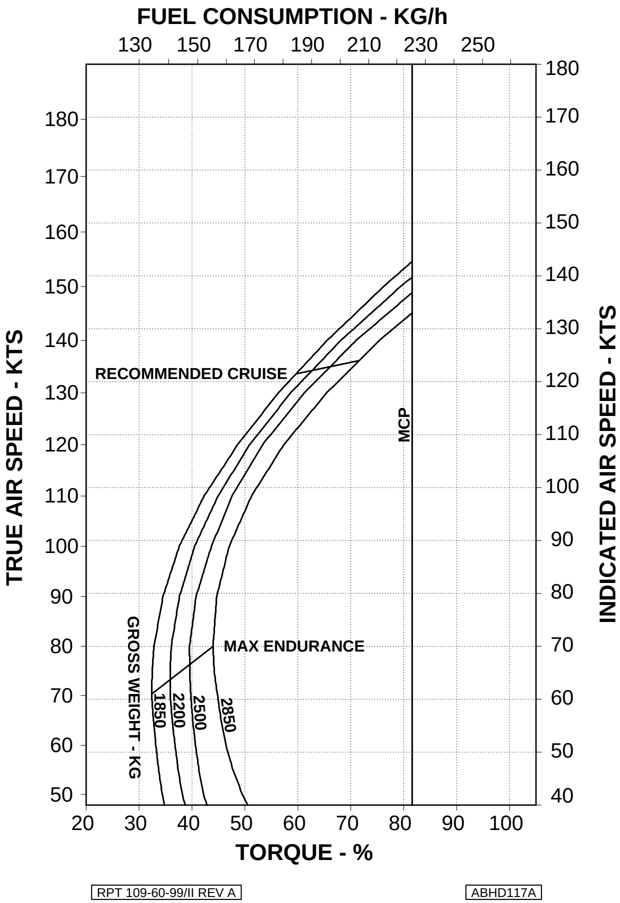


Figure 9-6. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 4000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -13°C

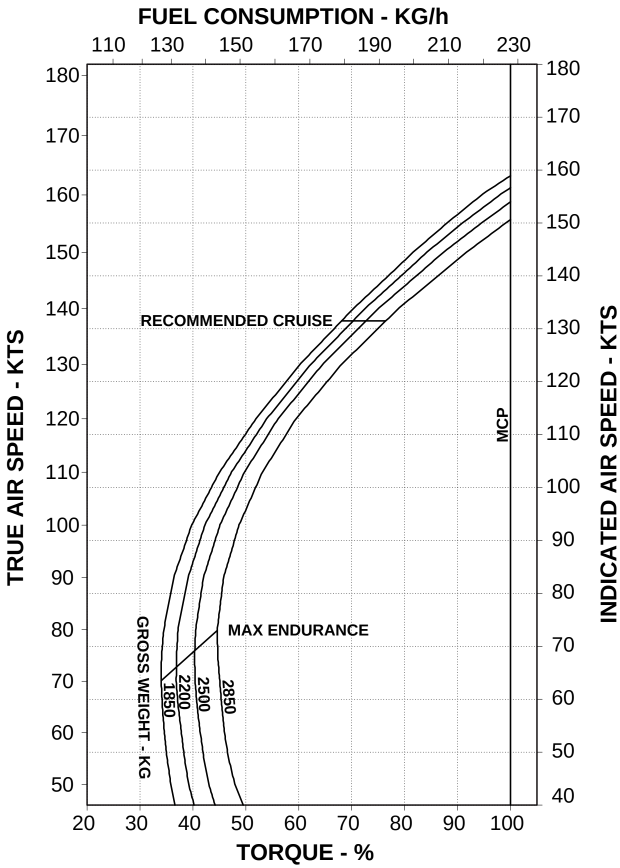


Figure 9-7. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 4000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 7°C

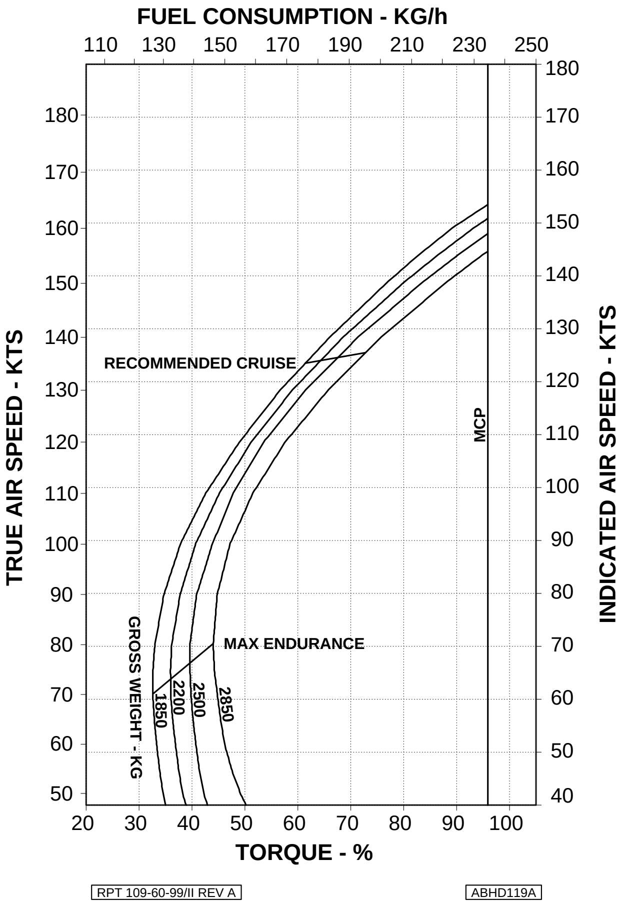


Figure 9-8. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 4000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 27°C

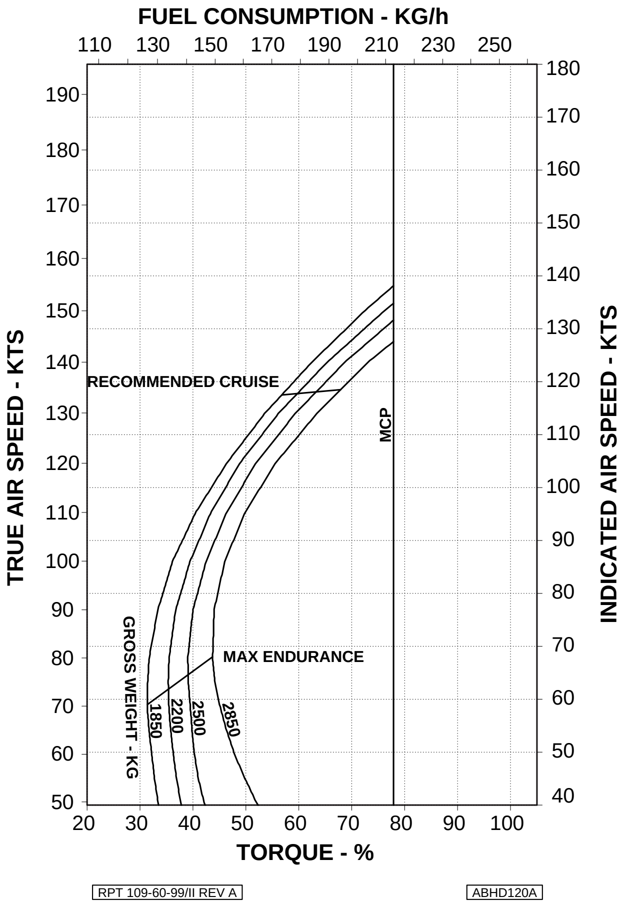


Figure 9-9. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 6000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -17°C

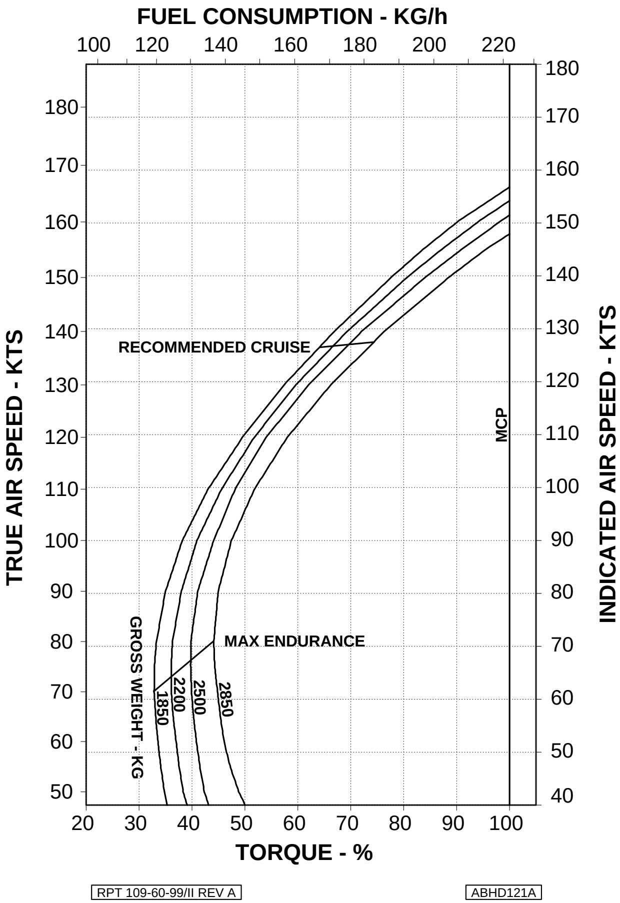


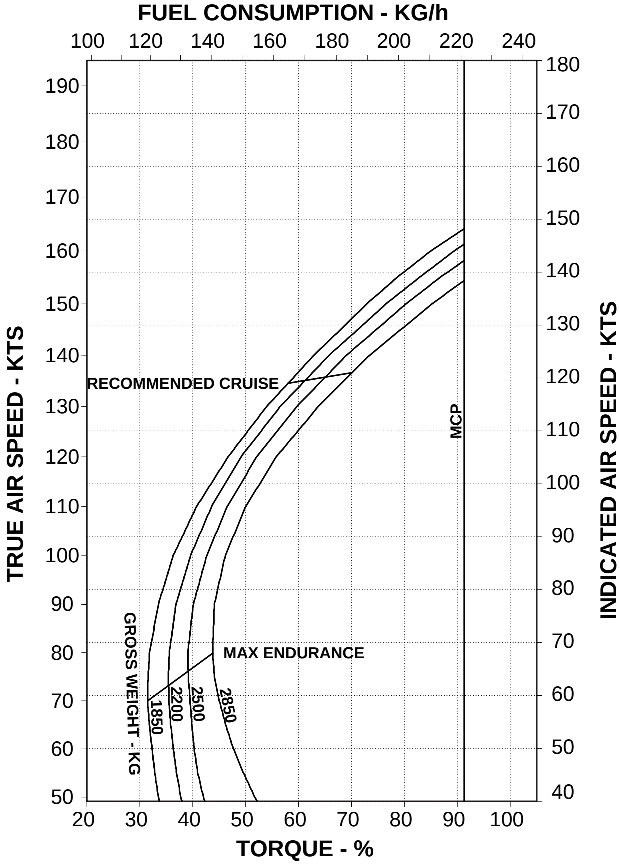
Figure 9-10. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 6000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 3°C



RPT 109-60-99/II REV A

ABHD122A

Figure 9-11. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 6000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 23°C

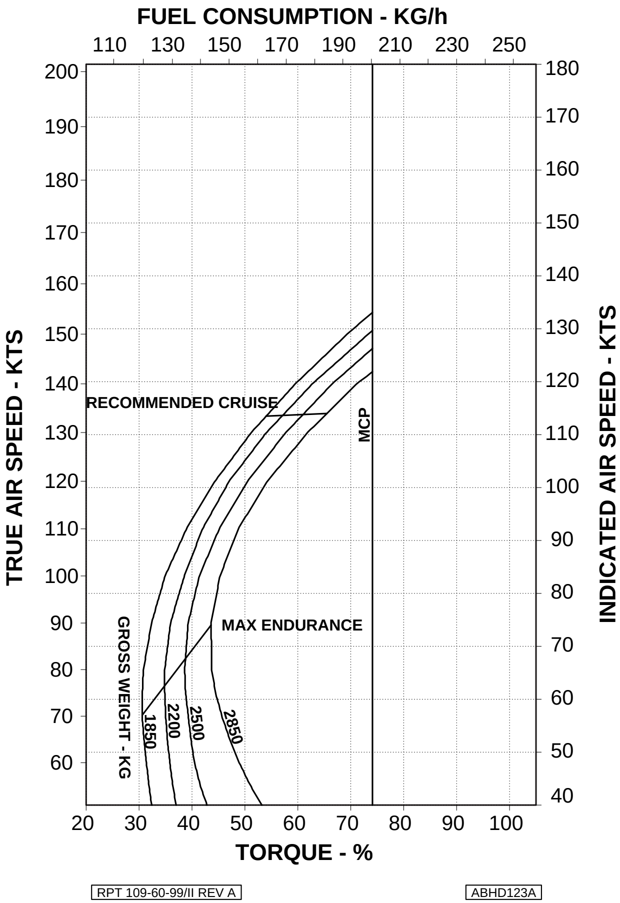


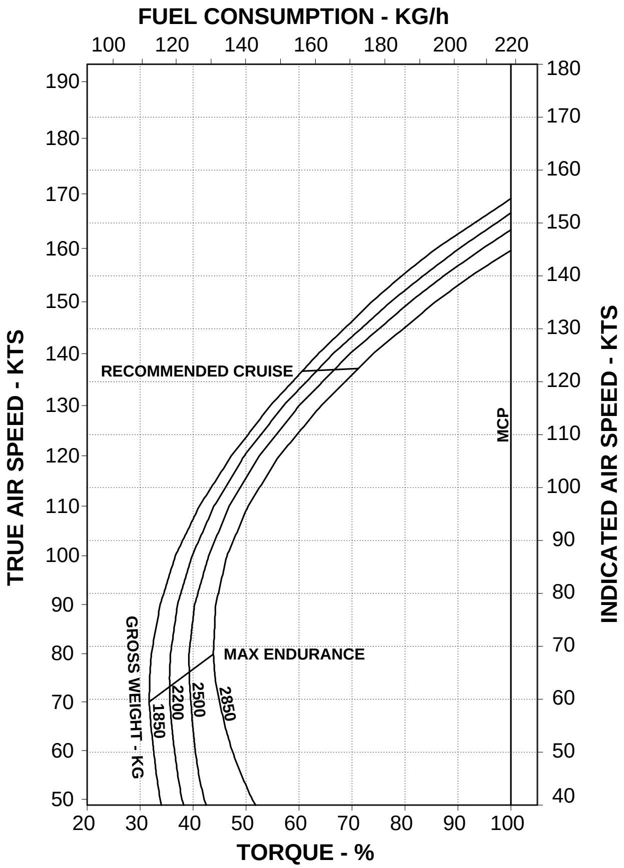
Figure 9-12. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 8000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -21°C



RPT 109-60-99/II REV A

ABHD124A

Figure 9-13. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 8000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -1°C

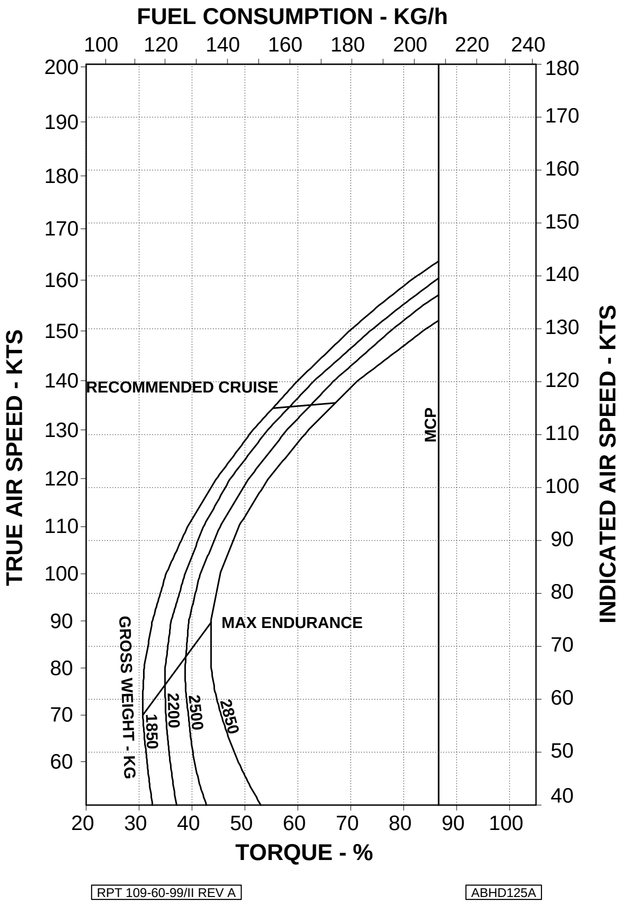


Figure 9-14. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 8000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 19˚C

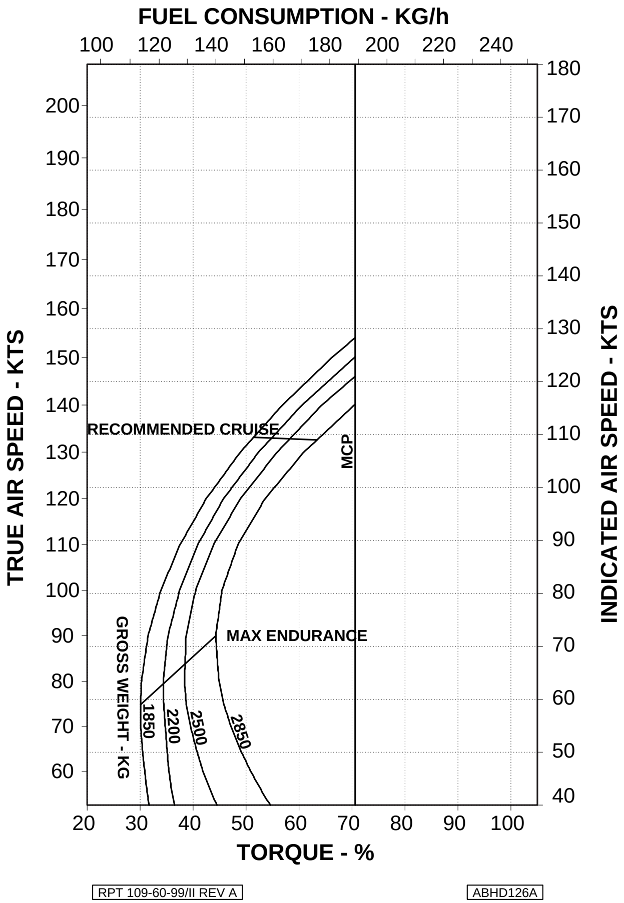


Figure 9-15. Cruise.

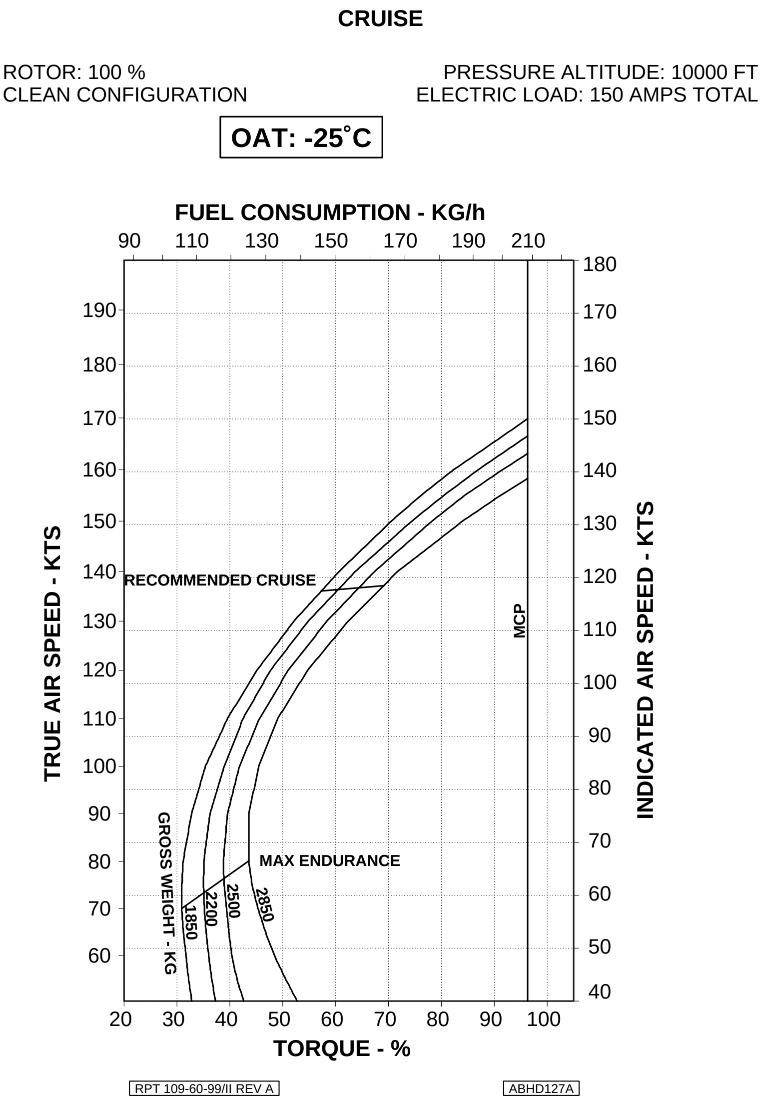


Figure 9-16. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 10000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: -5°C

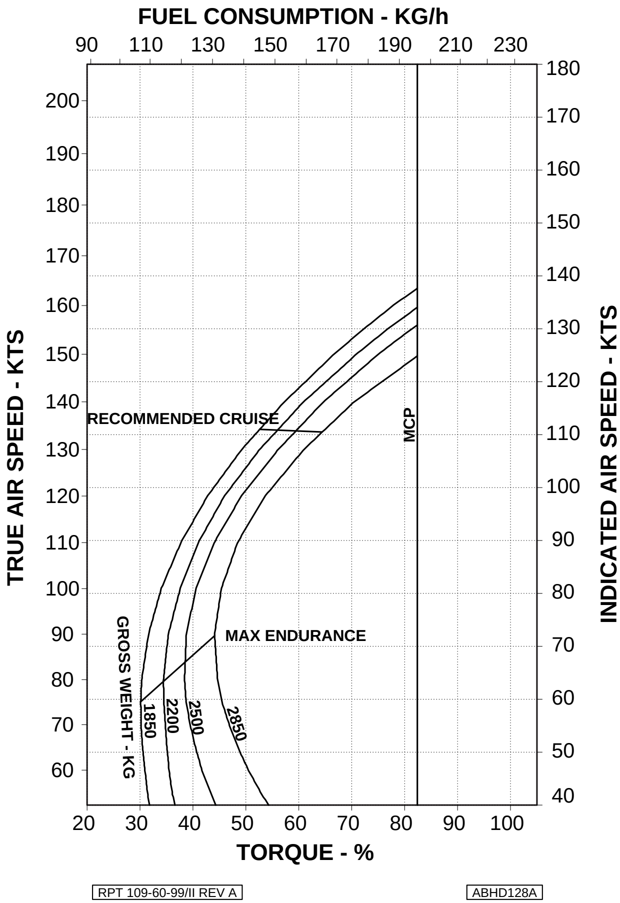


Figure 9-17. Cruise.

CRUISE

ROTOR: 100 %
CLEAN CONFIGURATION

PRESSURE ALTITUDE: 10000 FT
ELECTRIC LOAD: 150 AMPS TOTAL

OAT: 15°C

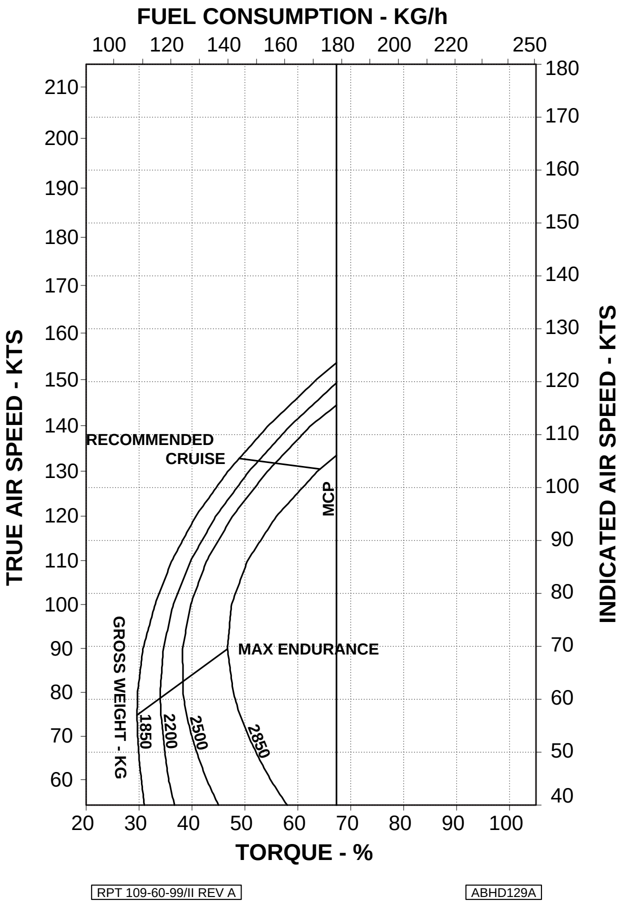


Figure 9-18. Cruise.

HOVERING CEILING - ONE ENGINE INOPERATIVE

The hovering charts are presented for the OEI (one engine inoperative) 2.5 minutes power rating in ground effect and out of ground effect.
The charts are based on flight test data.

HOVERING CEILING IN GROUND EFFECT
ONE ENGINE INOPERATIVE - 2.5 MINUTES POWER

ROTOR: 102 % | WHEEL HEIGHT: 3 FT
ZERO WIND | ELECTRIC LOAD: 150 AMPS TOTAL

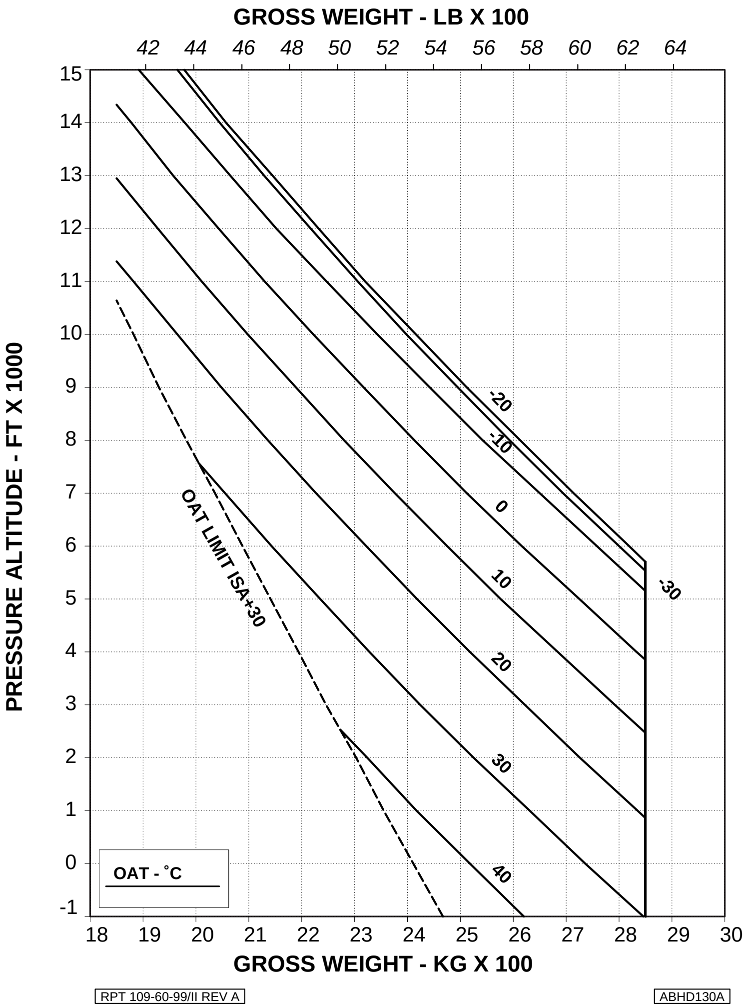


Figure 9-19. Hovering ceiling - IGE - OEI - 2.5 minutes power.

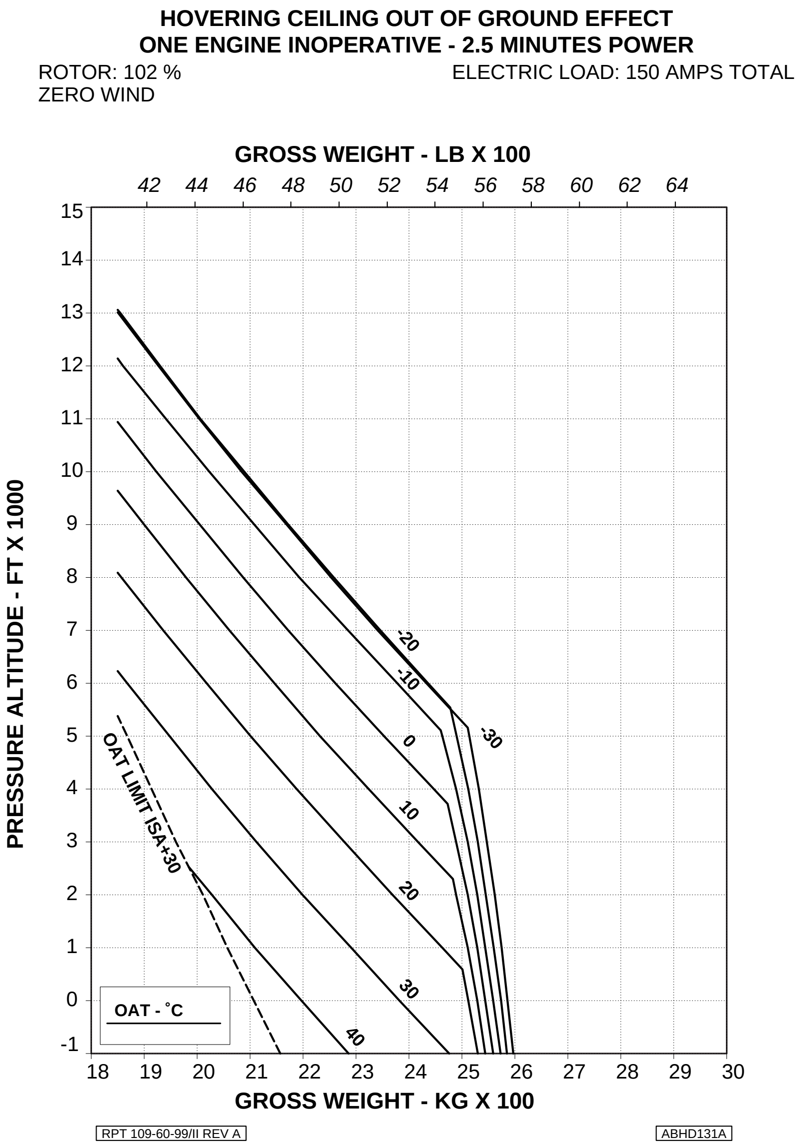


Figure 9-20. Hovering ceiling - OGE - OEI - 2.5 minutes power.